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(54) **COMPOUND, ORGANIC
ELECTROLUMINESCENCE DEVICE AND
ELECTRONIC APPARATUS**

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Related U.S. Application Data

(63) Continuation of application No. 18/891,132, filed on Sep. 20, 2024, which is a continuation of application No. 17/282,347, filed as application No. PCT/JP2019/039846 on Oct. 9, 2019, now Pat. No. 12,202,837, which is a continuation-in-part of application No. 16/593,743, filed on Oct. 4, 2019, now Pat. No. 10,804,474, and a continuation-in-part of application No. 16/557,675, filed on Aug. 30, 2019, now Pat. No. 10,763,444.

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H10K 85/60	(2023.01)
H10K 101/10	(2023.01)

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CPC **C07D 493/00** (2013.01); **H10K 85/40** (2023.02); **H10K 85/622** (2023.02); **H10K 85/623** (2023.02); **H10K 85/626** (2023.02); **H10K 85/633** (2023.02); **H10K 85/636** (2023.02); **H10K 85/6574** (2023.02); **H10K 85/6576** (2023.02); **H10K 50/11** (2023.02); **H10K 2101/10** (2023.02)

(58) **Field of Classification Search**

None

See application file for complete search history.

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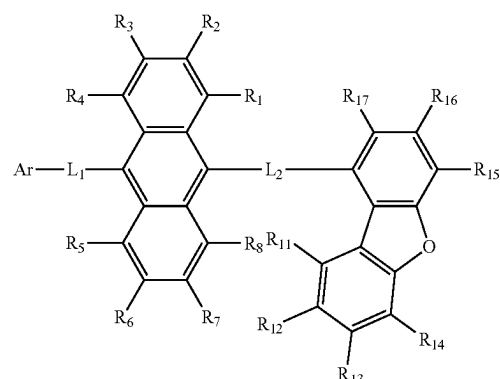
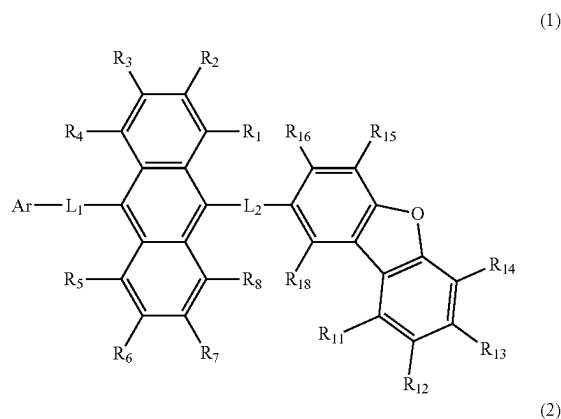
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(57)

ABSTRACT

A compound having structures represented by the following formulas (1) or (2): wherein in the formulas (1) and (2), at least one of R₁ to R₈ is a deuterium atom.



23 Claims, No Drawings

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COMPOUND, ORGANIC ELECTROLUMINESCENCE DEVICE AND ELECTRONIC APPARATUS

CROSS-REFERENCE TO RELATED APPLICATIONS

The present application is a Track One continuation of U.S. patent application Ser. No. 18/891,132, filed on Sep. 20, 2024, which is a continuation of U.S. patent application Ser. No. 17/282,347, filed on Apr. 1, 2021, which claims priority under 35 U.S.C. § 371 to International Patent Application No. PCT/JP2019/039846, filed Oct. 9, 2019, which claims priority to and the benefit of U.S. patent application Ser. No. 16/557,675, filed on Aug. 30, 2019 (issued as U.S. Pat. No. 10,763,444, on Sep. 1, 2020), and U.S. patent application Ser. No. 16/593,743, filed on Oct. 4, 2019 (issued as U.S. Pat. No. 10,804,474, on Oct. 4, 2019), and Japanese Patent Application No. 2019-101578, filed on May 30, 2019, and Japanese Patent Application No. 2019-101675, filed on May 30, 2019, and Japanese Patent Application No. 2018-191224, filed on Oct. 9, 2018, and Japanese Patent Application No. 2018-190838, filed on Oct. 9, 2018. The contents of these applications are hereby incorporated by reference in their entireties.

TECHNICAL FIELD

The invention relates to a new compound, an organic electroluminescence device and an electronic apparatus.

BACKGROUND ART

When a voltage is applied to an organic electroluminescence device (hereinafter may be referred to as an organic EL device), holes are injected to an emitting layer from an anode and electrons are injected to an emitting layer from a cathode. In the emitting layer, injected holes and electrons are re-combined and excitons are formed.

Although materials for an organic EL device are being improved gradually to increase the performances of the organic EL device (for example, Patent Documents 1 to 9), high performances are further offered. In particular, improvement in lifetime of an organic EL device is an important task relating to a lifetime of commercial products provided with the organic EL device, and thus a material enabling to realize long-lifetime organic EL device is required.

RELATED ART DOCUMENTS

Patent Documents

Patent Document 1: WO2017/023021
 Patent Document 2: WO2013/106041
 Patent Document 3: Publication of US Patent Application No. 20170025608
 Patent Document 4: WO2017/023021
 Patent Document 5: JP-A-2006-151844
 Patent Document 6: WO2010/099534
 Patent Document 7: WO2010/135395
 Patent Document 8: WO2010/071362
 Patent Document 9: WO2010/066830

SUMMARY OF THE INVENTION

An object of the invention is to provide a compound capable of producing an organic EL device having along

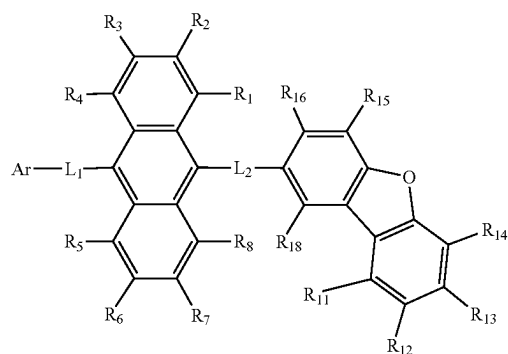
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lifetime, an organic EL device having along lifetime and an electronic apparatus provided with the organic EL device.

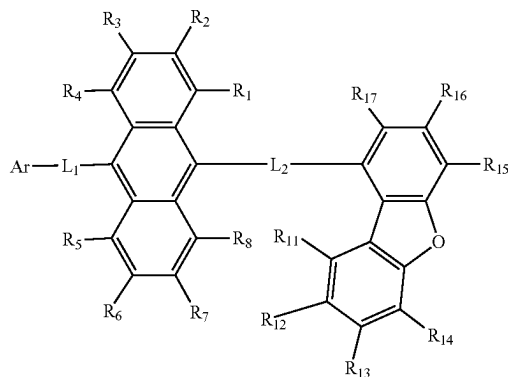
As a result of extensive studies, the inventors have found that an organic EL device having along lifetime can be obtained by using compounds having a specific structure represented by formula (1) or (2), and they have achieved the invention.

According to the invention, the following compound, organic EL device and electric apparatus can be provided.

1. A compound having structures represented by the following formulas (1) or (2):



(1)



(2)

wherein in the formulas (1) and (2),

R_1 to R_6 and R_{11} to R_{14} are independently

a hydrogen atom,

a substituted or unsubstituted alkyl group having 1 to 50 carbon atoms,

a substituted or unsubstituted alkenyl group having 2 to 50 carbon atoms,

a substituted or unsubstituted alkynyl group having 2 to 50 carbon atoms,

a substituted or unsubstituted cycloalkyl group having 3 to 50 ring carbon atoms,

$-\text{Si}(\text{R}_{901})(\text{R}_{902})(\text{R}_{903})$,

$-\text{O}(\text{R}_{904})$,

$-\text{S}(\text{R}_{905})$,

$-\text{N}(\text{R}_{906})(\text{R}_{907})$,

a halogen atom, a cyano group, a nitro group,

a substituted or unsubstituted aryl group having 6 to 50 ring carbon atoms, or

a substituted or unsubstituted monovalent heterocyclic group having 5 to 50 ring atoms;

R_{15} to R_{18} are hydrogen atoms;

R_{901} to R_{907} are independently

- a hydrogen atom,
 - a substituted or unsubstituted alkyl group having 1 to 50 carbon atoms,
 - a substituted or unsubstituted cycloalkyl group having 3 to 50 ring carbon atoms,
 - a substituted or unsubstituted aryl group having 6 to 50 ring carbon atoms, or
 - a substituted or unsubstituted monovalent heterocyclic group having 5 to 50 ring atoms;
when two or more of R₉₀₁ to R₉₀₇ exist, two or more of R₉₀₁ to R₉₀₇ may be the same with or different from each other;
 - at least one of R₉₀₁ to R₉₀₇ is a deuterium atom;
 - two or more adjacent groups of R₁ to R₄; two or more adjacent groups of R₅ to R₆ and two or more adjacent groups of R₁₁ to R₁₄ do not form a ring;
 - L₁ and L₂ are independently
 - a single bond,
 - a substituted or unsubstituted arylene group having 6 to 30 ring carbon atoms, or
 - a substituted or unsubstituted divalent heterocyclic group having 5 to 30 ring atoms; Ar is
 - a substituted or unsubstituted aryl group having 6 to 50 ring carbon atoms, or
 - a substituted or unsubstituted monovalent heterocyclic group having 5 to 50 ring atoms.
2. An organic electroluminescence device comprising:
 - a cathode, an anode, and one or two or more organic layer disposed between the cathode and the anode, wherein at least one organic layer comprises the compound according to the above 1.
 3. An electronic apparatus provided with the organic electroluminescence device according to the above 2.
- According to the invention, a compound capable of producing an organic EL device having along lifetime, an organic EL device having along lifetime, and an electronic apparatus provided with the organic EL device can be provided.

MODE FOR CARRYING OUT THE INVENTION

Definition

In this specification, a hydrogen atom means an atom including isotopes different in the number of neutrons, namely, a protium, a deuterium and a tritium.

In this specification, to a bondable position in which a symbol such as "R", or "D" representing a deuterium atom is not specified in a chemical formula, a hydrogen atom, that is, alight hydrogen atom, a deuterium atom, or a tritium atom is bonded thereto.

In this specification, a term "ring carbon atoms" represents the number of carbon atoms among atoms forming a subject ring itself of a compound having a structure in which atoms are bonded in a ring form (for example, a monocyclic compound, a fused ring compound, a cross-linked compound, a carbocyclic compound or a heterocyclic compound). When the subject ring is substituted by a substituent, the carbon contained in the substituent is not included in the number of ring carbon atoms. The same shall apply to the "ring carbon atoms" described below, unless otherwise noted. For example, abenzene ring has 6 ring carbon atoms, a naphthalene ring has 10 ring carbon atoms, a pyridine ring has 5 ring carbon atoms, and a furan ring has 4 ring carbon atoms. Further, for example, a 9,9-diphenylfluorenyl group has 13 ring carbon atoms, and a 9,9'-spirobifluorenyl group has 25 ring carbon atoms.

Further, when the benzene ring or the naphthalene ring is substituted by an alkyl group as a substituent, for example, the number of carbon atoms of the alkyl group is not included in the ring carbon atoms.

In this specification, a term "ring atoms" represents the number of atoms forming a subject ring itself of a compound having a structure in which atoms are bonded in a ring form (for example, a monocycle, a fused ring and a ring assembly) (for example, a monocyclic compound, a fused ring compound, a cross-linked compound, a carbocyclic compound or a heterocyclic compound). The term "ring atoms" does not include atoms which do not form the ring (for example, a hydrogen atom which terminates a bond of the atoms forming the ring) or atoms contained in a substituent when the ring is substituted by the substituent. The same shall apply to the "ring atoms" described below, unless otherwise noted. For example, a pyridine ring has 6 ring atoms, a quinazoline ring has 10 ring atoms, and a furan ring has 5 ring atoms. A hydrogen atom bonded with a carbon atom of the pyridine ring or the quinazoline ring or an atom forming the substituent is not included in the number of the ring atoms.

In this specification, a term "XX to YY carbon atoms" in an expression of "substituted or unsubstituted ZZ group including XX to YY carbon atoms" represents the number of carbon atoms when the ZZ group is unsubstituted. The number of carbon atoms of a substituent when the ZZ group is substituted is not included. Here, "YY" is larger than "XX", and "XX" and "YY" each mean an integer of 1 or more.

In this specification, a term "XX to YY atoms" in an expression of "substituted or unsubstituted ZZ group including XX to YY atoms" represents the number of atoms when the ZZ group is unsubstituted. The number of atoms of a substituent when the group is substituted is not included. Here, "YY" is larger than "XX", and "XX" and "YY" each mean an integer of 1 or more.

A term "unsubstituted" in the case of "substituted or unsubstituted ZZ group" means that the ZZ group is not substituted by a substituent, and a hydrogen atom is bonded therewith. Alternately, a term "substituted" in the case of "substituted or unsubstituted ZZ group" means that one or more hydrogen atoms in the ZZ group are substituted by a substituent. Similarly, a term "substituted" in the case of "BB group substituted by an AA group" means that one or more hydrogen atoms in the BB group are substituted by the AA group.

Hereinafter, the substituent described herein will be described.

The number of the ring carbon atoms of the "unsubstituted aryl group" described herein is 6 to 50, preferably 6 to 30, and more preferably 6 to 18, unless otherwise specified.

The number of the ring carbon atoms of the "unsubstituted heterocyclic group" described herein is 5 to 50, preferably 5 to 30, and more preferably 5 to 18, unless otherwise specified.

The number of the carbon atoms of the "unsubstituted alkyl group" described herein is 1 to 50, preferably 1 to 20, and more preferably 1 to 6, unless otherwise specified.

The number of the carbon atoms of the "unsubstituted alkenyl group" described herein is 2 to 50, preferably 2 to 20, and more preferably 2 to 6, unless otherwise specified.

The number of the carbon atoms of the "unsubstituted alkynyl group" described herein is 2 to 50, preferably 2 to 20, and more preferably 2 to 6, unless otherwise specified.

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The number of the ring carbon atoms of the “unsubstituted cycloalkyl group” described herein is 3 to 50, preferably 3 to 20, and more preferably 3 to 6, unless otherwise specified.

The number of the ring carbon atoms of the “unsubstituted arylene group” described herein is 6 to 50, preferably 6 to 30, and more preferably 6 to 18, unless otherwise specified.

The number of the ring atoms of the “unsubstituted divalent heterocyclic group” described herein is 5 to 50, preferably 5 to 30, and more preferably 5 to 18, unless otherwise specified.

The number of the carbon atoms of the “unsubstituted alkylene group” described herein is 1 to 50, preferably 1 to 20, and more preferably 1 to 6, unless otherwise specified.

Specific examples (specific example group G1) of the “substituted or unsubstituted aryl group” described herein include an unsubstituted aryl group and a substituted aryl group described below. (Here, a term “unsubstituted aryl group” refers to a case where the “substituted or unsubstituted aryl group” is the “unsubstituted aryl group,” and a term “substituted aryl group” refers to a case where the “substituted or unsubstituted aryl group” is the “substituted aryl group”. Hereinafter, a case of merely “aryl group” includes both the “unsubstituted aryl group” and the “substituted aryl group”.

The “substituted aryl group” refers to a case where the “unsubstituted aryl group” has a substituent, and specific examples thereof include a group in which the “unsubstituted aryl group” has the substituent, and a substituted aryl group described below. It should be noted that examples of the “unsubstituted aryl group” and examples of the “substituted aryl group” listed herein are only one example, and the “substituted aryl group” described herein also includes a group in which a group in which “unsubstituted aryl group” has a substituent further has a substituent, and a group in which “substituted aryl group” further has a substituent, and the like.

An unsubstituted aryl group:

a phenyl group,
a p-biphenyl group,
a m-biphenyl group,
an o-biphenyl group,
a p-terphenyl-4-yl group,
a p-terphenyl-3-yl group,
a p-terphenyl-2-yl group,
a m-terphenyl-4-yl group,
a m-terphenyl-3-yl group,
a m-terphenyl-2-yl group,
an o-terphenyl-4-yl group,
an o-terphenyl-3-yl group,
an o-terphenyl-2-yl group,
a 1-naphthyl group,
a 2-naphthyl group,
an anthryl group,
a benzanthryl group,
a phenanthryl group,
a benzophenanthryl group,
a phenalenyl group,
a pyrenyl group,
a chrysenyl group,
a benzo-chrysenyl group,
a triphenylenyl group,
a benzotriphenylenyl group,
a tetraceny group,
a pentaceny group,
a fluorenyl group,

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a 9,9-spirofluorenyl group,
a benzofluorenyl group,
a dibenzofluorenyl group,
a fluorantheny group,
a benzofluorantheny group, and
a perylenyl group.

A substituted aryl group:

an o-tolyl group,
a m-tolyl group,
a p-tolyl group,
a p-xylyl group,
a m-xylyl group,
an o-xylyl group,
a p-isopropyl phenyl group,
a m-isopropylphenyl group,
an o-isopropyl phenyl group,
a p-t-butylphenyl group,
a m-t-butylphenyl group,
an o-t-butylphenyl group,
a 3,4,5-trimethylphenyl group,
a 9,9-dimethylfluorenyl group,
a 9,9-diphenylfluorenyl group,
a 9,9-di(4-methylphenyl)fluorenyl group,
a 9,9-di(4-isopropylphenyl)fluorenyl group,
a 9,9-di(4-t-butylphenyl)fluorenyl group,
a cyanophenyl group,
a triphenylsilylphenyl group,
a trimethylsilylphenyl group,
a phenyl-naphthyl group, and
a naphthylphenyl group.

The “heterocyclic group” described herein is a ring group including at least one hetero atom in the ring atom. Specific examples of the hetero atom include a nitrogen atom, an oxygen atom, a sulfur atom, a silicon atom, a phosphorus atom and a boron atom.

The “heterocyclic group” described herein may be a monocyclic group, or a fused ring group.

The “heterocyclic group” described herein may be an aromatic heterocyclic group, or an aliphatic heterocyclic group.

Specific examples (specific example group G2) of the “substituted or unsubstituted heterocyclic group” include an unsubstituted heterocyclic group and a substituted heterocyclic group described below. (Here, the unsubstituted heterocyclic group refers to a case where the “substituted or unsubstituted heterocyclic group” is the “unsubstituted heterocyclic group,” and the substituted heterocyclic group refers to a case where the “substituted or unsubstituted heterocyclic group” is the “substituted heterocyclic group”. Hereinafter, the case of merely “heterocyclic group” includes both the “unsubstituted heterocyclic group” and the “substituted heterocyclic group”.

The “substituted heterocyclic group” refers to a case where the “unsubstituted heterocyclic group” has a substituent, and specific examples thereof include a group in which the “unsubstituted heterocyclic group” has a substituent, and a substituted heterocyclic group described below. It should be noted that examples of the “unsubstituted heterocyclic group” and examples of the “substituted heterocyclic group” listed herein are merely one example, and the “substituted heterocyclic group” described herein also includes a group in which “unsubstituted heterocyclic group” which has a substituent further has a substituent, and a group in which “substituted heterocyclic group” further has a substituent, and the like.

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An unsubstituted heterocyclic group including a nitrogen atom:

a pyrrolyl group,
 an imidazolyl group,
 a pyrazolyl group,
 a triazolyl group,
 a tetrazolyl group,
 an oxazolyl group,
 an isoxazolyl group,
 an oxadiazolyl group,
 a thiazolyl group,
 an isothiazolyl group,
 a thiadiazolyl group,
 a pyridyl group,
 a pyridazinyl group,
 a pyrimidinyl group,
 a pyrazinyl group,
 a triazinyl group,
 an indolyl group,
 an isoindolyl group,
 an indolizinyl group,
 a quinolizinyl group,
 a quinolyl group,
 an isoquinolyl group,
 a cinnolyl group,
 a phthalazinyl group,
 a quinazolinyl group,
 a quinoxalinyl group,
 a benzimidazolyl group,
 an indazolyl group,
 a phenanthrolinyl group,
 a phenanthridinyl group,
 an acidinyl group,
 a phenazinyl group,
 a carbazolyl group,
 a benzocarbazolyl group,
 a morpholino group,
 a phenoxazinyl group,
 a phenothiazinyl group,
 an azacarbazolyl group, and
 a diazacarbazolyl group.

An unsubstituted heterocyclic group including an oxygen atom:

a furyl group,
 an oxazolyl group,
 an isoxazolyl group,
 an oxadiazolyl group,
 a xanthenyl group,
 a benzofuranyl group,
 an isobenzofuranyl group,
 a dibenzofuranyl group,
 a naphthobenzofuranyl group,
 a benzoxazolyl group,
 a benzisoxazolyl group,
 a phenoxazinyl group,
 a morpholino group,
 a dinaphthofuranyl group,
 an azadibenzofuranyl group,
 a diazadibenzofuranyl group,
 an azanaphthobenzofuranyl group, and
 a diazanaphthobenzofuranyl group.

An unsubstituted heterocyclic group including a sulfur atom:

a thienyl group,
 a thiazolyl group,
 an isothiazolyl group,
 a thiadiazolyl group,
 a benzothiophenyl group,
 an isobenzothiophenyl group,

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a dibenzothiophenyl group,
 a naphthobenzothiophenyl group,
 a benzothiazolyl group,
 a benzisothiazolyl group,
 a phenothiazinyl group,
 a dinaphthothiophenyl group,
 an azadibenzothiophenyl group,
 a diazadibenzothiophenyl group,
 an azanaphthobenzothiophenyl group, and
 a diazanaphthobenzothiophenyl group.

A substituted heterocyclic group including a nitrogen atom:

a (9-phenyl)carbazolyl group,
 a (9-biphenyl)carbazolyl group,
 a (9-phenyl)phenylcarbazolyl group,
 a (9-naphthyl)carbazolyl group,
 a diphenylcarbazol-9-yl group,
 a phenylcarbazol-9-yl group,
 a methylbenzimidazolyl group,
 an ethylbenzimidazolyl group,
 a phenyltriazinyl group,
 a biphenyltriazinyl group,
 a diphenyltriazinyl group,
 a phenyquinazolinyl group, and
 a biphenylquinazolinyl group.

A substituted heterocyclic group including an oxygen atom:

a phenyldibenzofuranyl group,
 a methylbenzofuranyl group,
 a t-butylbenzofuranyl group, and
 a monovalent residue of spiro[9H-xanthene-9,9-[9H]fluorene].

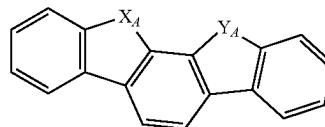
A substituted heterocyclic group including a sulfur atom:

a phenyldibenzothiophenyl group,
 a methylbenzothiophenyl group,
 a t-butylbenzothiophenyl group, and
 a monovalent residue of spiro[9H-thioxanthene-9,9-[9H]fluorene].

A monovalent group derived from the following unsubstituted heterocyclic ring containing at least one of a nitrogen atom, an oxygen atom and a sulfur atom by removal of one hydrogen atom bonded to the ring atoms thereof, and a monovalent group in which a monovalent group derived from the following unsubstituted heterocyclic ring has a substituent by removal of one hydrogen atom bonded to the ring atoms thereof:

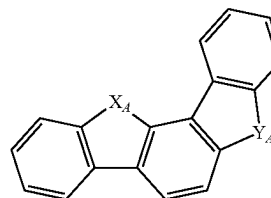
(XY-1)

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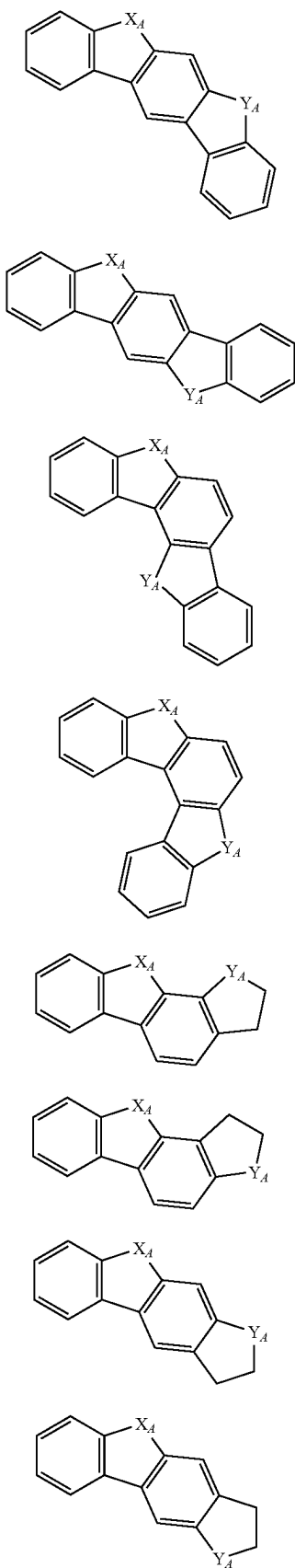
(XY-2)

65

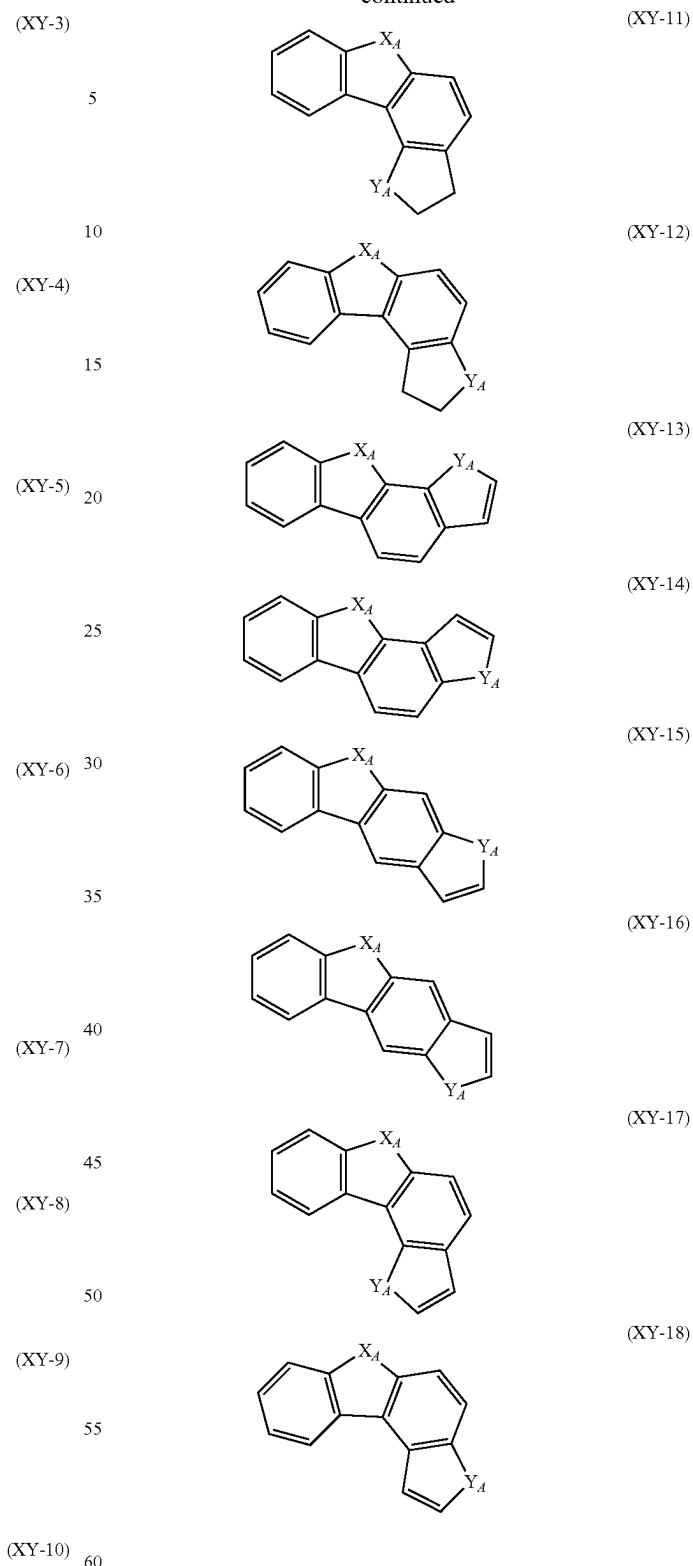


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-continued

**10**

-continued



In the formulas (XY-1) to (XY-18), X_A and Y_A are independently an oxygen atom, a sulfur atom, NH or CH_2 . However, at least one of X_A and Y_A is an oxygen atom, a sulfur atom or NH.

The heterocyclic ring represented by the formulas (XY-1) to (XY-18) becomes a monovalent heterocyclic group including a bond at an arbitrary position.

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An expression “the monovalent group derived from the unsubstituted heterocyclic ring represented by the formulas (XY-1) to (XY-18) has a substituent” refers to a case where the hydrogen atom bonded with the carbon atom which constitutes a skeleton of the formulas is substituted by a substituent, or a state in which X_A or Y_A is NH or CH_2 , and the hydrogen atom in the NH or CH_2 is replaced with a substituent.

Specific examples (specific example group G3) of the “substituted or unsubstituted alkyl group” include an unsubstituted alkyl group and a substituted alkyl group described below. (Here, the unsubstituted alkyl group refers to a case where the “substituted or unsubstituted alkyl group” is the “unsubstituted alkyl group,” and the substituted alkyl group refers to a case where the “substituted or unsubstituted alkyl group” is the “substituted alkyl group”). Hereinafter, the case of merely “alkyl group” includes both the “unsubstituted alkyl group” and the “substituted alkyl group”.

The “substituted alkyl group” refers to a case where the “unsubstituted alkyl group” has a substituent, and specific examples thereof include a group in which the “unsubstituted alkyl group” has a substituent, and a substituted alkyl group described below. It should be noted that examples of the “unsubstituted alkyl group” and examples of the “substituted alkyl group” listed herein are merely one example, and the “substituted alkyl group” described herein also includes a group in which “unsubstituted alkyl group” has a substituent further has a substituent, a group in which “substituted alkyl group” further has a substituent, and the like.

An unsubstituted alkyl group:

a methyl group,
an ethyl group,
a n-propyl group,
an isopropyl group,
a n-butyl group,
an isobutyl group,
a s-butyl group, and
a t-butyl group.

A substituted alkyl group:

a heptafluoropropyl group (including an isomer),
a pentafluoroethyl group,
a 2,2,2-trifluoroethyl group, and
a trifluoromethyl group.

Specific examples (specific example group G4) of the “substituted or unsubstituted alkenyl group” include an unsubstituted alkenyl group and a substituted alkenyl group described below. (Here, the unsubstituted alkenyl group refers to a case where the “substituted or unsubstituted alkenyl group” is the “unsubstituted alkenyl group,” and the substituted alkenyl group refers to a case where the “substituted or unsubstituted alkenyl group” is the “substituted alkenyl group”). Hereinafter, the case of merely “alkenyl group” includes both the “unsubstituted alkenyl group” and the “substituted alkenyl group”.

The “substituted alkenyl group” refers to a case where the “unsubstituted alkenyl group” has a substituent, and specific examples thereof include a group in which the “unsubstituted alkenyl group” has a substituent, and a substituted alkenyl group described below. It should be noted that examples of the “unsubstituted alkenyl group” and examples of the “substituted alkenyl group” listed herein are merely one example, and the “substituted alkenyl group” described herein also includes a group in which “unsubstituted alkenyl group” has a substituent further has a substituent, a group in which “substituted alkenyl group” further has a substituent, and the like.

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An unsubstituted alkenyl group and a substituted alkenyl group:

a vinyl group,
an allyl group,
a 1-butenyl group,
a 2-butenyl group,
a 3-butenyl group,
a 1,3-butanedieryl group,
a 1-methylvinyl group,
a 1-methylallyl group,
a 1,1-dimethylallyl group,
a 2-methylallyl group, and
a 1,2-dimethylallyl group.

Specific examples (specific example group G5) of the “substituted or unsubstituted alkynyl group” include an unsubstituted alkynyl group described below. (Here, the unsubstituted alkynyl group refers to a case where the “substituted or unsubstituted alkynyl group” is the “unsubstituted alkynyl group”). Hereinafter, a case of merely “alkynyl group” includes both the “unsubstituted alkynyl group” and the “substituted alkynyl group”.

The “substituted alkynyl group” refers to a case where the “unsubstituted alkynyl group” has a substituent, and specific examples thereof include a group in which the “unsubstituted alkynyl group” described below has a substituent.

An unsubstituted alkynyl group:

an ethynyl group.

Specific examples (specific example group G6) of the “substituted or unsubstituted cycloalkyl group” described herein include an unsubstituted cycloalkyl group and a substituted cycloalkyl group described below. (Here, the unsubstituted cycloalkyl group refers to a case where the “substituted or unsubstituted cycloalkyl group” is the “unsubstituted cycloalkyl group,” and the substituted cycloalkyl group refers to a case where the “substituted or unsubstituted cycloalkyl group” is the “substituted cycloalkyl group”). Hereinafter, a case of merely “cycloalkyl group” includes both the “unsubstituted cycloalkyl group” and the “substituted cycloalkyl group”.

The “substituted cycloalkyl group” refers to a case where the “unsubstituted cycloalkyl group” is the substituent, and specific examples thereof include a group in which the “unsubstituted cycloalkyl group” has a substituent, and a substituted cycloalkyl group described below. It should be noted that examples of the “unsubstituted cycloalkyl group” and examples of the “substituted cycloalkyl group” listed herein are merely one example, and the “substituted cycloalkyl group” described herein also includes a group in which “unsubstituted cycloalkyl group” has a substituent further has a substituent, a group in which “substituted cycloalkyl group” further has a substituent, and the like.

An unsubstituted aliphatic ring group:

a cyclopropyl group,
a cyclobutyl group,
a cyclopentyl group,
a cyclohexyl group,
a 1-adamantyl group,
a 2-adamantyl group,
a 1-norbornyl group, and
a 2-norbornyl group.

A substituted cycloalkyl group:

a 4-methylcyclohexyl group.

Specific examples (specific example group G7) of the group represented by $—Si(R901)(R902)(R903)$ described herein include

$—Si(G1)(G1)(G1)$,
 $—Si(G1)(G2)(G2)$,

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—Si(G1)(G1)(G2),
 —Si(G2)(G2)(G2),
 —Si(G3)(G3)(G3),
 —Si(G5)(G5)(G5) and
 —Si(G6)(G6)(G6).

In which,

G1 is the “aryl group” described in the specific example group G1.

G2 is the “heterocyclic group” described in the specific example group G2.

G3 is the “alkyl group” described in the specific example group G3.

G5 is the “alkynyl group” described in the specific example group G5.

G6 is the “cycloalkyl group” described in the specific example group G6.

Specific examples (specific example group G8) of the group represented by —O—(R904) described herein include

—O(G1),
 —O(G2),
 —O(G3) and
 —O(G6).

In which,

G1 is the “aryl group” described in the specific example group G1.

G2 is the “heterocyclic group” described in the specific example group G2.

G3 is the “alkyl group” described in the specific example group G3.

G6 is the “cycloalkyl group” described in the specific example group G6.

Specific examples (specific example group G9) of the group represented by —S—(R905) described herein include

—S(G1),
 —S(G2),
 —S(G3) and
 —S(G6).

In which,

G1 is the “aryl group” described in the specific example group G1.

G2 is the “heterocycle group” described in the specific example group G2.

G3 is the “alkyl group” described in the specific example group G3.

G6 is the “cycloalkyl group” described in the specific example group G6.

Specific examples (specific example group G10) of the group represented by —N(R₉₀₆)(R₉₀₇) described herein include

—N(G1)(G1),
 —N(G2)(G2),
 —N(G1)(G2),
 —N(G3)(G3) and
 —N(G6)(G6).

In which,

G1 is the “aryl group” described in the specific example group G1.

G2 is the “heterocycle group” described in the specific example group G2.

G3 is the “alkyl group” described in the specific example group G3.

G6 is the “cycloalkyl group” described in the specific example group G6.

Specific examples (specific example group G11) of the “halogen atom” described herein include a fluorine atom, a chlorine atom, a bromine atom and an iodine atom.

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Specific examples of the “alkoxy group” described herein include a group represented by —O(G3), where G3 is the “alkyl group” described in the specific example group G3. The number of carbon atoms of the “unsubstituted alkoxy group” are 1 to 50, preferably 1 to 30, and more preferably 1 to 18, unless otherwise specified.

Specific examples of the “alkylthiol group” described herein include a group represented by —S(G3), where G3 is the “alkyl group” described in the specific example group G3. The number of carbon atoms of the “unsubstituted alkylthio group” are 1 to 50, preferably 1 to 30, and more preferably 1 to 18, unless otherwise specified.

Specific examples of the “aryloxy group” described herein include a group represented by —O(G1), where G1 is the “aryl group” described in the specific example group G1. The number of ring carbon atoms of the “unsubstituted aryloxy group” are 6 to 50, preferably 6 to 30, and more preferably 6 to 18, unless otherwise specified.

Specific examples of the “arylthiol group” described herein include a group represented by —S(G1), where G1 is the “aryl group” described in the specific example group G1. The number of ring carbon atoms of the “unsubstituted arylthio group” are 6 to 50, preferably 6 to 30, and more preferably 6 to 18, unless otherwise specified.

Specific examples of the “aralkyl group” described herein include a group represented by —(G3)-(G1), where G3 is the “alkyl group” described in the specific example group G3, and G1 is the “aryl group” described in the specific example group G1. Accordingly, the “aralkyl group” is one embodiment of the “substituted alkyl group” substituted by the “aryl group”. The number of carbon atoms of the “unsubstituted aralkyl group,” which is the “unsubstituted alkyl group” substituted by the “unsubstituted aryl group,” are 7 to 50, preferably 7 to 30, and more preferably 7 to 18, unless otherwise specified.

Specific example of the “aralkyl group” include a benzyl group, a 1-phenylethyl group, a 2-phenylethyl group, a 1-phenylisopropyl group, a 2-phenylisopropyl group, a phenyl-*t*-butyl group, an α -naphthylmethyl group, a 1- α -naphthylethyl group, a 2- α -naphthylethyl group, a 1- α -naphthylisopropyl group, a 2- α -naphthylisopropyl group, a β -naphthylmethyl group, a 1- β -naphthylethyl group, a 2- β -naphthylethyl group, a 1- β -naphthylisopropyl group, and a 2- β -naphthylisopropyl group.

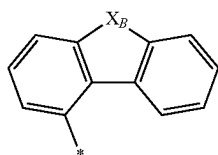
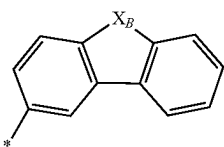
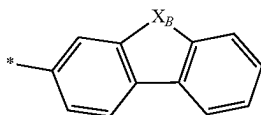
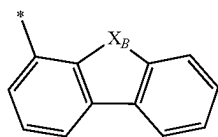
The substituted or unsubstituted aryl group described herein is, unless otherwise specified, preferably a phenyl group, a *p*-biphenyl group, *m*-biphenyl group, an *o*-biphenyl group, a *p*-terphenyl-4-yl group, a *p*-terphenyl-3-yl group, a *p*-terphenyl-2-yl group, *m*-terphenyl-4-yl group, *m*-terphenyl-3-yl group, *m*-terphenyl-2-yl group, an *o*-terphenyl-4-yl group, an *o*-terphenyl-3-yl group, an *o*-terphenyl-2-yl group, a 1-naphthyl group, a 2-naphthyl group, an anthryl group, a phenanthryl group, a pyrenyl group, a chrysenyl group, a triphenylenyl group, a fluorenyl group, a 9,9-spirobifluorenyl group, a 9,9-diphenylfluorenyl group, or the like.

The substituted or unsubstituted heterocyclic group described herein is, unless otherwise specified, preferably a pyridyl group, a pyrimidinyl group, a triazinyl group, a quinolyl group, an isoquinolyl group, a quinazolinyl group, a benzimidazolyl group, a phenanthrolinyl group, a carbazolyl group (a 1-carbazolyl group, a 2-carbazolyl group, a 3-carbazolyl group, a 4-carbazolyl group, or a 9-carbazolyl group), a benzocarbazolyl group, an azacarbazolyl group, a diazacarbazolyl group, a dibenzofuranyl group, a naphthobenzofuranyl group, an azadibenzofuranyl group, a diazadibenzofuranyl group, a dibenzothiophenyl group, an aph-

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thobenzothiophenyl group, an azadibenzothiophenyl group, a diazadibenzothiophenyl group, a (9-phenyl)carbazolyl group (a(9-phenyl)carbazol-1-yl group, a (9-phenyl)carbazol-2-yl group, a (9-phenyl)carbazol-3-yl group, or a (9-phenyl)carbazol-4-yl group), a (9-biphenyl)carbazolyl group, a (9-phenyl)phenylcarbazolyl group, a diphenylcarbazole-9-yl group, a phenylcarbazol-9-yl group, a phenyltriazinyl group, a biphenyltriazinyl group, diphenyltriazinyl group, a phenyldibenzofuranyl group, a phenyldibenzothiophenyl group, an indrocarbazolyl group, a pyrazinyl group, a pyridazinyl group, a quinazolinyl group, a cinnolinyl group, a phthalazinyl group, a quinoxalinyl group, a pyrrolyl group, an indolyl group, a pyrrolo[3,2,1-jk]carbazolyl group, a furanyl group, a benzofuranyl group, a thiophenyl group, a benzothiophenyl group, a pyrazolyl group, an imidazolyl group, a benzimidazolyl group, a triazolyl group, an oxazolyl group, a benzoxazolyl group, a thiazolyl group, a benzothiazolyl group, an isothiazolyl group, a benzisothiazolyl group, a thiadiazolyl group, an isoxazolyl group, a benzisoxazolyl group, a pyrrolidinyl group, a piperidinyl group, a piperazinyl group, a n imidazolidinyl group, an indro[3,2,1-jk]carbazolyl group, a dibenzothiophenyl group, or the like.

The dibenzofuranyl group and the dibenzothiophenyl group as described above are specifically any group described below, unless otherwise specified.



In the formulas (XY-76) to (XY-79), X_B is an oxygen atom or a sulfur atom.

The substituted or unsubstituted alkyl group described herein is, unless otherwise specified, preferably a methyl group, an ethyl group, a propyl group, an isopropyl group, a n-butyl group, an isobutyl group, a t-butyl group, or the like.

The “substituted or unsubstituted arylene group” described herein refers to a group in which the above-described “aryl group” is converted into divalence, unless otherwise specified. Specific examples (specific example group G12) of the “substituted or unsubstituted arylene group” include a group in which the “aryl group” described in the specific example

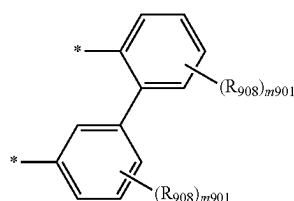
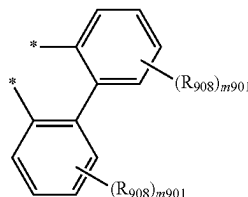
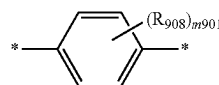
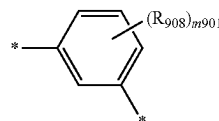
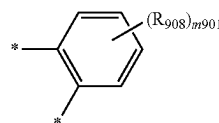
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group G1 is converted into divalence. Namely, specific examples (specific example group G12) of the “substituted or unsubstituted arylene group” refer to a group derived from the “aryl group” described in specific example group G1 by removal of one hydrogen atom bonded to the ring carbon atoms thereof.

Specific examples (specific example group G13) of the “substituted or unsubstituted divalent heterocyclic group” include a group in which the “heterocyclic group” described in the specific example group G2 is converted into divalence. Namely, specific examples (specific example group G13) of the “substituted or unsubstituted divalent heterocyclic group” refer to a group derived from the “heterocyclic group” described in specific example group G2 by removal of one hydrogen atom bonded to the ring atoms thereof.

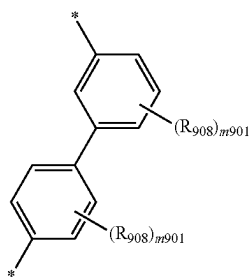
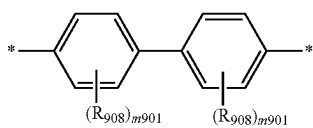
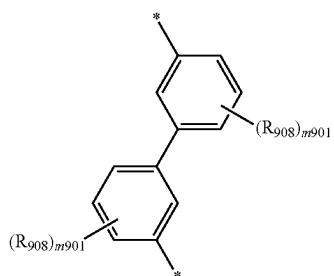
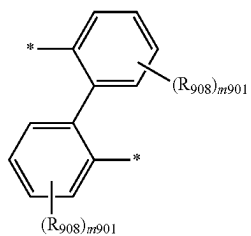
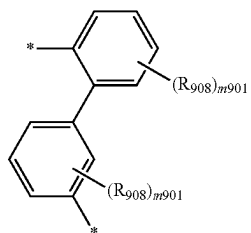
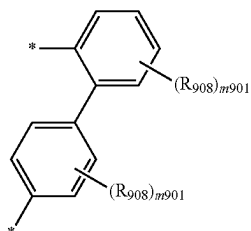
Specific examples (specific example group G14) of the “substituted or unsubstituted alkylene group” include a group in which the “alkyl group” described in the specific example group G3 is converted into divalence. Namely, specific examples (specific example group G14) of the “substituted or unsubstituted alkylene group” refer to a group derived from the “alkyl group” described in specific example group G3 by removal of one hydrogen atom bonded to the carbon atoms constituting the alkane structure thereof.

The substituted or unsubstituted arylene group described herein is any group described below, unless otherwise specified.



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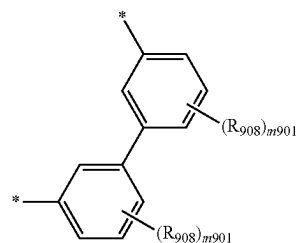
**18**

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(XY-25)

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(XY-84)

(XY-26)

15 In the formulas (XY-20) to (XY-29), (XY-83) and (XY-

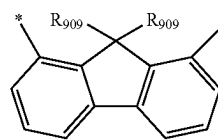
84), R_{908} is a substituent.

Then, m_{901} is an integer of 0 to 4, and when m_{901} is 2 or more, a plurality of R_{908} may be the same with or different from each other.

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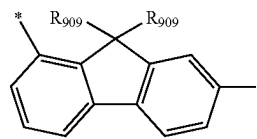
(XY-27)

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(XY-30)

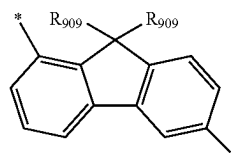
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(XY-31)

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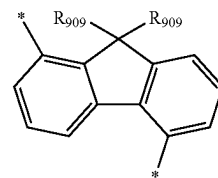
(XY-28)



(XY-32)

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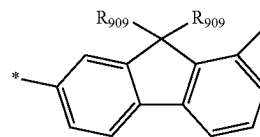
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(XY-33)

(XY-29)

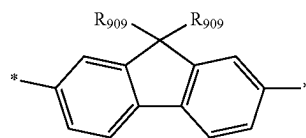
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(XY-34)

(XY-83)

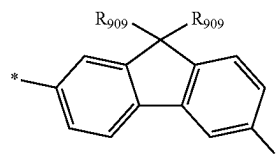
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(XY-35)

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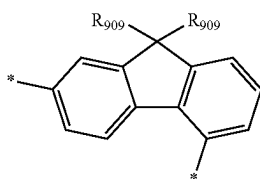
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(XY-36)

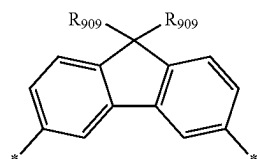
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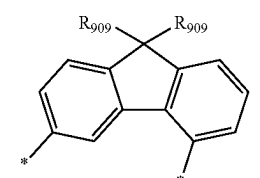
(XY-37)

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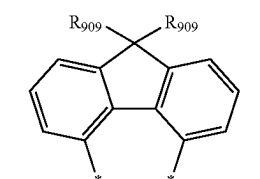


(XY-38)

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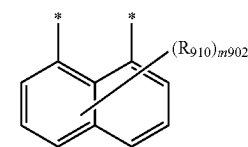
(XY-39)



(XY-40)

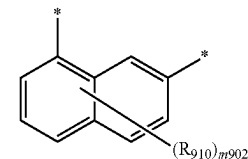
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In the formulas (XY-30) to (XY-40), R_{909} is independently a hydrogen atom or a substituent. Two of R_{909} may form a ring by bonding with each other through a single bond.



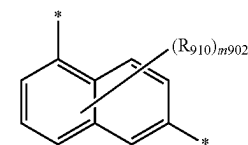
(XY-41)

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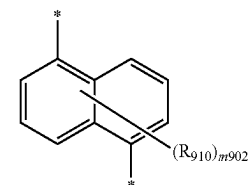
(XY-42)

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(XY-43)

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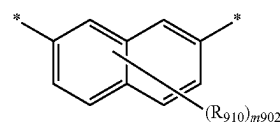


(XY-44)

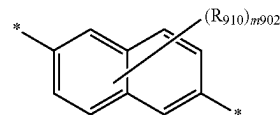
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(XY-45)

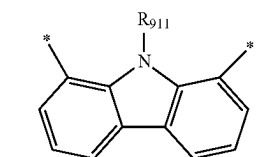


(XY-46)

15 In the formulas (XY-41) to (XY-46), R_{910} is a substituent.

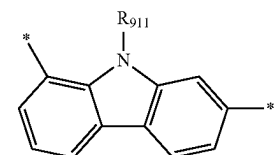
Then, $m902$ is an integer of 0 to 6. When $m902$ is 2 or more, a plurality of R_{910} may be the same with or different from each other.

20 The substituted or unsubstituted divalent heterocyclic group described herein is preferably any group described below, unless otherwise specified.

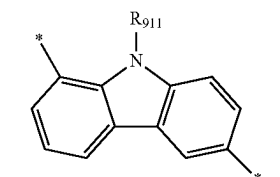


(XY-50)

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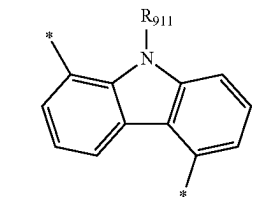


(XY-51)



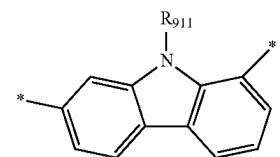
(XY-52)

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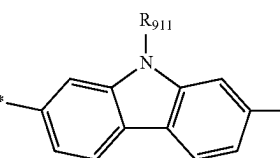
(XY-53)

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(XY-54)

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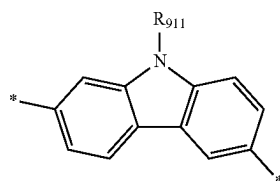


(XY-55)

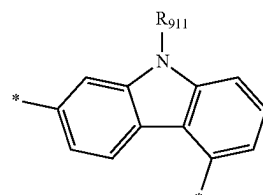
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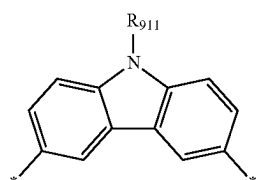
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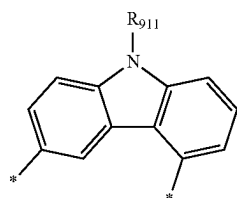
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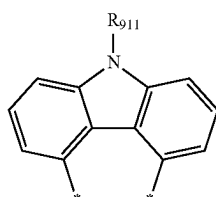
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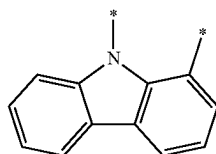
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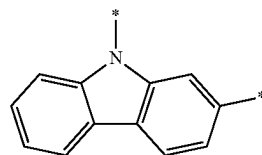
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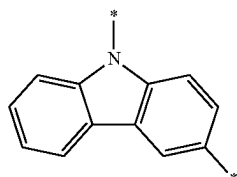
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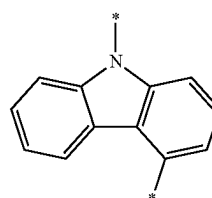
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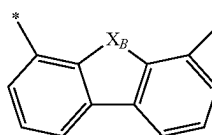
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-continued

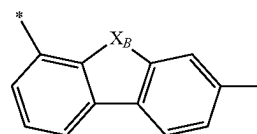


(XY-64)

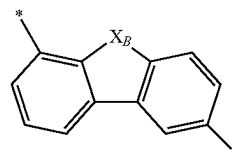
In the formulas (XY-50) to (XY-60), R_{911} is a hydrogen atom or a substituent.



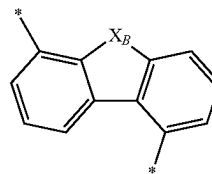
(XY-65)



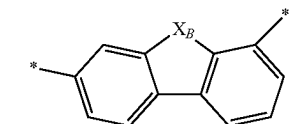
(XY-66)



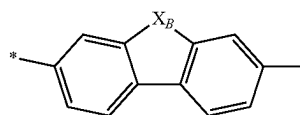
(XY-67)



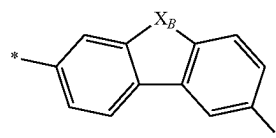
(XY-68)



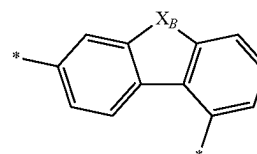
(XY-69)



(XY-70)



(XY-71)

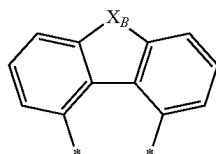
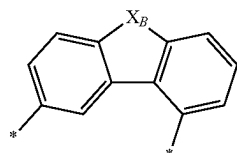
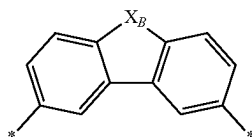


(XY-72)

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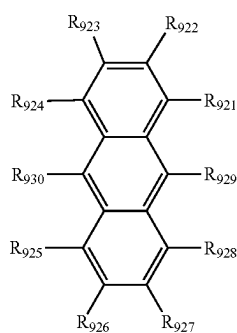
23

-continued



In the formulas (XY-65) to (XY-75), X_B is an oxygen atom or a sulfur atom.

Herein, a case where “one or more sets of two or more groups adjacent to each other form a substituted or unsubstituted and saturated or unsaturated ring by bonding with each other” will be described by taking, as an example, a case of an anthracene compound represented by the following formula (XY-80) in which a mother skeleton is an anthracene ring.



For example, two adjacent to each other into one set when “one or more sets of two or more groups adjacent to each other form the ring by bonding with each other” among R_{921} to R_{930} include R_{921} and R_{922} , R_{922} and R_{923} , R_{923} and R_{924} , R_{924} and R_{930} , R_{930} and R_{925} , R_{925} and R_{926} , R_{926} and R_{927} , R_{927} and R_{928} , R_{928} and R_{929} , and R_{929} and R_{921} .

The above-described “one or more sets” means that two or more sets of two groups adjacent to each other may simultaneously form the ring. For example, a case where R_{921} and R_{922} form a ring A by bonding with each other, and simultaneously R_{925} and R_{926} form a ring B by bonding with each other is represented by the following formula (XY-81).

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(XY-73)

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(XY-74)

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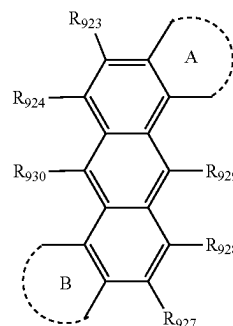
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(XY-75)

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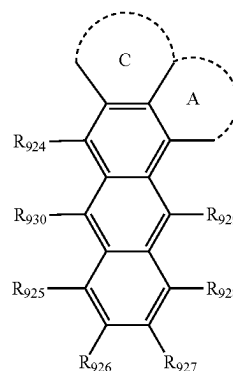
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(XY-81)



A case where “two or more groups adjacent to each other” form a ring means that, for example, R_{921} and R_{922} form a ring A by bonding with each other, and R_{922} and R_{923} form a ring C by bonding with each other. A case where the ring A and ring C sharing R_{922} are formed, in which the ring A and the ring C are fused to the anthracene mother skeleton by three of R_{921} to R_{923} adjacent to each other, is represented by the following (XY-82).

(XY-82)



The rings A to C formed in the formulas (XY81) and (XY82) are saturated or unsaturated ring.

A term “unsaturated ring” means an aromatic hydrocarbon ring or an aromatic heterocyclic ring. A term “saturated ring” means an aliphatic hydrocarbon ring or an aliphatic heterocyclic ring.

For example, the ring A formed by R_{921} and R_{922} being bonded with each other, represented by the formula (XY-81), means a ring formed by a carbon atom of the anthracene skeleton bonded with R_{921} , a carbon atom of the anthracene skeleton bonded with R_{922} , and one or more arbitrary elements. Specific examples include, when the ring A is formed by w and R_{922} , a case where an unsaturated ring is formed of a carbon atom of an anthracene skeleton bonded with R_{921} , a carbon atom of the anthracene skeleton bonded with R_{922} , and four carbon atoms, in which a ring formed by R_{921} and R_{922} is formed into an abenzene ring. Further, when a saturated ring is formed, the ring is formed into a cyclohexane ring.

Here, “arbitrary elements” are preferably a C element, a N element, an O element and a S element. In the arbitrary elements (for example, a case of the C element or the N element), the bond(s) that is(are) not involved in the formation of the ring may be terminated by a hydrogen atom, or may be substituted by an arbitrary substituent. When the ring

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contains the arbitrary elements other than the C element, the ring to be formed is a heterocyclic ring.

The number of "one or more arbitrary elements" forming the saturated or unsaturated ring is preferably 2 or more and 15 or less, more preferably 3 or more and 12 or less, and further preferably 3 or more and 5 or less.

As specific examples of the aromatic hydrocarbon ring, a structure in which the aryl group described in specific example group G1 is terminated with a hydrogen atom may be mentioned.

As specific examples of the aromatic heterocyclic ring, a structure in which the aromatic heterocyclic group described in specific example group G2 is terminated with a hydrogen atom may be mentioned.

As specific examples of the aliphatic hydrocarbon ring, a structure in which the cycloalkyl group described in specific example group G6 is terminated with a hydrogen atom may be mentioned.

When the above-described "saturated or unsaturated ring" has a substituent, the substituent is an "arbitrary substituent" as described below, for example. When the above-mentioned "saturated or unsaturated ring" has a substituent, specific examples of the substituent refer to the substituents described in above-mentioned "the substituent described herein".

In one embodiment of this specification, the substituent (hereinafter; referred to as an "arbitrary substituent" in several cases) in the case of the "substituted or unsubstituted" is a group selected from the group consisting of

- an unsubstituted alkyl group including 1 to 50 carbon atoms,
- an unsubstituted alkenyl group including 2 to 50 carbon atoms,
- an unsubstituted alkynyl group including 2 to 50 carbon atoms,
- an unsubstituted cycloalkyl group including 3 to 50 ring carbon atoms,
- Si(R₉₀₁)(R₉₀₂)(R₉₀₃),
- O—(R₉₀₄),
- S—(R₉₀₅),
- N(R₉₀₆)(R₉₀₇)

wherein,

R₉₀₁ to R₉₀₇ are independently

a hydrogen atom,

a substituted or unsubstituted alkyl group including 1 to 50 carbon atoms,

a substituted or unsubstituted cycloalkyl group including 3 to 50 ring carbon atoms,

a substituted or unsubstituted aryl group including 6 to 50 ring carbon atoms, or

a substituted or unsubstituted monovalent heterocyclic group including 5 to 50 ring atoms; and when two or more of R₉₀₁ to R₉₀₇ exist, two or more of R₉₀₁ to R₉₀₇ may be the same with or different from each other,

a halogen atom, a cyano group, a nitro group, an unsubstituted aryl group including 6 to 50 ring carbon atoms, and

an unsubstituted monovalent heterocyclic group including 5 to 50 ring atoms.

In one embodiment, the substituent in the case of "substituted or unsubstituted" is a group selected from the group consisting of

- an alkyl group including 1 to 50 carbon atoms,
- an aryl group including 6 to 50 ring carbon atoms, and
- a monovalent heterocyclic group including 5 to 50 ring atoms.

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In one embodiment, the substituent in the case of "substituted or unsubstituted" is a group selected from the group consisting of

- an alkyl group including 1 to 18 carbon atoms,
- an aryl group including 6 to 18 ring carbon atoms, and
- a monovalent heterocyclic group including 5 to 18 ring atoms.

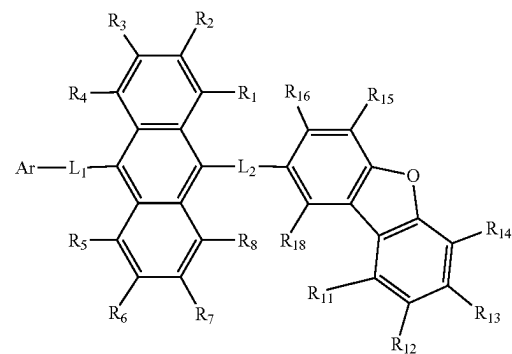
Specific examples of each group of the arbitrary substituent described above are as described above.

Herein, unless otherwise specified, the saturated or unsaturated ring (preferably substituted or unsubstituted and saturated or unsaturated five-membered or six-membered ring, more preferably abenzene ring) may be formed by the arbitrary substituents adjacent to each other.

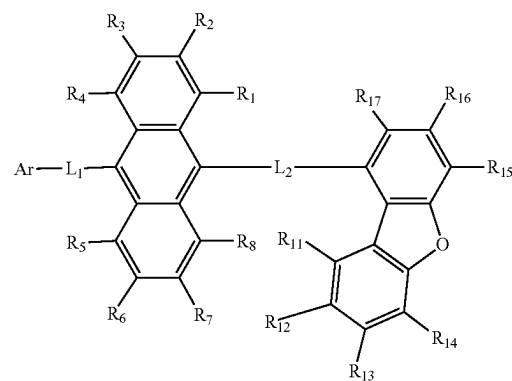
Herein, unless otherwise specified, the arbitrary substituent may further have the substituent. Specific examples of the substituent that the arbitrary substituent further has include to the ones same as the arbitrary substituent described above.

[New Compound]

The compound according to one aspect of the invention is represented by the following formula (1) or (2).



(1)



(2)

In the formulas (1) and (2),

R₁ to R₈ and R₁₁ to R₁₄ are independently

a hydrogen atom,

a substituted or unsubstituted alkyl group having 1 to 50 carbon atoms,

a substituted or unsubstituted alkenyl group having 2 to 50 carbon atoms,

a substituted or unsubstituted alkynyl group having 2 to 50 carbon atoms,

a substituted or unsubstituted cycloalkyl group having 3 to 50 ring carbon atoms,

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—Si(R₉₀₁)(R₉₀₂)(R₉₀₃),
 —O—(R₉₀₄),
 —S—(R₉₀₅),
 —N(R₉₀₆)(R₉₀₇),
 a halogen atom, a cyano group, a nitro group,
 a substituted or unsubstituted aryl group having 6 to 50
 ring carbon atoms, or
 a substituted or unsubstituted monovalent heterocyclic
 group having 5 to 50 ring atoms;
 R₁₅ to R₁₈ are hydrogen atoms;
 R₉₀₁ to R₉₀₇ are independently
 a hydrogen atom,
 a substituted or unsubstituted alkyl group having 1 to 50
 carbon atoms,
 a substituted or unsubstituted cycloalkyl group having 3
 to 50 ring carbon atoms,
 a substituted or unsubstituted aryl group having 6 to 50
 ring carbon atoms, or
 a substituted or unsubstituted monovalent heterocyclic
 group having 5 to 50 ring atoms;
 when two or more of R₉₀₁ to R₉₀₇ exist, two or more of
 R₉₀₁ to R₉₀₇ may be the same with or different from
 each other;
 at least one of R₁ to R₈ is a deuterium atom;
 two or more adjacent groups of R₁ to R₄, two or more
 adjacent groups of R₅ to R₆ and two or more adjacent
 groups of R₁₁ to R₁₄ do not form a ring;
 L₁ and L₂ are independently
 a single bond,
 a substituted or unsubstituted arylene group having 6 to
 30 ring carbon atoms, or
 a substituted or unsubstituted divalent heterocyclic group
 having 5 to 30 ring atoms;
 Ar is
 a substituted or unsubstituted aryl group having 6 to 50
 ring carbon atoms, or
 a substituted or unsubstituted monovalent heterocyclic
 group having 5 to 50 ring atoms.

By having the above structure, the compound according to
 one aspect of the invention can enhance the device
 performance when the compound is used in an organic EL
 device. Specifically, it is possible to provide an organic EL
 device with longer life.

According to one aspect of the present invention, a
 method for improving a performance of an organic EL
 device can also be provided. the method is characterized in
 that the above compound is used in the emitting layer of the
 organic EL device. Specifically, the method can improve an
 organic EL device performance as compared with the case
 where a compound having the same structure as formula (1)
 or (2) except that only protium atoms are contained as
 hydrogen atoms (hereinafter also referred to as “protium
 compound”) is used as a host material. The case where the
 protium compound is used means that a host material in an
 emitting layer consists essentially of the protium compound
 (the ratio of the protium compound to the sum of the protium
 compound and the compound represented by formula (1) or
 (2) is 90 mol % or more, 95 mol % or more, or 99 mol %
 or more).

That is, it is possible to increase a performance of an
 organic EL device by, instead of a protium compound or in
 addition to a protium compound, using a compound
 obtained by replacing at least one protium atoms on an
 anthracene skeleton of the protium compound with a deu-
 terium atom (a compound represented by formula (1) or (2))
 as a host material.

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All of R₁ to R₈ may be deuterium atoms or apart of them
 (e.g., one or two or more of R₁ to R₈) may be deuterium
 atoms.

R₁ to R₈ that are not deuterium atoms are preferably
 hydrogen atoms (protium atoms).

In one embodiment, at least one hydrogen atom contained
 in one or more groups selected from a group consisting of L₁
 and L₂ is a deuterium atom. In more detail, in one embodi-
 ment, one or more groups selected from the group consisting
 of L₁ and L₂ are an unsubstituted arylene group having 6 to
 30 ring carbon atoms in which at least one hydrogen atom
 is a deuterium atom, or an unsubstituted divalent heterocy-
 clic group having 5 to 30 ring atoms in which at least one
 hydrogen atom is a deuterium atom.

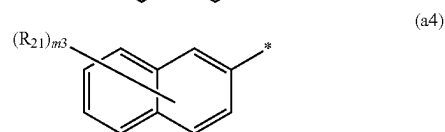
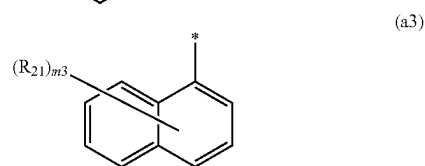
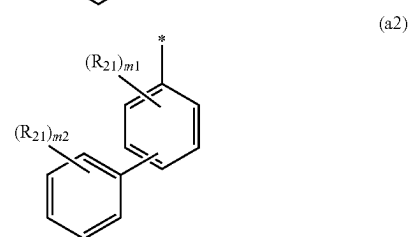
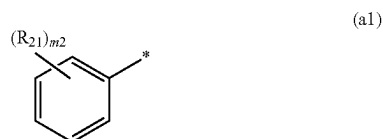
In one embodiment, L₁ and L₂ are independently a single
 bond, or a substituted or unsubstituted arylene group having
 6 to 14 ring carbon atoms. It is preferable that at least one
 of L₁ and L₂ is a single bond.

In one embodiment, R₁₁ to R₁₄ are hydrogen atoms.

In one embodiment, at least one R₁₁ to R₁₈ is a deuterium
 atom.

In one embodiment, at least one hydrogen atom contained
 in one or more Ar is a deuterium atom. In more detail, in one
 embodiment, Ar is an unsubstituted aryl group having 6 to
 50 ring carbon atoms in which at least one hydrogen atom
 is a deuterium atom, or an unsubstituted monovalent het-
 erocyclic group having 5 to 50 ring atoms in which at least
 one hydrogen atom is a deuterium atom.

Ar is preferably a substituted or unsubstituted aryl group
 having 6 to 50 ring carbon atoms, more preferably selected
 from groups represented by the following formulas (a1) to
 (a4).



In the formulas (a1) to (a4), * is a single bond bonding to
 L₁; R₂₁ is independently

a halogen atom, a cyano group, a nitro group,
 a substituted or unsubstituted alkyl group having 1 to 50
 carbon atoms,

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a substituted or unsubstituted alkenyl group having 2 to 50 carbon atoms,
 a substituted or unsubstituted alkynyl group having 2 to 50 carbon atoms,
 a substituted or unsubstituted cycloalkyl group having 3 to 50 ring carbon atoms,
 $-\text{Si}(\text{R}_{901})(\text{R}_{902})(\text{R}_{903})$,
 $-\text{O}-(\text{R}_{904})$,
 $-\text{S}-(\text{R}_{905})$,
 $-\text{N}(\text{R}_{906})(\text{R}_{907})$,
 a substituted or unsubstituted aryl group having 6 to 50 ring carbon atoms, or
 a substituted or unsubstituted monovalent heterocyclic group having 5 to 50 ring atoms;
 R_{901} to R_{907} are as defined in the formula (1);
 $m1$ is independently an integer of 0 to 4;
 $m2$ is independently an integer of 0 to 5;
 $m3$ is independently an integer of 0 to 7;
 when each of $m1$ to $m3$ is 2 or more, the plural R_{21s} may be the same or different; and
 when each of $m1$ to $m3$ is 2 or more, adjacent plural R_{21s} are bonded with each other to form a substituted or unsubstituted, saturated or unsaturated ring, or do not form a substituted or unsubstituted, saturated or unsaturated ring.

A bond position on L_2 side on the dibenzofuran skeleton, that is, at positions 1, 3, and 4 of dibenzofuran in formula (1), and at positions 2, 3, and 4 of dibenzofuran in formula (2), all of which are bonded to hydrogen atom, may be deuterium atom, or may be one or more (for example, positions 1 and 4 in formula (1), and positions 2 and 4 in formula (2)).

Existence of a deuterium atom in the compound is confirmed by Mass Spectrometry or ^1H -NMR Spectrometry. The bonding position of a deuterium atom in the compound is identified by ^1H -NMR Spectrometry. In concrete terms, it is confirmed as follows.

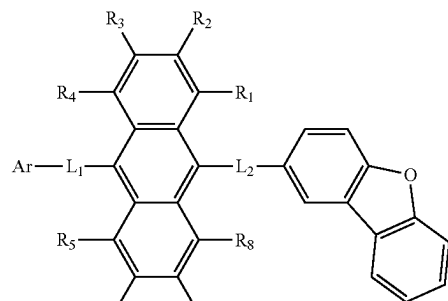
If it is identified that, by Mass Spectrometry, a molecular weight of a target compound is greater by "one" than a molecular weight of a corresponding compound in which all hydrogen atoms are protium atoms, it is confirmed that one deuterium atom exists in the target compound. Further, the number of deuterium atoms in a molecule can be confirmed by an integration value obtained by ^1H -NMR analysis on the target compound, since no signal is observed by performing ^1H -NMR analysis on a deuterium atom. The bonding position of a deuterium can be identified by performing ^1H -NMR analysis on the target compound and assigning signals.

The composition according to one aspect of the invention contains the compound represented by formula (1) or (2), and the content ratio of the protium compound to the total of the compound represented by formula (1) or (2) and the protium compound is preferably 99 mol % or less. The content ratio of the protium compound is confirmed by Mass Spectrometry.

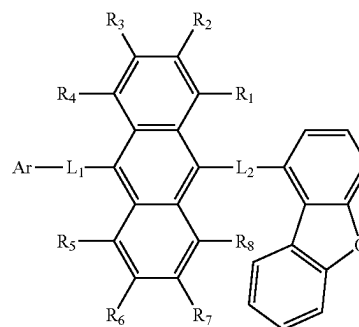
In the composition according to one aspect of the invention, the content ratio of the compound represented by formula (1) or (2) to the total of the compound represented by formula (1) or (2) and the protium compound is 30 mol % or more, 50 mol % or more, 70 mol % or more, 90 mol % or more, 95 mol % or more, 99 mol % or more, or 100 mol %.

In one embodiment, the compound represented by the formula (1) is a compound represented by following formula (1-1) and the compound represented by the formula (2) is a compound represented by following formula (2-1).

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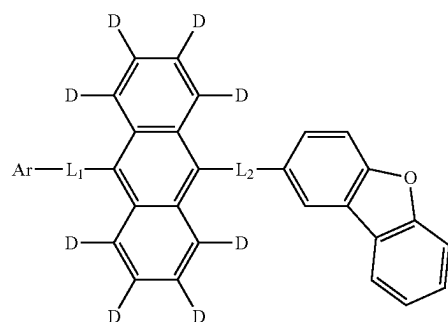
(1-1)



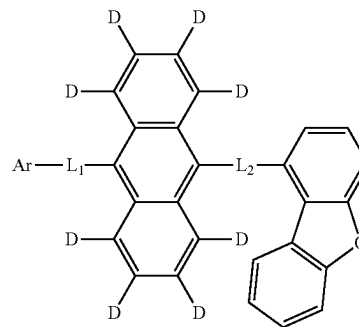
(2-1)

In the formulas (1-1) and (2-1), R_1 to R_8 , Ar, L_1 and L_2 are as defined in the formula (1).

In one embodiment, the compound represented by the formula (1) is a compound represented by following formula (1-2) and the compound represented by the formula (2) is a compound represented by following formula (2-2).



(1-2)



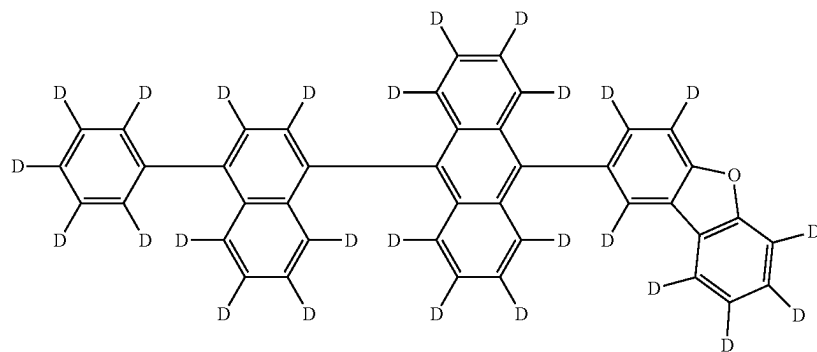
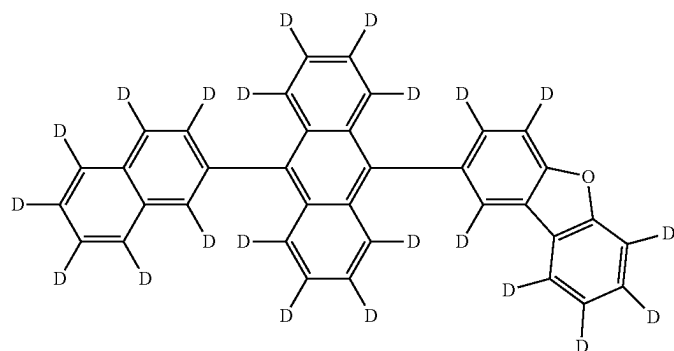
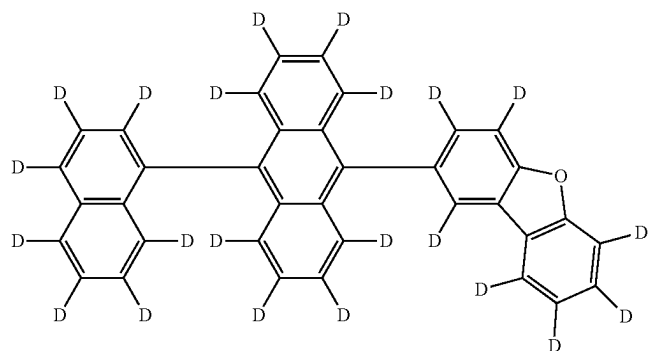
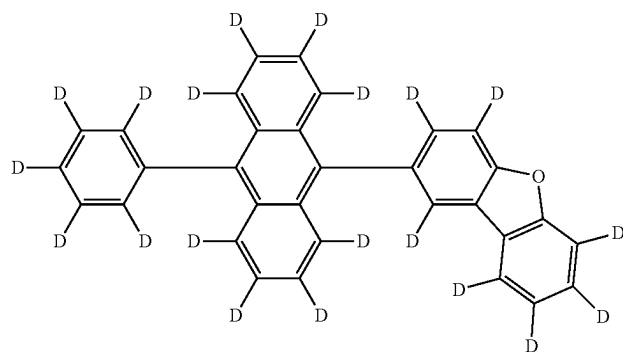
(2-2)

In the formulas (1-2) and (2-2), Ar, L_1 and L_2 are as defined in the formula (1).

The compound represented by the formula (1) or (2) can be synthesized in accordance with the synthesis process

described in Examples by using publicly known alternative reactions or materials corresponding to a target compound.

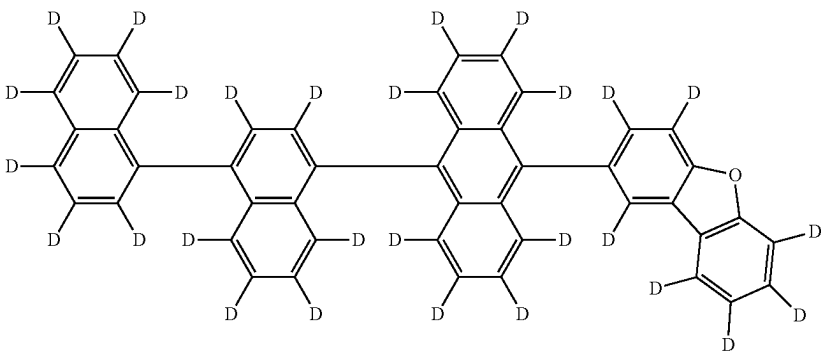
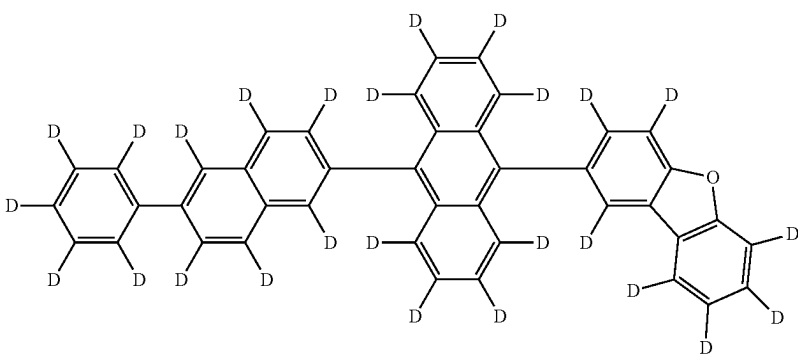
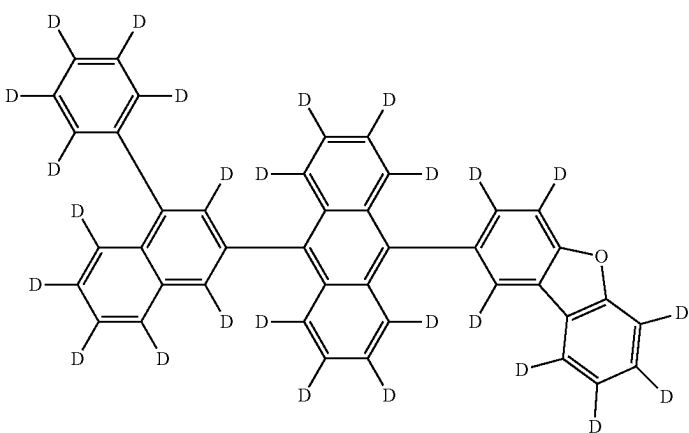
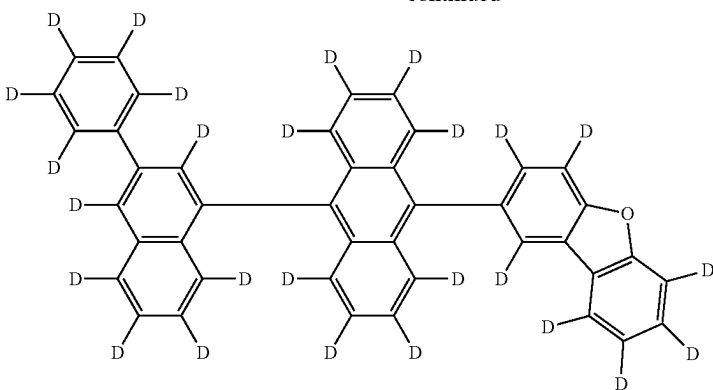
Examples of the compound according to one aspect of the invention include the following compounds. In the specific examples, D represents deuterium atom.



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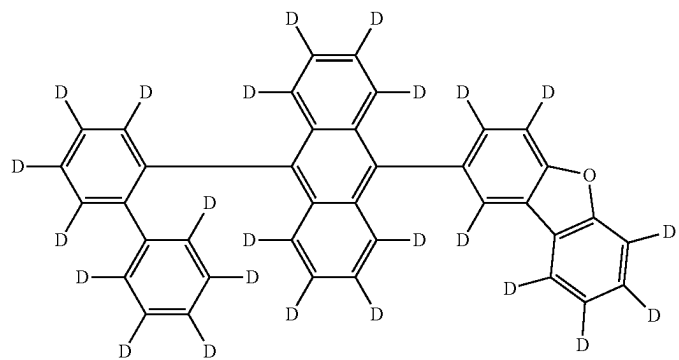
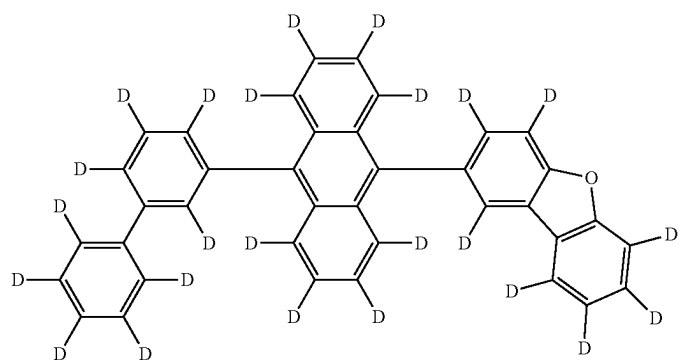
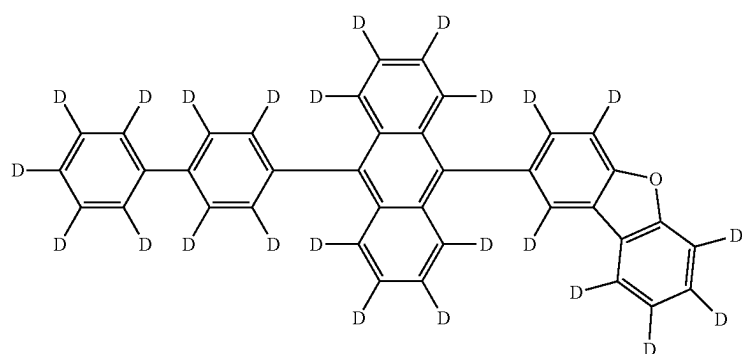
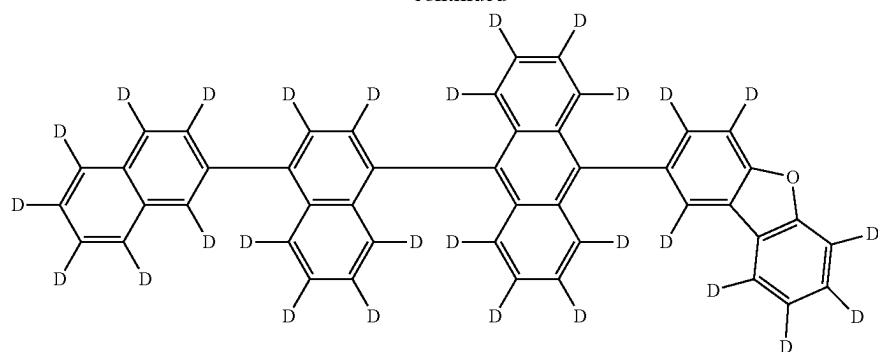
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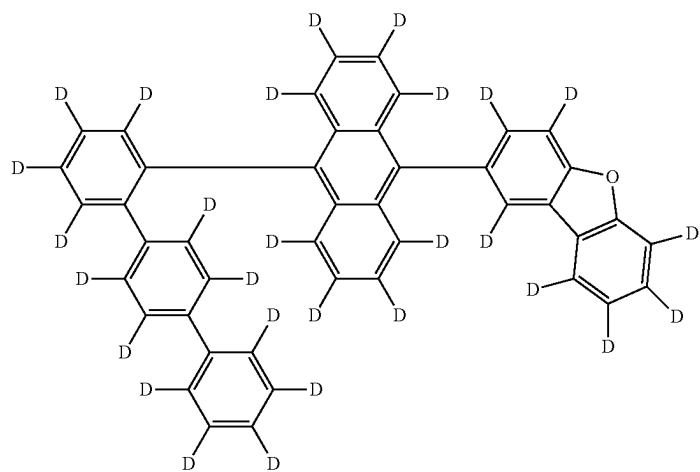
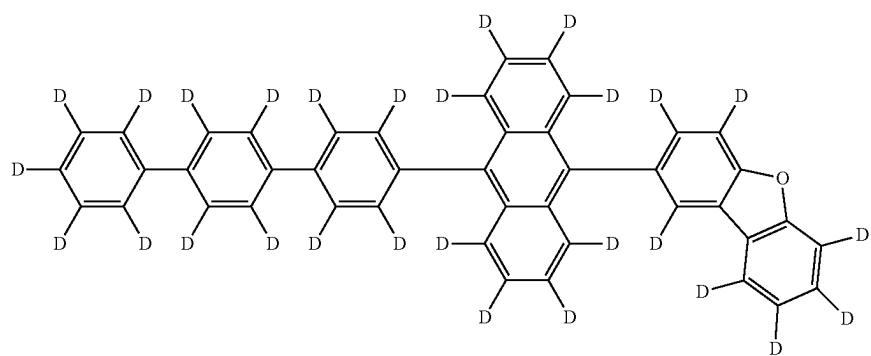
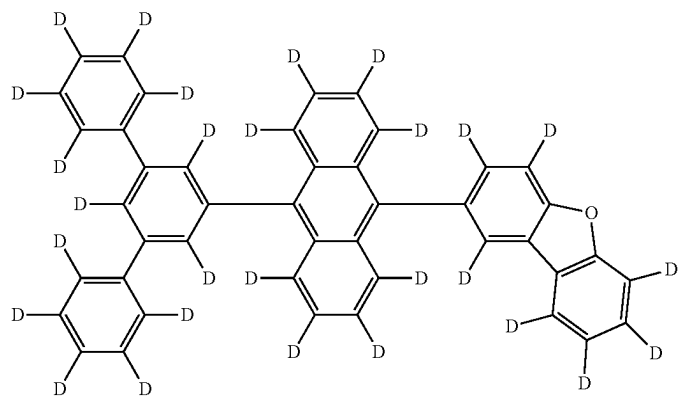
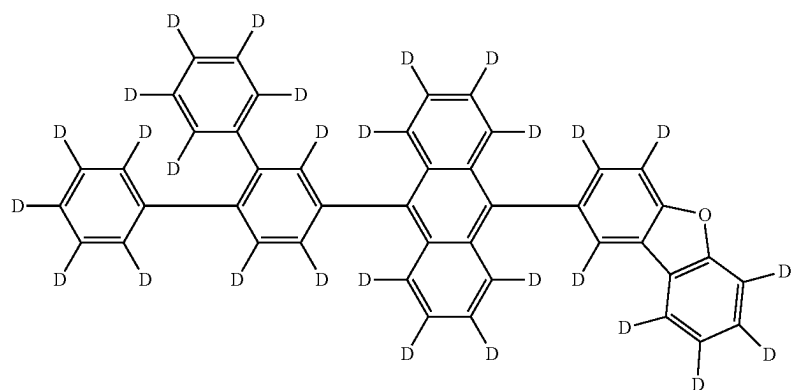
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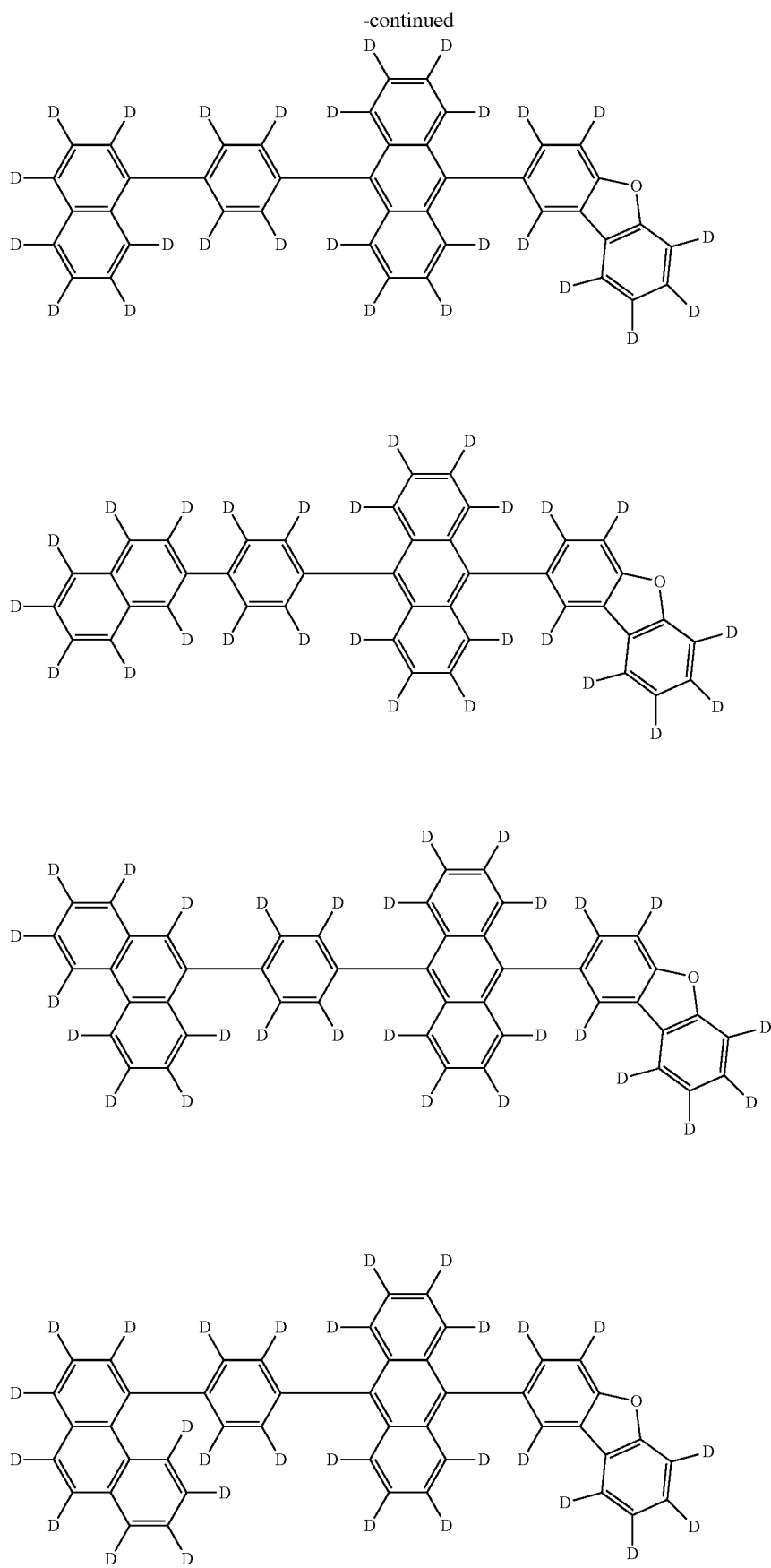
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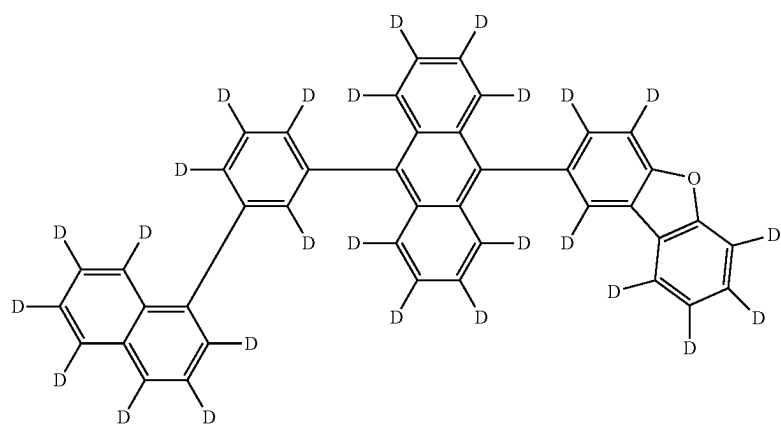
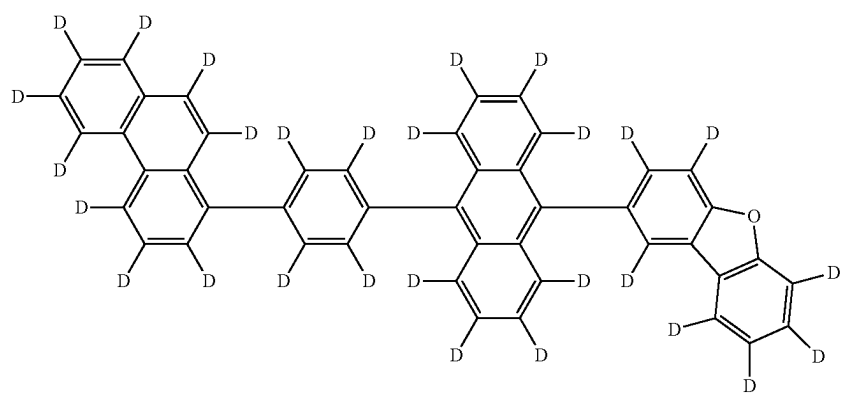
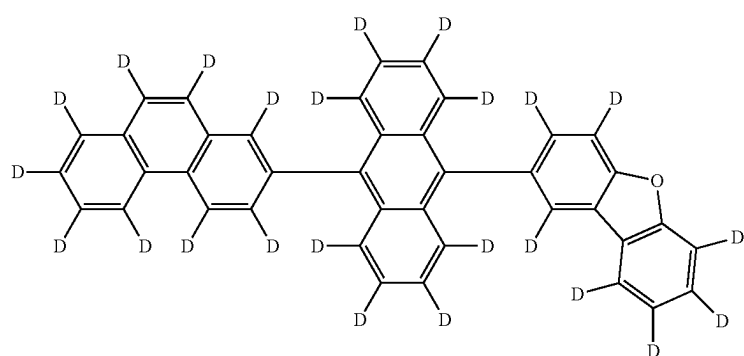
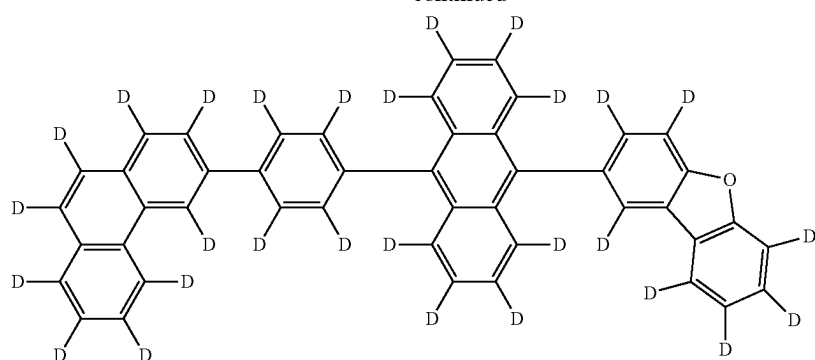
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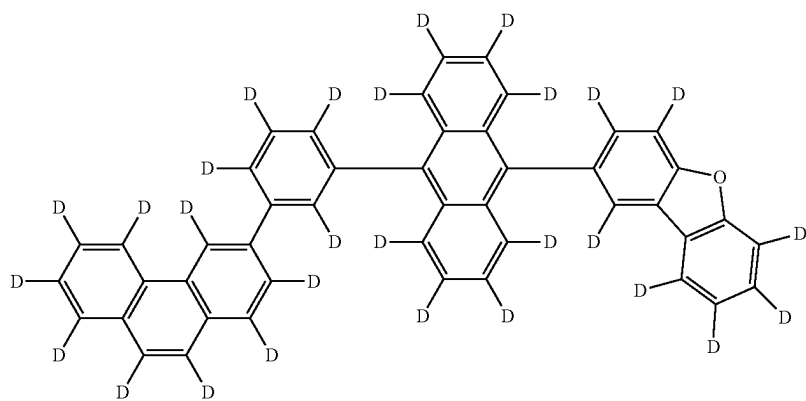
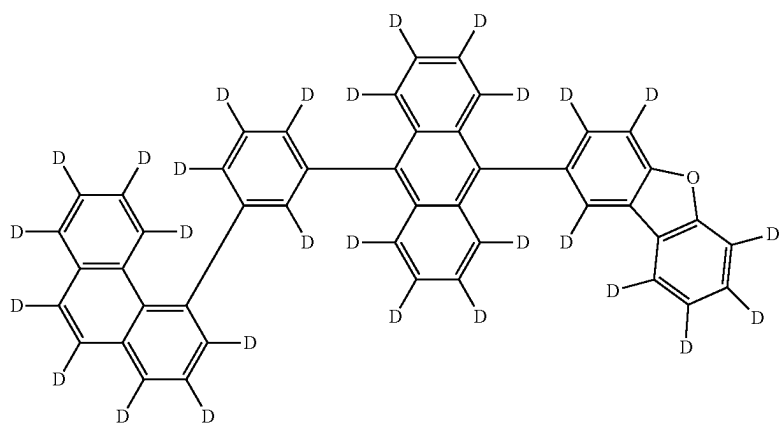
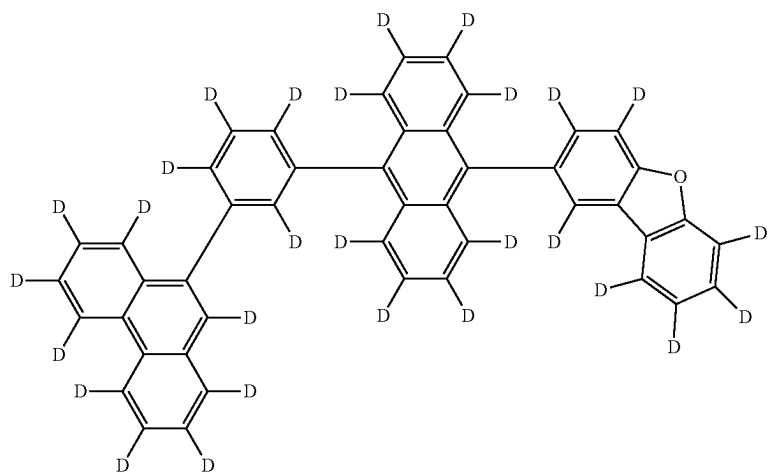
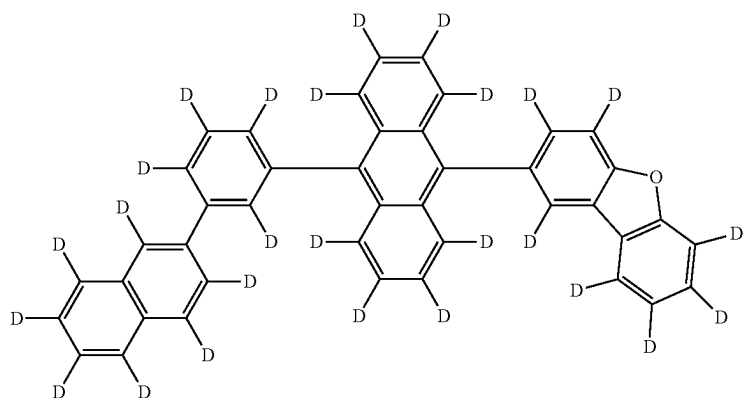
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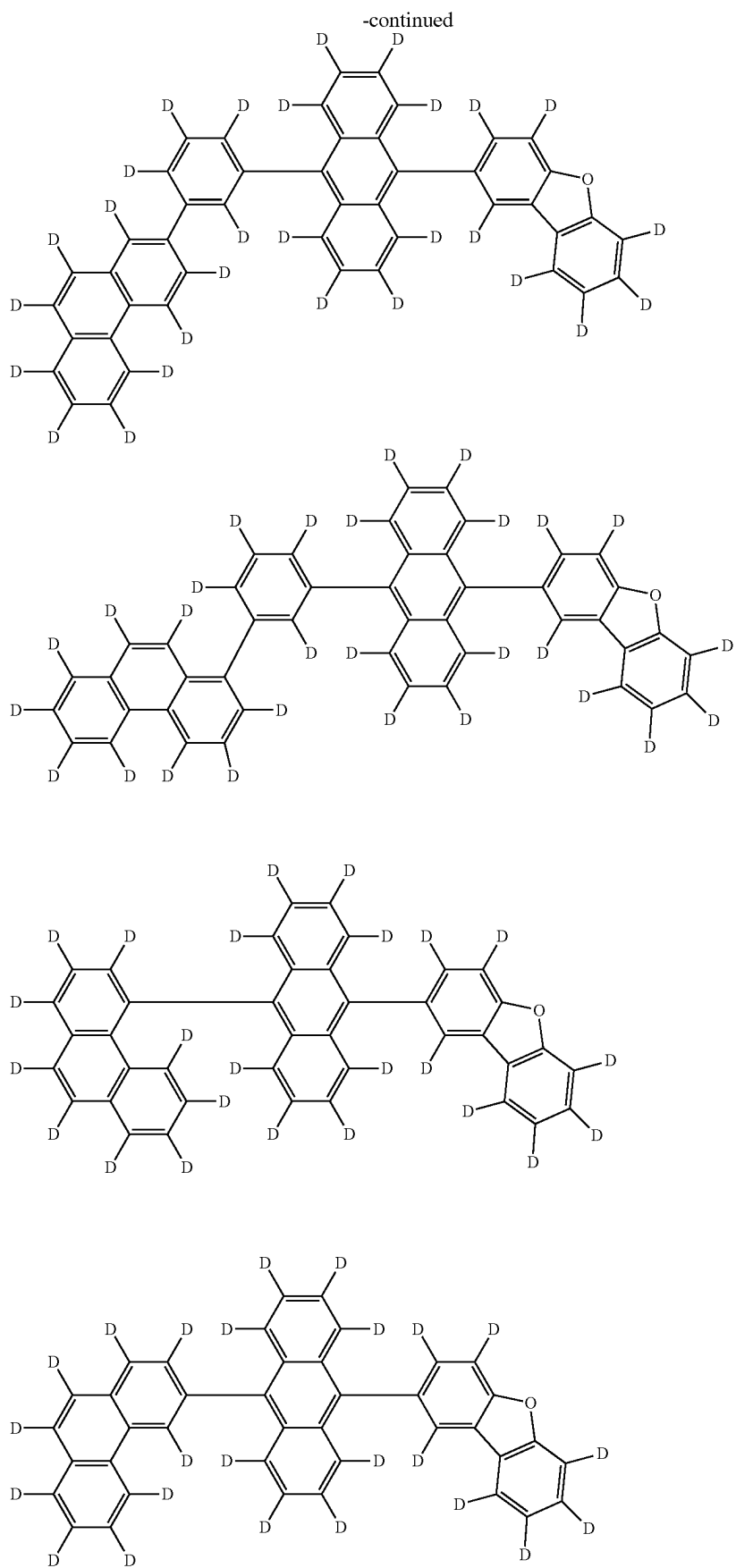


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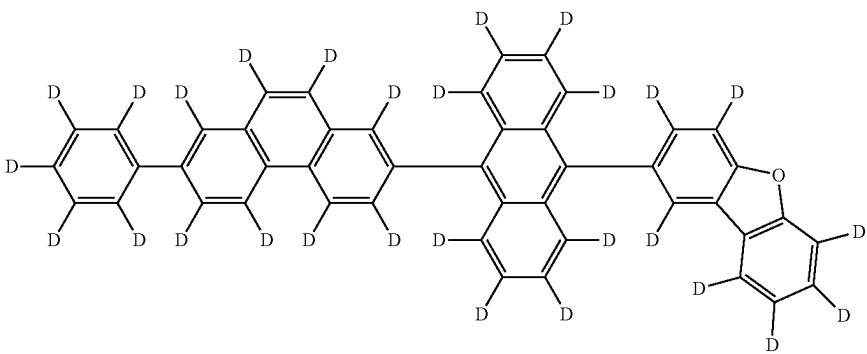
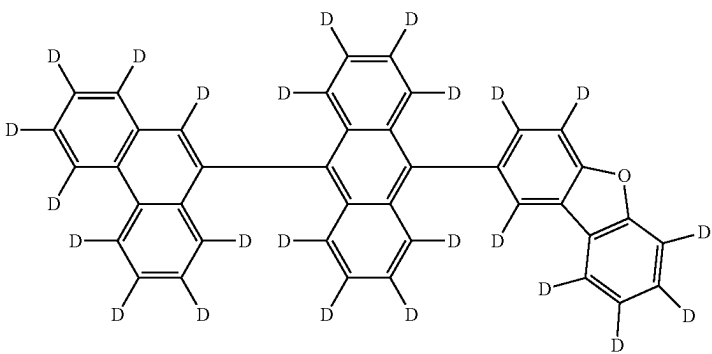
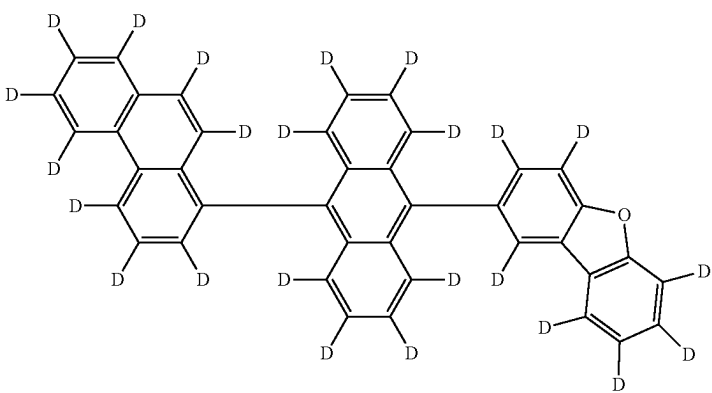
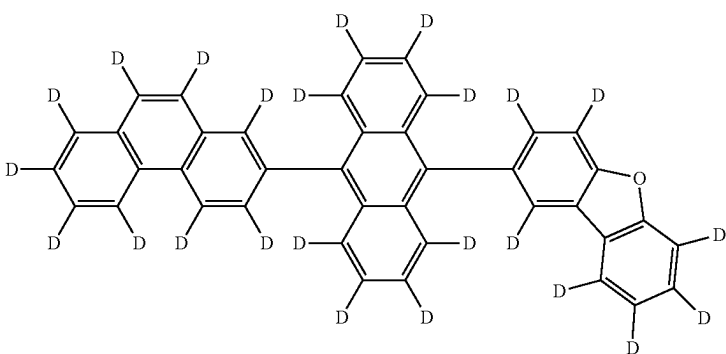
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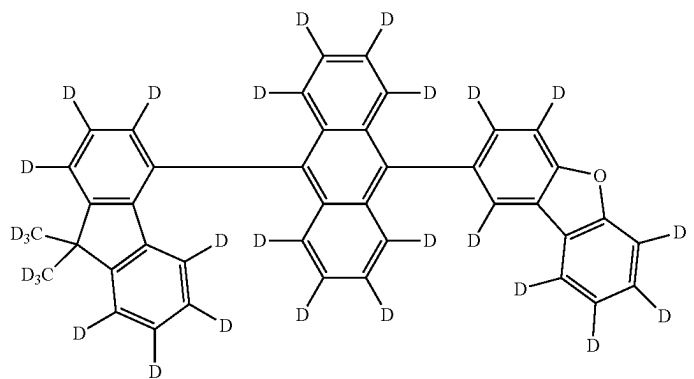
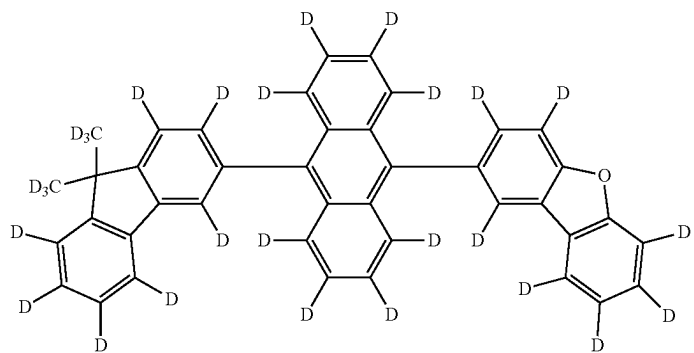
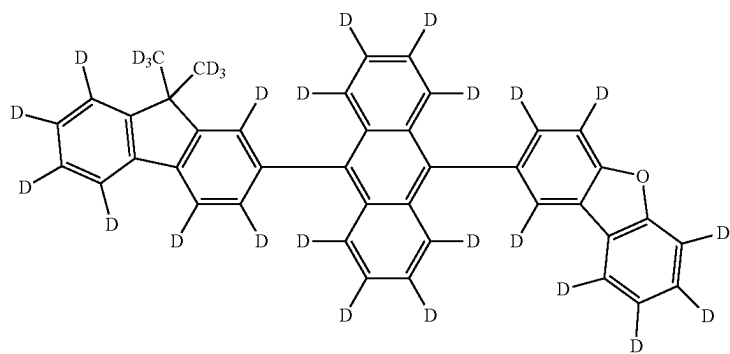
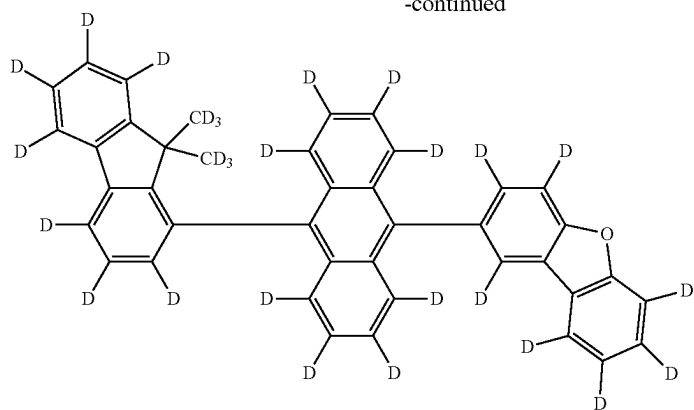
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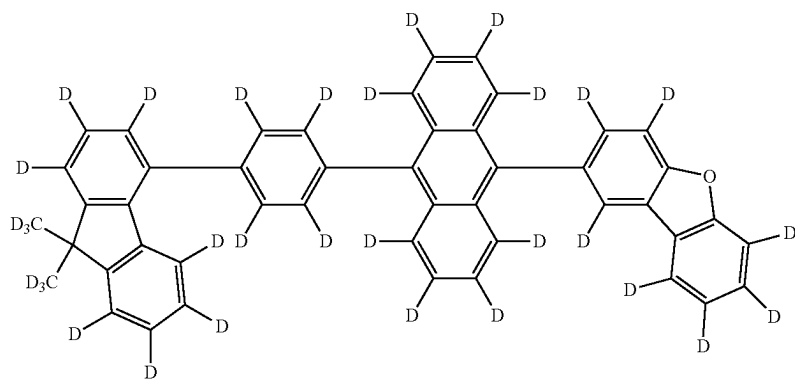
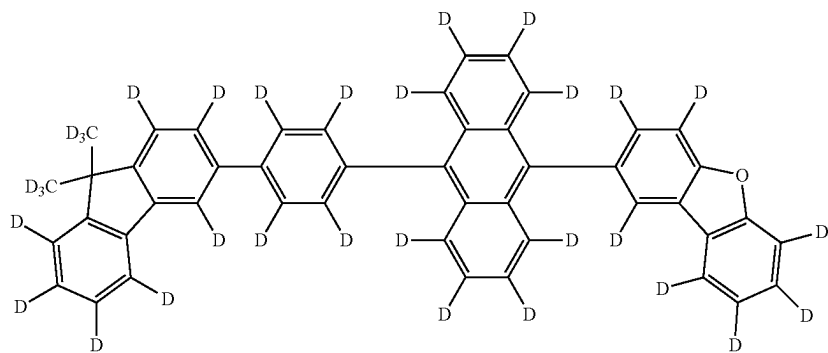
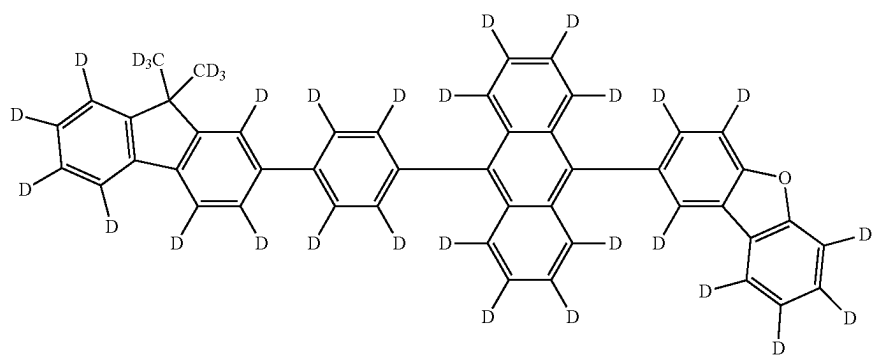
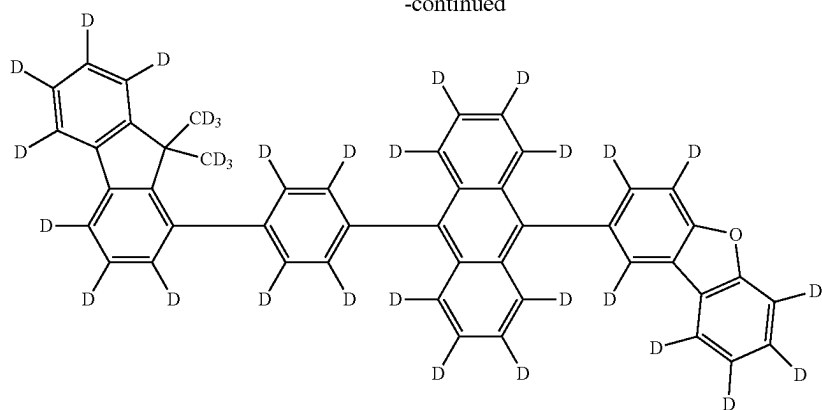
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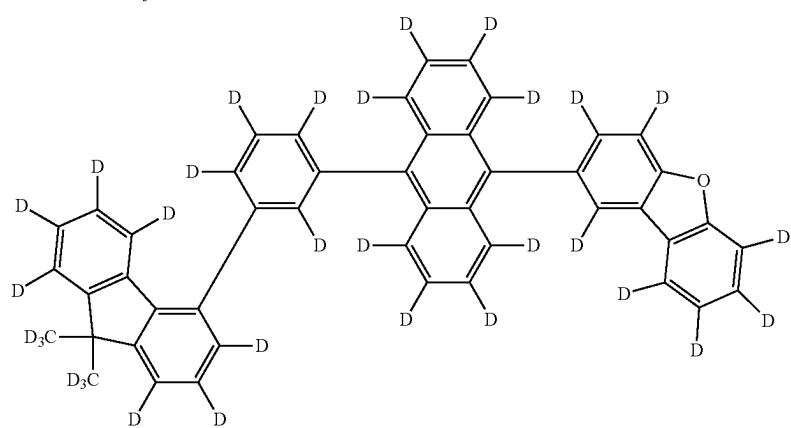
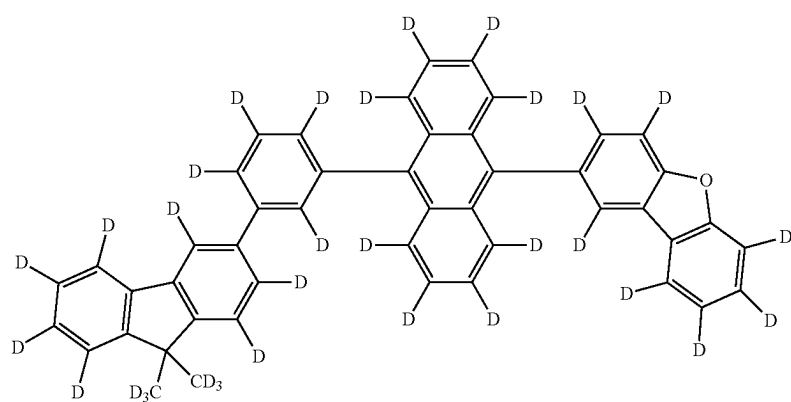
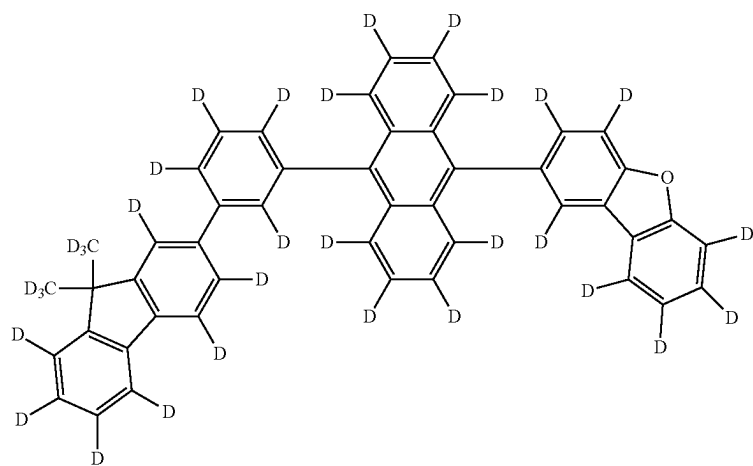
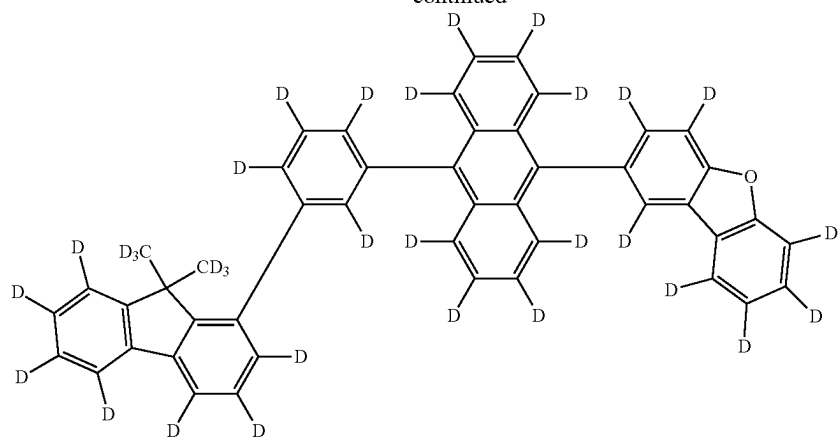
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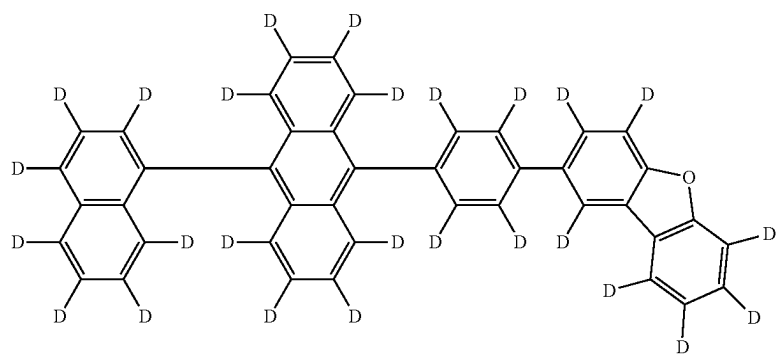
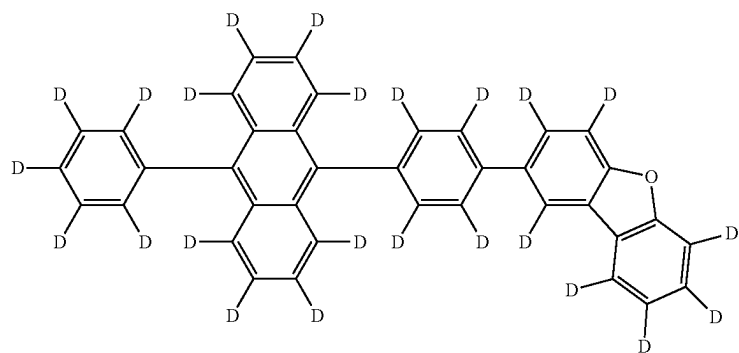
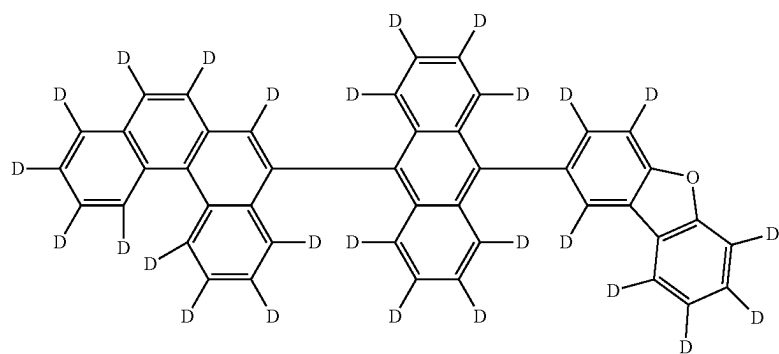
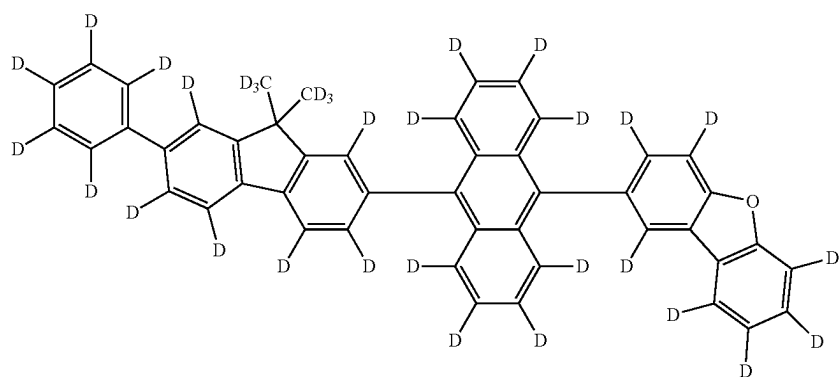
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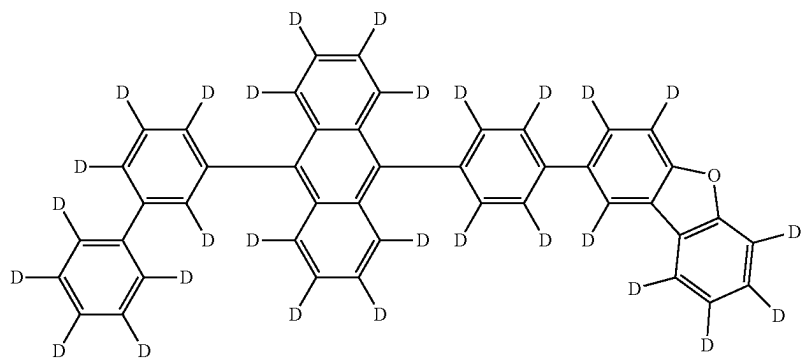
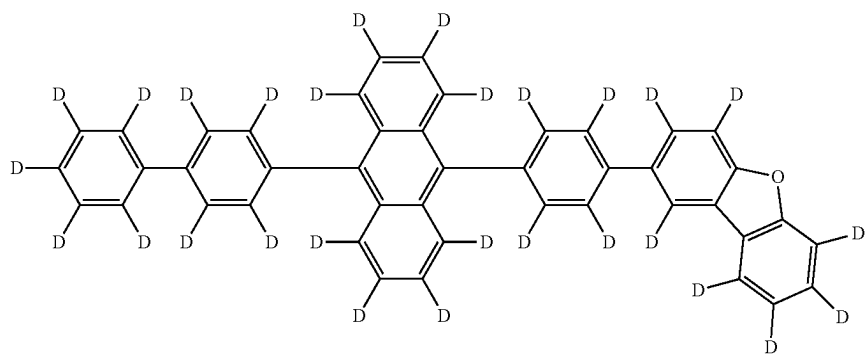
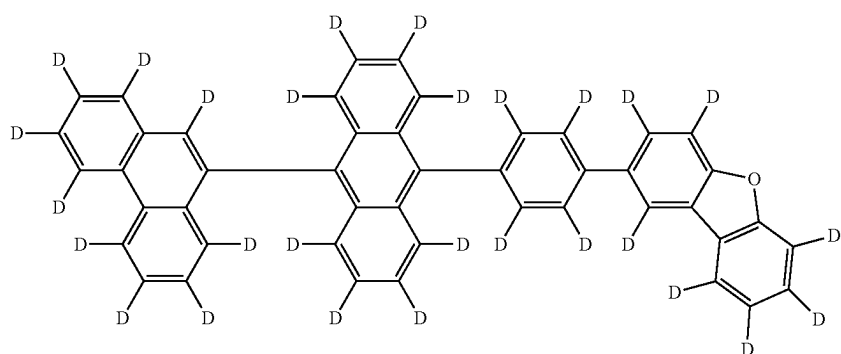
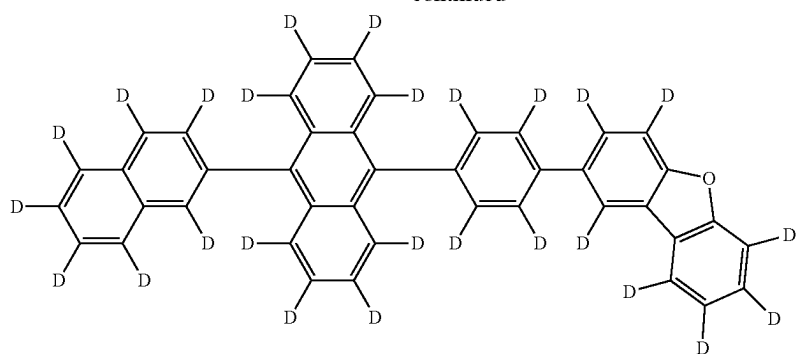
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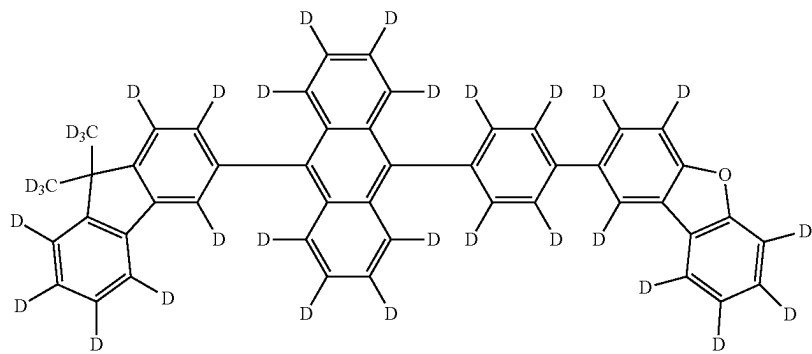
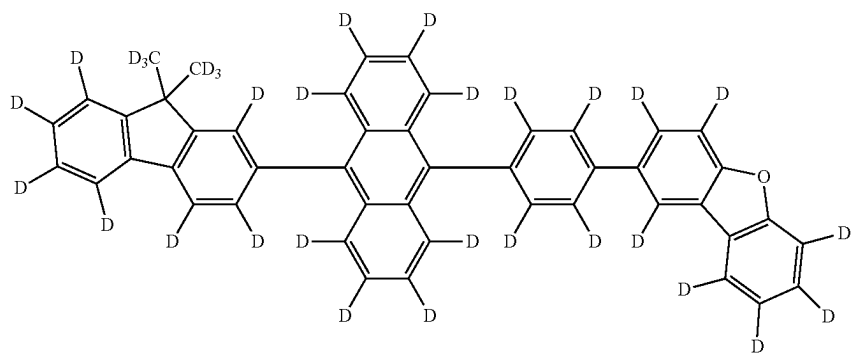
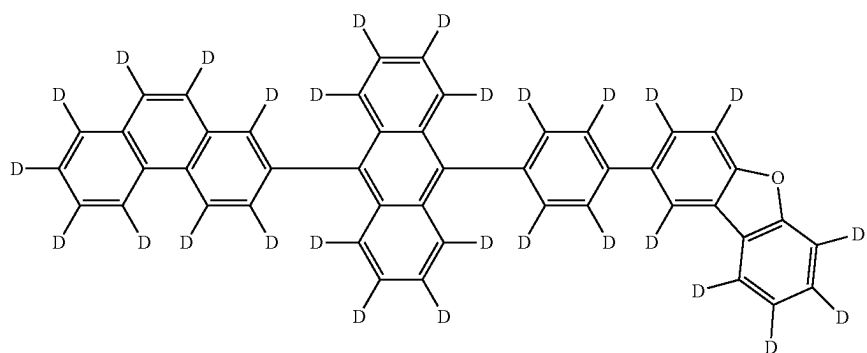
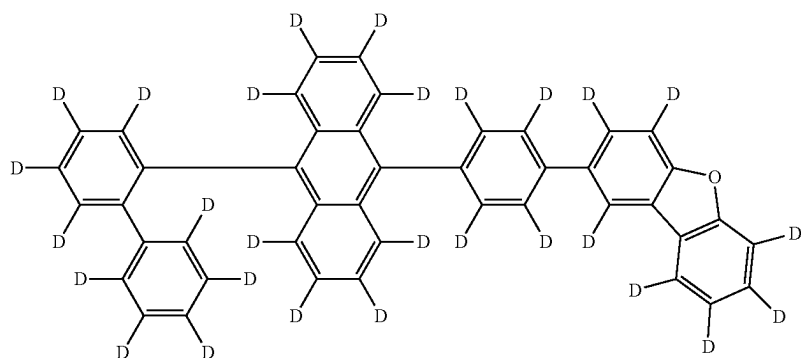
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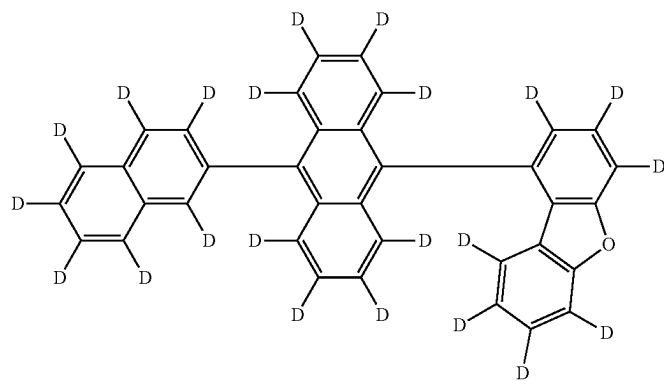
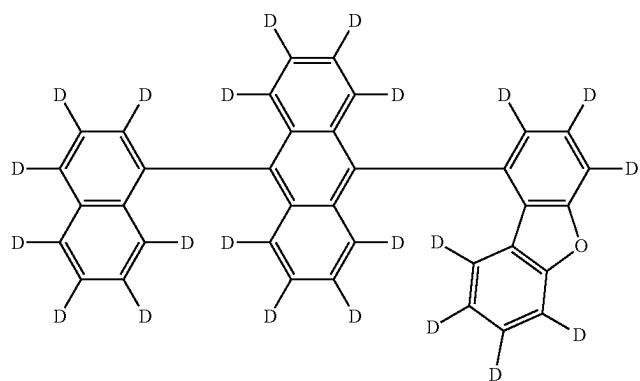
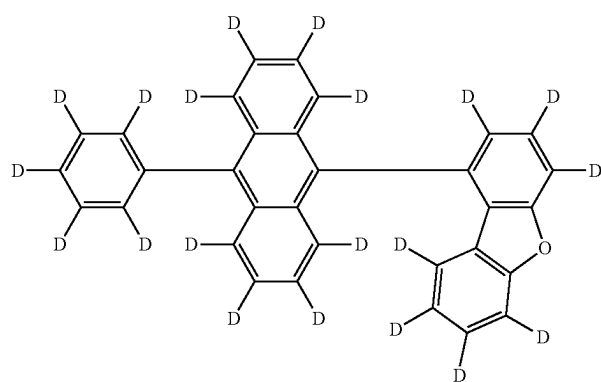
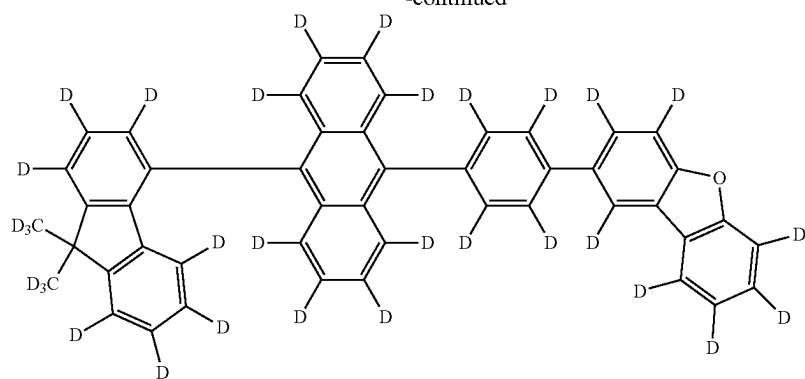
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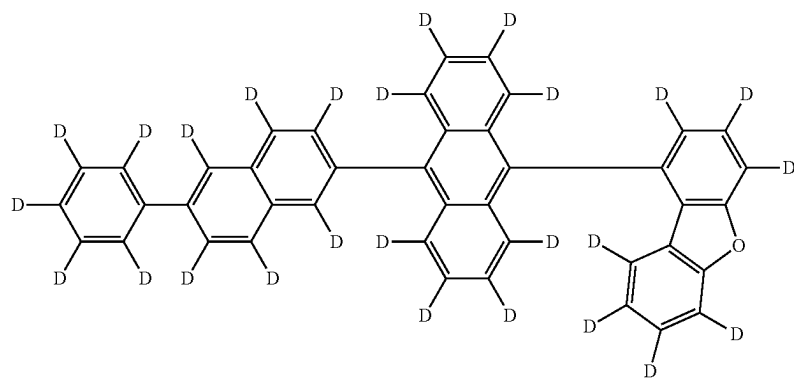
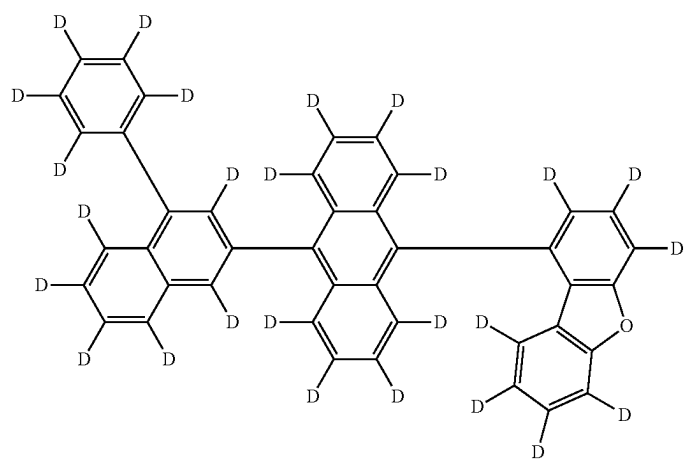
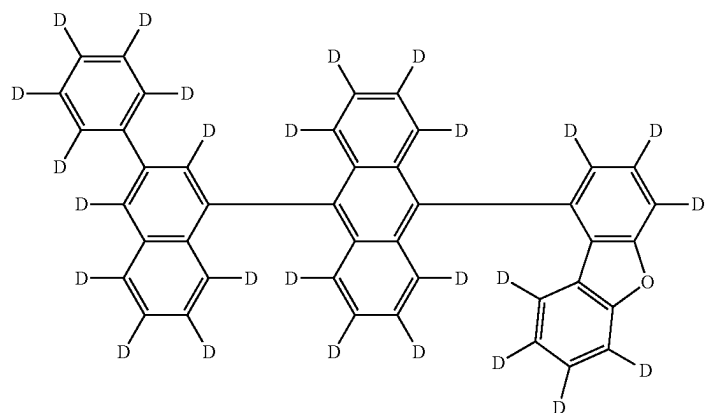
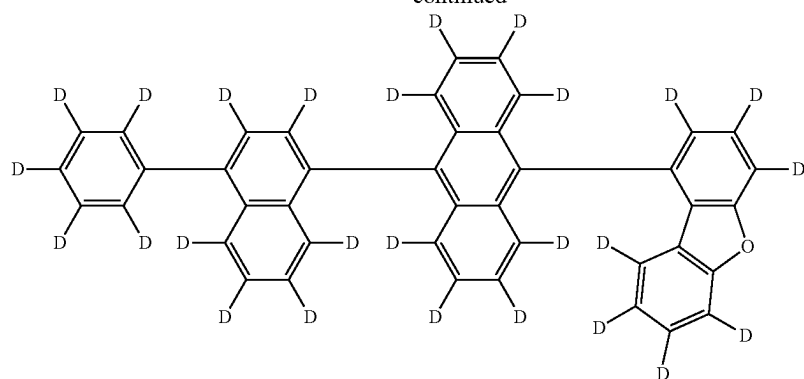
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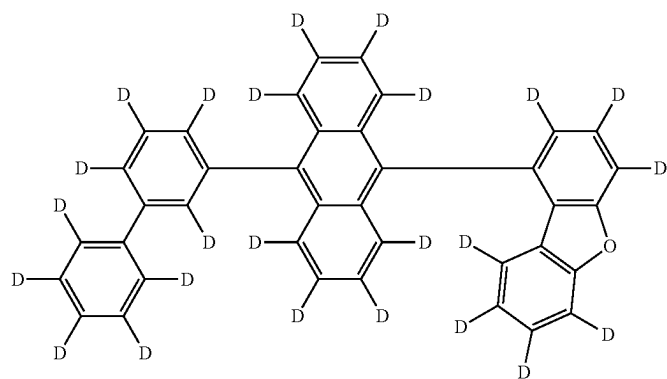
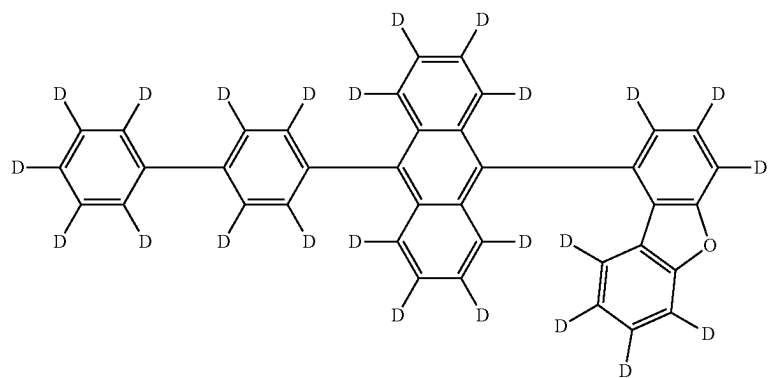
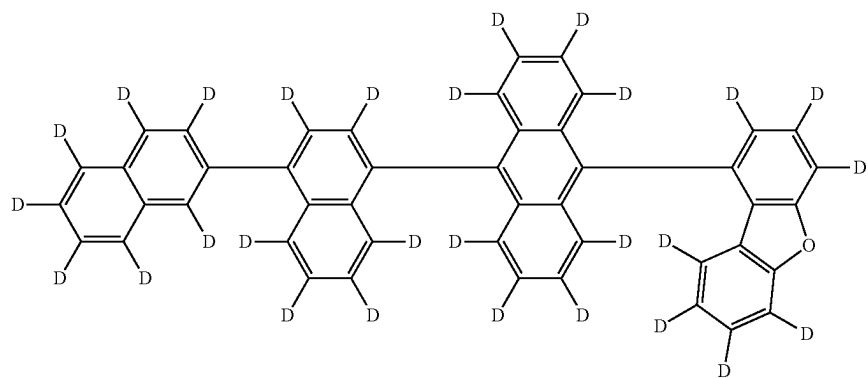
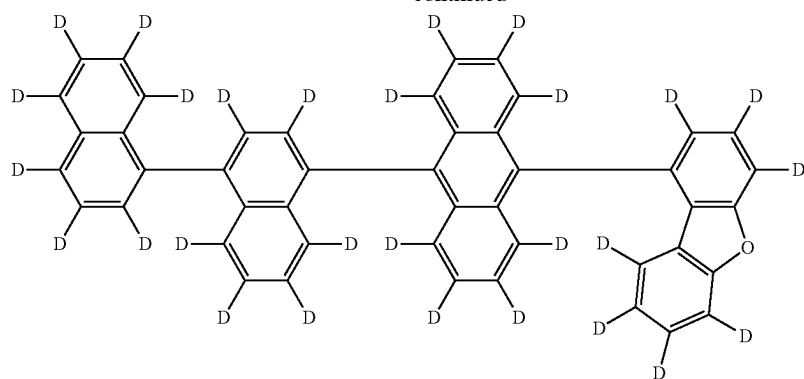
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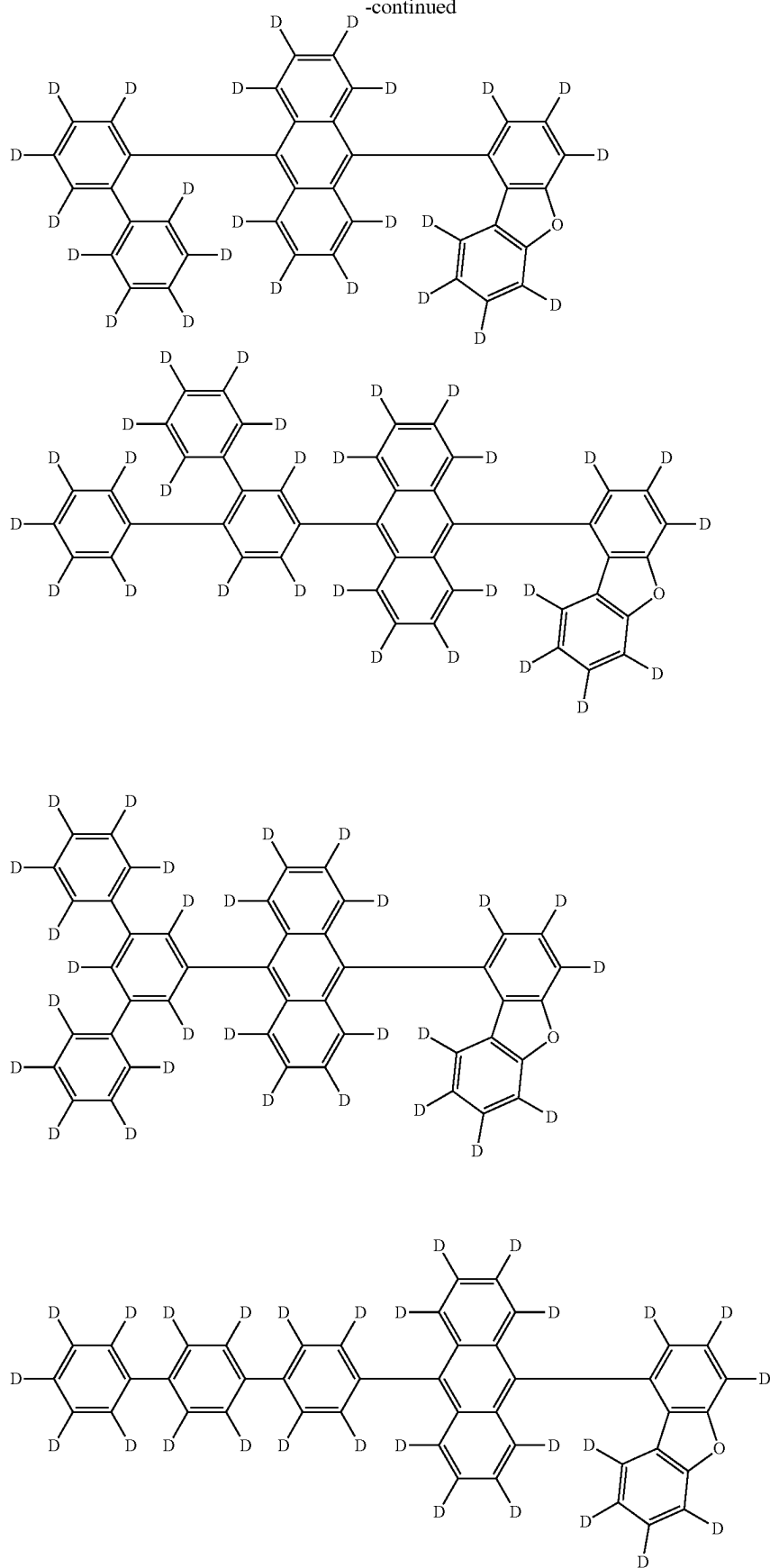
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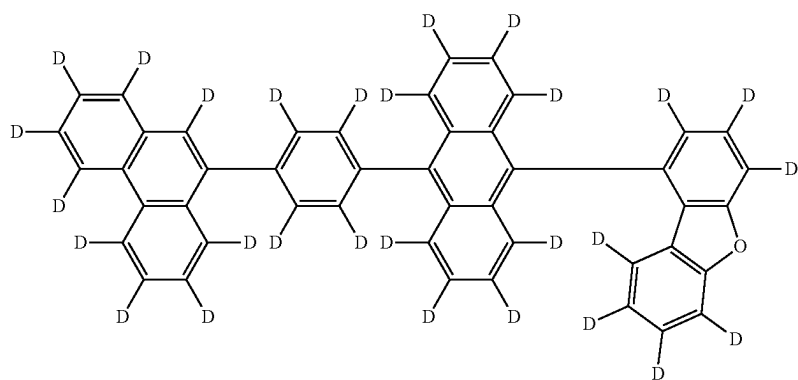
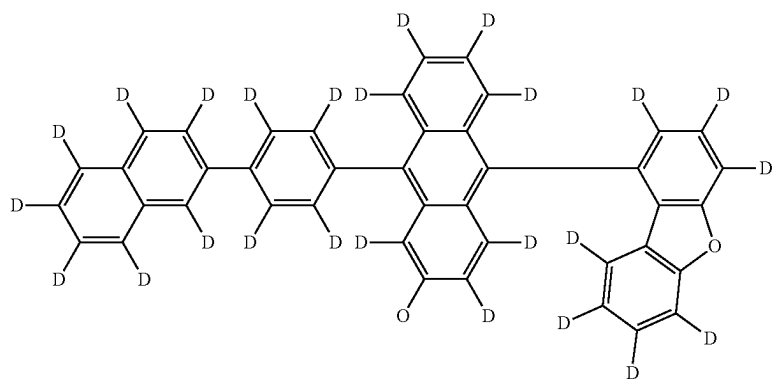
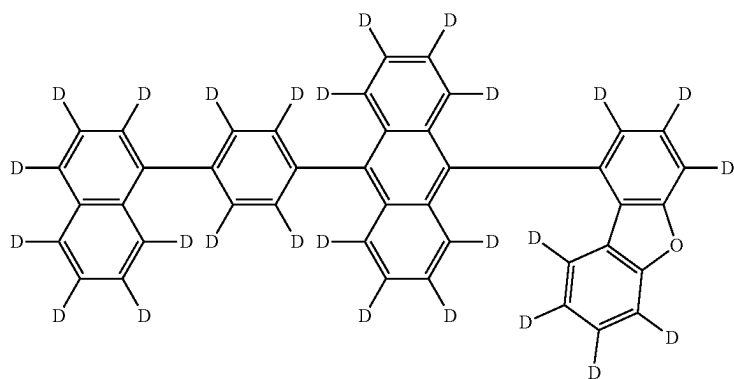
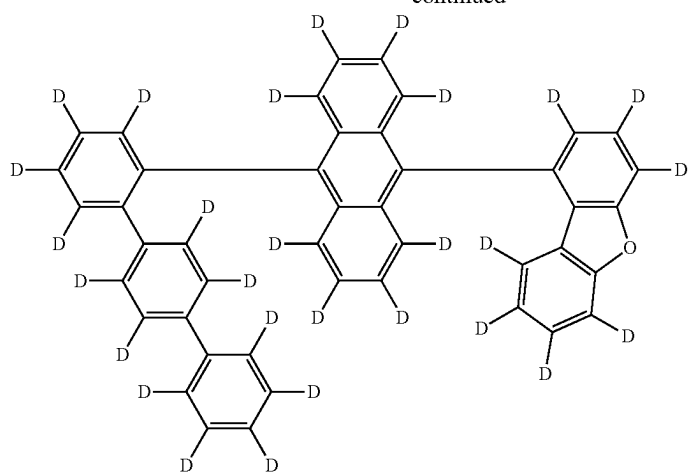
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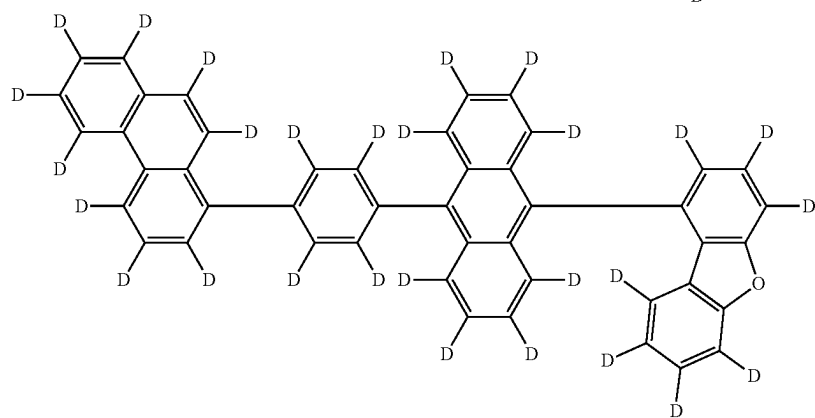
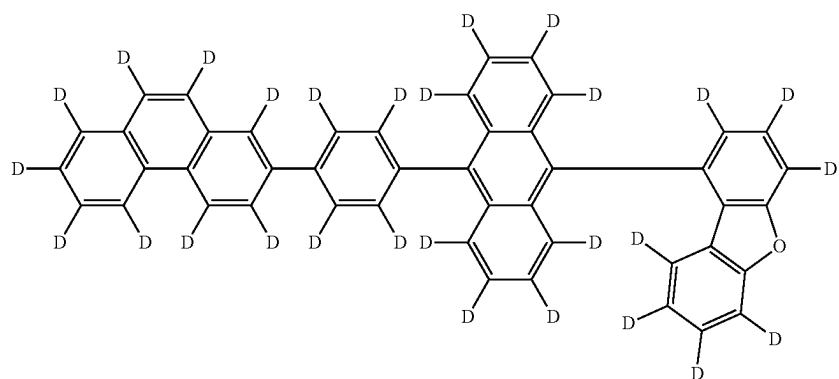
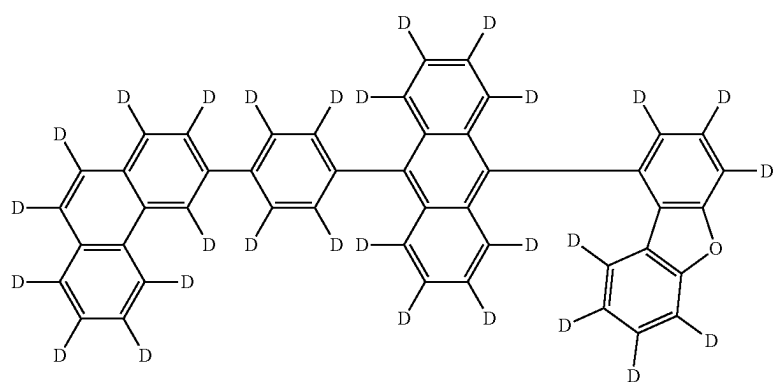
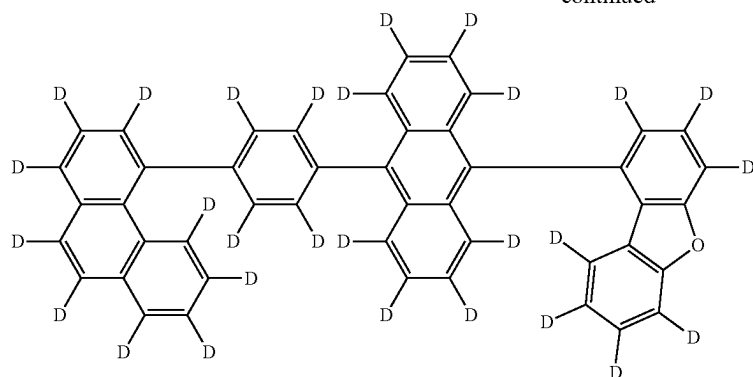
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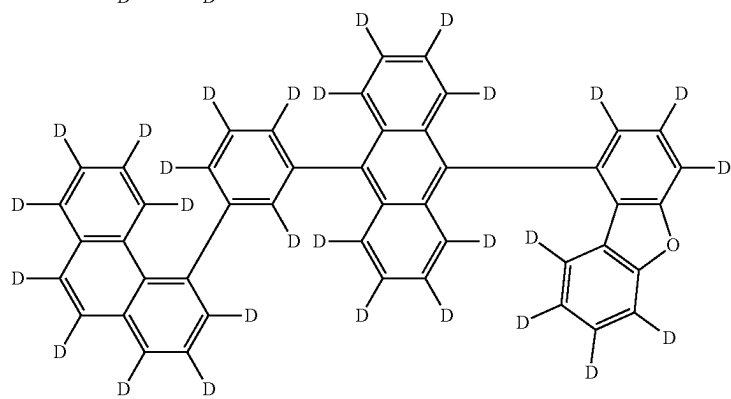
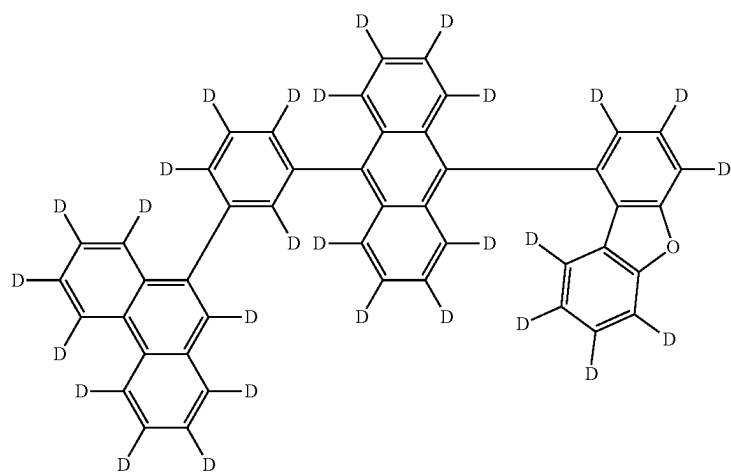
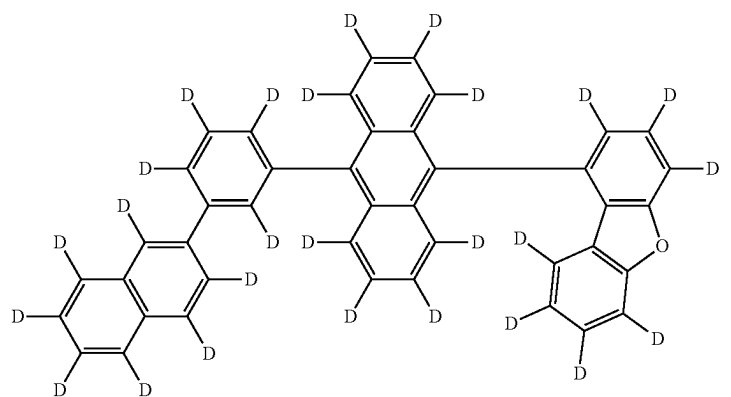
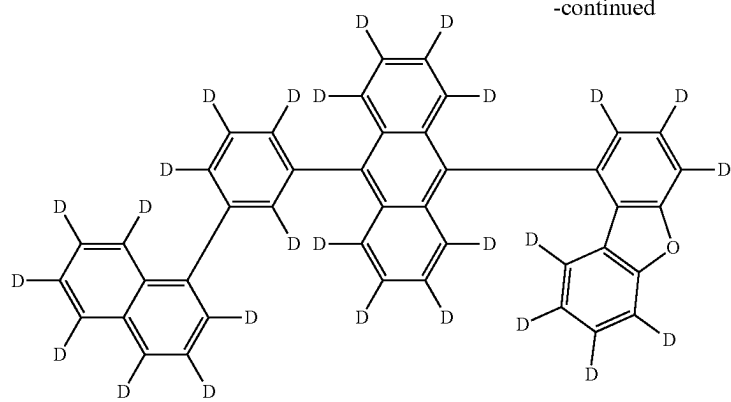
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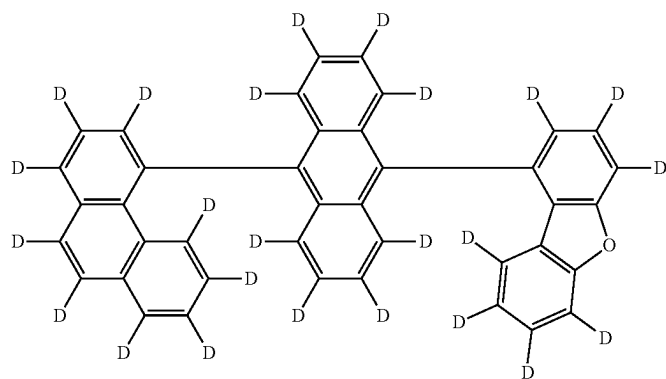
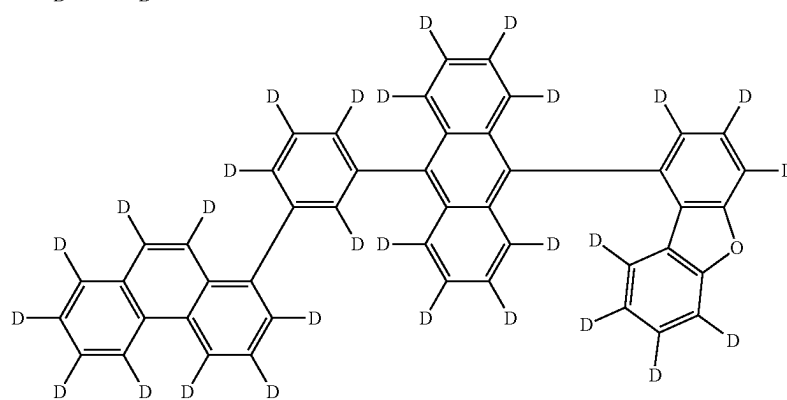
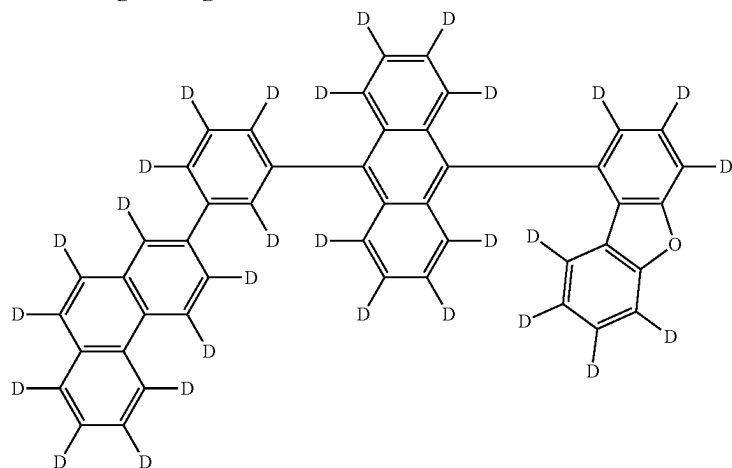
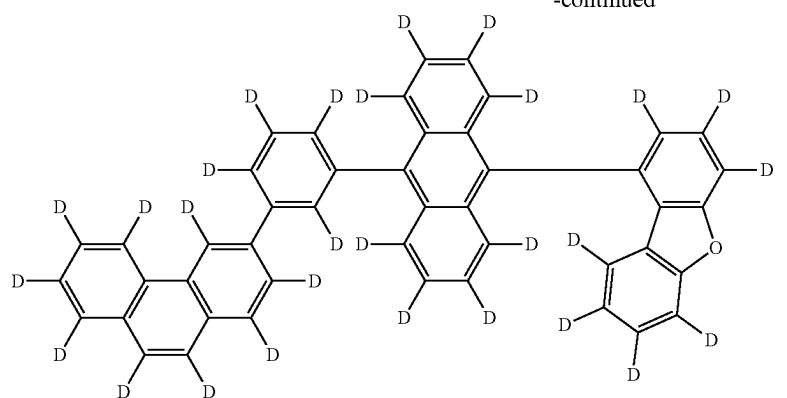
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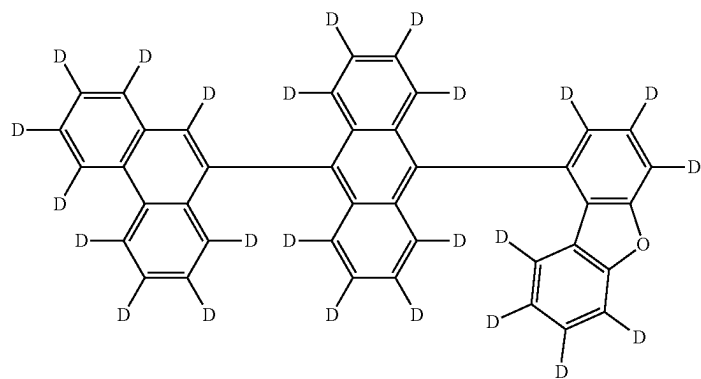
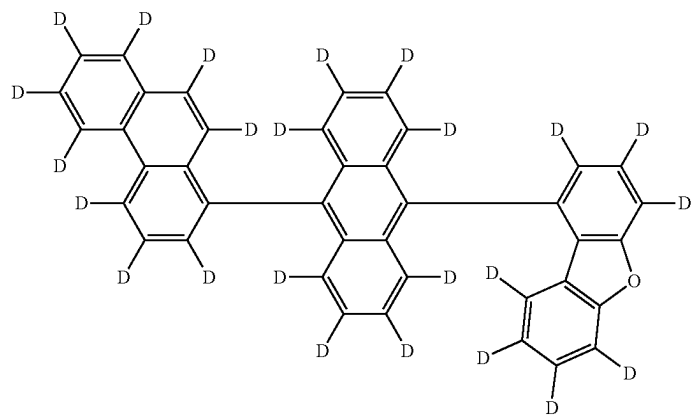
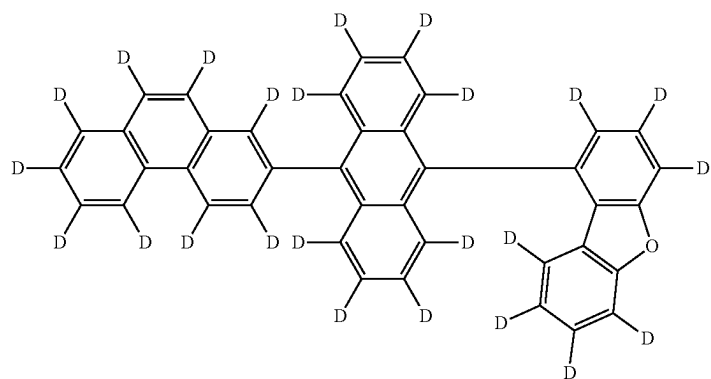
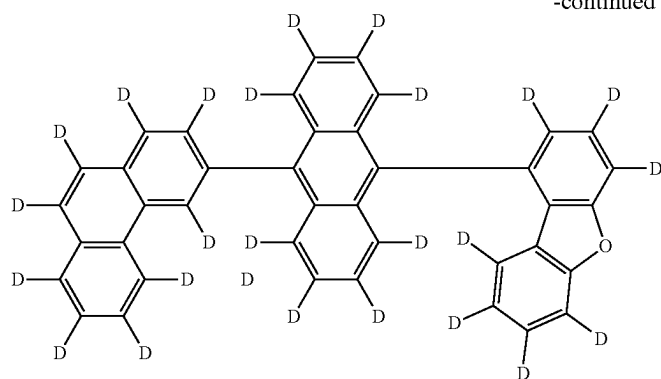
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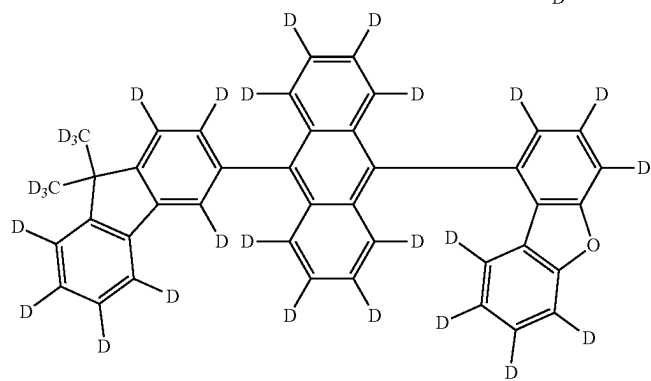
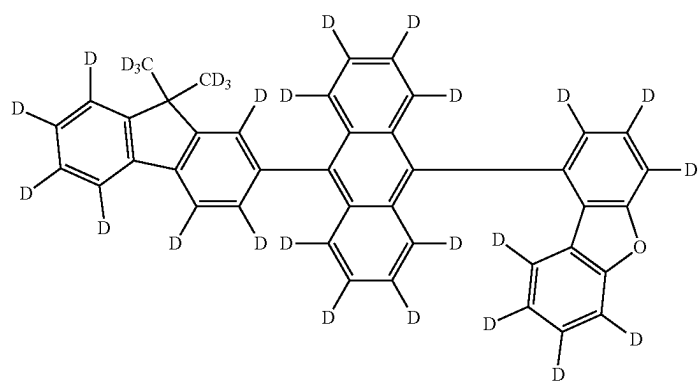
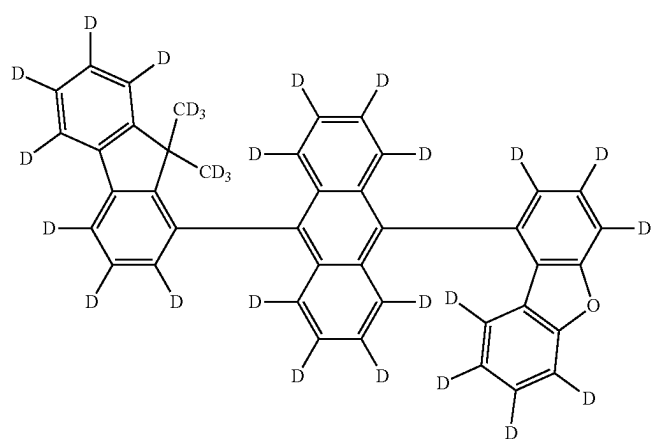
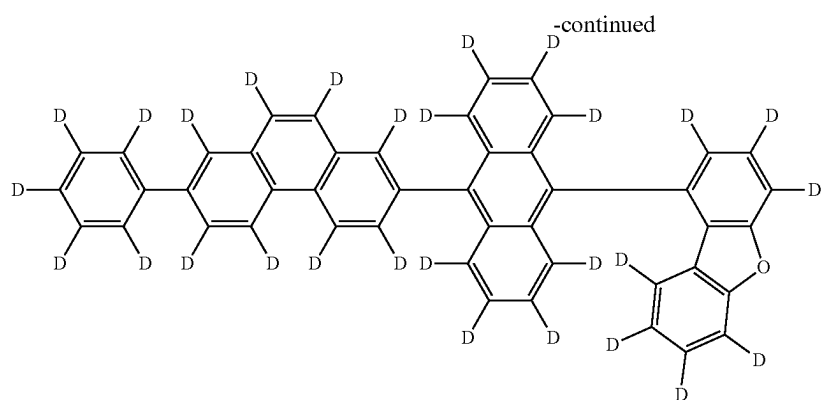
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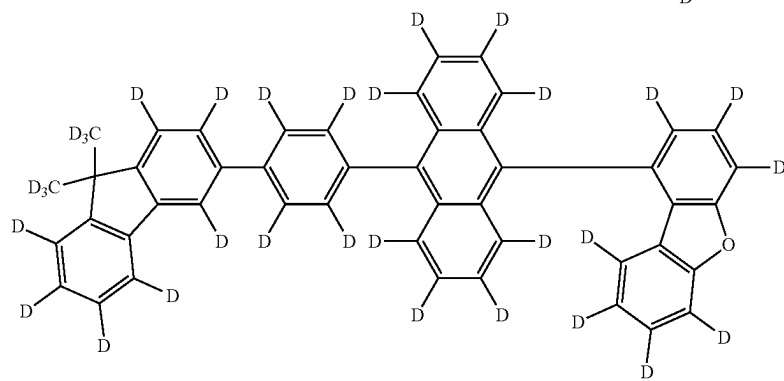
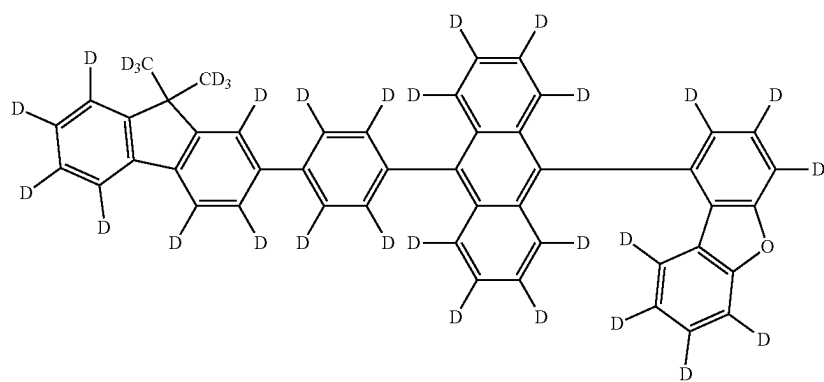
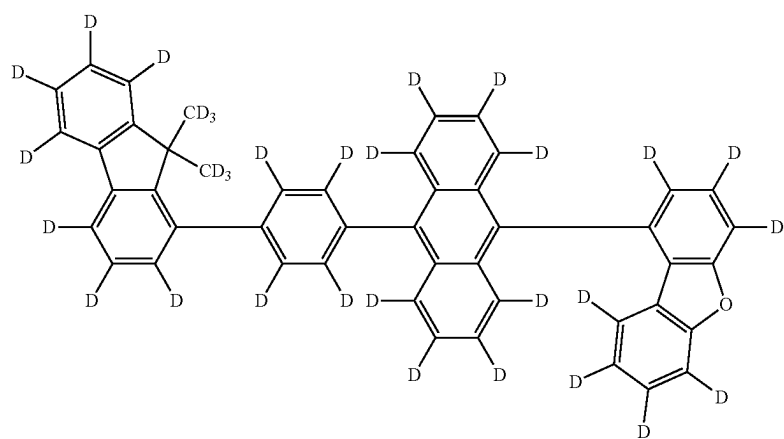
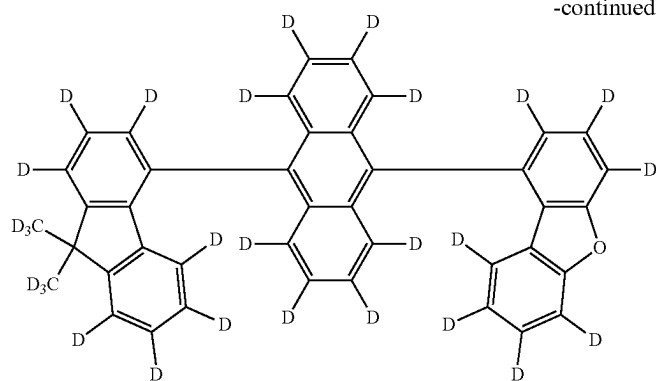
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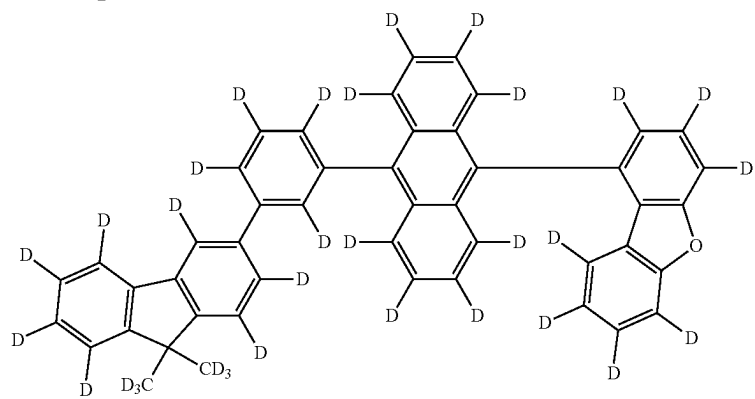
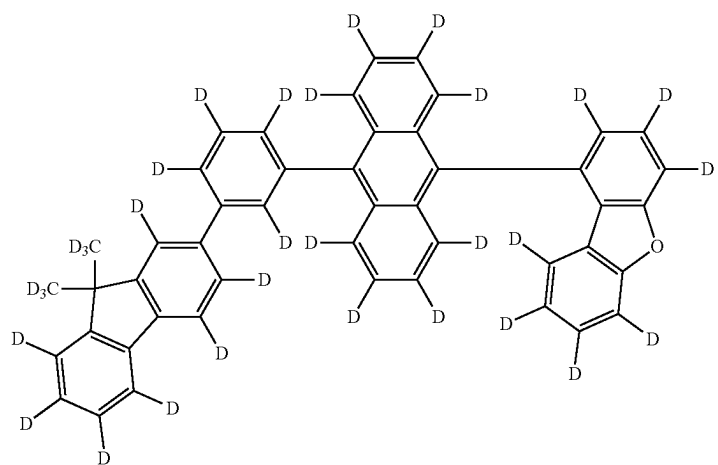
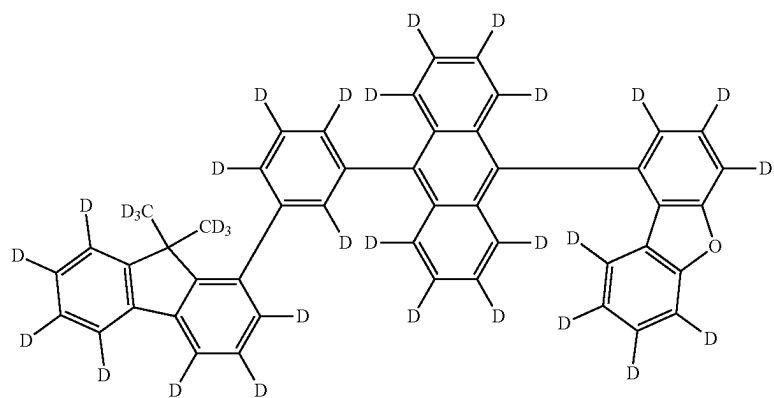
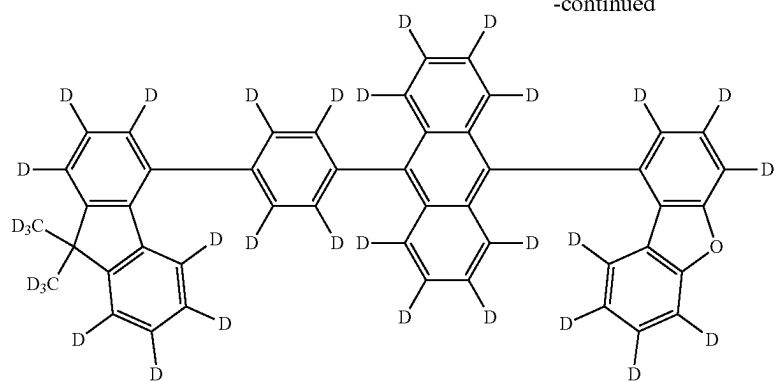
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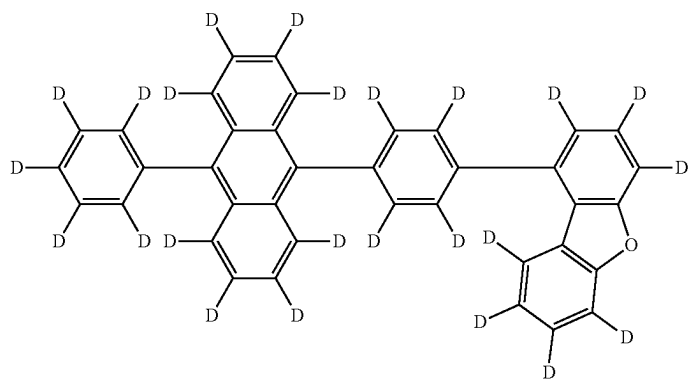
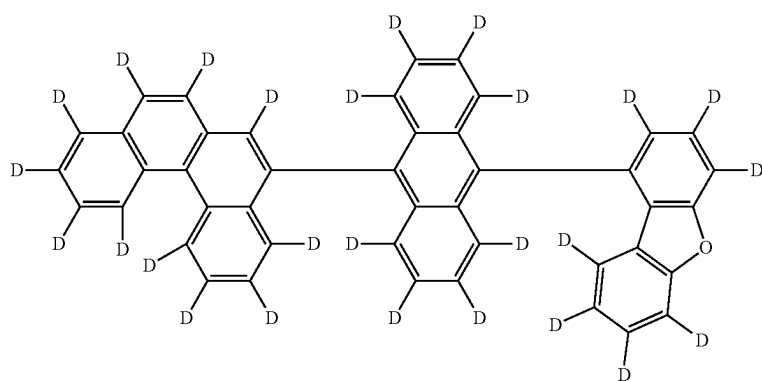
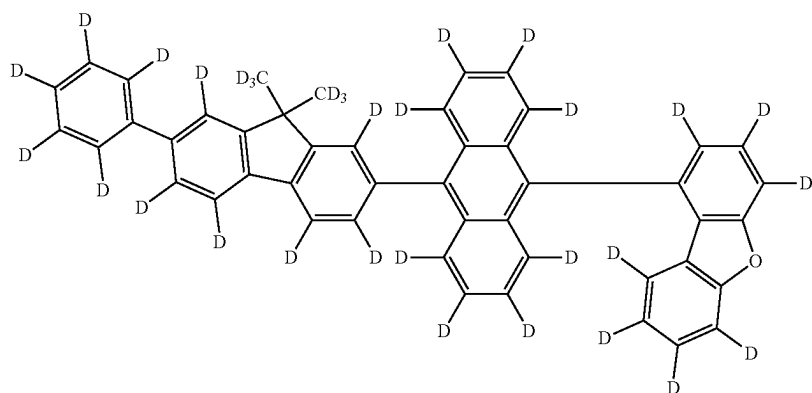
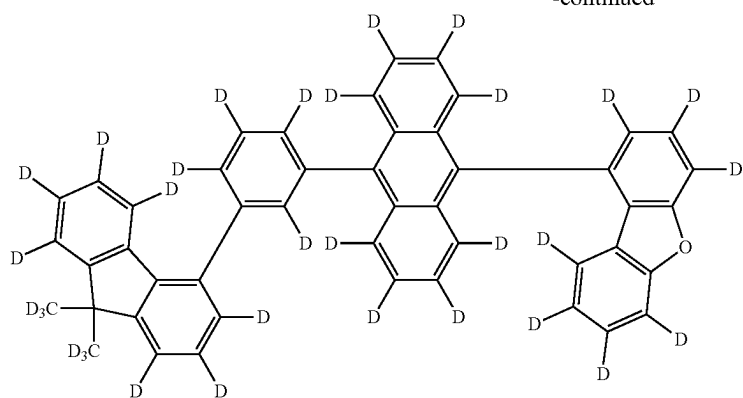
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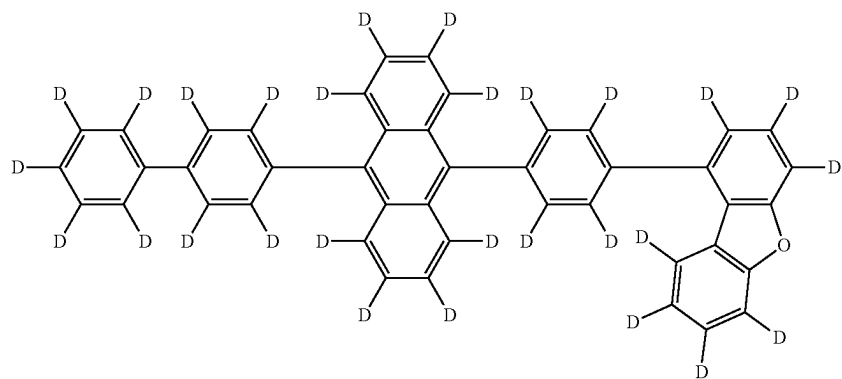
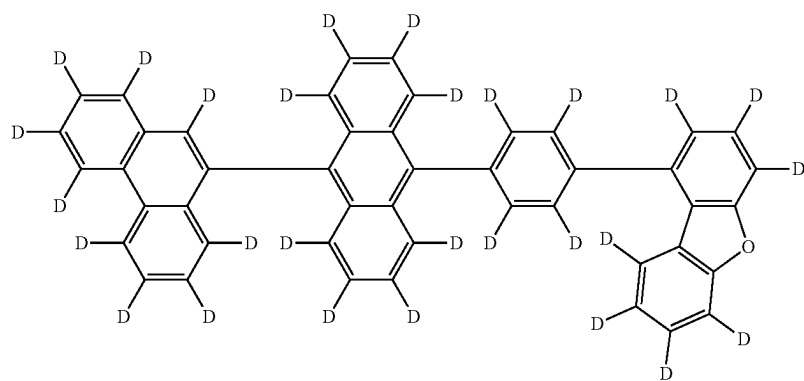
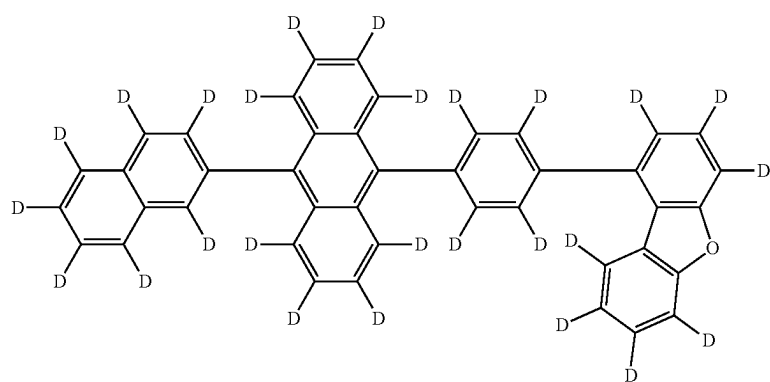
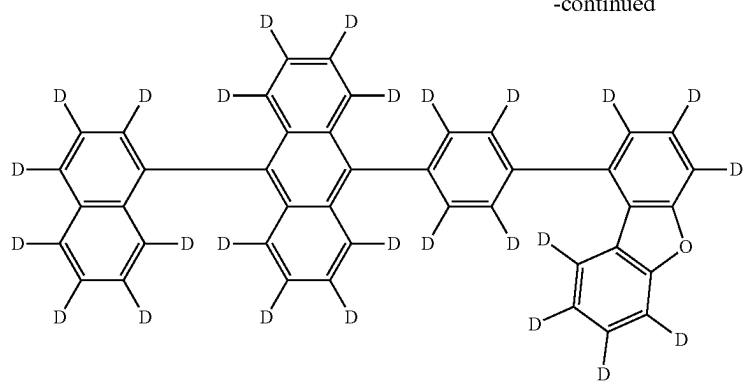
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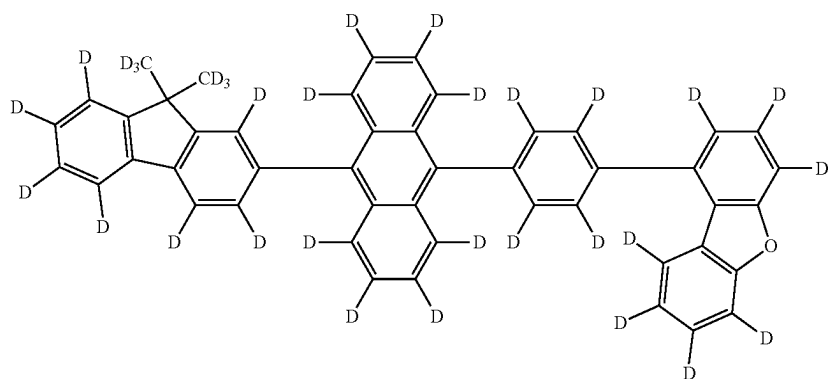
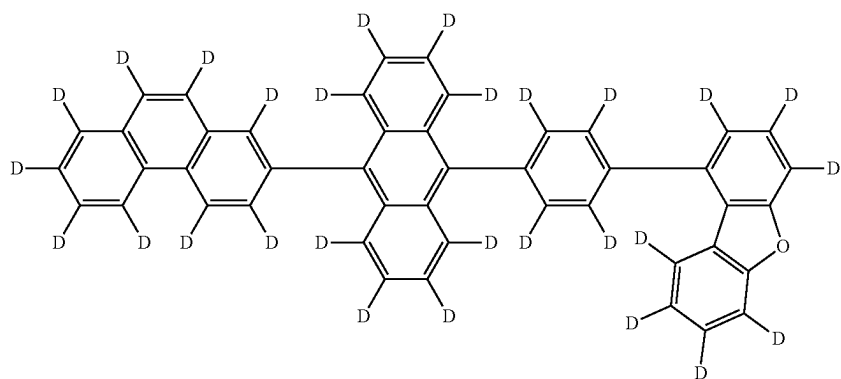
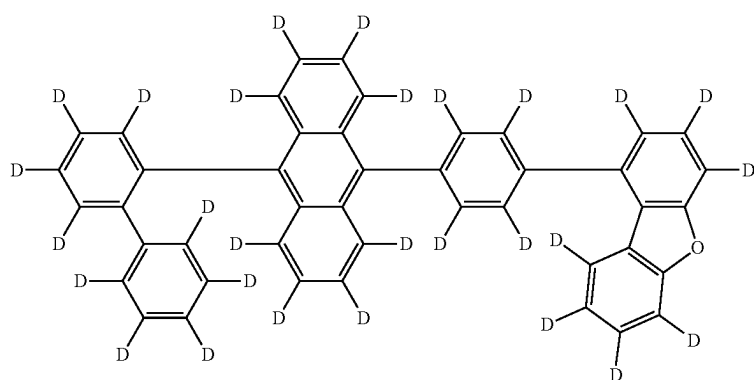
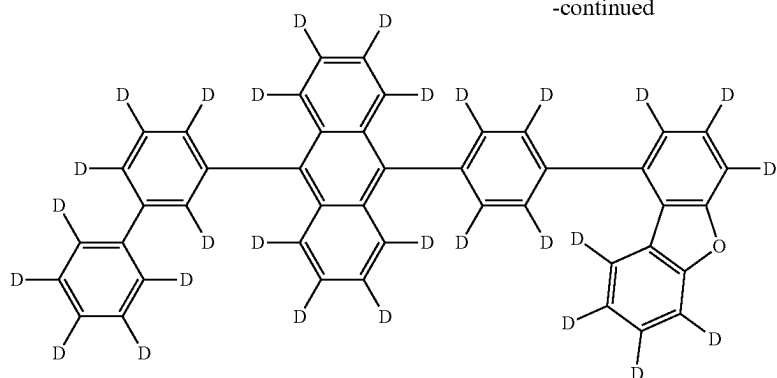
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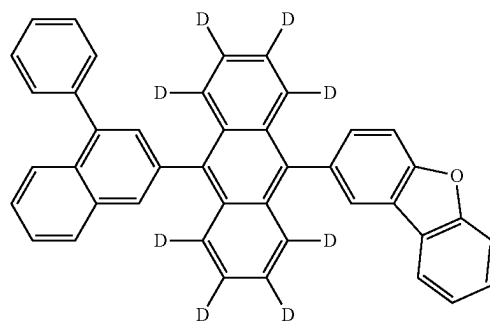
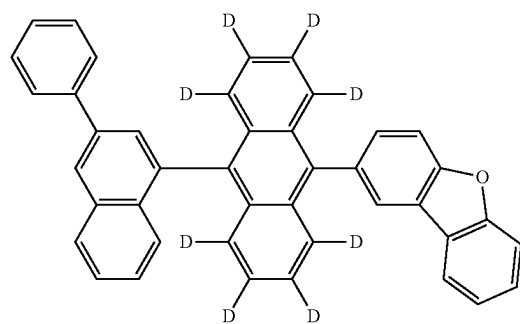
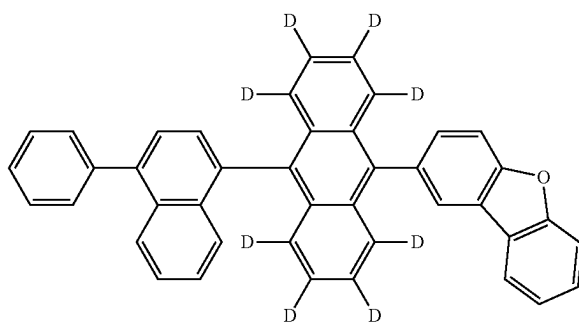
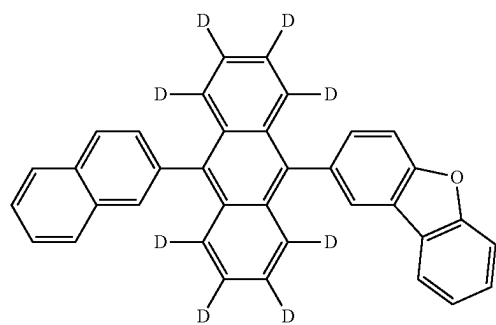
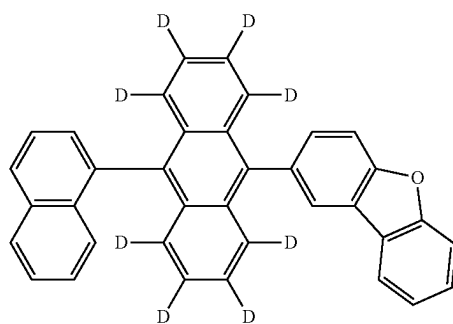
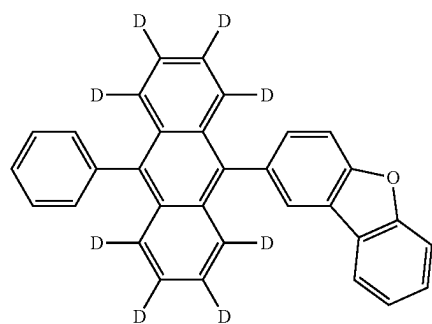
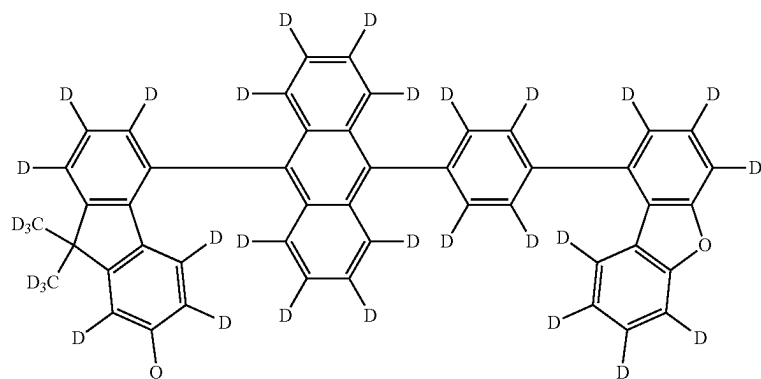
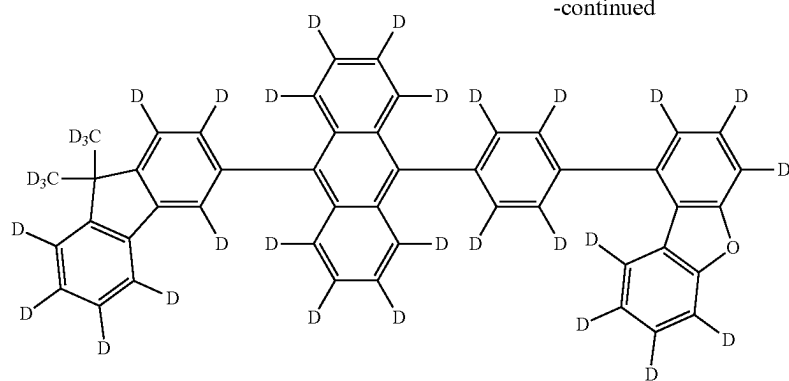
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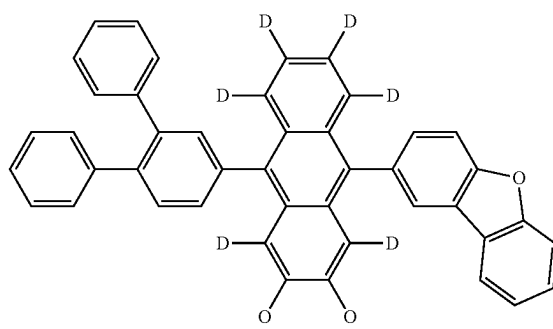
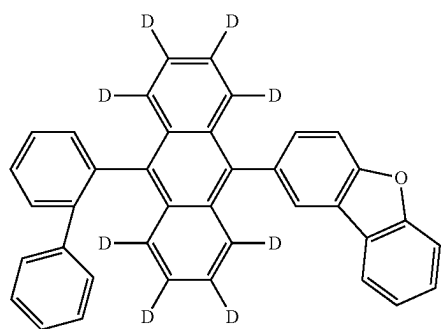
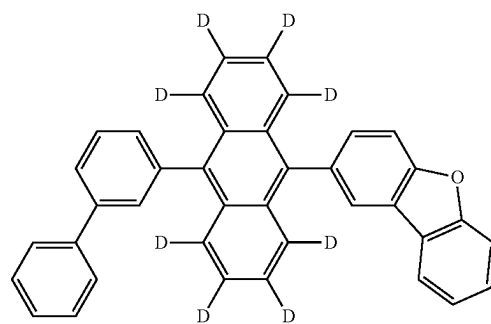
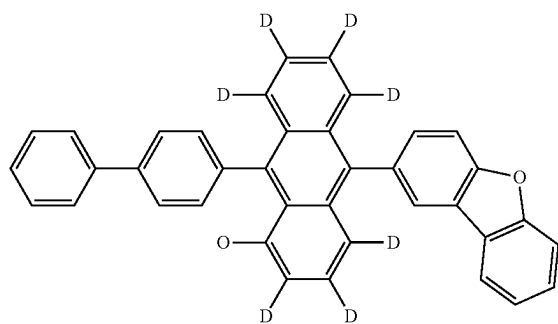
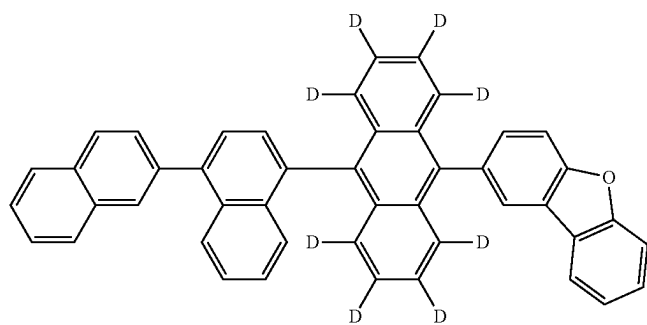
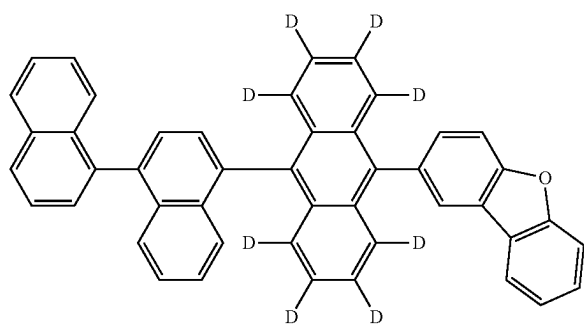
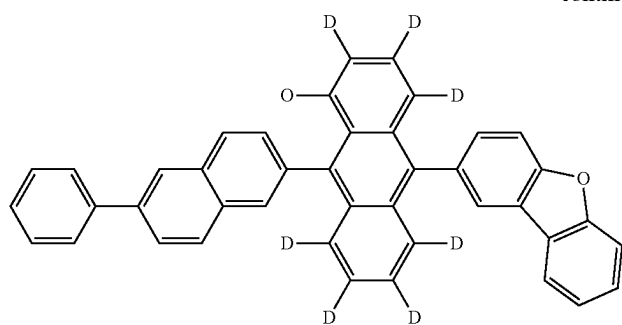
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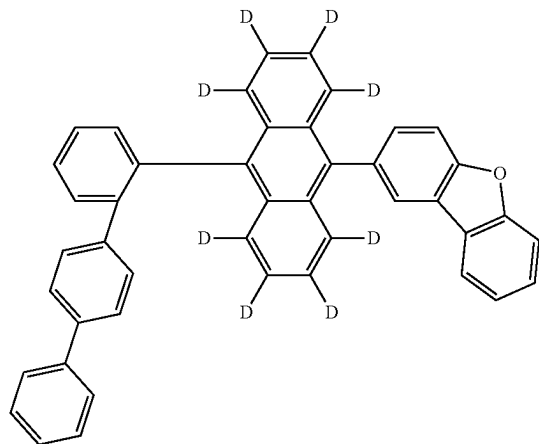
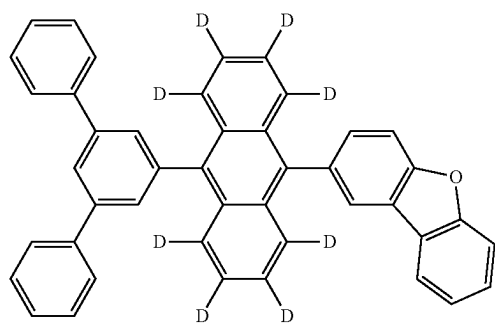


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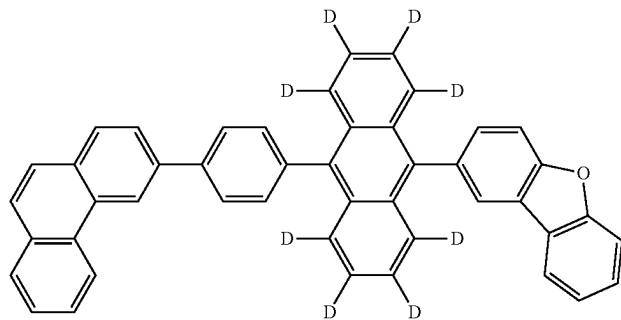
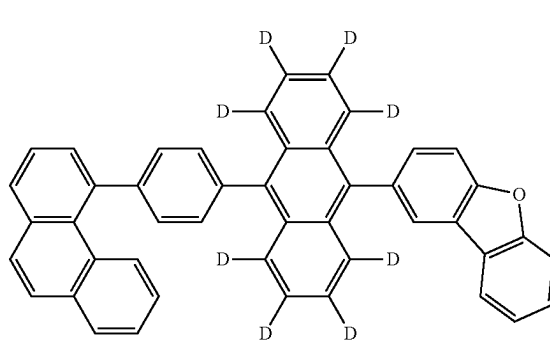
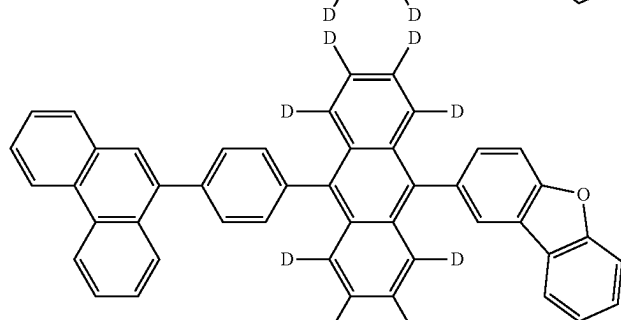
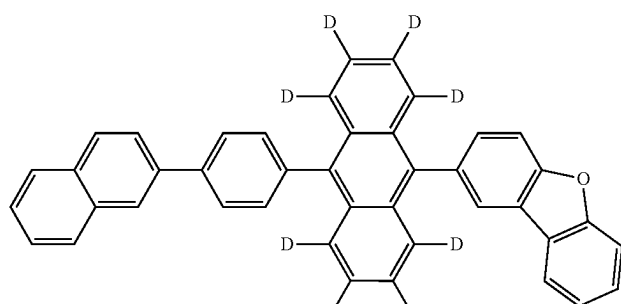
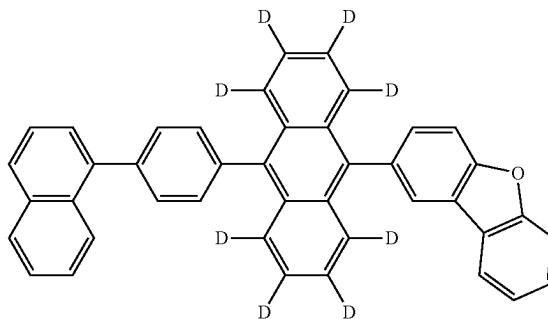
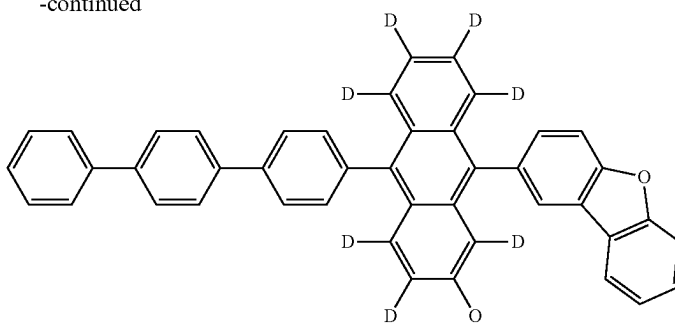
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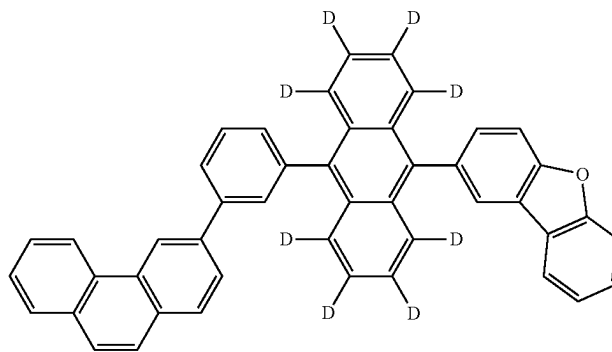
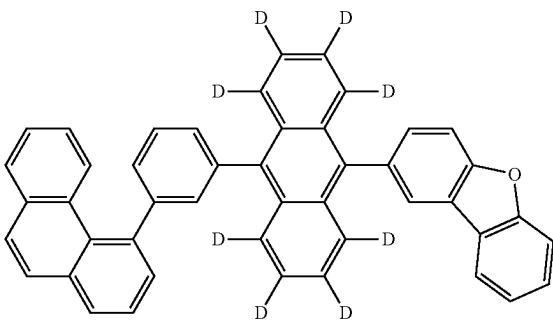
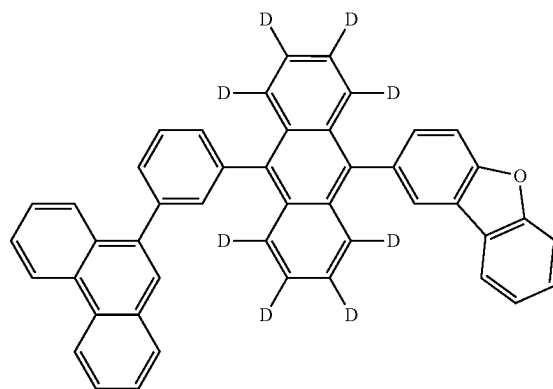
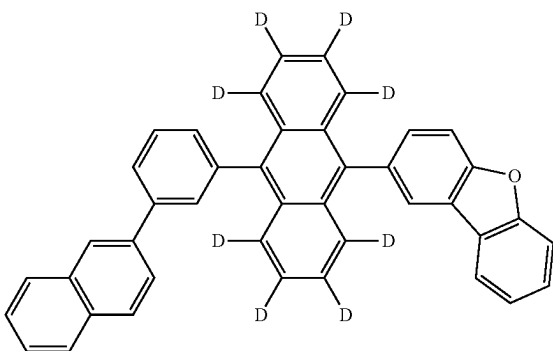
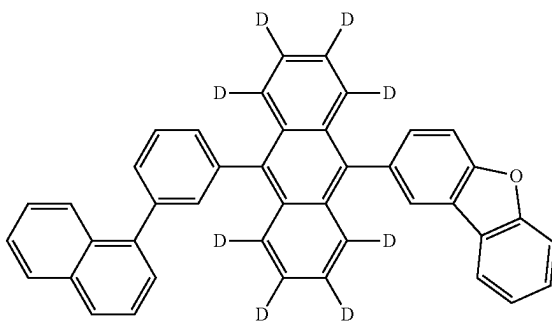
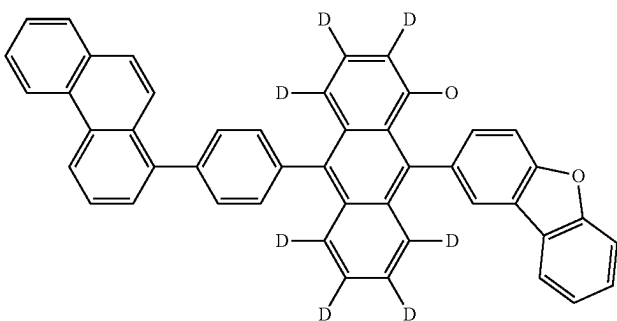
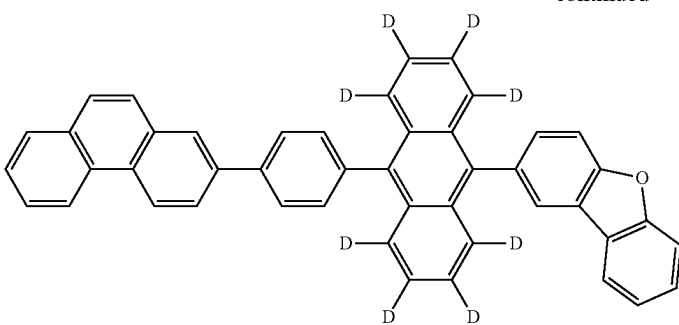
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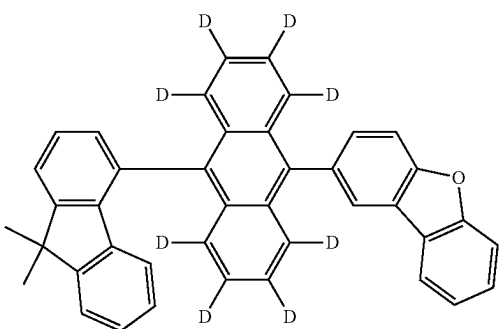
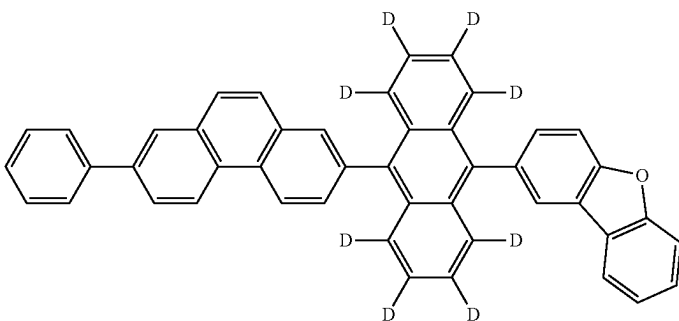
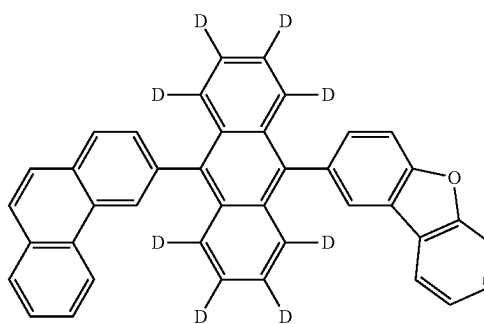
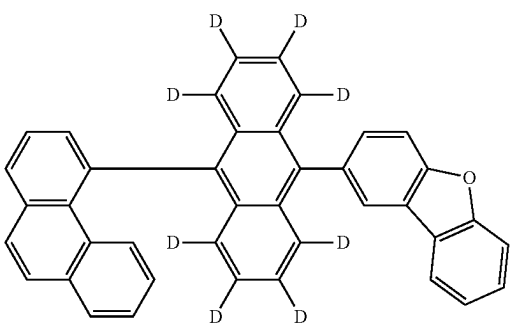
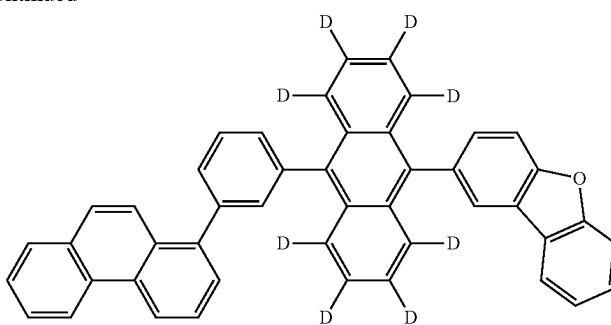
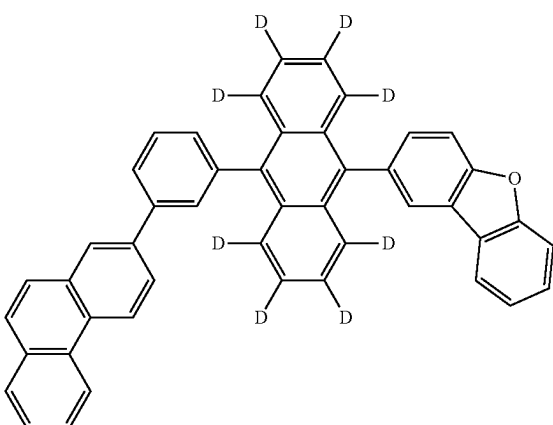
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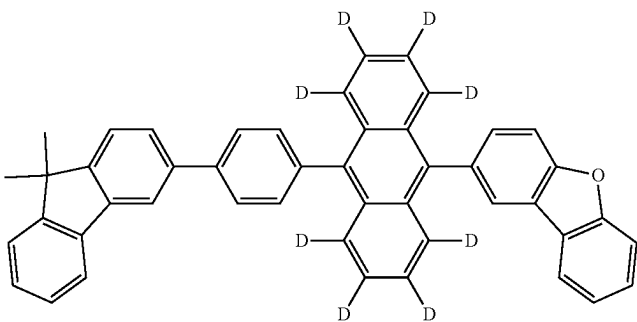
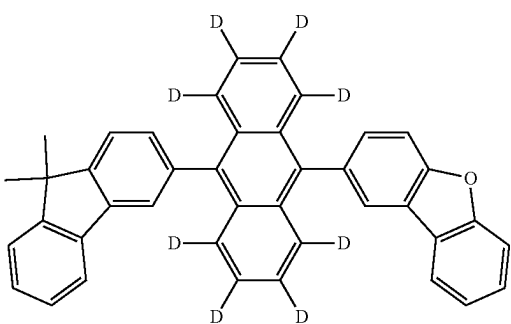
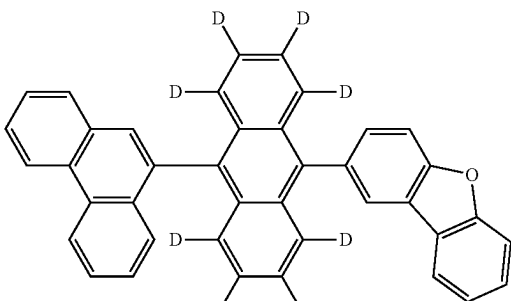
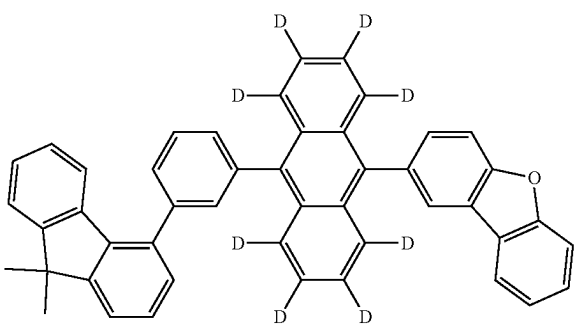
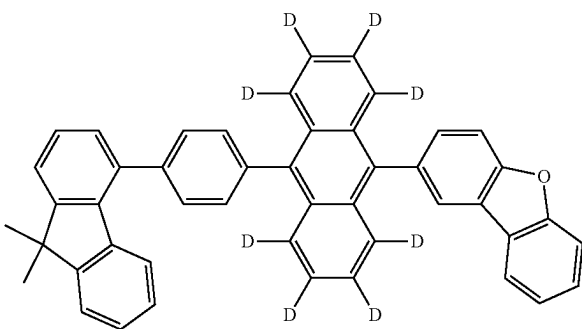
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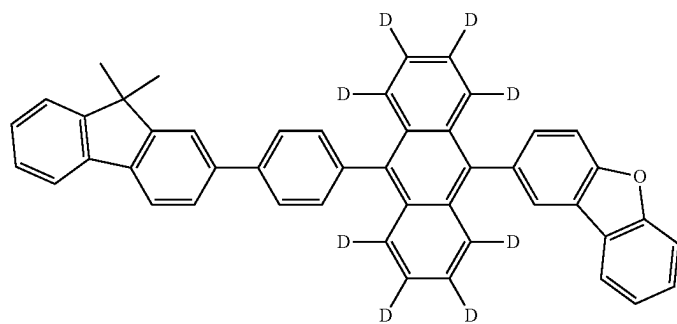
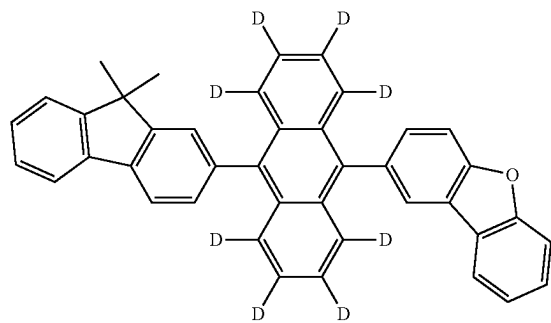
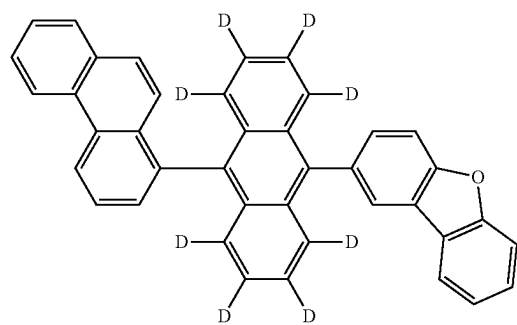
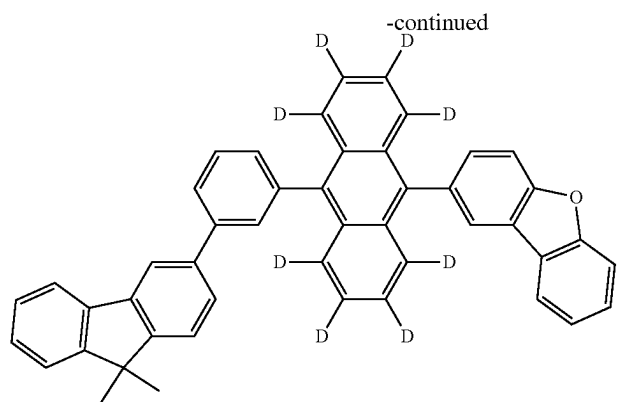
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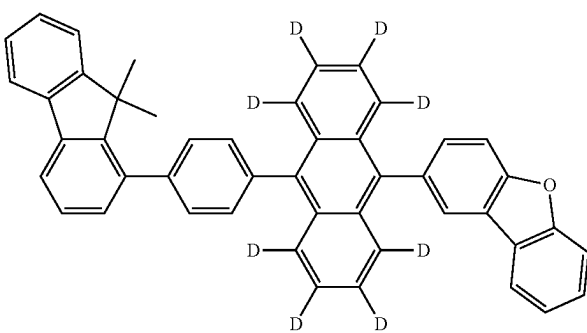
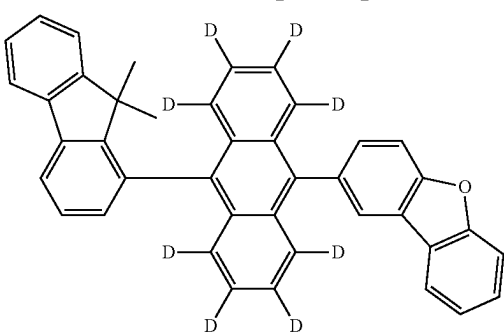
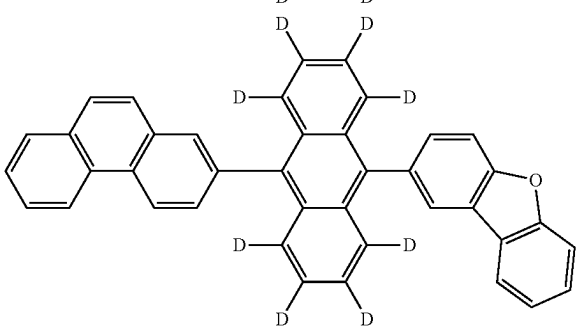
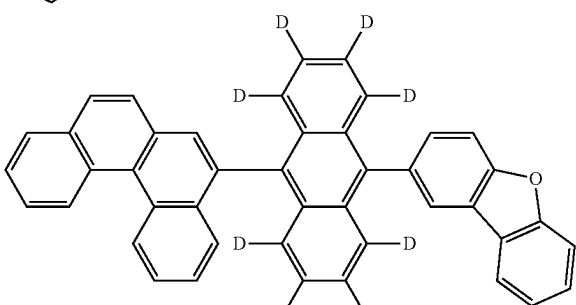
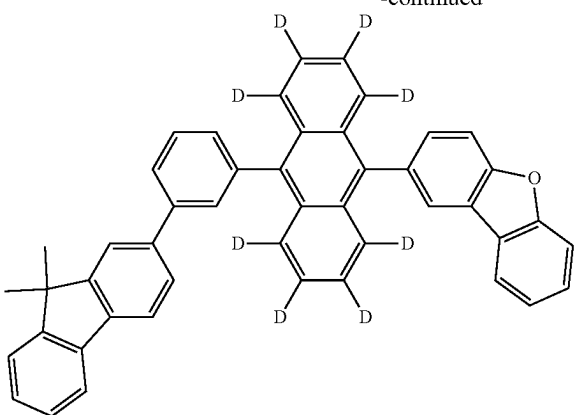
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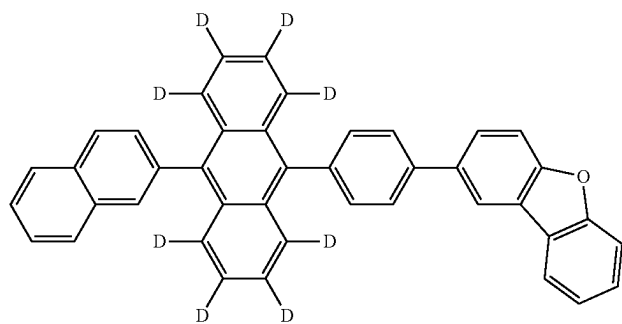
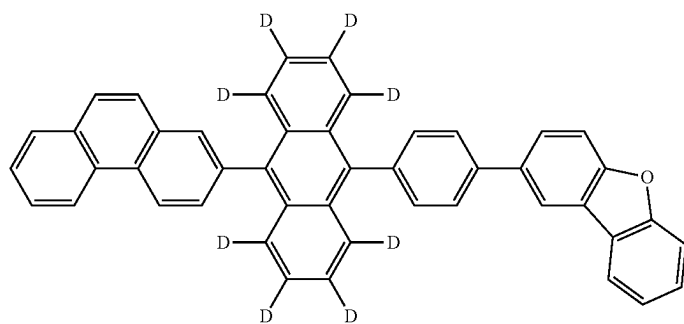
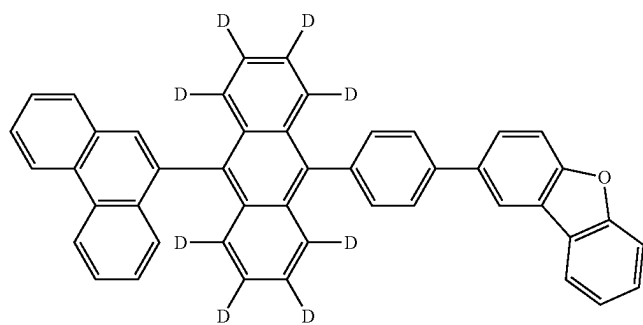
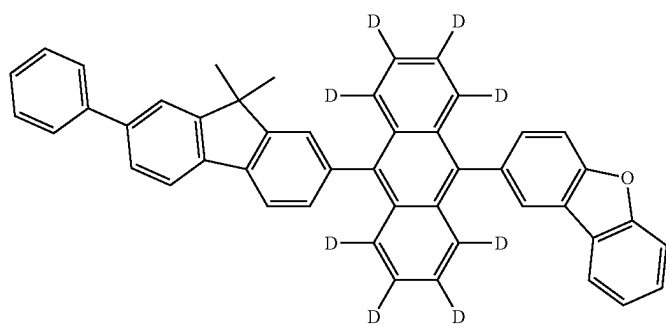
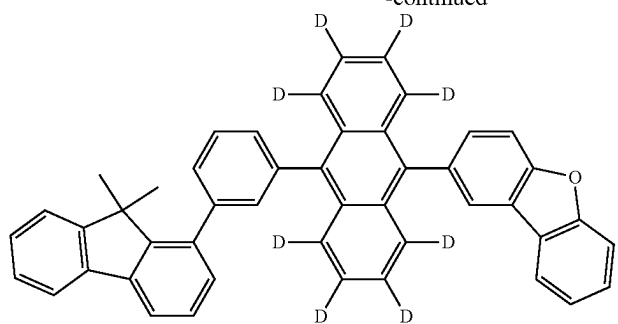
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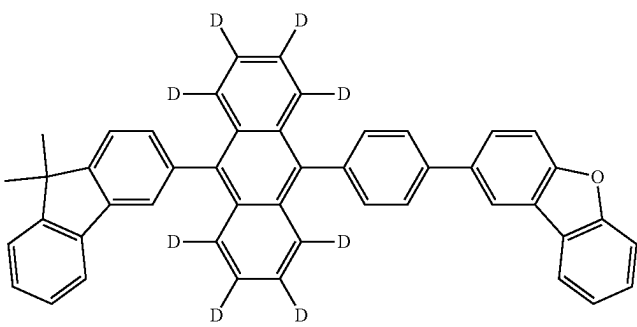
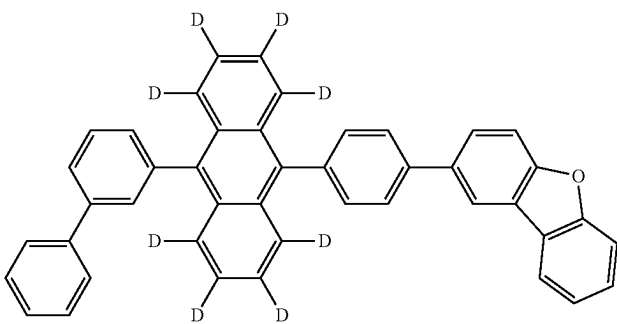
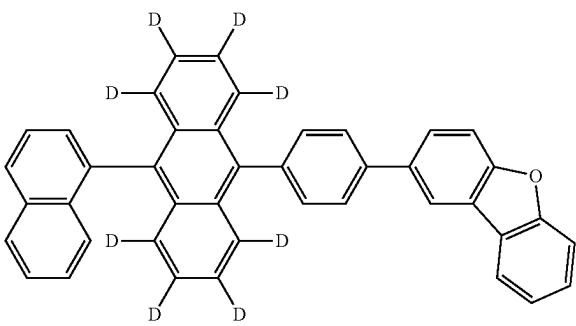
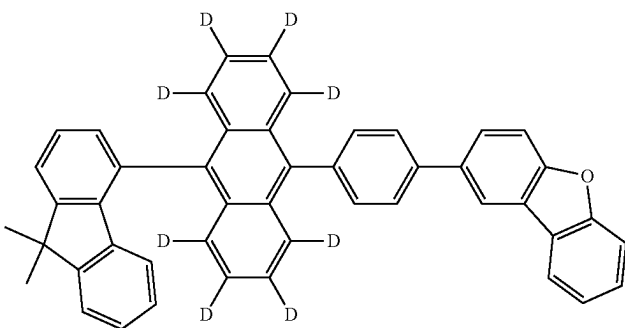
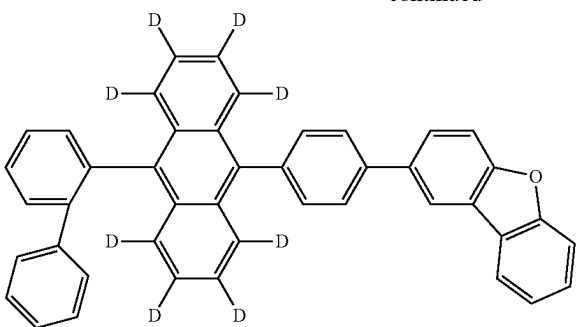
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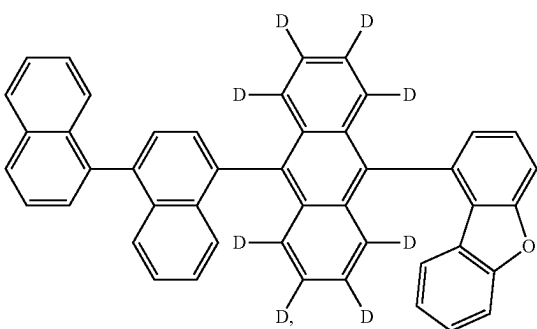
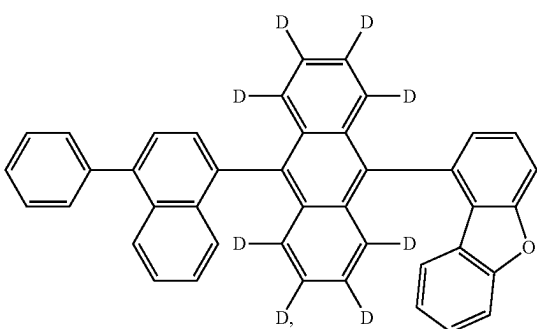
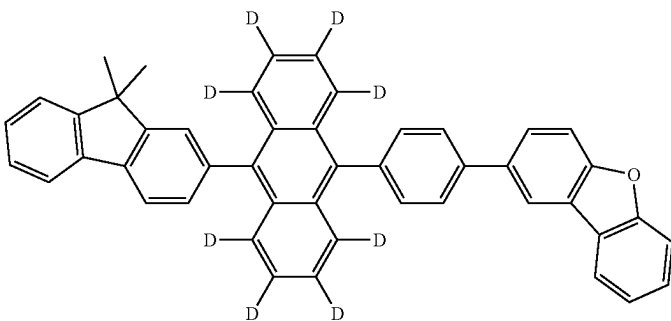
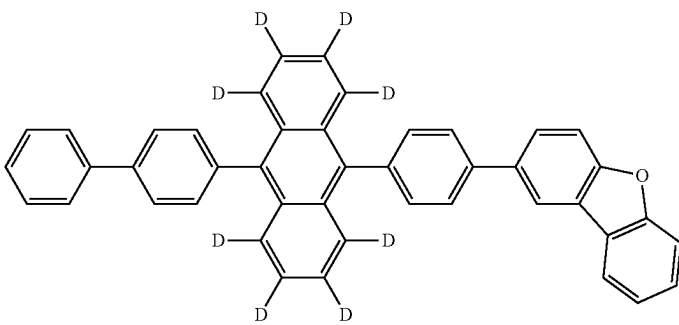
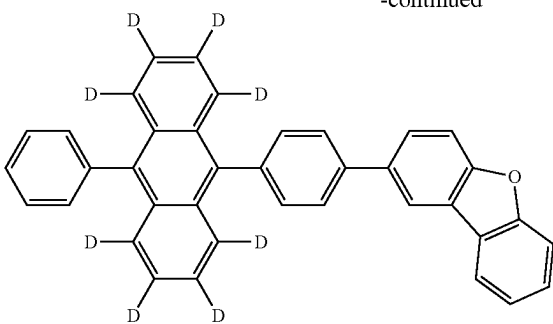
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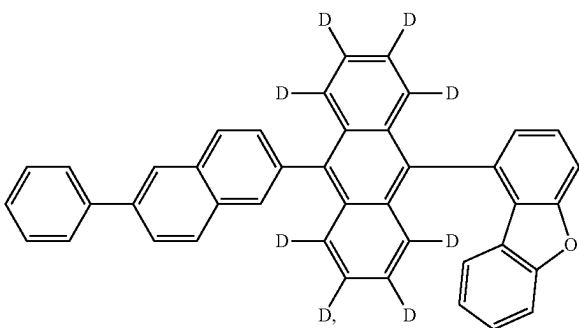
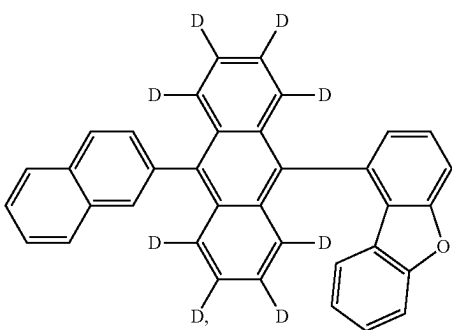
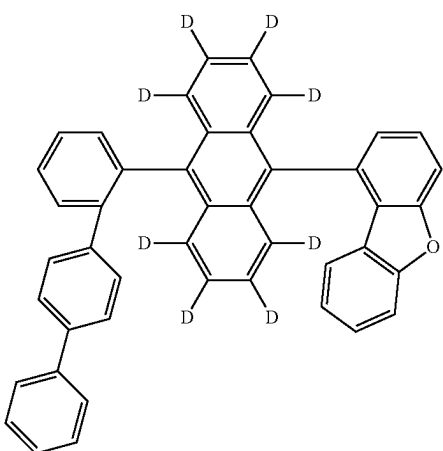
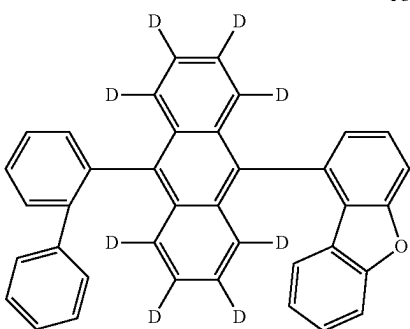
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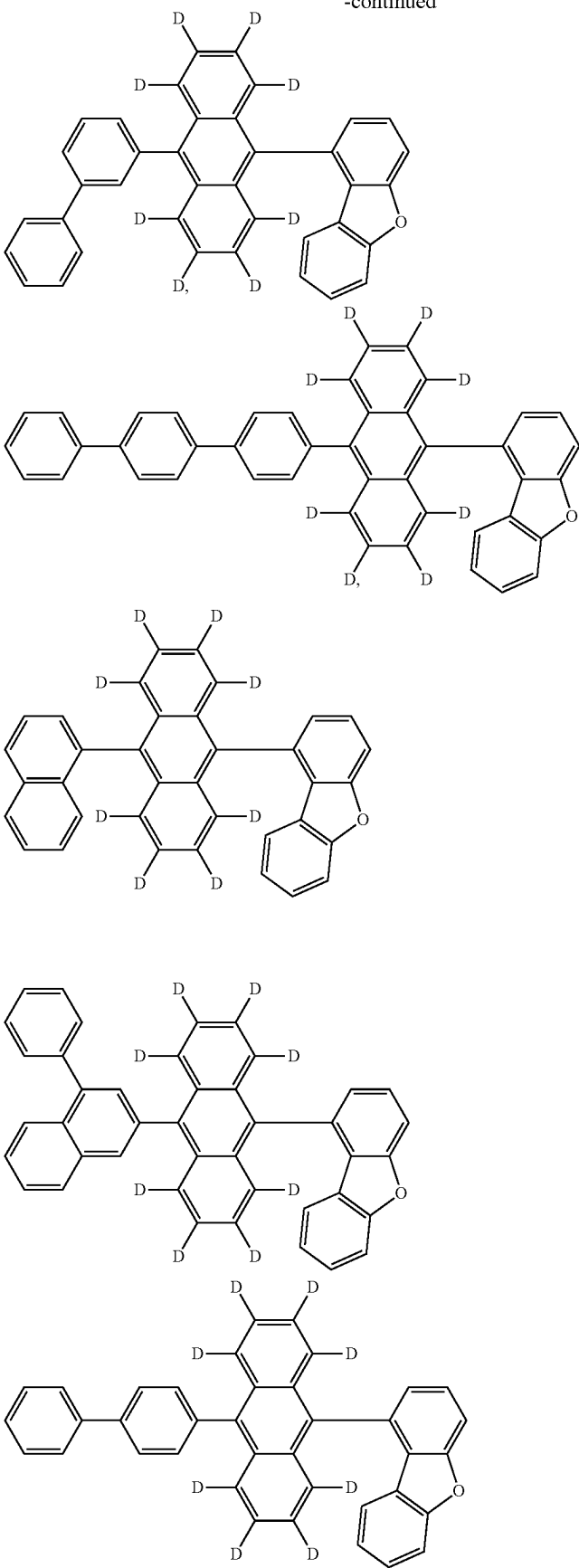
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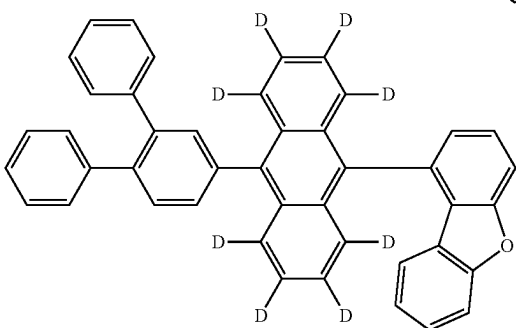
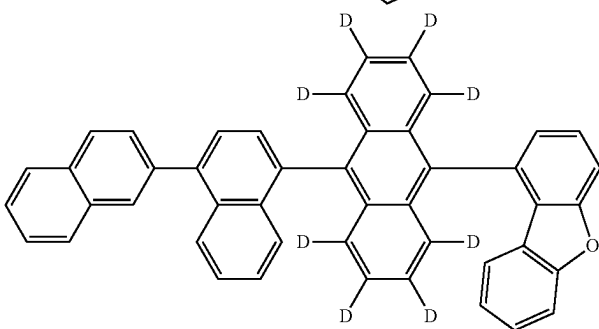
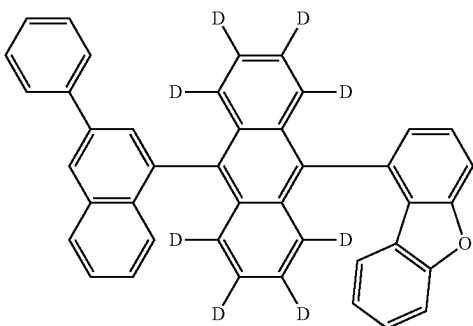
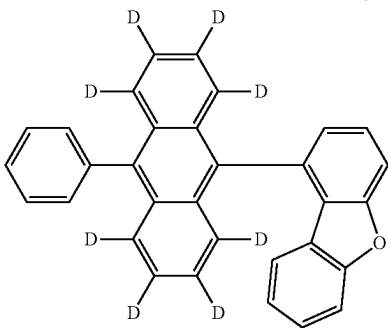
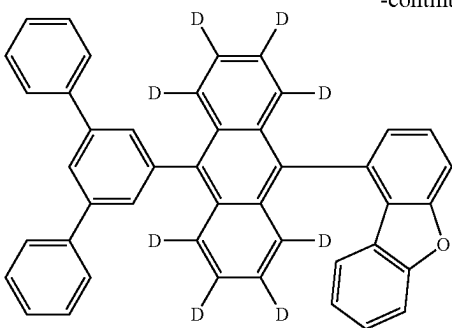
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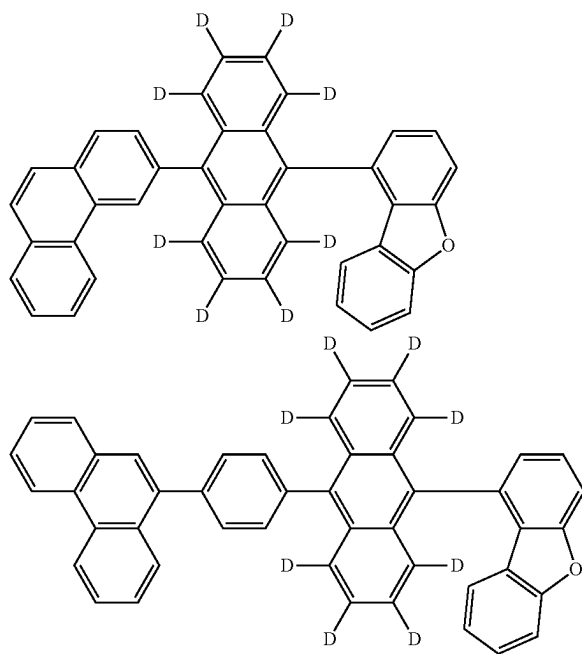
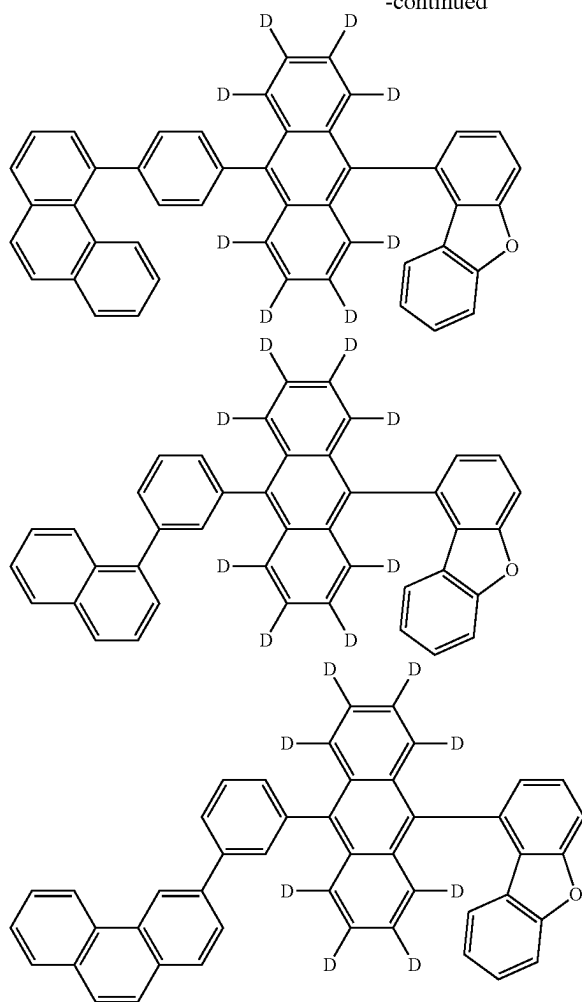
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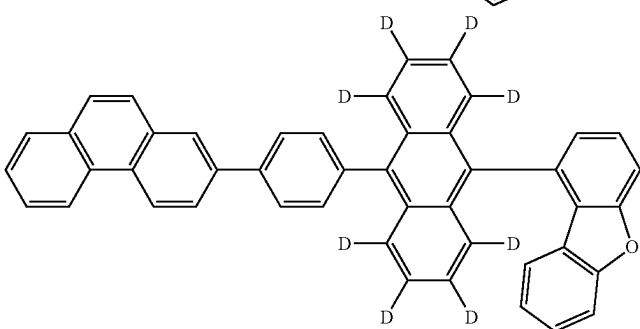
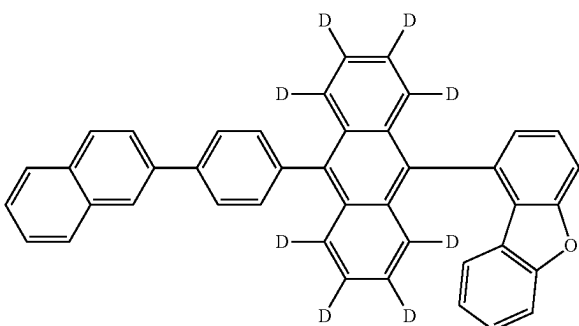
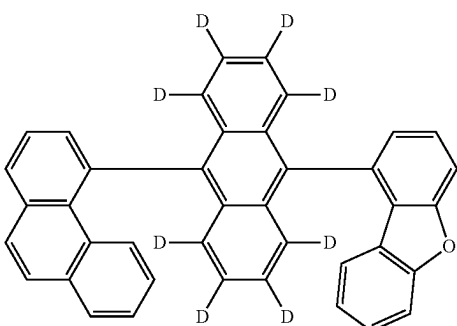
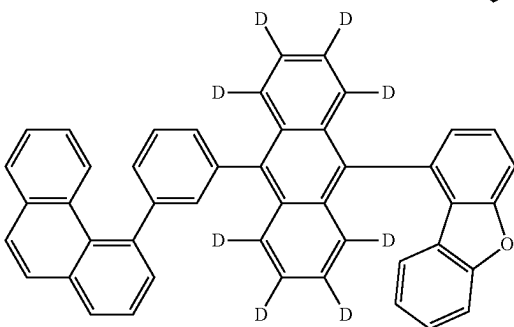
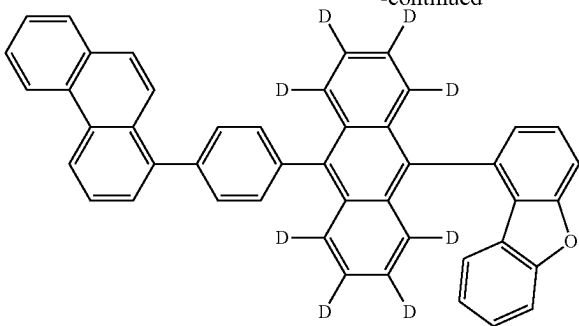
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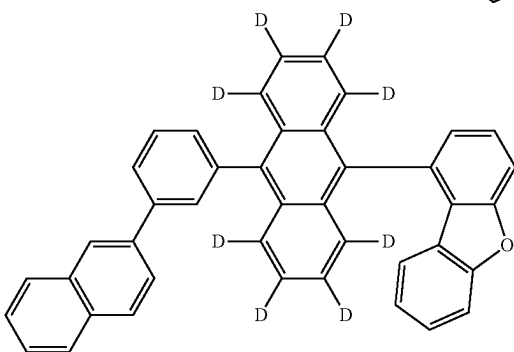
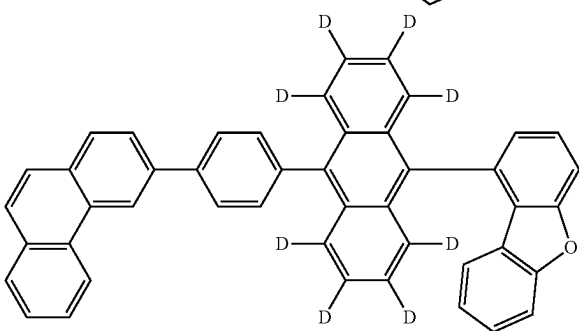
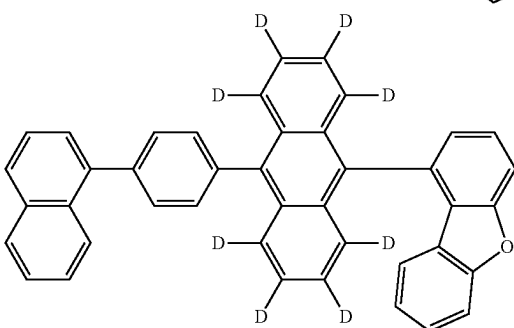
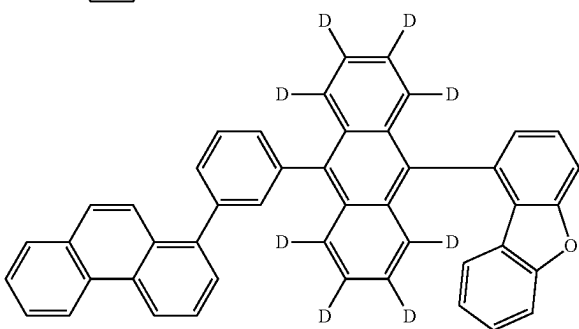
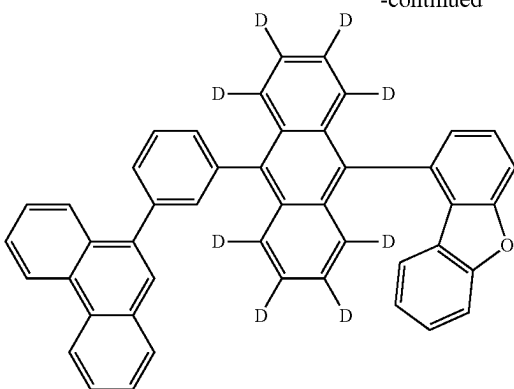
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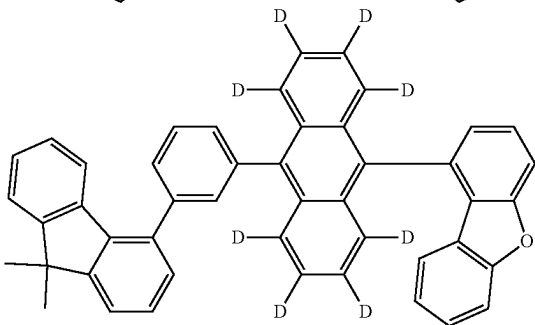
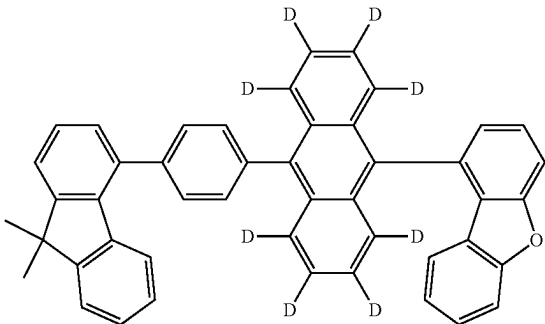
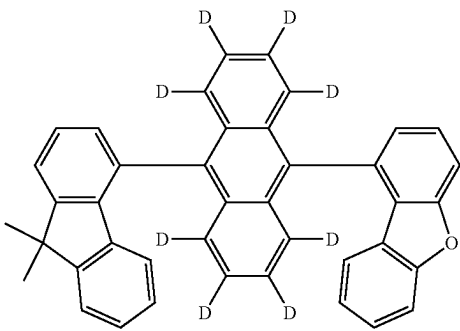
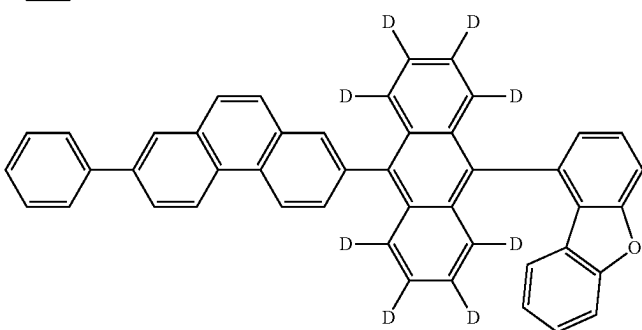
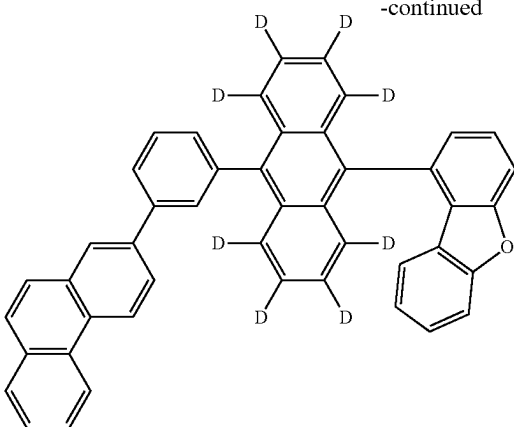
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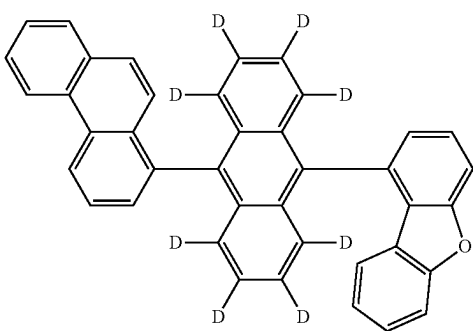
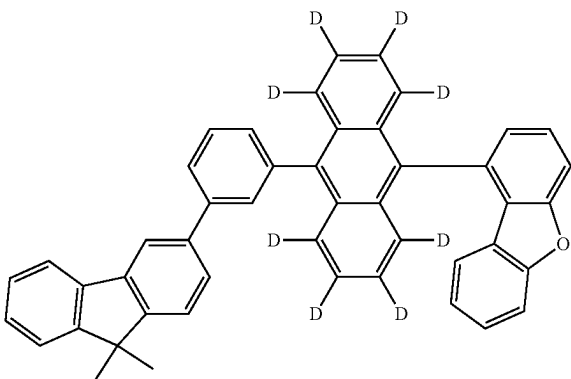
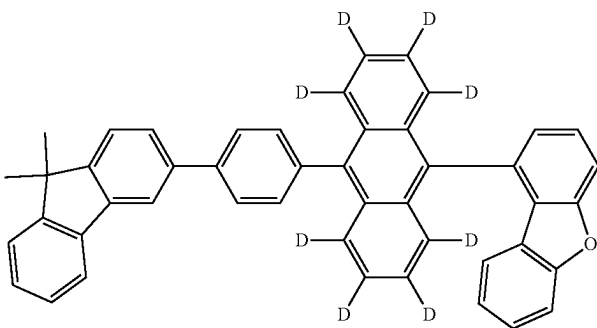
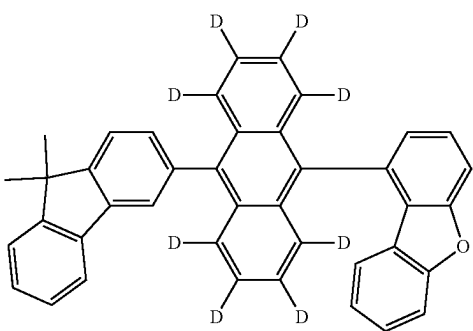
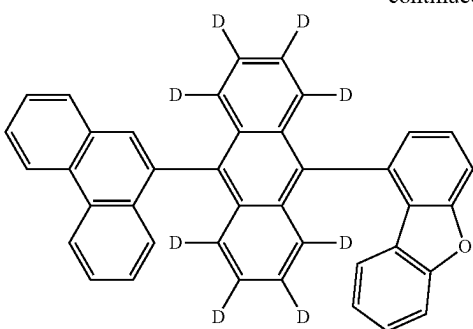
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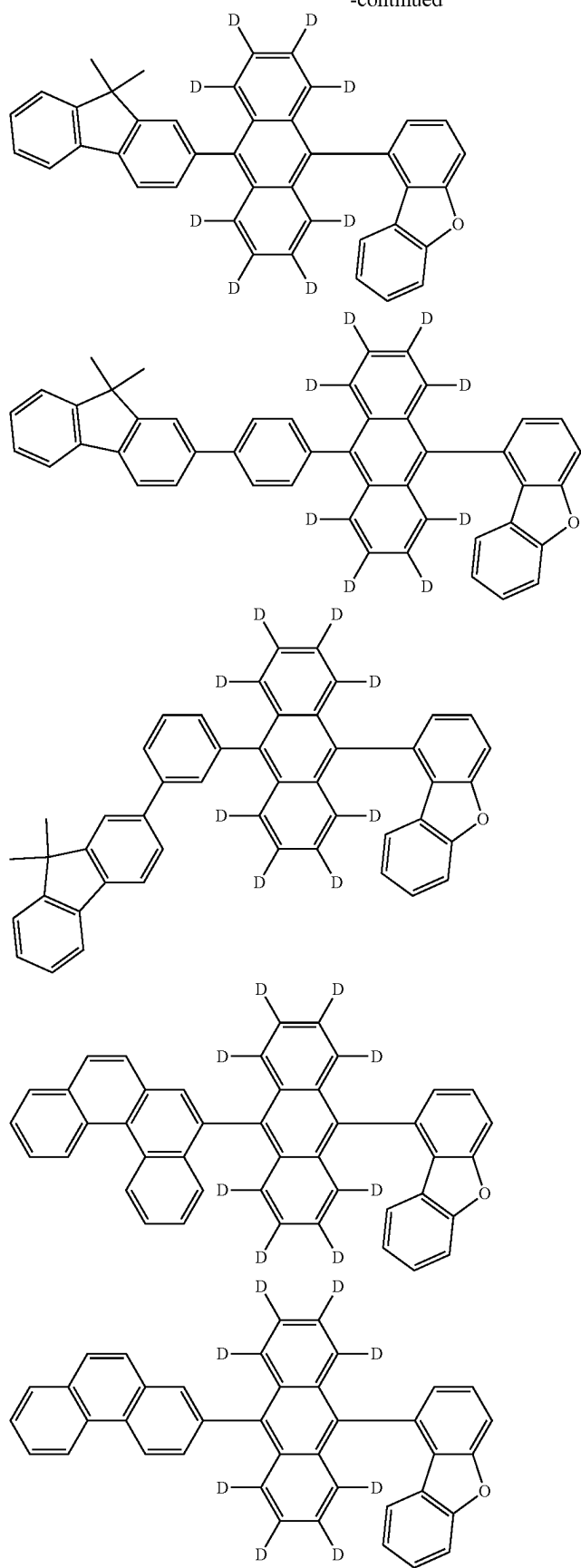
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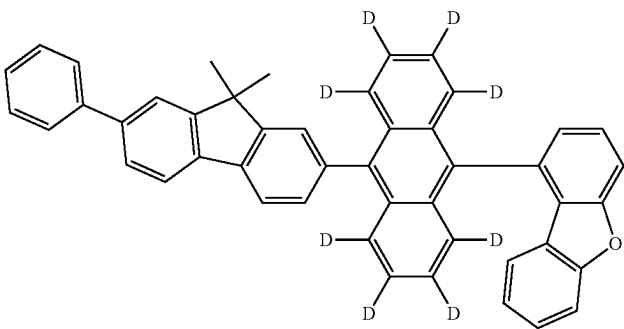
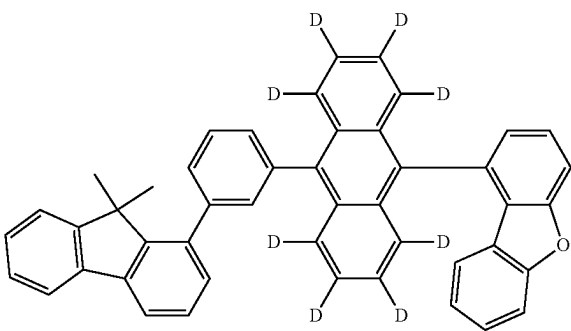
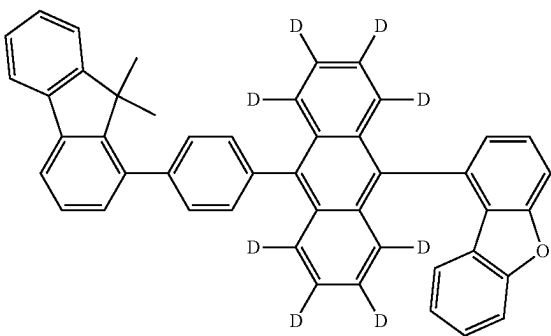
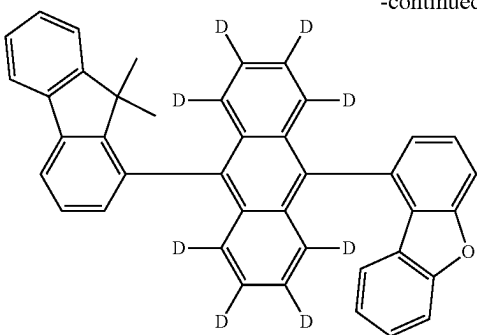
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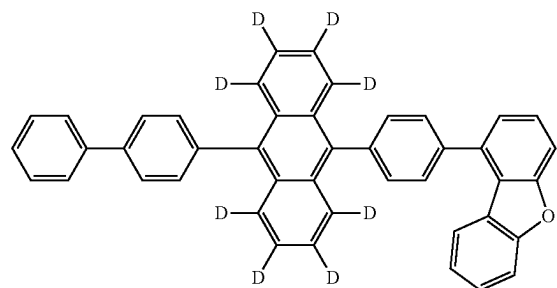
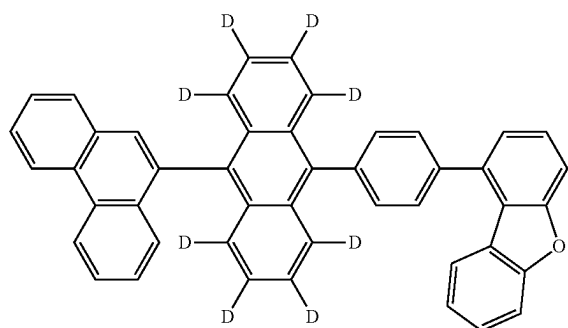
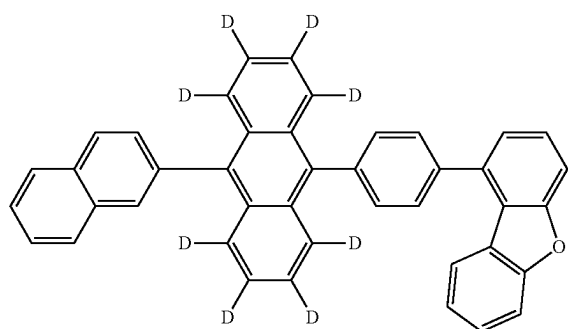
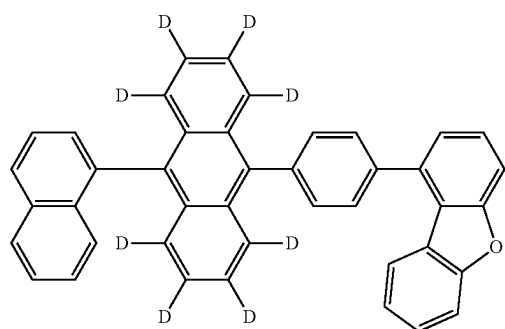
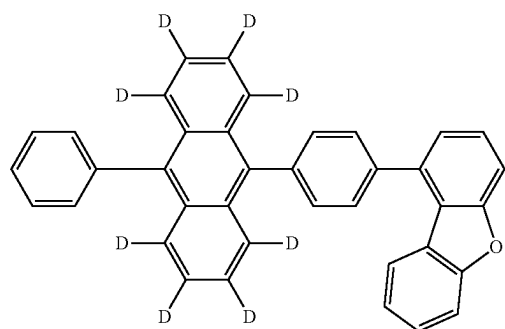
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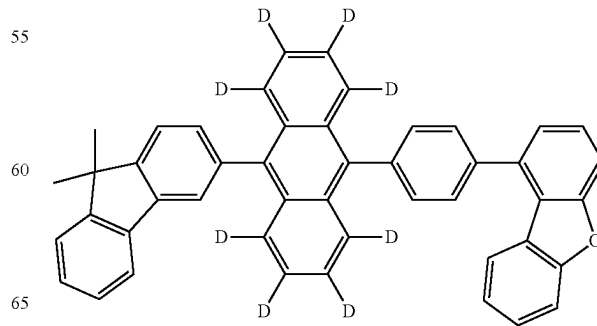
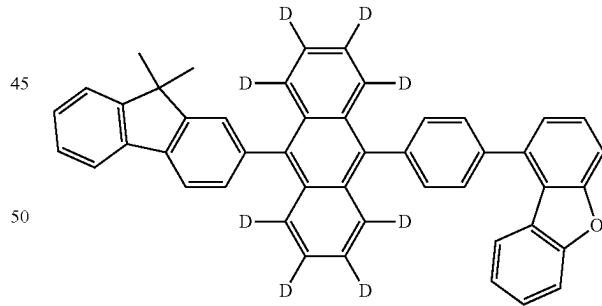
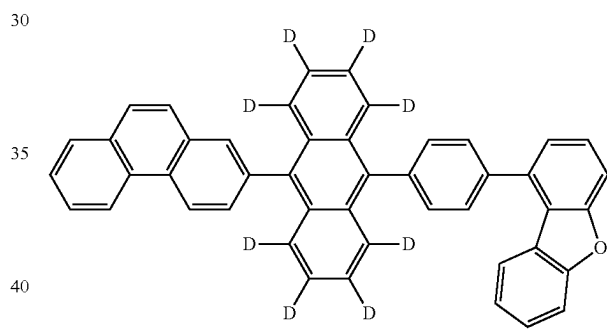
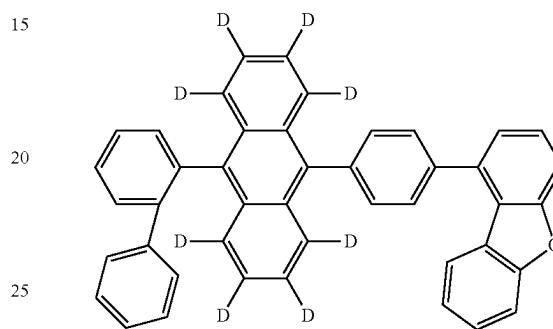
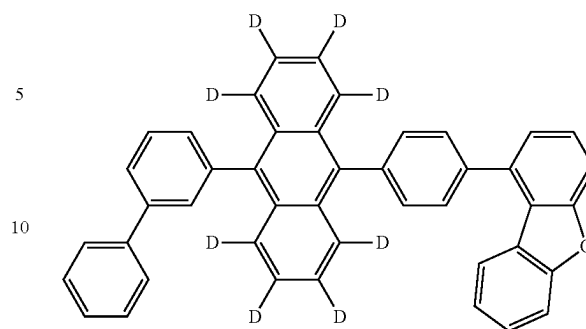
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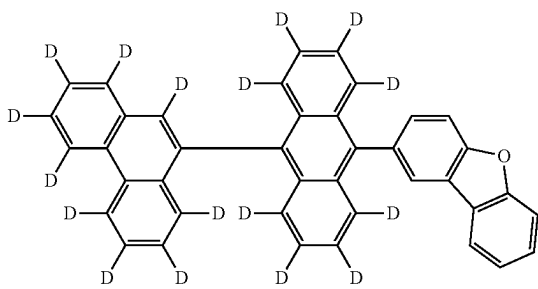
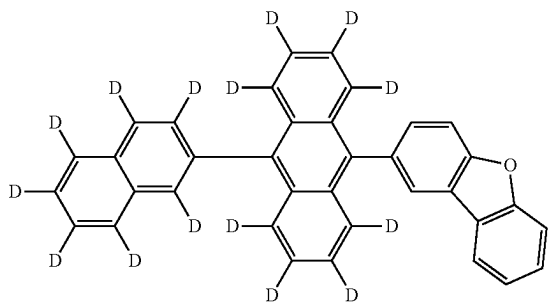
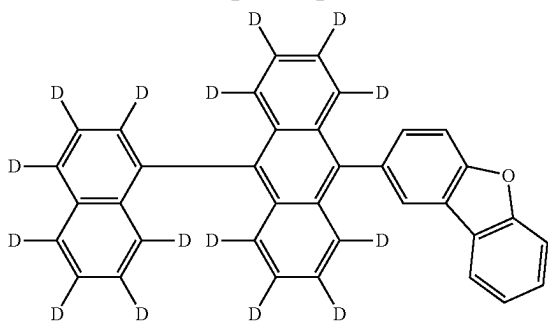
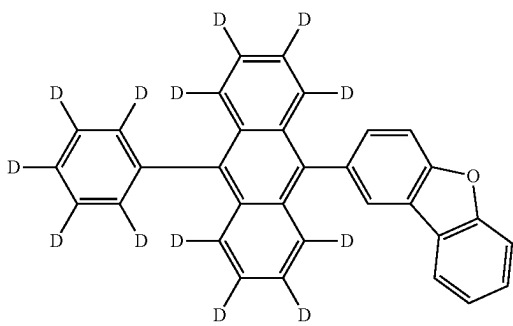
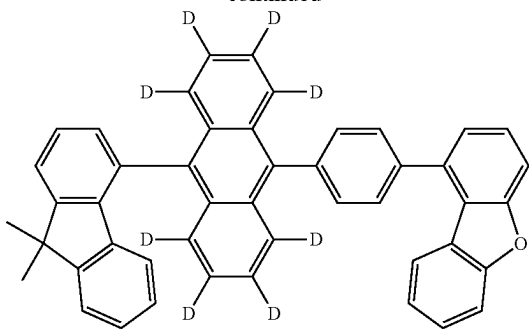
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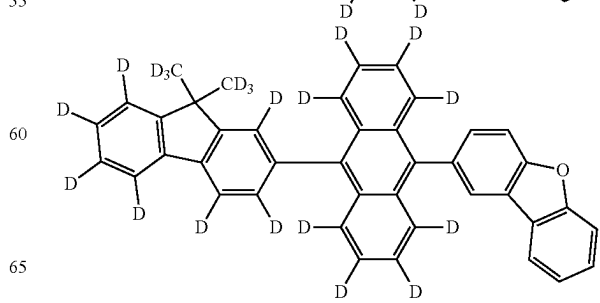
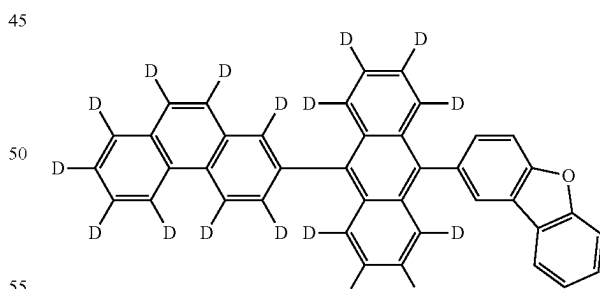
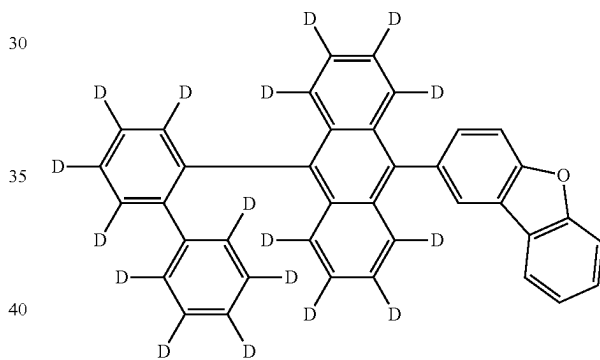
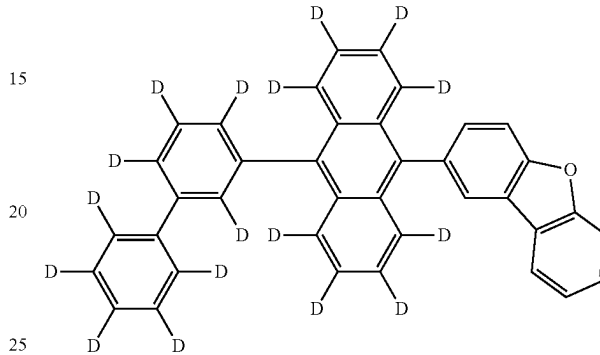
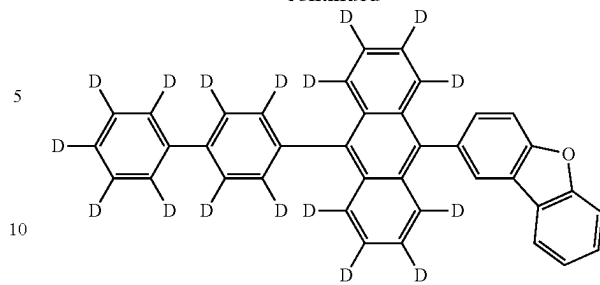
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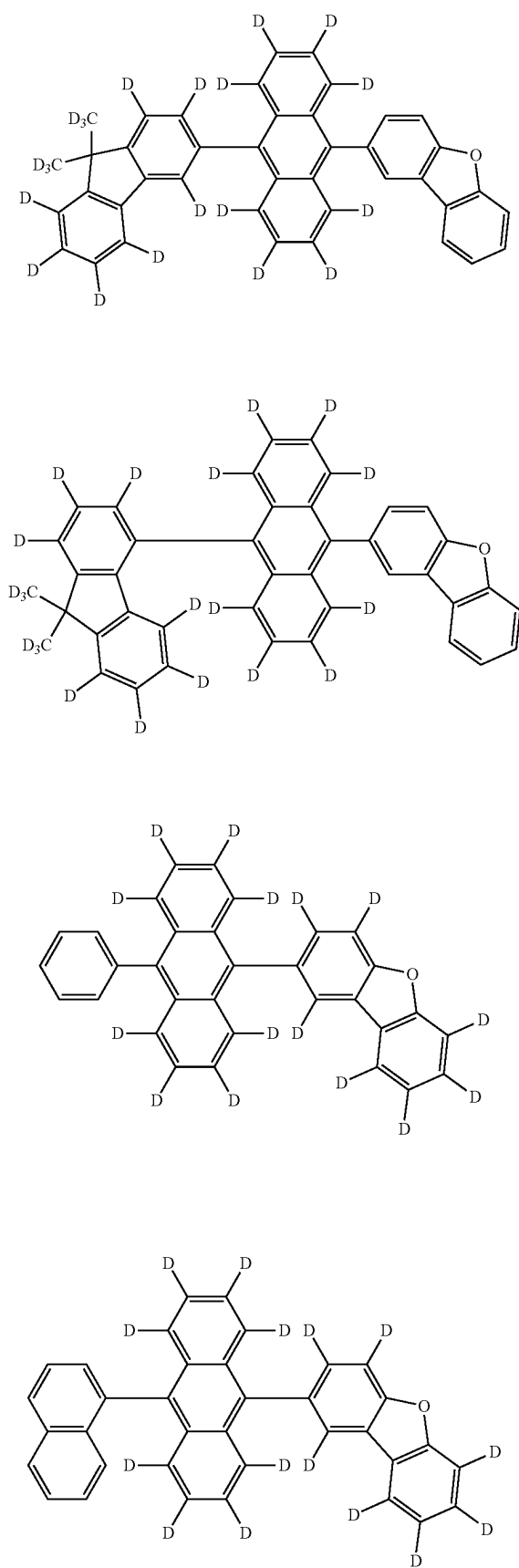
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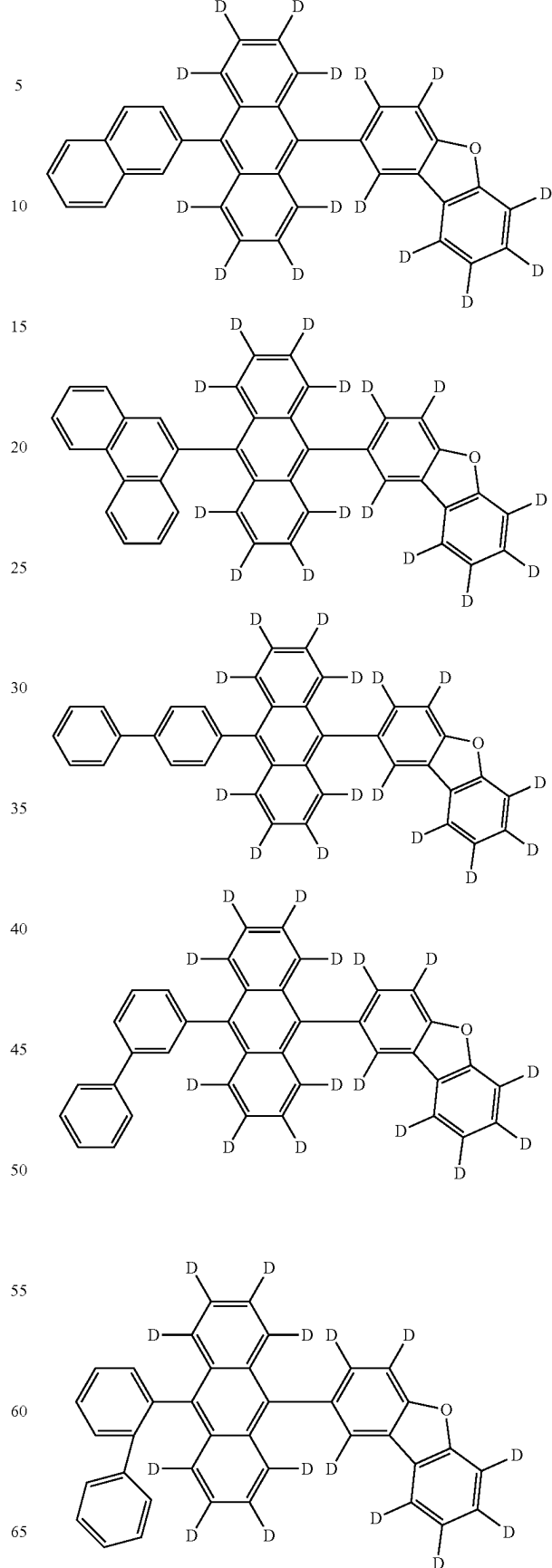


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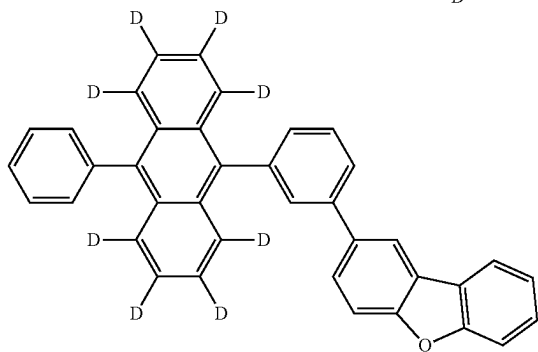
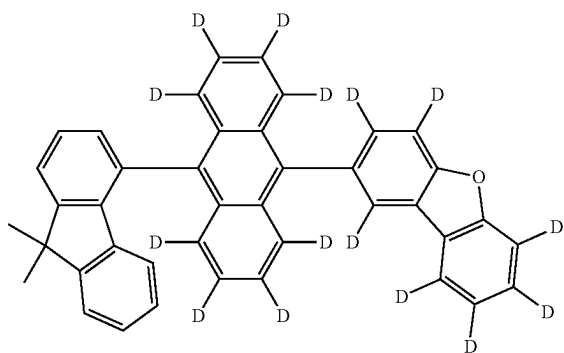
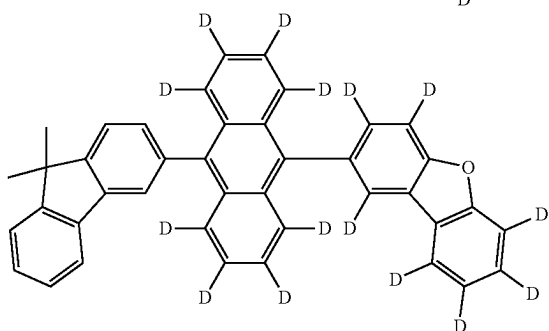
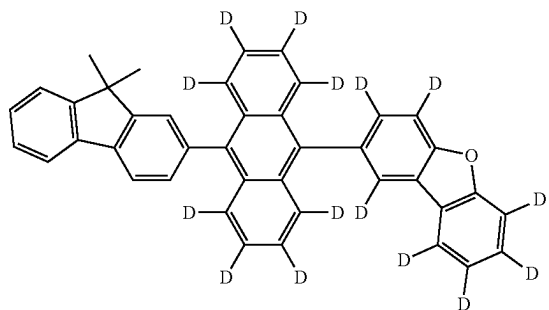
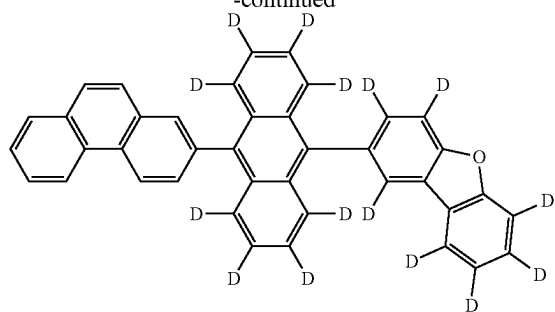
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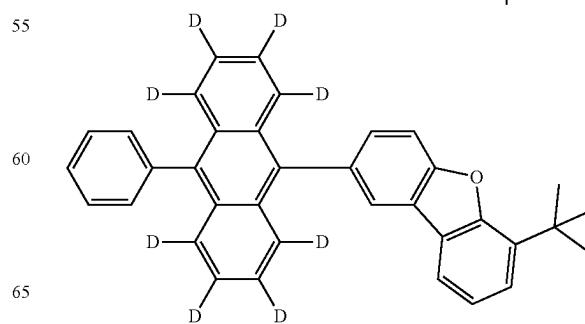
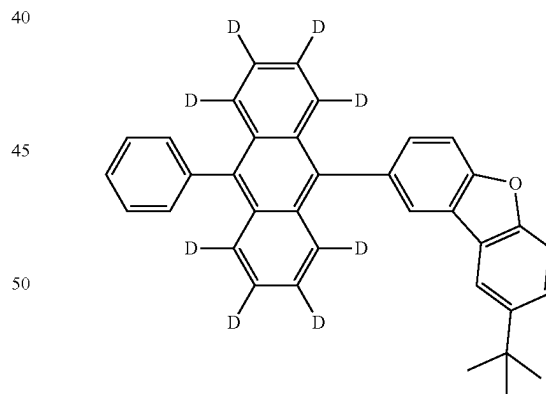
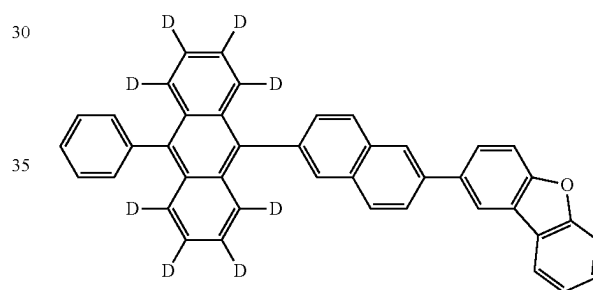
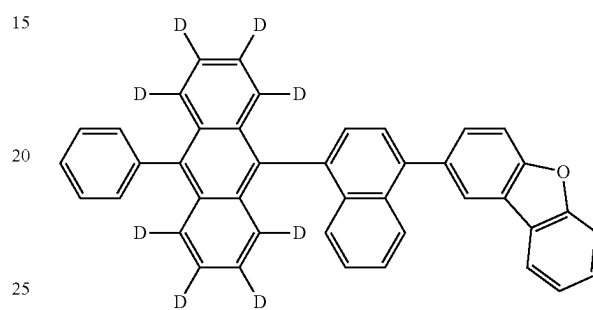
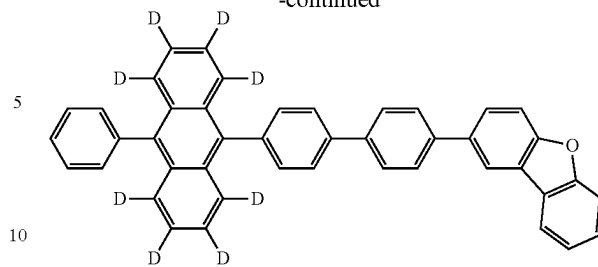


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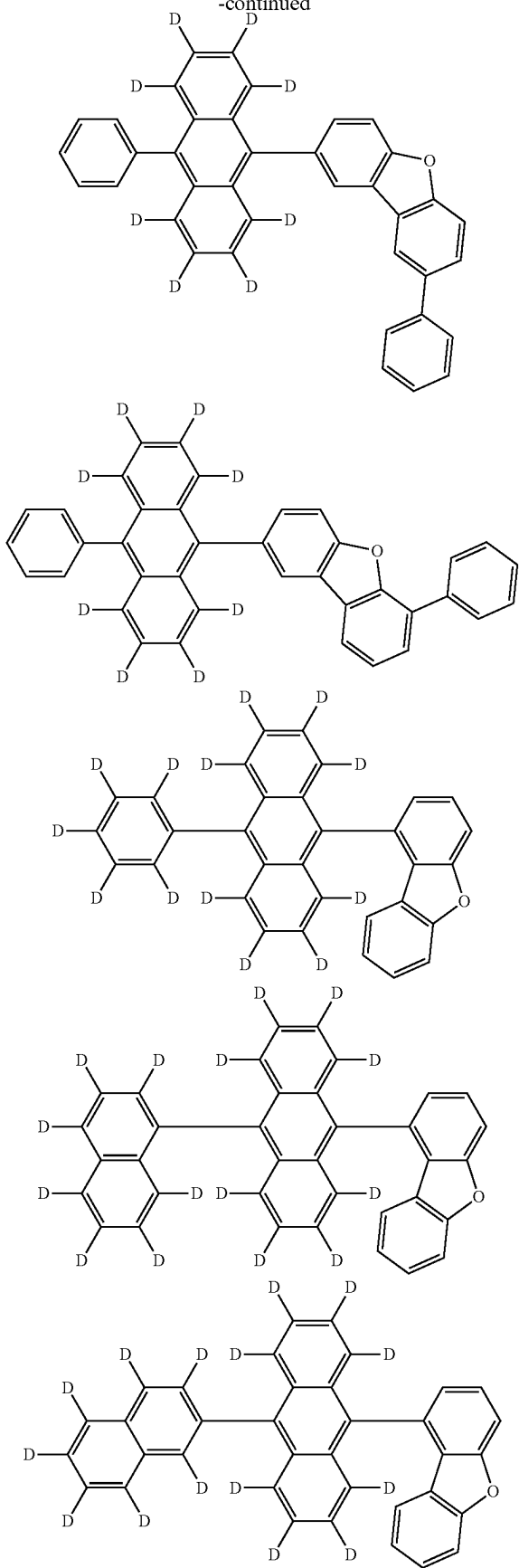
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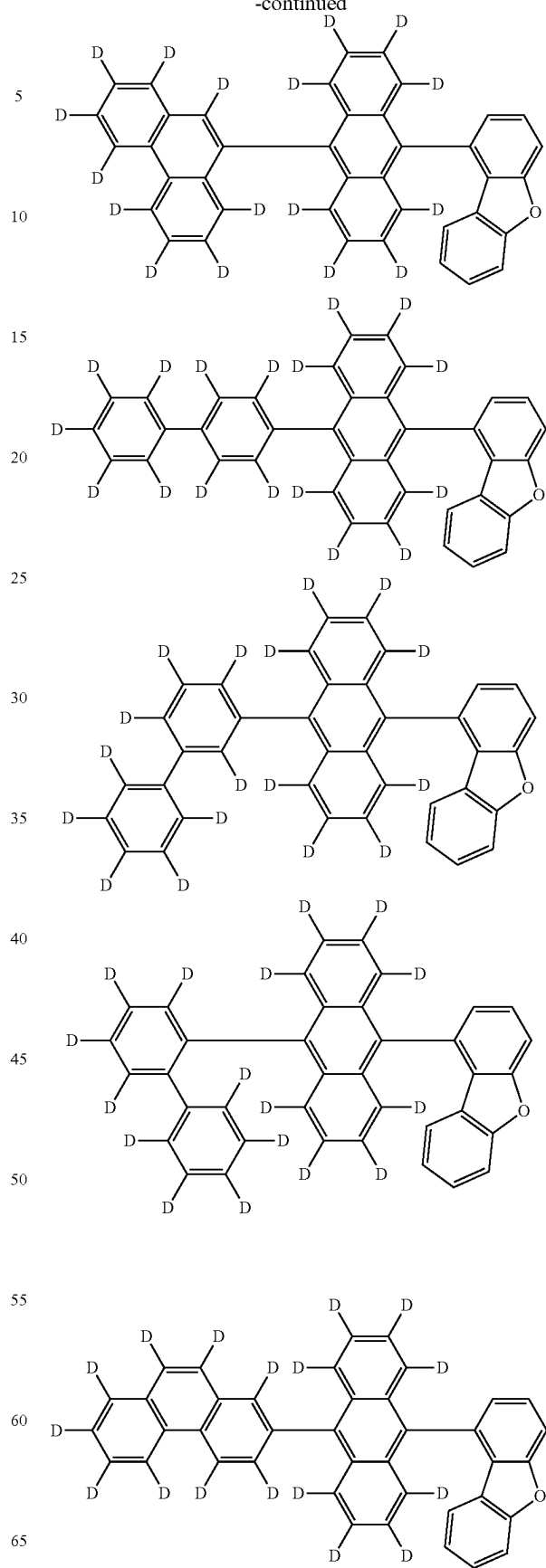


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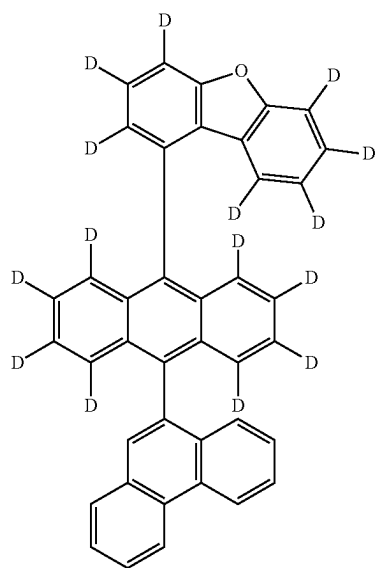
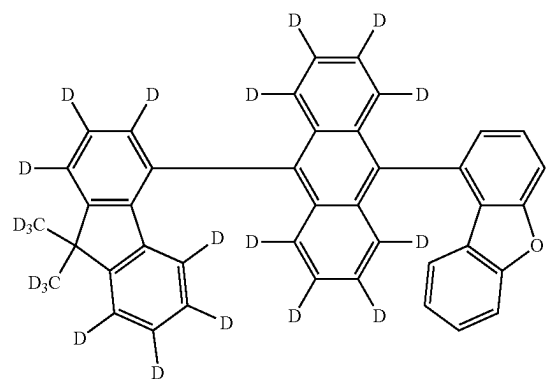
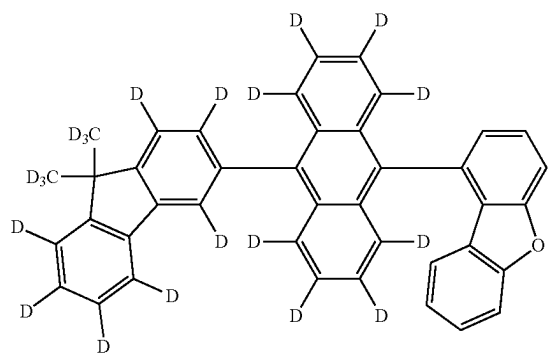
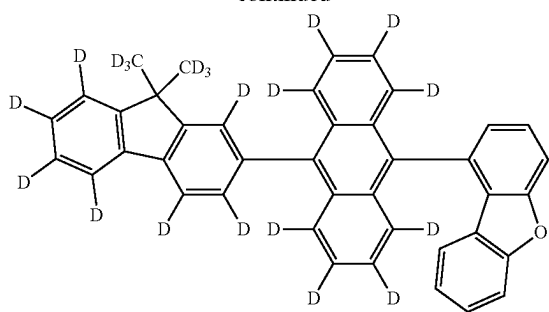
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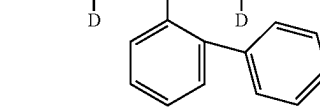
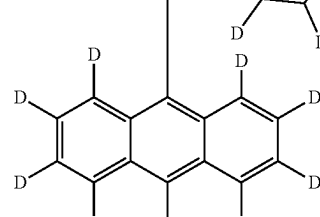
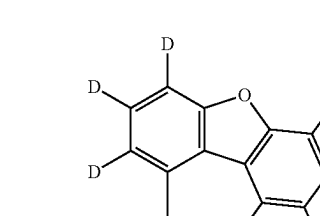
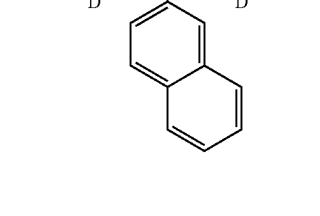
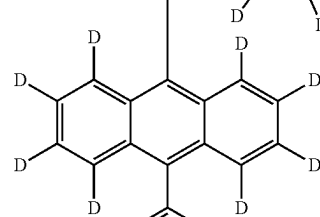
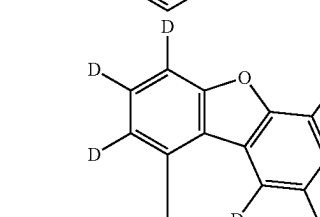
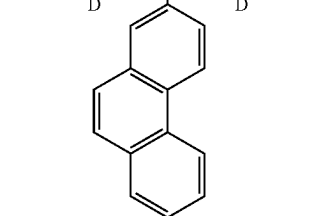
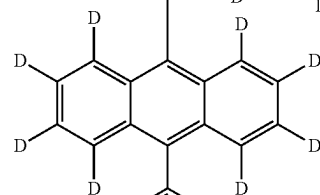
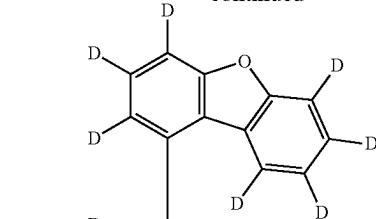


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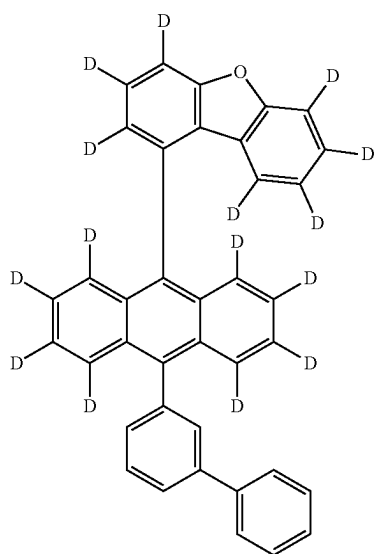
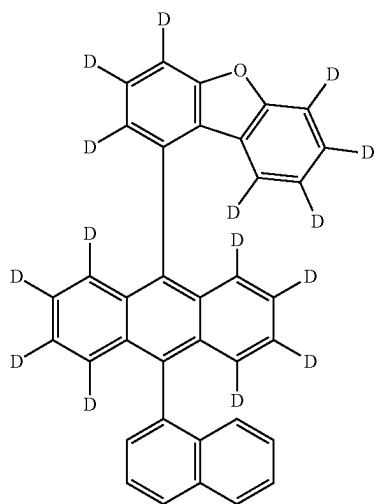
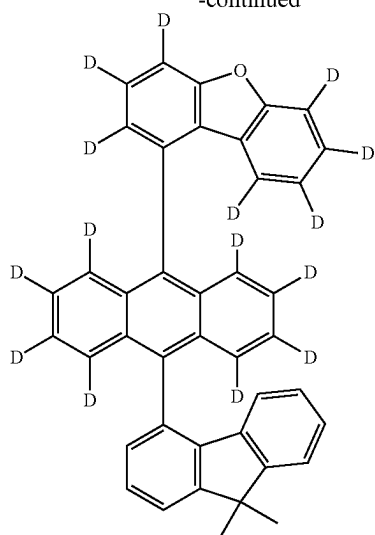
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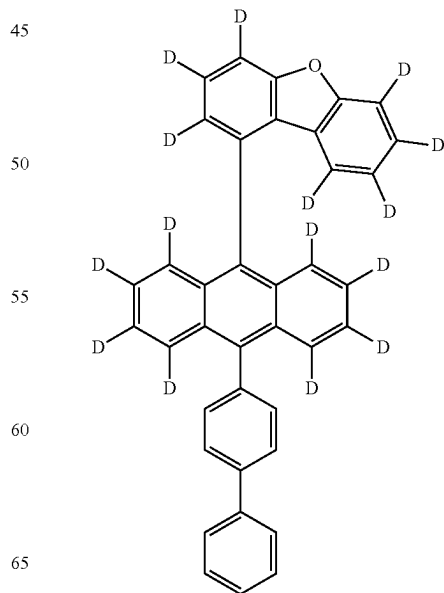
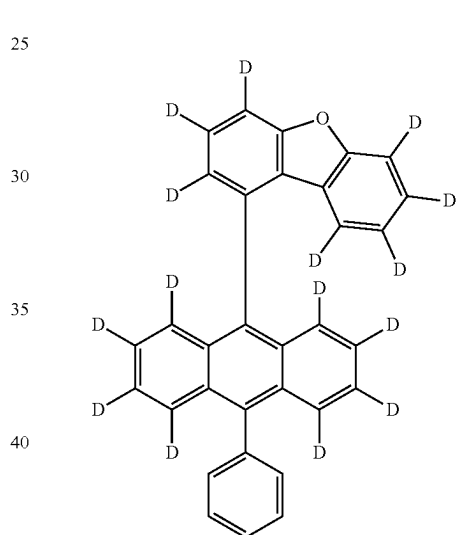
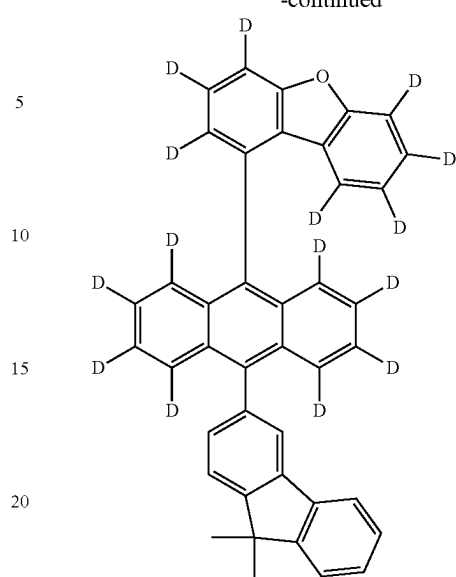


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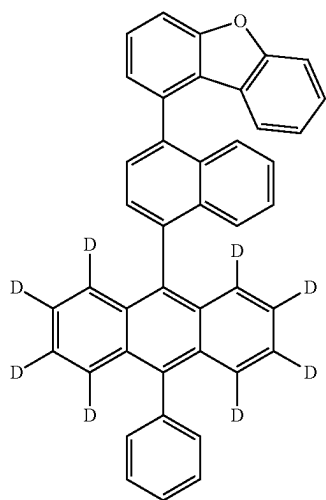
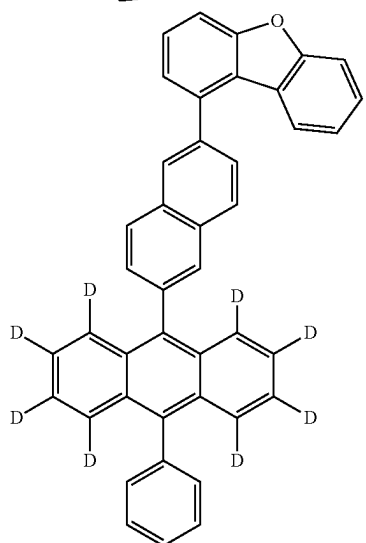
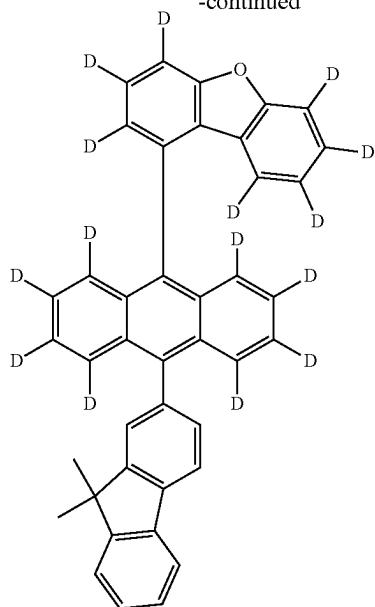
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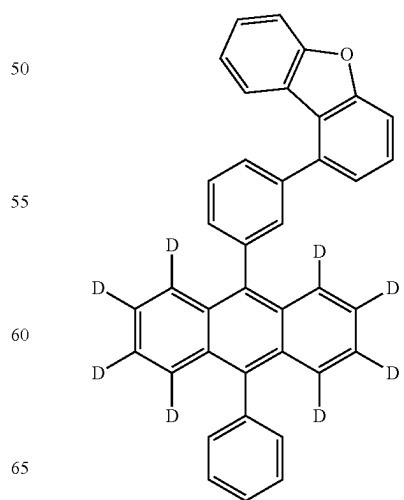
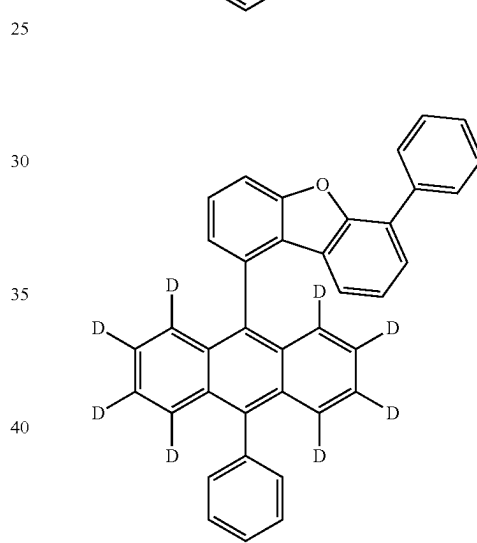
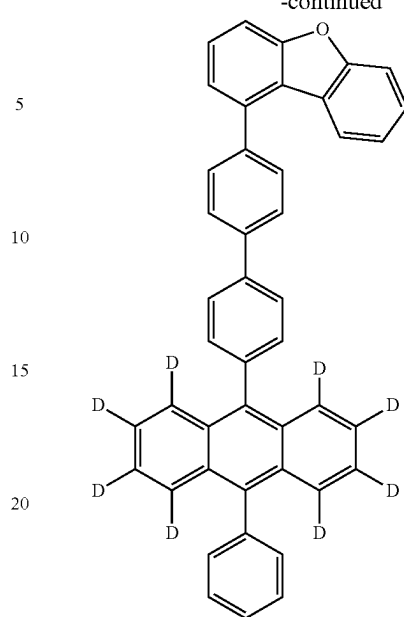


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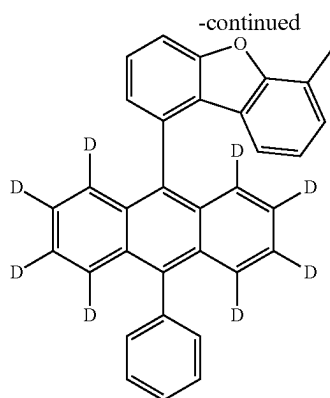
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The above compound can be used as a material for organic EL device, preferably a host material for an emitting layer.

[Organic EL Device]

The organic EL device according to one aspect of the invention comprises a cathode, an anode and one or two or more organic layer disposed between the cathode and the anode, and at least one organic layer comprises the compound represented by formula (1) or (2), or the composition of the invention described above. A conventionally known material, device configuration can be applied, except that at least one of organic layer comprises the compound represented by formula (1) or (2), or the composition of the invention described above, as long as the advantageous effect of the Invention is not impaired. In the above organic EL device, one or two or more organic layers preferably comprise emitting layer and the emitting layer comprises the compound represented by formula (1) or (2), or the composition of the invention described above.

In one embodiment, the emitting layer of the organic EL device according to one aspect of the invention contains the compound represented by formula (1) or (2) and the protium compound, and the content ratio of the protium compound to the total of the compound represented by formula (1) or (2) and the protium compound is 99 mol % or less.

In one embodiment, the emitting layer of the organic EL device according to one aspect of the invention contains the compound represented by formula (1) or (2) and the protium compound, and the content ratio of the compound represented by formula (1) or (2) to the total of the compound represented by formula (1) or (2) and the protium compound is 30 mol % or more, 50 mol % or more, 70 mol % or more, 90 mol % or more, 95 mol % or more, 99 mol % or more, or 100 mol %.

One embodiment of the organic EL device preferably has the hole-transporting layer between the anode and the emitting layer.

One embodiment of the organic EL device preferably has the electron-transporting layer between the cathode and the emitting layer.

Specific examples of a typified device configuration of the organic EL device of the invention include structures such as

- (1) an anode/an emitting layer/a cathode,
- (2) an anode/a hole-injecting layer/an emitting layer/a cathode,
- (3) an anode/an emitting layer/an electron-injecting-transporting layer/a cathode,
- (4) an anode/a hole-injecting layer/an emitting layer/an electron-injecting-transporting layer/a cathode,

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(5) an anode/an organic semiconductor layer/an emitting layer/a cathode,

(6) an anode/an organic semiconductor layer/an electron barrier layer/an emitting layer/a cathode,

5 (7) an anode/an organic semiconductor layer/an emitting layer/an adhesion improving layer/a cathode,

(8) an anode/a hole-injecting-transporting layer/an emitting layer/an electron-injecting-transporting layer/a cathode,

10 (9) an anode/an insulating layer/an emitting layer/an insulating layer/a cathode,

(10) an anode/an inorganic semiconductor layer/an insulating layer/an emitting layer/an insulating layer/a cathode,

15 (11) an anode/an organic semiconductor layer/an insulating layer/an emitting layer/an insulating layer/a cathode,

(12) an anode/an insulating layer a hole-injecting-transporting layer/an emitting layer/an insulating layer/a cathode, and

(13) an anode/an insulating layer a hole-injecting-transporting layer/an emitting layer an electron-injecting-transporting layer/a cathode.

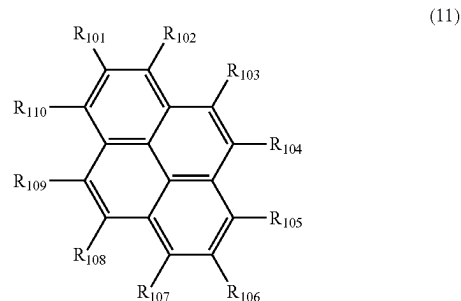
25 Among the above-described structures, a configuration of (8) is preferably used, but the configuration is not limited thereto.

In this specification, the term “hole-injecting-transporting layer” herein means “at least one of the hole-injecting layer and the hole-transporting layer”, and the term “electron-injecting-transporting layer” herein means “at least one of the electron-injecting layer and the electron-transporting layer”.

The emitting layer preferably comprises one or more compounds (dopant material) selected from the group consisting of compounds represented by the following formulas (11), (21), (31), (41), (51), (61), (71) and (81) in addition to the compound (host material) represented by the formula (1) or (2).

(Compound Represented by Formula (11))

The compound represented by the formula (11) is explained below.



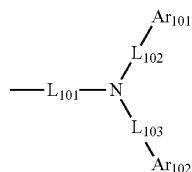
In the formula (11), one or more pairs of two or more adjacent groups of R₁₀₁ to R₁₁₀ are bonded with each other to form a substituted or unsubstituted, saturated or unsaturated ring, or do not form a substituted or unsubstituted, saturated or unsaturated ring;

at least one of R₁₀₁ to R₁₁₀ is a monovalent group represented by the formula (12);

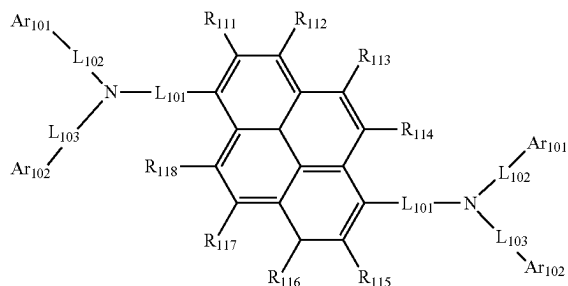
R₁₀₁ to R₁₁₀ that do not form the substituted or unsubstituted, saturated or unsaturated ring and that are not a

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monovalent group represented by the following formula (12) are independently
 a hydrogen atom,
 a substituted or unsubstituted alkyl group having 1 to 50 carbon atoms,
 a substituted or unsubstituted alkenyl group having 2 to 50 carbon atoms,
 a substituted or unsubstituted alkynyl group having 2 to 50 carbon atoms,
 a substituted or unsubstituted cycloalkyl group having 3 to 50 ring carbon atoms,
 $\text{—Si(R}_{901}\text{)(R}_{902}\text{)(R}_{903}\text{)}$,
 $\text{—O—(R}_{904}\text{)}$,
 $\text{—S—(R}_{905}\text{)}$,
 $\text{—N(R}_{906}\text{)(R}_{907}\text{)}$,
 a halogen atom, a cyano group, a nitro group,
 a substituted or unsubstituted aryl group having 6 to 50 ring carbon atoms, or
 a substituted or unsubstituted monovalent heterocyclic group having 5 to 50 ring atoms;
 R_{901} to R_{907} are as defined in the formula (1);



wherein, in the formula (12), Ar_{101} and Ar_{102} are independently
 a substituted or unsubstituted aryl group having 6 to 50 ring carbon atoms, or
 a substituted or unsubstituted monovalent heterocyclic group having 5 to 50 ring atoms; L_{101} to L_{103} are independently
 a single bond,
 a substituted or unsubstituted arylene group having 6 to 30 ring carbon atoms, or
 a substituted or unsubstituted divalent heterocyclic group having 5 to 30 ring atoms.
 In the formula (11), it is preferable that two of R_{101} to R_{110} are the group represented by the formula (12).
 In one embodiment, the compound represented by the formula (11) is represented by the following formula (13).



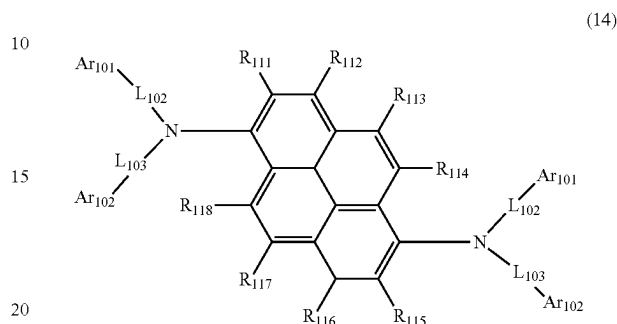
In the formula (13), R_{111} to R_{118} are the same as R_{101} to R_{110} that is not a monovalent group represented by the

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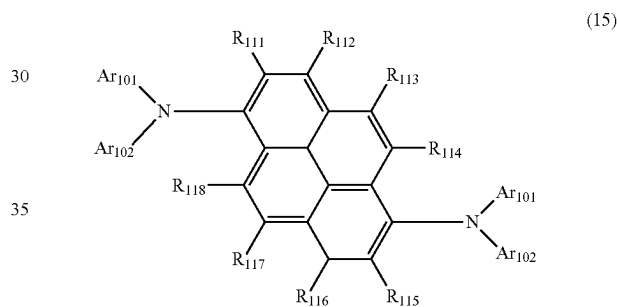
formula (12) in the formula (11). Ar_{901} , Ar_{902} , L_{101} , L_{102} and L_{103} are as defined in the formula (12).

In the formula (11), L_{101} is preferably a single bond and L_{102} and L_{103} are preferably a single bond.

In one embodiment, the compound represented by the formula (11) is represented by the formula (14) or (15).

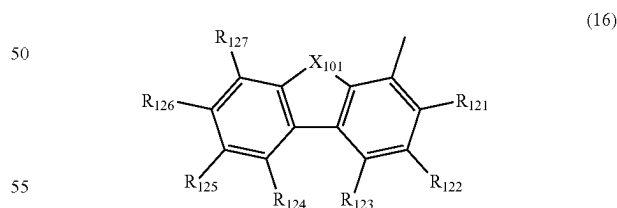


In the formula (14), R_{111} to R_{118} are as defined in the formula (13). Ar_{101} , Ar_{102} , L_{102} and L_{103} are as defined in the formula (12).



In the formula (15), R_{111} to R_{118} are as defined in the formula (13). Ar_{101} and Ar_{102} are as defined in the formula (12).

In the formula (11) and formula (12), it is preferable that at least one of Ar_{101} and Ar_{102} is the group represented by the following formula (16).



In the formula (16),

X_{101} is an oxygen atom or a sulfur atom;

one or more pairs of two or more adjacent groups of R_{121} to R_{127} are bonded with each other to form a substituted or unsubstituted, saturated or unsaturated ring, or do not form a substituted or unsubstituted, saturated or unsaturated ring;

R_{121} to R_{127} that do not form the substituted or unsubstituted, saturated or unsaturated ring are independently a hydrogen atom,

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a substituted or unsubstituted alkyl group having 1 to 50 carbon atoms,
 a substituted or unsubstituted alkenyl group having 2 to 50 carbon atoms,
 a substituted or unsubstituted alkynyl group having 2 to 50 carbon atoms,
 a substituted or unsubstituted cycloalkyl group having 3 to 50 ring carbon atoms,
 $-\text{Si}(\text{R}_{901})(\text{R}_{902})(\text{R}_{903})$,
 $-\text{O}-(\text{R}_{904})$,
 $-\text{S}-(\text{R}_{905})$,
 $-\text{N}(\text{R}_{906})(\text{R}_{907})$,
 a halogen atom, a cyano group, a nitro group,
 a substituted or unsubstituted aryl group having 6 to 50 ring carbon atoms, or
 a substituted or unsubstituted monovalent heterocyclic group having 5 to 50 ring atoms; and
 R_{901} to R_{907} are as defined in the formula (1).
 It is preferable that X_{101} is an oxygen atom.
 It is preferable that at least one of R_{121} to R_{127} is a substituted or unsubstituted alkyl group having 1 to 50 carbon atoms,
 a substituted or unsubstituted alkenyl group having 2 to 50 carbon atoms,
 a substituted or unsubstituted alkynyl group having 2 to 50 carbon atoms,
 a substituted or unsubstituted cycloalkyl group having 3 to 50 ring carbon atoms,
 a substituted or unsubstituted aryl group having 6 to 50 ring carbon atoms, or
 a substituted or unsubstituted monovalent heterocyclic group having 5 to 50 ring atoms.
 It is preferable that in the formula (11) and formula (12), Ar_{101} is a group represented by the formula (16) and Ar_{102} is a substituted or unsubstituted aryl group having 6 to 50 ring carbon atoms.
 In one embodiment, the compound represented by the formula (11) is represented by the following formula (17).

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In the formula (17), R_{111} to R_{118} are as defined in the formula (13), and R_{121} to R_{127} are as defined in the formula (16);

R_{131} to R_{135} are independently

a hydrogen atom,

a substituted or unsubstituted alkyl group having 1 to 50 carbon atoms,

a substituted or unsubstituted alkenyl group having 2 to 50 carbon atoms,

a substituted or unsubstituted alkynyl group having 2 to 50 carbon atoms,

a substituted or unsubstituted cycloalkyl group having 3 to 50 ring carbon atoms,

$-\text{Si}(\text{R}_{901})(\text{R}_{902})(\text{R}_{903})$,

$-\text{O}-(\text{R}_{904})$,

$-\text{S}-(\text{R}_{905})$,

$-\text{N}(\text{R}_{906})(\text{R}_{907})$,

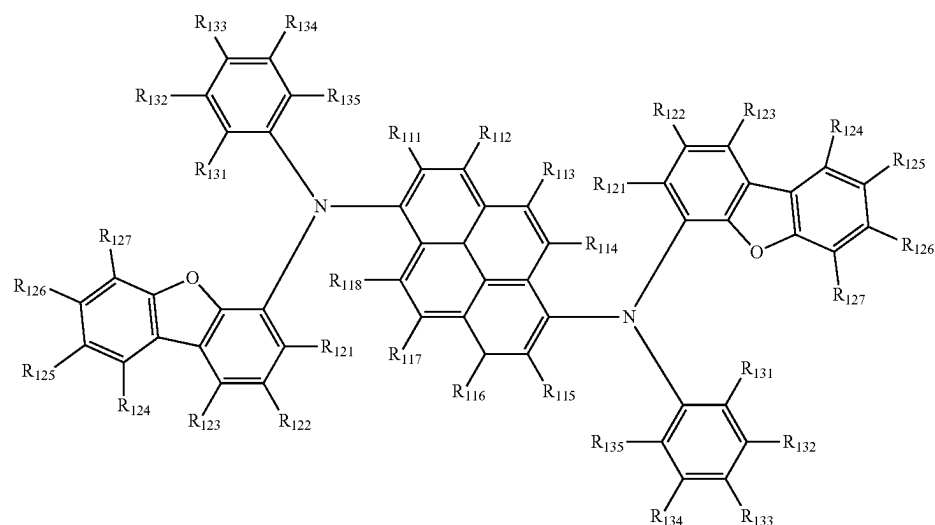
a halogen atom, a cyano group, a nitro group,

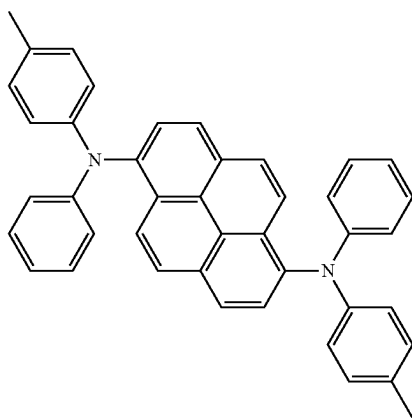
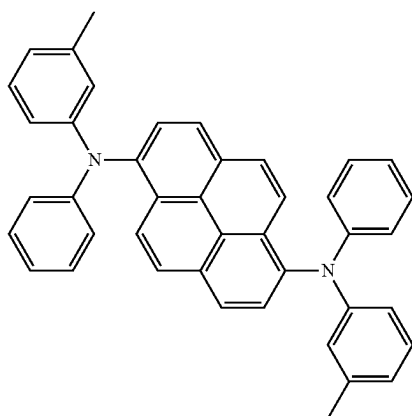
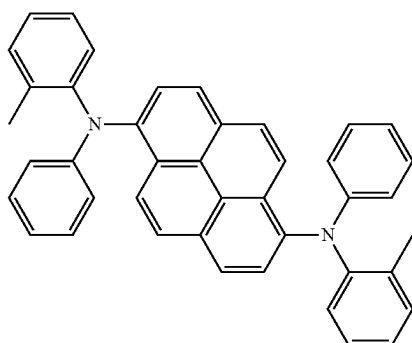
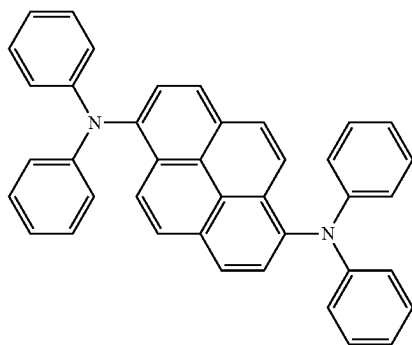
a substituted or unsubstituted aryl group having 6 to 50 ring carbon atoms, or

a substituted or unsubstituted monovalent heterocyclic group having 5 to 50 ring atoms; and

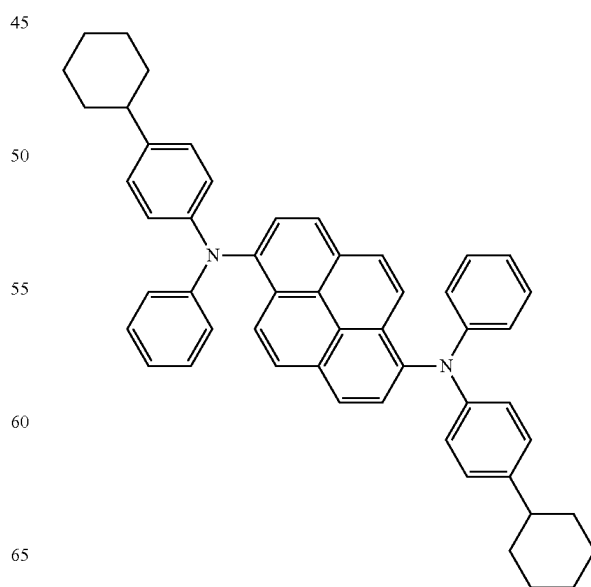
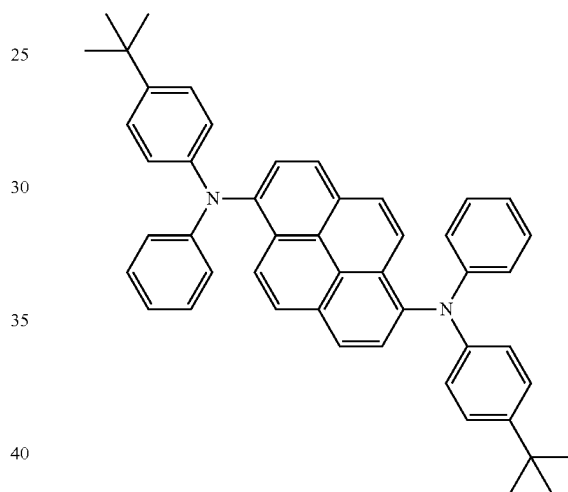
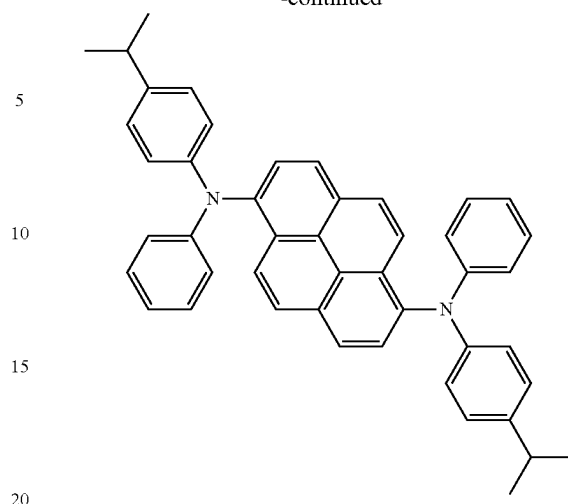
R_{901} to R_{907} are as defined in the formula (1).

As the compound represented by the formula (11), the following compounds can be given as specific examples, for example. In the following example compounds, Me represents methyl group.



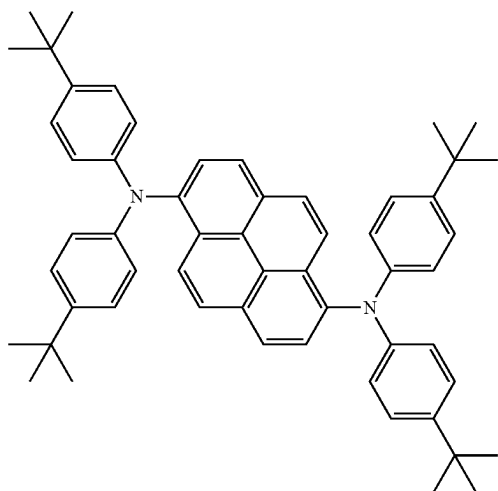
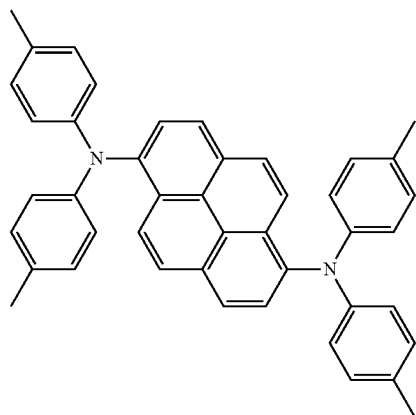
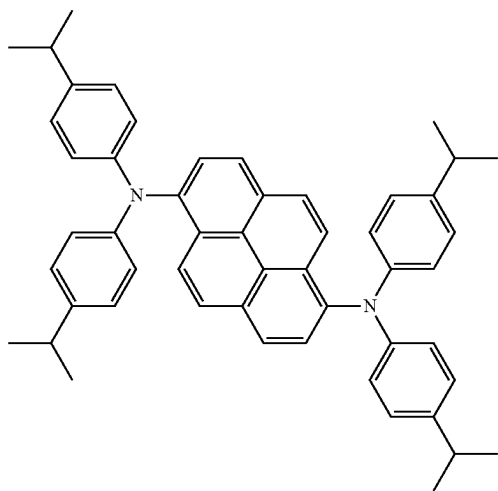
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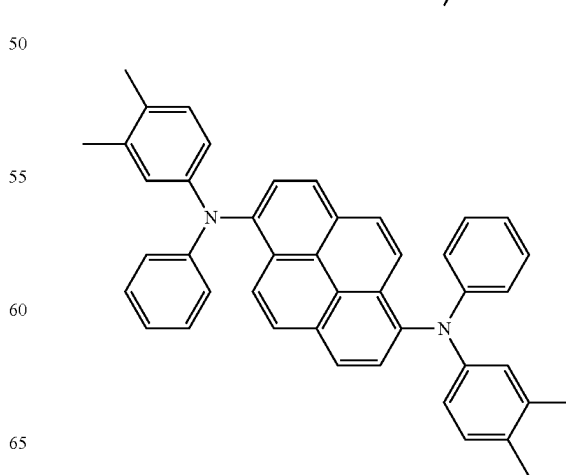
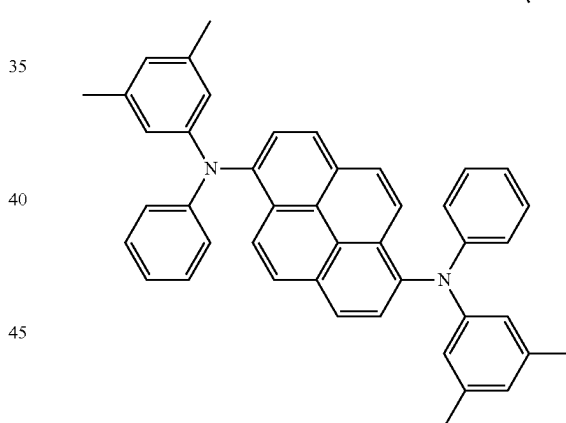
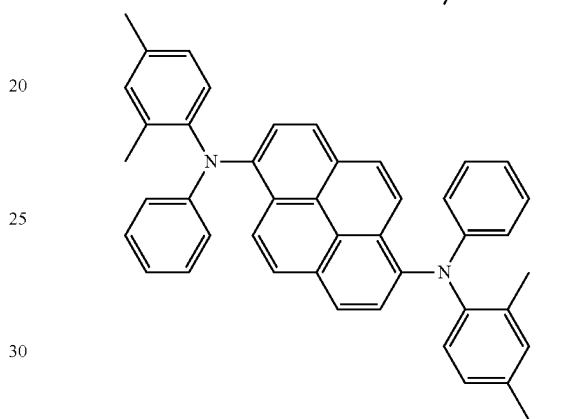
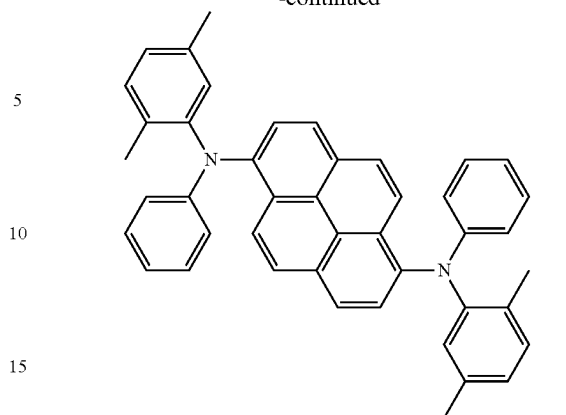


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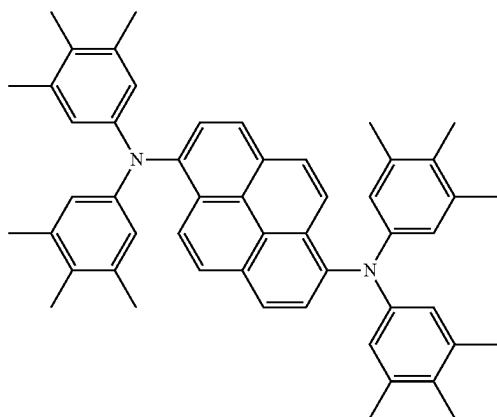
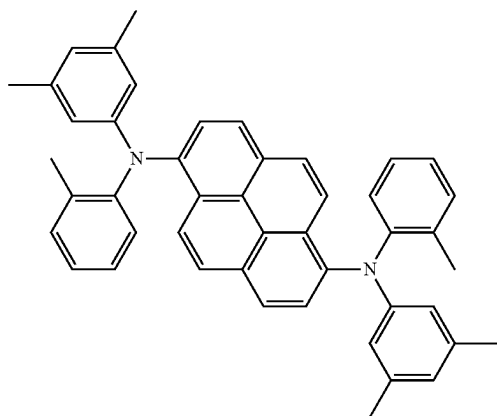
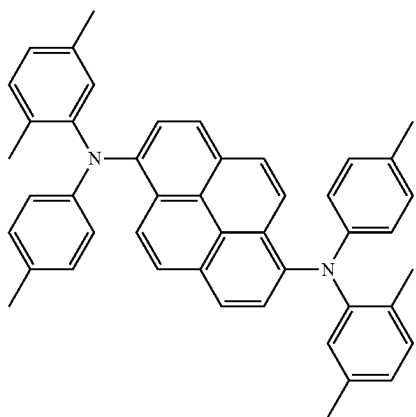
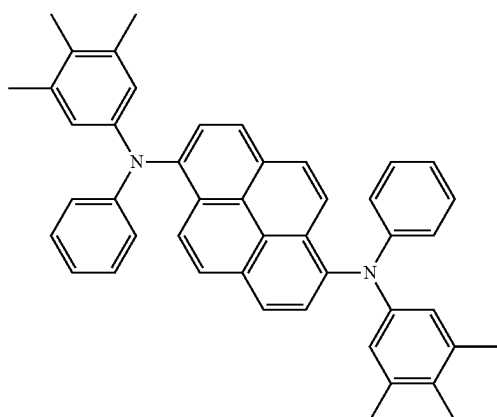
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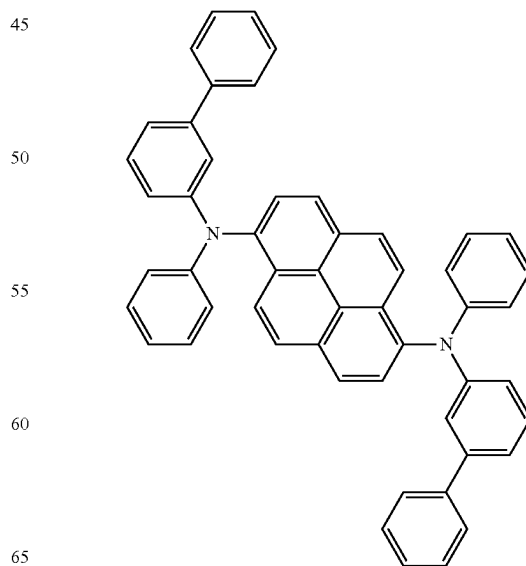
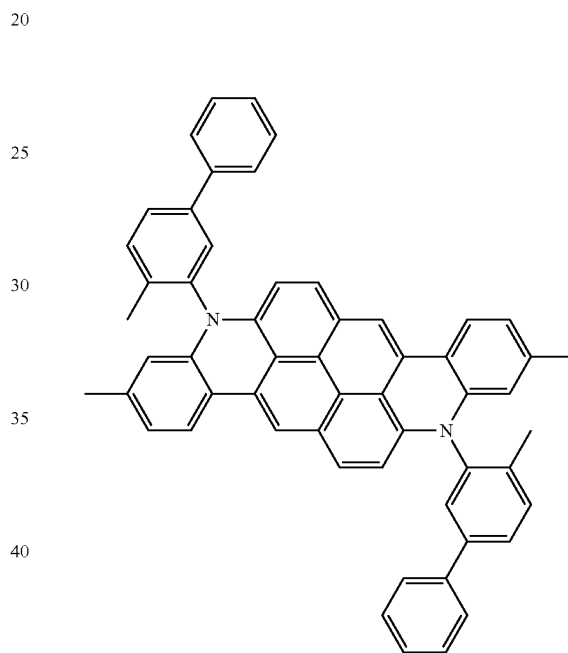
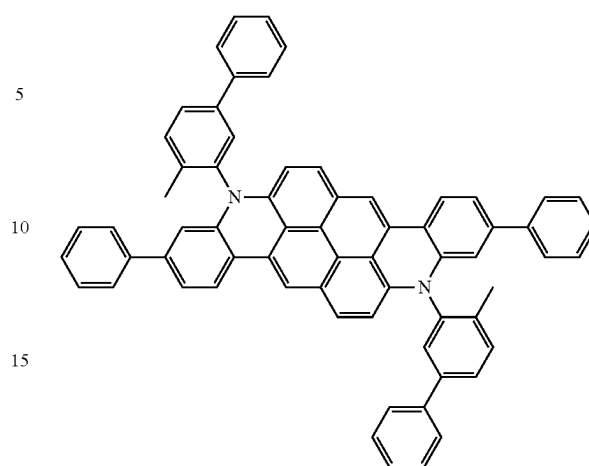


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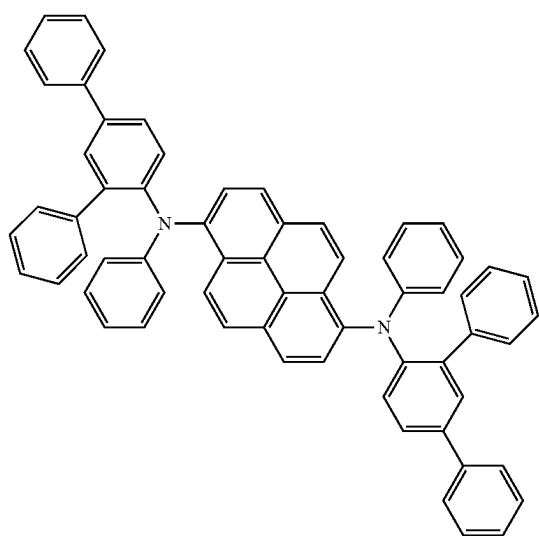
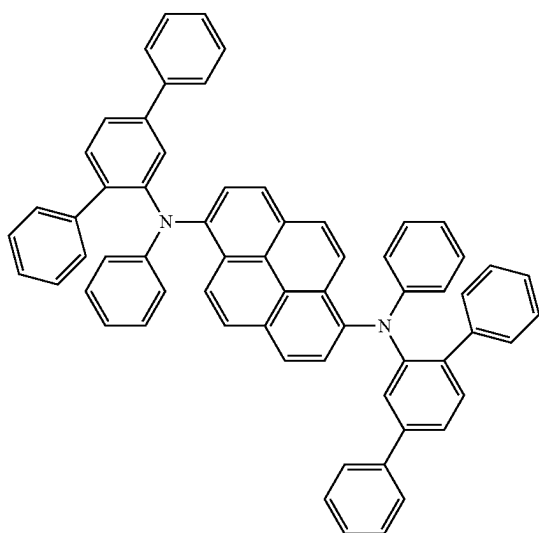
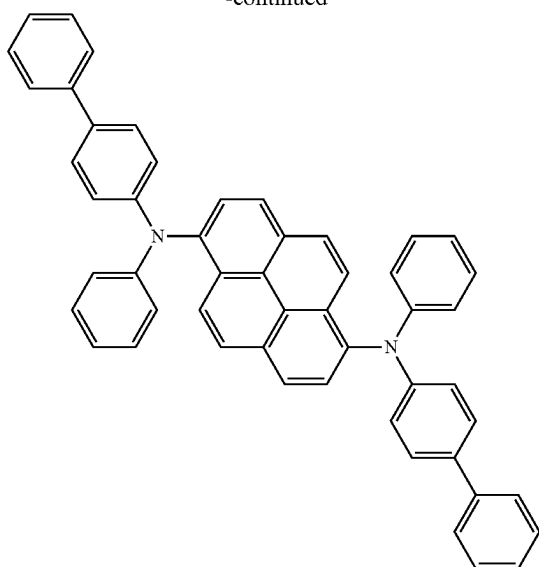
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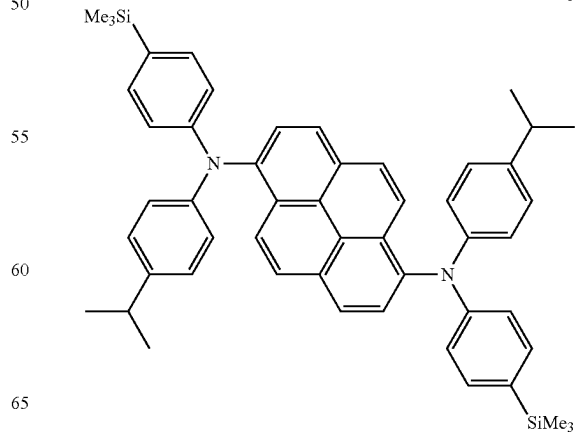
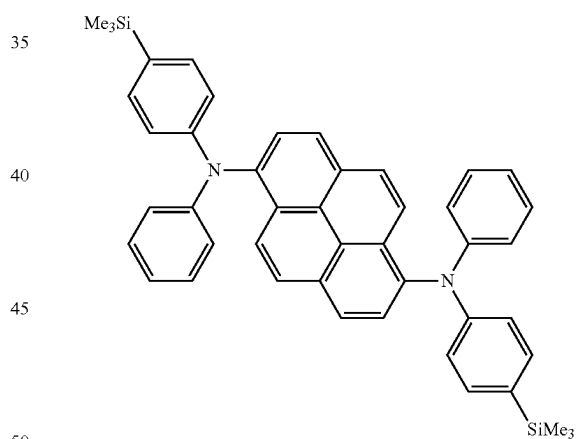
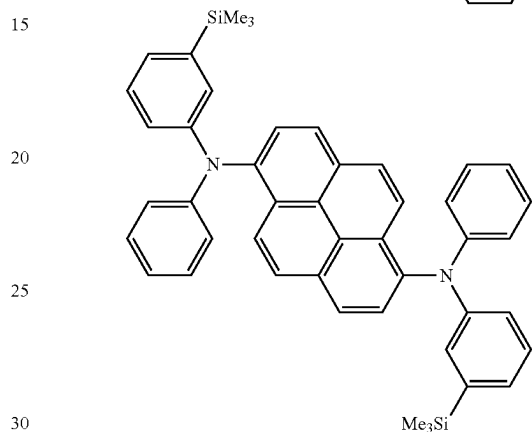
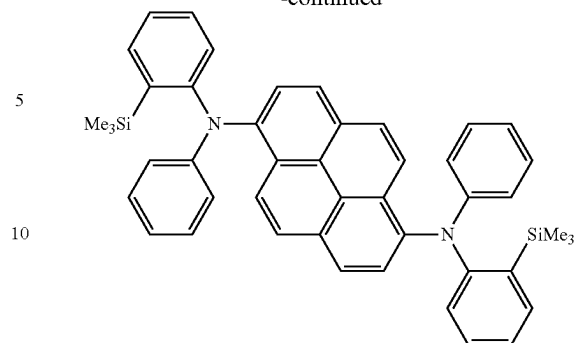


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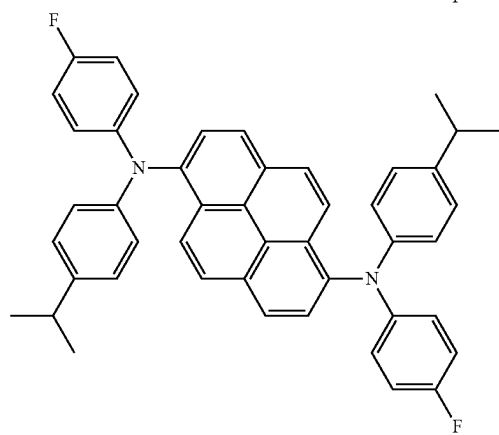
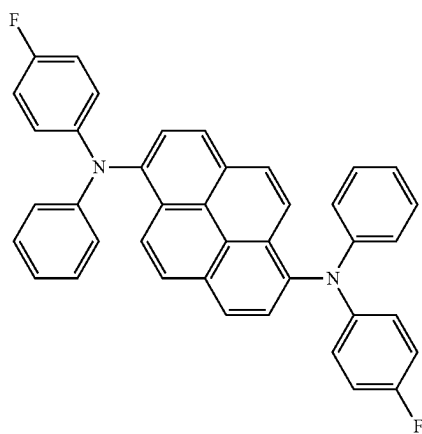
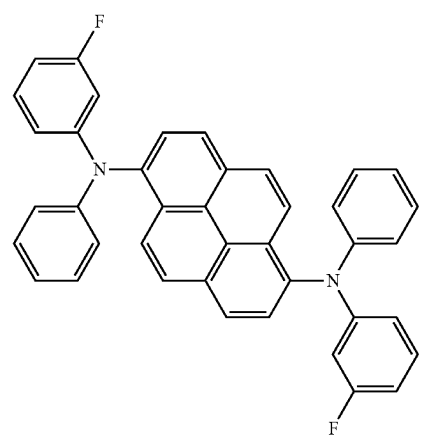
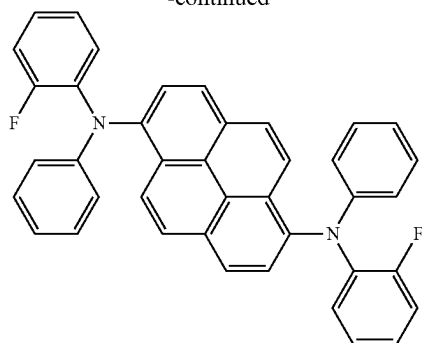
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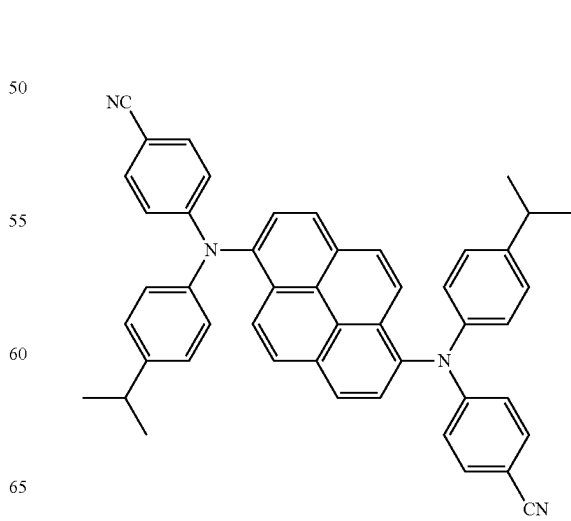
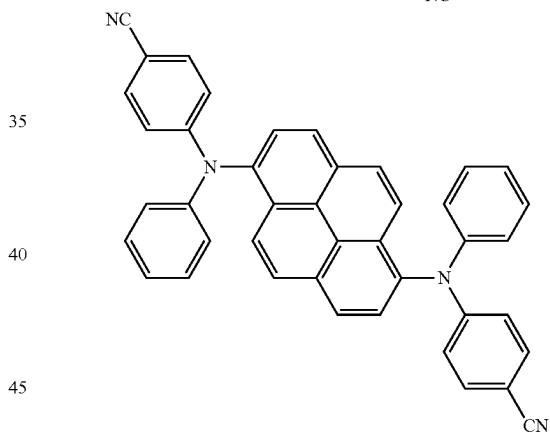
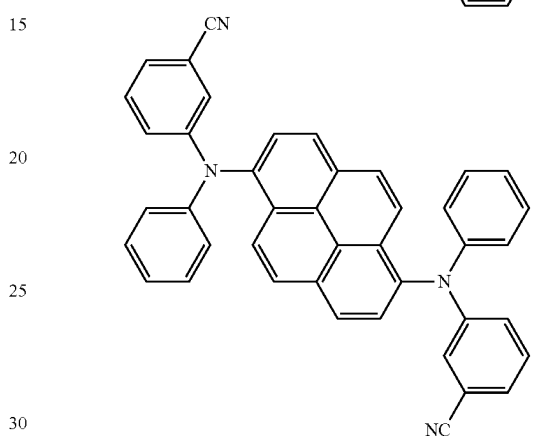
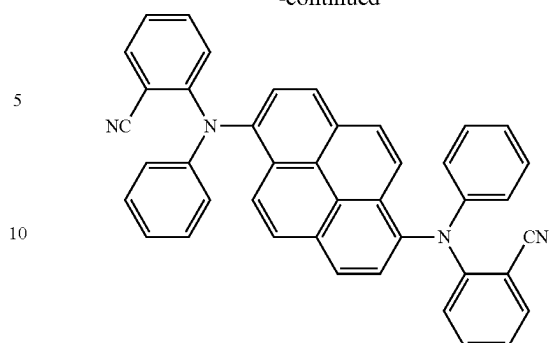


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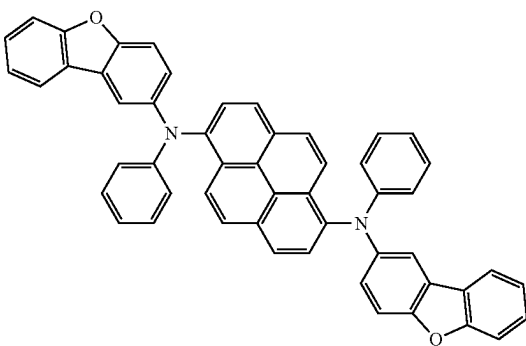
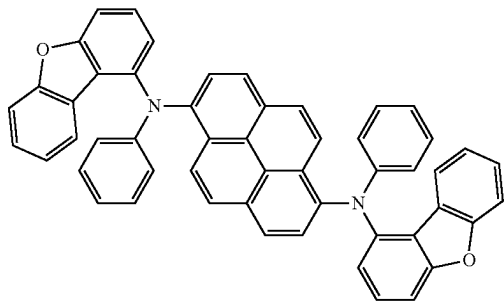
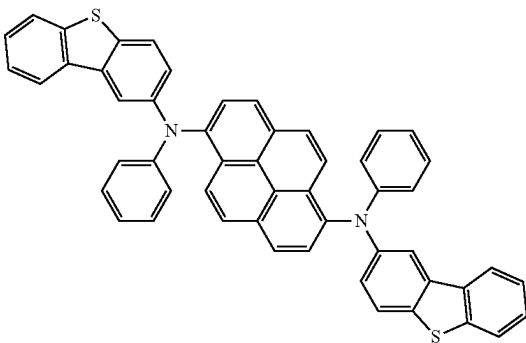
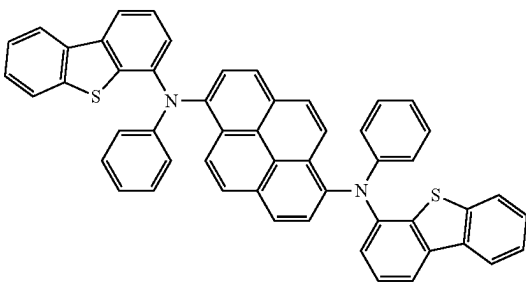
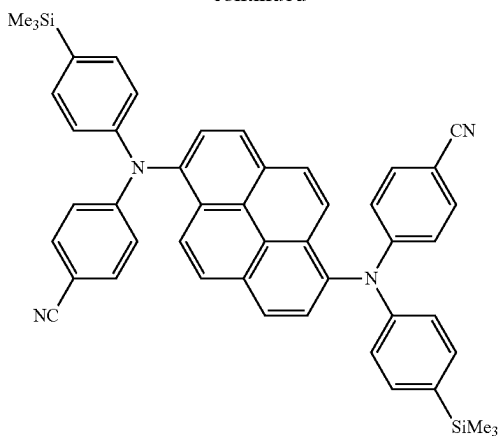
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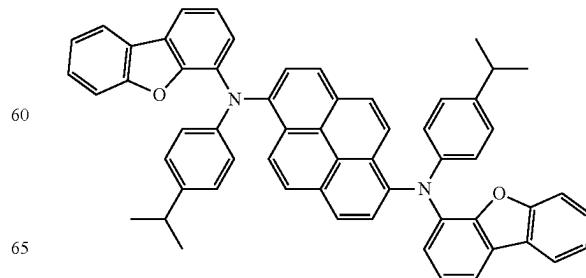
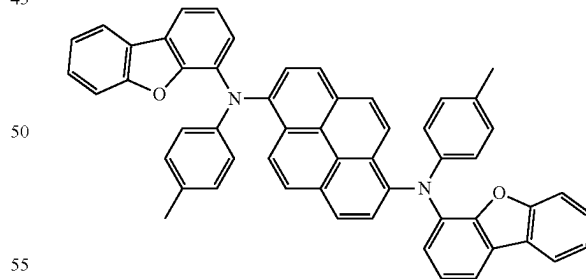
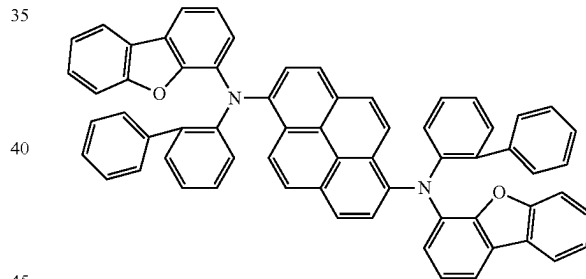
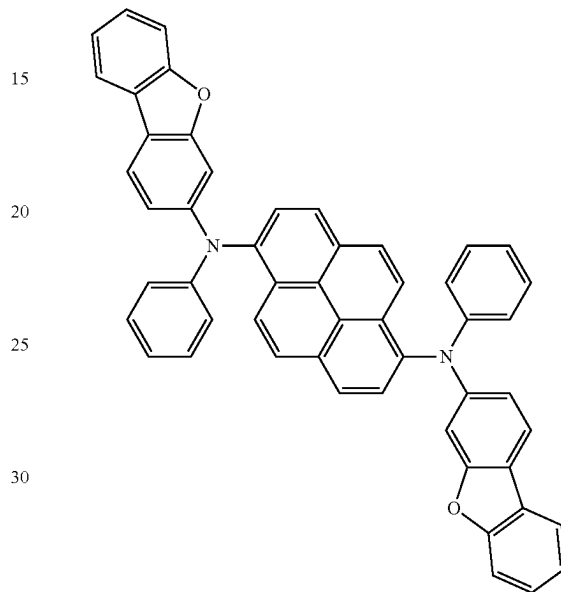
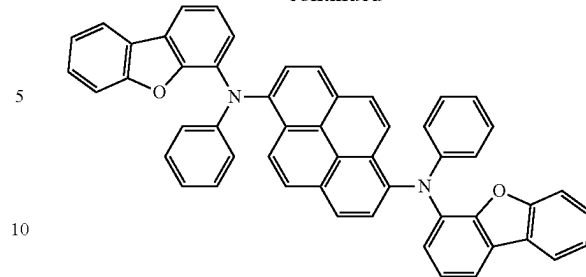
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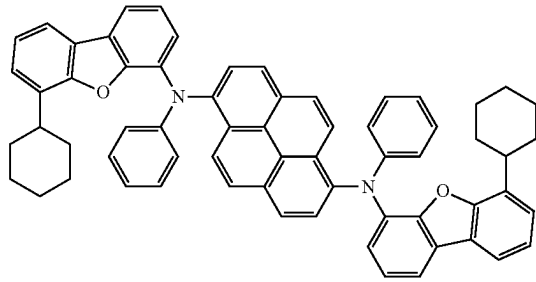
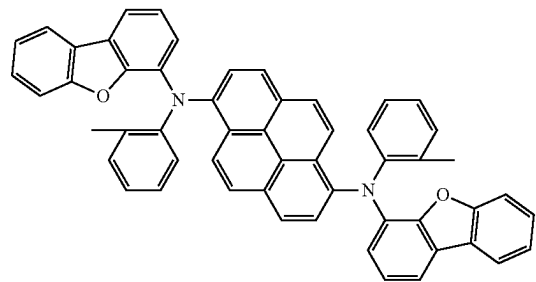
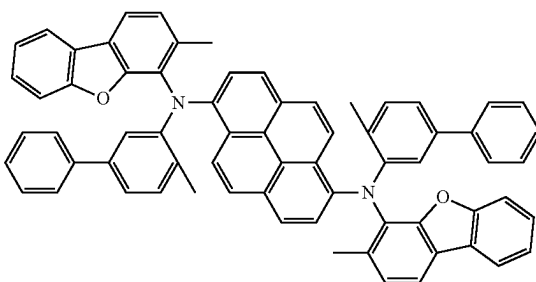
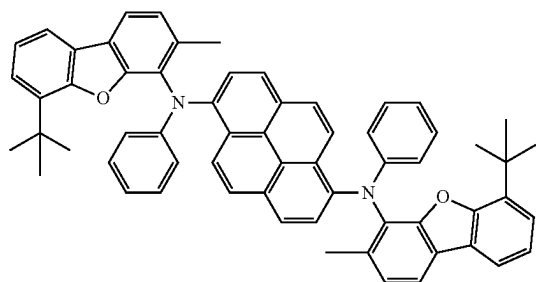
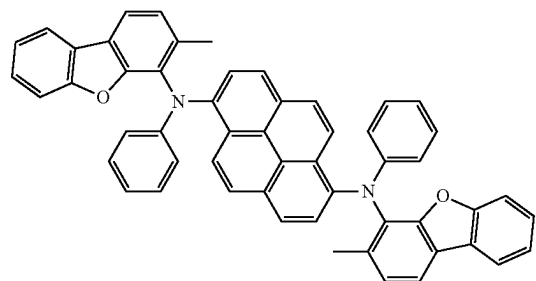
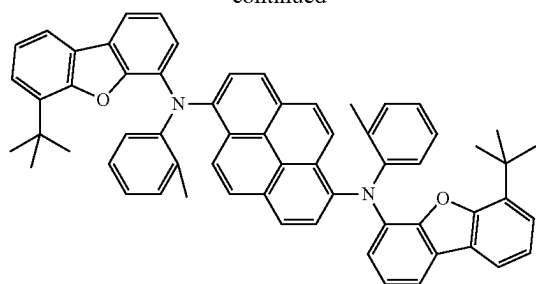
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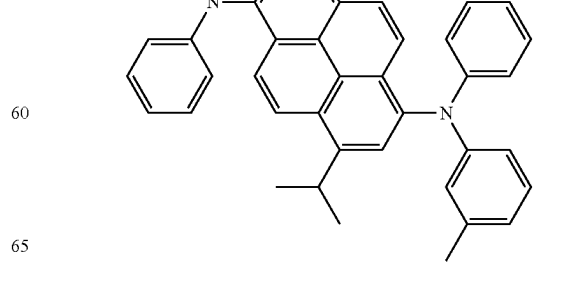
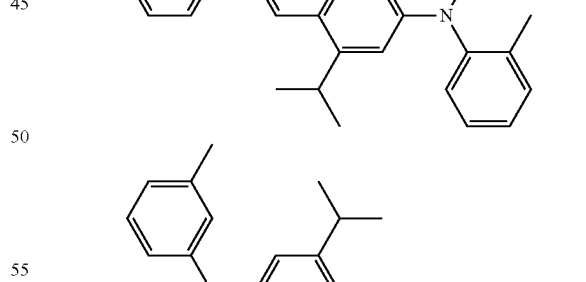
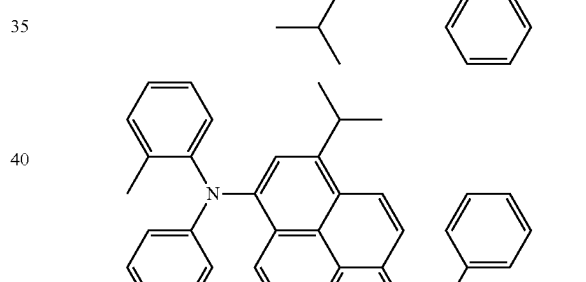
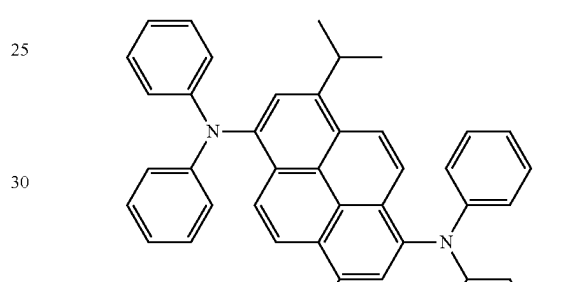
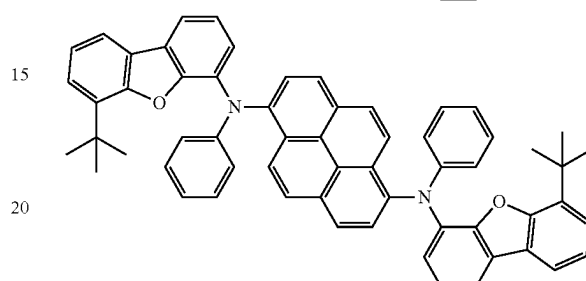
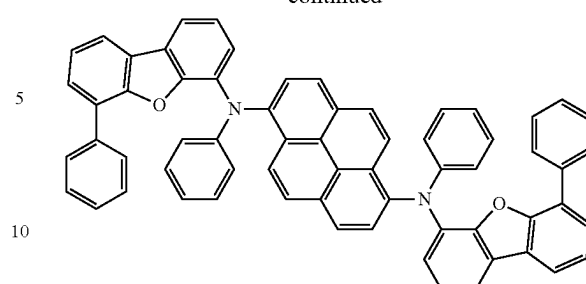


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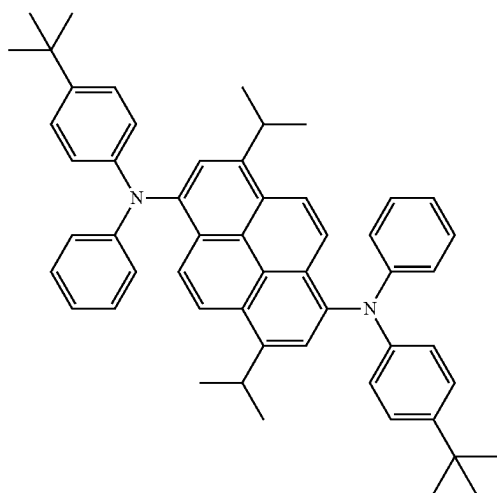
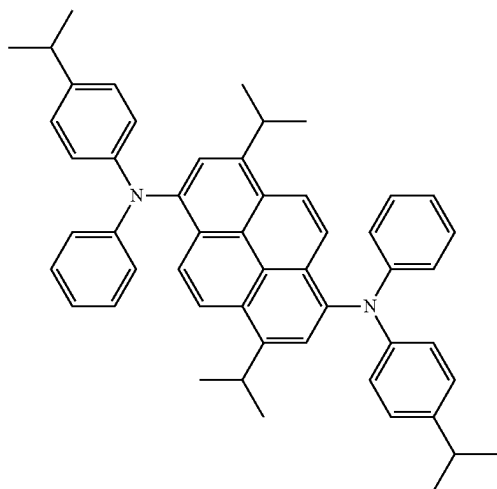
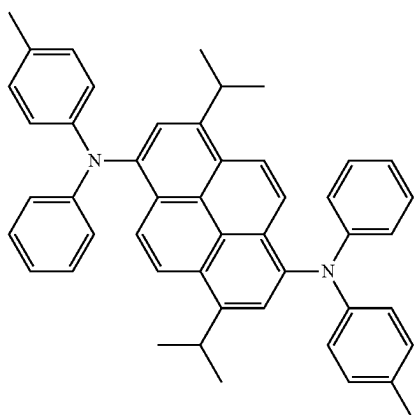
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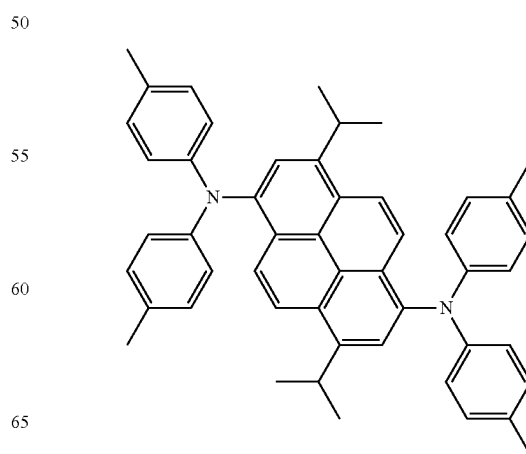
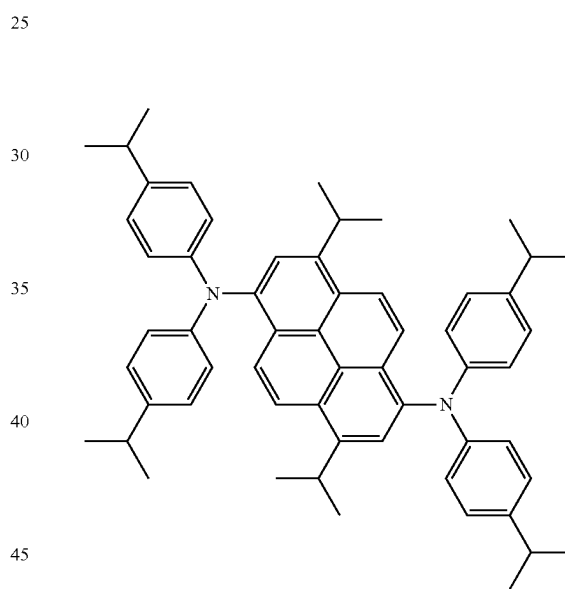
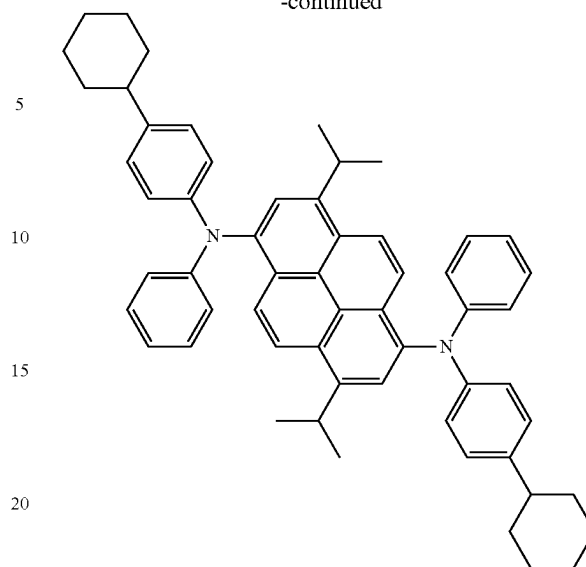


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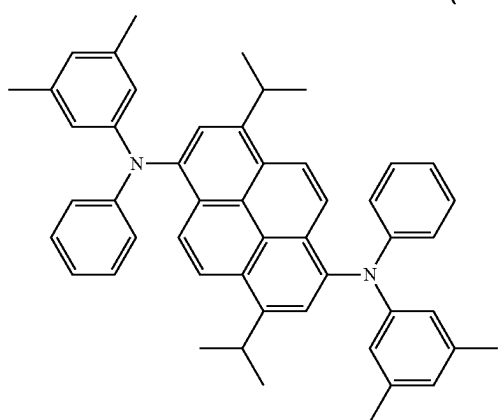
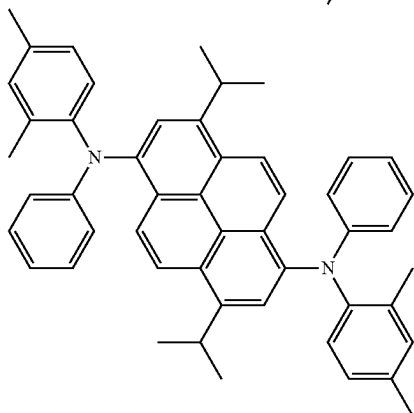
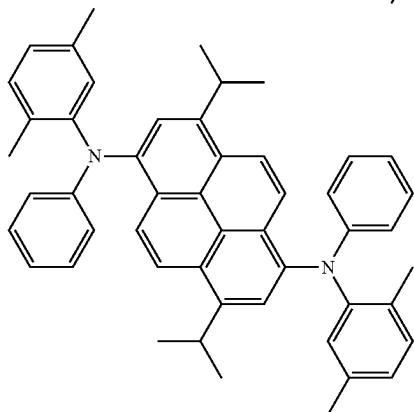
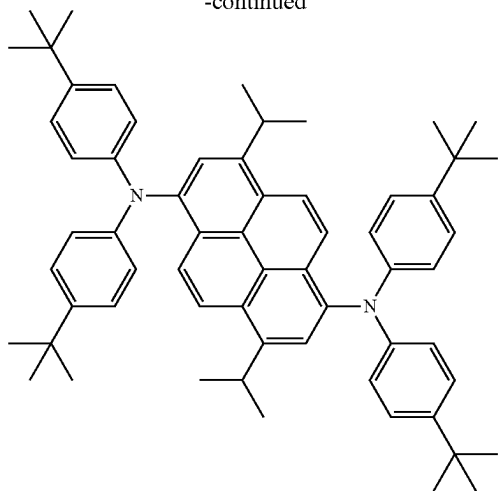
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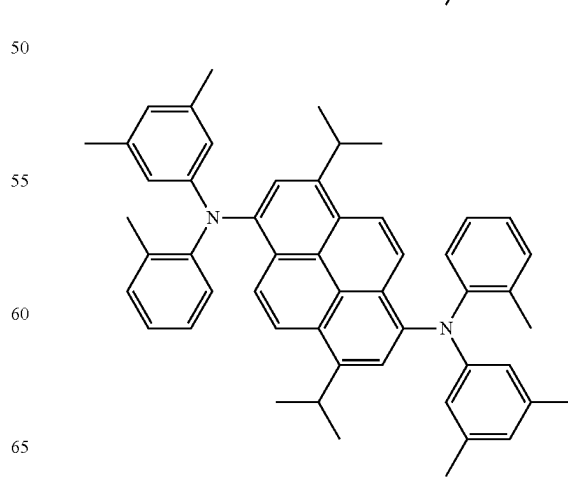
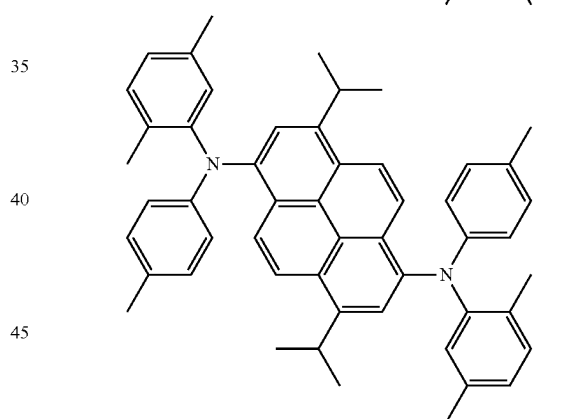
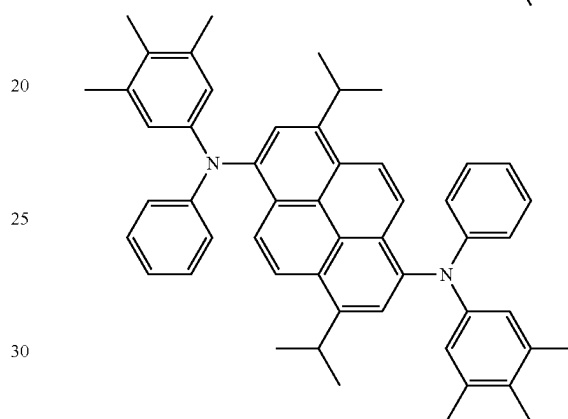
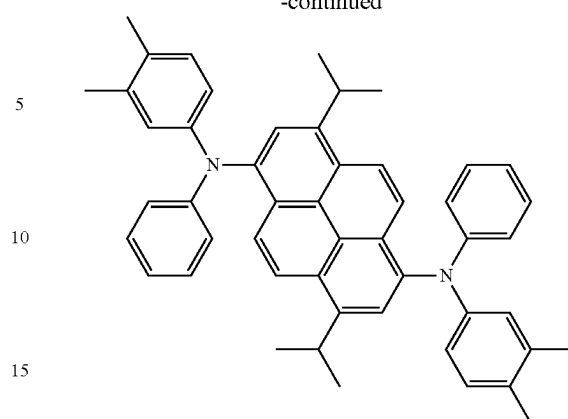


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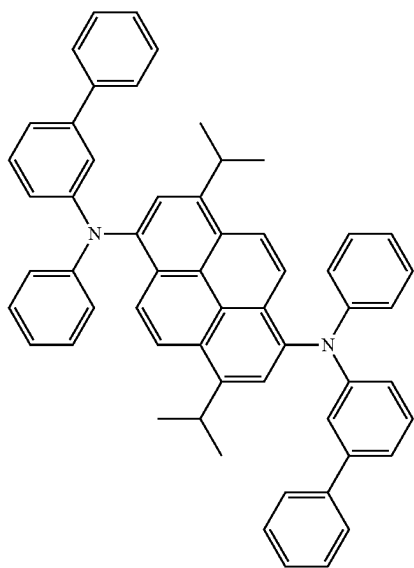
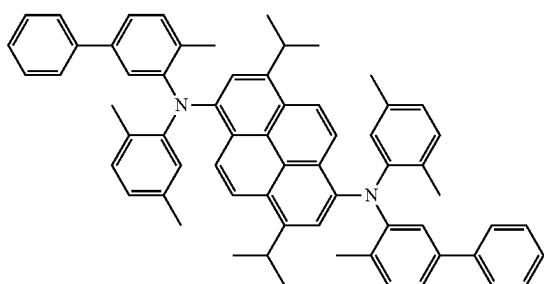
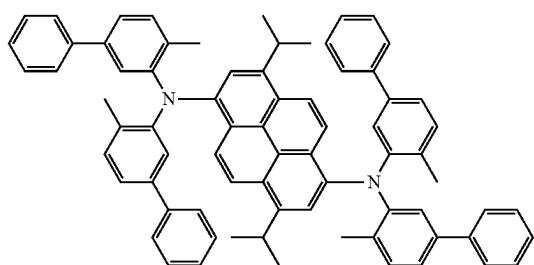
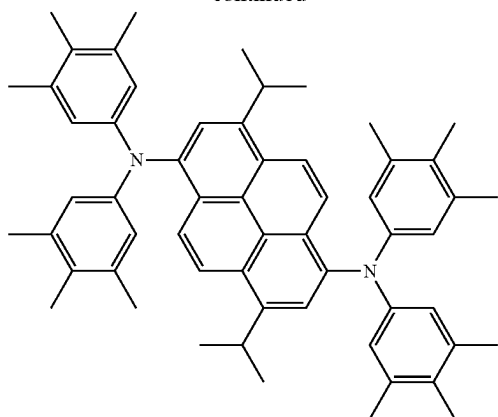
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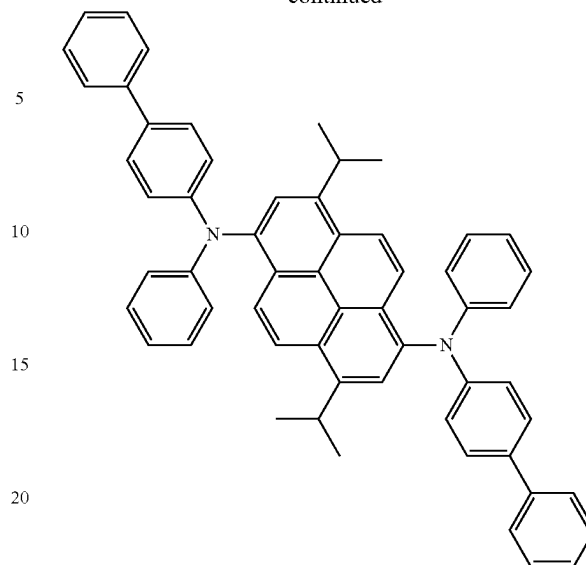


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**174**

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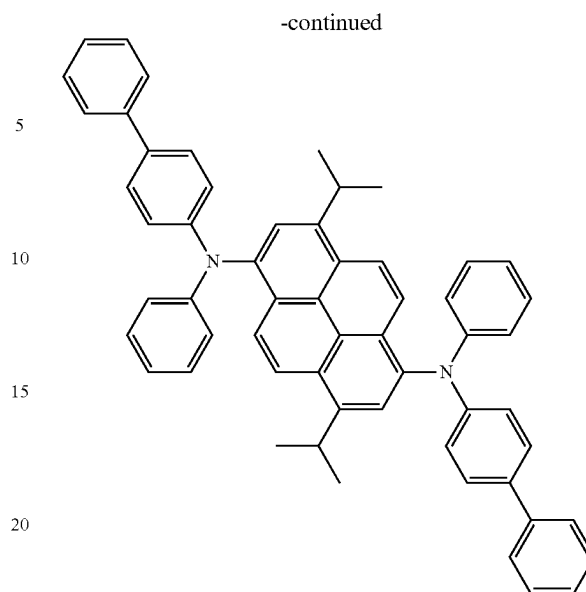
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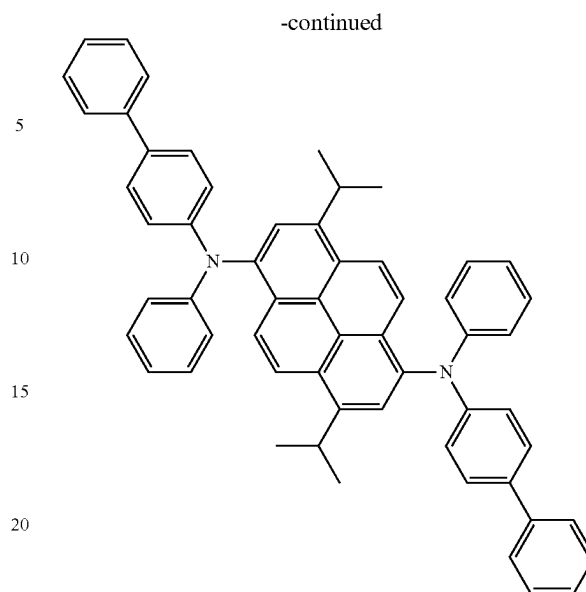
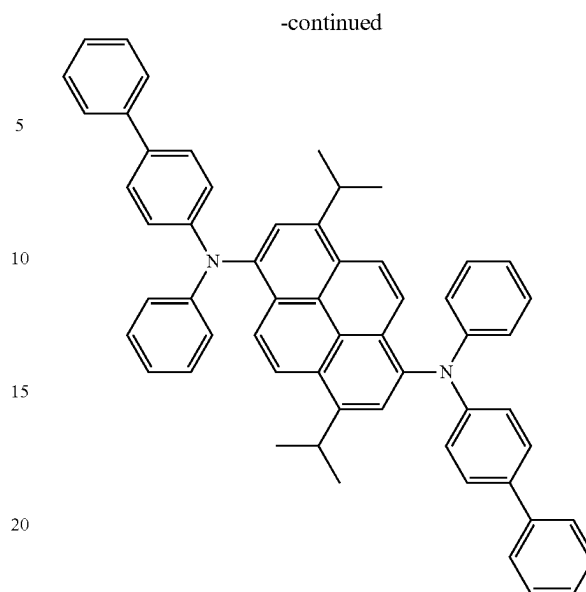
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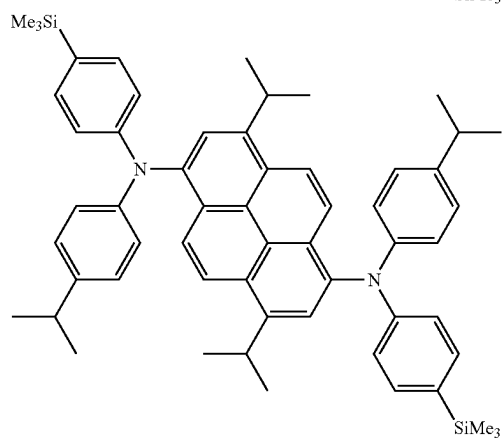
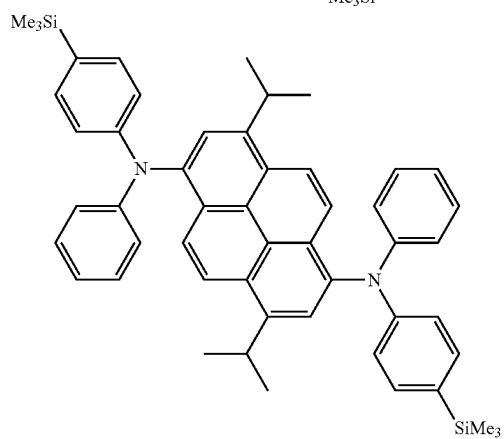
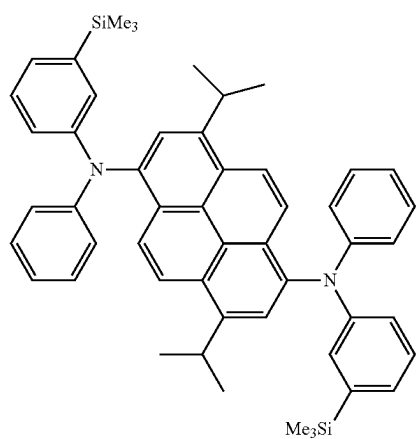
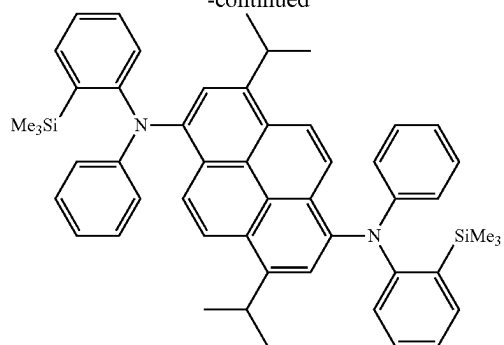
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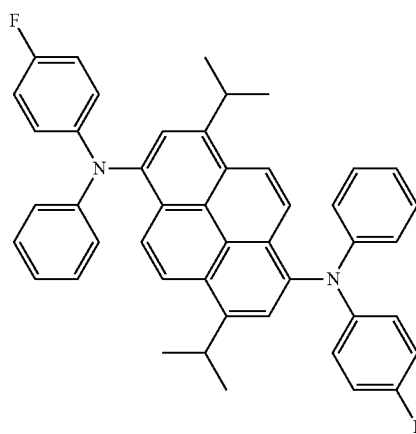
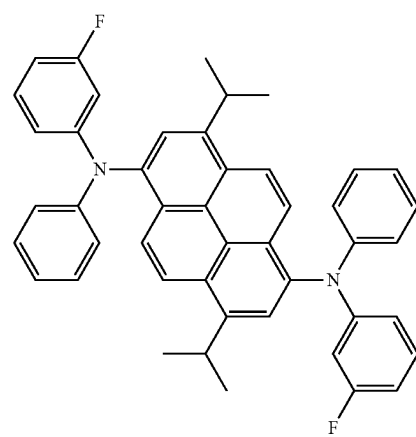
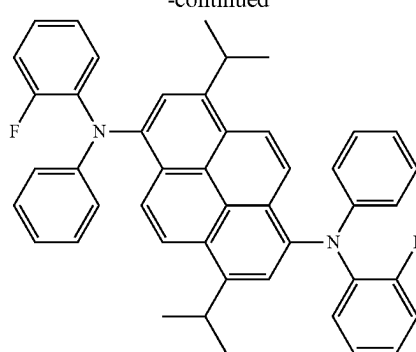


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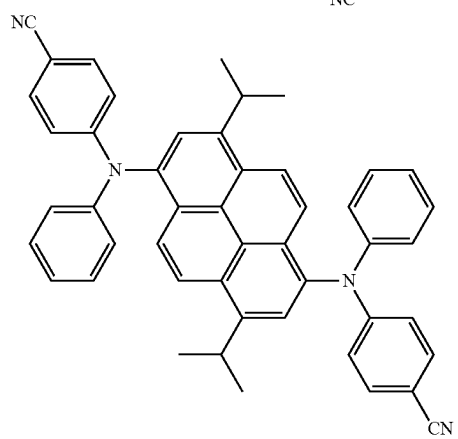
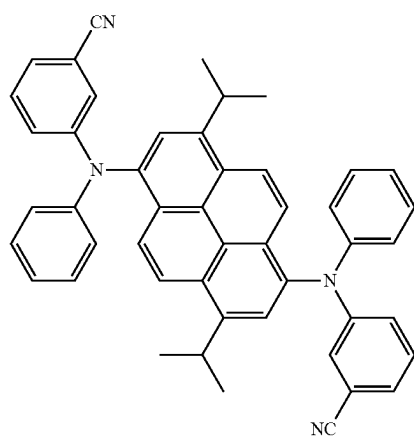
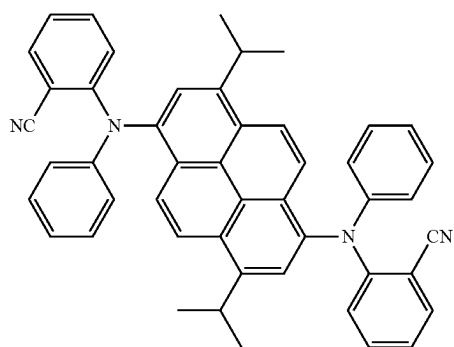
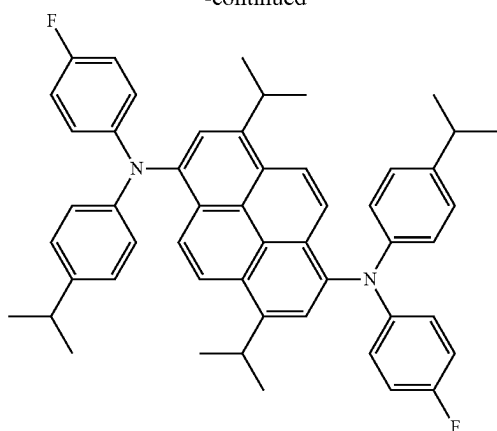
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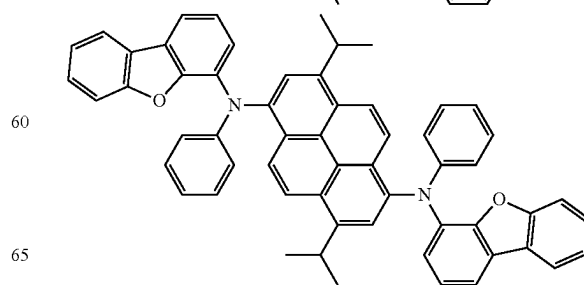
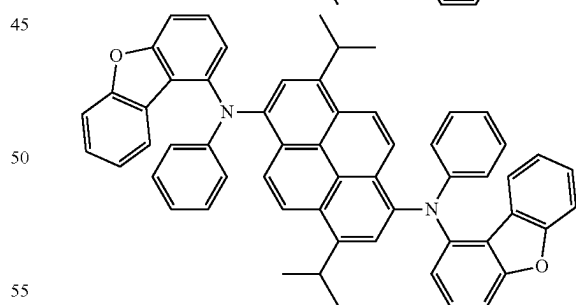
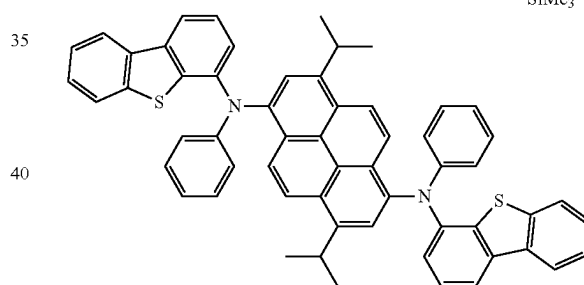
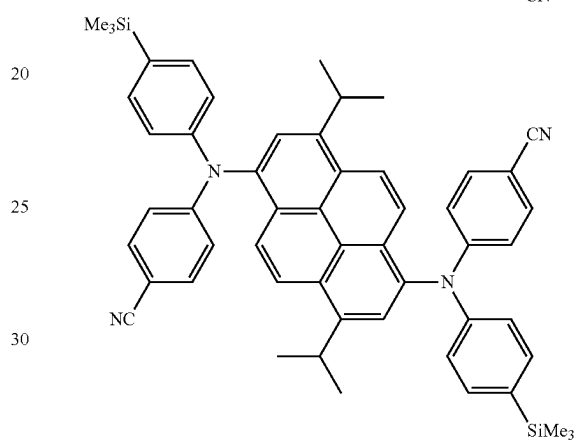
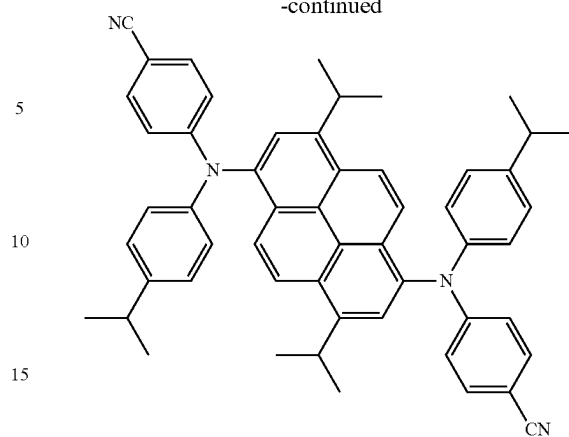


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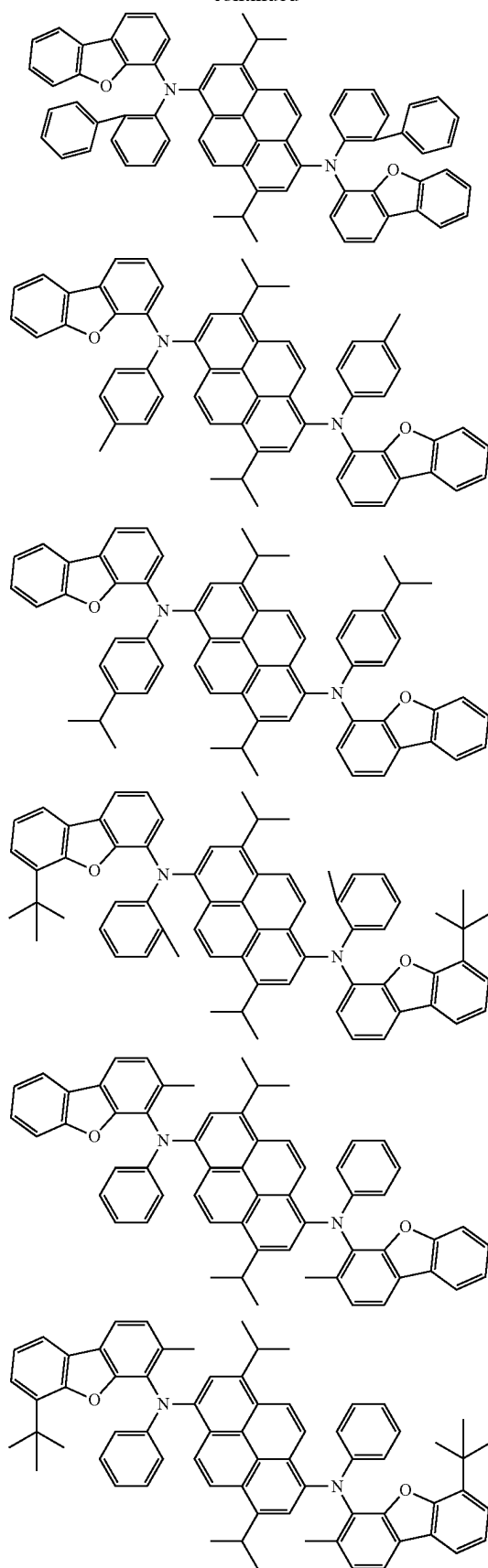
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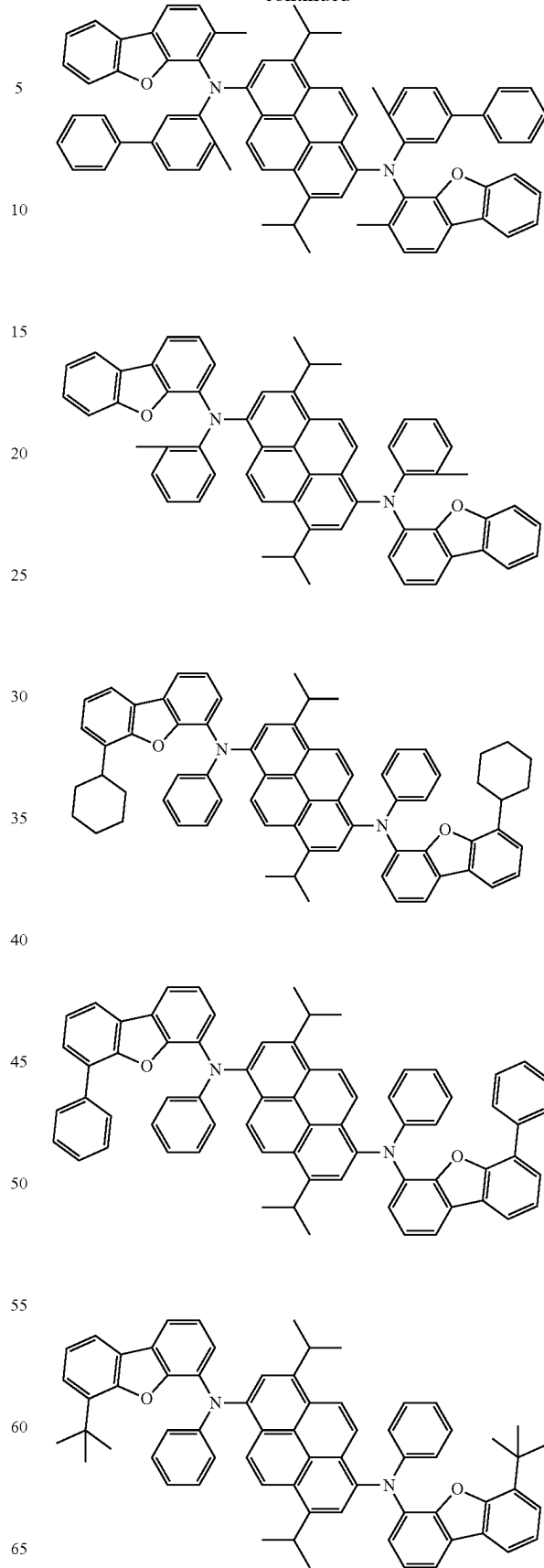


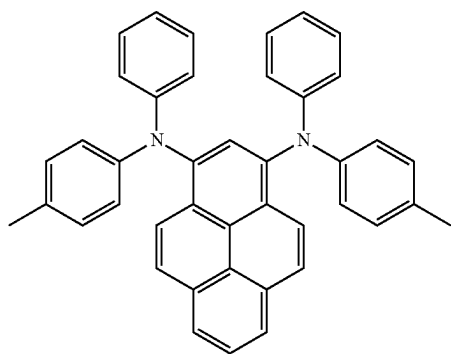
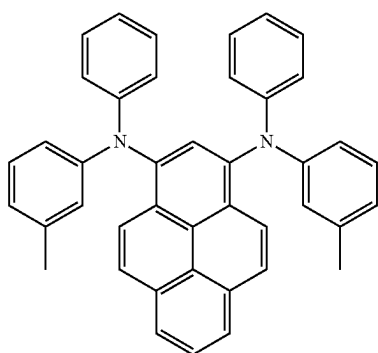
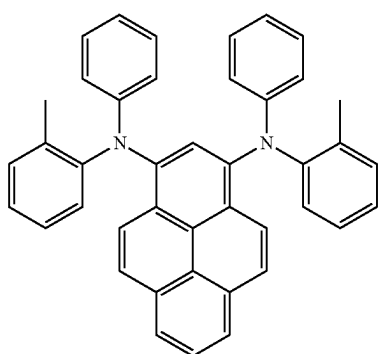
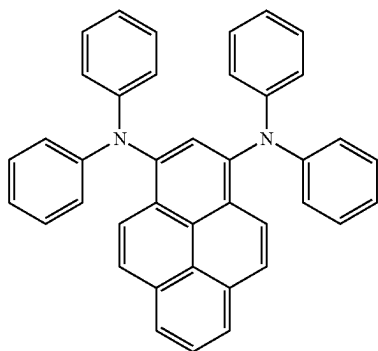
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**180**

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181**182**

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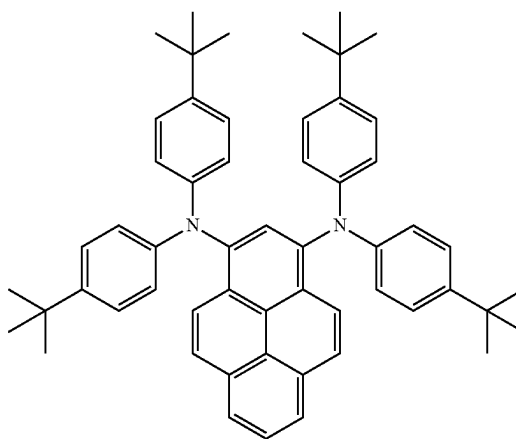
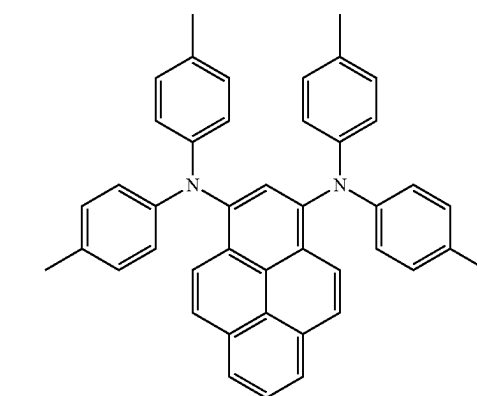
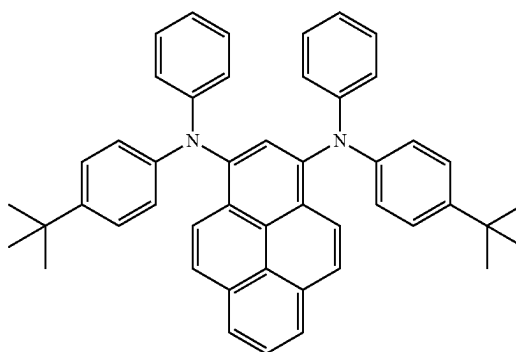
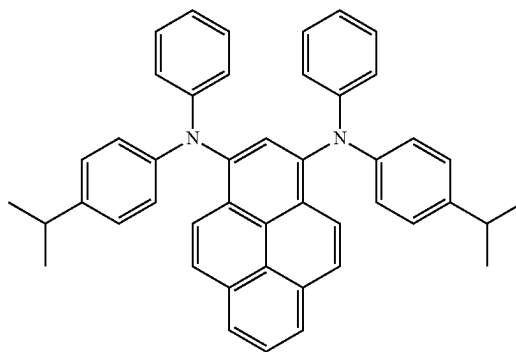
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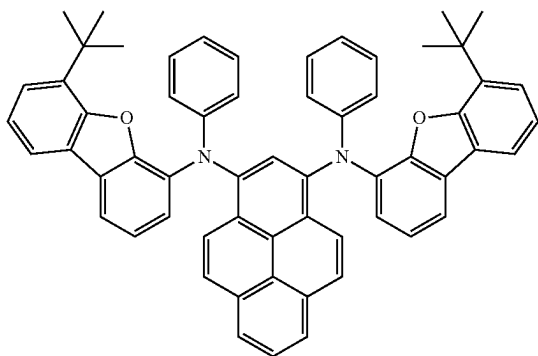
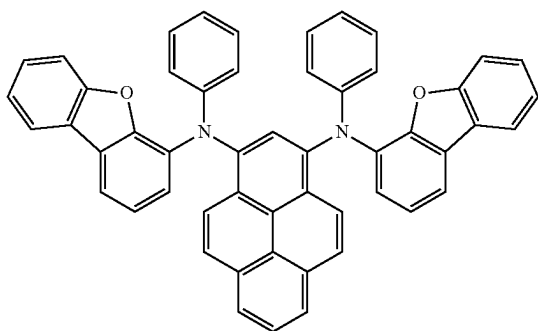
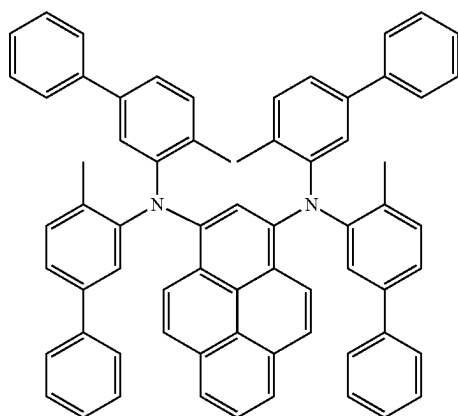
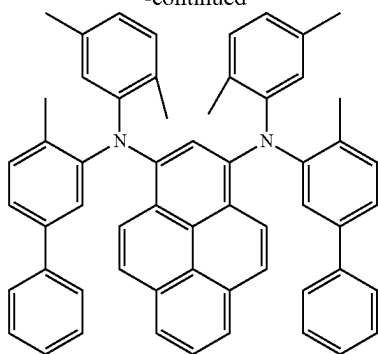
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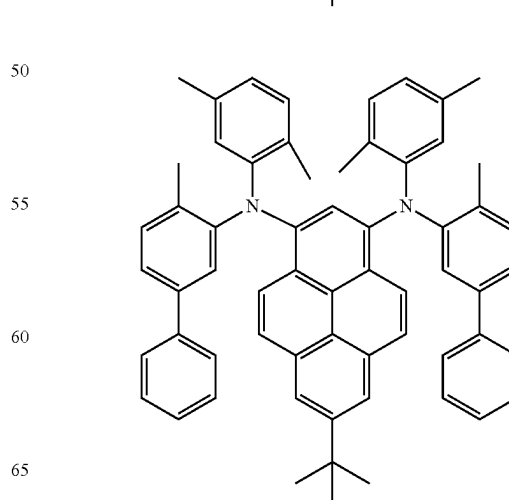
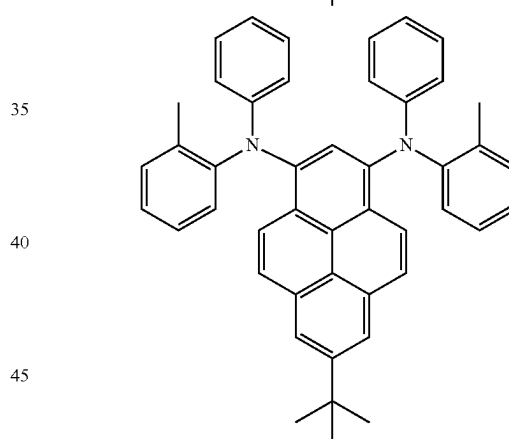
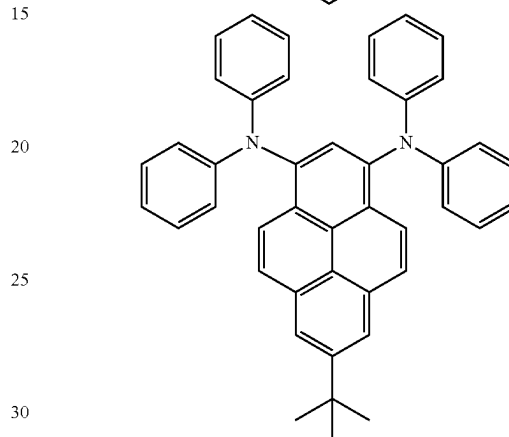
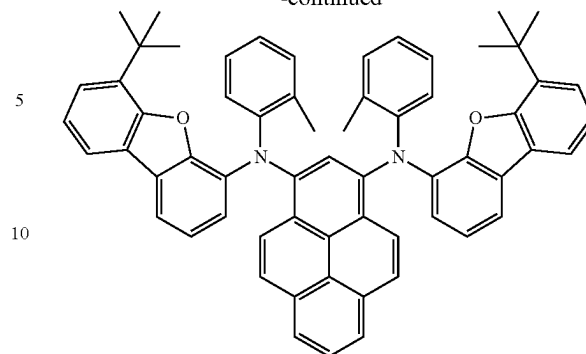


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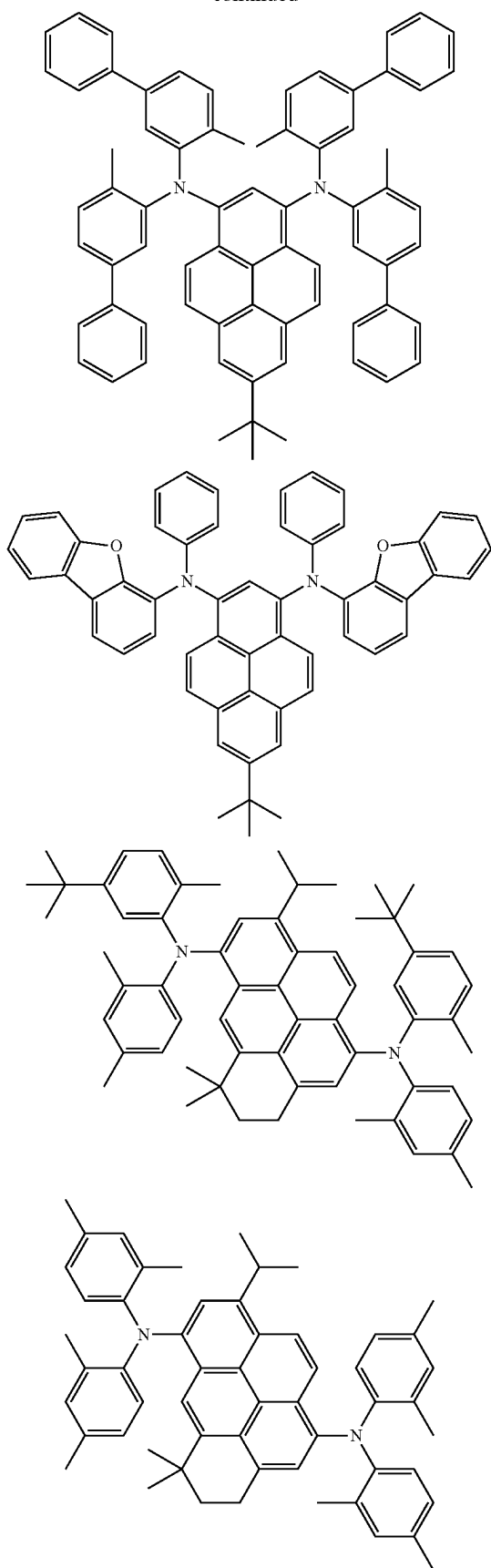
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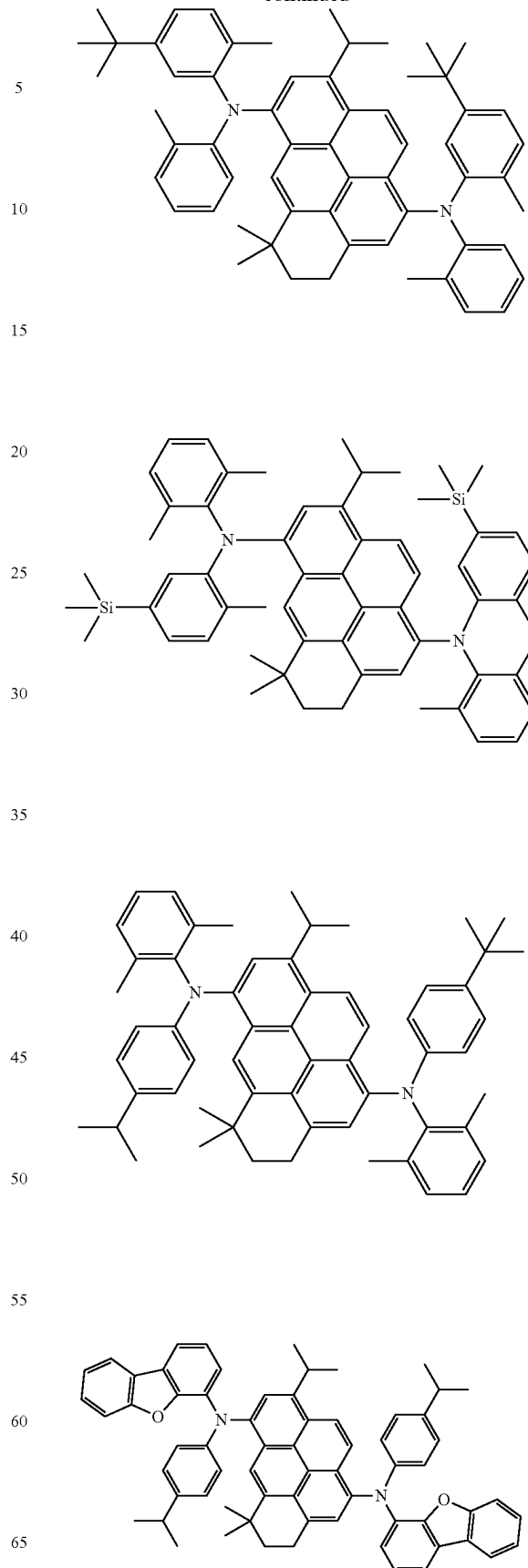


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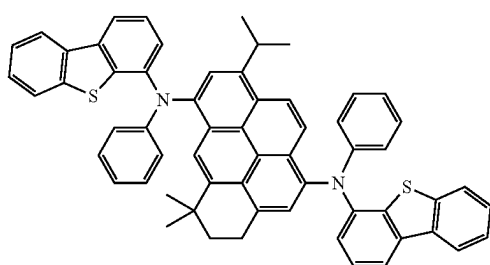
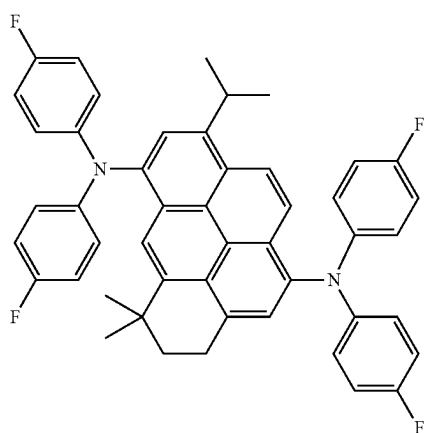
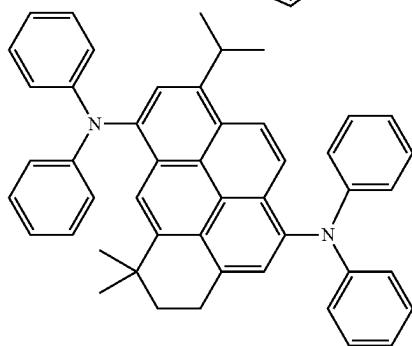
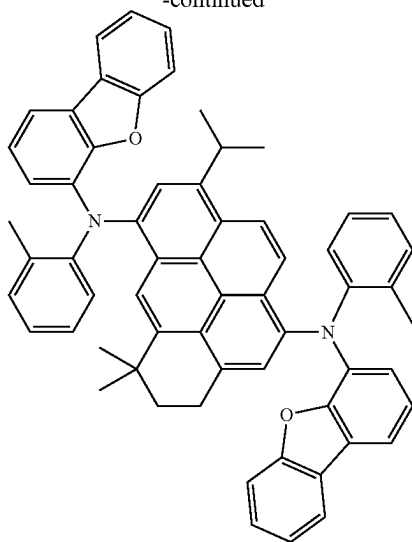
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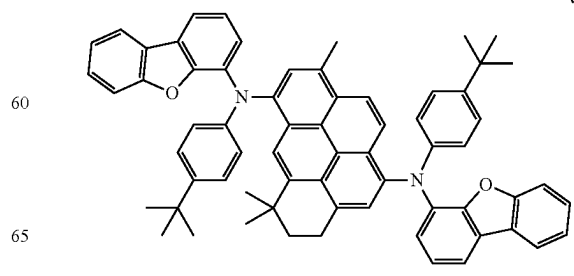
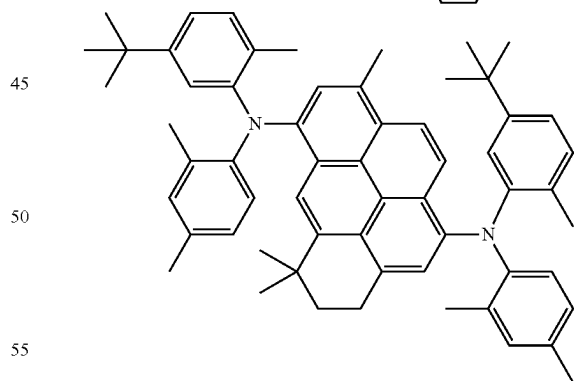
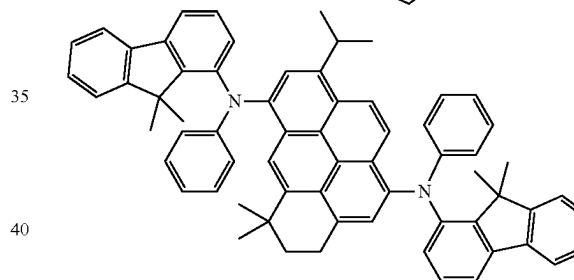
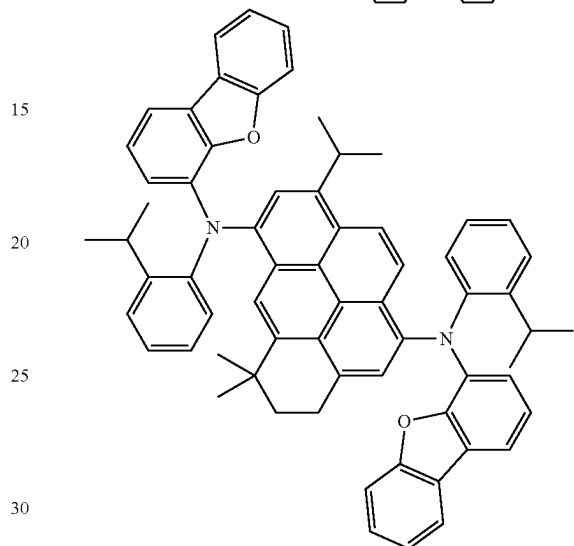
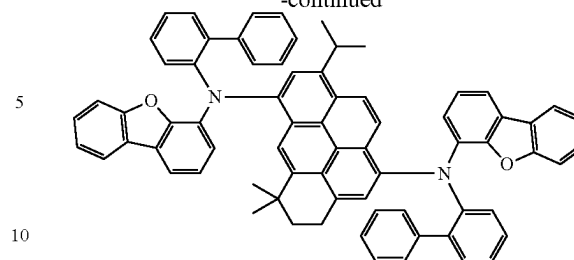


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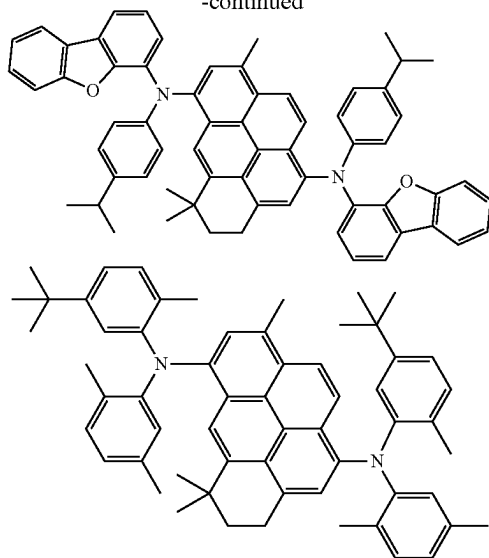
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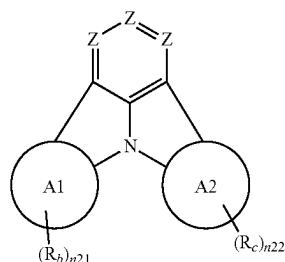
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(Compound Represented by Formula (21))

The compound represented by the formula (21) is explained below.



In the formula (21),

Zs are independently CR_a or N;

A1 ring and A2 ring are independently a substituted or unsubstituted aromatic hydrocarbon ring having 6 to 50 ring carbon atoms, or a substituted or unsubstituted heterocyclic ring having 5 to 50 ring atoms;

when plural R_as exist, one or more pairs of two or more adjacent groups of R_a are bonded with each other to form a substituted or unsubstituted, saturated or unsaturated ring, or do not form a substituted or unsubstituted, saturated or unsaturated ring;

when plural R_bs exist, one or more pairs of two or more adjacent groups of R_b are bonded with each other to form a substituted or unsubstituted, saturated or unsaturated ring, or do not form a substituted or unsubstituted, saturated or unsaturated ring;

when plural R_cs exist, one or more pairs of two or more adjacent groups of R_c are bonded with each other to form a substituted or unsubstituted, saturated or unsaturated ring, or do not form a substituted or unsubstituted, saturated or unsaturated ring;

n21 and n22 are independently an integer of 0 to 4;

R_a to R_c that do not form the substituted or unsubstituted, saturated or unsaturated ring are independently

a hydrogen atom,

a substituted or unsubstituted alkyl group having 1 to 50 carbon atoms,

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a substituted or unsubstituted alkenyl group having 2 to 50 carbon atoms,

a substituted or unsubstituted alkynyl group having 2 to 50 carbon atoms,

5 a substituted or unsubstituted cycloalkyl group having 3 to 50 ring carbon atoms,

—Si(R₉₀₁)(R₉₀₂)(R₉₀₃),

10 —O—(R₉₀₄),

—S—(R₉₀₅),

—N(R₉₀₆)(R₉₀₇),

a halogen atom, a cyano group, a nitro group,

15 a substituted or unsubstituted aryl group having 6 to 50 ring carbon atoms, or

a substituted or unsubstituted monovalent heterocyclic group having 5 to 50 ring atoms;

R₉₀₁ to R₉₀₇ are as defined in the formula (1).

20 The “aromatic hydrocarbon ring” of A1 ring and A2 ring has the same structure as the compound obtained by introducing a hydrogen atom into the “aryl group” described above.

The “aromatic hydrocarbon ring” of the A1 ring and the A2 ring contains to carbon atoms in the fused bicyclic structure at the center of the formula (21) as ring atoms. Examples of “substituted or unsubstituted aromatic hydrocarbon ring having 6 to 50 ring carbon atoms” include compounds in which a hydrogen atom is introduced into the “aryl group” described in the example group G1.

30 The “heterocyclic ring” of A1 ring and A2 ring has the same structure as the compound obtained by introducing a hydrogen atom into the “heterocyclic group” described above. The “heterocyclic ring” of the A1 ring and the A2 ring contains two carbon atoms in the fused bicyclic structure at the center of the formula (21) as ring atoms. Examples of “substituted or unsubstituted heterocyclic ring having 51 to 50 ring atoms” include compounds in which a hydrogen atom is introduced into the “heterocyclic group” described in the example group G2.

R_b is bonded to one of carbon atoms which form the aromatic hydrocarbon ring of A1 ring, or one of atoms which form the heterocycle of A1 ring.

35 R_c is bonded to one of carbon atoms which form the aromatic hydrocarbon ring of A2 ring, or one of atoms which form the heterocycle of A2 ring.

It is preferable that at least one (preferably two) of R_a to R_c is a group represented by the following formula (21a).



In the formula (21a),

L₂₀₁ is

a single bond,

a substituted or unsubstituted arylene group having 6 to 30 ring carbon atoms, or

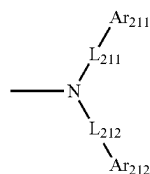
a substituted or unsubstituted bivalent heterocyclic group having 5 to 30 ring atoms;

Ar₂₀₁ is

a substituted or unsubstituted arylene group having 6 to 50 ring carbon atoms,

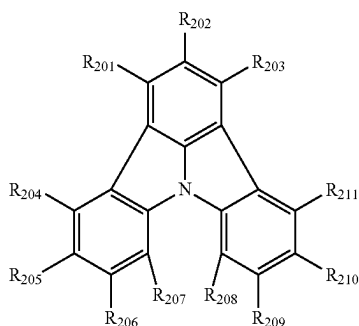
65 a substituted or unsubstituted monovalent heterocyclic group having 5 to 50 ring atoms, or a group represented by the following formula (21b):

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wherein in the formula (21b),
 L_{211} and L_{212} are independently
 a single bond,
 a substituted or unsubstituted arylene group having 6 to
 30 ring carbon atoms, or
 a substituted or unsubstituted divalent heterocyclic group
 having 5 to 30 ring atoms;
 Ar_{212} and Ar_{22} are bonded with each other to form a
 substituted or unsubstituted, saturated or unsaturated
 ring, or do not form a substituted or unsubstituted,
 saturated or unsaturated ring; and
 Ar_{212} and Ar_{22} that do not form a substituted or unsub-
 stituted, saturated or unsaturated ring are independ-
 ently
 a substituted or unsubstituted aryl group having 6 to 50
 ring carbon atoms or
 a substituted or unsubstituted monovalent heterocyclic
 group having 5 to 50 ring atoms.

In one embodiment, the compound represented by the
 formula (21) is represented by the following formula (22).



In the formula (22),

one or more pairs of two or more adjacent groups of R_{201}
 to R_{211} are bonded with each other to form a substituted
 or unsubstituted, saturated or unsaturated ring, or do
 not form a substituted or unsubstituted saturated or
 unsaturated ring;

R_{201} to R_{211} that do not form the substituted or unsubsti-
 tuted, saturated or unsaturated ring are independently
 a hydrogen atom,
 a substituted or unsubstituted alkyl group having 1 to 50
 carbon atoms,
 a substituted or unsubstituted alkenyl group having 2 to
 50 carbon atoms,
 a substituted or unsubstituted alkynyl group having 2 to
 50 carbon atoms,
 a substituted or unsubstituted cycloalkyl group having 3
 to 50 ring carbon atoms,

$—Si(R_{901})(R_{902})(R_{903})$,

$—O—(R_{904})$,

$—S—(R_{905})$,

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$—N(R_{906})(R_{907})$,

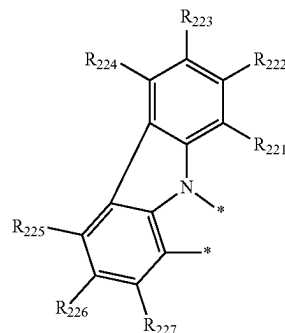
a halogen atom, a cyano group, a nitro group,
 a substituted or unsubstituted aryl group having 6 to 50
 ring carbon atoms, or

5 a substituted or unsubstituted monovalent heterocyclic
 group having 5 to 50 ring atoms; and

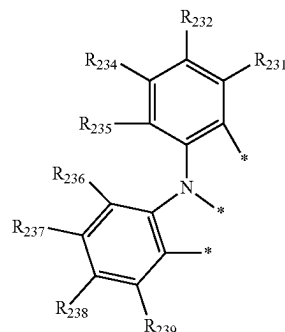
R_{901} to R_{907} are as defined in the formula (1).

It is preferable that at least one (preferably two) of R_{201}
 to R_{211} is the group represented by the formula (21a). It is
 preferable that R_{204} and R_{211} are the group represented by
 the formula (21a).

In one embodiment, the compound represented by the
 formula (21) is a compound obtained by bonding the struc-
 ture represented by the following formula (21-1) or (21-2) to
 A1 ring. In one embodiment, the compound represented by
 the formula (22) is a compound obtained by bonding the
 structure represented by the following formula (21-1) or
 (21-2) to the ring to which R_{204} to R_{207} bonds to.



21-1



21-2

In the formula (21-1), two bonds shown by * independ-
 ently bond to a ring carbon atom in the aromatic hydro-
 carbon ring or a ring atom in the heterocyclic group in A1
 ring in the formula (21), or bond to one of R_{204} to R_{207} in the
 formula (22);

wherein in the formula (21-2), three bonds shown by *
 independently bond to a ring carbon atom in the
 aromatic hydrocarbon ring or a ring atom in the het-
 erocyclic group in A1 ring in the formula (21), or bond
 to one of R_{204} to R_{207} in the formula (22);

One or more pairs of two or more adjacent groups of R_{221}
 to R_{227} and R_{232} to R_{239} are bonded with each other to
 form a substituted or unsubstituted, saturated or unsatu-
 rated ring, or do not form a substituted or unsubstituted,
 saturated or unsaturated ring;

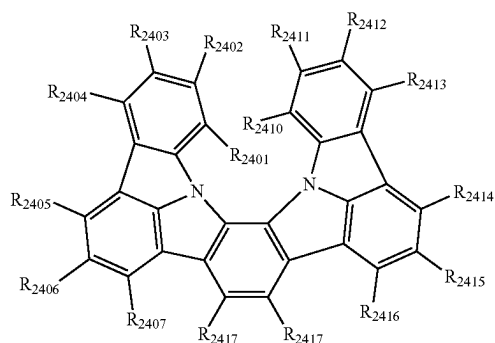
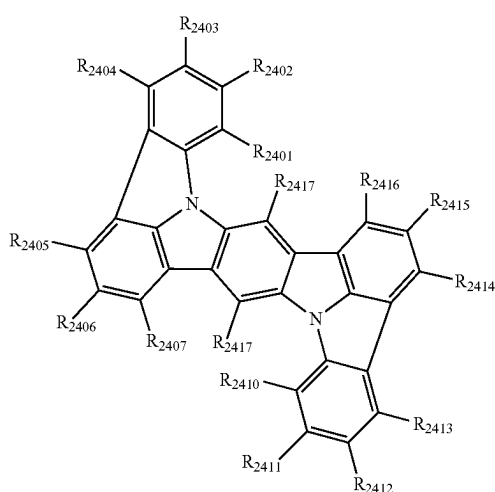
R_{221} to R_{227} and R_{231} to R_{239} that do not form the
 substituted or unsubstituted, saturated or unsaturated
 ring are independently

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- a hydrogen atom,
 a substituted or unsubstituted alkyl group having 1 to 50 carbon atoms,
 a substituted or unsubstituted alkenyl group having 2 to 50 carbon atoms,
 a substituted or unsubstituted alkynyl group having 2 to 50 carbon atoms,
 a substituted or unsubstituted cycloalkyl group having 3 to 50 ring carbon atoms,
 $-\text{Si}(\text{R}_{901})(\text{R}_{902})(\text{R}_{903})$,
 $-\text{O}-(\text{R}_{904})$,
 $-\text{S}-(\text{R}_{905})$,
 $-\text{N}(\text{R}_{906})(\text{R}_{907})$,
 a halogen atom, a cyano group, a nitro group,
 a substituted or unsubstituted aryl group having 6 to 50 ring carbon atoms, or
 a substituted or unsubstituted monovalent heterocyclic group having 5 to 50 ring atoms; and

R_{901} to R_{907} are as defined in the formula (1).

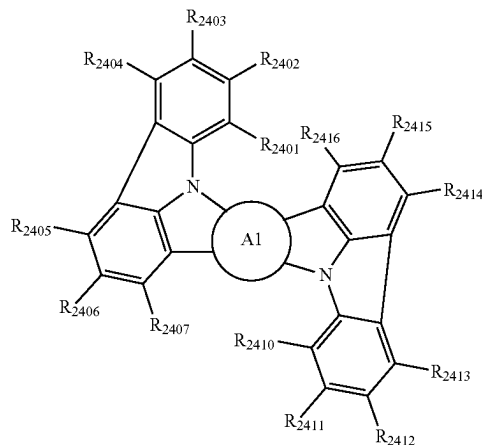
In one embodiment, the compound represented by the formula (21) is a compound represented by the following formula (21-3), (21-4), or (21-5).



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(21-5)



In the formulas (21-3), (21-4) and (21-5),

A1 ring is as defined in the formula (21);

R_{2401} to R_{2407} are the same as R_{221} to R_{277} in the formulas (21-1) and (21-2);

R_{2410} to R_{2417} are the same as R_{201} to R_{211} in the formula (22); and

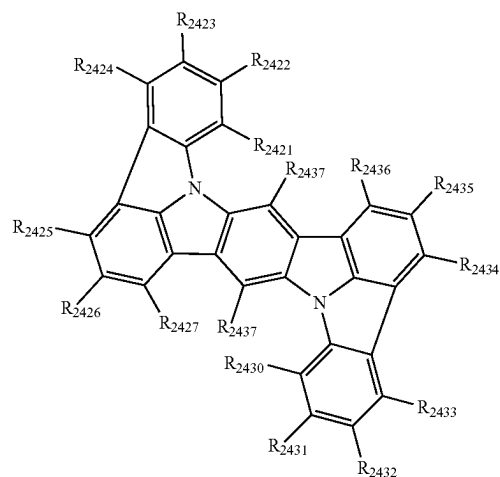
two R_{2417} may be the same with or different from each other.

In one embodiment, the substituted or unsubstituted aromatic hydrocarbon ring having 6 to 50 ring carbon atoms of A1 ring in the formula (21-5) is a substituted or unsubstituted naphthalene ring, or a substituted or unsubstituted fluorene ring.

In one embodiment, the substituted or unsubstituted heterocycle having 5 to 50 ring atoms of A1 ring in the formula (21-5) is a substituted or unsubstituted dibenzofuran ring, a substituted or unsubstituted carbazole ring, or a substituted or unsubstituted dibenzothiophene ring.

In one embodiment, the compound represented by the formula (21) or (22) is selected from the group consisting of the compounds represented by the following formulas (21-6-1) to (21-6-7).

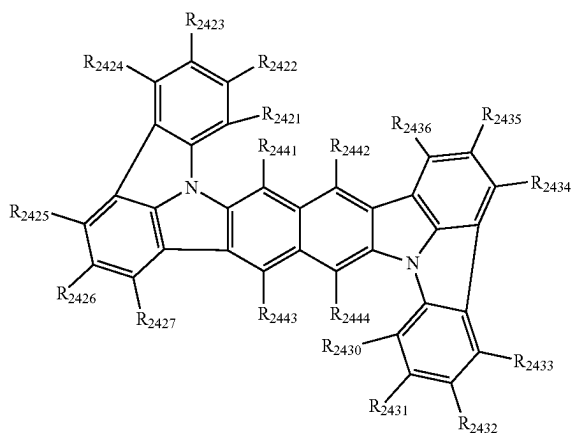
(21-6-1)



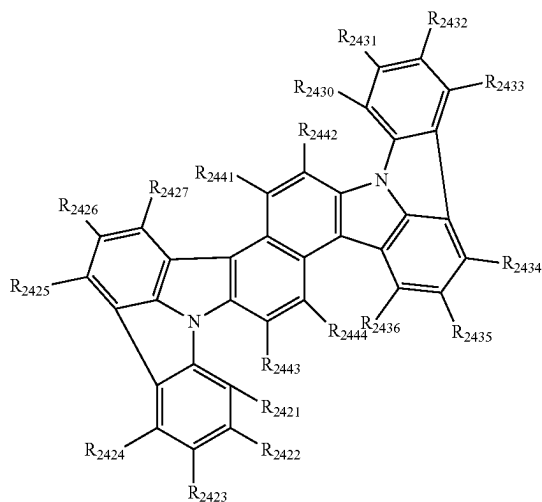
195

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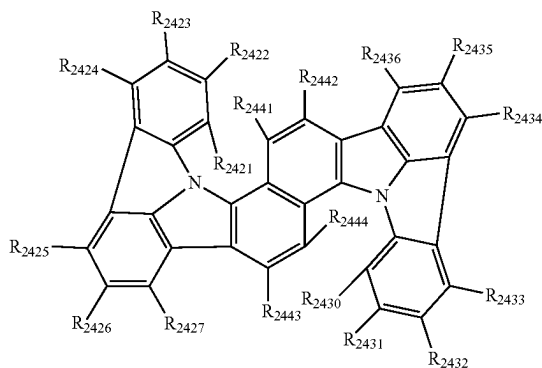
(21-6-2)



(21-6-3)

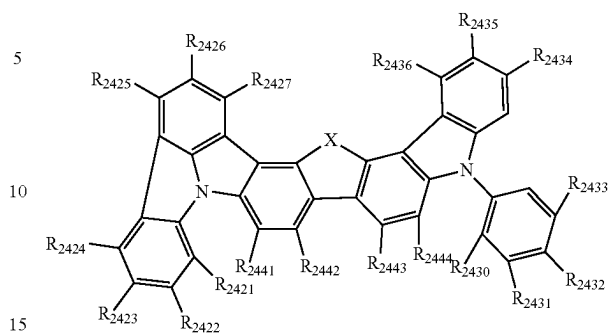


(21-6-4)

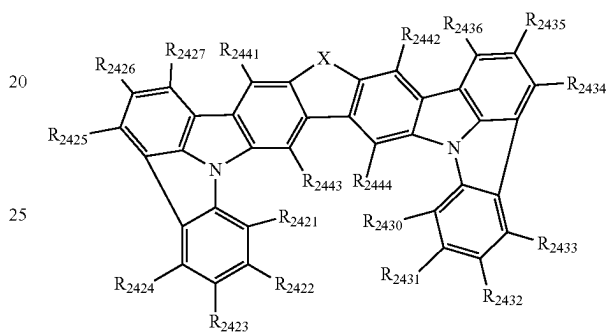
**196**

-continued

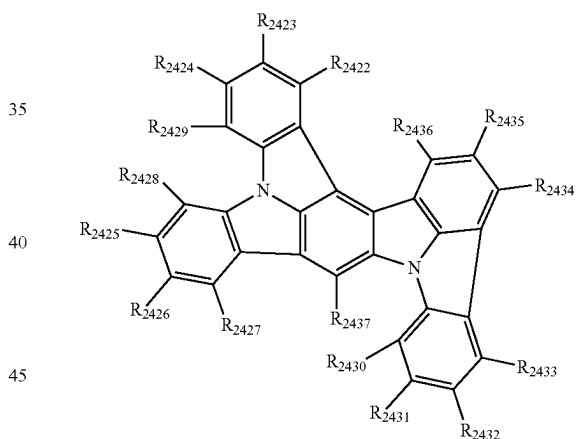
(21-6-5)



(21-6-6)



(21-6-7)



In the formulas (21-6-1) to (21-6-7),

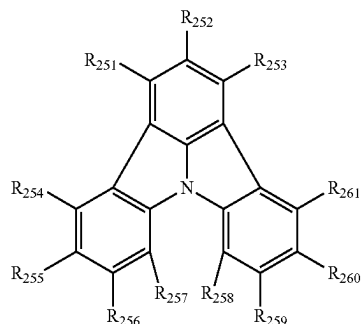
R₂₄₂₁ to R₂₄₂₇ are the same as R₂₂₁ to R₂₂₇ in the formulas (21-1) and (21-2);R₂₄₃₀ to R₂₄₃₇ and R₂₄₄₁ to R₂₄₄₄ are the same as R₂₀₁ to R₂₁₁ in the formula (22);X is O, NR₉₀₁, or C(R₉₀₂) (R₉₀₃); andR₉₀₁ to R₉₀₃ are as defined in the formula (1).

In one embodiment, in the compound represented by the formula (22), one or more pairs of two or more adjacent groups of R₂₀₁ to R₂₁₁ are bonded with each other to form a substituted or unsubstituted, saturated or unsaturated ring. This embodiment is described in the following formula (25).

(Compound Represented by Formula (25))

The compound represented by the formula (25) is explained below.

197



In the formula (25),

two or more pairs selected from a group consisting of R_{251} and R_{252} , R_{252} and R_{253} , R_{254} and R_{255} , R_{255} and R_{256} , R_{256} and R_{257} , R_{258} and R_{259} , R_{259} and R_{260} , and R_{260} and R_{261} bond with each other to form a substituted or unsubstituted, saturated or unsaturated ring;

Provided that the pair of R_{251} and R_{252} and the pair of R_{252} and R_{253} do not form a ring simultaneously; the pair of R_{254} and R_{255} and the pair of R_{255} and R_{256} do not form a ring simultaneously; the pair of R_{256} and R_{257} do not form a ring simultaneously; the pair of R_{258} and R_{259} and the pair of R_{259} and R_{260} do not form a ring simultaneously; and the pair of R_{259} and R_{260} and the pair of R_{260} and R_{261} do not form a ring simultaneously;

When two or more rings are formed by R_{251} to R_{261} , the rings may be the same or different;

R_2 , to R_2 , that do not form the substituted or unsubstituted, saturated or unsaturated ring are independently

a hydrogen atom,

a substituted or unsubstituted alkyl group having 1 to 50 carbon atoms,

a substituted or unsubstituted alkenyl group having 2 to 50 carbon atoms,

a substituted or unsubstituted alkynyl group having 2 to 50 carbon atoms,

a substituted or unsubstituted cycloalkyl group having 3 to 50 ring carbon atoms, or

$-\text{Si}(\text{R}_{901})(\text{R}_{902})(\text{R}_{903})$,

$-\text{O}-(\text{R}_{904})$,

$-\text{S}-(\text{R}_{905})$,

$-\text{N}(\text{R}_{906})(\text{R}_{907})$,

a halogen atom, a cyano group, a nitro group,

a substituted or unsubstituted aryl group having 6 to 50 ring carbon atoms, or

a substituted or unsubstituted monovalent heterocyclic group having 5 to 50 ring atoms; and

R_{901} to R_{907} are as defined in the formula (1).

In the formula (25), R_n and R_{n+1} (n is an integer selected from 251, 252, 254 to 256 and 258 to 260) bond with each other to form a substituted or unsubstituted, saturated or unsaturated ring together with two ring carbon atoms to which R_n and R_{n+1} bond with. The ring is preferably configured with atoms selected from C atom, O atom, S atom and N atom, and the number of atoms is preferably 3 to 7, more preferably 5 or 6.

The number of the above-described ring structures in the compound represented by the formula (25) is, for example, 2, 3 or 4. Two or more ring structures may exist in the same benzene ring of the main skeleton in the formula (25), or may exist indifferent benzene rings. For example, the com-

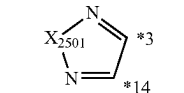
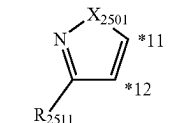
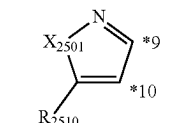
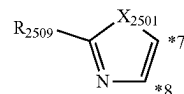
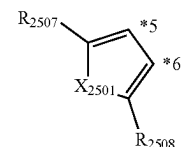
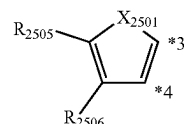
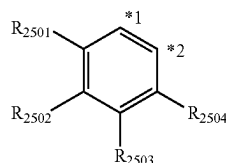
198

pound has three ring structures, one ring structure may exist in each of the three benzene rings in the formula (25).

As the above-mentioned ring structure in the compound represented by the formula (25), structures represented by the following formulas (251) to (260) can be given, for example.

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In the formulas (251) to (257),

each of *1 and *2, *3 and *4, *5 and *6, *7 and *8, *9 and *10, *11 and *12, and *13 and

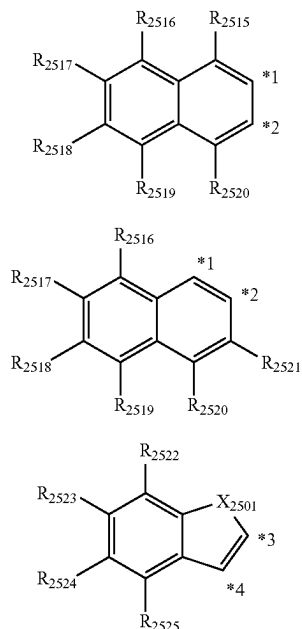
*14 represents two ring carbon atoms to which R_n and R_{n+1} bond, and R_n may bond to either one of the two ring carbon atoms of *1 and *2, *3 and *4, *5 and *6, *7 and *8, *9 and *10, *11 and *12, and *13 and *14;

X_{2501} is $\text{C}(\text{R}_{2512})$ (R_{2513}), NR_{2514} , O or S;

One or more pairs of two or more adjacent groups of R_{2501} to R_{2506} and R_{2512} to R_{2513} are bonded with each other to form a substituted or unsubstituted, saturated or unsaturated ring, or do not form a substituted or unsubstituted saturated or unsaturated ring; and

R_{2501} to R_{2514} that do not form a substituted or unsubstituted saturated or unsaturated ring are the same as R_{251} to R_{261} .

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In the formulas (258) to (260),

each of *1 and *2, and *3 and *4 represents two ring carbon atoms to which R_n and R_{n+1} bond, and R_n may bond to either one of the two ring carbon atoms of *1 and *2, or *3 and *4;

X_{2501} is $C(R_{2512})(R_{2513})$, NR_{2514} , O or S;

One or more pairs of two or more adjacent groups of R_{2515} to R_{2525} bond to each other to form a substituted or unsubstituted, saturated or unsaturated ring, or do not form a substituted or unsubstituted saturated or unsaturated ring; and

R_{2515} to R_{2521} and R_{2522} to R_{2525} that do not form a substituted or unsubstituted saturated or unsaturated ring are the same as R_{251} to R_{261} .

In the formula (25), it is preferable that at least one of R_{252} , R_{254} , R_{255} , R_{260} and R_{261} (preferably at least one of R_{252} , R_{256} , and R_{260} , more preferably R_{252}) is a group which does not form a ring.

(i) Substituent in the case where the ring structure formed by R_n and R_{n+1} has a substituent in the formula (25),

(ii) R_{251} to R_{261} that do not form a ring structure in the formula (25), and

(iii) R_{2501} to R_{2514} and R_{2515} to R_{2525} in the formulas (251) to (260) are preferably independently

a hydrogen atom,

a substituted or unsubstituted alkyl group having 1 to 50 carbon atoms,

a substituted or unsubstituted alkenyl group having 2 to 50 carbon atoms,

a substituted or unsubstituted alkynyl group having 2 to 50 carbon atoms,

a substituted or unsubstituted cycloalkyl group having 3 to 50 ring carbon atoms,

$-N(R_{901})(R_{907})$,

a substituted or unsubstituted aryl group having 6 to 50 ring carbon atoms,

a substituted or unsubstituted monovalent heterocyclic group having 5 to 50 ring atoms, or

a group selected from the following groups.

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(258)

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(259) 10

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(260)

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(261)

(262)

(263)

(264)

In the formulas (261) to (264),

R_d s are independently

a hydrogen atom,

a substituted or unsubstituted alkyl group having 1 to 50 carbon atoms,

a substituted or unsubstituted alkenyl group having 2 to 50 carbon atoms,

a substituted or unsubstituted alkynyl group having 2 to 50 carbon atoms,

a substituted or unsubstituted cycloalkyl group having 3 to 50 ring carbon atoms,

$-Si(R_{901})(R_{902})(R_{903})$,

$-O-(R_{904})$,

$-S-(R_{905})$,

$-N(R_{906})(R_{907})$,

a halogen atom, a cyano group, a nitro group,

a substituted or unsubstituted aryl group having 6 to 50 ring carbon atoms, or

a substituted or unsubstituted monovalent heterocyclic group having 5 to 50 ring atoms;

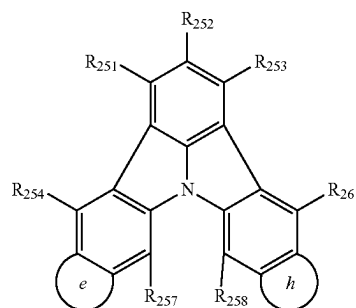
X is $C(R_{901})(R_{902})$, NR_{903} , O, or S;

R_{901} to R_{907} are as defined in the formula (1); and

$p1$ is independently an integer of 0 to 5, $p2$ is independently an integer of 0 to 4, $p3$ is an integer of 0 to 3, and $p4$ is an integer of 0 to 7.

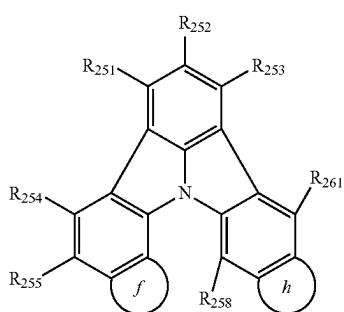
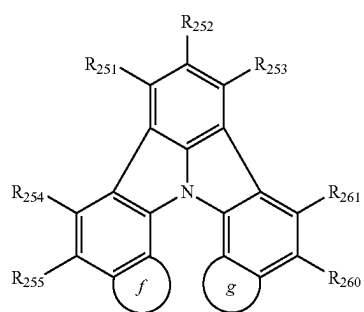
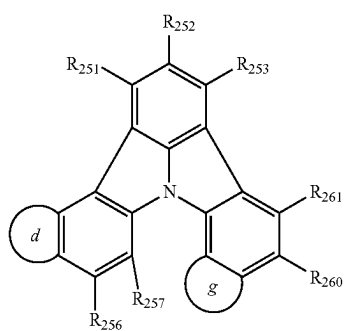
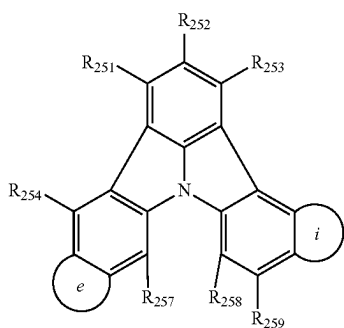
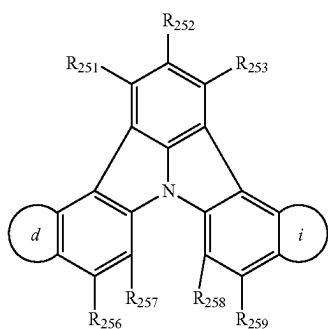
In one embodiment, the compound represented by the formula (25) is represented by the following formulas (25-1) to (25-6).

(25-1)



201

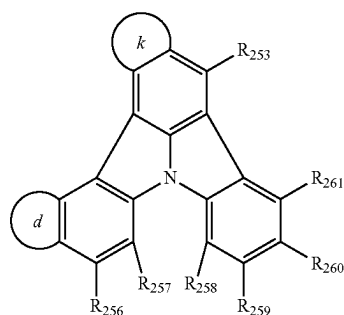
-continued

**202**

(25-2) In the formulas (25-1) to (25-6), ring d to ring i are independently a substituted or unsubstituted, saturated or unsaturated ring; and R₂₅₁ to R₂₆₁ are the same as defined in the formula (25).

5 In one embodiment, the compound represented by the formula (25) is represented by the following formulas (25-7) to (25-12).

10 (25-7)



(25-3) 15

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(25-4)

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(25-5)

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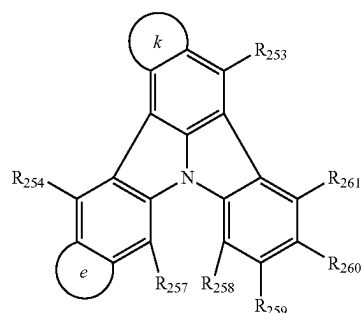
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(25-6)

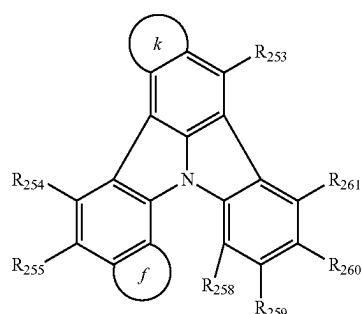
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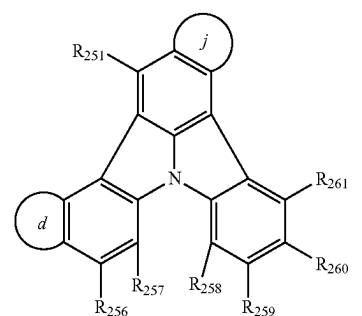
65



(25-8)



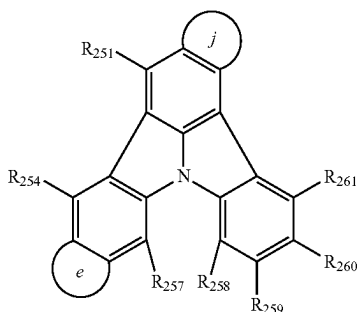
(25-9)



(25-10)

203

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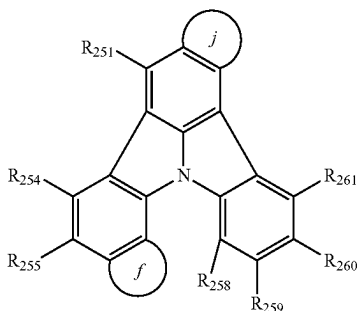


(25-11)

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(25-12) 15

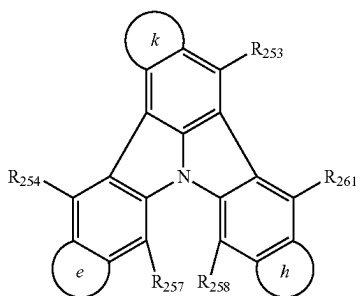


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25

In the formulas (25-7) to (25-12), ring d to ring f, ring k, and ring j are independently a substituted or unsubstituted, saturated or unsaturated ring; and R₂₅₁ to R₂₆₁ are the same as defined in the formula (25).

In one embodiment, the compound represented by the formula (25) is represented by the following formulas (25-13) to (25-21).

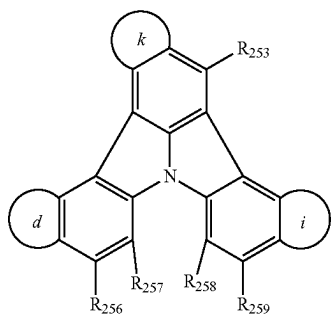


(25-13)

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(25-14)

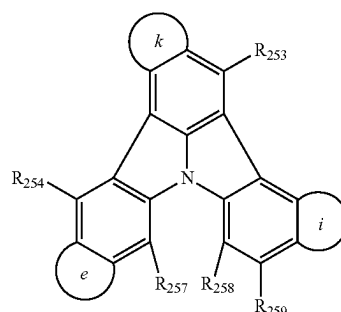
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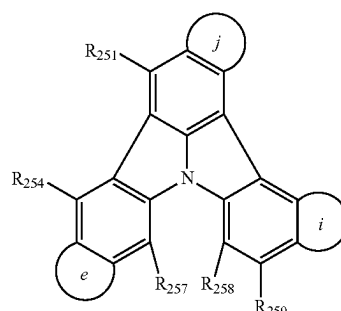
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204

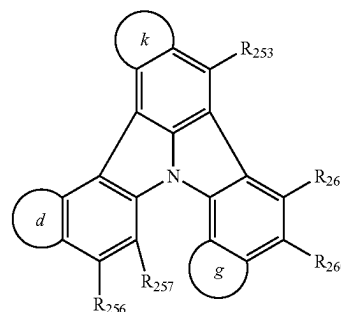
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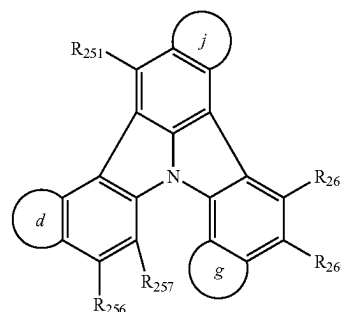
(25-15)



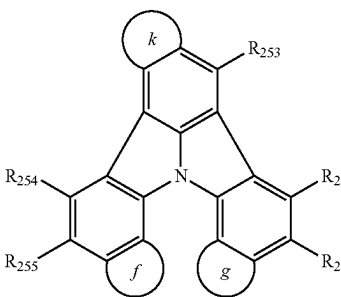
(25-16)



(25-17)



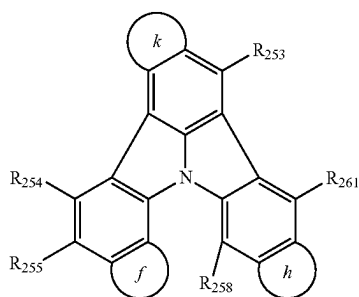
(25-18)



(25-19)

205

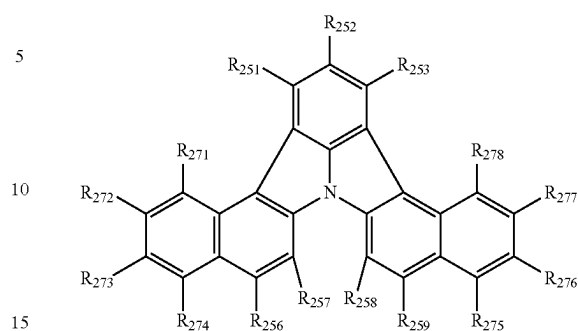
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(25-20)

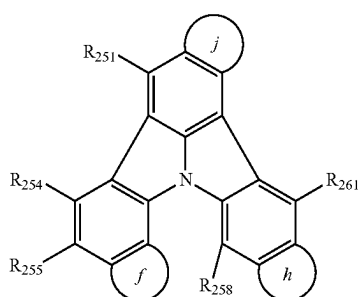
206

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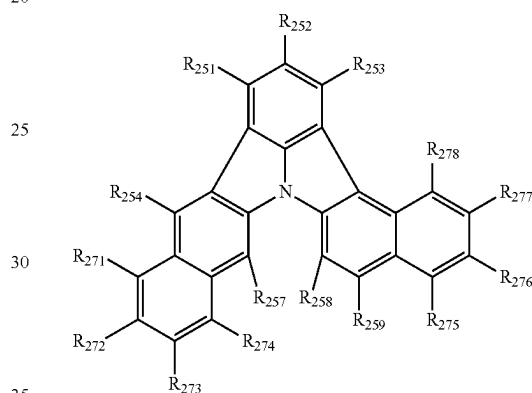
(25-23)

(25-21)



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(25-24)



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In the formulas (25-13) to (25-21), ring d to ring k are independently a substituted or unsubstituted, saturated or unsaturated ring; and R₂₅₁ to R₂₆₁ are the same as defined in the formula (25).

As a substituent in the case where the ring g or ring h further has a substituent,

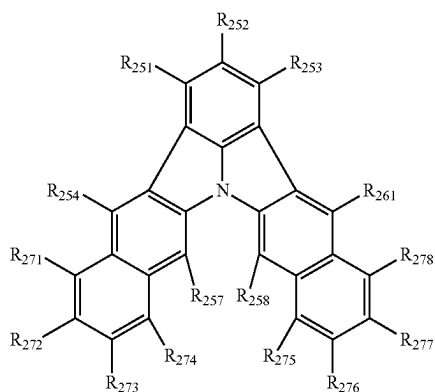
a substituted or unsubstituted alkyl group having 1 to 50 carbon atoms,

a substituted or unsubstituted aryl group having 6 to 50 ring carbon atoms, or

a group represented by the formula (261), (263) or (264) can be given for example.

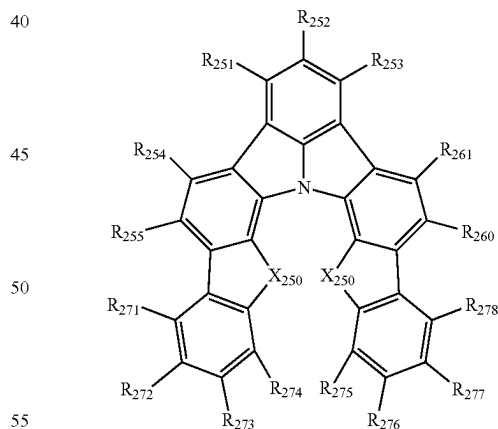
In one embodiment, the compound represented by the formula (25) is represented by one of the following formulas (25-22) to (25-25).

(25-22)



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(25-25)



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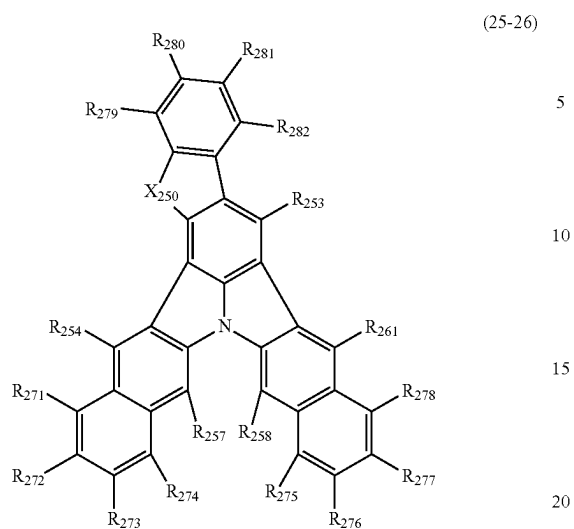
In the formulas (25-22) to (25-25), X₂₅₀ is independently C(R₉₀₁)(R₉₀₂), NR₉₀₃, O or S; R₂₅₁ to R₂₆₁, and R₂₇₁ to R₂₇₈ are the same as R₂₅₁ to R₂₆₁ in the formula (25); and R₉₀₁ to R₉₀₃ are as defined in the formula (1).

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In one embodiment, the compound represented by the formula (25) is represented by the following formula (25-26).

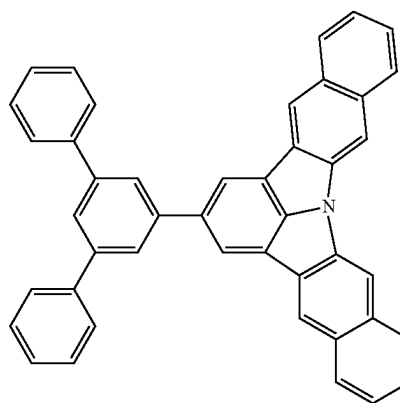
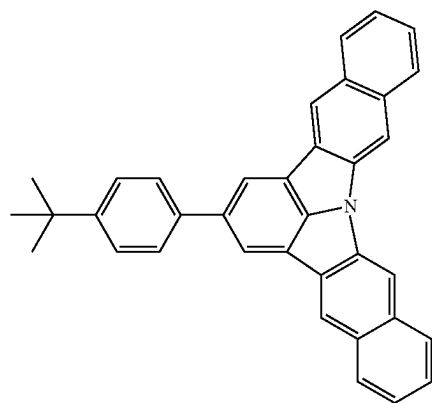
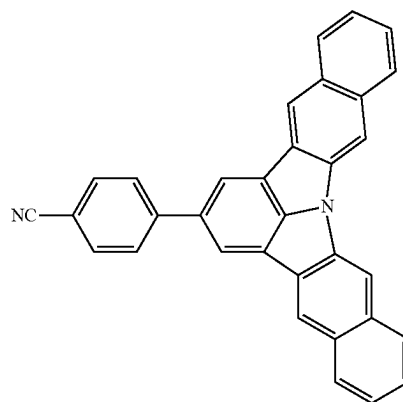
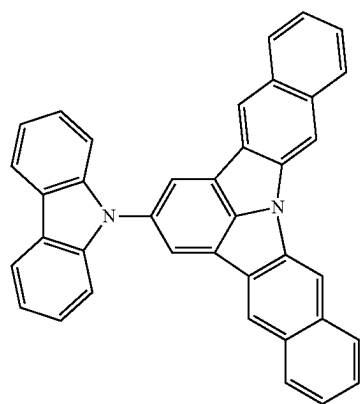
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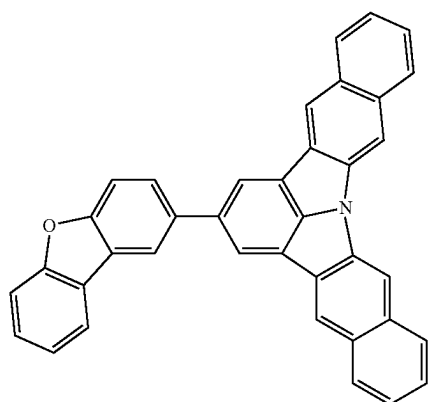


In the formula (25-26), X_{250} is $C(R_{901})(R_{902})$, NR_{903} , O or S; R_{253} , R_{254} , R_{257} , R_{258} , R_{261} , and R_{271} to R_{282} are the same as R_{251} to R_{261} in the formula (25); and R_{901} to R_{903} are as defined in the formula (1).

As the compound represented by the formula (21), the following compounds can be shown for example. In the following example compounds, Me represents methyl group.

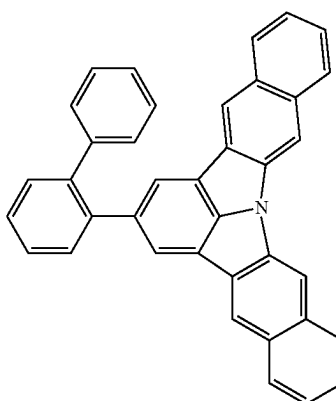
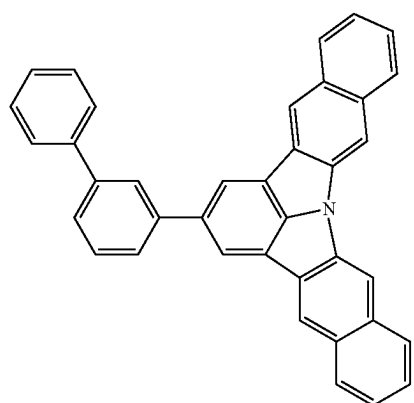
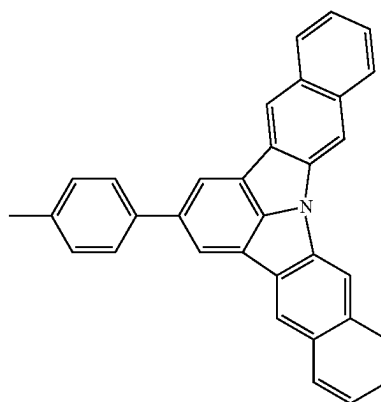
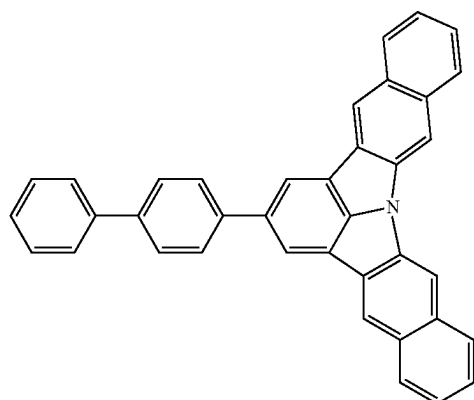
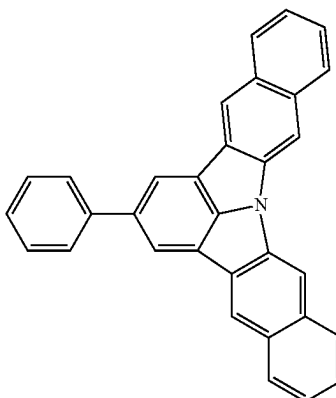


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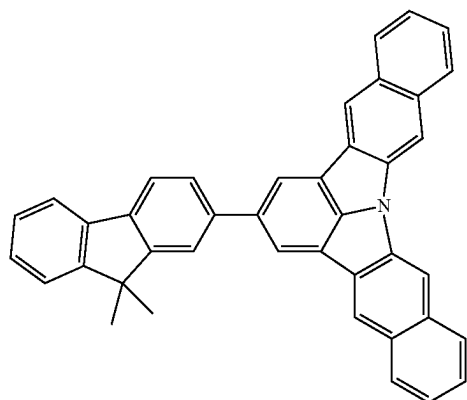


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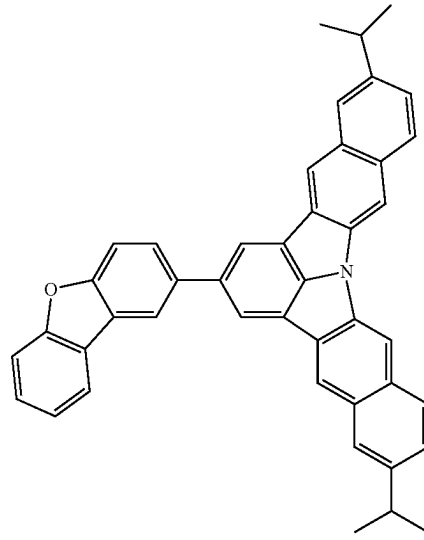
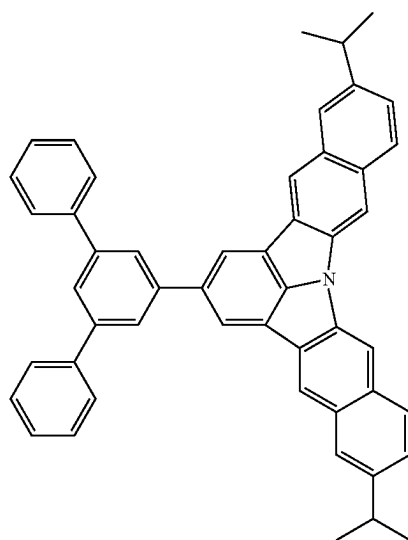
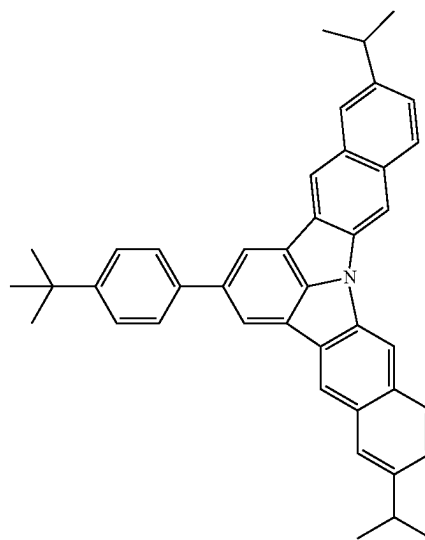
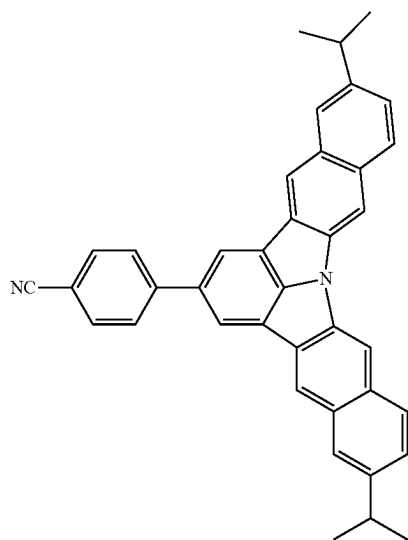
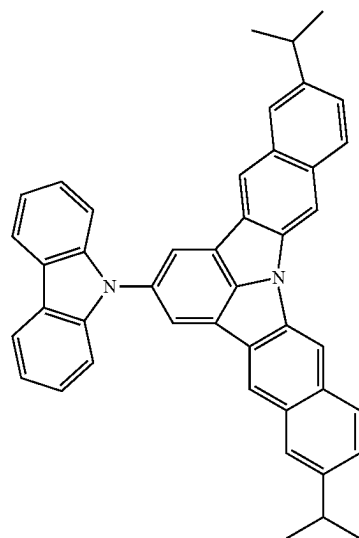


211

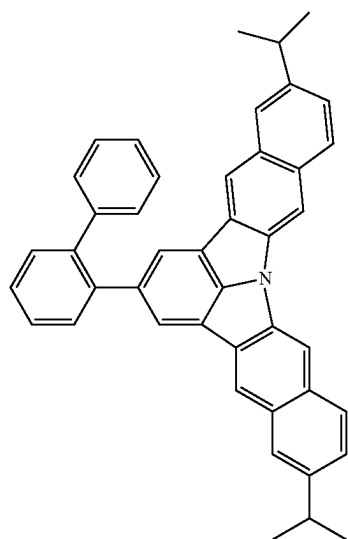
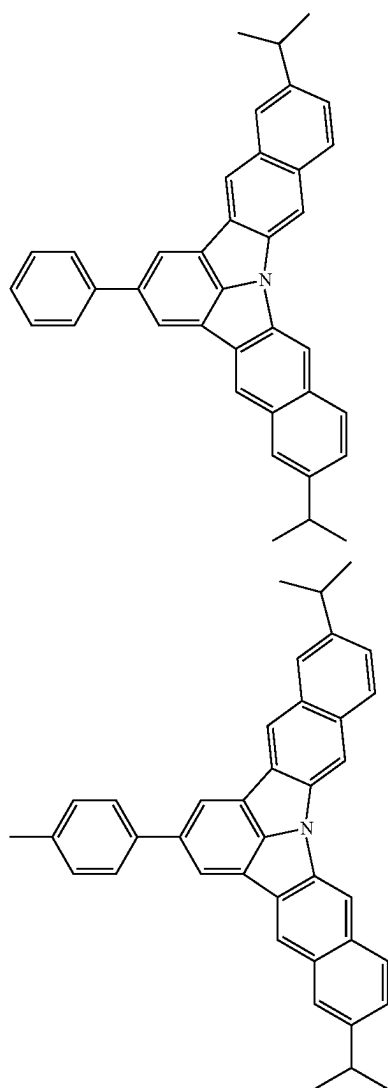


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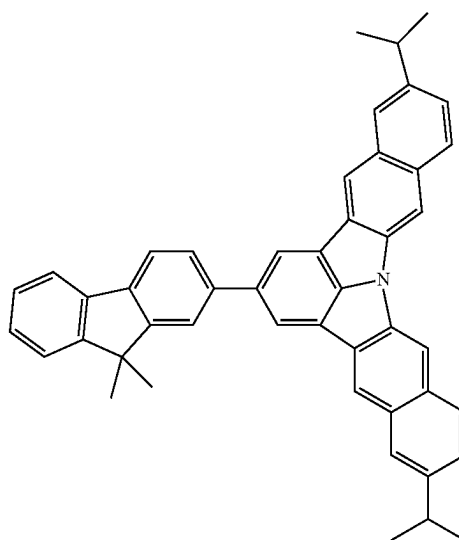
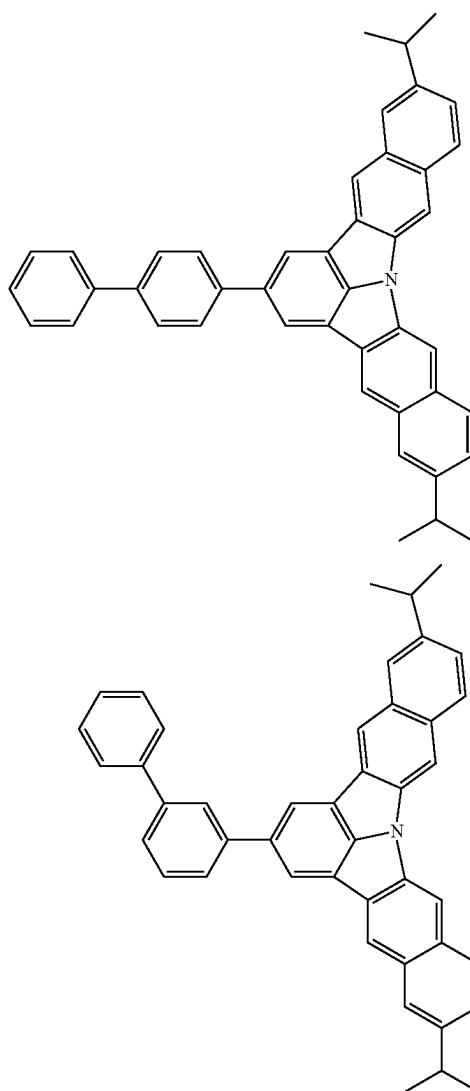


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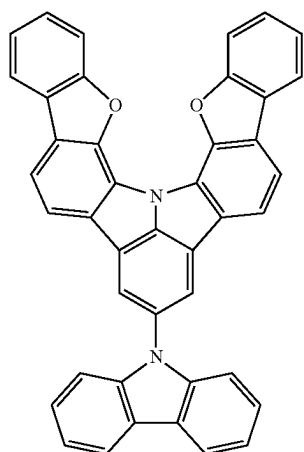


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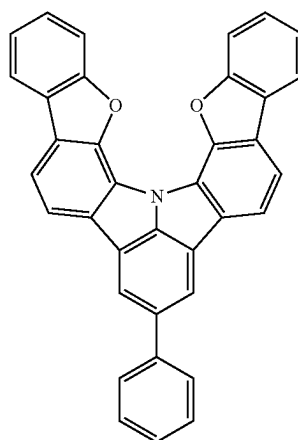
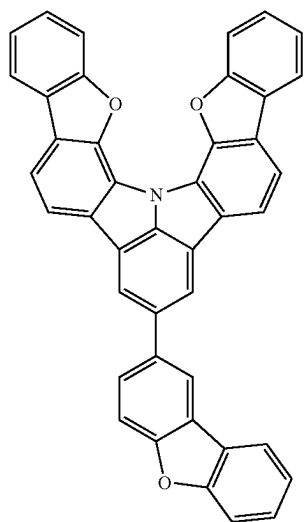
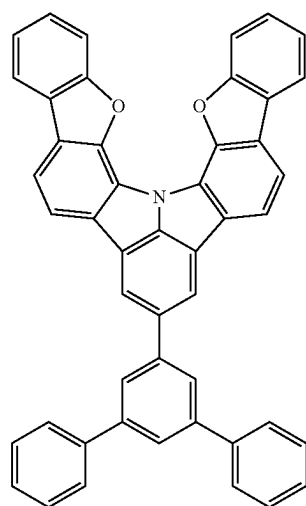
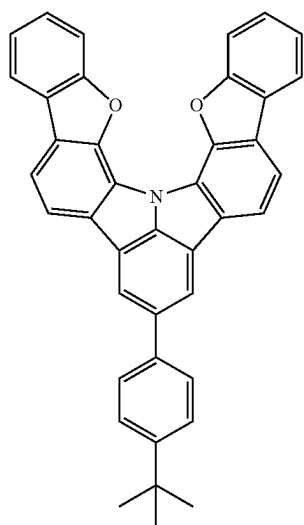
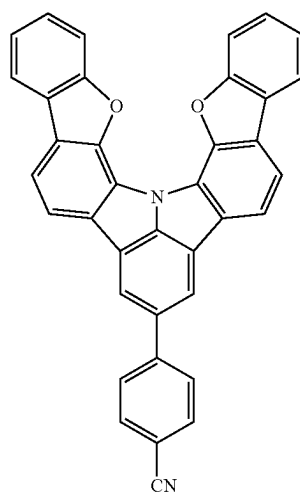


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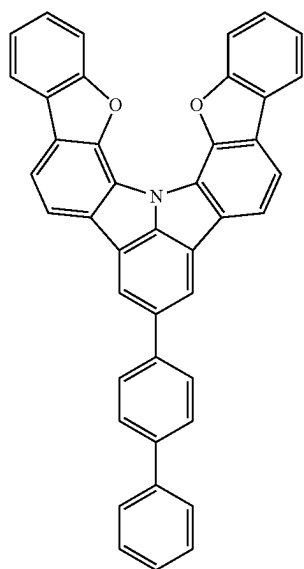


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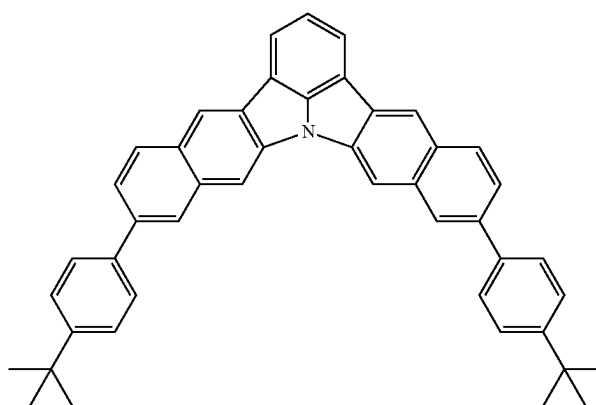
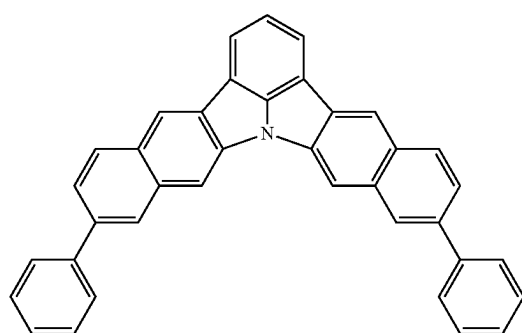
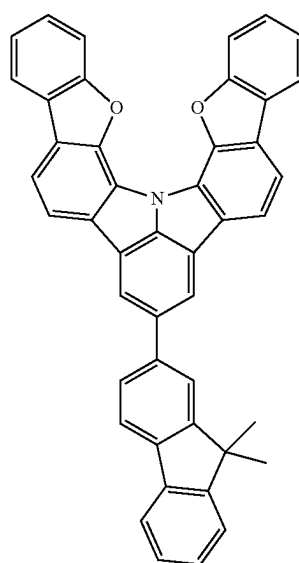
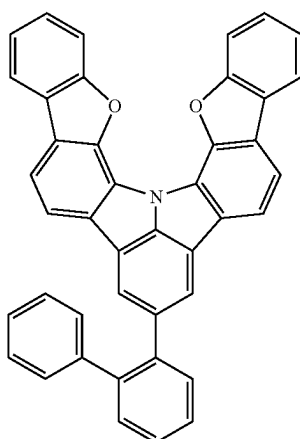
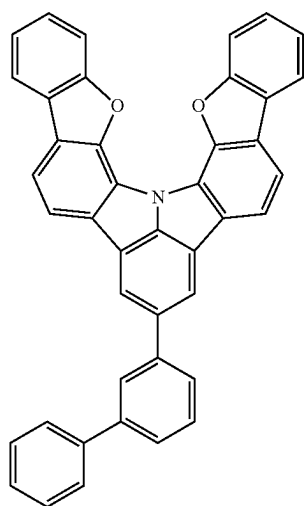
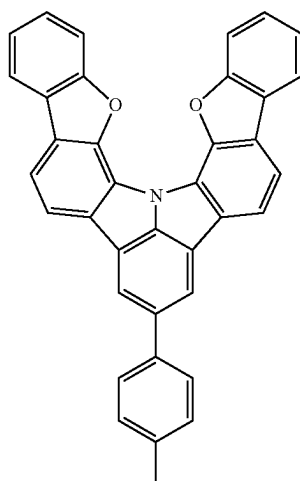


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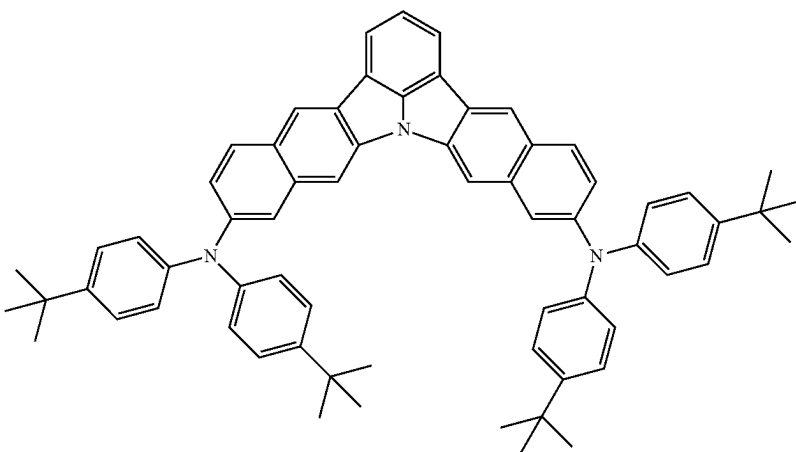
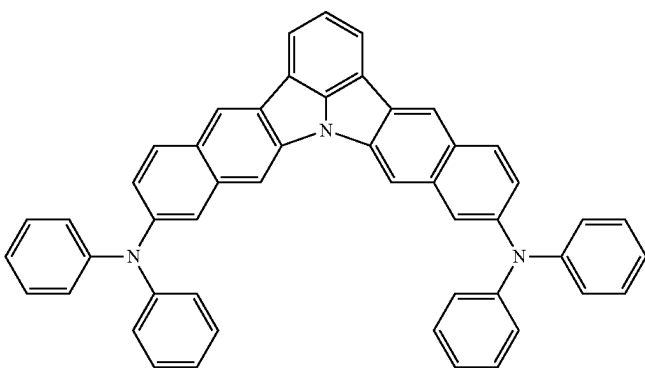
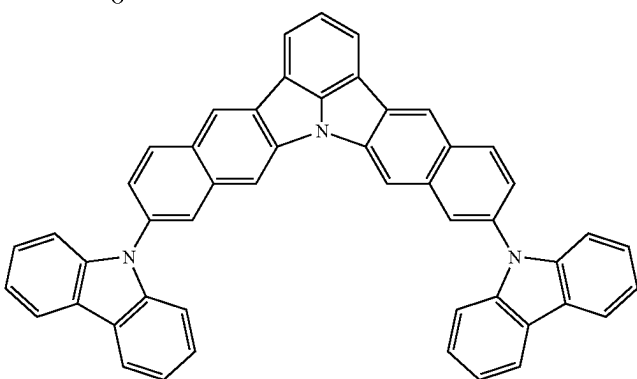
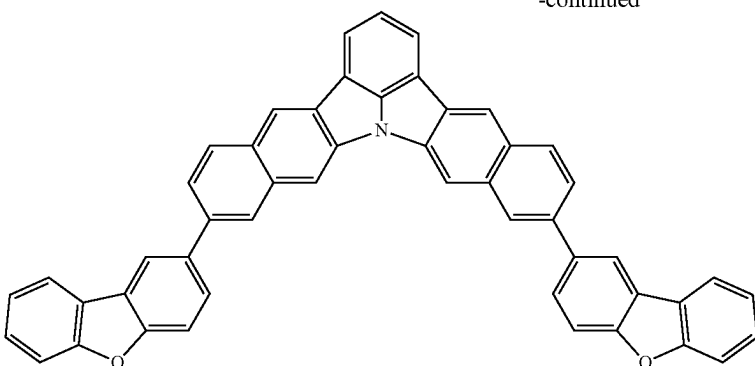
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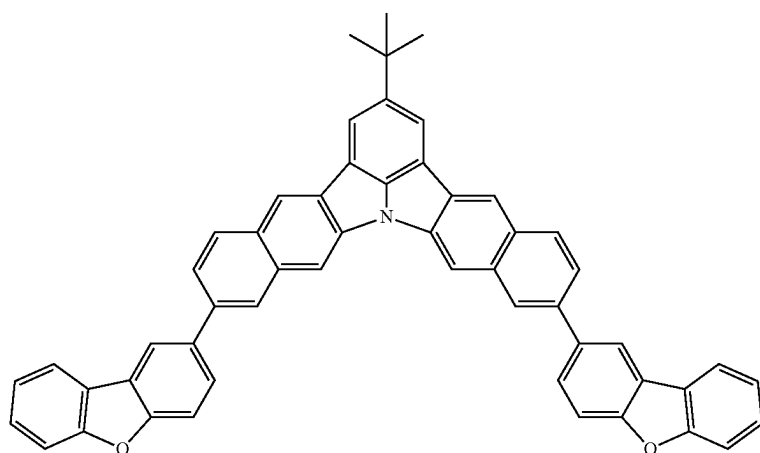
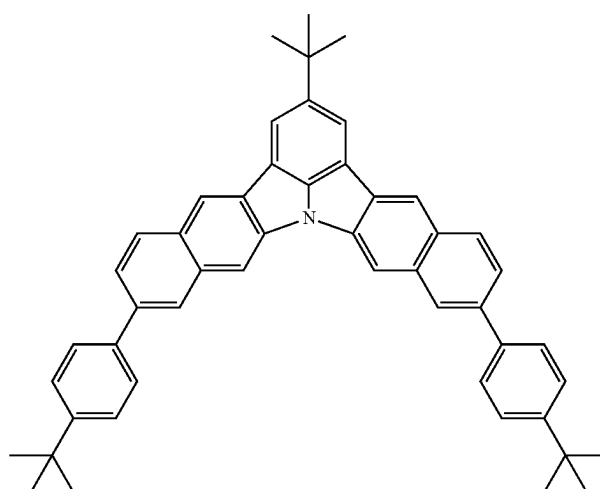
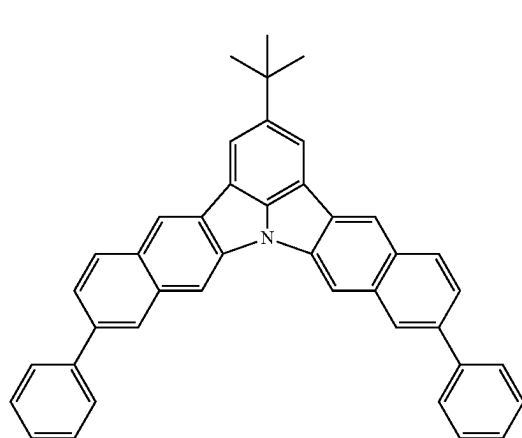
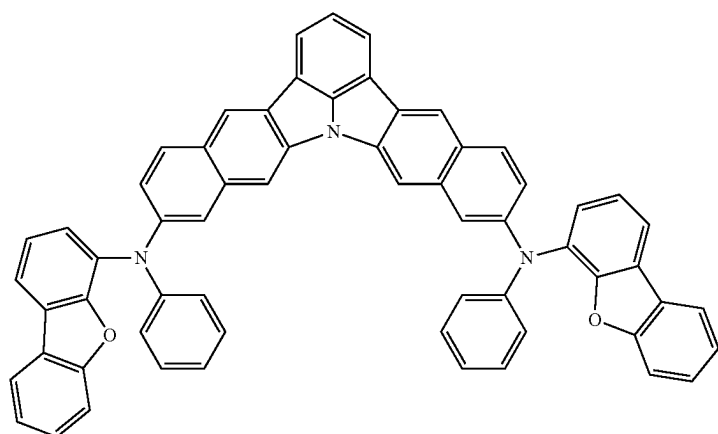
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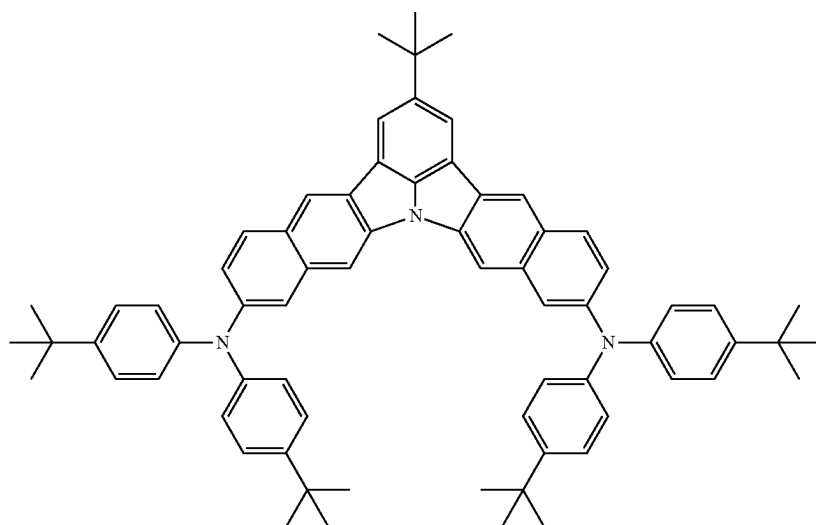
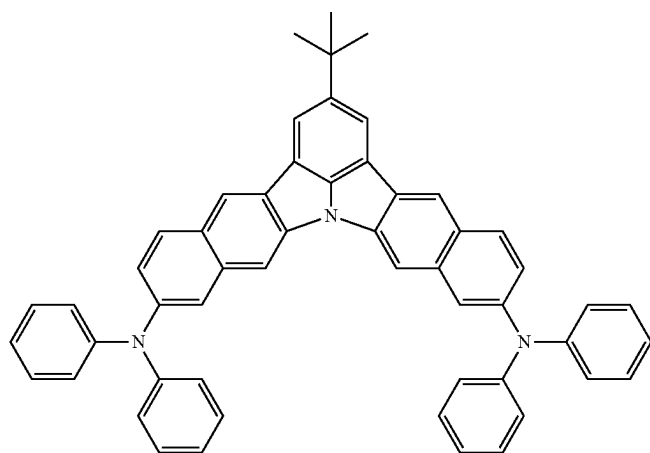
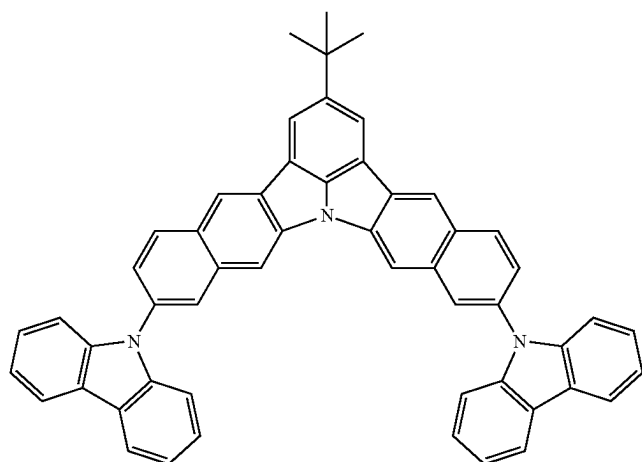
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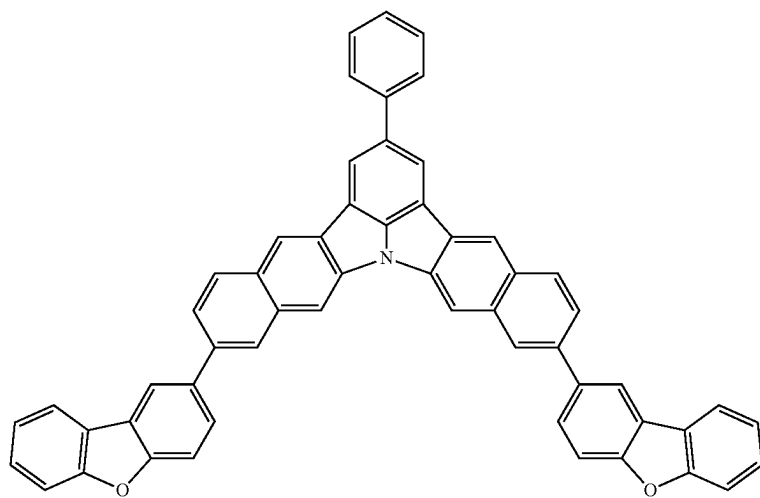
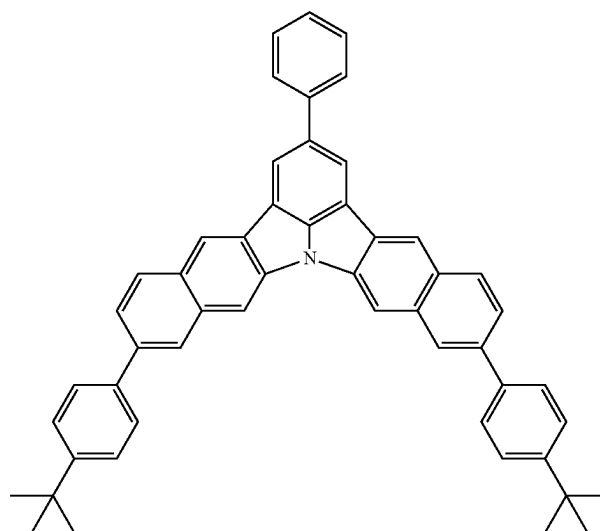
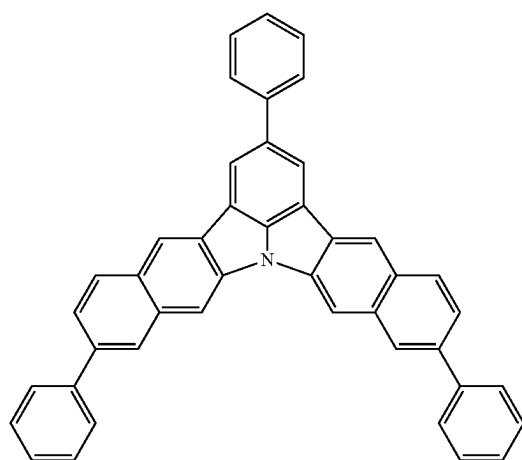
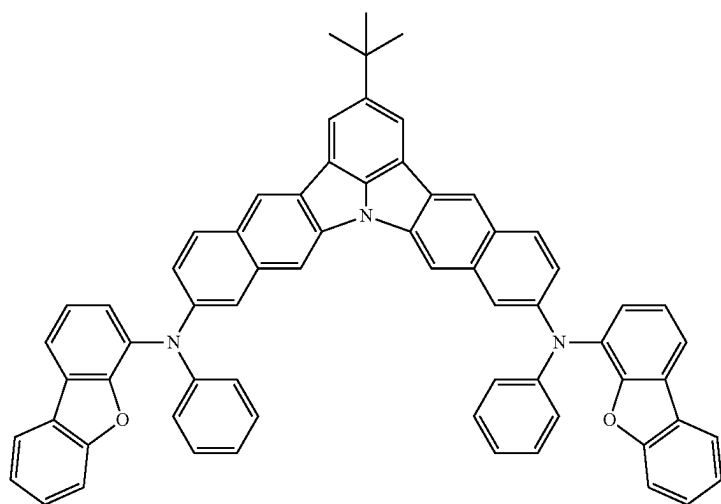
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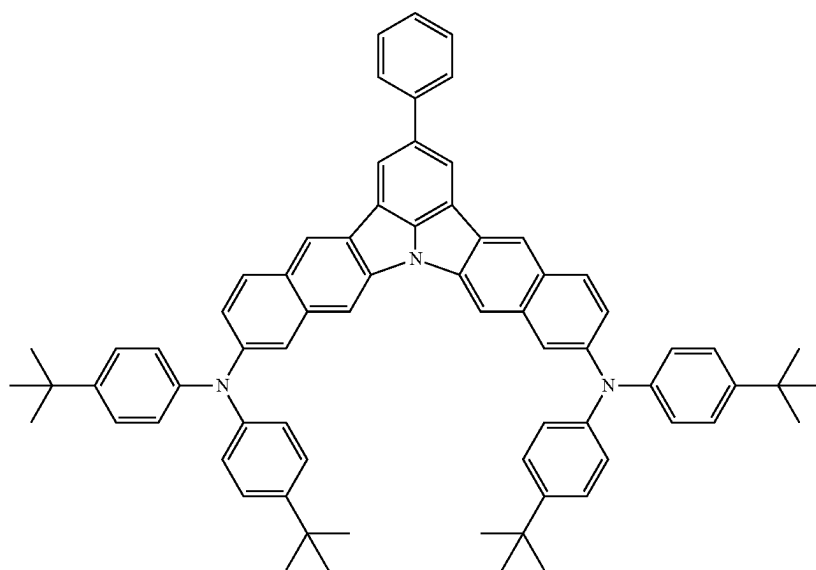
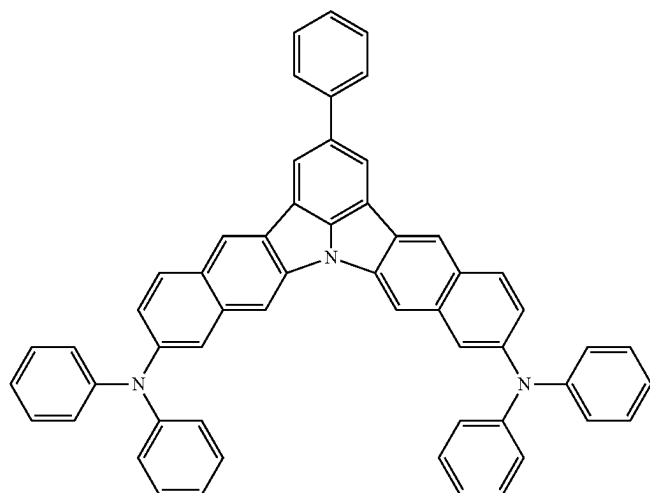
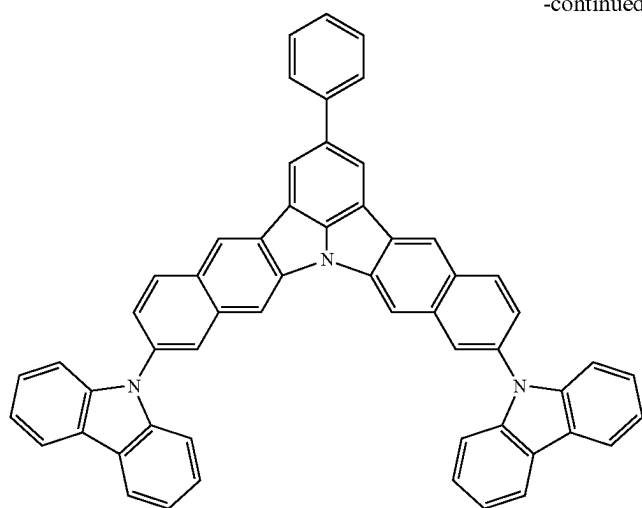
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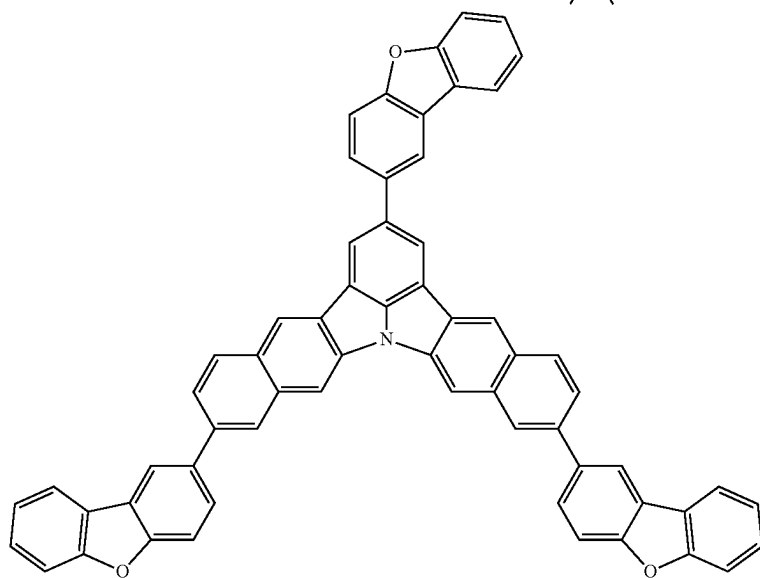
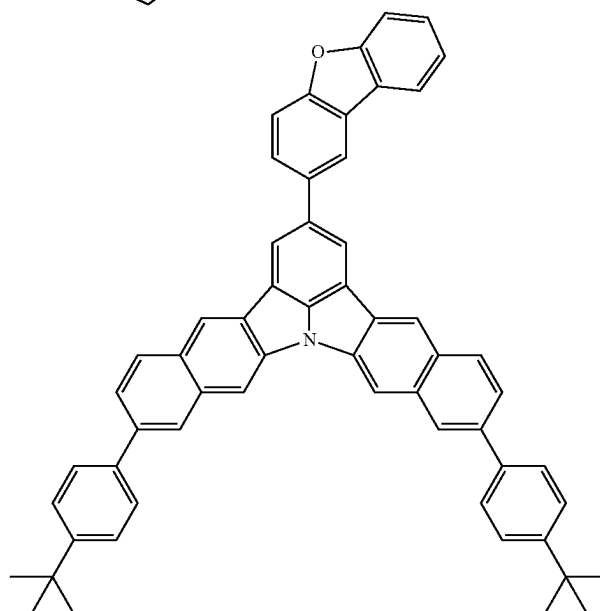
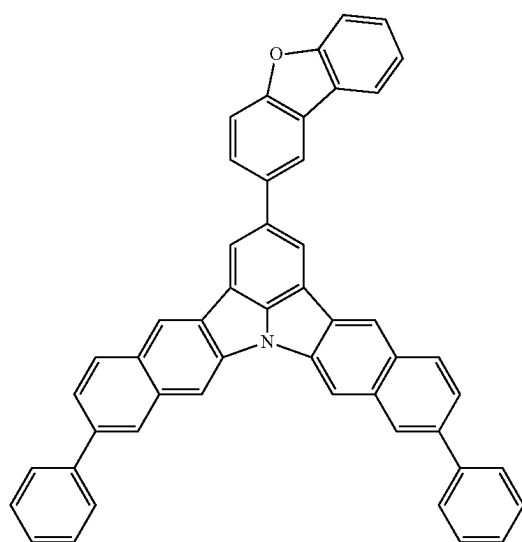
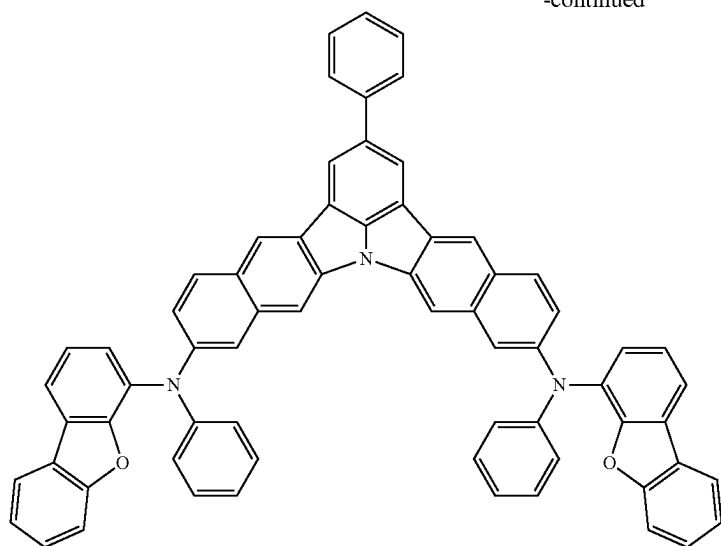
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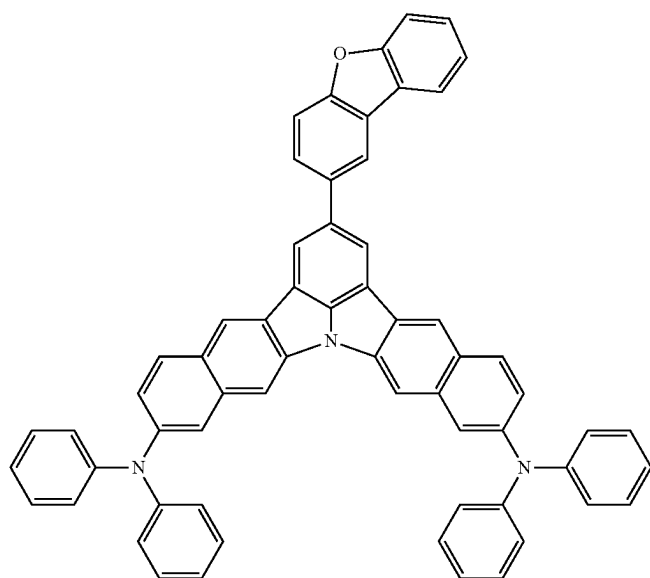
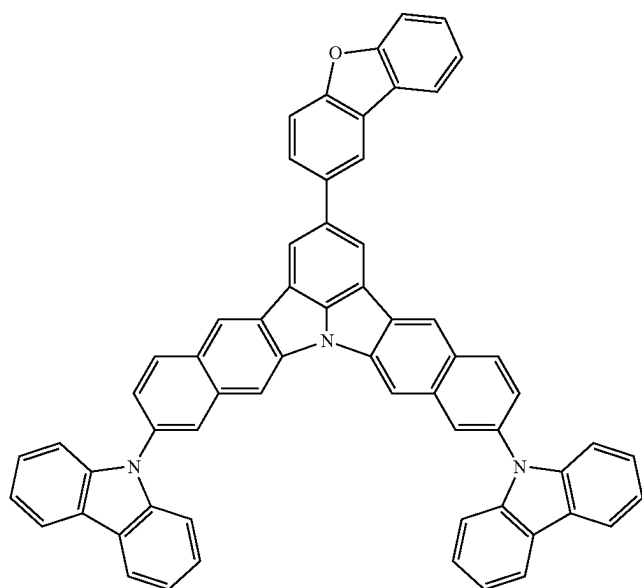
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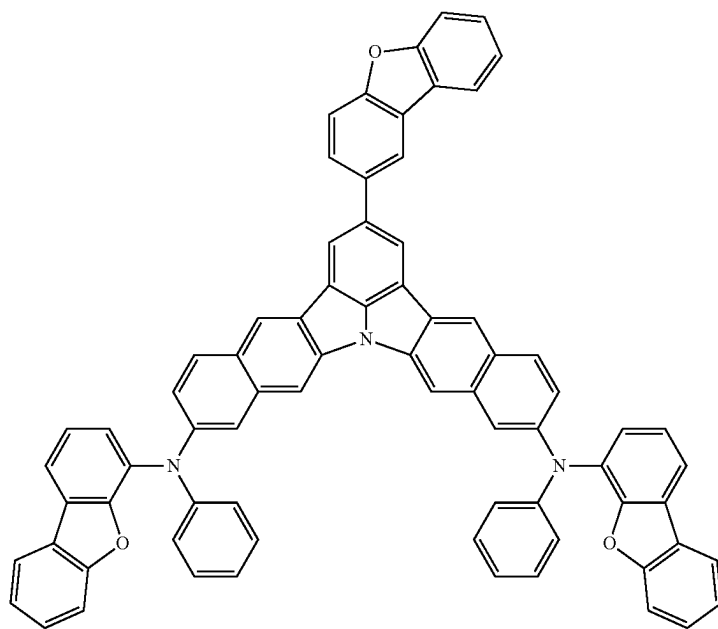
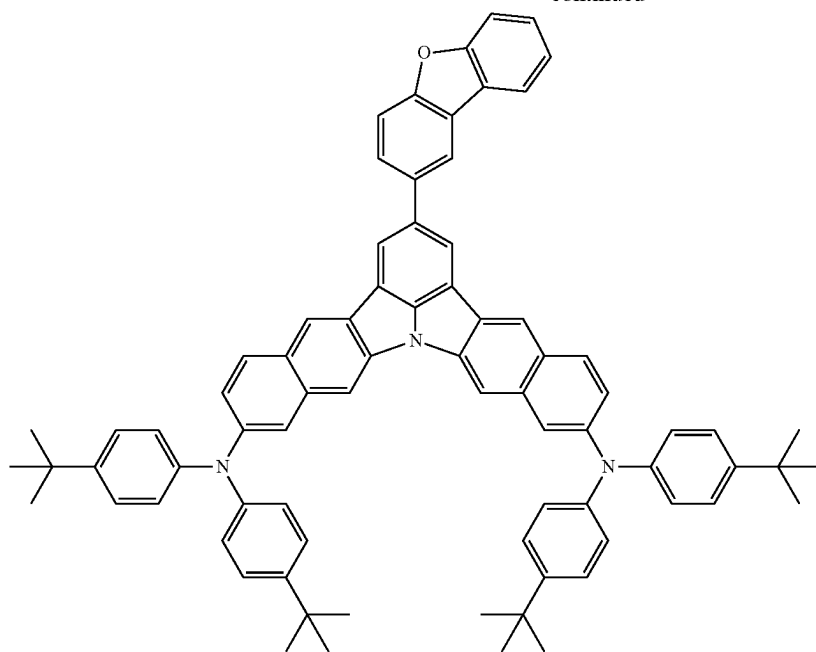
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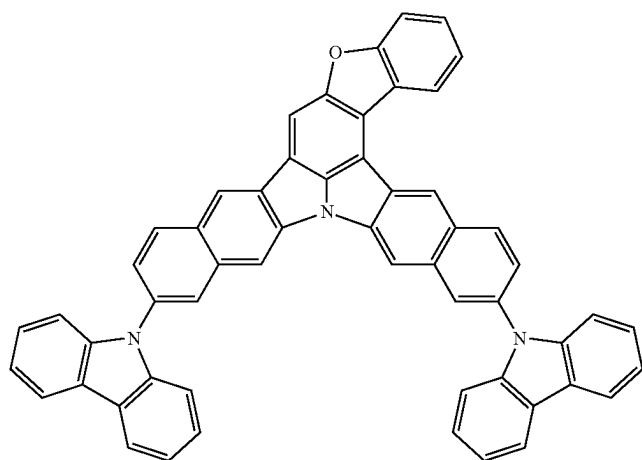
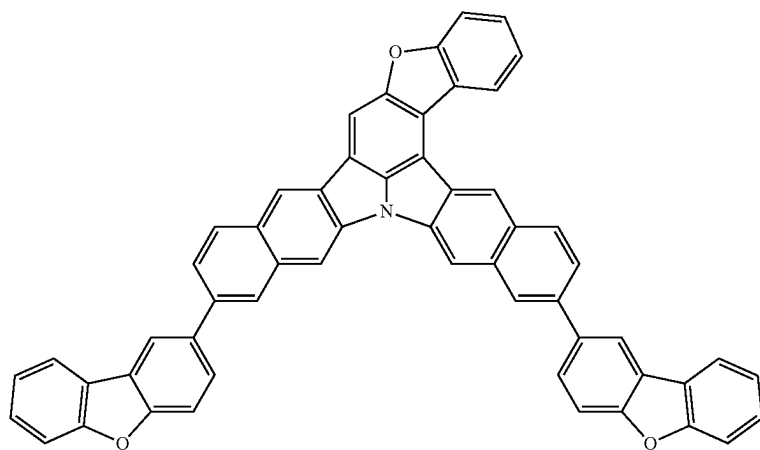
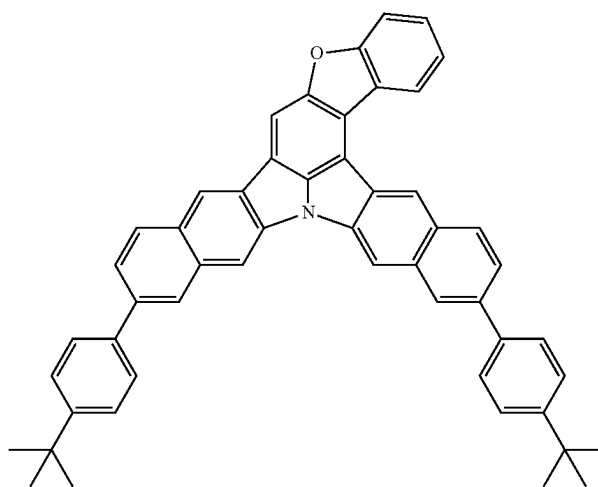
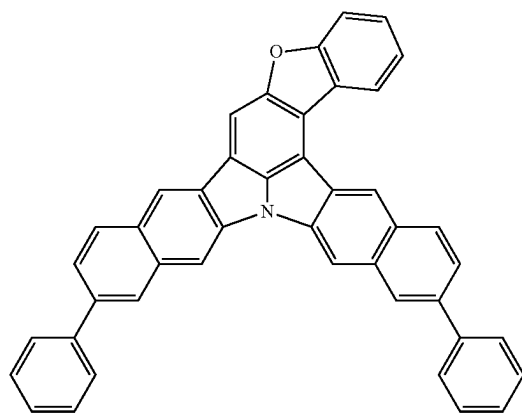
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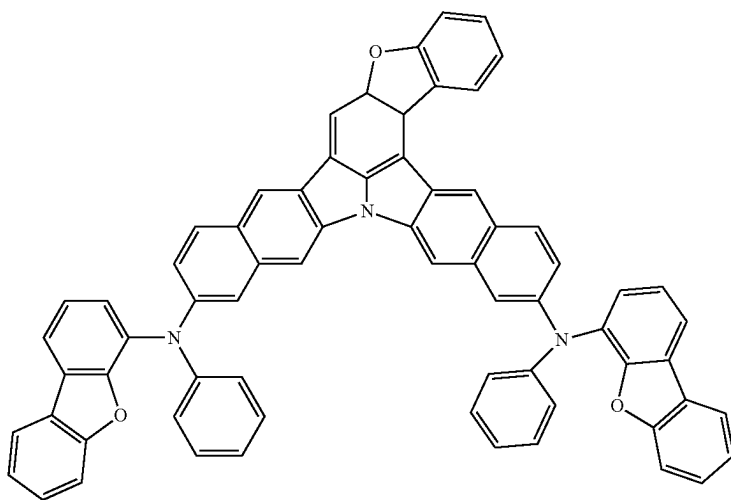
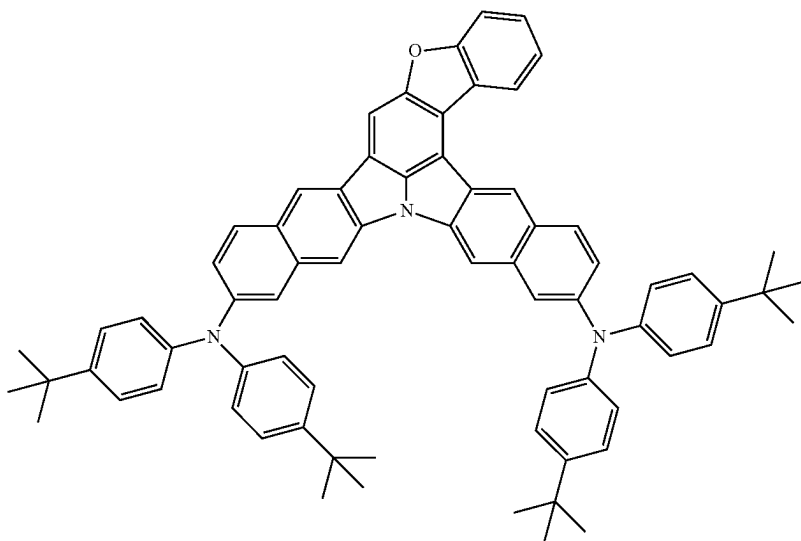
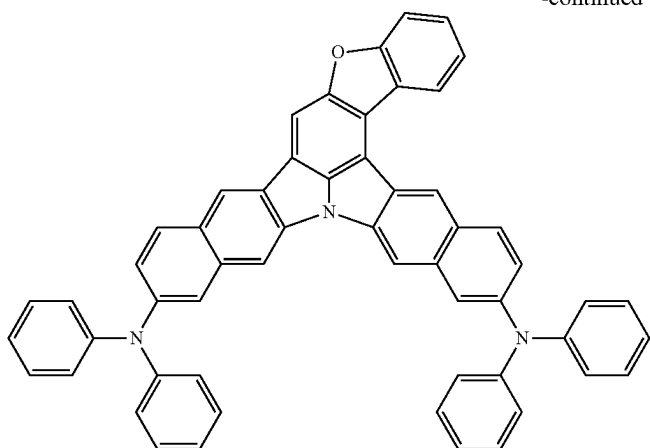
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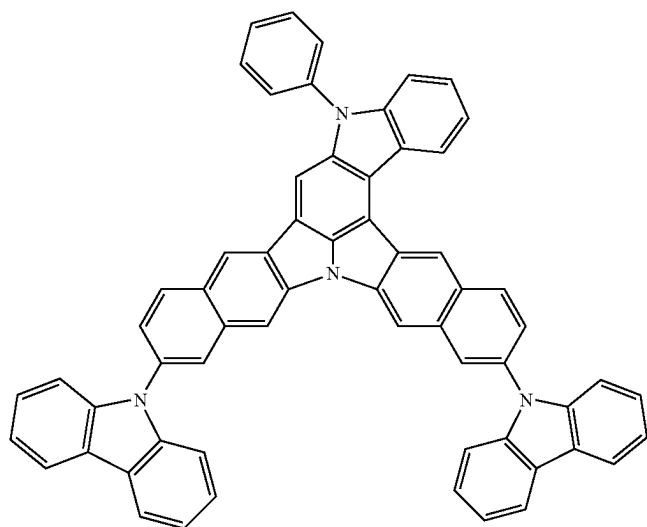
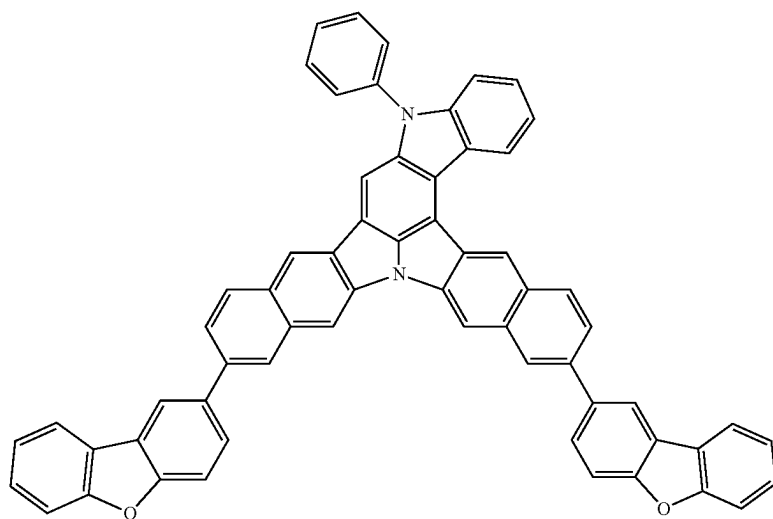
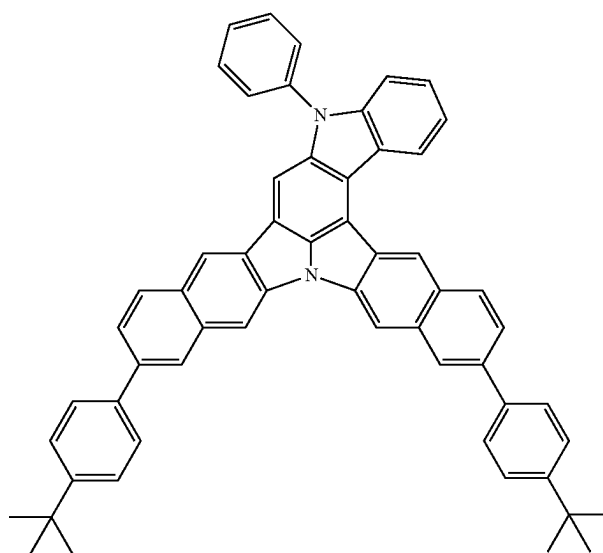
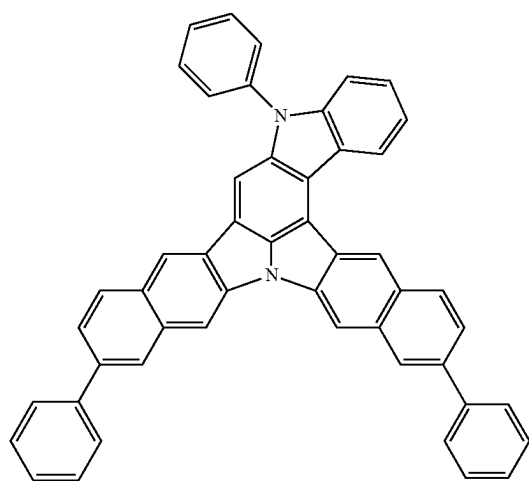
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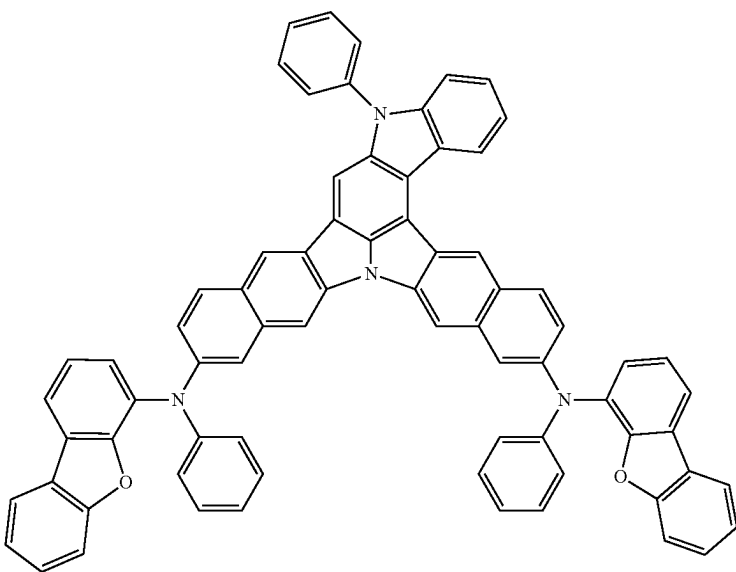
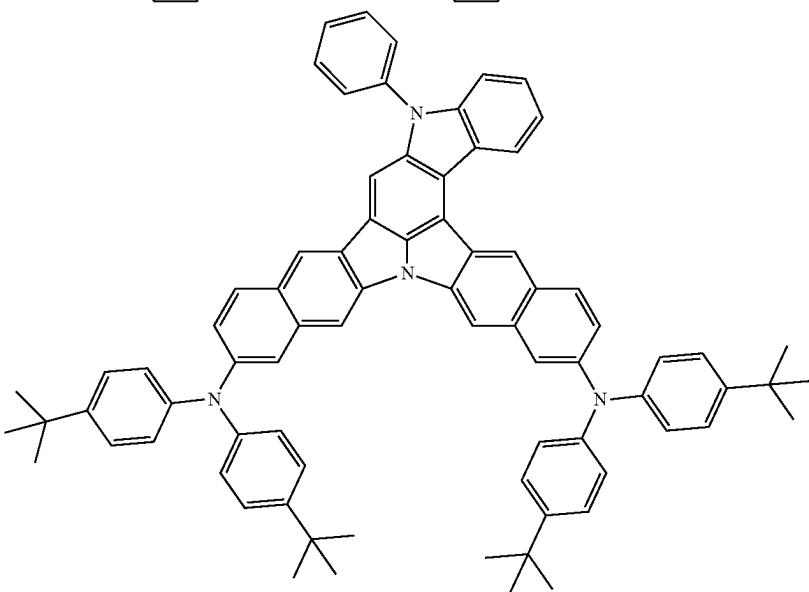
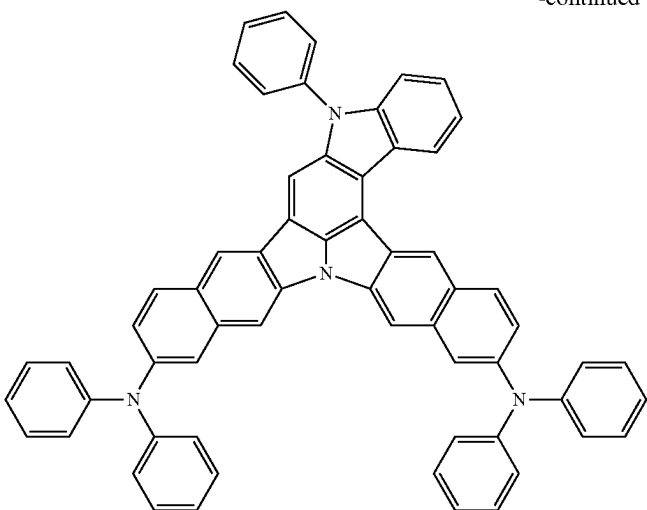
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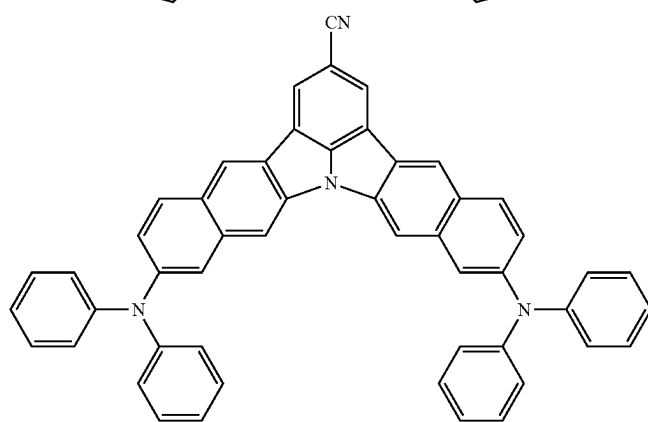
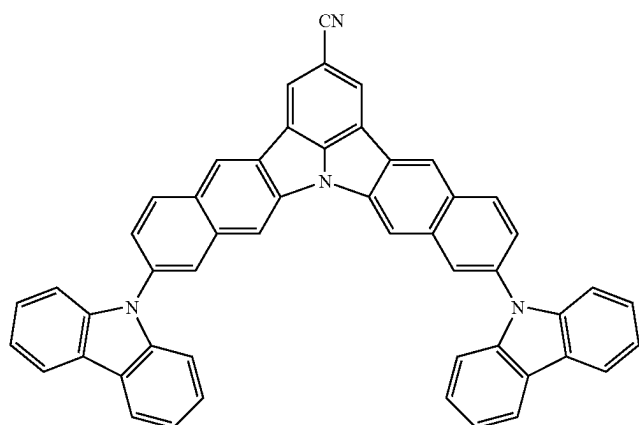
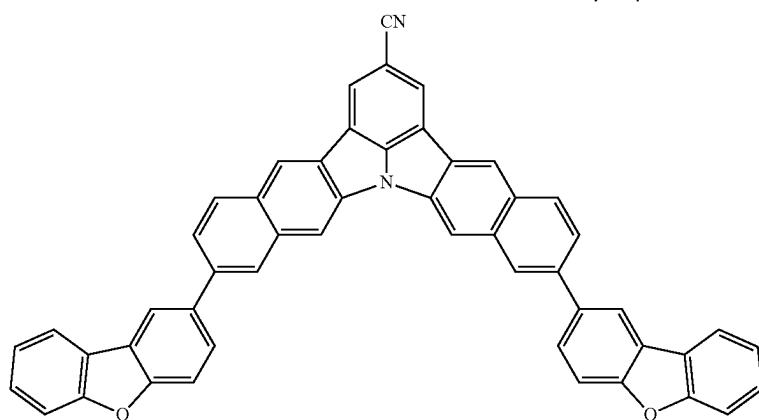
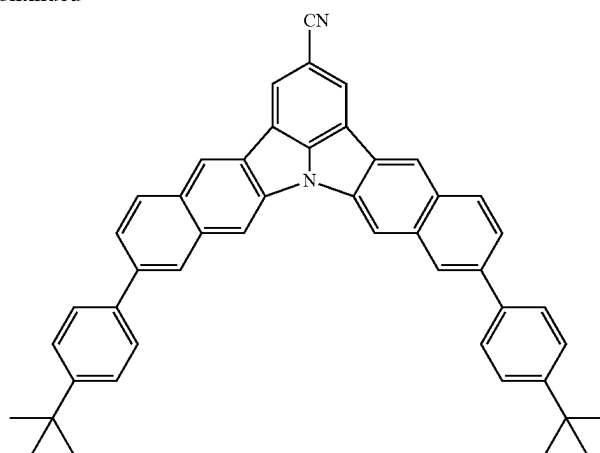
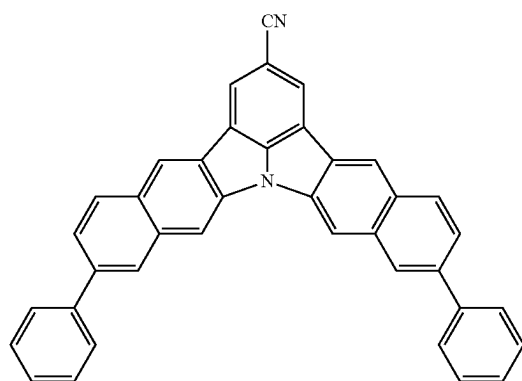
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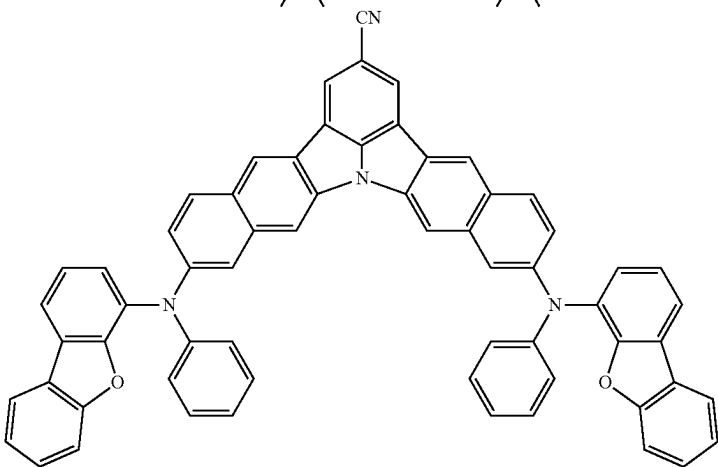
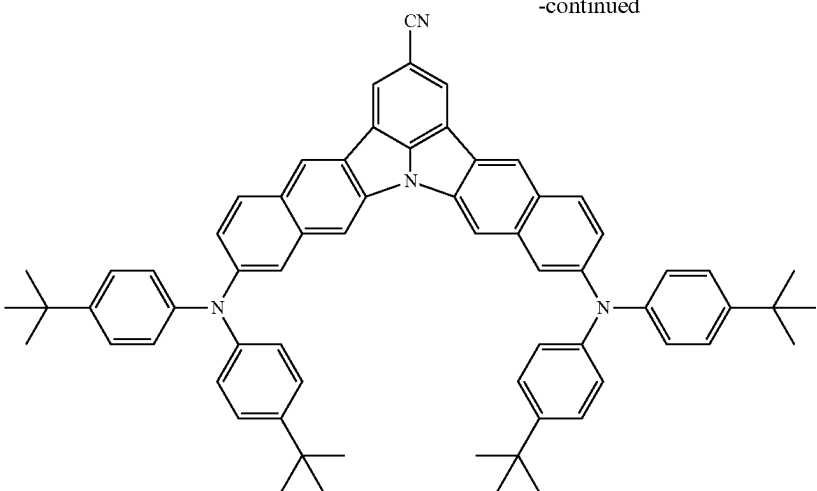
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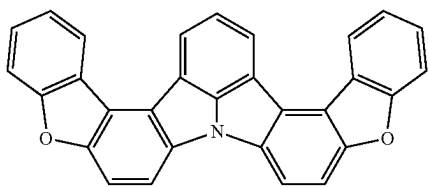
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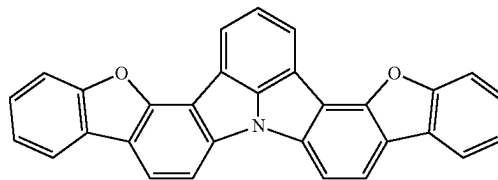


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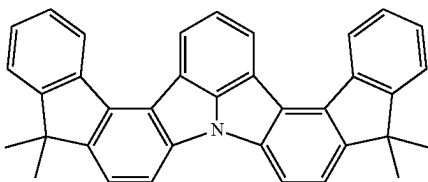
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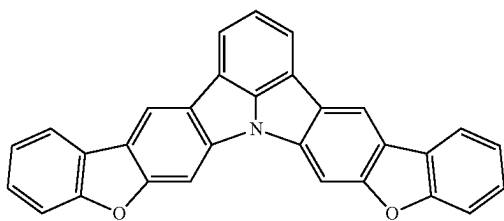
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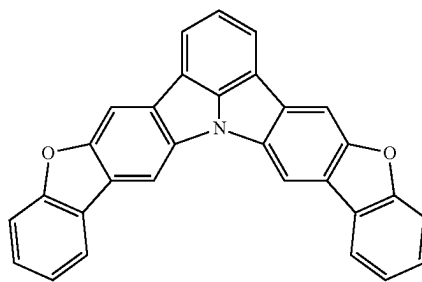
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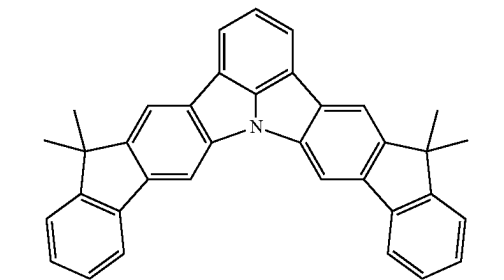
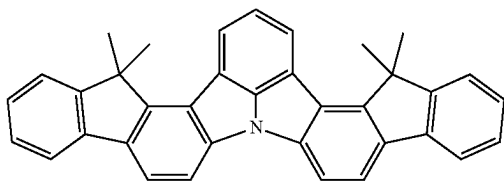
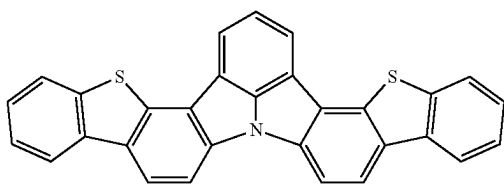
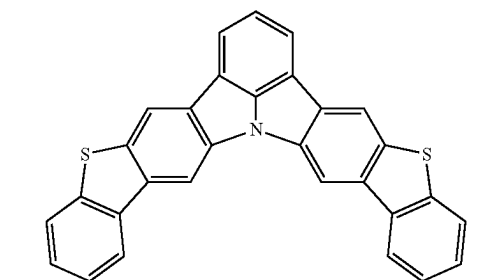
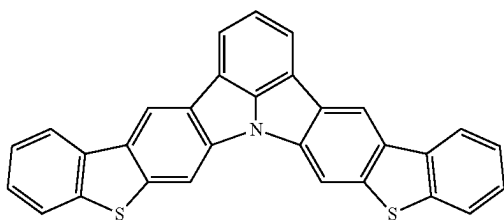
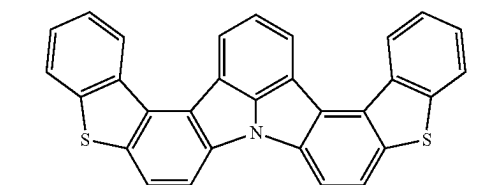
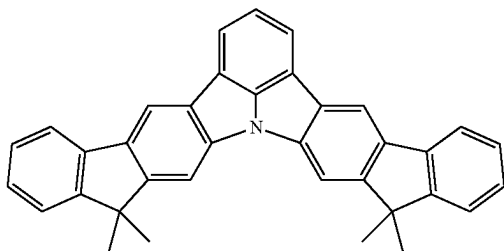
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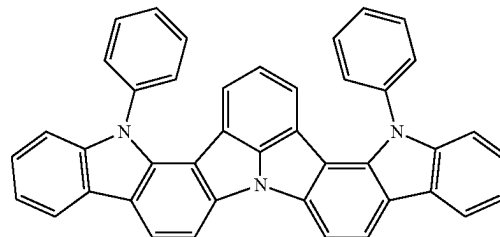
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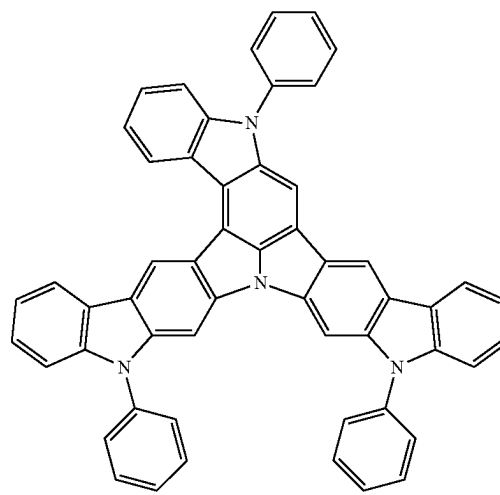
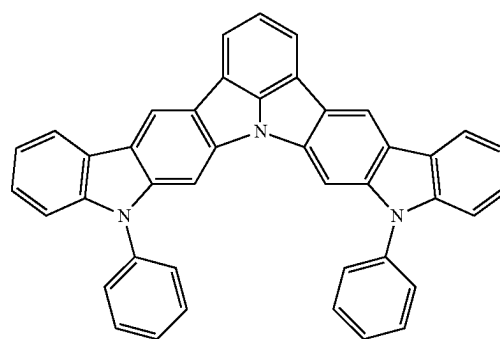
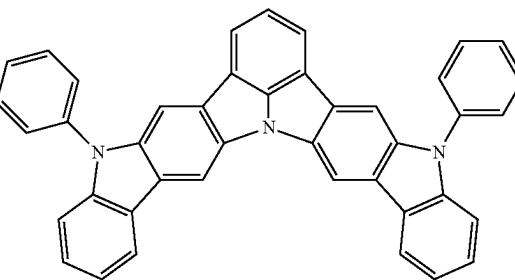
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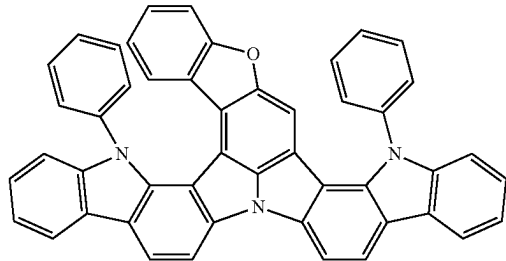
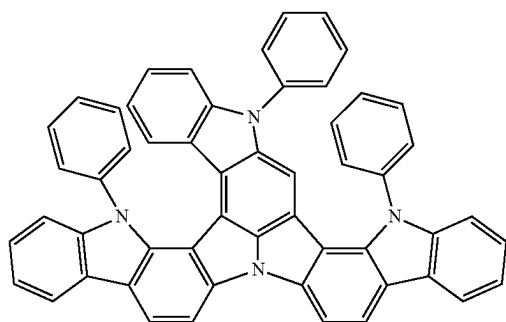
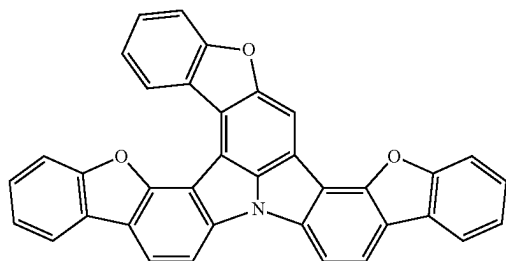
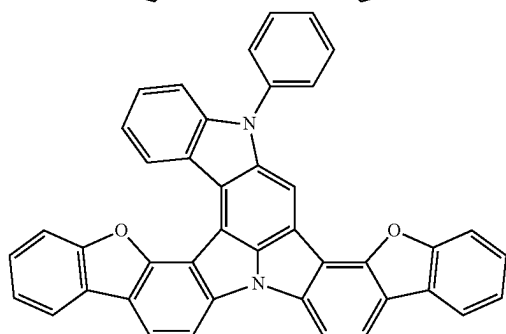
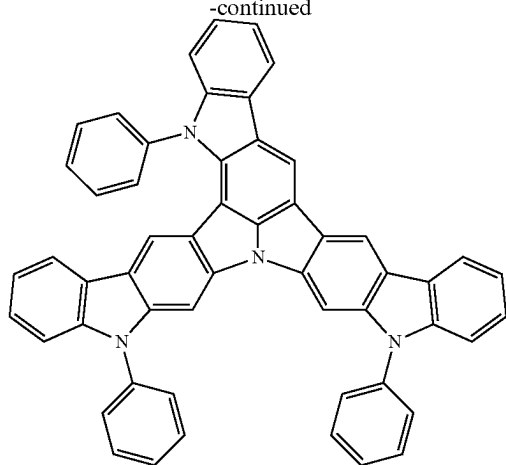
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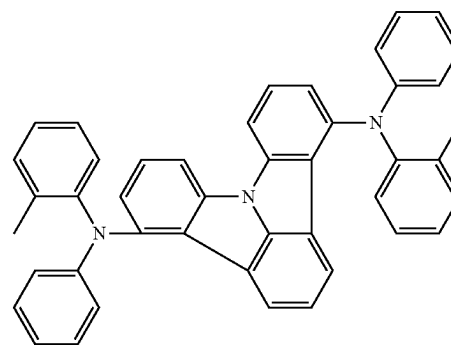
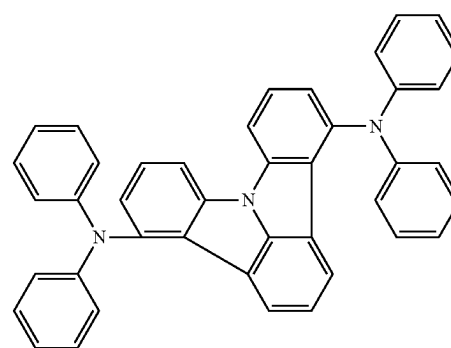
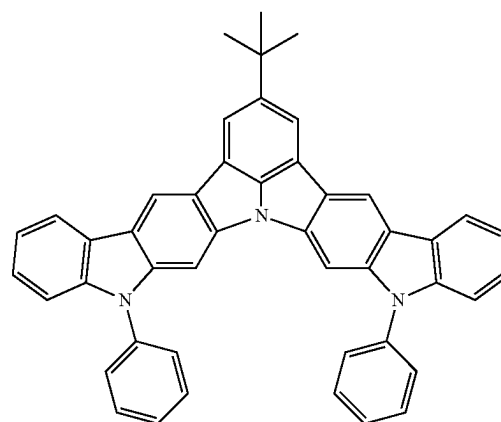
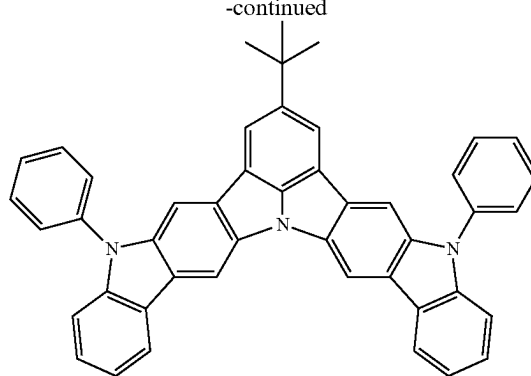
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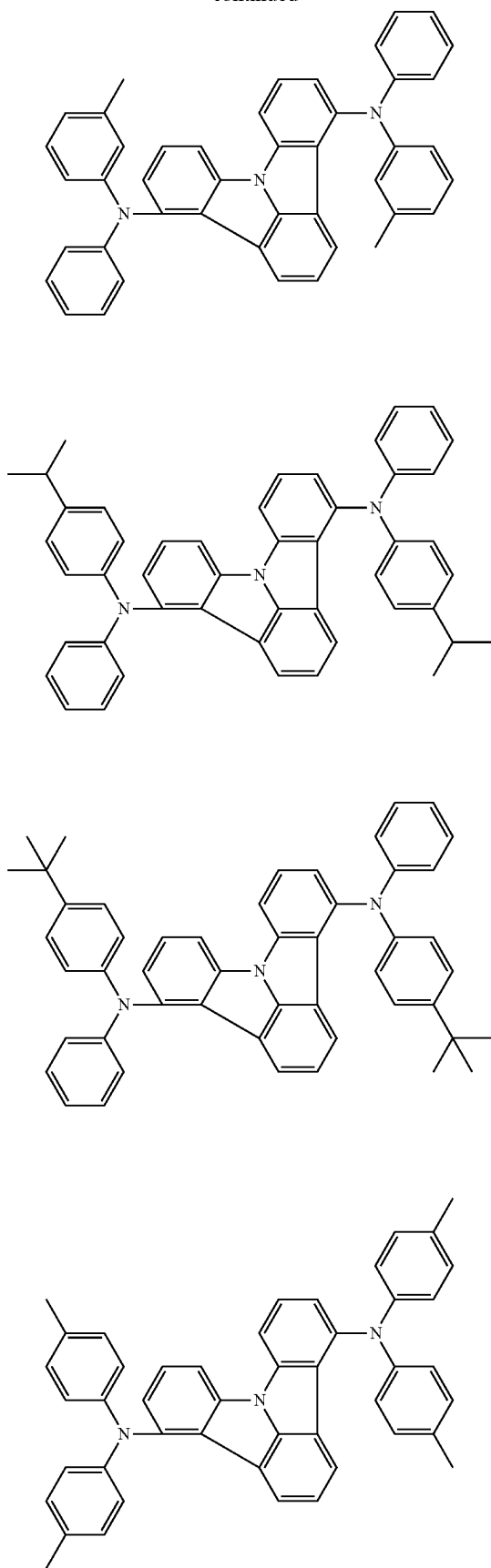
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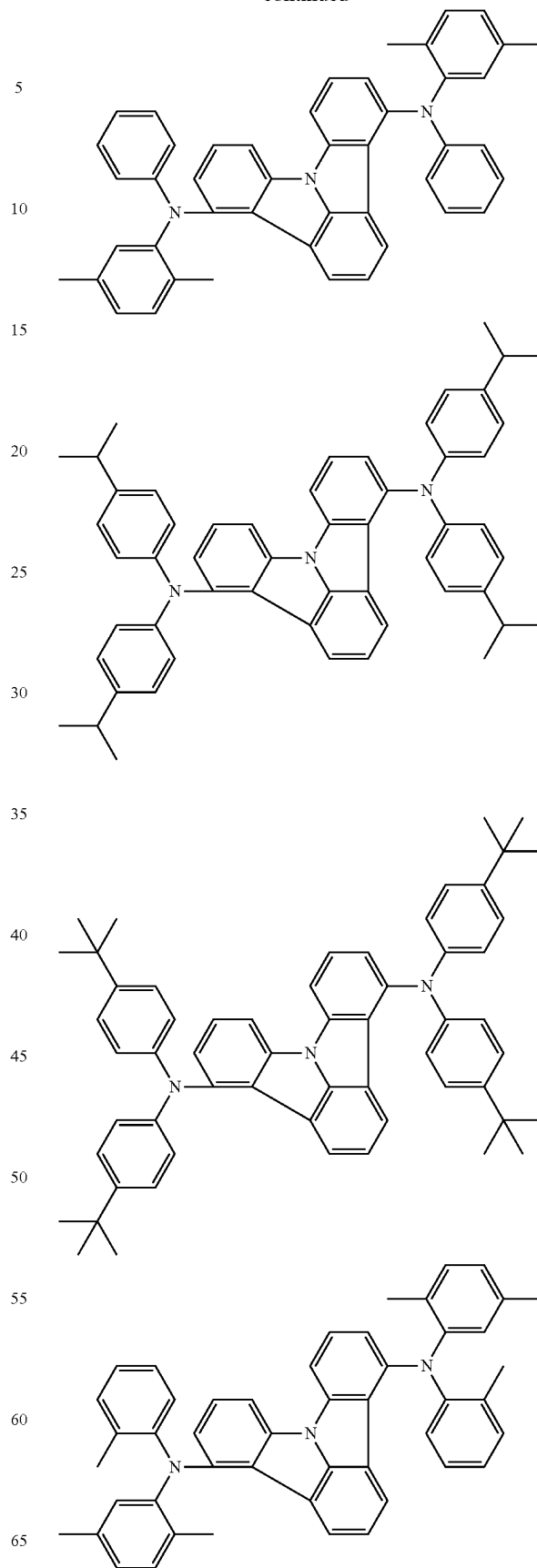


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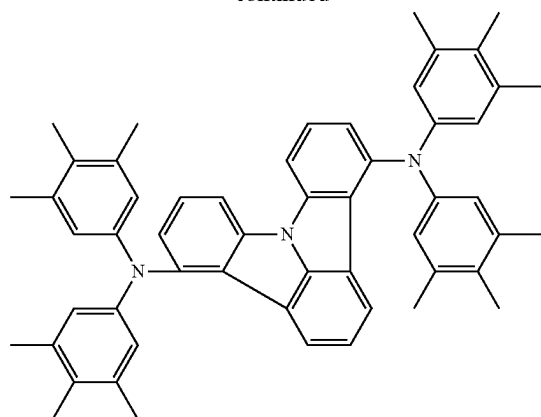
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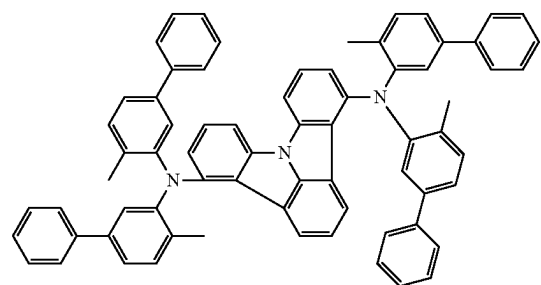
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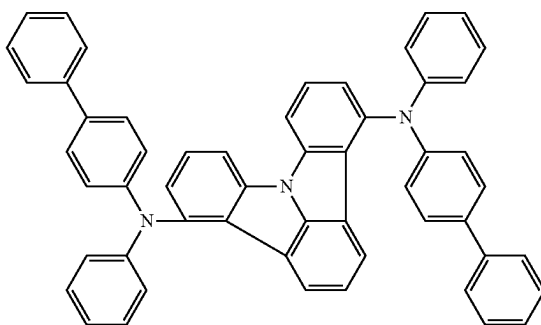
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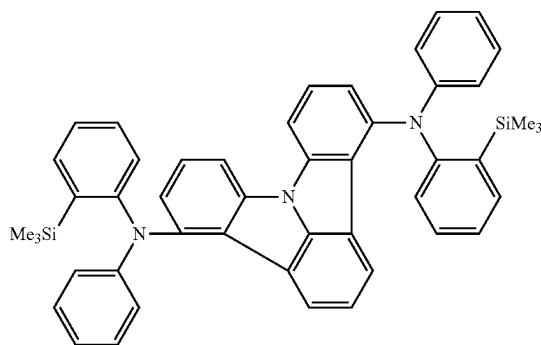
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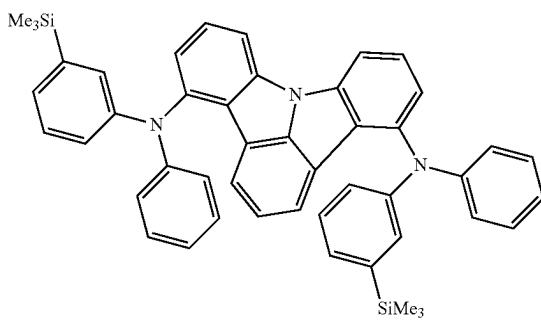
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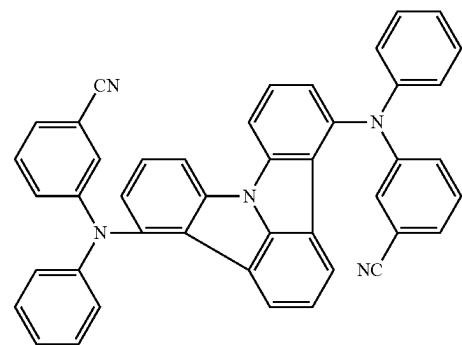
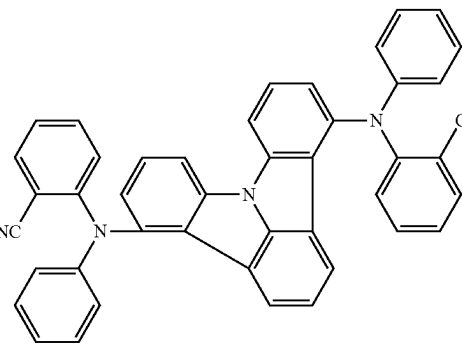
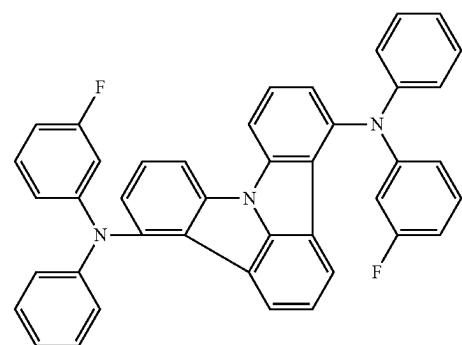
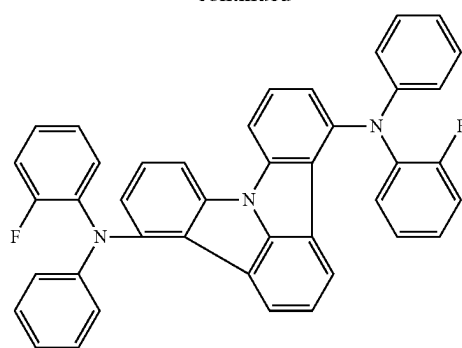
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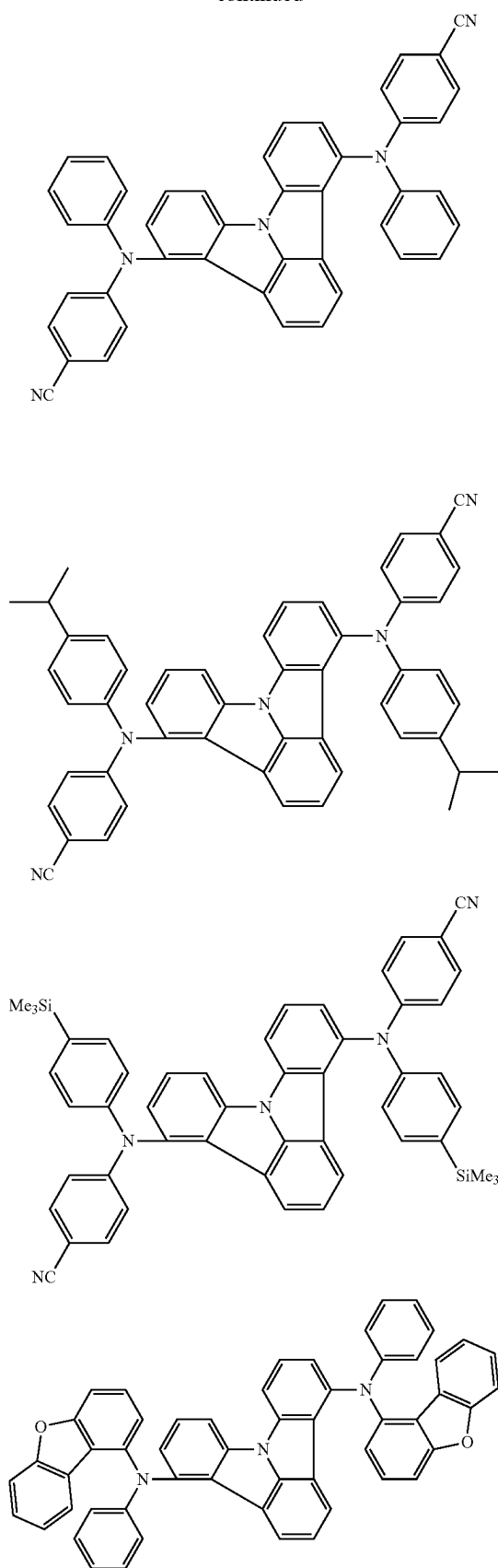
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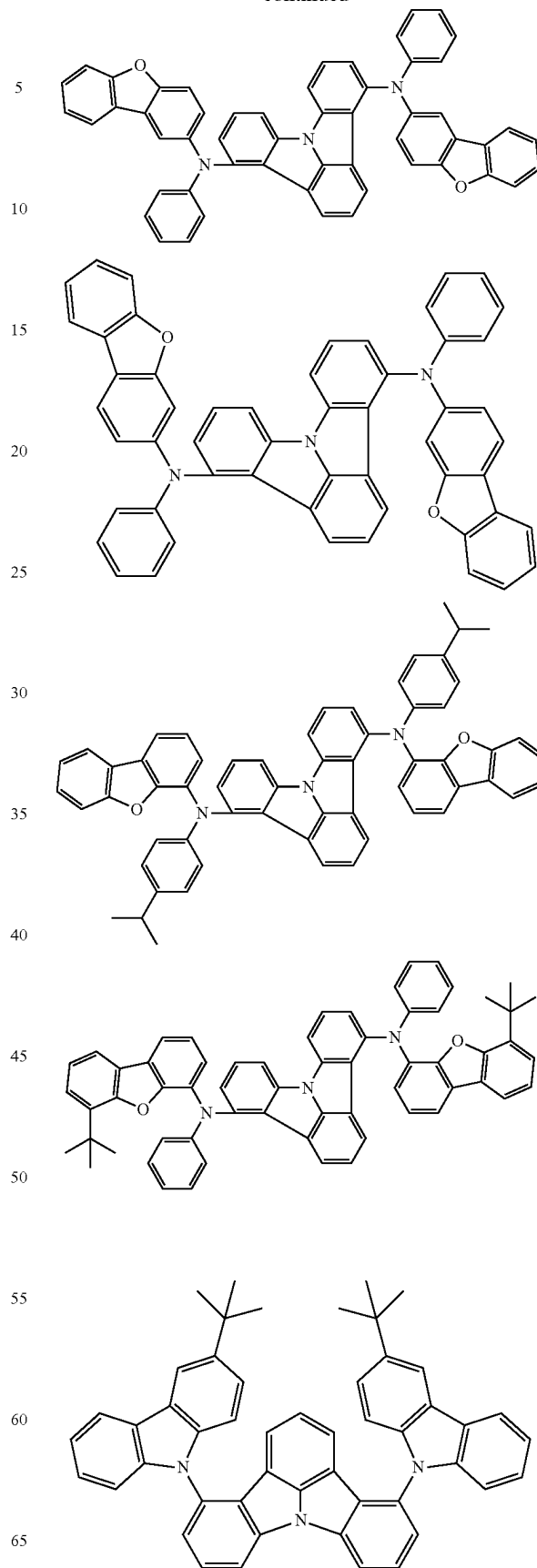


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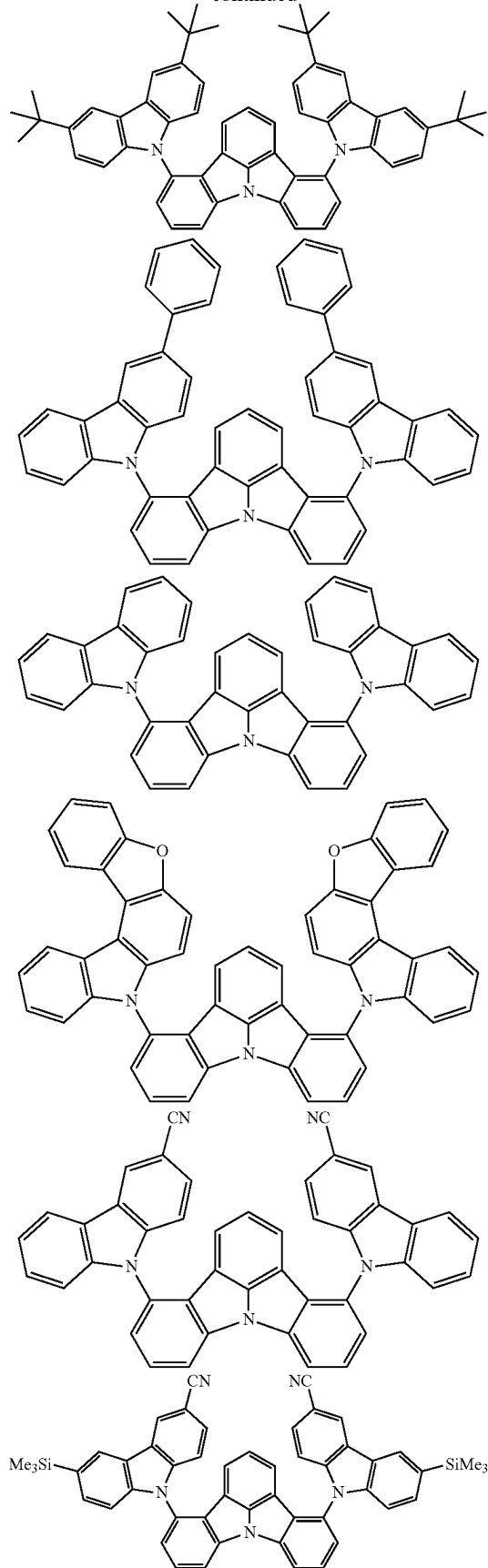
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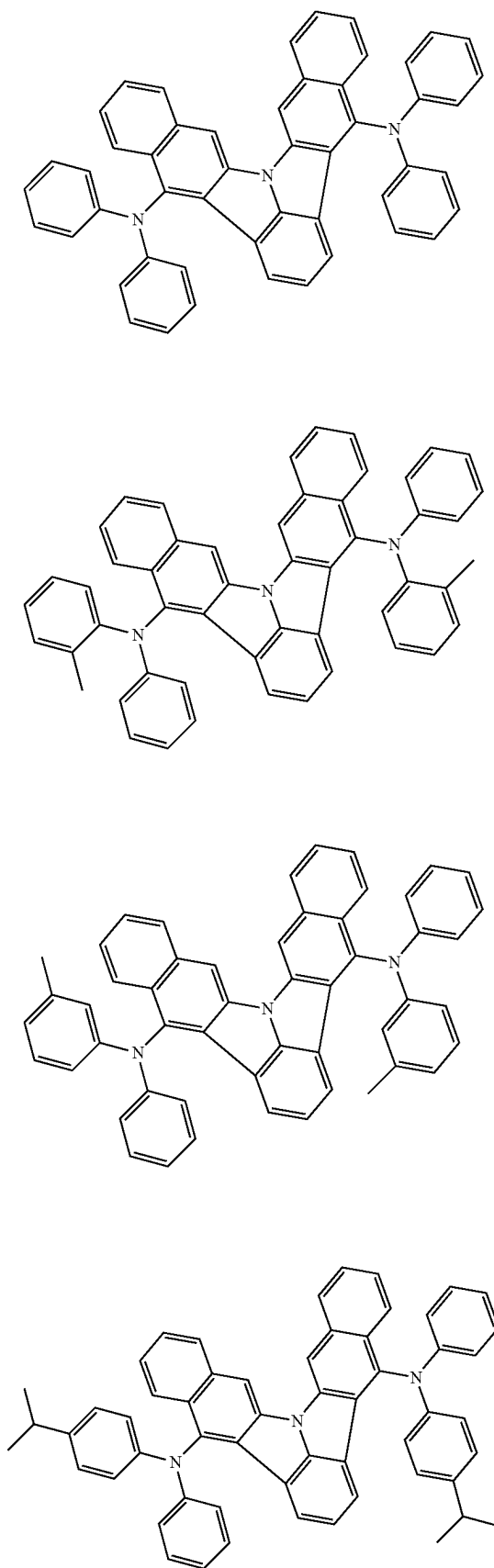
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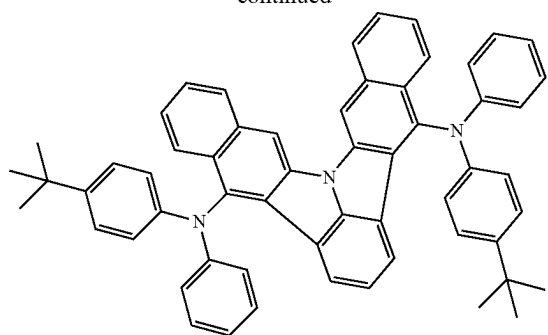
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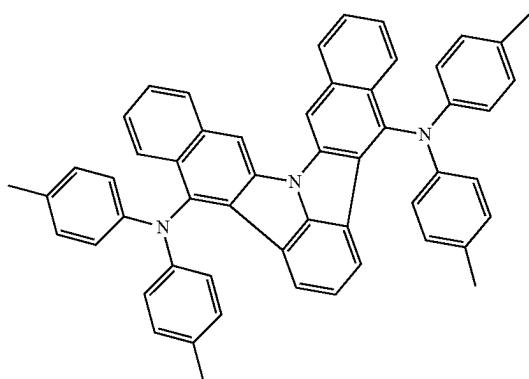
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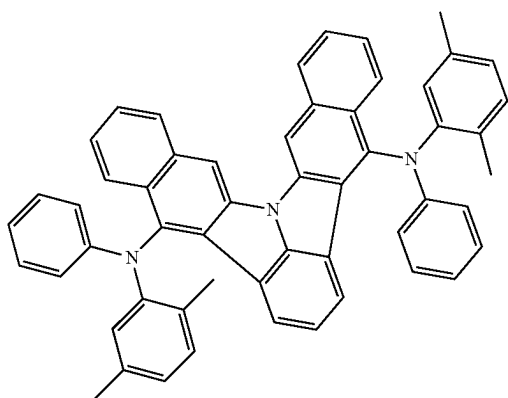
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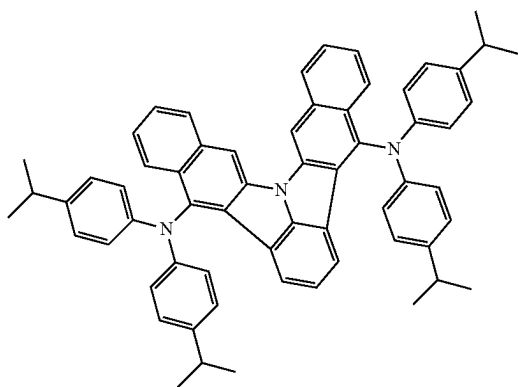
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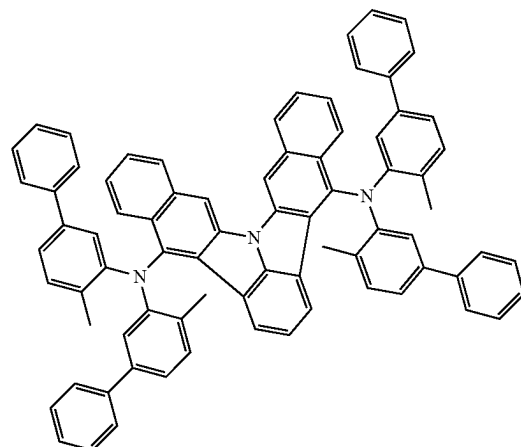
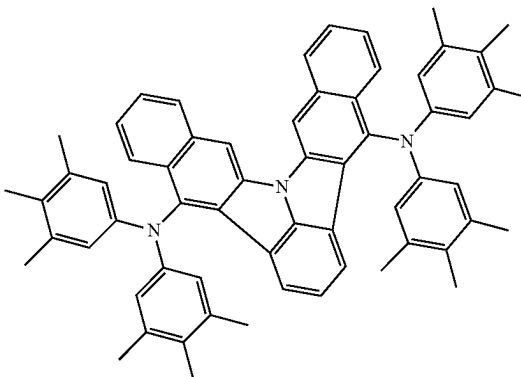
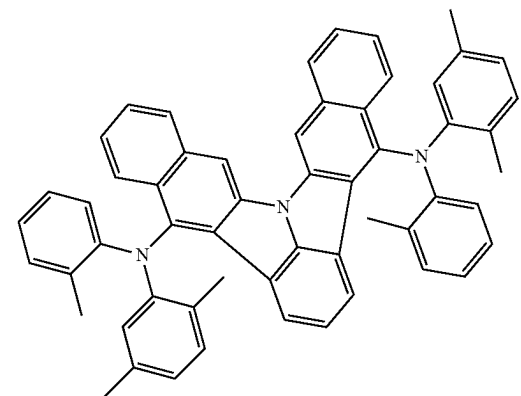
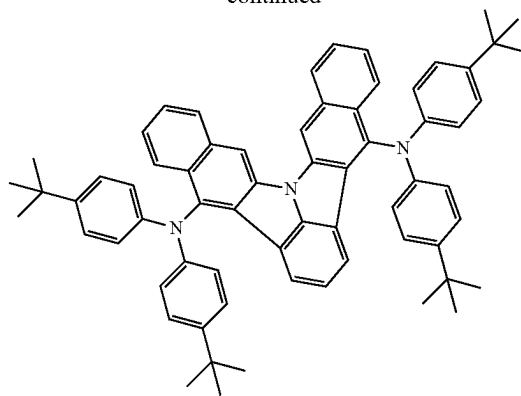
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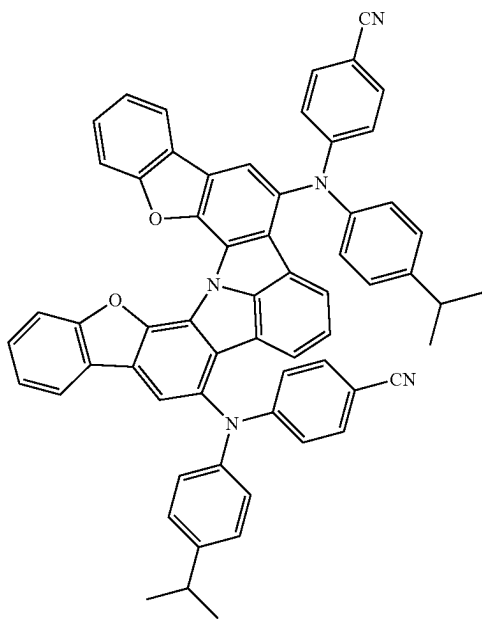
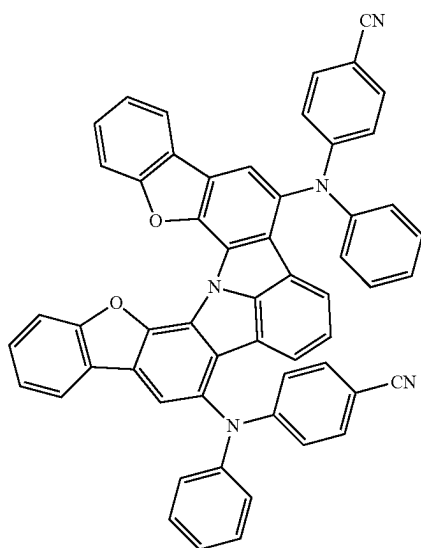
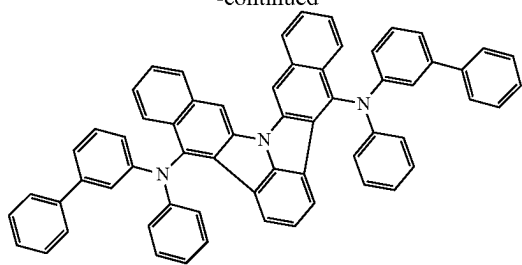
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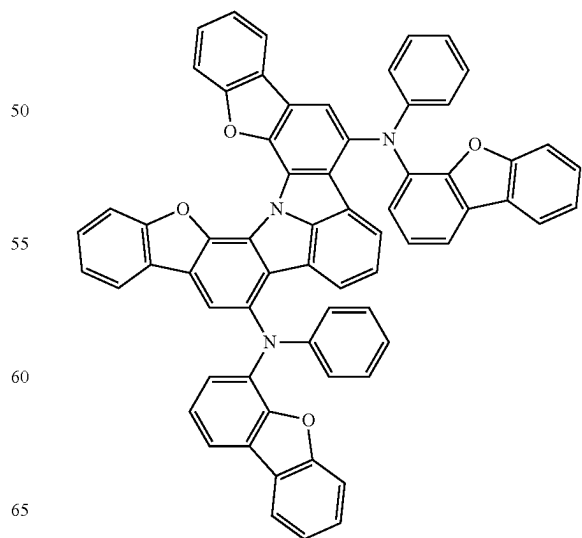
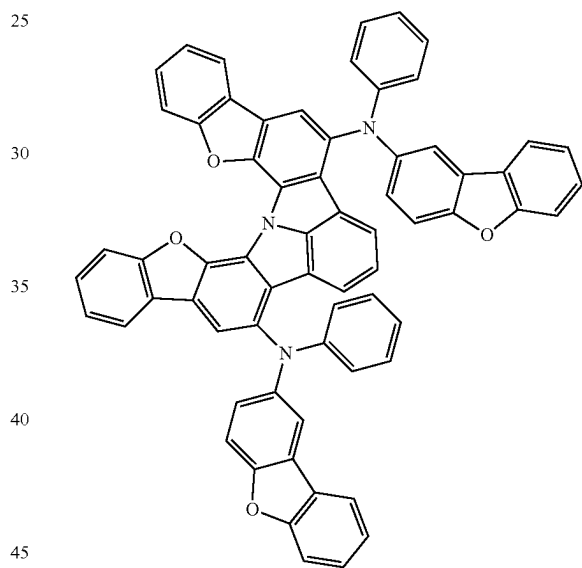
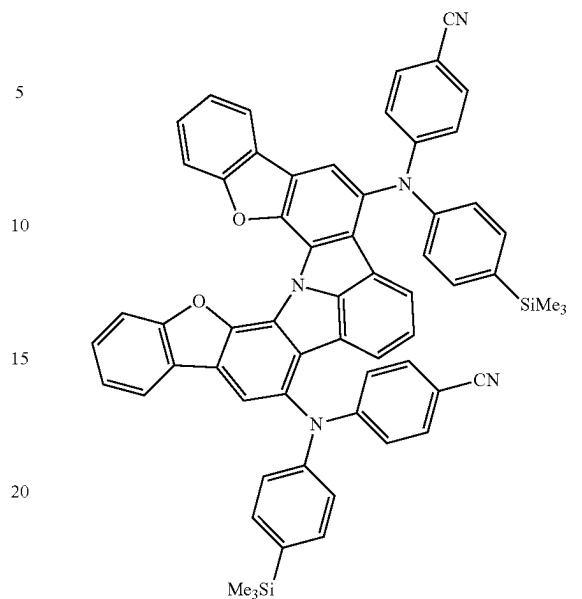


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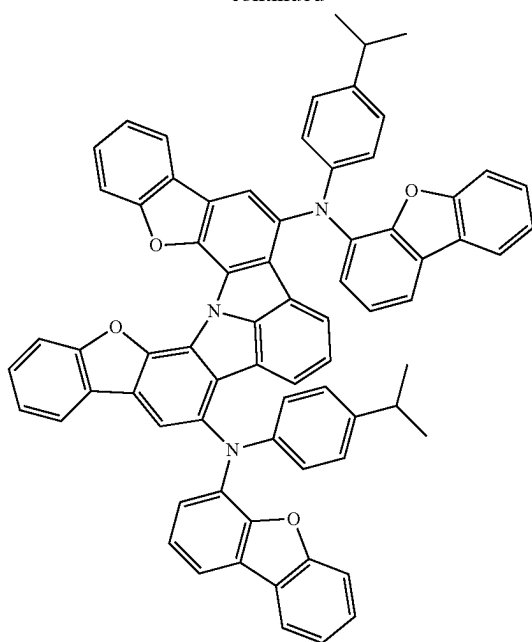
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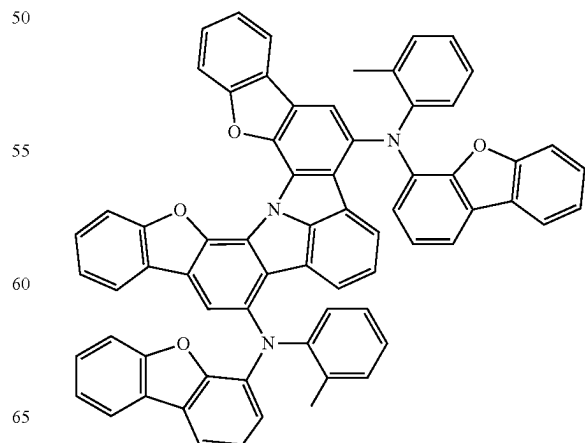
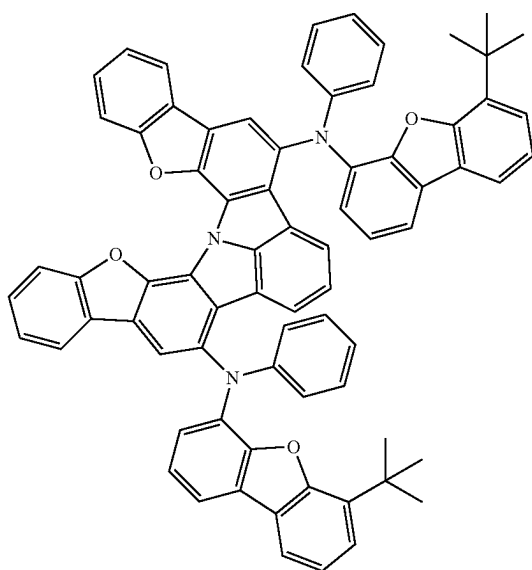
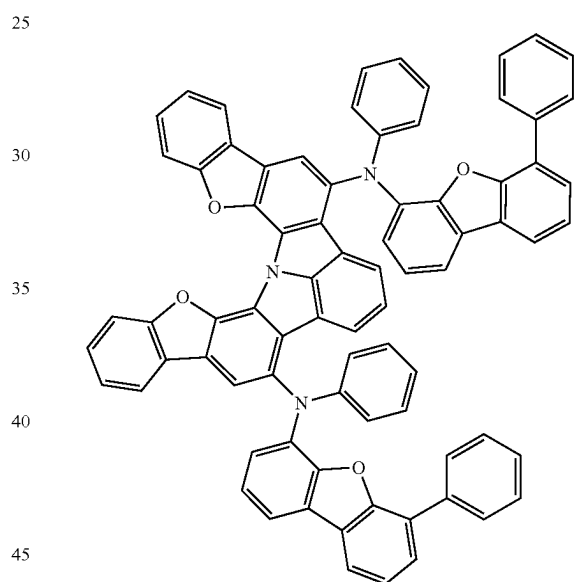
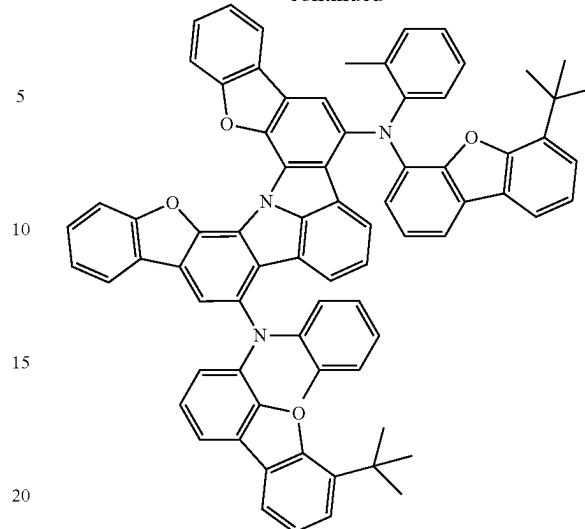


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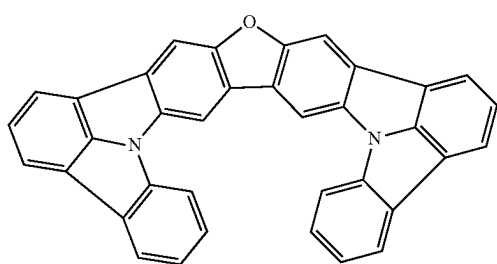
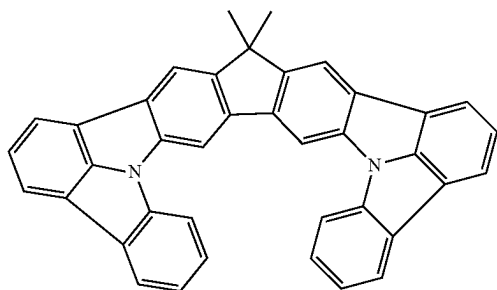
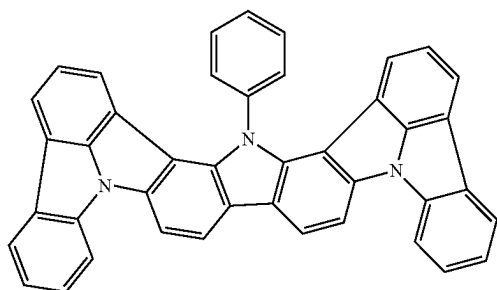
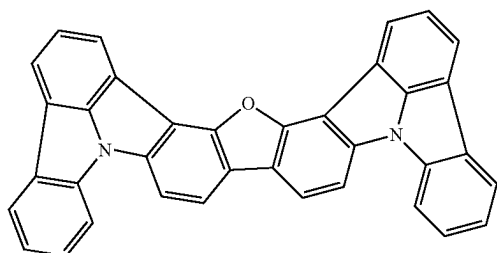
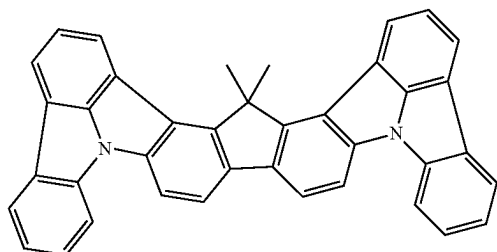
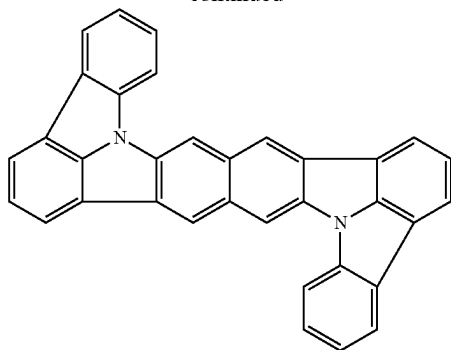
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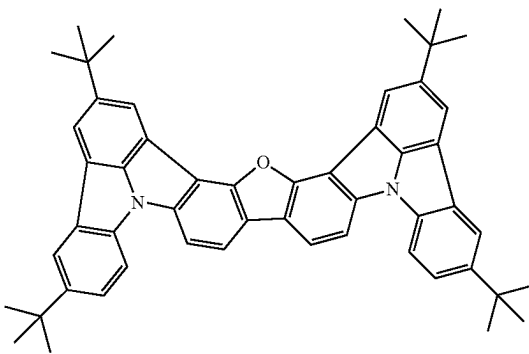
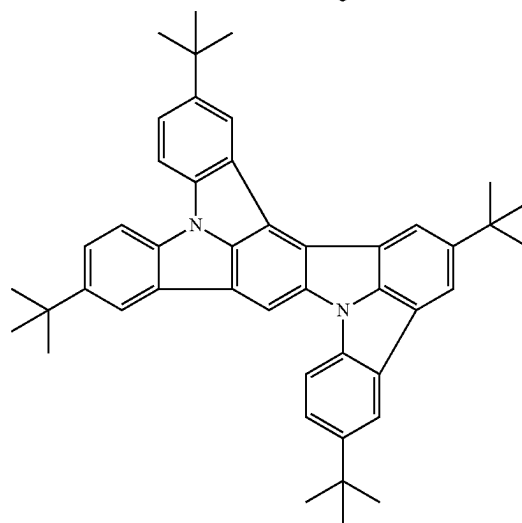
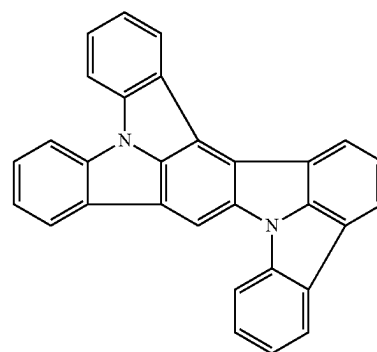
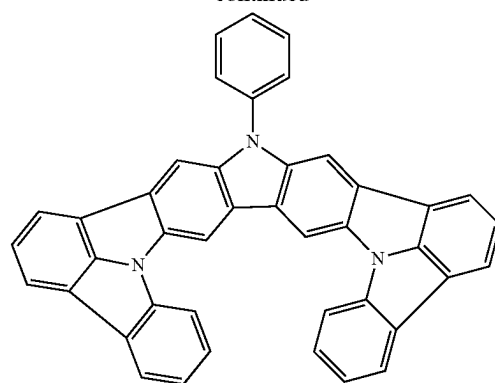
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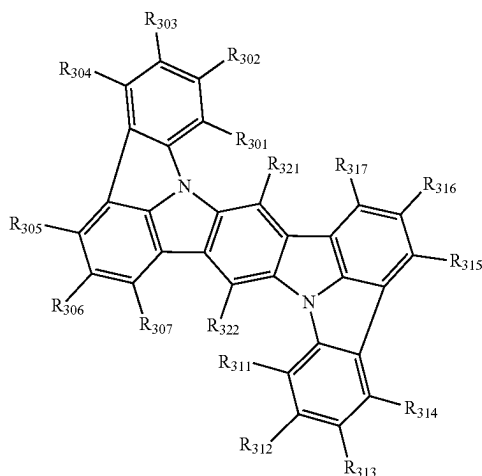


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(Compound Represented by Formula (31))

The compound represented by the formula (31) is explained below.

The compound represented by formula (31) is a compound corresponding to the compound represented by the formula (21-3).



In the formula (31),

one or more pairs of two or more adjacent groups of R₃₀₁ to R₃₀₇ and R₃₁₁ to R₃₁₇ form a substituted or unsubstituted, saturated or unsaturated ring, or do not form a substituted or unsubstituted, saturated or unsaturated ring;

R₃₀₁ to R₃₀₇ and R₃₁₁ to R₃₁₇ that do not form the substituted or unsubstituted, saturated or unsaturated ring are independently

a hydrogen atom,

a substituted or unsubstituted alkyl group having 1 to 50 carbon atoms,

a substituted or unsubstituted alkenyl group having 2 to 50 carbon atoms,

a substituted or unsubstituted alkynyl group having 2 to 50 carbon atoms,

a substituted or unsubstituted cycloalkyl group having 3 to 50 ring carbon atoms,

—Si(R₉₀₁)(R₉₀₂)(R₉₀₃),

—O—(R₉₀₄),

—S—(R₉₀₅),

—N(R₉₀₆)(R₉₀₇),

a halogen atom, a cyano group, a nitro group, a substituted or unsubstituted aryl group having 6 to 50 ring carbon atoms, or

a substituted or unsubstituted monovalent heterocyclic group having 5 to 50 ring atoms;

R₃₂₁ and R₃₂₂ are independently

a hydrogen atom,

a substituted or unsubstituted alkyl group having 1 to 50 carbon atoms,

a substituted or unsubstituted alkenyl group having 2 to 50 carbon atoms,

a substituted or unsubstituted alkynyl group having 2 to 50 carbon atoms,

a substituted or unsubstituted cycloalkyl group having 3 to 50 ring carbon atoms,

—Si(R₉₀₁)(R₉₀₂)(R₉₀₃),

—O—(R₉₀₄),

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—S—(R₉₀₅),

—N(R₉₀₆)(R₉₀₇),

a halogen atom, a cyano group, a nitro group,

a substituted or unsubstituted aryl group having 6 to 50 ring carbon atoms, or

a substituted or unsubstituted monovalent heterocyclic group having 5 to 50 ring atoms; and

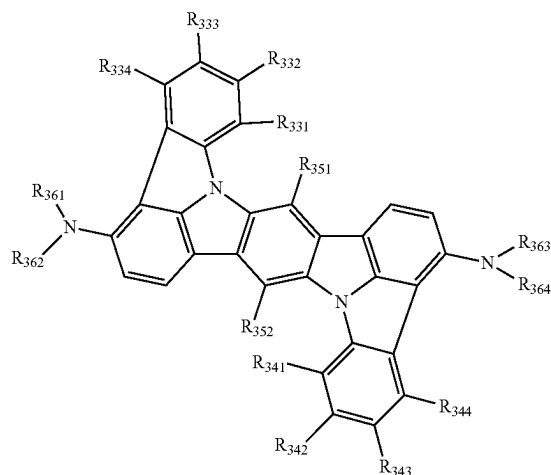
R₉₀₁ to R₉₀₇ are as defined in the formula (1).

Example of "One pair of two or more adjacent groups of R₃₀₁ to R₃₀₇ and R₃₁₁ to R₃₁₇" is pairs of R₃₀₁ and R₃₀₂, R₃₀₂ and R₃₀₃, R₃₀₃ and R₃₀₄, R₃₀₅ and R₃₀₆, R₃₀₆ and R₃₀₇, and R₃₀₁, R₃₀₂ and R₃₀₃, and the like.

In one embodiment, at least one of R₃₀₁ to R₃₀₇ and R₃₁₁ to R₃₁₇, preferably two of R₃₀₁ to R₃₀₇ and R₃₁₁ to R₃₁₇ is a group represented by —N(R₉₀₆)(R₉₀₇).

In one embodiment, R₃₀₁ to R₃₀₇ and R₃₁₁ to R₃₁₇ are independently a hydrogen atom, a substituted or unsubstituted aryl group having 6 to 50 ring carbon atoms, or a substituted or unsubstituted monovalent heterocyclic group having 5 to 50 ring atoms.

In one embodiment, the compound represented by the formula (31) is a compound represented by the following formula (32):



wherein in the formula (32),

one or more pairs of two or more adjacent groups of R₃₃₁ to R₃₃₄ and R₃₄₁ to R₃₄₄ form a substituted or unsubstituted, saturated or unsaturated ring, or do not form a substituted or unsubstituted saturated or unsaturated ring;

R₃₃₁ to R₃₃₄ and R₃₄₁ to R₃₄₄ that do not form the substituted or unsubstituted, saturated or unsaturated ring and R₃₅₁ and R₃₅₂ are independently

a hydrogen atom,

a substituted or unsubstituted aryl group having 6 to 50 ring carbon atoms, or

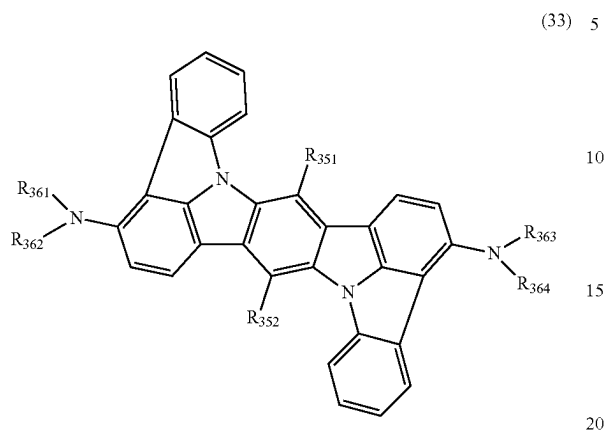
a substituted or unsubstituted monovalent heterocyclic group having 5 to 50 ring atoms; and R₃₆₁ to R₃₆₄ are independently

a substituted or unsubstituted aryl group having 6 to 50 ring carbon atoms, or

a substituted or unsubstituted monovalent heterocyclic group having 5 to 50 ring atoms.

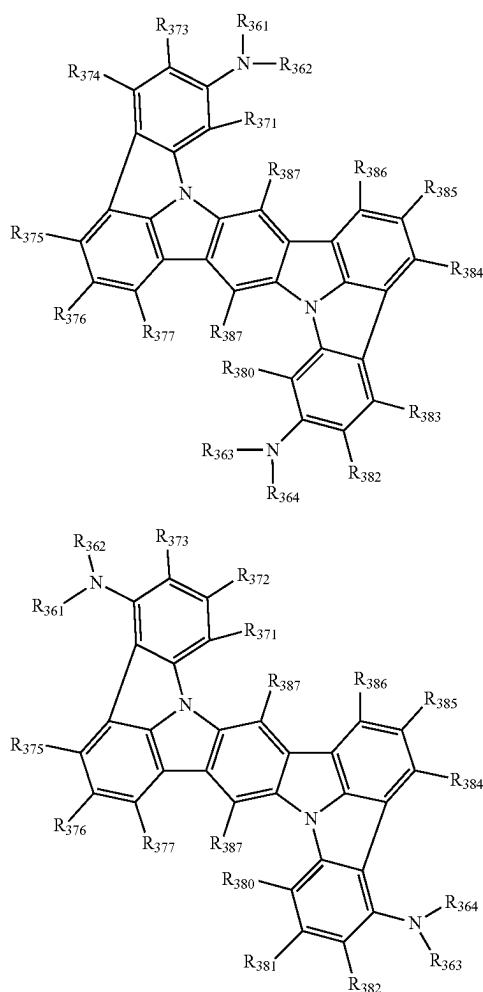
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In one embodiment, the compound represented by the formula (31) is a compound represented by the formula (33):



wherein in the formula (33), R₃₅₁, R₃₆₂, and R₃₆₁ to R₃₆₄ are as defined in the formula (32).

In one embodiment, the compound represented by the formula (31) is a compound represented by the formula (34) or (35):



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wherein in the formulas (34) and (35), R₃₆₁ to R₃₆₄ are as defined in the formula (32); one or more pairs of two or more adjacent groups of R₃₇₁ to R₃₇₇ and R₃₈₀ to R₃₈₆ form a substituted or unsubstituted, saturated or unsaturated ring, or do not form a substituted or unsubstituted saturated or unsaturated ring; and

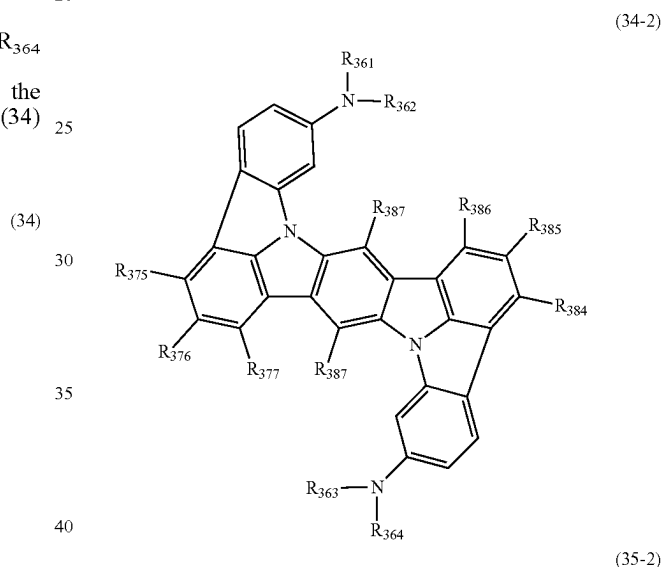
R₃₇₁ to R₃₇₇ and R₃₈₀ to R₃₈₆ that do not form the substituted or unsubstituted, saturated or unsaturated ring and R₃₈₇ are independently

a hydrogen atom,

a substituted or unsubstituted aryl group having 6 to 50 ring carbon atoms, or

a substituted or unsubstituted monovalent heterocyclic group having 5 to 50 ring atoms, two R₃₈₇ or may be the same with or different from each other.

In one embodiment, the compound represented by the formula (31) is a compound represented by the formula (34-2) or (35-2):



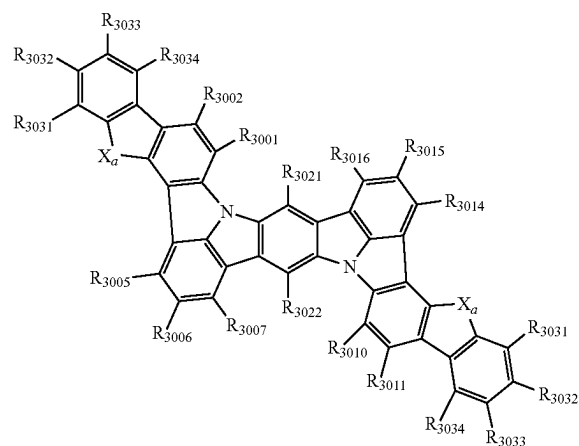
wherein in the formulas (34-2) and (35-2), R₃₆₁ to R₃₆₄, R₃₇₅ to R₃₇₇ and R₃₈₄ to R₃₈₇ are as defined in the formulas (34) and (35).

In one embodiment, R₃₆₁ to R₃₆₄ the formulas (32), (33), (34), (35), (34-2) and (35-2) are independently a substituted or unsubstituted aryl group having 6 to 50 ring carbon atoms (preferably a phenyl group).

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In one embodiment, R_{321} and R_{322} in the formula (31), and R_{351} and R_{352} the formulas (32), (33), (34), (35), (34-2) and (35-2) are independently a hydrogen atom or a substituted or unsubstituted aryl group having 6 to 50 ring carbon atoms (preferably a phenyl group).

In one embodiment, the compound represented by the formula (31) is the compound represented by the following formula (36).



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In the formula (36),

one or more pairs of two or more adjacent groups of R_{3001} , R_{3002} , R_{3005} to R_{3007} , R_{3010} , R_{3011} , R_{3014} to R_{3016} and R_{3031} to R_{3034} bond with each other to form a substituted or unsubstituted, saturated or unsaturated ring or do not form the ring;

X_a s are independently selected from O, S and N(R_{35});

R_{35} and R_{3031} bond with each other to form a substituted or unsubstituted, saturated or unsaturated ring or do not form the ring; and

R_{3001} , R_{3002} , R_{3005} to R_{3007} , R_{3010} , R_{3011} , R_{3014} to R_{3016} and R_{3031} to R_{3035} that do not form the ring and R_{3021} and R_{3022} are independently

a hydrogen atom,

a substituted or unsubstituted aryl group having 6 to 50 ring carbon atoms, or

a substituted or unsubstituted monovalent heterocyclic group having 5 to 50 ring atoms.

In one embodiment, a substituent in the case of "substituted" in the formulas (31) to (36) is

a substituted or unsubstituted alkyl group having 1 to 50 carbon atoms,

a substituted or unsubstituted alkenyl group having 2 to 50 carbon atoms,

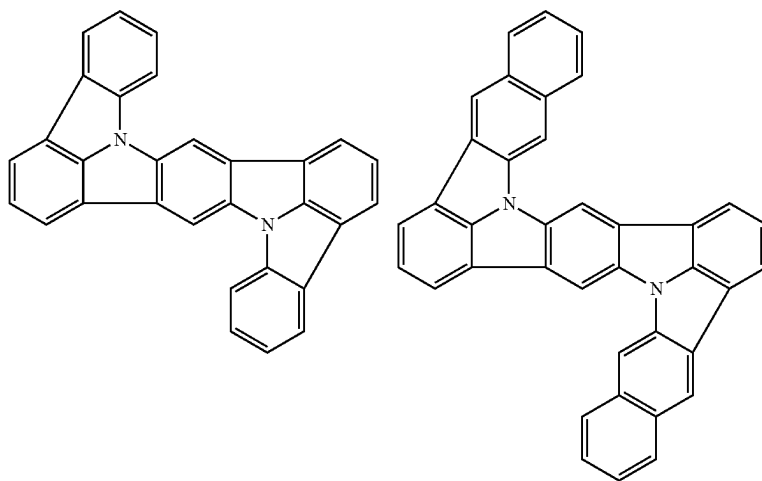
a substituted or unsubstituted alkynyl group having 2 to 50 carbon atoms,

a substituted or unsubstituted cycloalkyl group having 3 to 50 ring carbon atoms,

a substituted or unsubstituted aryl group having 6 to 50 ring carbon atoms, or

a substituted or unsubstituted monovalent heterocyclic group having 5 to 50 ring atoms.

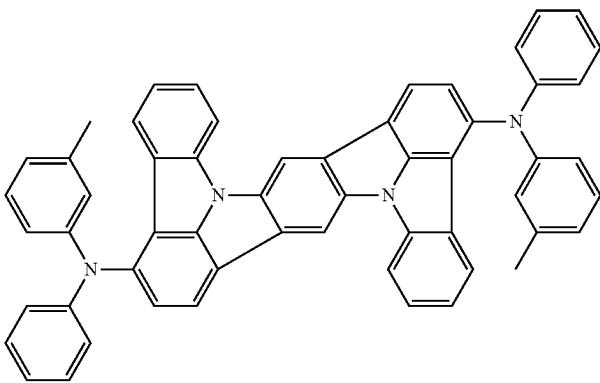
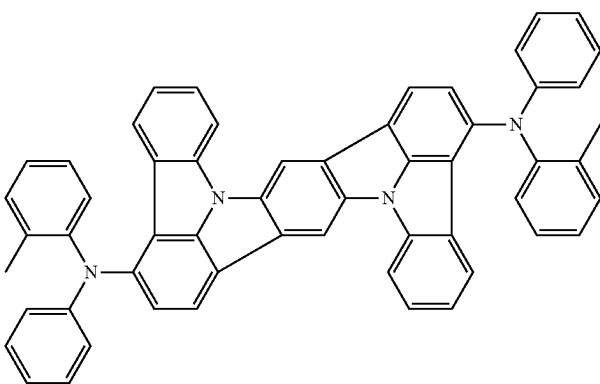
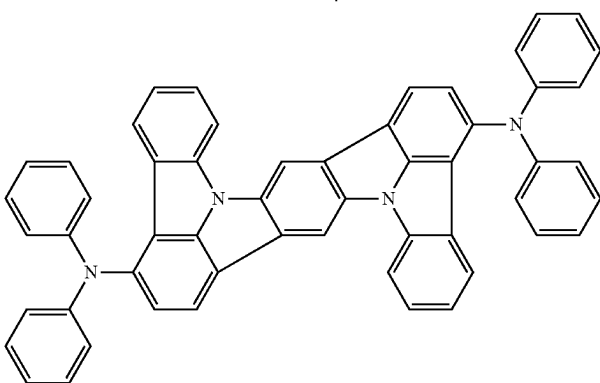
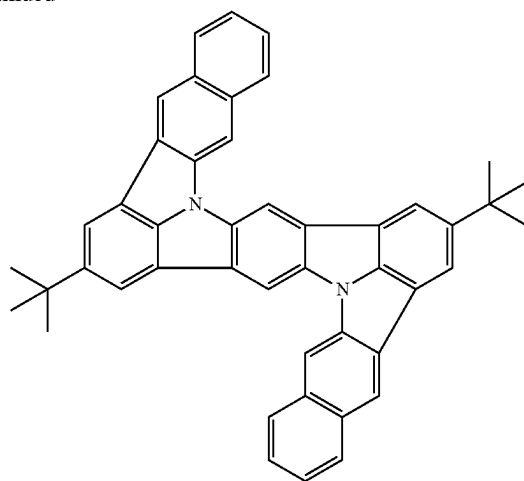
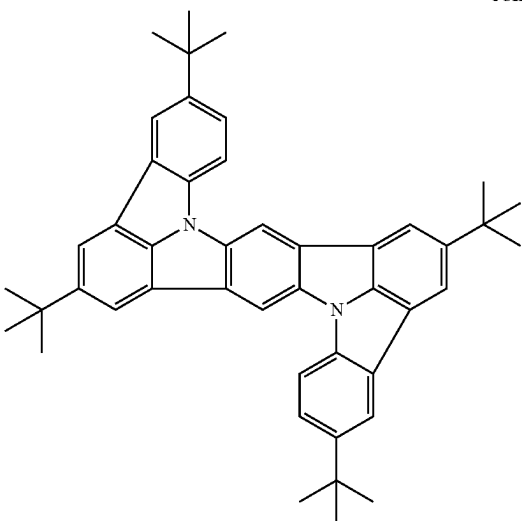
As the compound represented by the formula (31), the following compounds can be given for example. In the following example compounds, Me represents methyl group.



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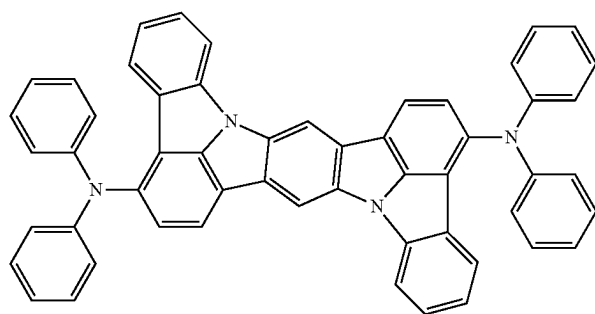
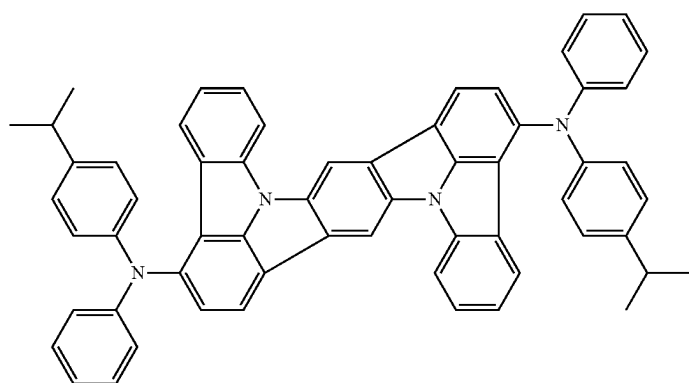
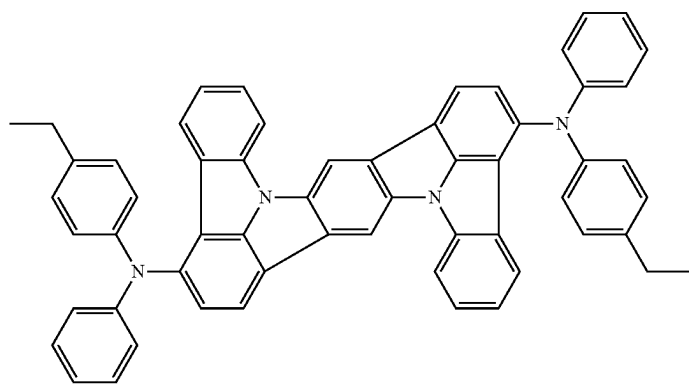
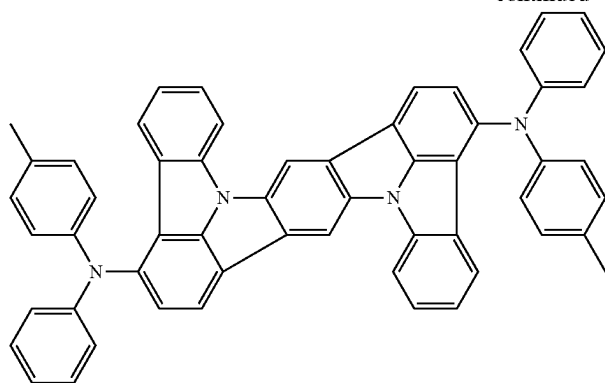
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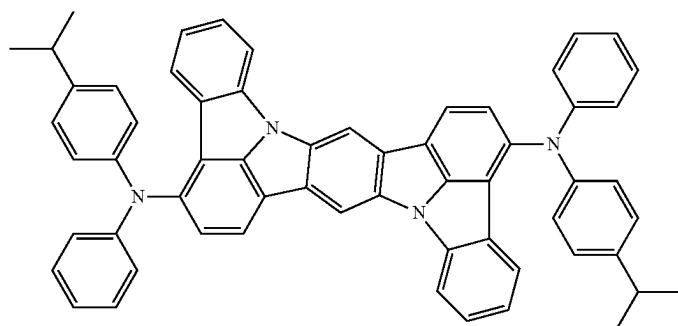
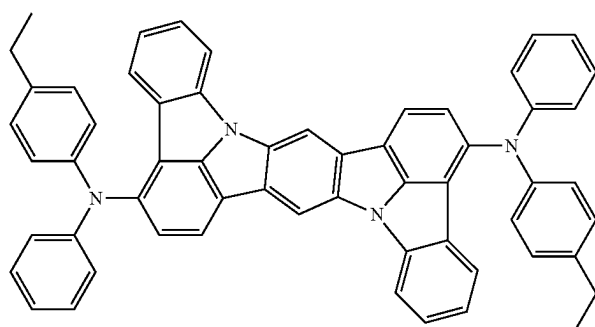
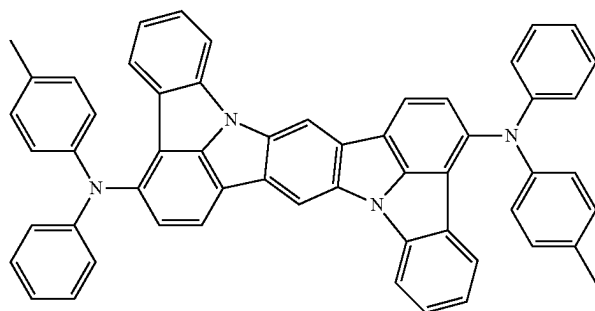
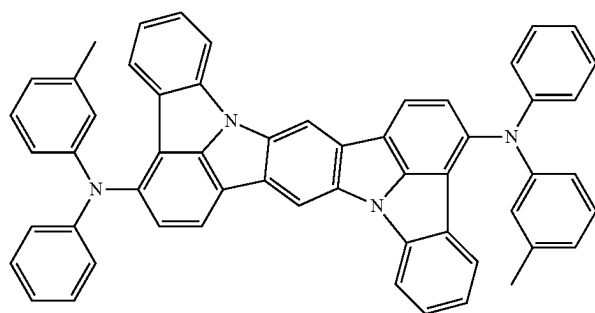
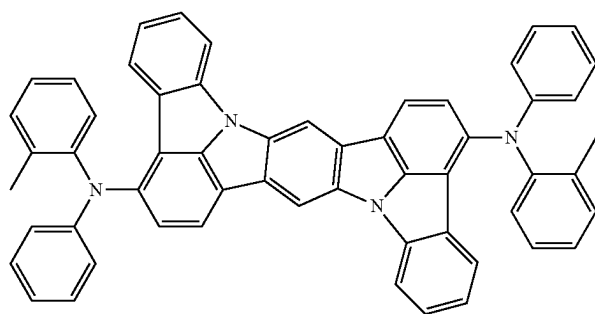
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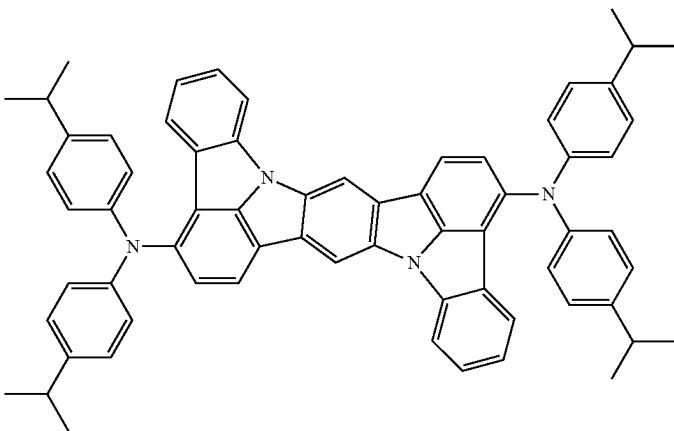
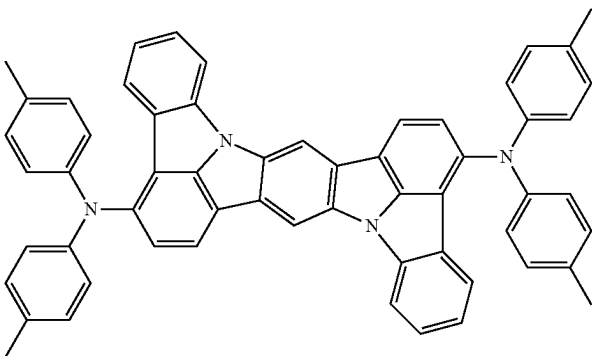
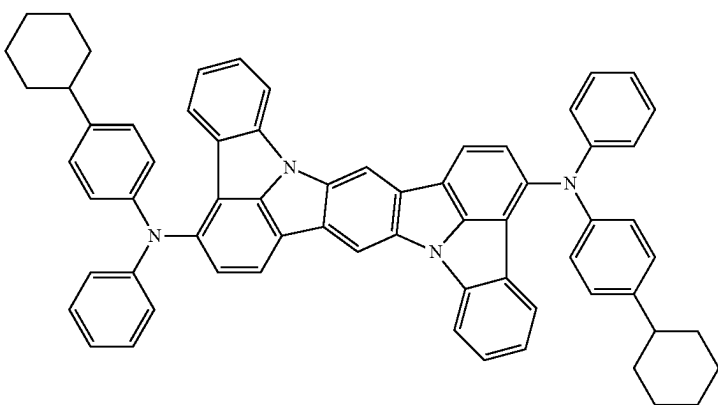
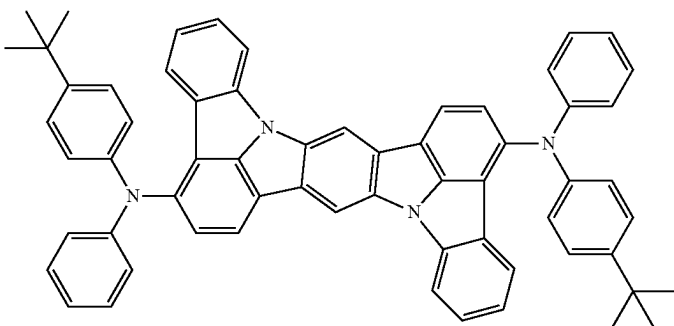
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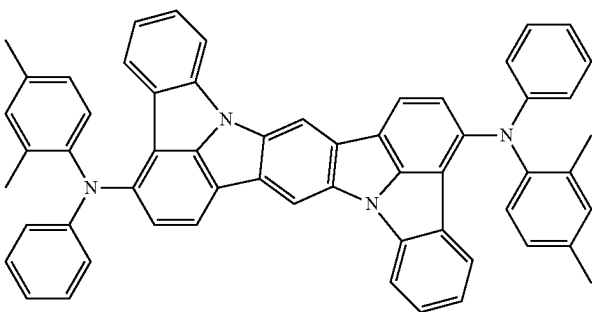
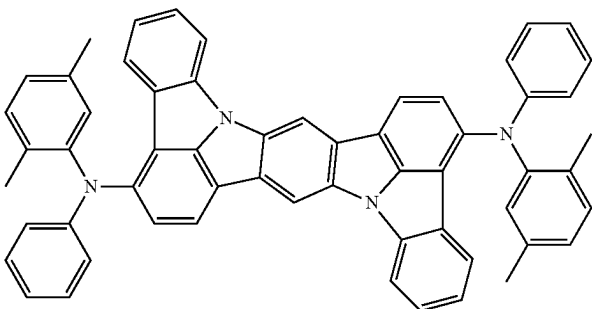
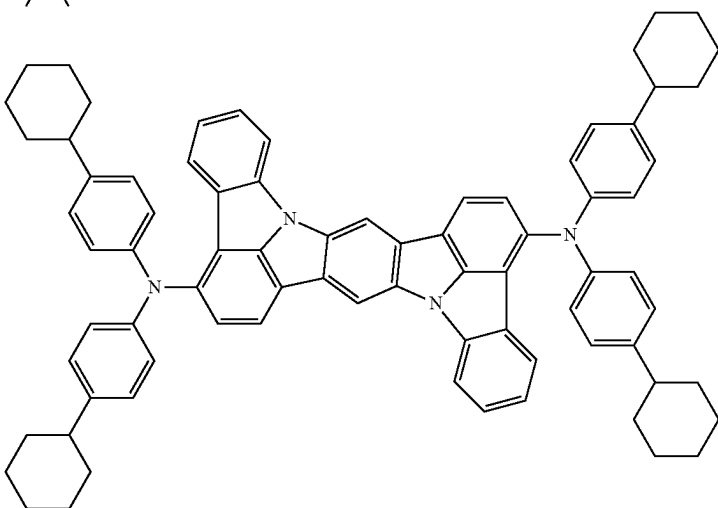
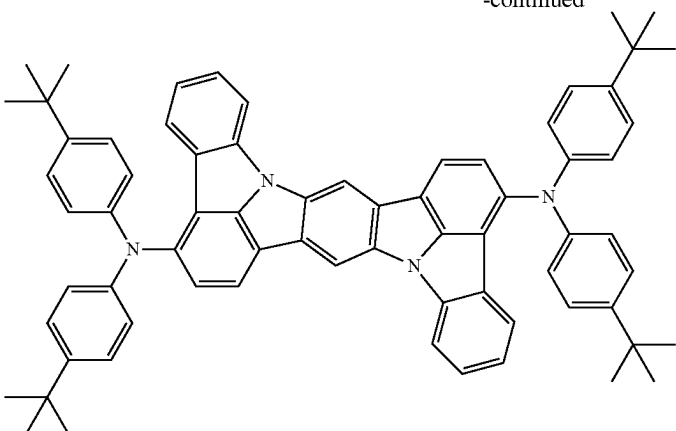
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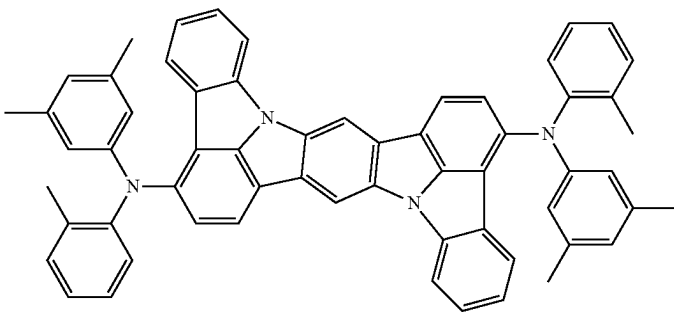
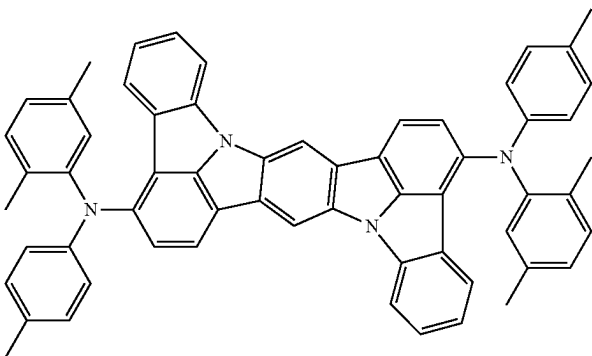
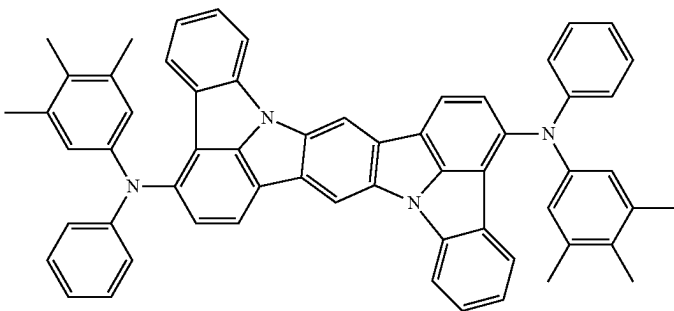
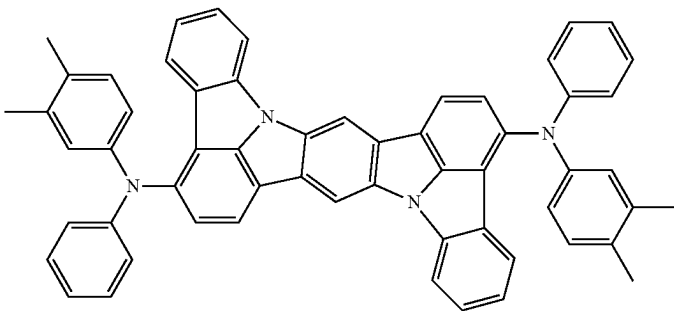
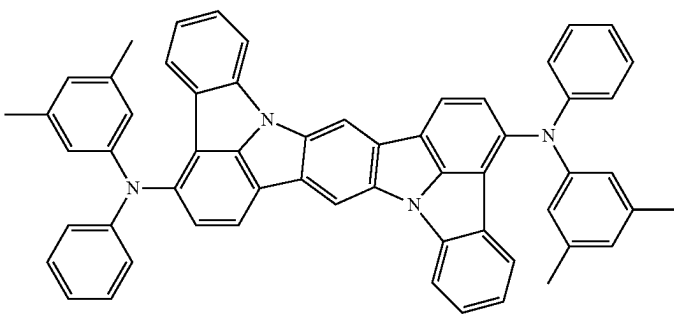
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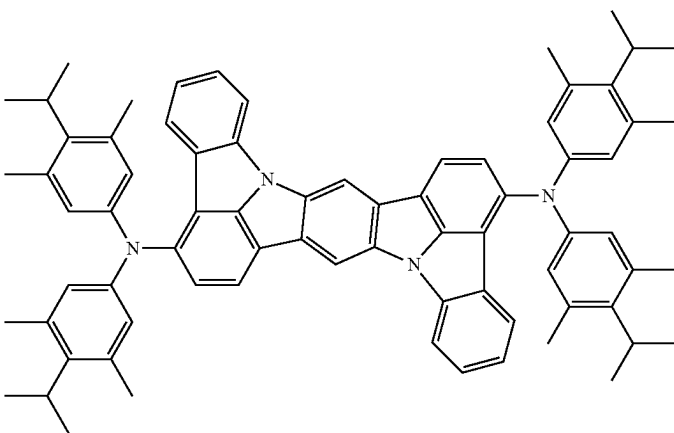
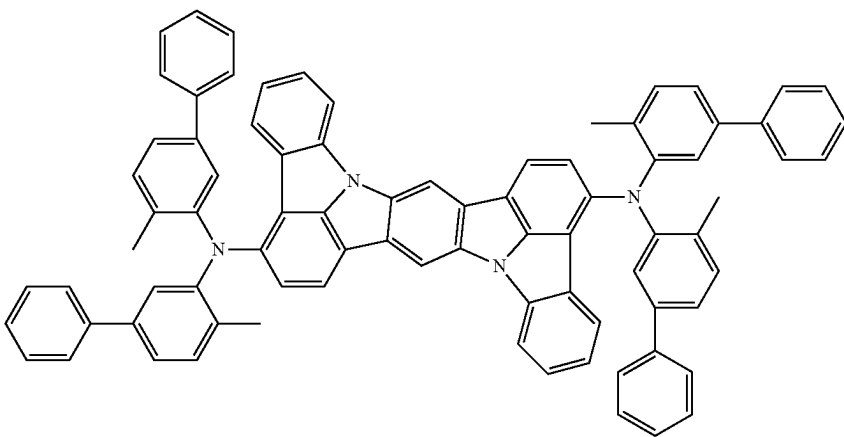
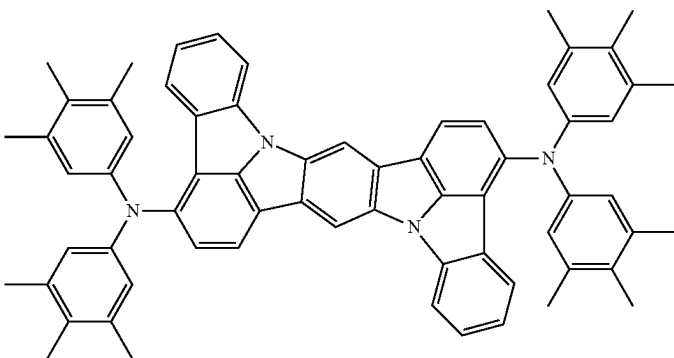
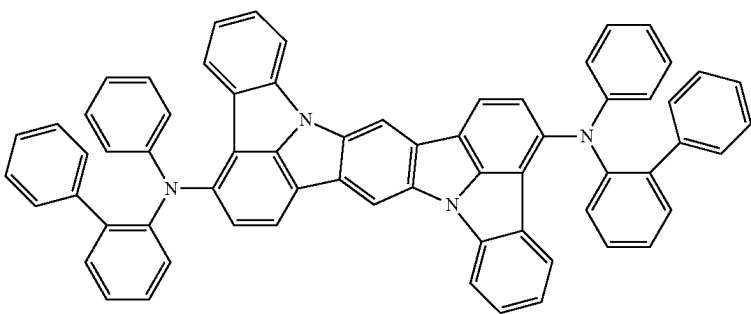
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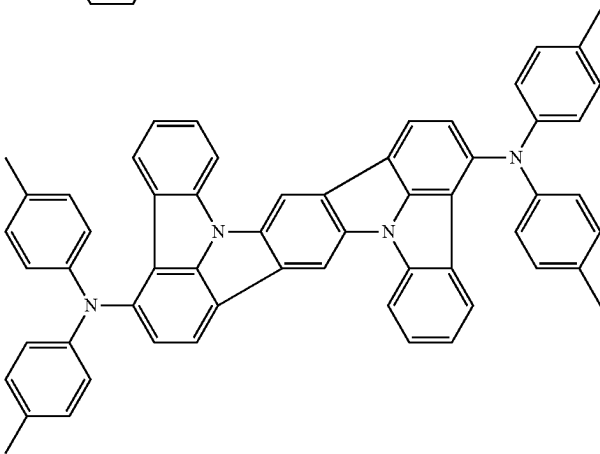
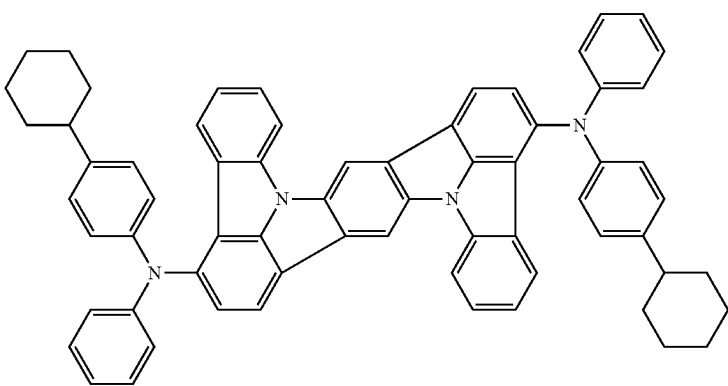
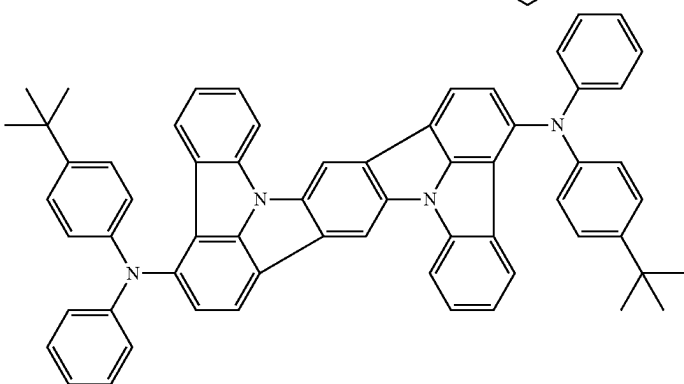
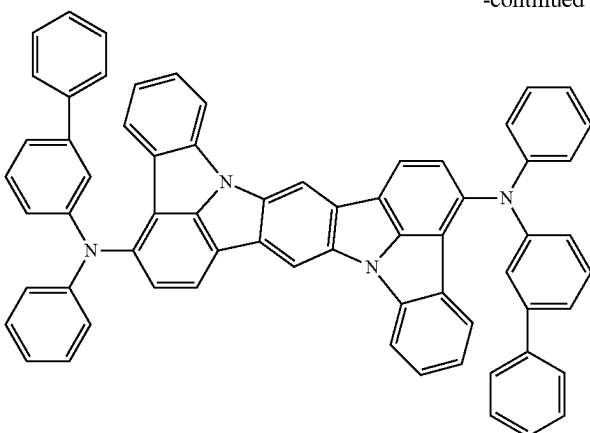
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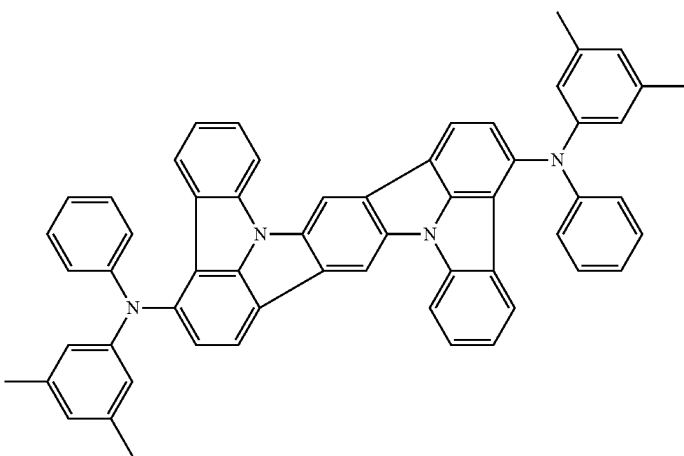
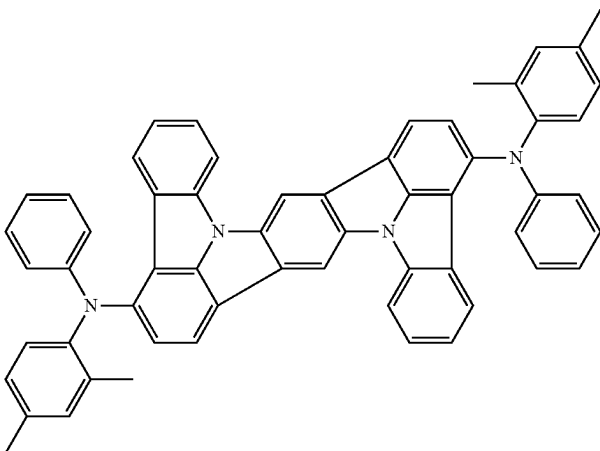
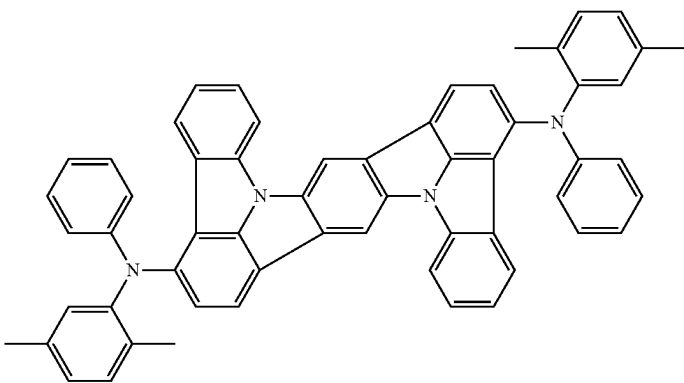
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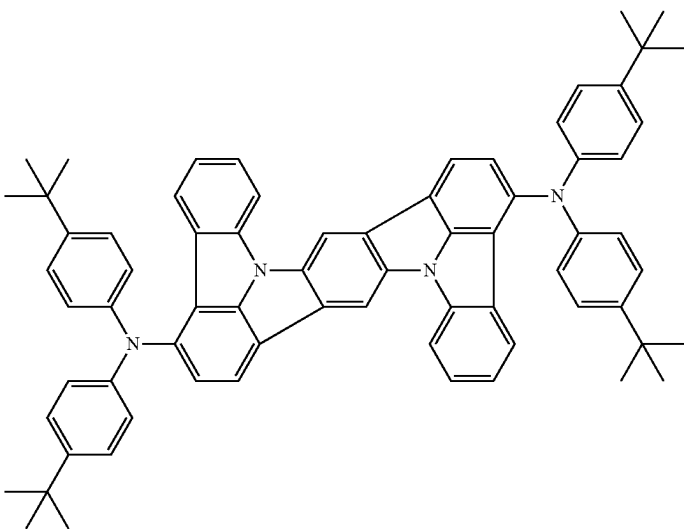
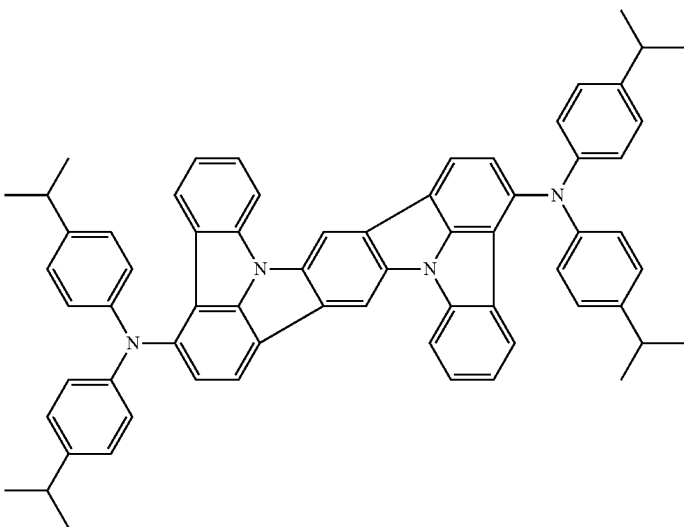
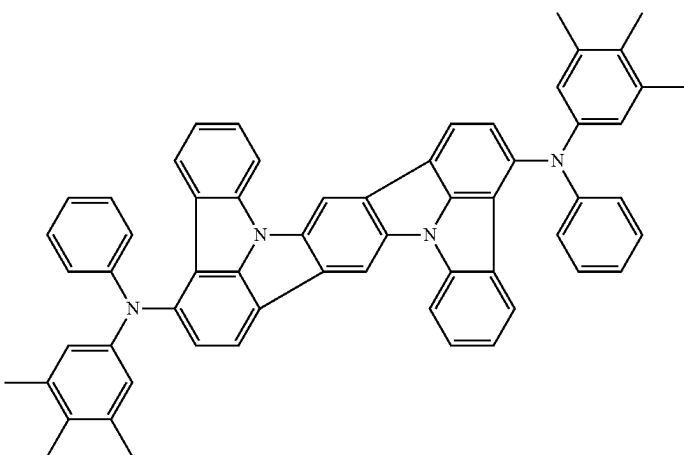
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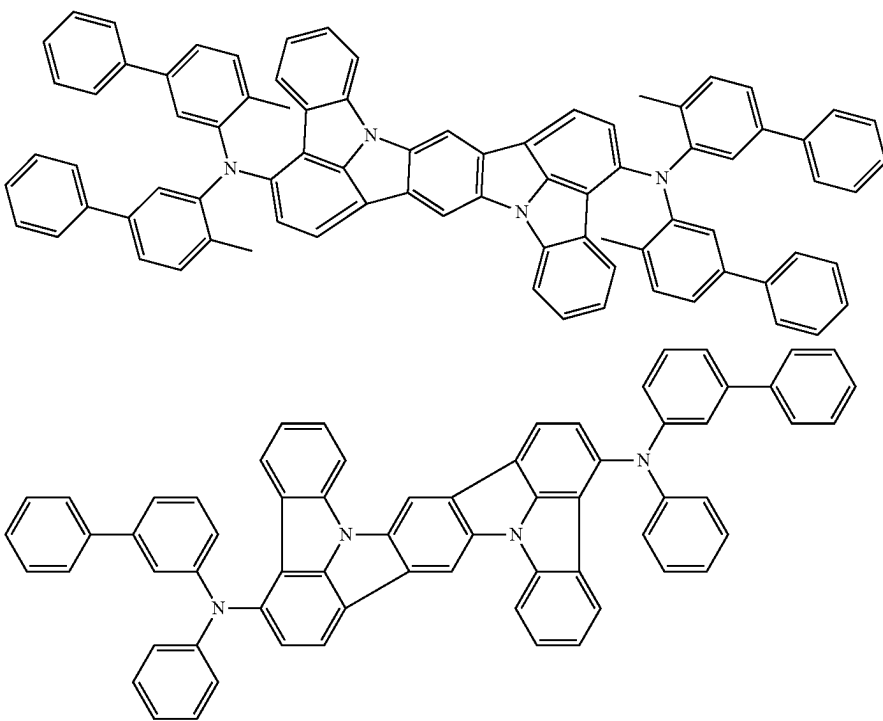
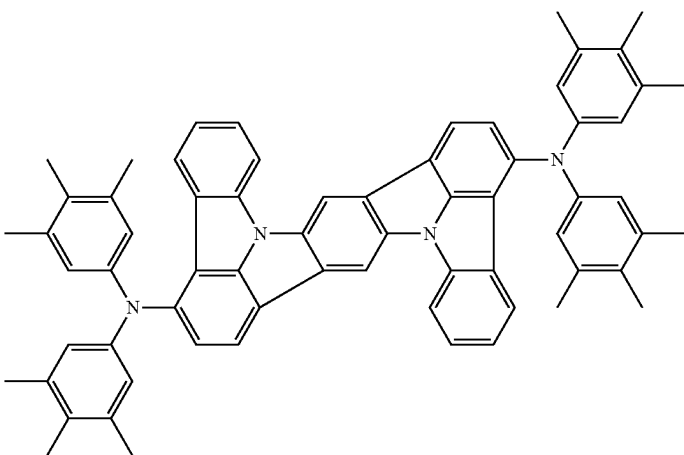
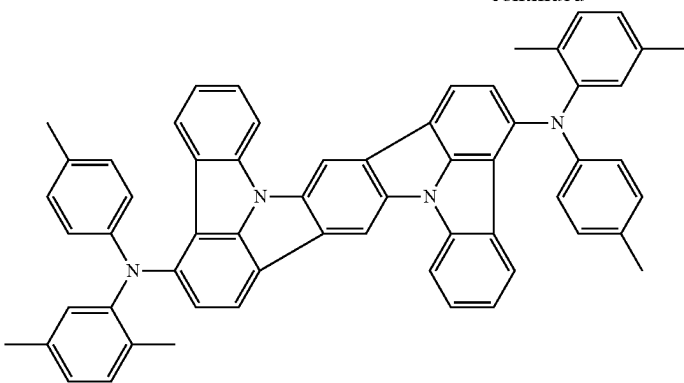
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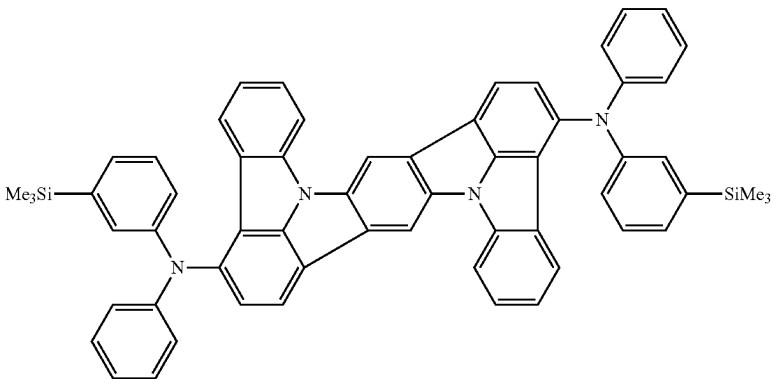
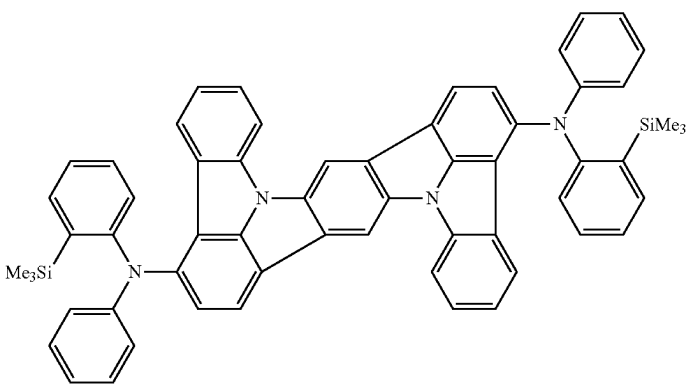
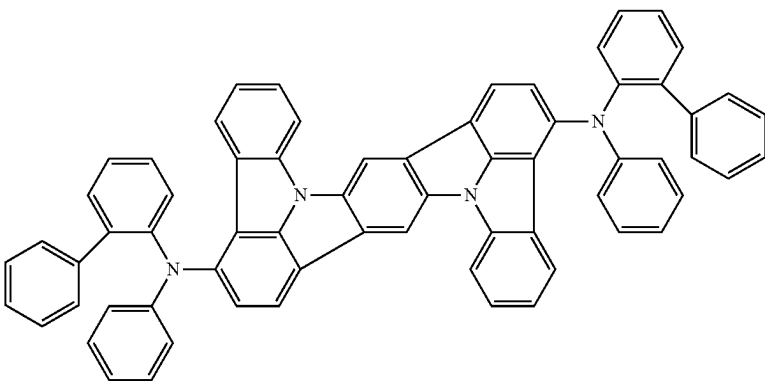
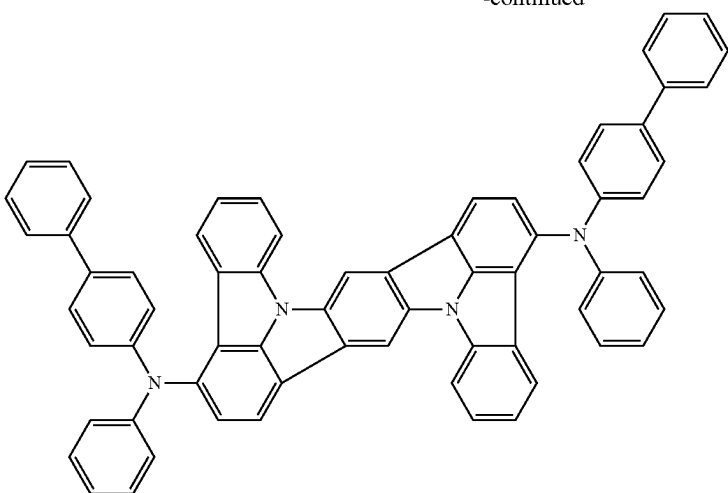
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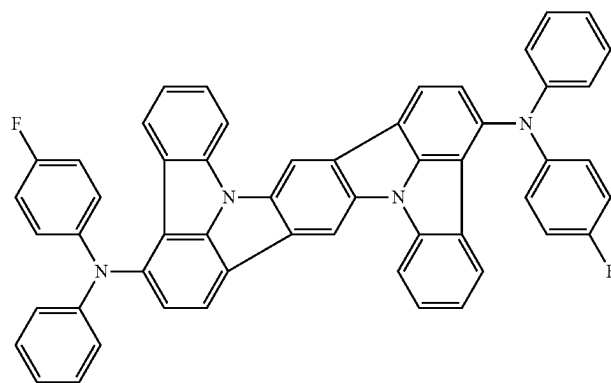
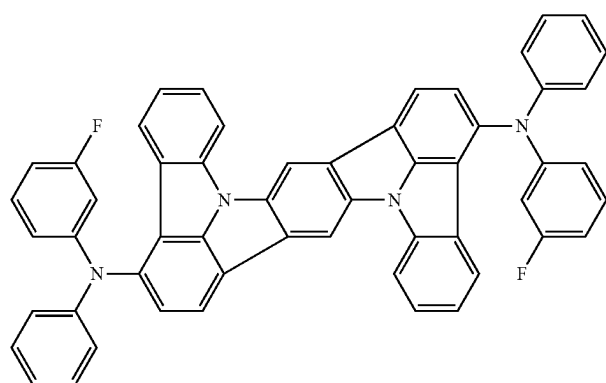
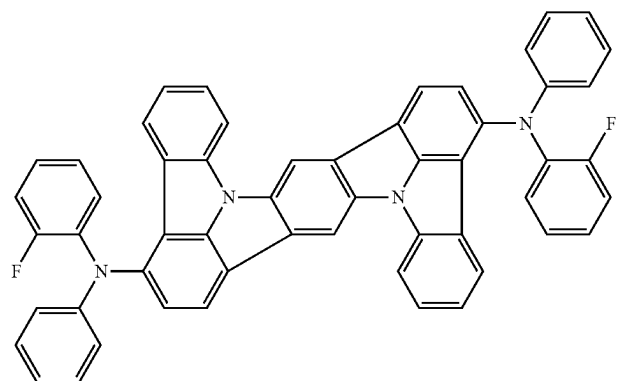
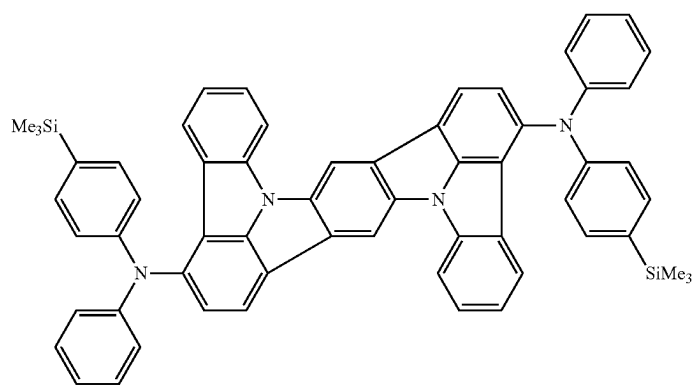
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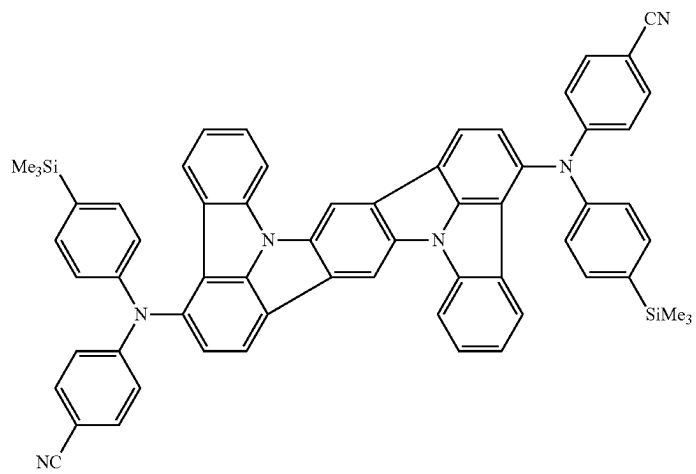
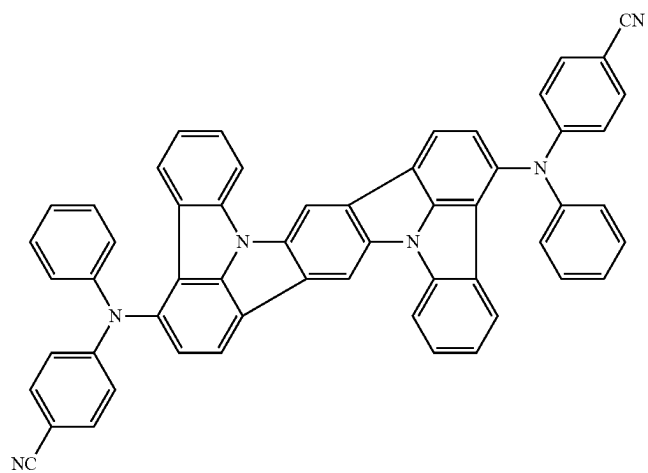
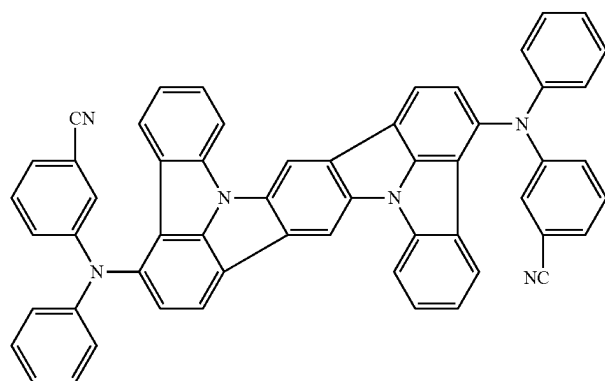
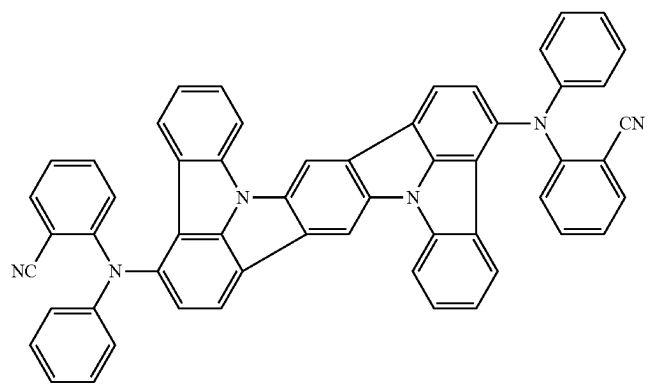
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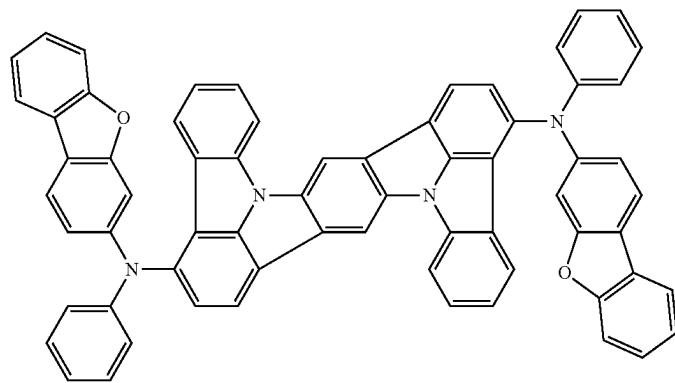
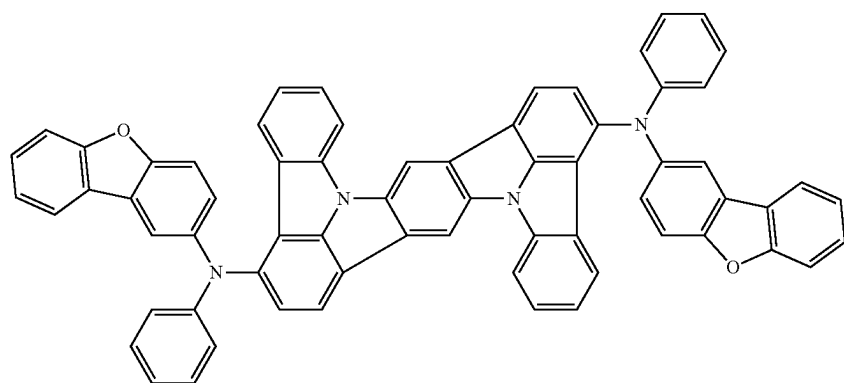
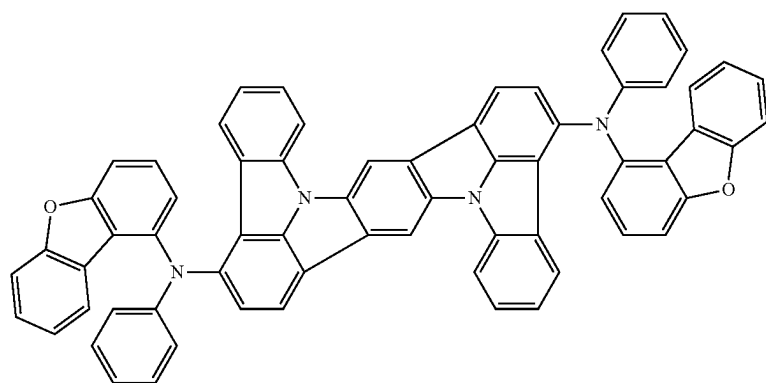
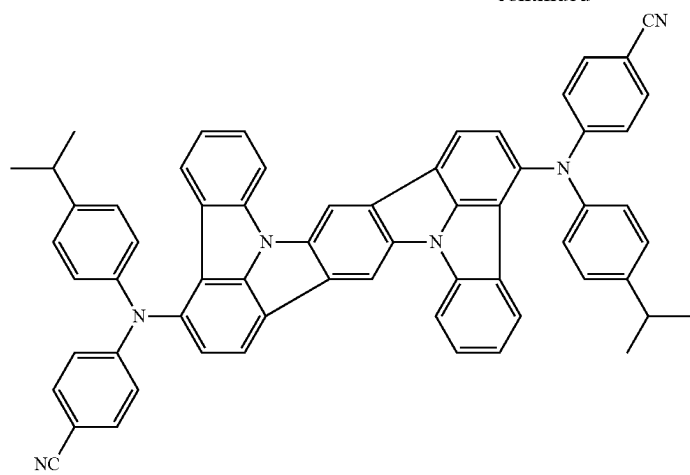
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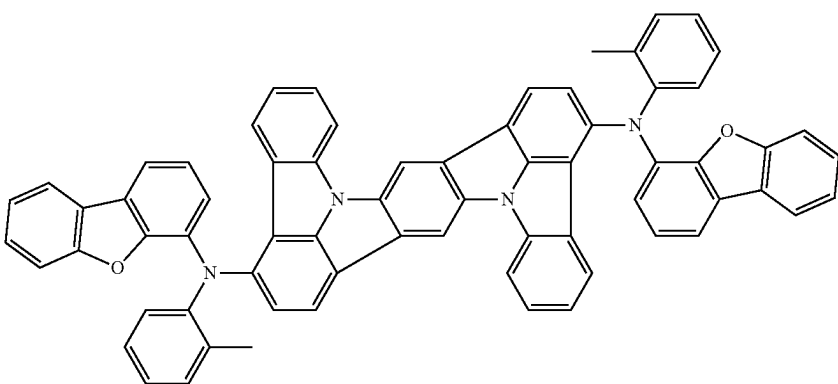
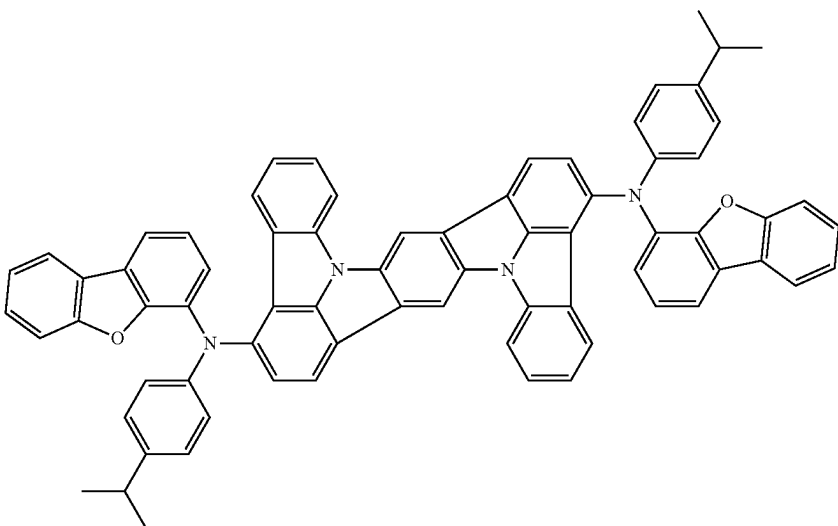
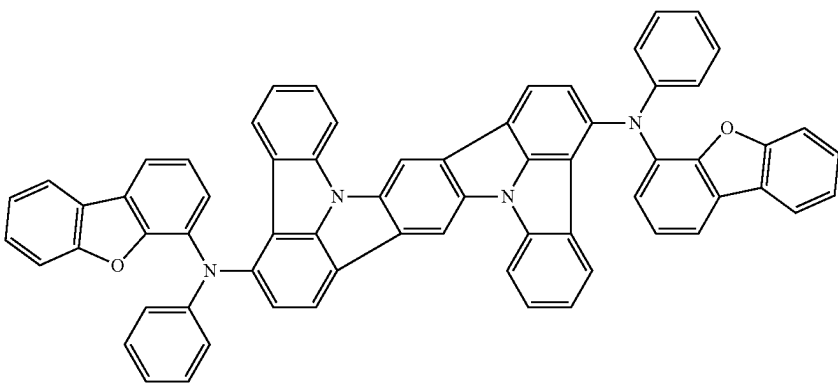
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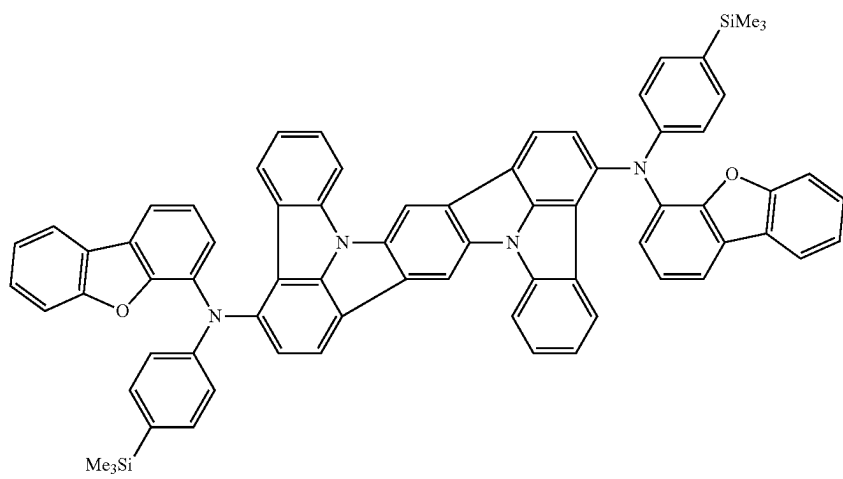
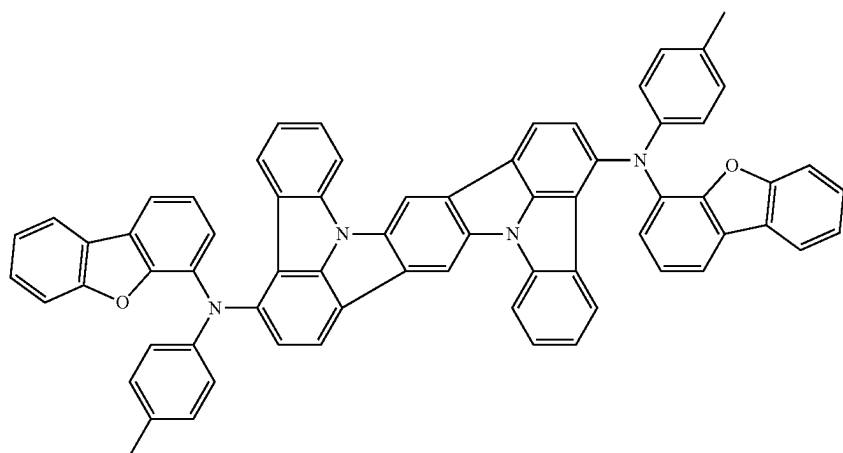
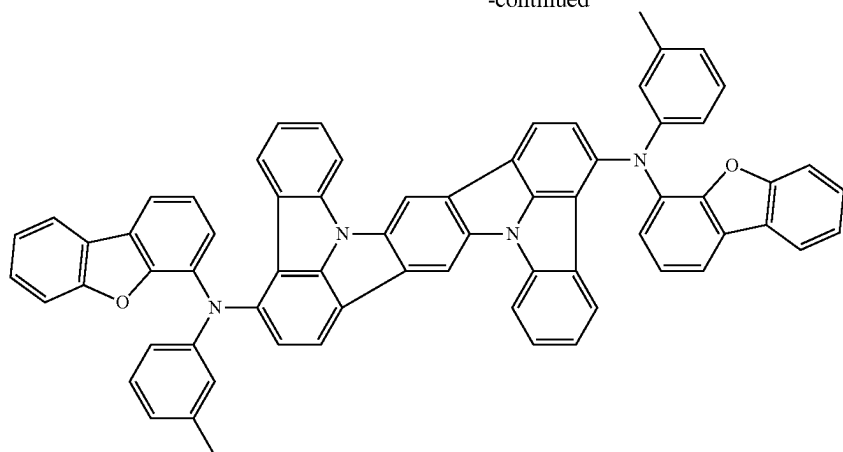
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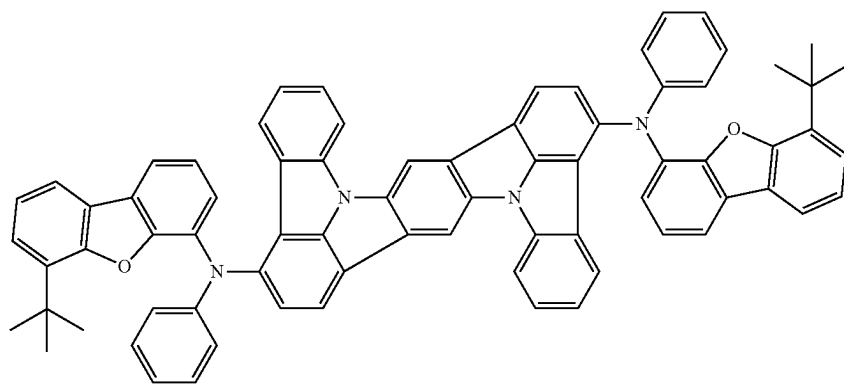
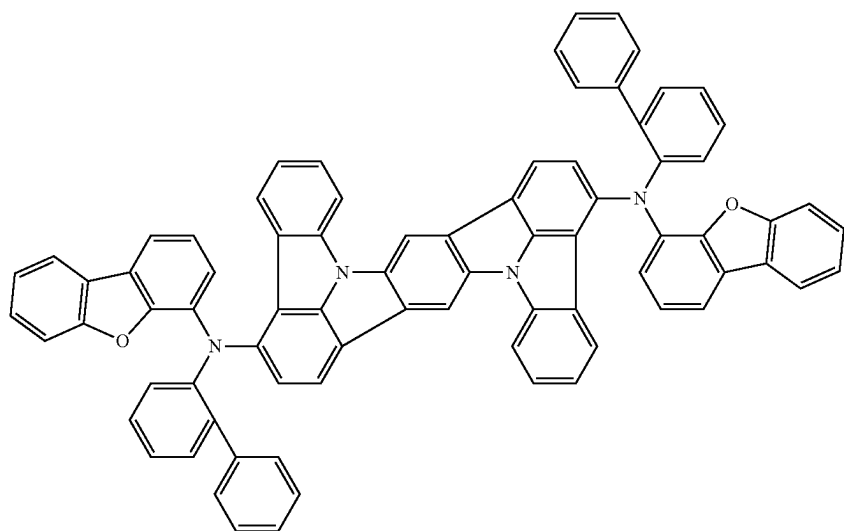
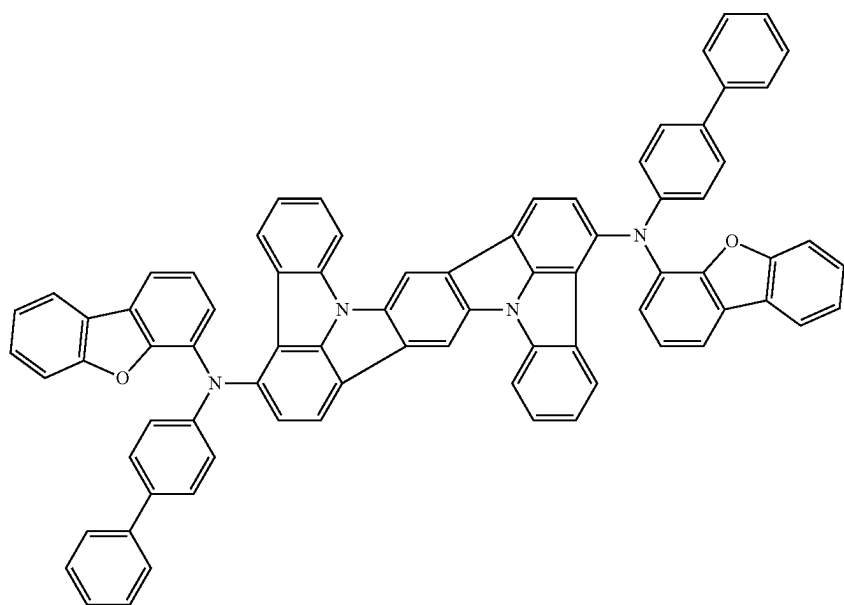
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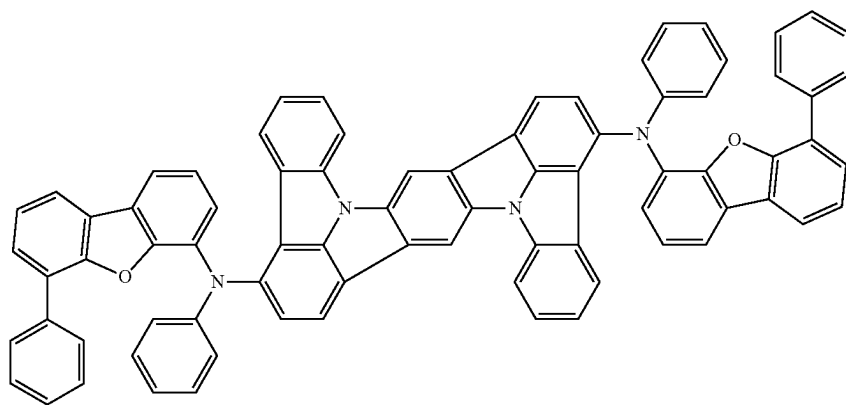
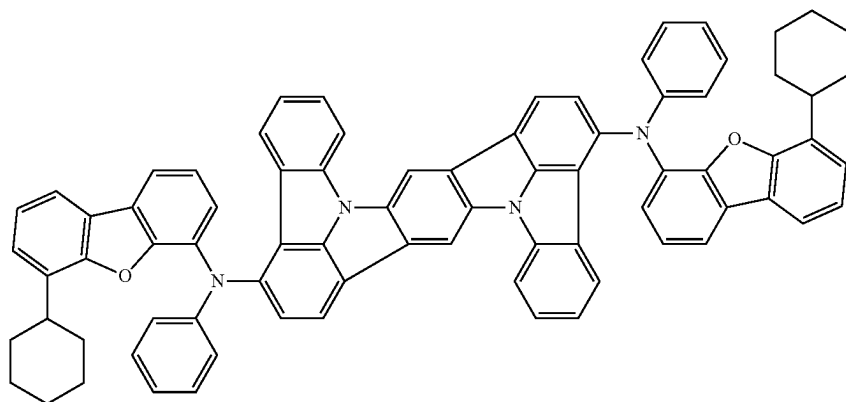
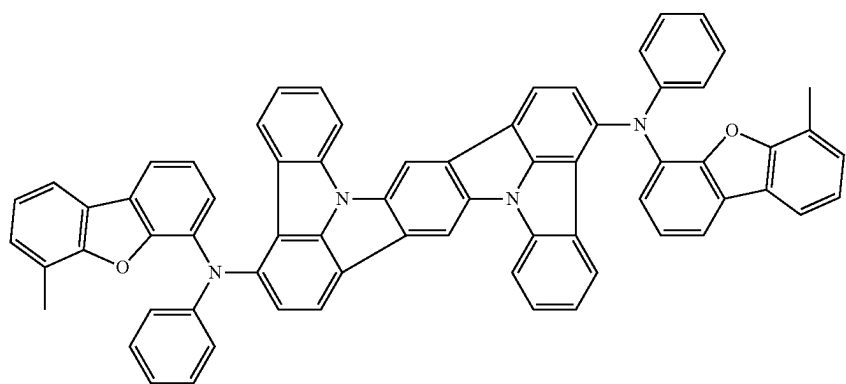
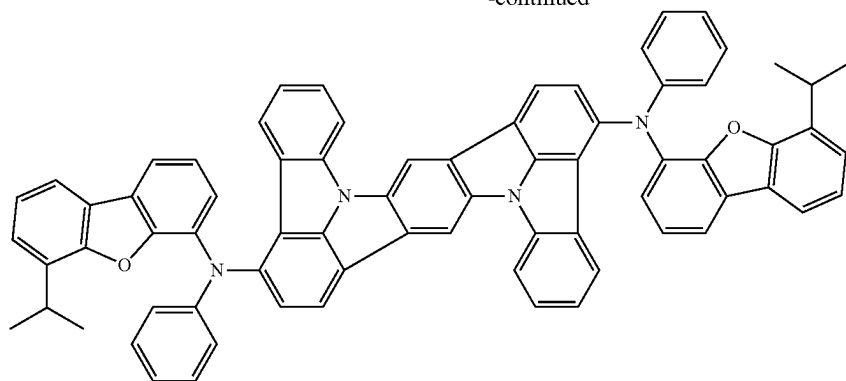
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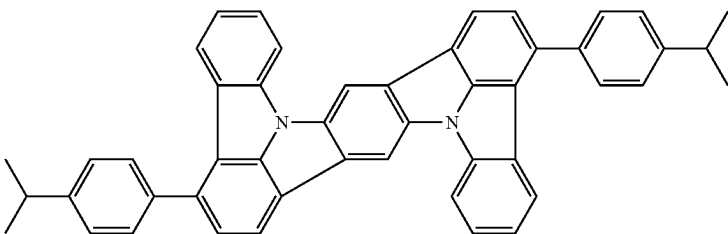
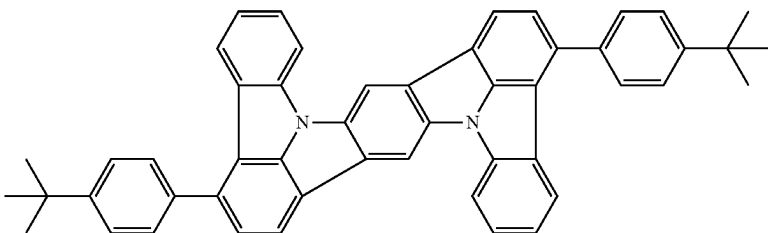
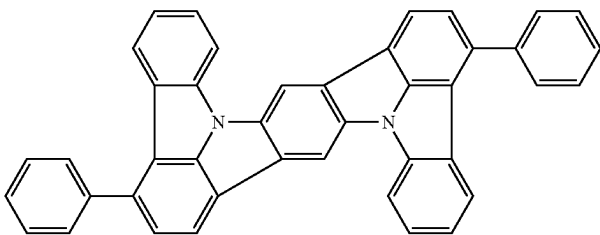
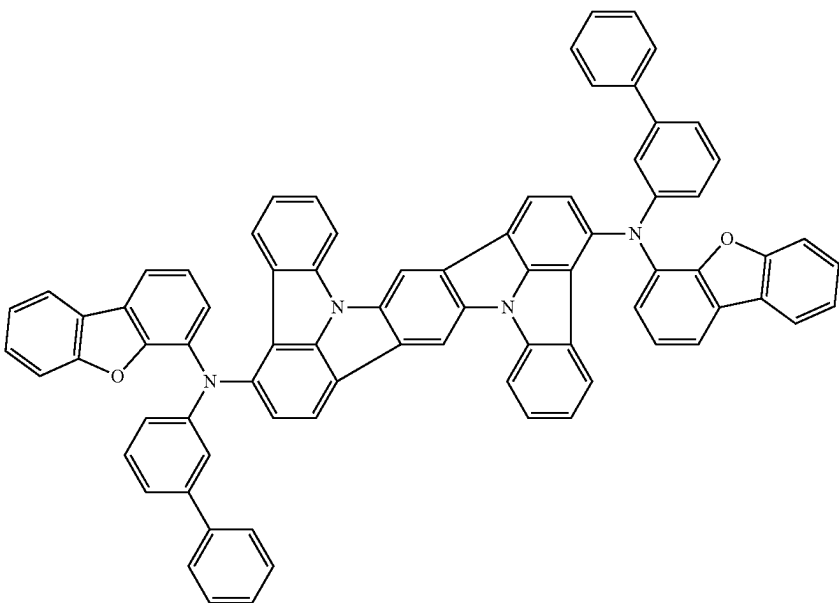
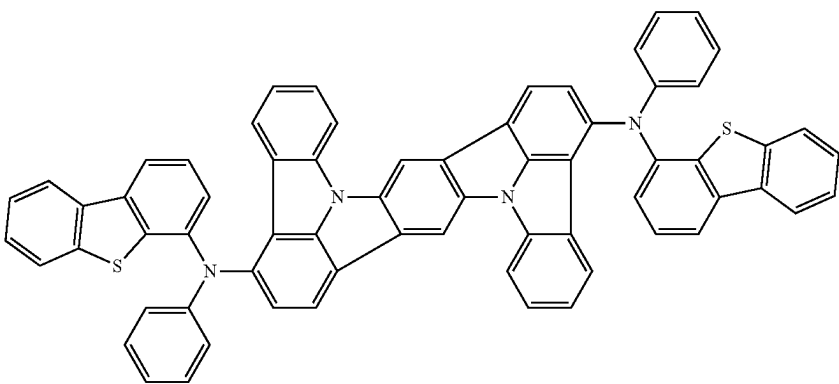
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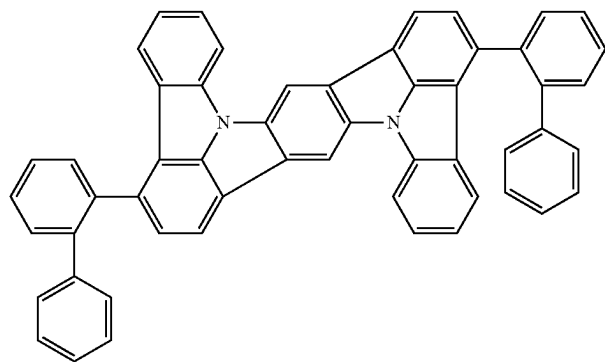
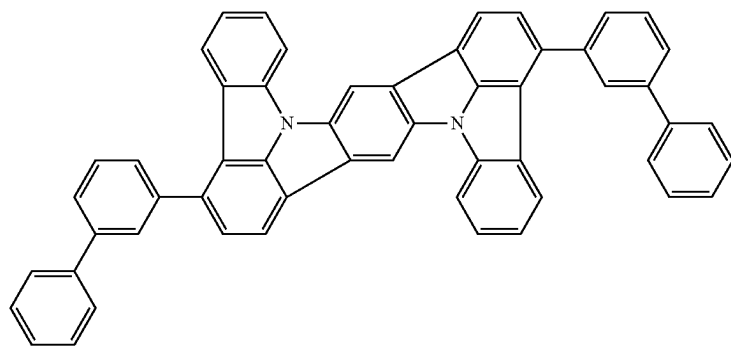
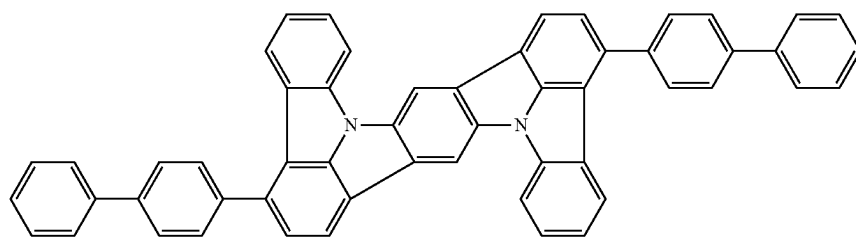
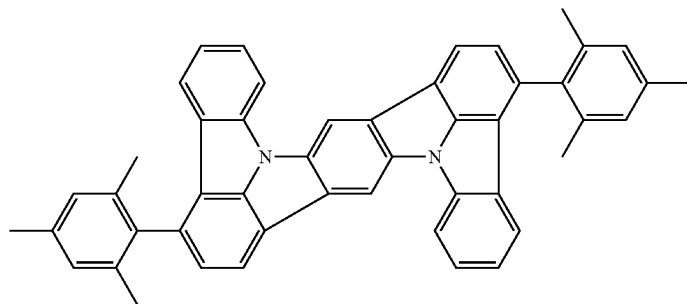
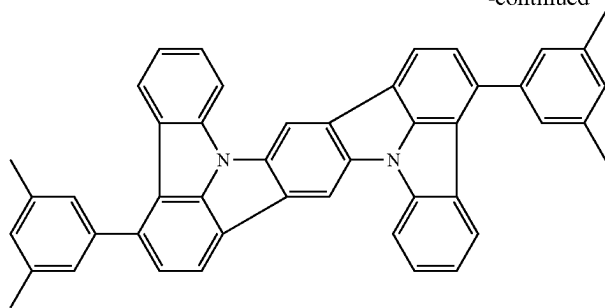
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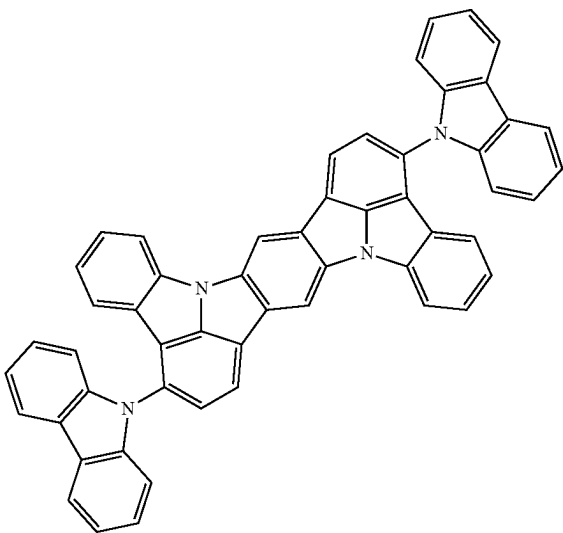
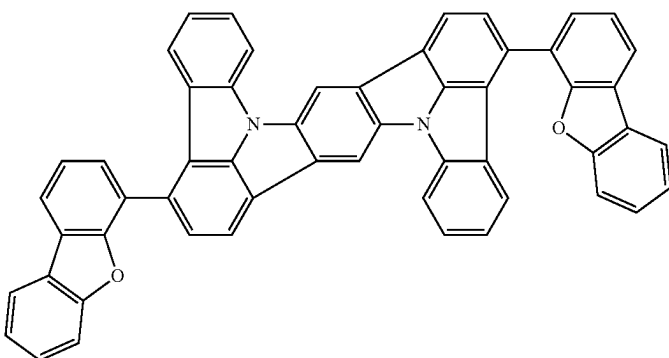
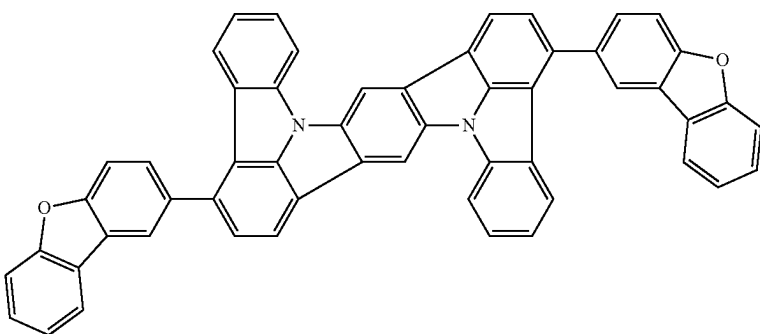
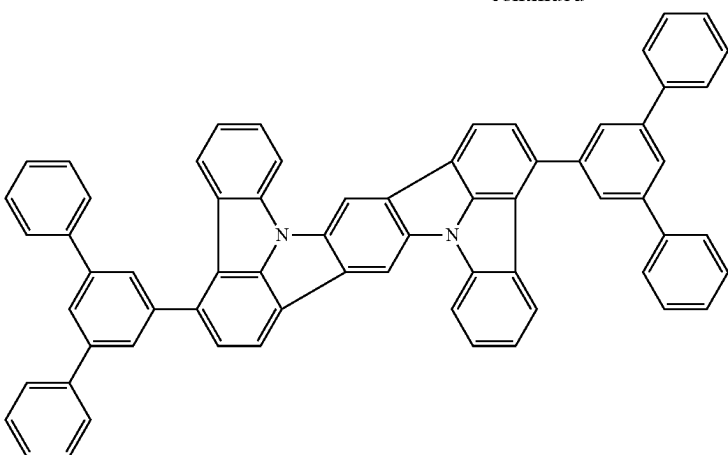
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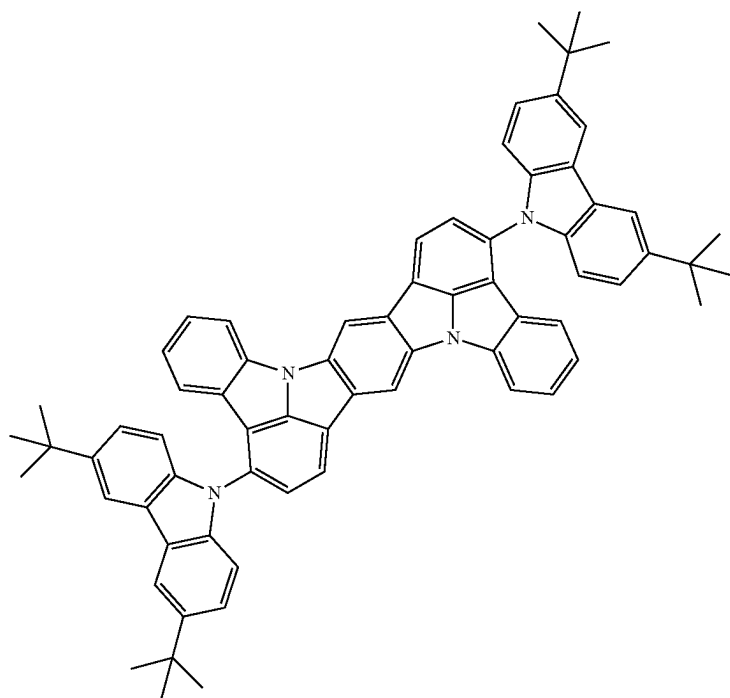
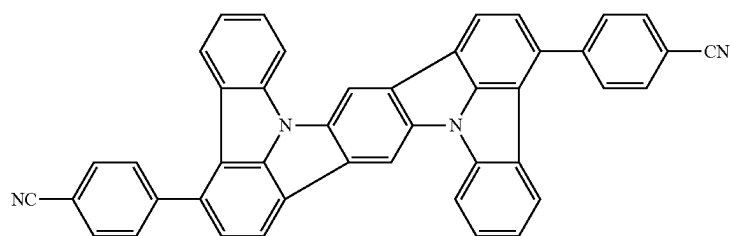
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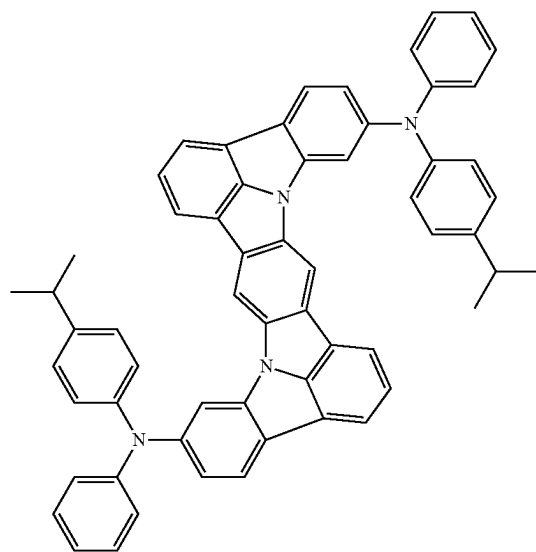
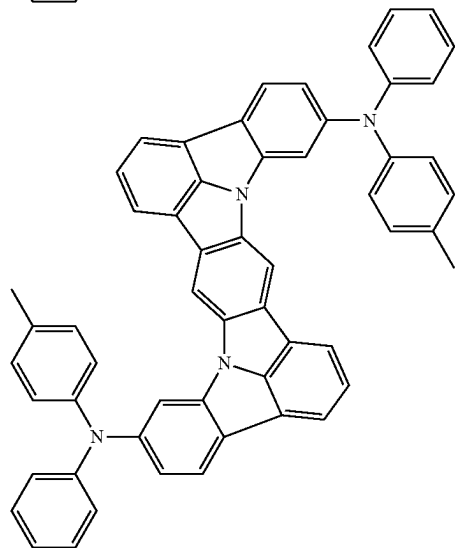
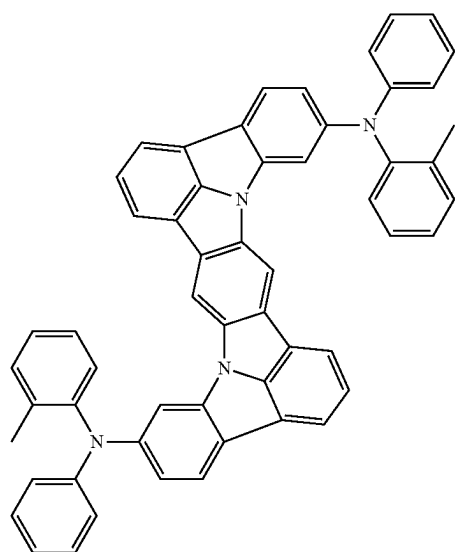


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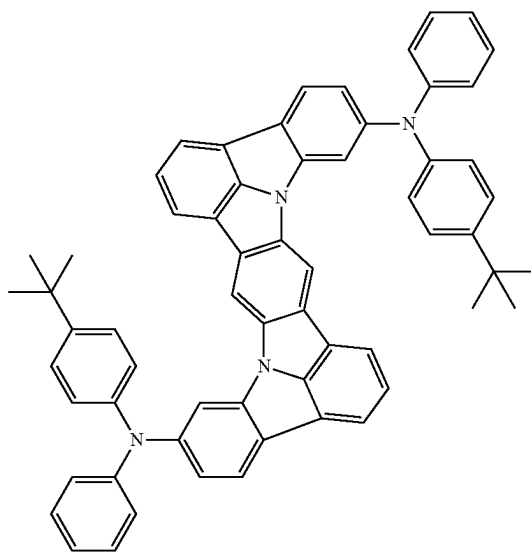
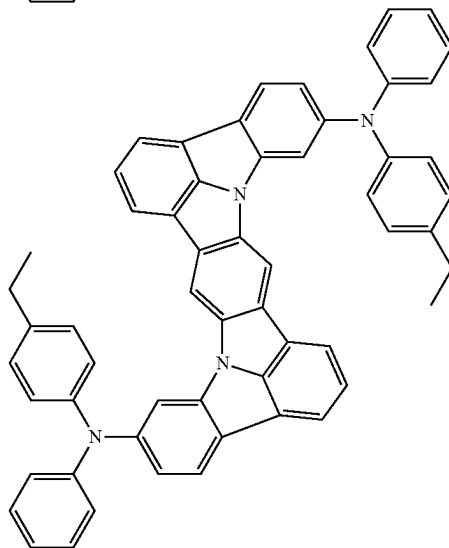
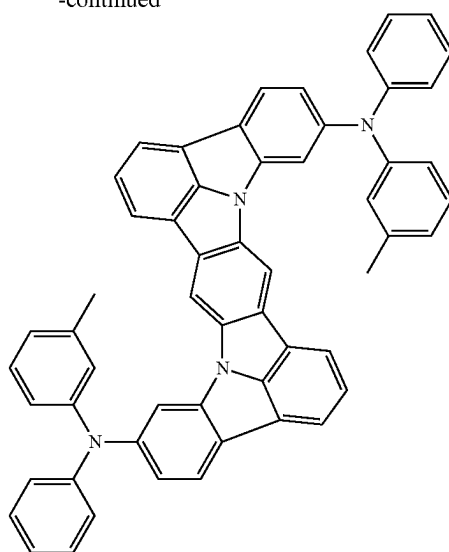


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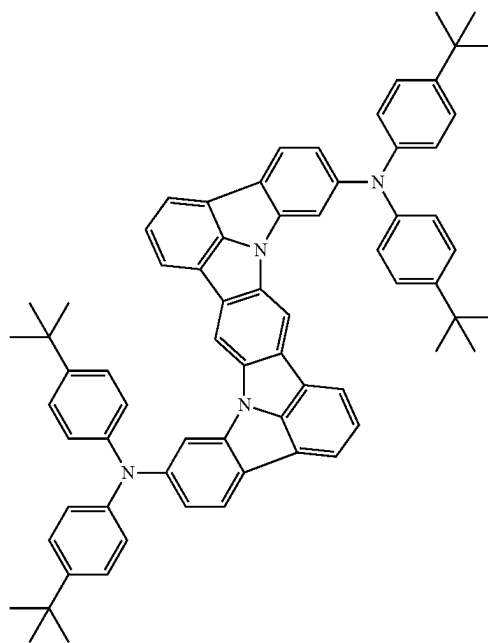
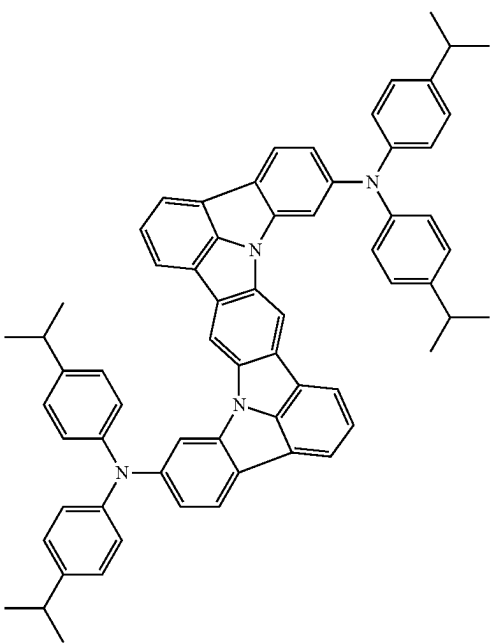
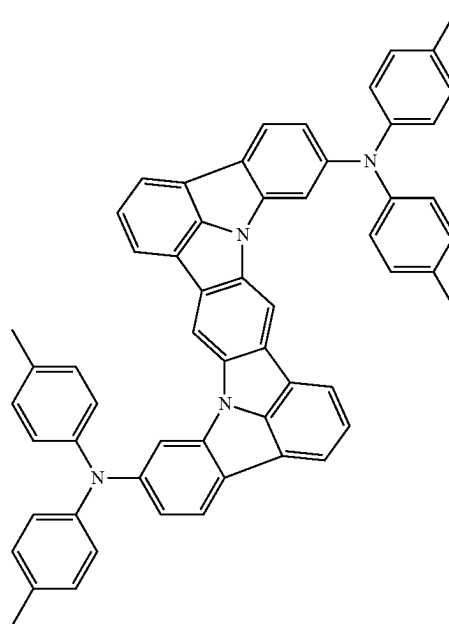
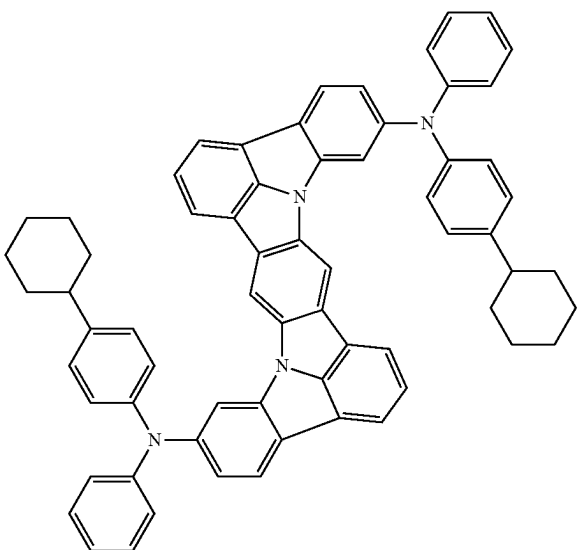
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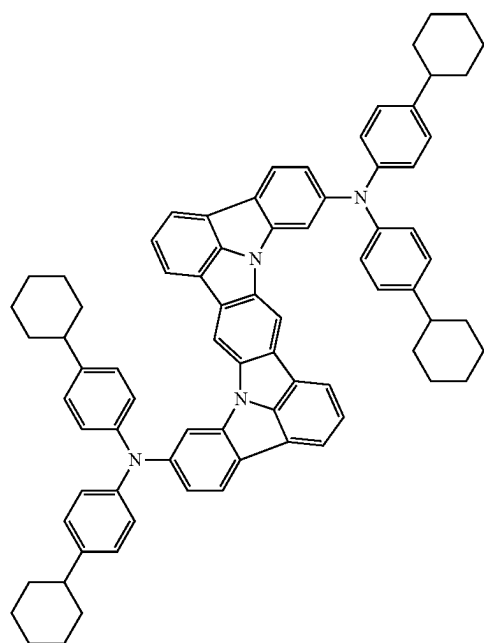
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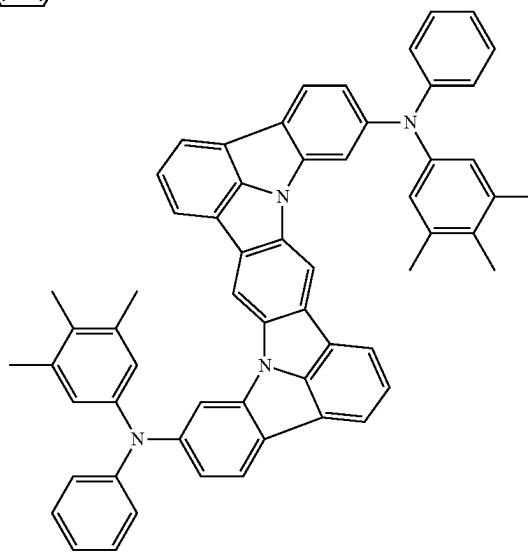
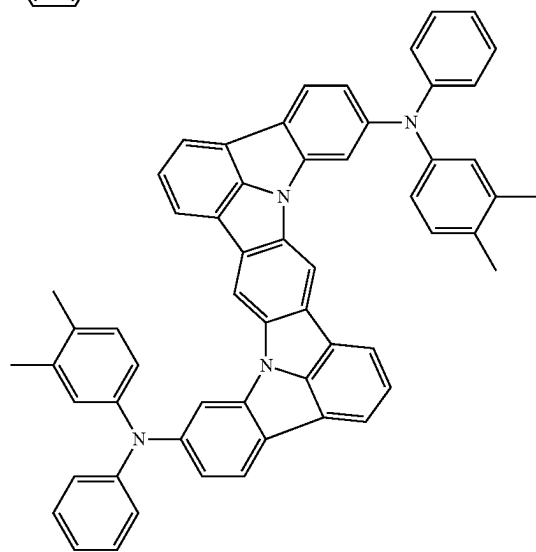
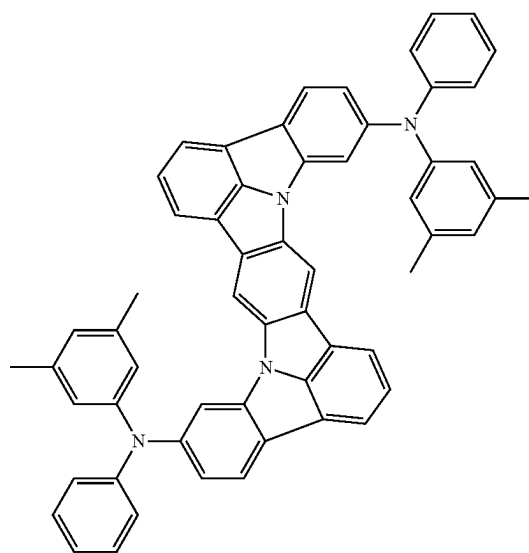
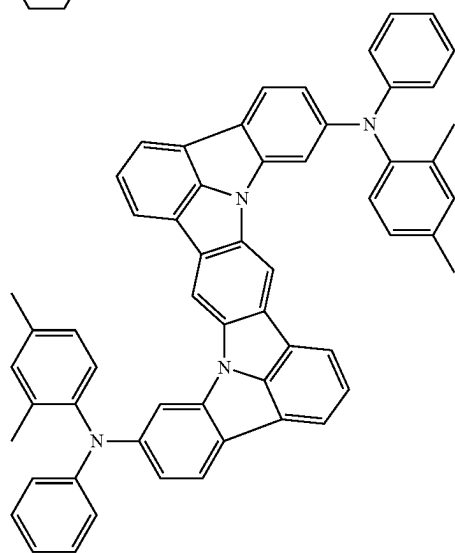
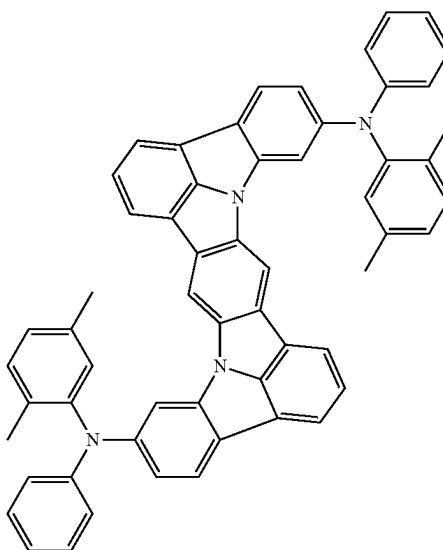


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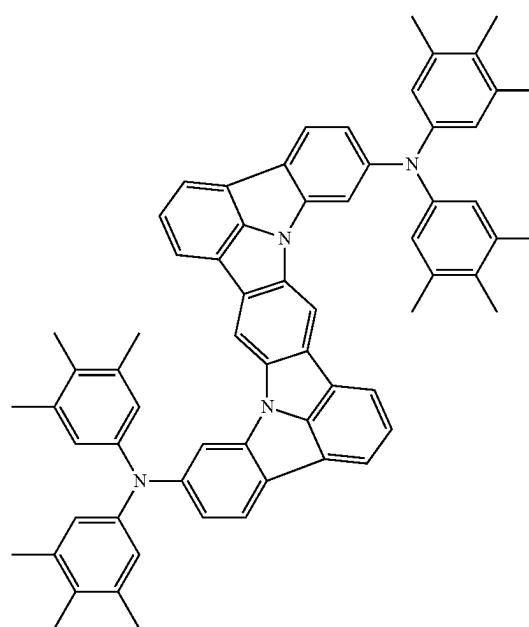
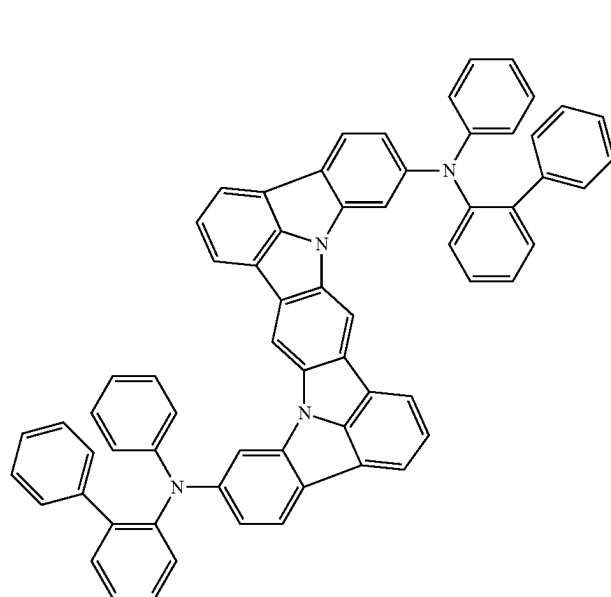
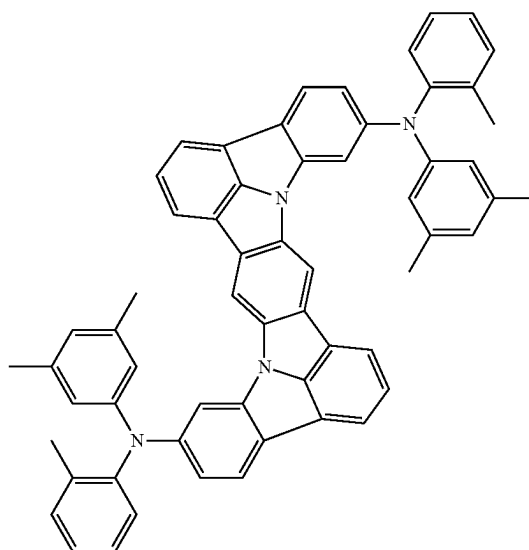
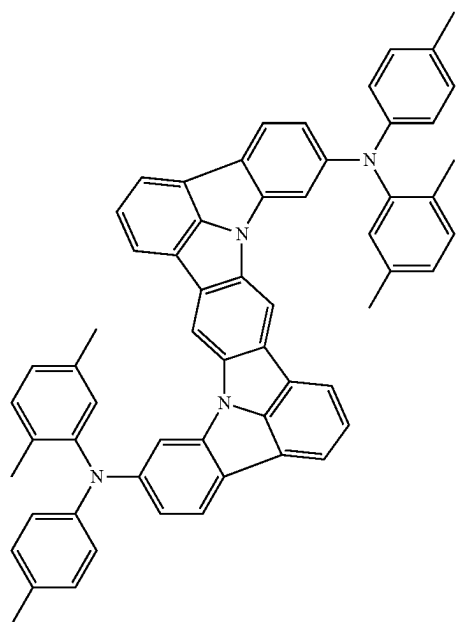
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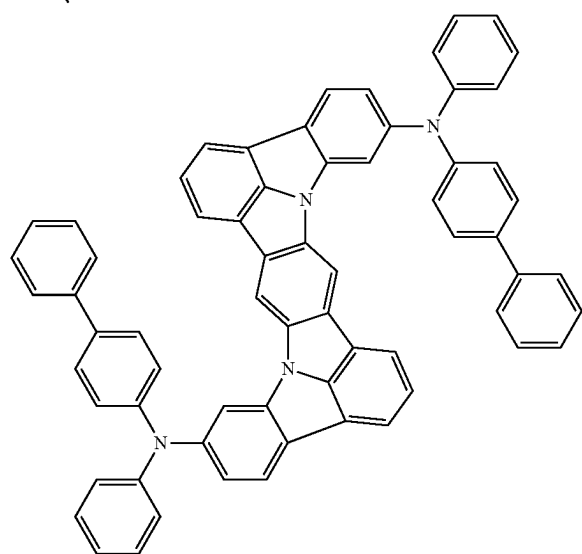
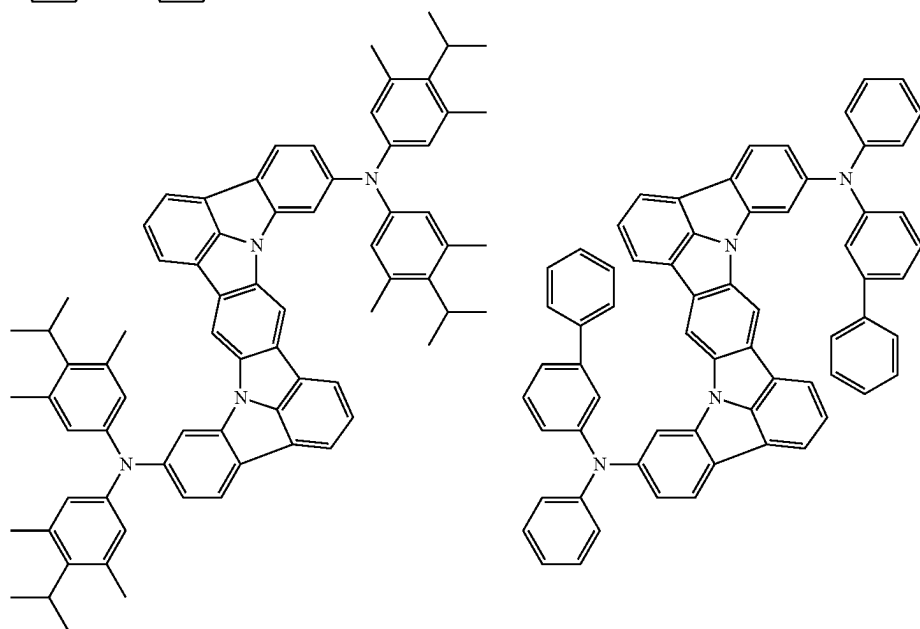
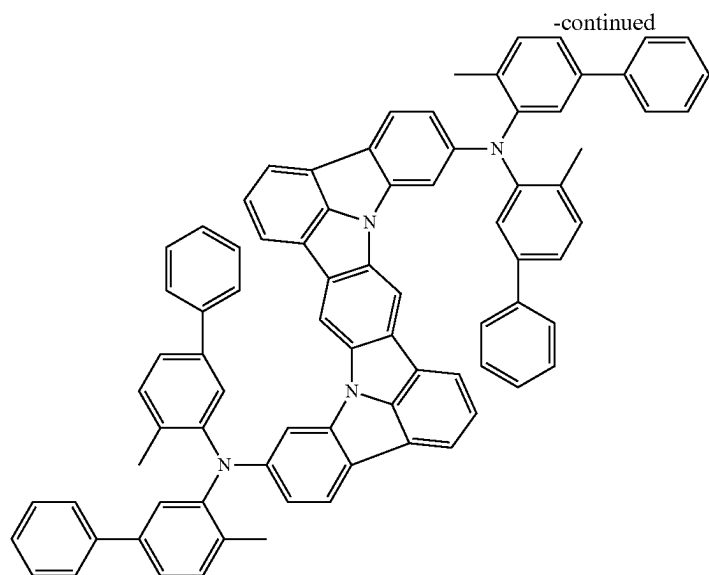
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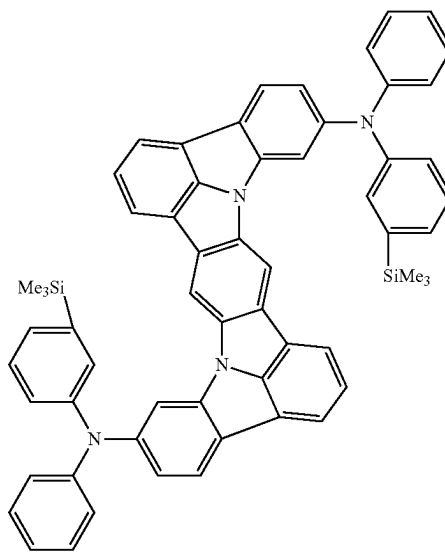
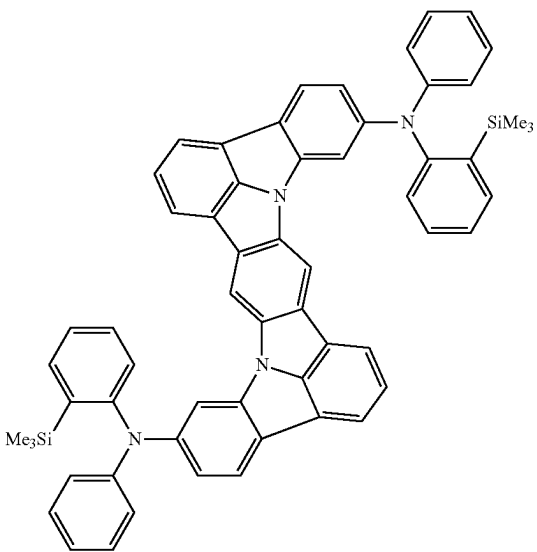
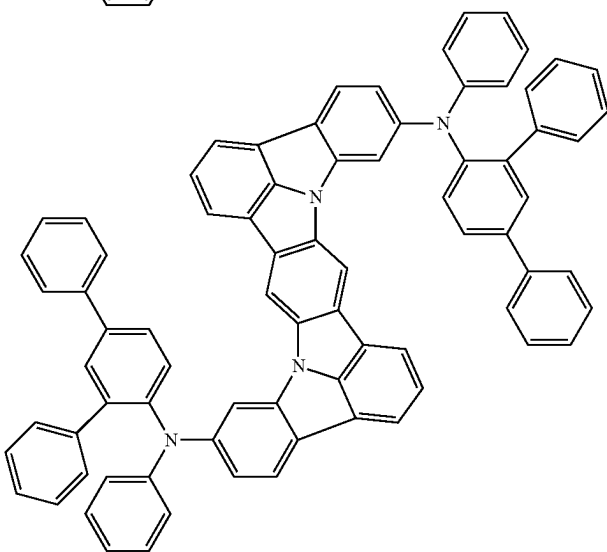
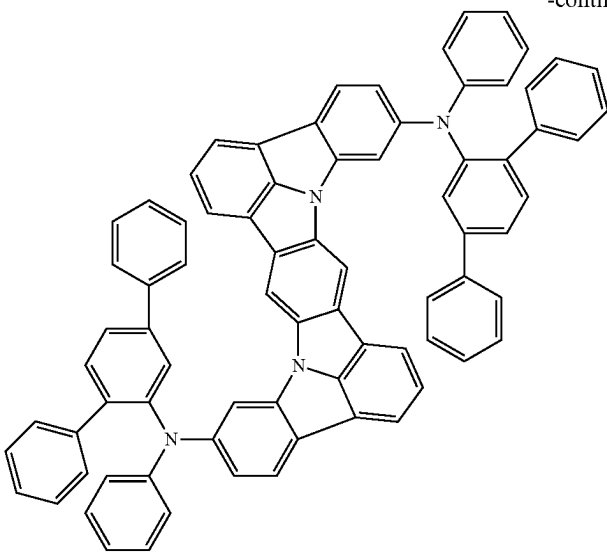
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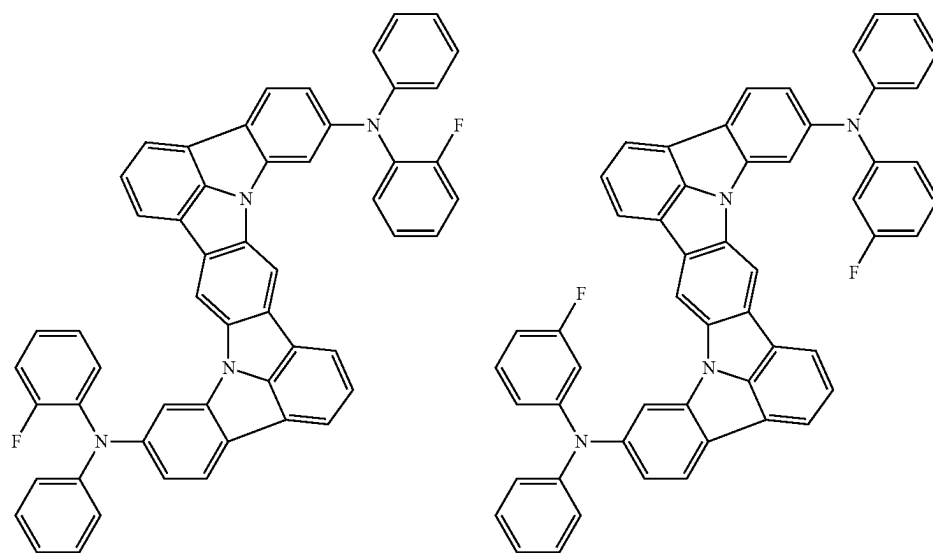
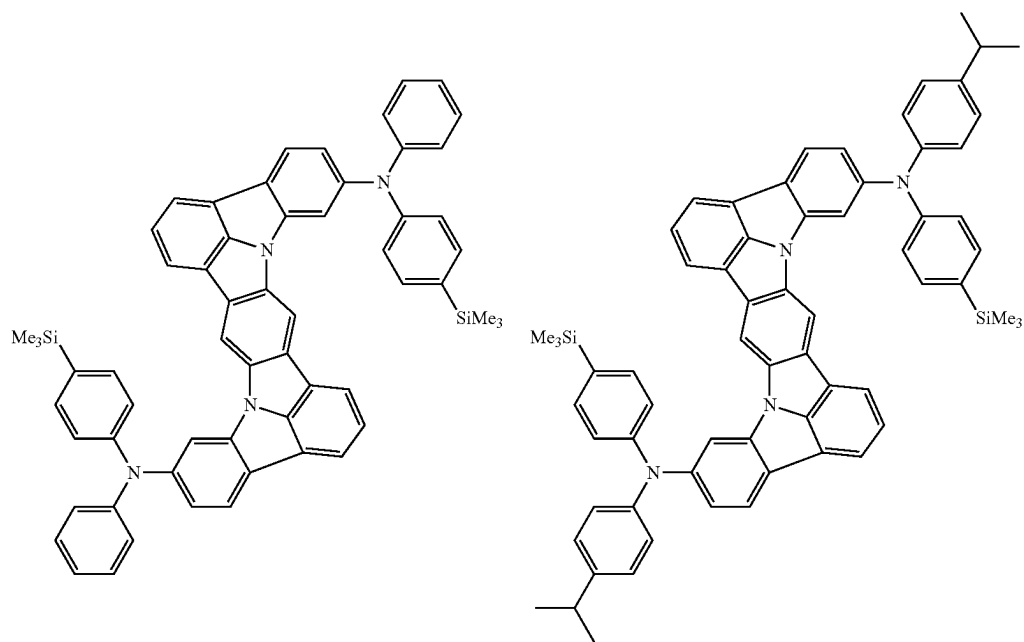
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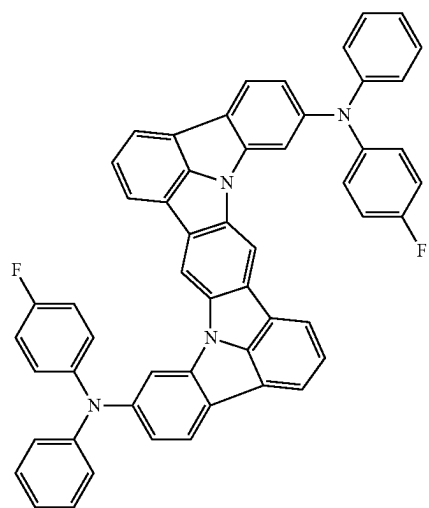
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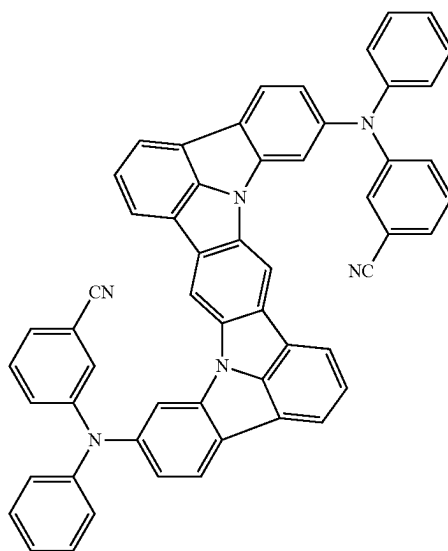
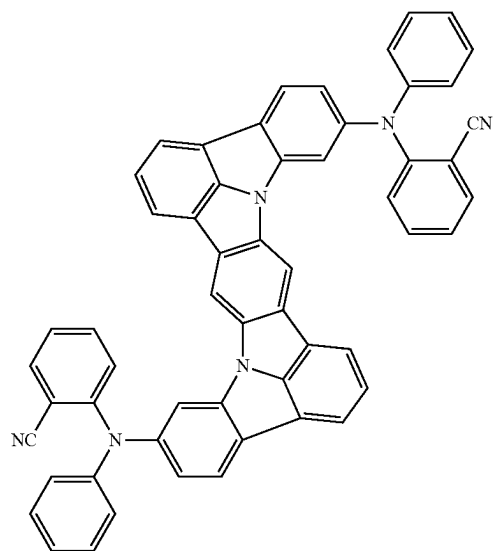
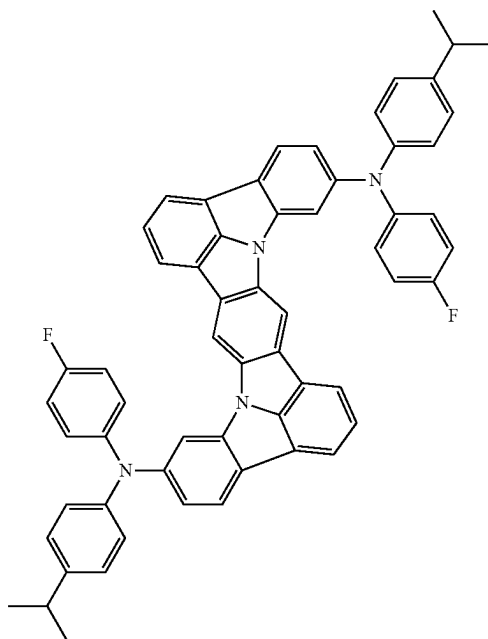


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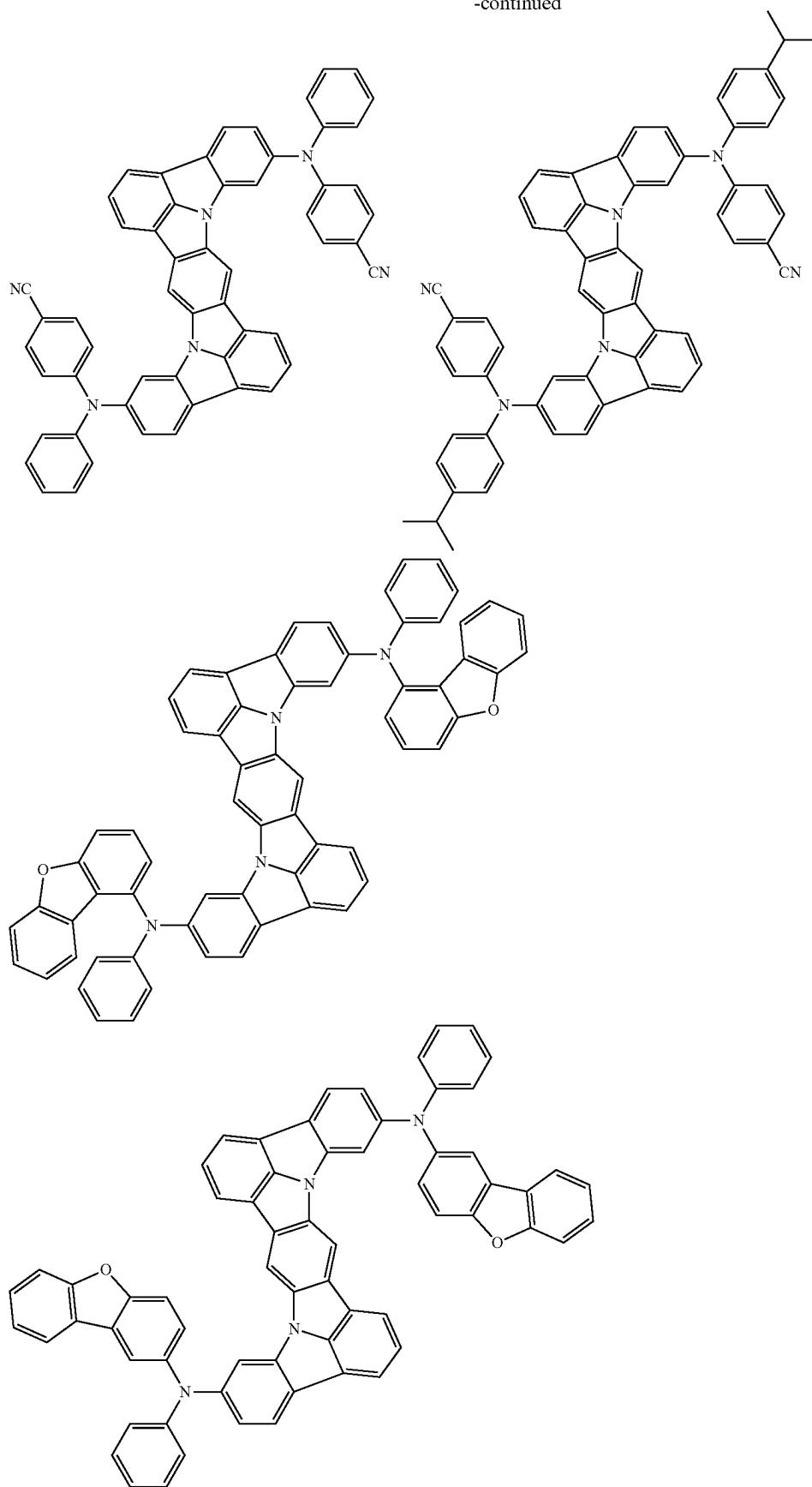
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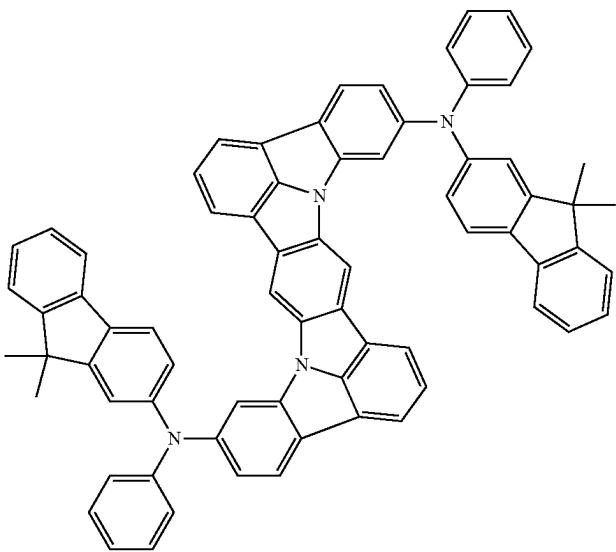
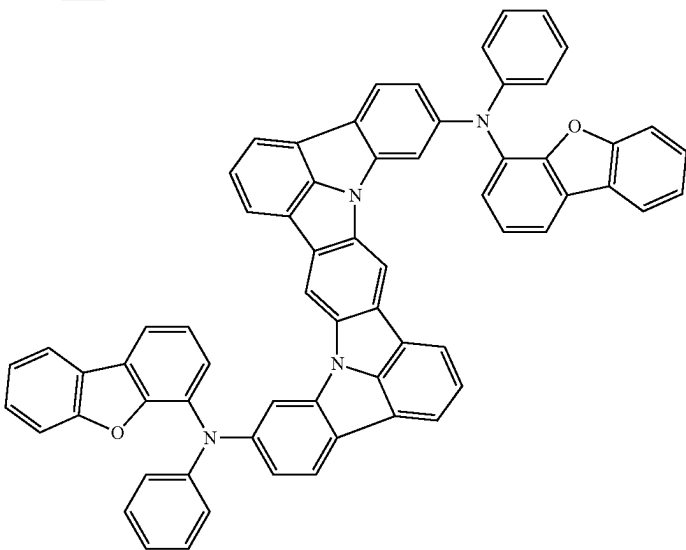
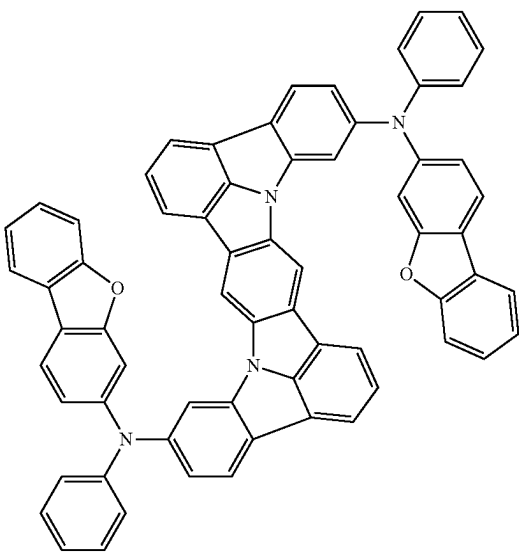
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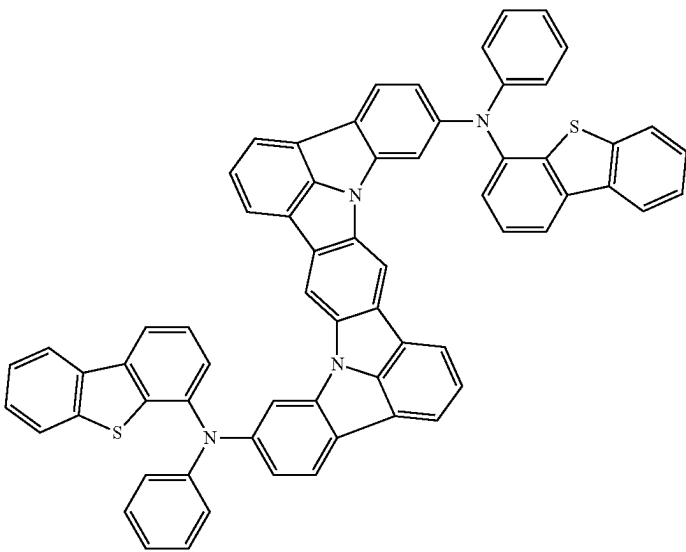
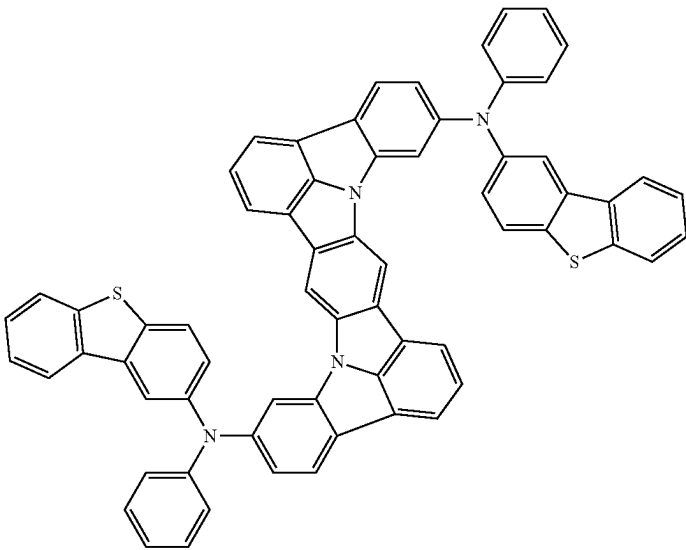
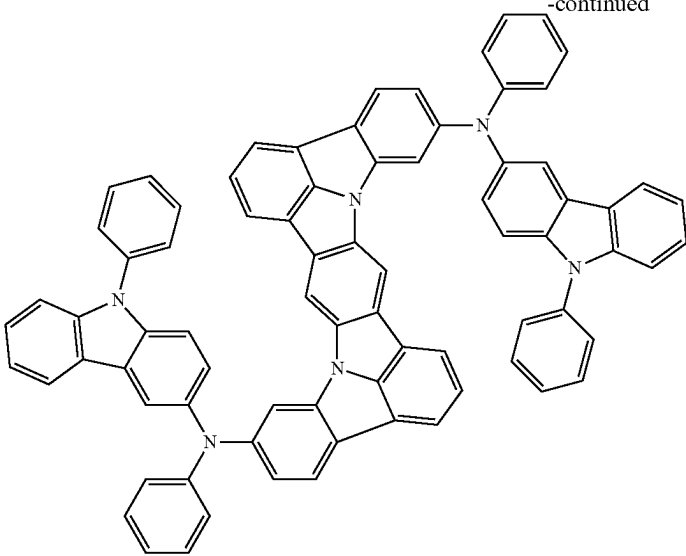
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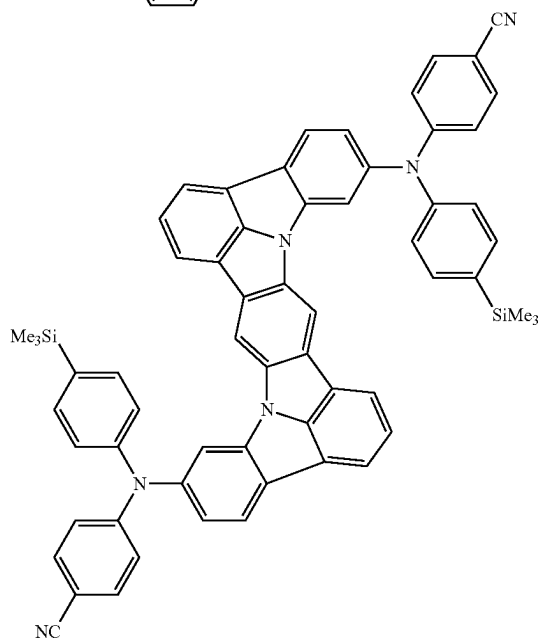
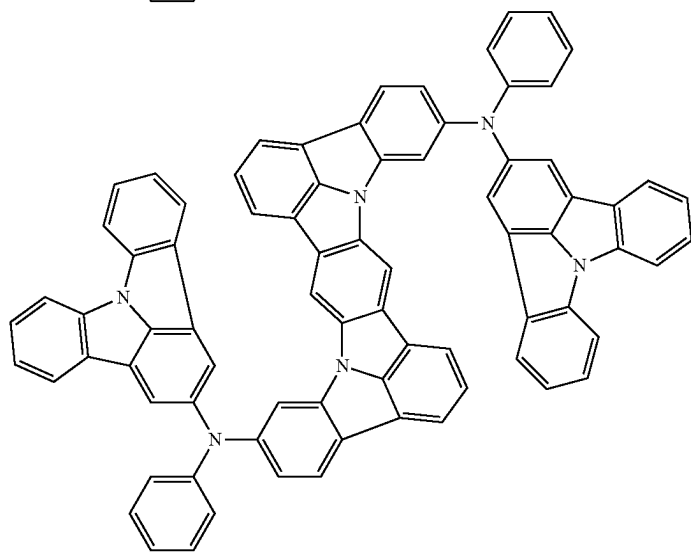
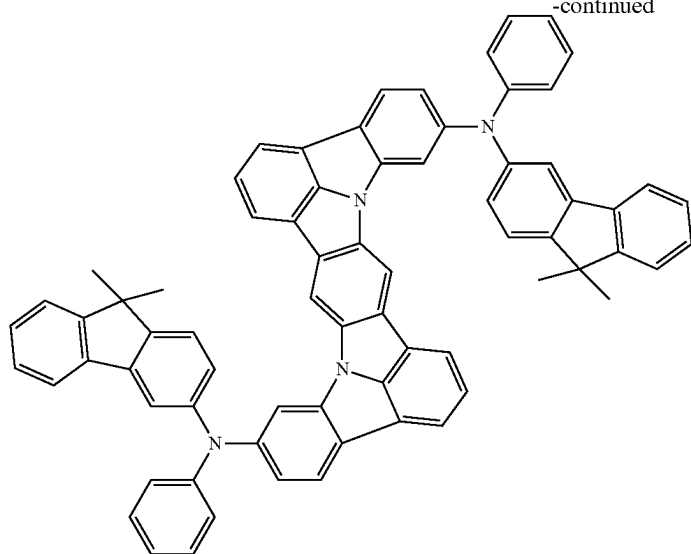
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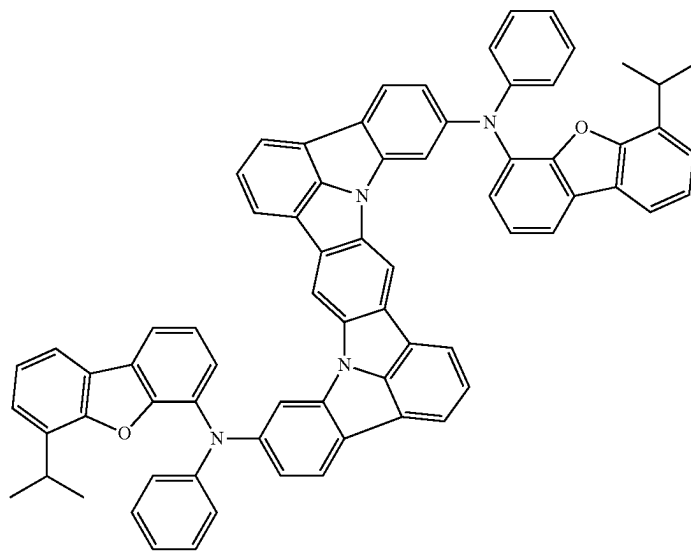
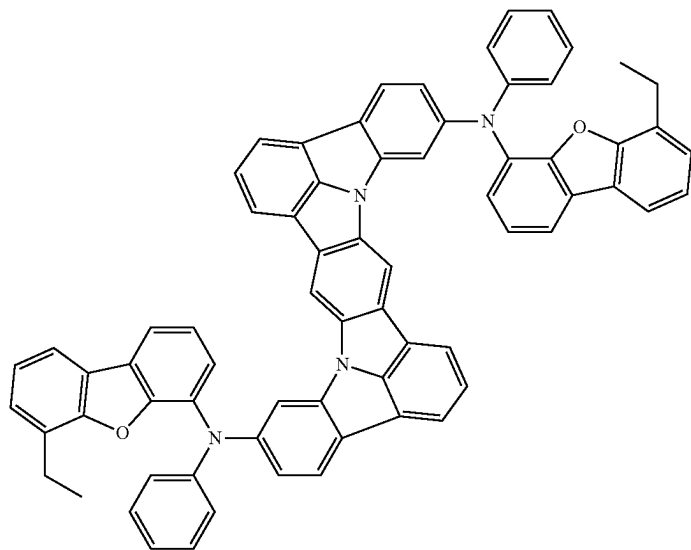
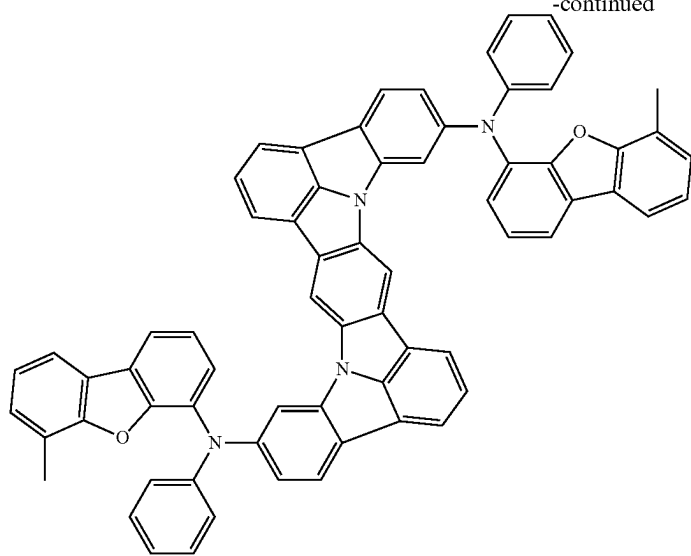
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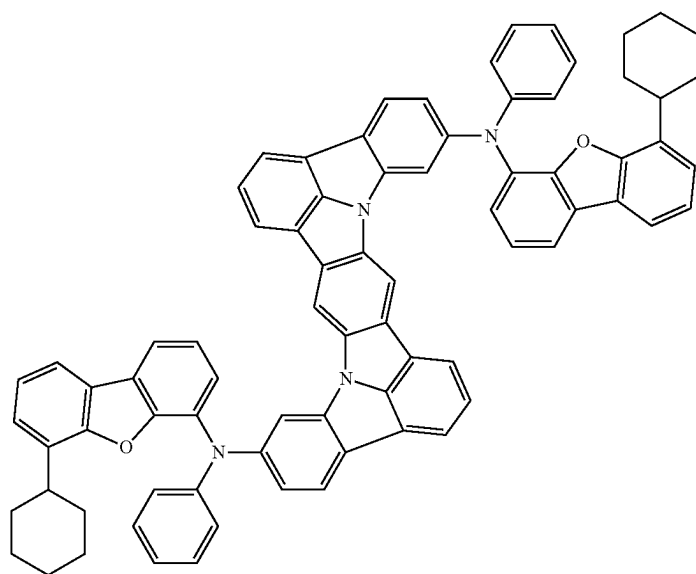
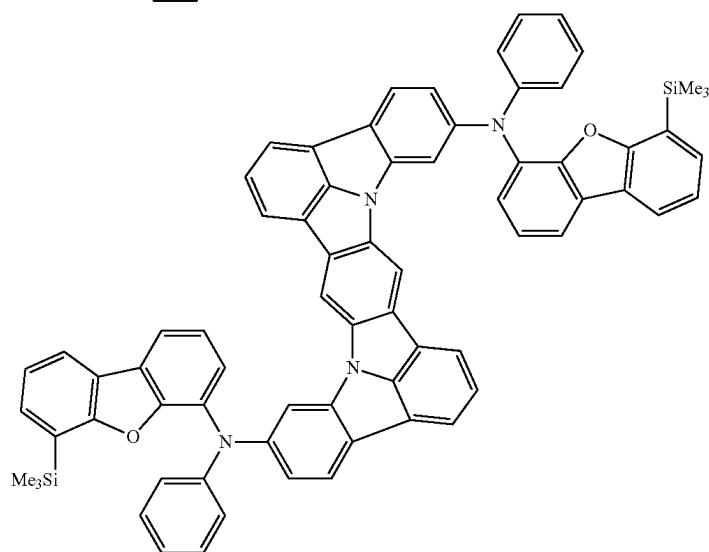
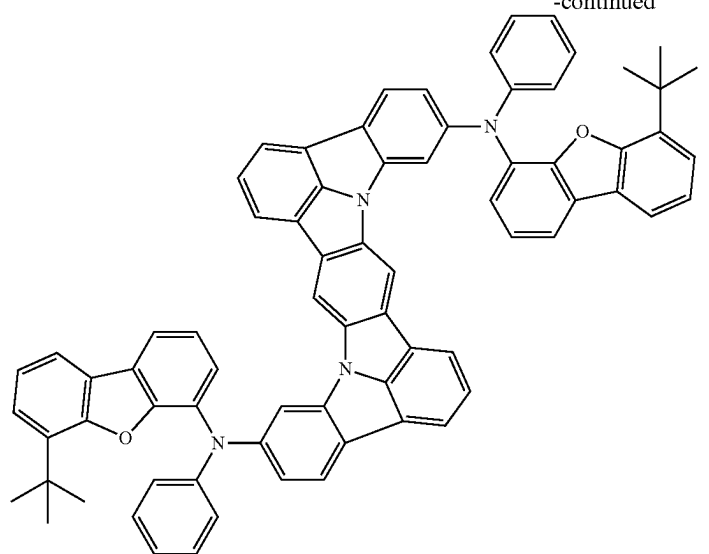
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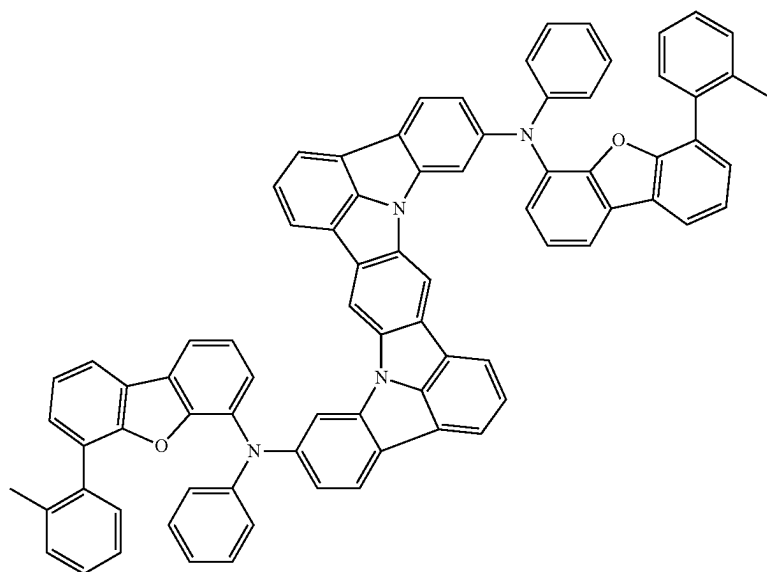
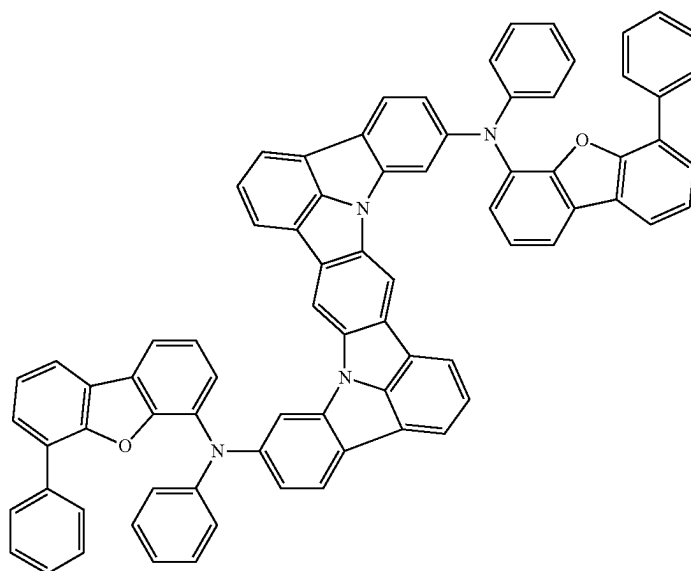
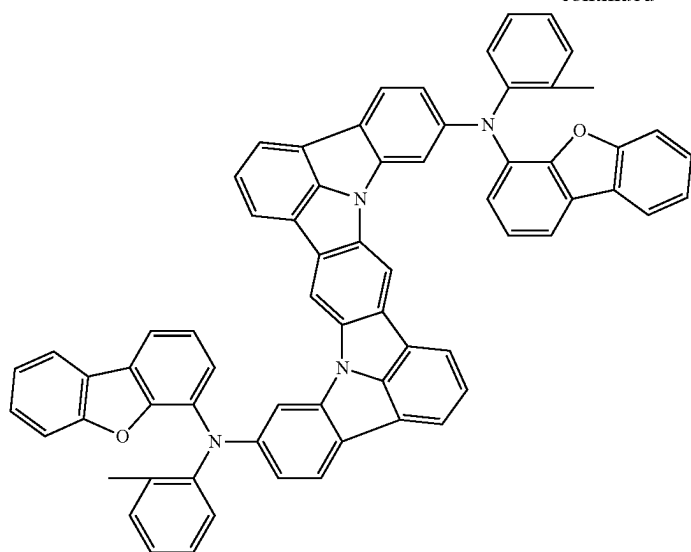
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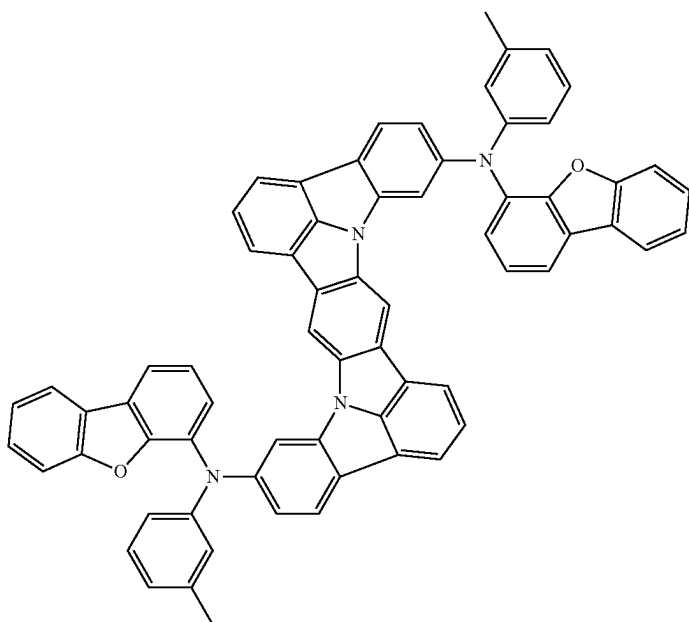
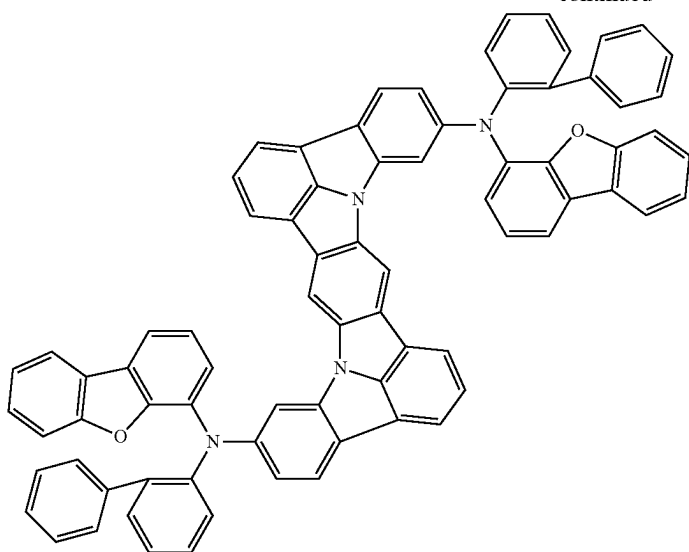
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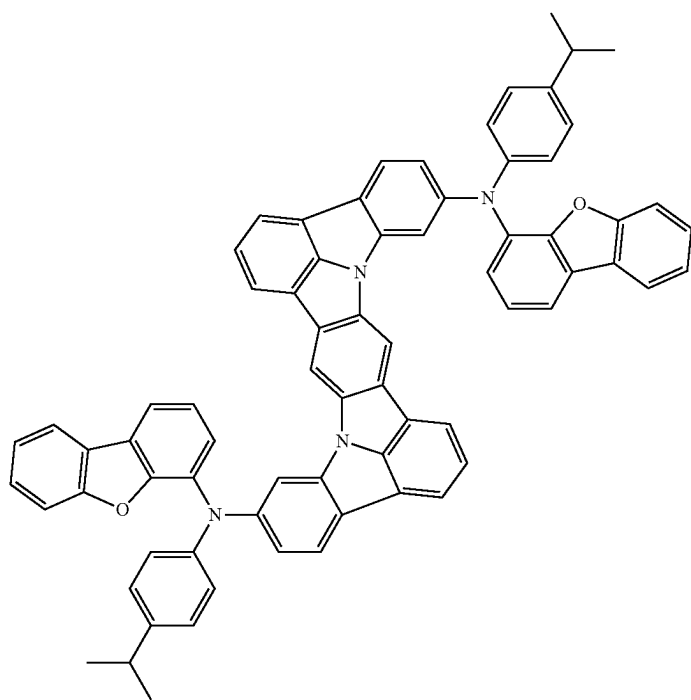
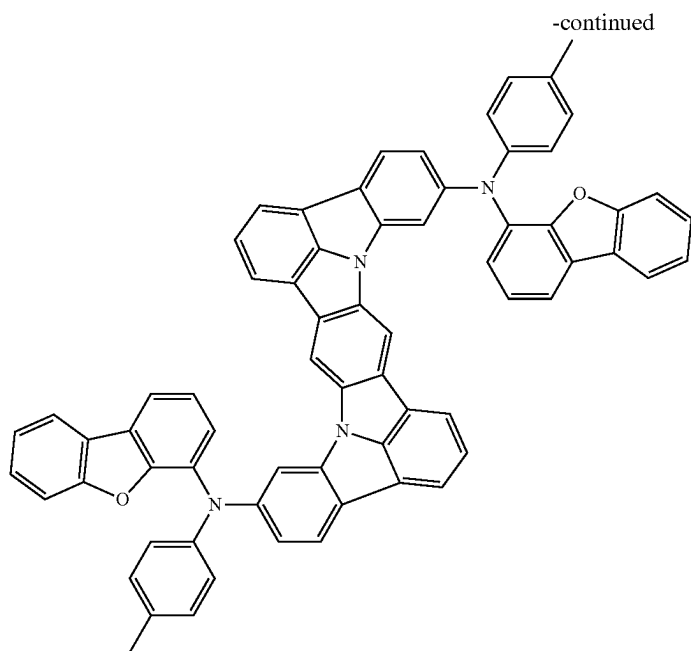
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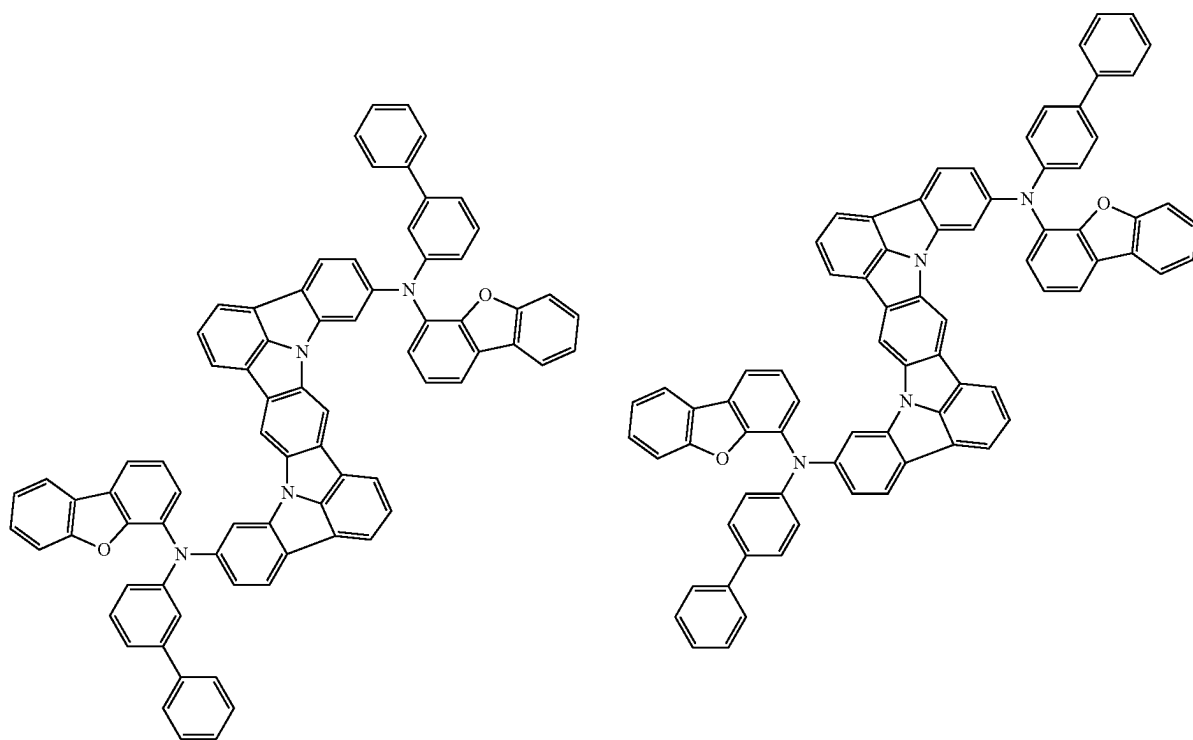
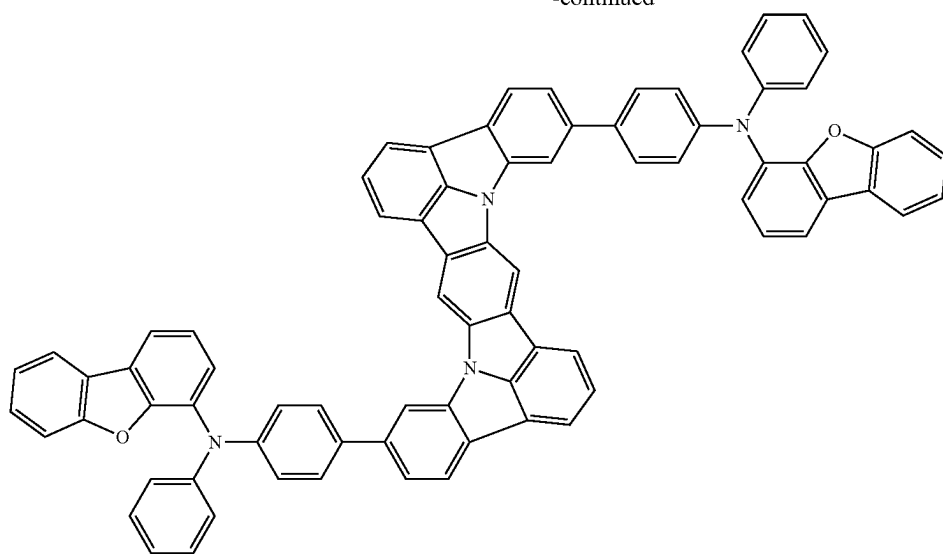
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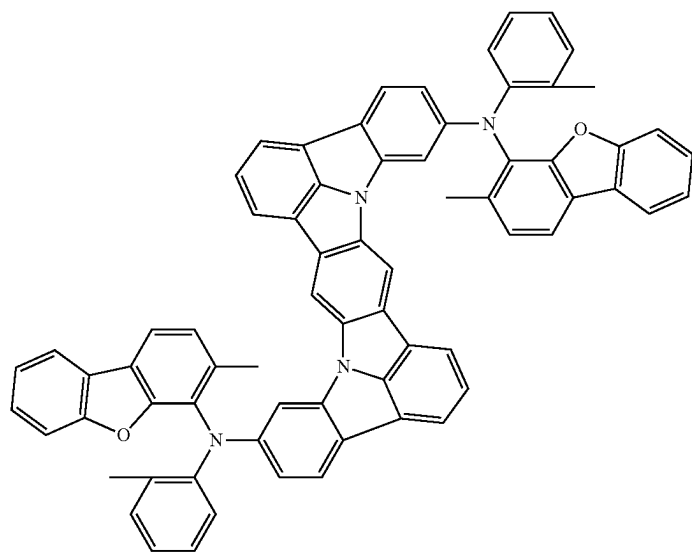
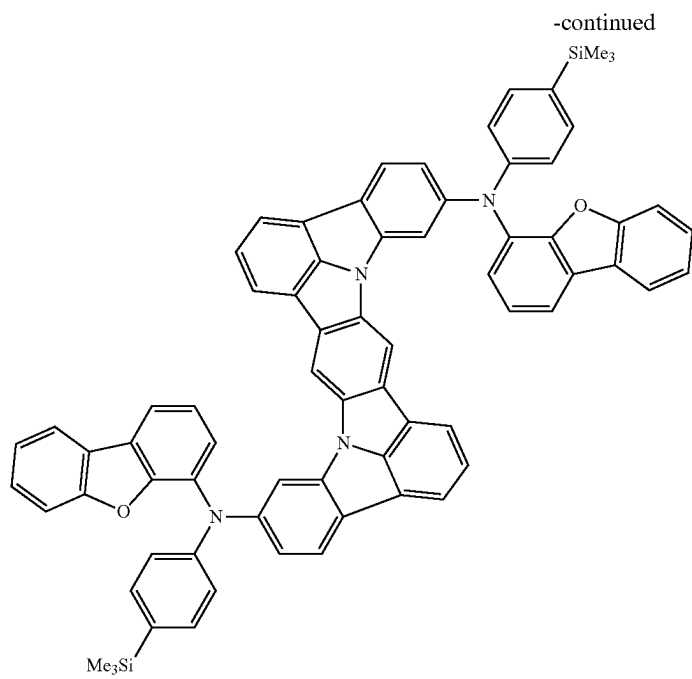
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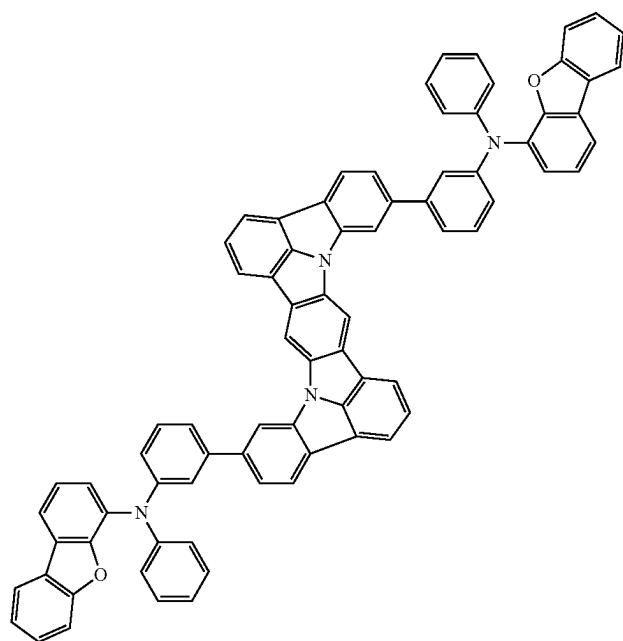
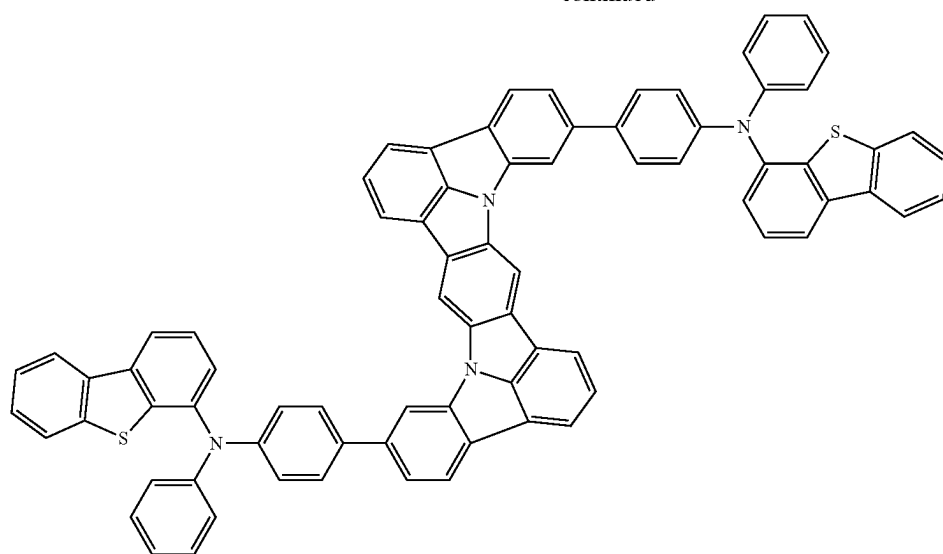
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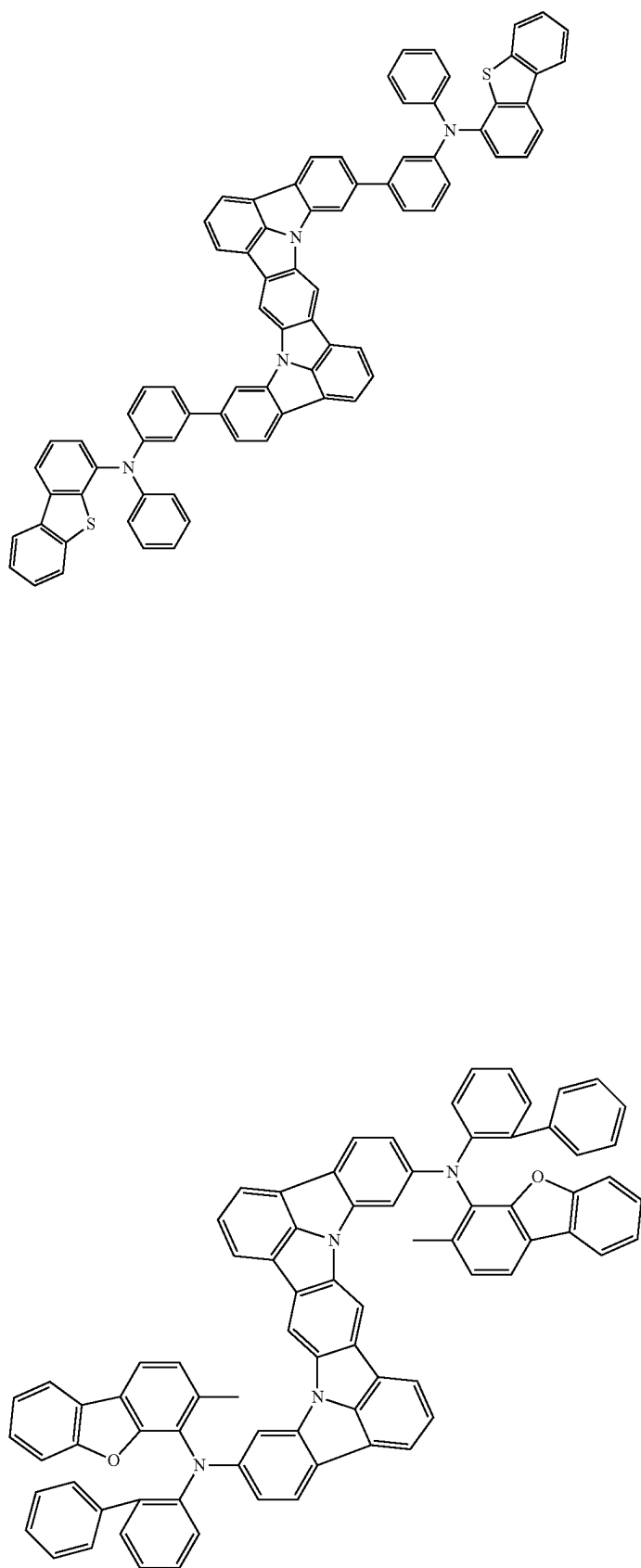
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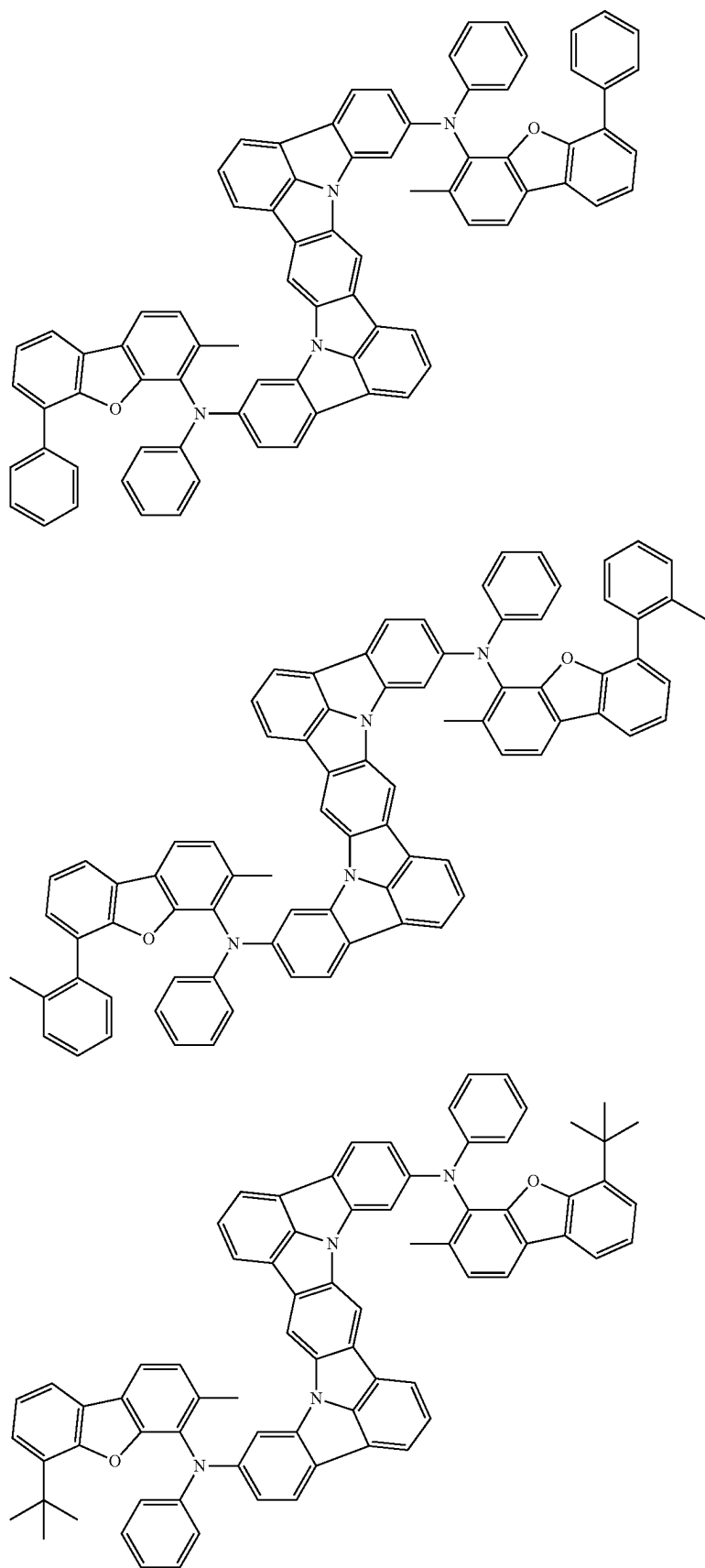
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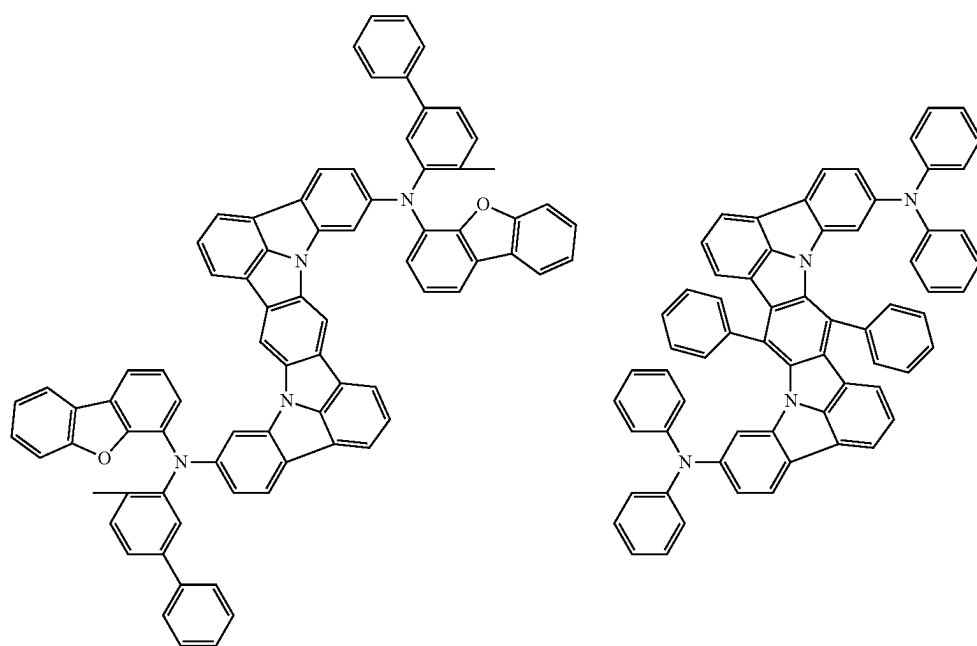
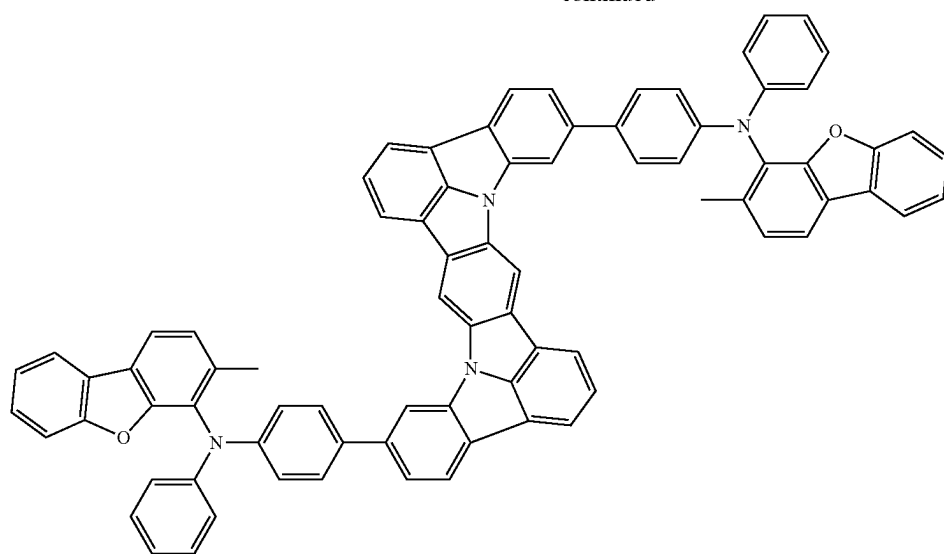
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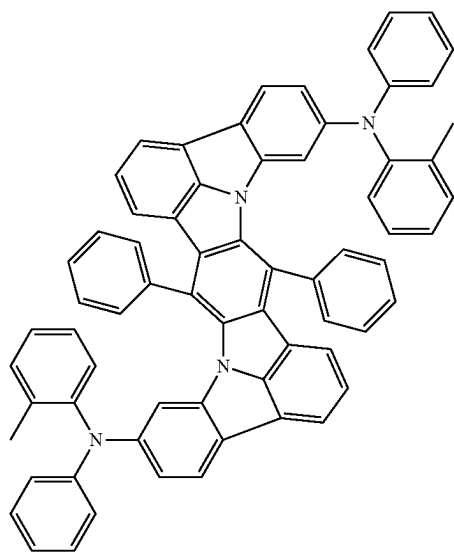
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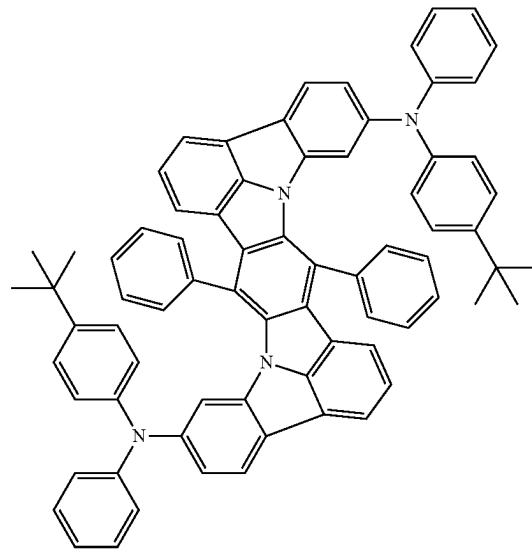
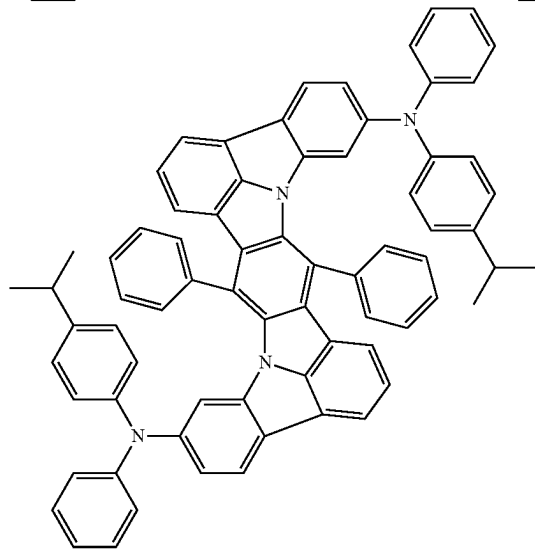
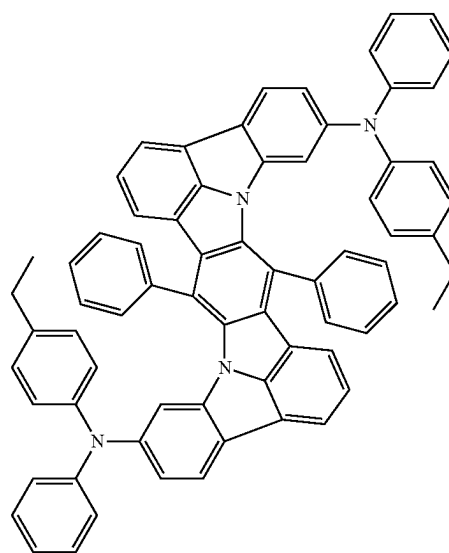
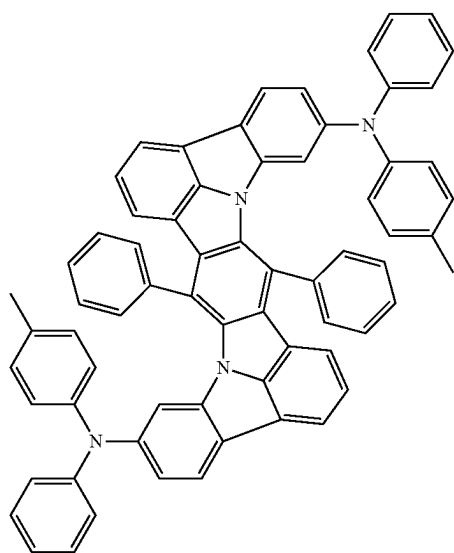
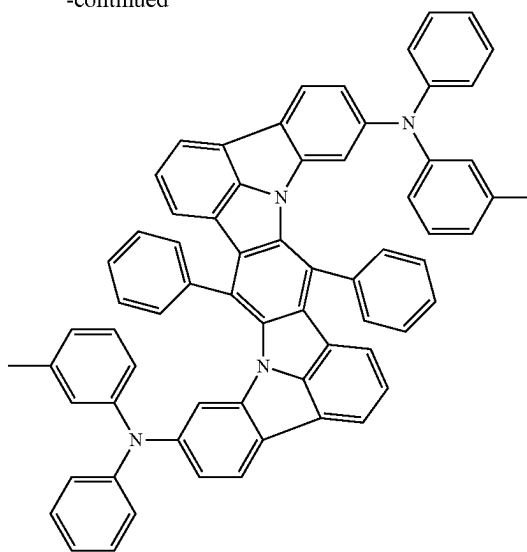


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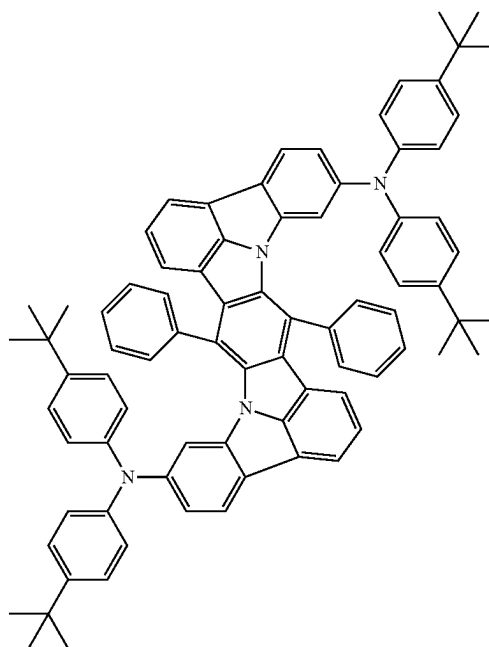
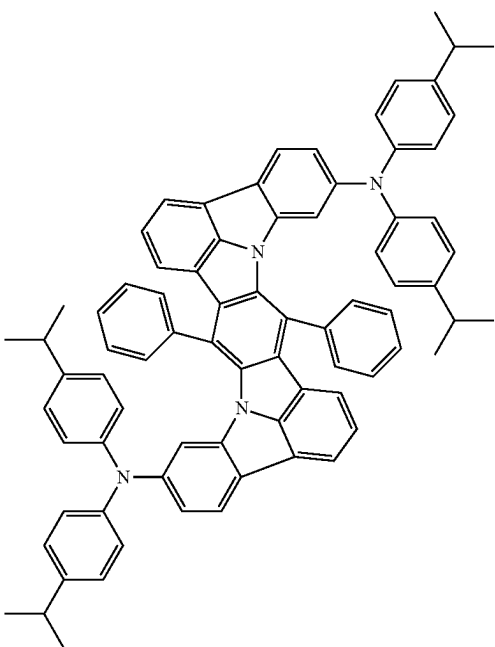
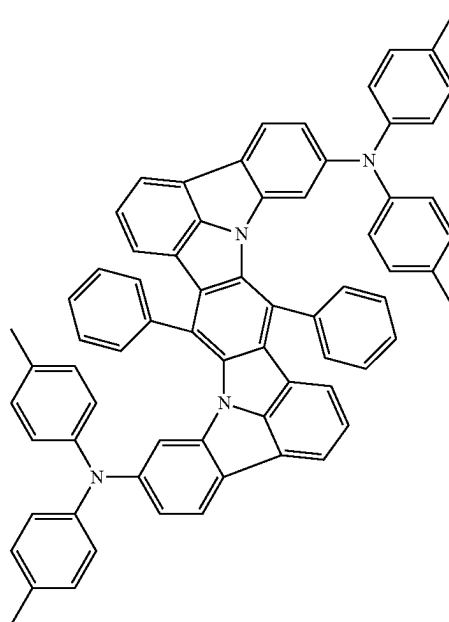
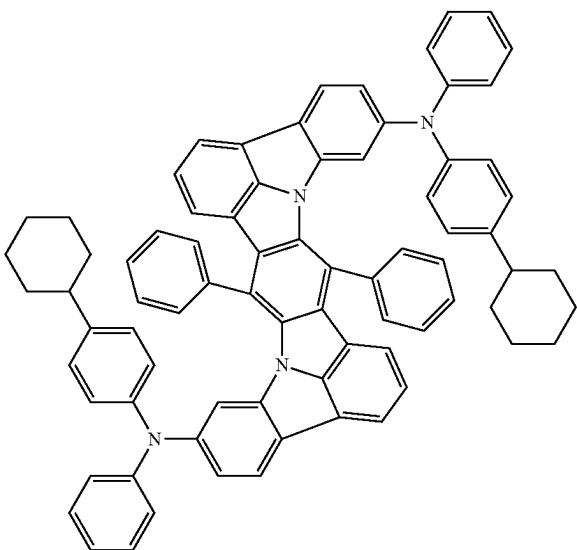
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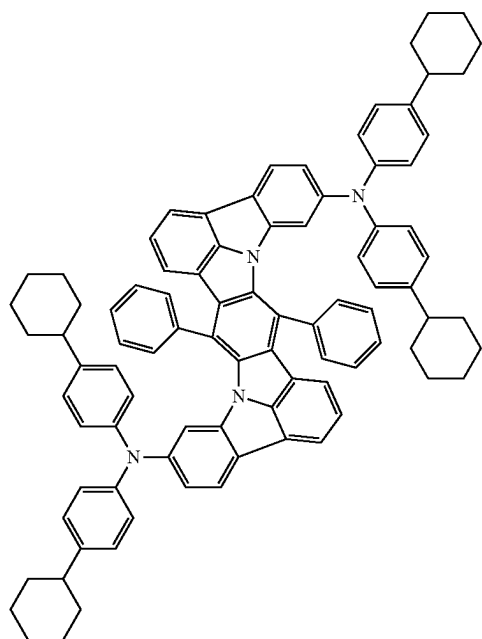
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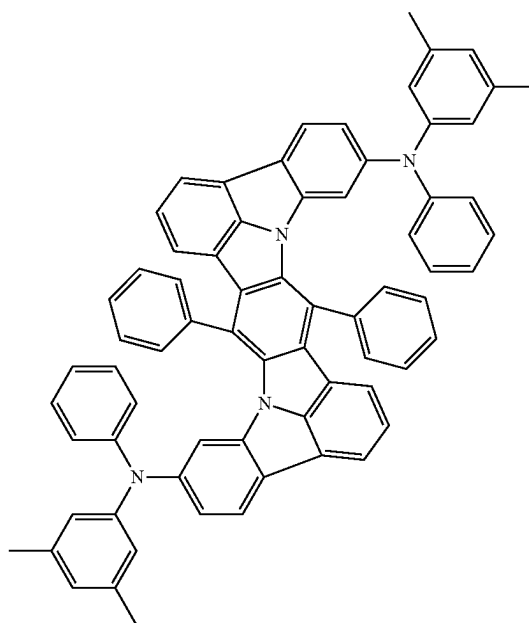
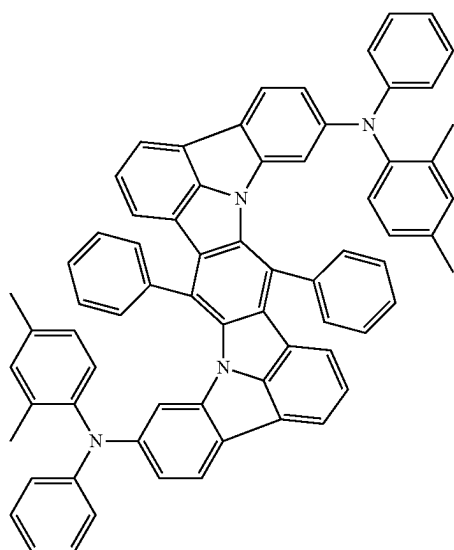
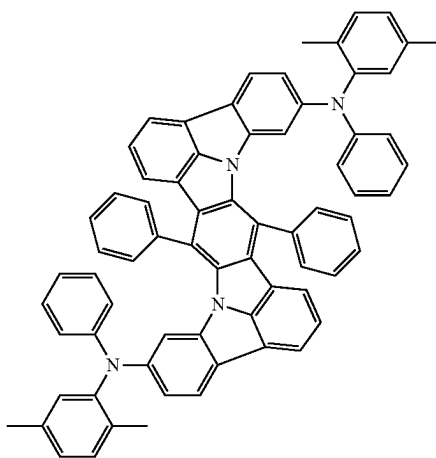


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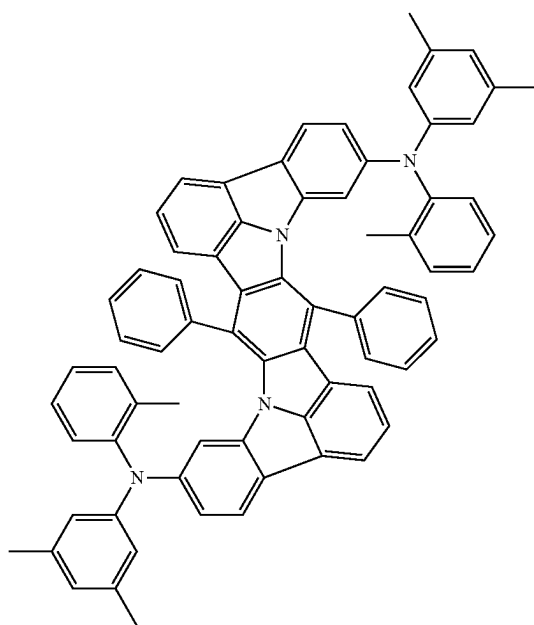
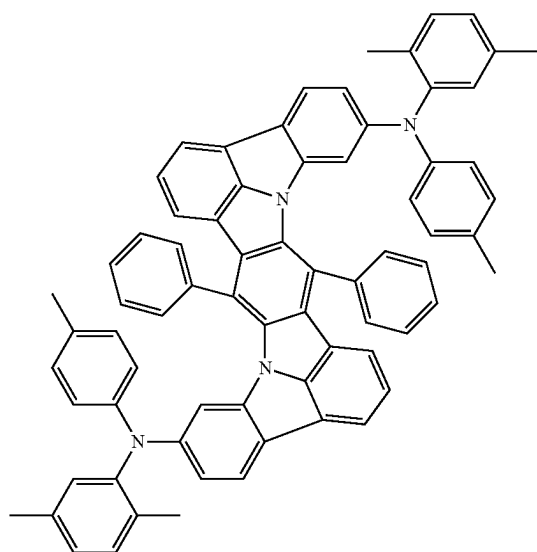
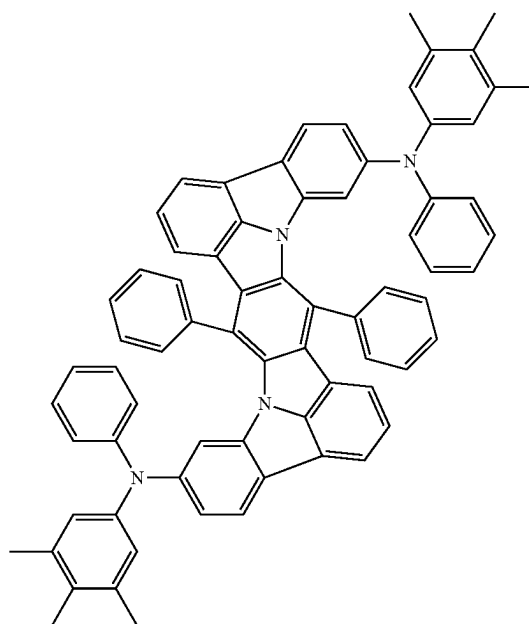
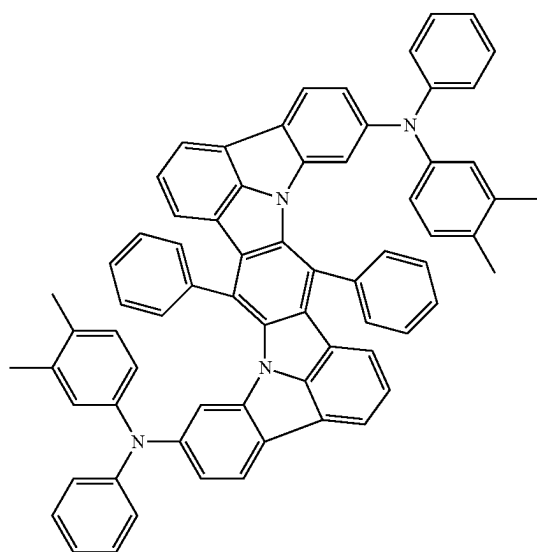
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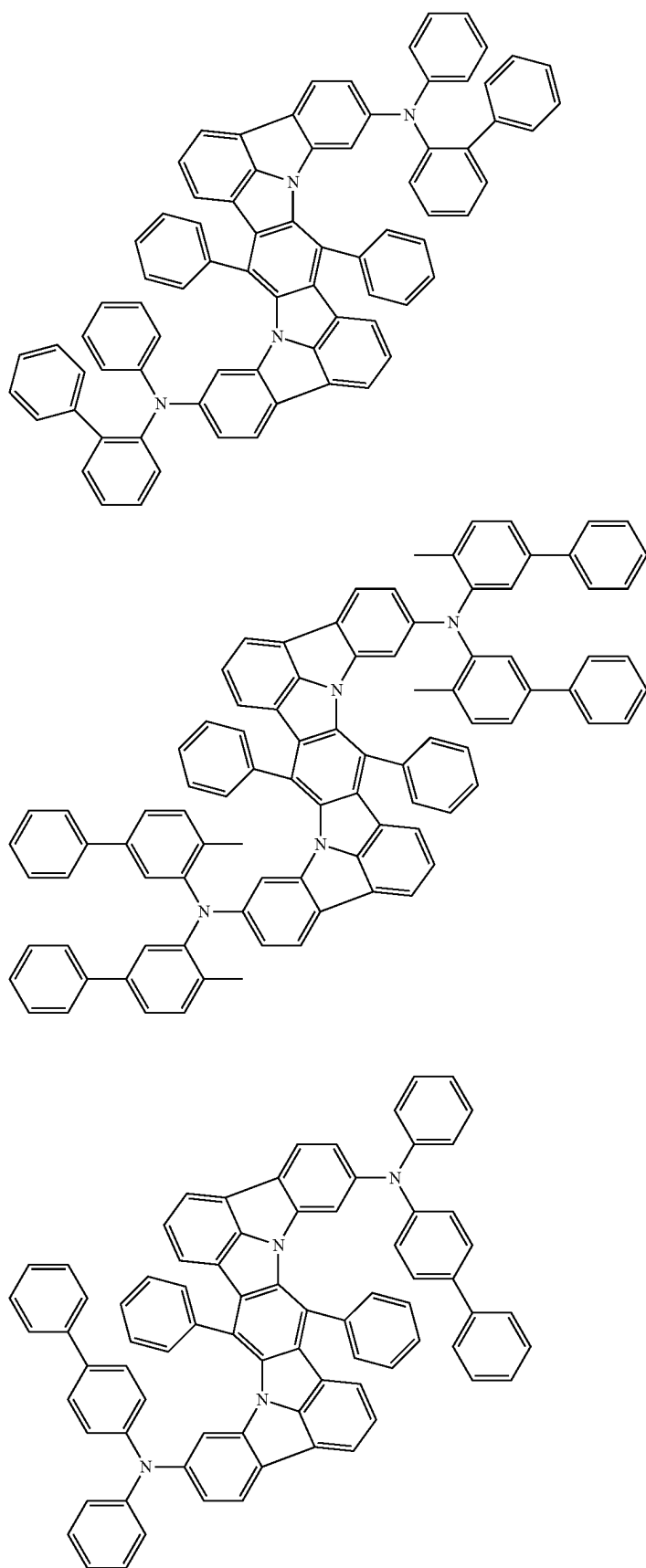
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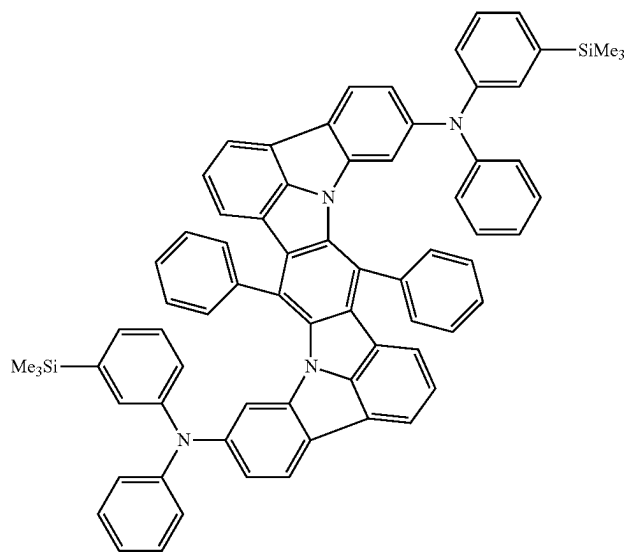
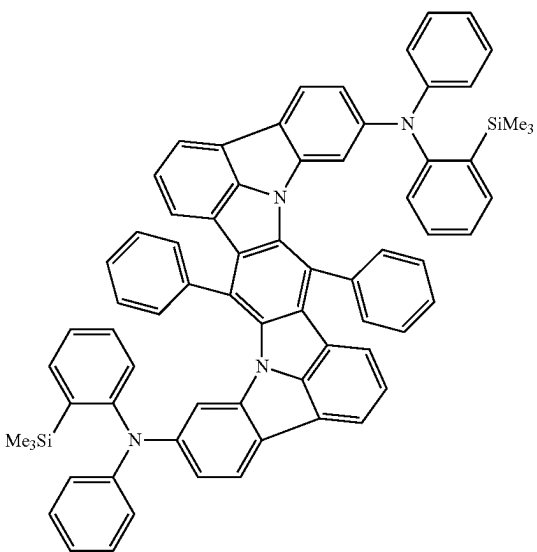
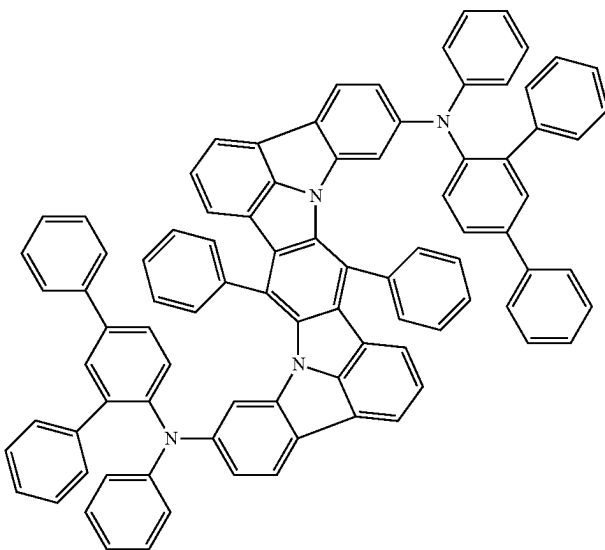
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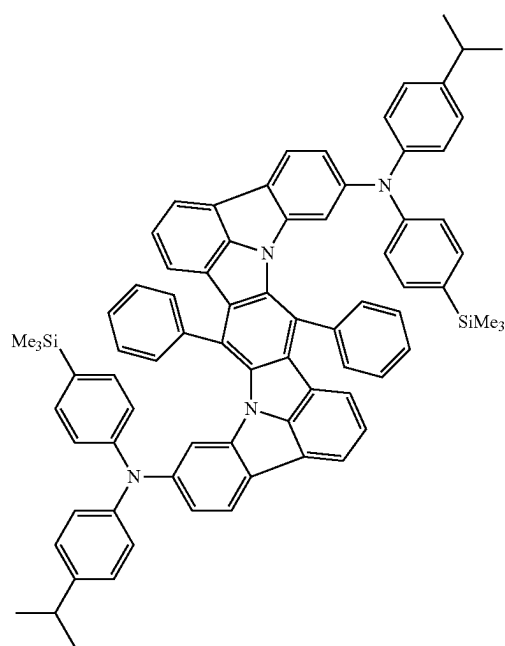
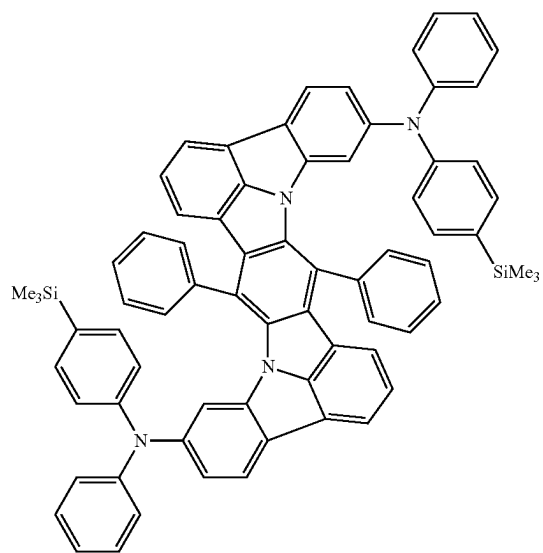
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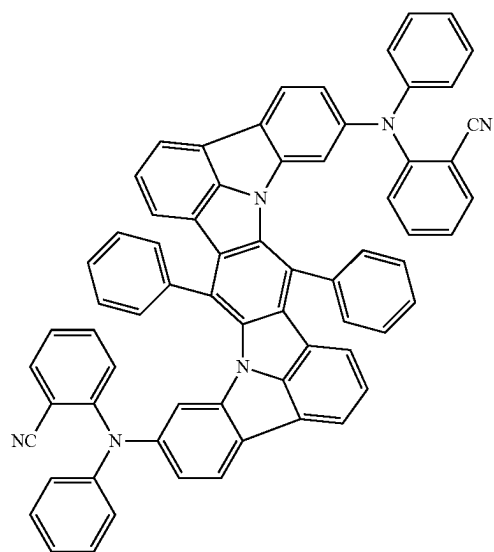
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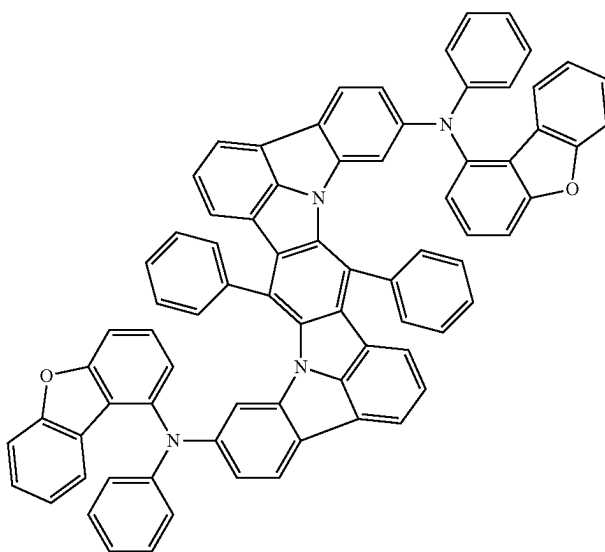
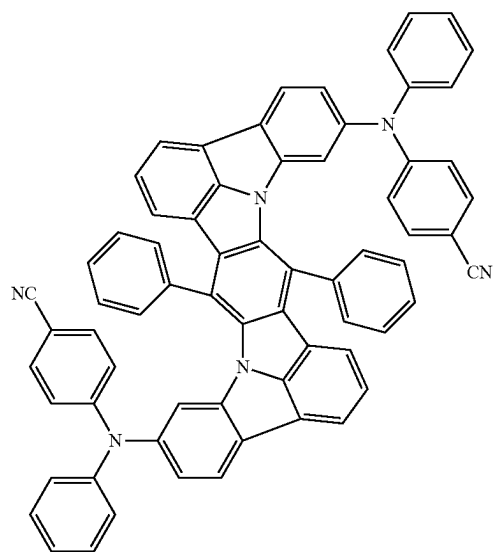
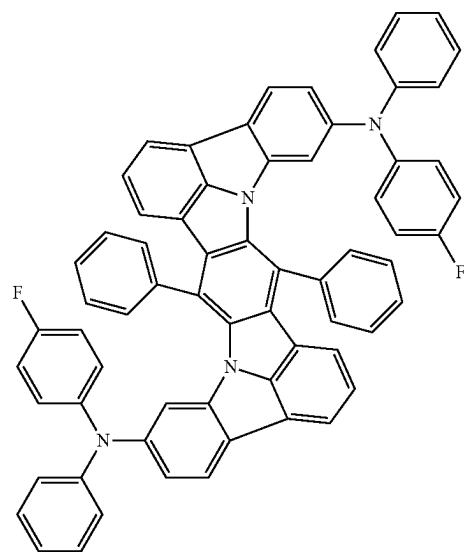
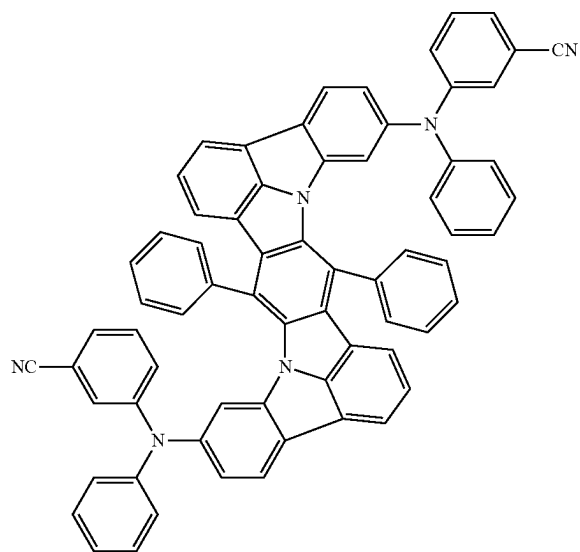
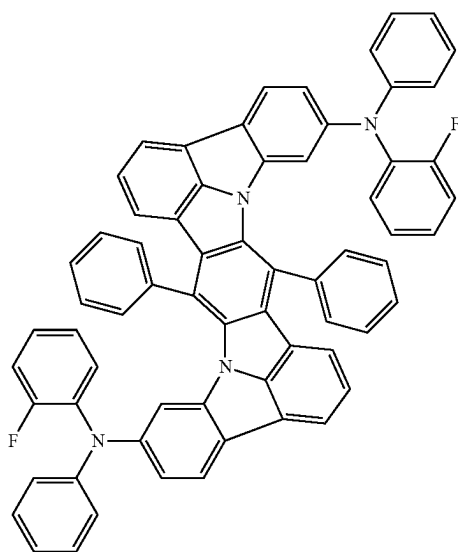


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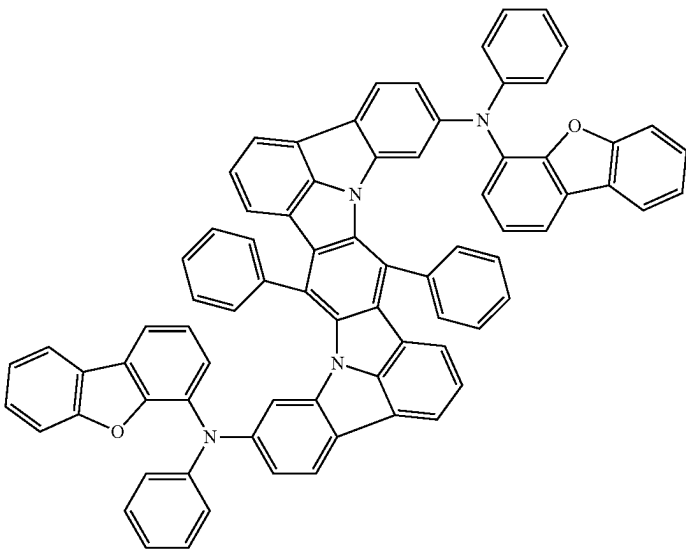
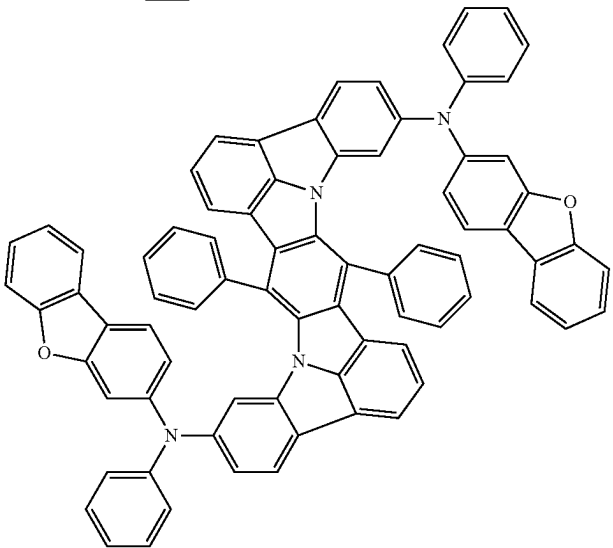
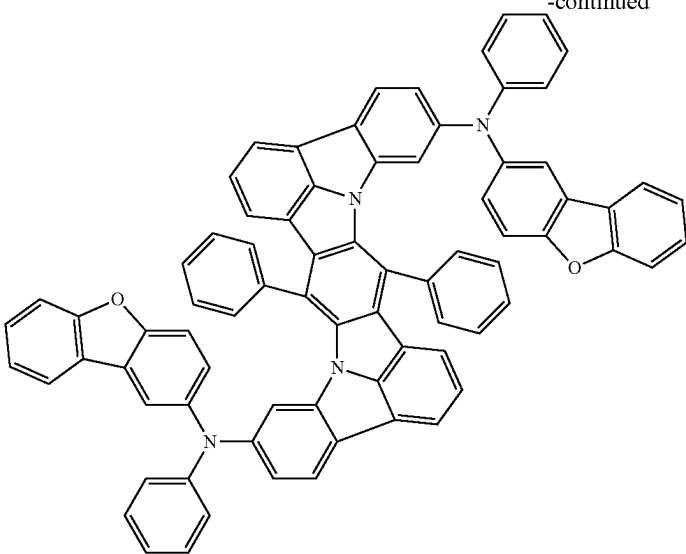
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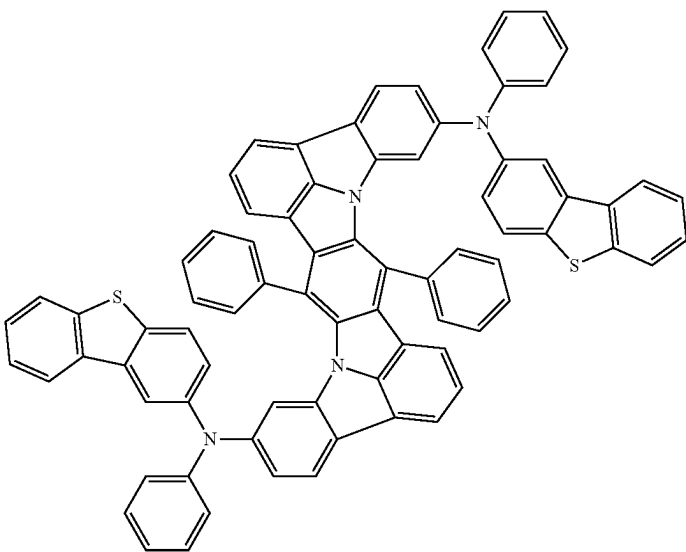
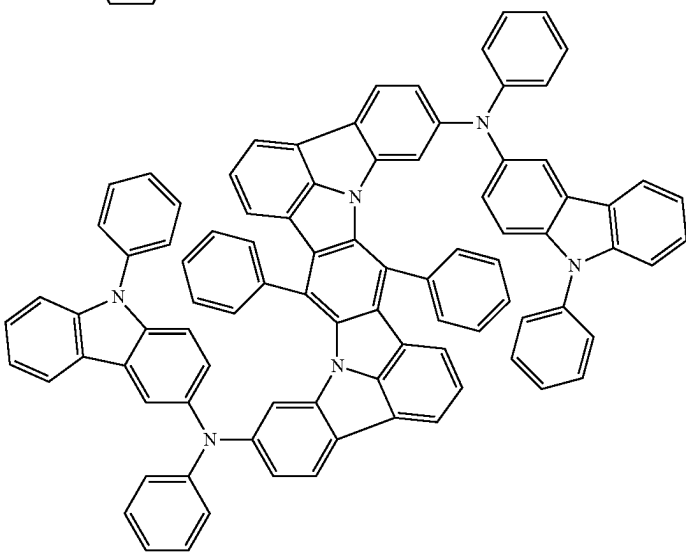
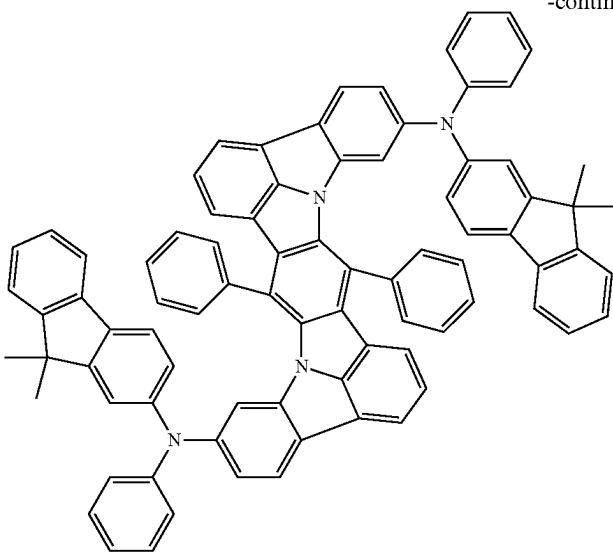
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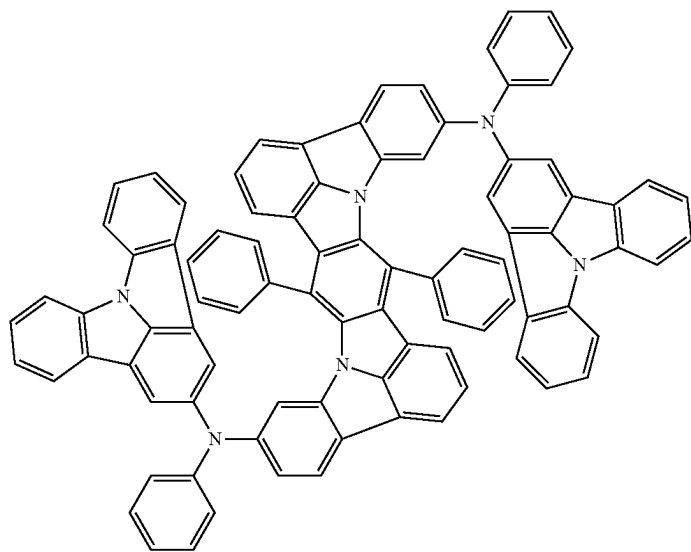
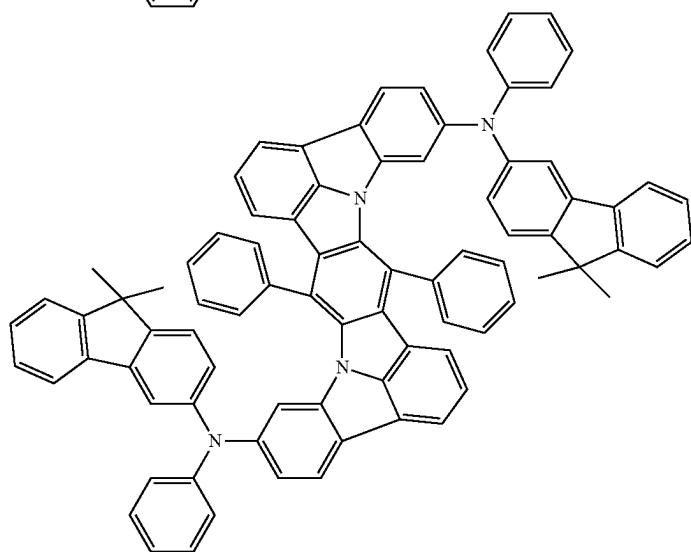
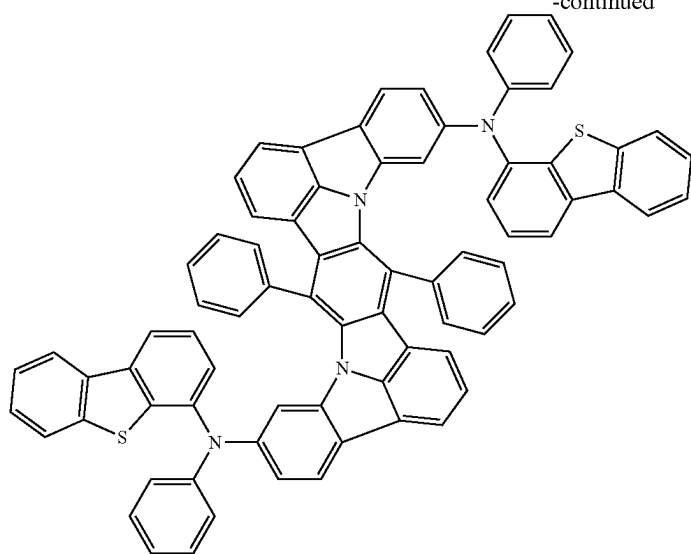
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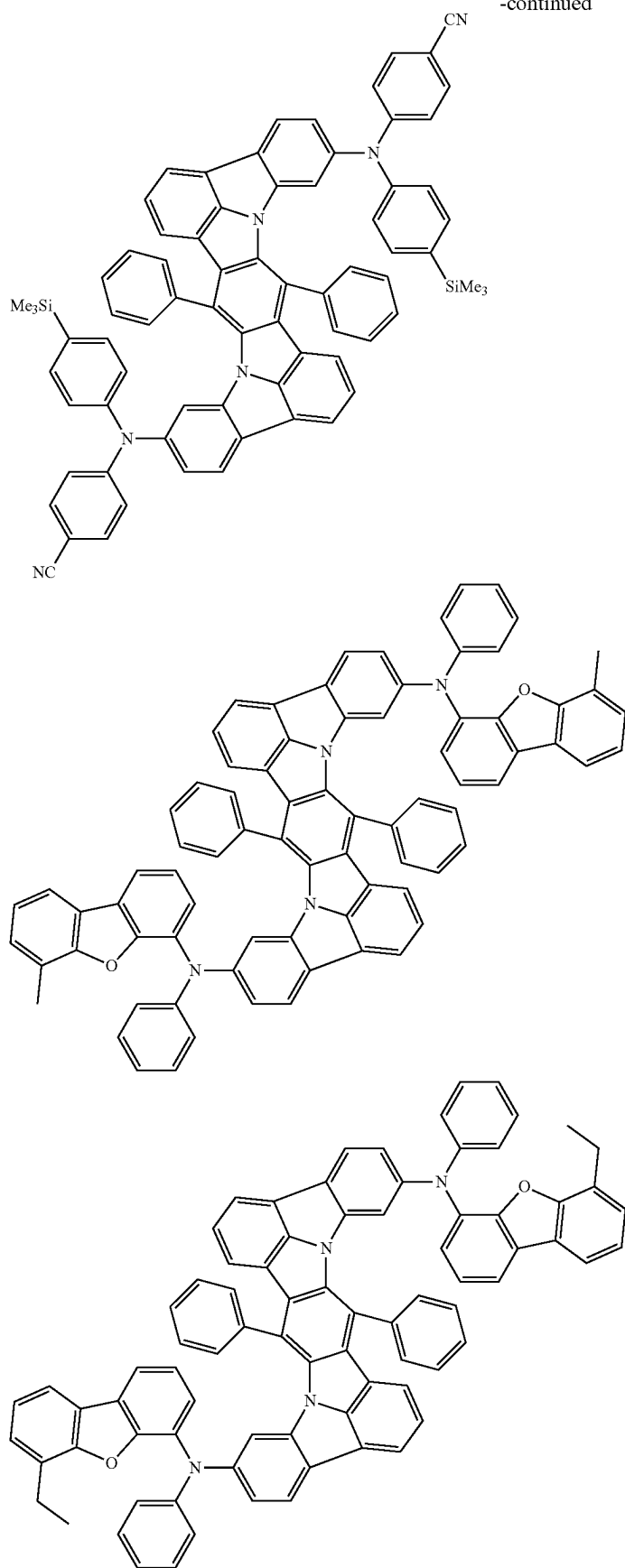
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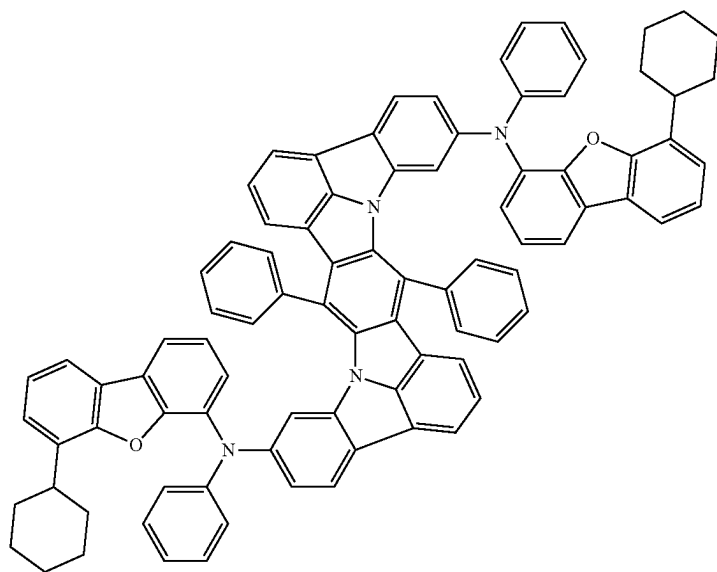
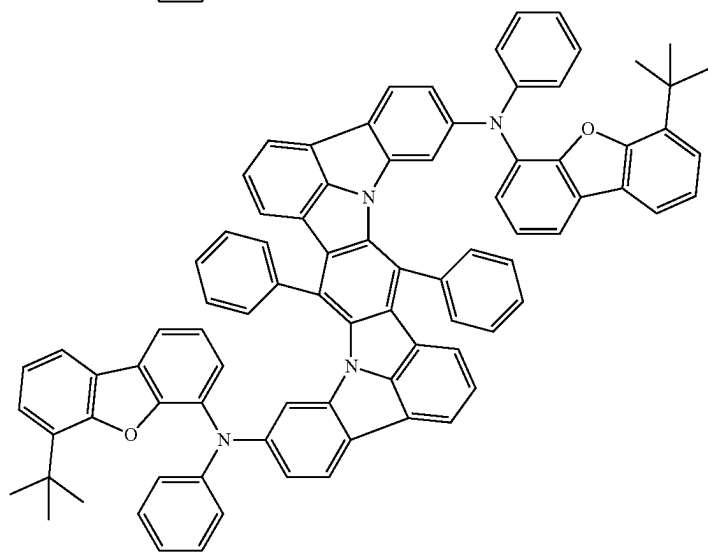
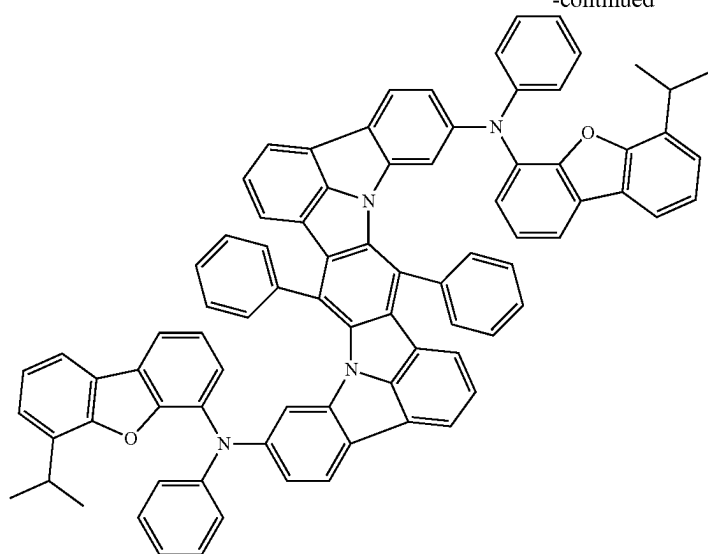
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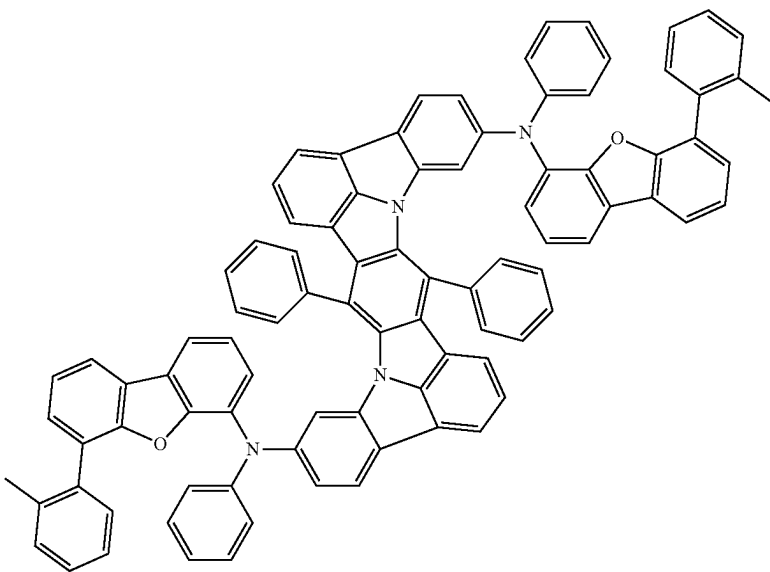
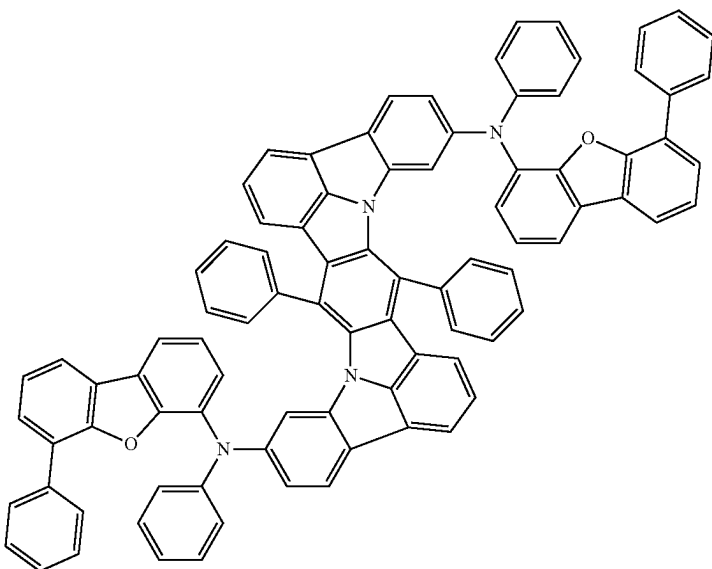
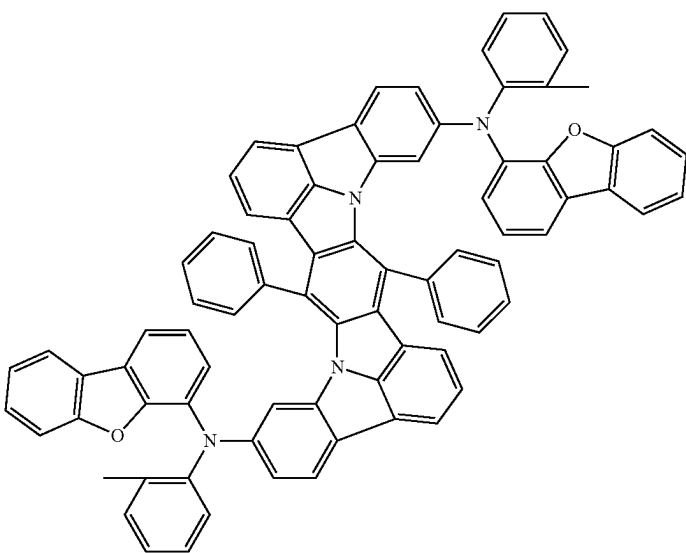
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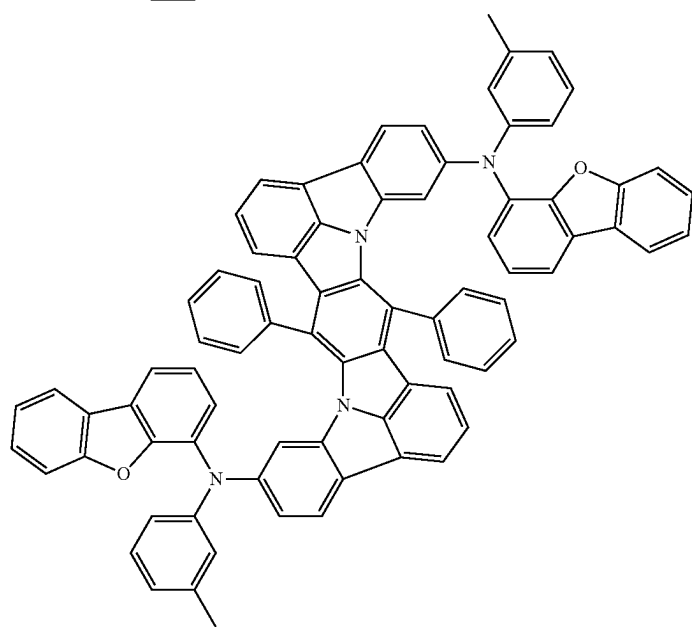
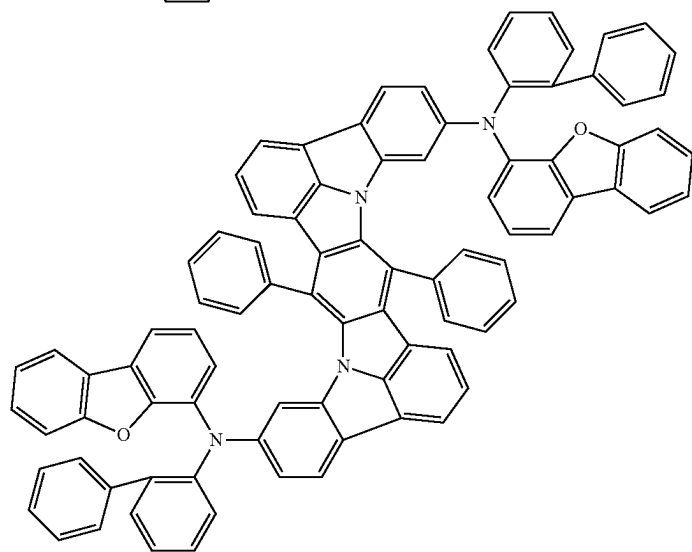
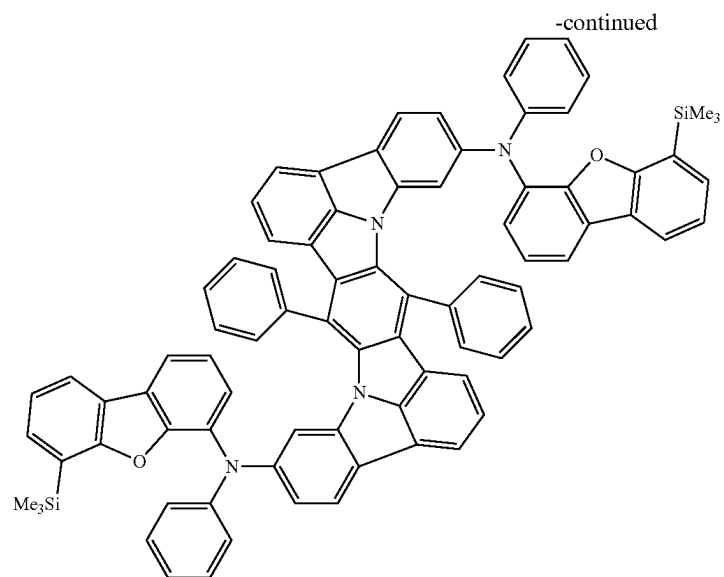
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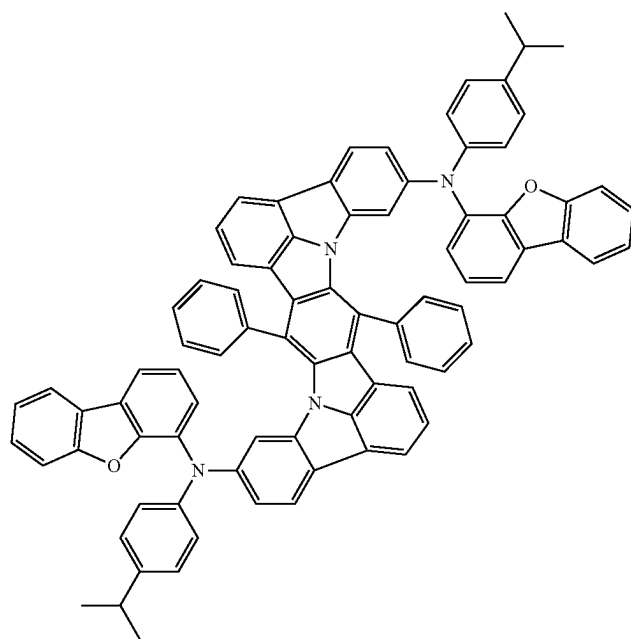
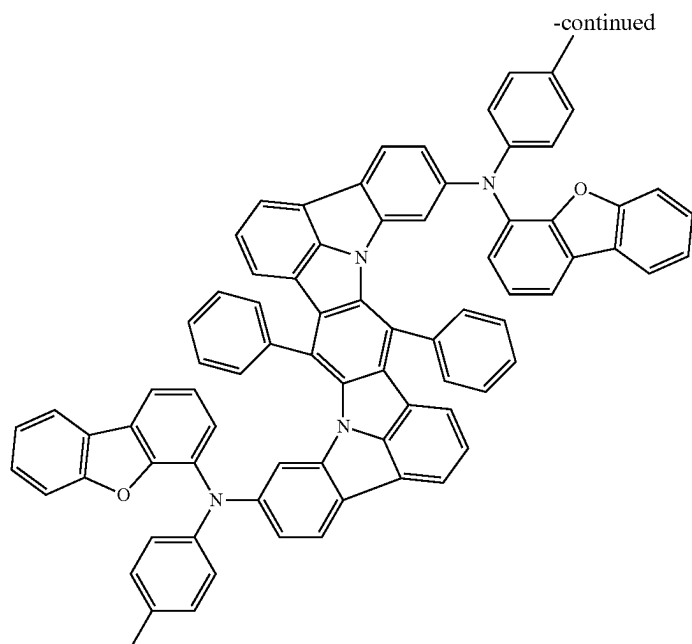
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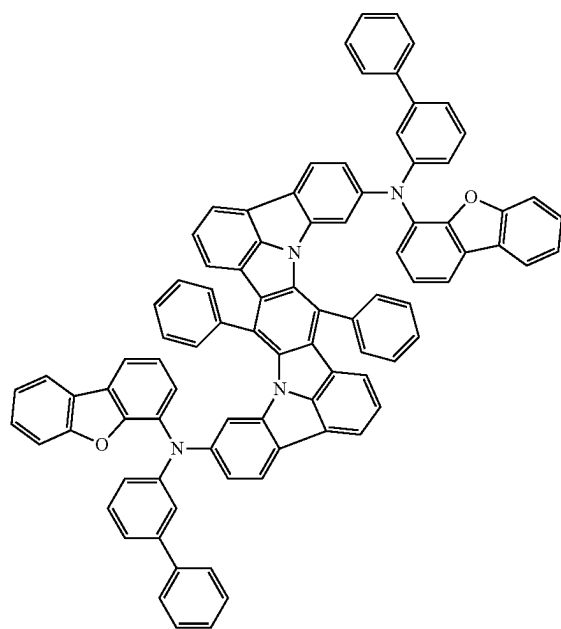
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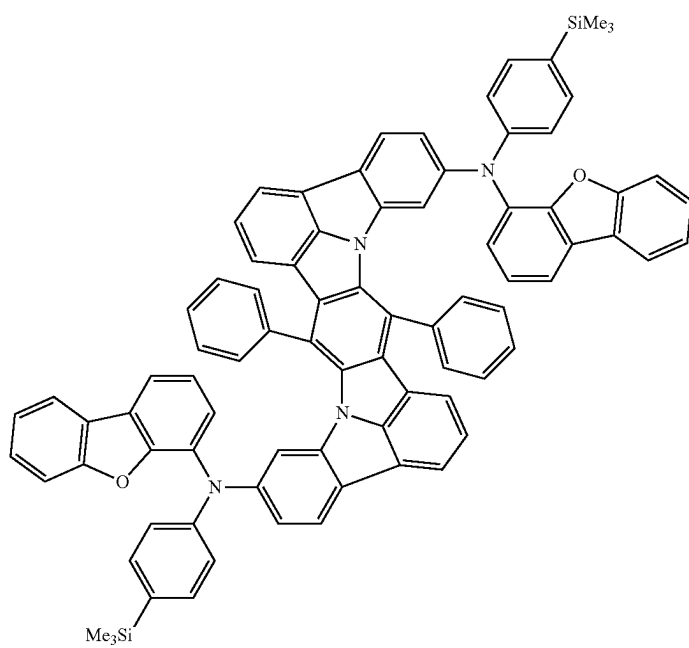
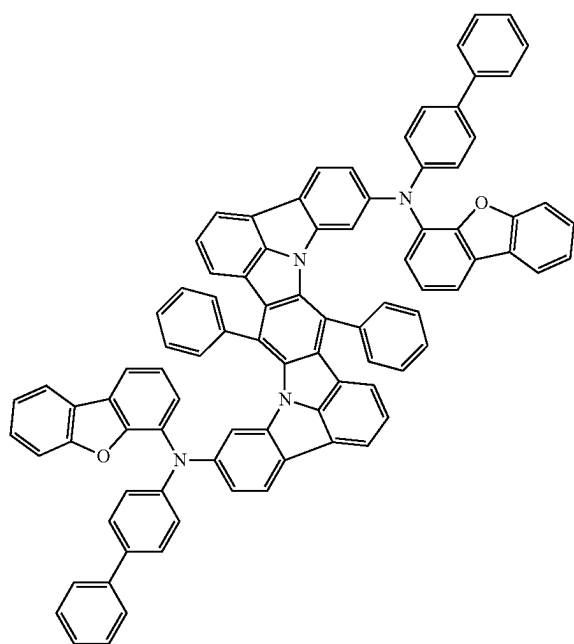
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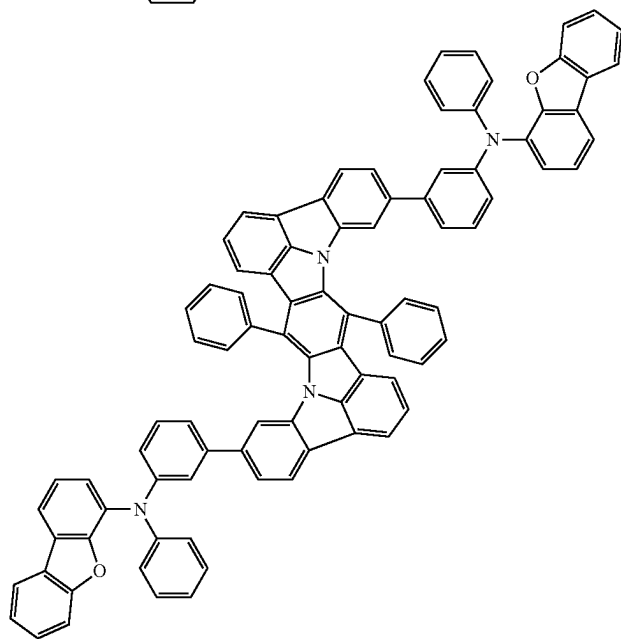
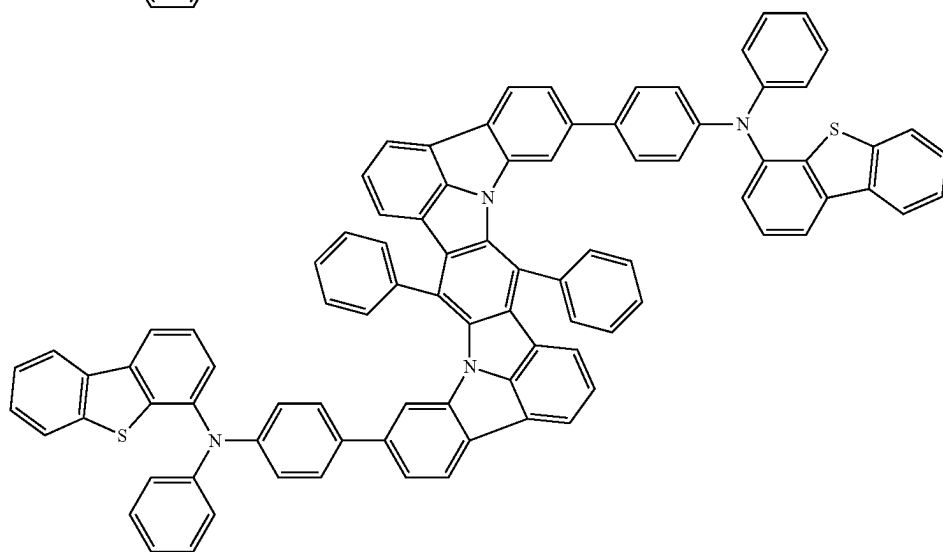
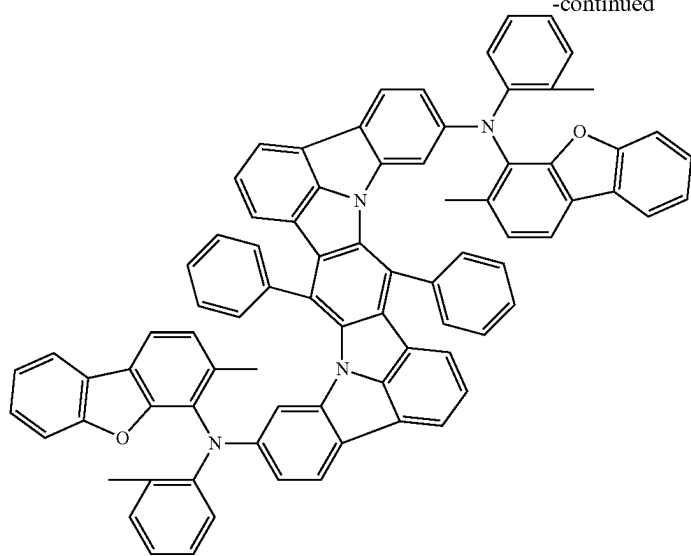
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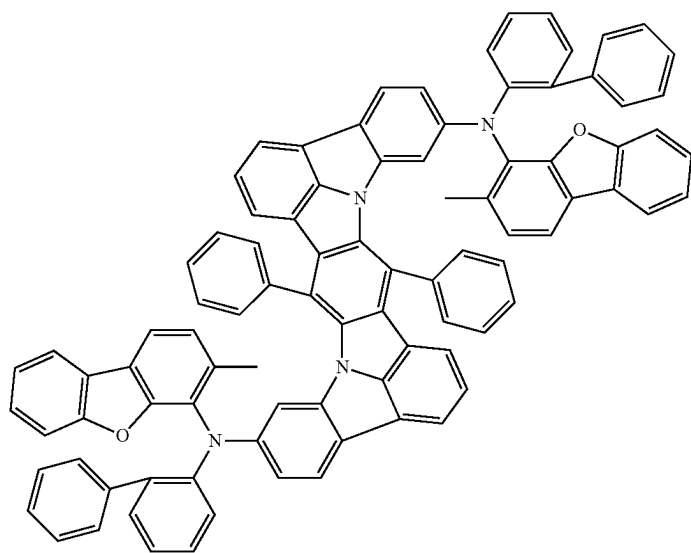
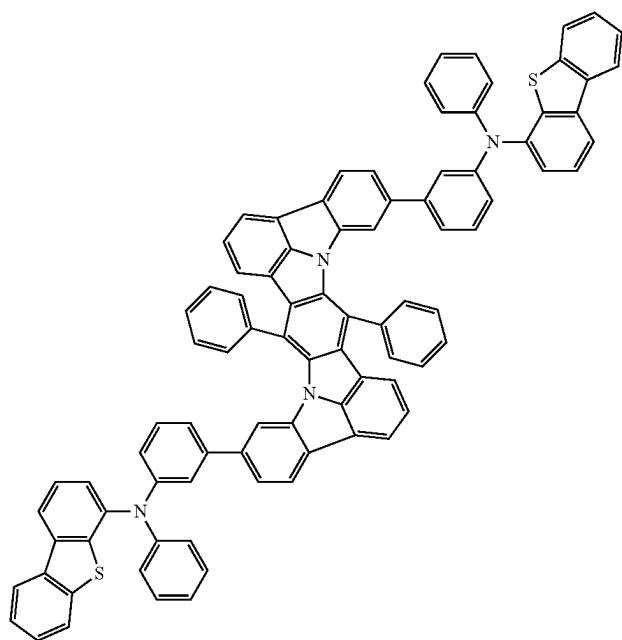
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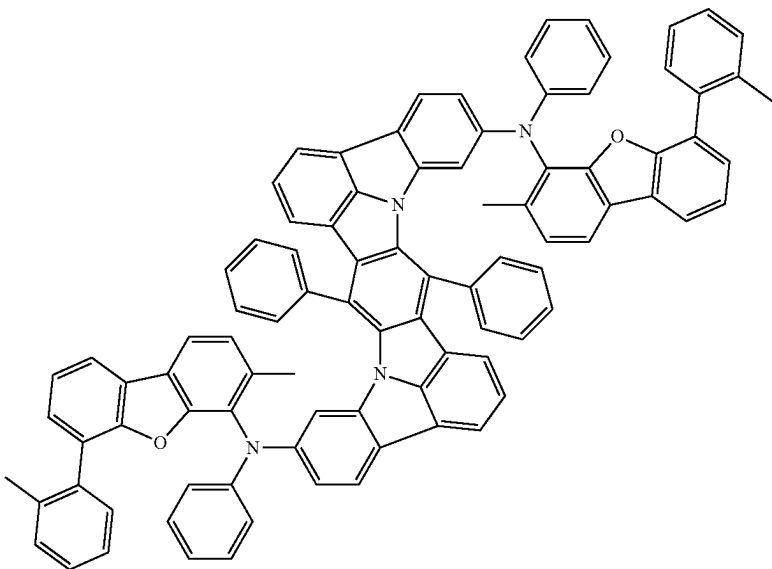
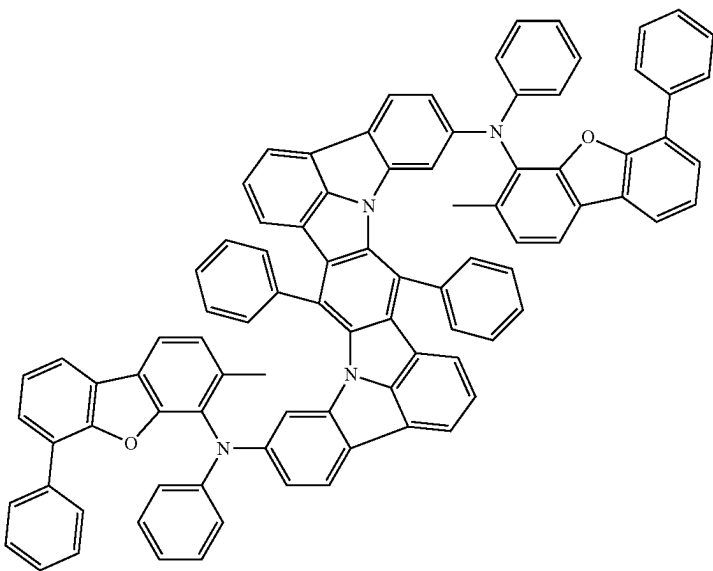
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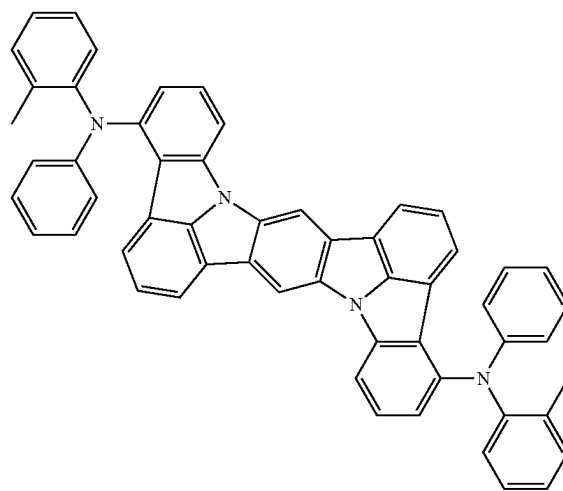
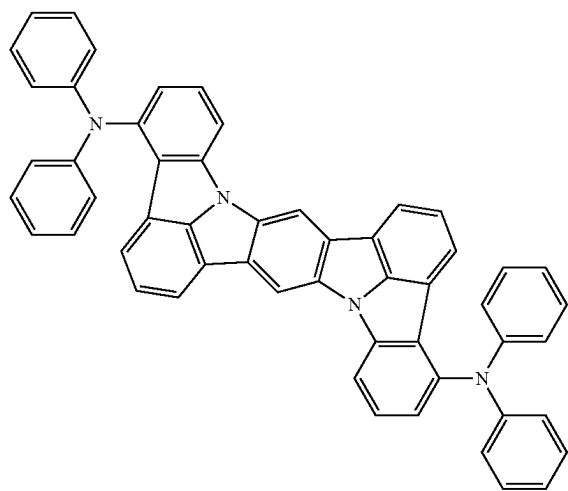
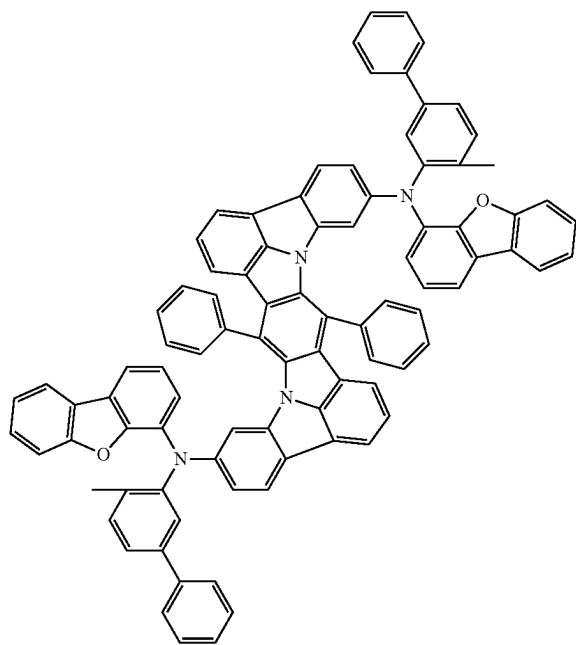
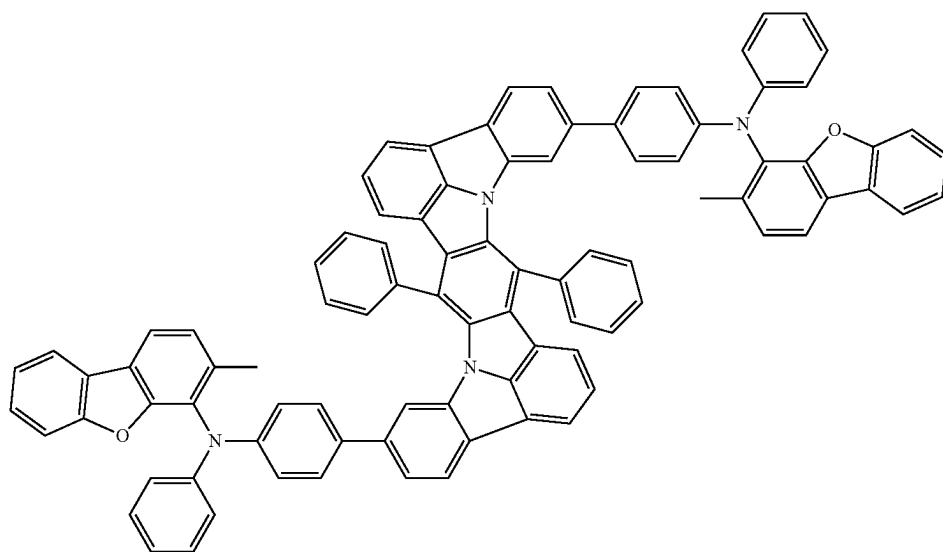
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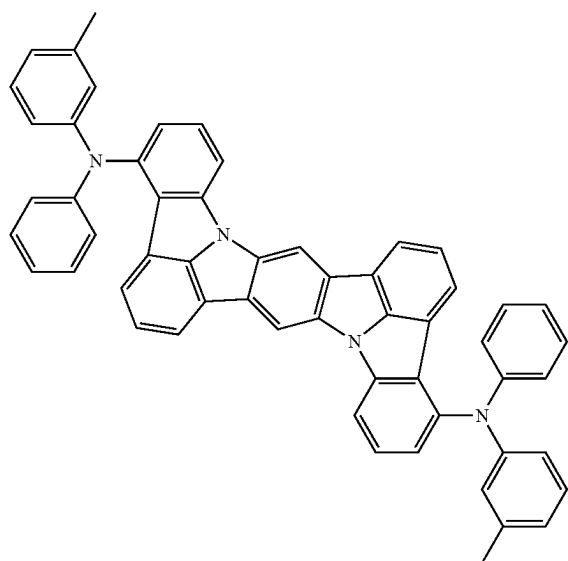
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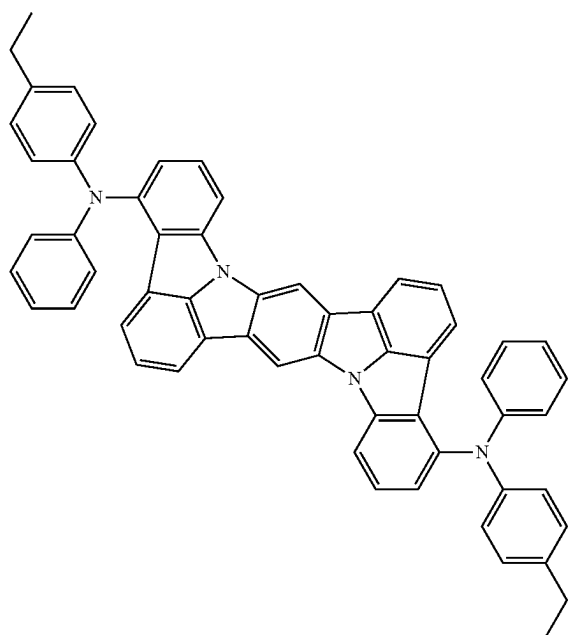
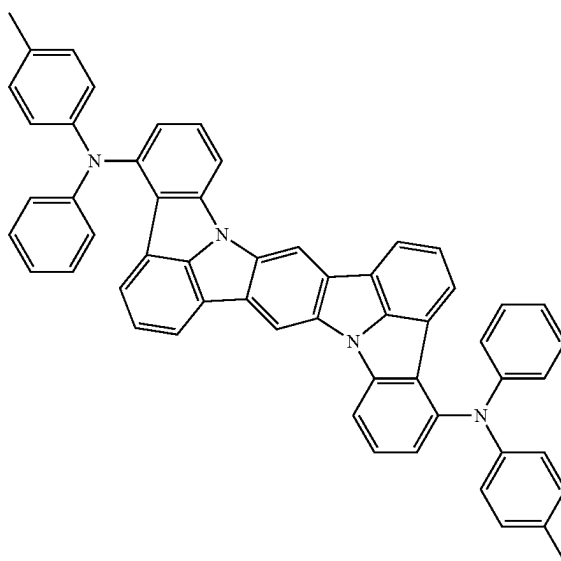


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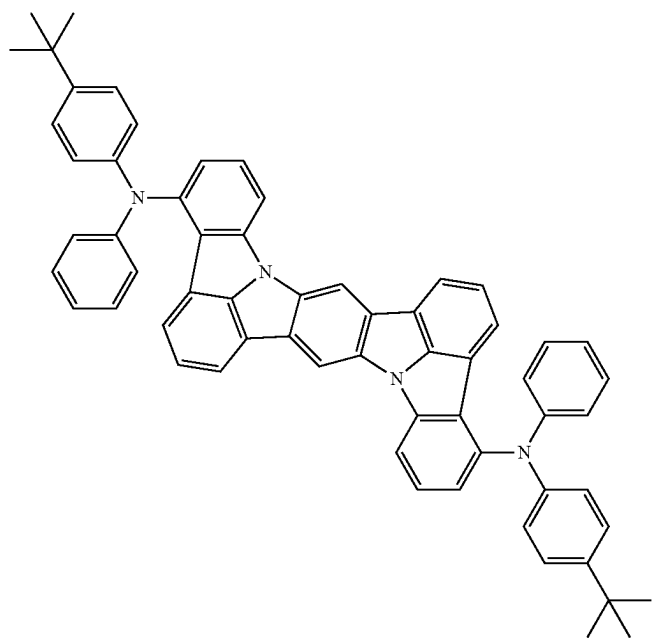
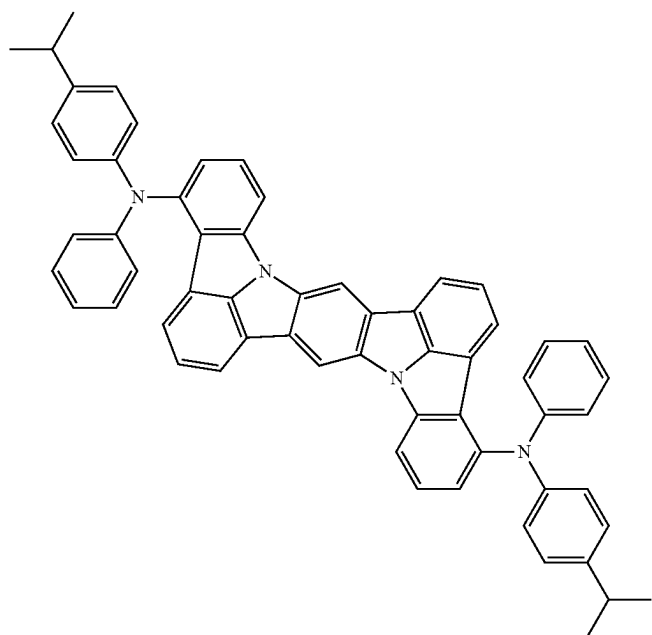
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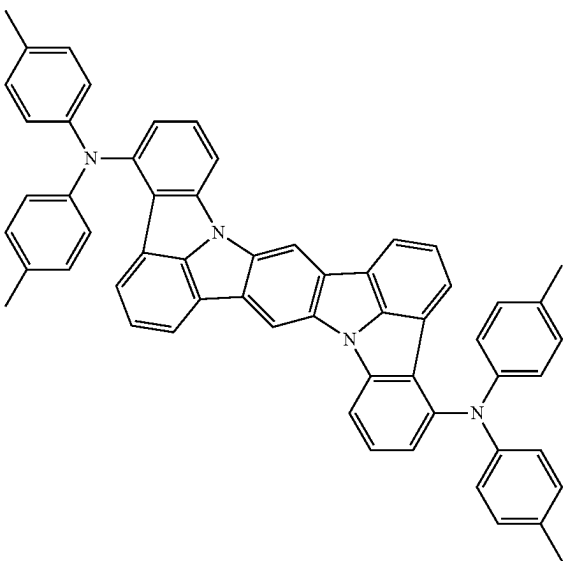
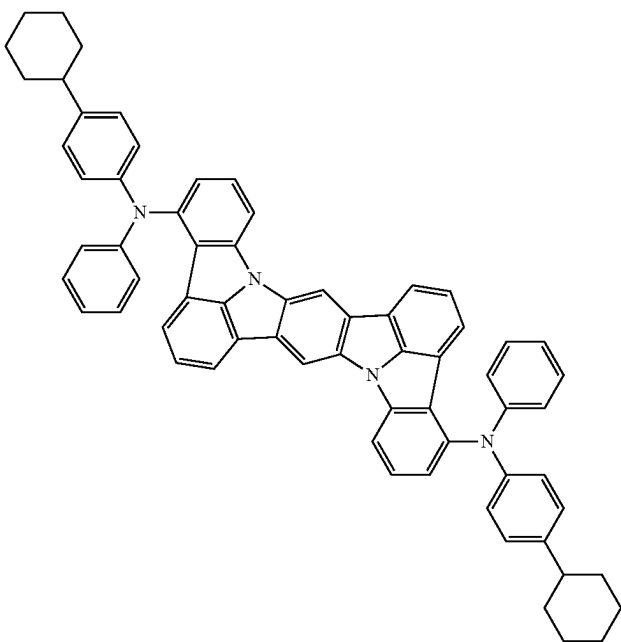
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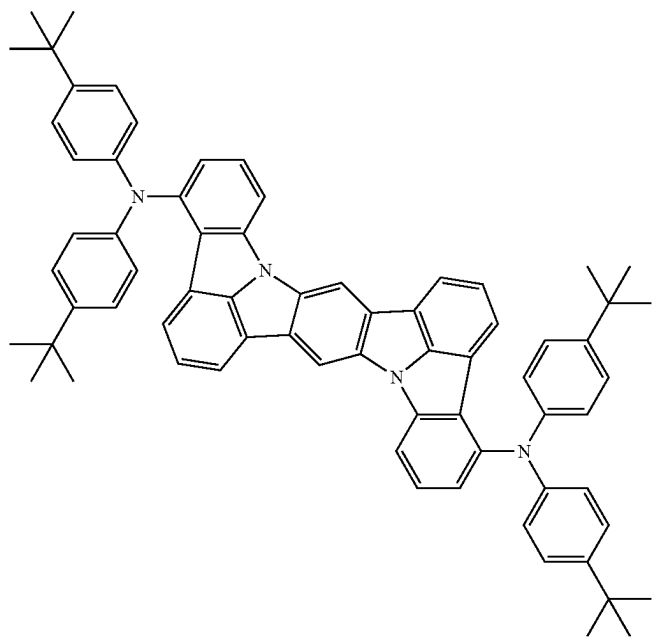
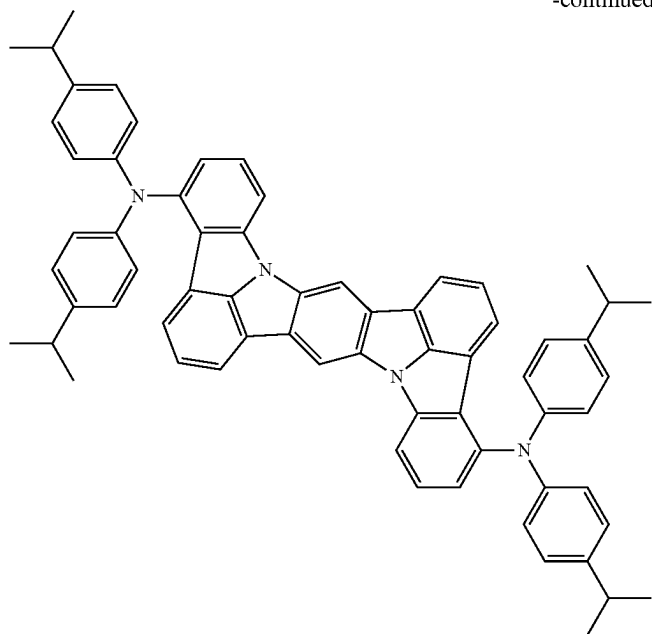
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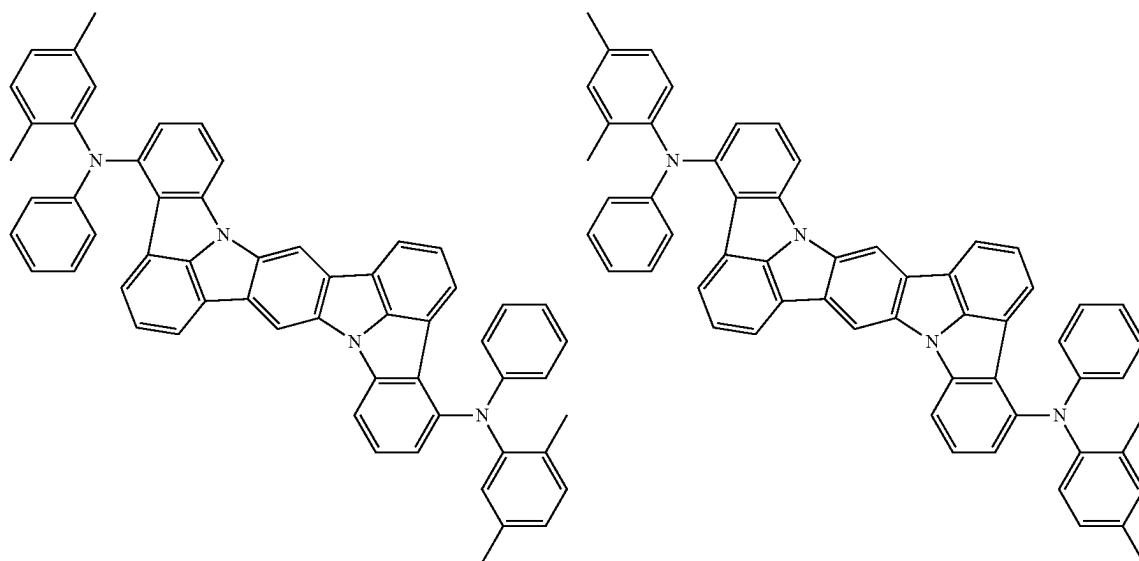
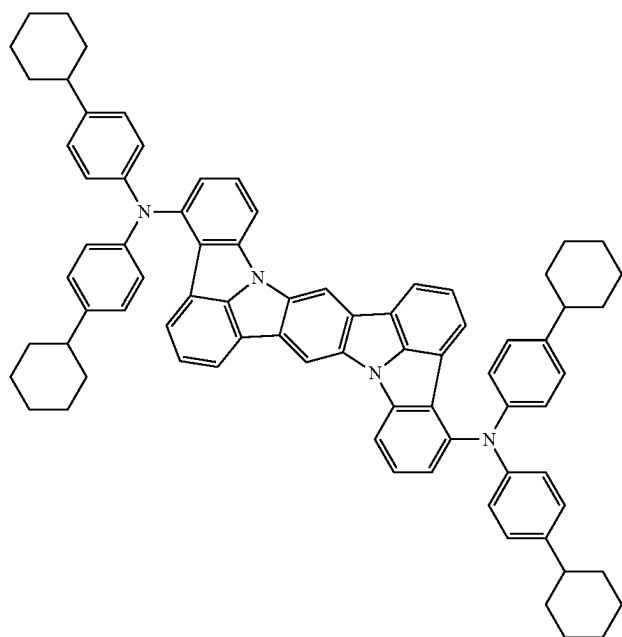
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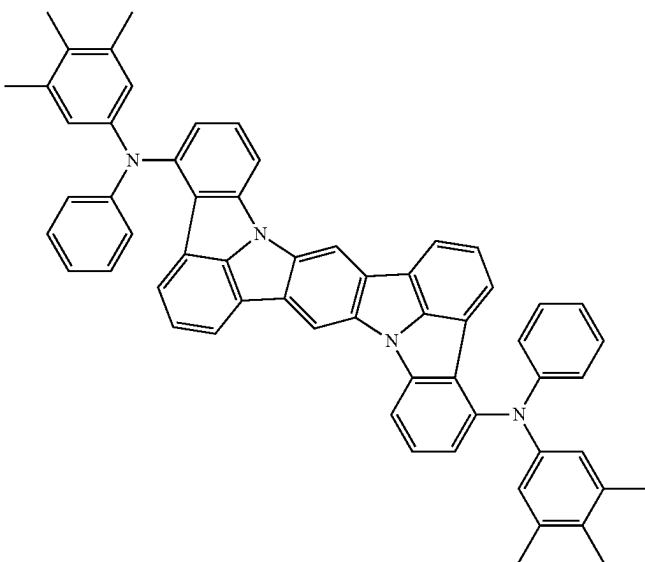
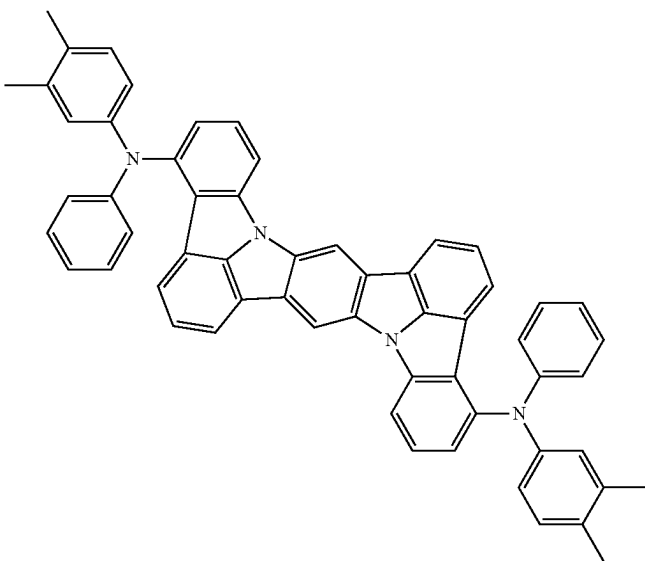
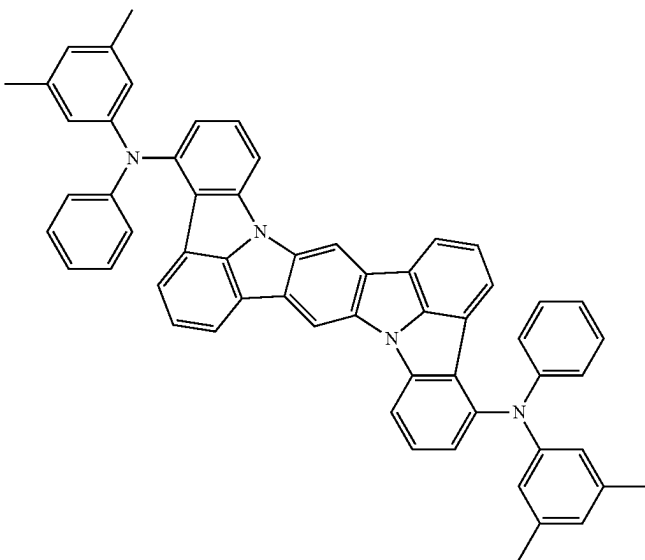
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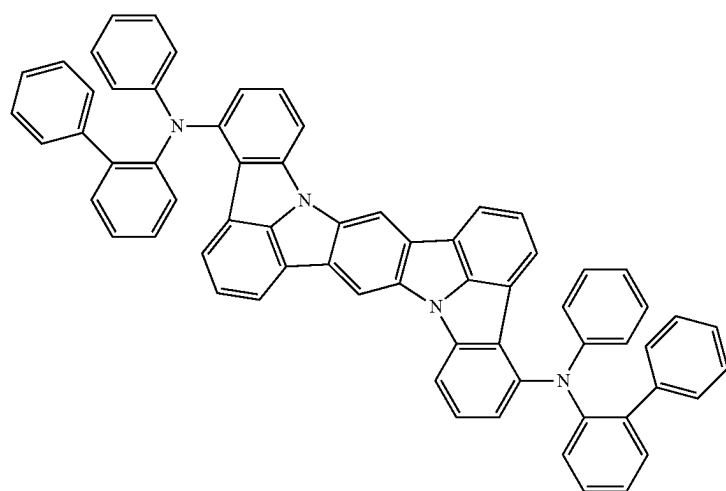
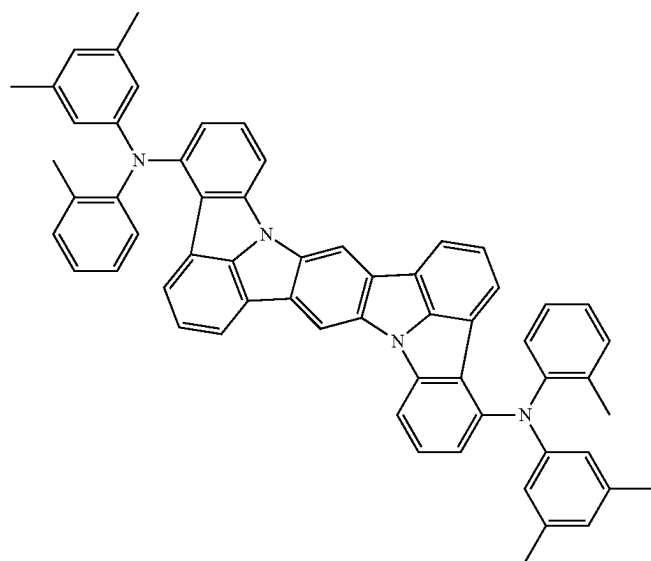
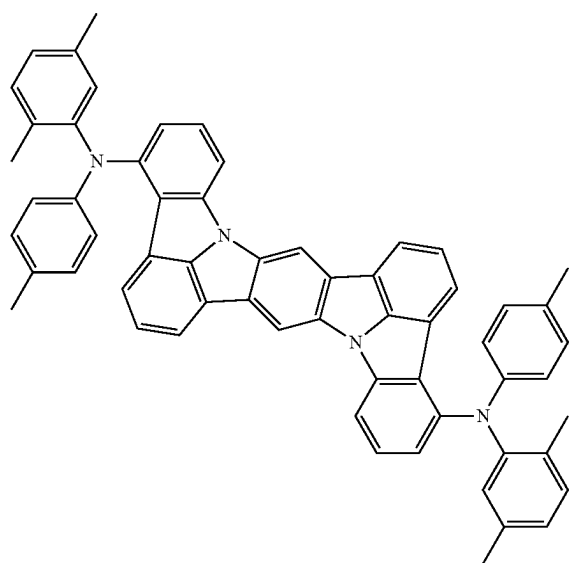
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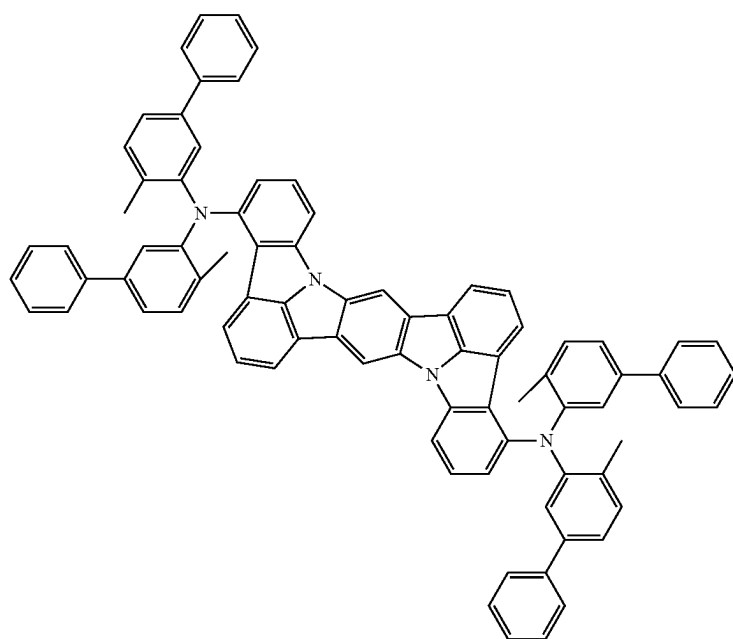
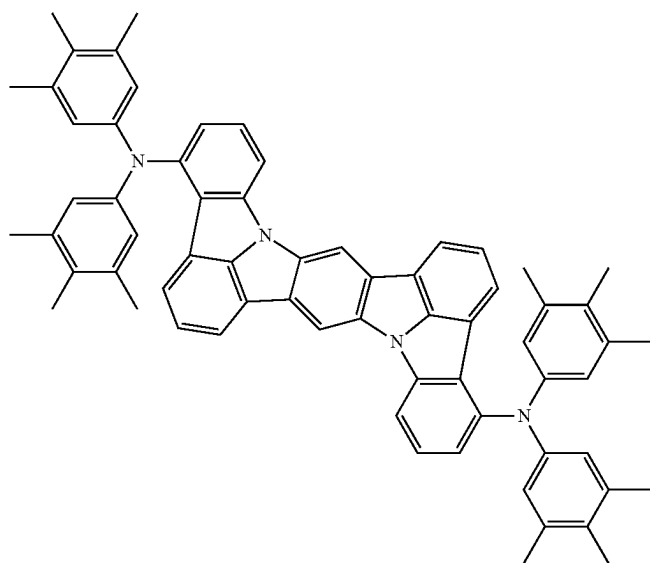
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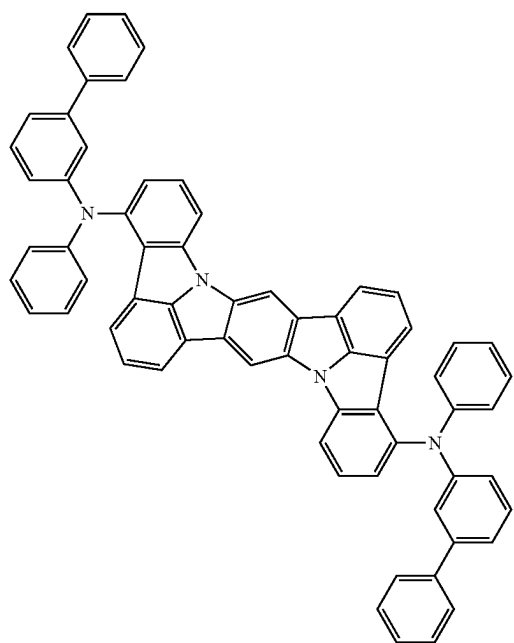
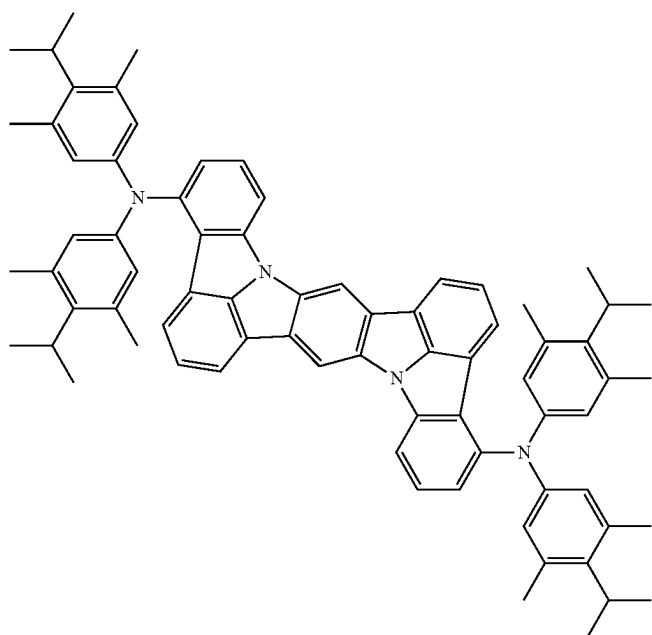
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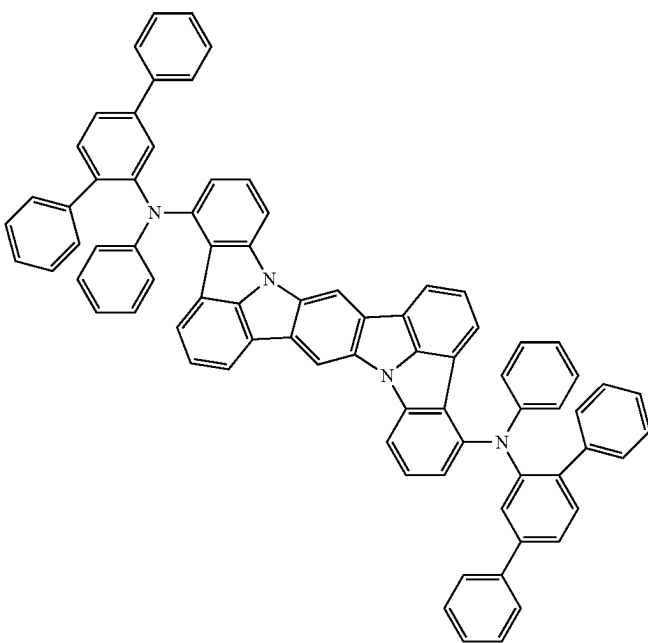
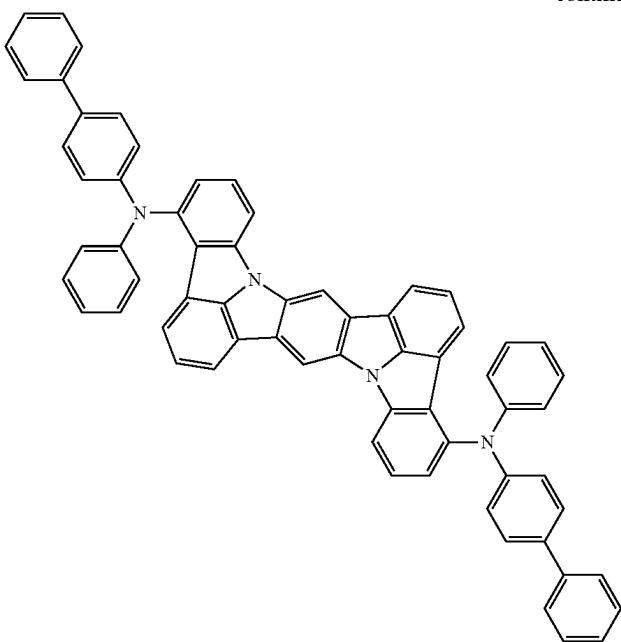
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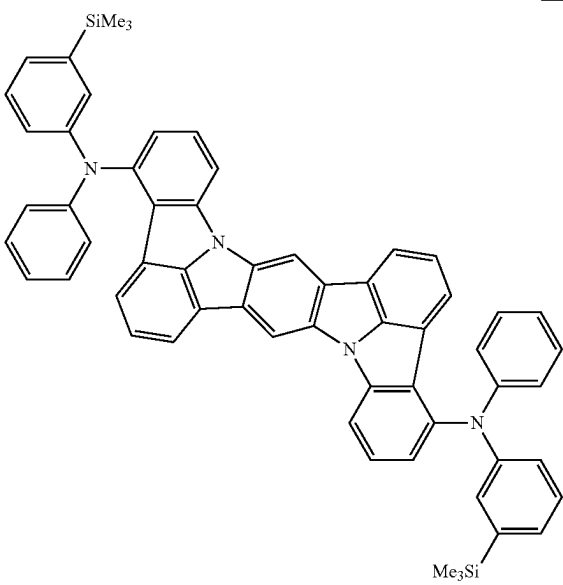
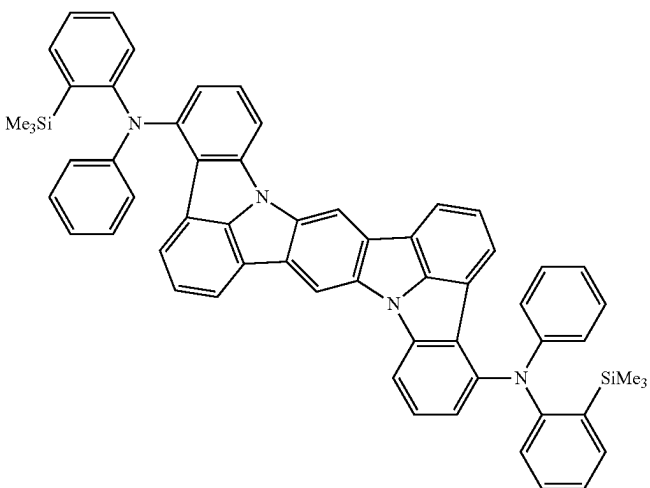
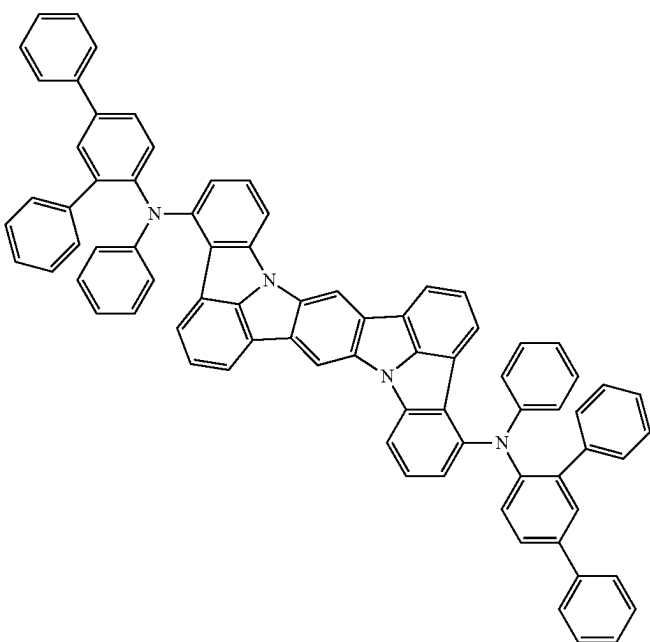
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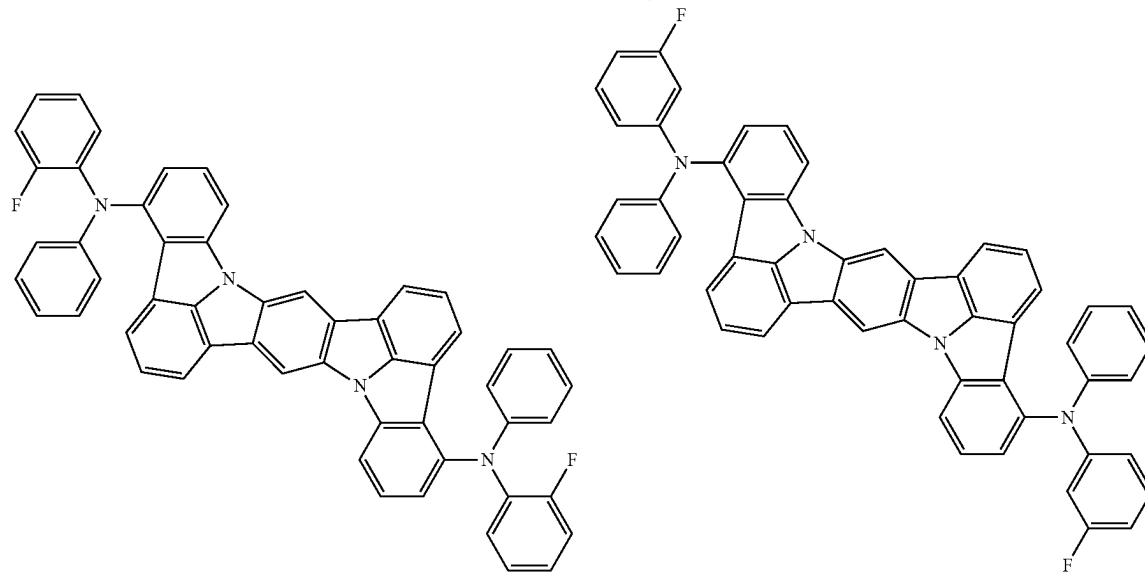
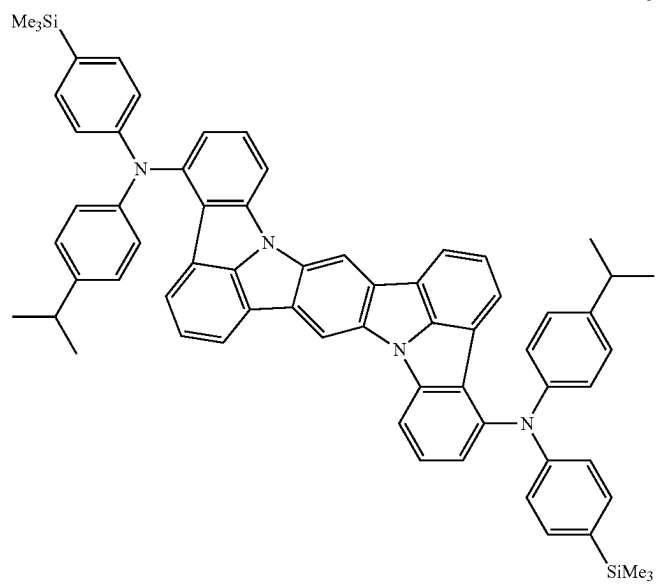
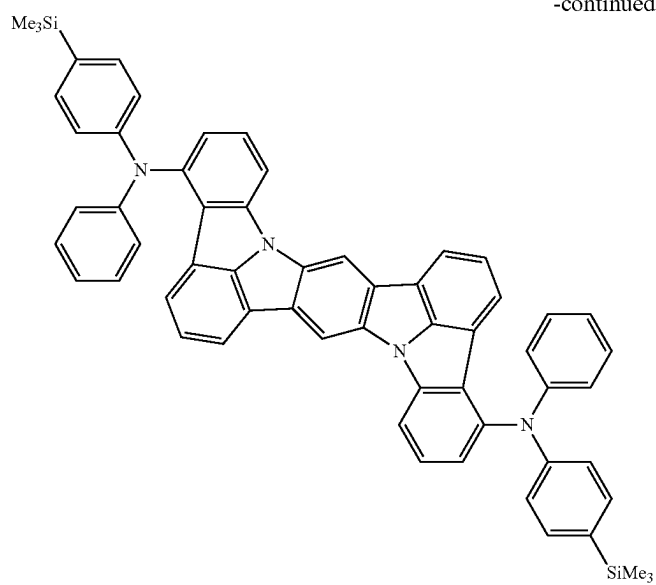
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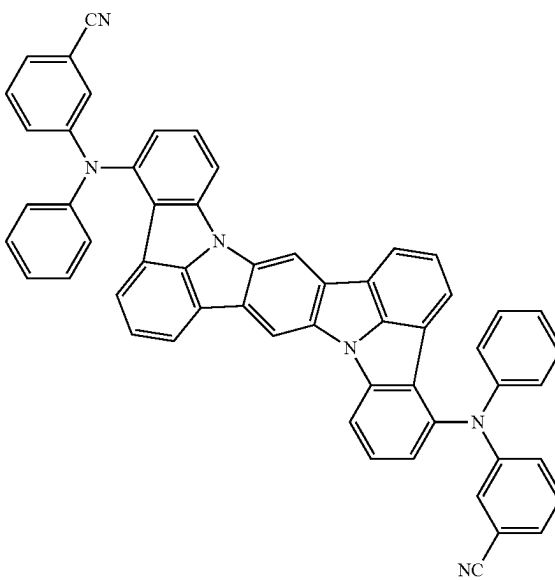
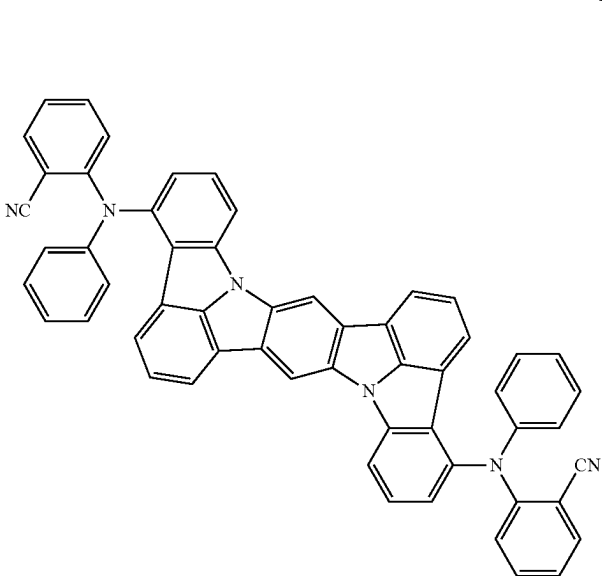
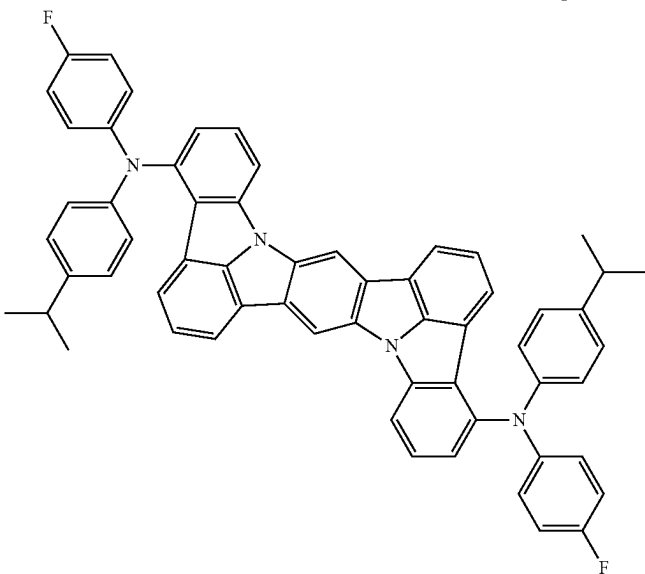
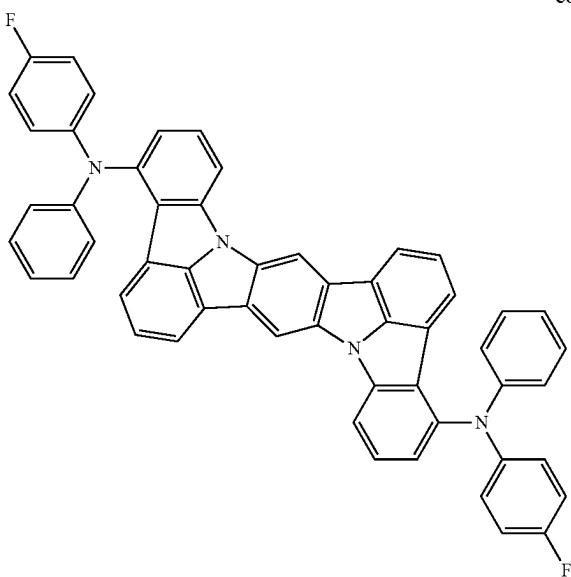
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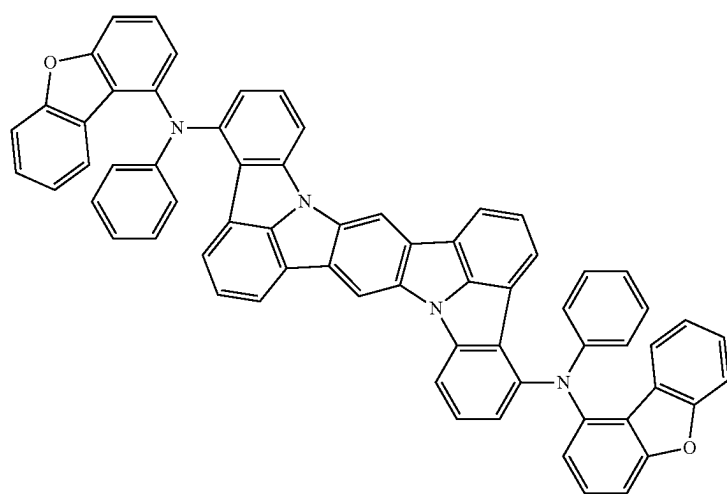
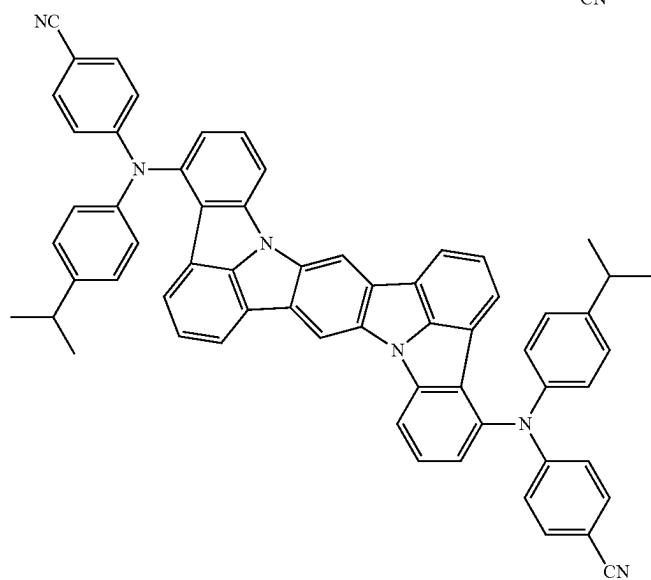
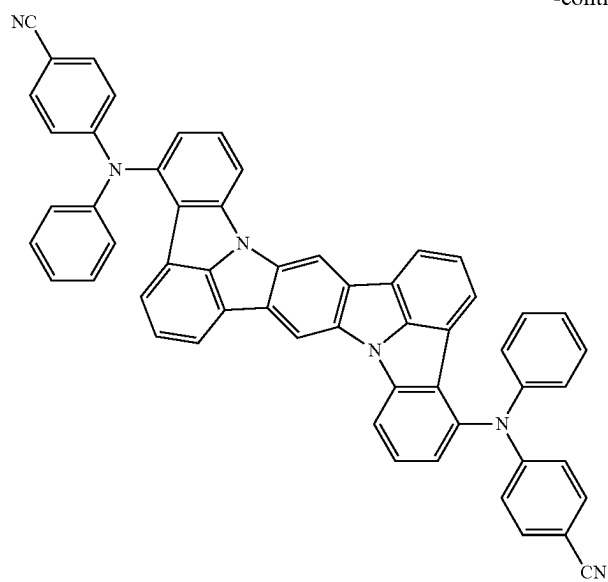
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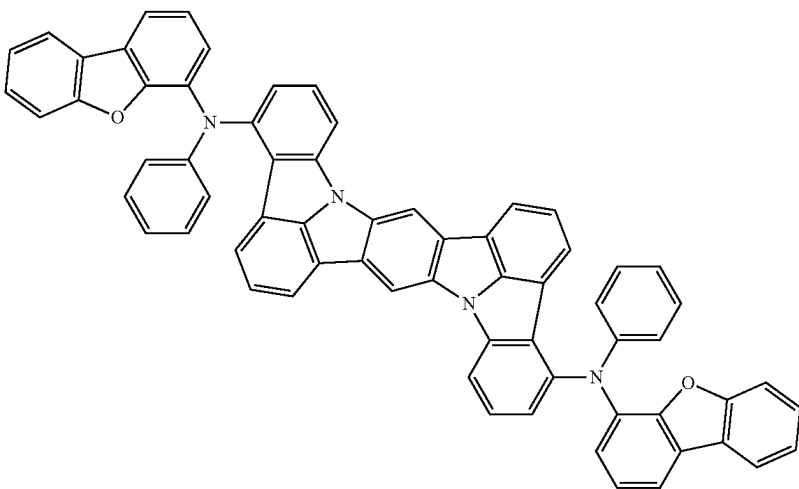
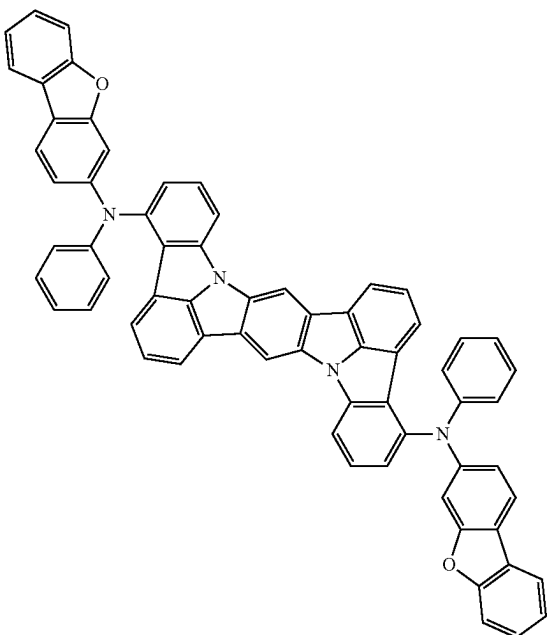
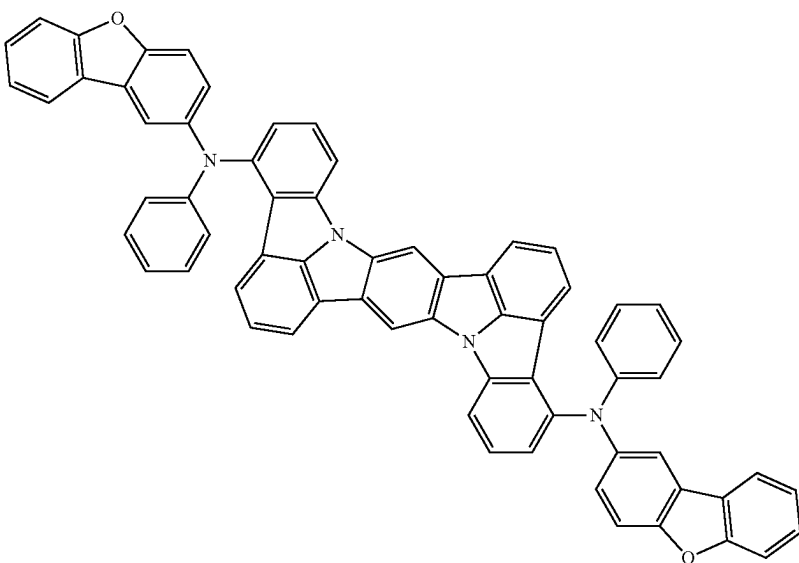
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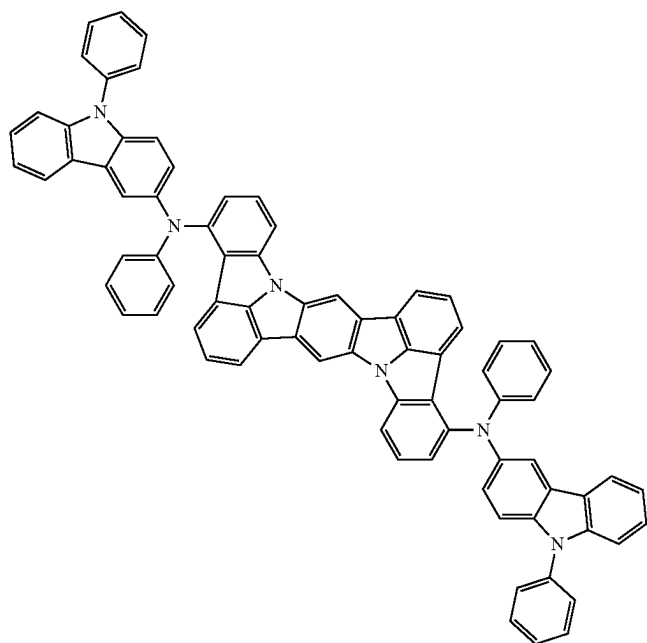
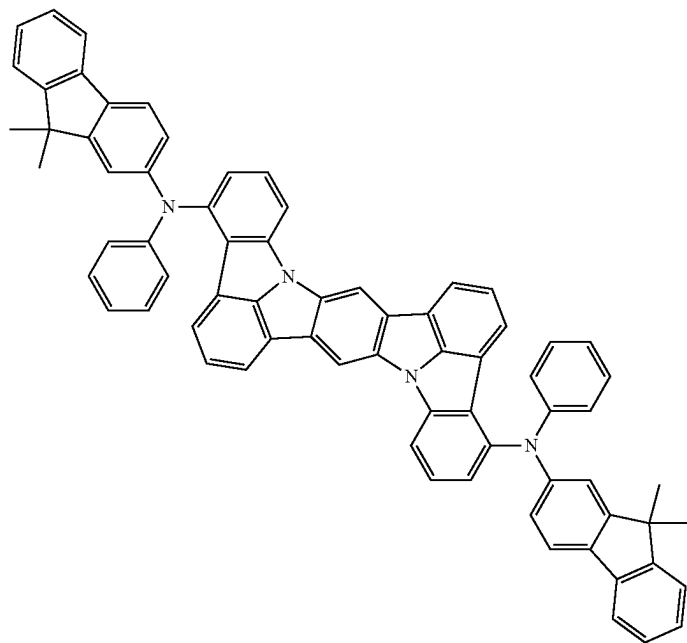
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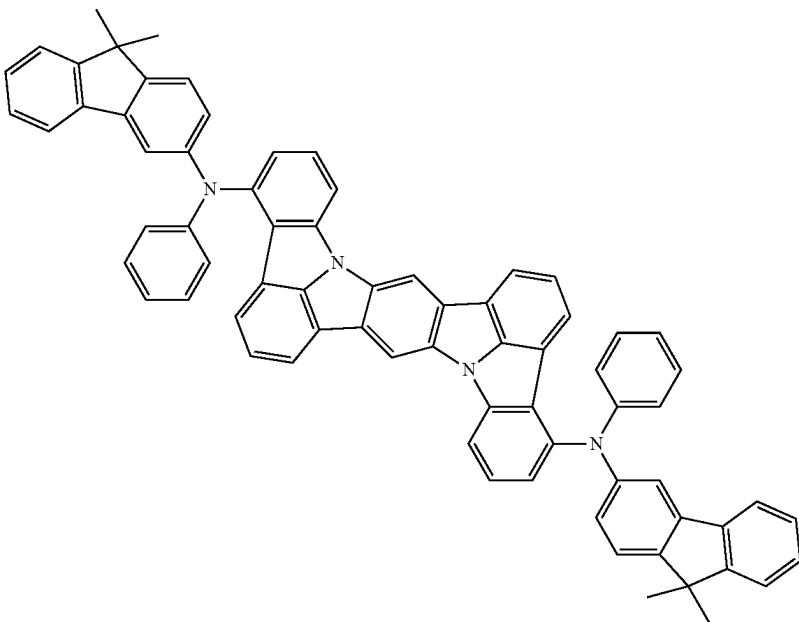
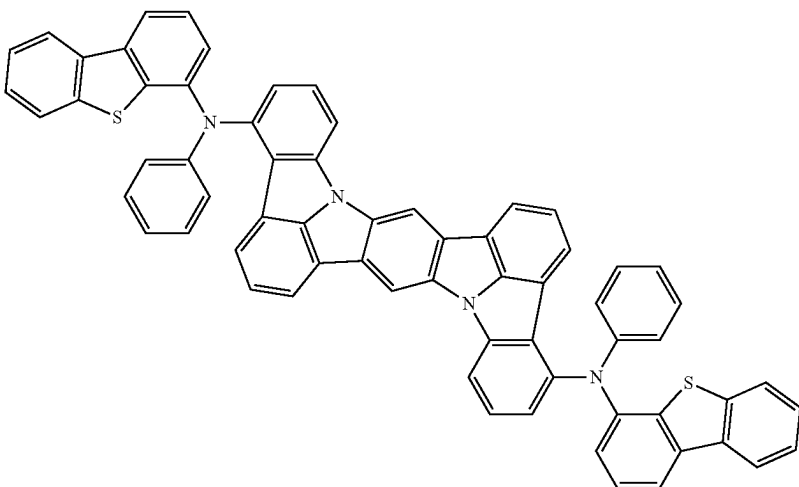
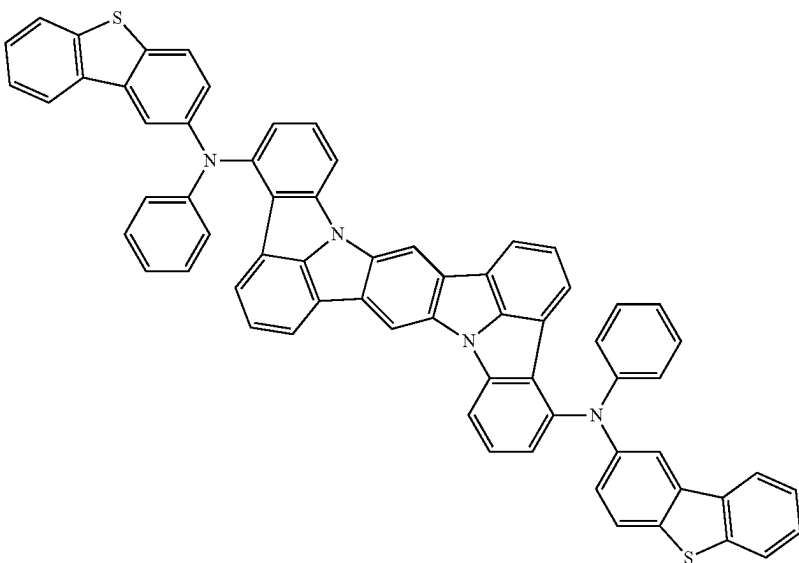
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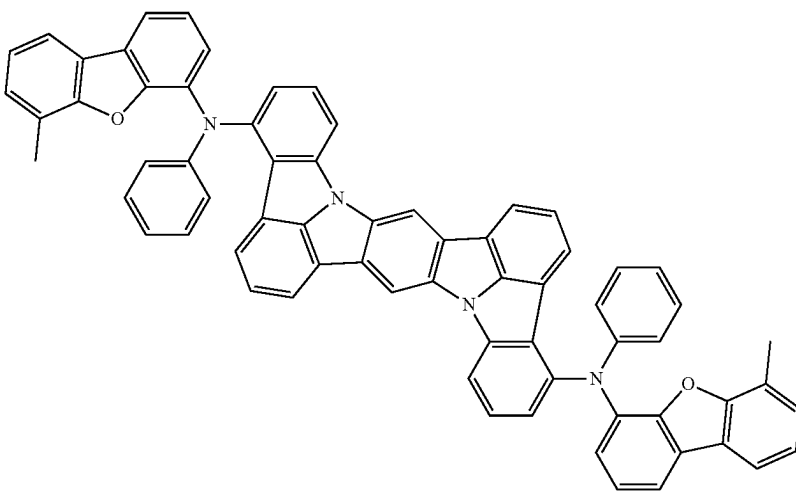
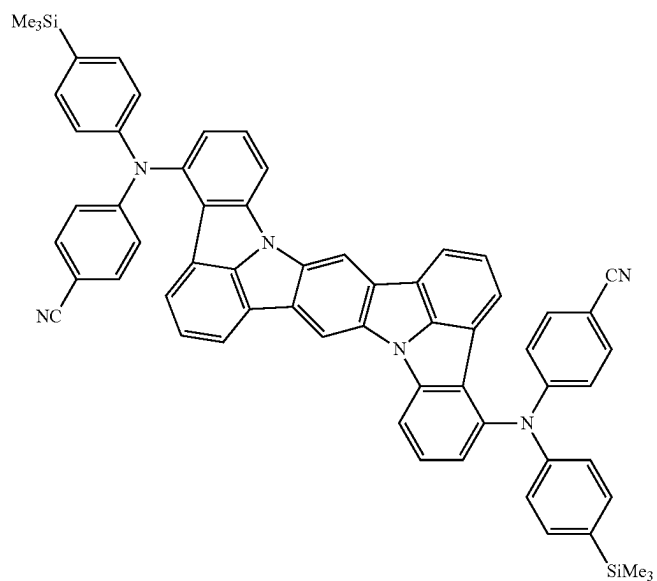
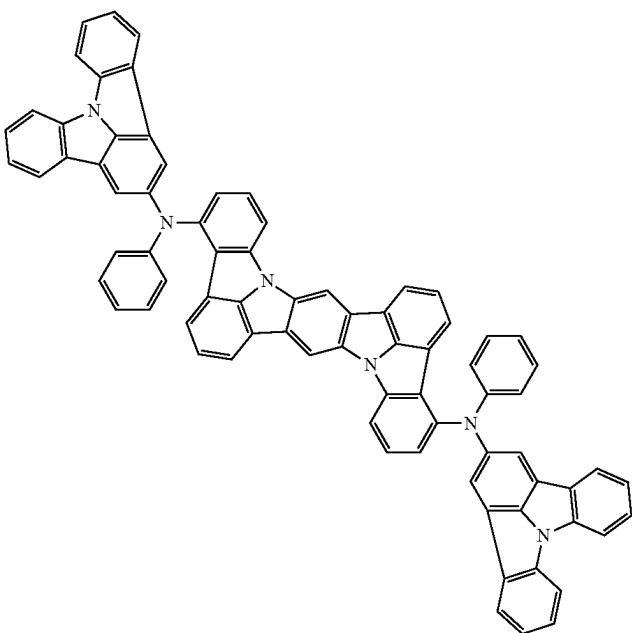
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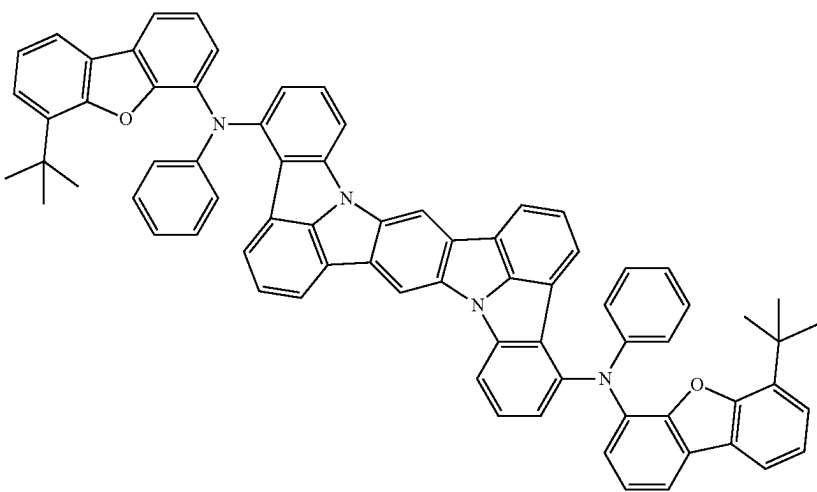
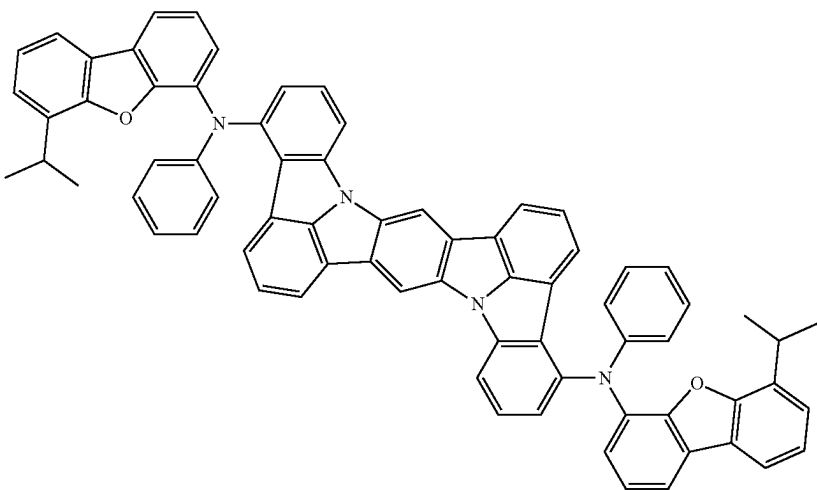
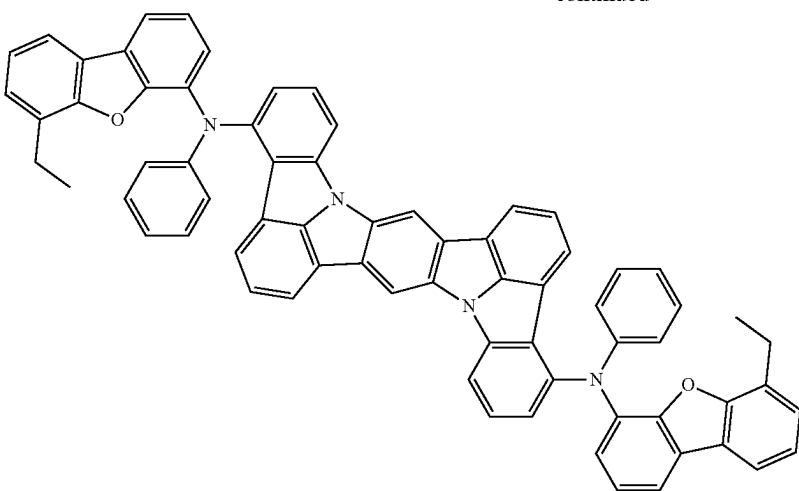
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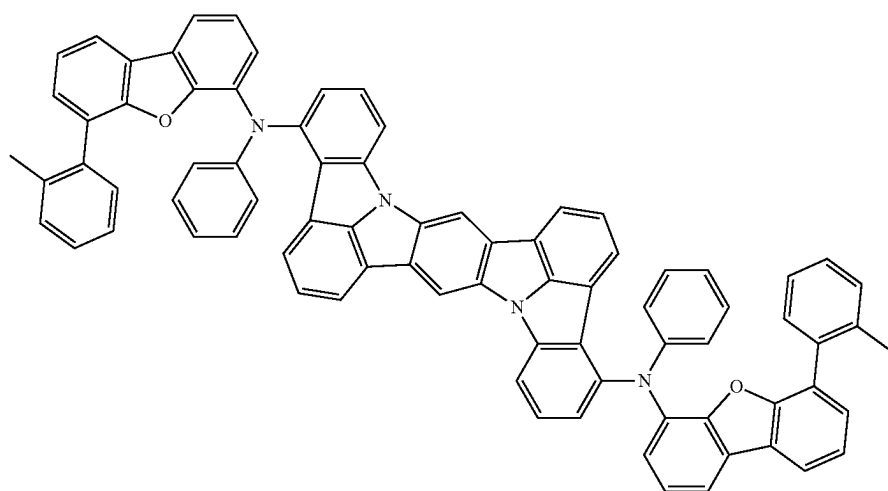
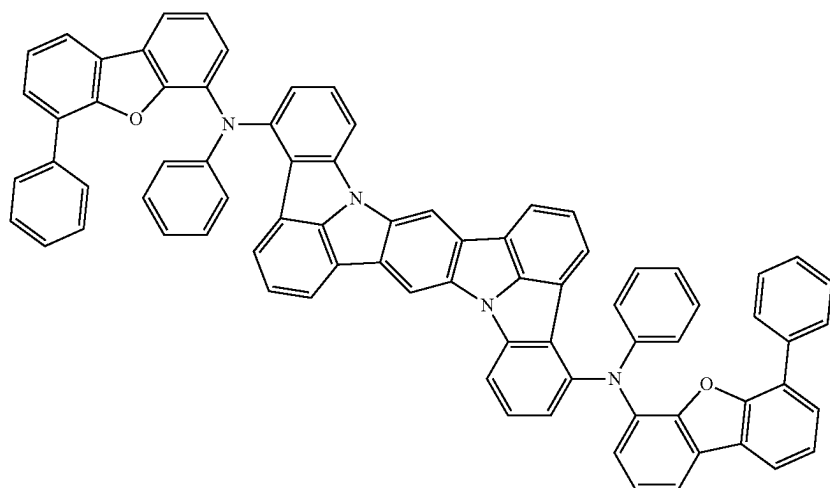
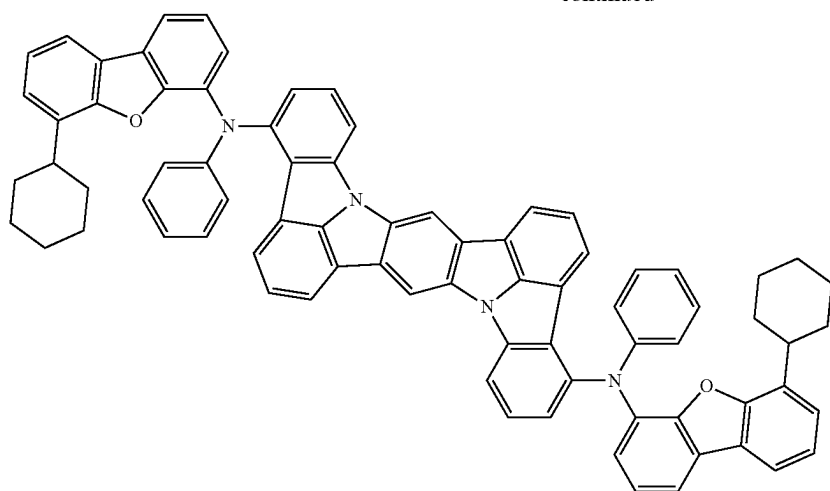
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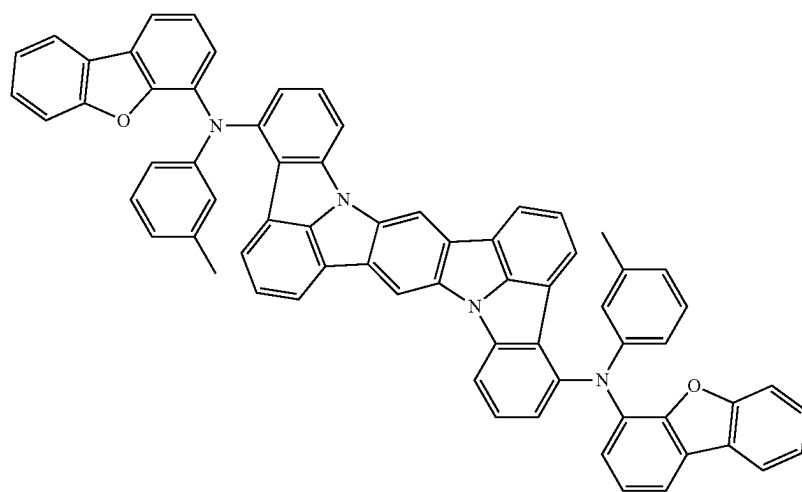
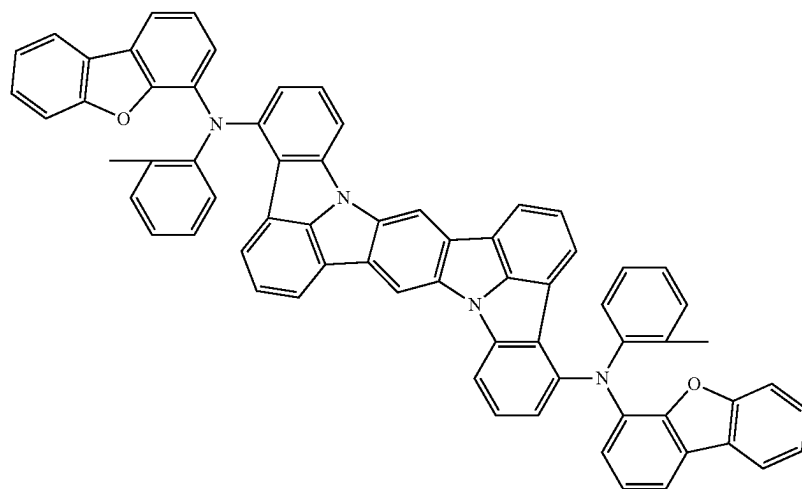
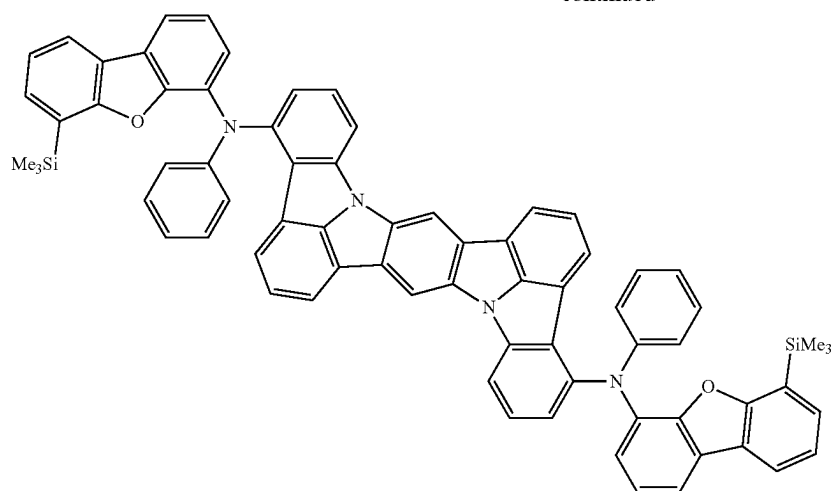
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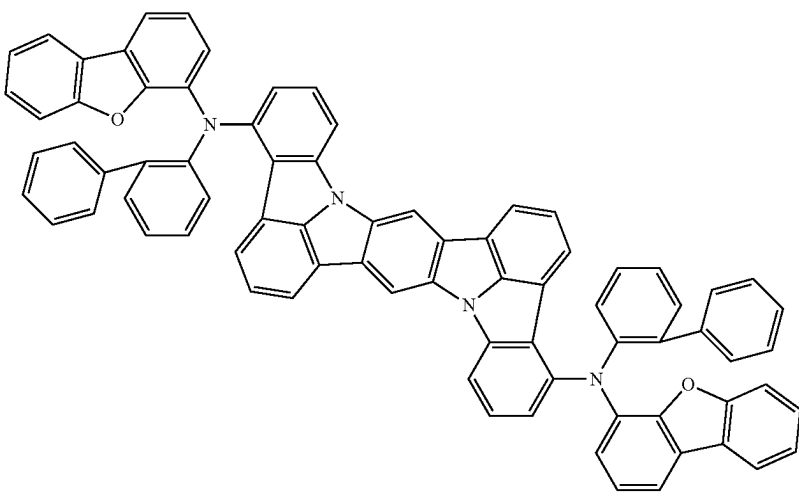
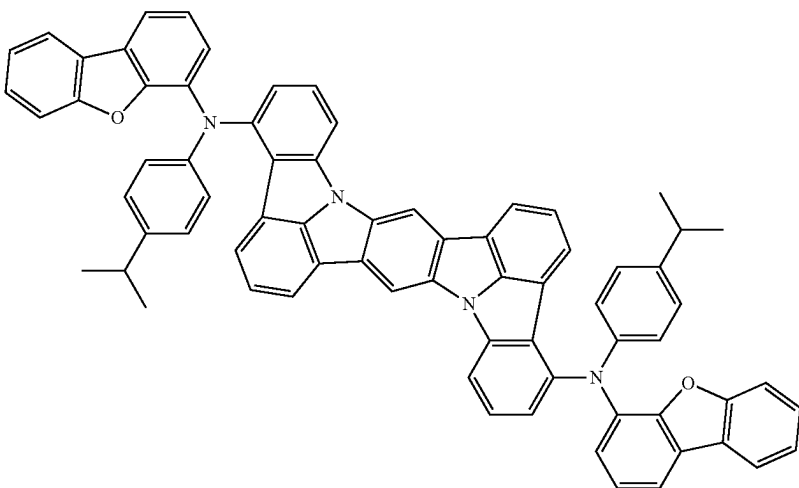
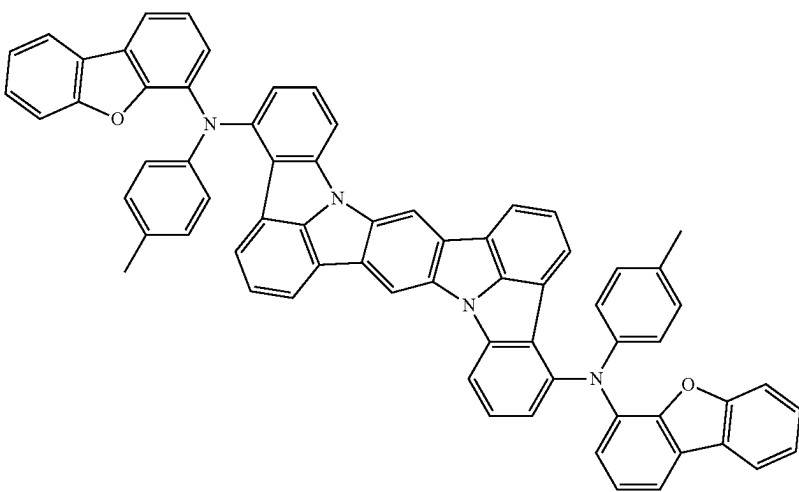
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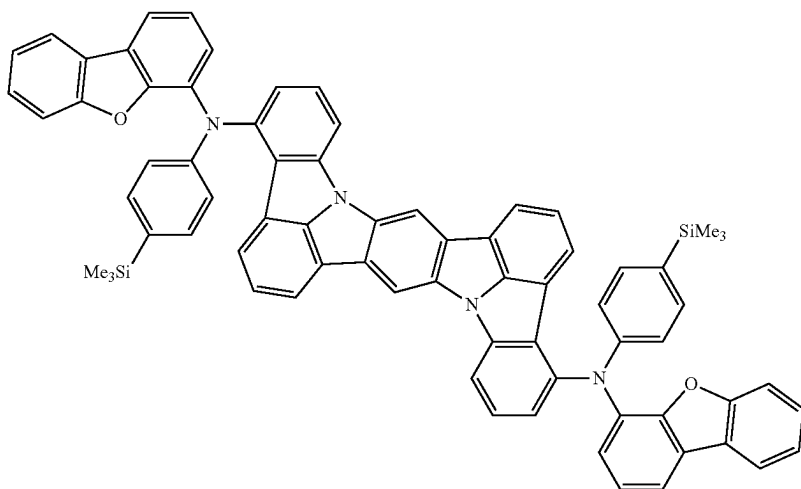
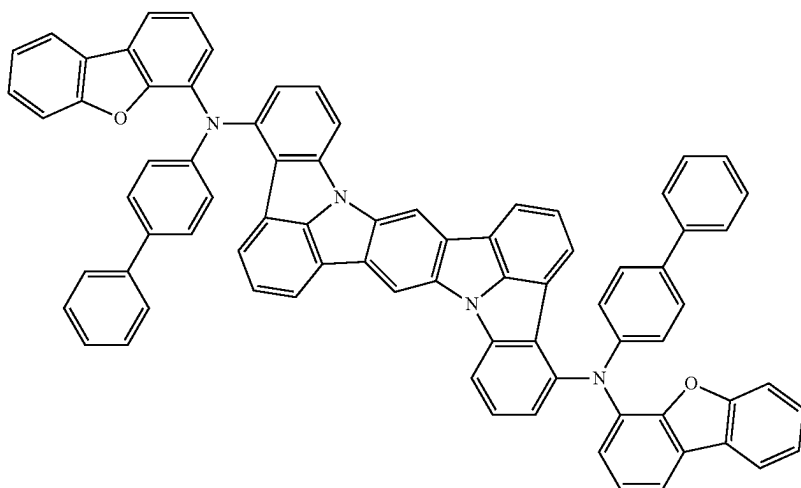
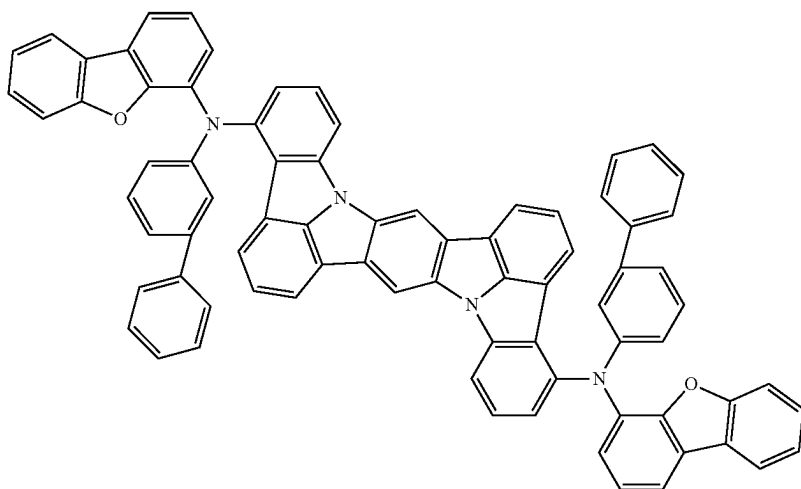
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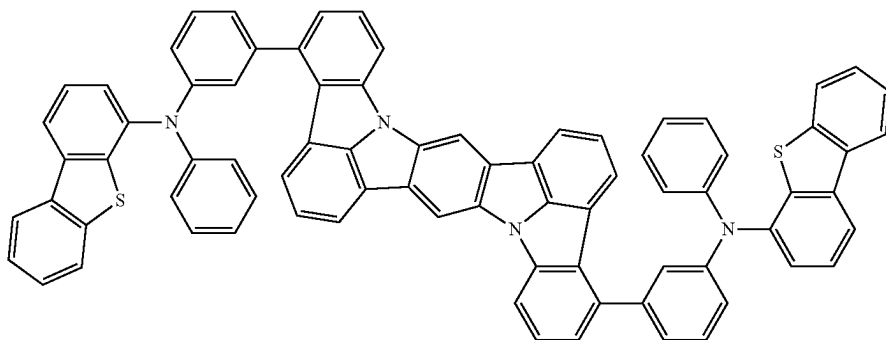
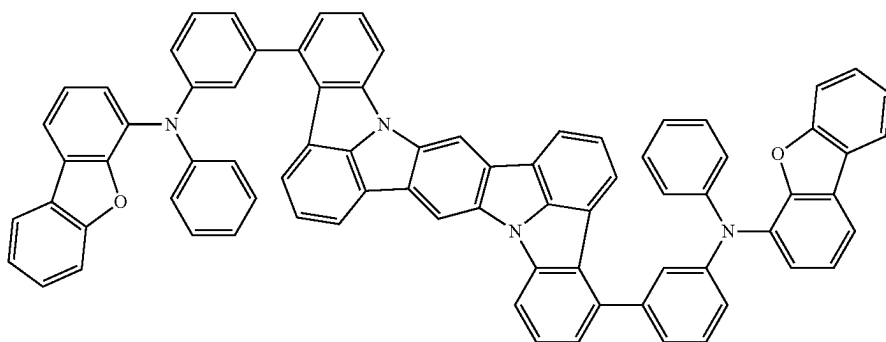
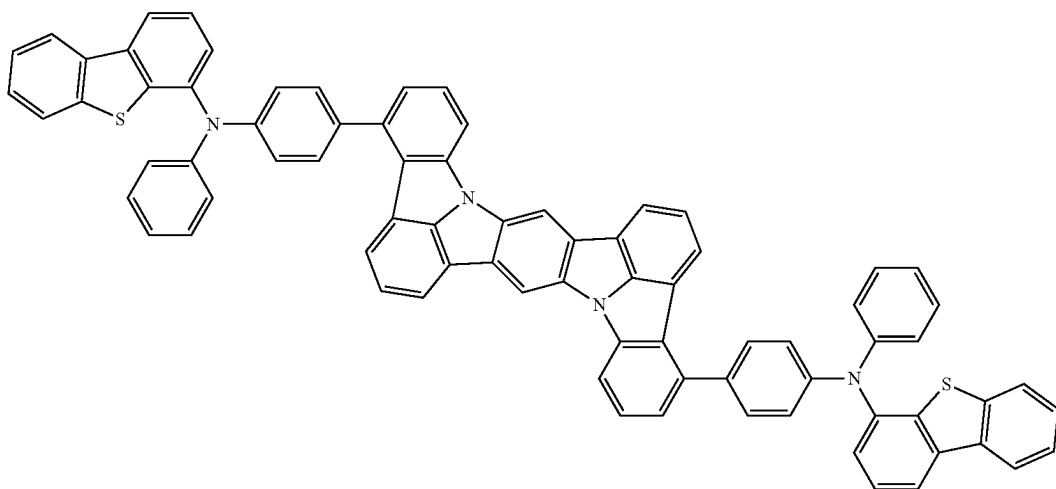
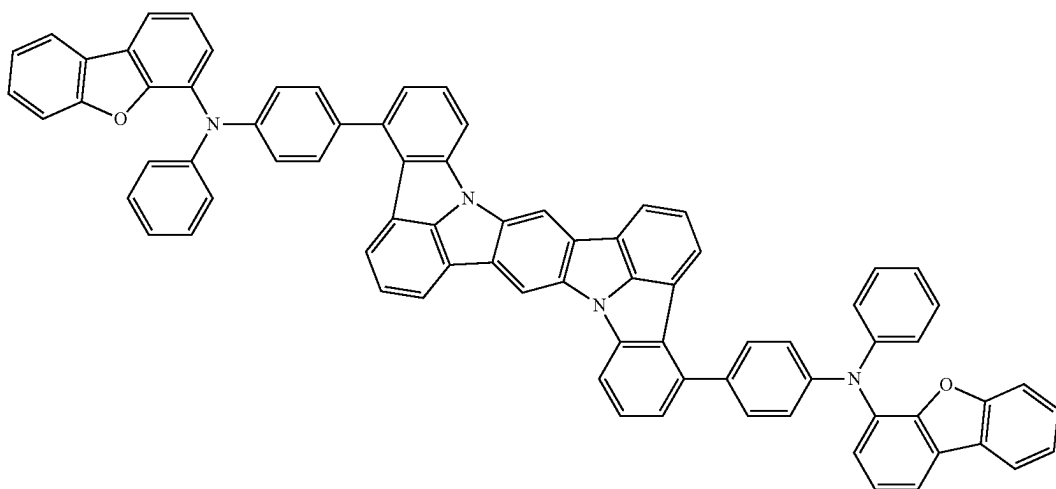
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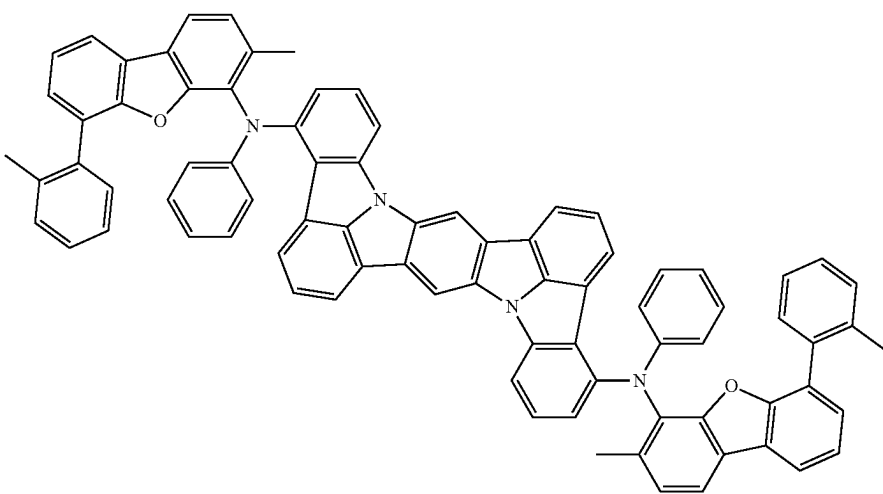
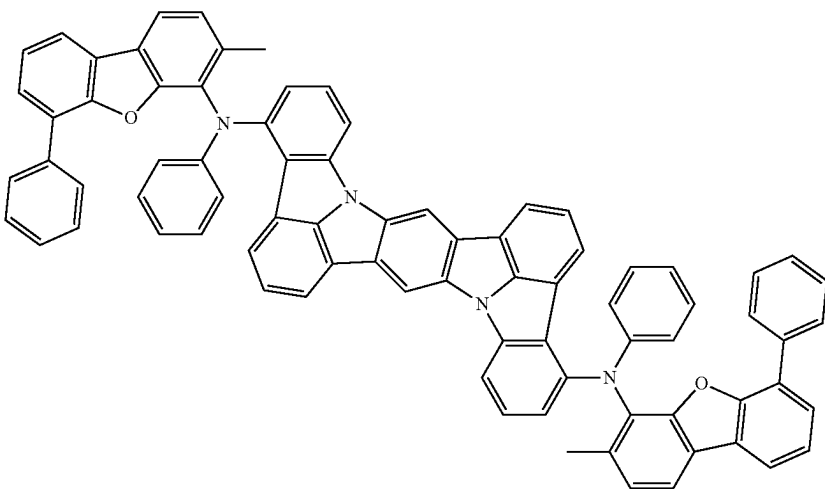
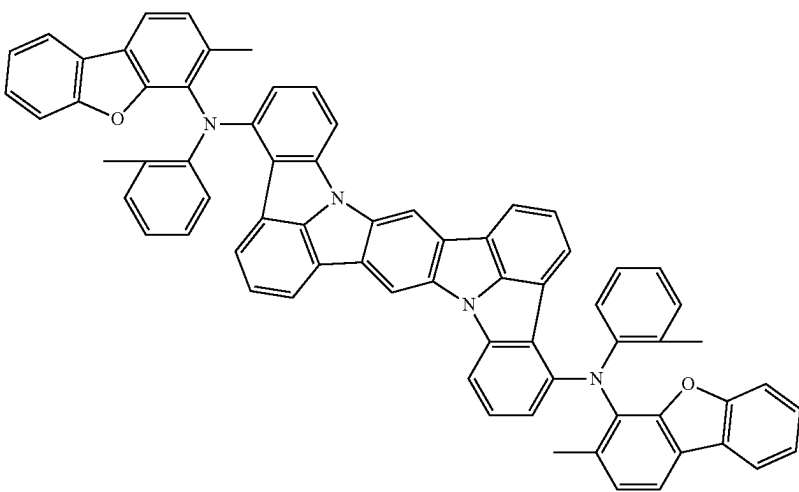
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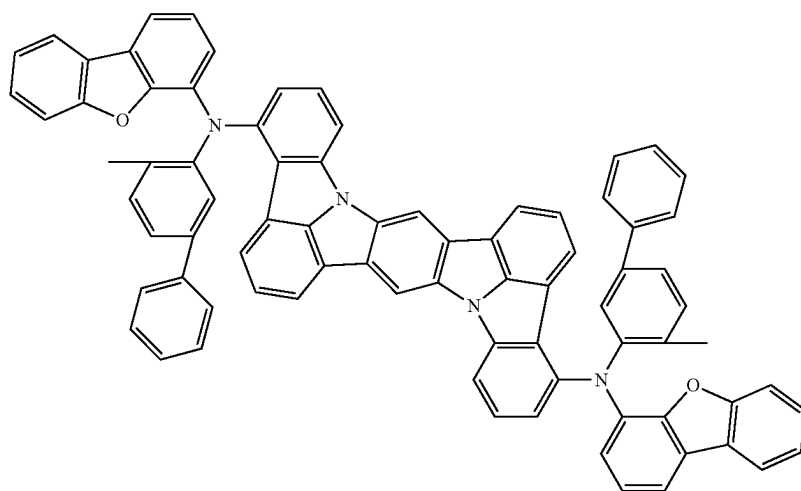
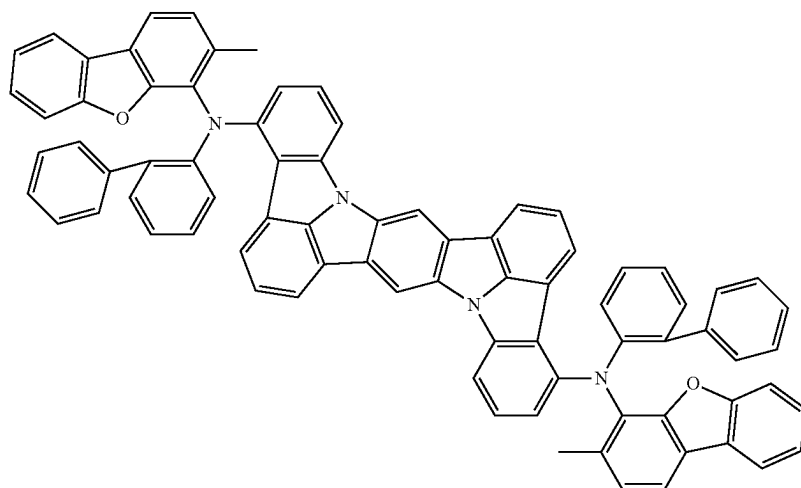
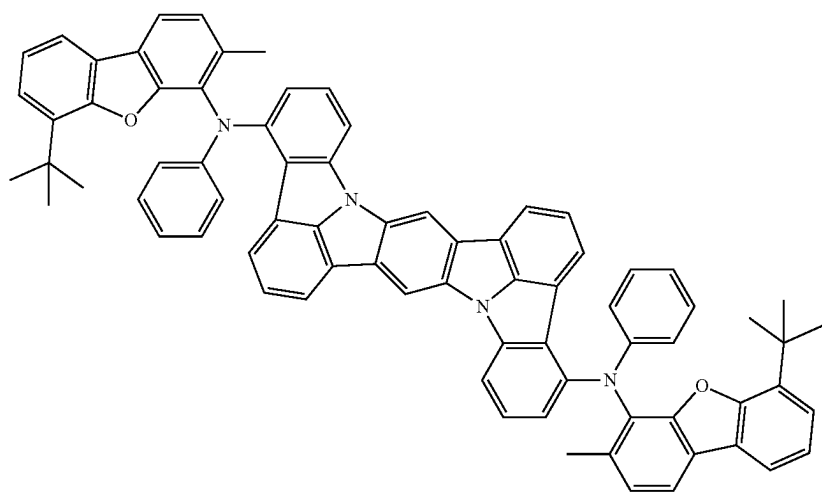
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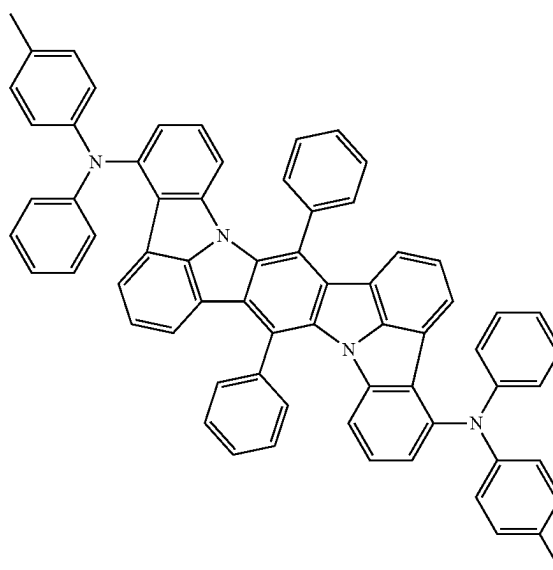
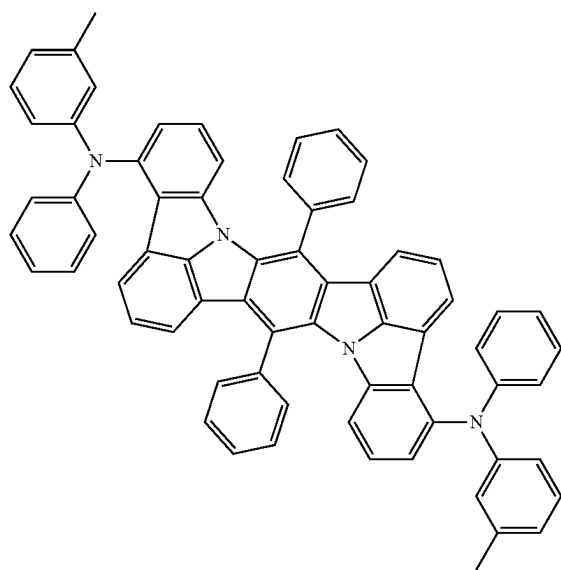
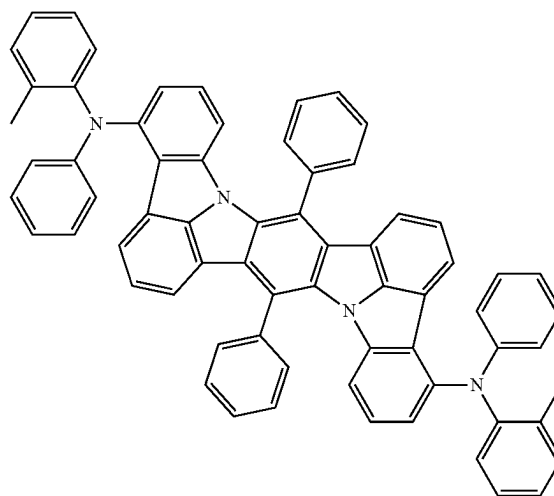
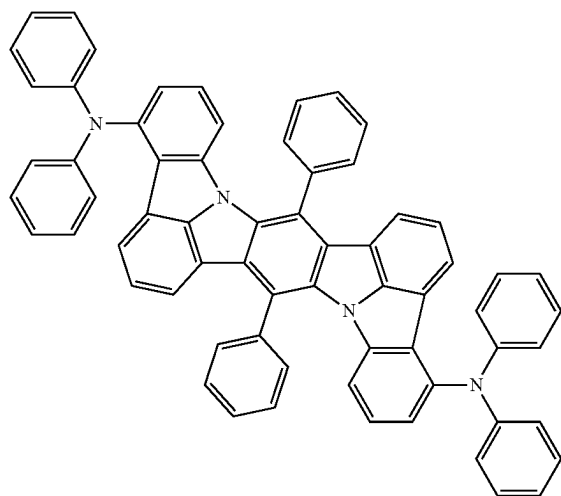
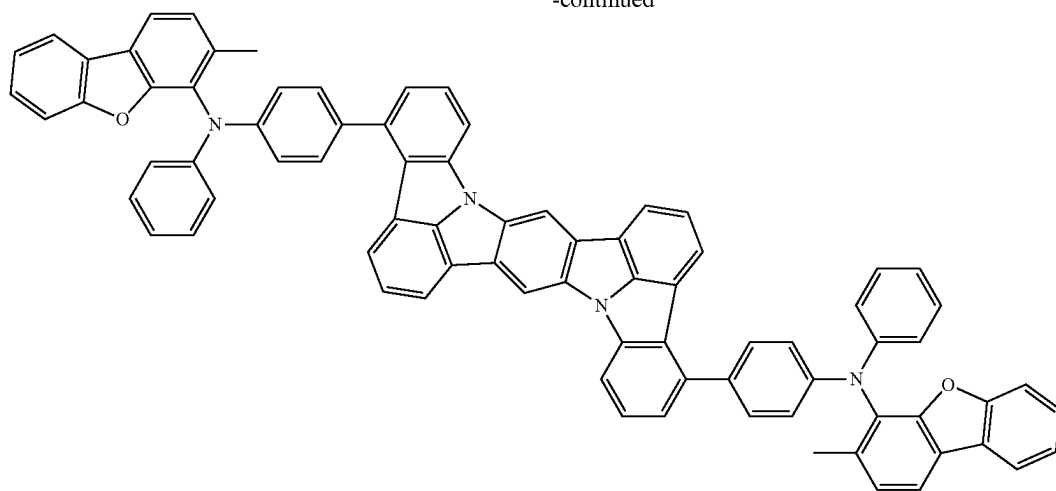
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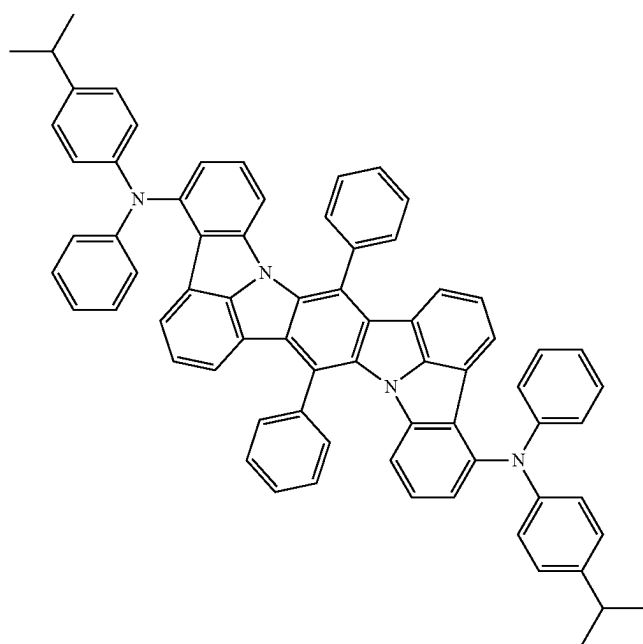
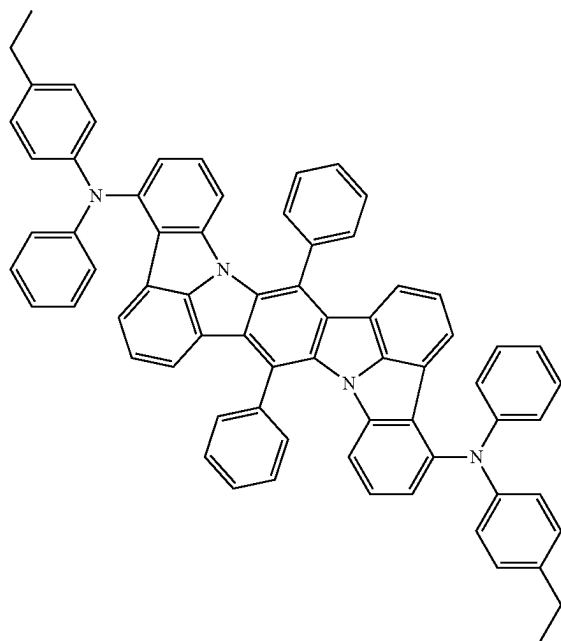
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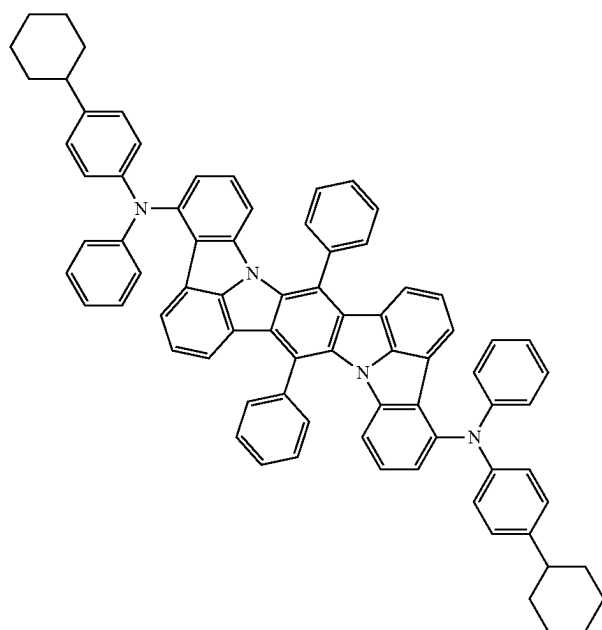
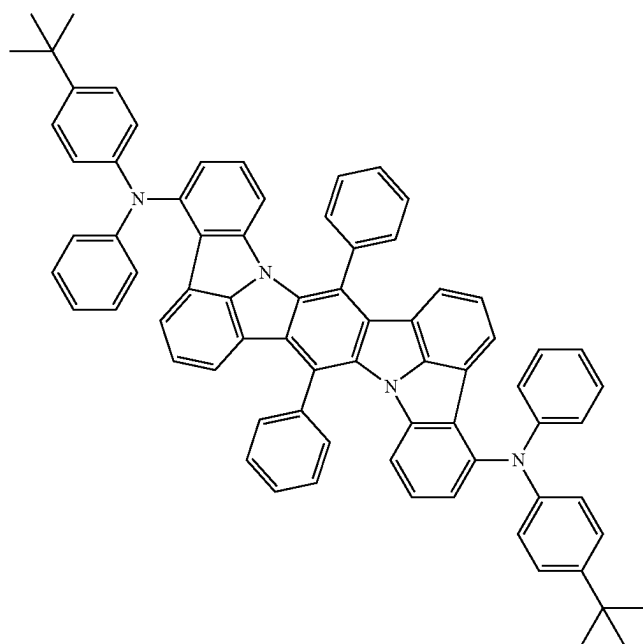
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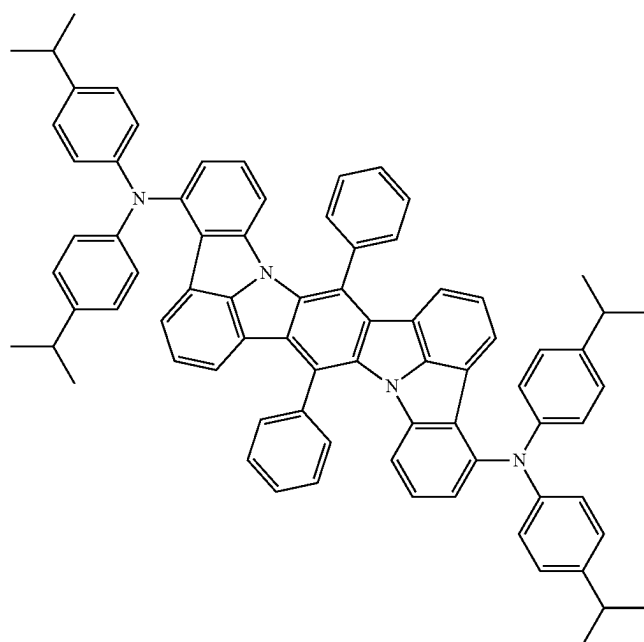
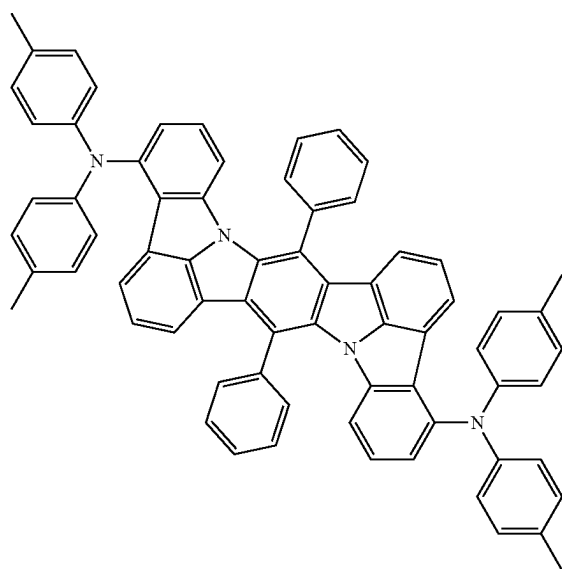


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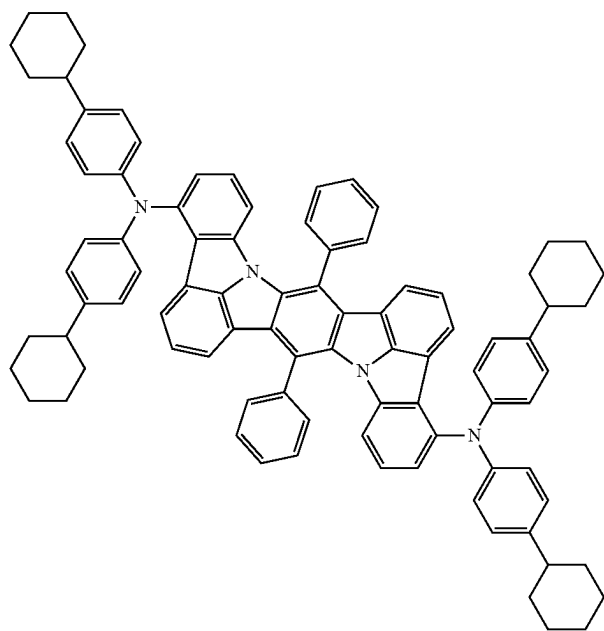
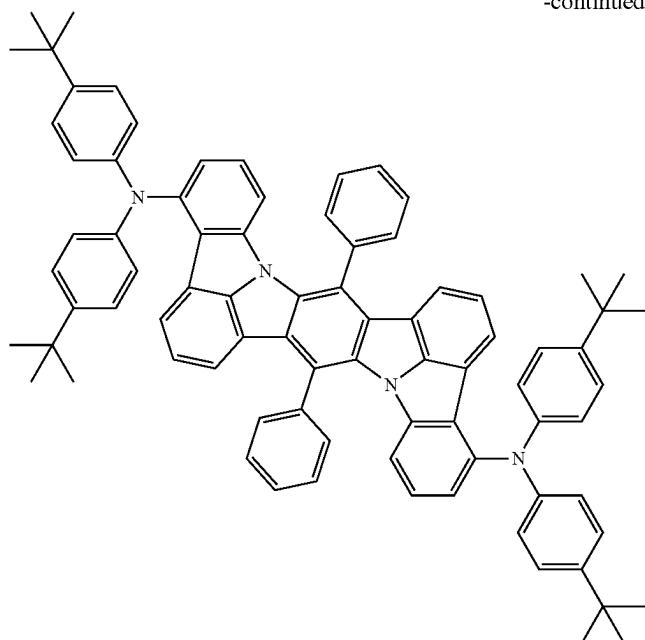




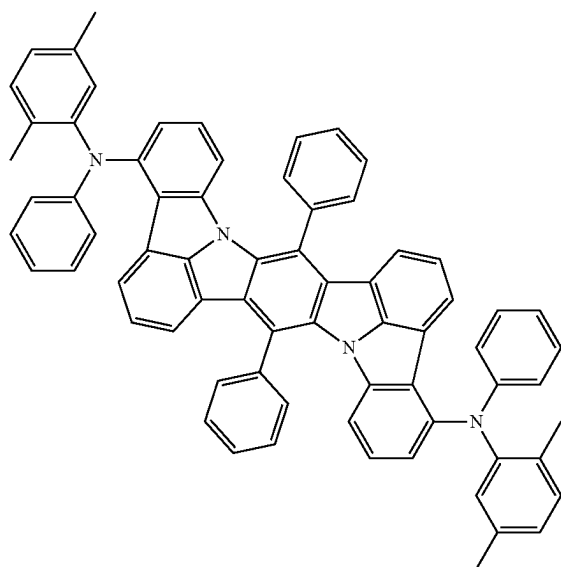
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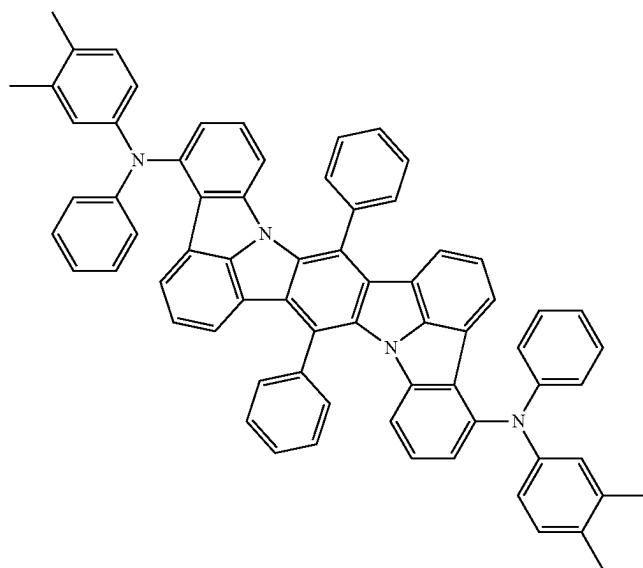
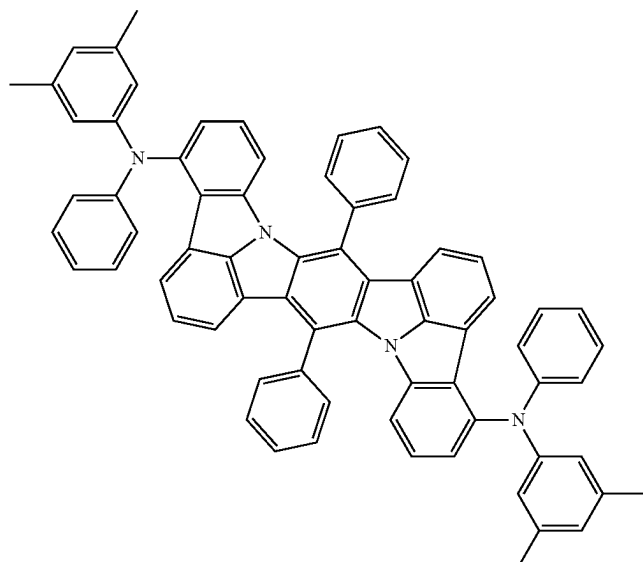
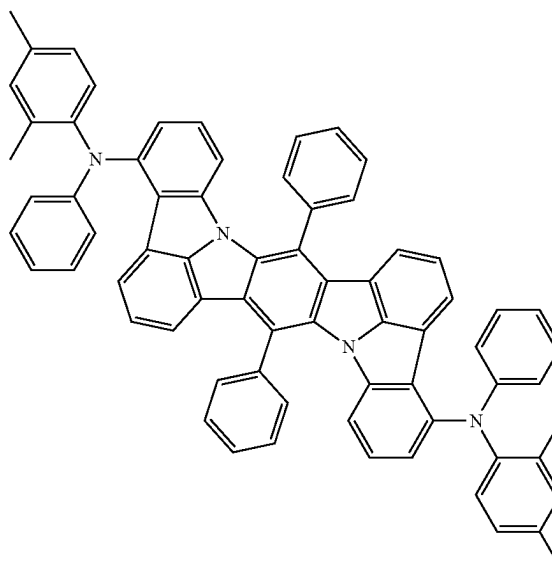


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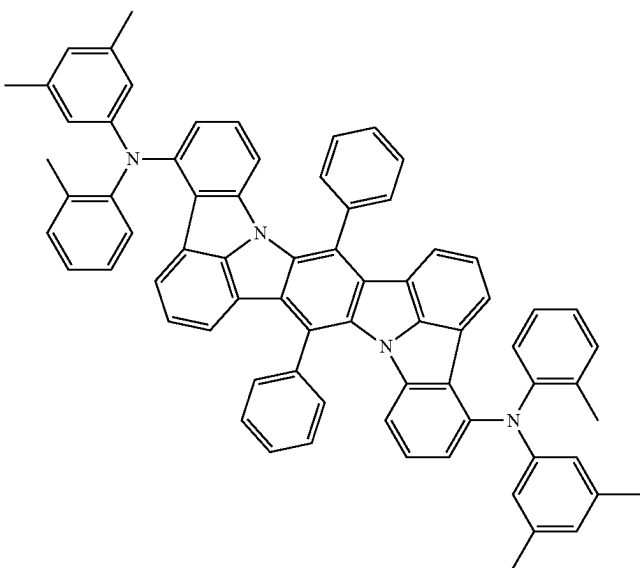
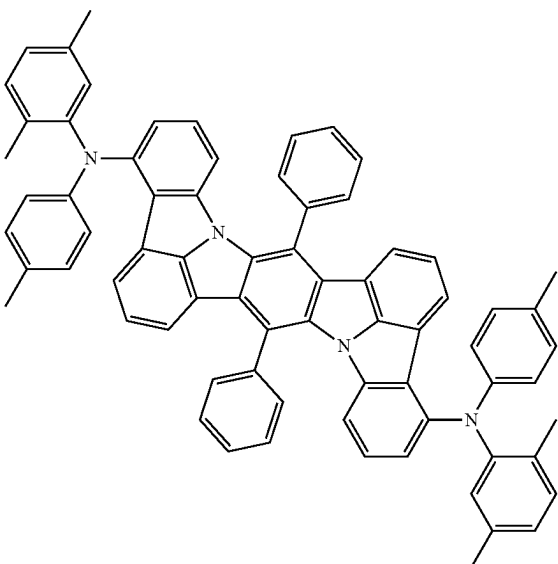
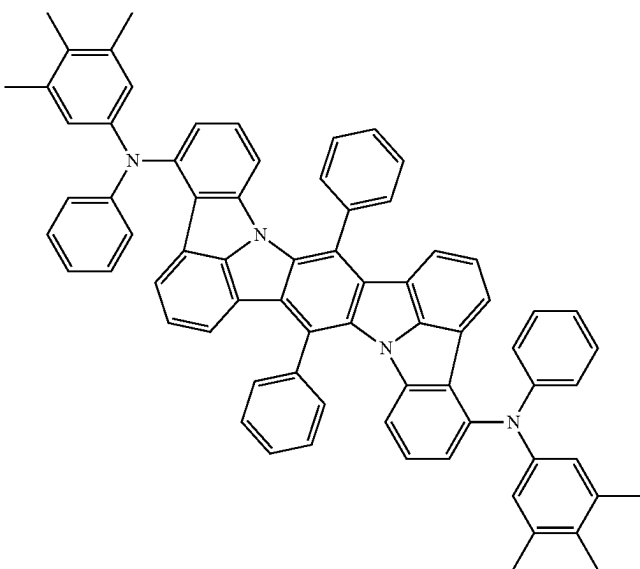
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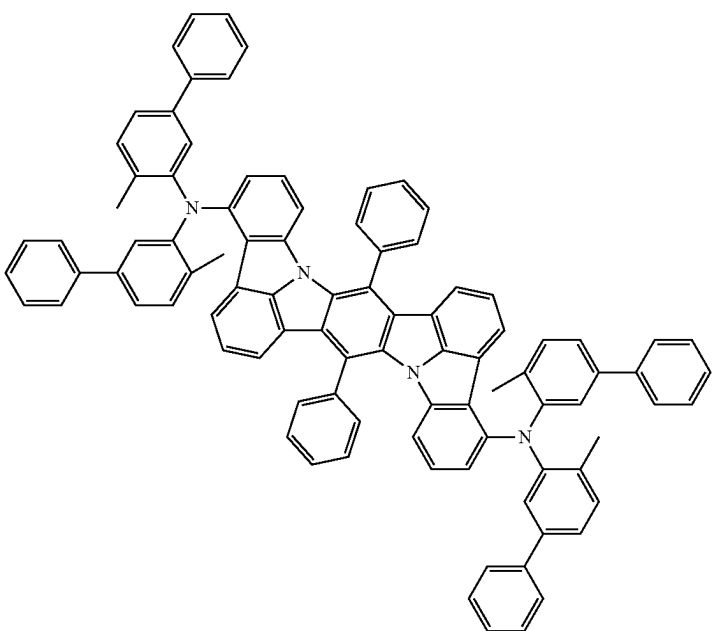
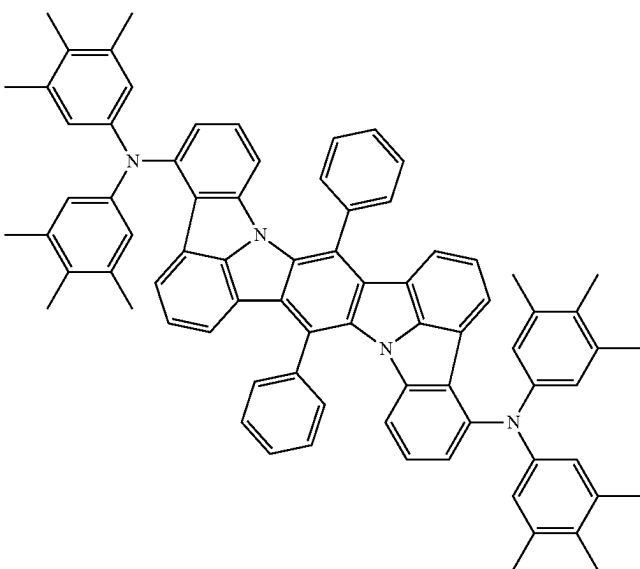
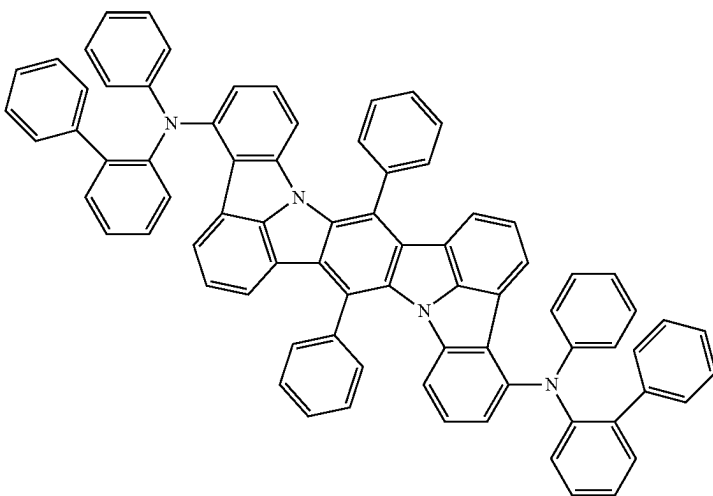
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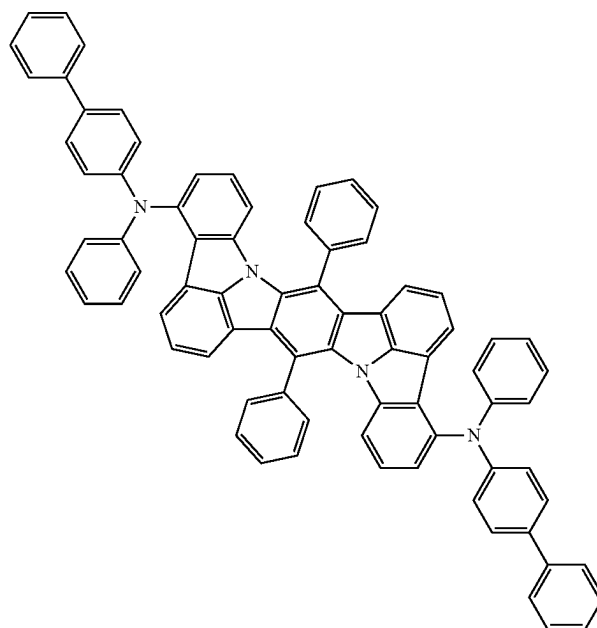
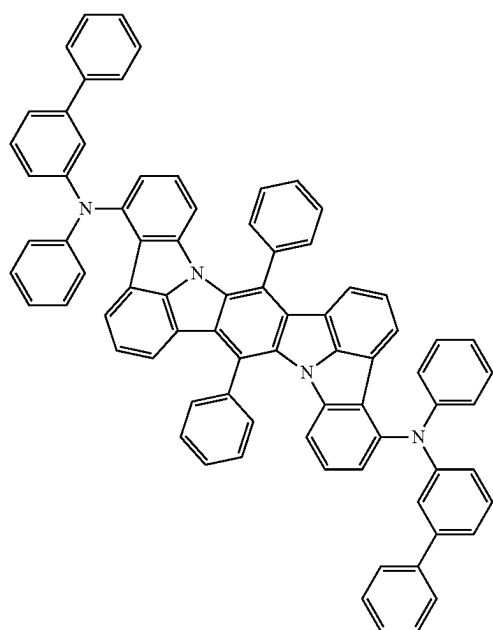
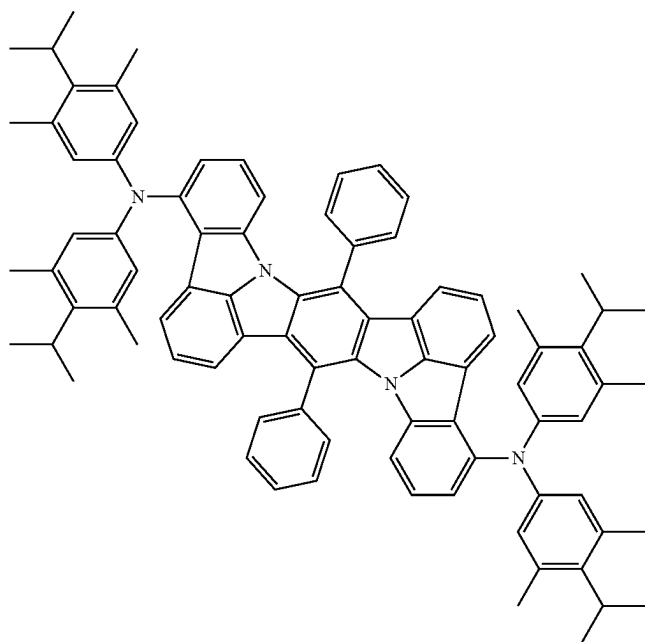
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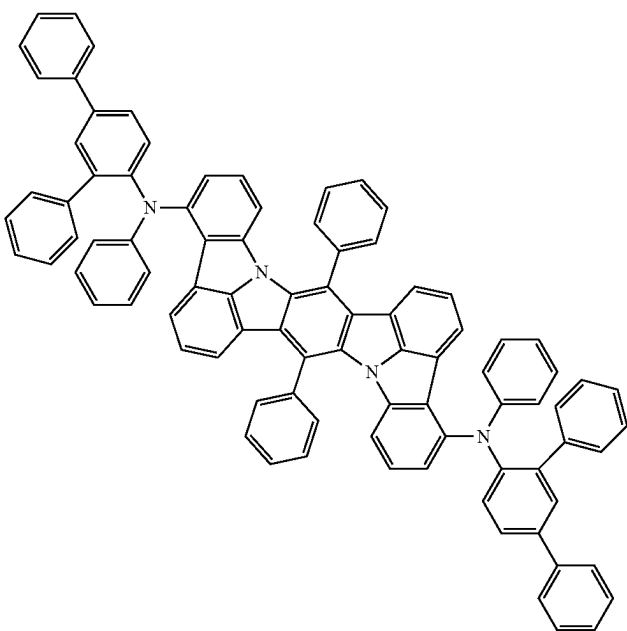
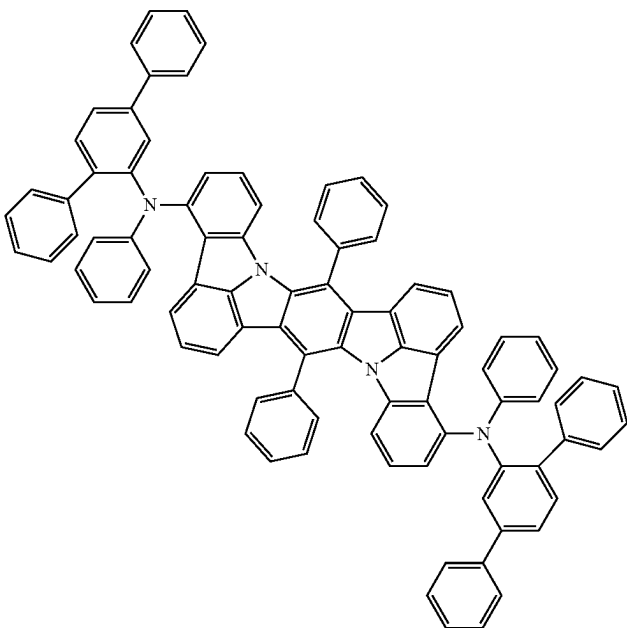
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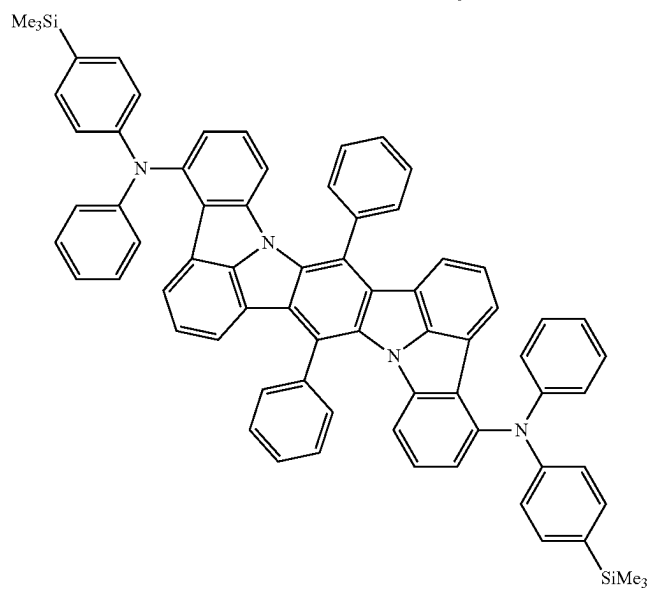
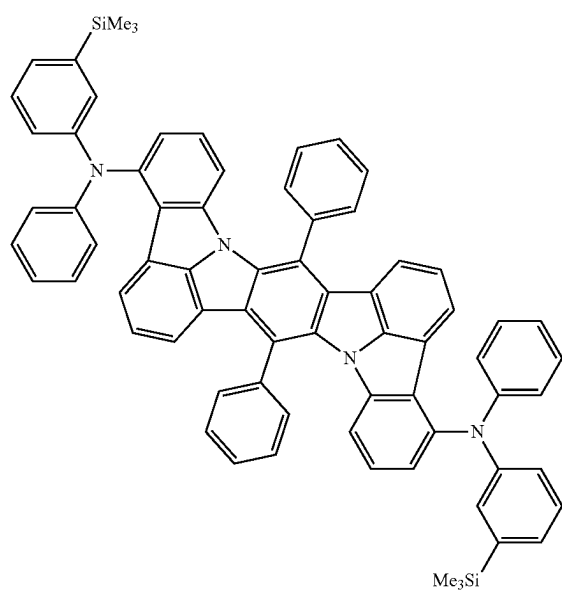
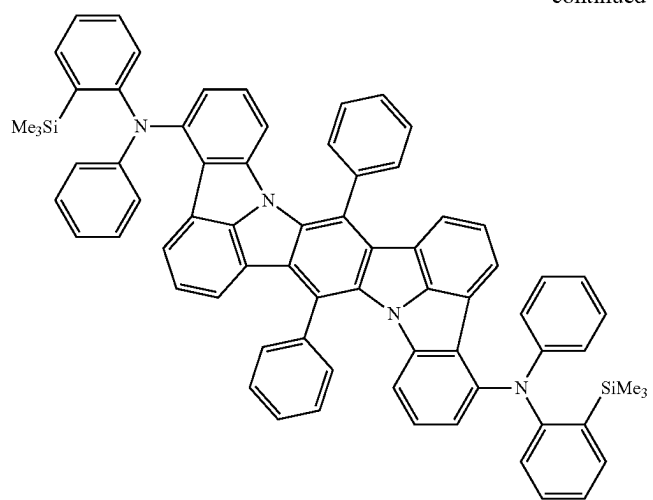
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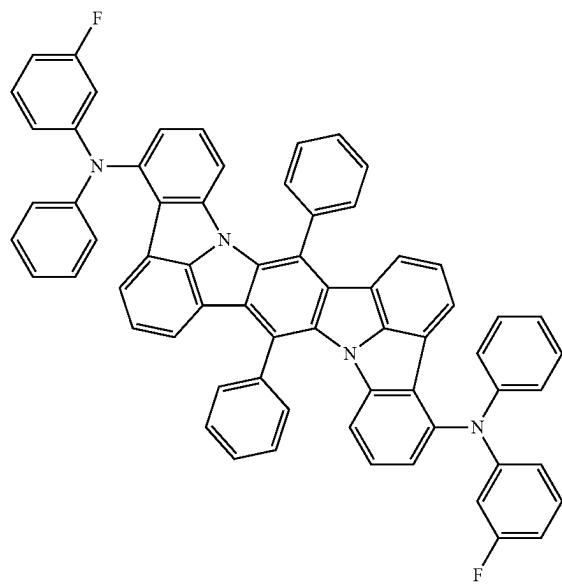
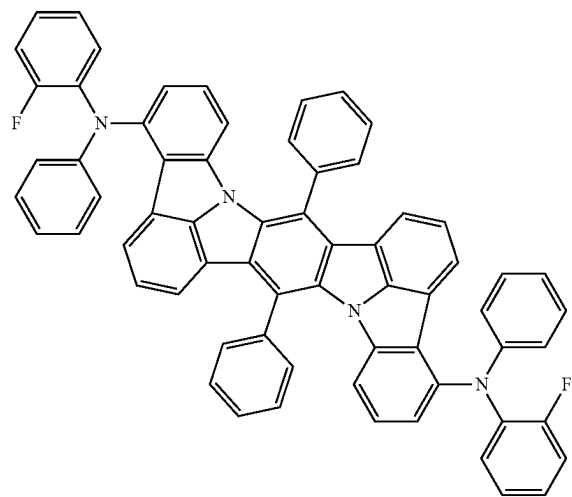
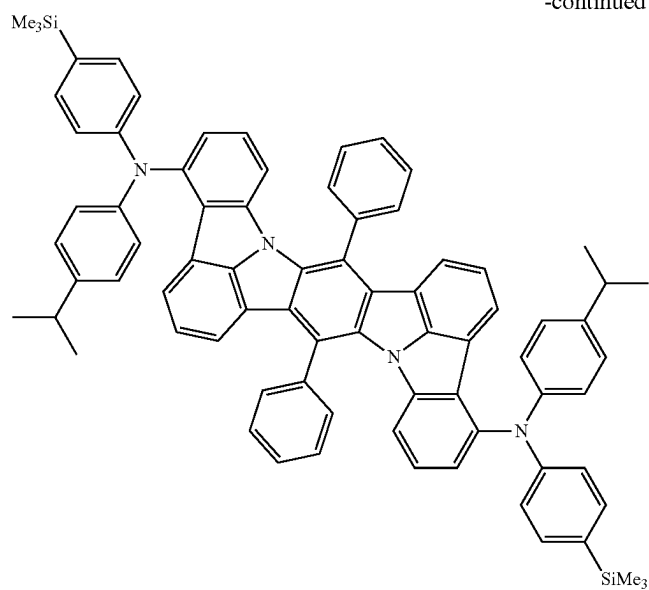
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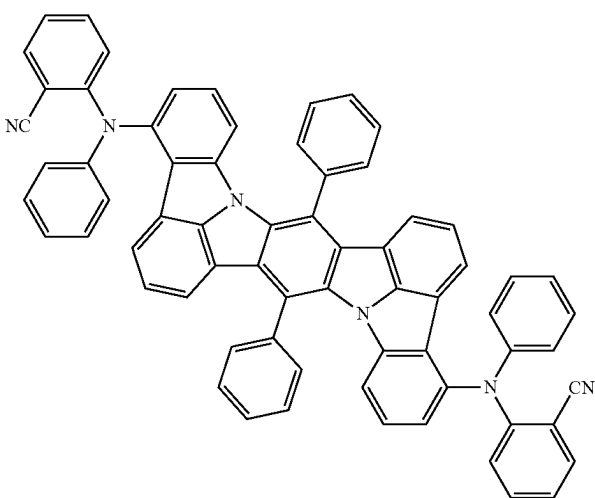
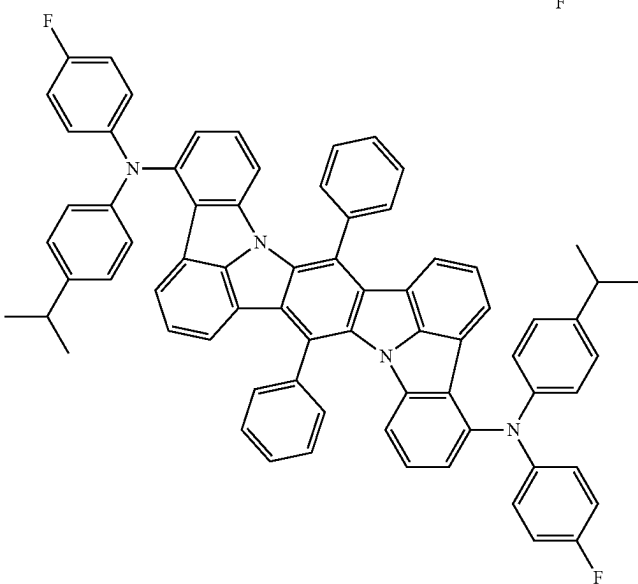
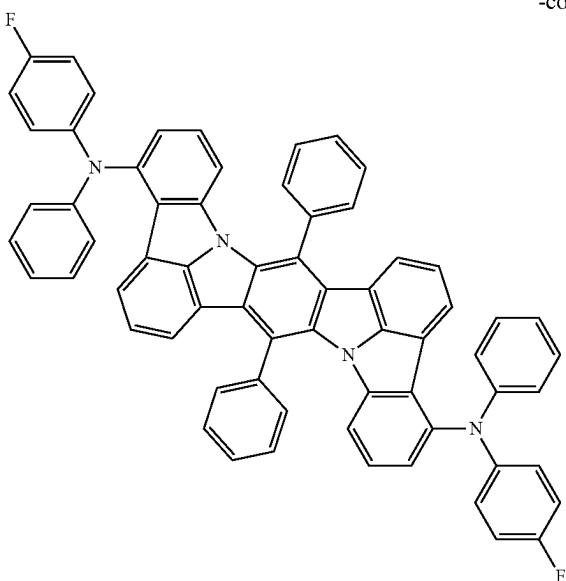
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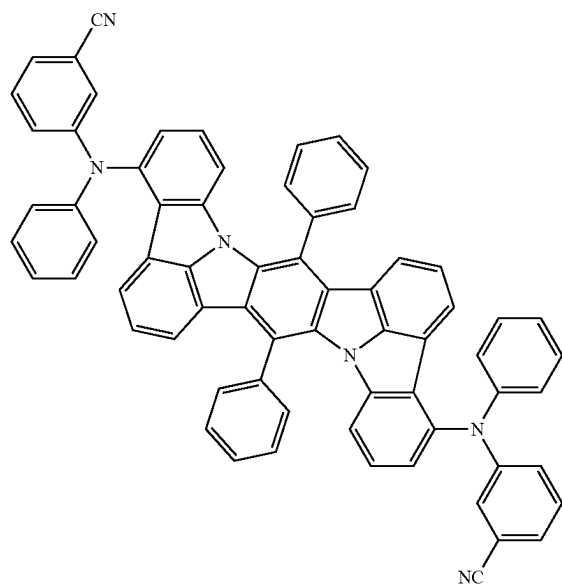
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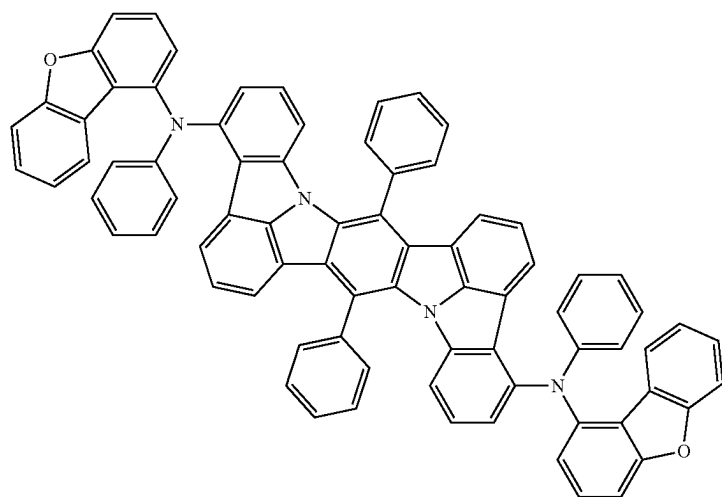
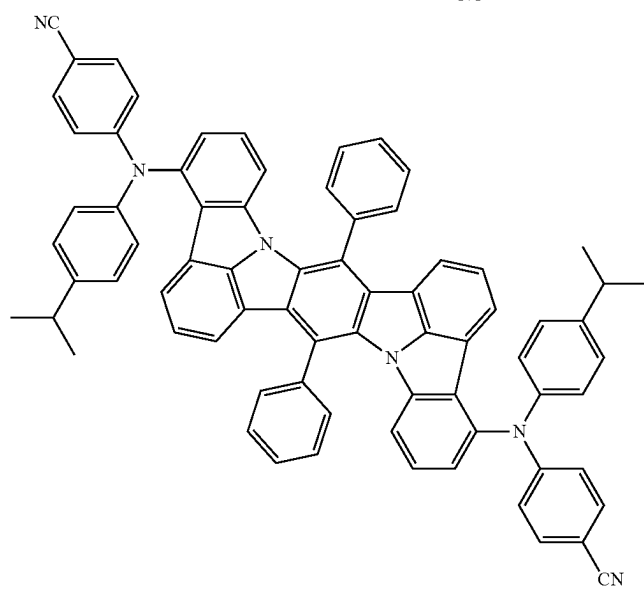
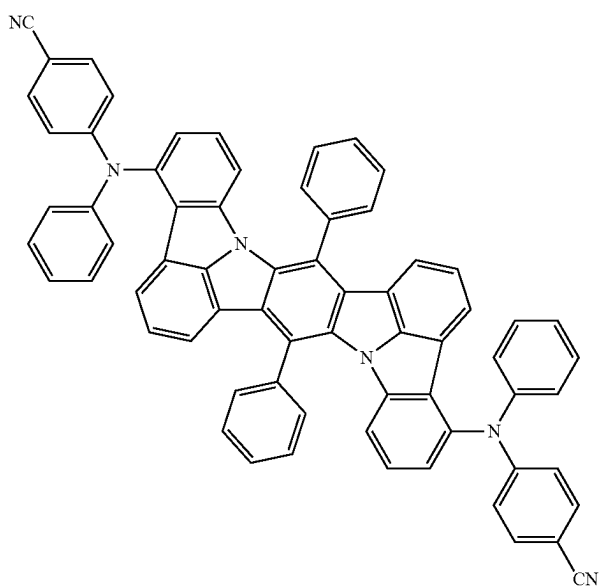


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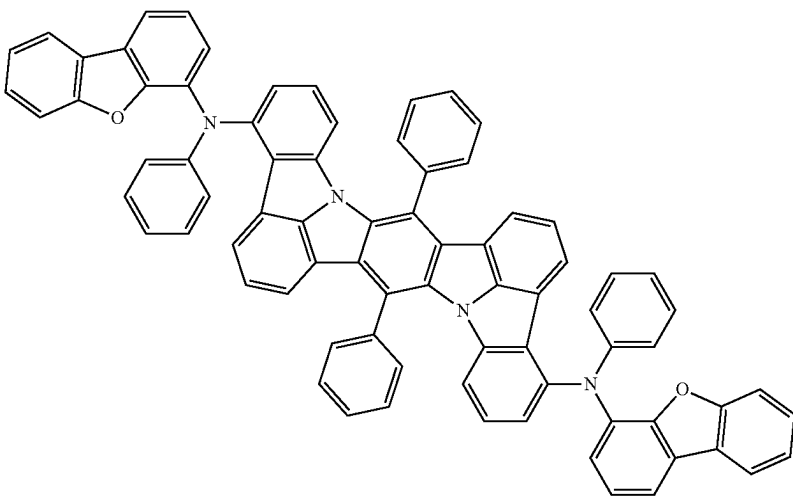
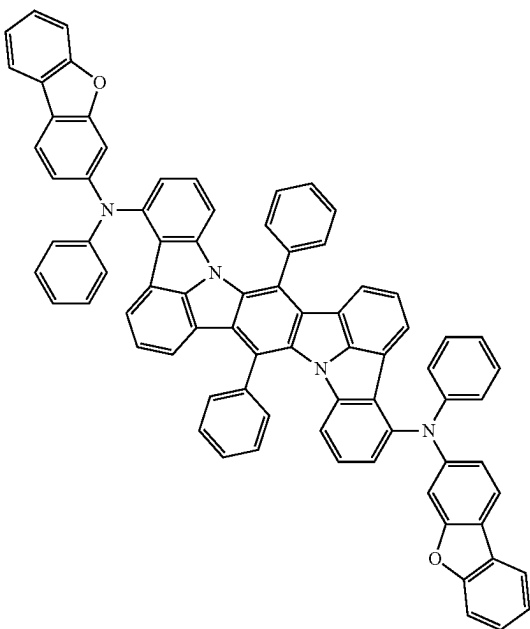
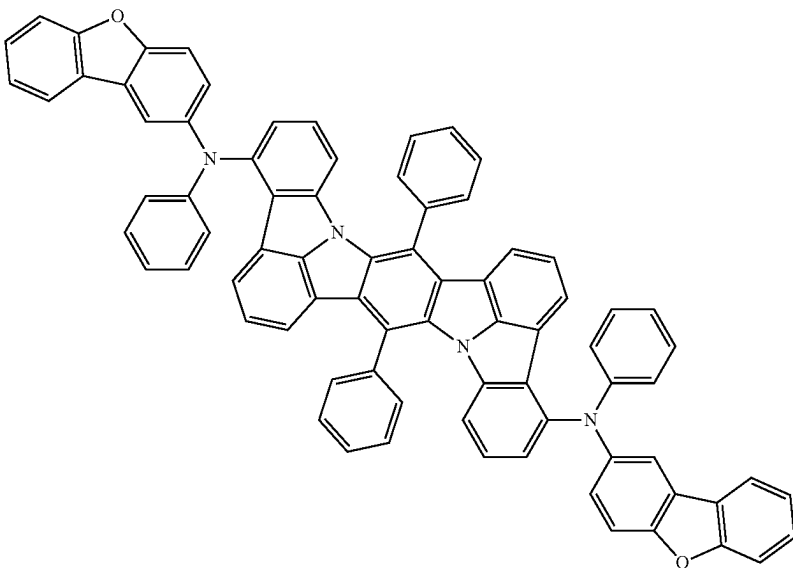
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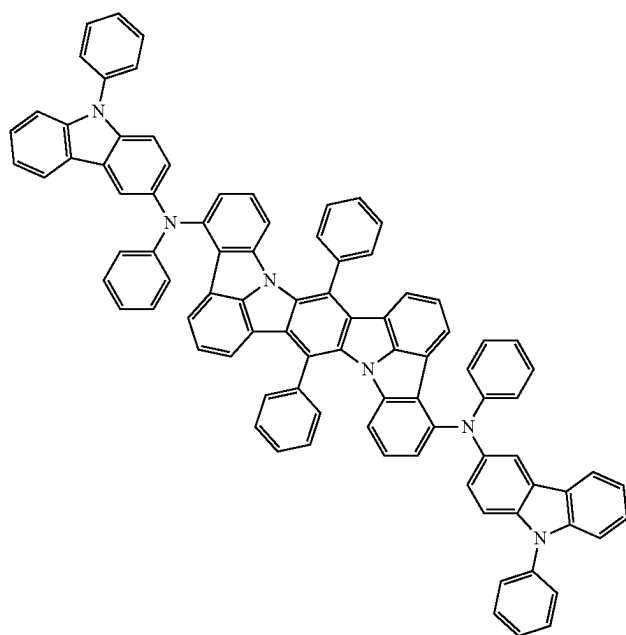
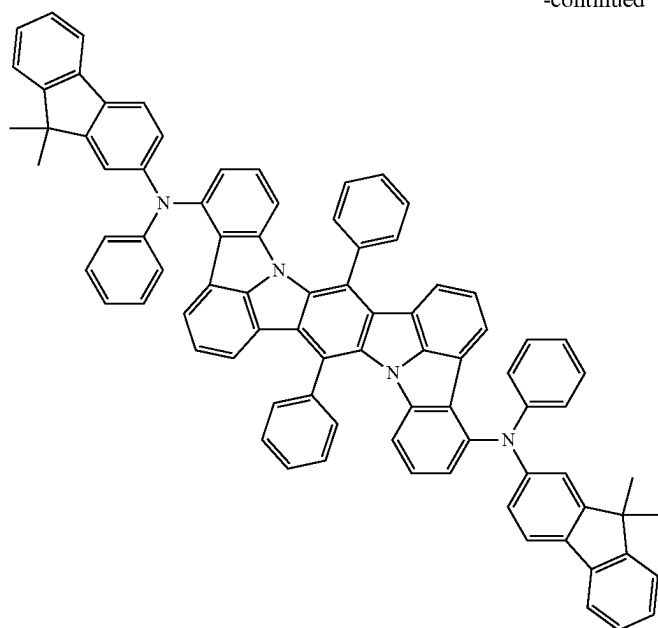
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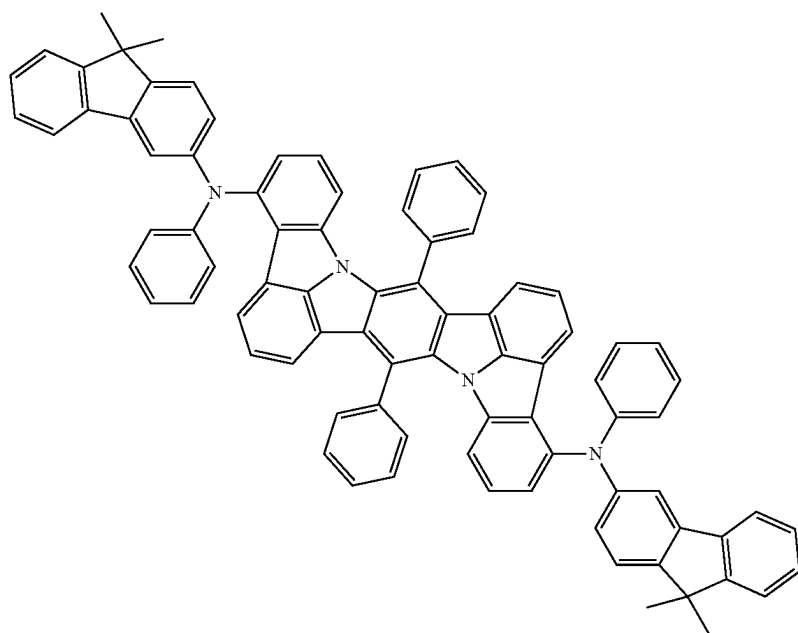
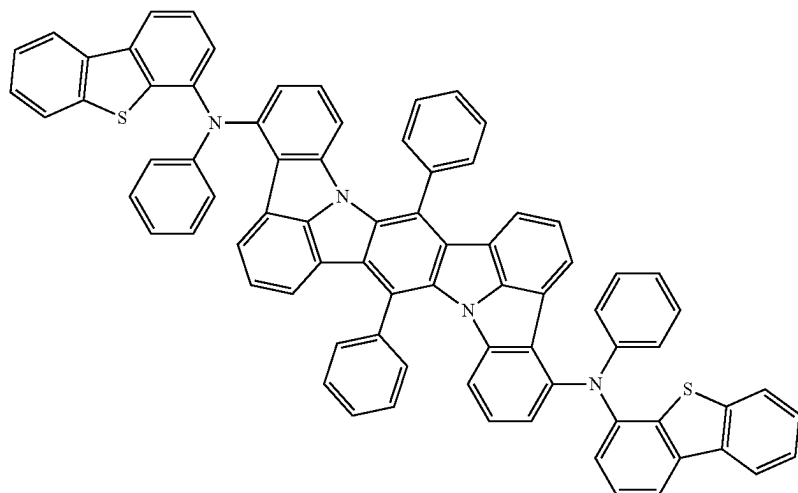
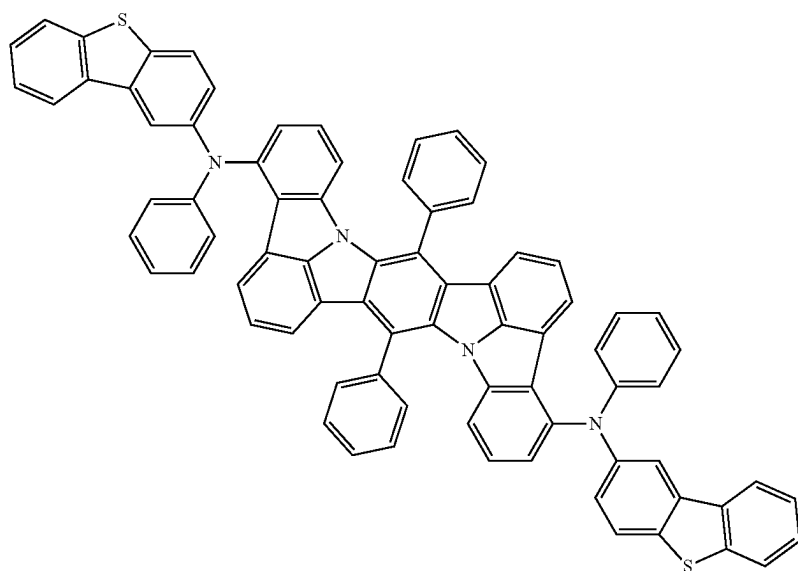
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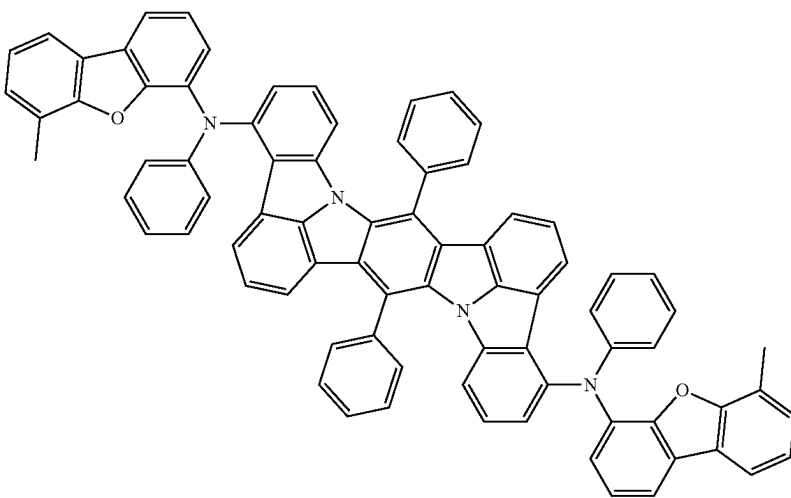
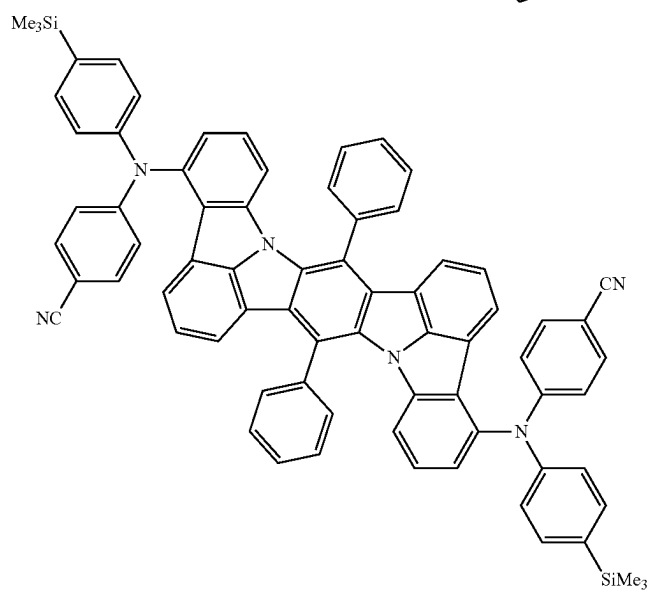
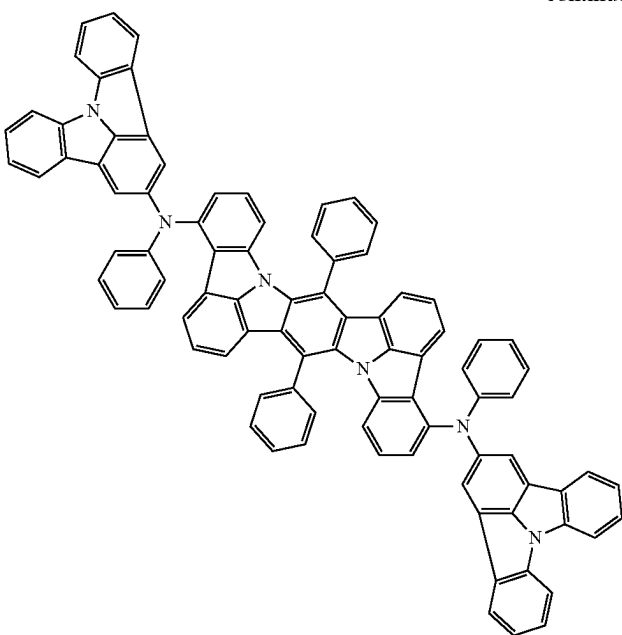
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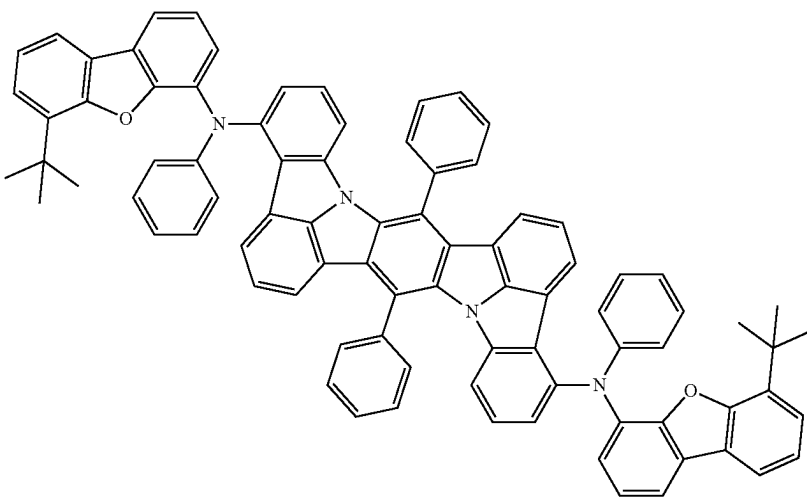
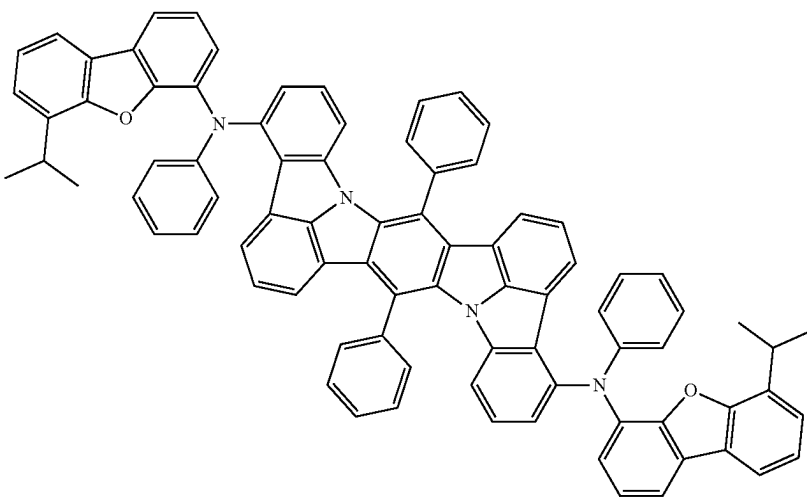


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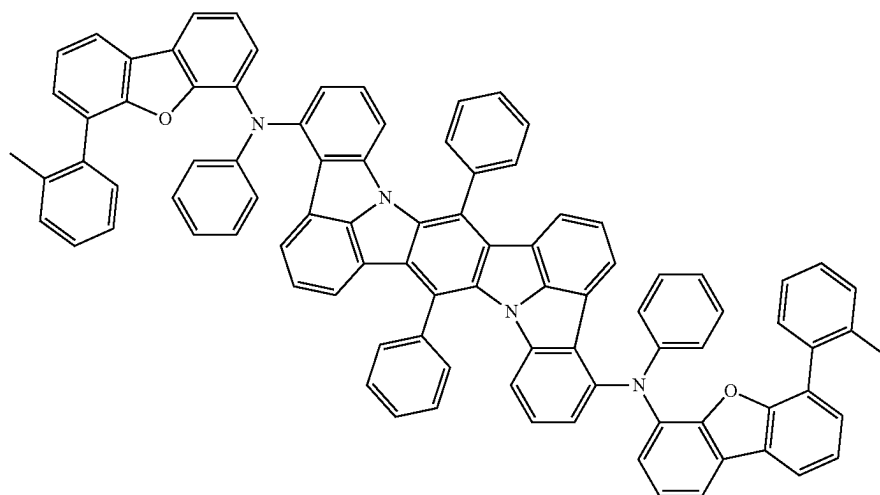
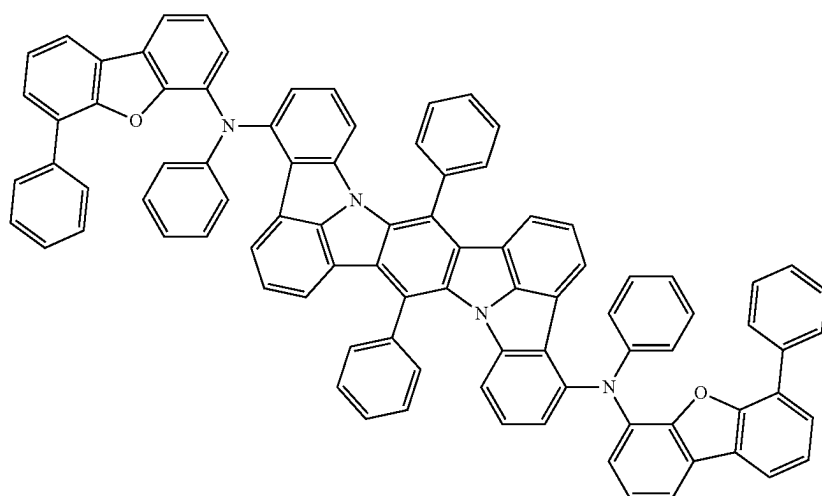
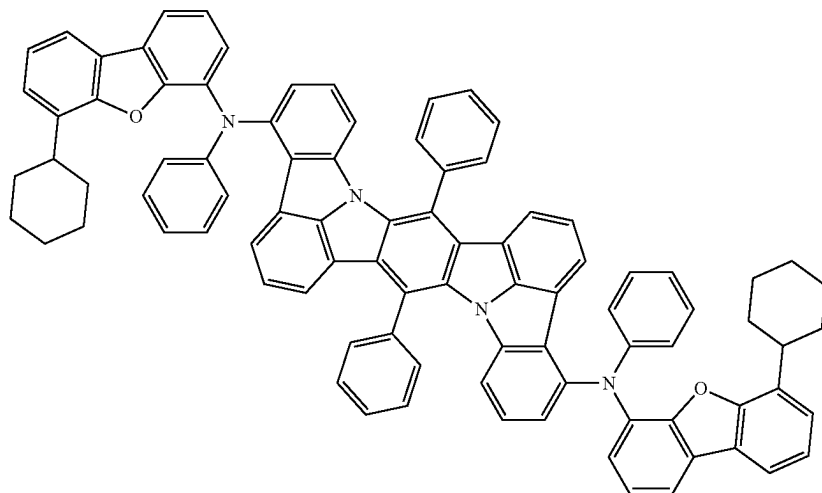


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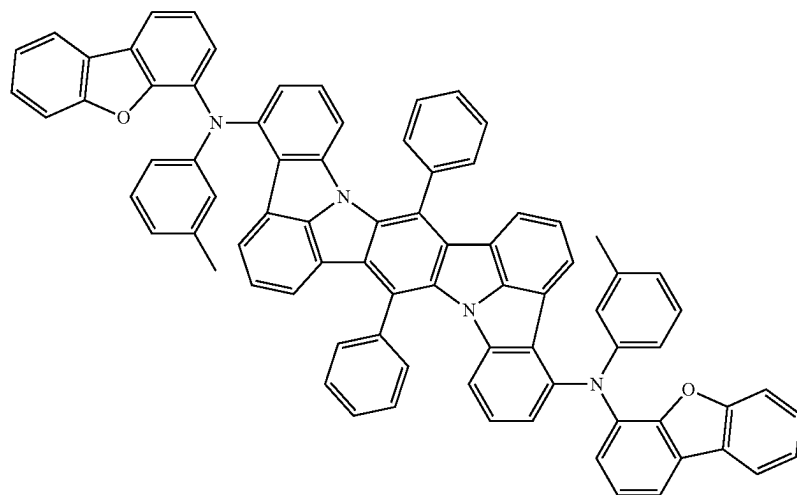
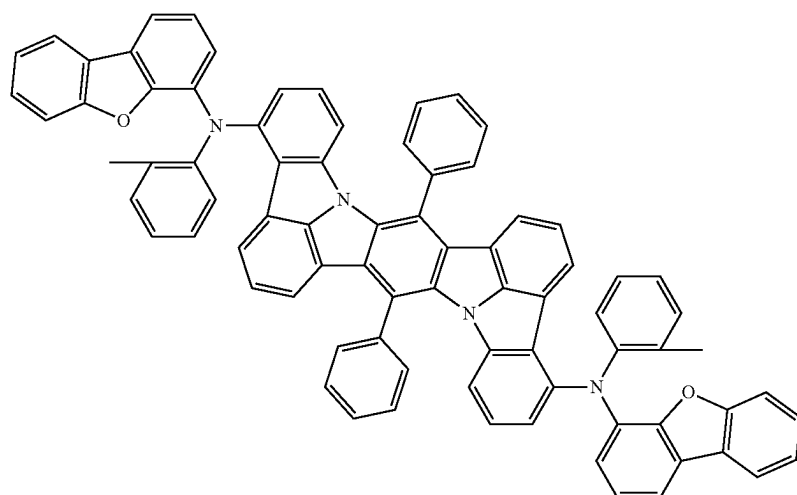
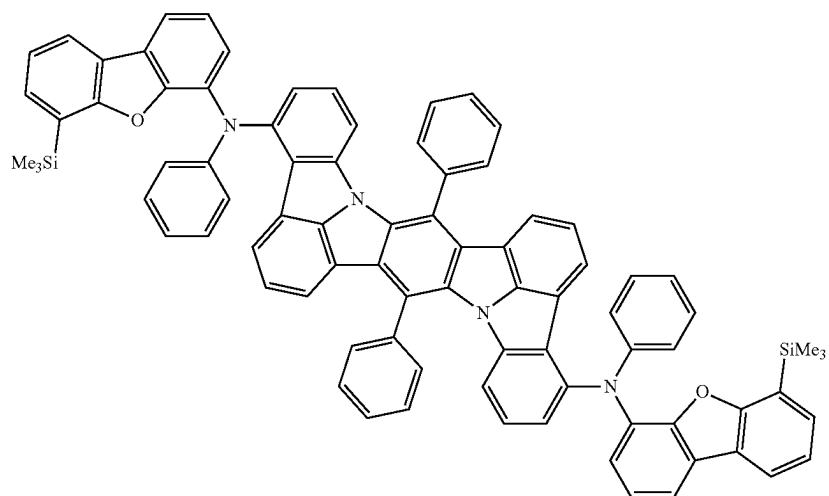
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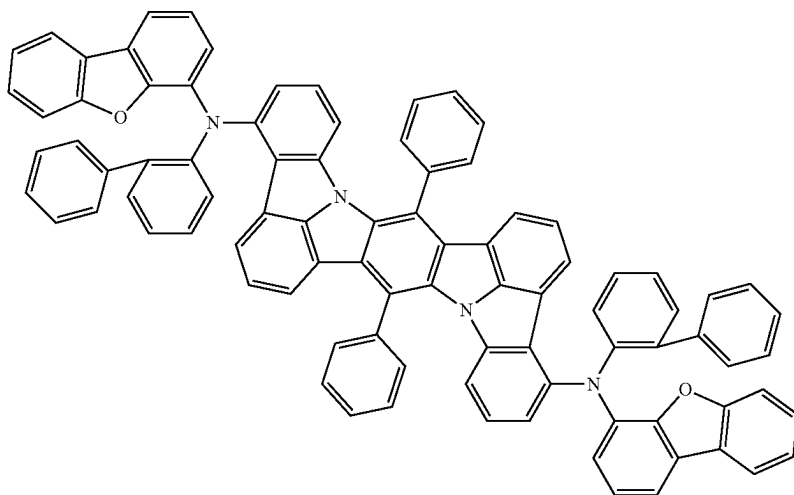
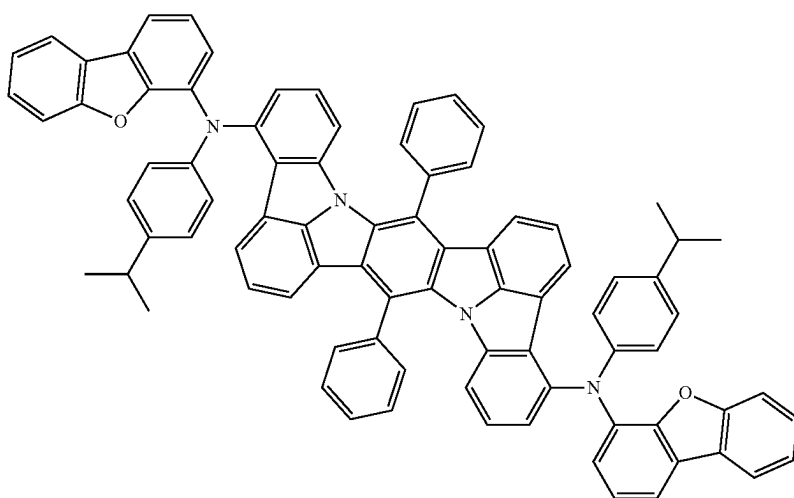
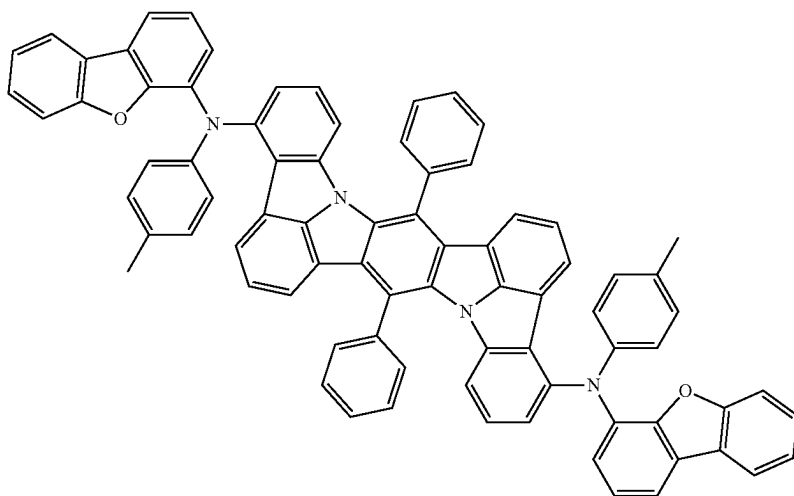
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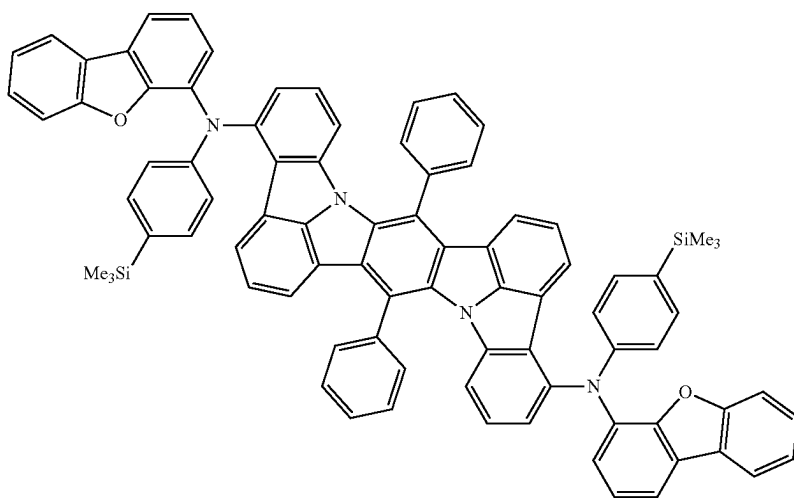
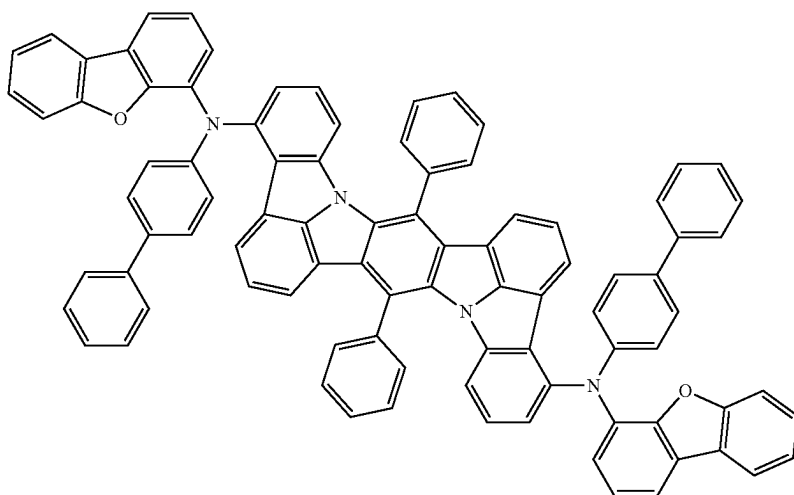
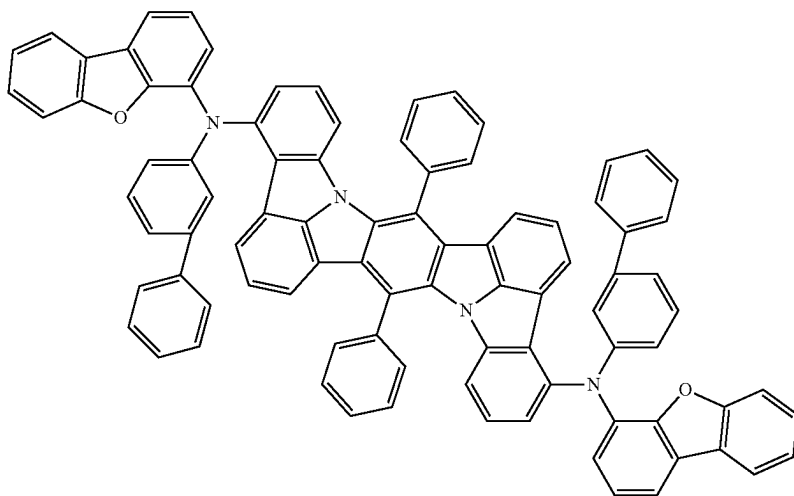
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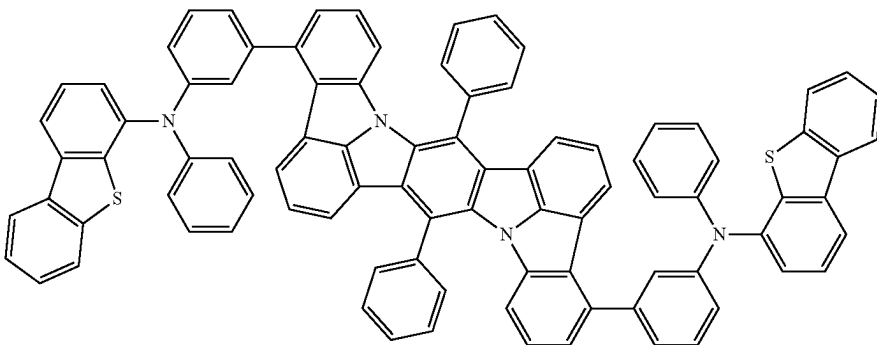
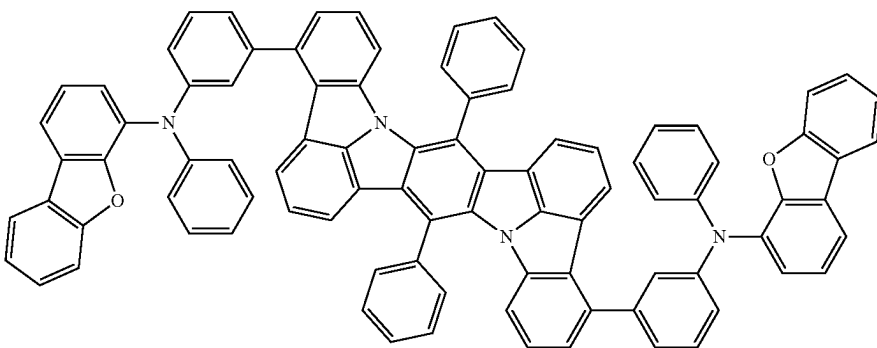
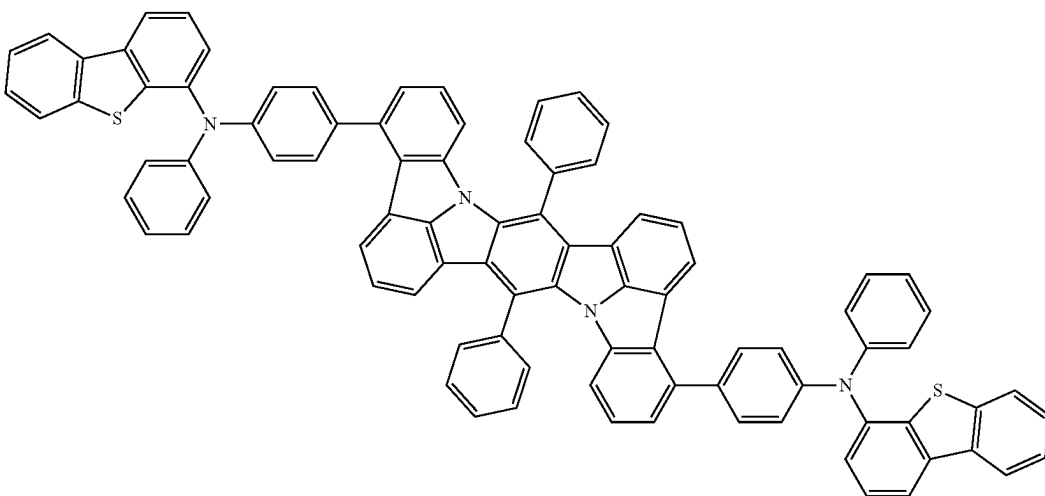
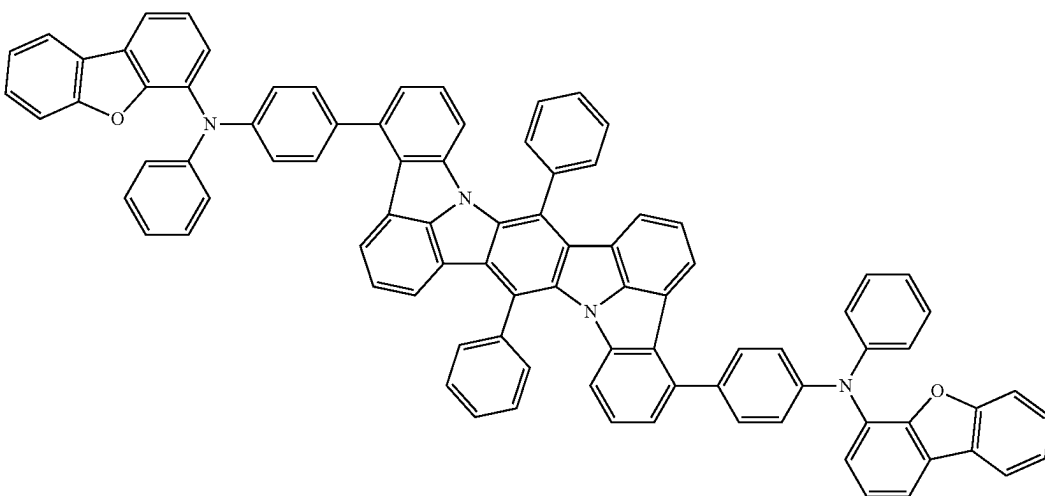
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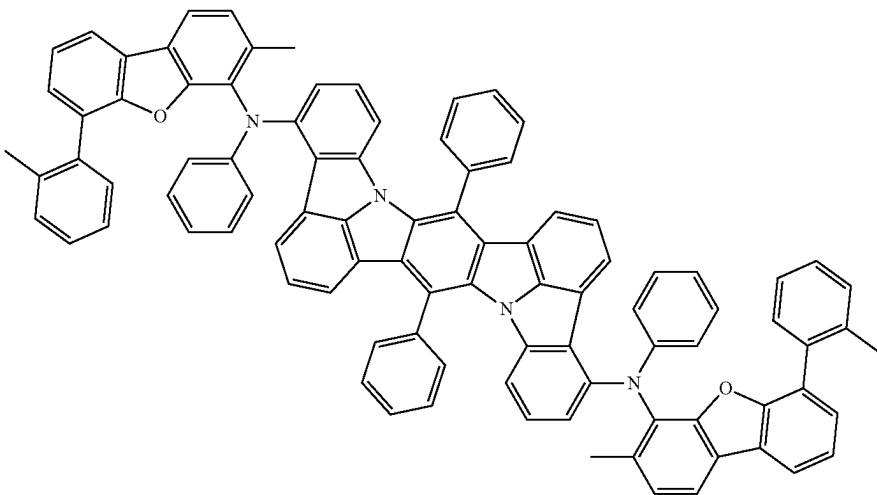
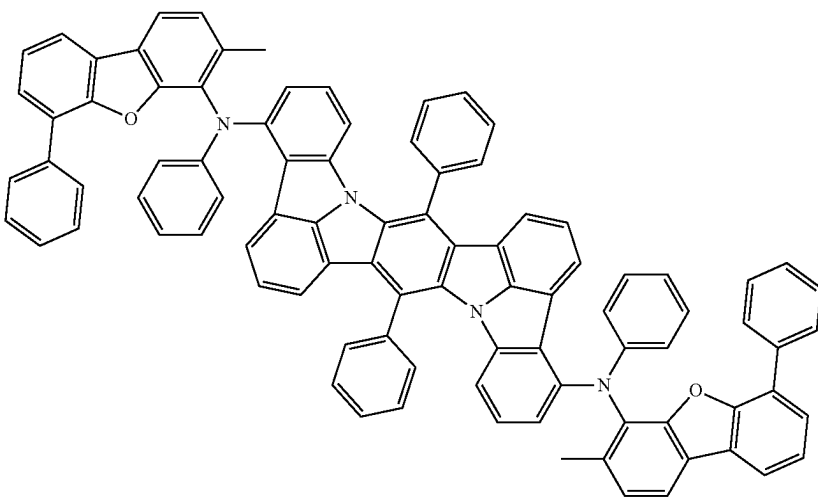
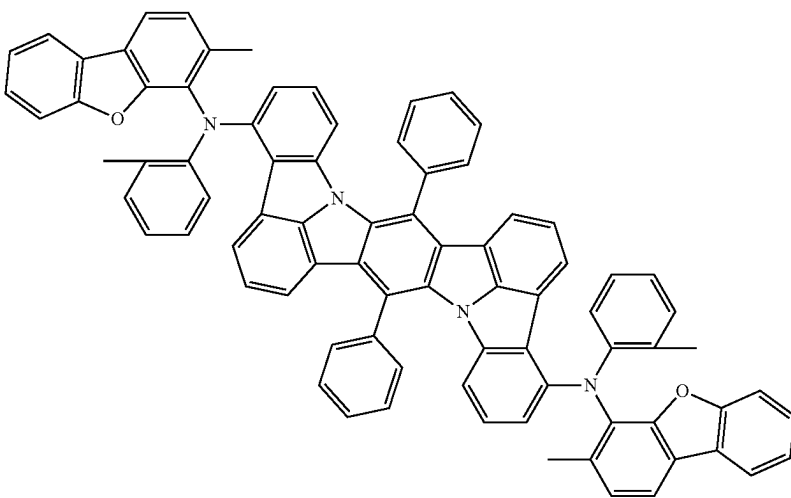
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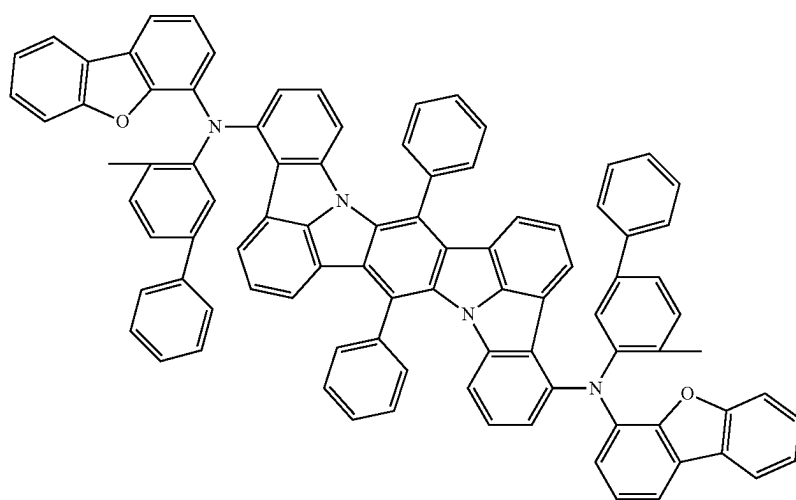
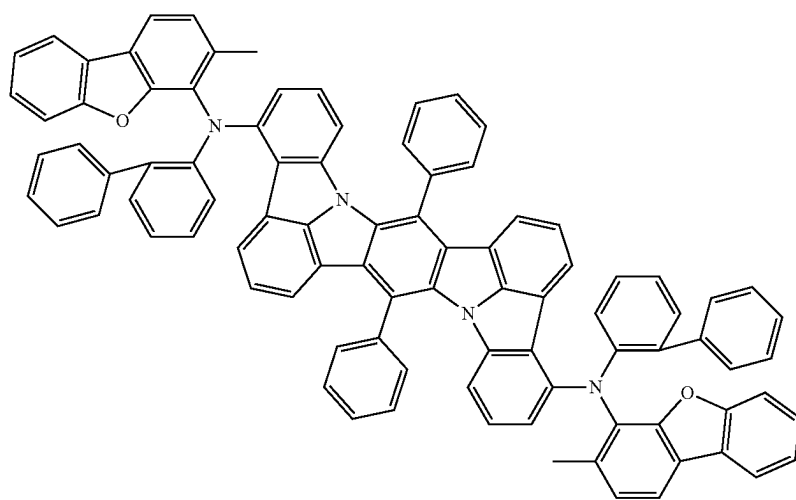
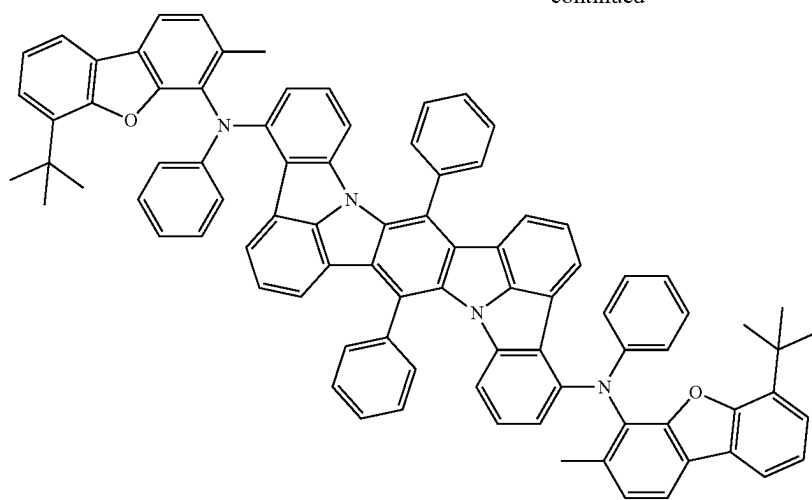
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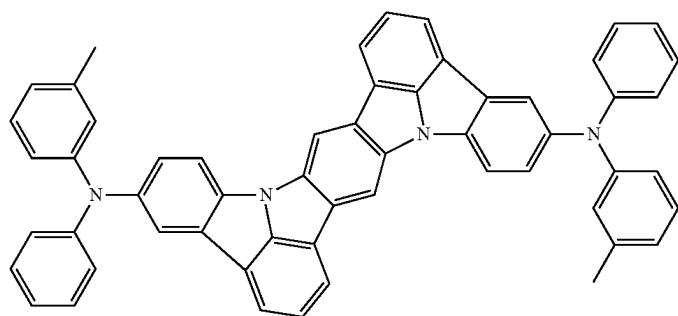
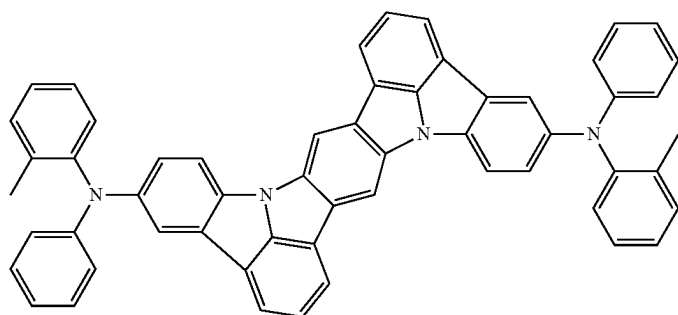
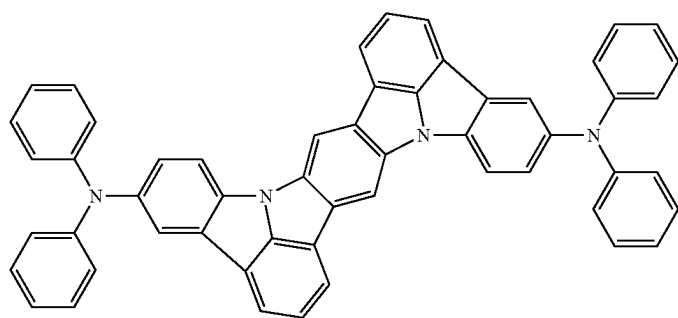
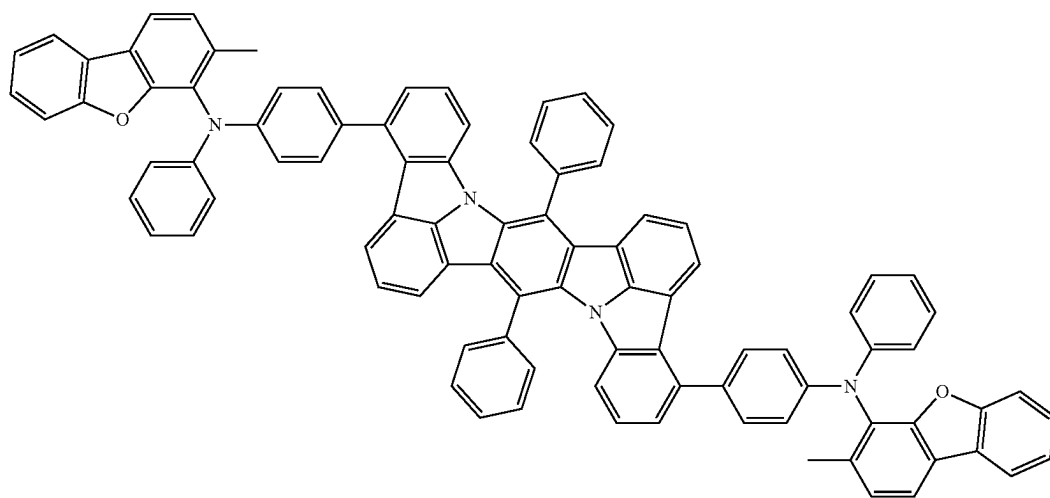
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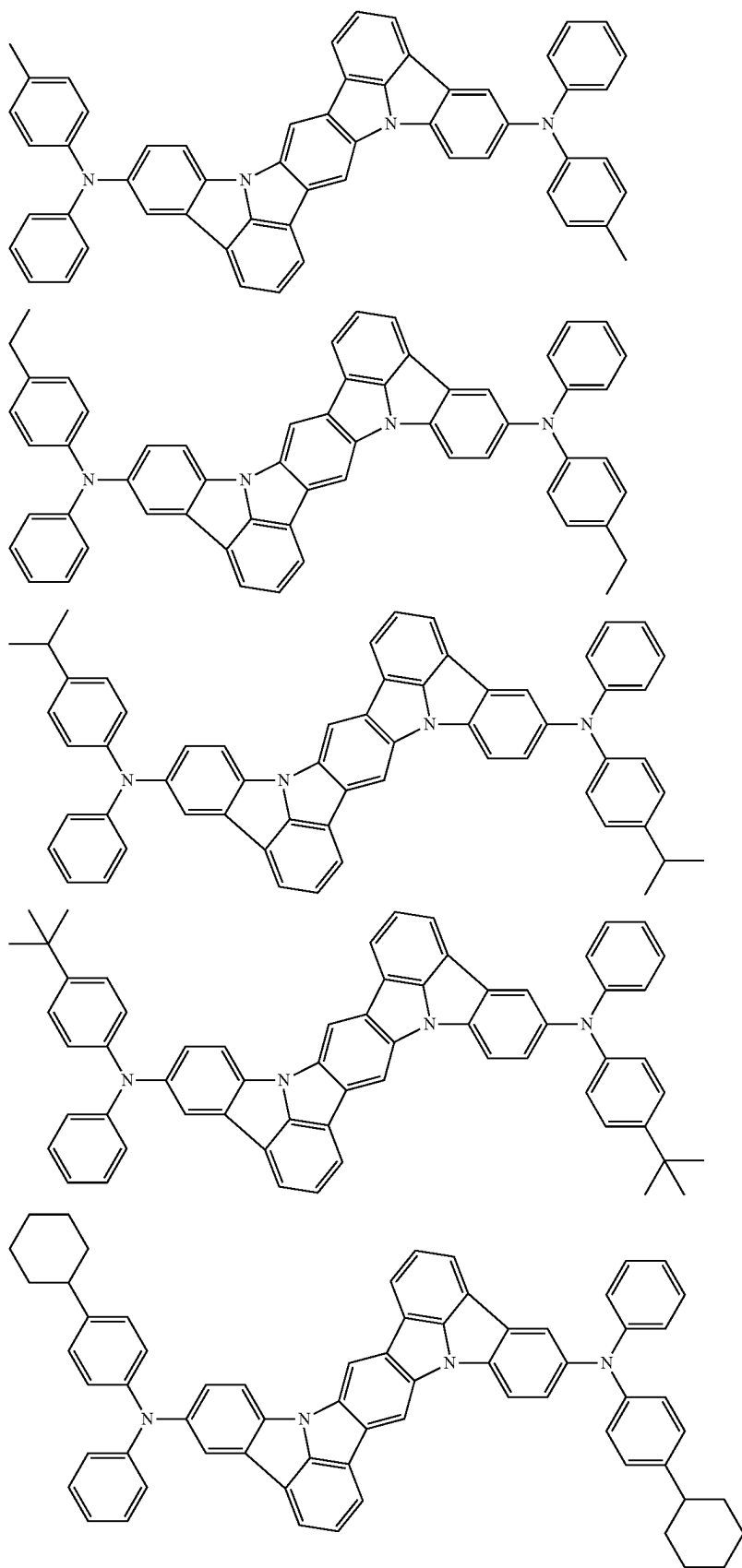
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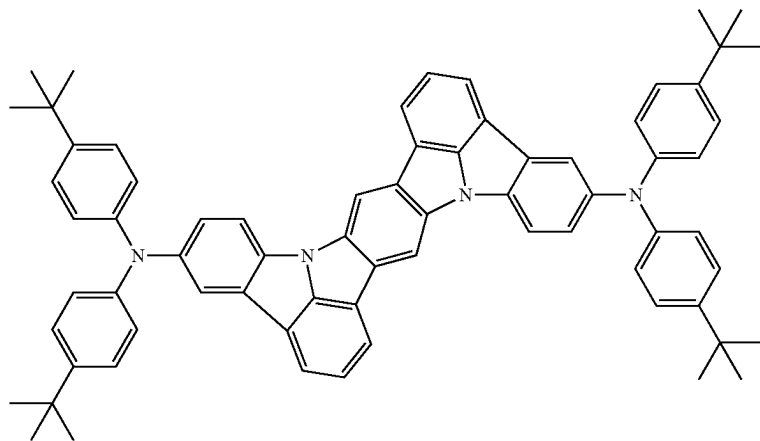
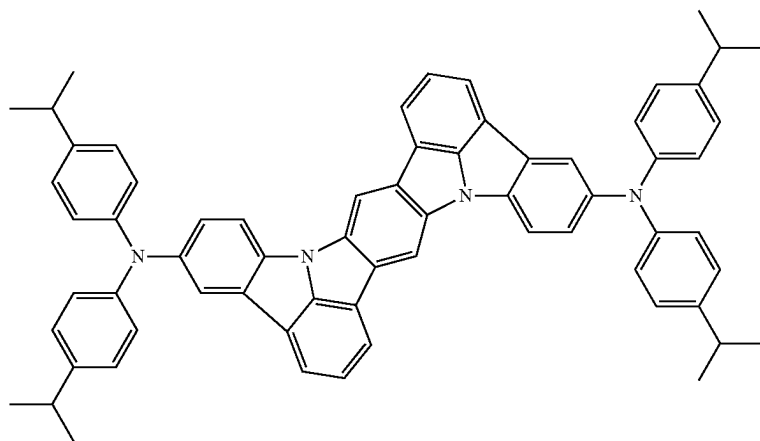
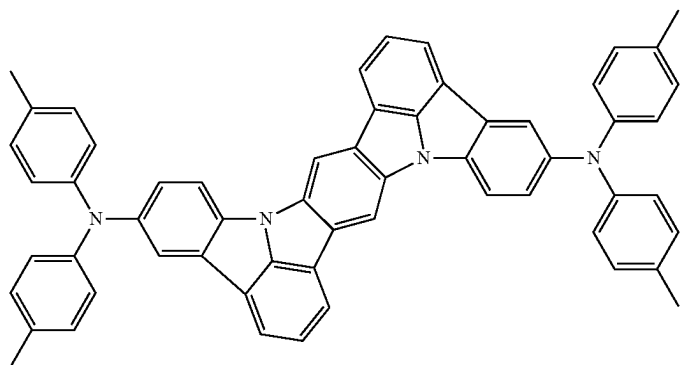
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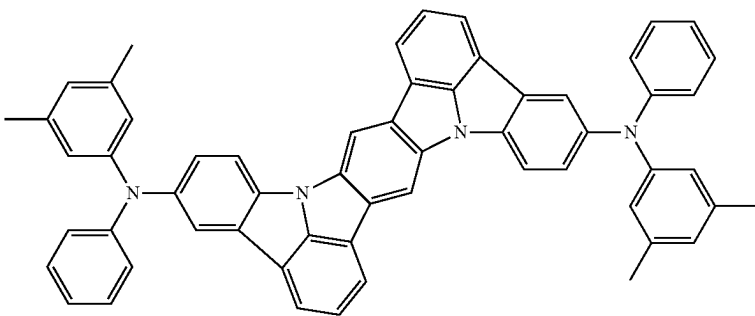
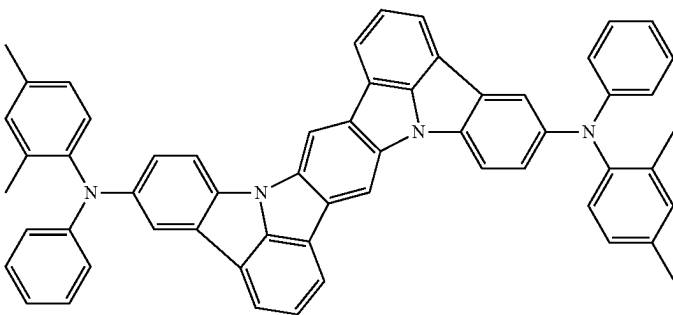
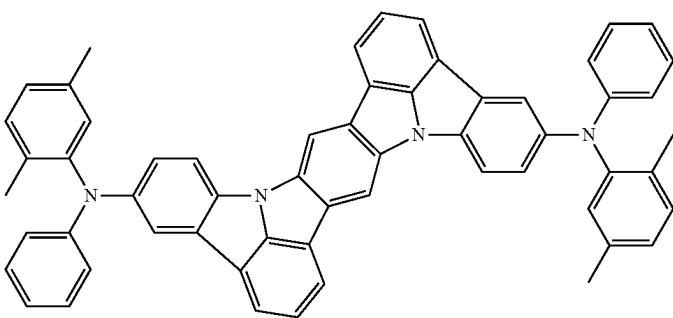
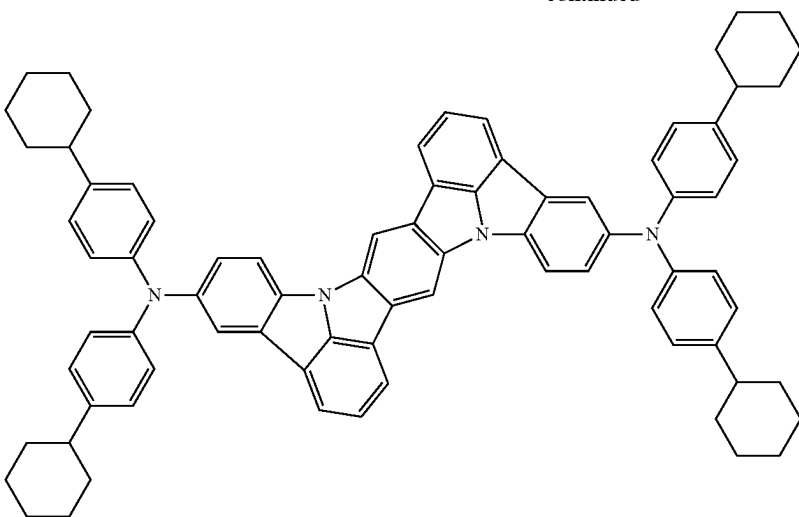
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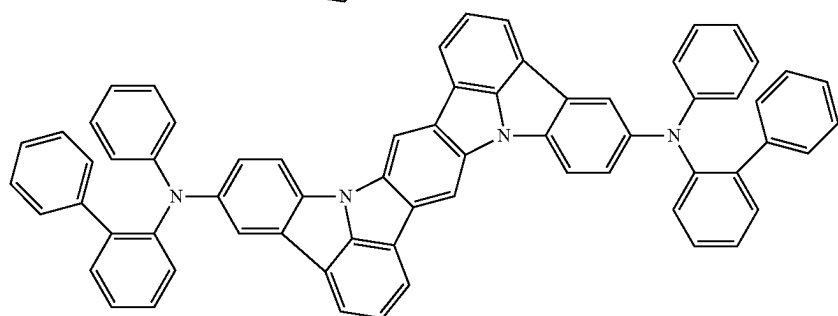
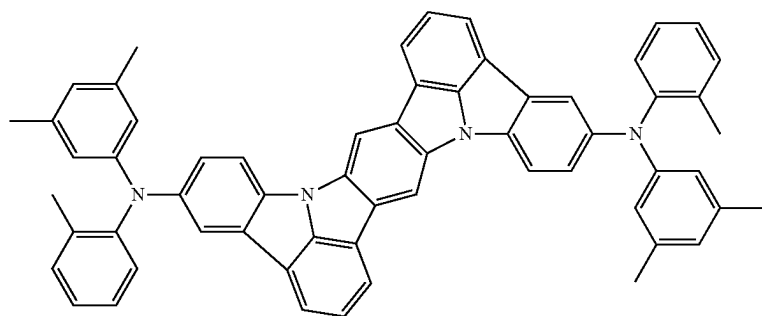
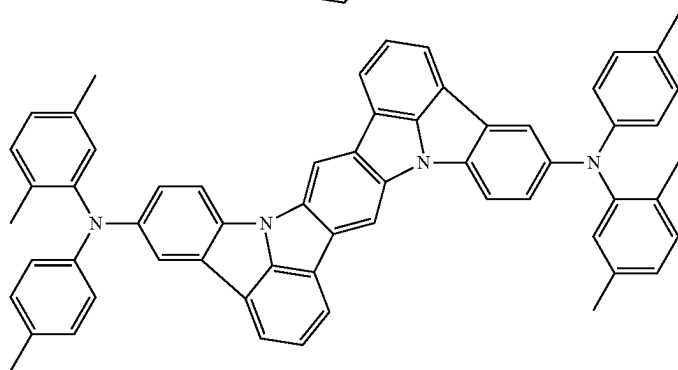
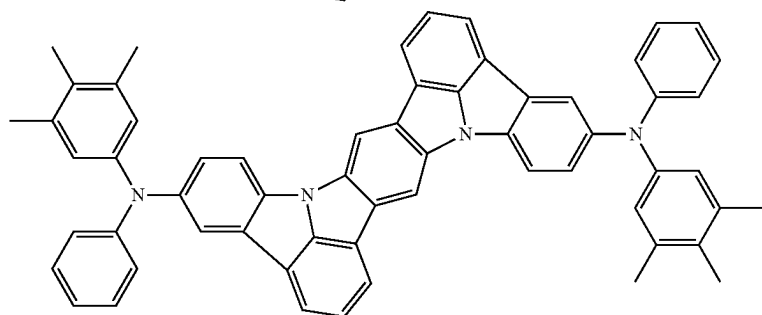
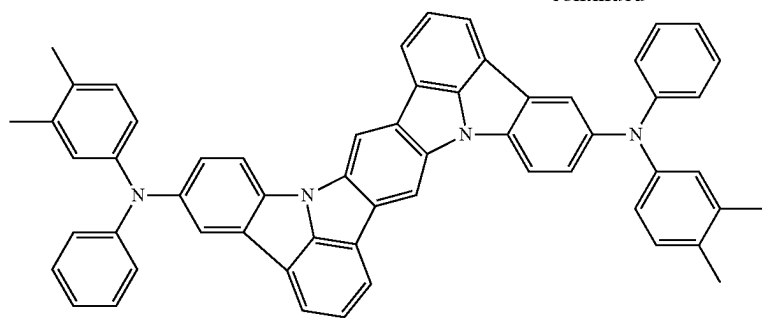
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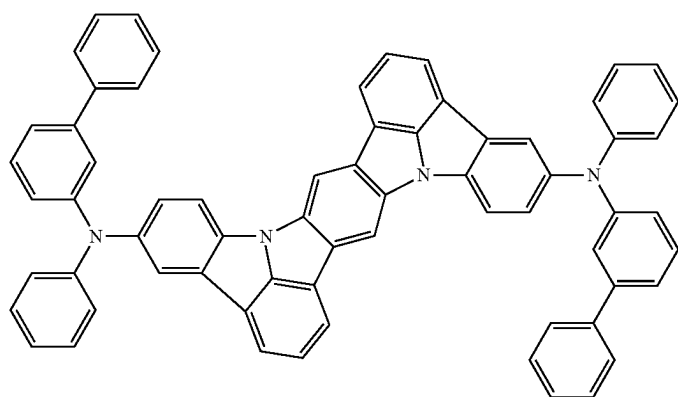
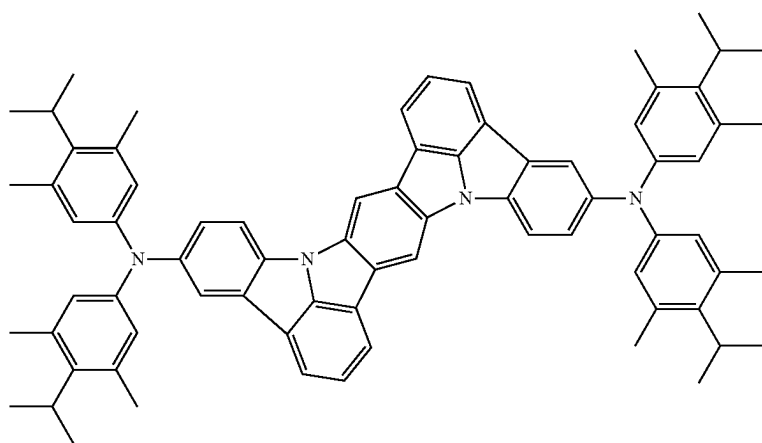
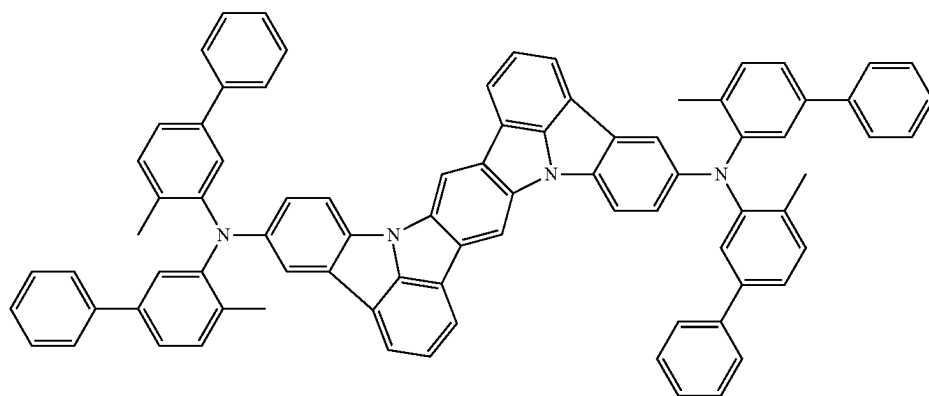
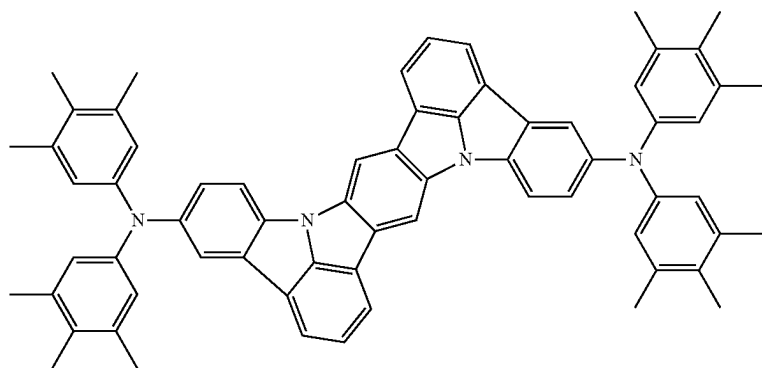
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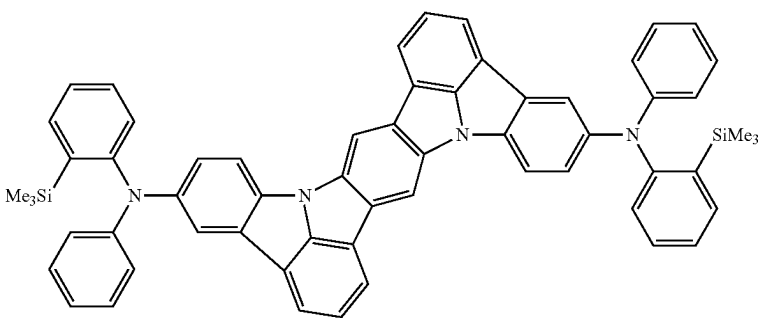
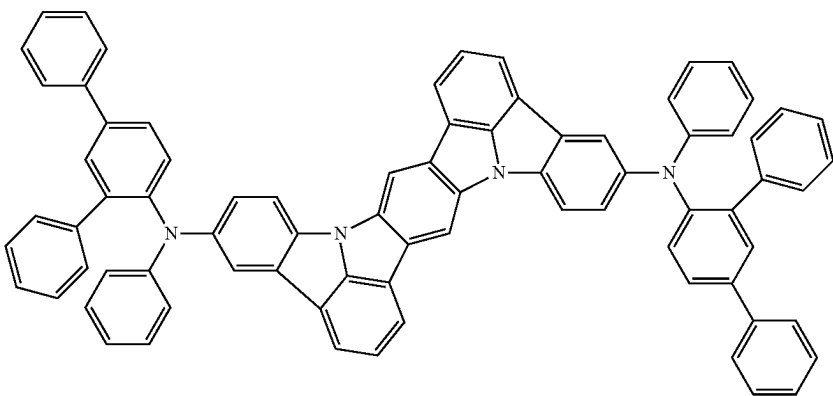
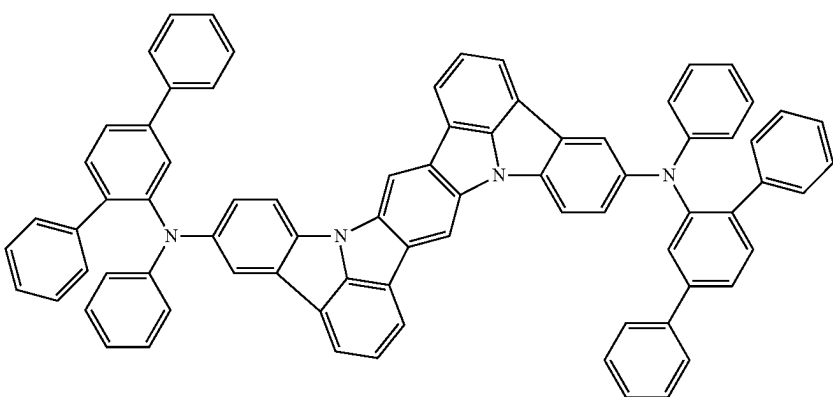
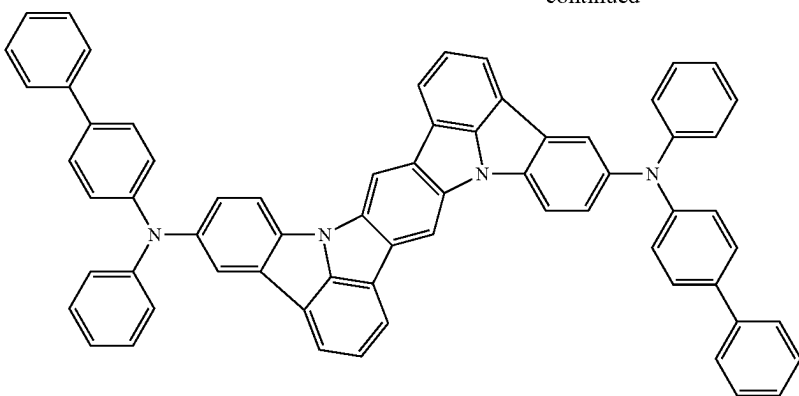
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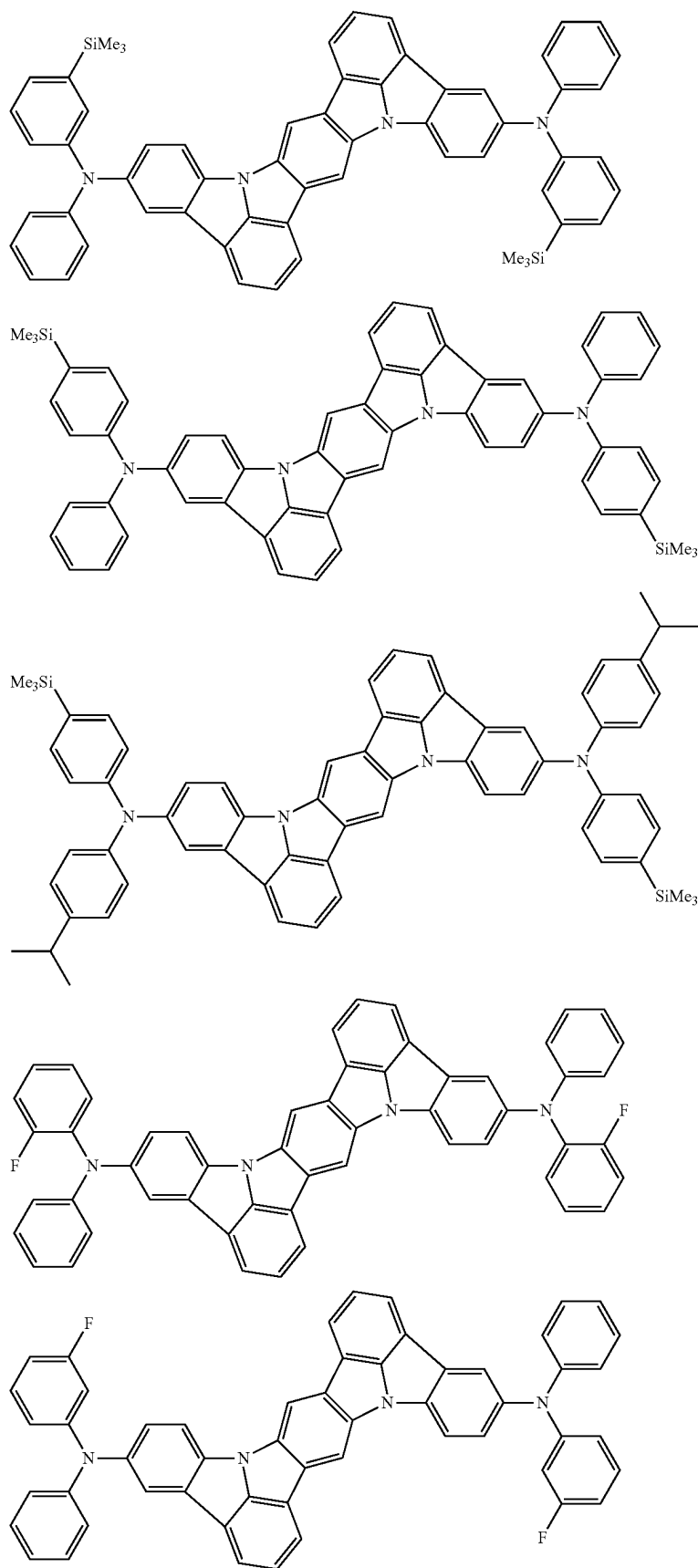
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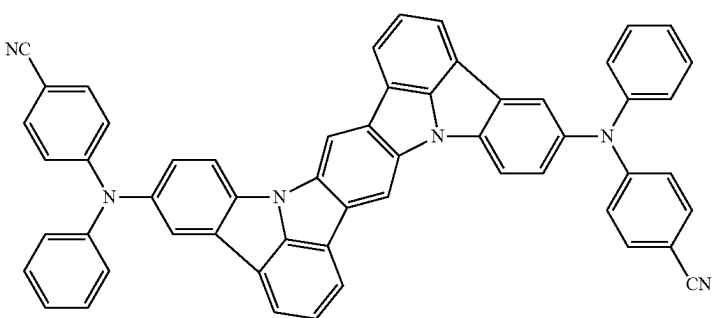
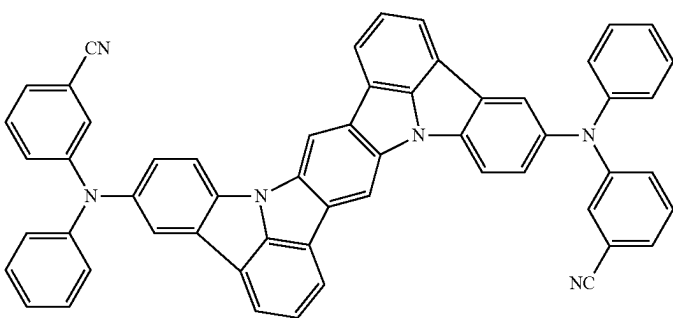
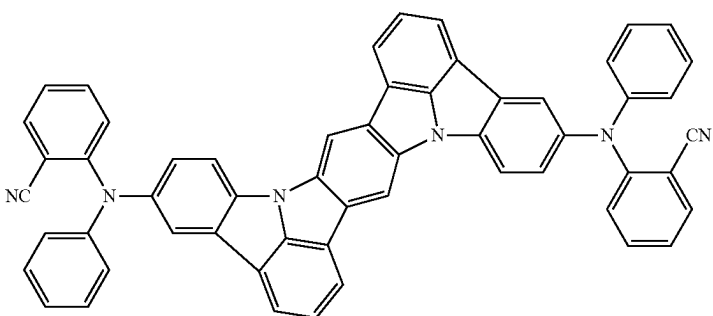
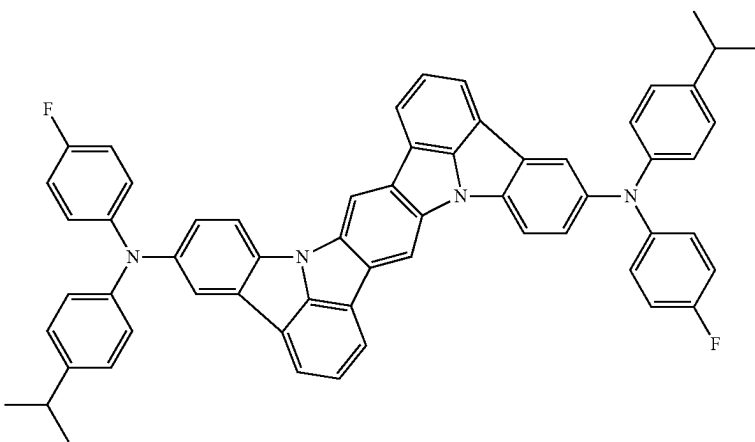
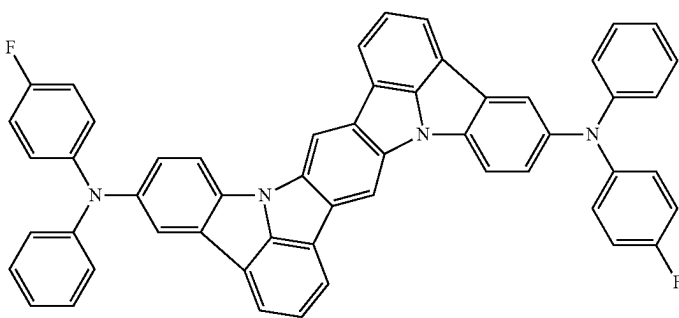
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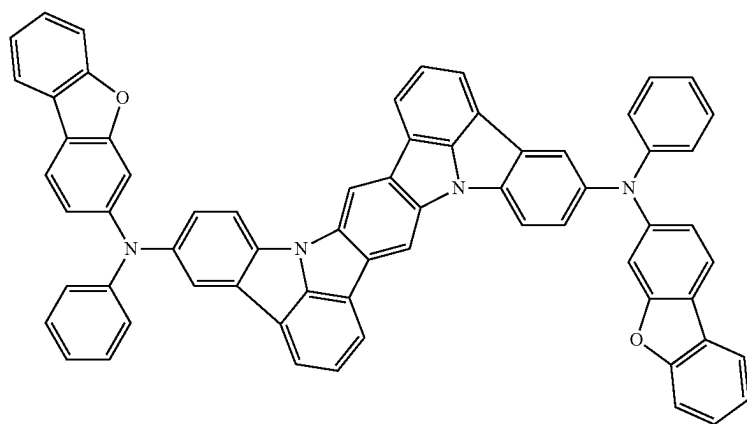
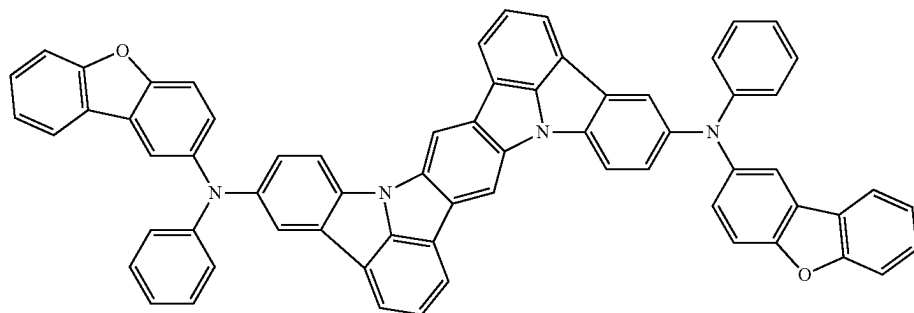
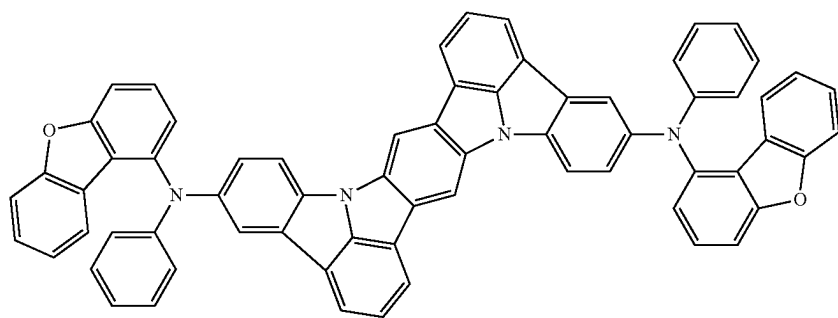
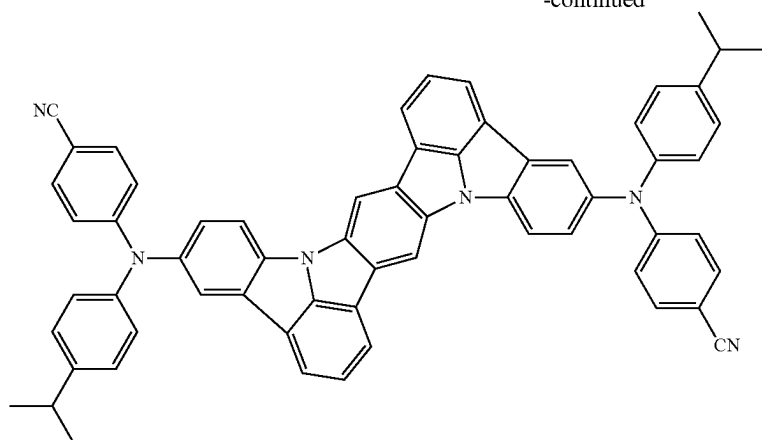
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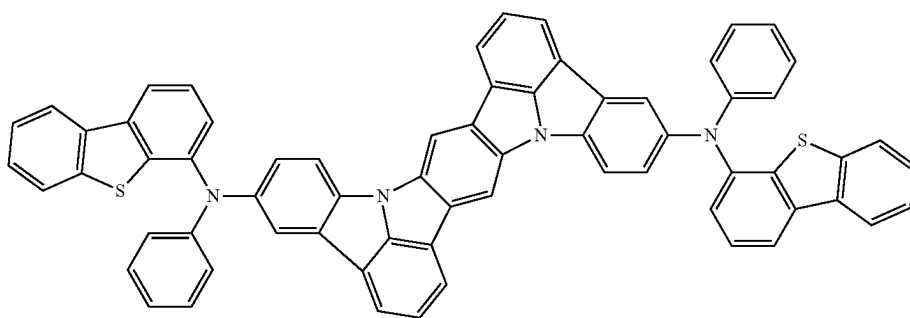
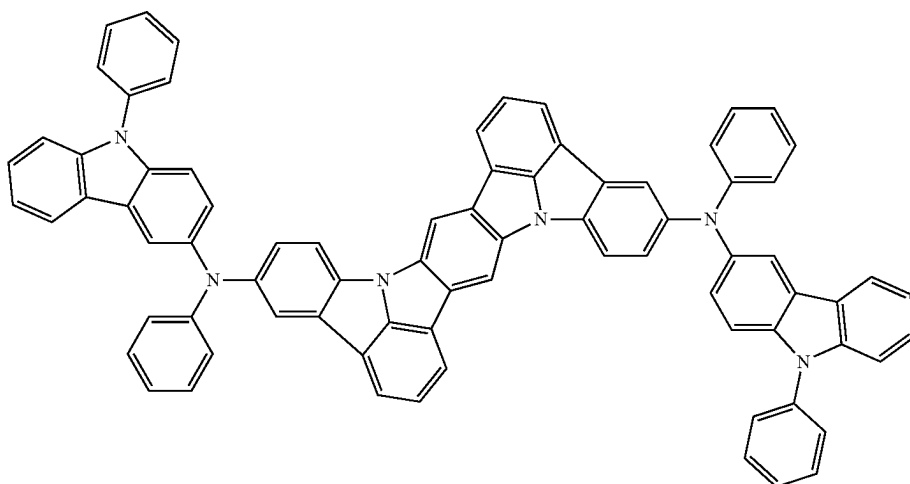
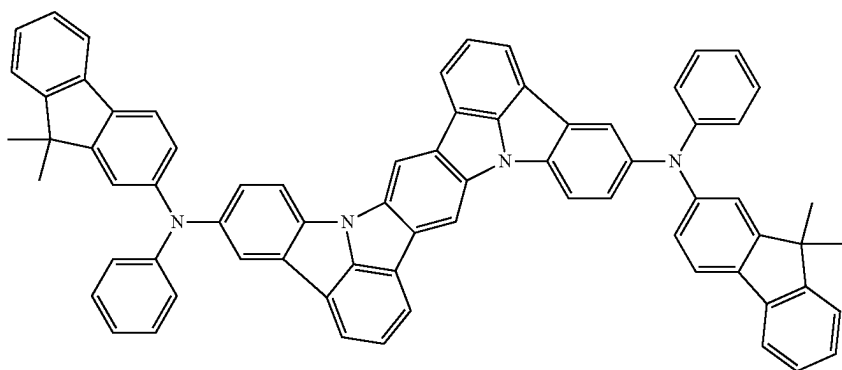
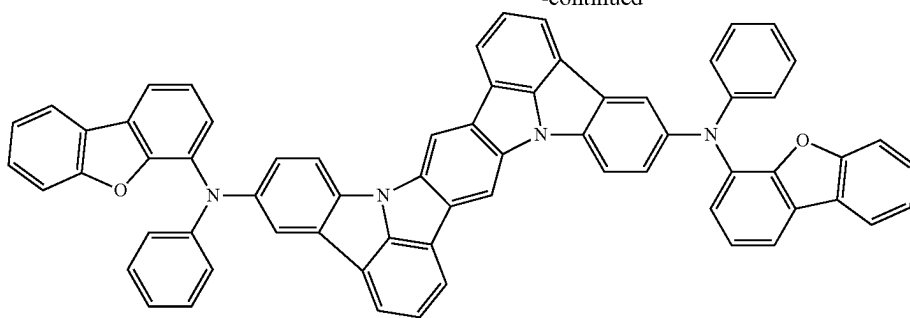
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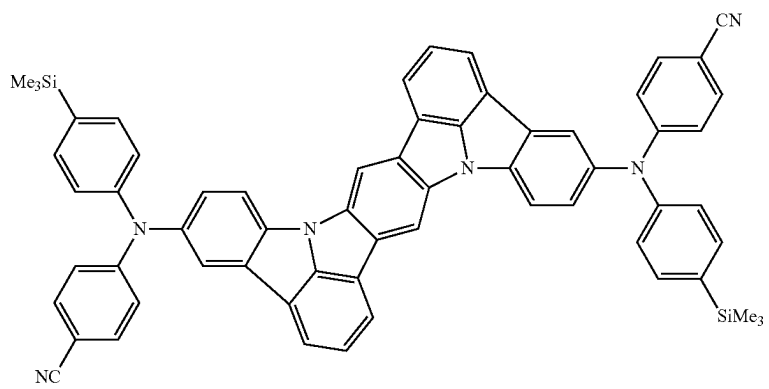
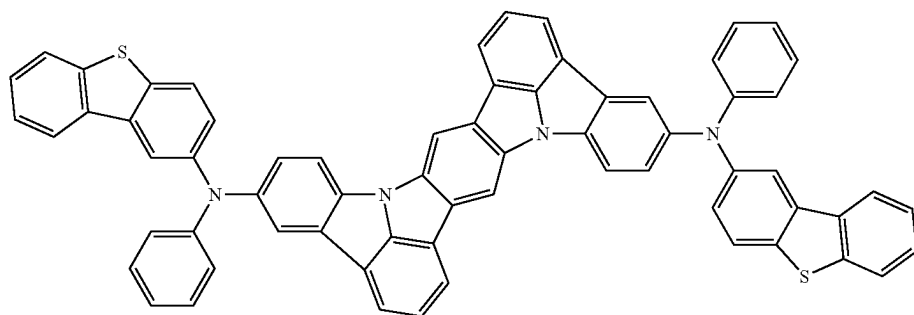
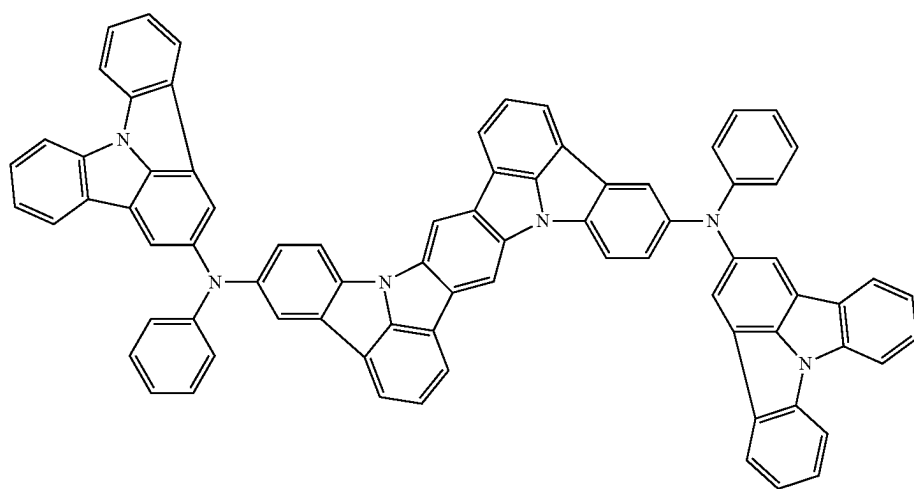
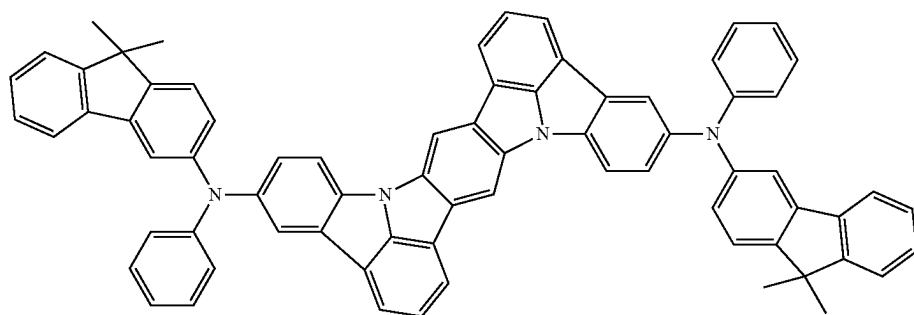
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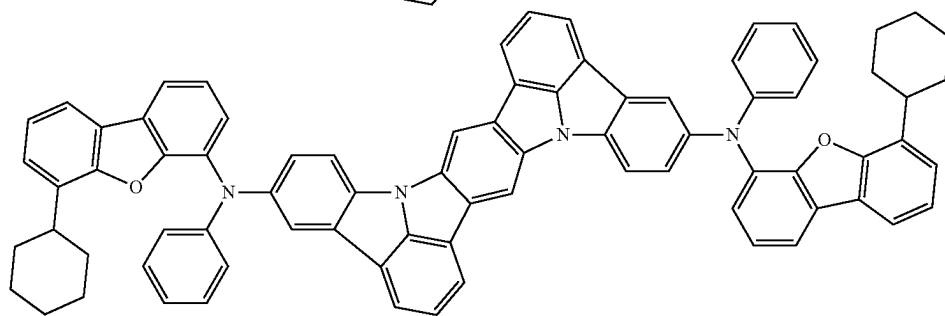
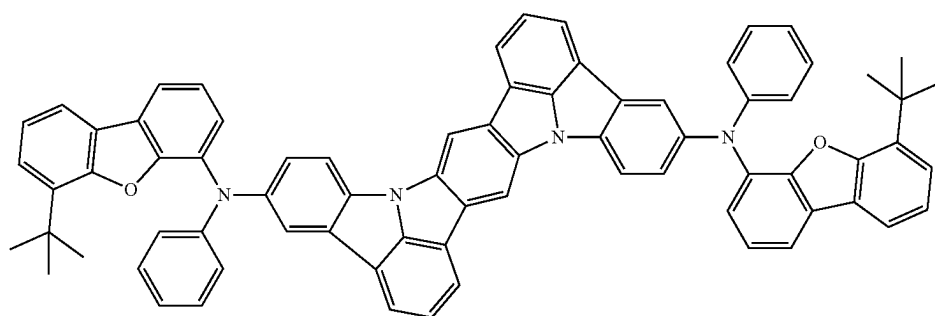
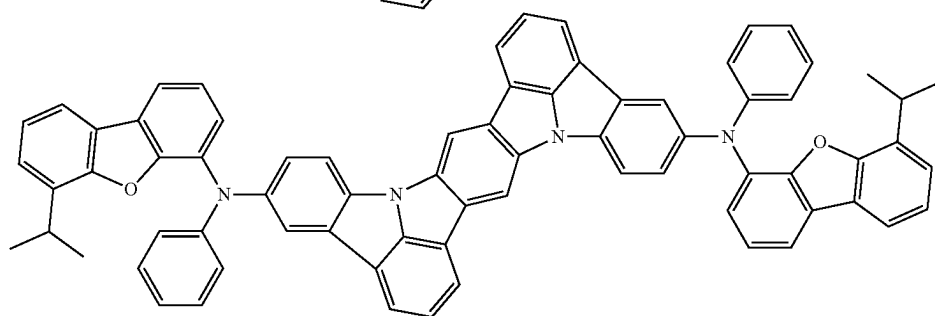
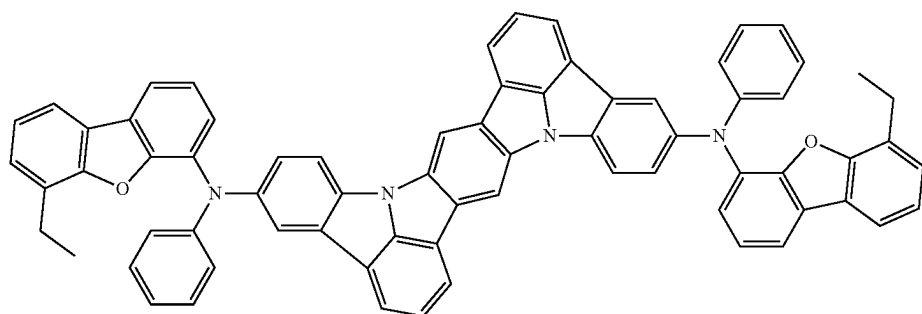
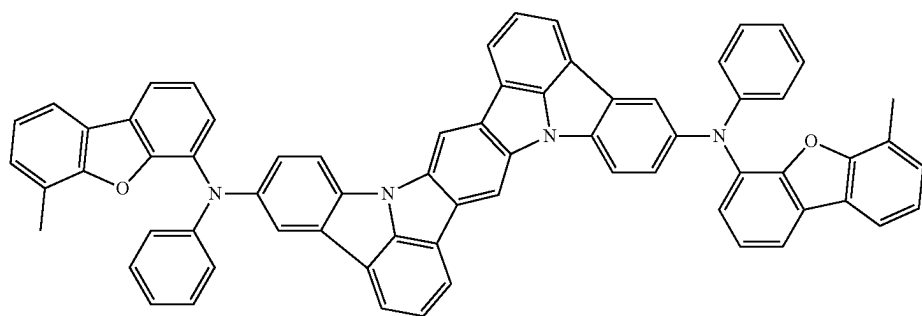
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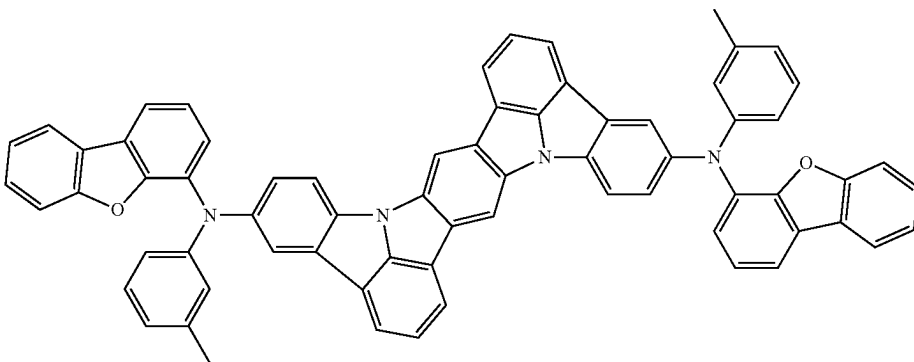
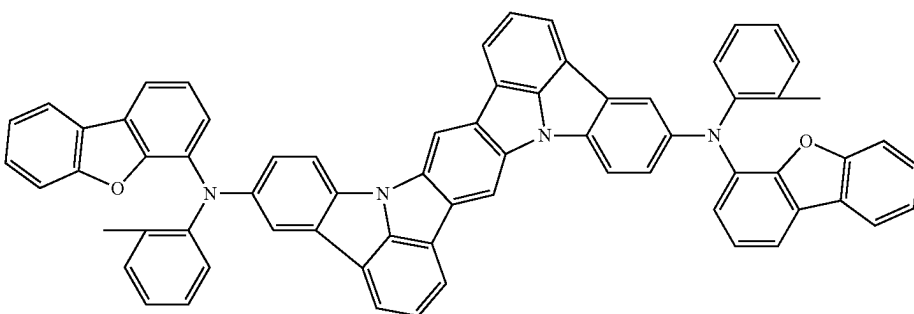
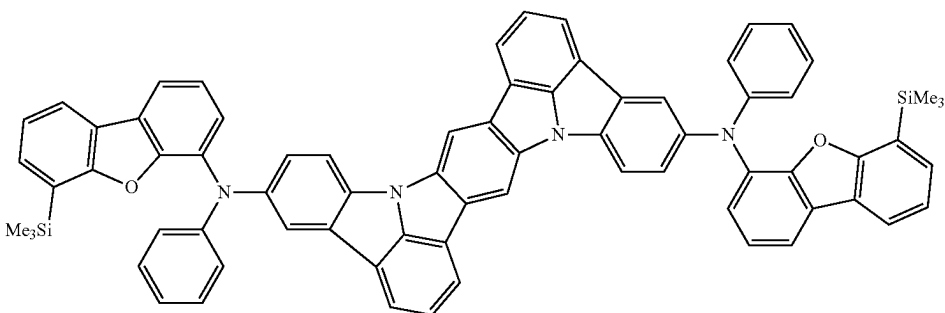
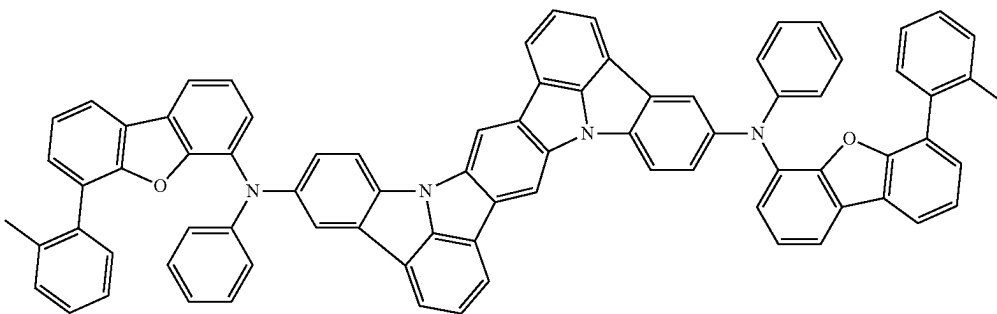
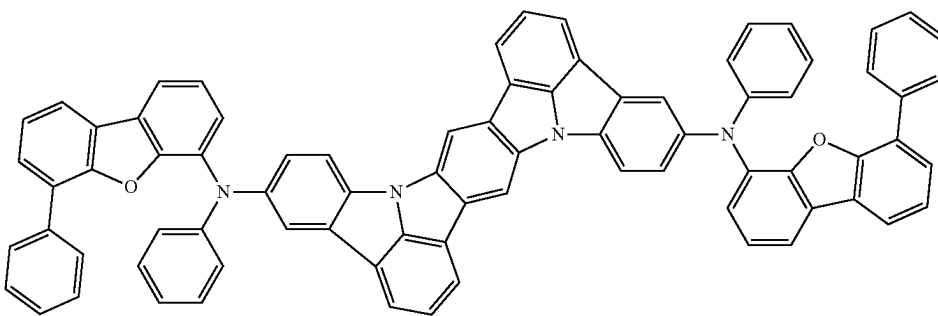
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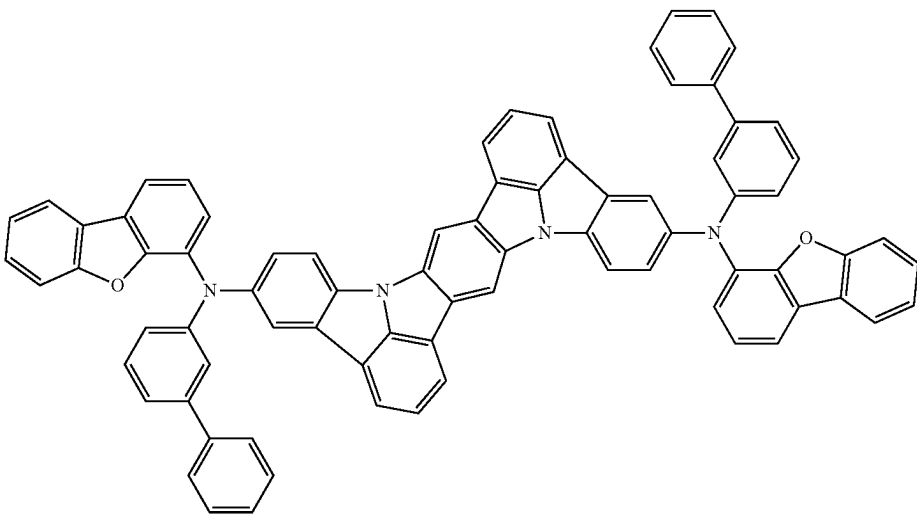
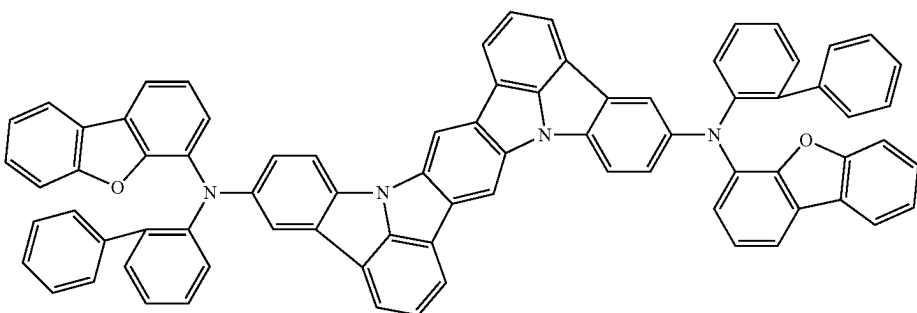
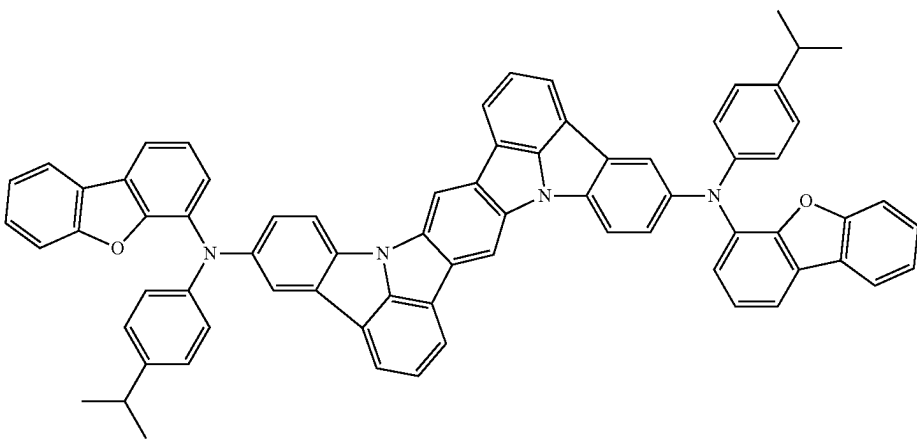
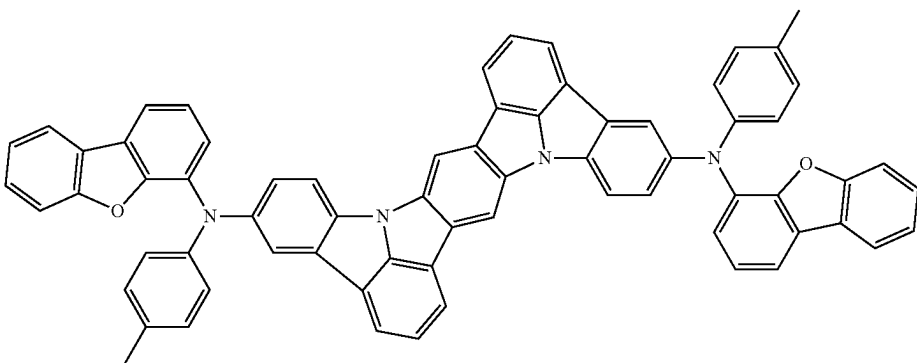
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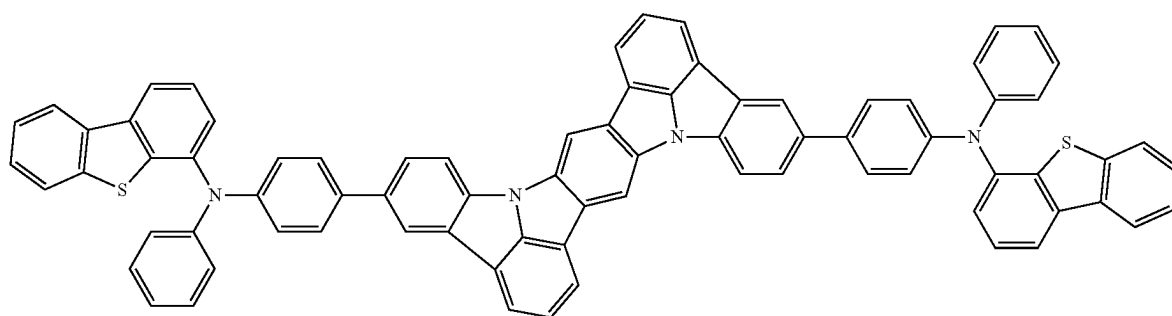
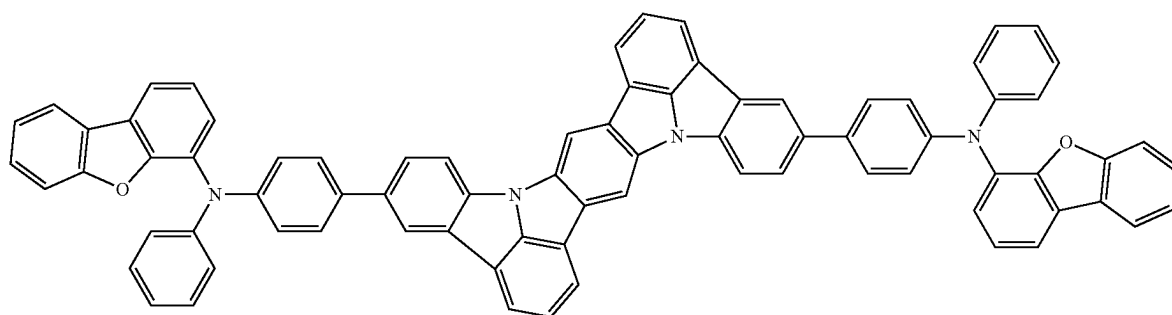
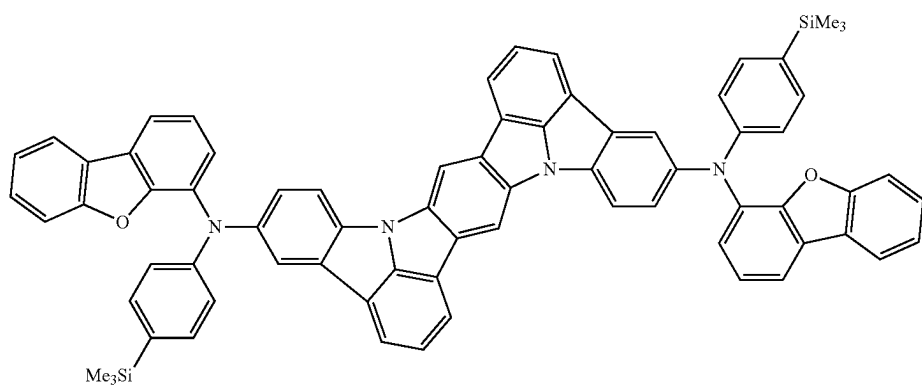
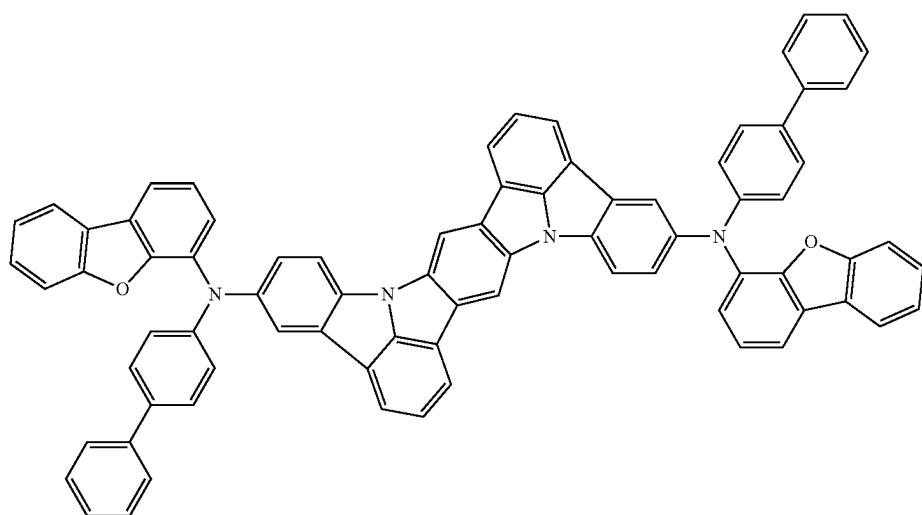
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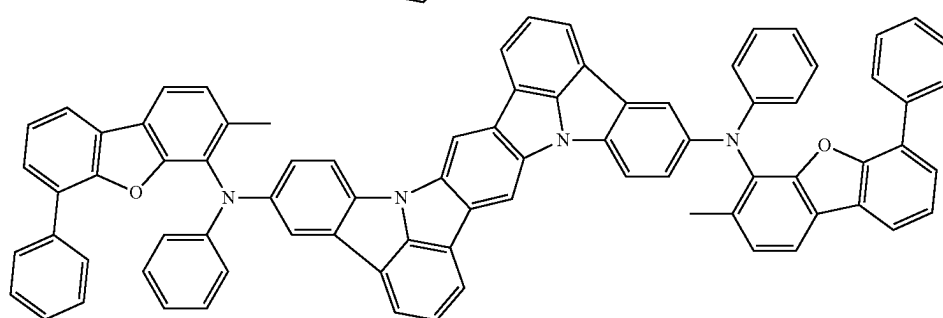
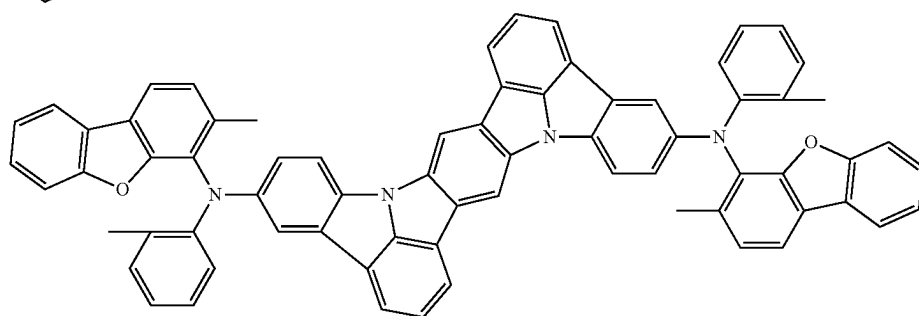
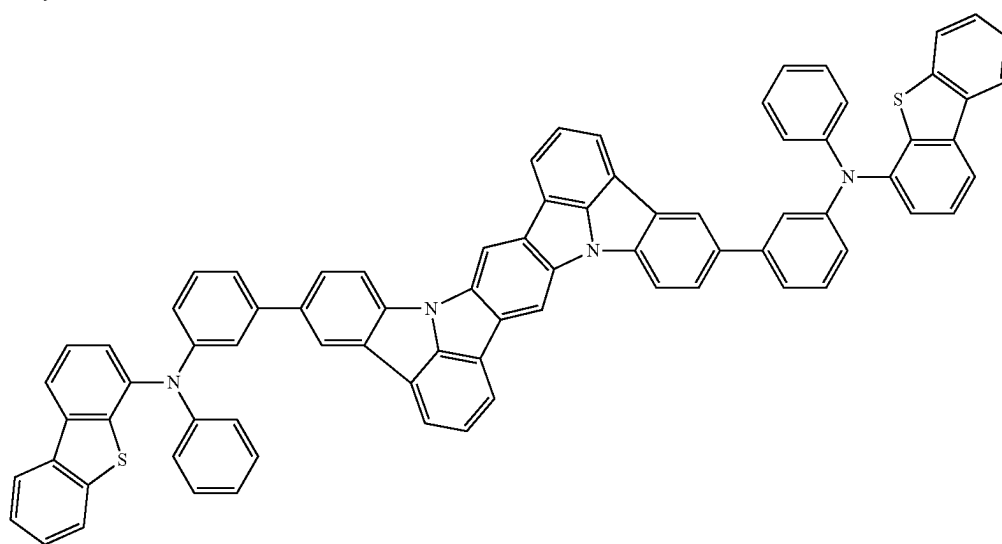
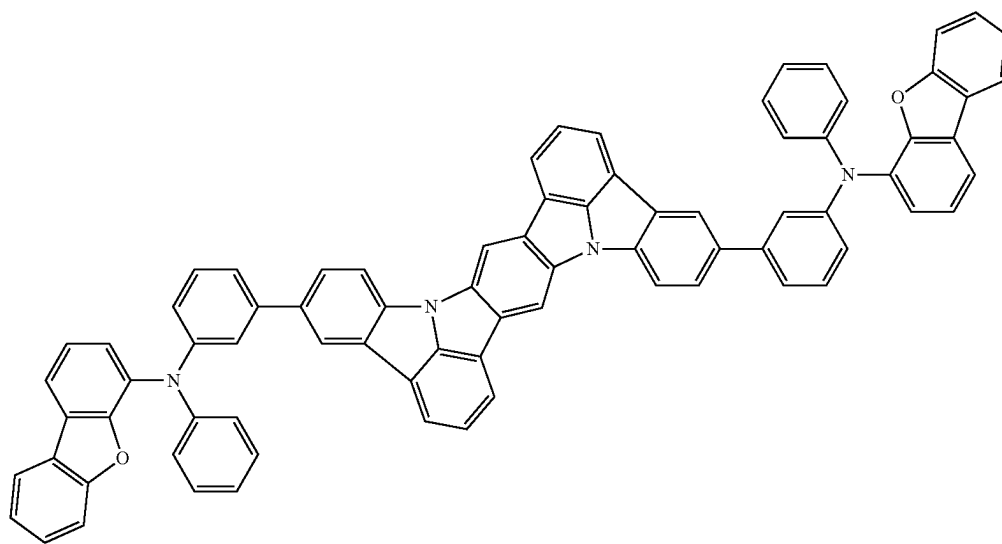
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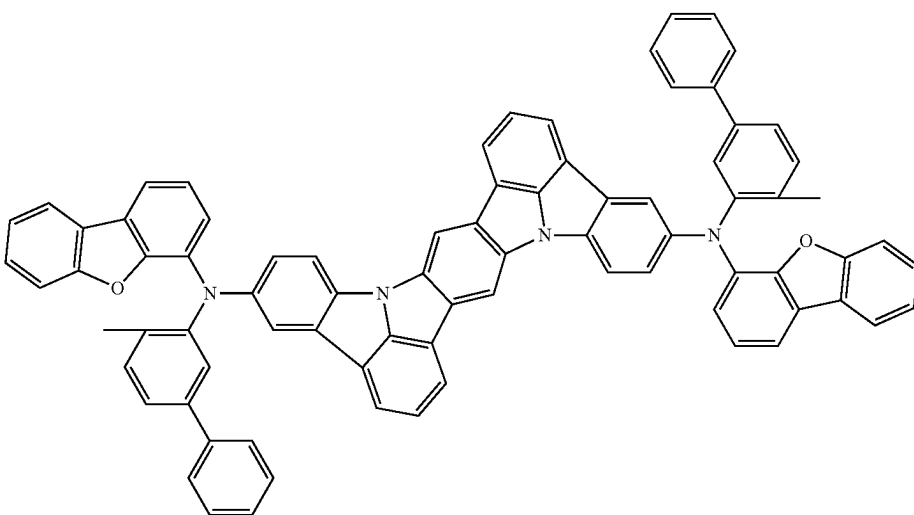
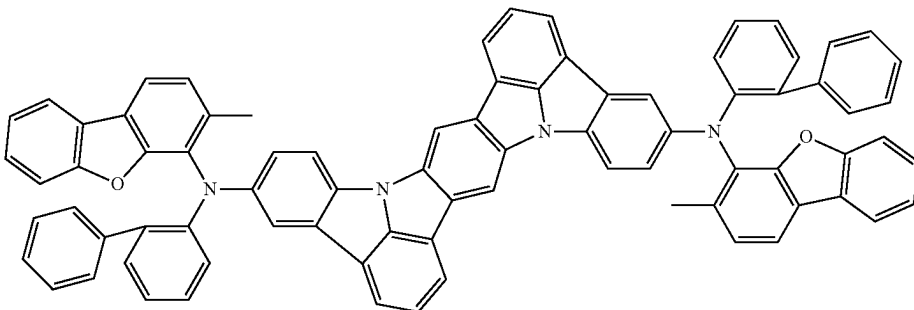
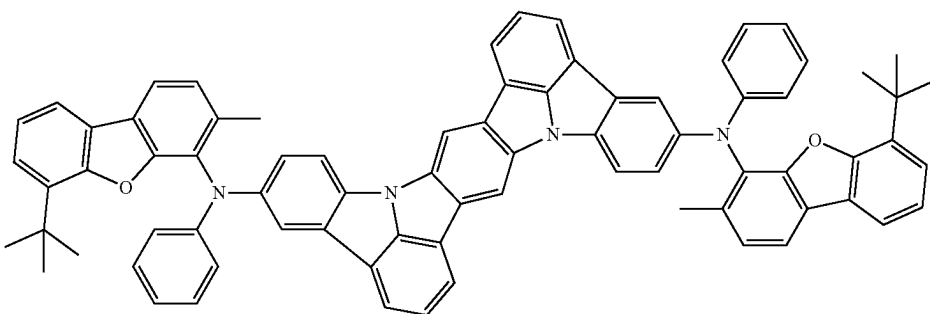
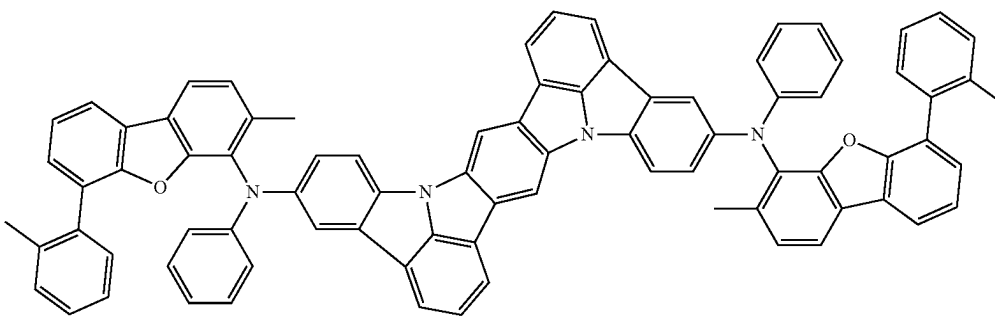
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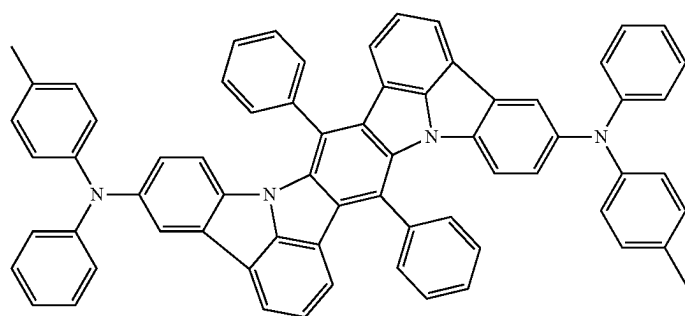
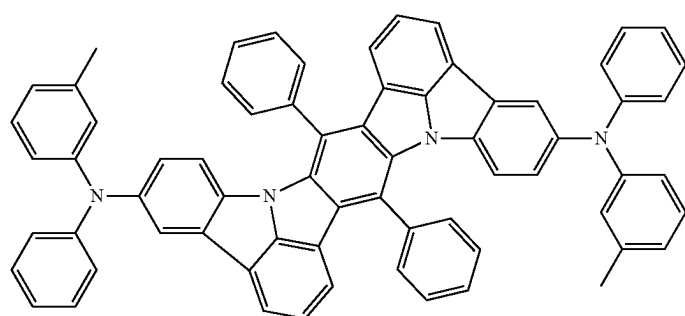
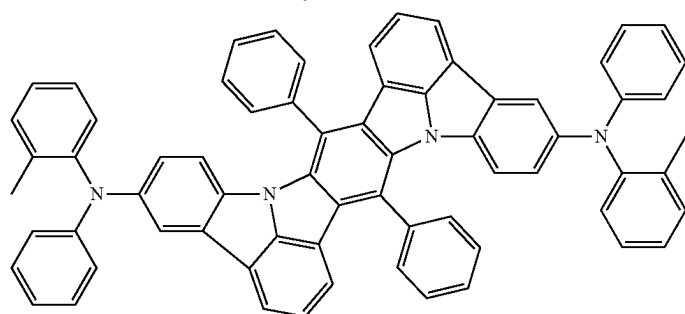
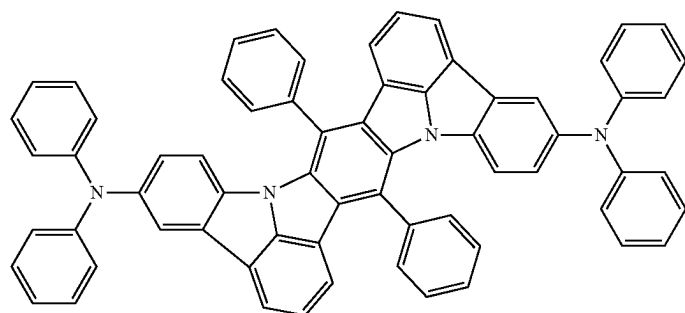


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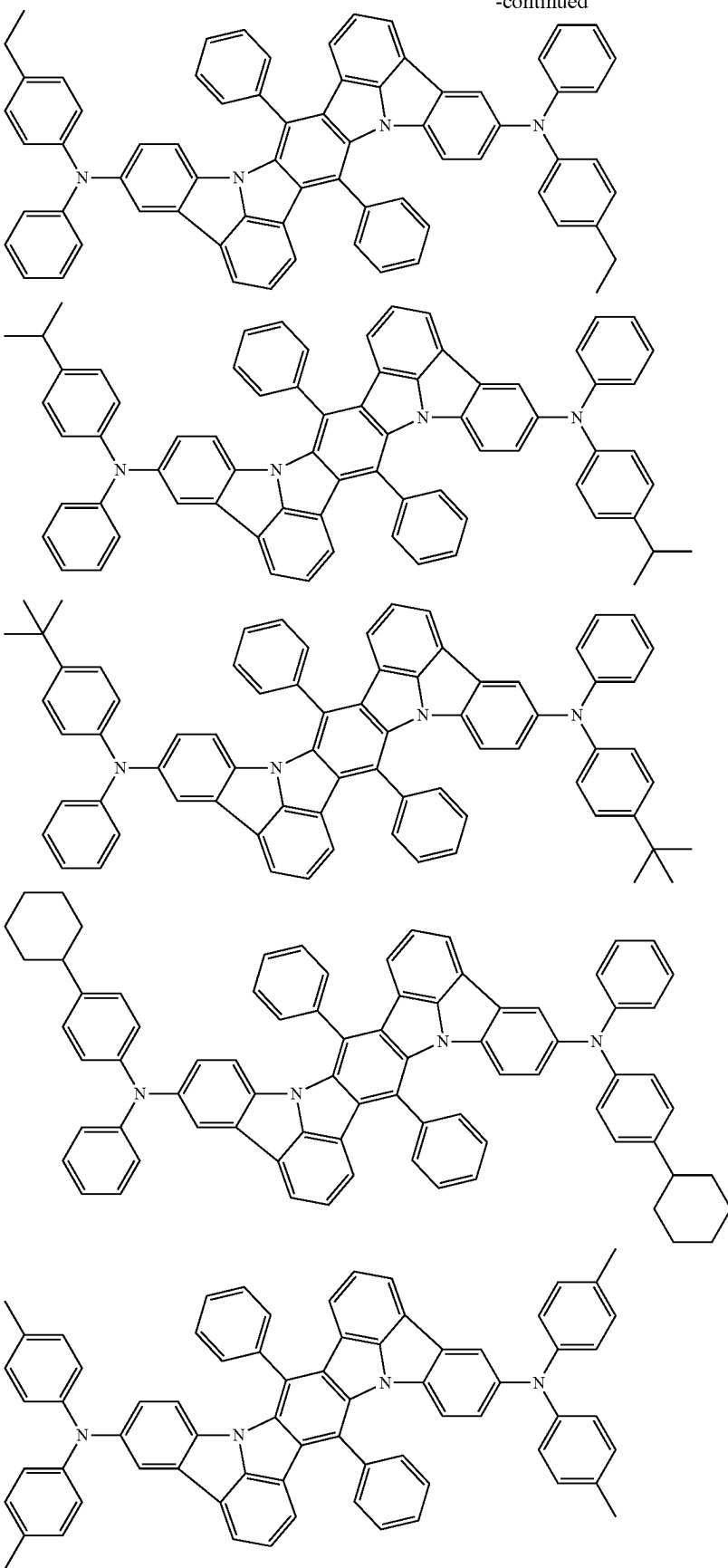


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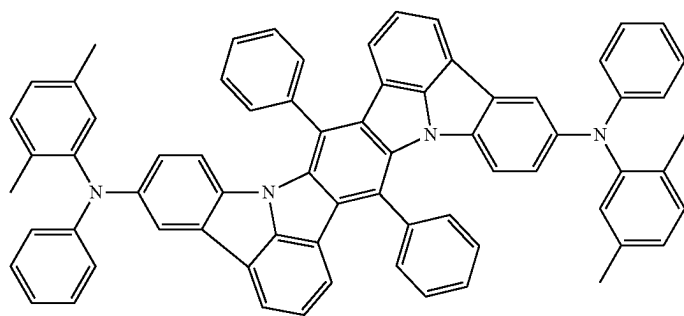
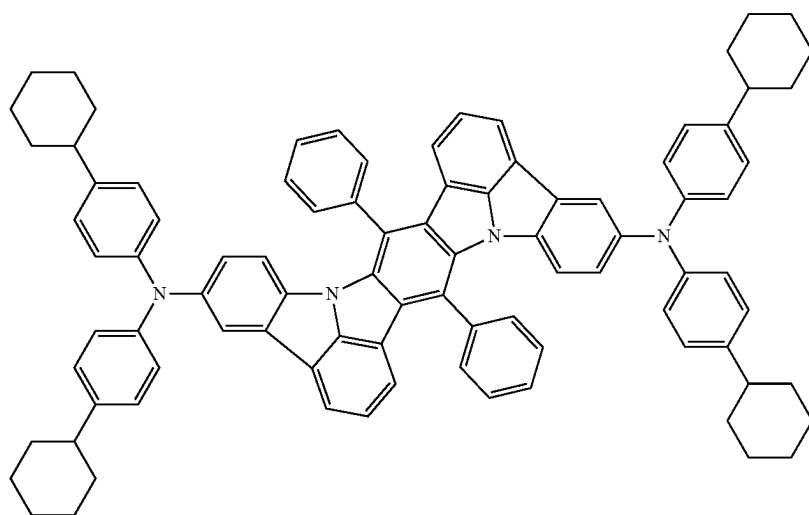
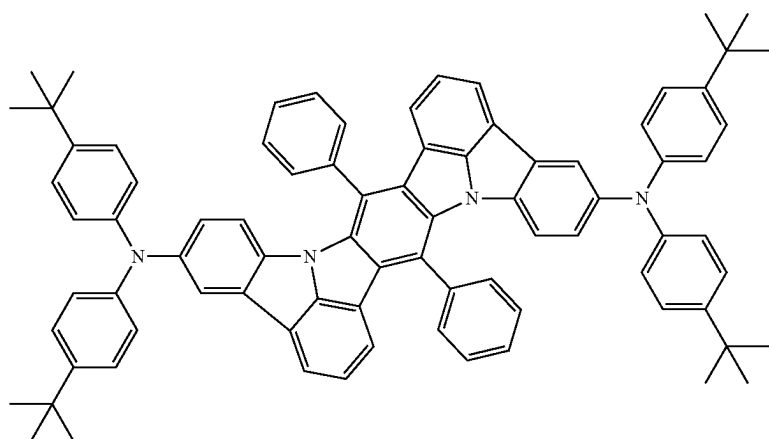
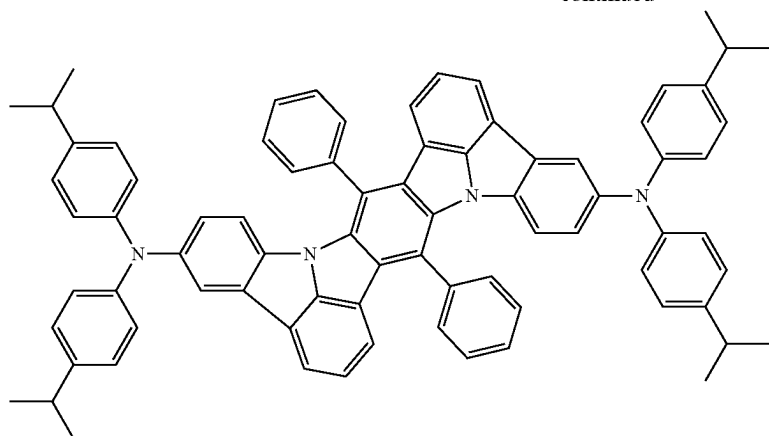
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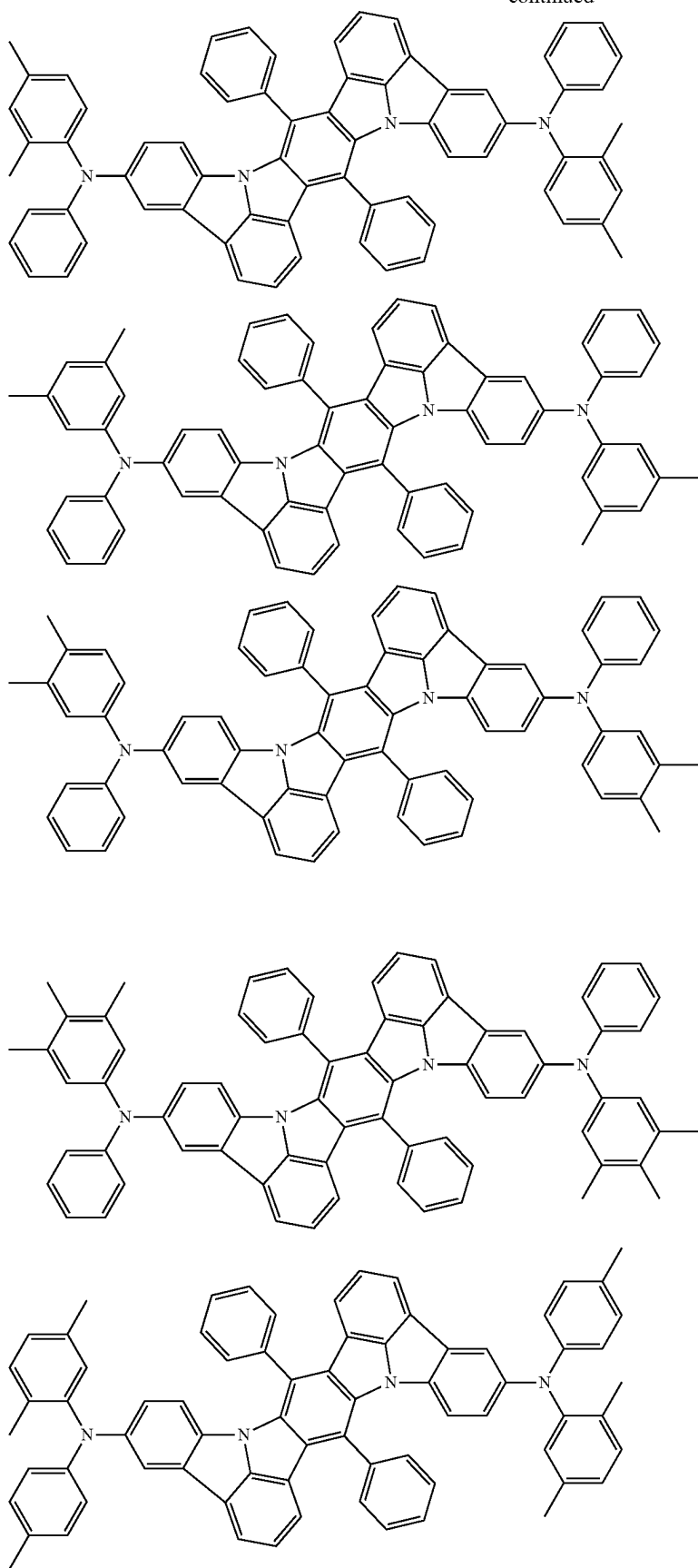
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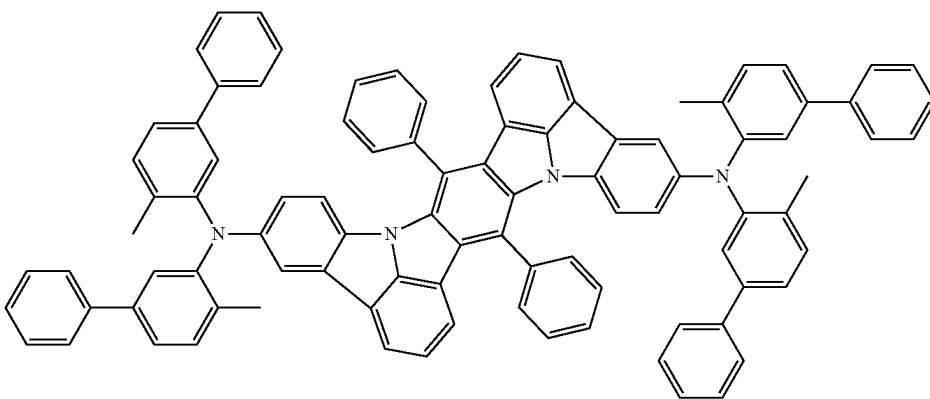
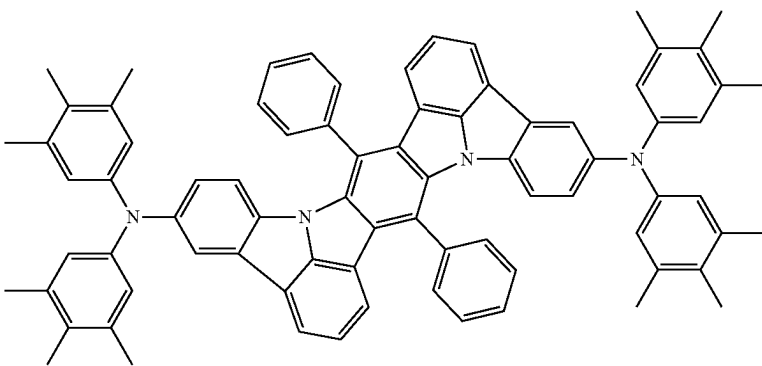
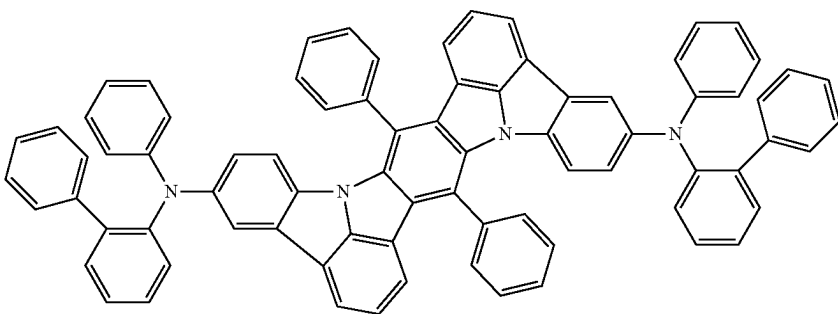
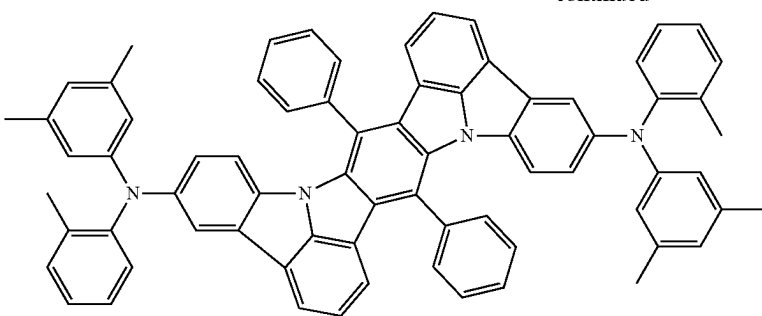
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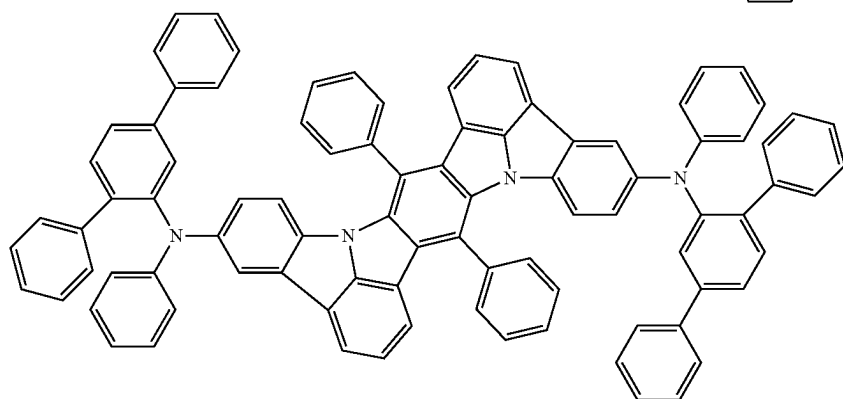
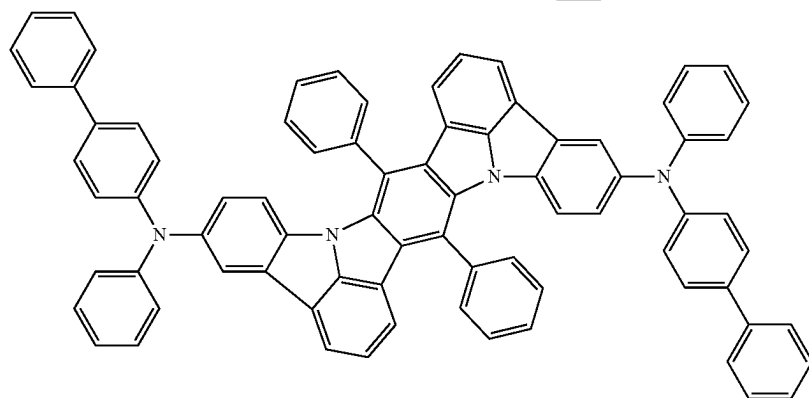
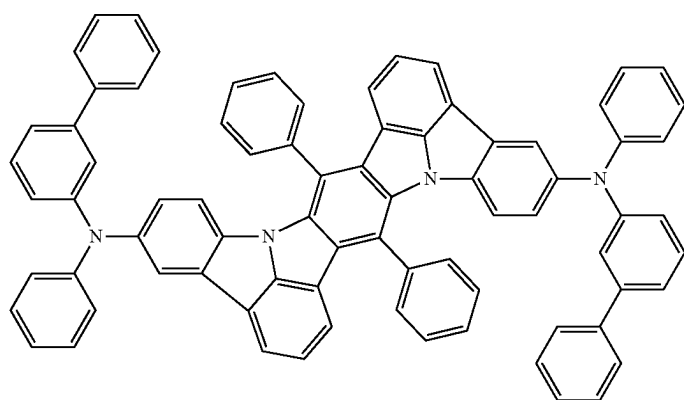
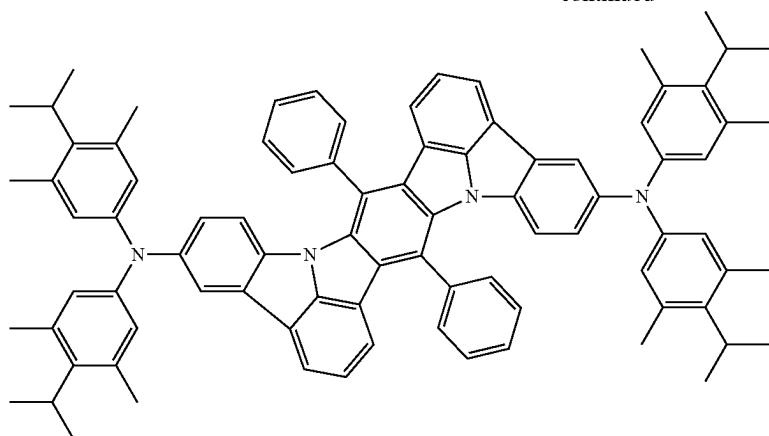
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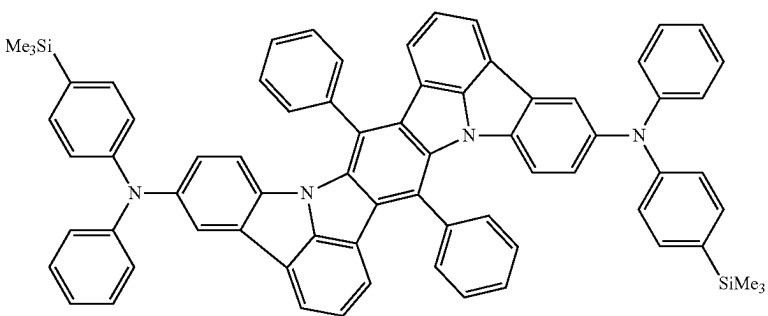
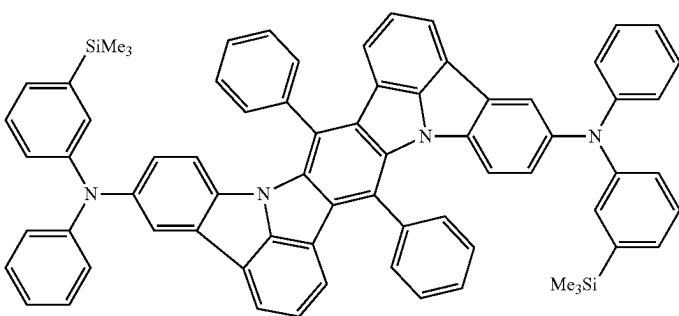
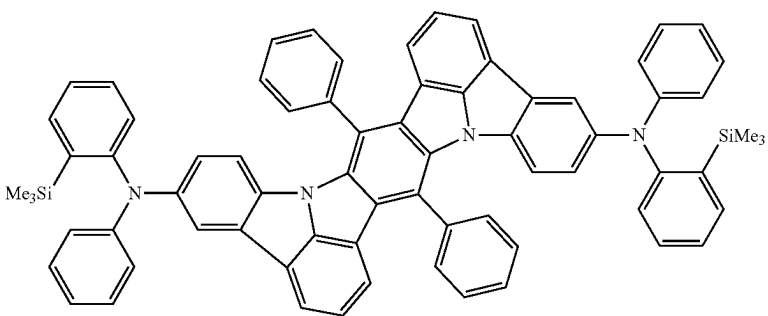
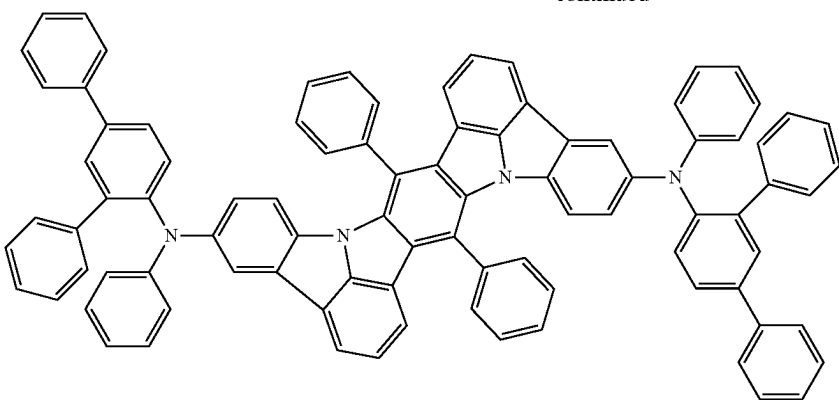
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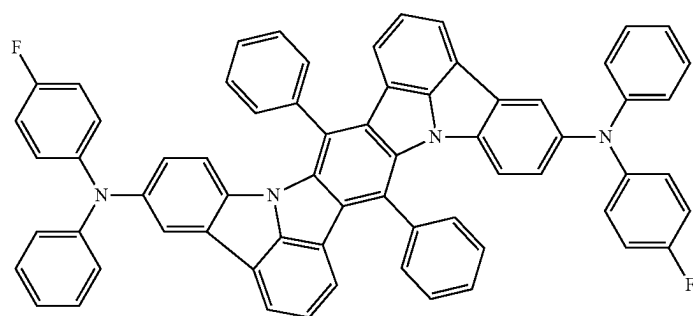
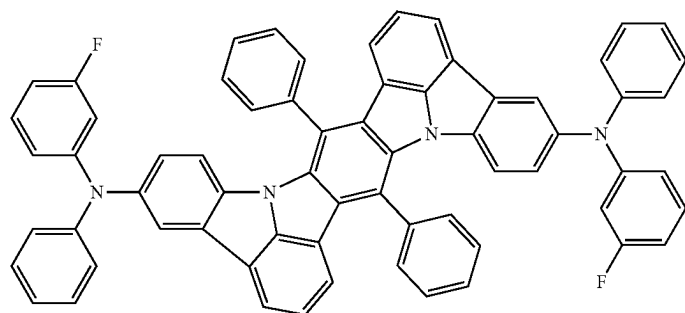
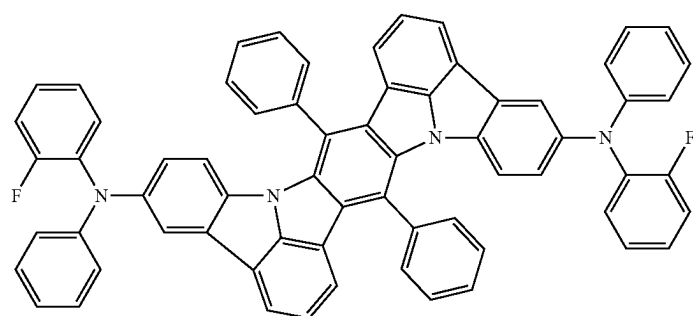
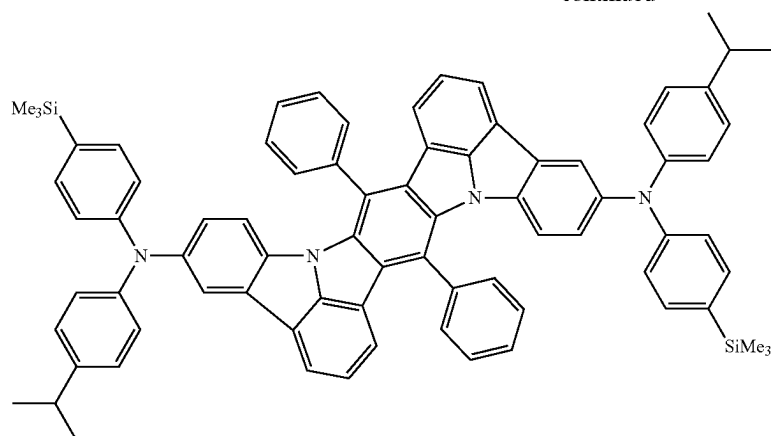
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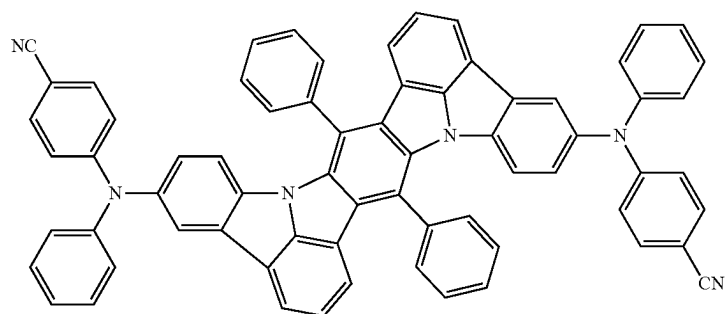
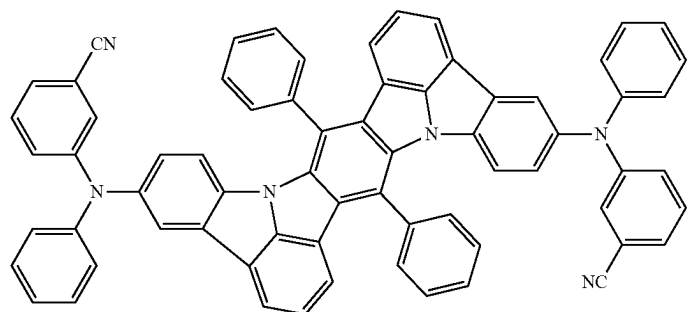
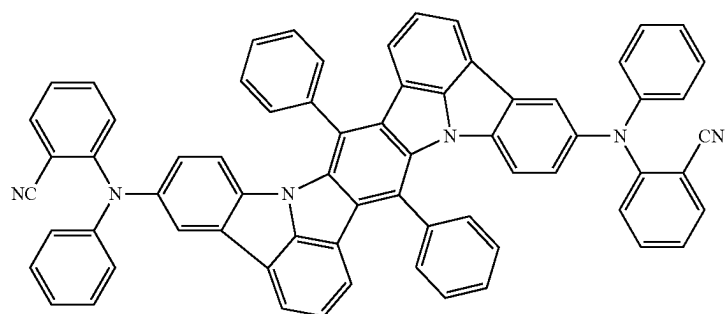
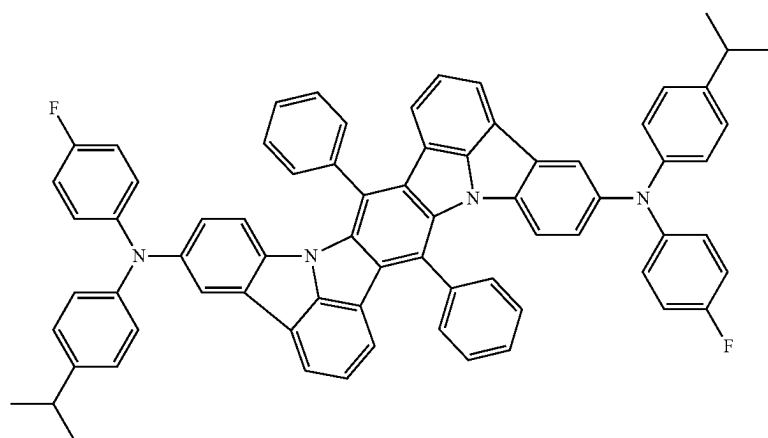
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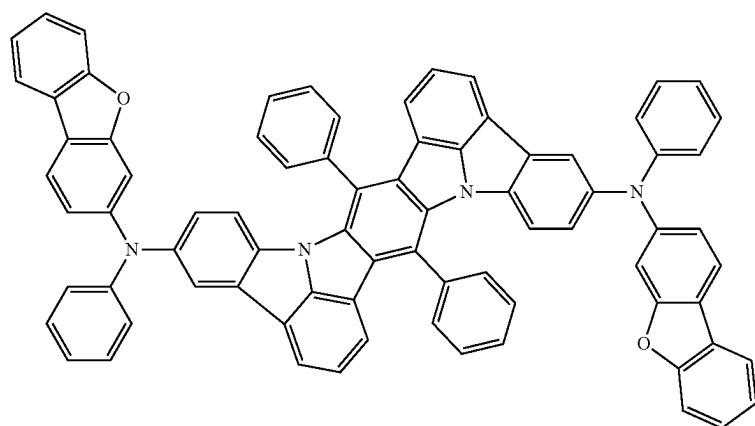
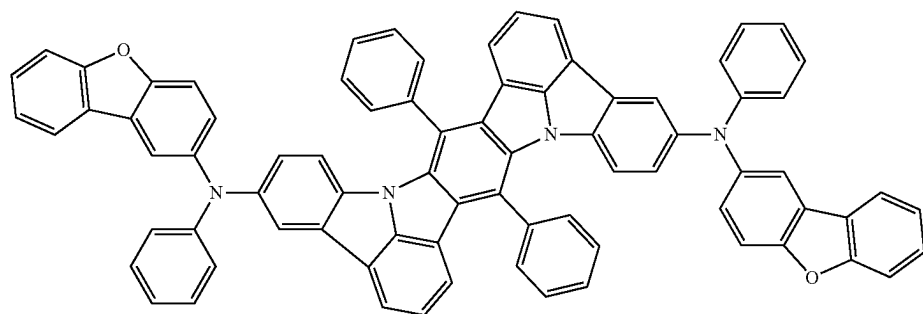
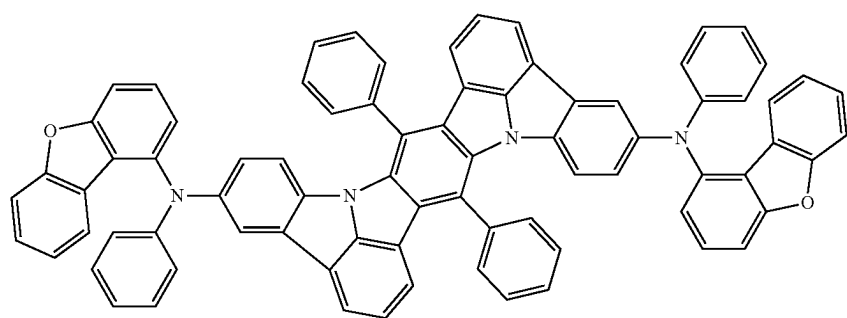
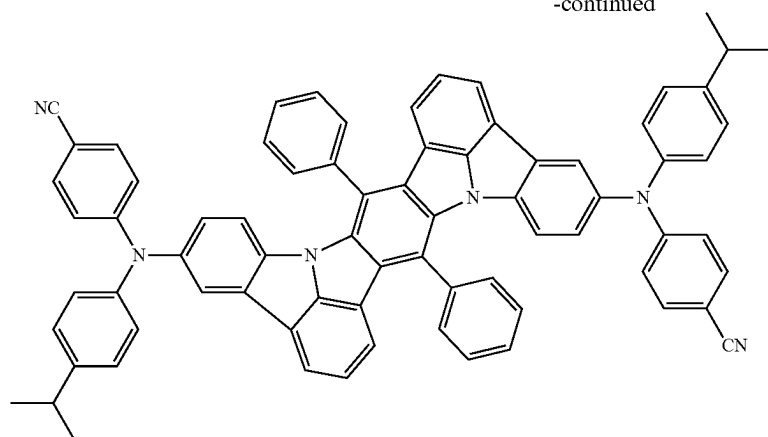
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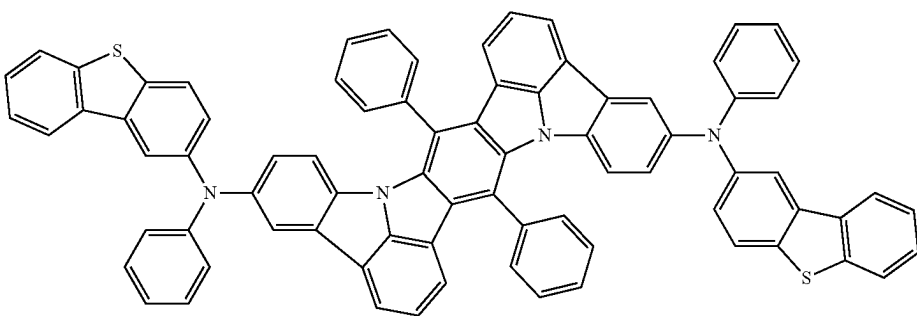
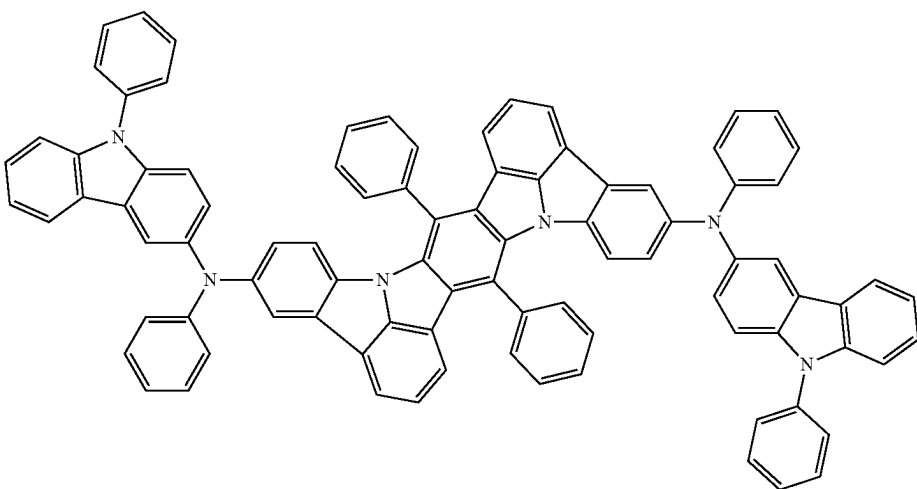
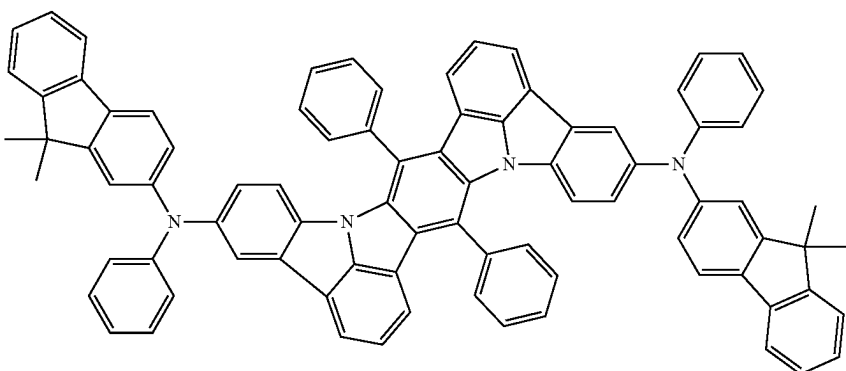
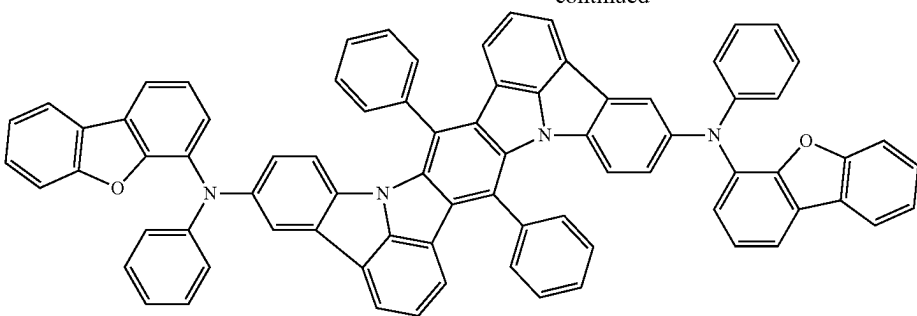
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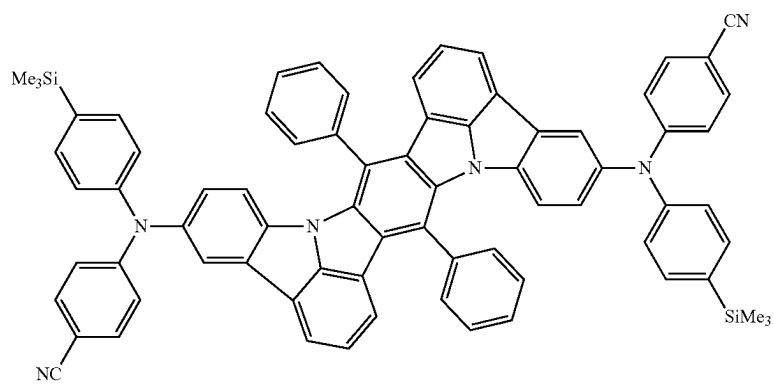
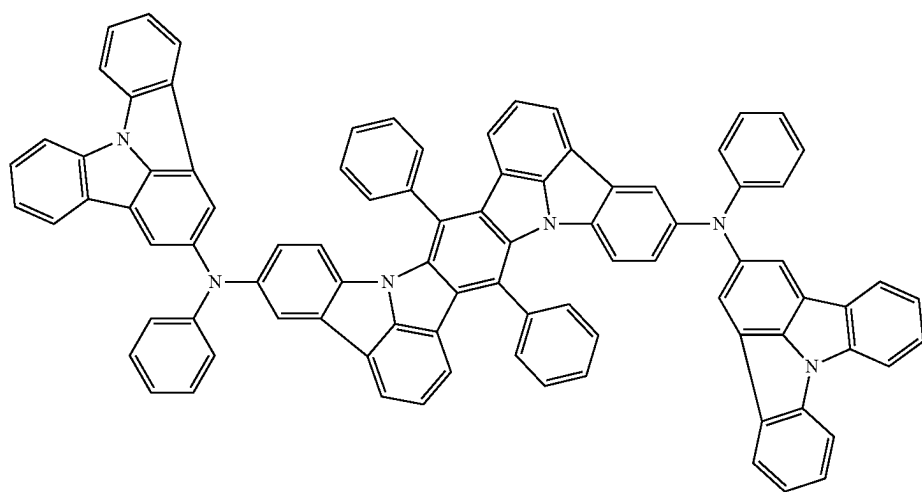
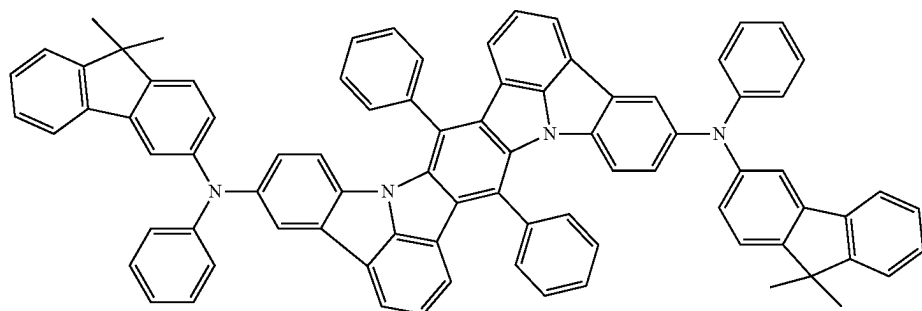
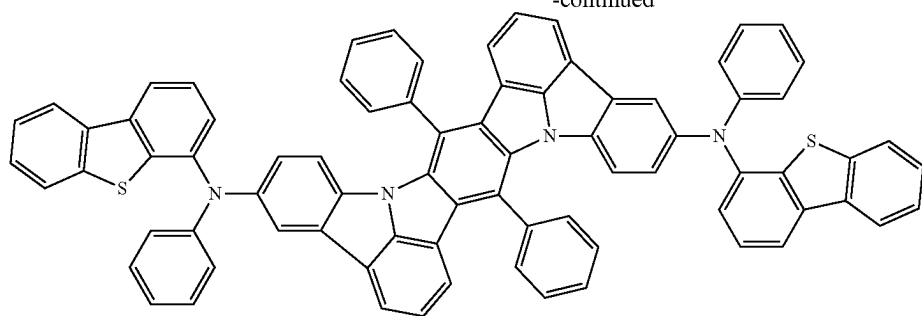
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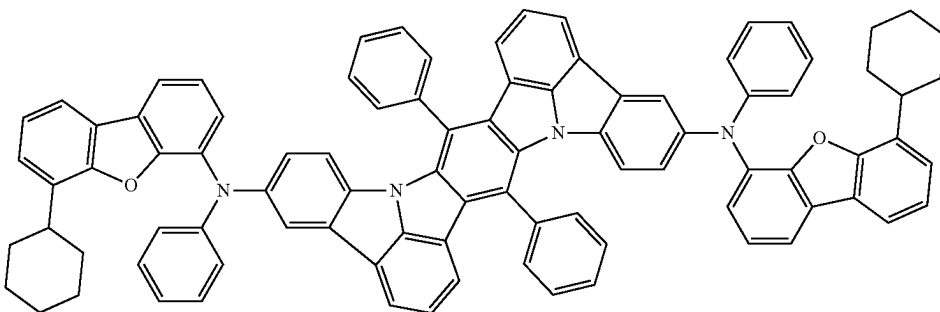
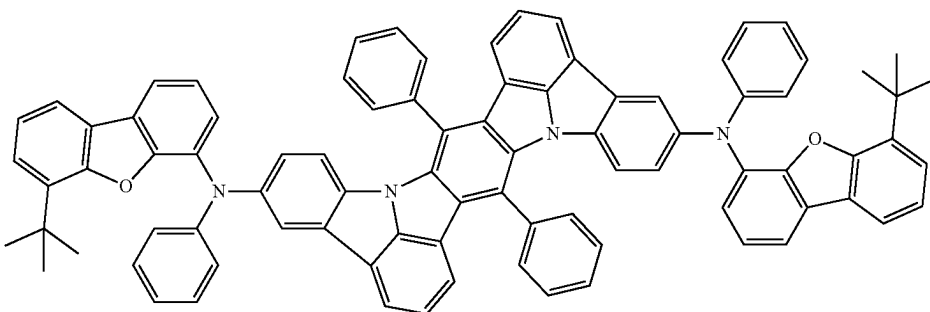
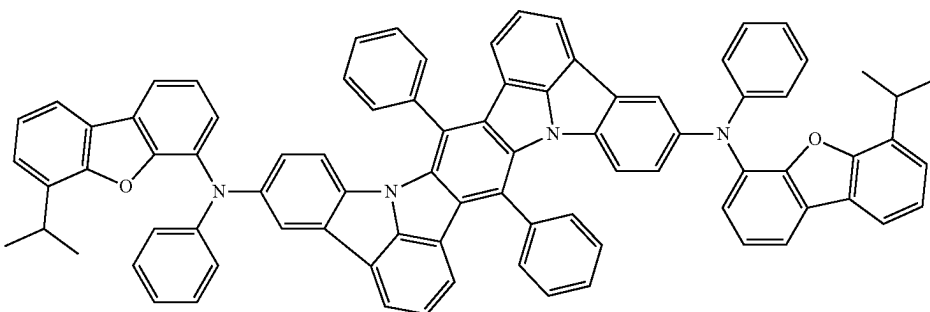
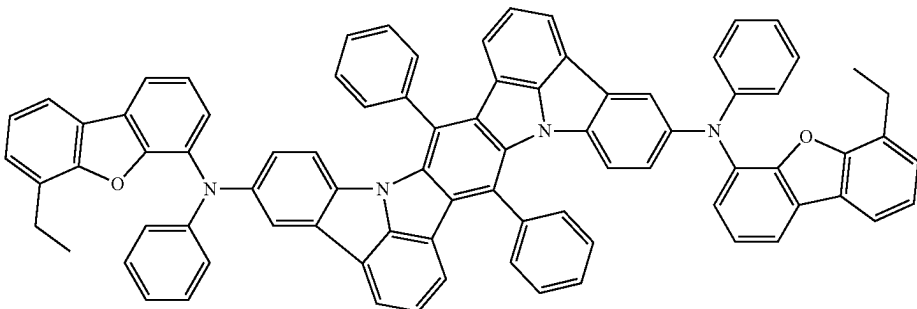
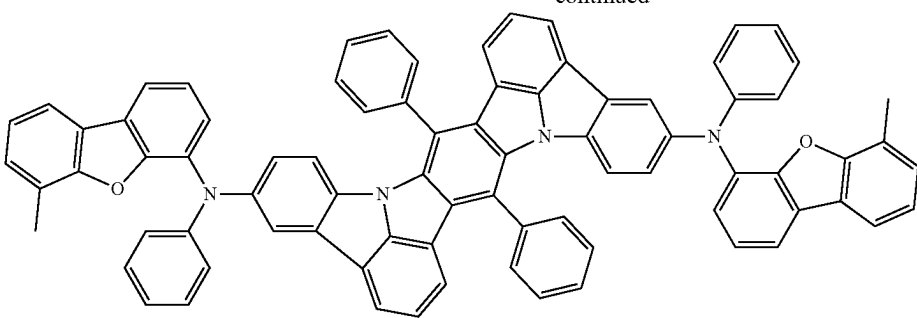
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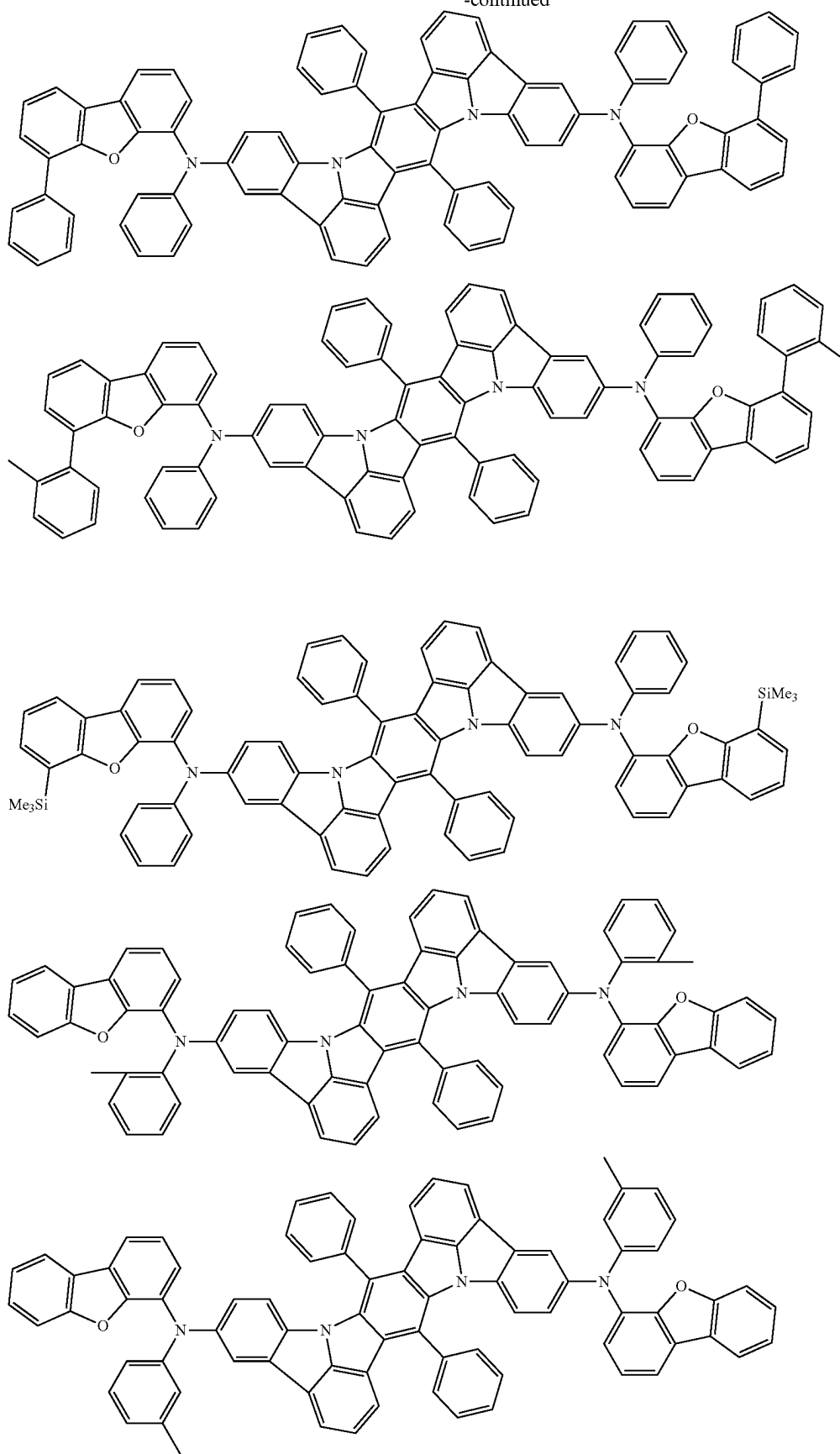
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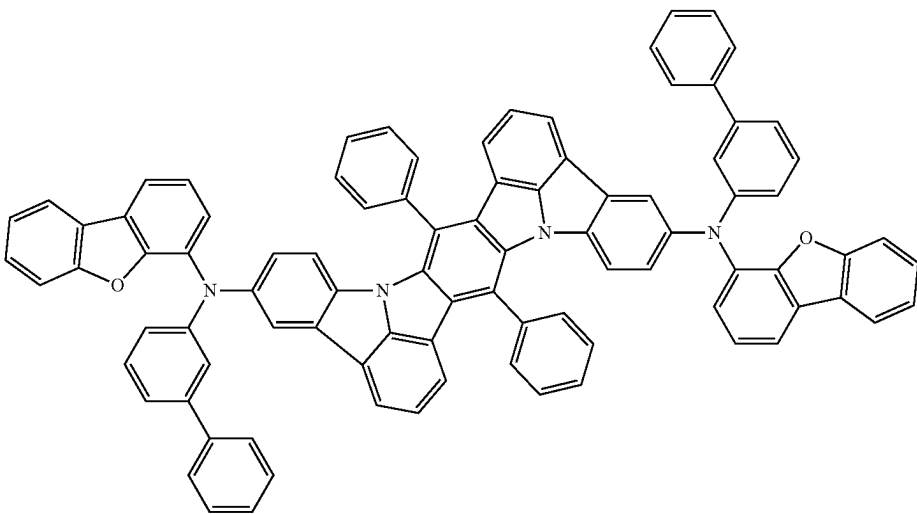
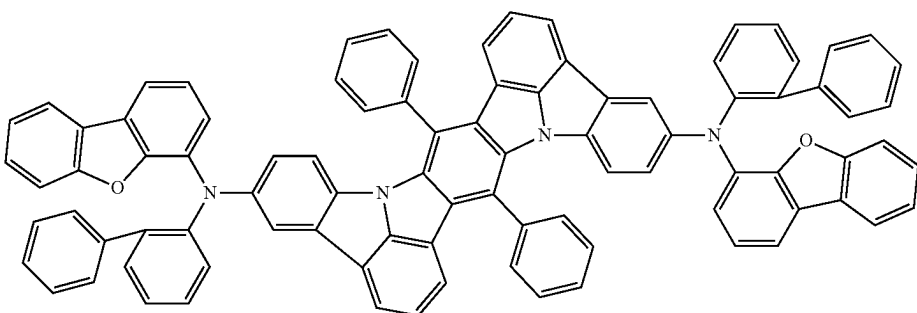
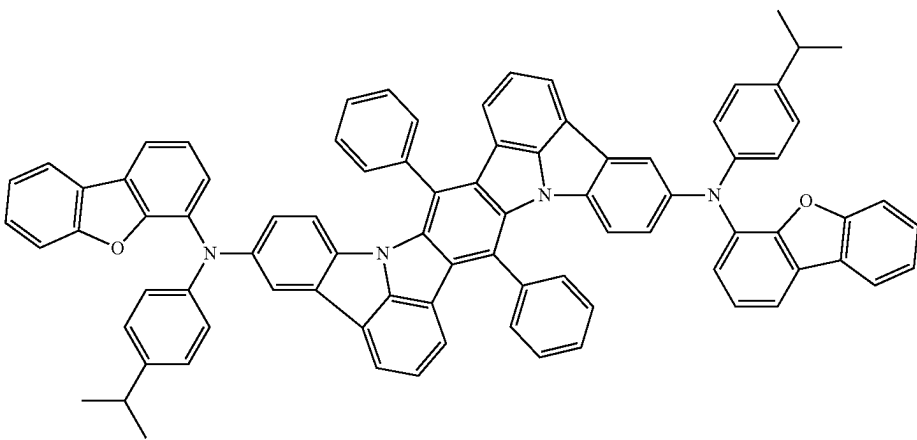
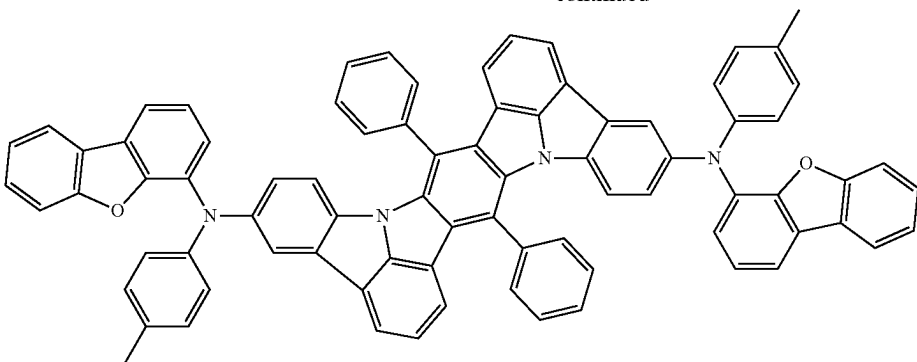
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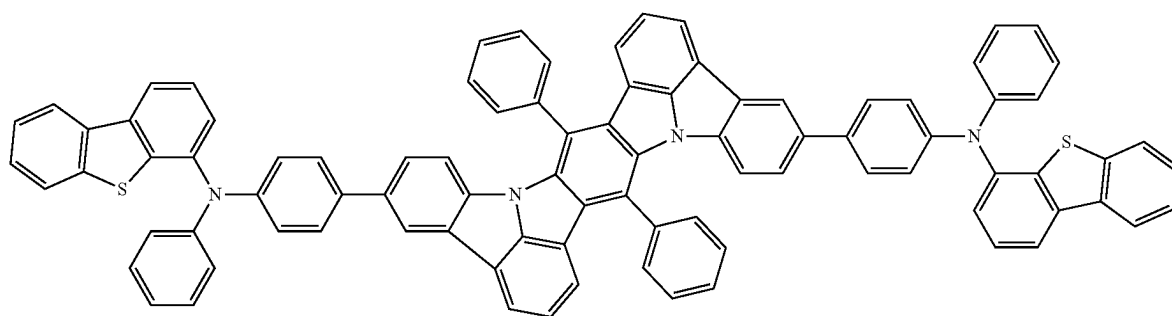
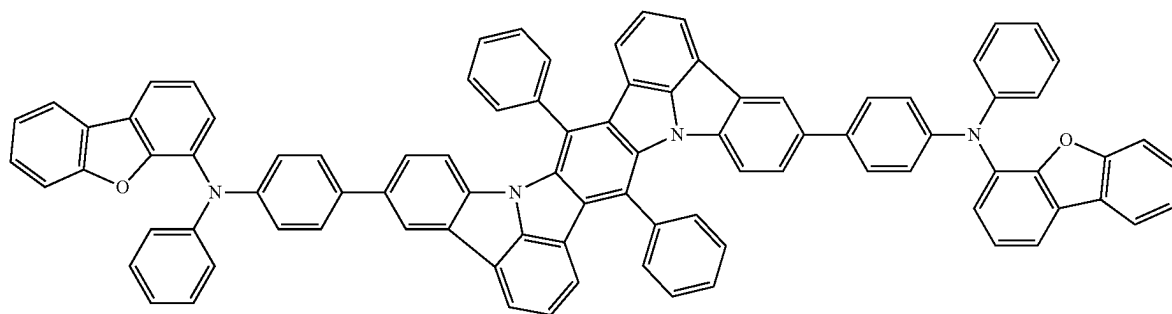
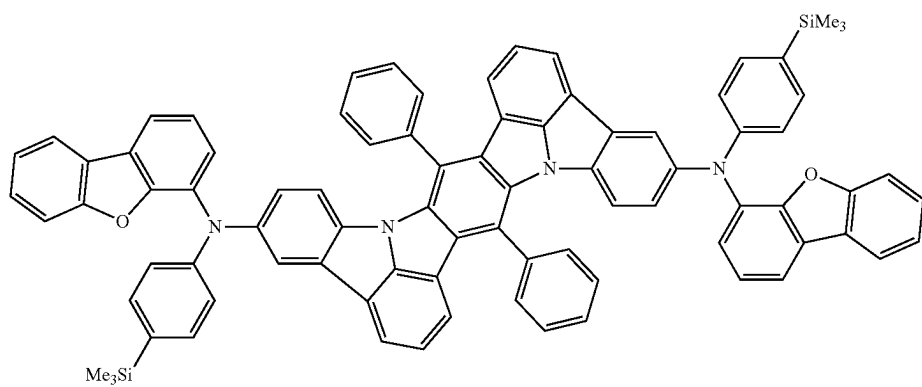
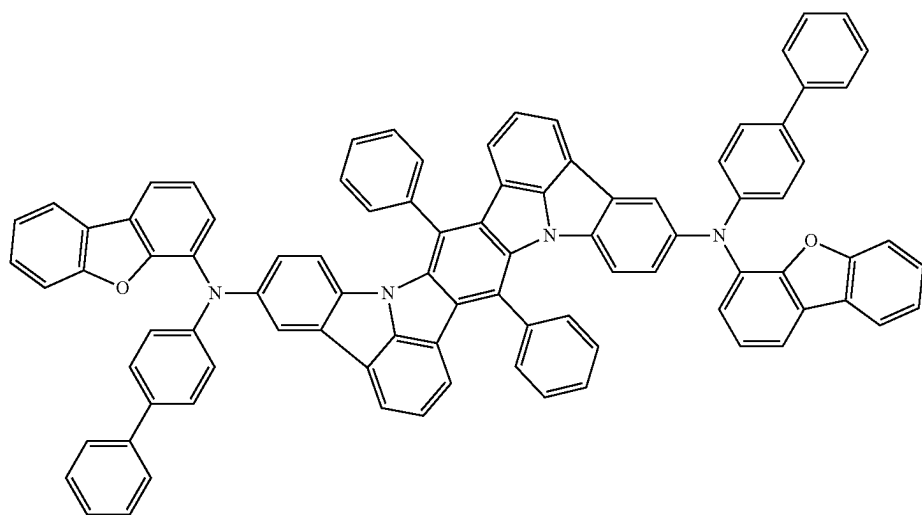
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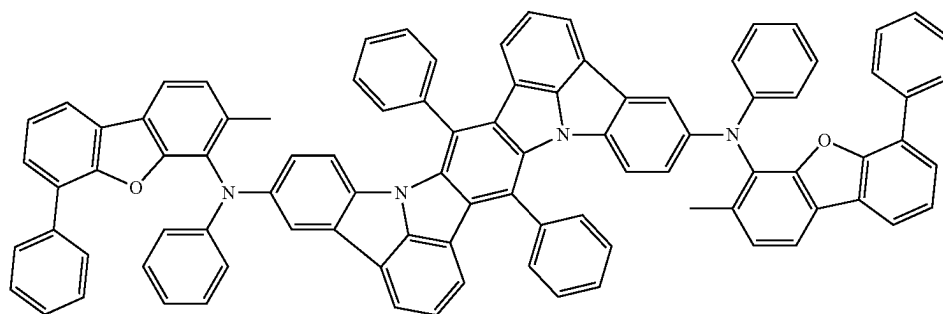
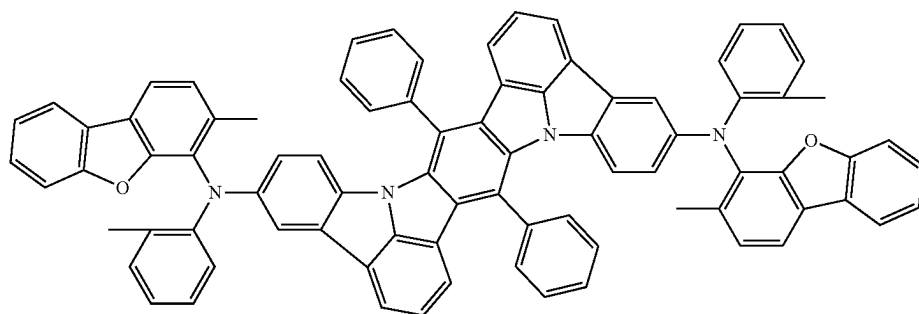
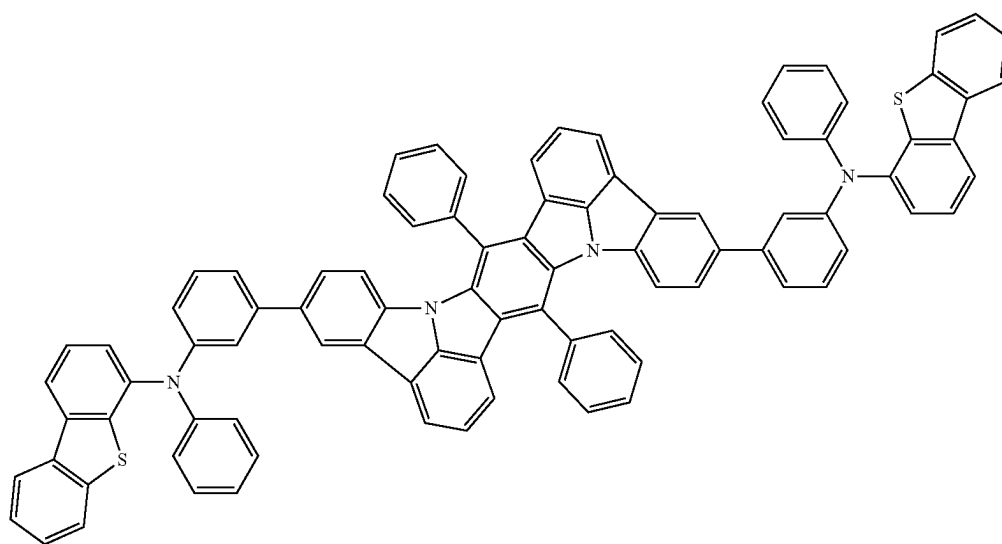
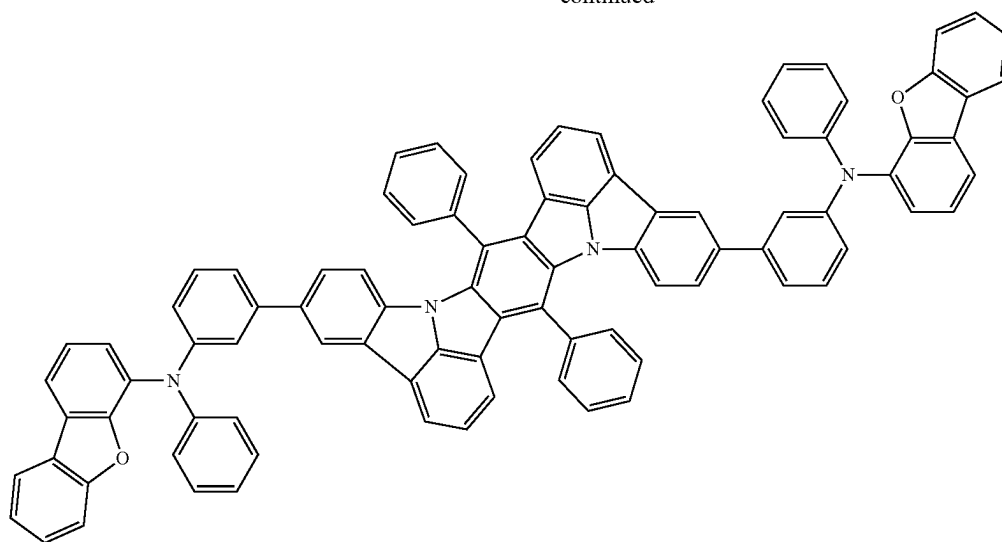
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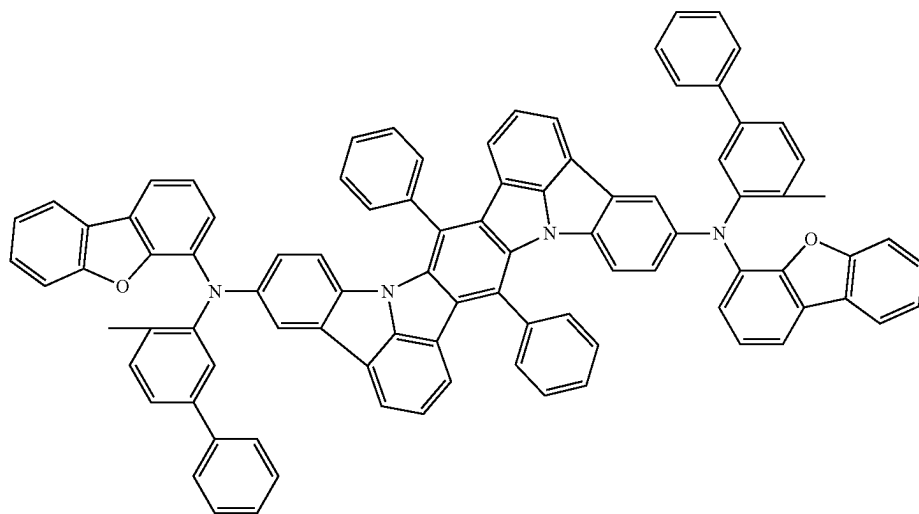
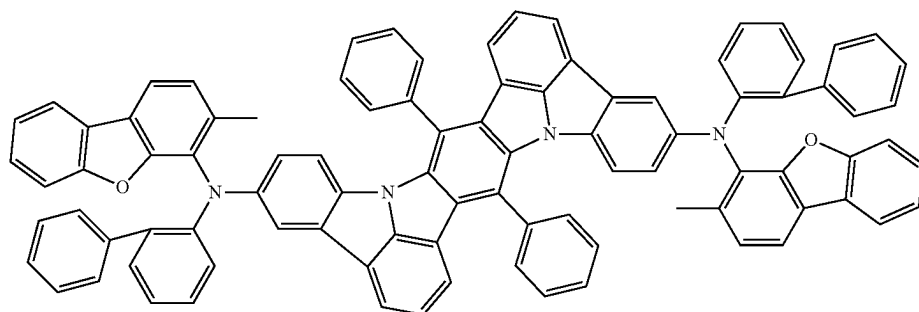
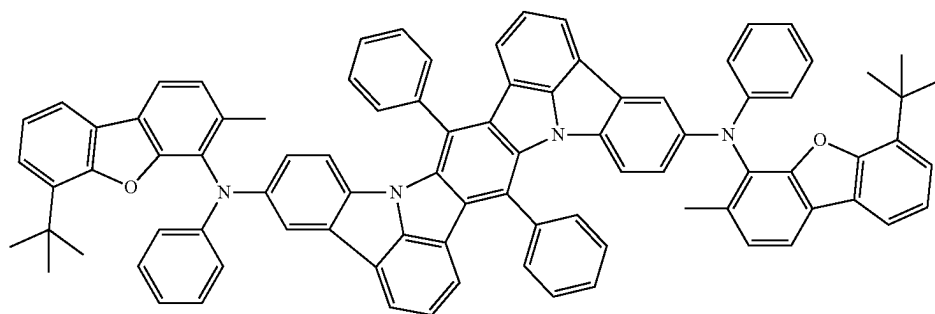
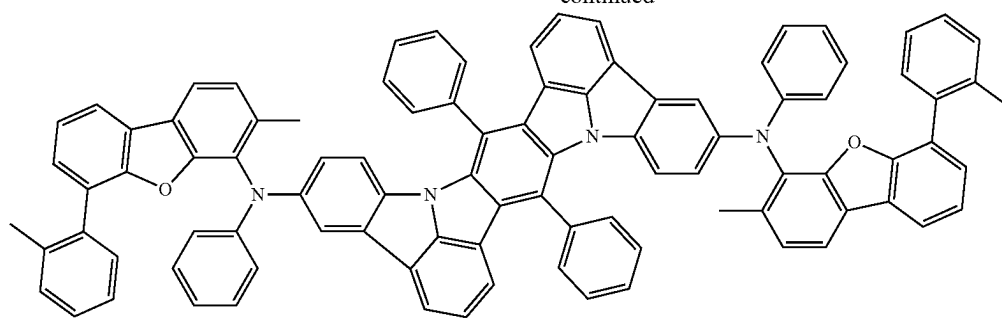
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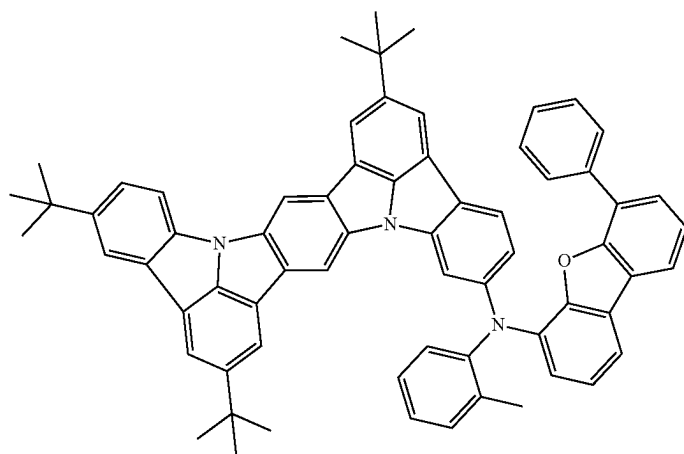
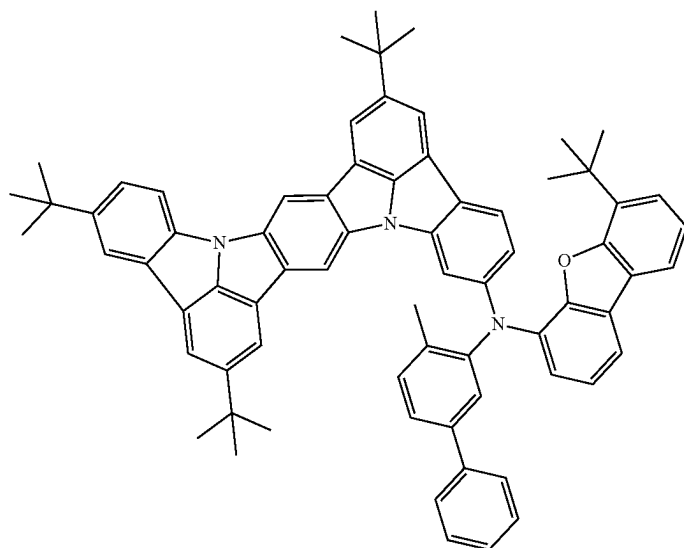
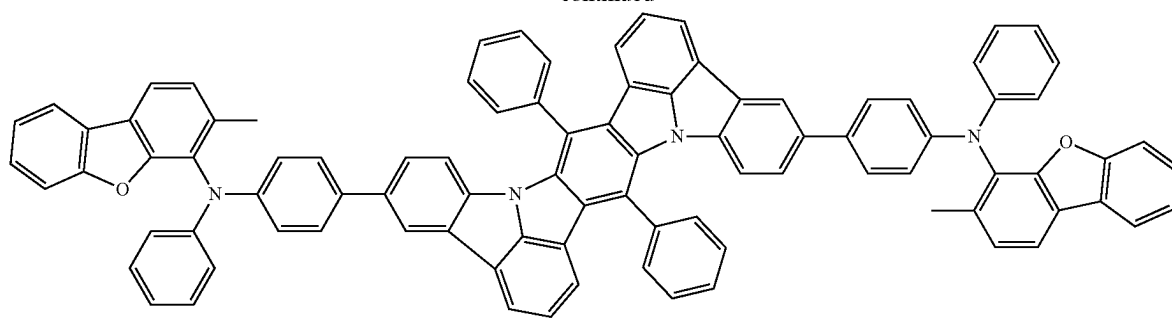
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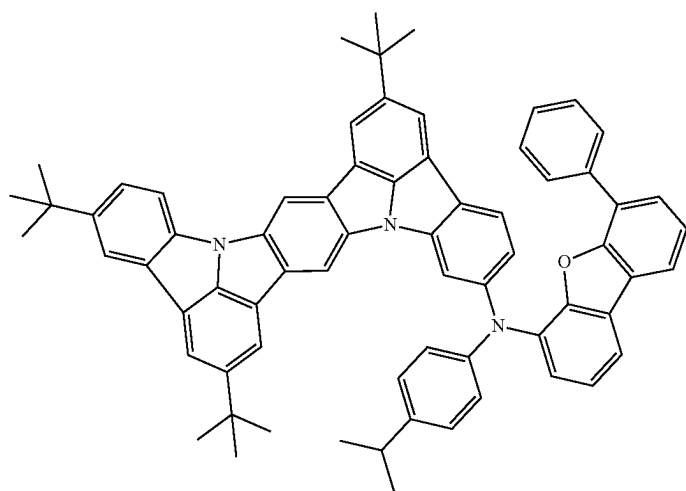
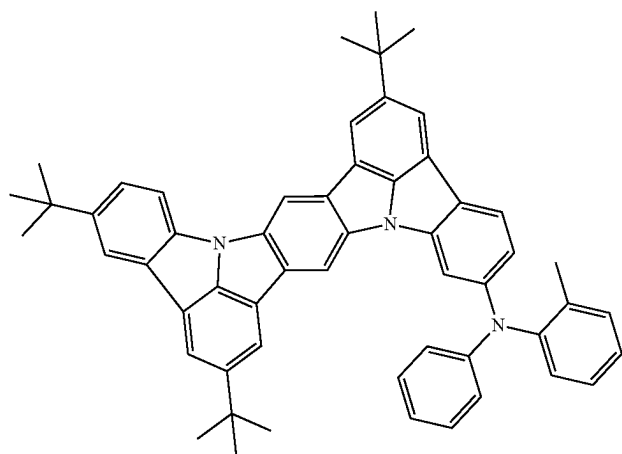
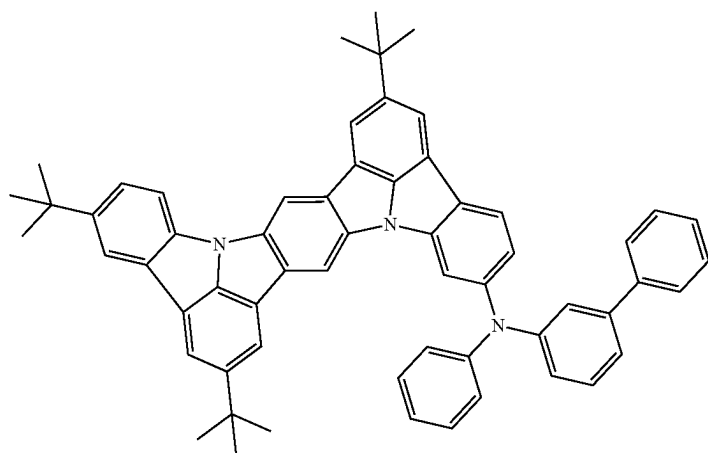
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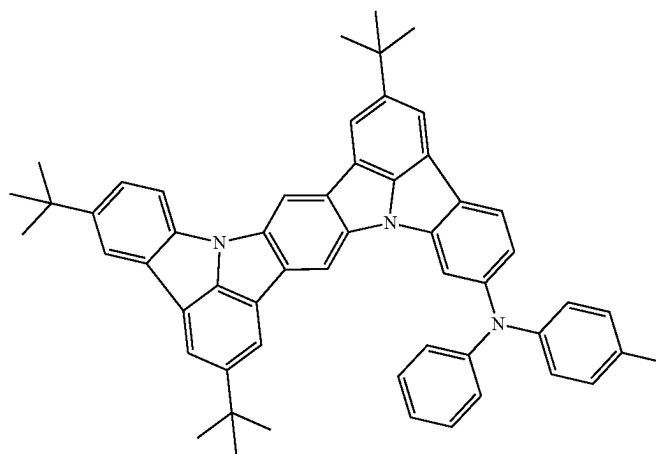
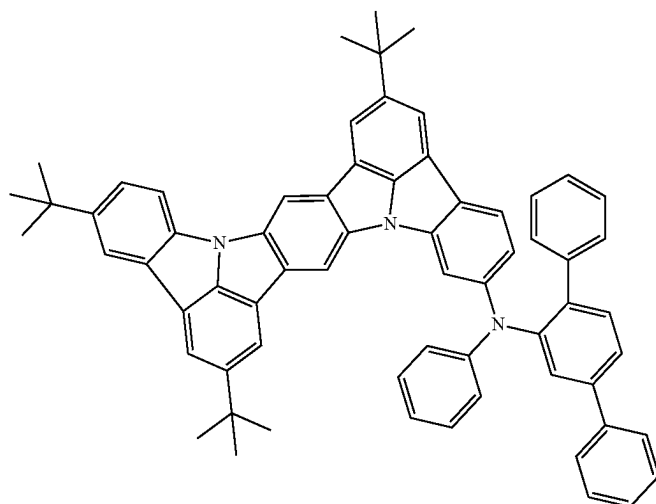
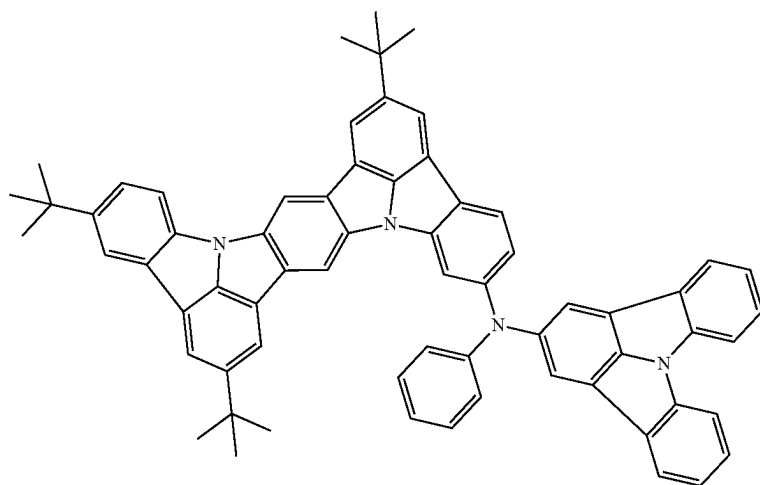
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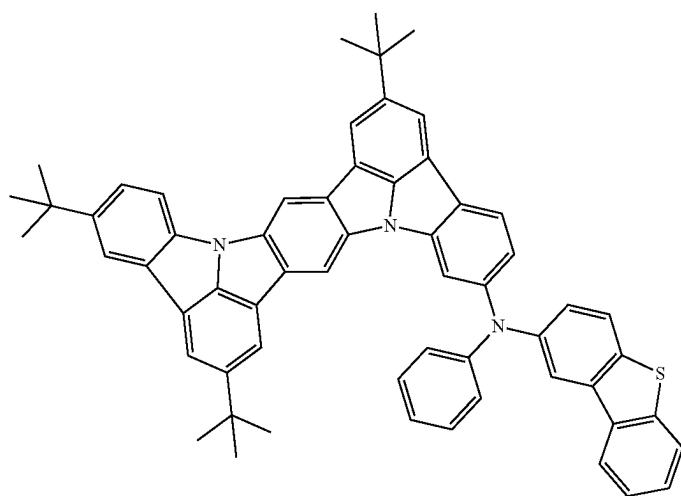
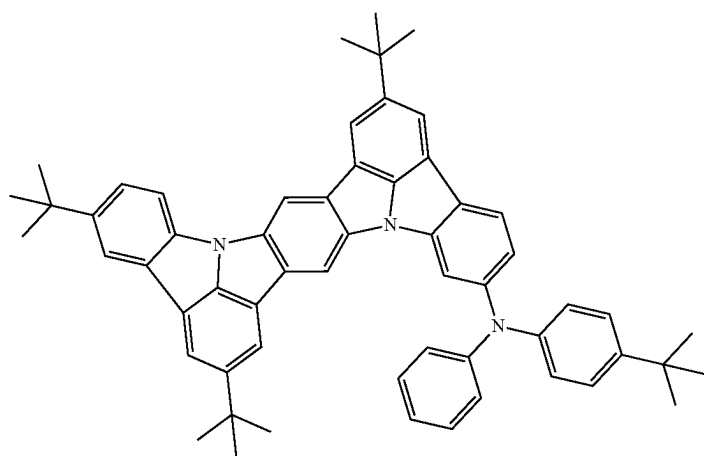
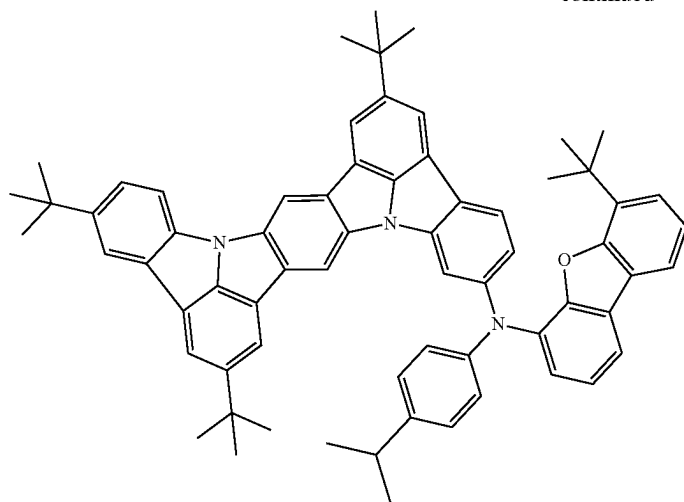
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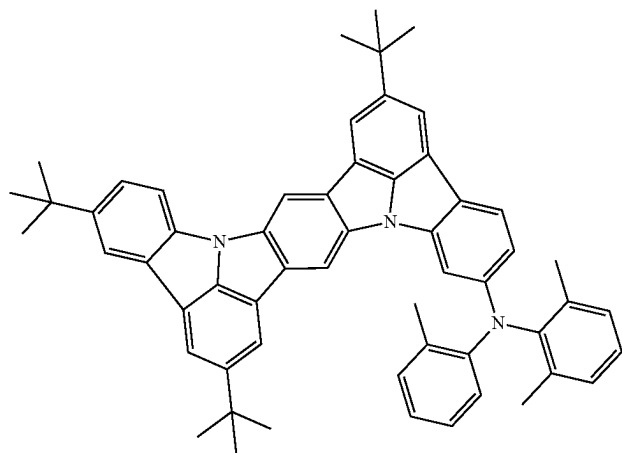
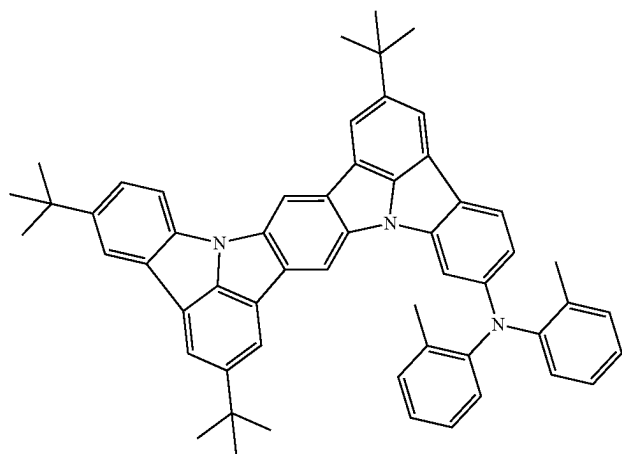
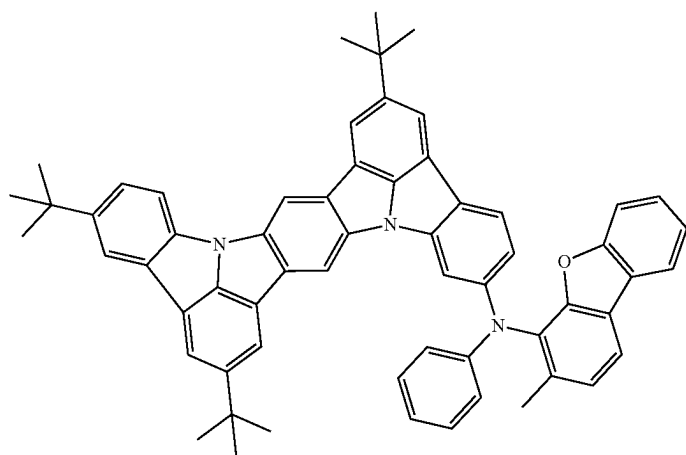
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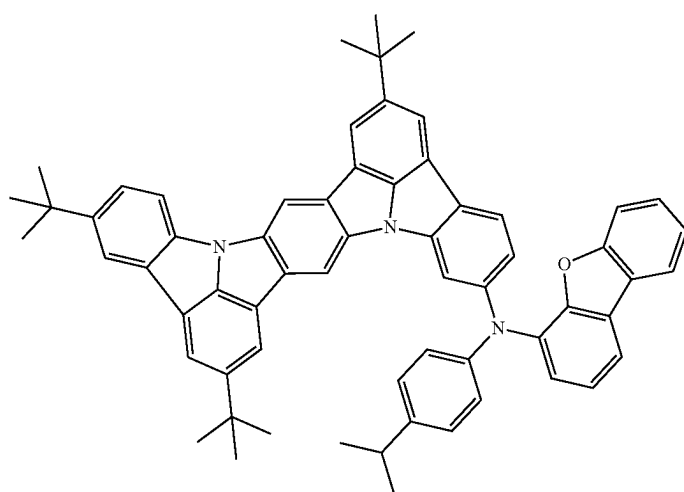
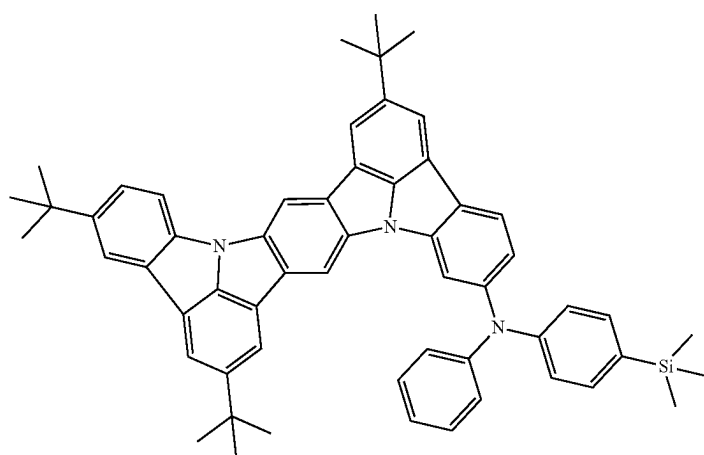
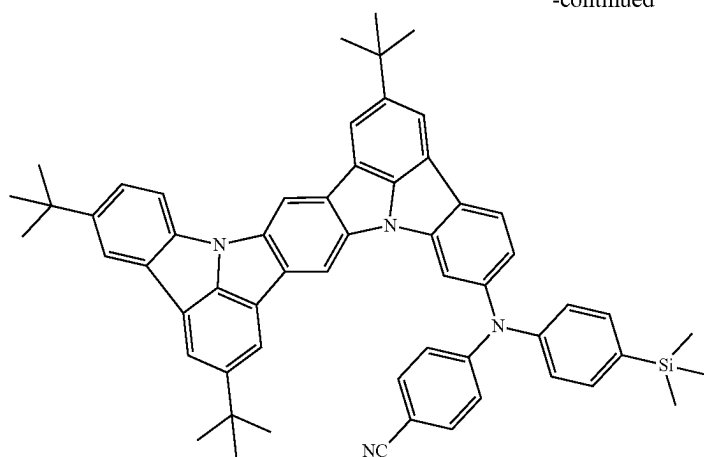
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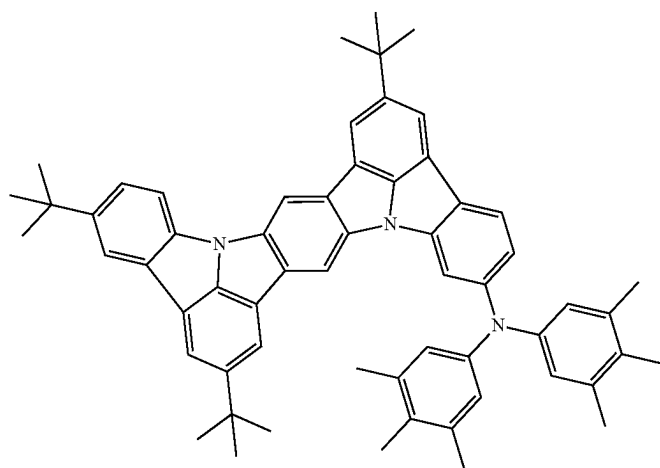
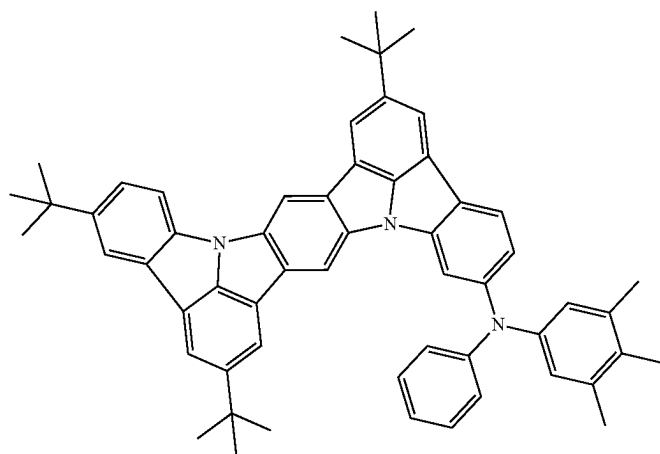
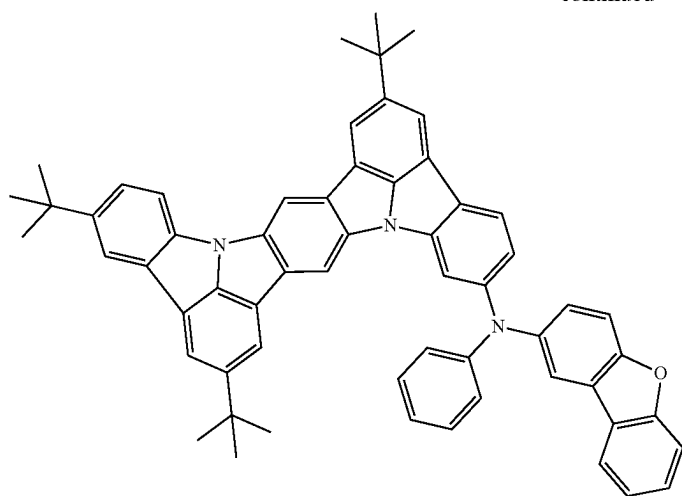
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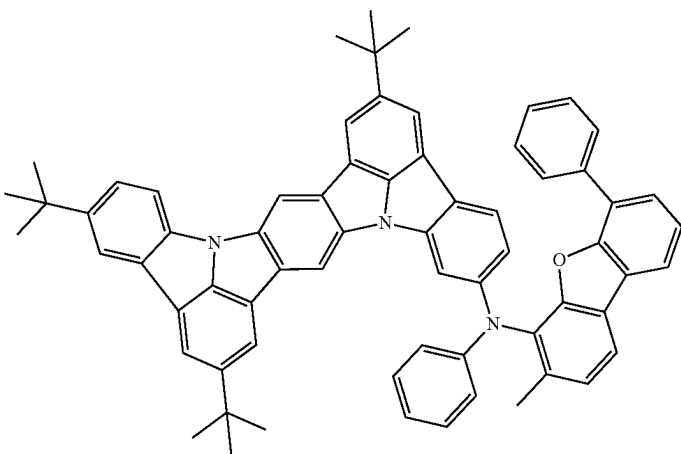
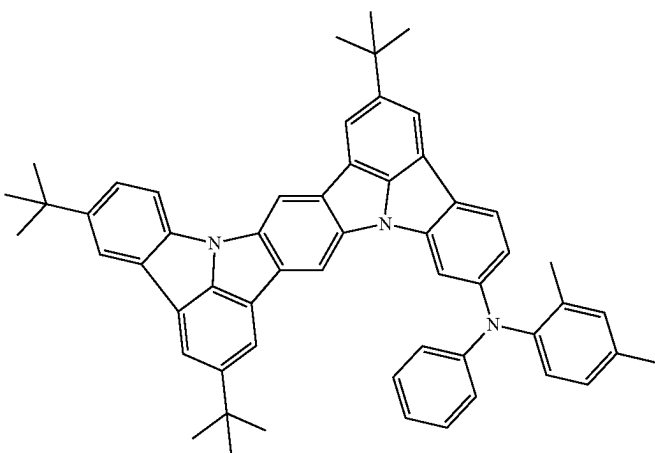
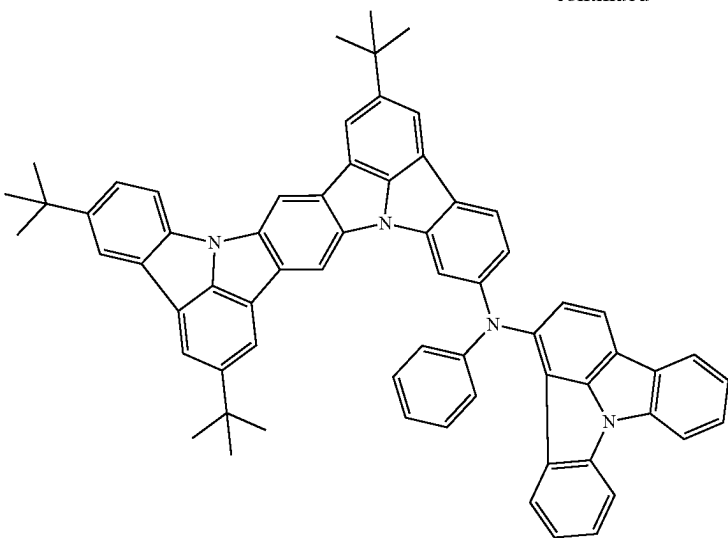
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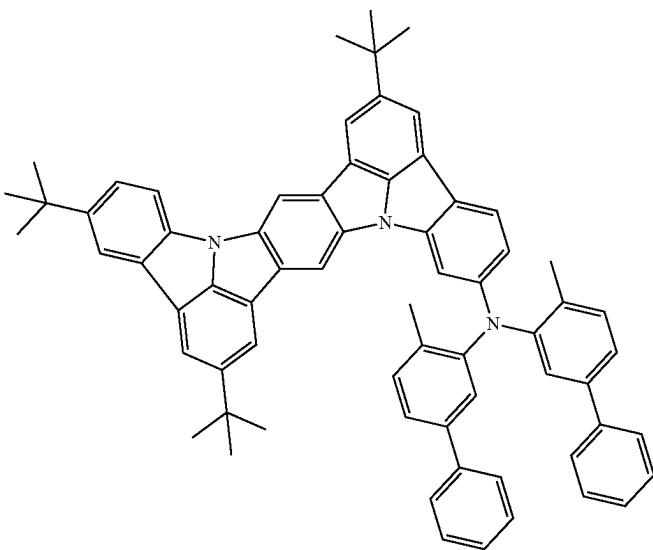
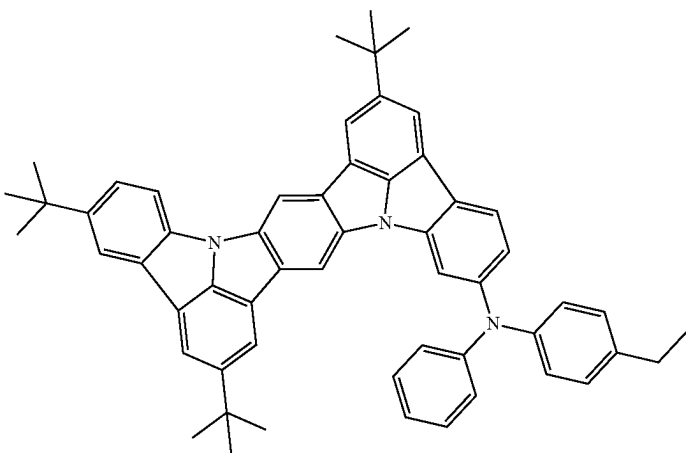
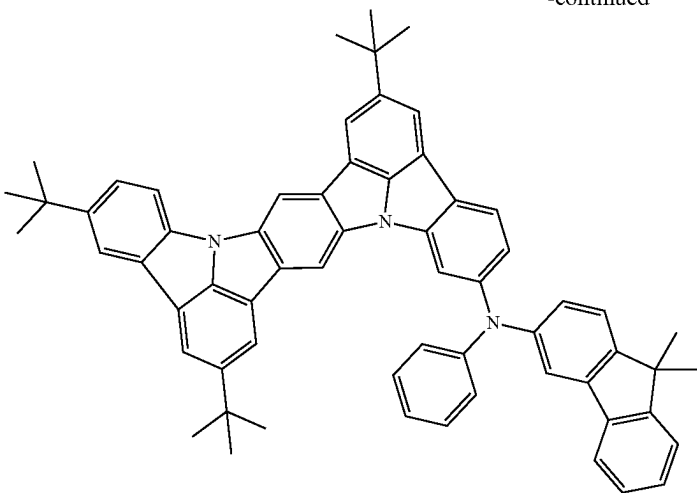
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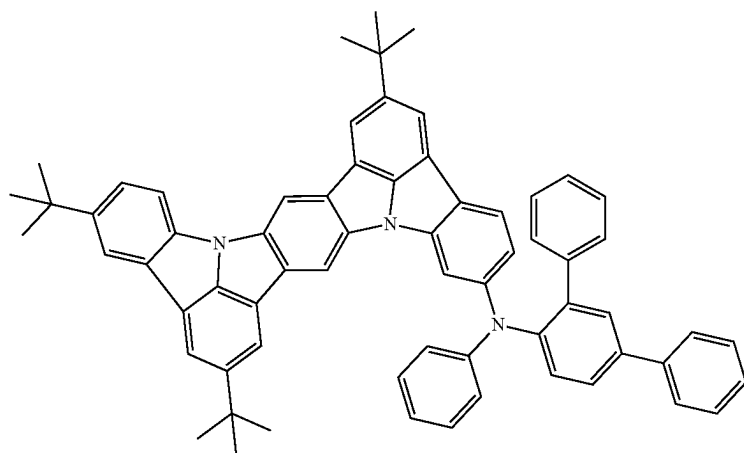
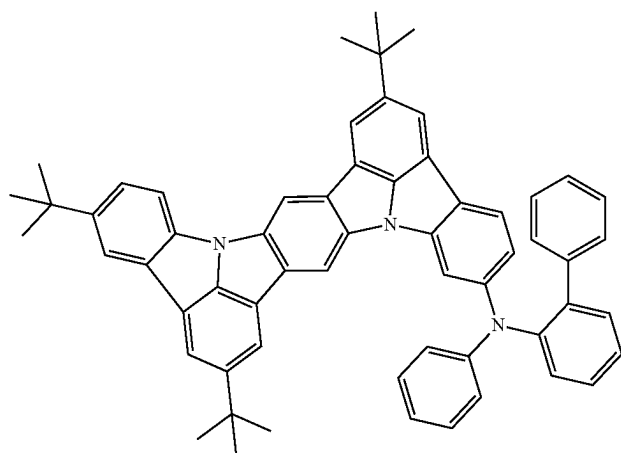
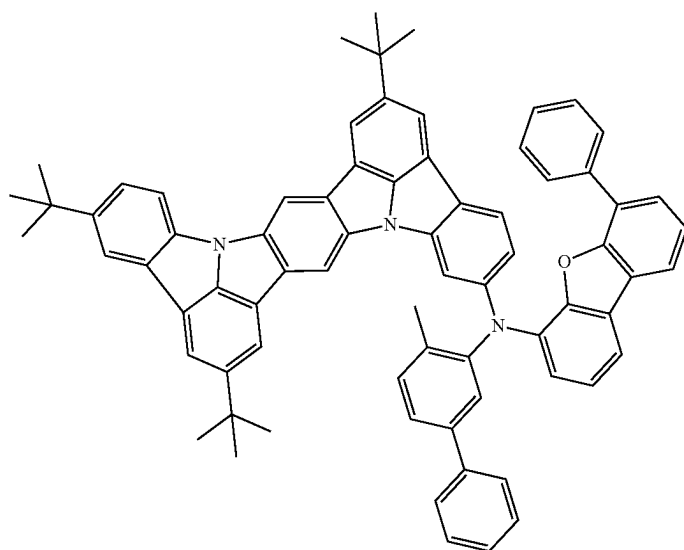
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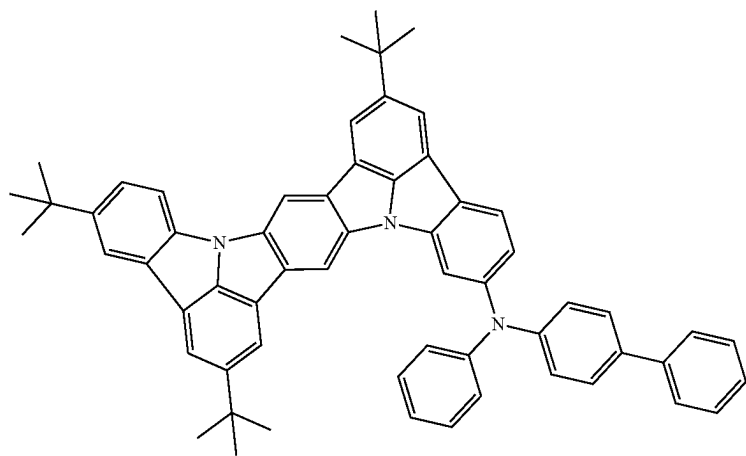
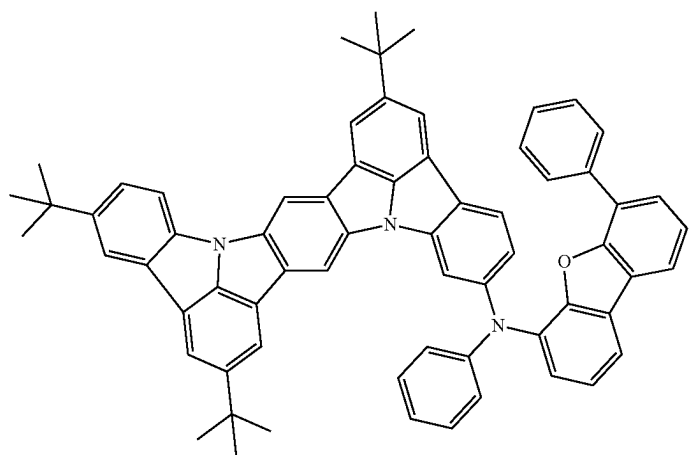
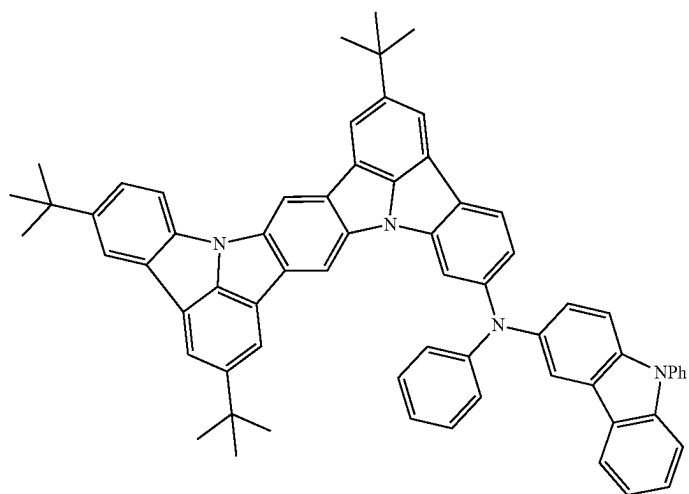
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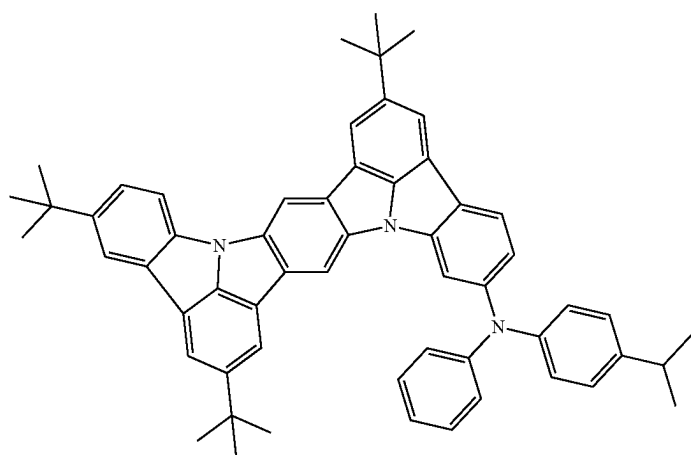
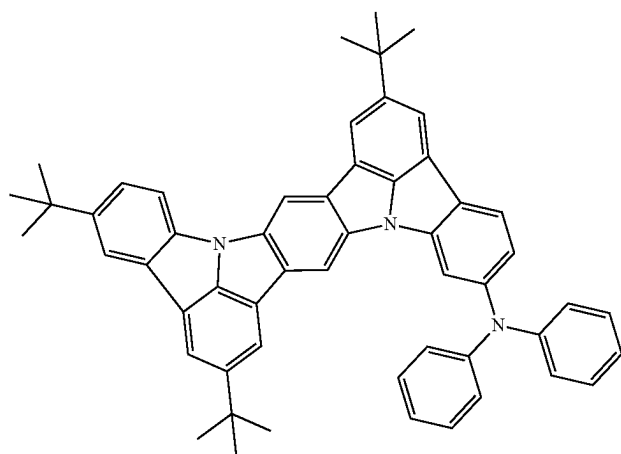
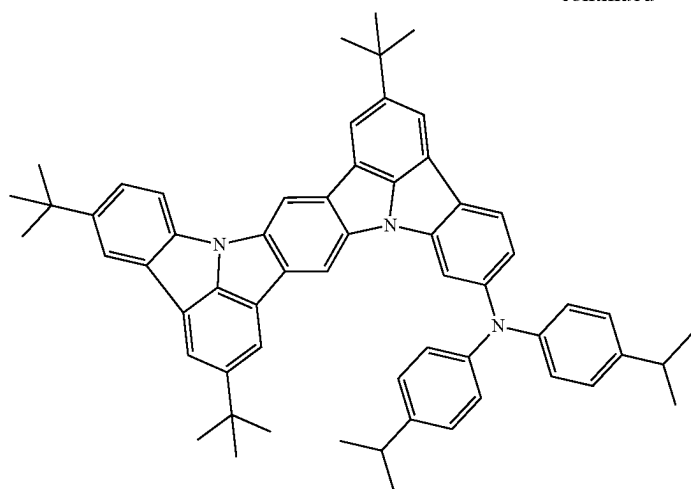
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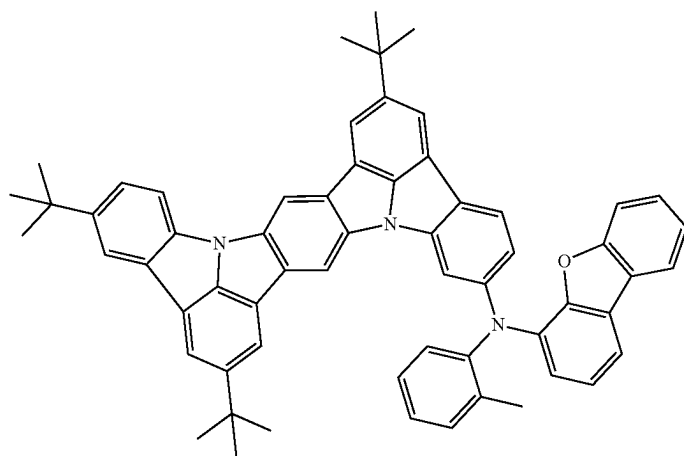
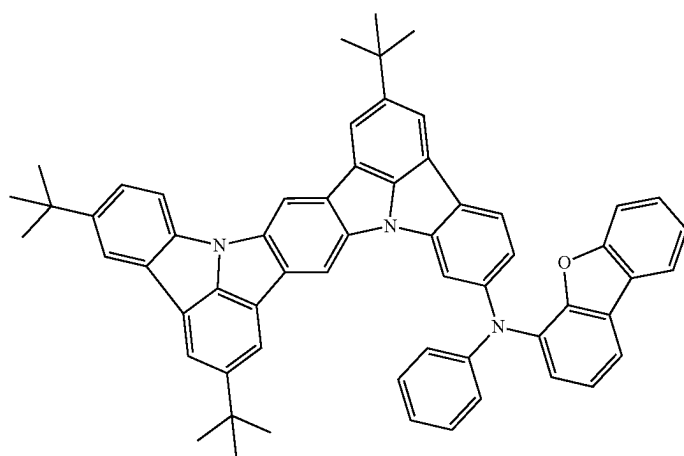
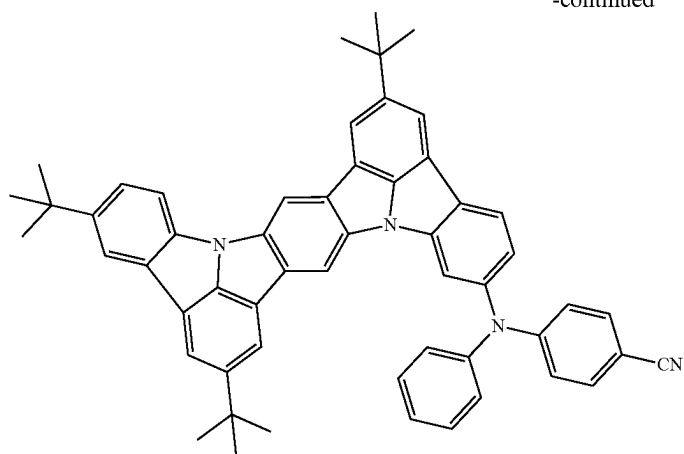
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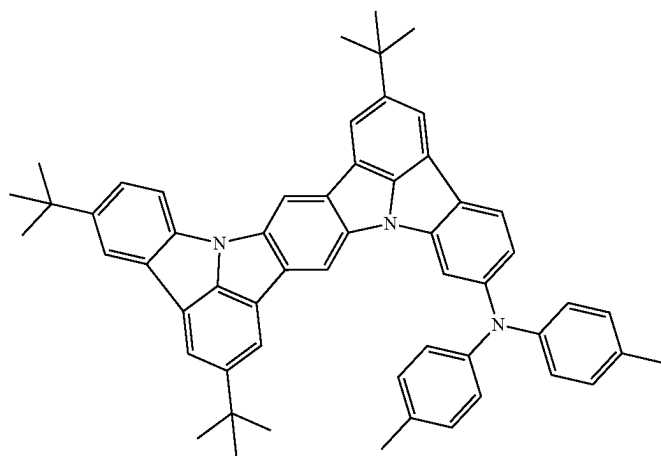
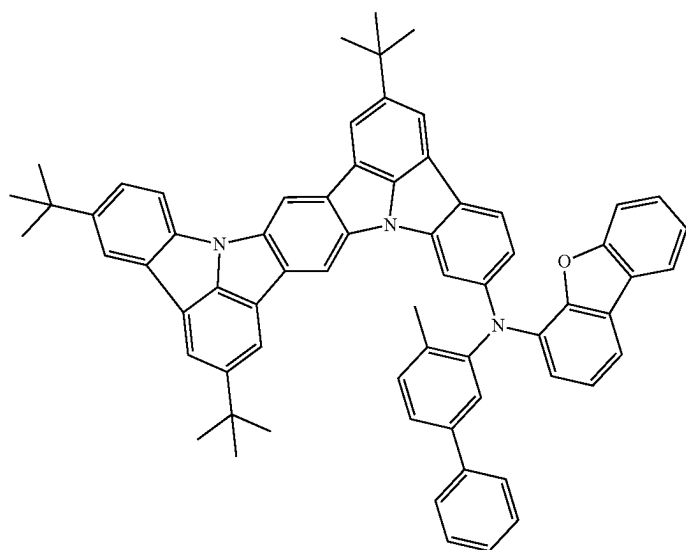
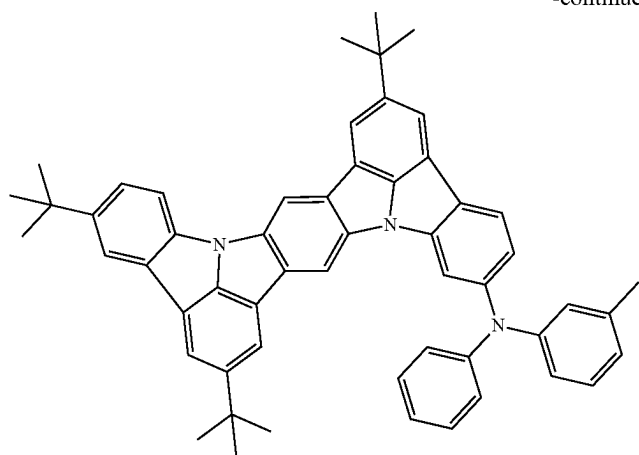
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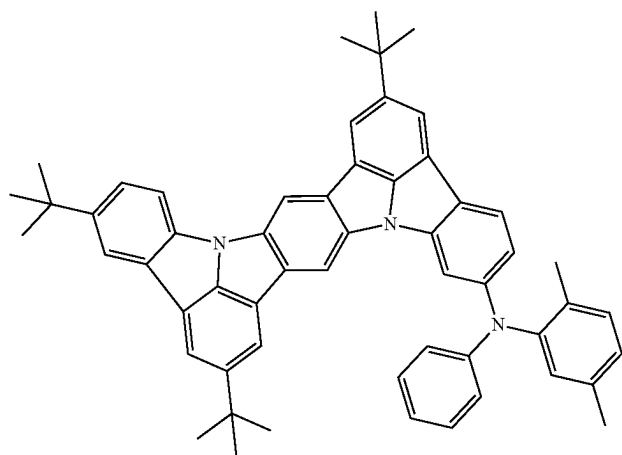
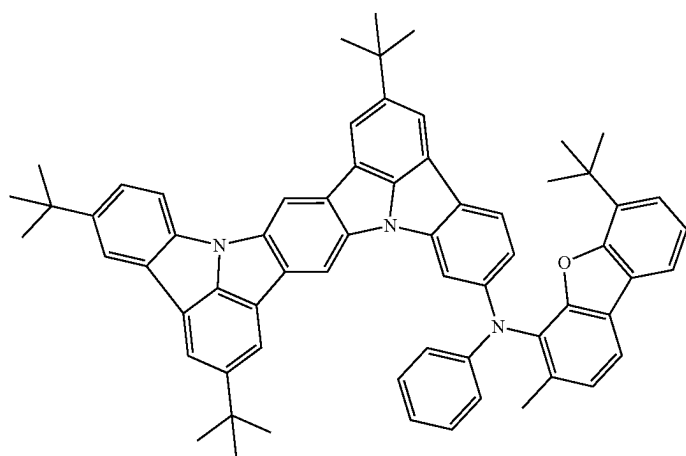
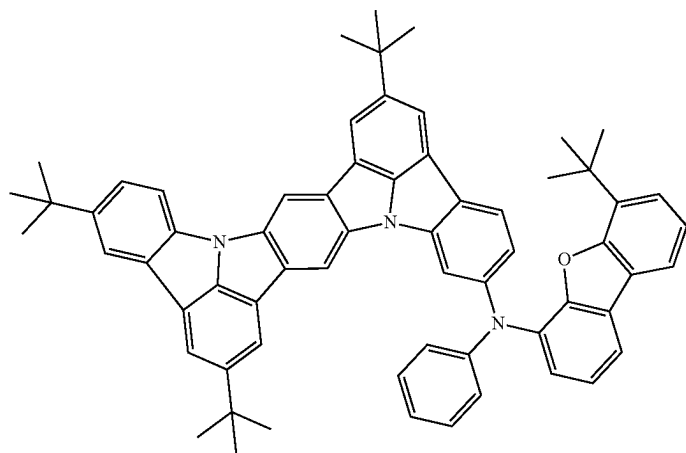
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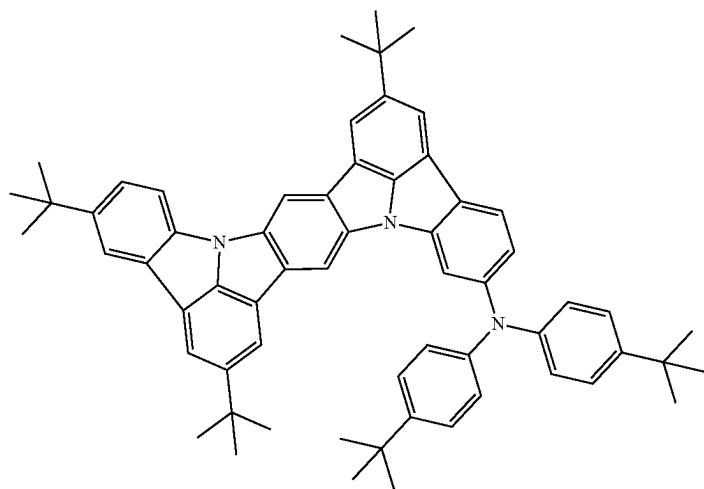
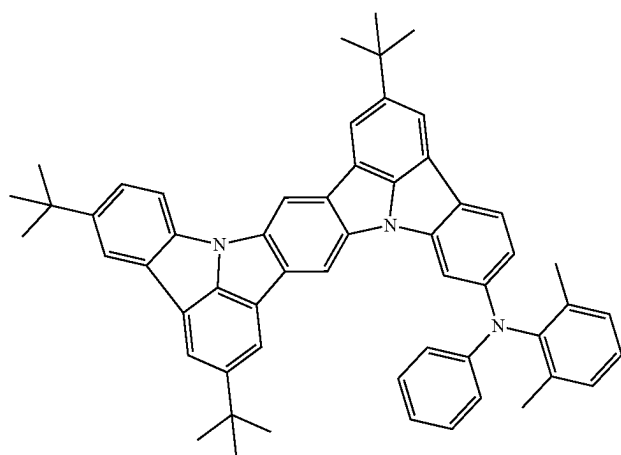
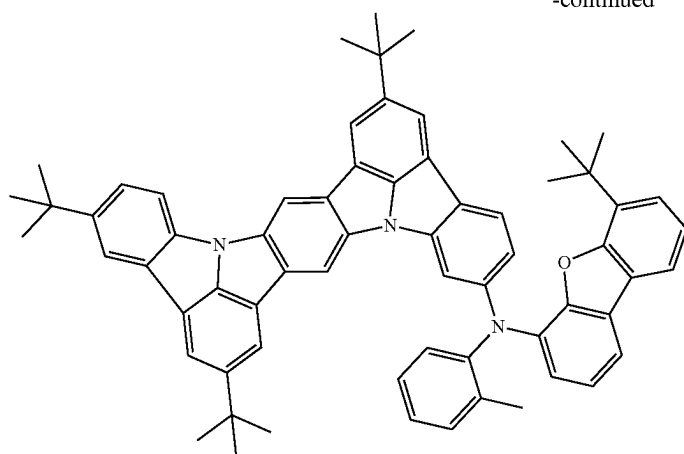
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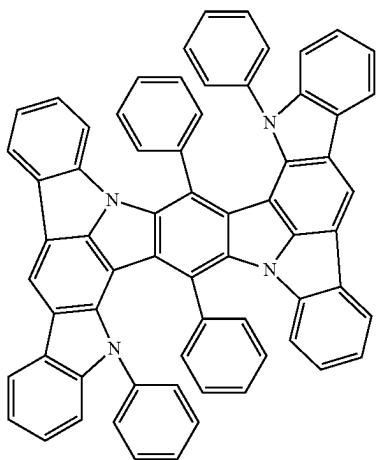
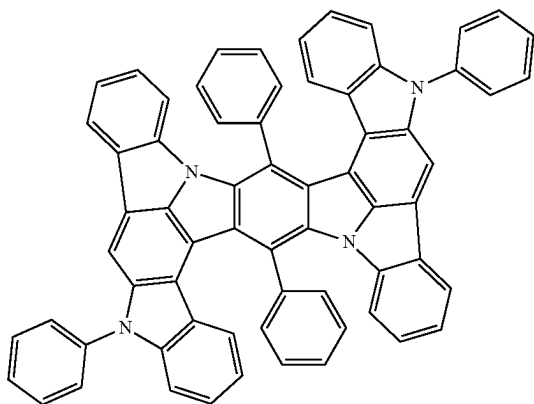
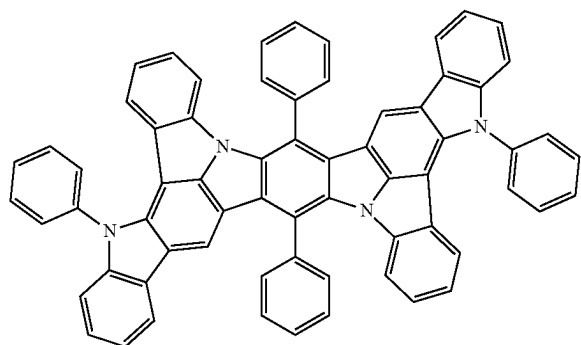
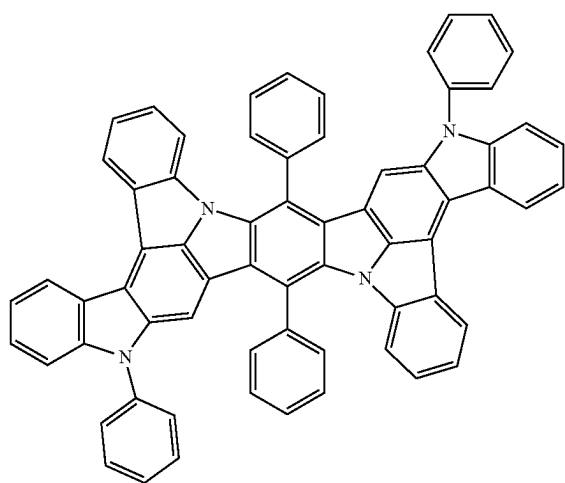


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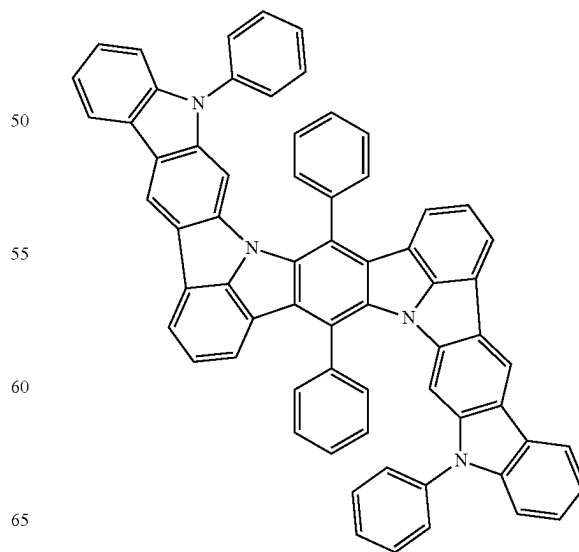
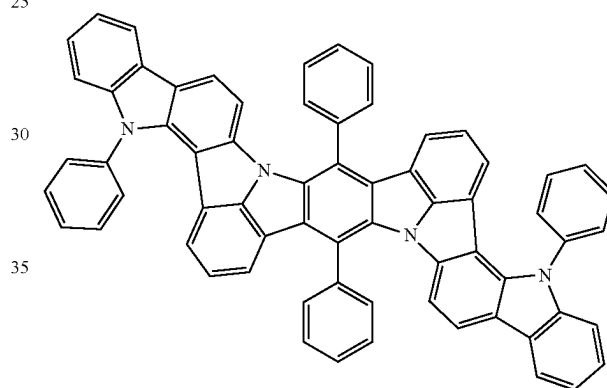
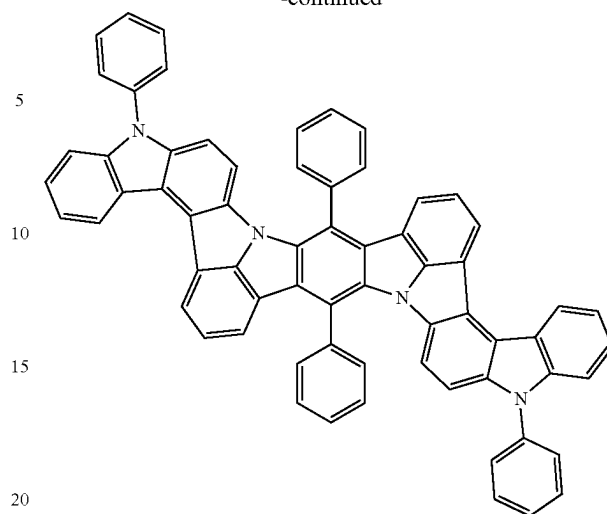
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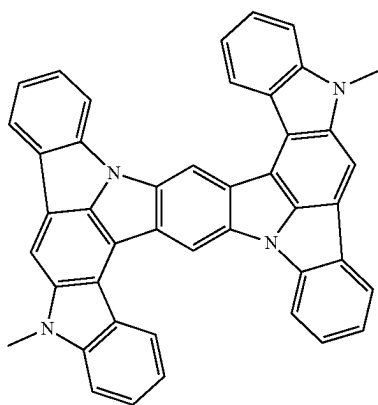
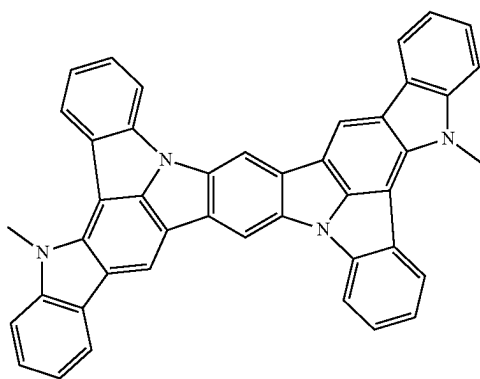
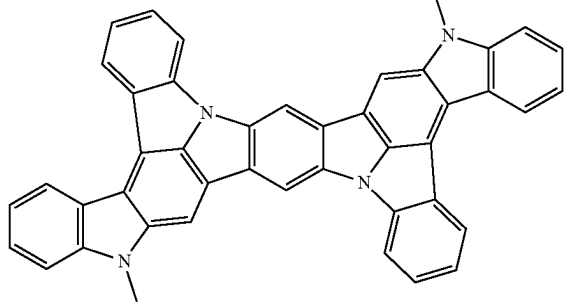
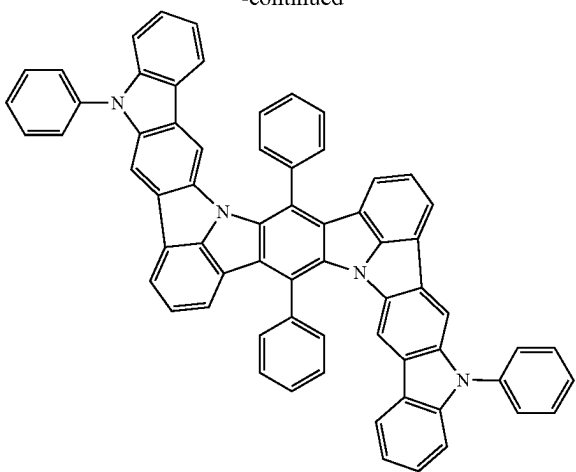
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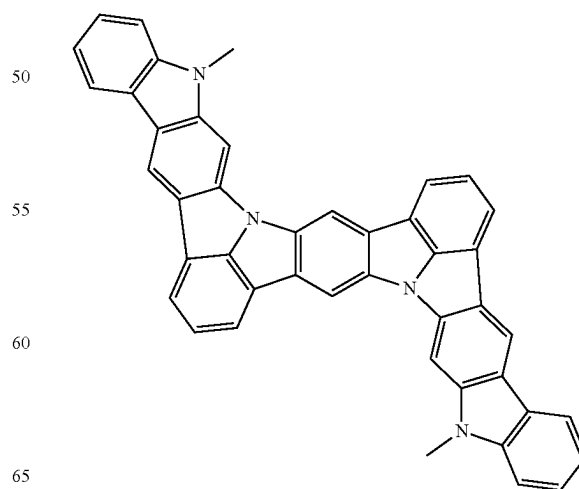
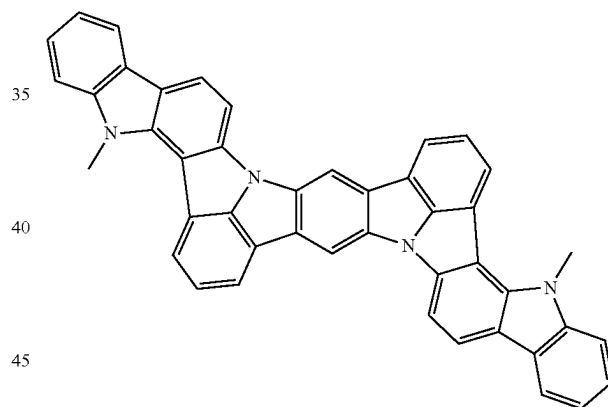
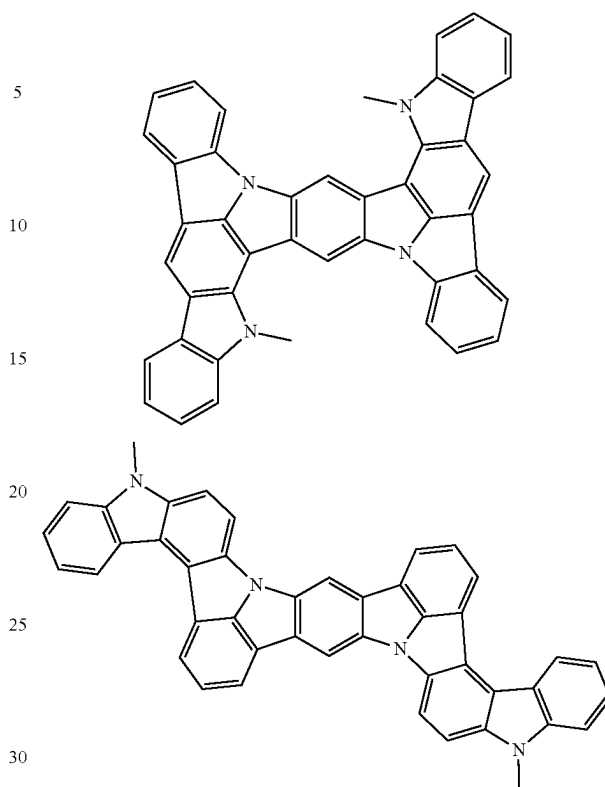


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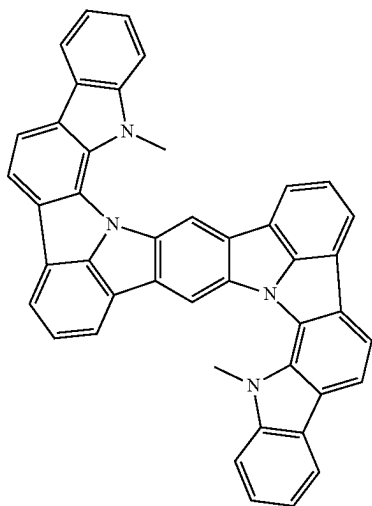
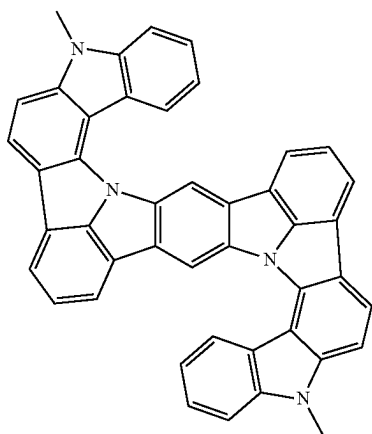
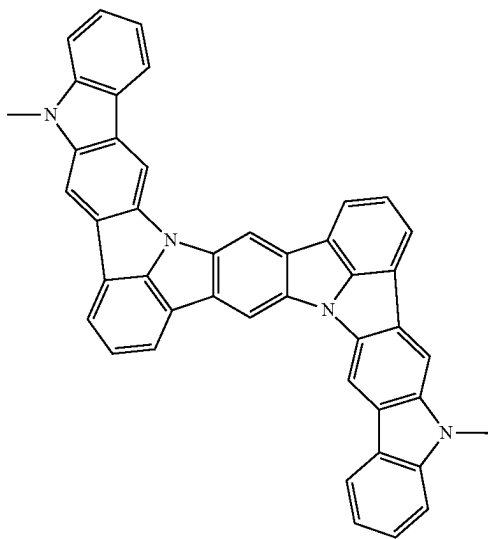
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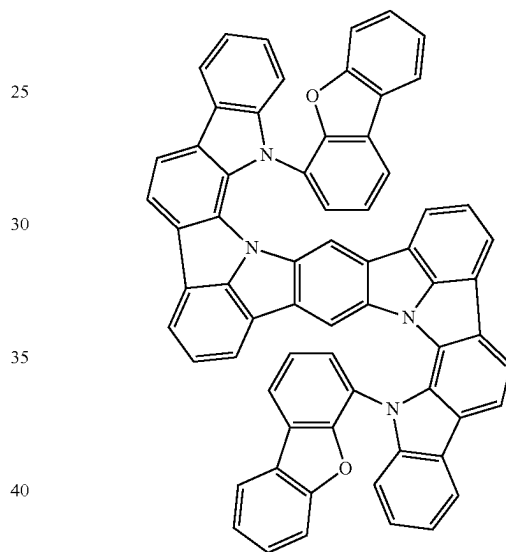
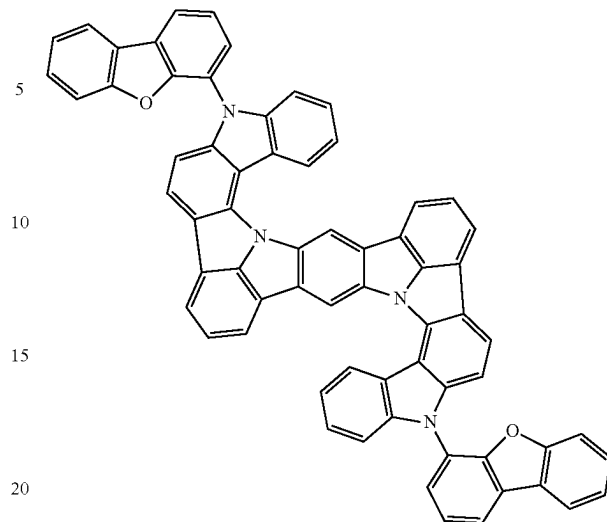


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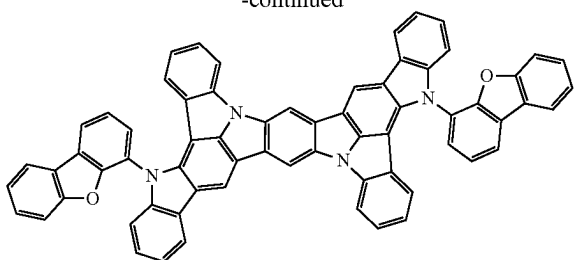
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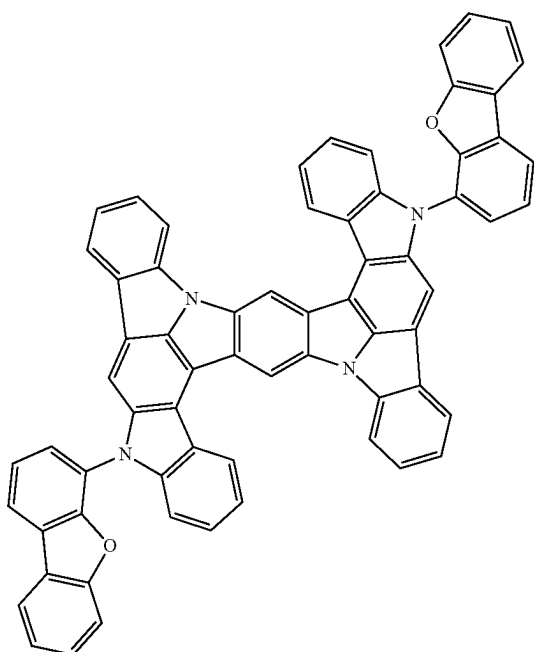
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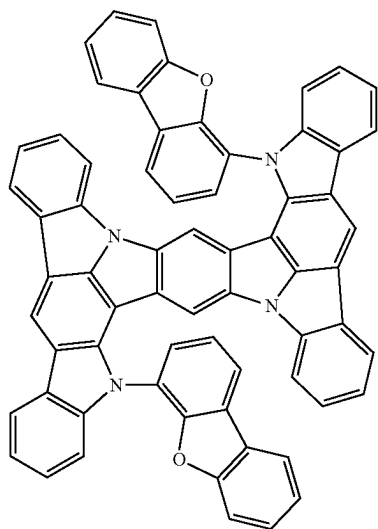
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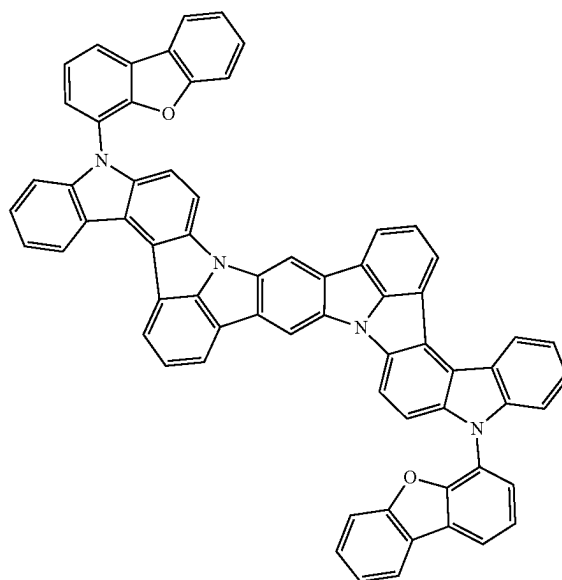
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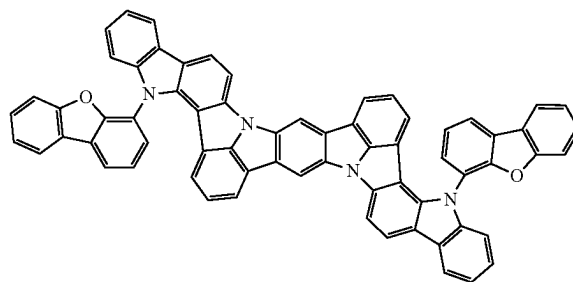
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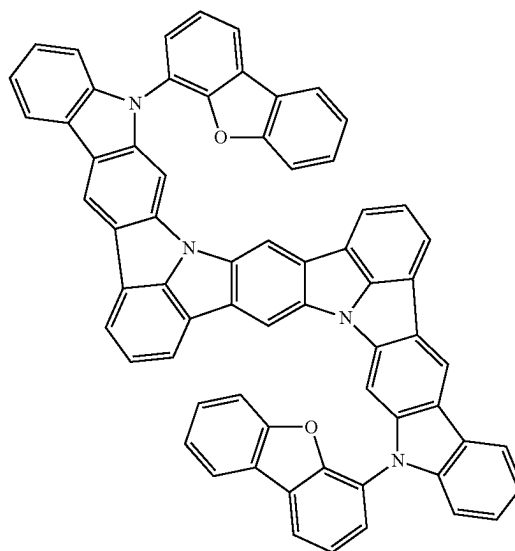
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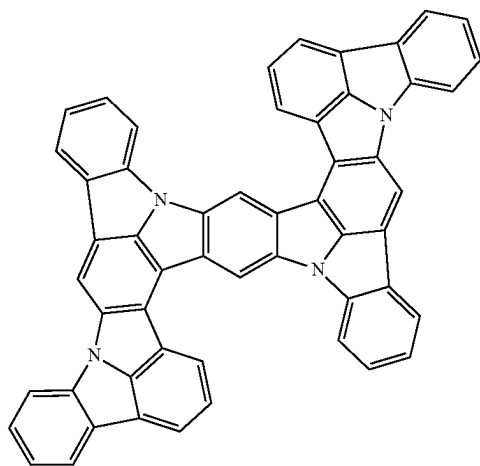
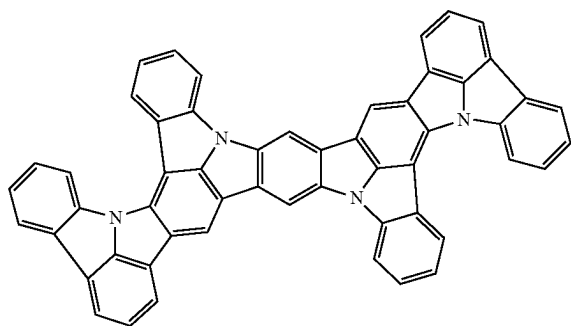
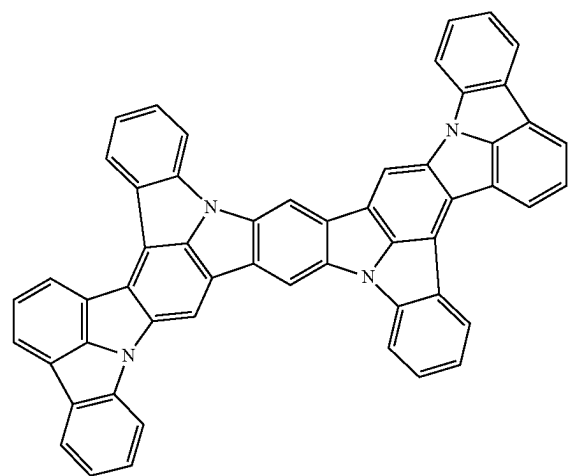
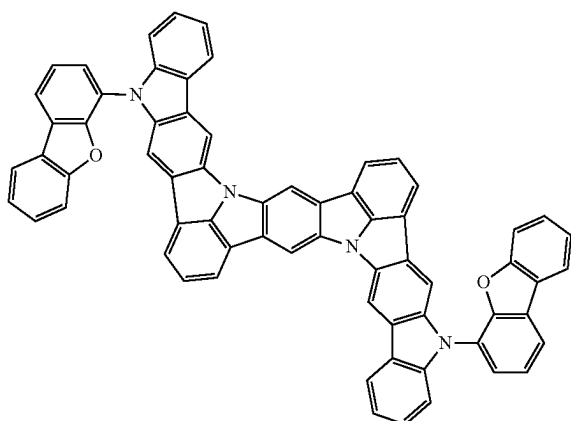
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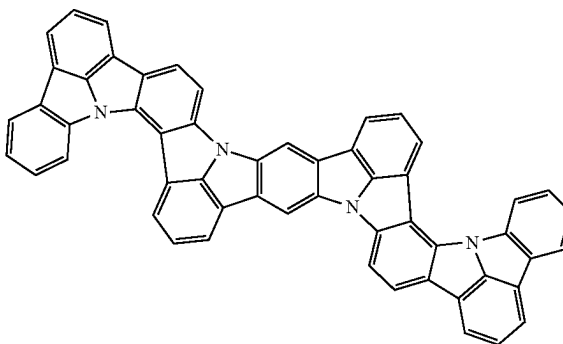
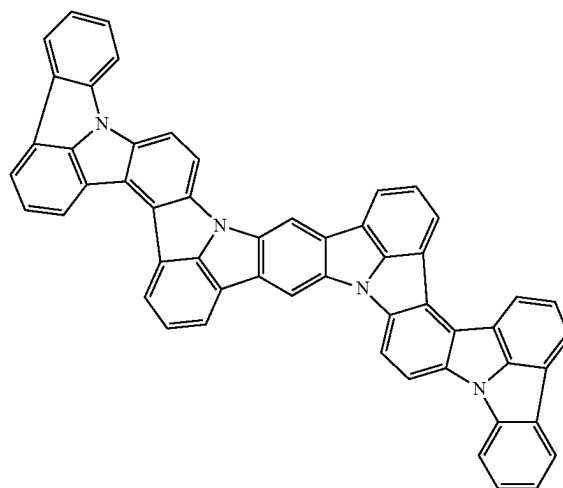
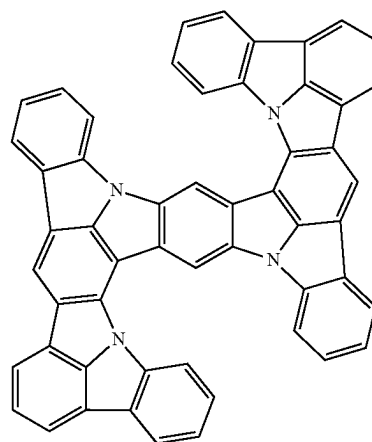


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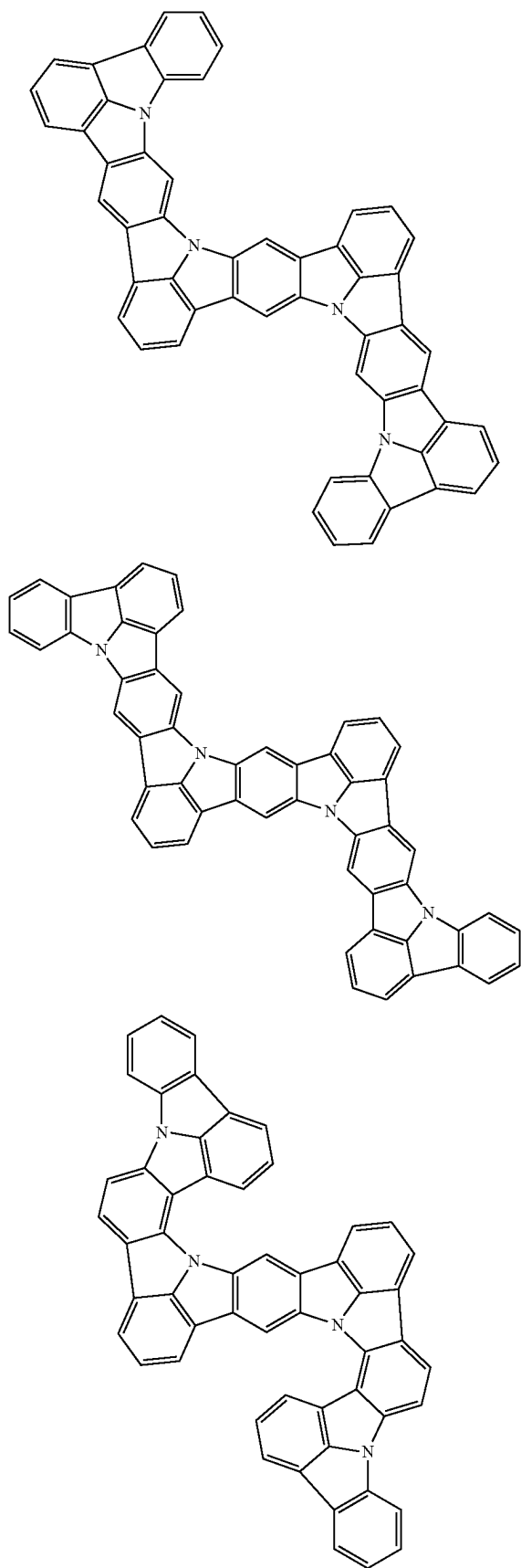
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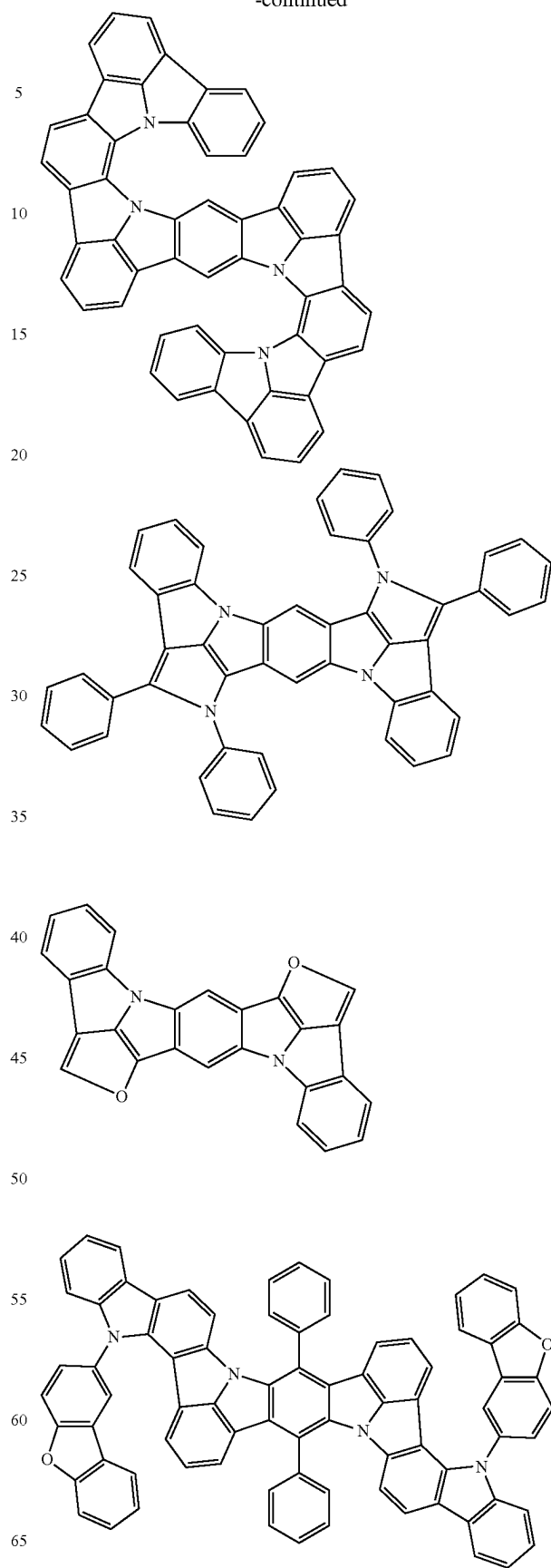


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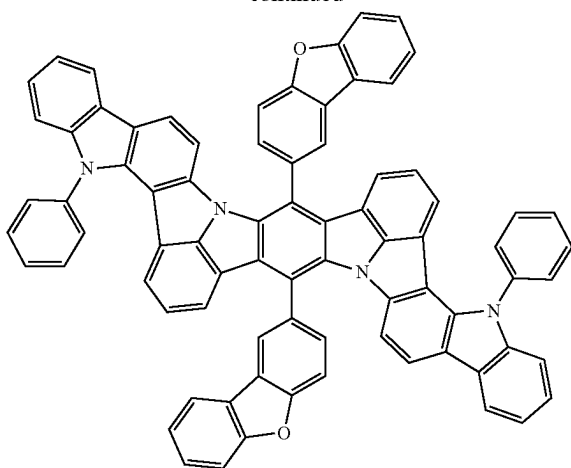
**628**

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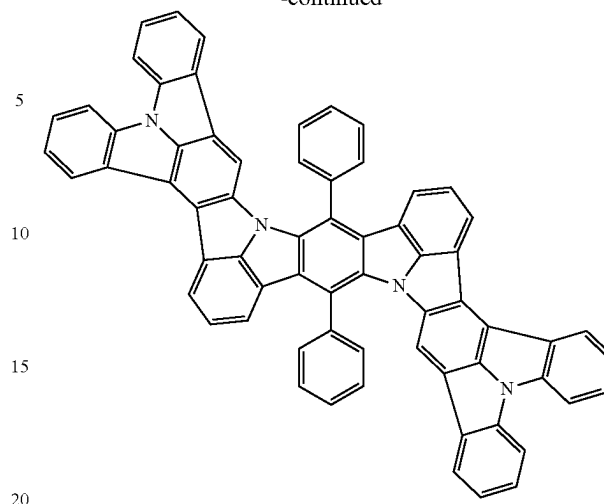


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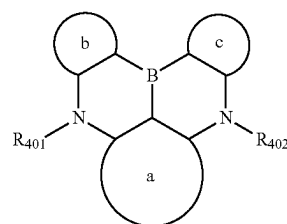
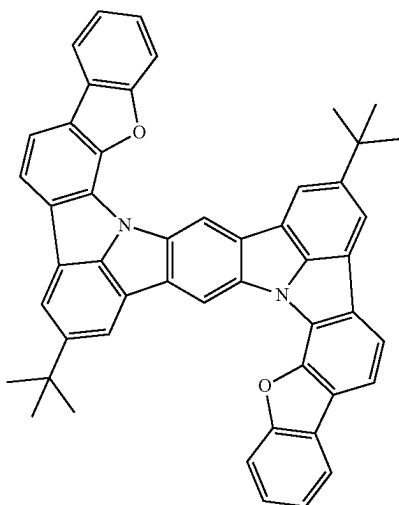
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(Compound Represented by Formula (41))

The compound represented by the formula (41) is explained below.



In the formula (41),

a ring, b ring and c ring are independently

a substituted or unsubstituted aromatic hydrocarbon ring

having 6 to 50 ring carbon atoms, or

a substituted or unsubstituted heterocyclic ring having 5

to 50 ring atoms;

R₄₀₁ and R₄₀₂ are independently bonded to the a ring, the

b ring or the c ring to form a substituted or unsubstituted

heterocyclic ring or do not form a substituted or

unsubstituted heterocyclic ring;

R₄₀₁ and R₄₀₂ that do not form the substituted or unsubstituted heterocyclic ring are independently

a substituted or unsubstituted alkyl group having 1 to 50 carbon atoms,

a substituted or unsubstituted alkenyl group having 2 to 50 carbon atoms,

a substituted or unsubstituted alkynyl group having 2 to 50 carbon atoms,

a substituted or unsubstituted cycloalkyl group having 3 to 50 ring carbon atoms,

a substituted or unsubstituted aryl group having 6 to 50 ring carbon atoms, or

a substituted or unsubstituted monovalent heterocyclic group having 5 to 50 ring atoms.

The a ring, b ring and c ring are rings (a substituted or unsubstituted aromatic hydrocarbon ring having 6 to 50 ring carbon atoms or a substituted or unsubstituted heterocyclic ring having 5 to 50 ring atoms) fuse to the fused bicyclic structure composed of B atom and two N atoms in the center of the formula (41).

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The “aromatic hydrocarbon ring” of the a ring, the b ring and the c ring has the same structure as the compound obtained by introducing a hydrogen atom into the “aryl group” described above. The “aromatic hydrocarbon ring” of the a ring contains three carbon atoms in the fused bicyclic structure in the center of the formula (41) as ring atoms. The “aromatic hydrocarbon ring” of the b ring and the c ring contain two carbon atoms in the fused bicyclic structure in the center of the formula (41) as ring atoms. As examples of “substituted or unsubstituted aromatic hydrocarbon ring having 6 to 50 ring carbon atoms”, compounds in which a hydrogen atom is introduced into the “aryl group” described in the group G1 and the like can be given.

The “heterocyclic ring” of the a ring, the b ring and the c ring has the same structure as the compound obtained by introducing a hydrogen atom into the “heterocyclic group” described above. The “heterocyclic ring” of the a ring contains three carbon atoms in the fused bicyclic structure in the center of the formula (41) as ring atoms. The “heterocyclic ring” of the b ring and the c ring contain two carbon atoms in the fused bicyclic structure in the center of the formula (41) as ring atoms. As examples of “substituted or unsubstituted heterocyclic ring having 5 to 50 ring atoms”, compounds in which a hydrogen atom is introduced into the “heterocyclic group” described in the group G2.

R_{401} and R_{402} may be independently bonded to the a ring, the b ring or the c ring to form a substituted or unsubstituted heterocyclic ring. This heterocyclic ring contains the nitrogen atom in the fused bicyclic structure in the center of the formula (41). This heterocyclic ring may contain a heteroatom other than the nitrogen atom. “ R_{401} and R_{402} are bonded to the a ring, the b ring or the c ring” means, specifically, an atom forming the a ring, the b ring or the c ring is bonded to an atom forming R_{401} and R_{402} . For example, it is possible that R_{401} is bonded to the a ring to form a nitrogen-containing heterocyclic ring having a two-ring fused structure (or three or more rings fused structure) in which a ring containing R_{401} and the a ring are fused. Specific examples of the nitrogen-containing heterocyclic ring include compound and the like corresponding to a heterocyclic group of 2 ring condensation or more containing nitrogen among specific example groups G2.

The same applies to the case where R_{401} is bonded to the b ring, R_{402} is bonded to the a ring, and R_{402} is bonded to the c ring.

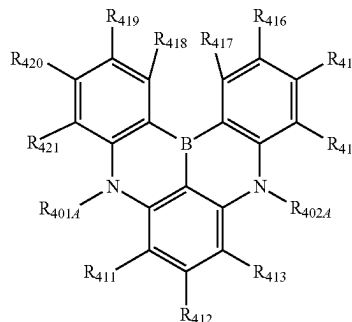
In one embodiment, the a ring, the b ring and the c ring in the formula (41) are independently a substituted or unsubstituted aromatic hydrocarbon ring having 6 to 50 ring carbon atoms.

In one embodiment, the a ring, the b ring and the c ring in the formula (41) are independently a substituted or unsubstituted benzene ring or a substituted or unsubstituted naphthalene ring.

In one embodiment, R_{401} and R_{402} in the formula (41) are independently a substituted or unsubstituted aryl group having 6 to 50 ring carbon atoms, or a substituted or unsubstituted monovalent heterocyclic group having 5 to 50 ring atoms, and preferably a substituted or unsubstituted aryl group having 6 to 50 ring carbon atoms.

In one embodiment, the compound represented by the formula (41) is a compound represented by the following formula (42):

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(42)

wherein in the formula (42),

R_{401A} is bonded with one or more groups selected from R_{411} or R_{421} to form a substituted or unsubstituted heterocyclic ring, or does not form a substituted or unsubstituted heterocyclic ring; R_{402A} is bonded with one or more group selected from R_{413} or R_{414} to form a substituted or unsubstituted heterocyclic ring, or does not form a substituted or unsubstituted heterocyclic ring;

R_{401A} and R_{402A} that do not form a substituted or unsubstituted heterocyclic ring are independently a substituted or unsubstituted alkyl group having 1 to 50 carbon atoms, a substituted or unsubstituted alkenyl group having 2 to 50 carbon atoms, a substituted or unsubstituted alkynyl group having 2 to 50 carbon atoms, a substituted or unsubstituted cycloalkyl group having 3 to 50 ring carbon atoms, a substituted or unsubstituted aryl group having 6 to 50 ring carbon atoms, or a substituted or unsubstituted monovalent heterocyclic group having 5 to 50 ring atoms;

One or more pairs of two or more adjacent groups of R_{411} to R_{421} are bonded with each other to form a substituted or unsubstituted, saturated or unsaturated ring, or do not form a substituted or unsubstituted, saturated or unsaturated ring;

R_{411} to R_{421} that do not form the substituted or unsubstituted heterocyclic ring or the substituted or unsubstituted, saturated or unsaturated ring are independently a hydrogen atom,

a substituted or unsubstituted alkyl group having 1 to 50 carbon atoms,

a substituted or unsubstituted alkenyl group having 2 to 50 carbon atoms,

a substituted or unsubstituted alkynyl group having 2 to 50 carbon atoms,

a substituted or unsubstituted cycloalkyl group having 3 to 50 ring carbon atoms,

—Si(R_{901})(R_{902})(R_{903}),

—O—(R_{904}),

—S—(R_{905}),

—N(R_{906})(R_{907}),

a halogen atom, a cyano group, a nitro group,

a substituted or unsubstituted aryl group having 6 to 50 ring carbon atoms, or

a substituted or unsubstituted monovalent heterocyclic group having 5 to 50 ring atoms; and

R_{901} to R_{907} are as defined in the formula (1).

R_{401A} and R_{402A} in the formula (42) correspond to R_{401} and R_{402} in the formula (41).

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R_{401A} and R_{411} may be bonded with each other to form a nitrogen-containing heterocyclic ring having two-ring fused structure (or three or more rings fused structure) which is a fused ring of a ring containing R_{401A} and R_{411} and the benzene ring of the a ring, for example. As examples of the nitrogen-containing heterocyclic ring, compounds correspond to nitrogen-containing heterocyclic group having two or more ring fused structure in the group G2 can be given. The same applies to the cases where R_{401A} and R_{412} are bonded, R_{402A} and R_{413} are bonded, and R_{402A} and R_{414} are bonded.

One or more pairs of two or more adjacent groups of R_{411} to R_{421} are bonded with each other to form a substituted or unsubstituted, saturated or unsaturated ring. For example, R_{411} and R_{412} are bonded to form a benzene ring, an indole ring, a pyrrole ring, a benzofuran ring, a benzothiophene ring or the like which fuses to the six-membered ring to which R_{411} and R_{412} bond, and the formed fused ring is a naphthalene ring, a carbazole ring, an indole ring, a dibenzofuran ring or a dibenzothiophene ring.

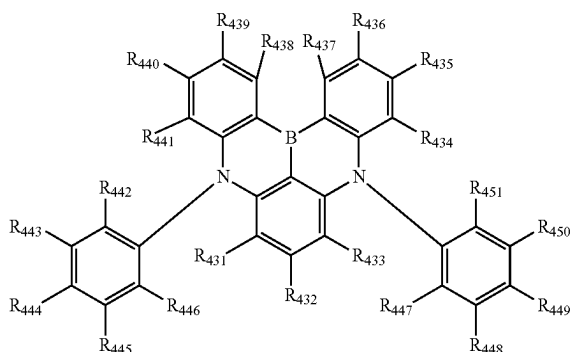
In one embodiment, R_{411} to R_{421} that do not contribute to form a ring are independently a hydrogen atom, a substituted or unsubstituted alkyl group having 1 to 50 carbon atoms, a substituted or unsubstituted aryl group having 6 to 50 ring carbon atoms, or a substituted or unsubstituted monovalent heterocyclic group having 5 to 50 ring atoms.

In one embodiment, R_{411} to R_{421} that do not contribute to form a ring are independently a hydrogen atom, a substituted or unsubstituted aryl group having 6 to 50 ring carbon atoms, or a substituted or unsubstituted monovalent heterocyclic group having 5 to 50 ring atoms.

In one embodiment, R_{411} to R_{421} that do not contribute to form a ring are independently a hydrogen atom or a substituted or unsubstituted alkyl group having 1 to 50 carbon atoms.

In one embodiment, R_{411} to R_{421} that do not contribute to form a ring are independently a hydrogen atom, a substituted or unsubstituted alkyl group having 1 to 50 carbon atoms, and at least one of R_{411} to R_{421} is a substituted or unsubstituted alkyl group having 1 to 50 carbon atoms.

In one embodiment, the compound represented by the formula (42) is a compound represented by the following formula (43):



wherein in the formula (43),

R_{431} is bonded with R_{446} to form a substituted or unsubstituted heterocyclic ring, or does not form a substituted or unsubstituted heterocyclic ring; R_{433} is bonded with R_{447} to form a substituted or unsubstituted heterocyclic

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ring, or does not form a substituted or unsubstituted heterocyclic ring; R_{441} is bonded with R_{442} to form a substituted or unsubstituted heterocyclic ring, or does not form a substituted or unsubstituted heterocyclic ring; R_{441} is bonded with R_{442} to form a substituted or unsubstituted heterocyclic ring, or does not form a substituted or unsubstituted heterocyclic ring;

One or more pairs of two or more adjacent groups of R_{431} to R_{451} are bonded with each other to form a substituted or unsubstituted, saturated or unsaturated ring, or do not form a substituted or unsubstituted, saturated or unsaturated ring; R_{431} to R_{451} that do not form a substituted or unsubstituted heterocyclic ring are independently

a hydrogen atom,

a substituted or unsubstituted alkyl group having 1 to 50 carbon atoms,

a substituted or unsubstituted alkenyl group having 2 to 50 carbon atoms,

a substituted or unsubstituted alkynyl group having 2 to 50 carbon atoms,

a substituted or unsubstituted cycloalkyl group having 3 to 50 ring carbon atoms,

$-\text{Si}(\text{R}_{901})(\text{R}_{902})(\text{R}_{903})$,

$-\text{O}-(\text{R}_{904})$,

$-\text{S}-(\text{R}_{905})$,

$-\text{N}(\text{R}_{906})(\text{R}_{907})$,

a halogen atom, a cyano group, a nitro group,

a substituted or unsubstituted aryl group having 6 to 50 ring carbon atoms, or

a substituted or unsubstituted monovalent heterocyclic group having 5 to 50 ring atoms; and

R_{901} , R_{906} to R_{907} are as defined in the formula (1).

R_{431} may bond to R_{446} to form a substituted or unsubstituted heterocyclic ring. For example, R_{431} may bond with R_{446} to form a nitrogen-containing heterocyclic ring with three or more fused rings of the benzene ring to which R_{446} bond, a nitrogen-containing ring and the benzene ring of the a ring. As examples of the nitrogen-containing heterocyclic ring, compounds correspond to nitrogen-containing heterocyclic group having three or more ring fused structure in the group G2 can be given. The same applies to the cases where R_{433} and R_{447} are bonded, R_{434} and R_{451} are bonded, and R_{441} and R_{442} are bonded.

In one embodiment, R_{431} to R_{451} that do not contribute to form a ring are independently, a hydrogen atom, a substituted or unsubstituted alkyl group having 1 to 50 carbon atoms, a substituted or unsubstituted aryl group having 6 to 50 carbon atoms, or a substituted or unsubstituted monovalent heterocyclic group having 5 to 50 ring atoms.

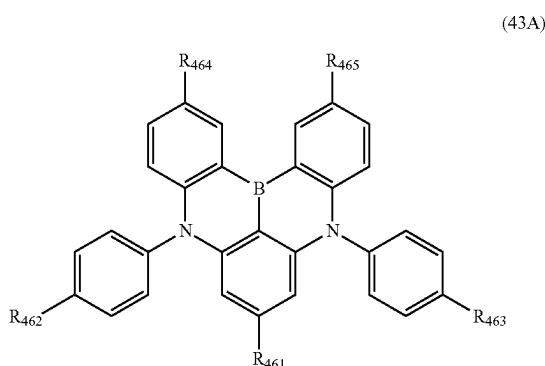
In one embodiment, R_{431} to R_{451} that do not contribute to form a ring are independently, a hydrogen atom, a substituted or unsubstituted aryl group having 6 to 50 carbon atoms, or a substituted or unsubstituted monovalent heterocyclic group having 5 to 50 ring atoms.

In one embodiment, R_{431} to R_{451} that do not contribute to form a ring are independently a hydrogen atom or a substituted or unsubstituted alkyl group having 1 to 50 carbon atoms.

In one embodiment, R_{431} to R_{451} that do not contribute to form a ring are independently a hydrogen atom, a substituted or unsubstituted alkyl group having 1 to 50 carbon atoms, and at least one of R_{431} to R_{451} is a substituted or unsubstituted alkyl group having 1 to 50 carbon atoms.

In one embodiment, the compound represented by the formula (43) is a compound represented by the following formula (43A):

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wherein in the formula (43A),

R₄₆₁ is

a hydrogen atom,

a substituted or unsubstituted alkyl group having 1 to 50 carbon atoms,

a substituted or unsubstituted alkenyl group having 2 to 50 carbon atoms,

a substituted or unsubstituted alkynyl group having 2 to 50 carbon atoms,

a substituted or unsubstituted cycloalkyl group having 3 to 50 ring carbon atoms, or

a substituted or unsubstituted aryl group having 6 to 50 ring carbon atoms; and

R₄₆₂ to R₄₆₅ are independently

a substituted or unsubstituted alkyl group having 1 to 50 carbon atoms,

a substituted or unsubstituted alkenyl group having 2 to 50 carbon atoms,

a substituted or unsubstituted alkynyl group having 2 to 50 carbon atoms,

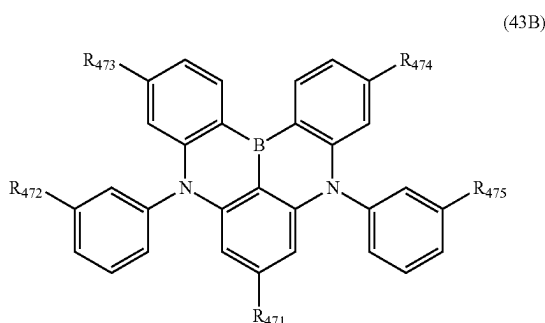
a substituted or unsubstituted cycloalkyl group having 3 to 50 ring carbon atoms, or

a substituted or unsubstituted aryl group having 6 to 50 ring carbon atoms.

In one embodiment, R₄₆₁ to R₄₆₅ are independently a substituted or unsubstituted alkyl group having 1 to 50 carbon atoms or a substituted or unsubstituted aryl group having 6 to 50 ring carbon atoms.

In one embodiment, R₄₆₁ and R₄₆₅ are independently a substituted or unsubstituted alkyl group having 1 to 50 carbon atoms.

In one embodiment, the compound represented by the formula (43) is a compound represented by the following formula (43B):



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wherein in the formula (43B),

R₄₇₁ and R₄₇₂ are independently,

a hydrogen atom,

a substituted or unsubstituted alkyl group having 1 to 50 carbon atoms,

a substituted or unsubstituted alkenyl group having 2 to 50 carbon atoms,

a substituted or unsubstituted alkynyl group having 2 to 50 carbon atoms,

a substituted or unsubstituted cycloalkyl group having 3 to 50 ring carbon atoms,

—N(R₉₀₆) (R₉₀₇), or

a substituted or unsubstituted aryl group having 6 to 50 ring carbon atoms;

R₄₇₃ to R₄₇₅ are independently,

a substituted or unsubstituted alkyl group having 1 to 50 carbon atoms,

a substituted or unsubstituted alkenyl group having 2 to 50 carbon atoms,

a substituted or unsubstituted alkynyl group having 2 to 50 carbon atoms,

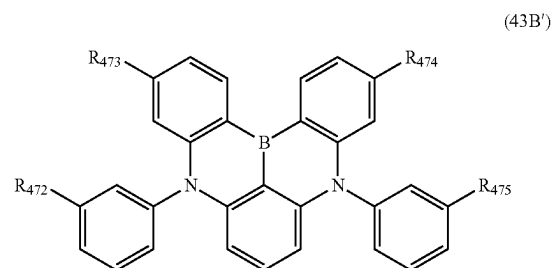
a substituted or unsubstituted cycloalkyl group having 3 to 50 ring carbon atoms,

—N(R₉₀₆) (R₉₀₇), or

a substituted or unsubstituted aryl group having 6 to 50 ring carbon atoms; and

R₉₀₆ and R₉₀₇ are as defined in the formula (1).

In one embodiment, the compound represented by the formula (43) is the compound represented by the following formula (43B'):



wherein in the formula (43B'), R₄₇₂ to R₄₇₅ are as defined in the formula (43B).

In one embodiment, at least one of R₄₇₁ to R₄₇₅ is a substituted or unsubstituted alkyl group having 1 to 50 carbon atoms,

a substituted or unsubstituted alkenyl group having 2 to 50 carbon atoms,

a substituted or unsubstituted alkynyl group having 2 to 50 carbon atoms,

a substituted or unsubstituted cycloalkyl group having 3 to 50 ring carbon atoms,

—N(R₉₀₆) (R₉₀₇), or

a substituted or unsubstituted aryl group having 6 to 50 ring carbon atoms.

In one embodiment,

R₄₇₂ is

a hydrogen atom,

a substituted or unsubstituted alkyl group having 1 to 50 carbon atoms,

—N(R₉₀₆) (R₉₀₇), or

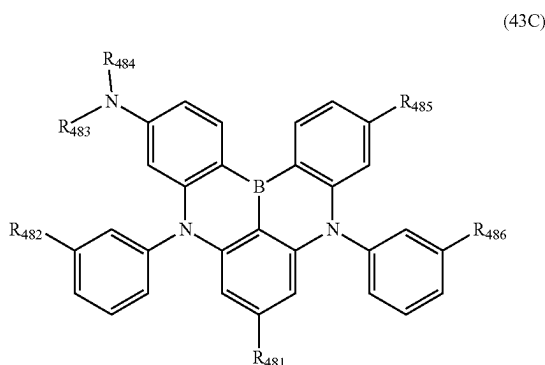
a substituted or unsubstituted aryl group having 6 to 50 ring carbon atoms; and

R₄₇₁ and R₄₇₃ to R₄₇₅ are independently

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a substituted or unsubstituted alkyl group having 1 to 50 carbon atoms,
 $\text{—N(R}_{906}\text{)(R}_{907}\text{)}$, or
 a substituted or unsubstituted aryl group having 6 to 50 ring carbon atoms.

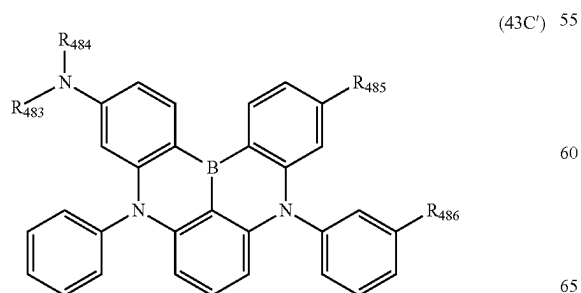
In one embodiment, the compound represented by the formula (43) is a compound represented by the formula (43C):



wherein in the formula (43C),
 R₄₈₁ and R₄₈₂ are independently
 a hydrogen atom,

a substituted or unsubstituted alkyl group having 1 to 50 carbon atoms,
 a substituted or unsubstituted alkenyl group having 2 to 50 carbon atoms,
 a substituted or unsubstituted alkynyl group having 2 to 50 carbon atoms,
 a substituted or unsubstituted cycloalkyl group having 3 to 50 ring carbon atoms, or
 a substituted or unsubstituted aryl group having 6 to 50 ring carbon atoms; and
 R₄₈₃ to R₄₈₆ are independently
 a substituted or unsubstituted alkyl group having 1 to 50 carbon atoms,
 a substituted or unsubstituted alkenyl group having 2 to 50 carbon atoms,
 a substituted or unsubstituted alkynyl group having 2 to 50 carbon atoms,
 a substituted or unsubstituted cycloalkyl group having 3 to 50 ring carbon atoms, or
 a substituted or unsubstituted aryl group having 6 to 50 ring carbon atoms.

In one embodiment, the compound represented by the formula (43) is the compound represented by the following formula (43C'):



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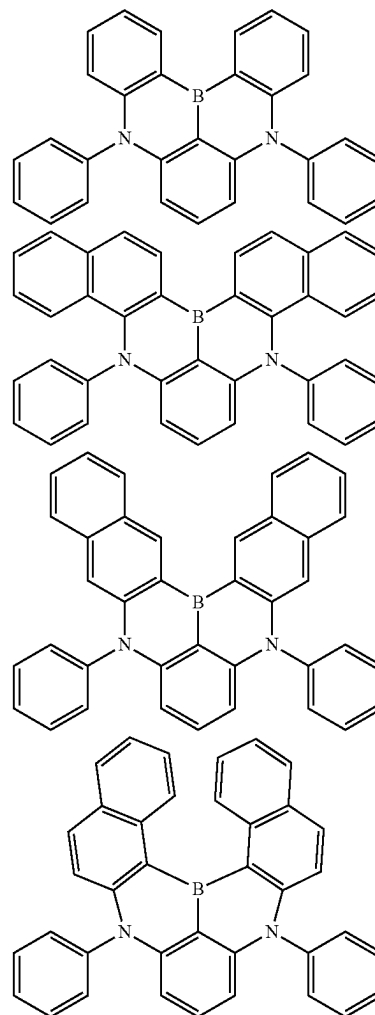
wherein in the formula (43C'), R₄₈₃ to R₄₈₆ are as defined in the formula (43C).

In one embodiment, R₄₈₁ to R₄₈₆ are independently a substituted or unsubstituted alkyl group having 1 to 50 carbon atoms or a substituted or unsubstituted aryl group having 6 to 50 ring carbon atoms.

In one embodiment, R₄₈₁ to R₄₈₆ are independently a substituted or unsubstituted aryl group having 6 to 50 ring carbon atoms.

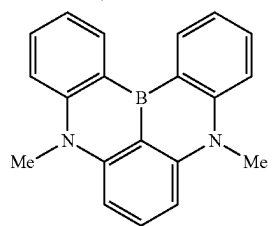
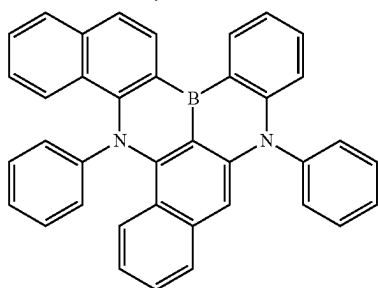
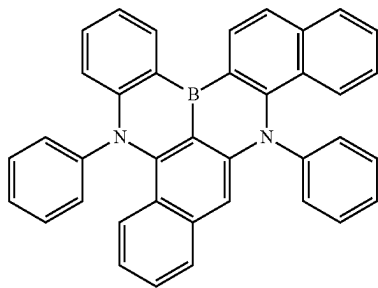
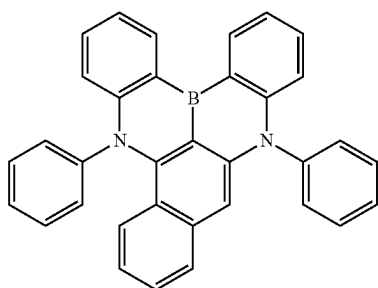
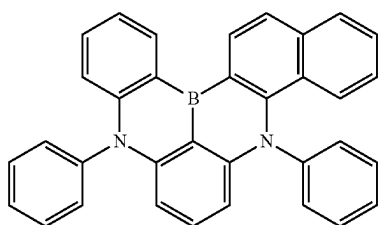
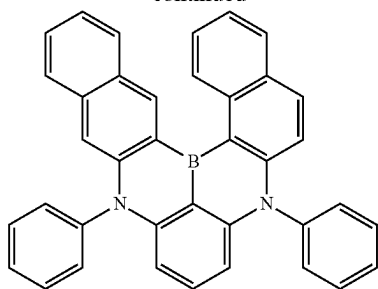
The compound represented by the formula (41) can be synthesized by the following method: An intermediate is obtained by bonding the a ring, the b ring and the c ring with linking groups (a group containing N—R₁ and a group containing N—R₂) (first reaction), and a final compound is obtained by bonding the a ring, the b ring and the c ring with a linking group (a group containing B) (second reaction). In the first reaction, an amination reaction such as Buchwald-Hartwig reaction can be applied. In the second reaction, tandem hetero-Friedel-Crafts reaction or the like can be applied.

Examples of the compound represented by the formula (41) are described below. They are just exemplified compounds and the compound represented by the formula (41) is not limited to the following examples. In the following example compounds, Me represents methyl group and tBu represents tert-butyl group.

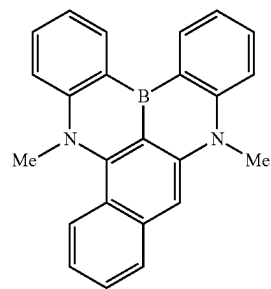
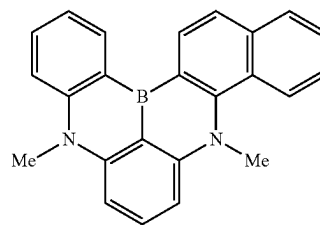
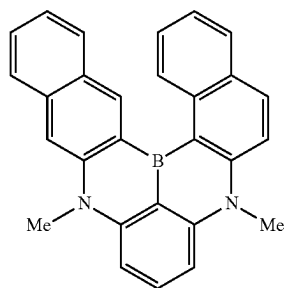
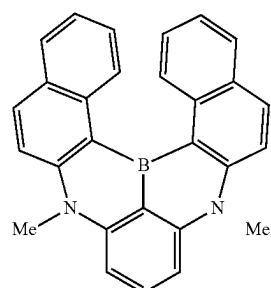
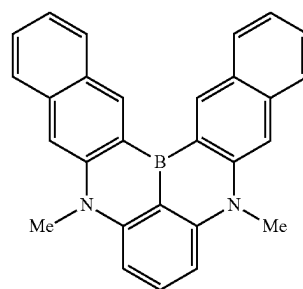
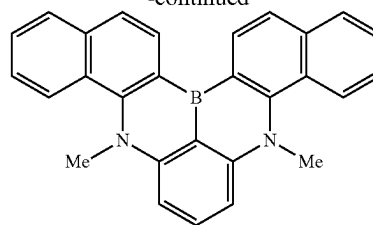


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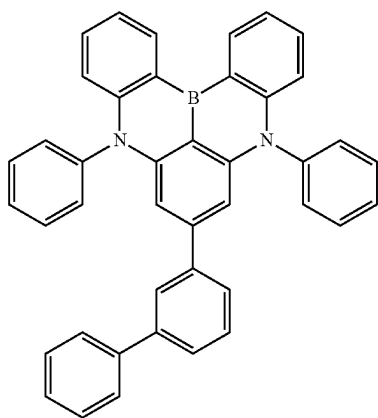
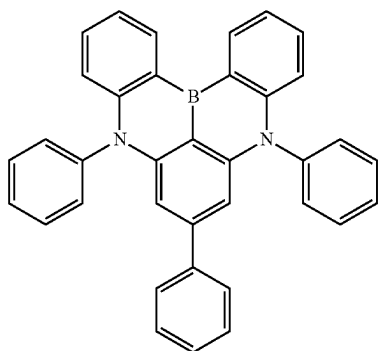
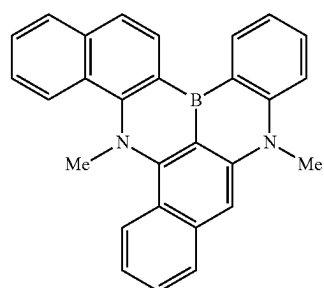
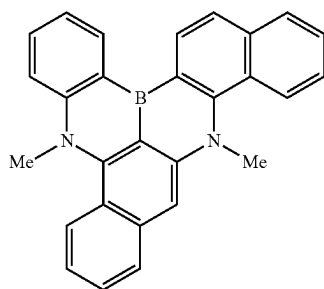
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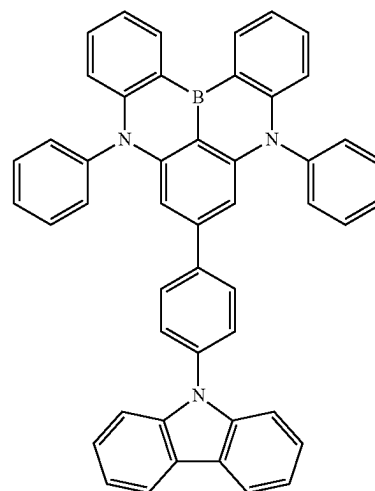
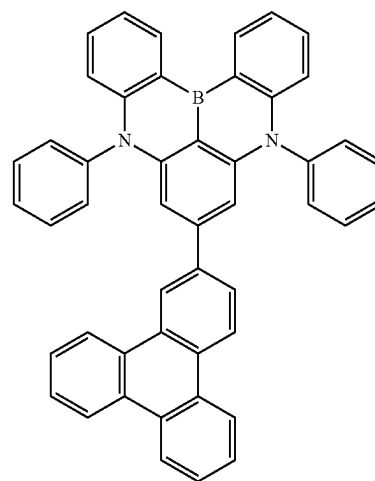
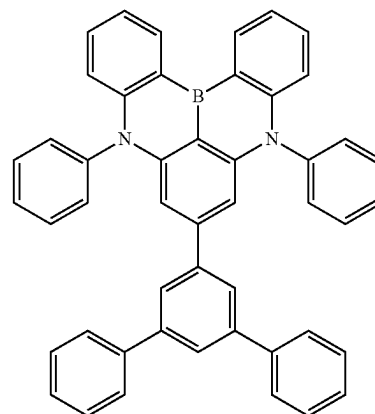
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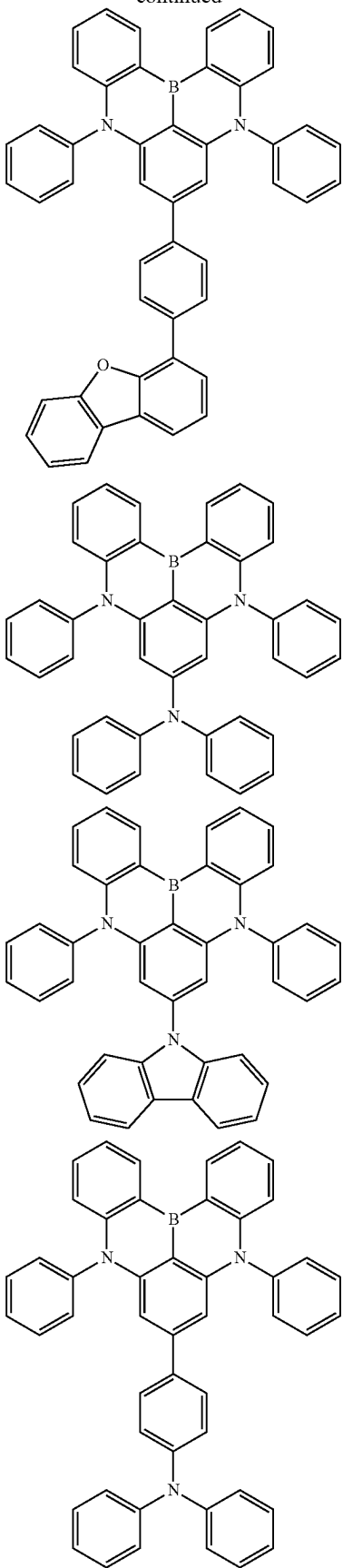
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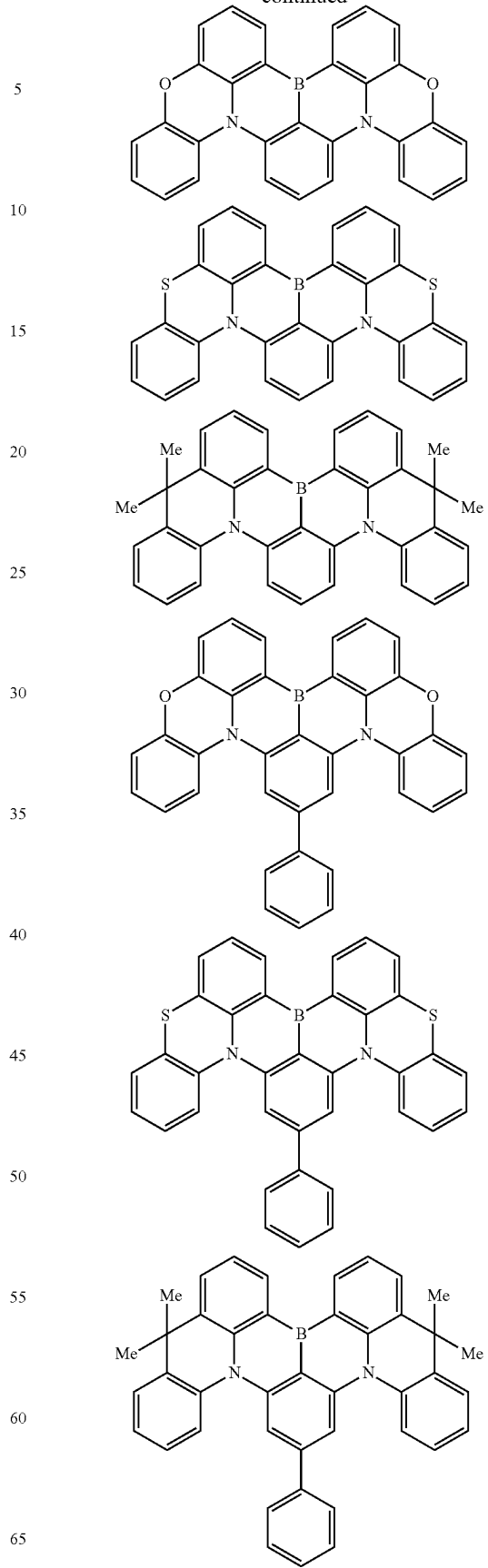
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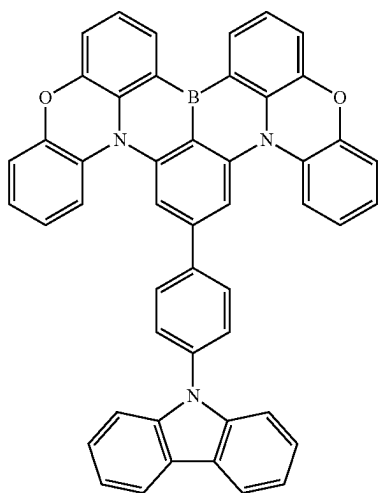
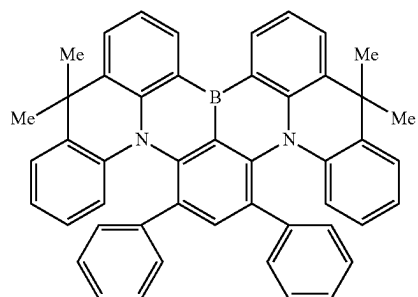
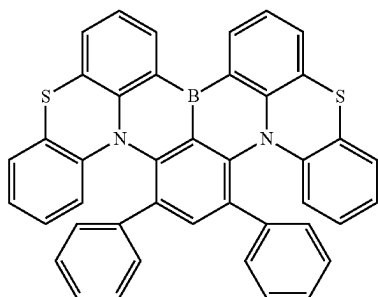
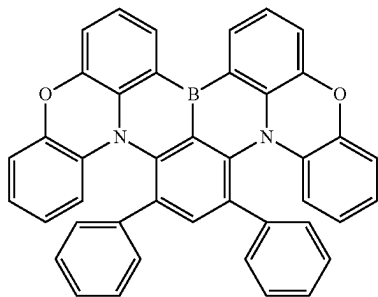
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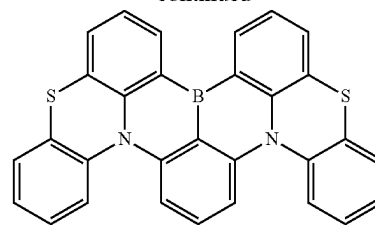
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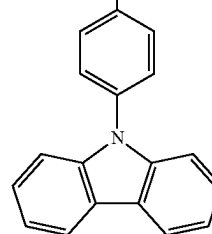
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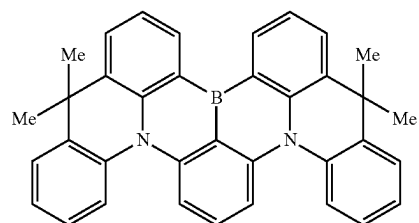
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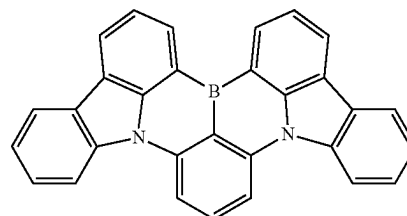


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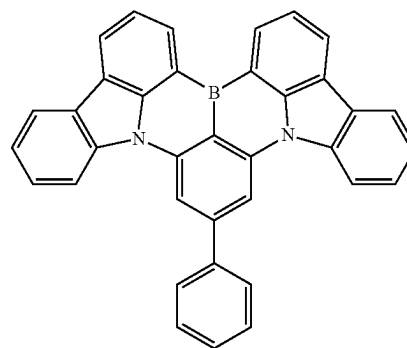


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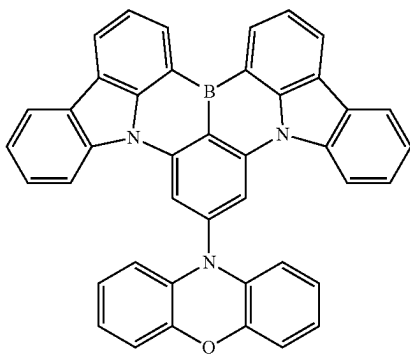
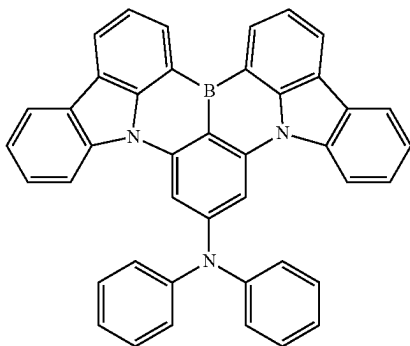
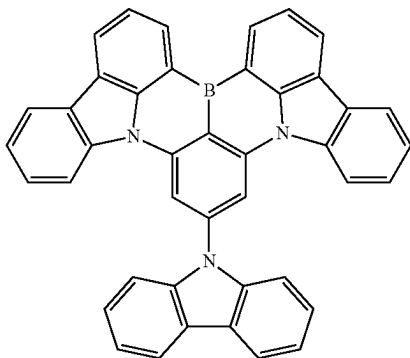
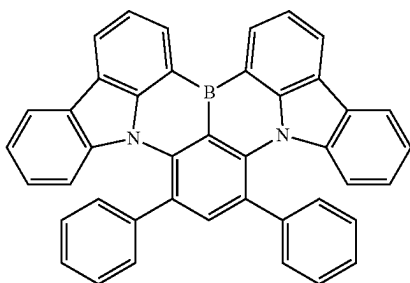
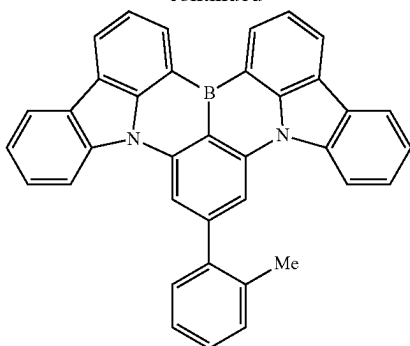
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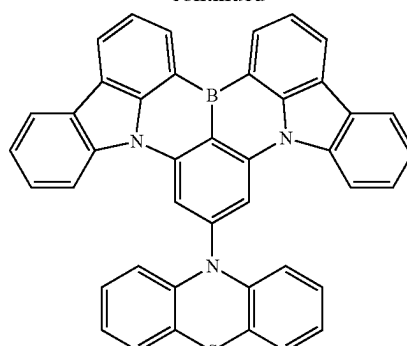
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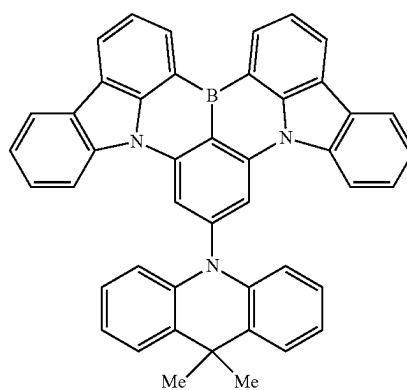
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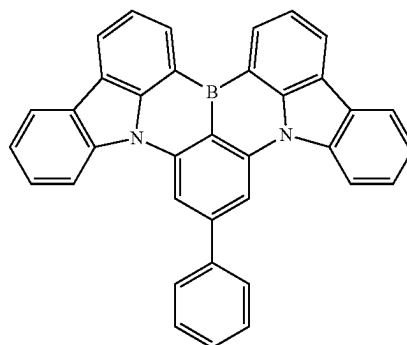


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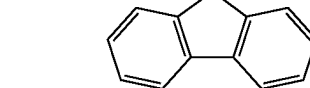
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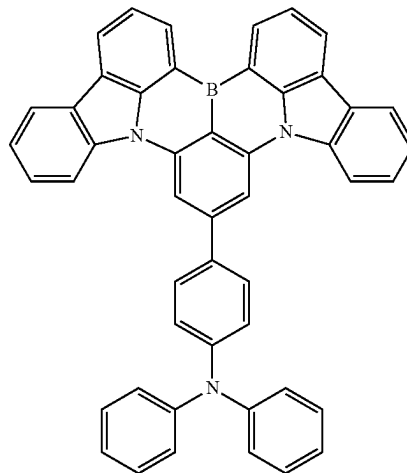
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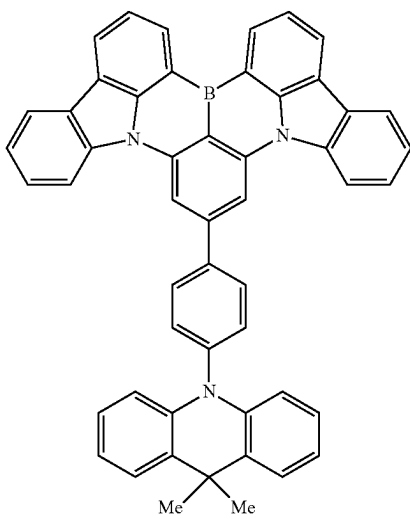
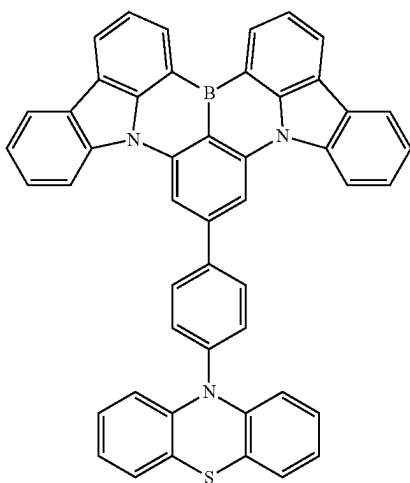
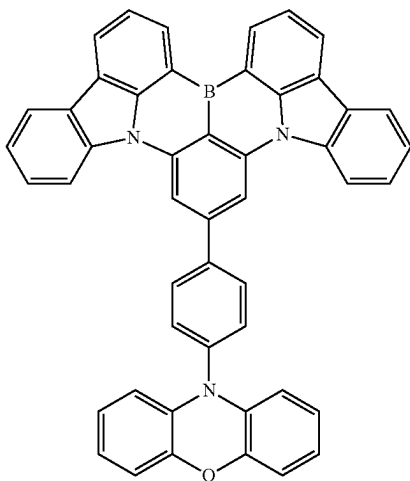


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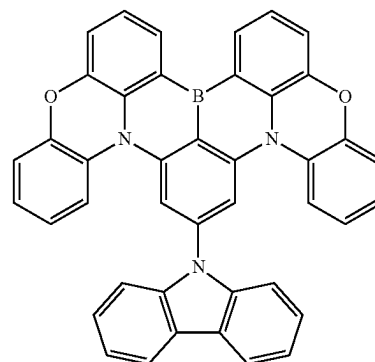
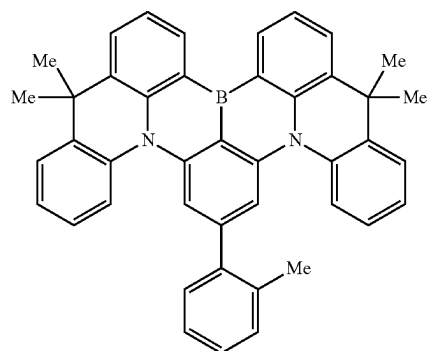
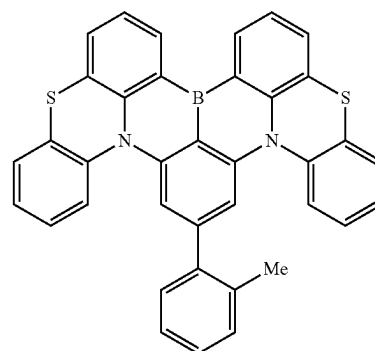
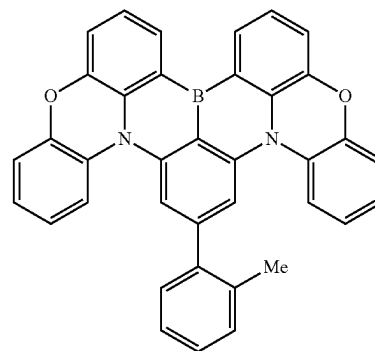
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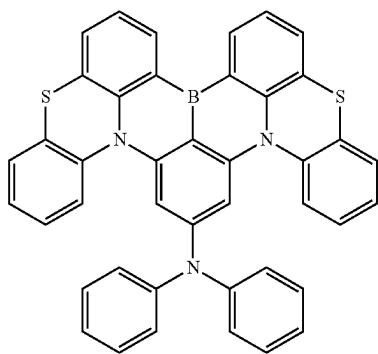
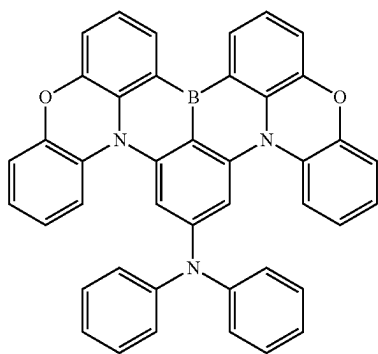
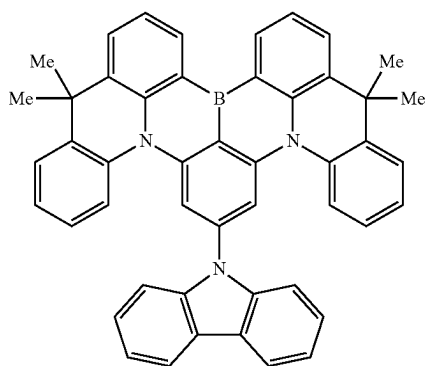
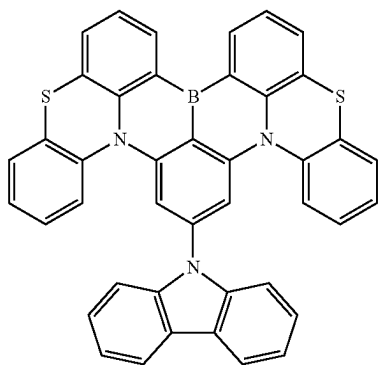
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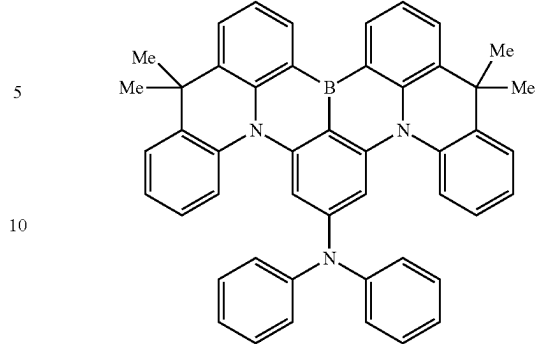


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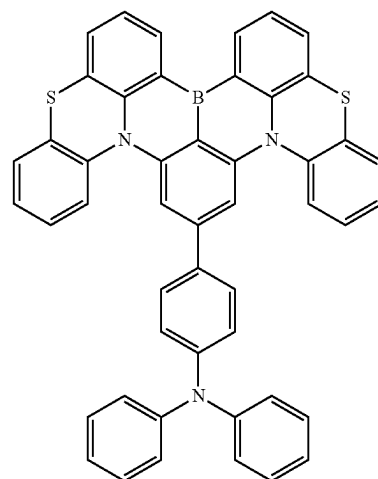
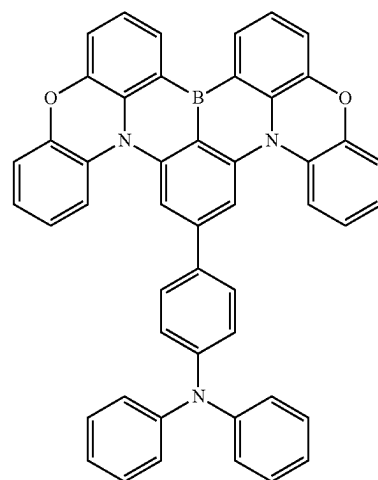
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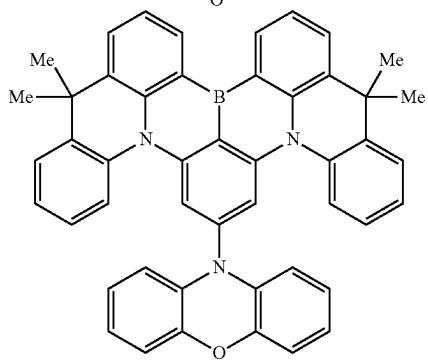
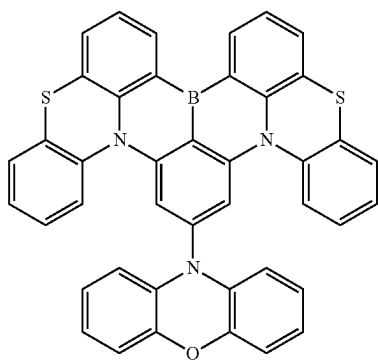
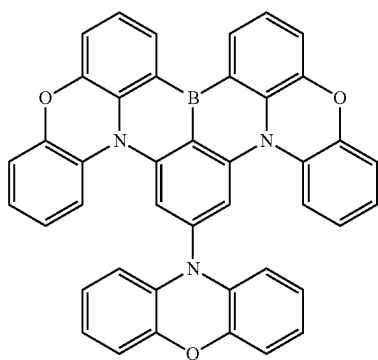
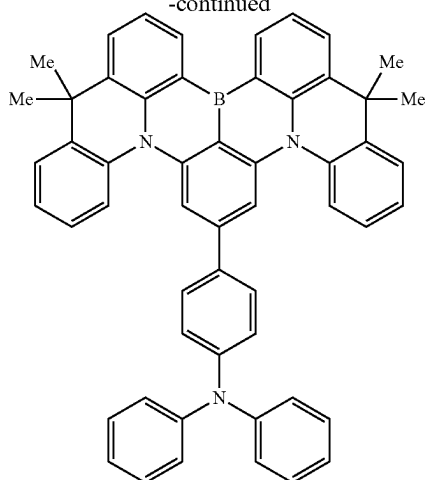
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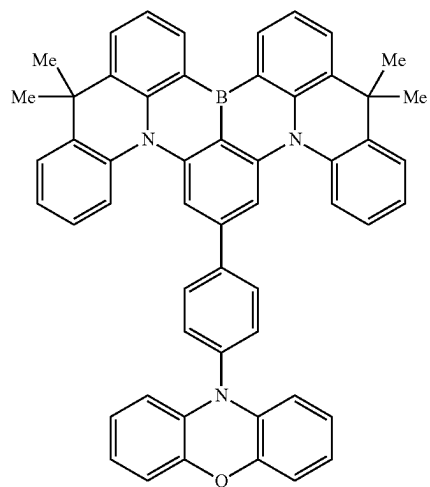
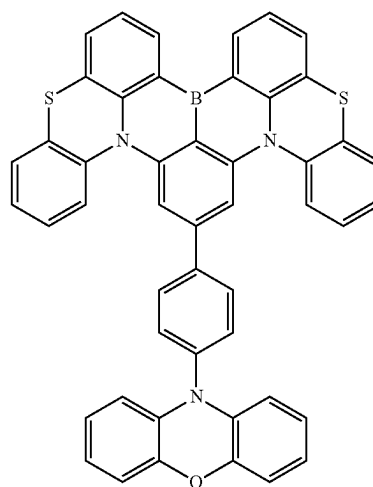
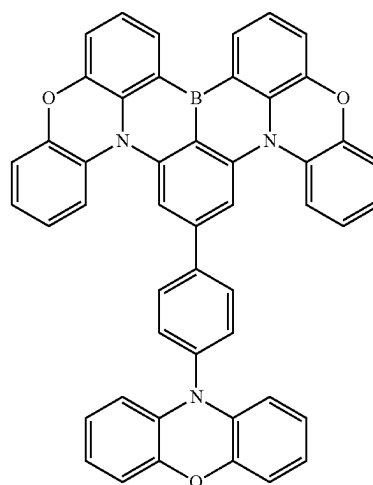
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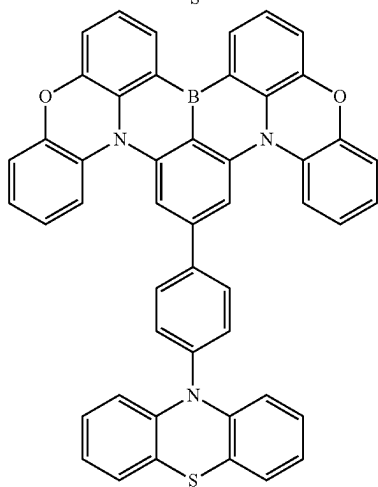
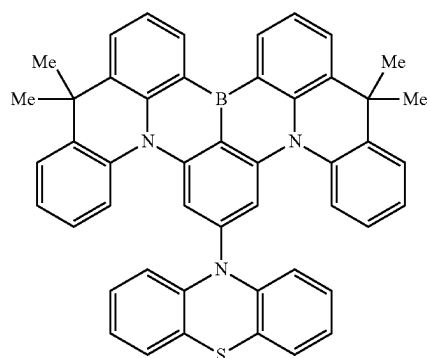
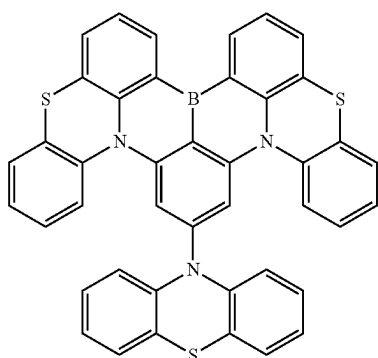
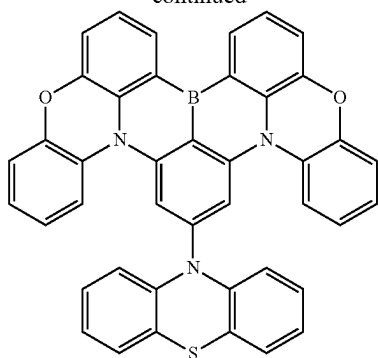
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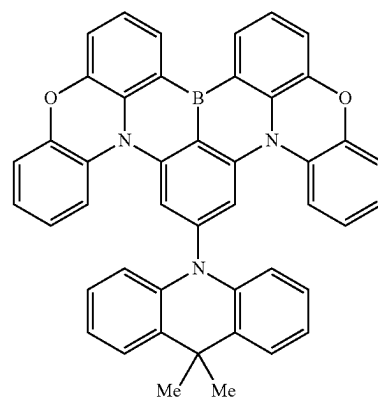
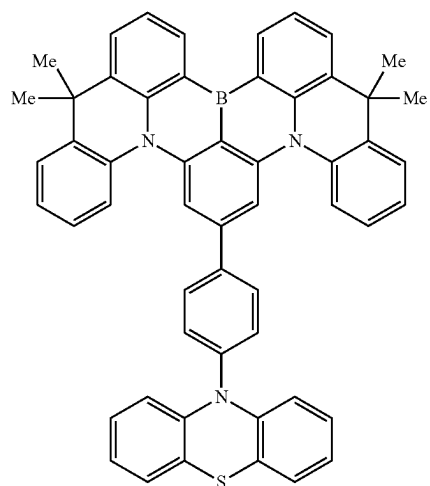
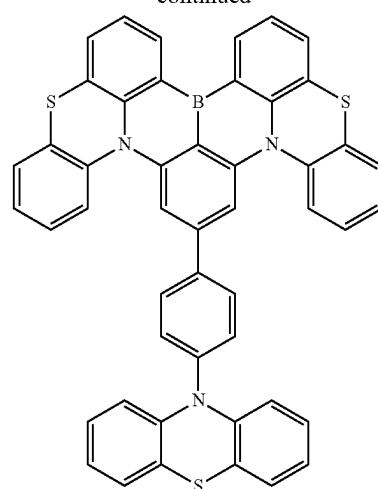
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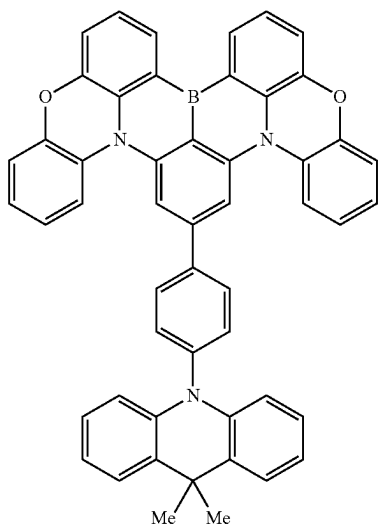
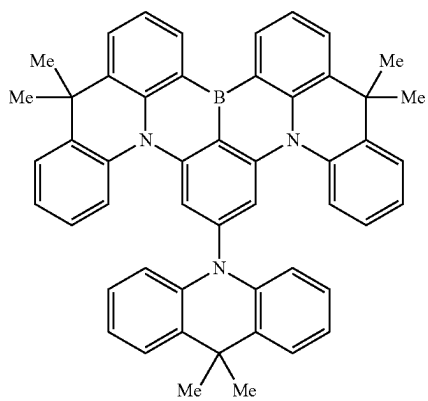
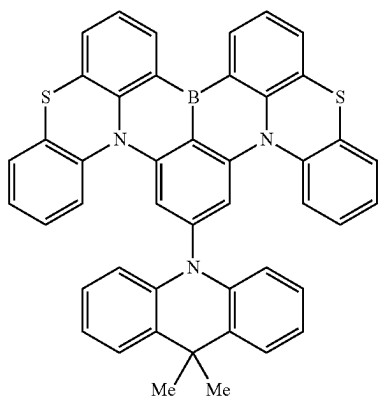
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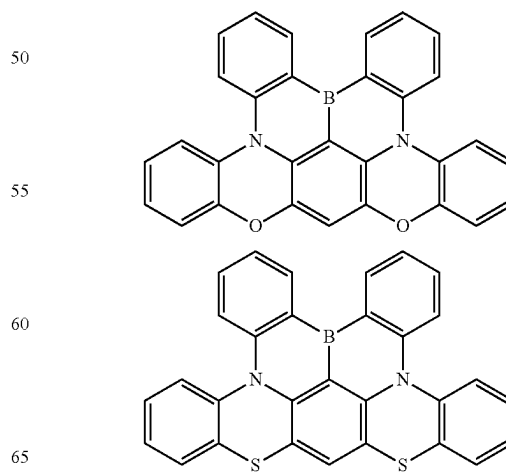
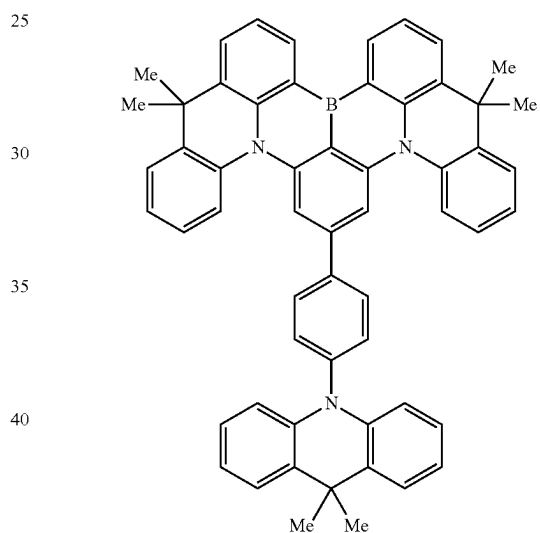
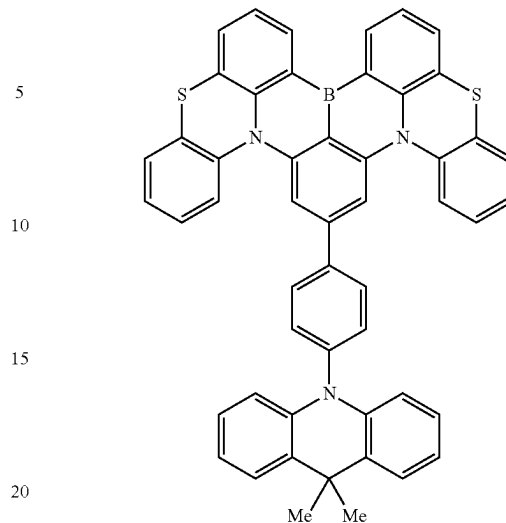


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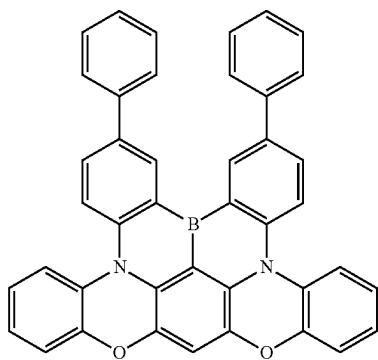
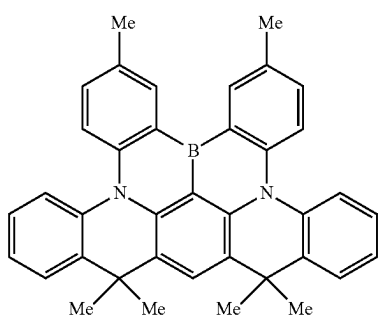
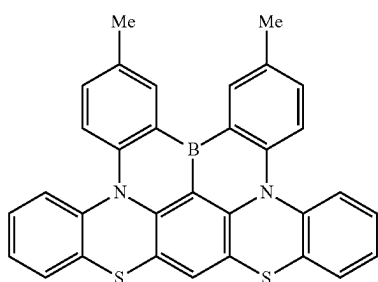
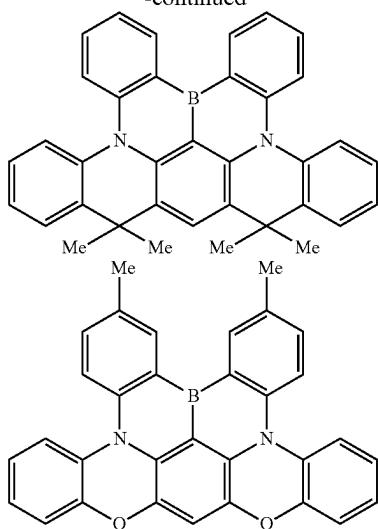
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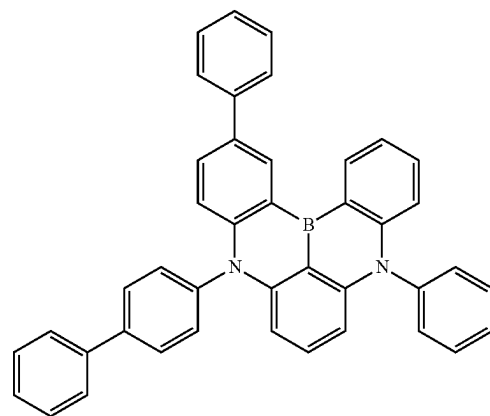
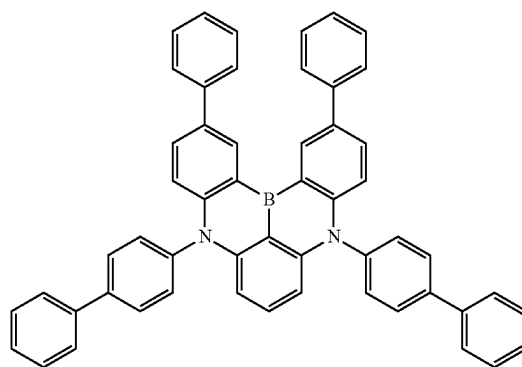
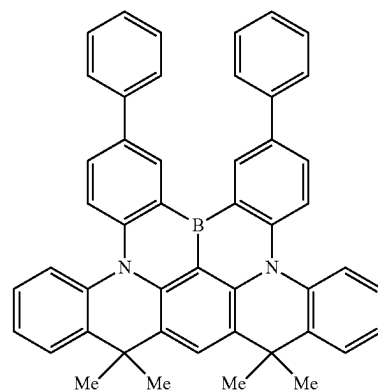
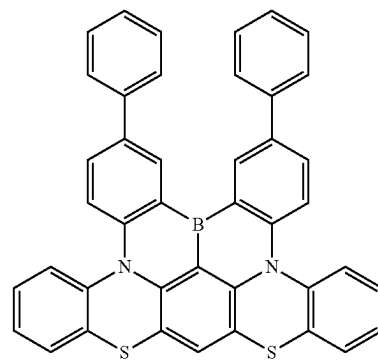
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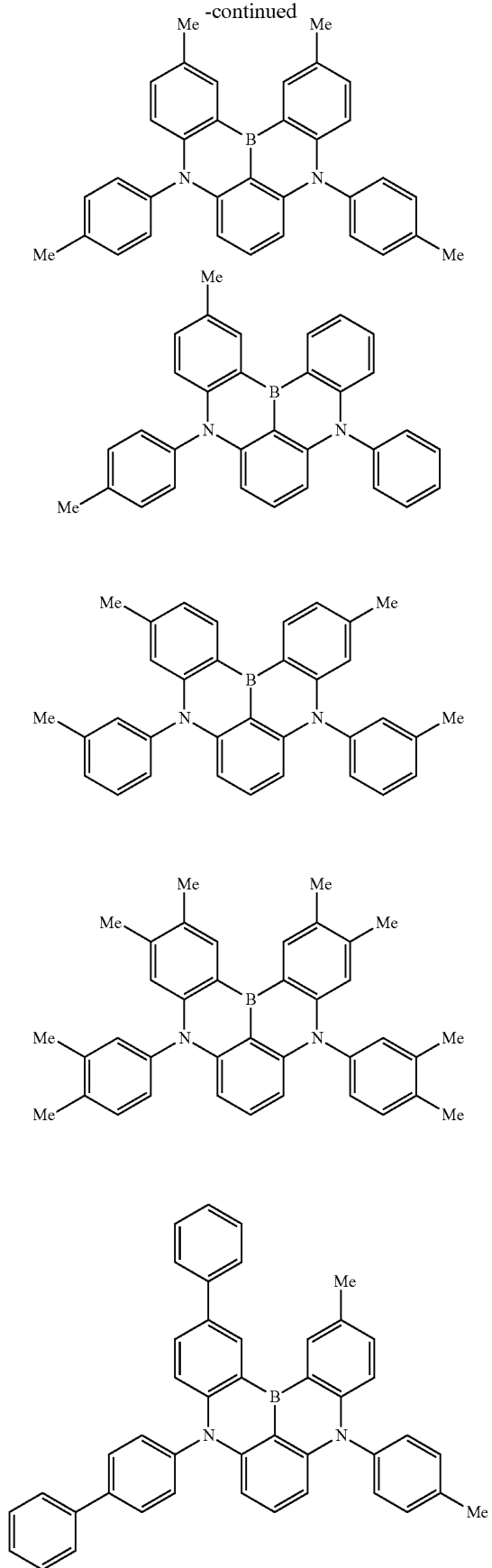
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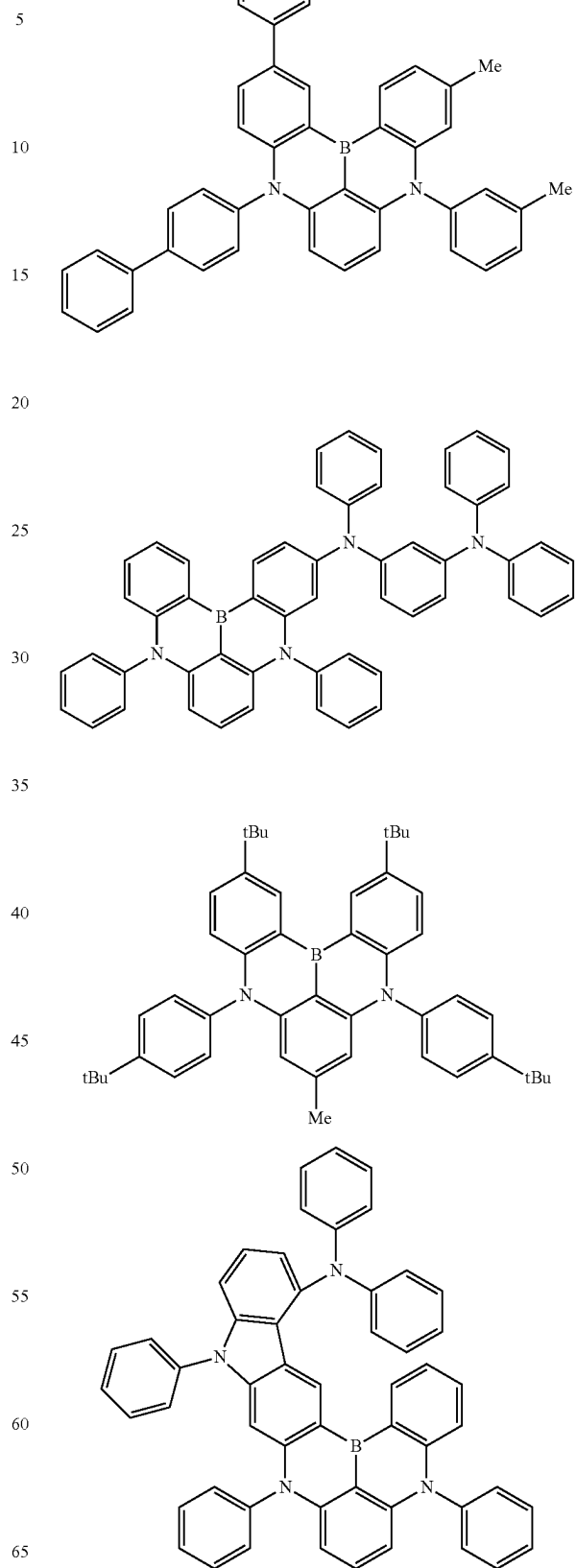


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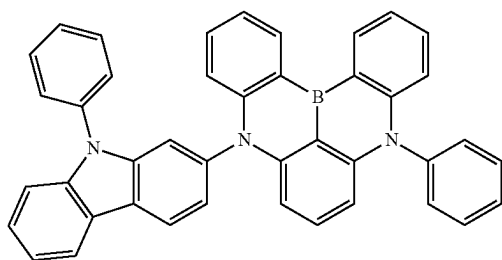
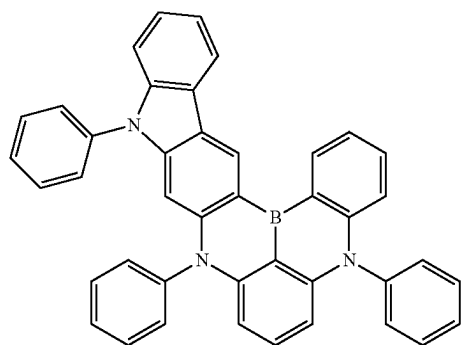
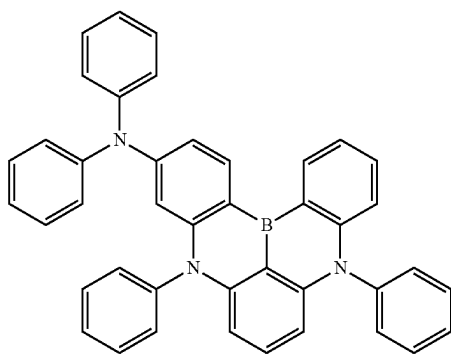
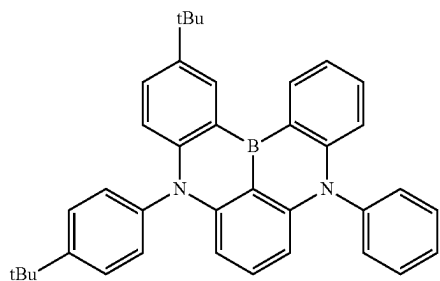
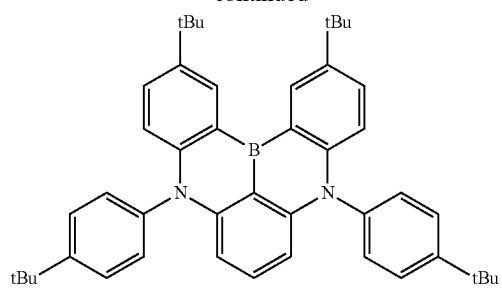
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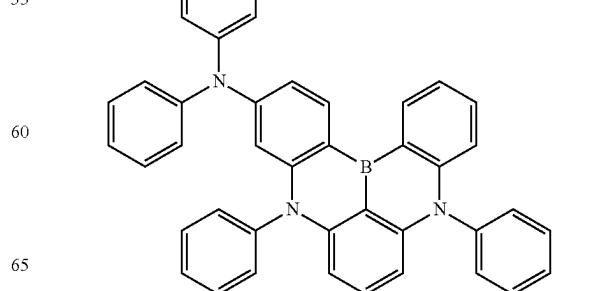
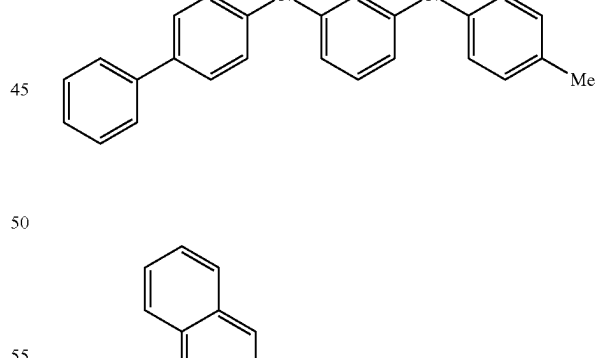
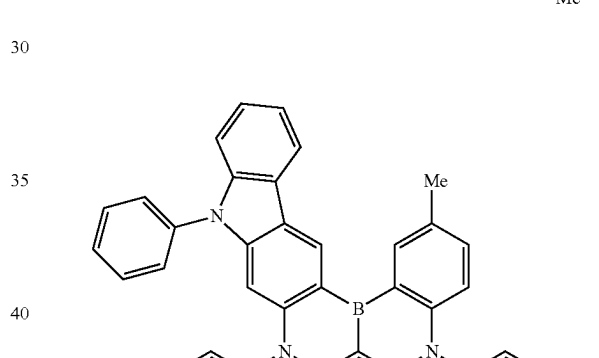
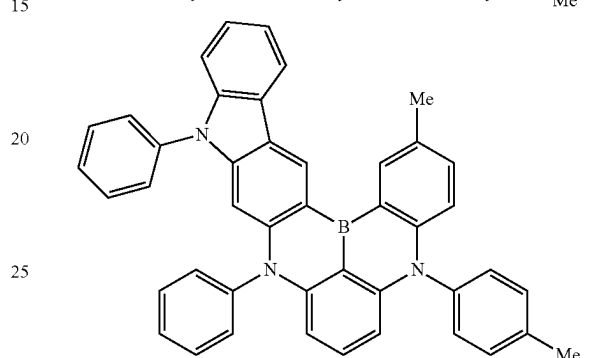
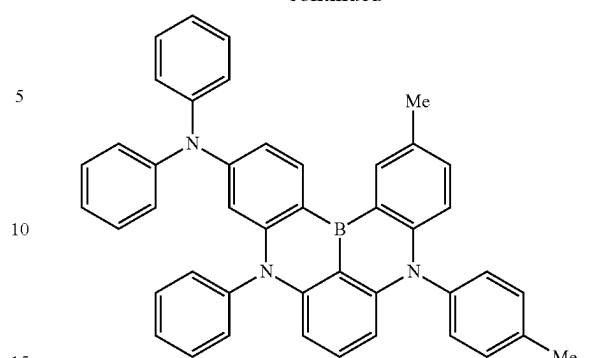


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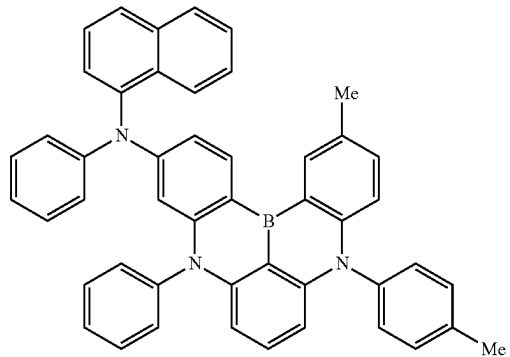
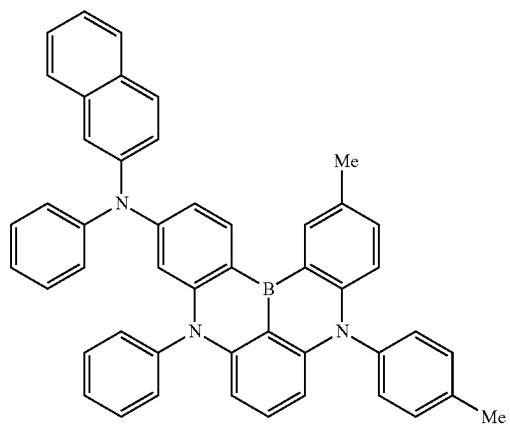
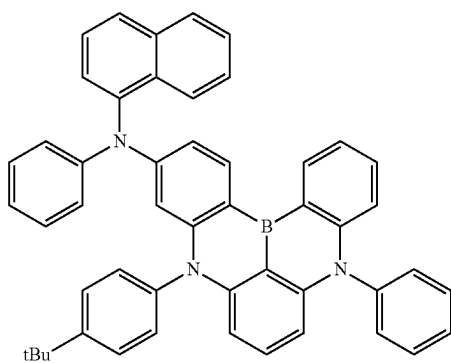
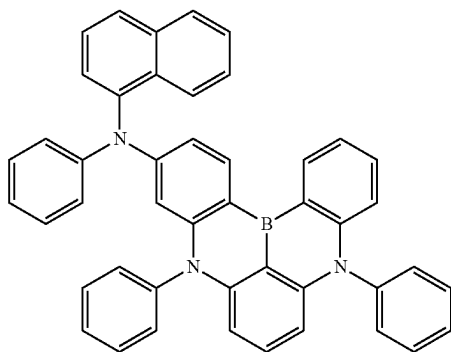
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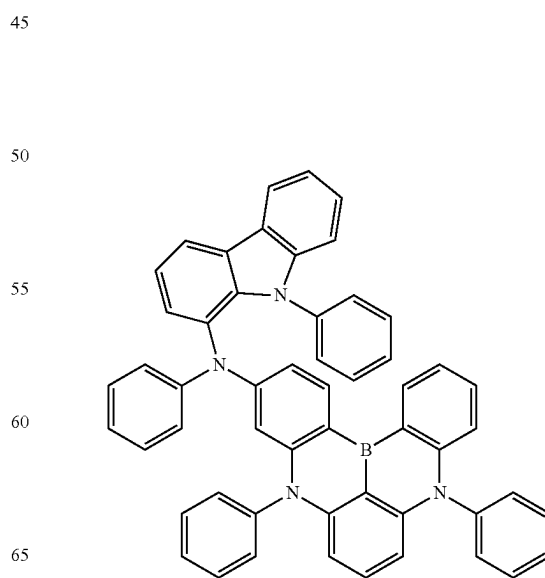
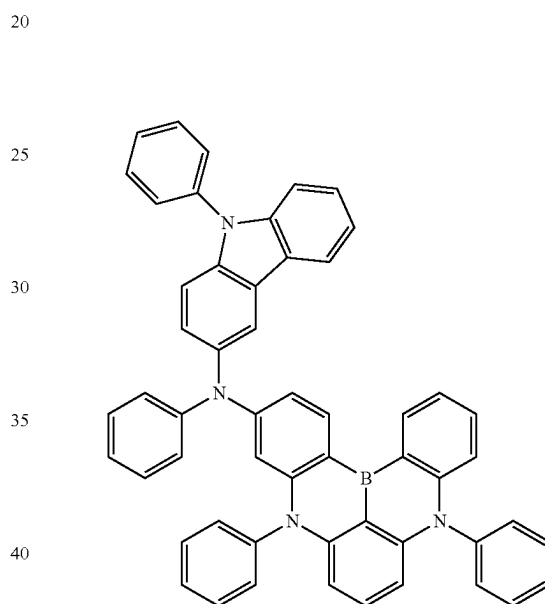
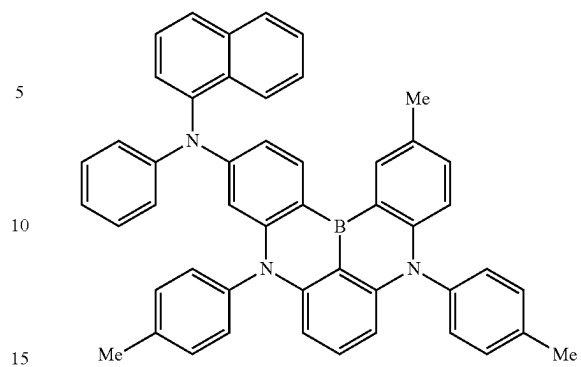


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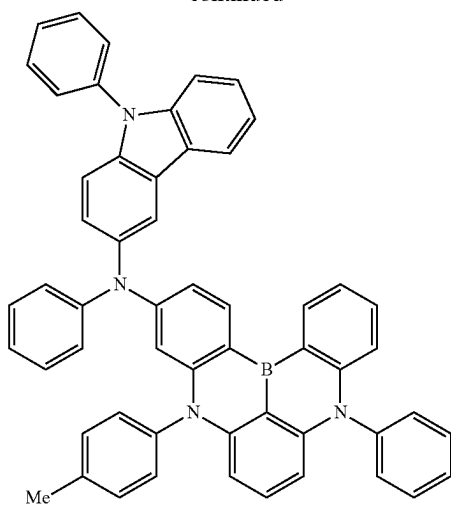
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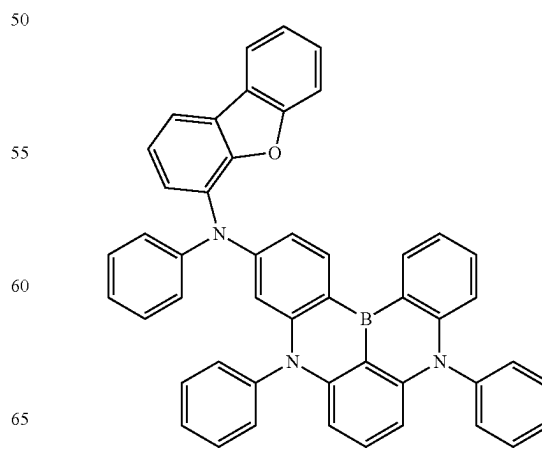
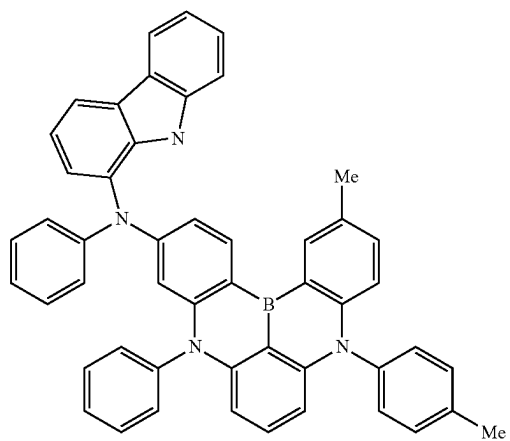
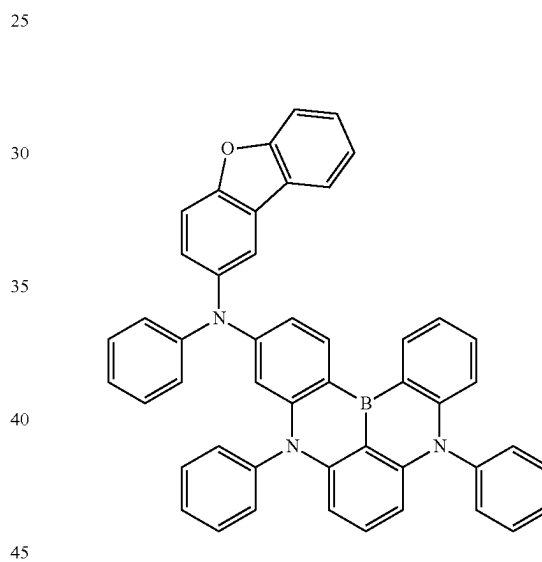
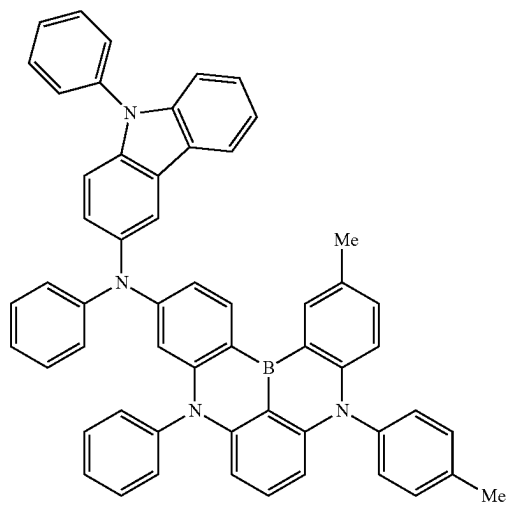
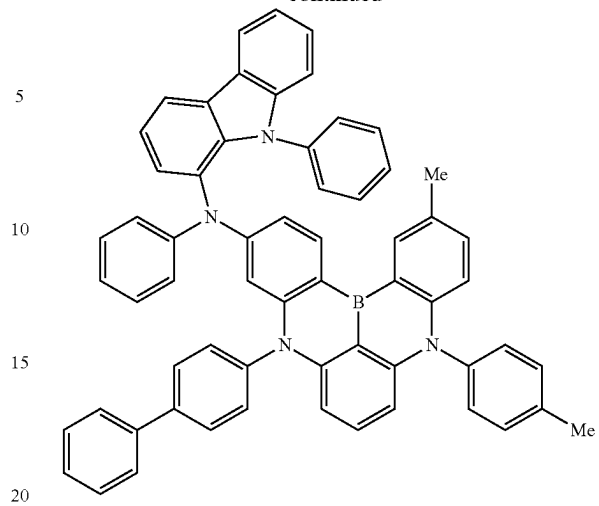


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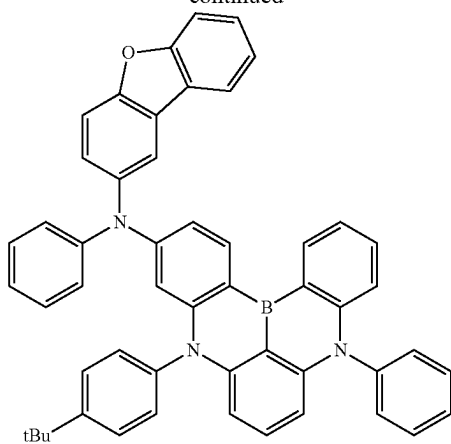
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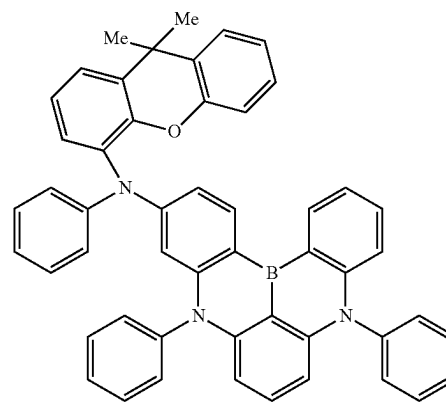
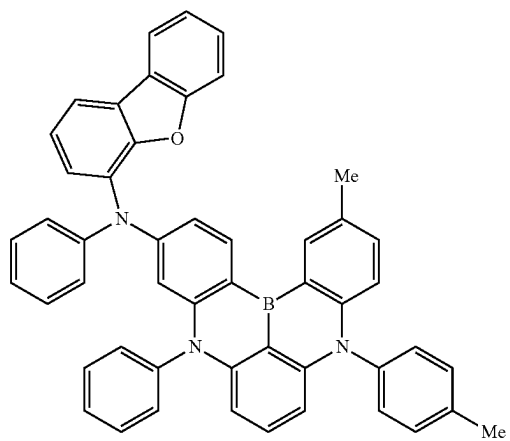
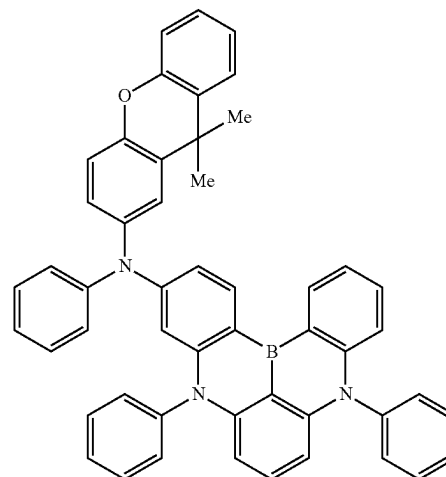
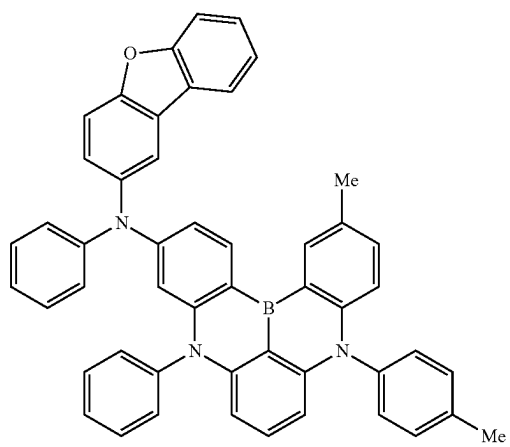
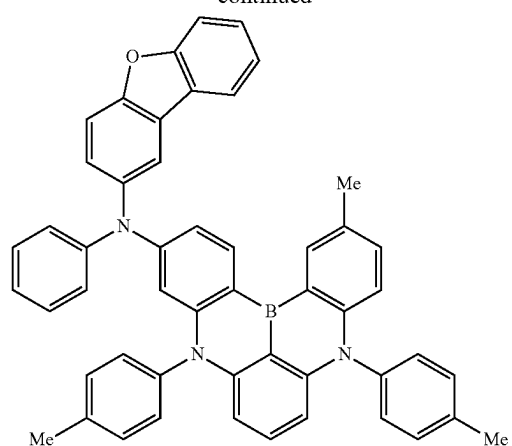
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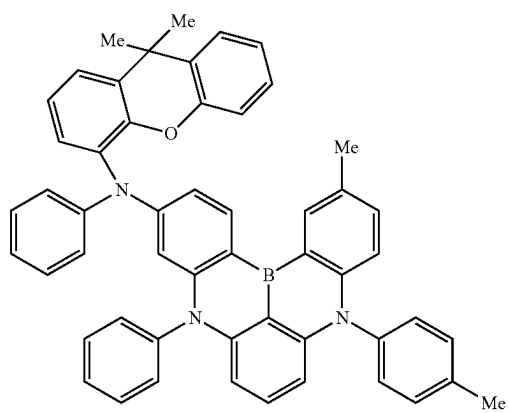
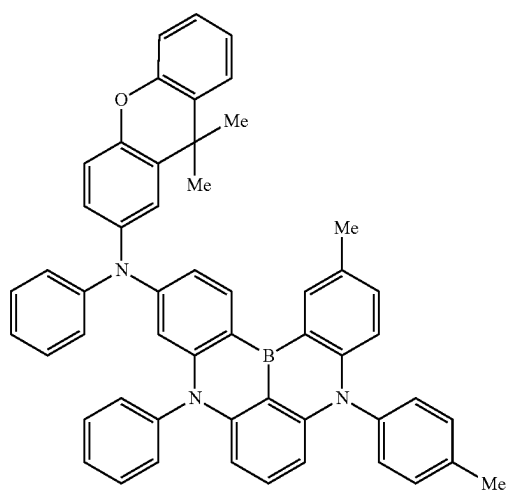
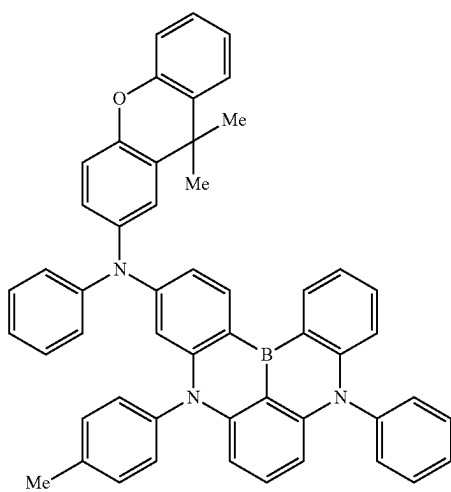
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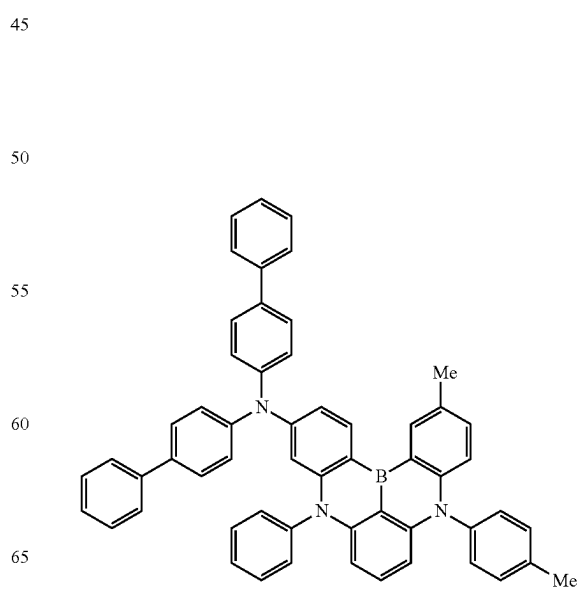
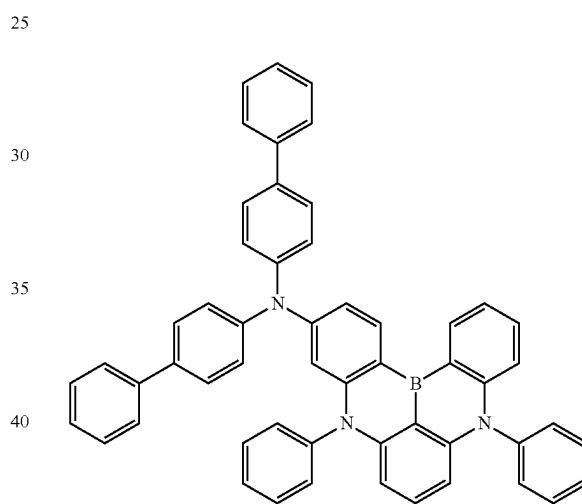
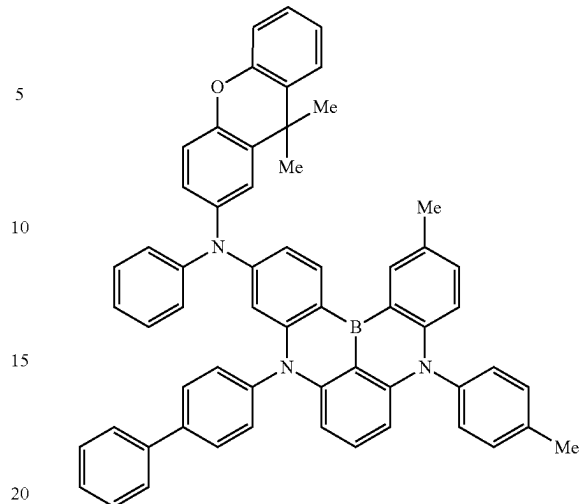


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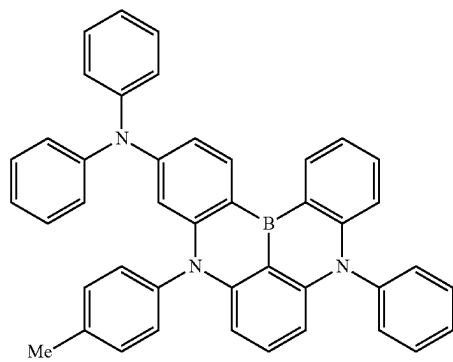
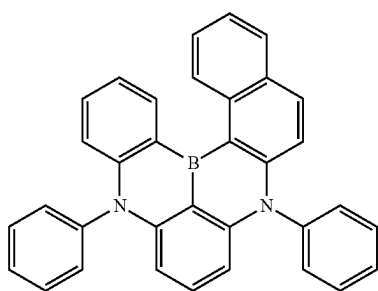
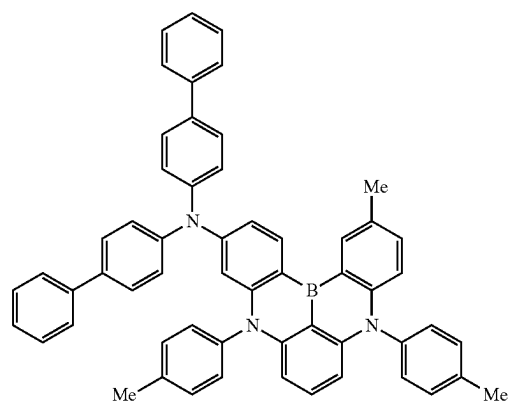
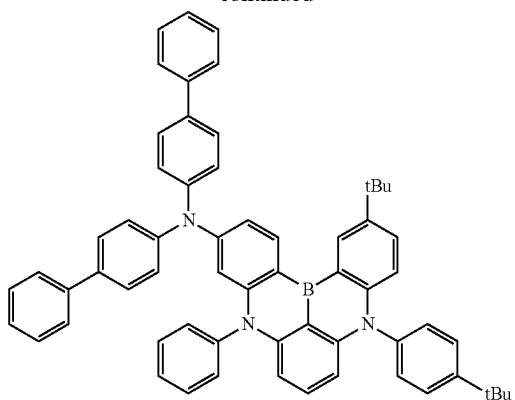
**672**

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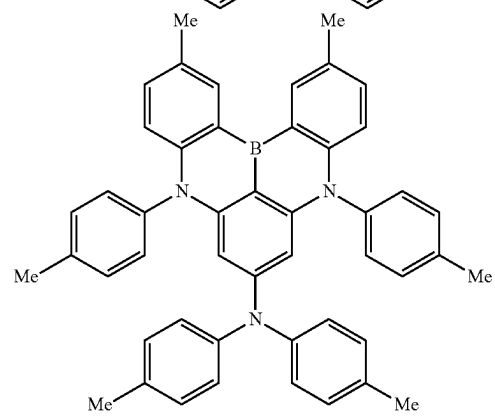
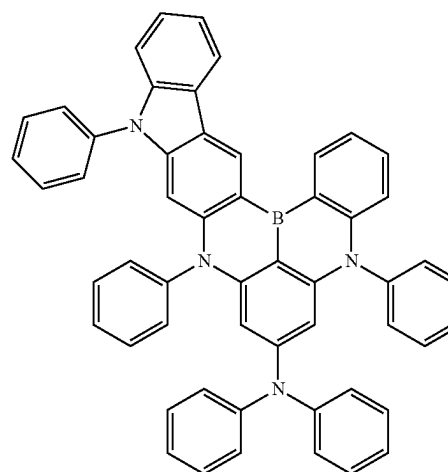
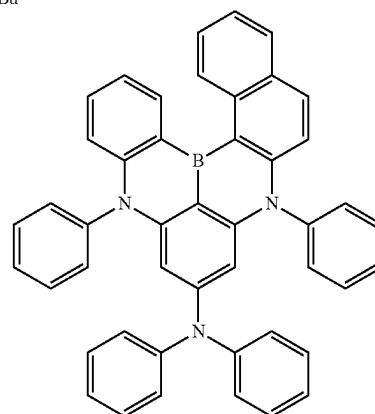
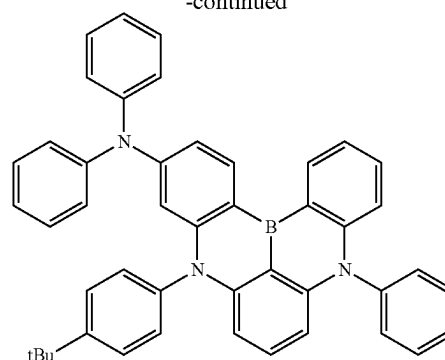
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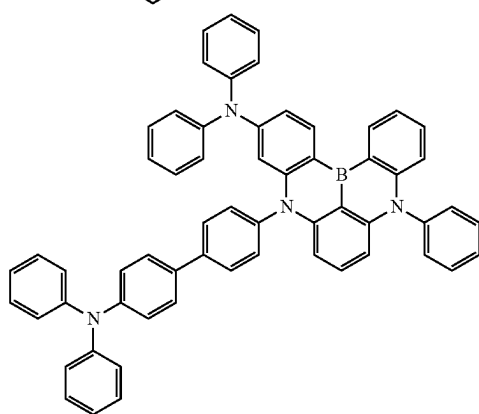
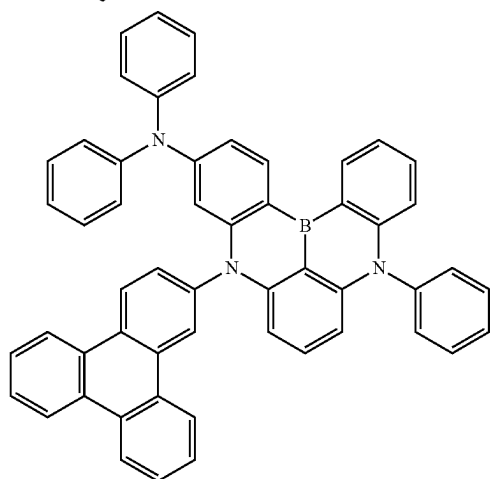
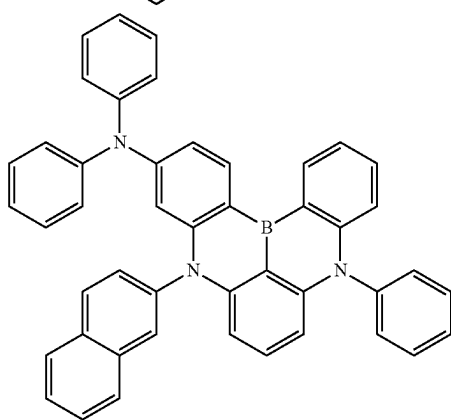
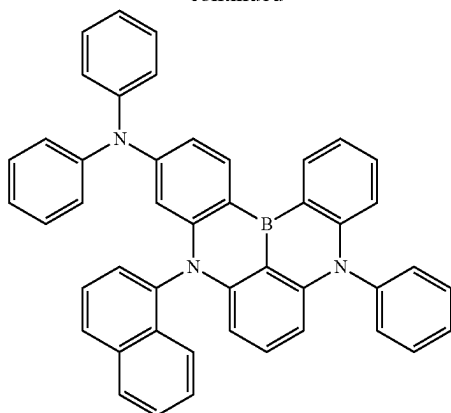
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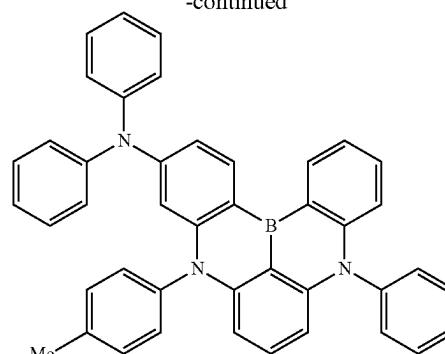
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**676**

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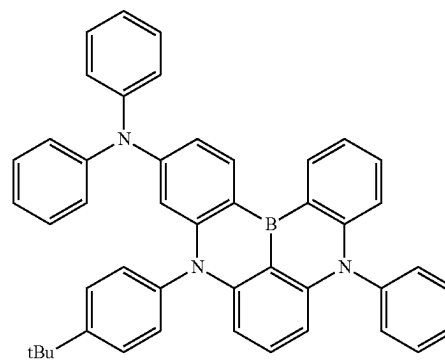
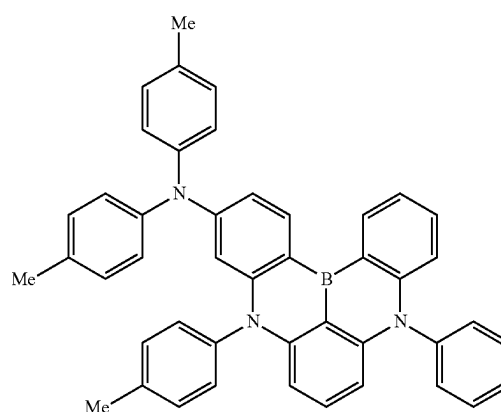
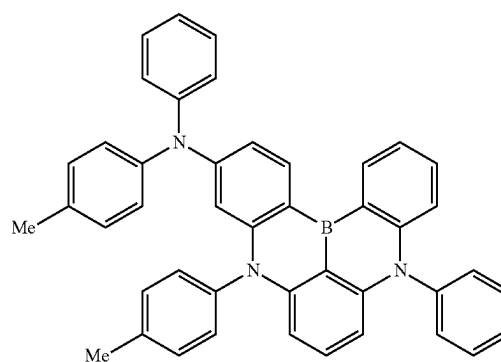
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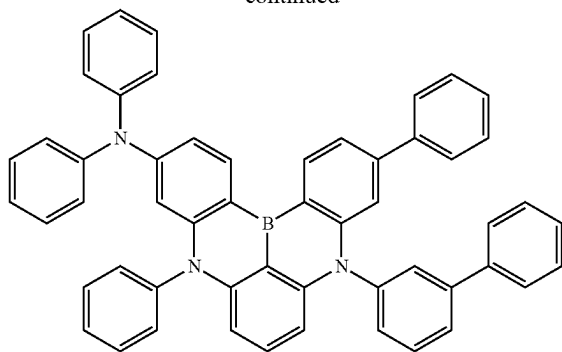
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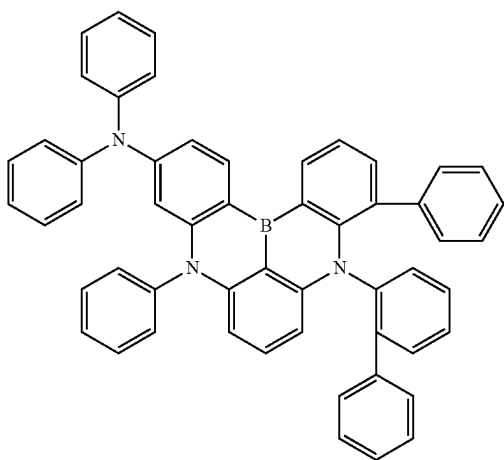
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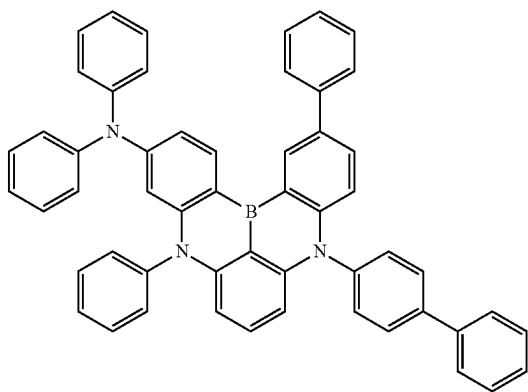


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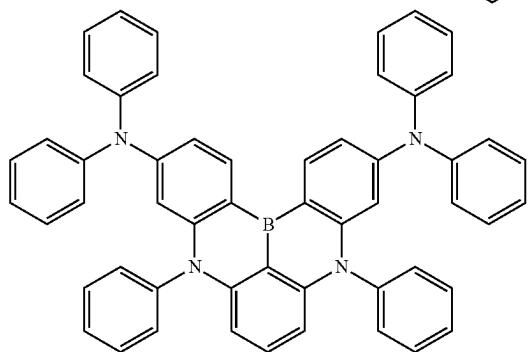
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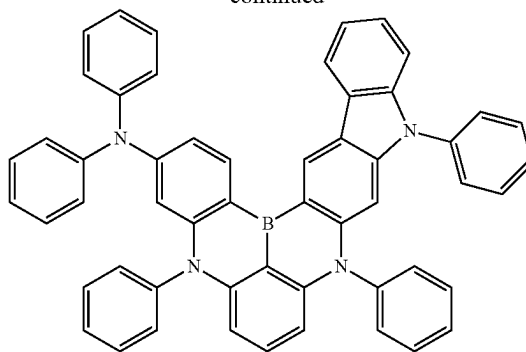
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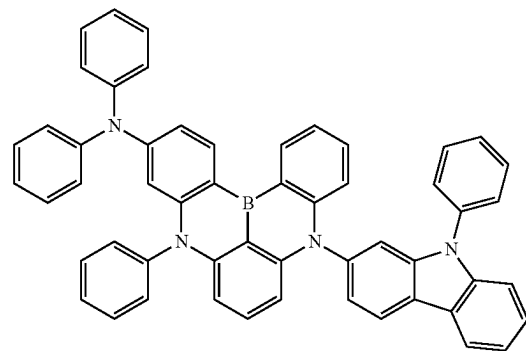
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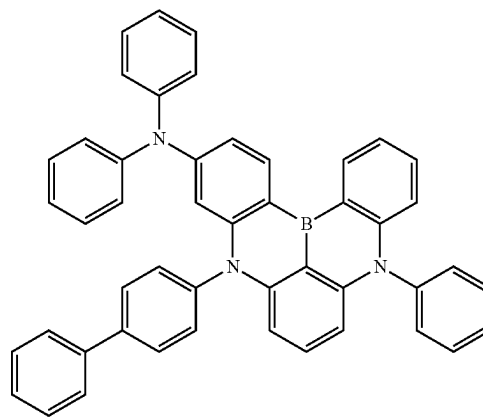


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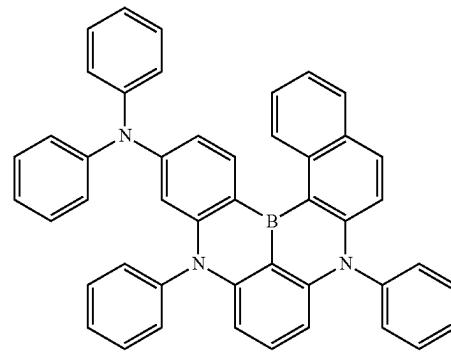
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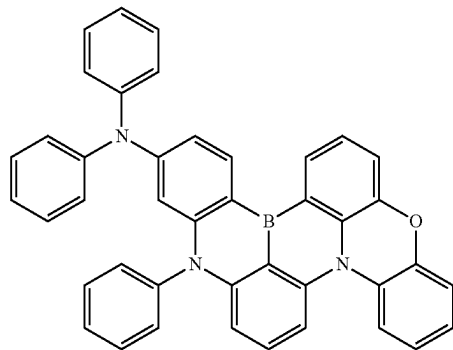
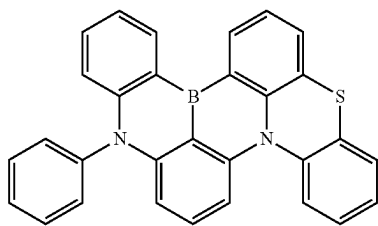
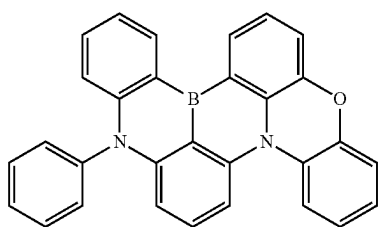
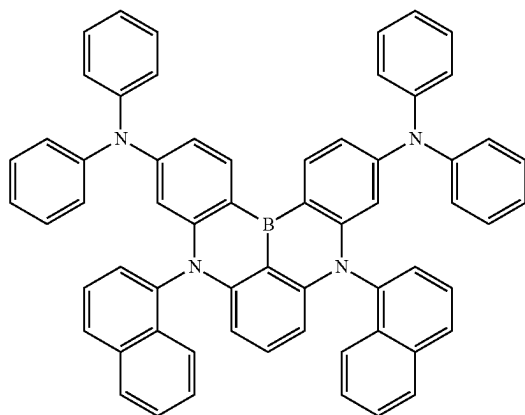
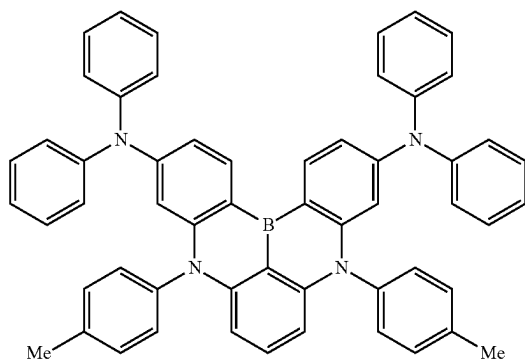
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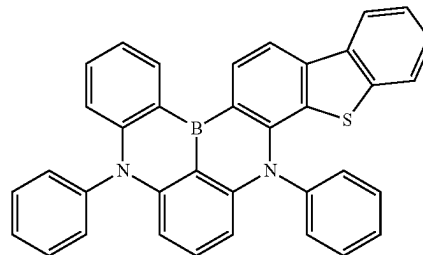
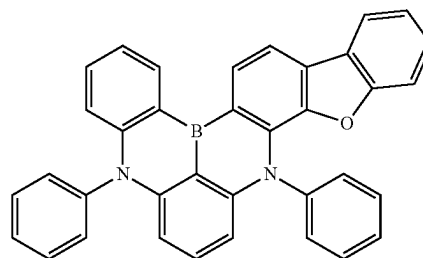
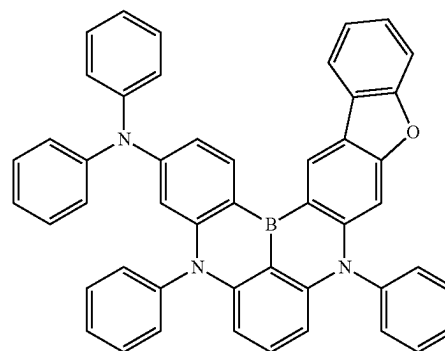
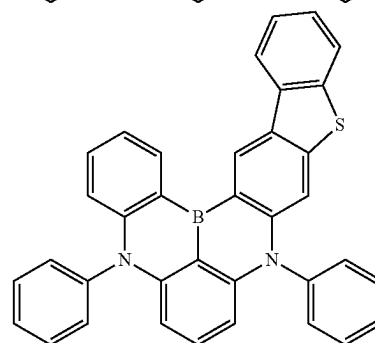
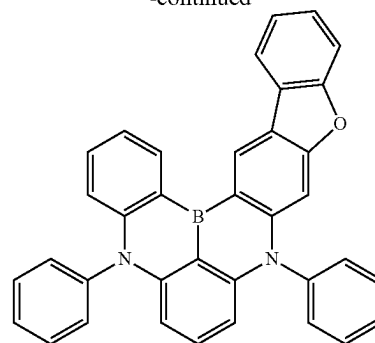
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679

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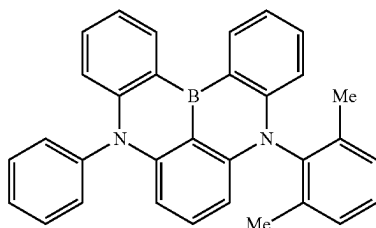
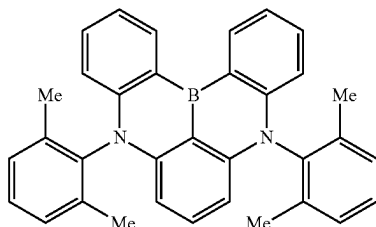
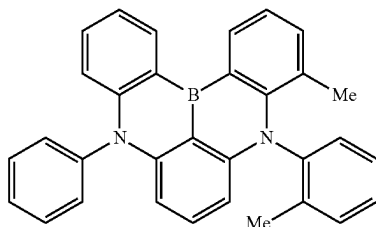
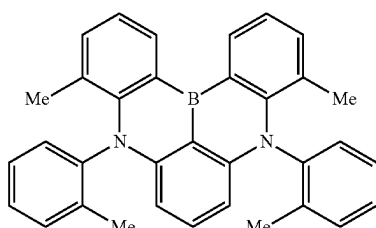
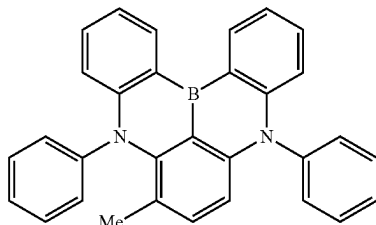
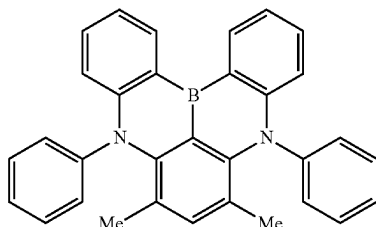
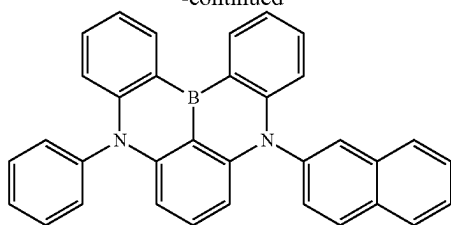
**680**

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681

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**682**

(Compound Represented by Formula (51))

The compound represented by the formula (51) is explained below

5 (51)

10 (52)

(53)

15 (54)

20 (55)

25 (56)

30 (56)

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In the formula (51),

r ring is a ring represented by the formula (52) or formula (53) which is fused to an adjacent ring at an arbitrary position;

q ring and s ring are independently a ring represented by the formula (54) which is fused to an adjacent ring at an arbitrary position;

p ring and t ring are independently a ring represented by the formula (55) or the formula (56) which is fused to an adjacent ring at an arbitrary position;

when plural R_{501} s exist, adjacent plural R_{501} s are bonded with each other to form a substituted or unsubstituted, saturated or unsaturated ring, or do not form a substituted or unsubstituted, saturated or unsaturated ring; X_{501} is an oxygen atom, a sulfur atom, or NR_{502} ; R_{901} and R_{502} that do not form the substituted or unsubstituted saturated or unsaturated ring are

a hydrogen atom,

a substituted or unsubstituted alkyl group having 1 to 50 carbon atoms,

a substituted or unsubstituted alkenyl group having 2 to 50 carbon atoms,

a substituted or unsubstituted alkynyl group having 2 to 50 carbon atoms,

a substituted or unsubstituted cycloalkyl group having 3 to 50 ring carbon atoms,

 $-Si(R_{901})(R_{902})(R_{903})$, $-O-(R_{904})$, $-S-(R_{905})$, $-N(R_{906})(R_{907})$,

a halogen atom, a cyano group, a nitro group,

a substituted or unsubstituted aryl group having 6 to 50 ring carbon atoms, or

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a substituted or unsubstituted monovalent heterocyclic group having 5 to 50 ring atoms;

R_{901} to R_{907} are as defined in the formula (1);

Ar_{501} and Ar_{502} are independently

a substituted or unsubstituted alkyl group having 1 to 50 carbon atoms,

a substituted or unsubstituted alkenyl group having 2 to 50 carbon atoms,

a substituted or unsubstituted alkynyl group having 2 to 50 carbon atoms,

a substituted or unsubstituted cycloalkyl group having 3 to 50 ring carbon atoms,

a substituted or unsubstituted aryl group having 6 to 50 ring carbon atoms, or

a substituted or unsubstituted monovalent heterocyclic group having 5 to 50 ring atoms;

L_{501} is

a substituted or unsubstituted alkylene group having 1 to 50 carbon atoms,

a substituted or unsubstituted alkenylene group having 2 to 50 carbon atoms,

a substituted or unsubstituted alkynylene group having 2 to 50 carbon atoms,

a substituted or unsubstituted cycloalkylene group having 3 to 50 ring carbon atoms,

a substituted or unsubstituted arylene group having 6 to 50 ring carbon atoms, or

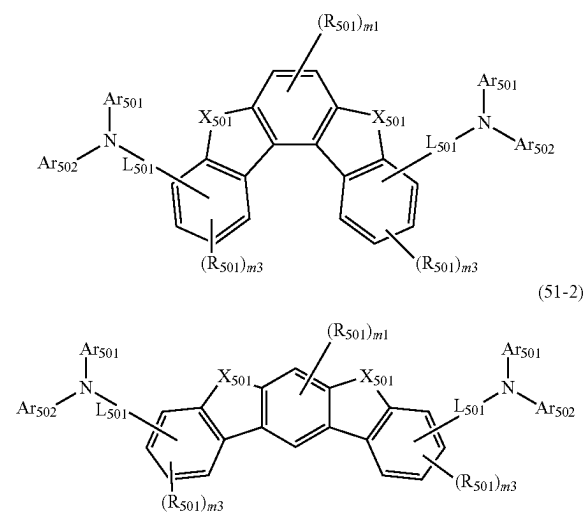
a substituted or unsubstituted divalent heterocyclic group having 5 to 50 ring atoms;

$m1$ is independently an integer of 0 to 2, $m2$ is independently an integer of 0 to 4, $m3$ is independently an integer of 0 to 3, and $m4$ is independently an integer of 0 to 5; when plural R_{501} s exist, the plural R_{501} s may be the same or different;

In the formula (51), each of the p ring to the t ring is fused to an adjacent ring by sharing two carbon atoms. The position and direction of fusing are not limited, and condensation is possible at any position and direction.

In one embodiment, in the formula (52) or (53) of the r ring, R_{501} is a hydrogen atom.

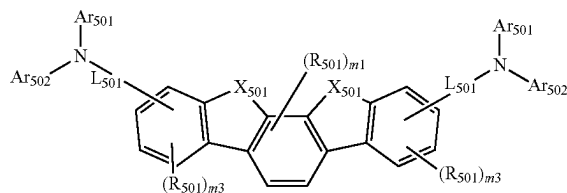
In one embodiment, the compound represented by the formula (51) is represented by any one of the following formulas (51-1) to (51-6):



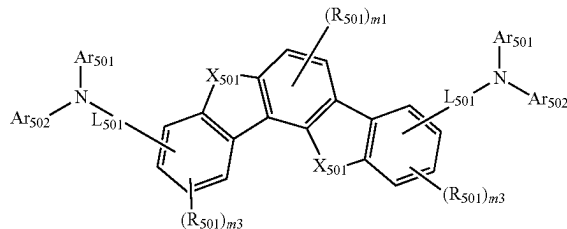
684

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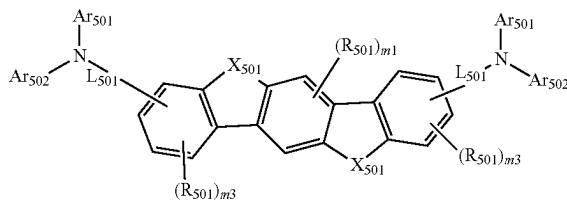
(51-3)



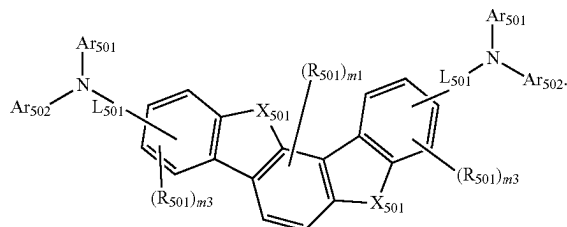
(51-4)



(51-5)

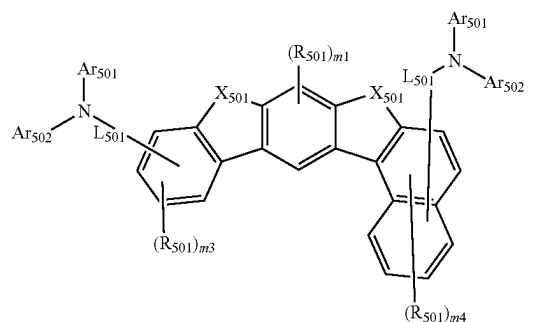


(51-6)



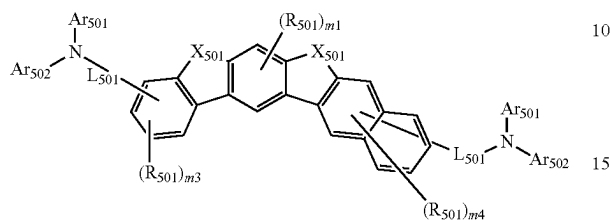
wherein in the formulas (51-1) to (51-6), R_{501} , X_{501} , Ar_{501} , Ar_{502} , L_{501} , $m1$ and $m3$ are as defined in the formula (51).

In one embodiment, the compound represented by the formula (51) is represented by any one of the following formulas (51-11) to (51-13):

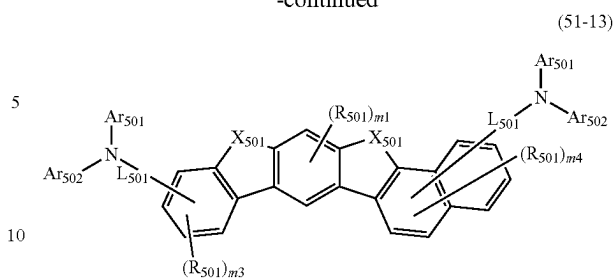


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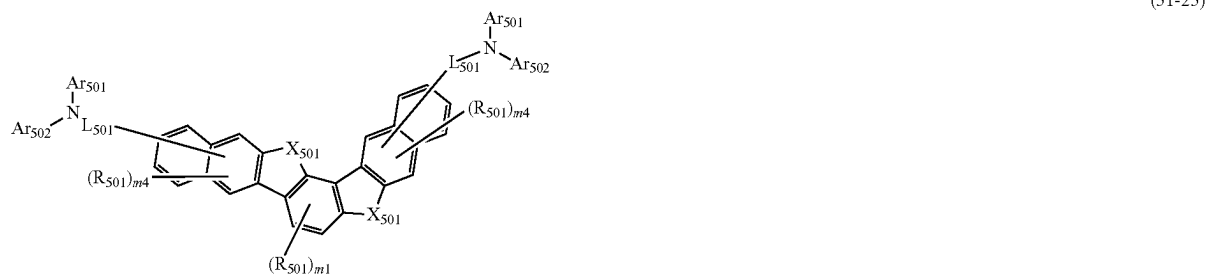
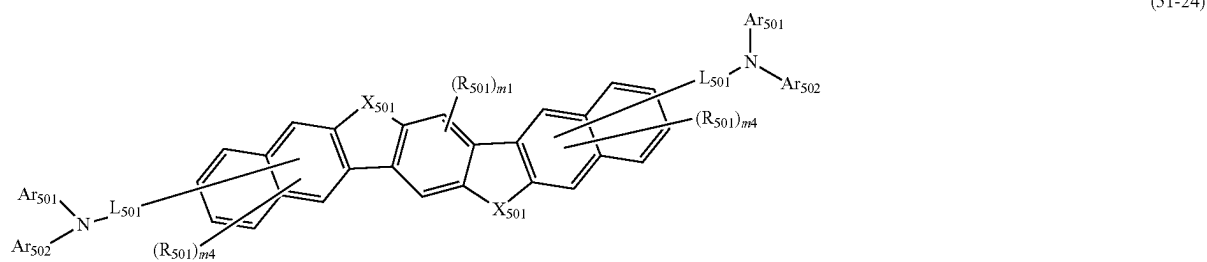
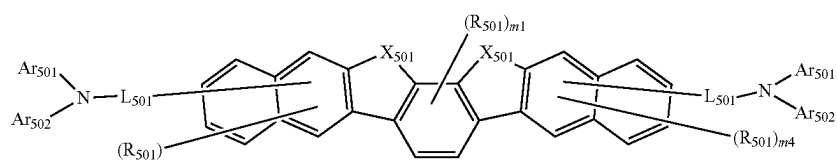
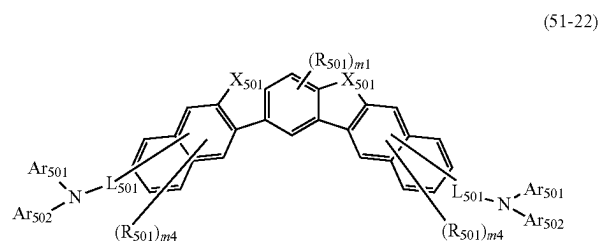
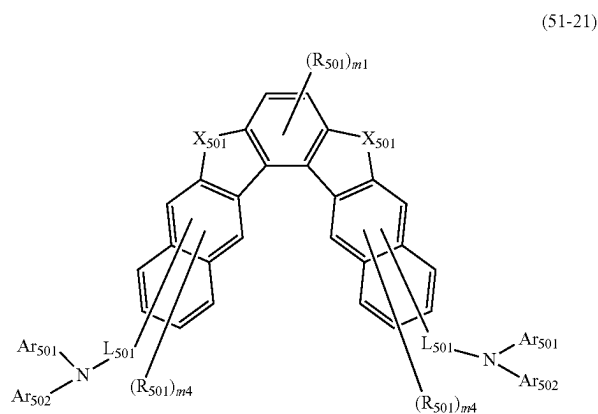
**686**

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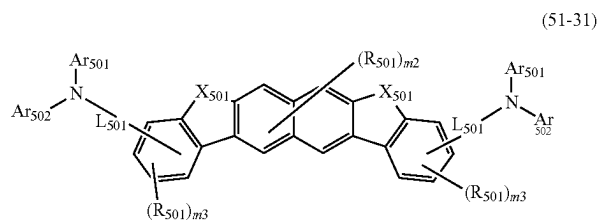
wherein in the formulas (51-11) to (51-13), R_{501} , X_{501} , Ar_{501} , Ar_{502} , L_{501} , $m1$, $m3$ and $m4$ are as defined in the formula (51).

In one embodiment, the compound represented by the formula (51) is represented by any one of the following formulas (51-21) to (51-25):

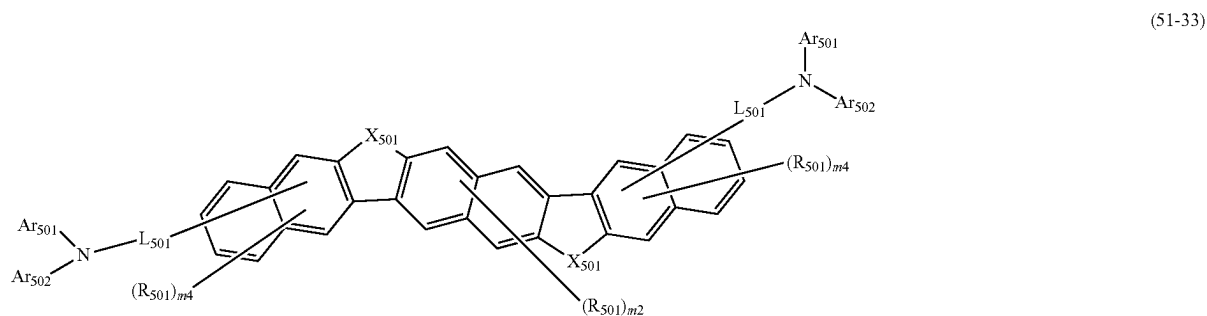
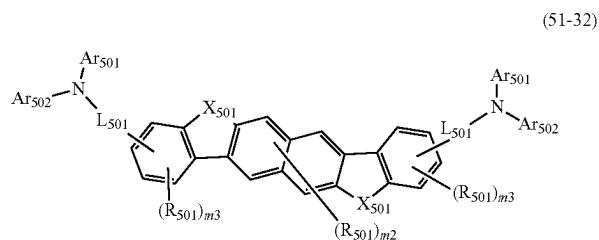


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wherein in the formulas (51-21) to (51-25), R_{501} , X_{501} , Ar_{501} , Ar_{502} , L_{501} , $m1$ and $m4$ are as defined in the formula (51).

**688**

In one embodiment, the compound represented by the formula (51) is represented by anyone of the following formulas (51-31) to (51-33):

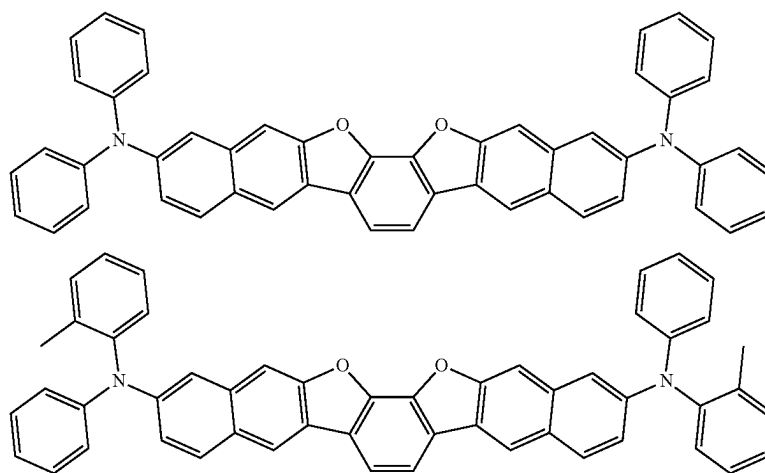


wherein in the formulas (51-31) to (51-33), R_{501} , X_{501} , Ar_{501} , Ar_{502} , L_{501} , $m2$ to $m4$ are as defined in the formula (51).

In one embodiment, Ar_{501} and Ar_{502} are independently a substituted or unsubstituted aryl group having 6 to 50 ring carbon atoms.

In one embodiment, one of Ar_{501} and Ar_{502} is a substituted or unsubstituted aryl group having 6 to 50 ring carbon atoms and the other is a substituted or unsubstituted monovalent heterocyclic ring having 5 to 50 ring atoms.

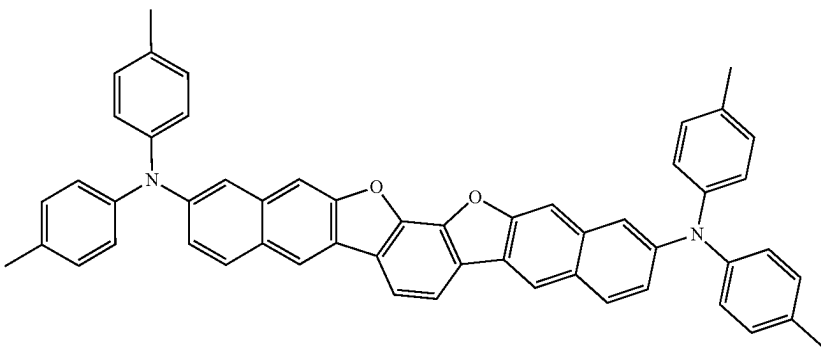
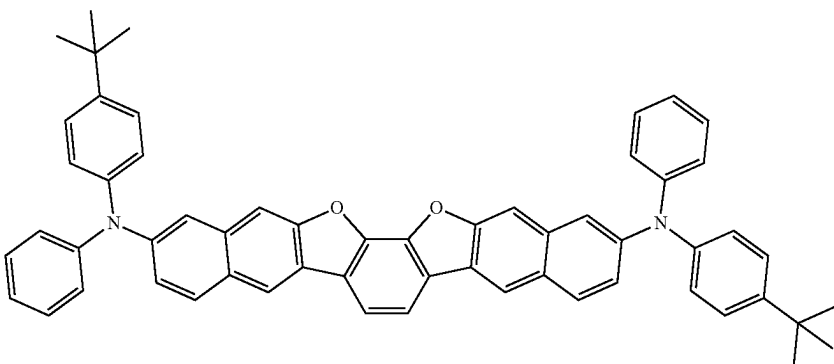
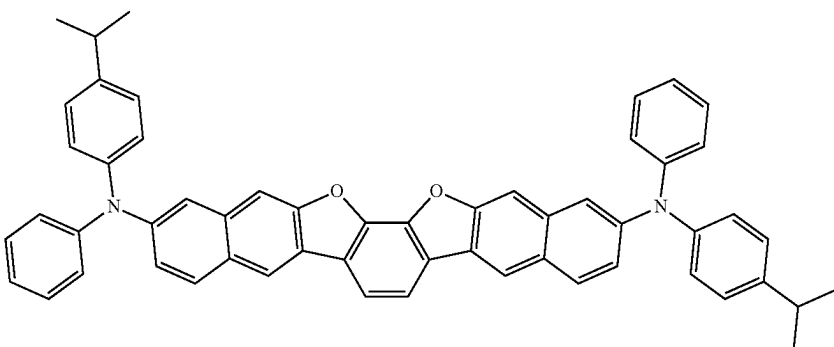
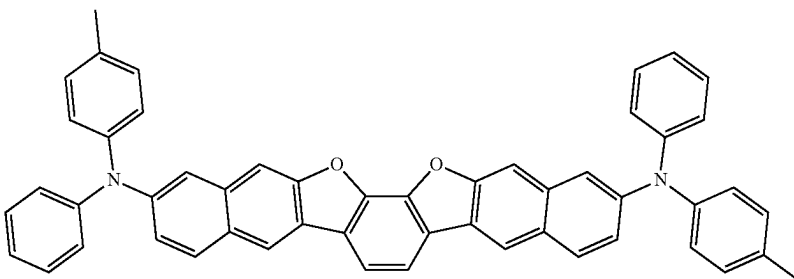
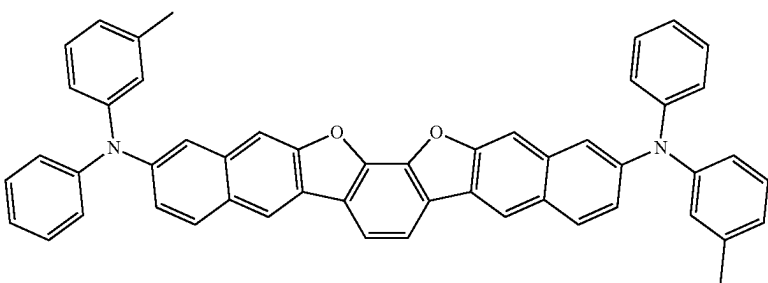
As examples of the compound represented by the formula (51), the following compounds can be given, for example. In the following example compounds, Me represents methyl group.



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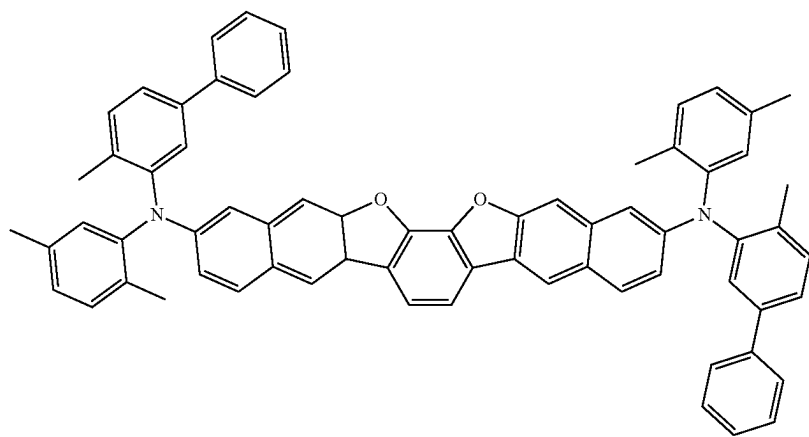
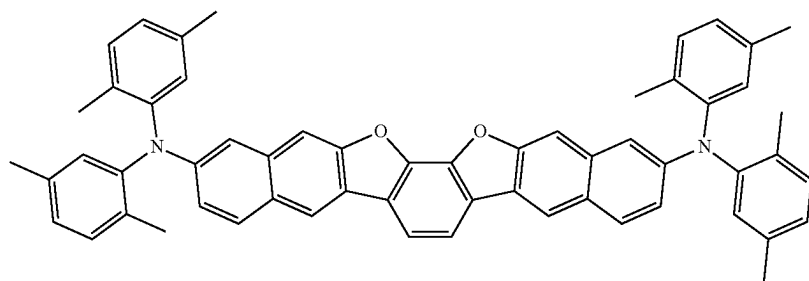
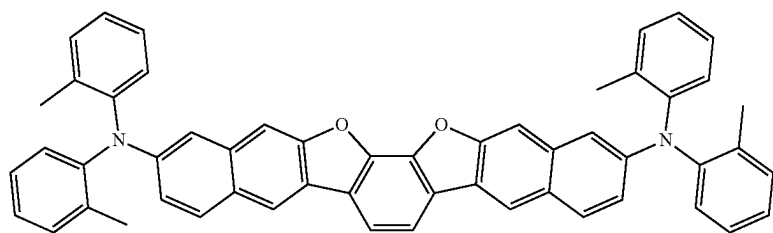
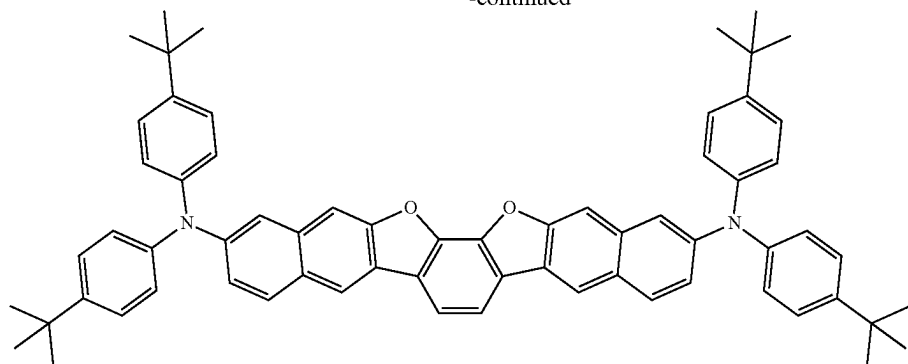
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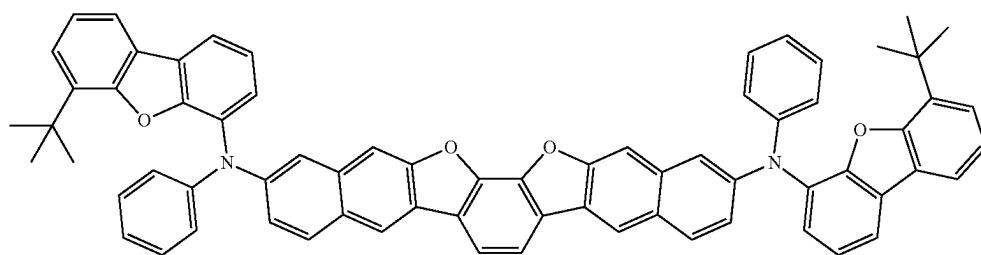
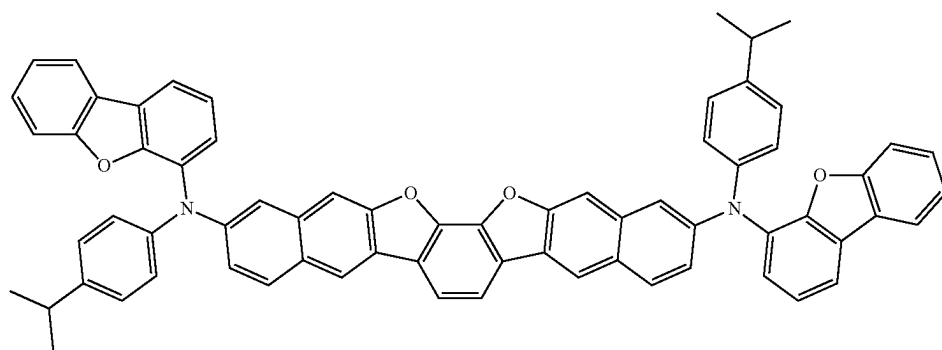
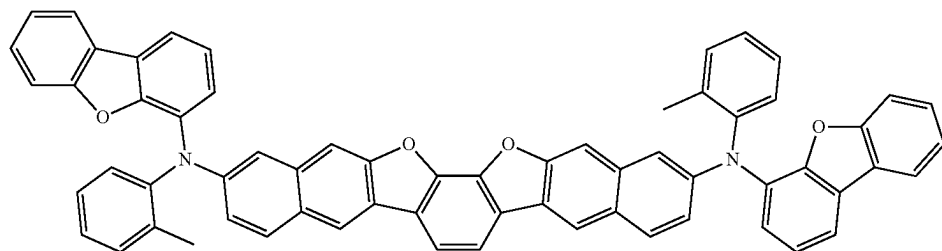
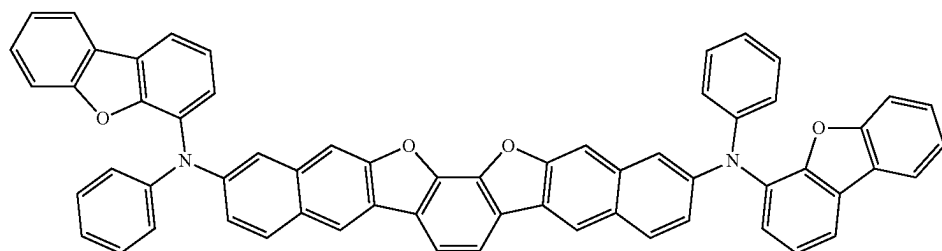
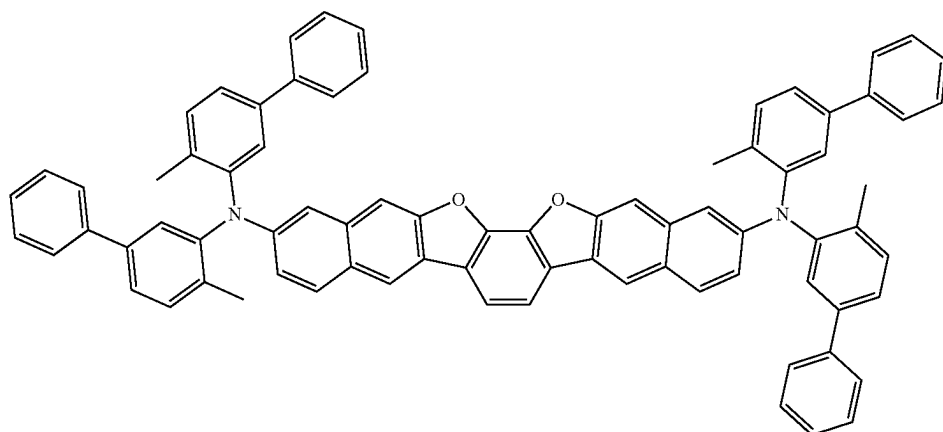
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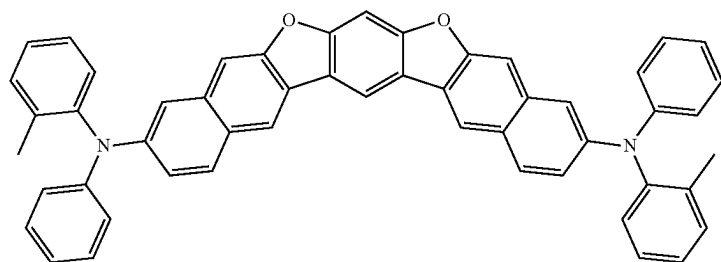
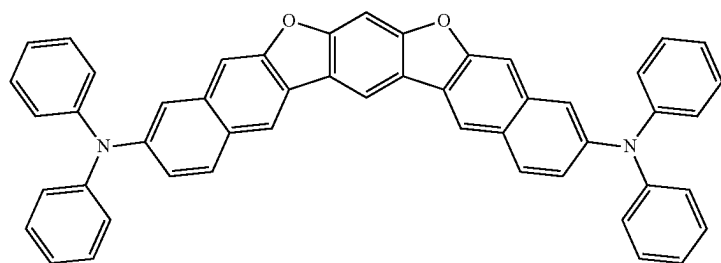
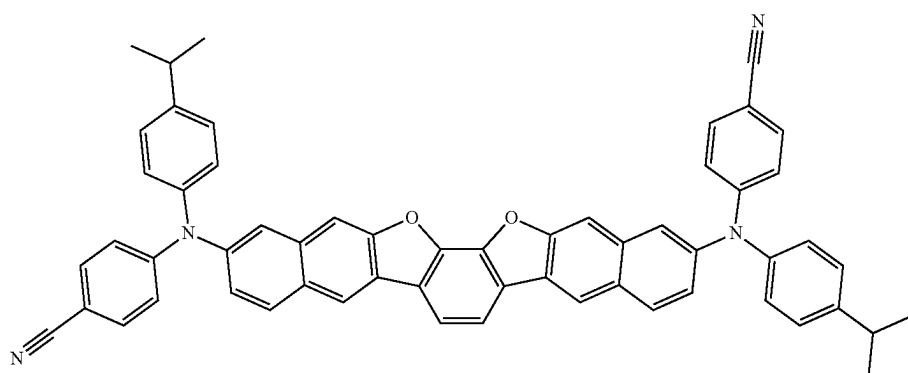
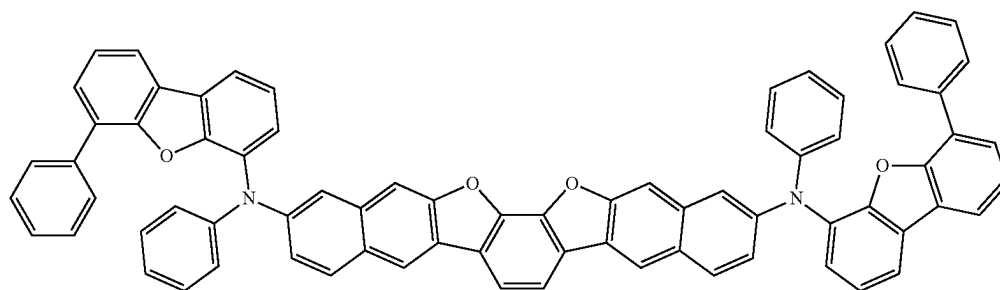
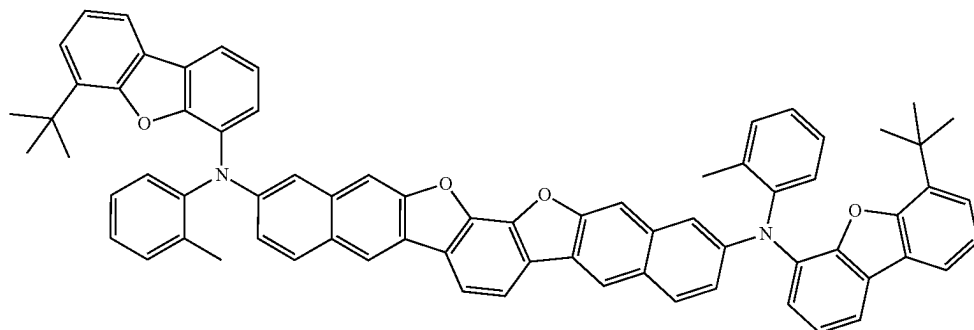
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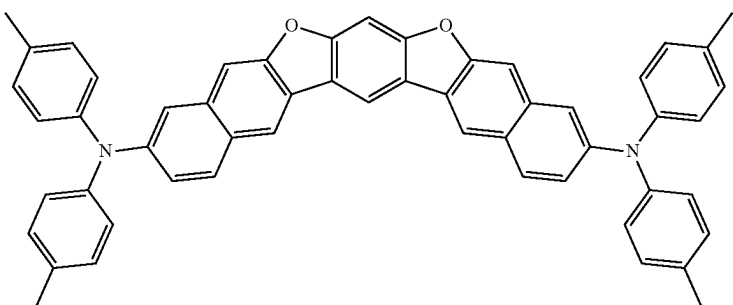
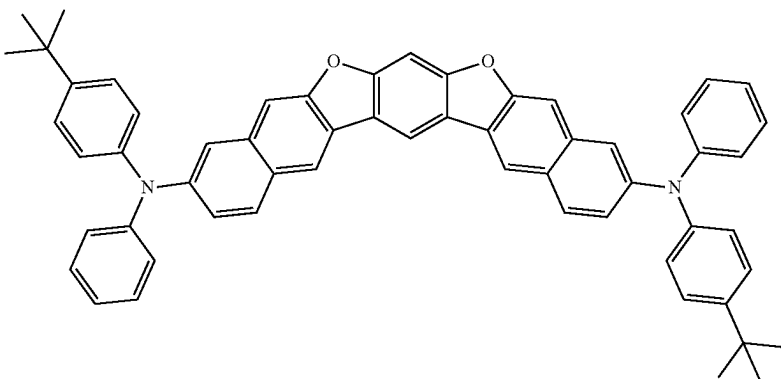
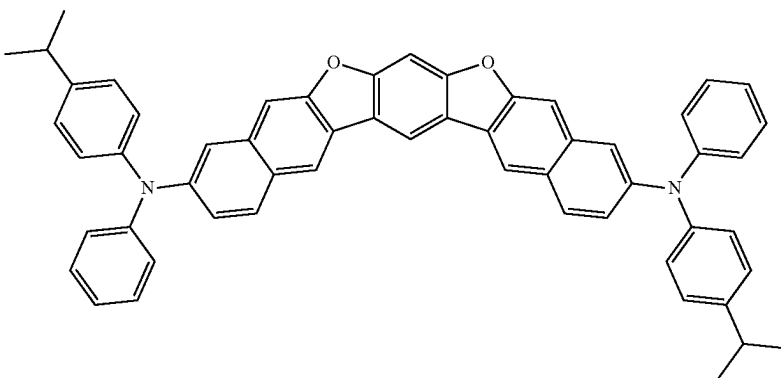
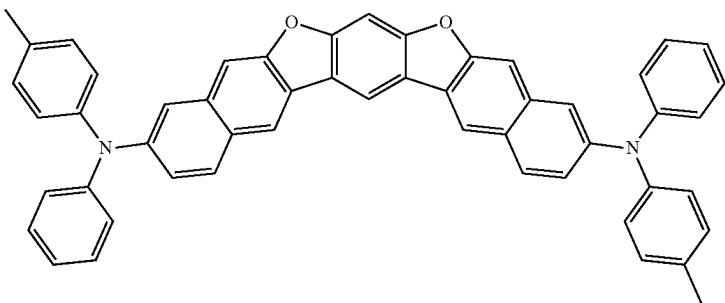
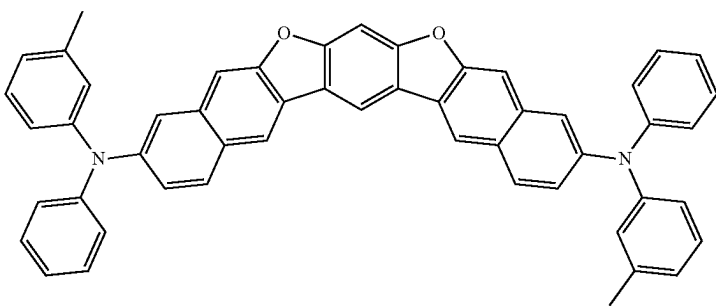
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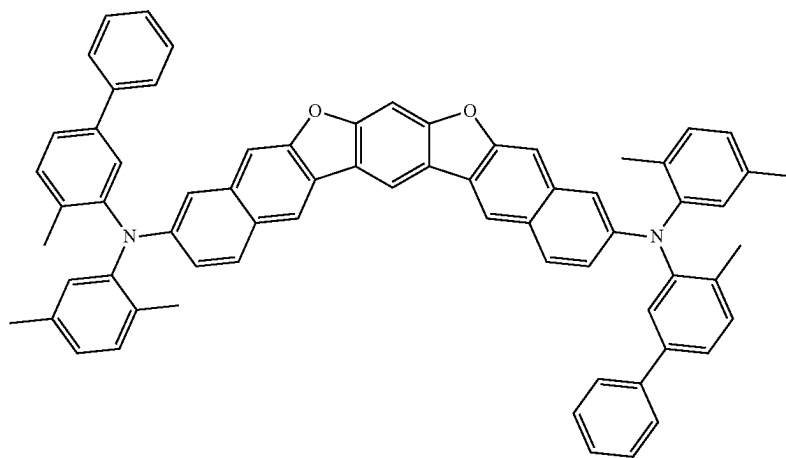
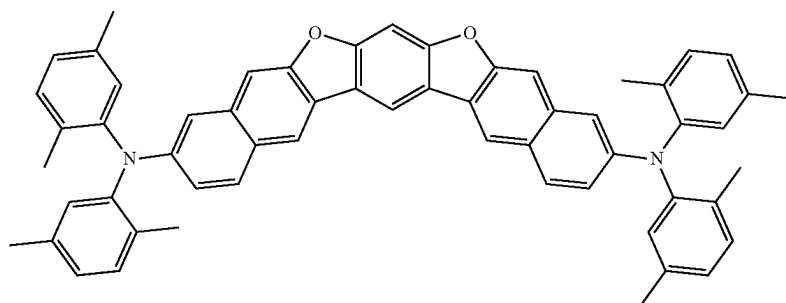
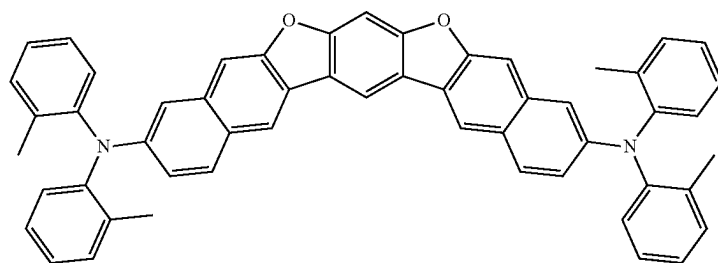
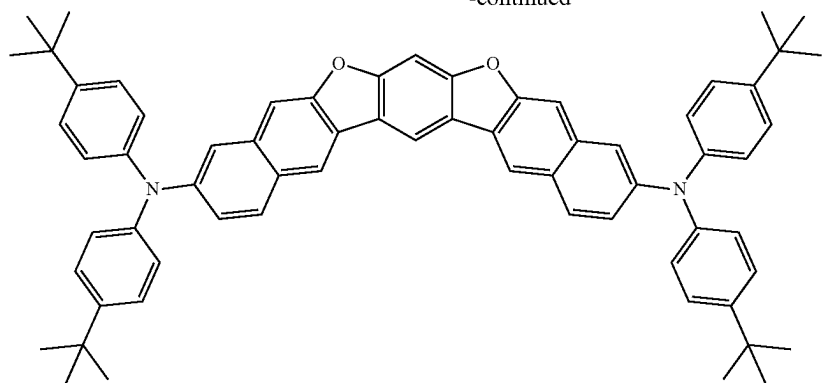
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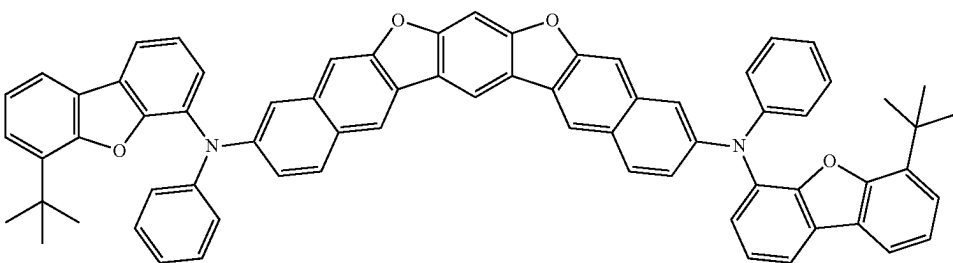
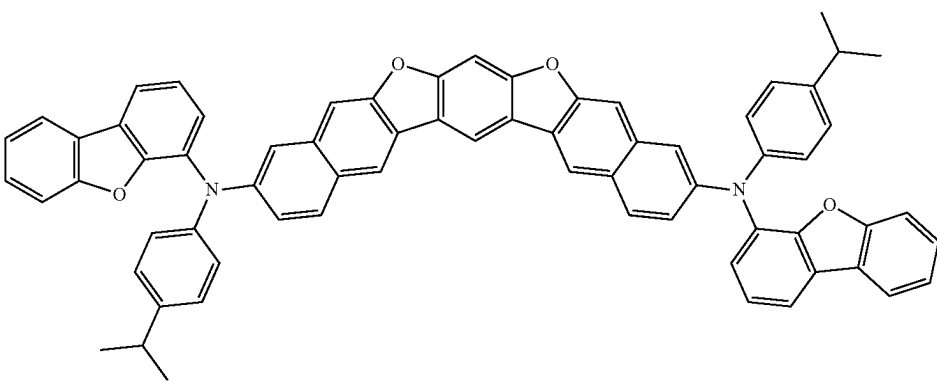
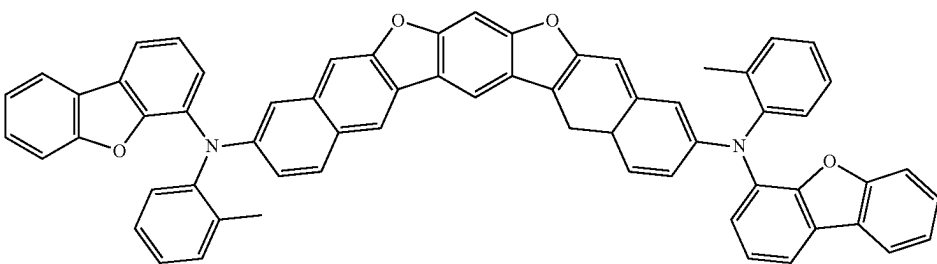
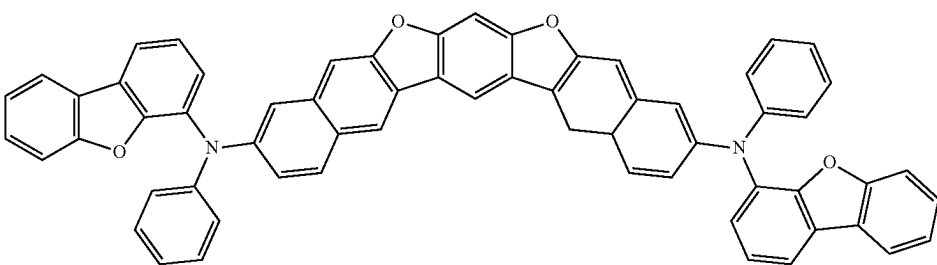
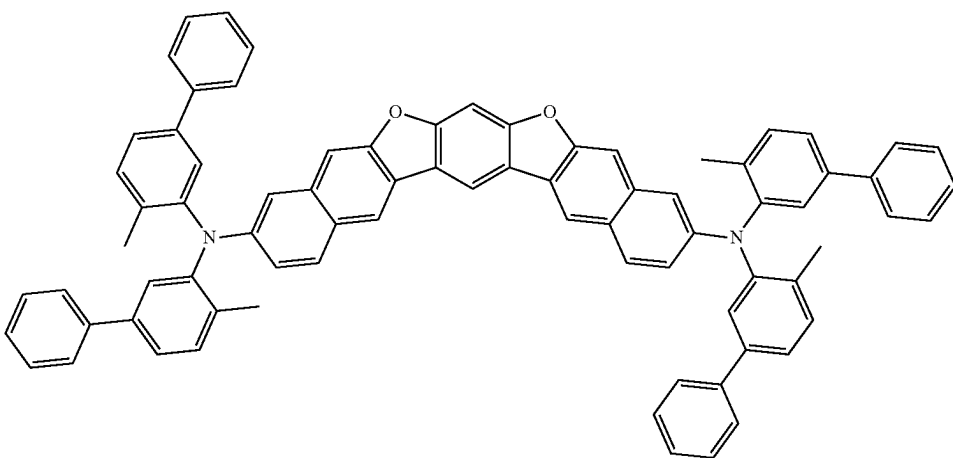
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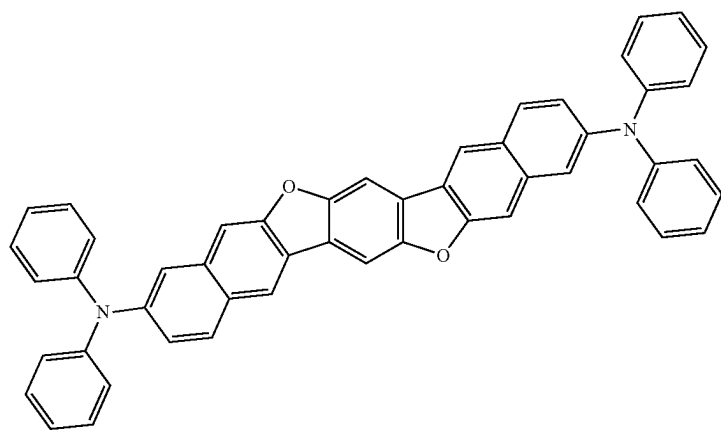
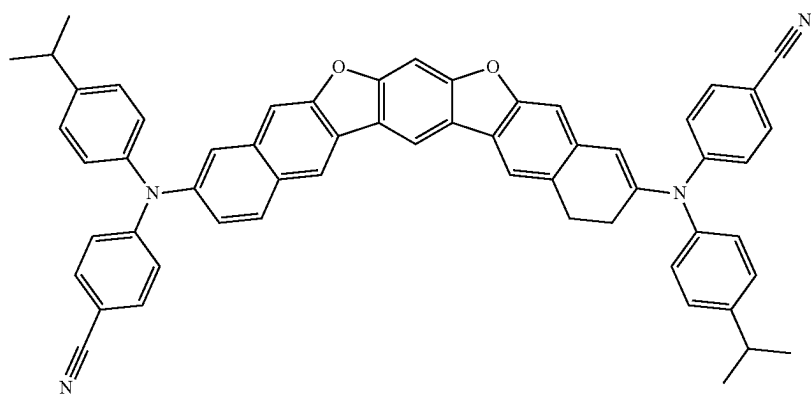
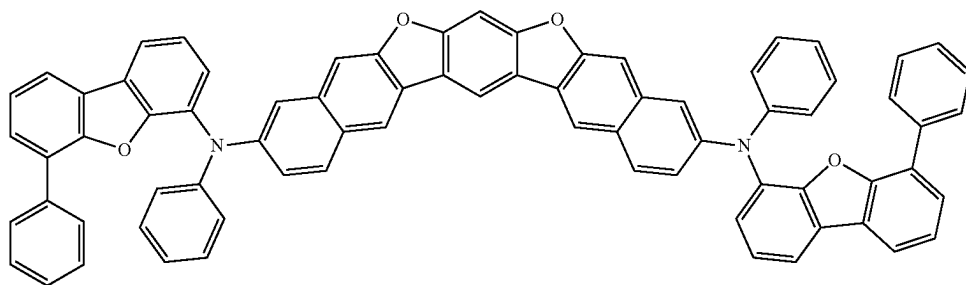
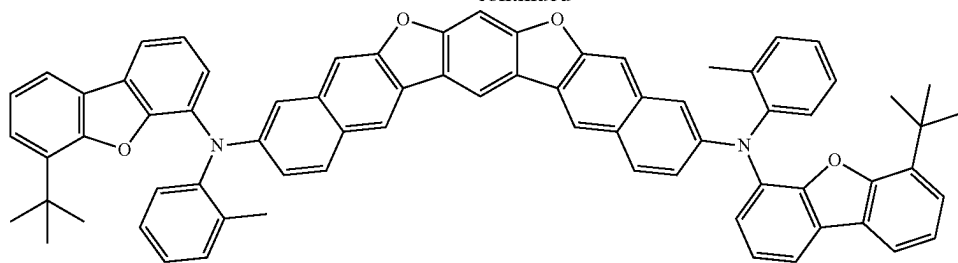
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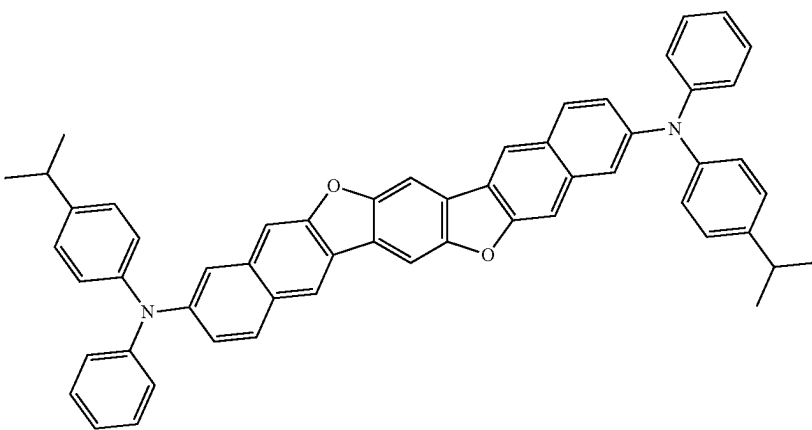
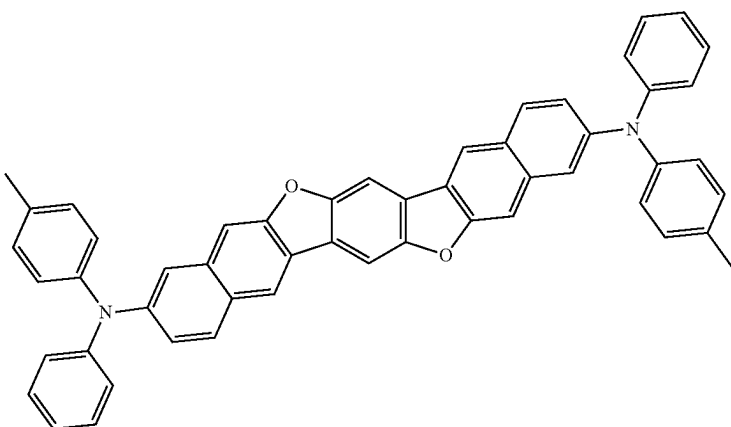
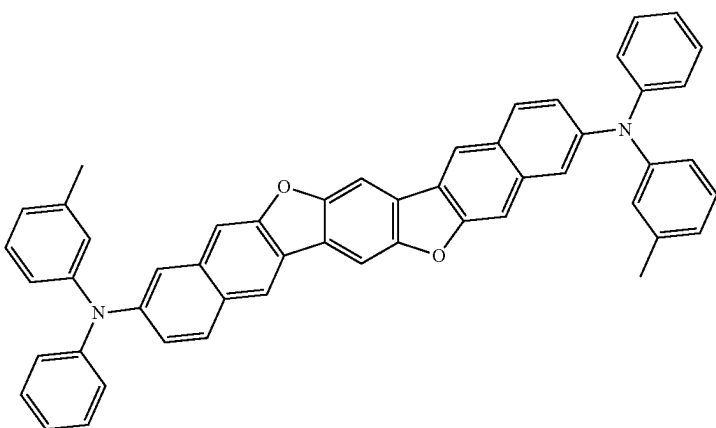
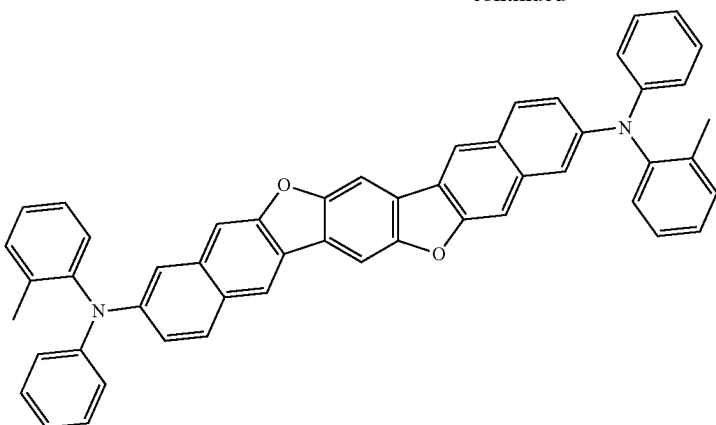
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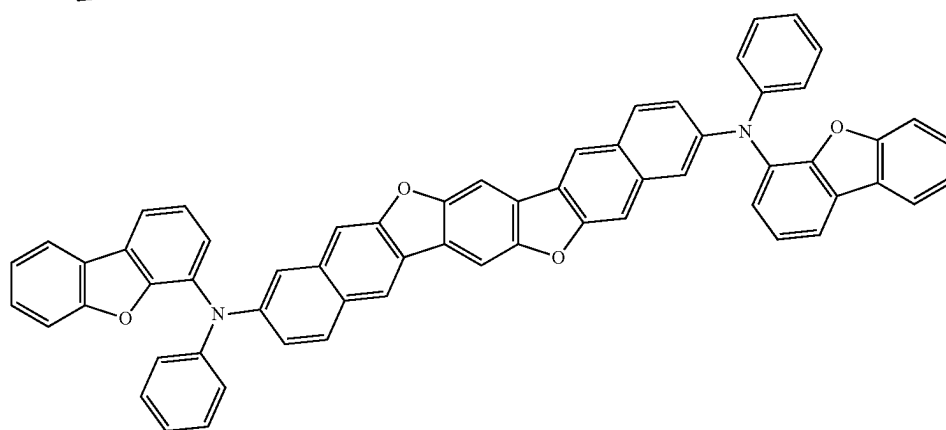
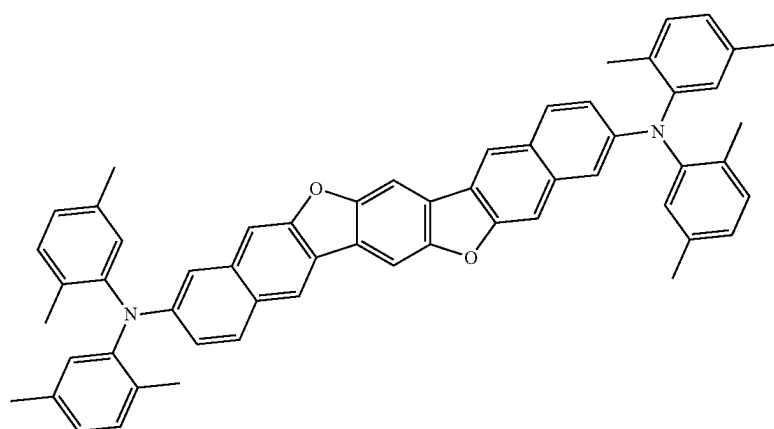
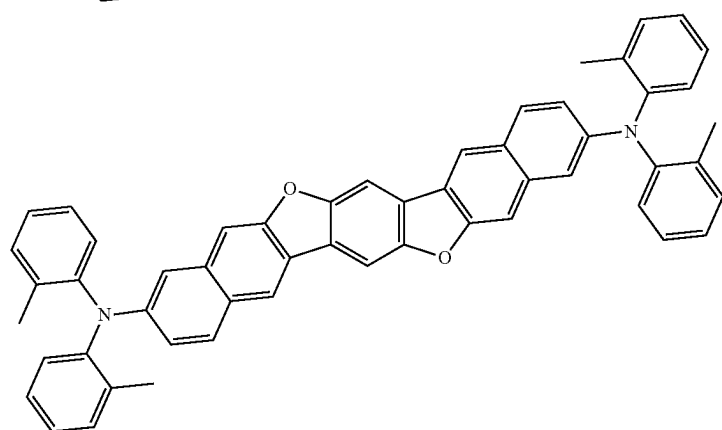
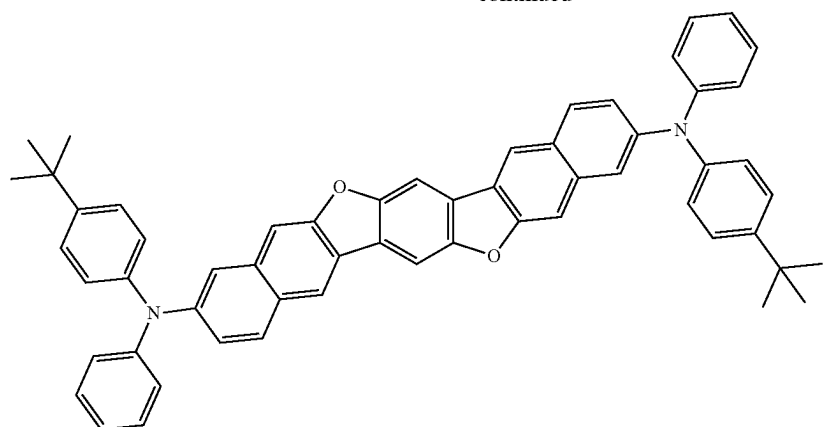
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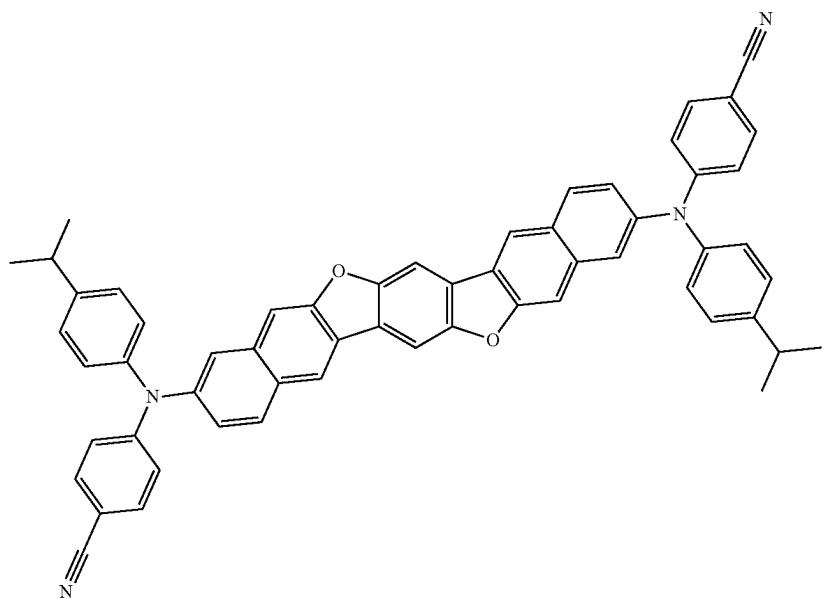
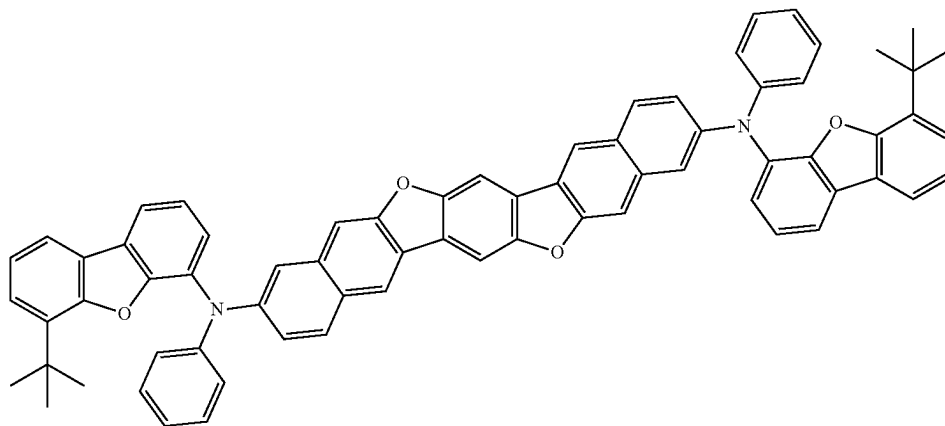
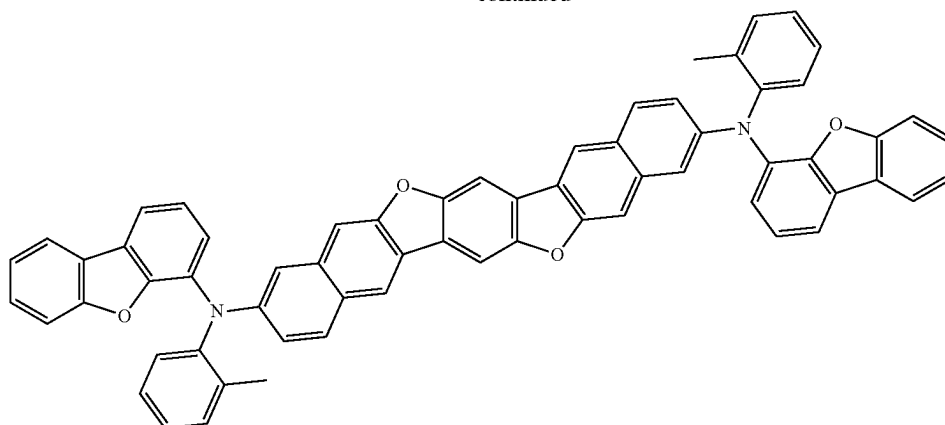
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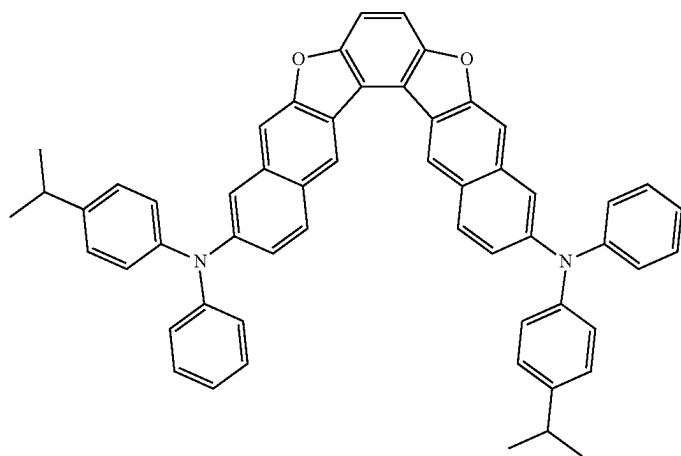
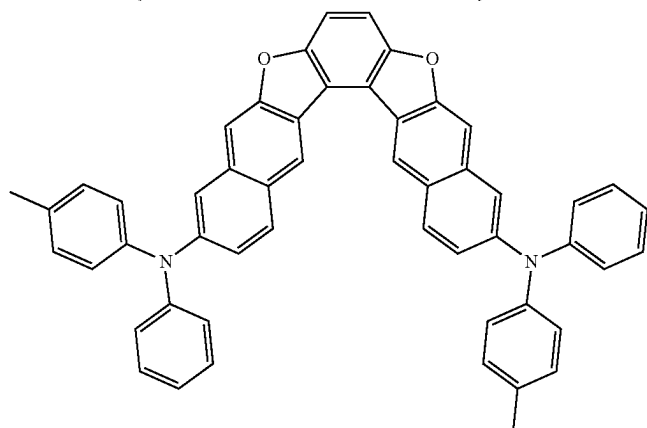
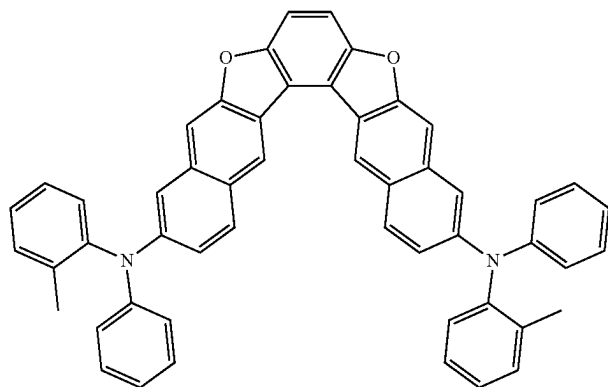
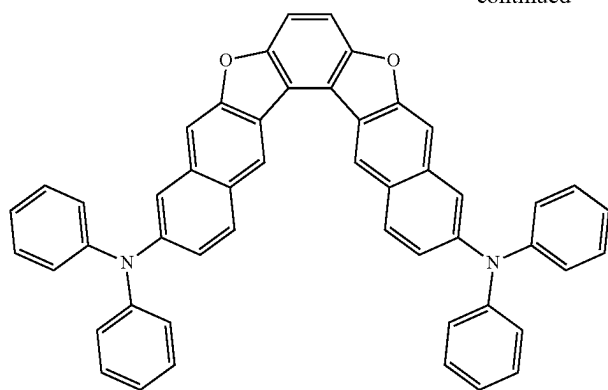
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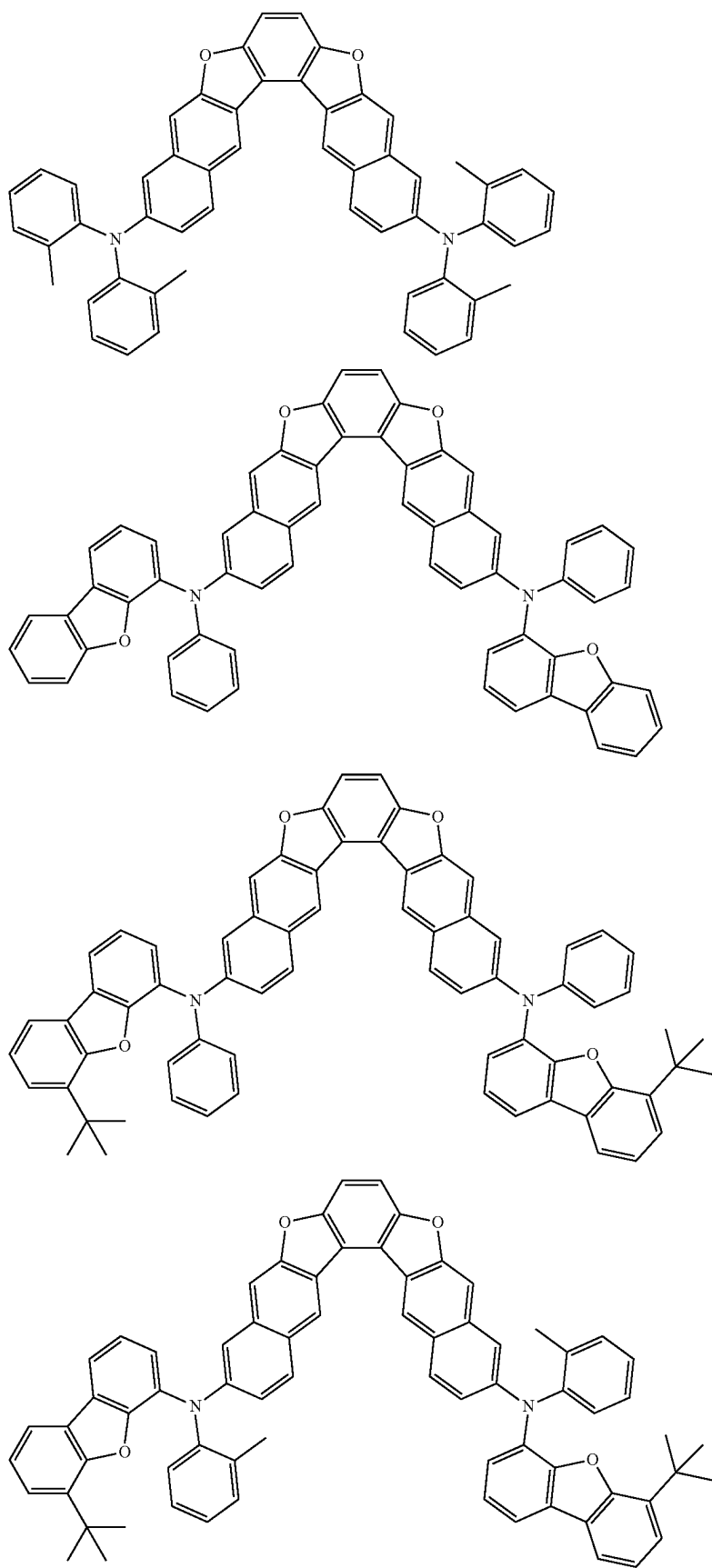
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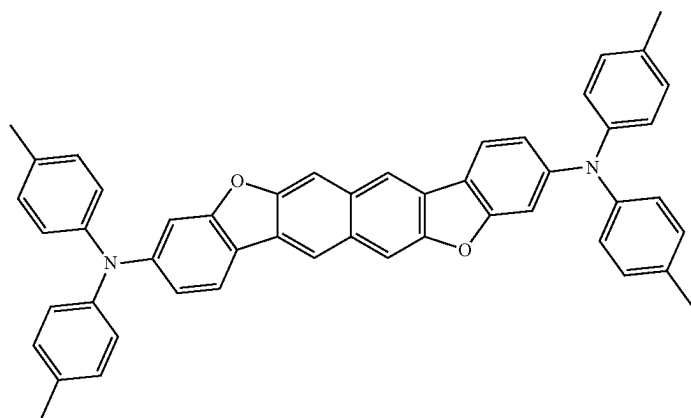
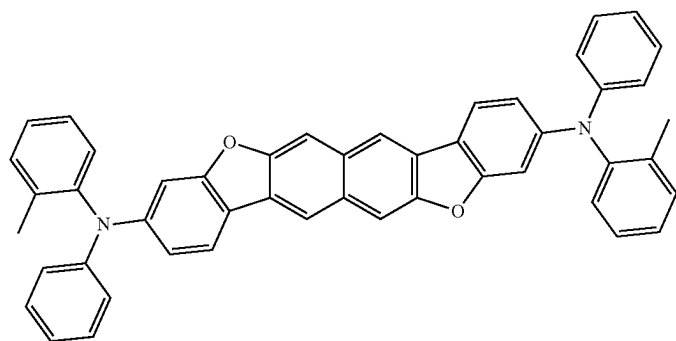
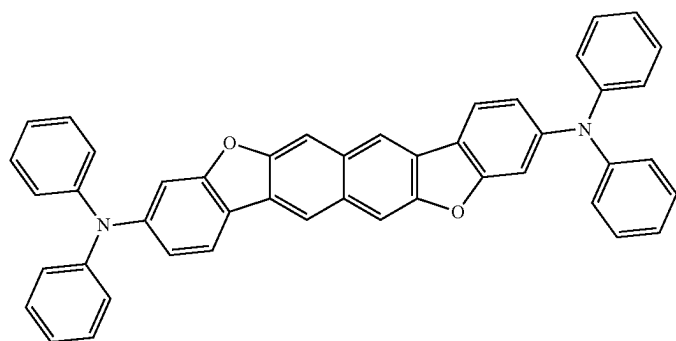
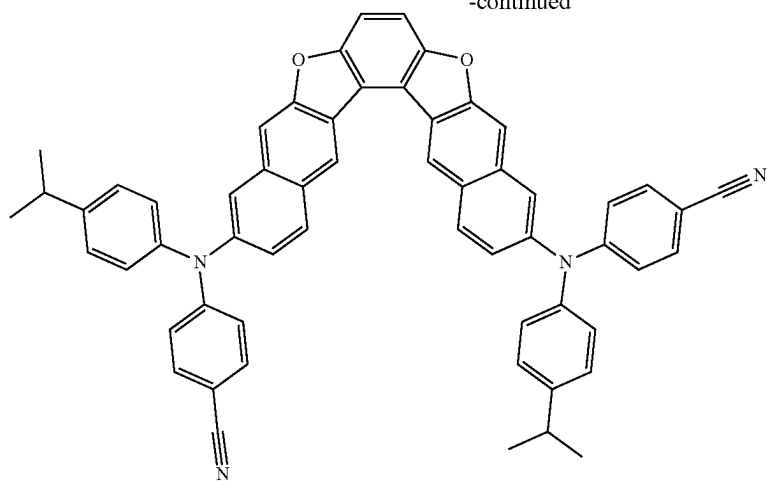
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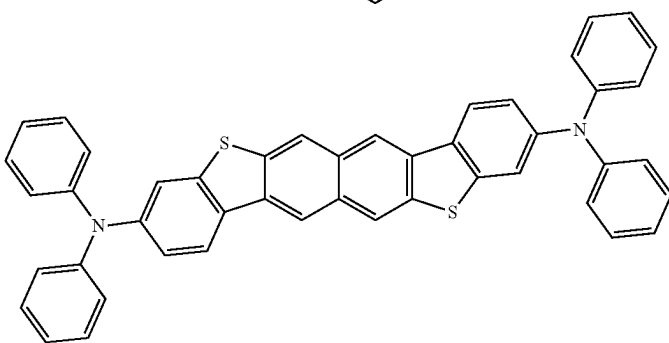
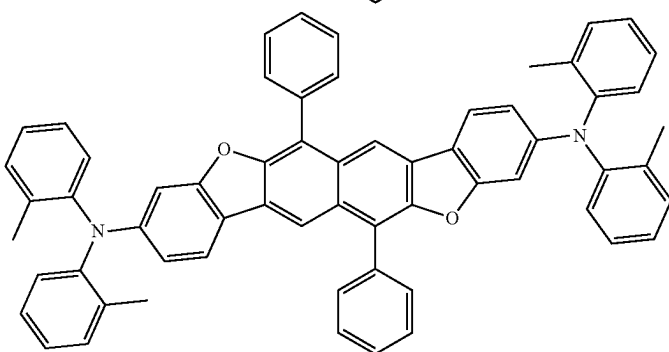
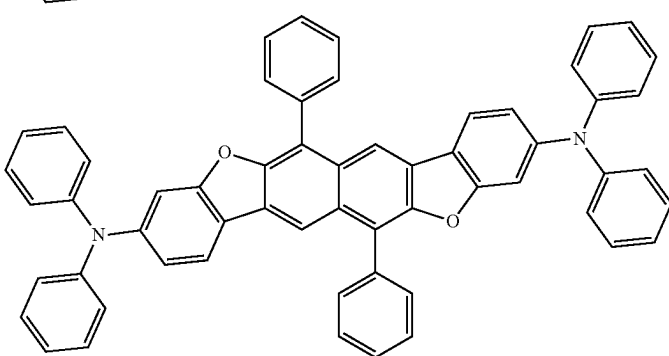
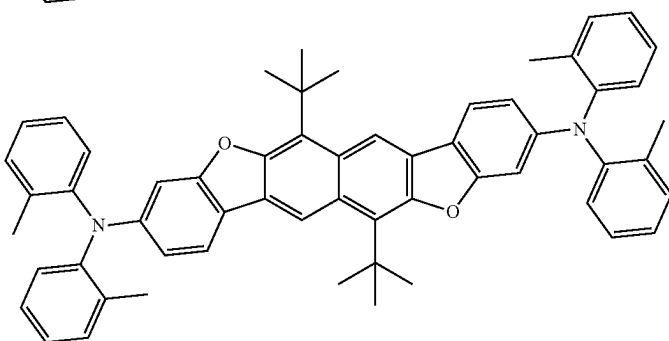
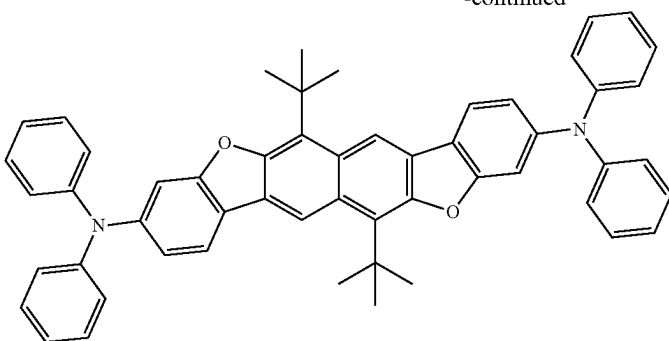
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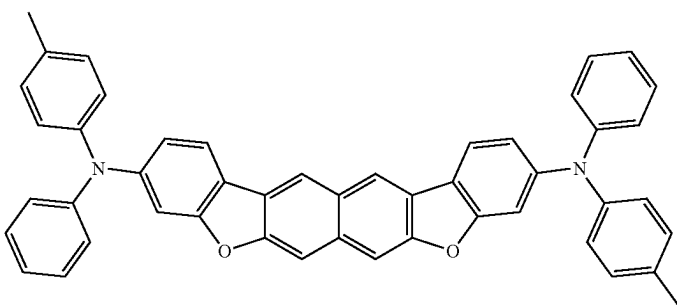
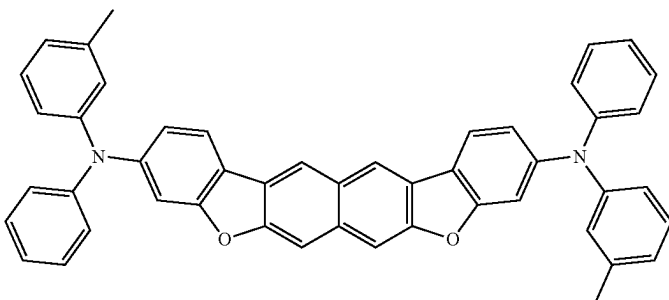
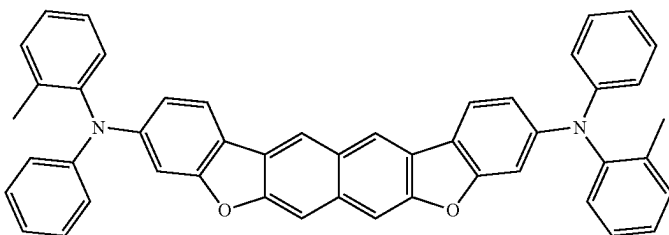
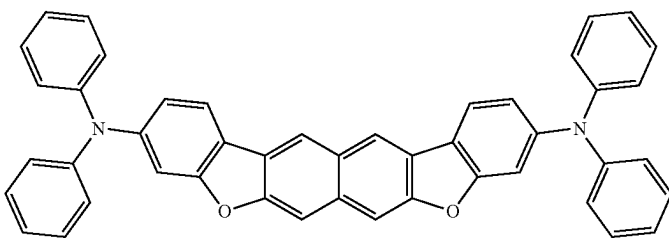
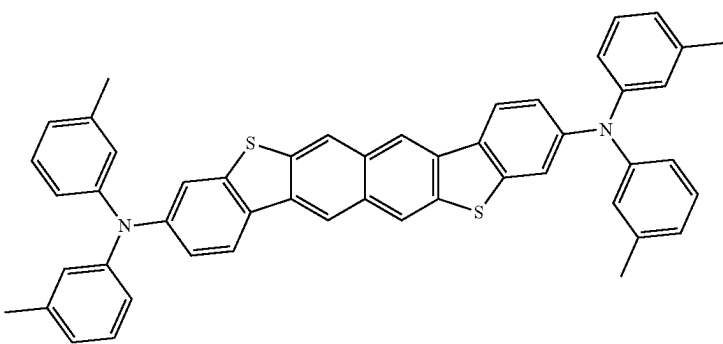
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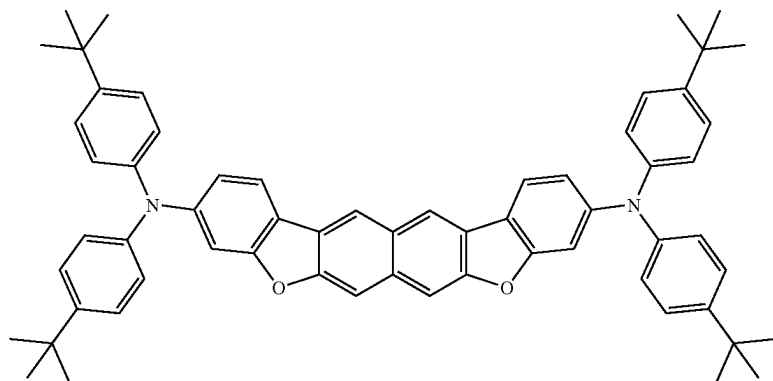
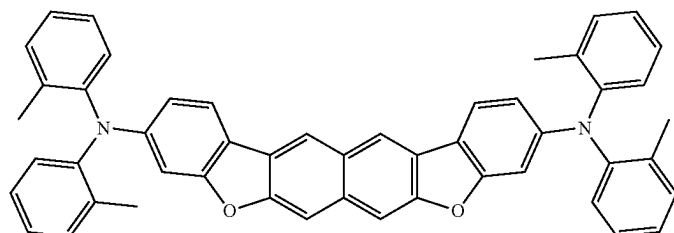
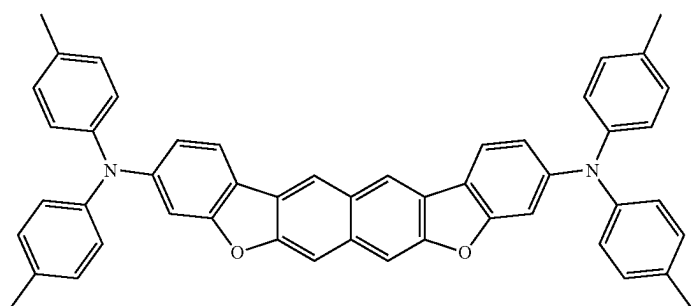
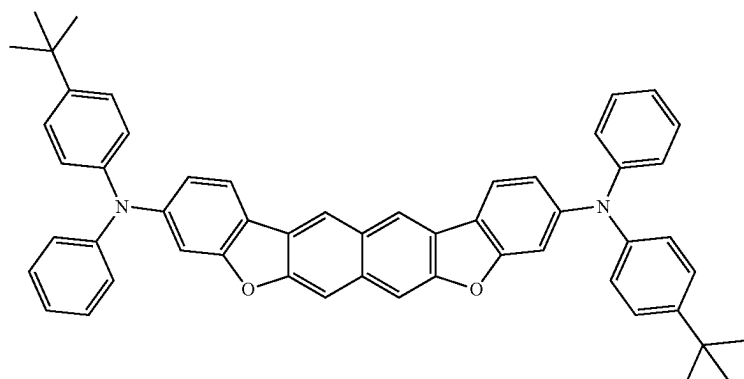
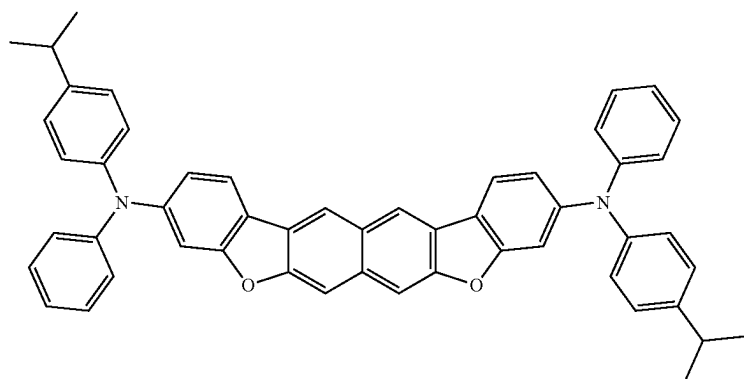
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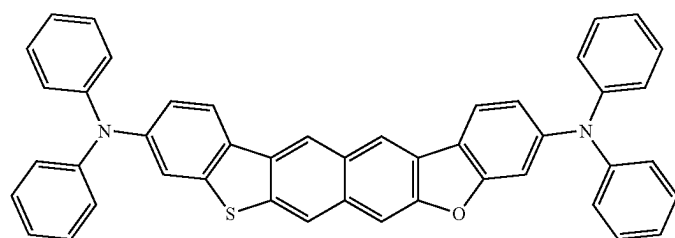
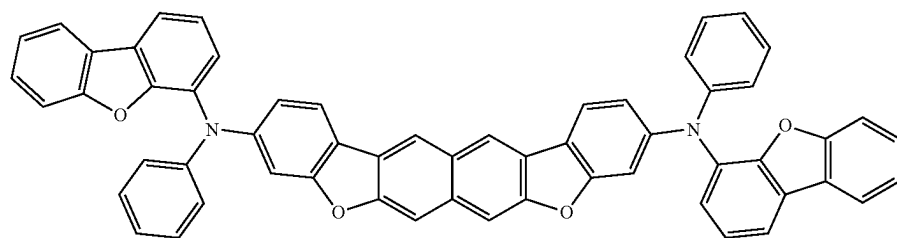
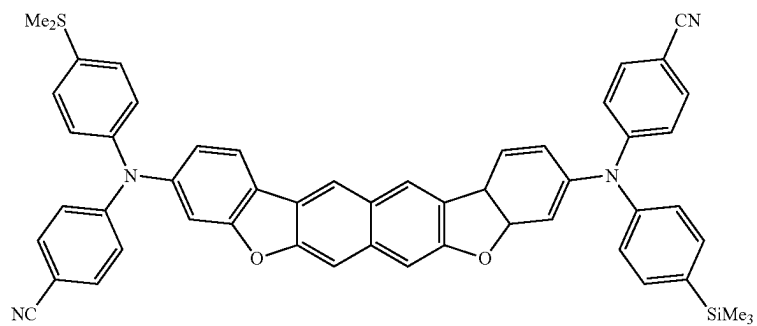
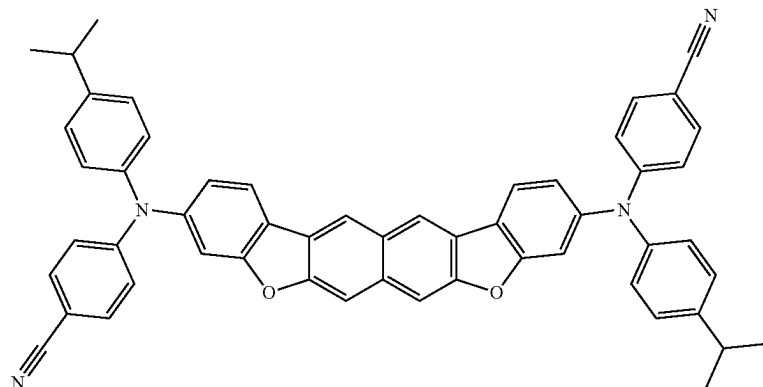
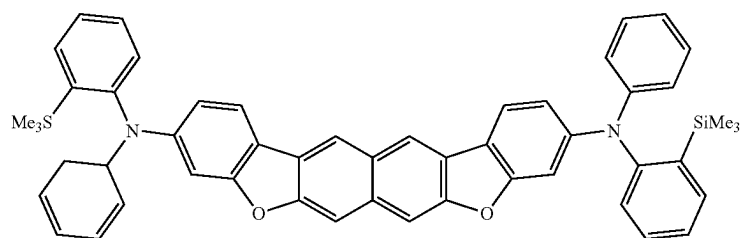
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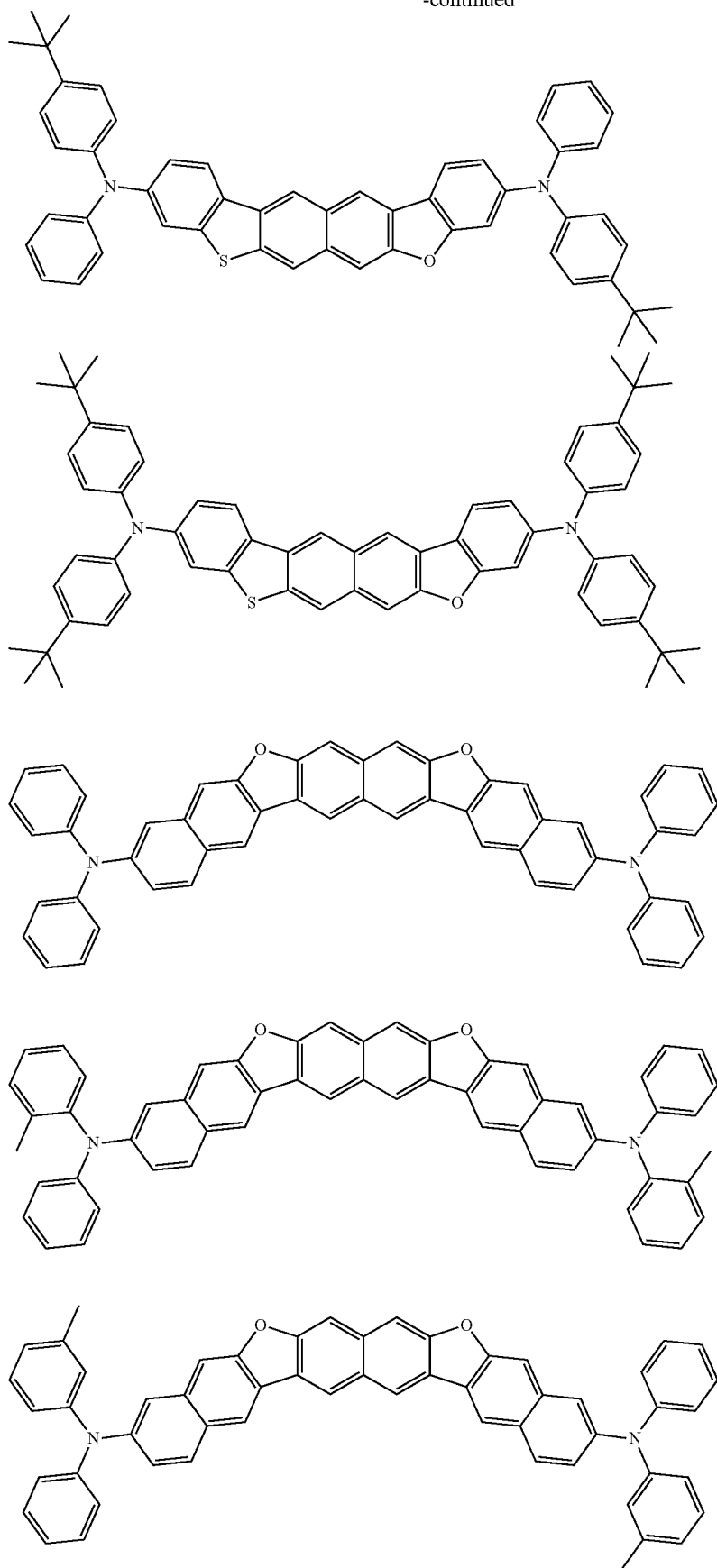
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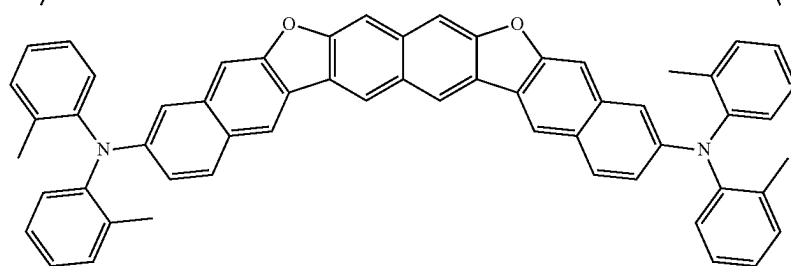
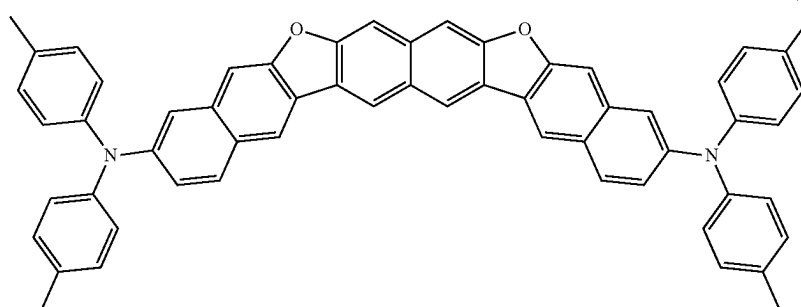
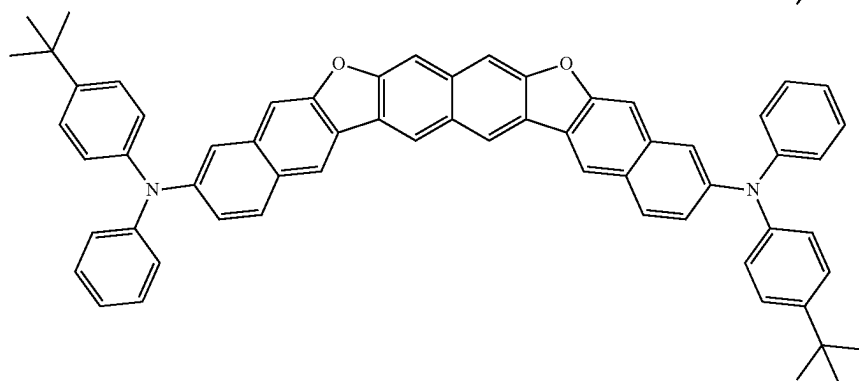
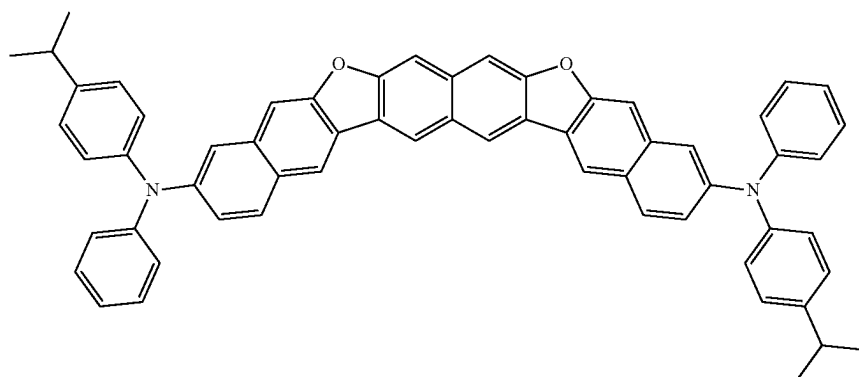
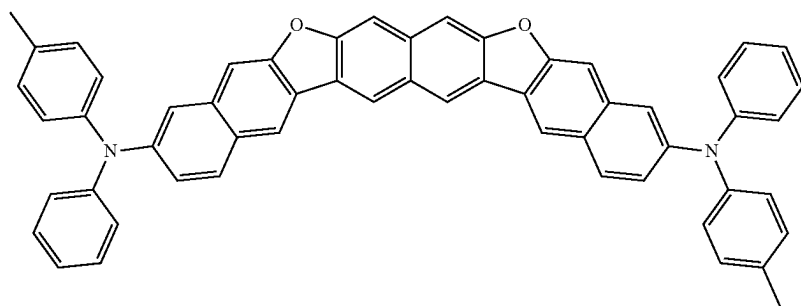
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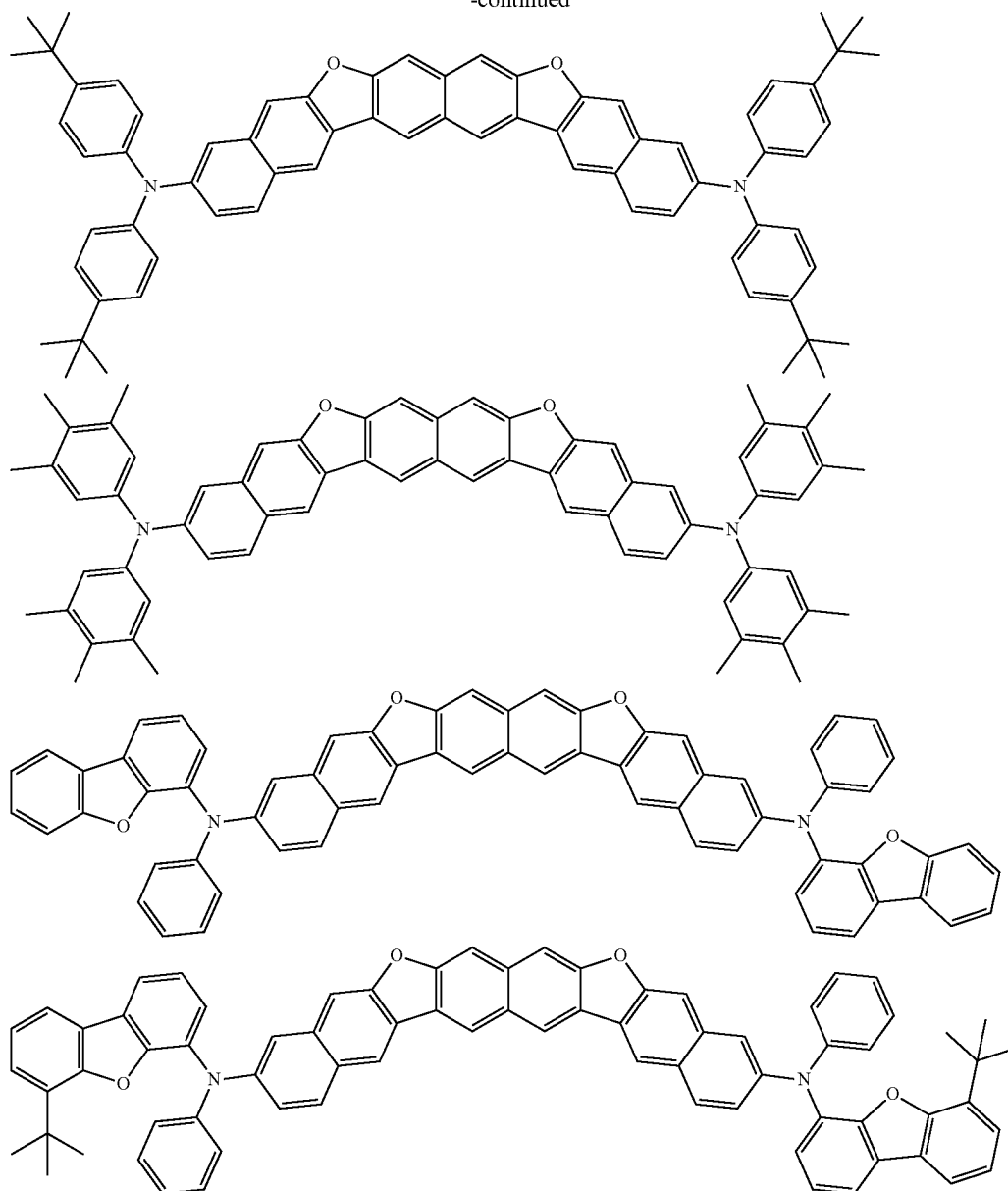
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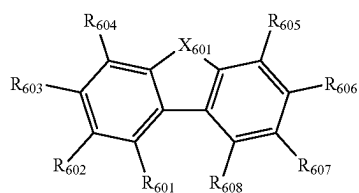
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(Compound Represented by Formula (61))

The compound represented by the formula (61) is explained below.

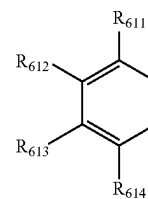
at least one pair of R_{605} and R_{606} , R_{606} and R_{607} , and R_{607} and R_{608} are bonded with each other to form a divalent group represented by formula (62);



(61)

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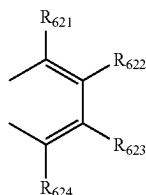
In the formula (61),
at least one pair of R_{601} and R_{602} , R_{602} and R_{603} , and R_{603} and R_{604} are bonded with each other to form a divalent group represented by the formula (62);



(62)

731

-continued



(63)

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at least one of R_{601} to R_{604} that does not form the divalent group represented by the formula (62), and R_{611} to R_{614} is a monovalent group represented by the following formula (64);

at least one of R_{605} to R_{608} that do not form the divalent group represented by the formula (63), and R_{621} to R_{624} is a monovalent group represented by the following formula (64);

X_{601} is an oxygen atom, a sulfur atom, or NR_{609} ;

R_{601} to R_{608} that do not form the divalent group represented by the formulas (62) and (63) and that is not the monovalent group represented by the formula (64), R_{611} to R_{614} and R_{621} to R_{624} that are not the monovalent group represented by the formula (64), and R_{609} are independently

a hydrogen atom,

a substituted or unsubstituted alkyl group having 1 to 50 carbon atoms,

a substituted or unsubstituted alkenyl group having 2 to 50 carbon atoms,

a substituted or unsubstituted alkynyl group having 2 to 50 carbon atoms,

a substituted or unsubstituted cycloalkyl group having 3 to 50 ring carbon atoms,

$-\text{Si}(R_{901})(R_{902})(R_{903})$,

$-\text{O}- (R_{904})$,

$-\text{S}- (R_{905})$,

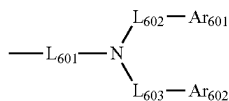
$-\text{N}(R_{906})(R_{907})$,

a halogen atom, a cyano group, a nitro group,

a substituted or unsubstituted aryl group having 6 to 50 ring carbon atoms, or

a substituted or unsubstituted monovalent heterocyclic group having 5 to 50 ring atoms;

R_{901} to R_{907} are as defined in the formula (1);



(64)

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wherein, in the formula (64), Ar_{601} and Ar_{602} are independently a substituted or unsubstituted aryl group having 6 to 50 ring carbon atoms, or

a substituted or unsubstituted monovalent heterocyclic group having 5 to 50 ring atoms;

L_{601} to L_{603} are independently

a single bond,

a substituted or unsubstituted arylene group having 6 to 30 ring carbon atoms,

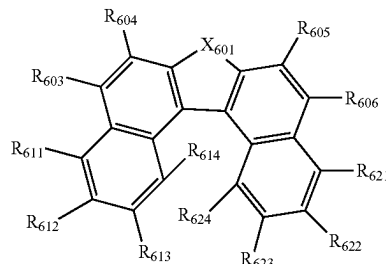
a substituted or unsubstituted divalent heterocyclic group having 5 to 30 ring atoms, or

a divalent linking group formed by bonding 2 to 4 above mentioned groups;

732

In the formula (61), positions at which the divalent group represented by the formula (62) and the divalent group represented by the formula (63) are formed are not limited, and said groups can be formed at possible positions in R_{601} to R_{608} .

In one embodiment, the compound represented by the formula (61) is represented by any one of the following formulas (61-1) to (61-6):



(61-1)

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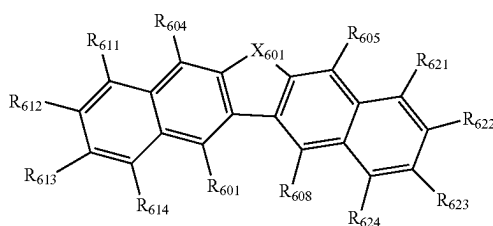
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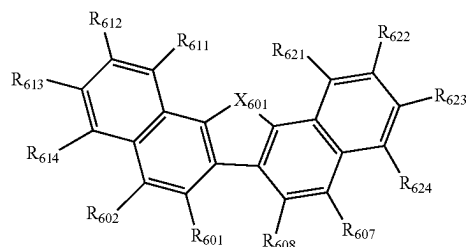
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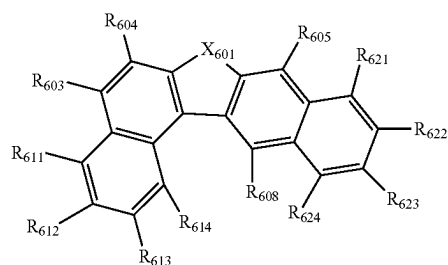
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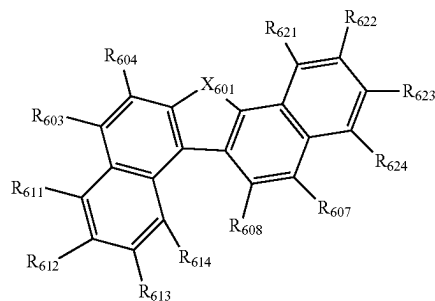
(61-2)



(61-3)



(61-4)

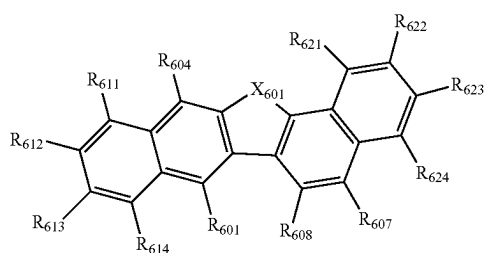


(61-5)

733

-continued

(61-6)



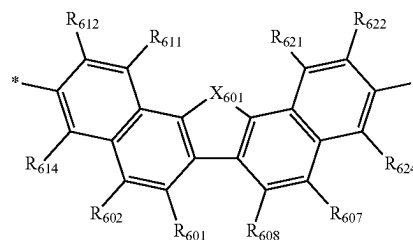
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734

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(61-9)



wherein in the formulas (61-1) to (61-6), X_{601} is as

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defined in the formula (61);

at least two of R_{601} to R_{624} are monovalent groups represented by the formula (64);

R_{601} to R_{624} that are not monovalent groups represented by the formula (64) are independently

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a hydrogen atom,

a substituted or unsubstituted alkyl group having 1 to 50 carbon atoms,

a substituted or unsubstituted alkenyl group having 2 to 50 carbon atoms,

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a substituted or unsubstituted alkynyl group having 2 to 50 carbon atoms,

a substituted or unsubstituted cycloalkyl group having 3 to 50 ring carbon atoms,

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—Si(R_{901})(R_{902})(R_{903}),

—O—(R_{904}),

—S—(R_{905}),

—N(R_{906})(R_{907}),

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a halogen atom, a cyano group, a nitro group,

a substituted or unsubstituted aryl group having 6 to 50 ring carbon atoms, or

a substituted or unsubstituted monovalent heterocyclic group having 5 to 50 ring atoms; and

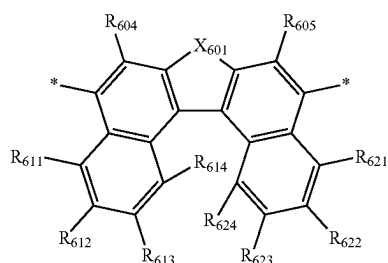
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R_{901} to R_{907} are as defined in the formula (1).

In one embodiment, the compound represented by the formula (61) is represented by any one of the following formulas (61-7) to (61-18):

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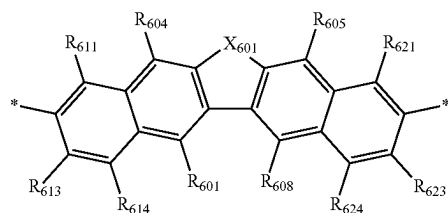
(61-7)



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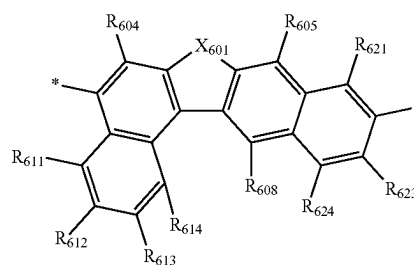
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(61-8)

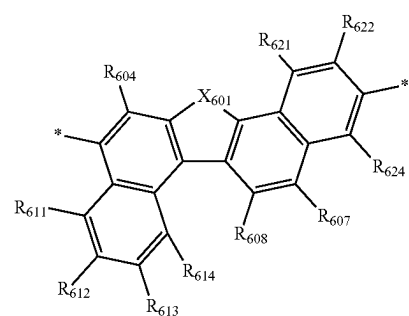


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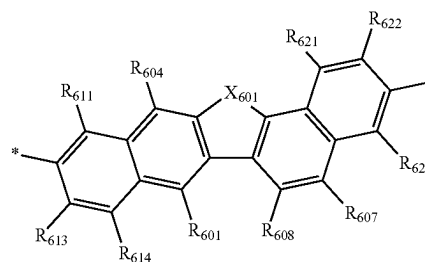
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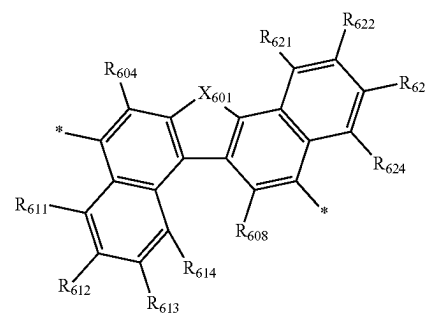
(61-11)



(61-12)

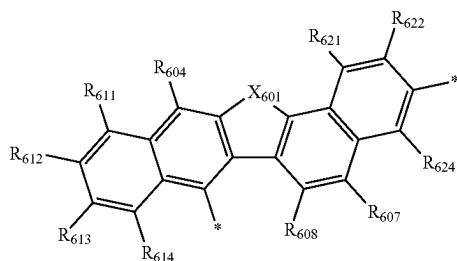


(61-13)

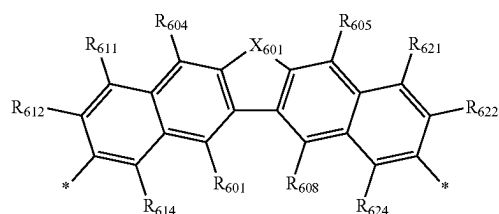


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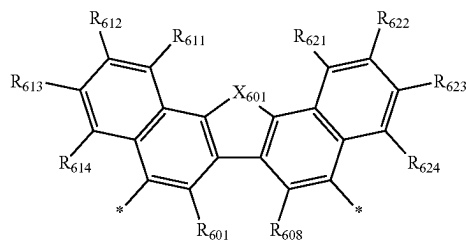
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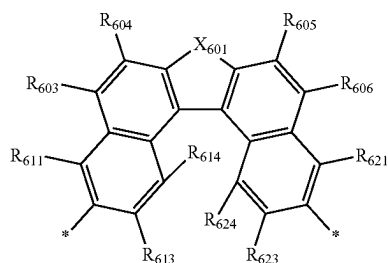
(61-14)



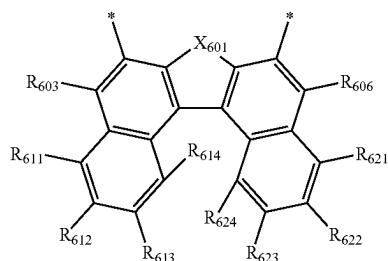
(61-15)



(61-16)



(61-17)



(61-18)

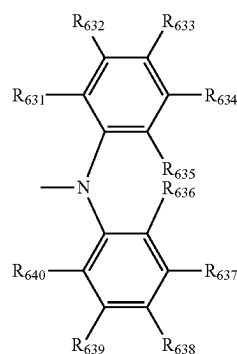
wherein in the formulas (61-7) to (61-18), X_{601} is as defined in the formula (61); * is a single bond bonding to the monovalent group represented by the formula (64); and R_{601} to R_{624} are the same as R_{601} to R_{624} that are not monovalent groups represented by the formula (64).

R_{601} to R_{608} which do not form the divalent group represented by the formula (62) and (63) and are not monovalent groups represented by the formula (64), and R_{611} to R_{614} and R_{621} to R_{624} which are not monovalent groups represented by the formula (64) are preferably independently

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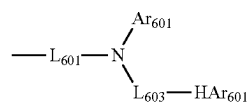
a hydrogen atom,
a substituted or unsubstituted alkyl group having 1 to 50 carbon atoms,
a substituted or unsubstituted alkenyl group having 2 to 50 carbon atoms,
a substituted or unsubstituted alkynyl group having 2 to 50 carbon atoms,
a substituted or unsubstituted cycloalkyl group having 3 to 50 ring carbon atoms,
a substituted or unsubstituted aryl group having 6 to 50 ring carbon atoms, or
a substituted or unsubstituted monovalent heterocyclic group having 5 to 50 ring atoms.

The monovalent group represented by the formula (64) is preferably represented by the following formulas (65) or (66):



(65)

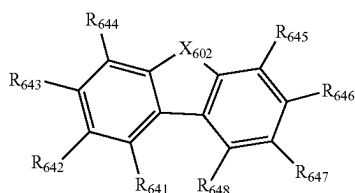
wherein in the formula (65), R_{631} to R_{640} are independently
a hydrogen atom,
a substituted or unsubstituted alkyl group having 1 to 50 carbon atoms,
a substituted or unsubstituted alkenyl group having 2 to 50 carbon atoms,
a substituted or unsubstituted alkynyl group having 2 to 50 carbon atoms,
a substituted or unsubstituted cycloalkyl group having 3 to 50 ring carbon atoms,
 $-\text{Si}(\text{R}_{901})(\text{R}_{902})(\text{R}_{903})$,
 $-\text{O}-(\text{R}_{904})$,
 $-\text{S}-(\text{R}_{905})$,
 $-\text{N}(\text{R}_{906})(\text{R}_{907})$,
a halogen atom, a cyano group, a nitro group,
a substituted or unsubstituted aryl group having 6 to 50 ring carbon atoms, or
a substituted or unsubstituted monovalent heterocyclic group having 5 to 50 ring atoms; and
 R_{901} to R_{907} are as defined in the formula (1);



(66)

wherein in the formula (66), Ar_{601} , L_{601} and L_{603} are as defined in the formula (64); and HAr_{601} is a structure represented by the following formula (67);

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wherein in the formula (67) X_{602} is an oxygen atom or a sulfur atom;

any one of R_{641} to R_{648} is a single bond bonding to L_{603} ; R_{641} to R_{648} which are not single bonds are independently a hydrogen atom,

a substituted or unsubstituted alkyl group having 1 to 50 carbon atoms,

a substituted or unsubstituted alkenyl group having 2 to 50 carbon atoms,

a substituted or unsubstituted alkynyl group having 2 to 50 carbon atoms,

a substituted or unsubstituted cycloalkyl group having 3 to 50 ring carbon atoms,

$-\text{Si}(\text{R}_{901})(\text{R}_{902})(\text{R}_{903})$,

$-\text{O}-(\text{R}_{904})$,

$-\text{S}-(\text{R}_{905})$,

$-\text{N}(\text{R}_{906})(\text{R}_{907})$,

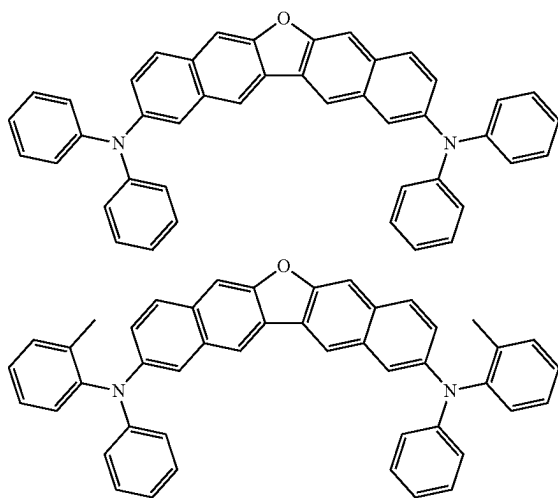
a halogen atom, a cyano group, a nitro group,

a substituted or unsubstituted aryl group having 6 to 50 ring carbon atoms, or

a substituted or unsubstituted monovalent heterocyclic group having 5 to 50 ring atoms; and

R_{907} , to R_{907} are as defined in the formula (1).

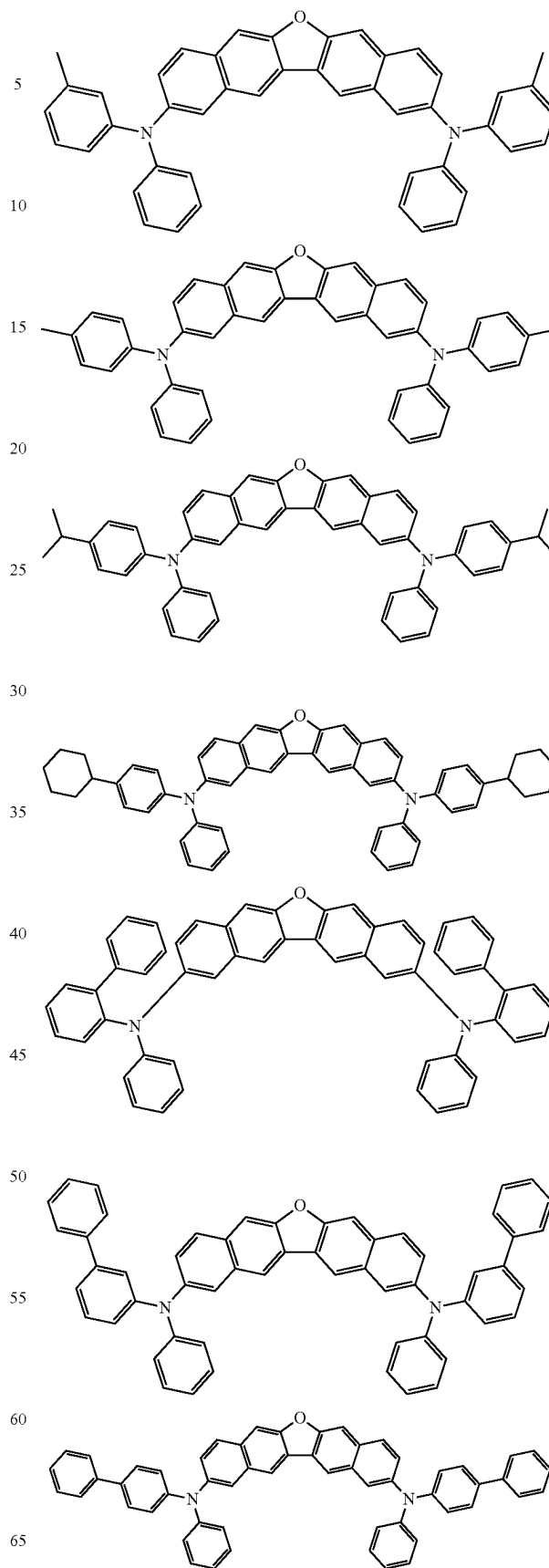
As specific example of the compound represented by the formula (61), in addition to the compounds described in WO2014/104144, the following compounds can be given, for example. In the following example compounds, Me represents methyl group.



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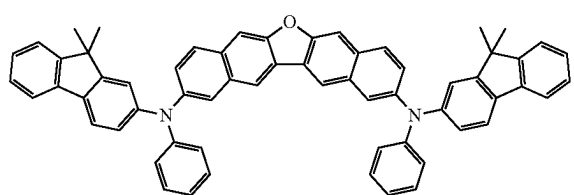
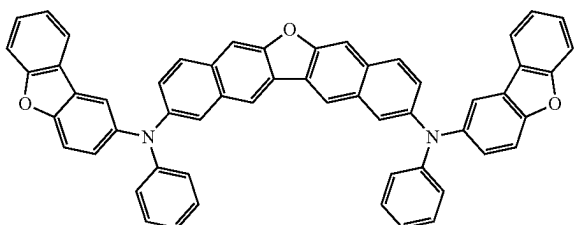
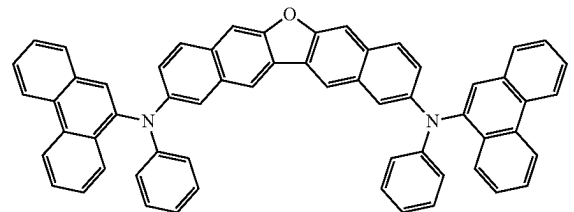
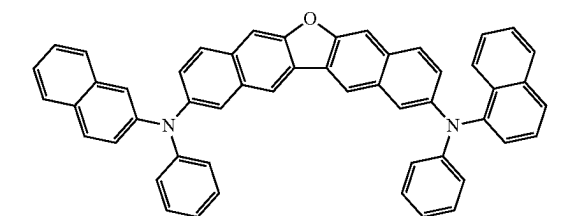
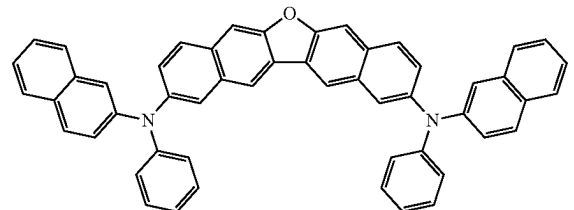
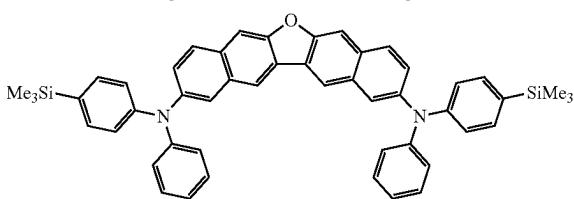
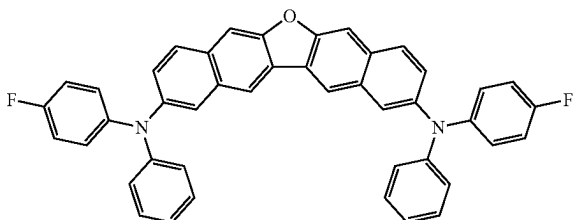
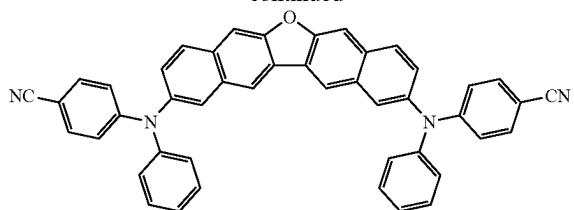
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(67)

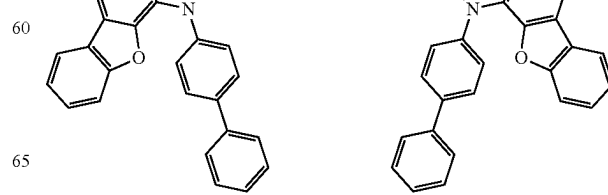
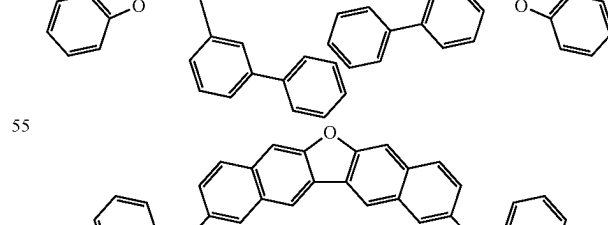
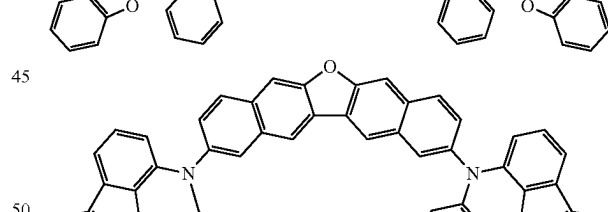
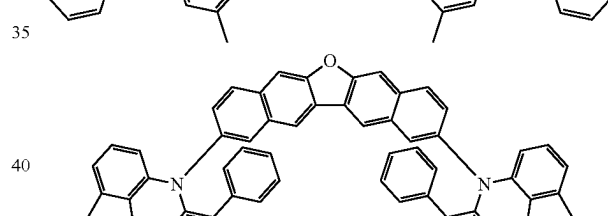
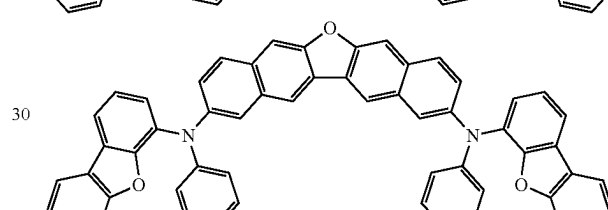
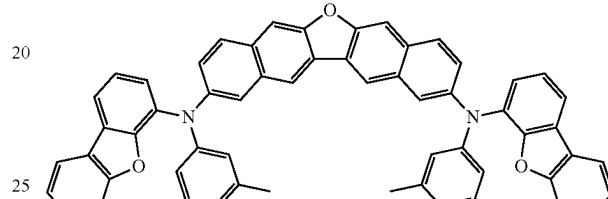
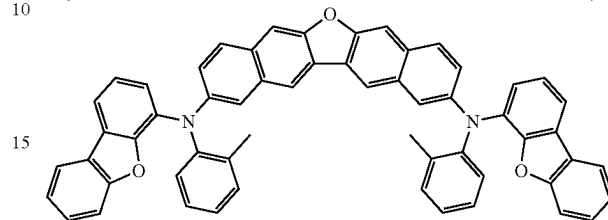
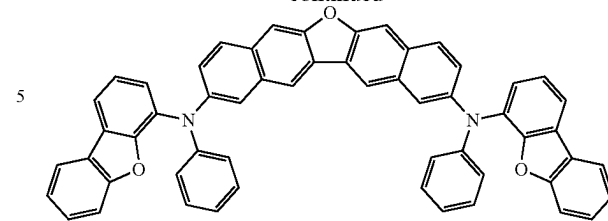


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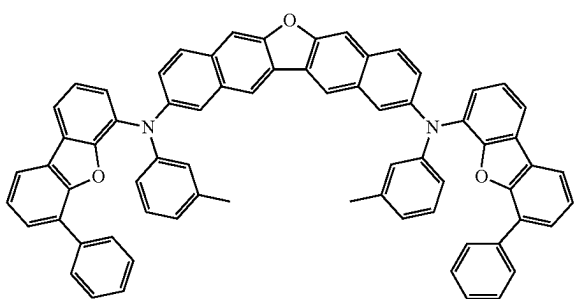
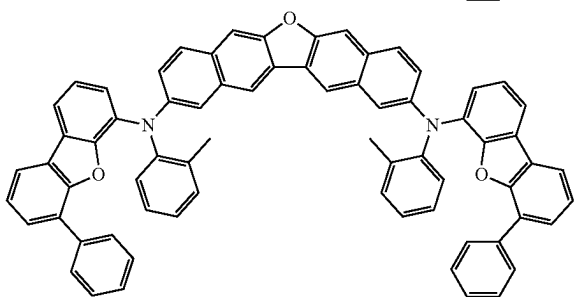
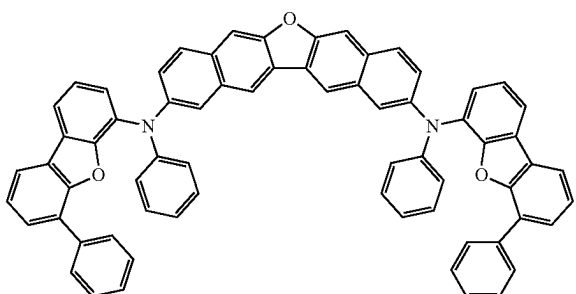
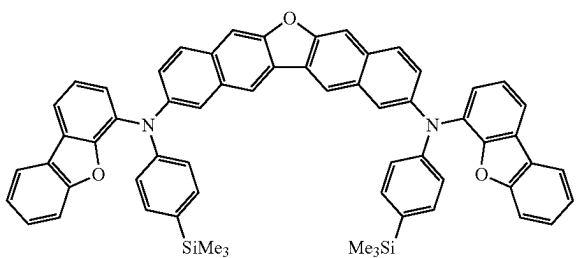
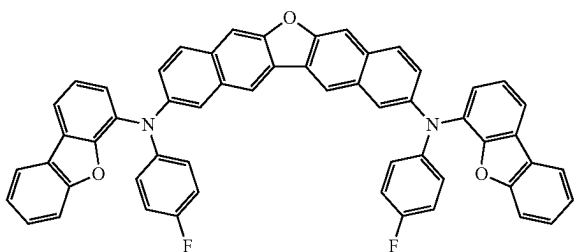
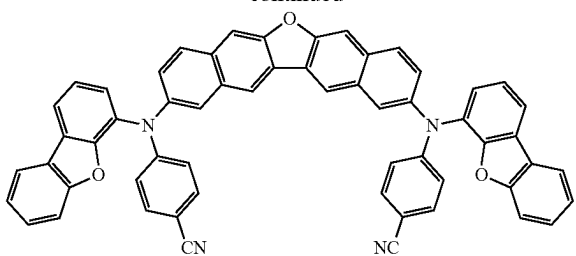
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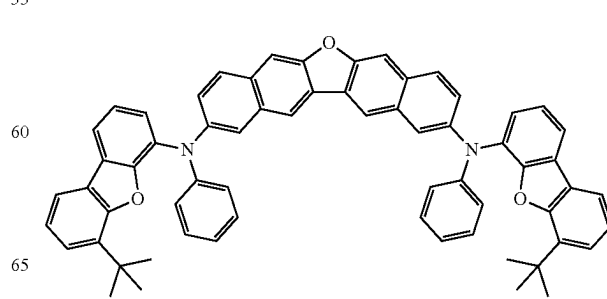
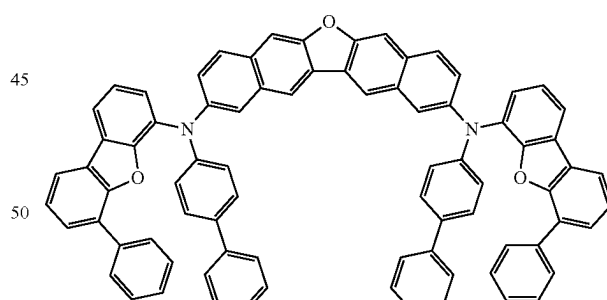
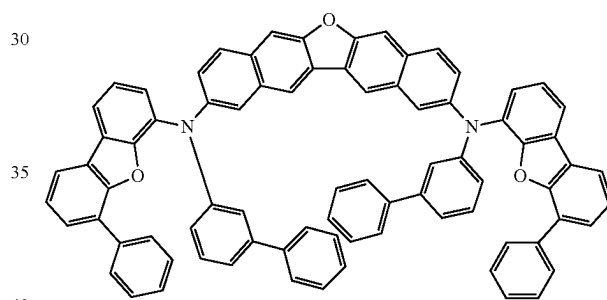
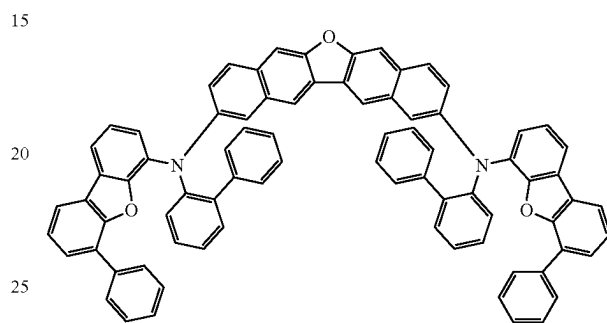
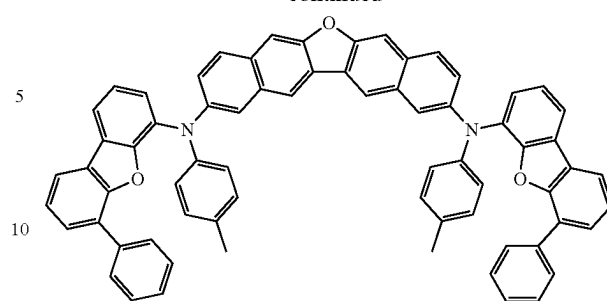


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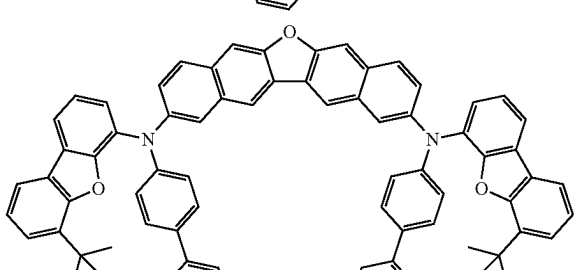
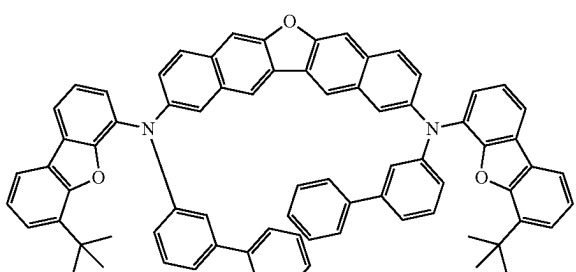
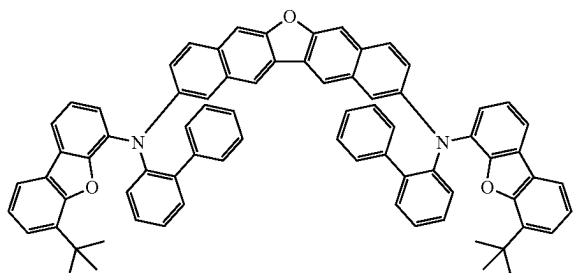
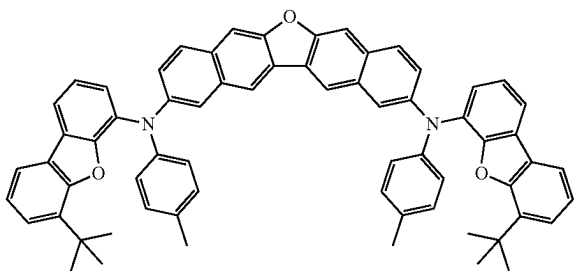
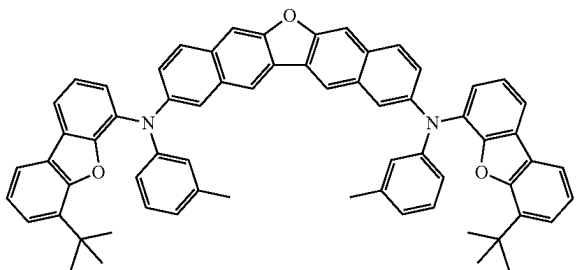
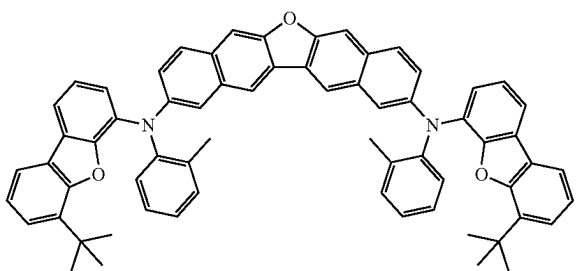
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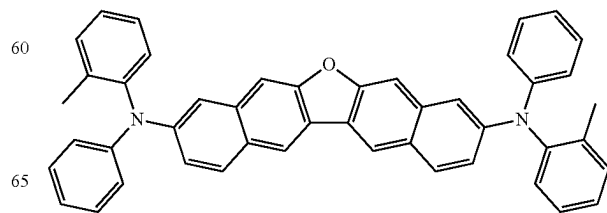
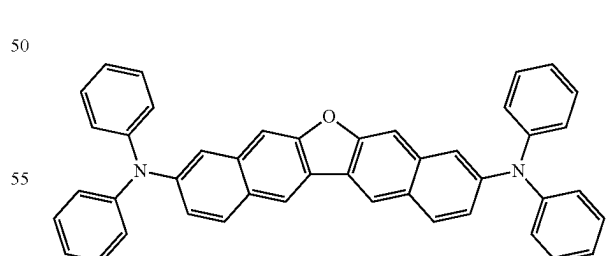
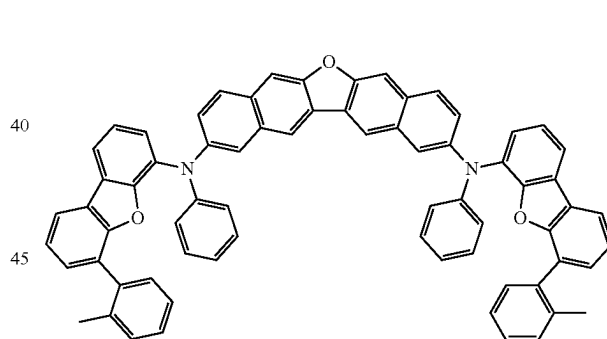
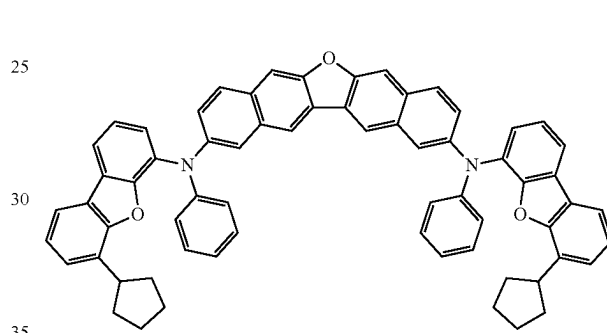
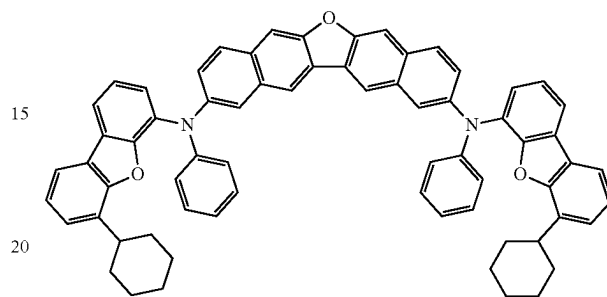
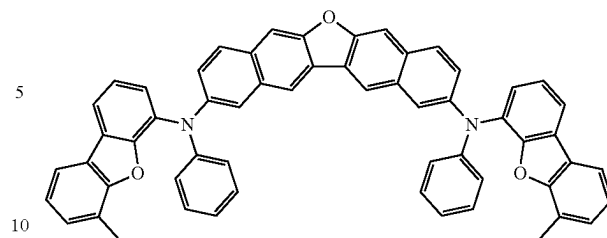


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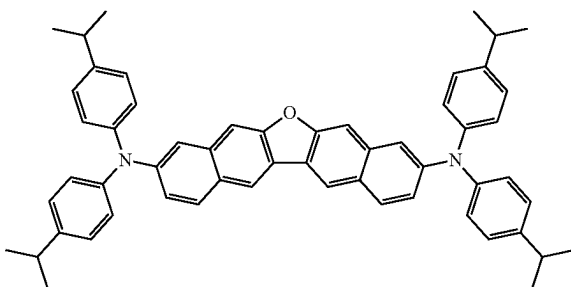
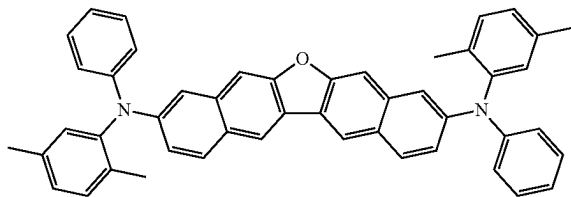
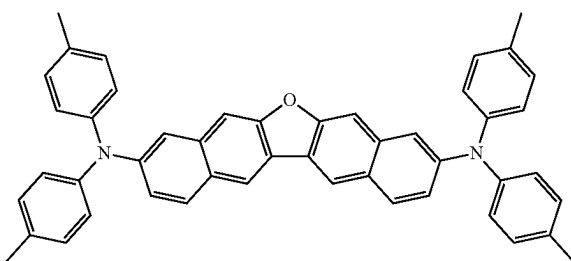
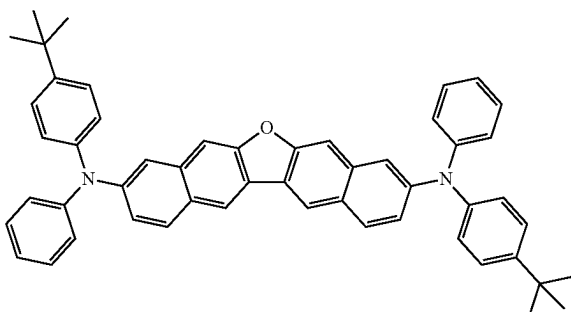
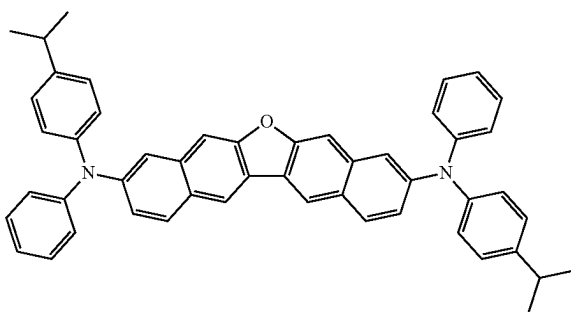
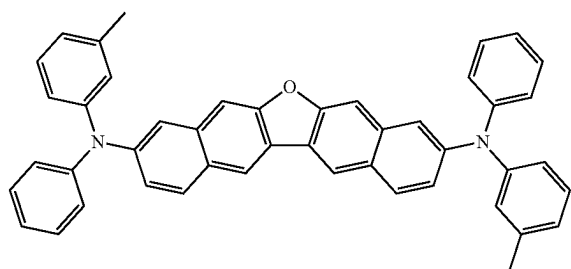
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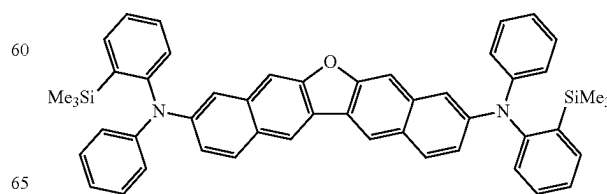
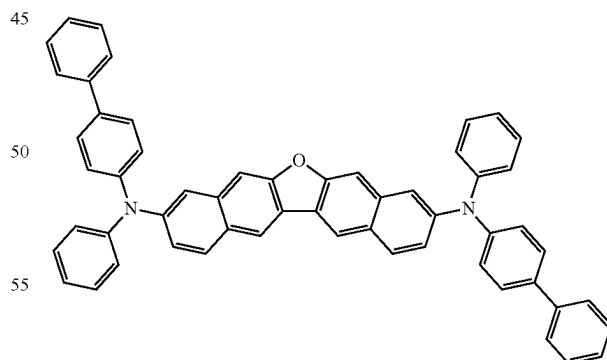
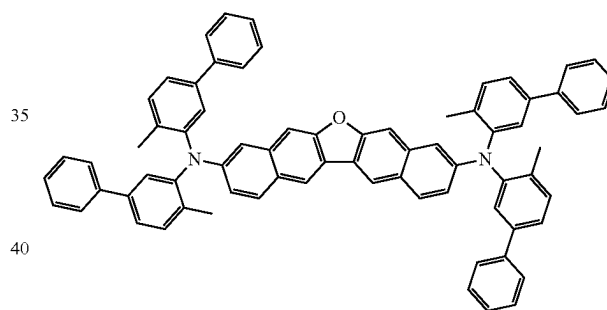
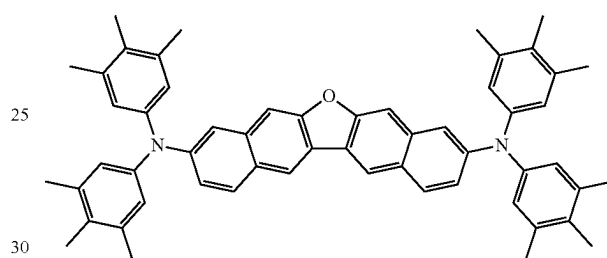
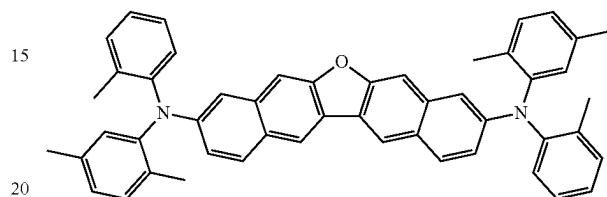
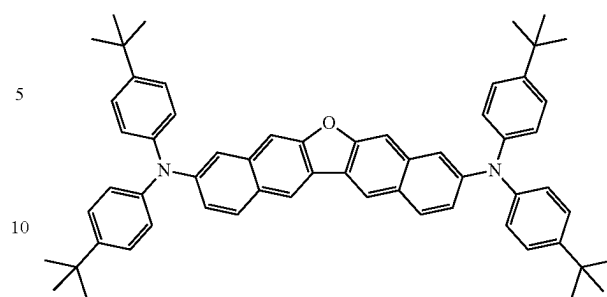


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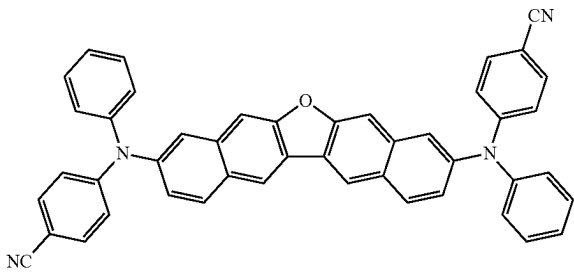
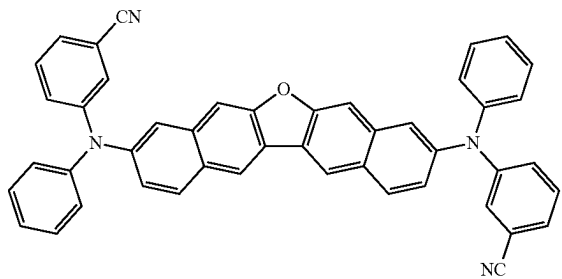
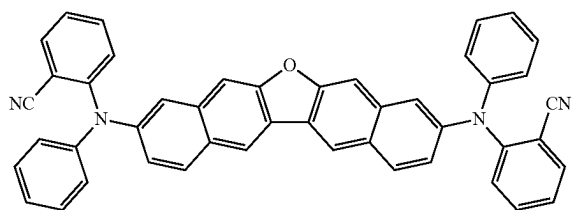
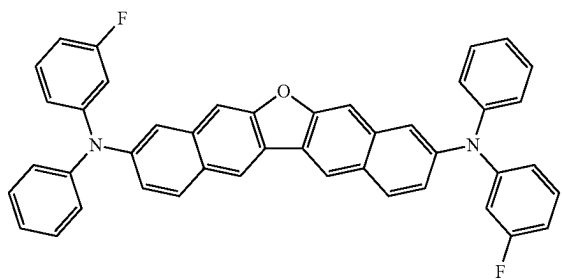
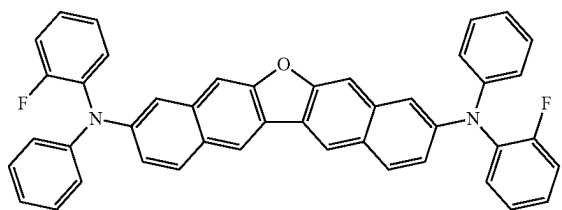
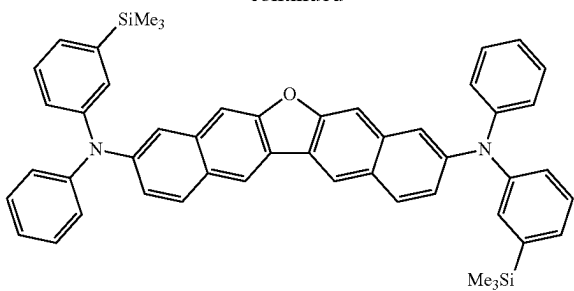
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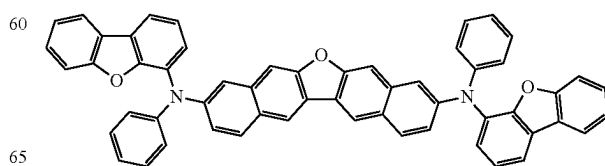
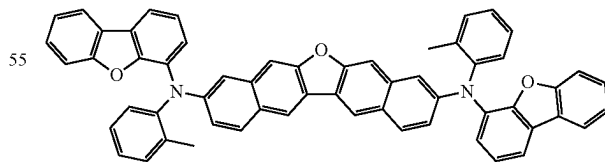
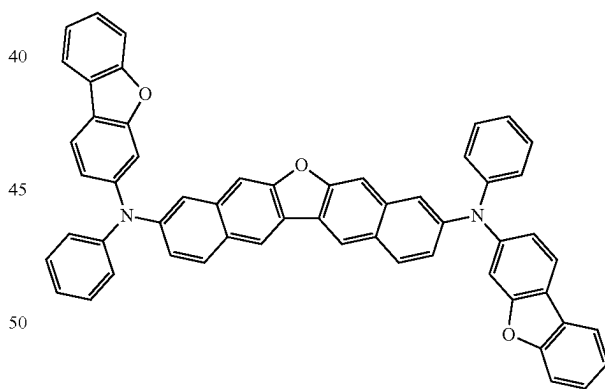
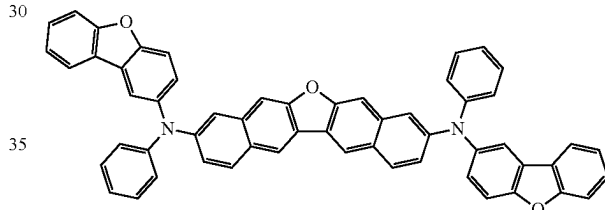
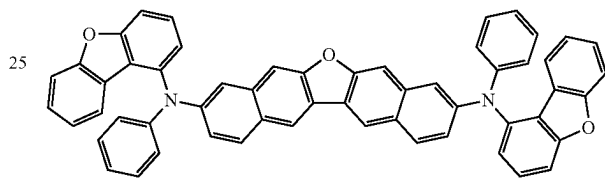
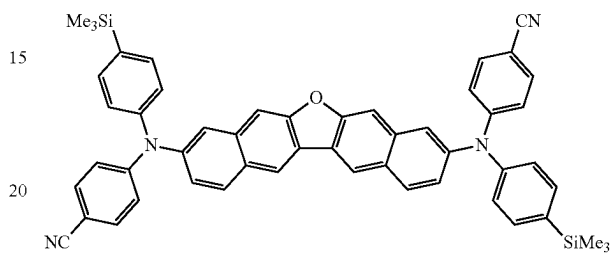
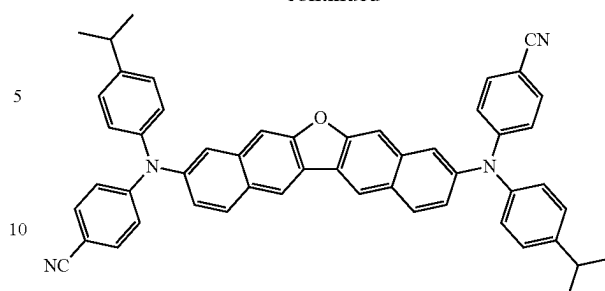


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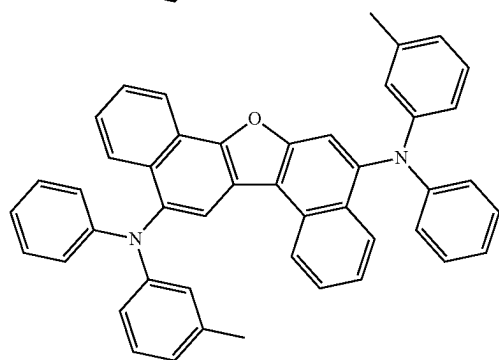
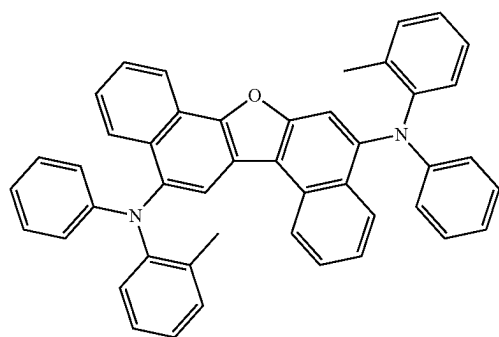
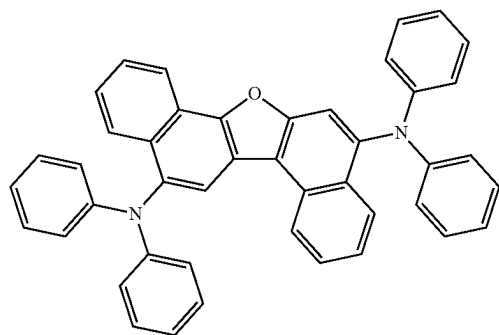
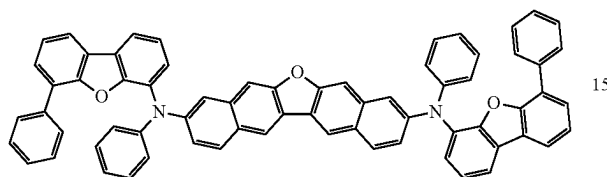
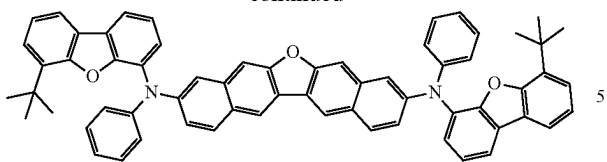
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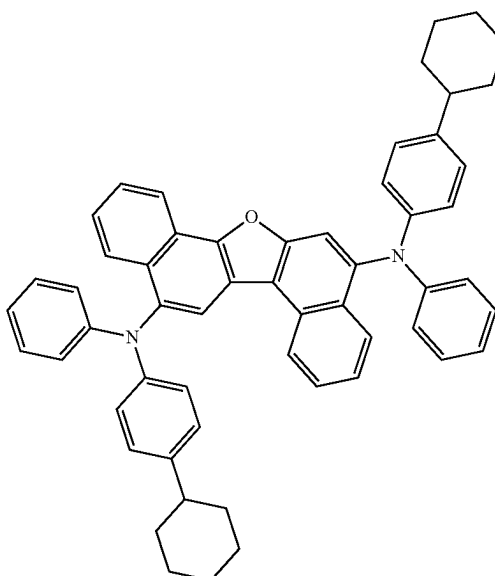
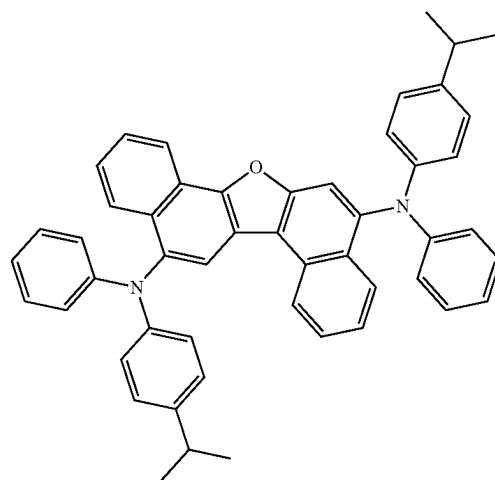
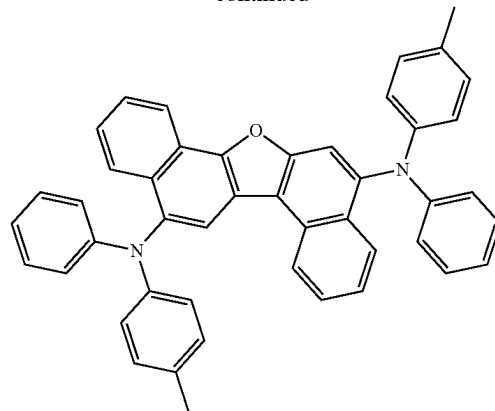


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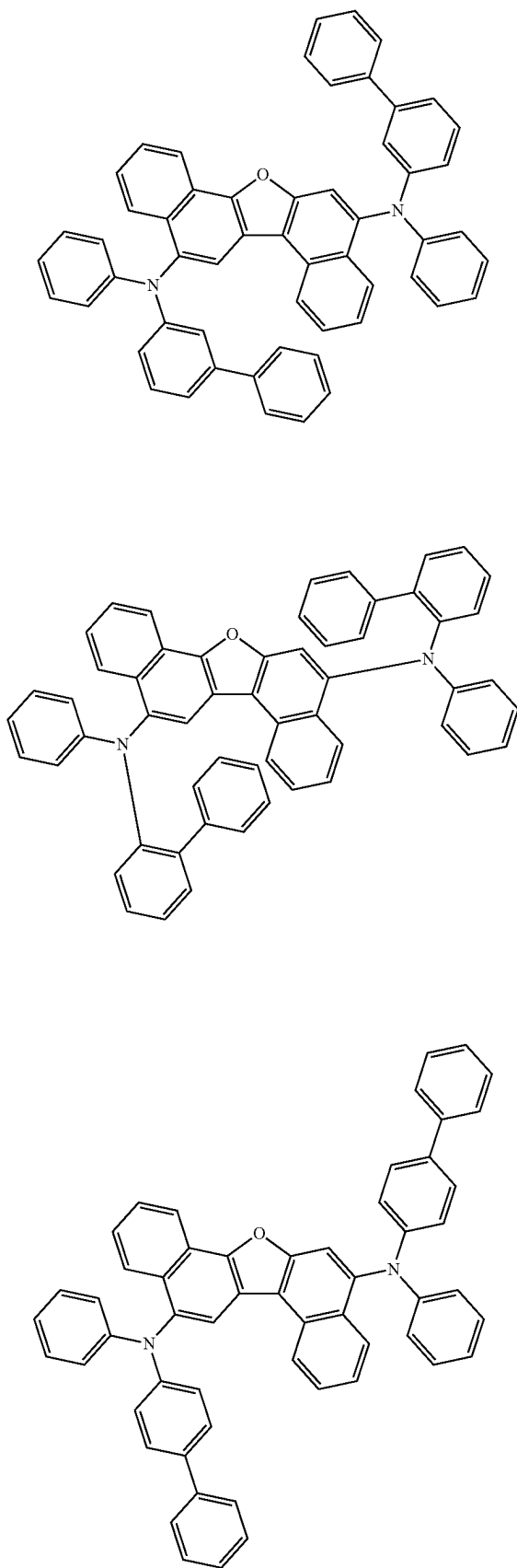
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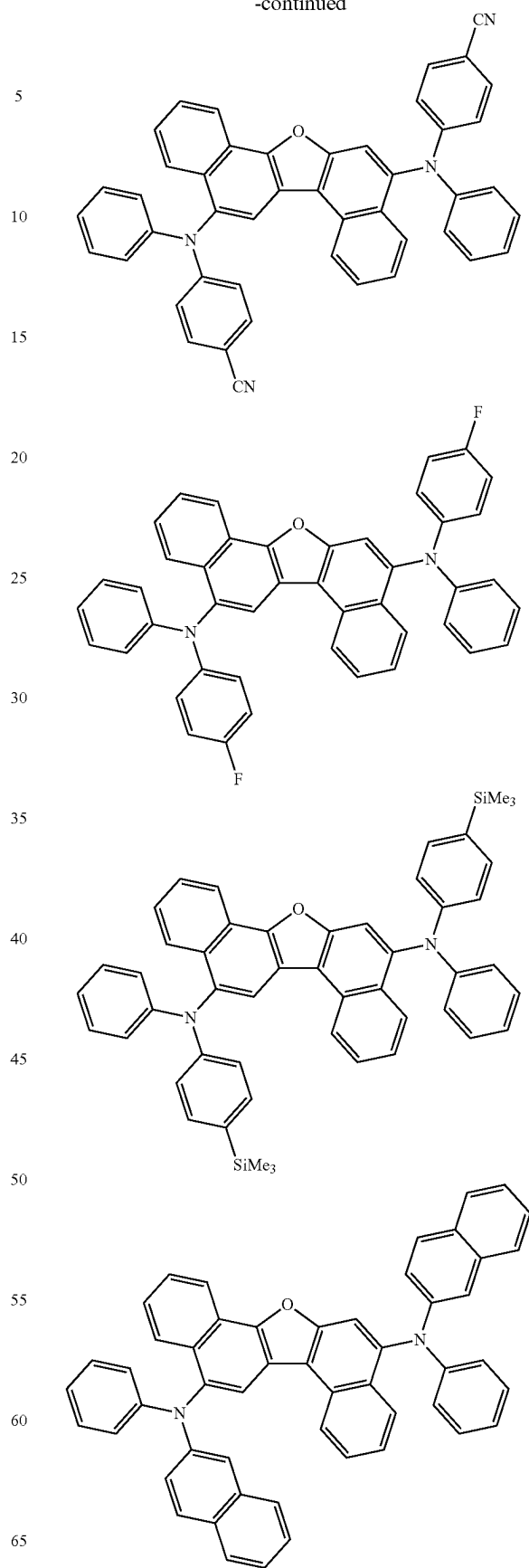


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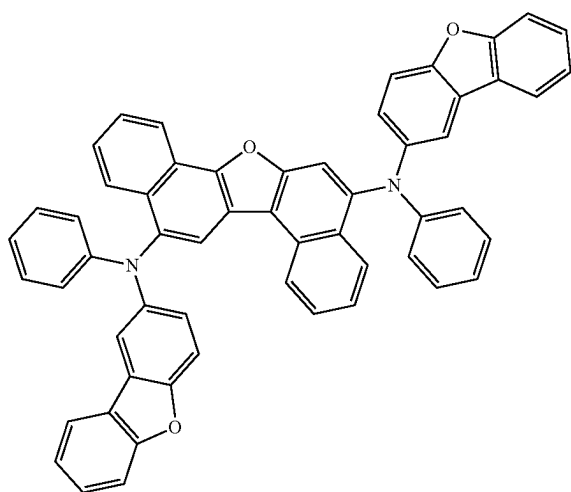
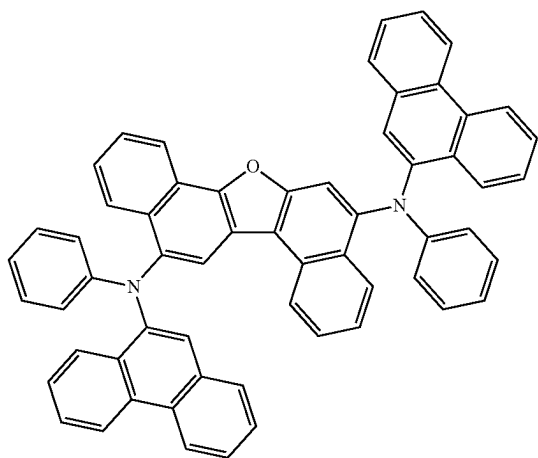
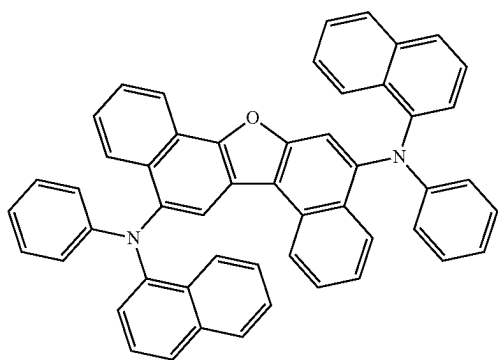
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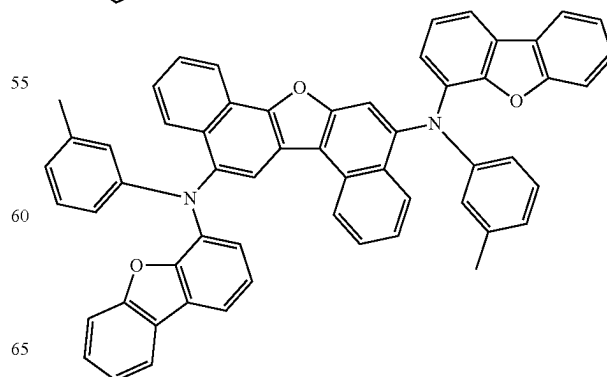
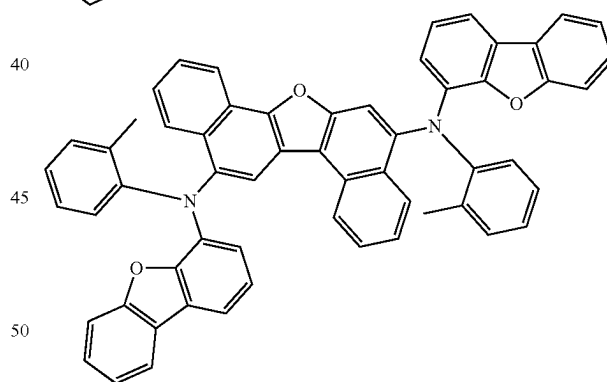
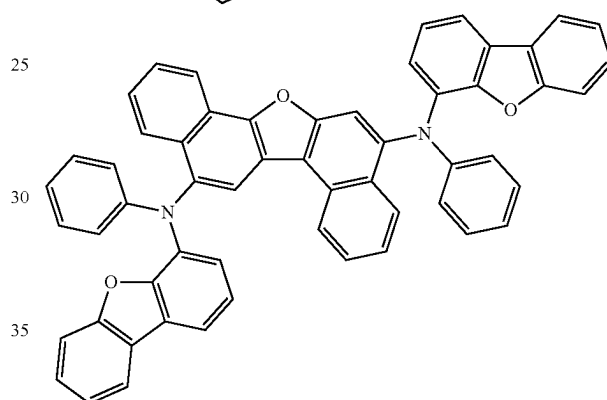
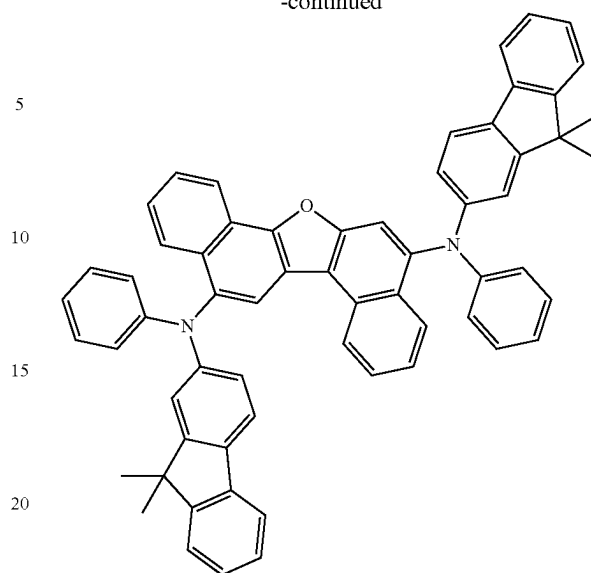


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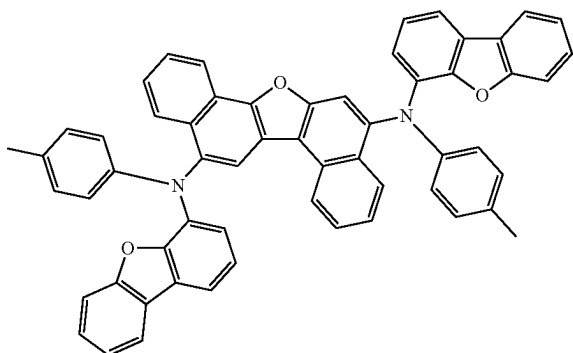
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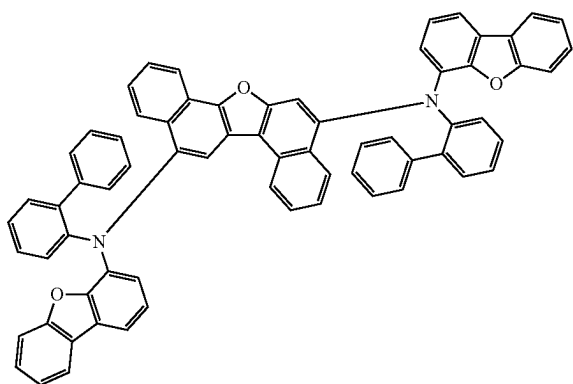
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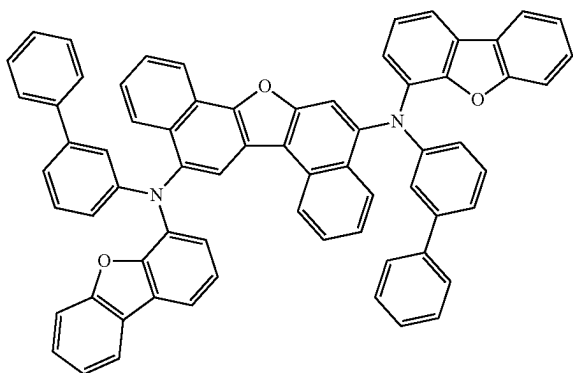
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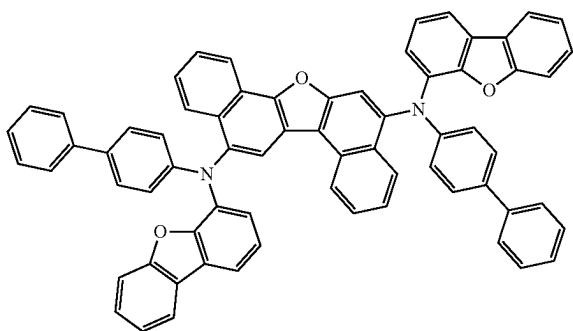
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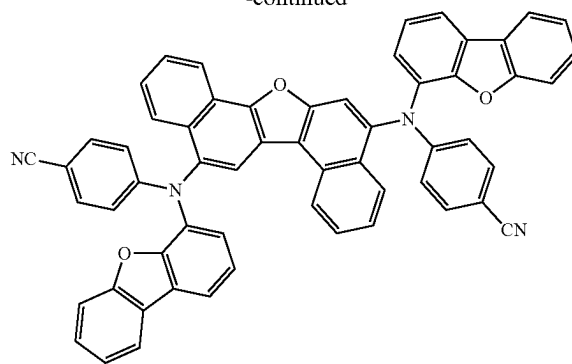
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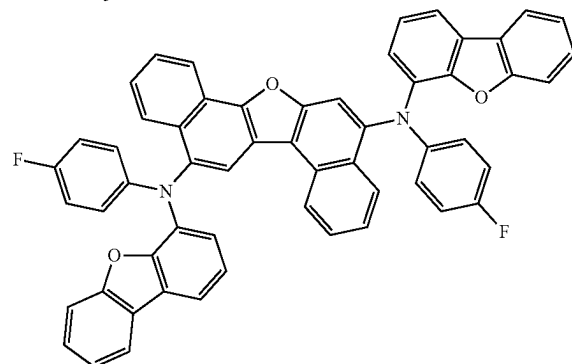
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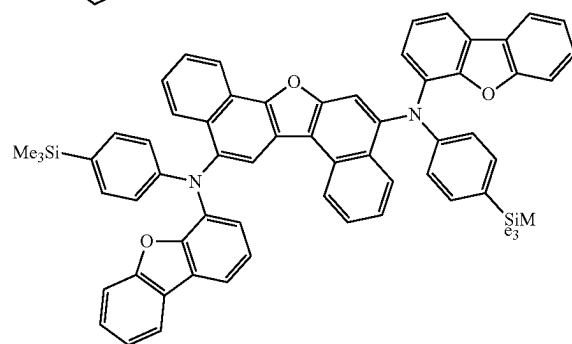
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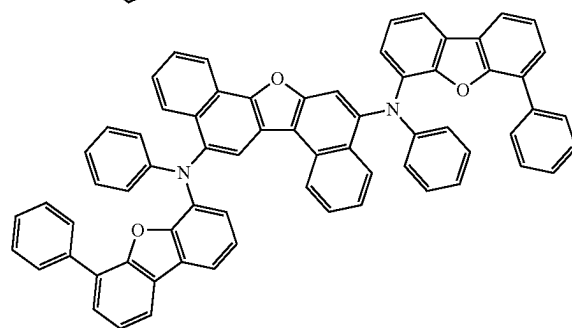
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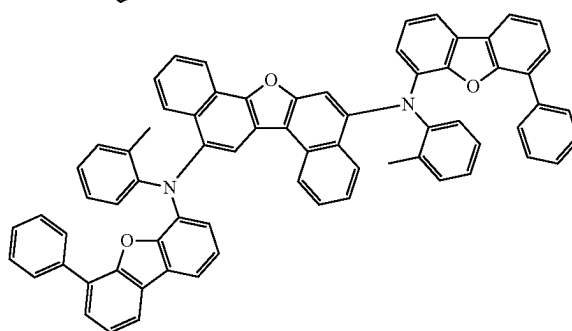
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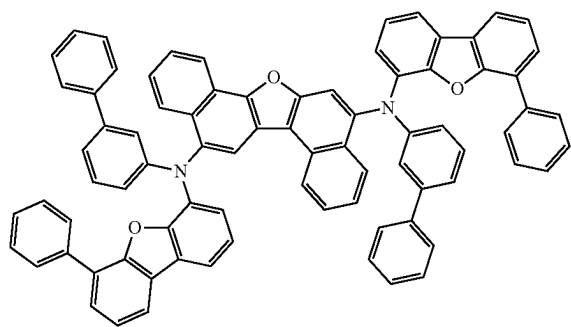
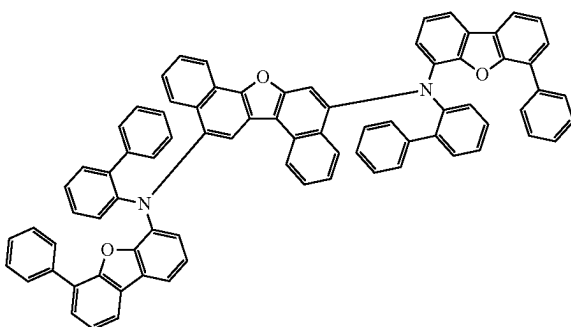
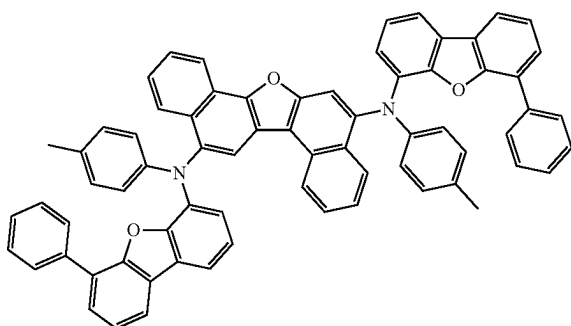
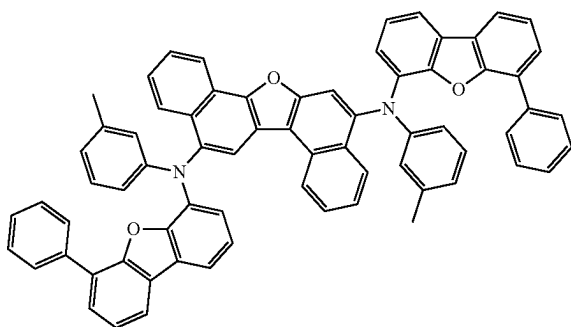
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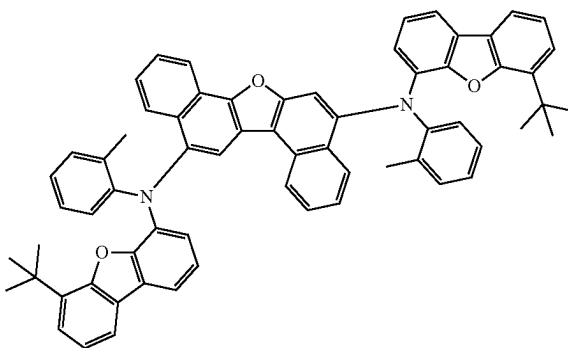
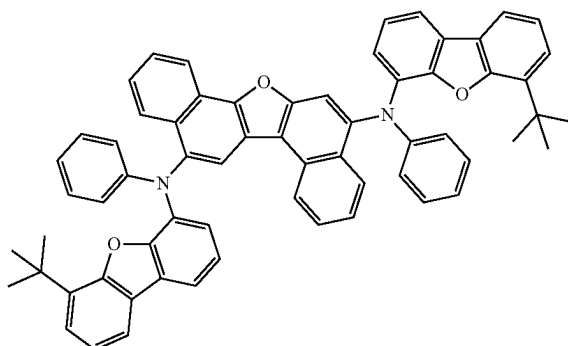
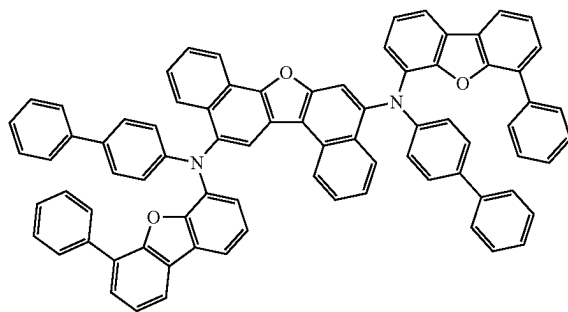
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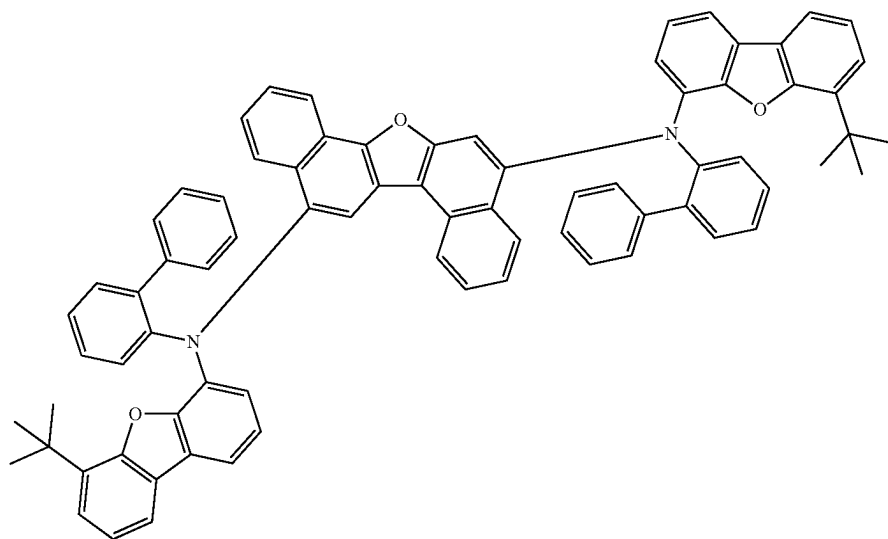
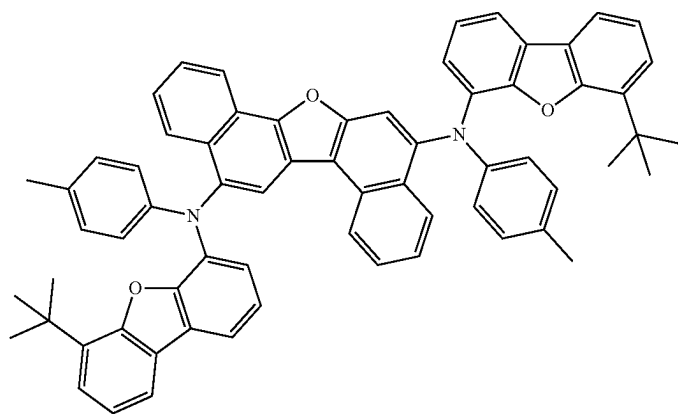
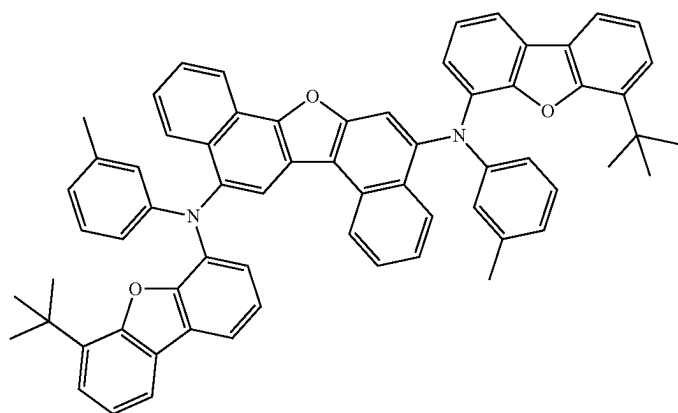
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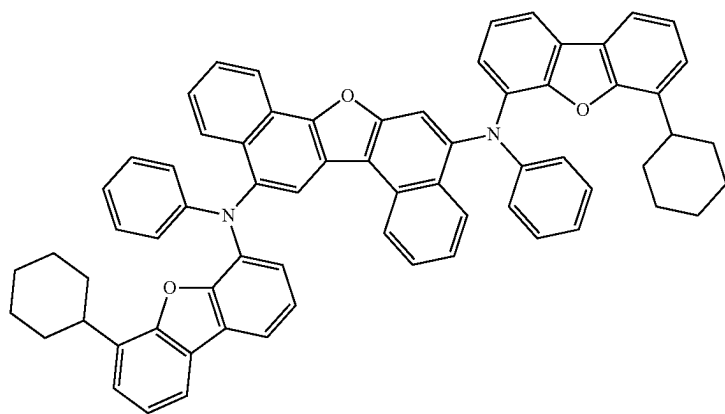
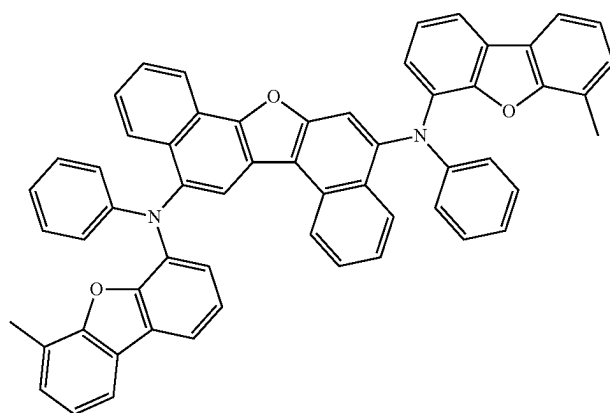
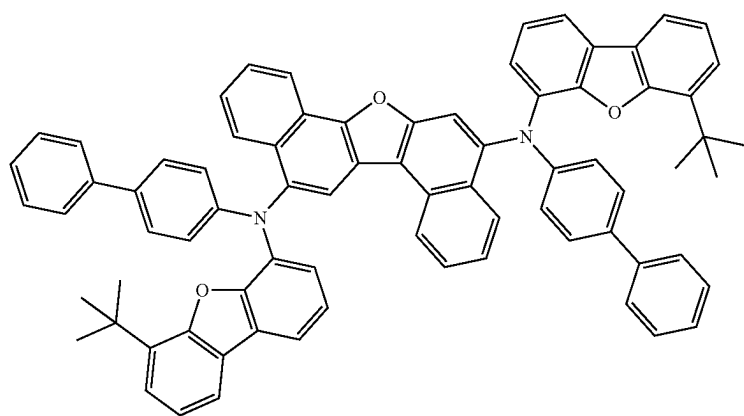
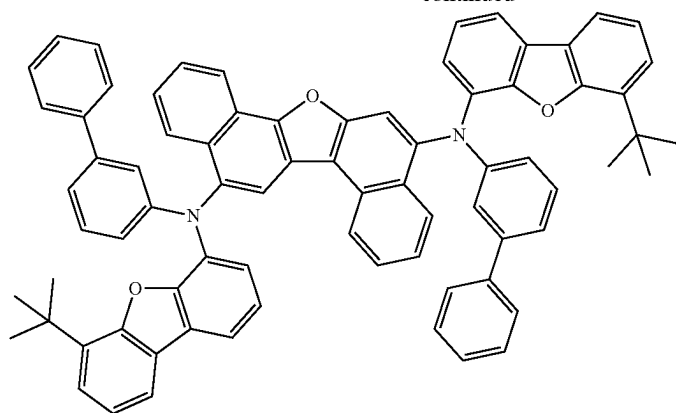
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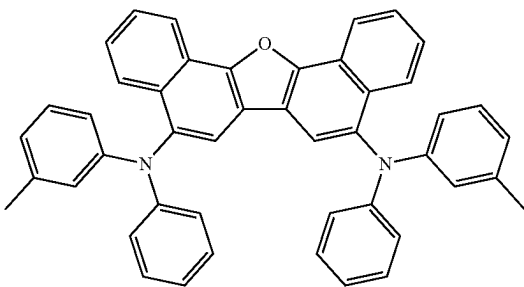
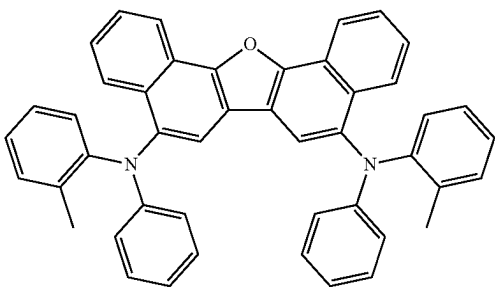
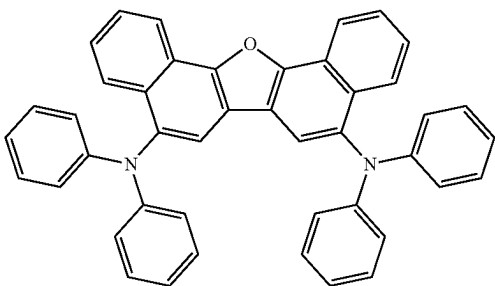
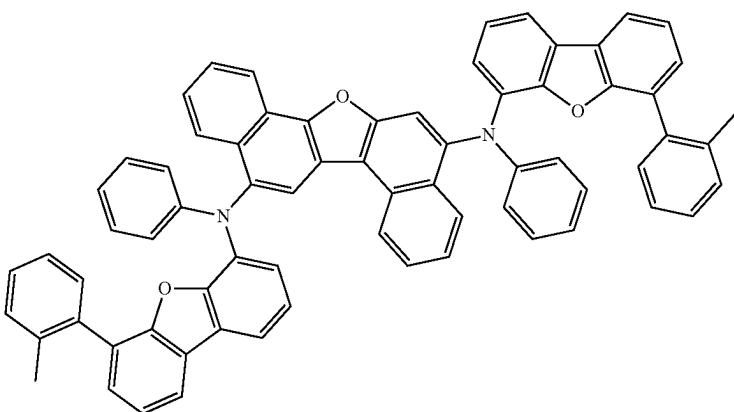
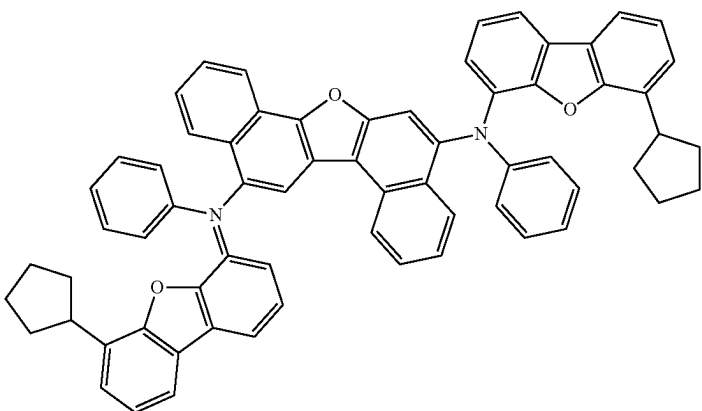


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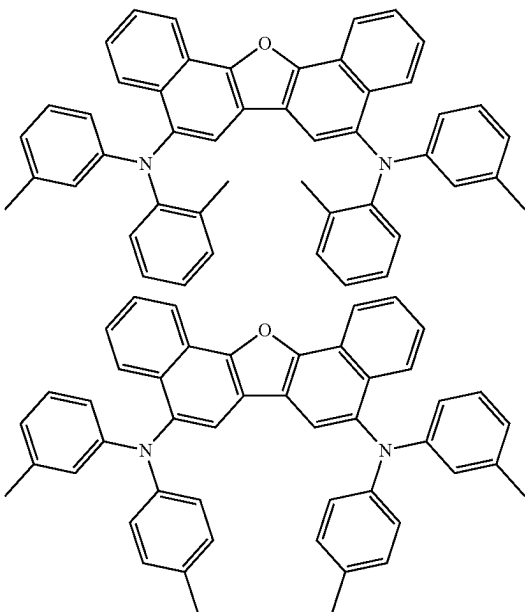
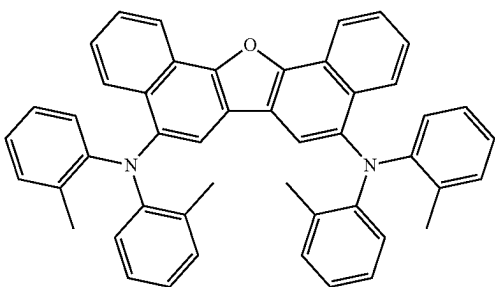
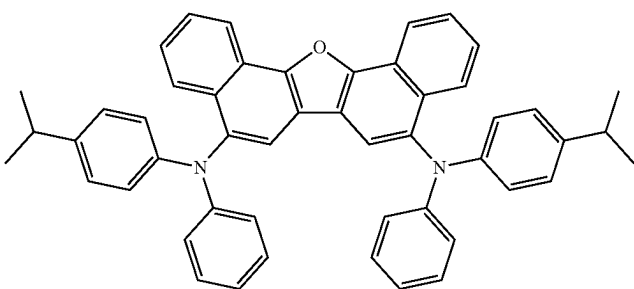
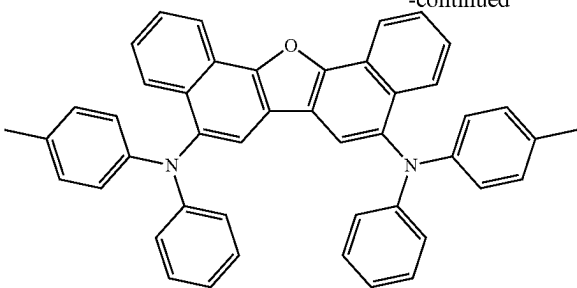




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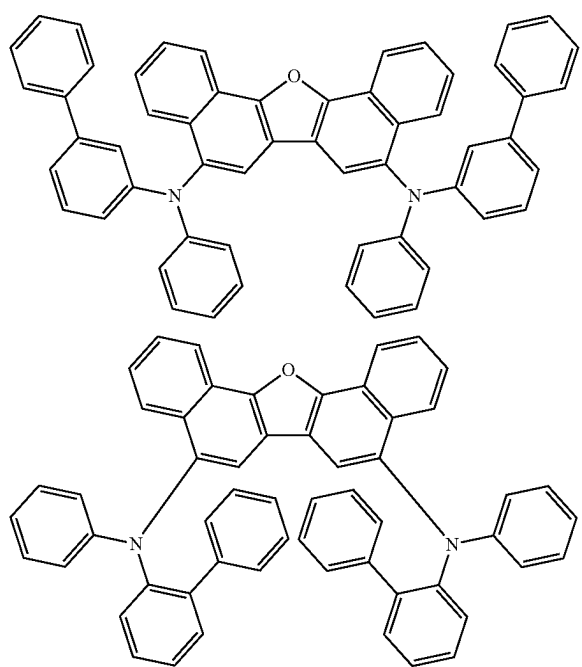
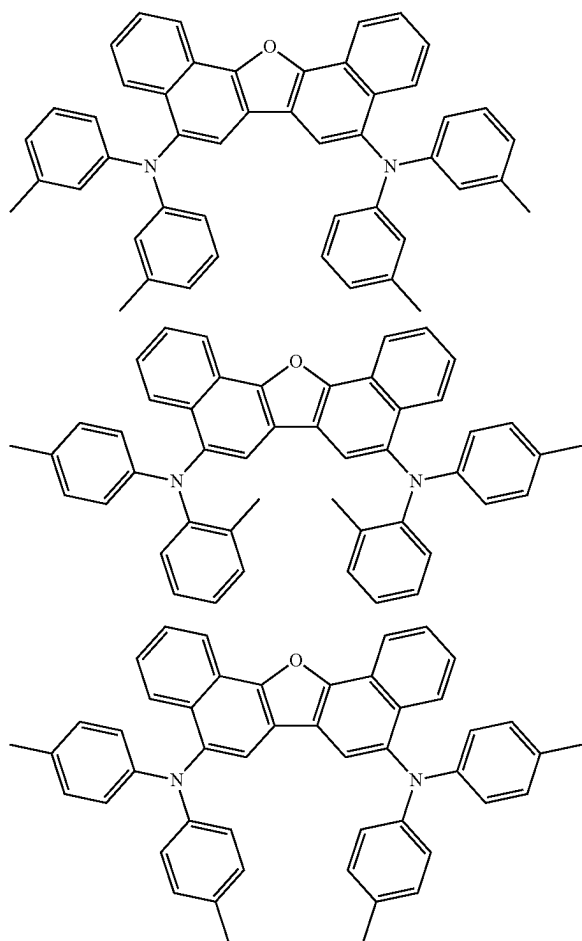
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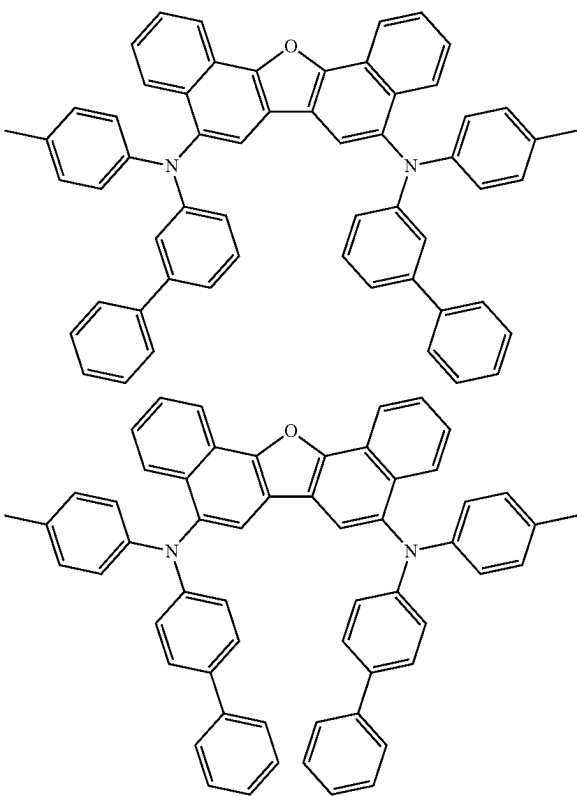
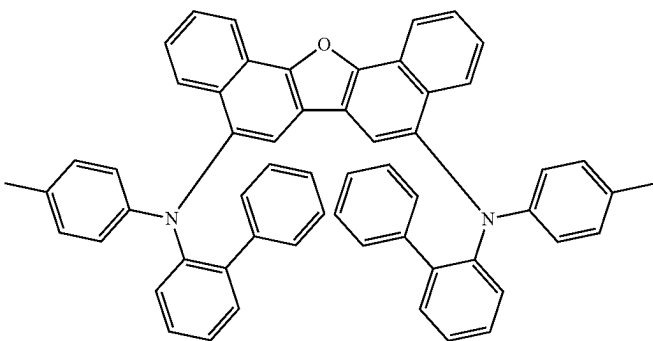
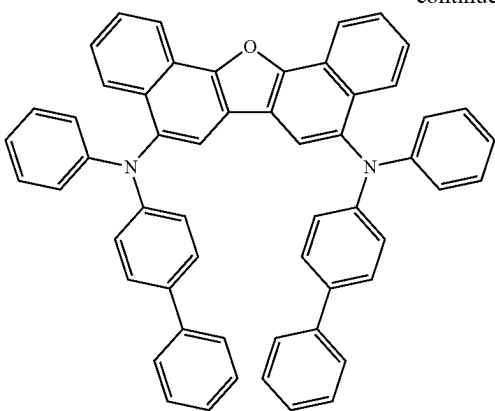
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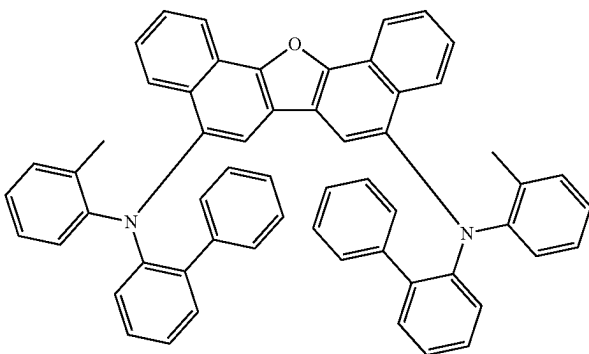
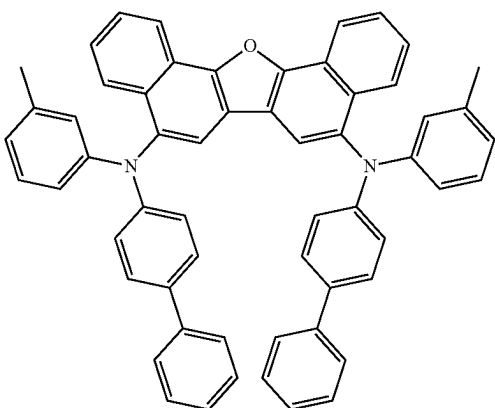
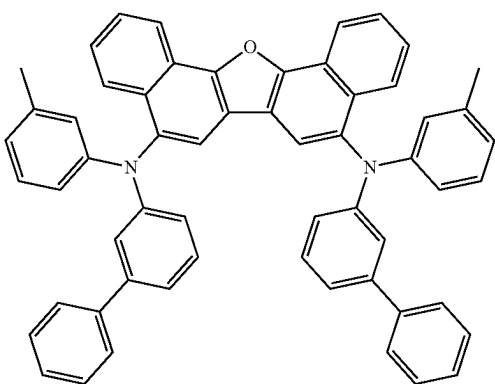
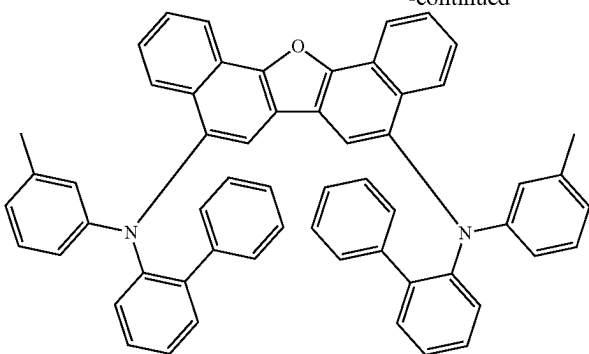
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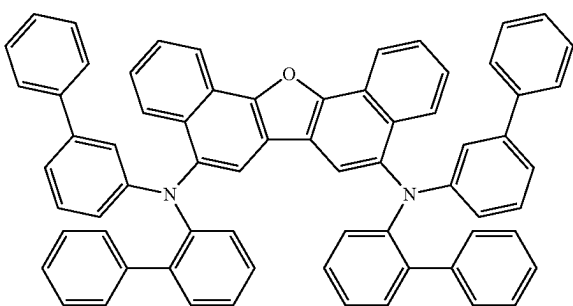
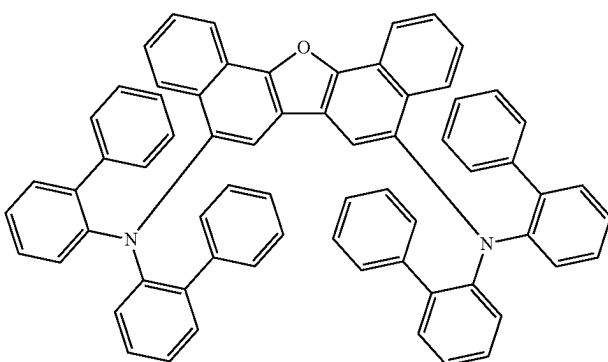
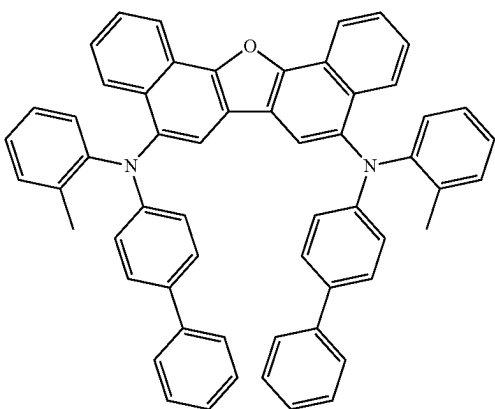
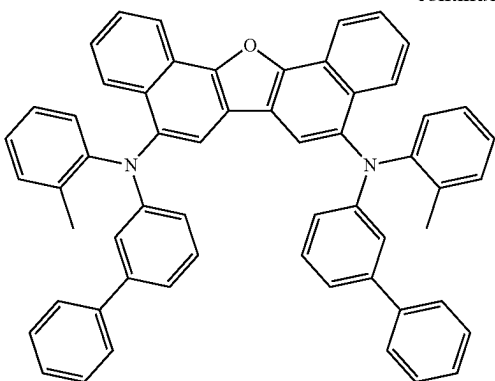
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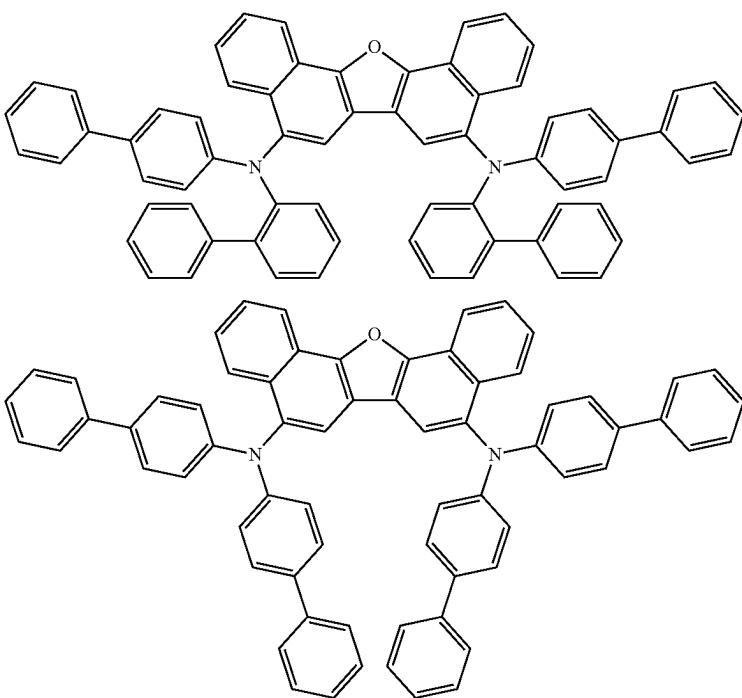
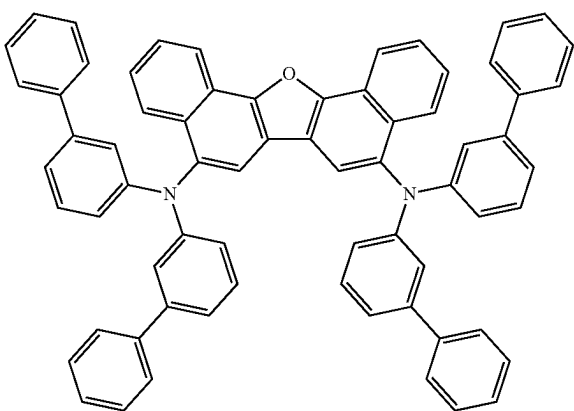
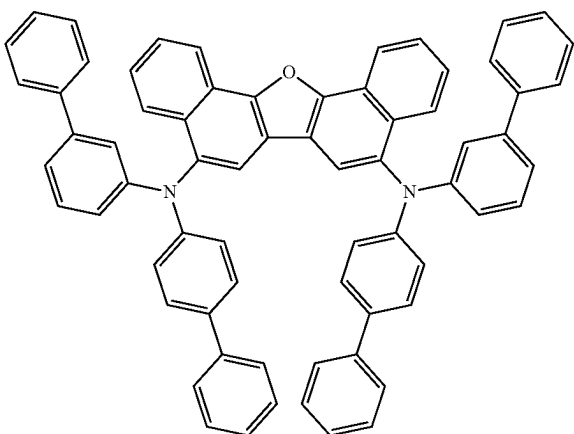
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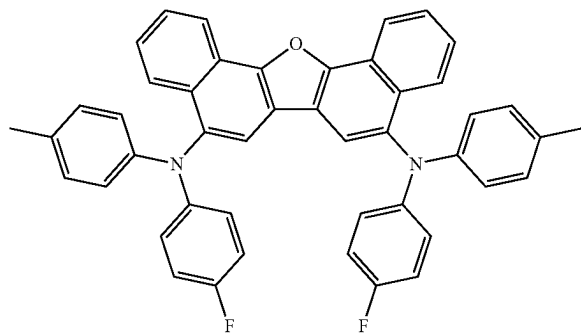
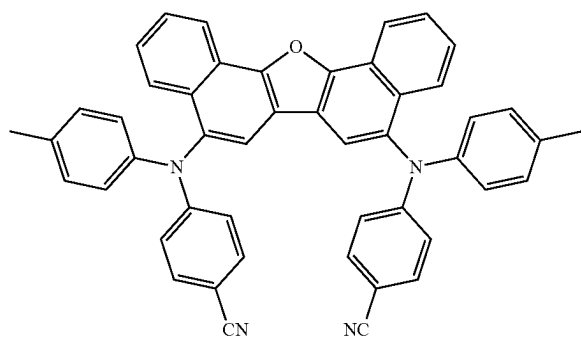
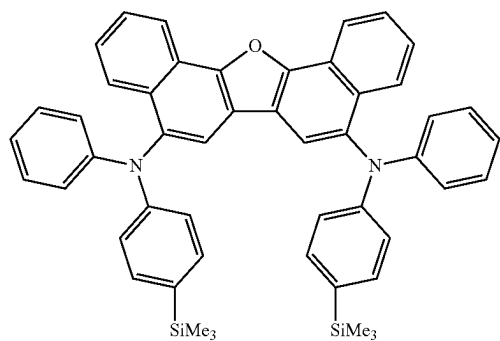
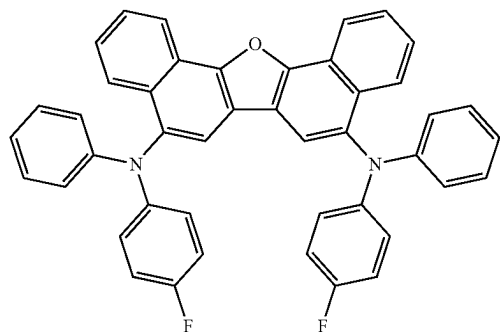
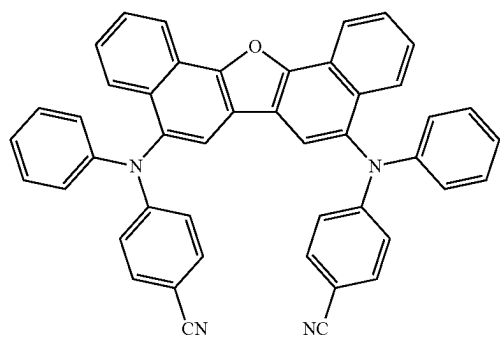
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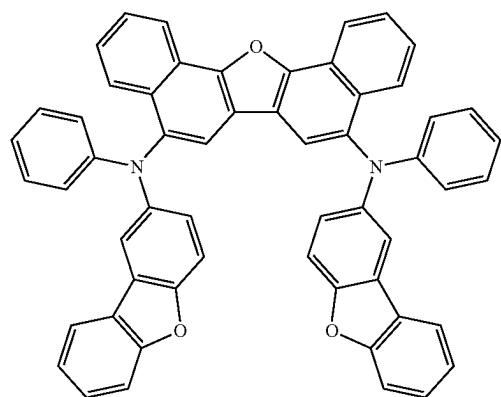
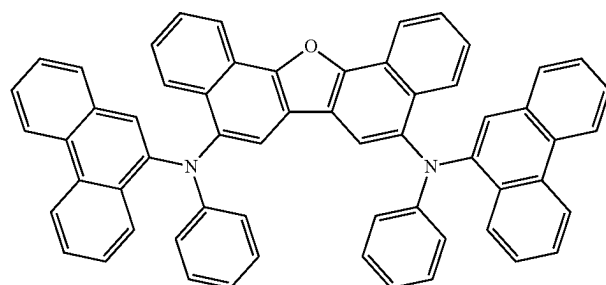
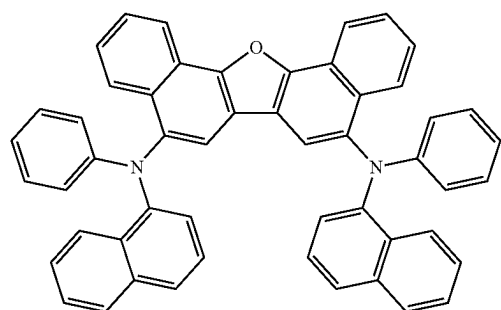
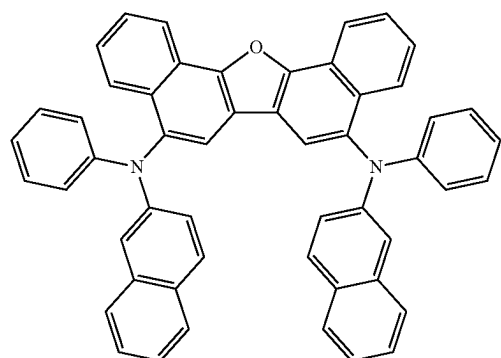
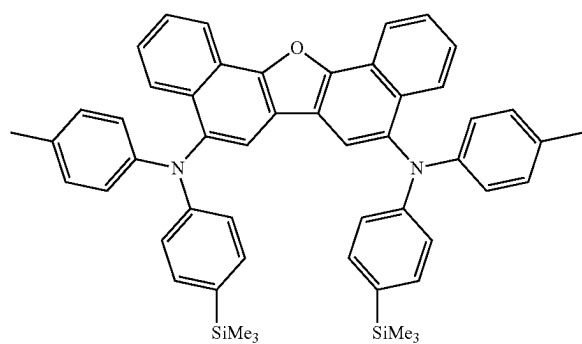
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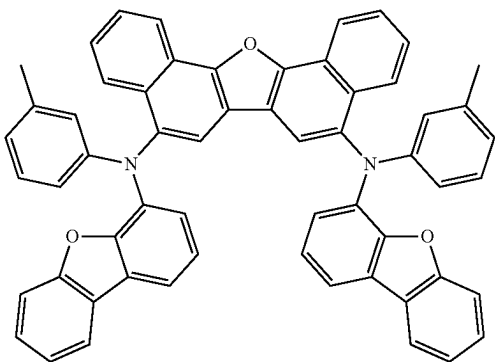
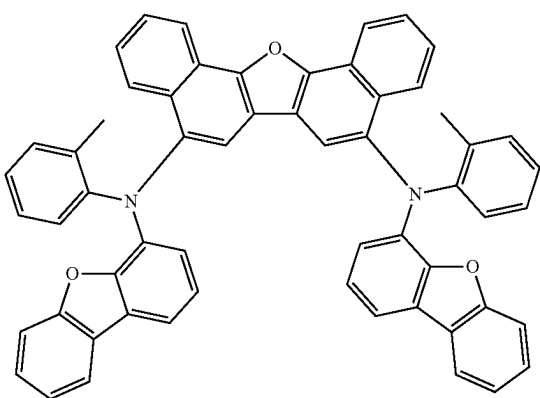
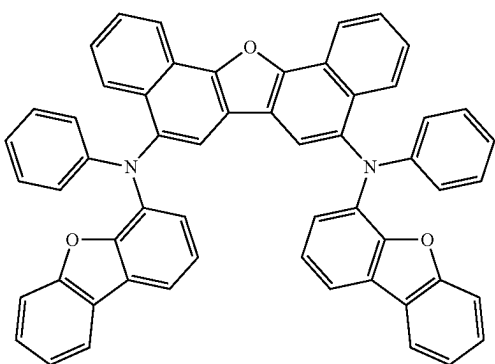
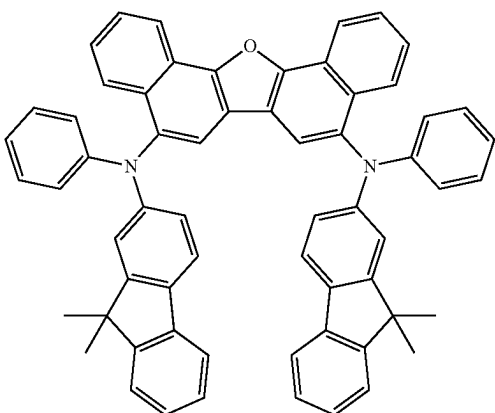
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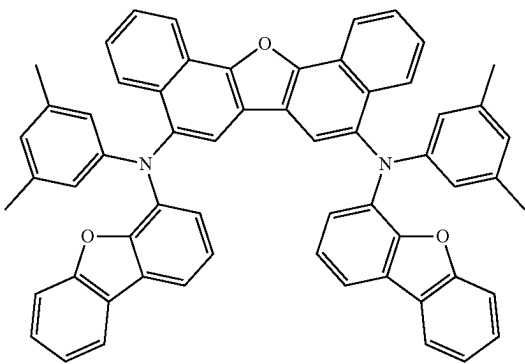
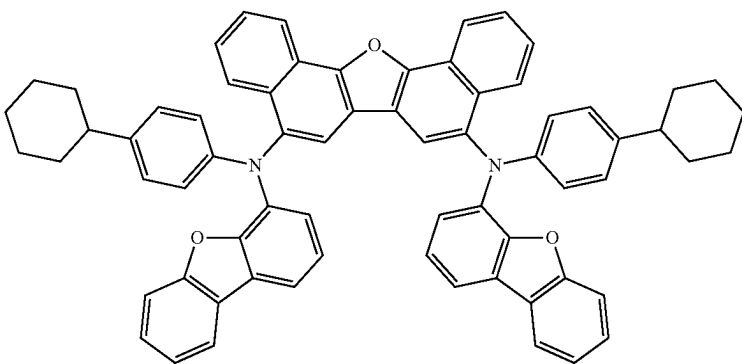
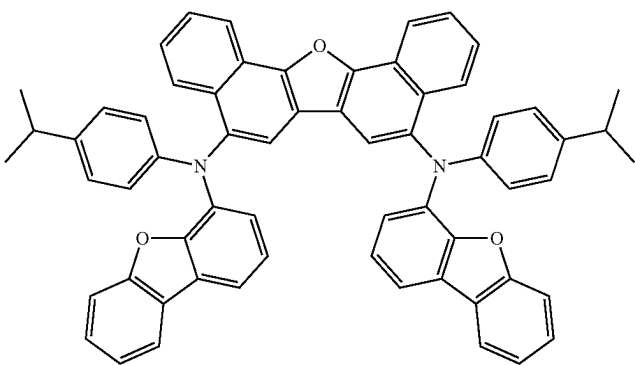
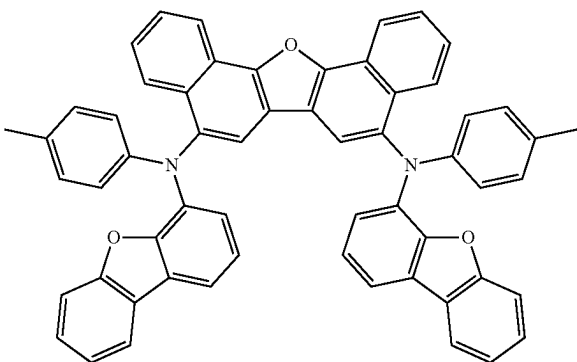
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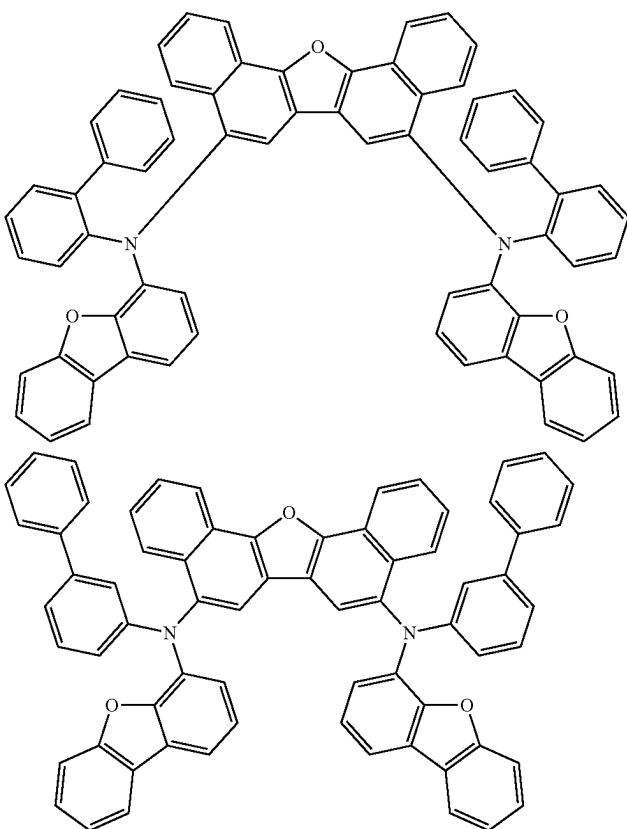
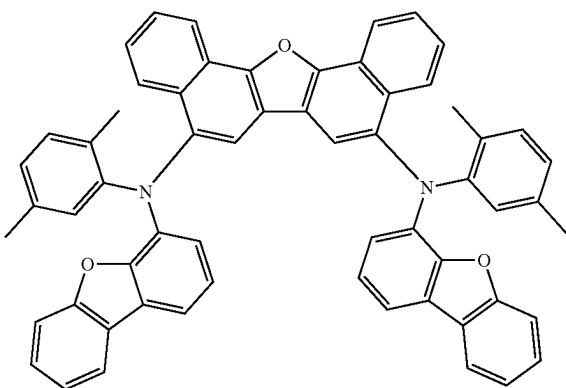
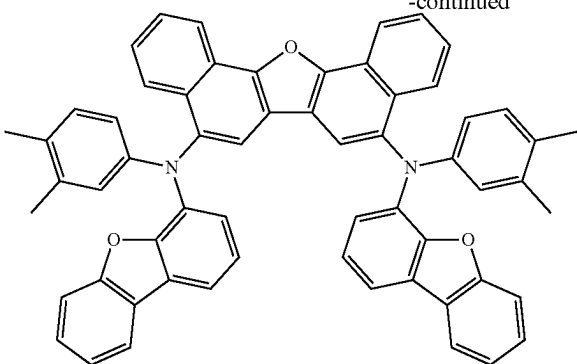
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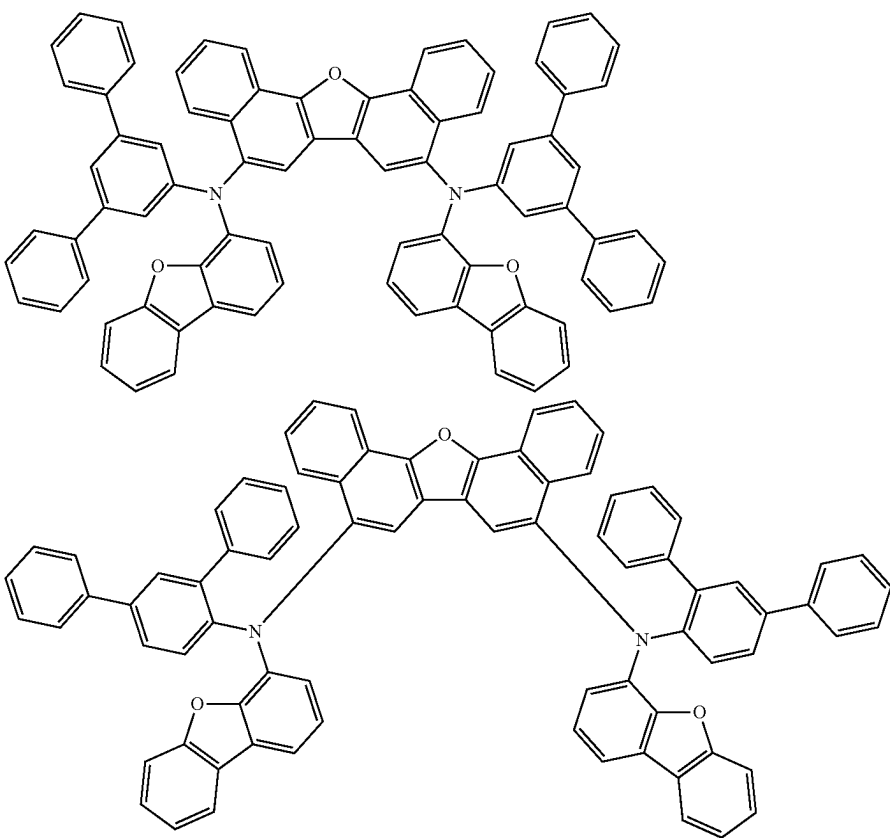
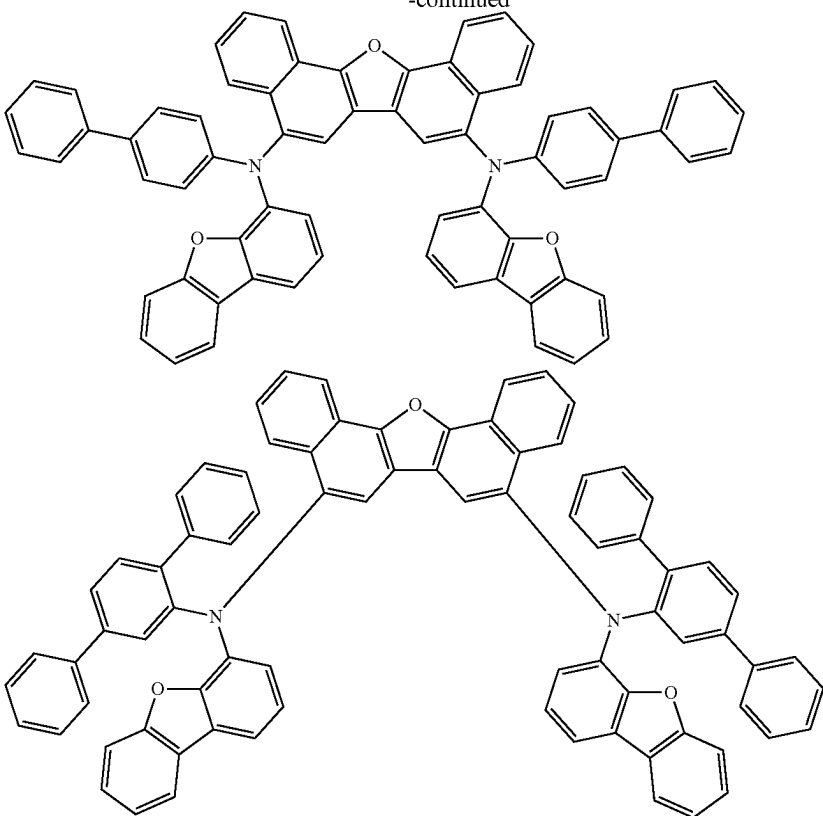
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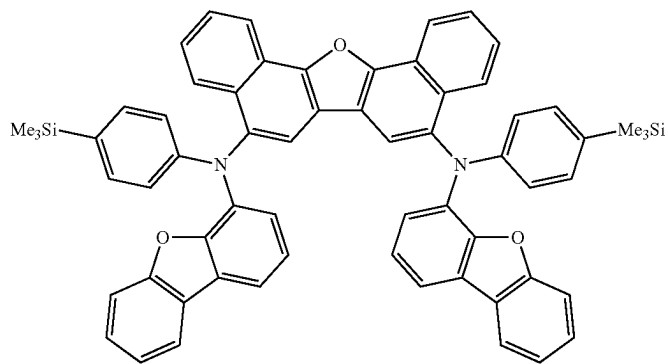
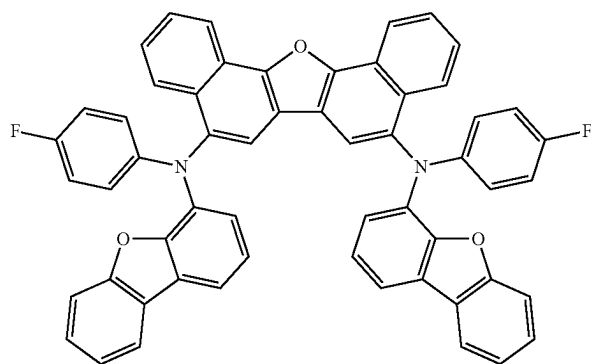
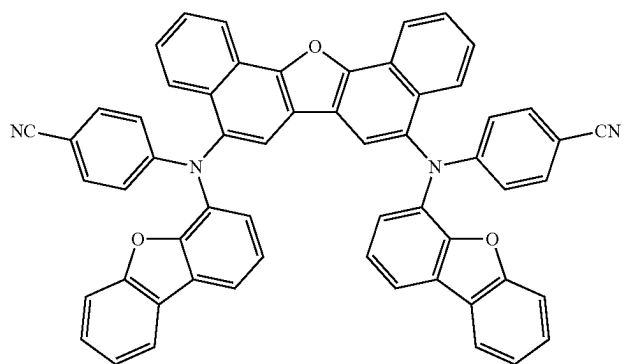
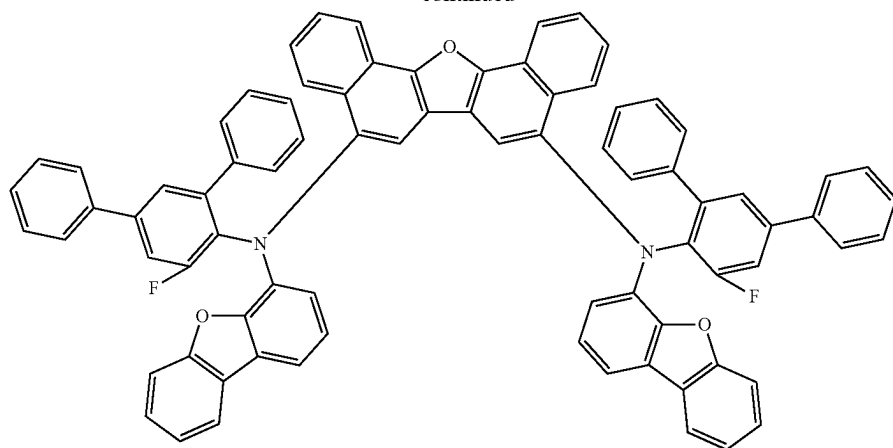
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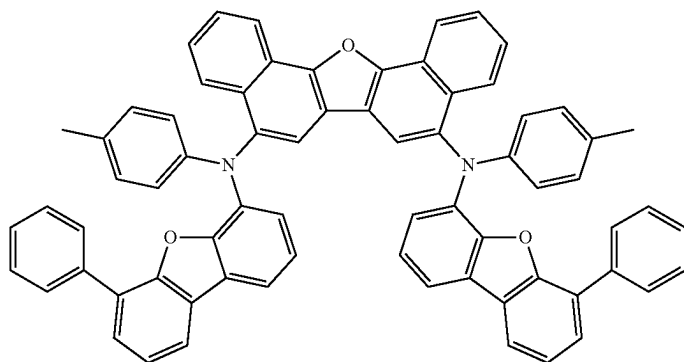
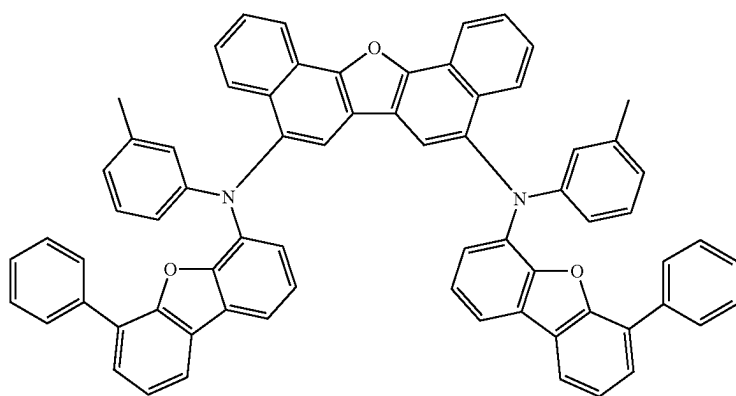
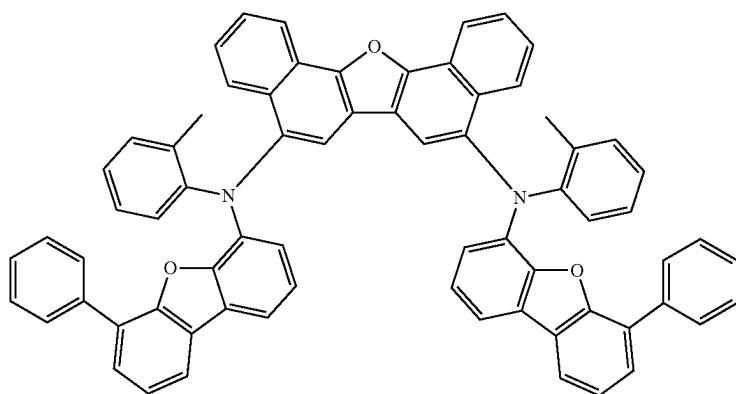
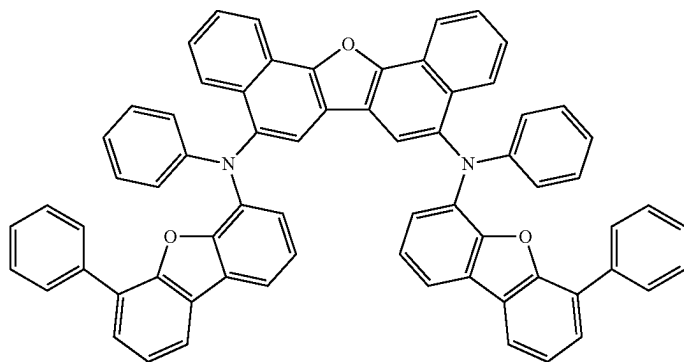
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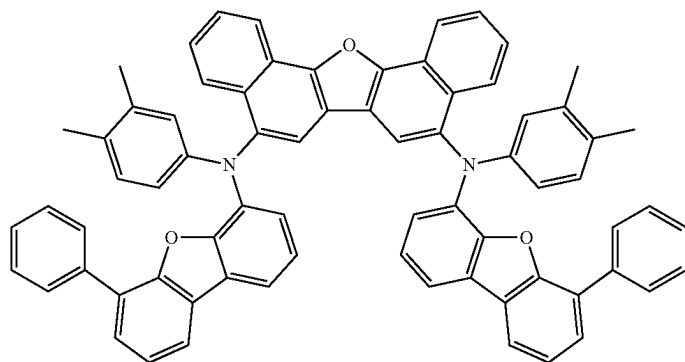
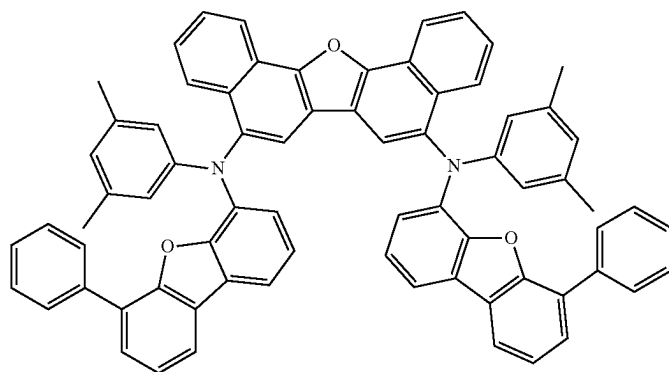
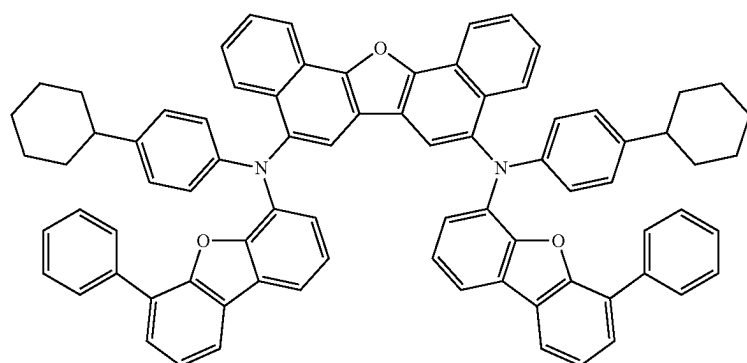
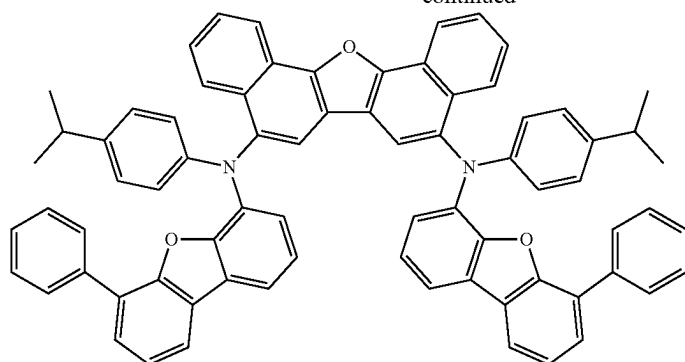
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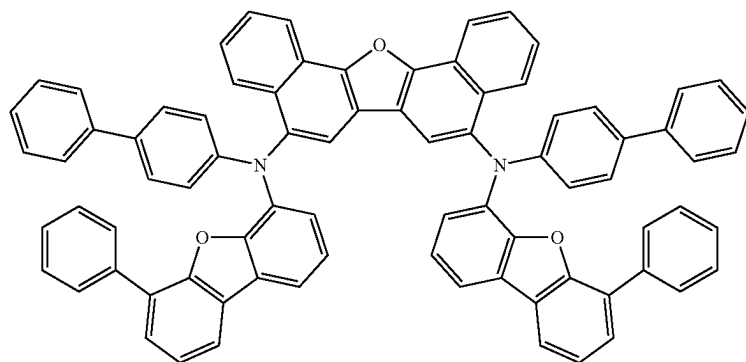
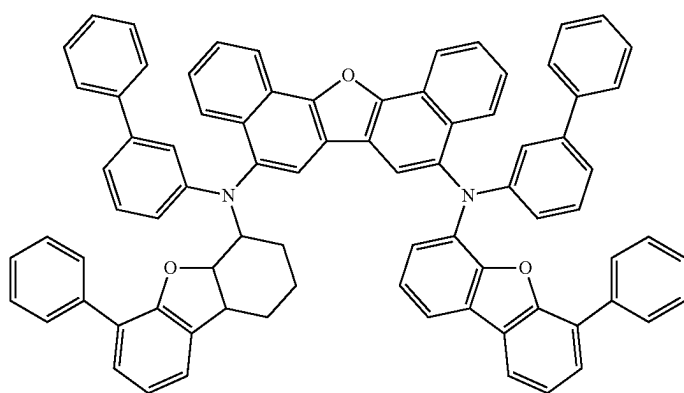
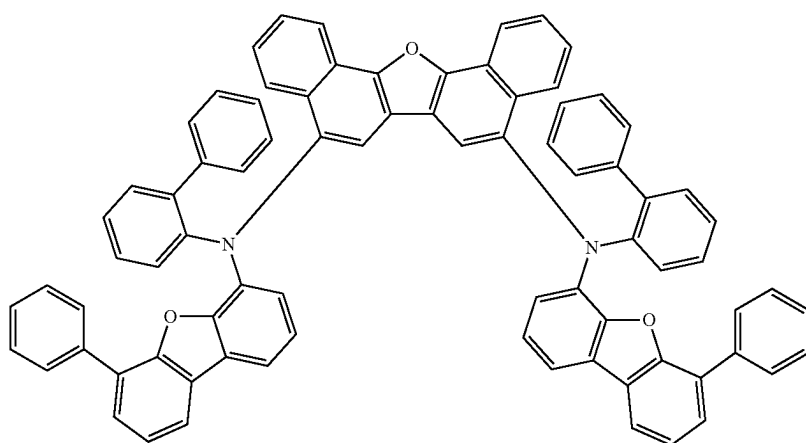
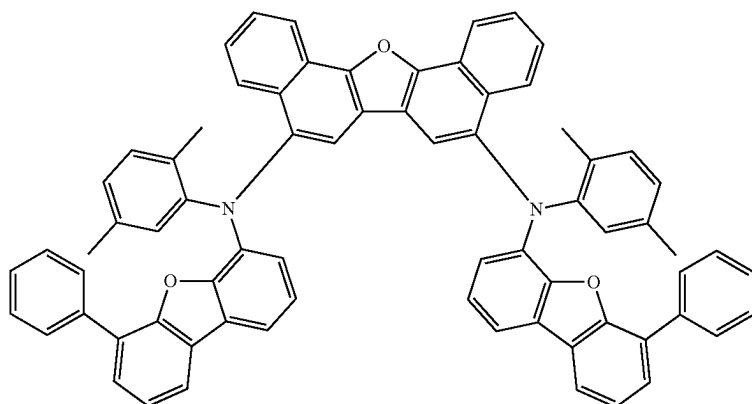
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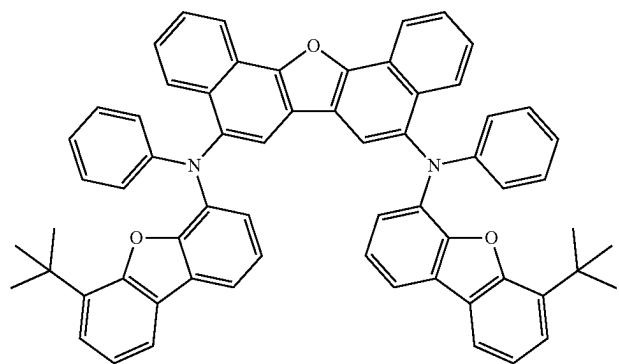
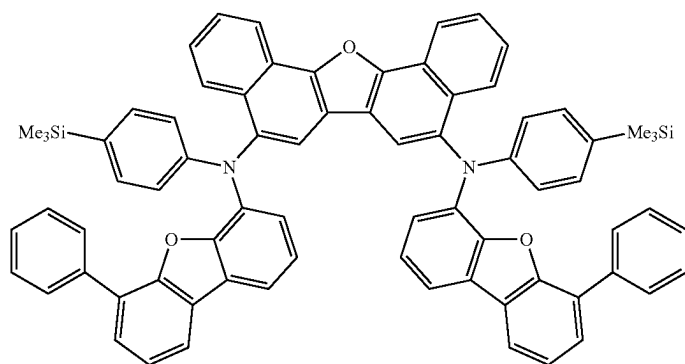
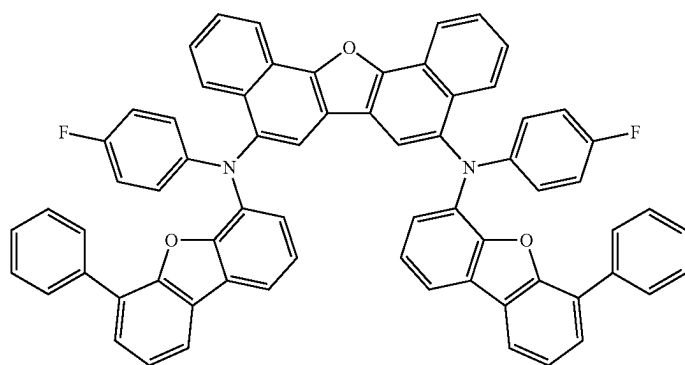
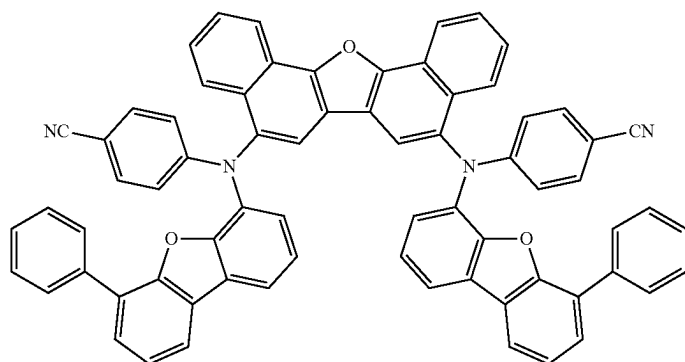
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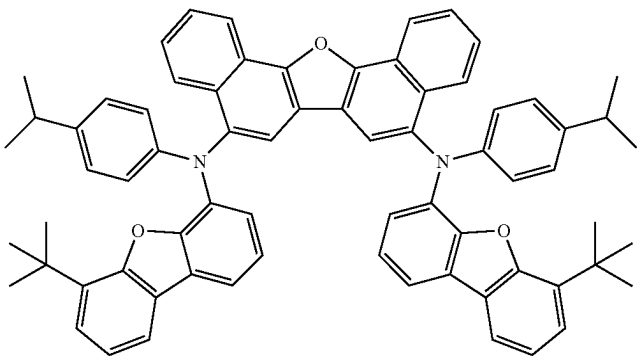
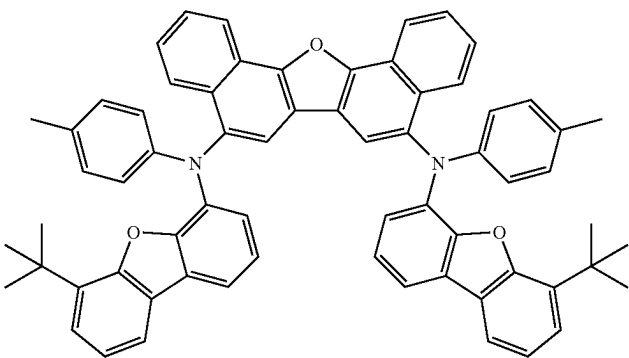
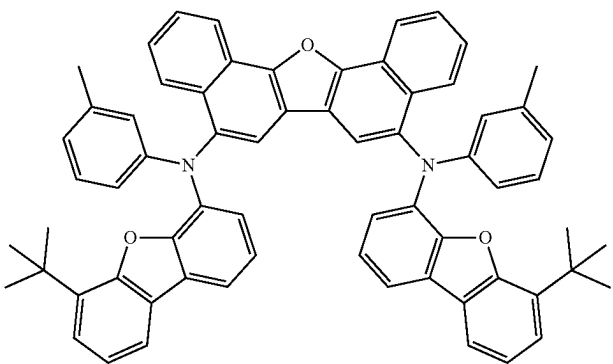
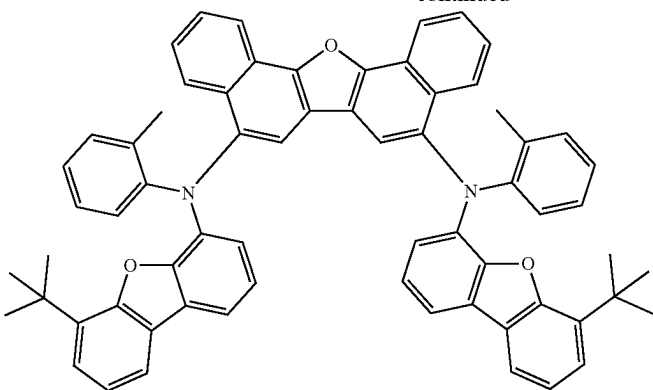
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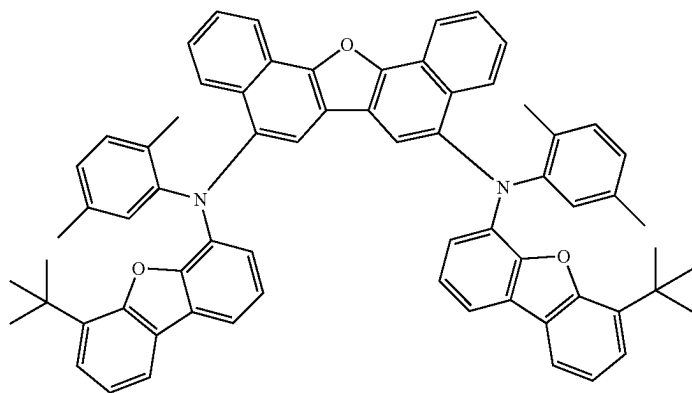
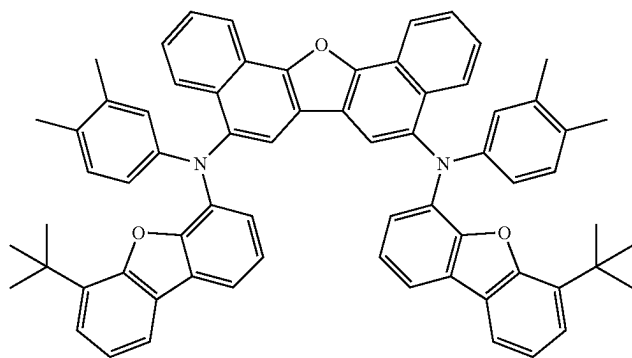
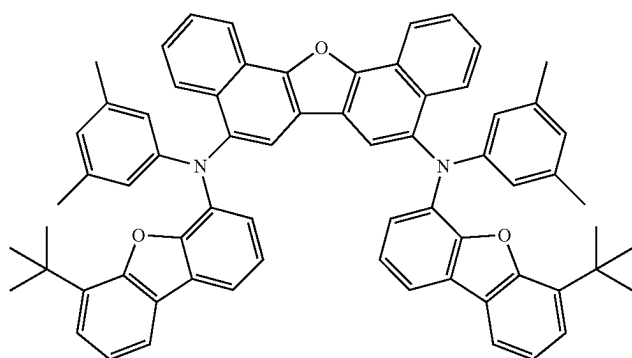
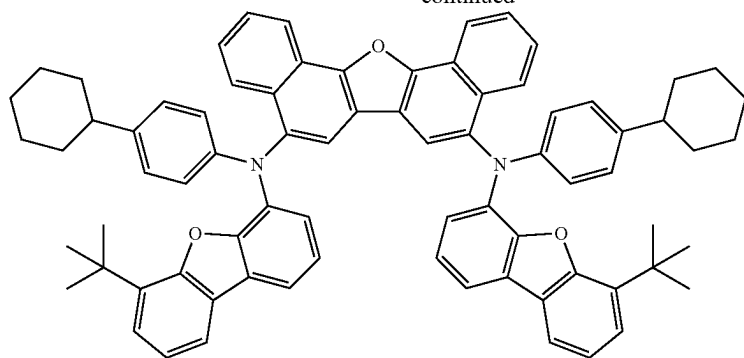
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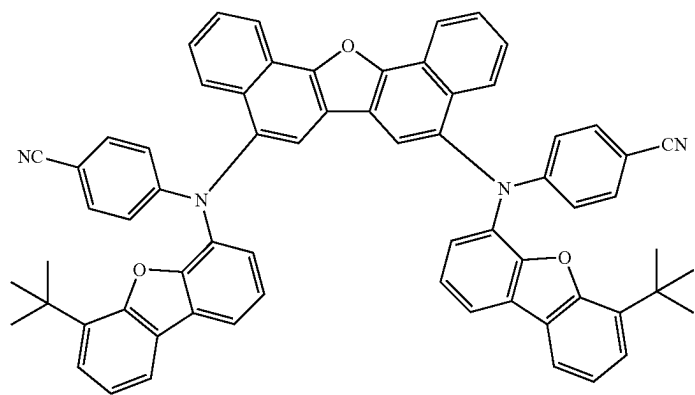
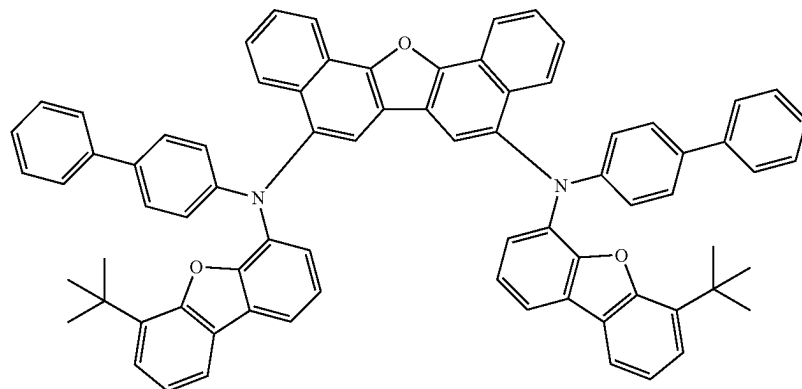
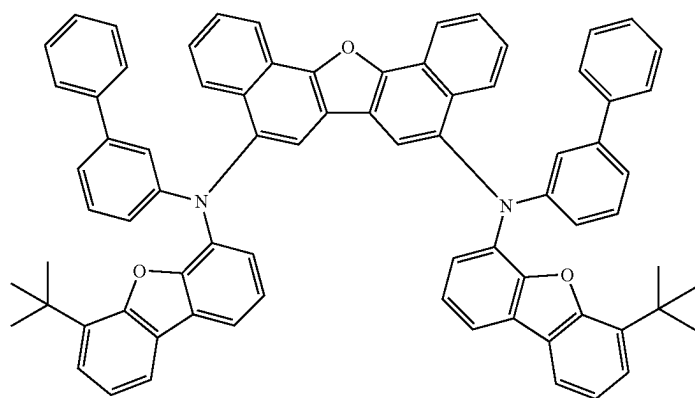
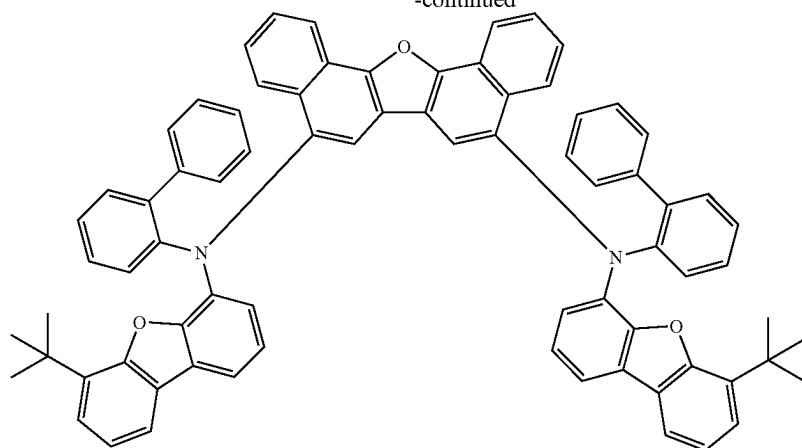
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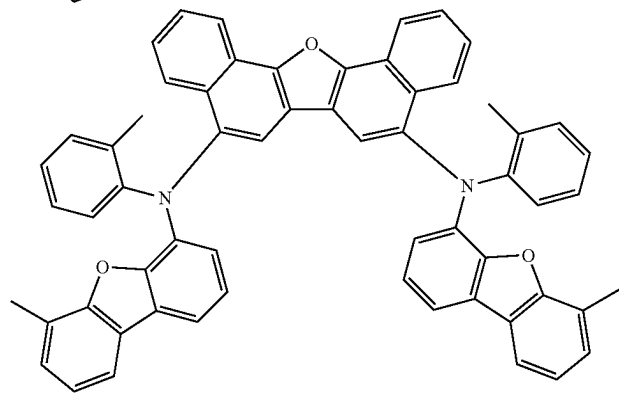
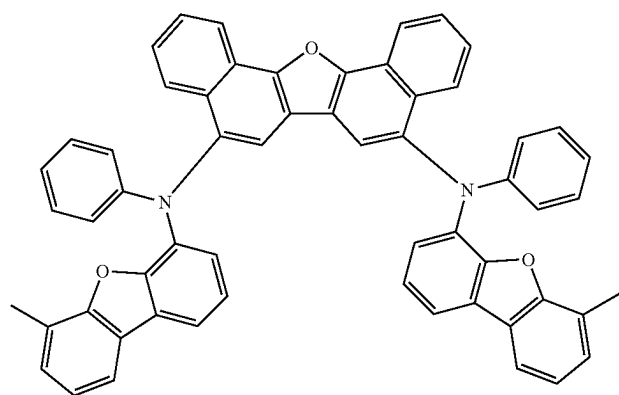
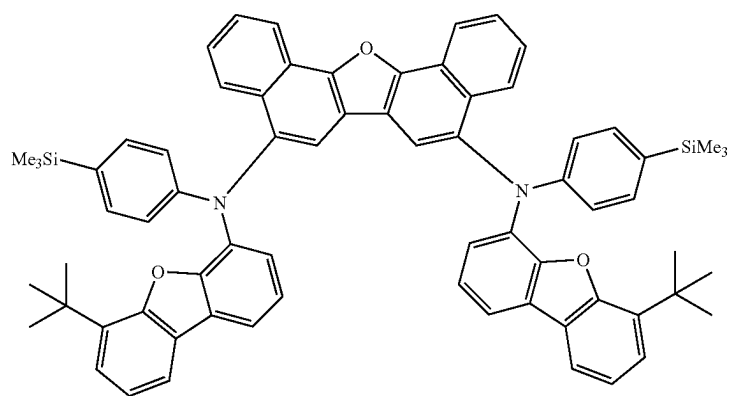
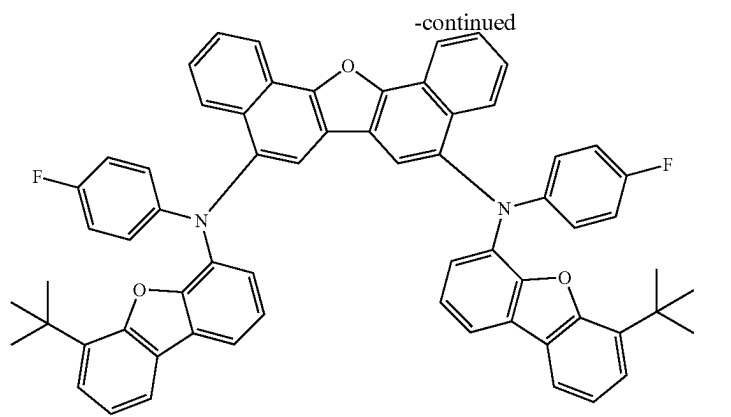
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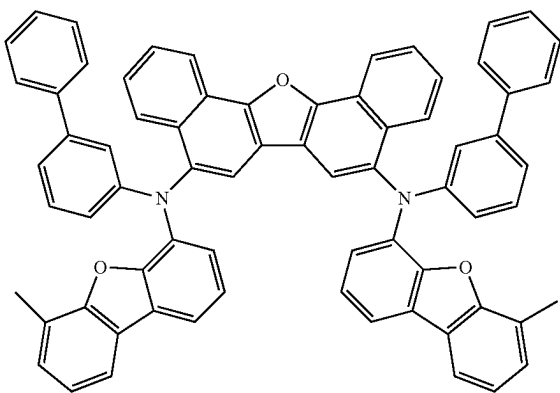
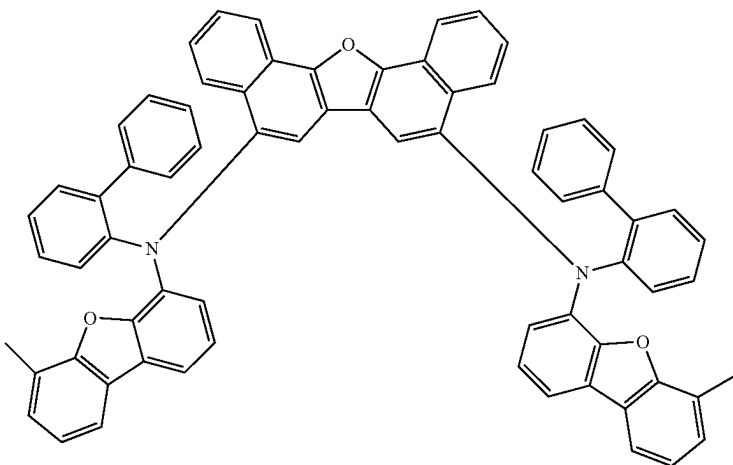
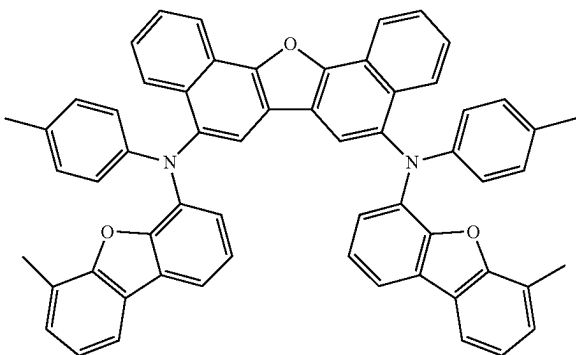
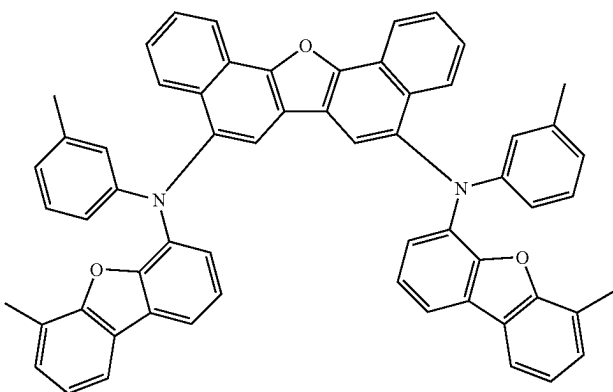
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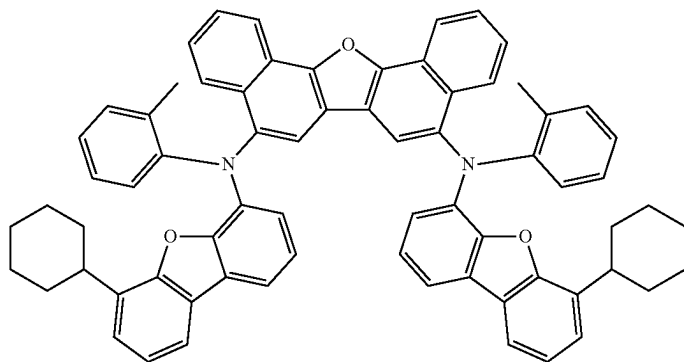
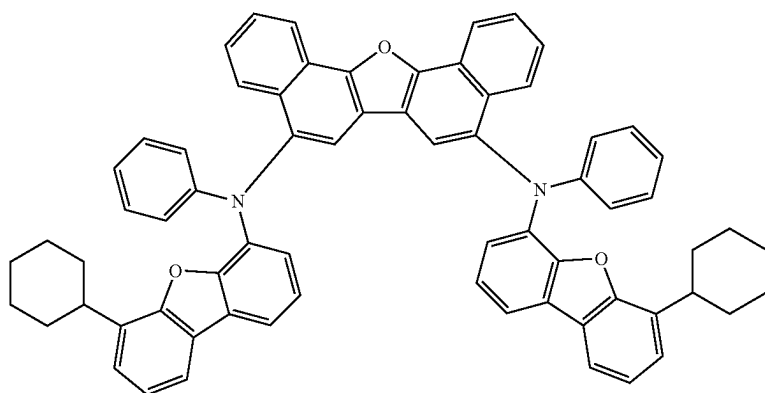
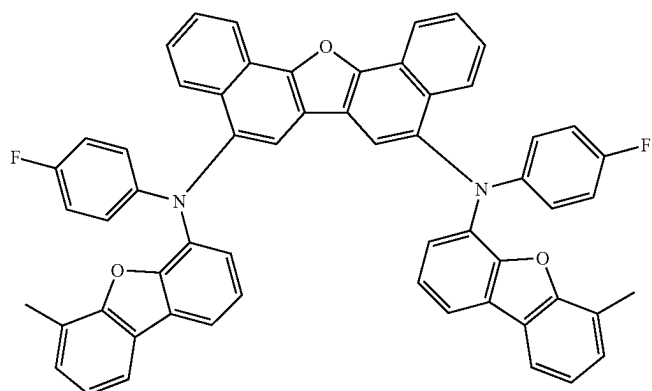
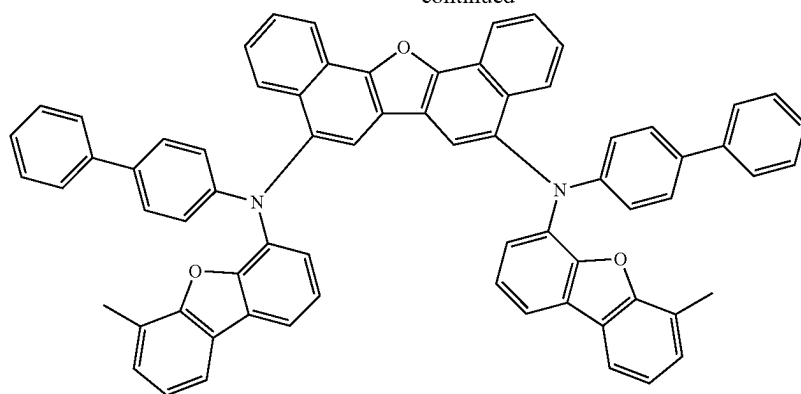
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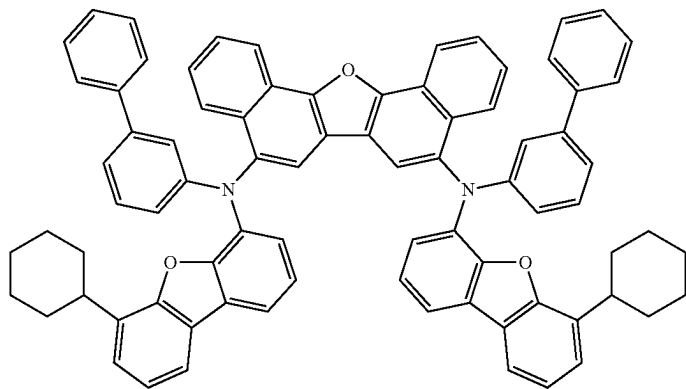
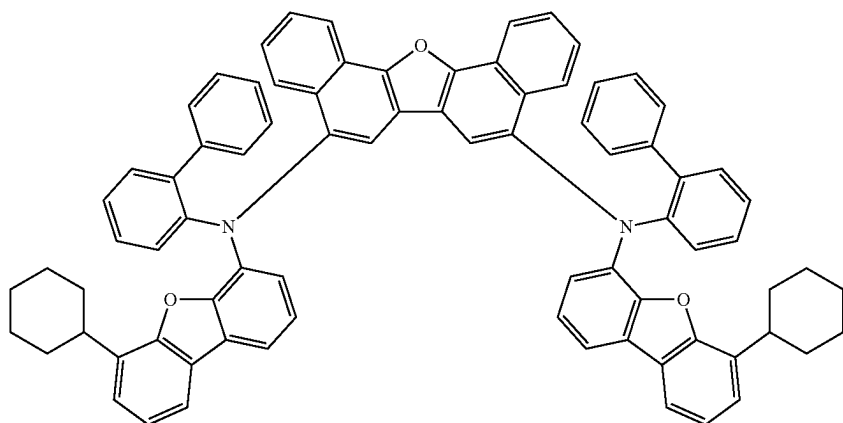
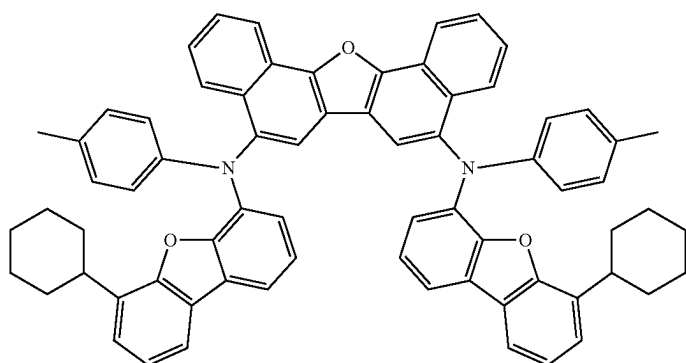
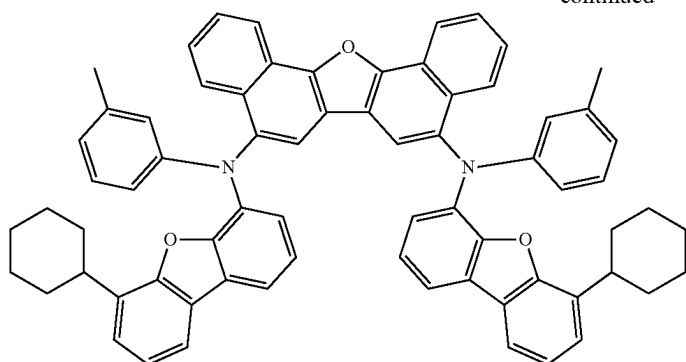
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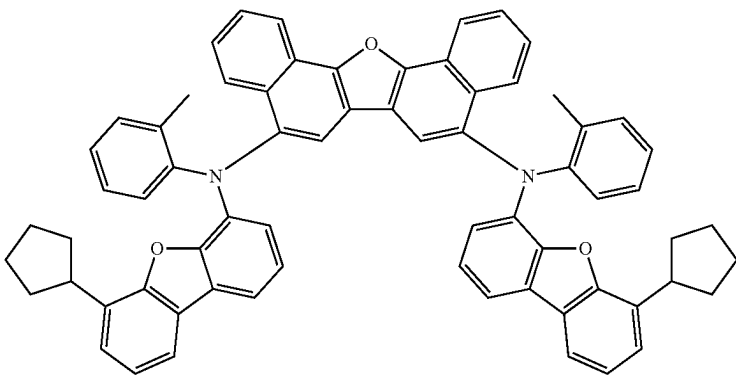
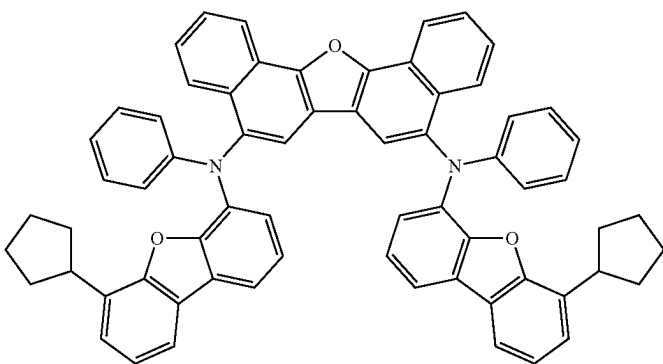
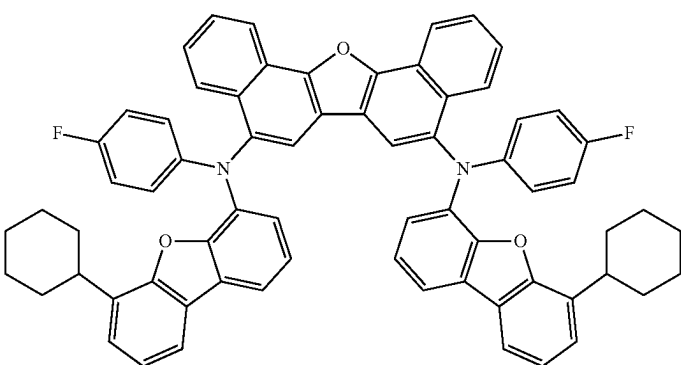
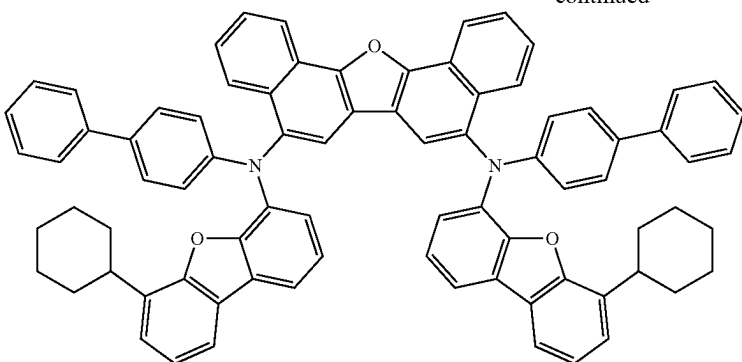
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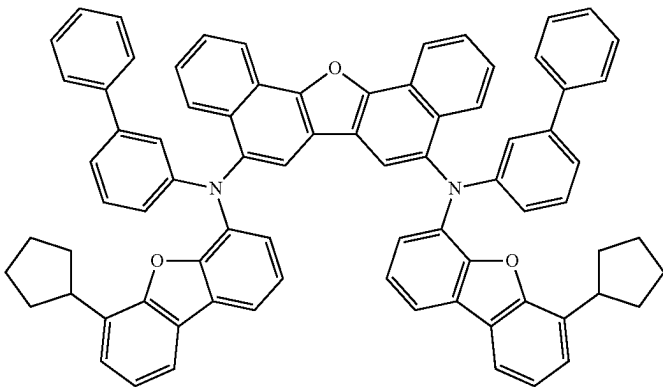
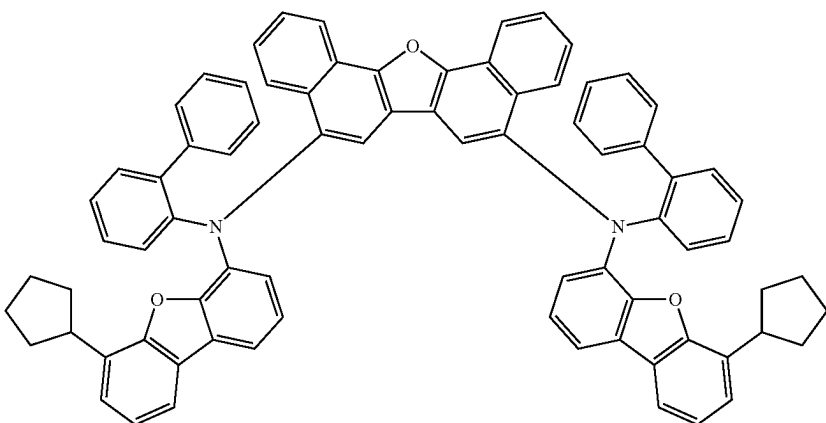
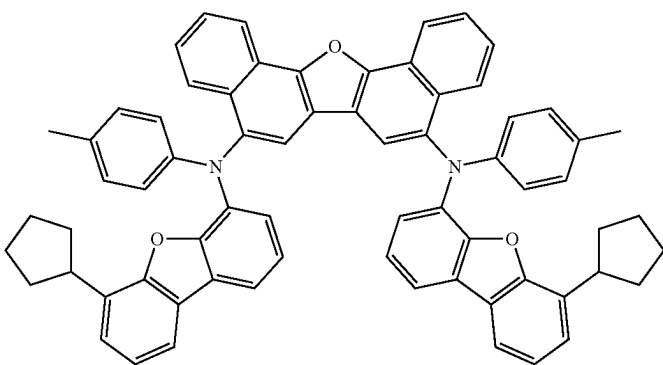
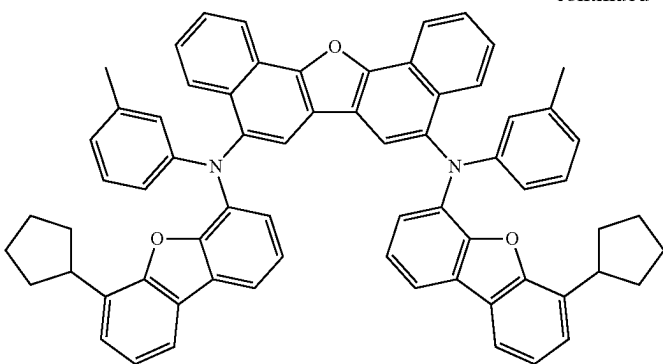
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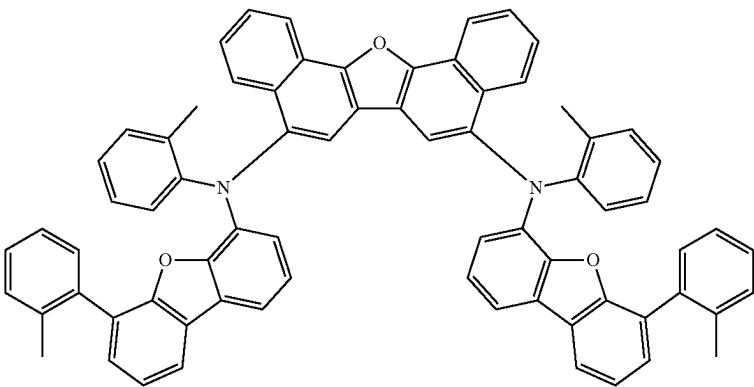
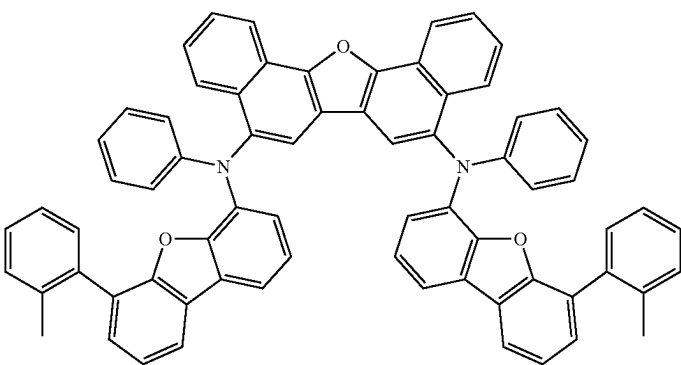
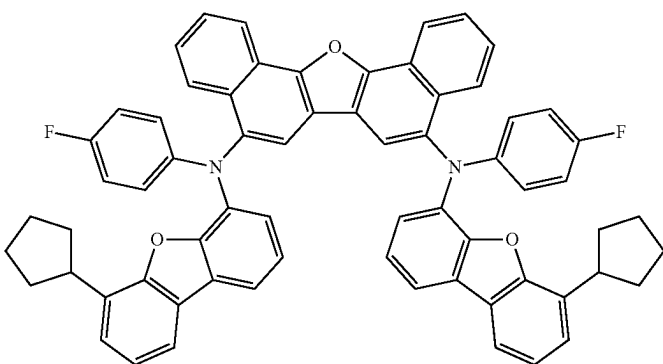
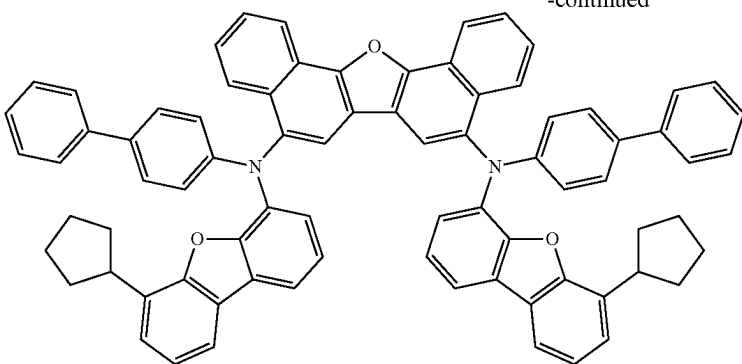
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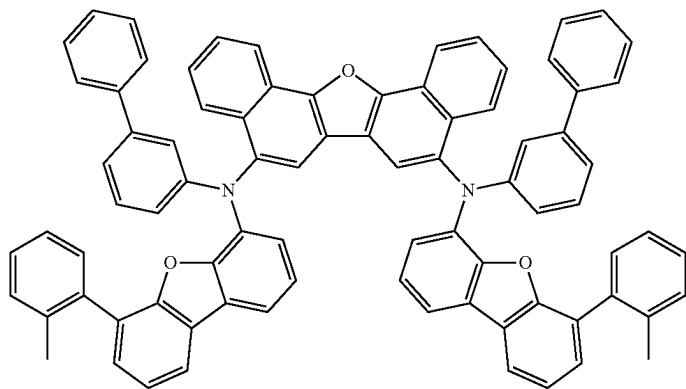
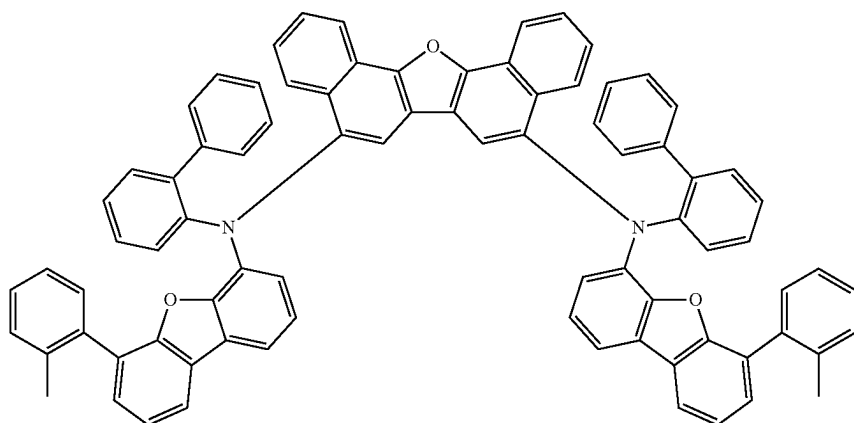
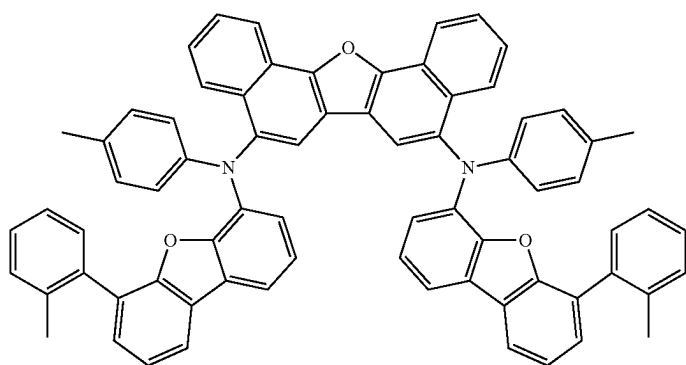
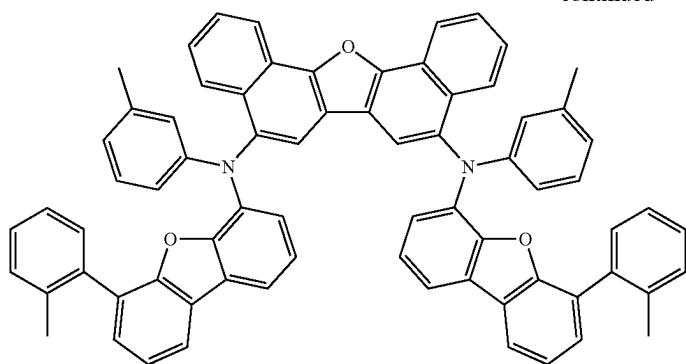
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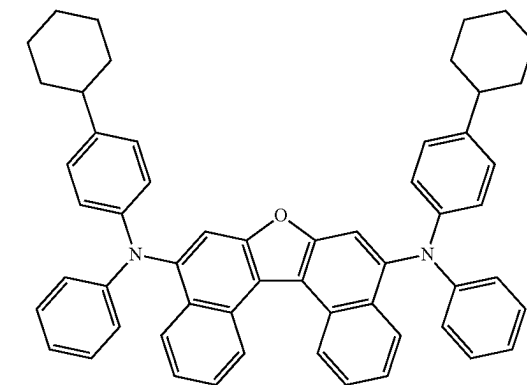
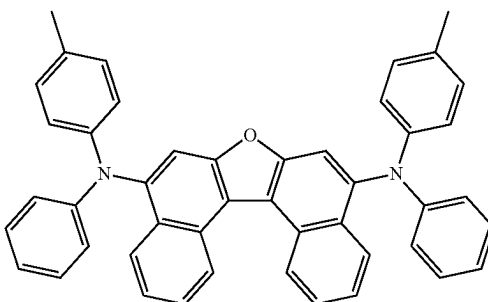
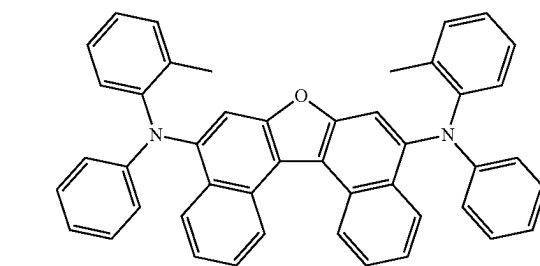
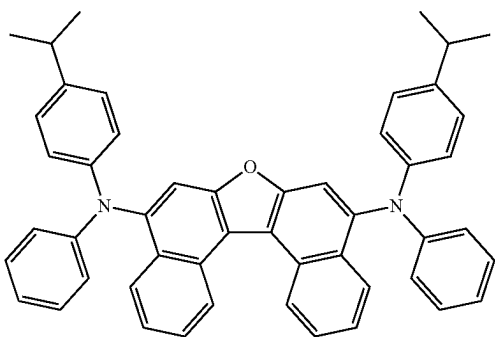
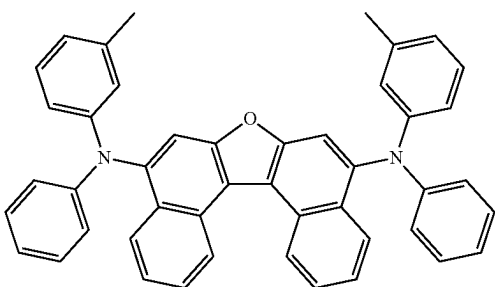
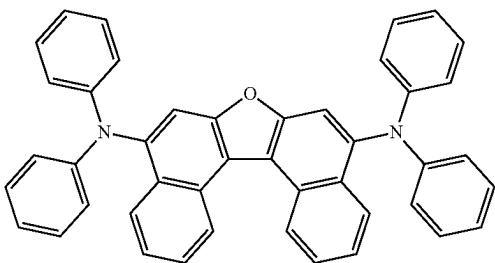
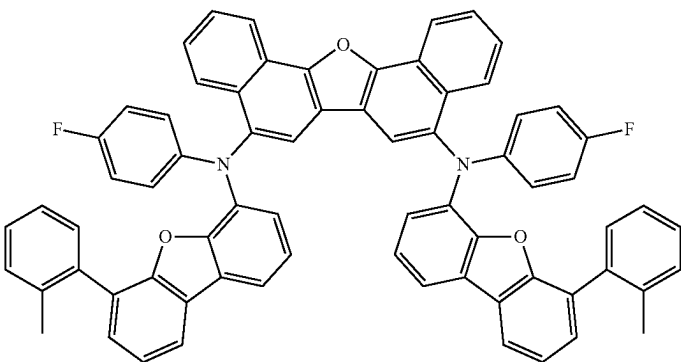
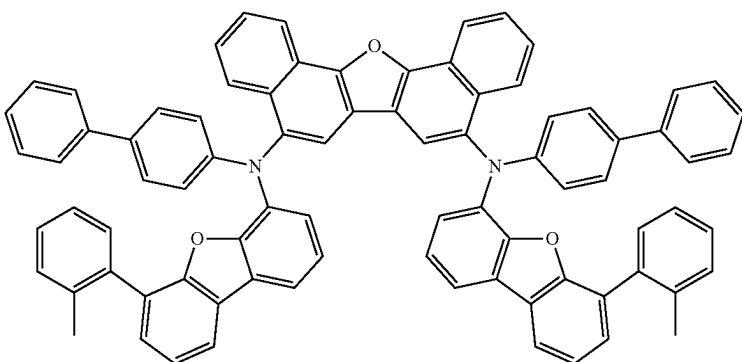
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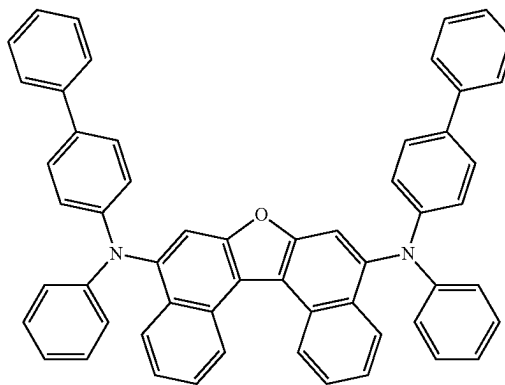
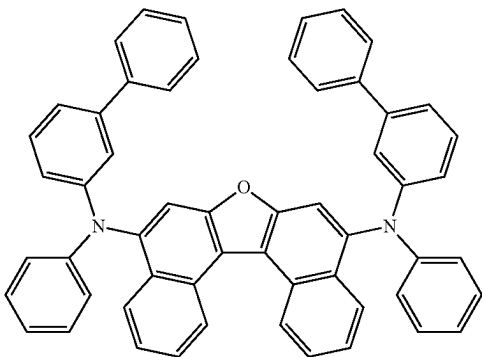
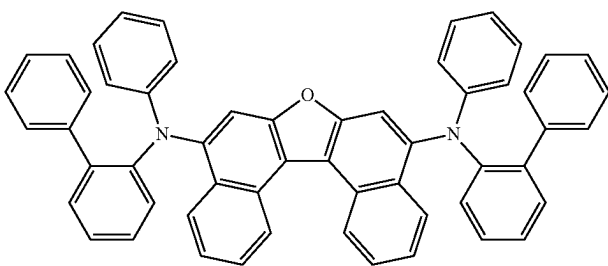
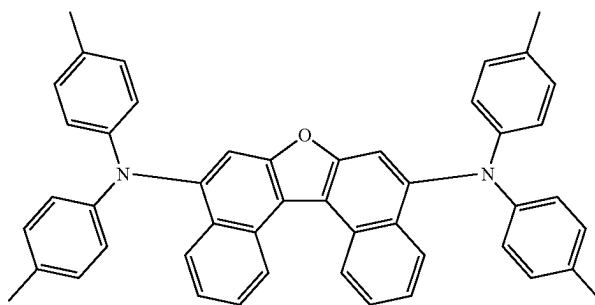
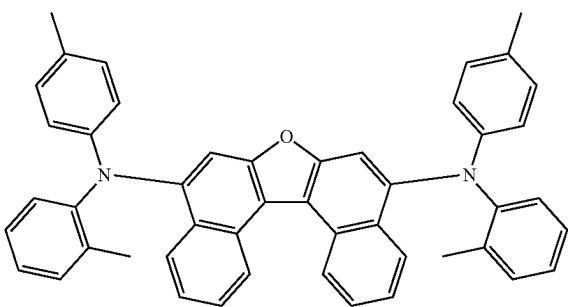
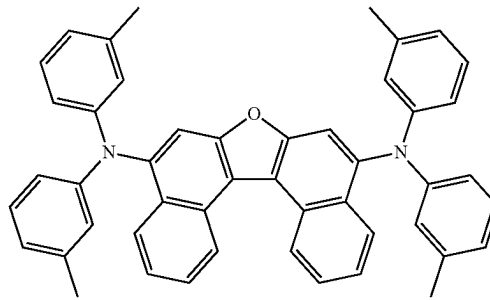
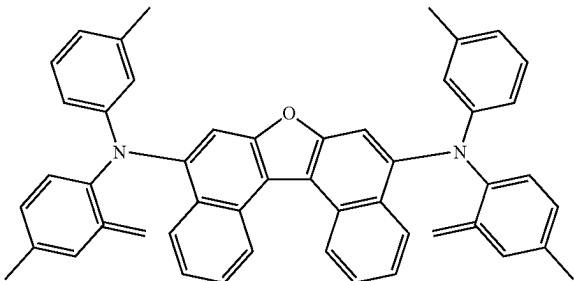
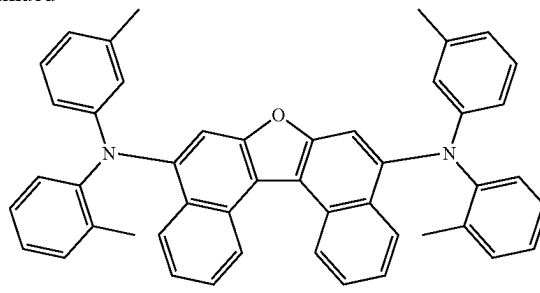
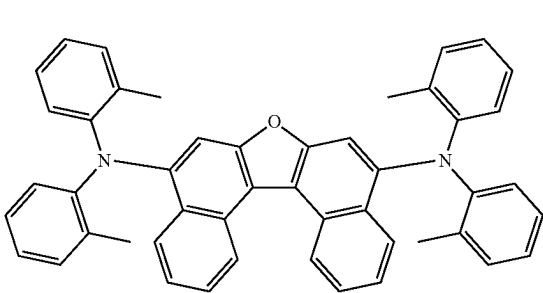
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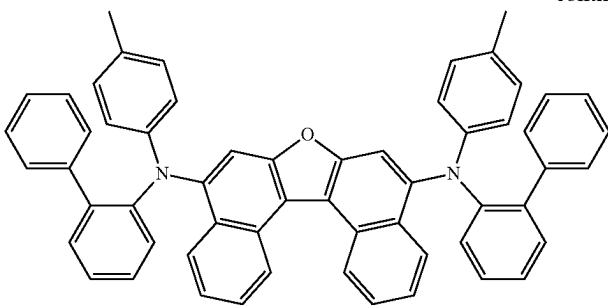
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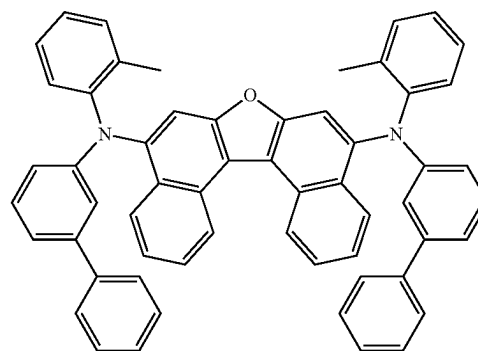
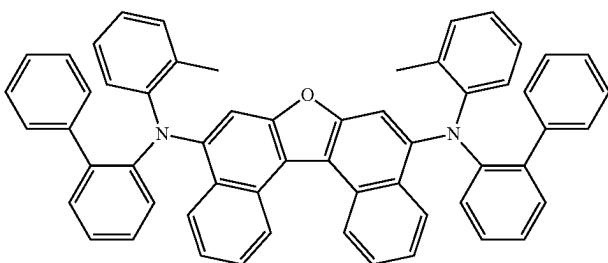
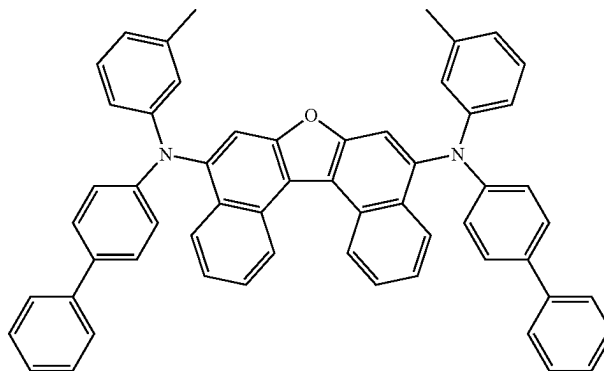
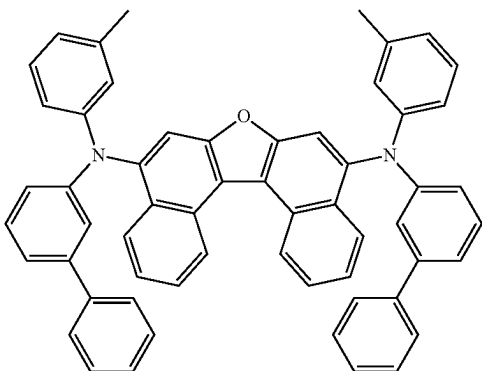
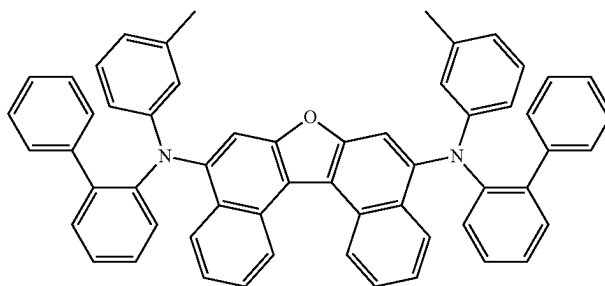
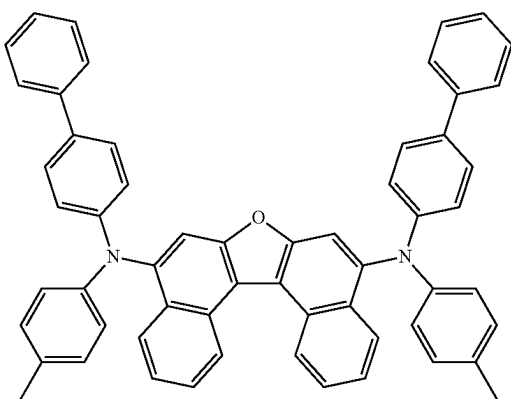
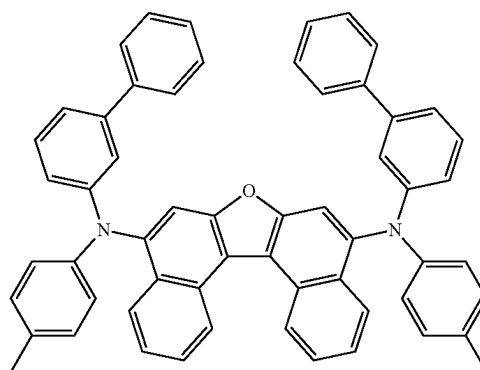


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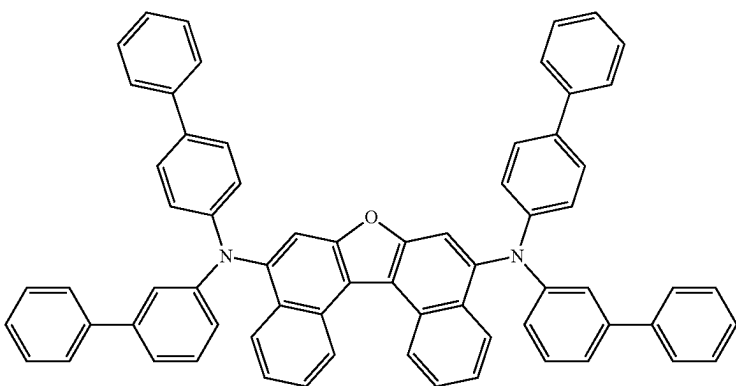
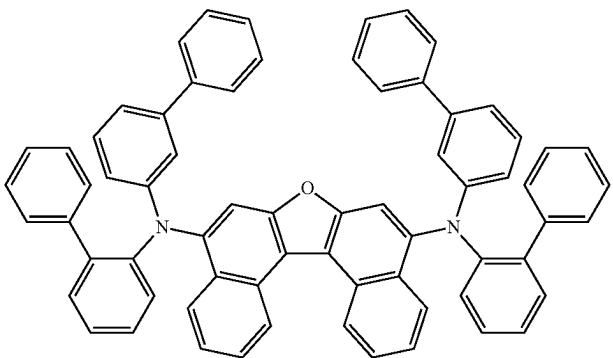
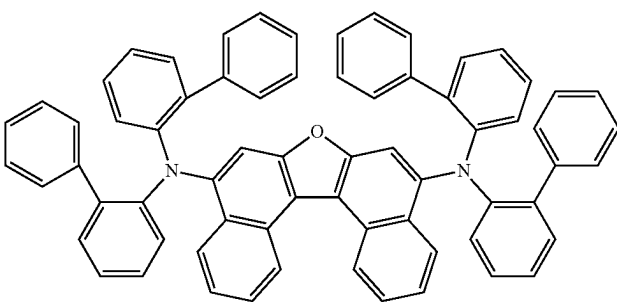
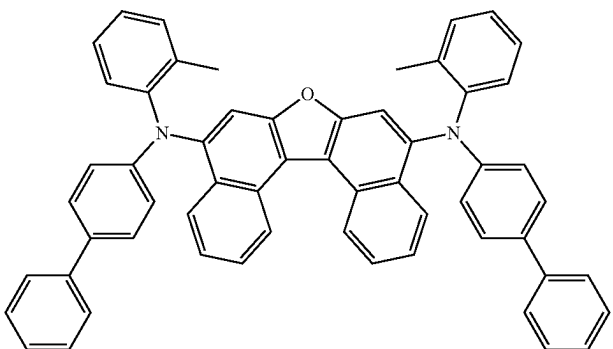
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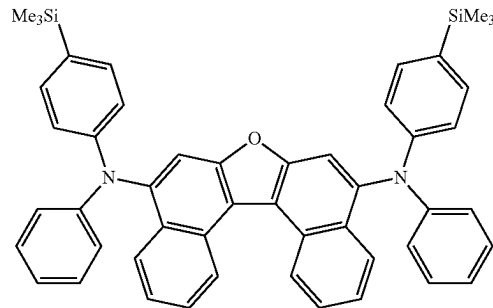
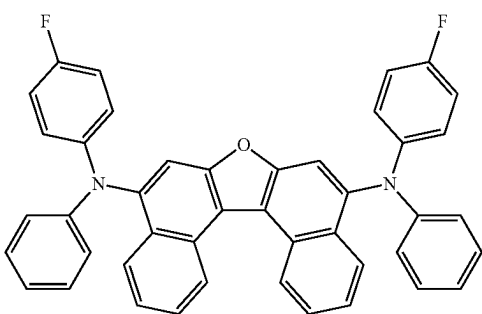
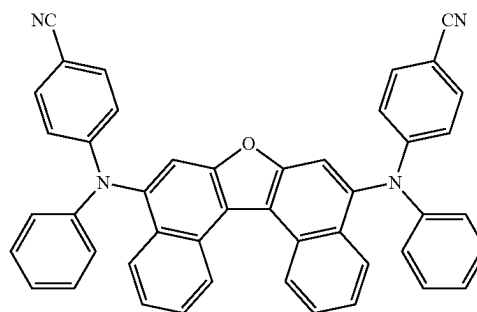
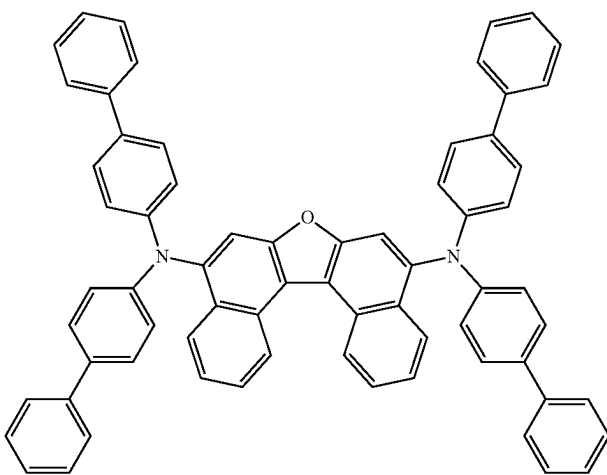
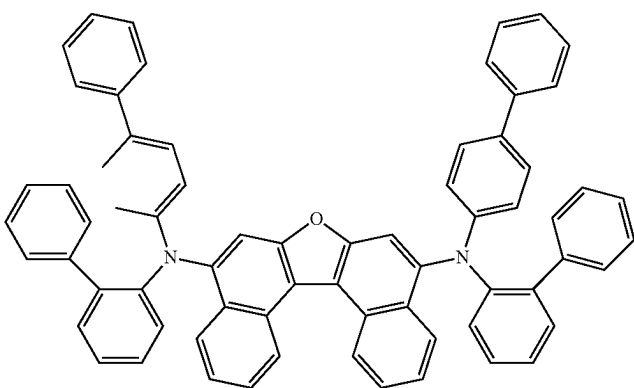
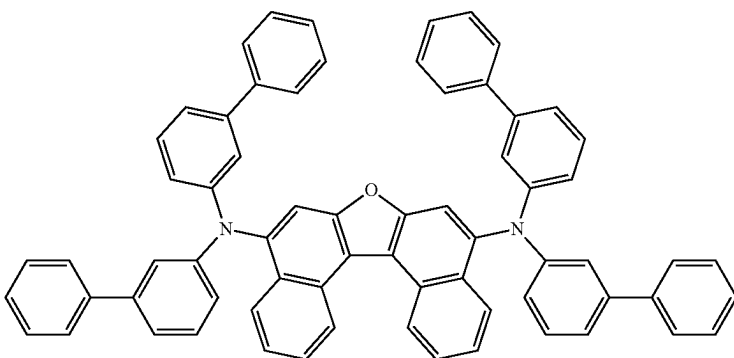
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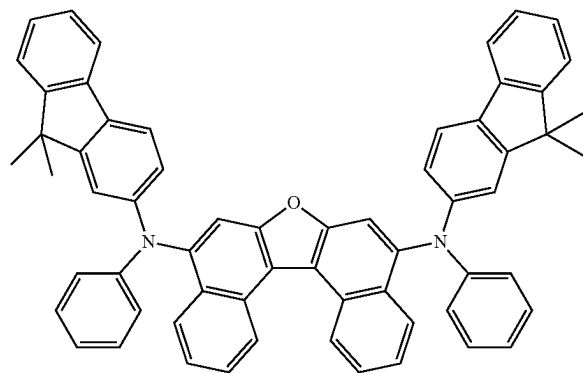
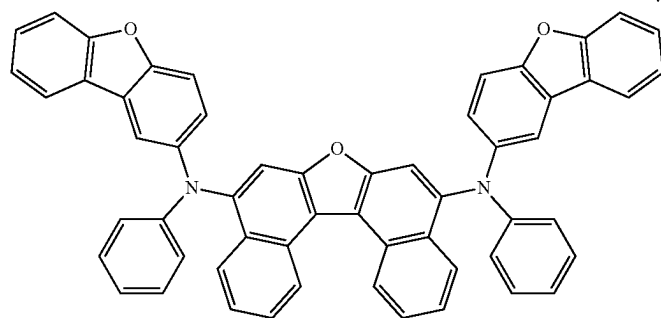
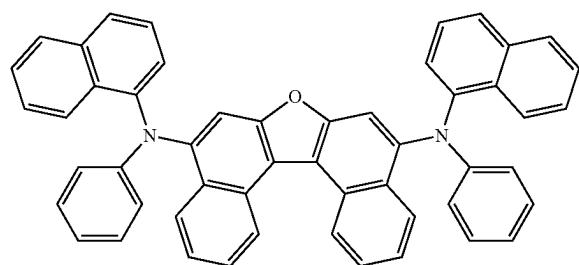
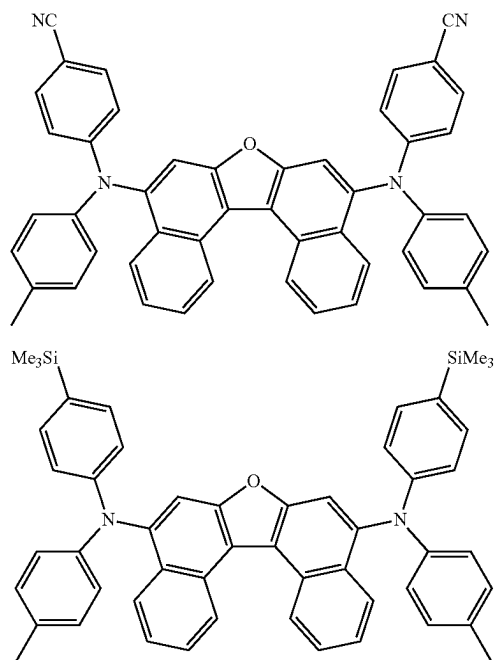
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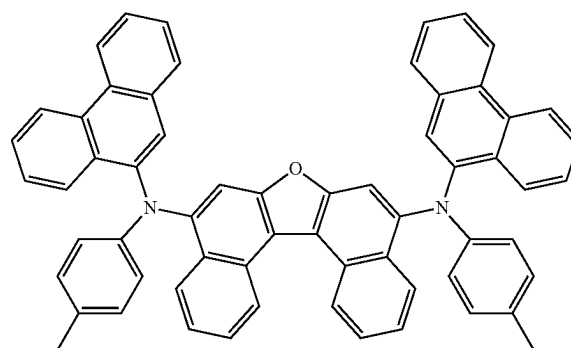
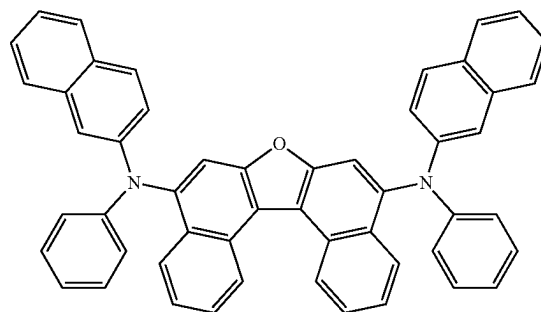
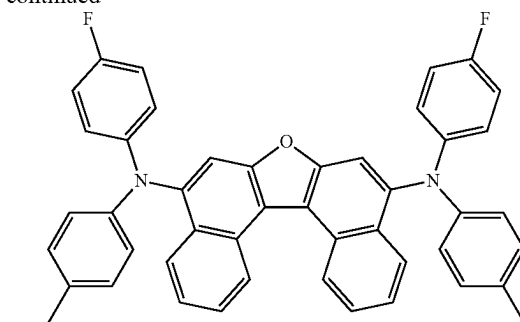


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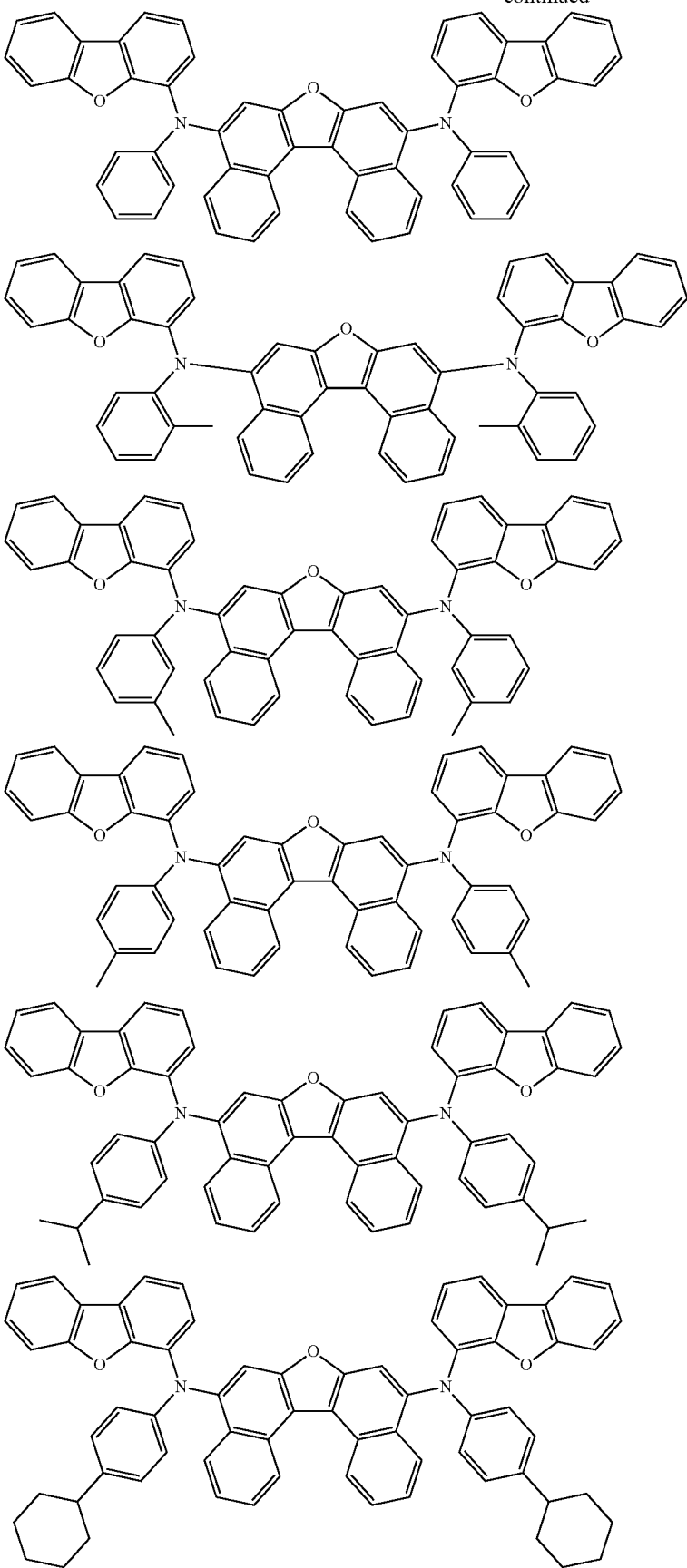
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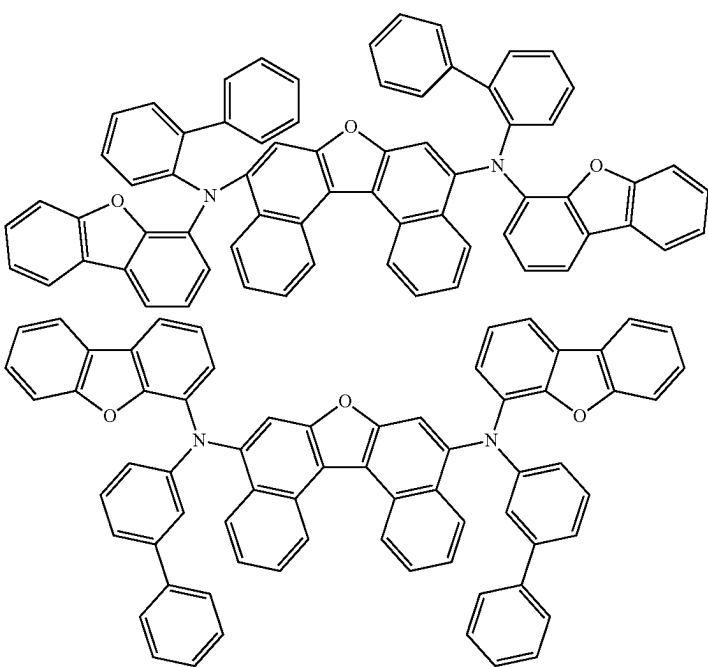
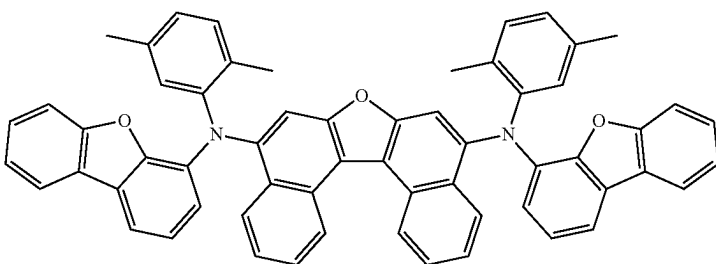
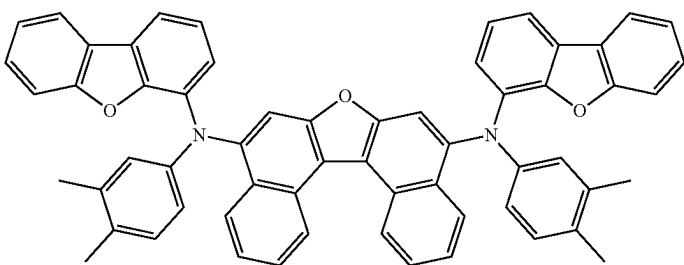
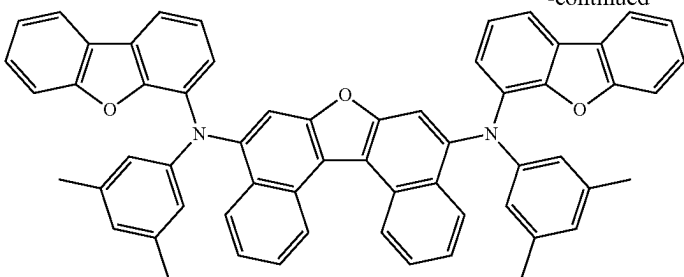
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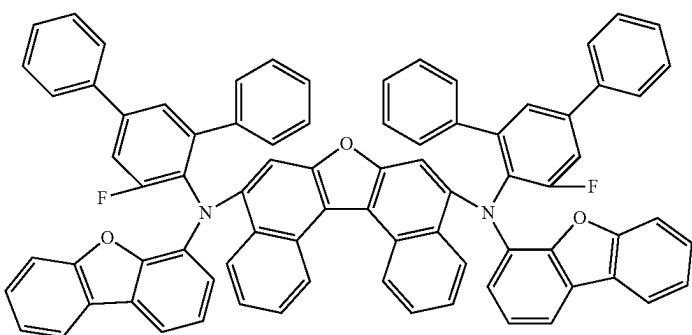
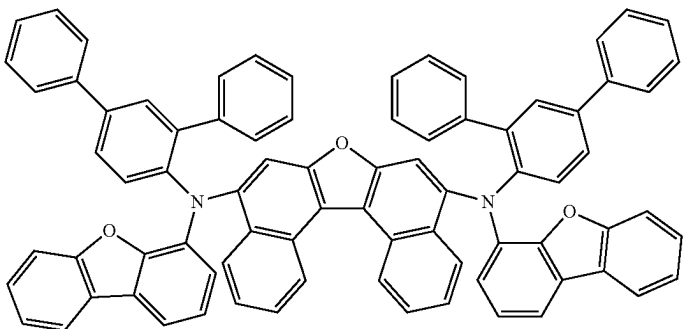
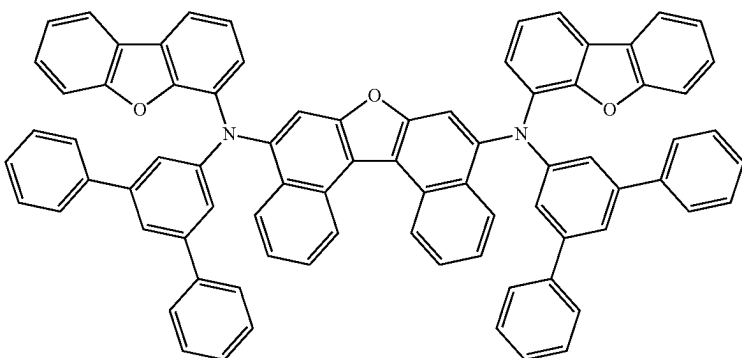
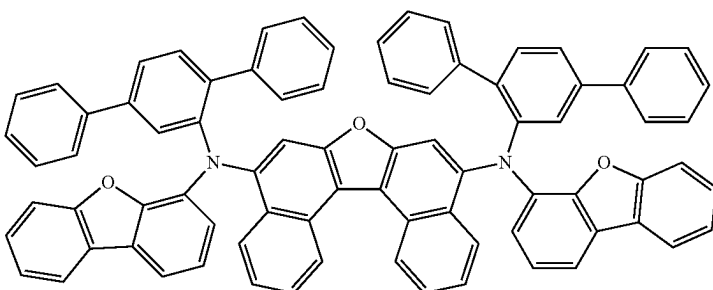
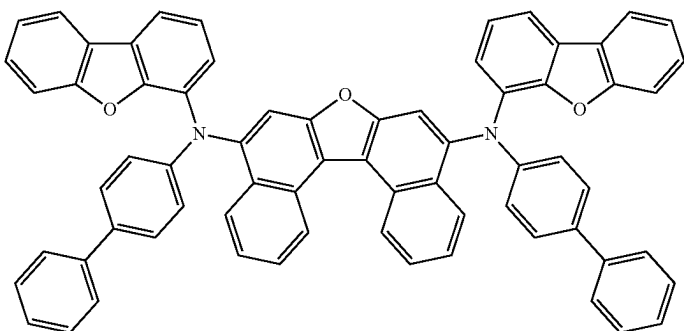
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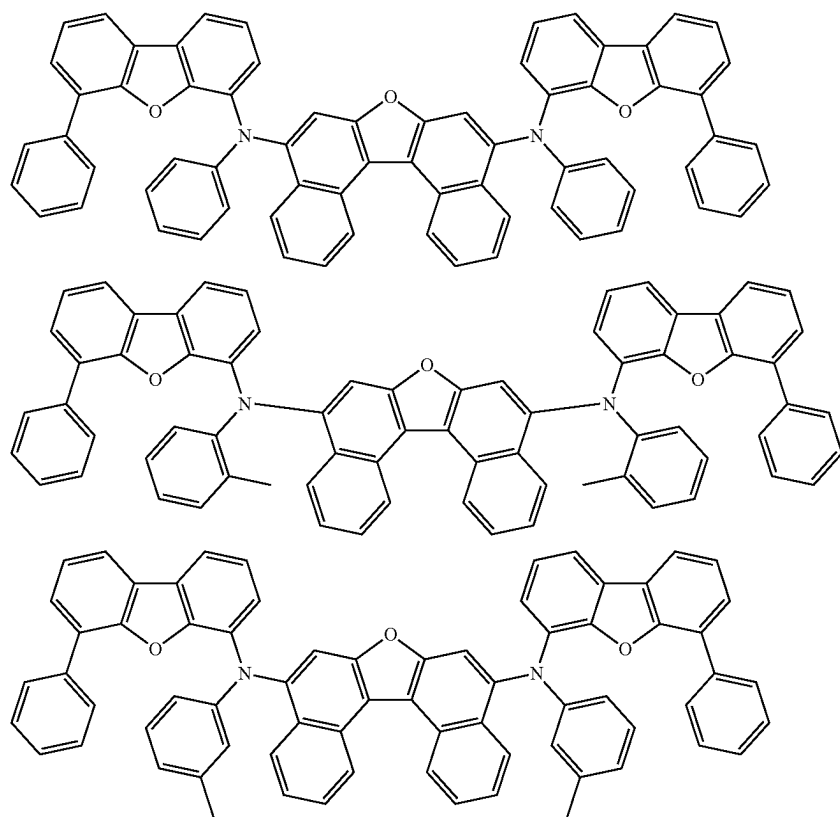
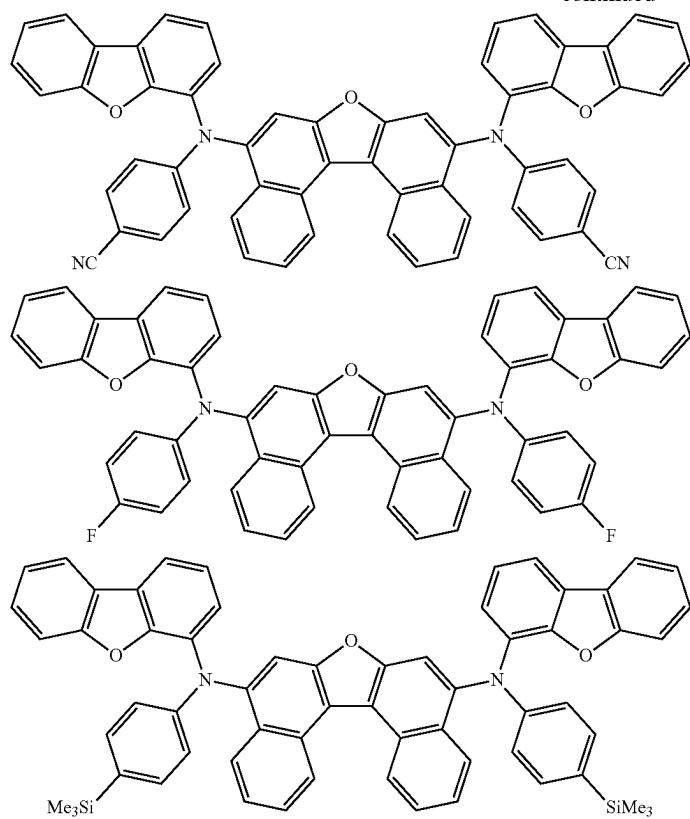
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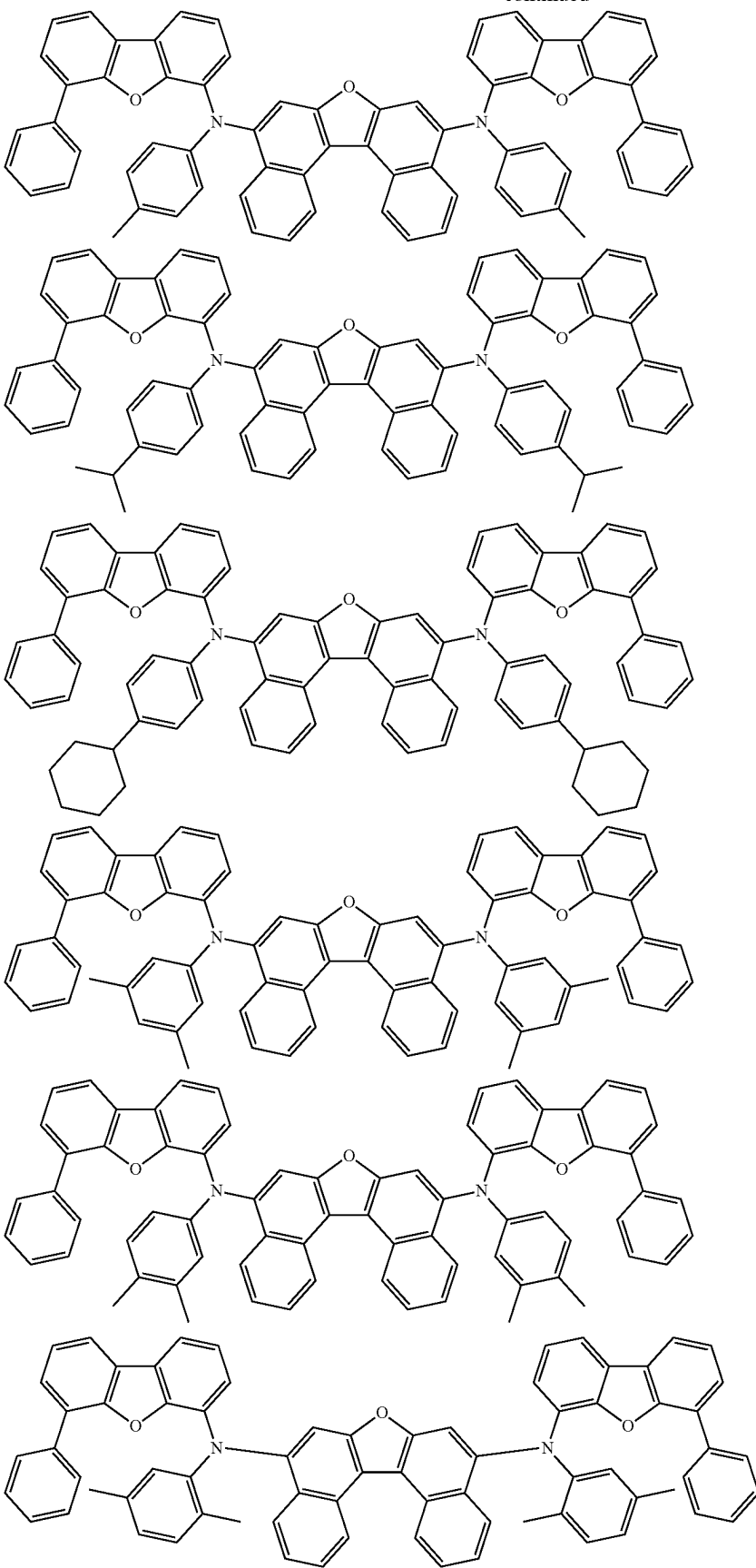
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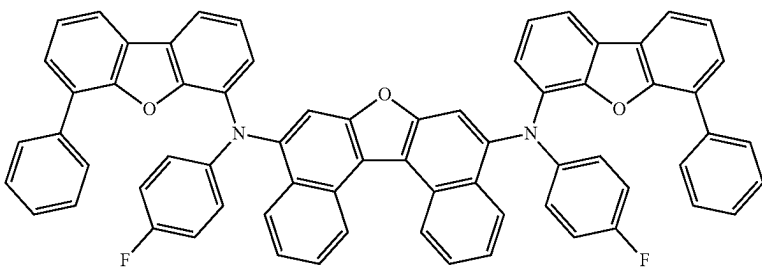
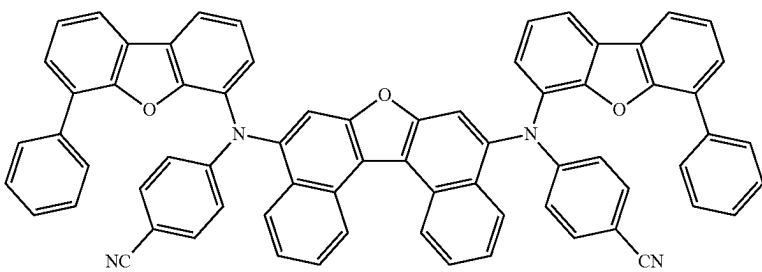
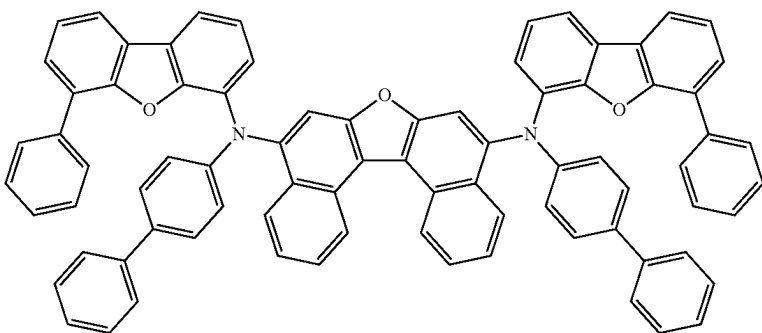
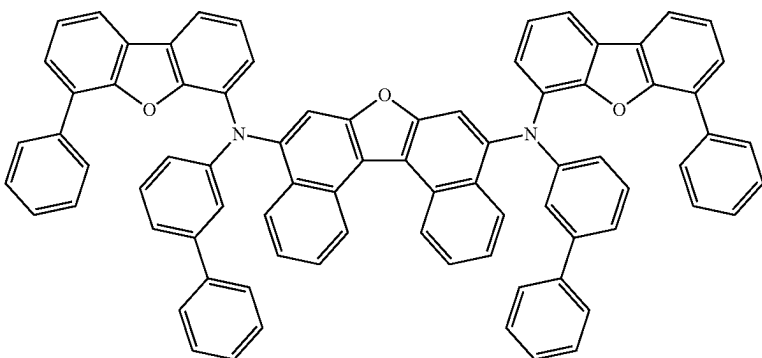
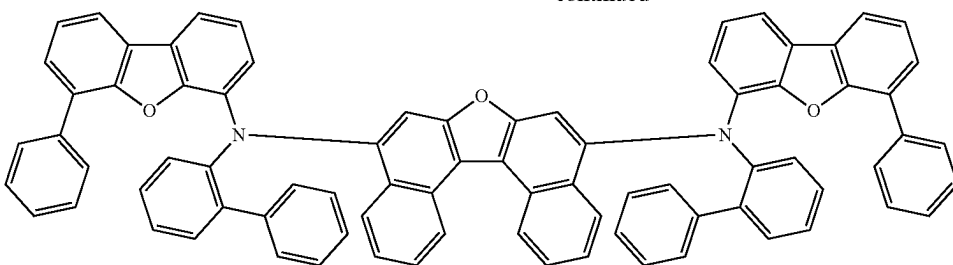
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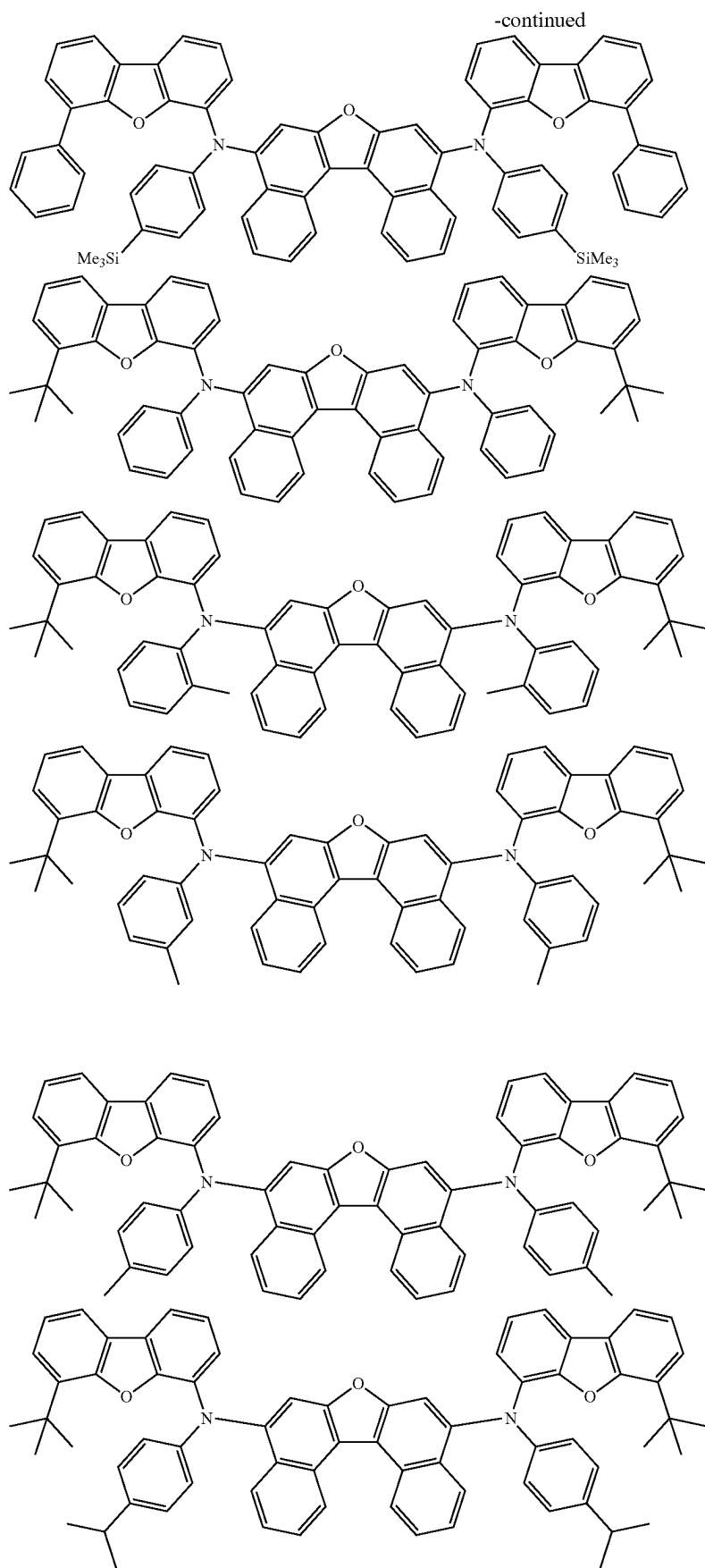


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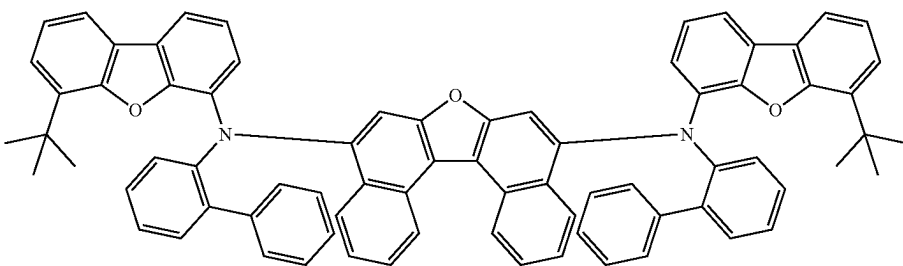
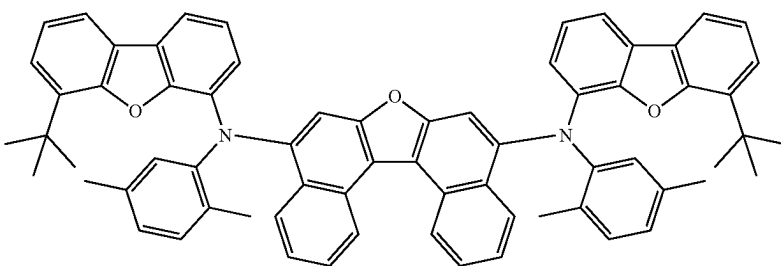
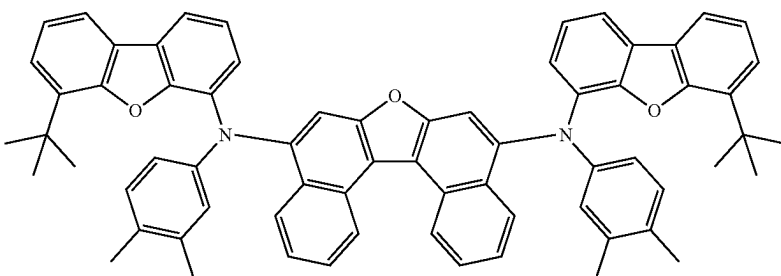
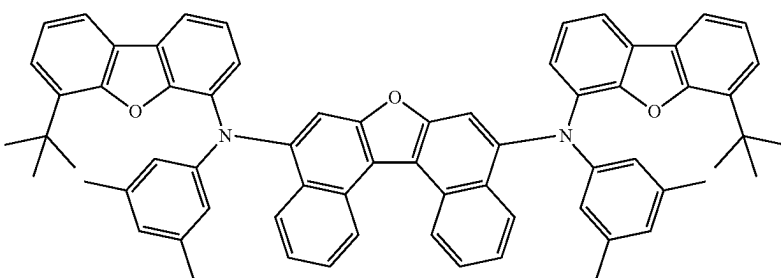
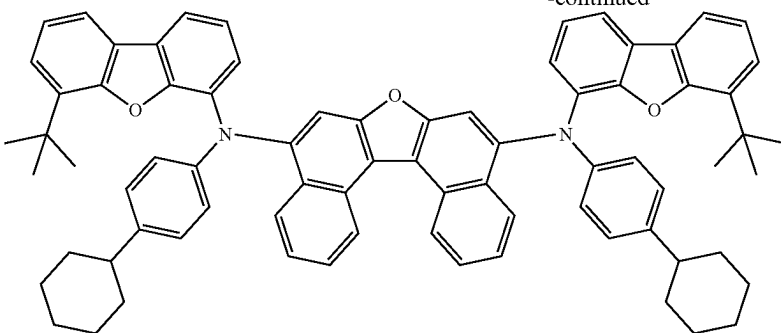




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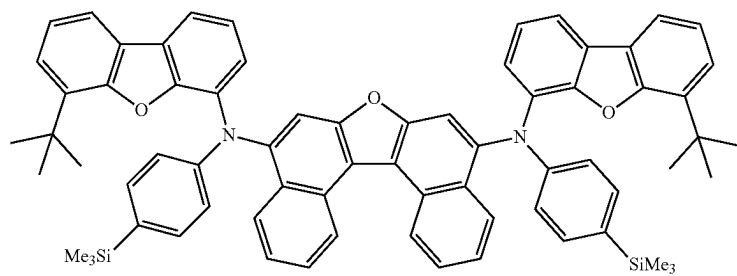
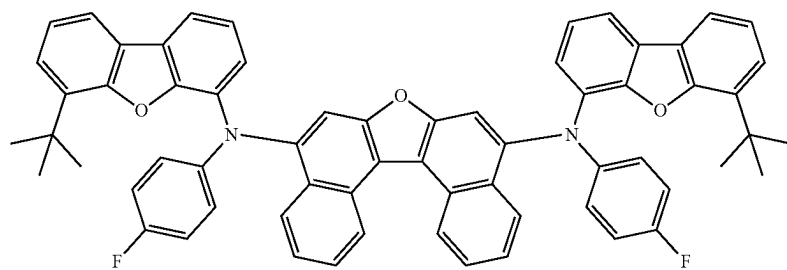
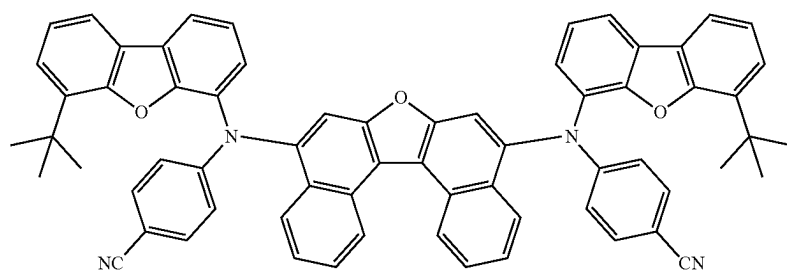
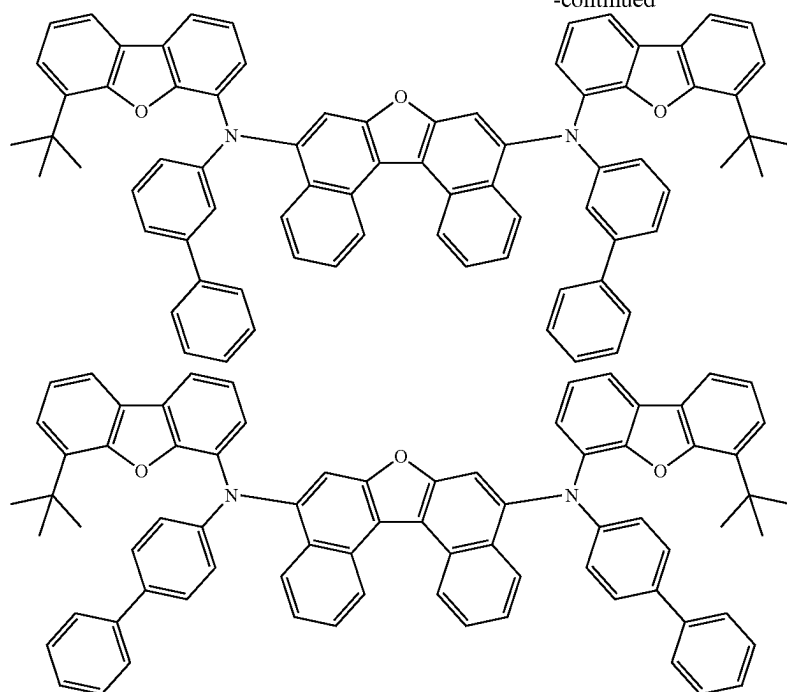
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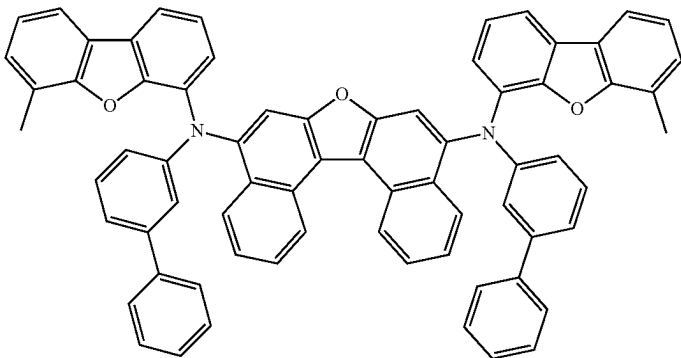
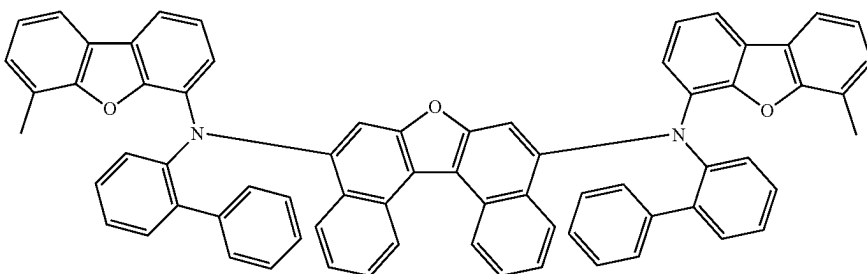
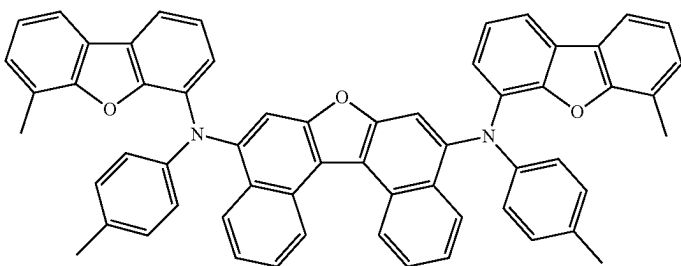
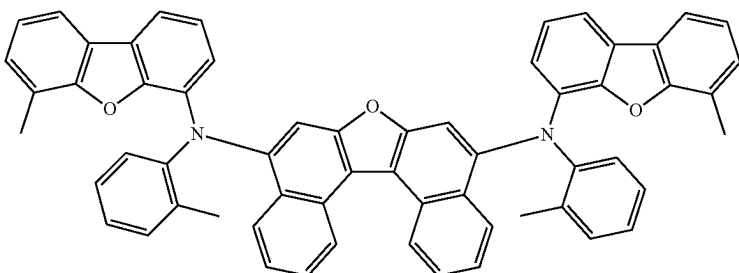
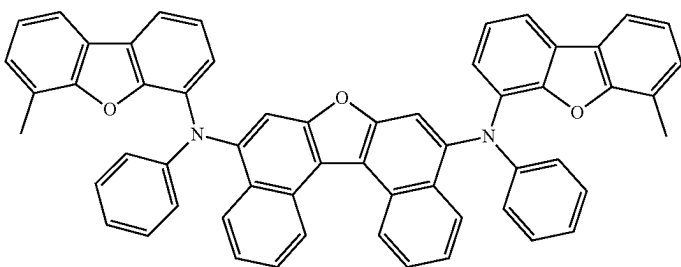
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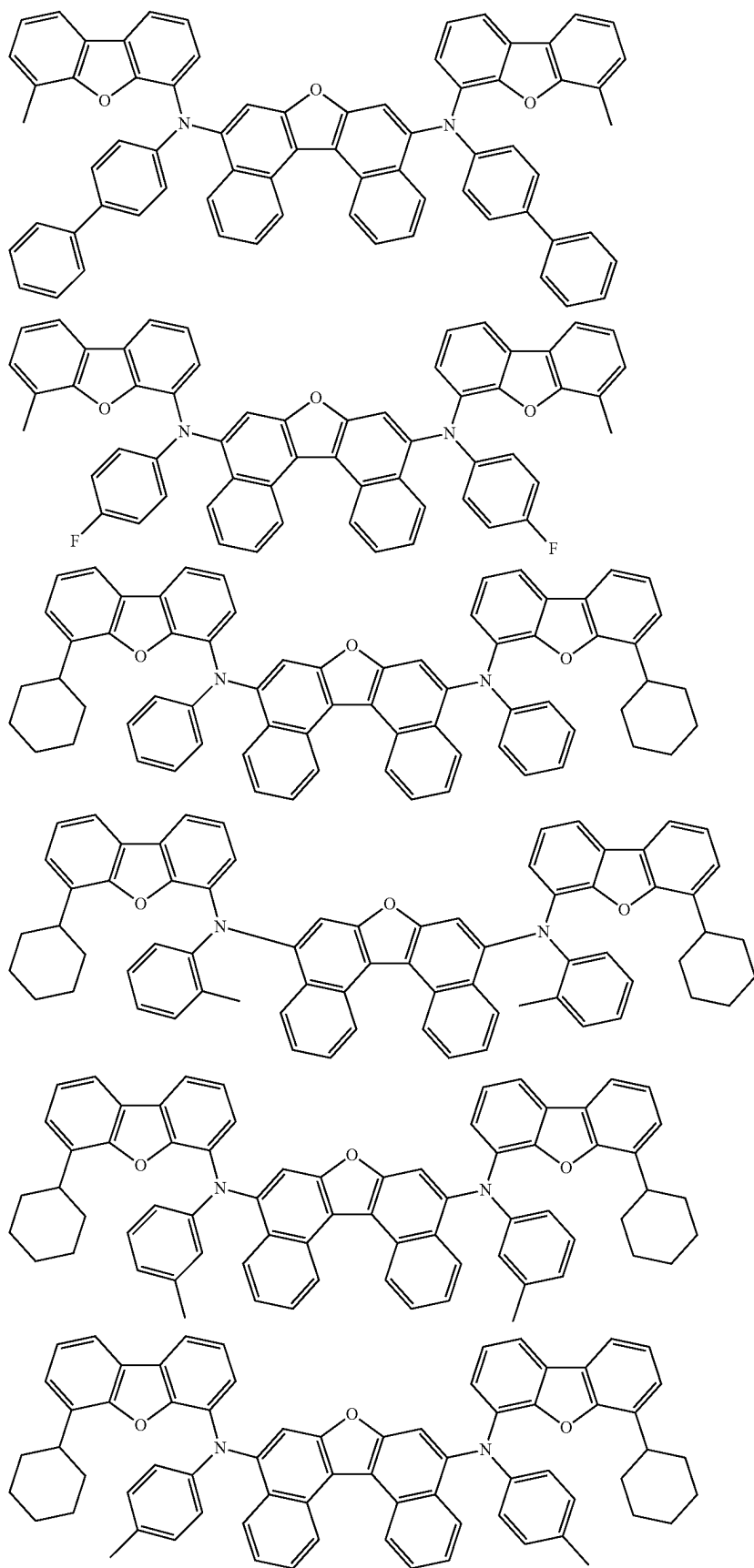
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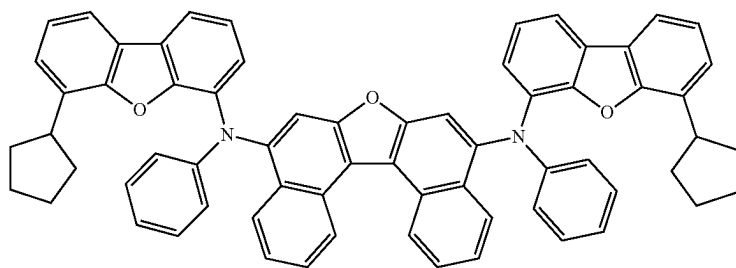
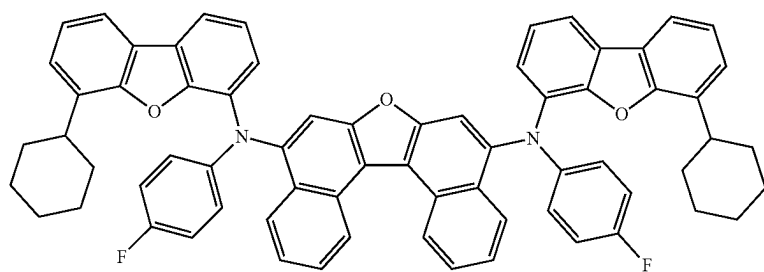
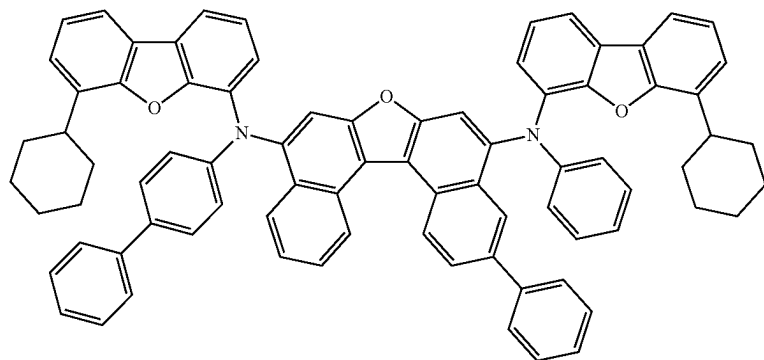
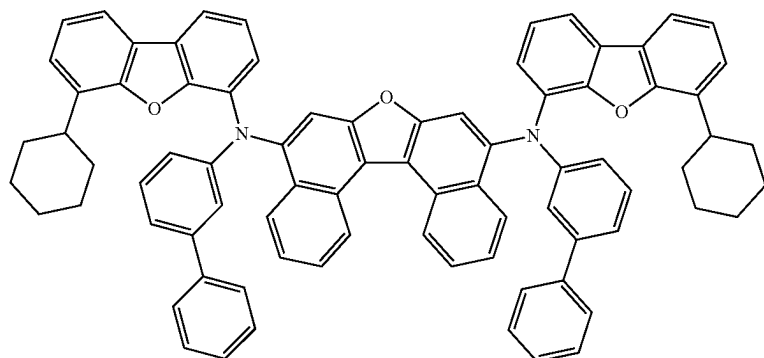
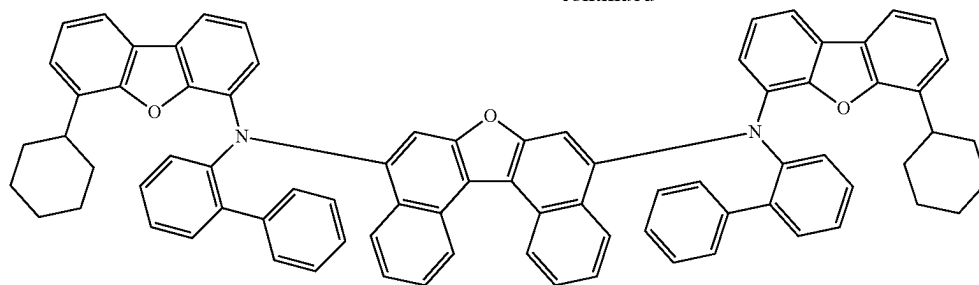
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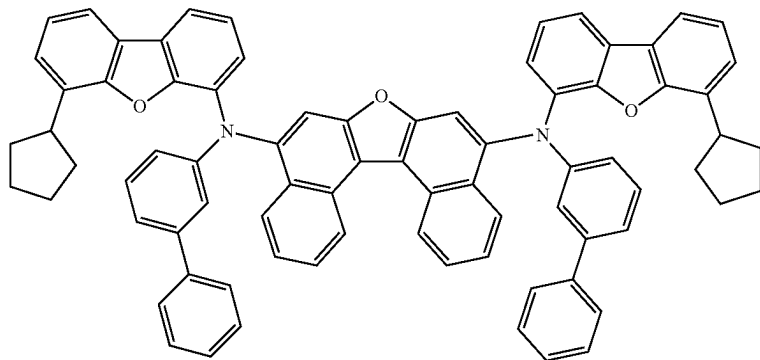
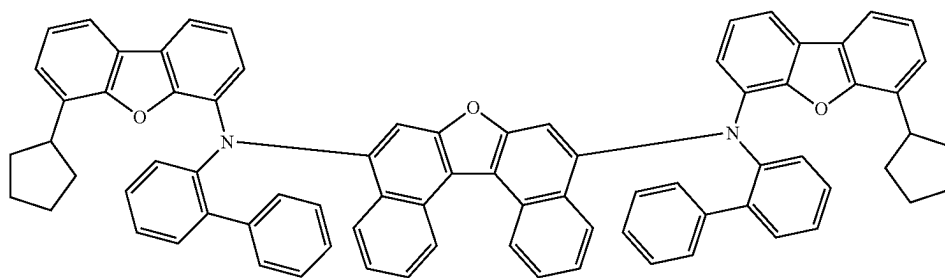
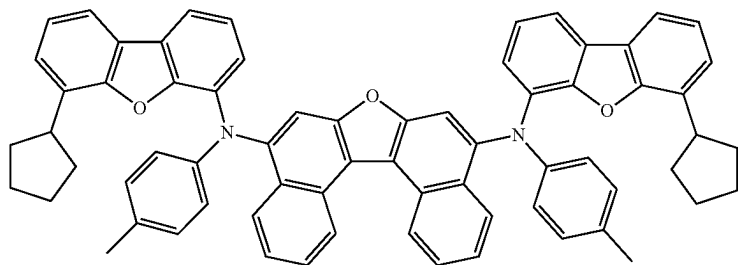
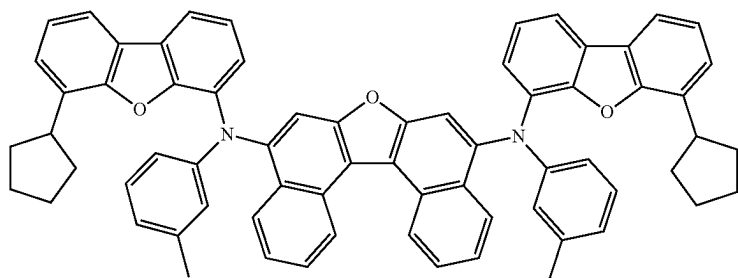
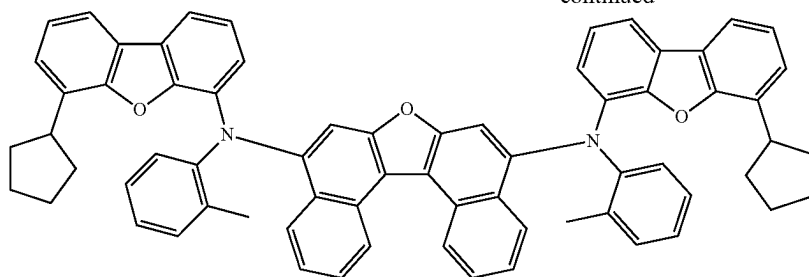
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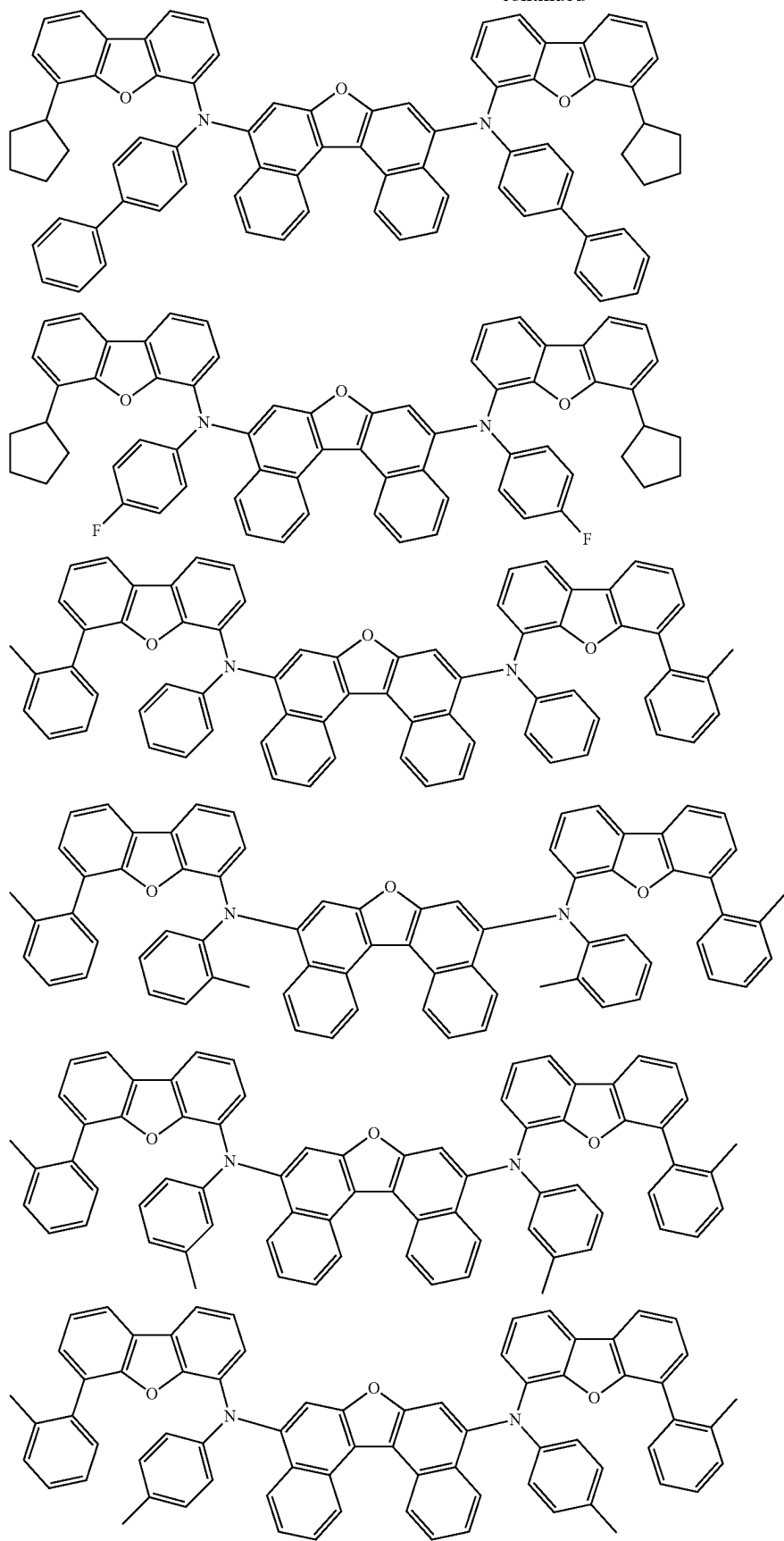
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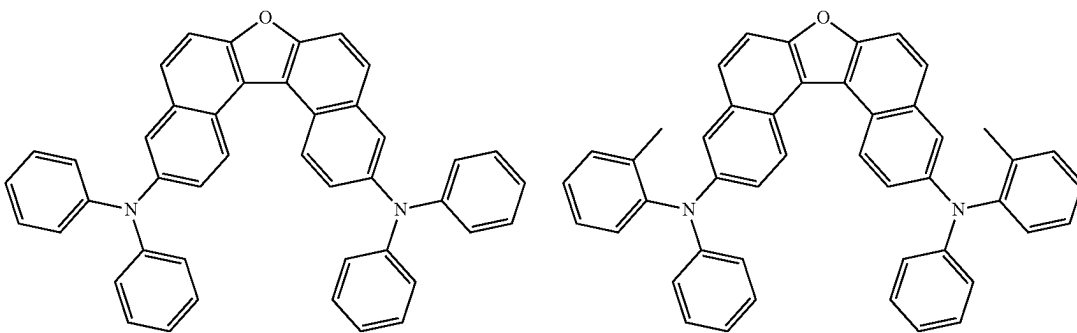
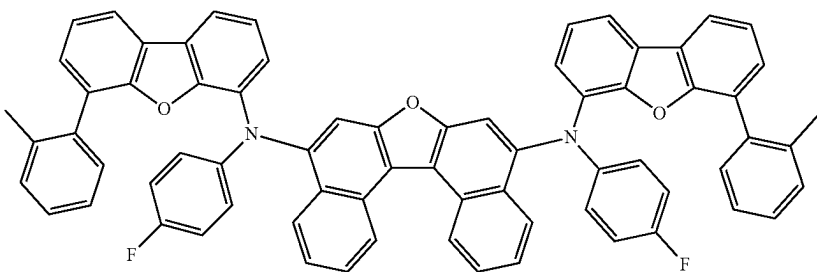
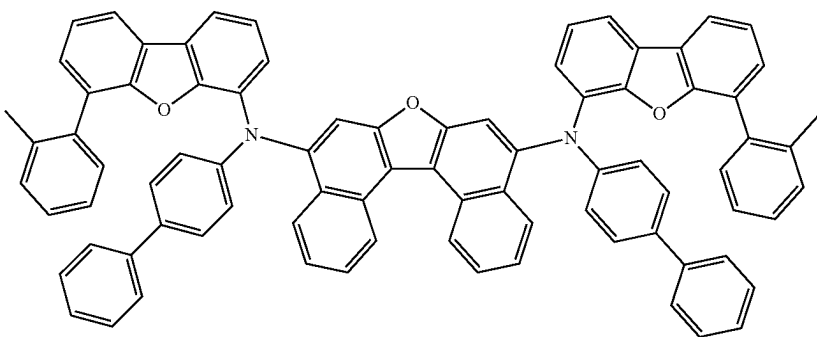
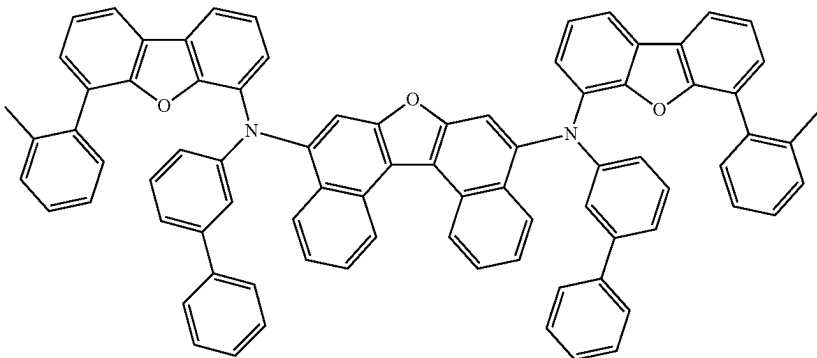
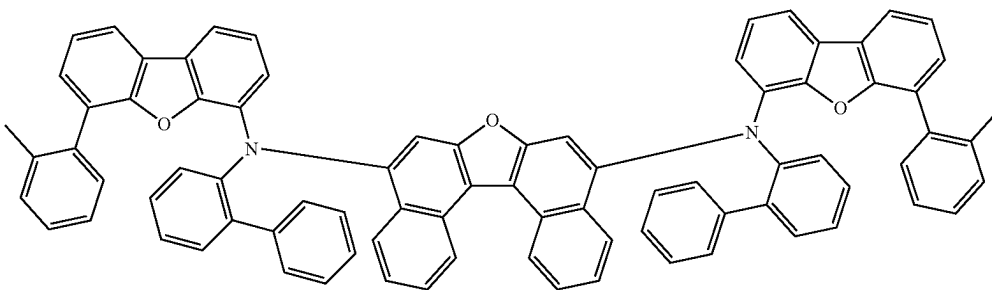
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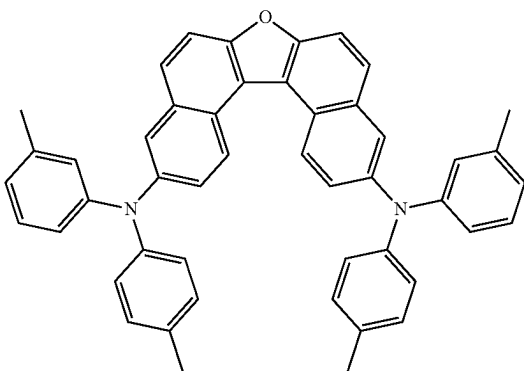
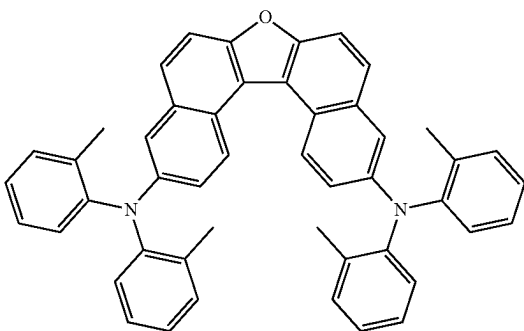
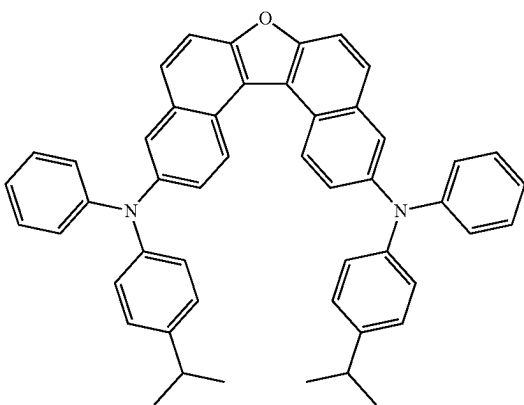
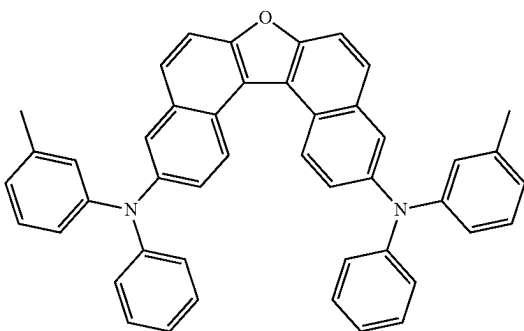
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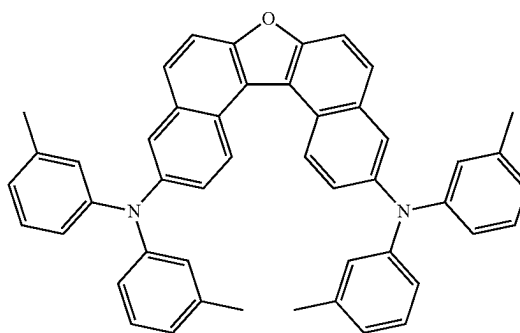
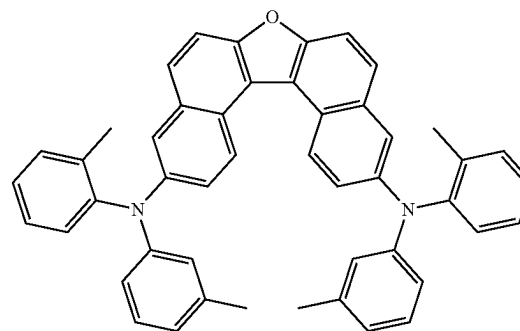
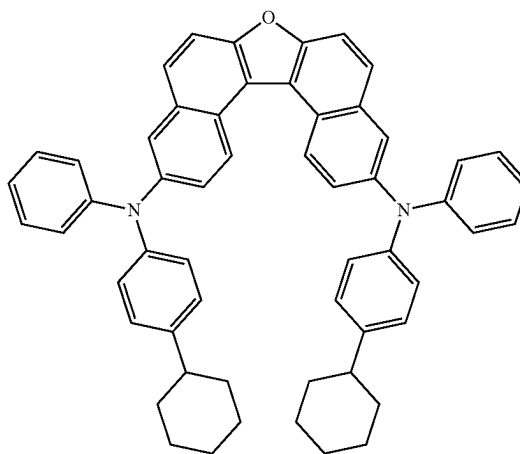
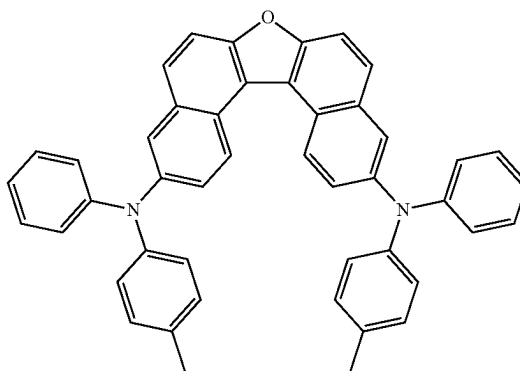


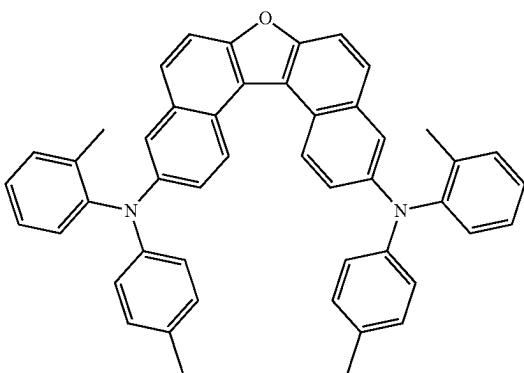
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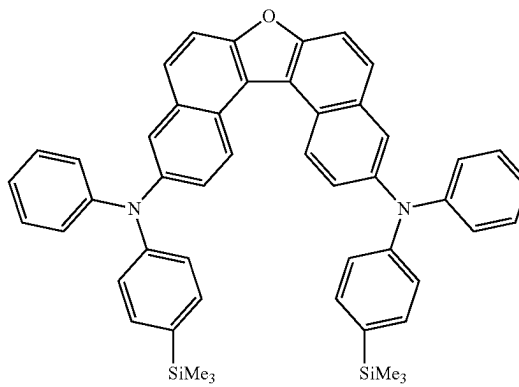
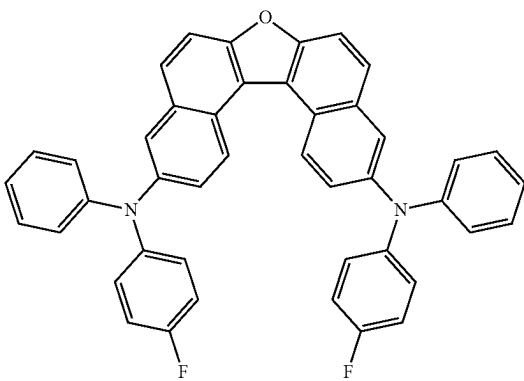
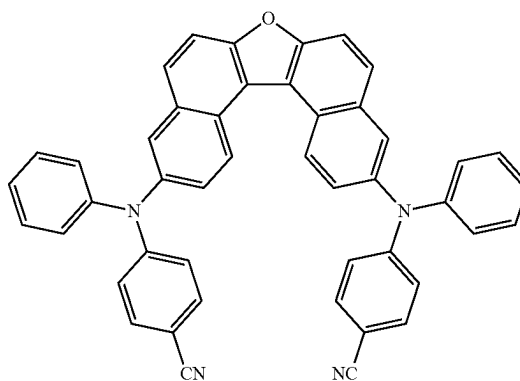
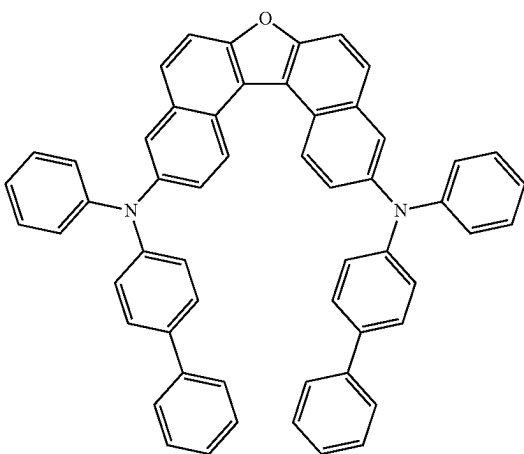
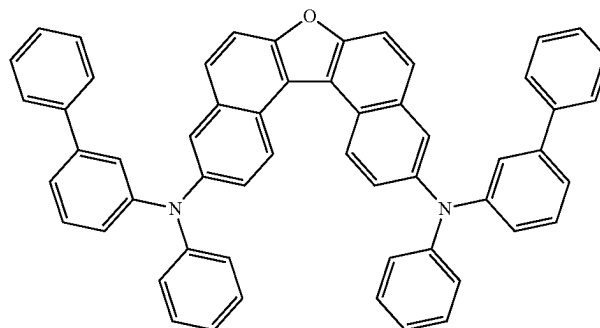
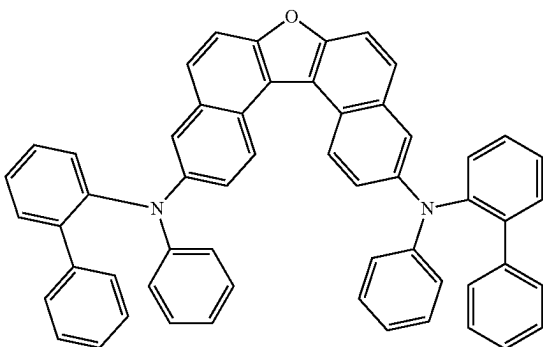
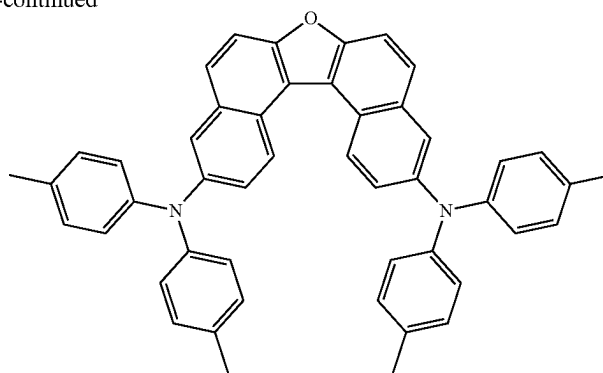
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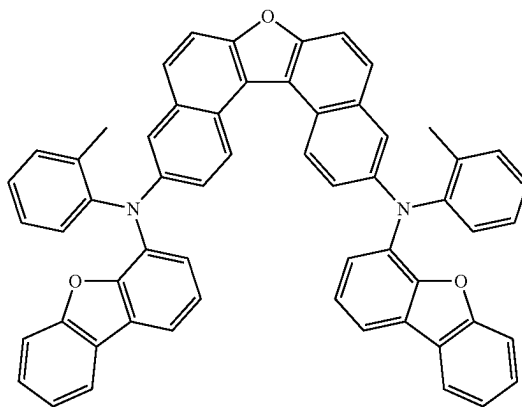
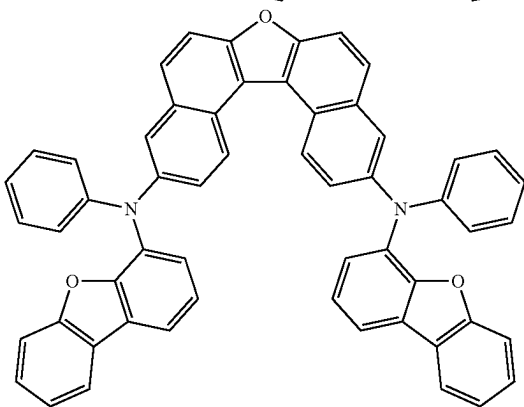
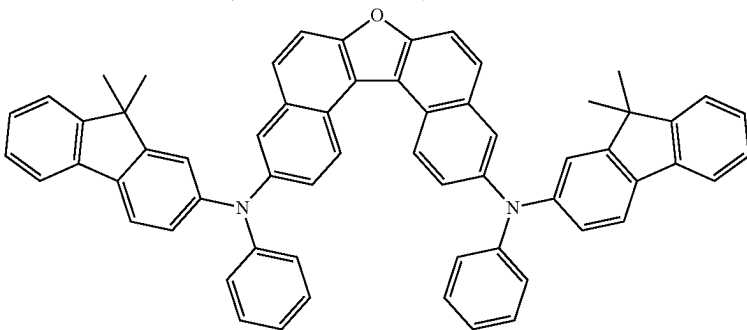
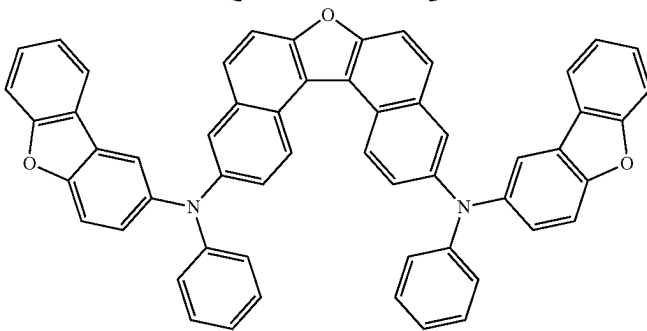
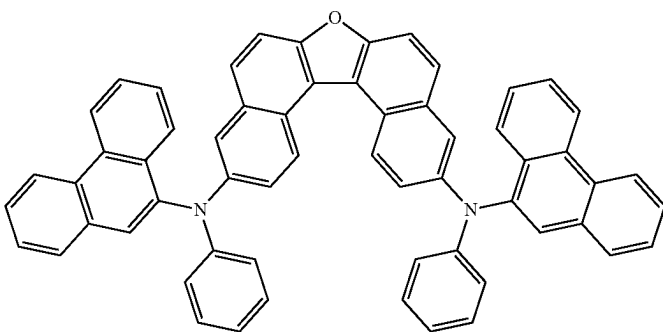
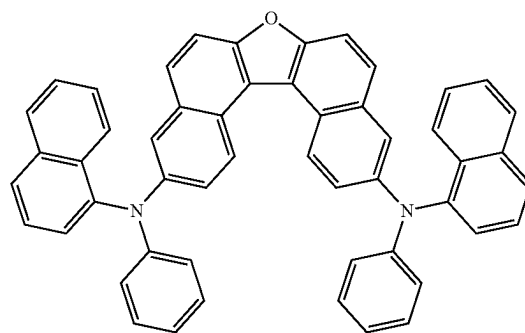
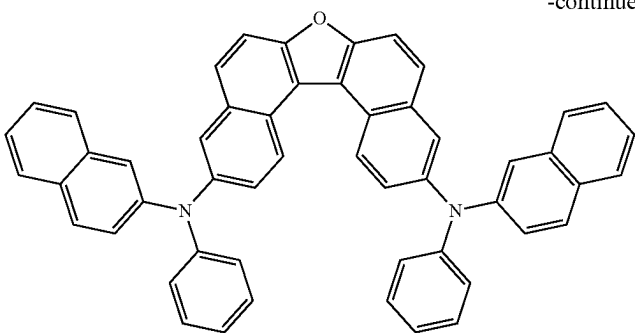
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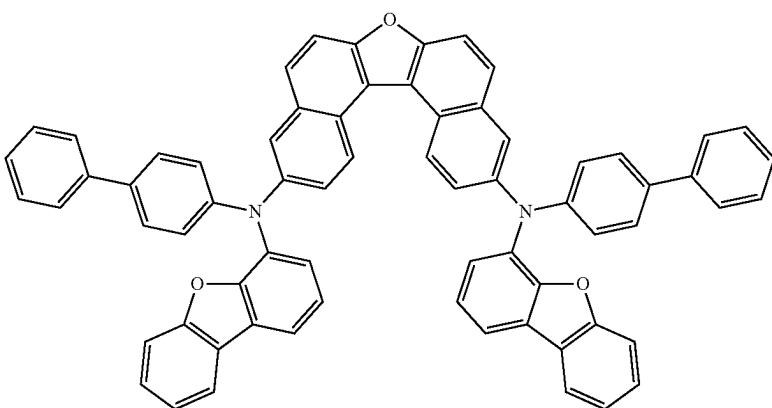
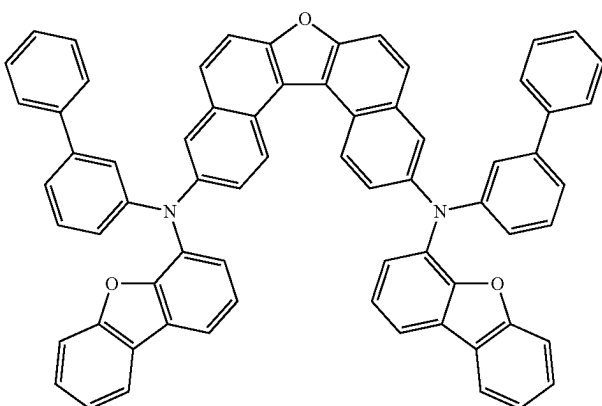
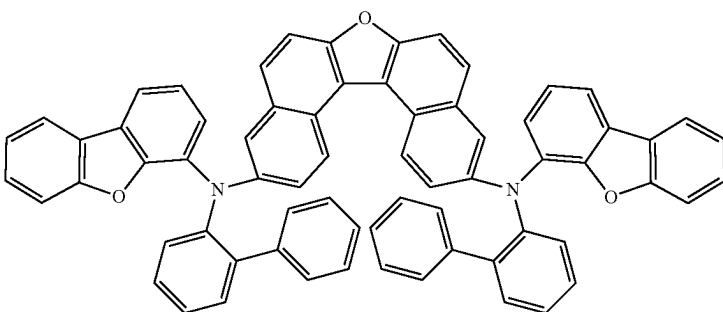
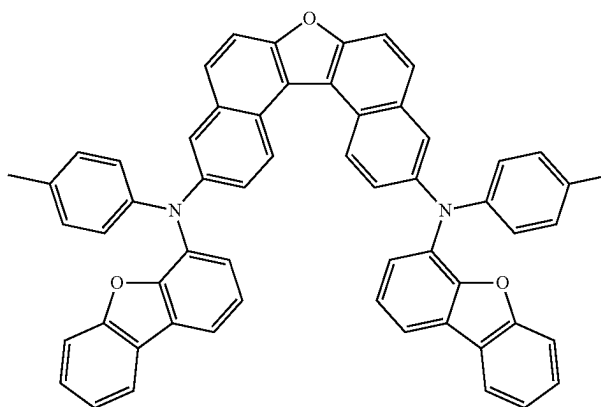
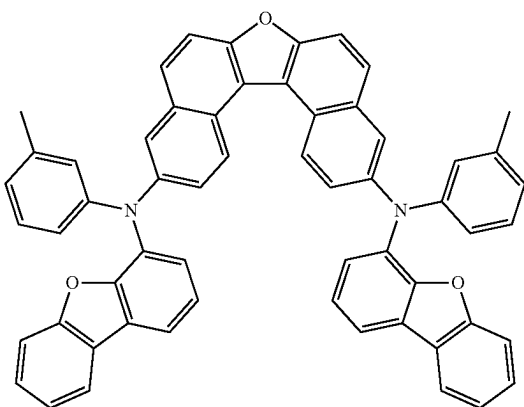
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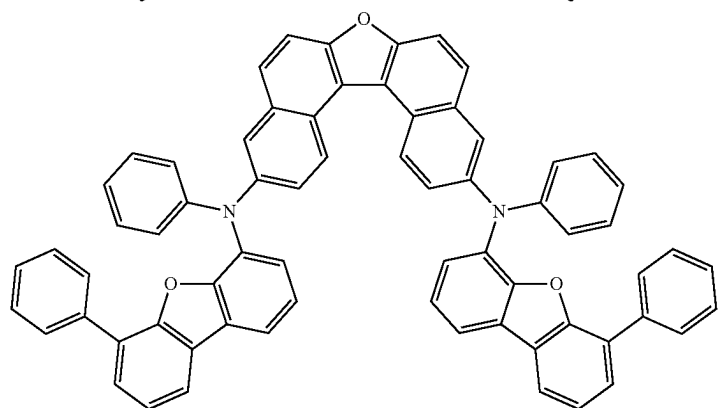
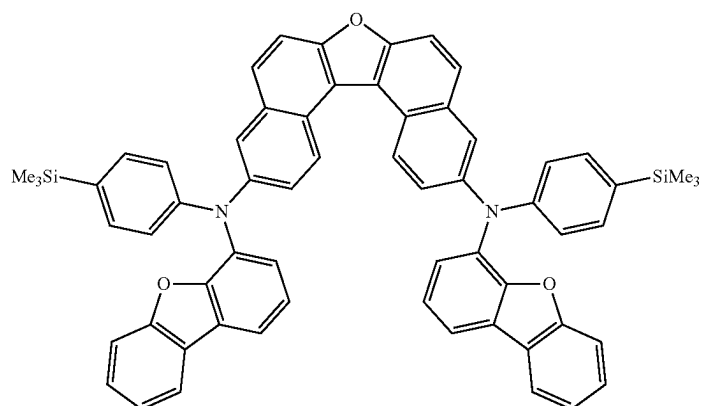
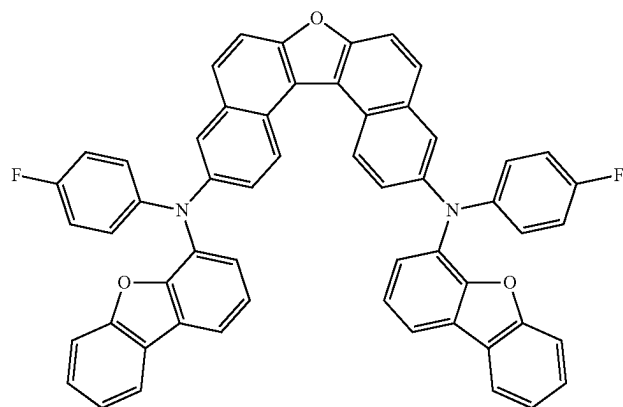
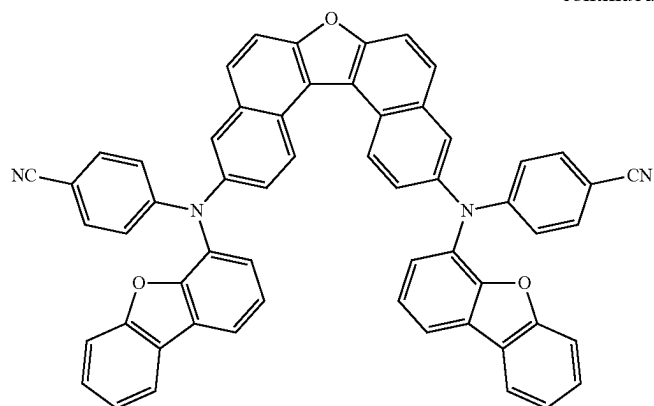
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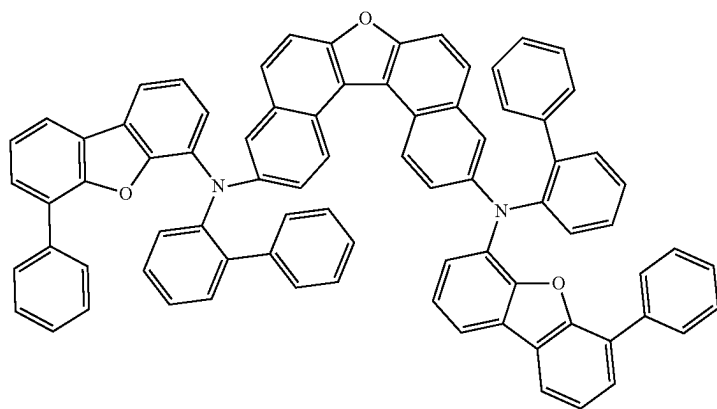
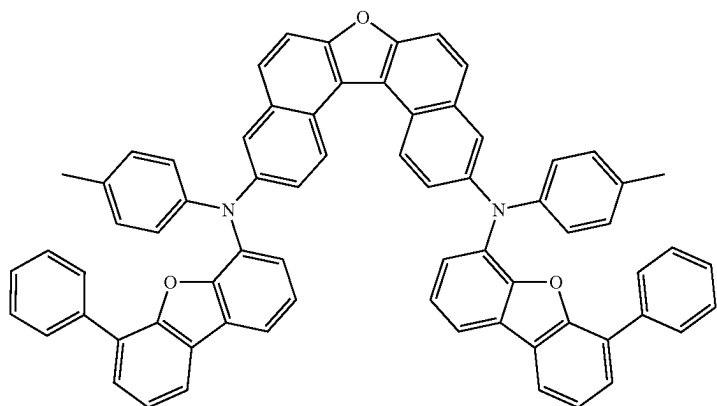
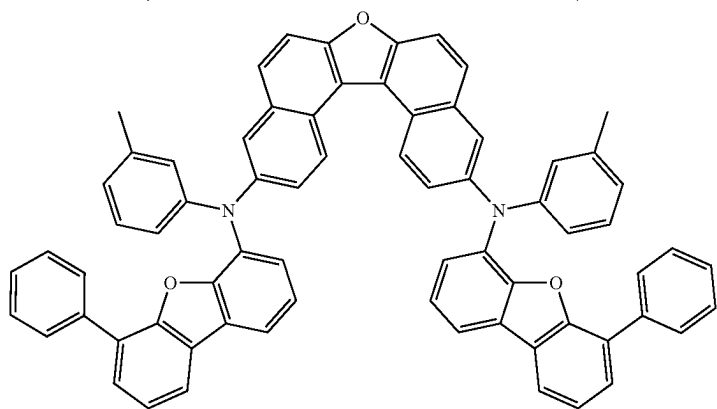
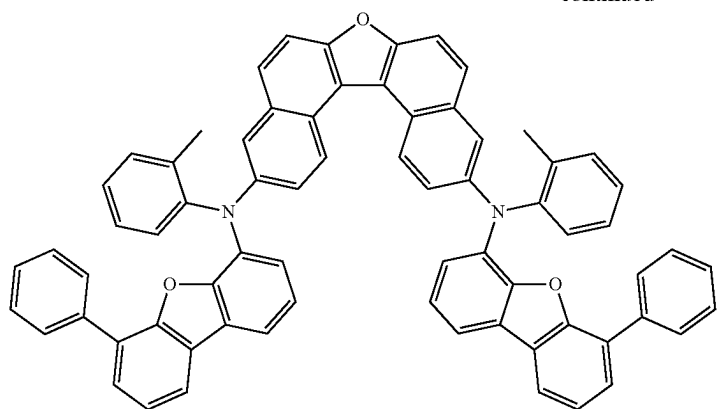
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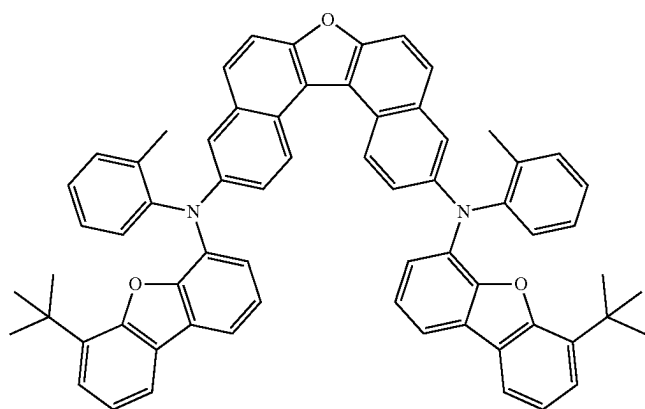
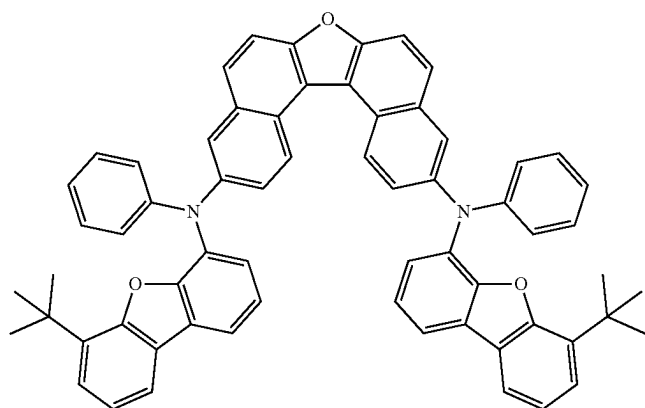
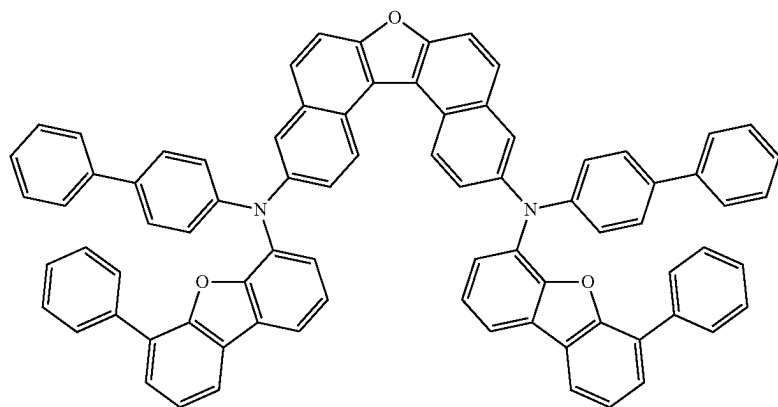
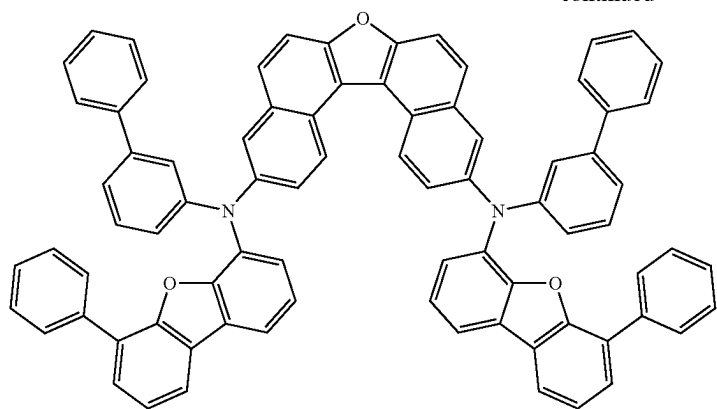
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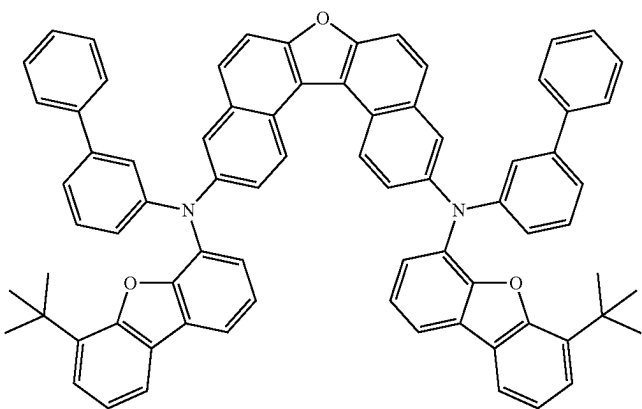
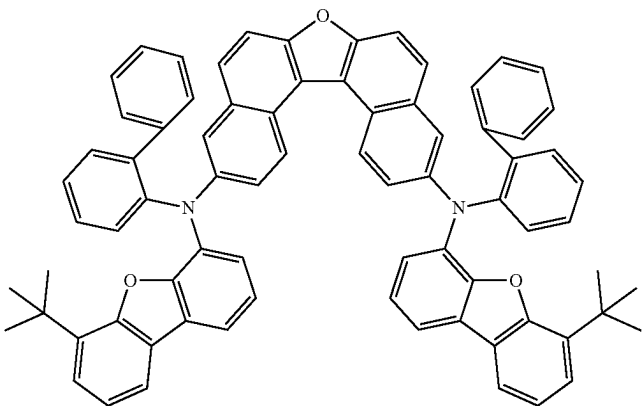
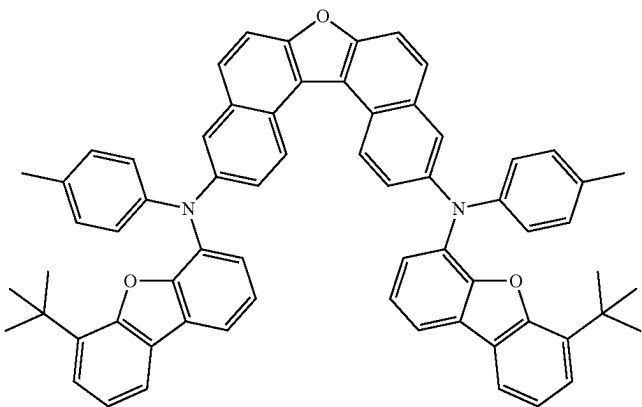
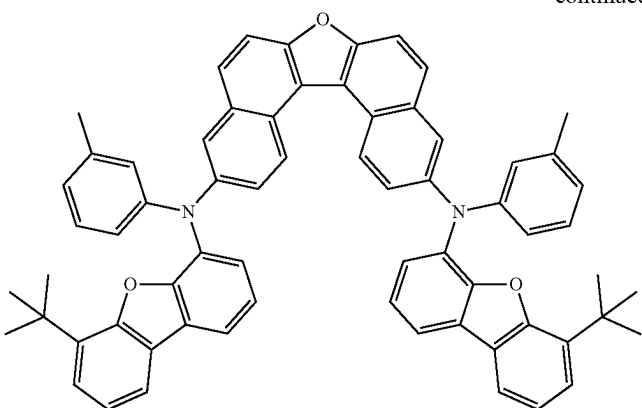
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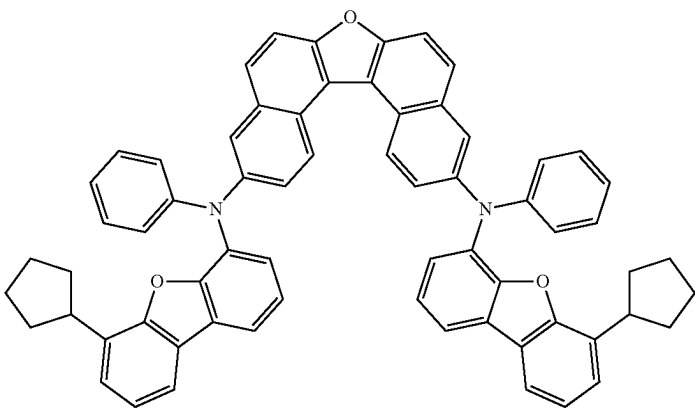
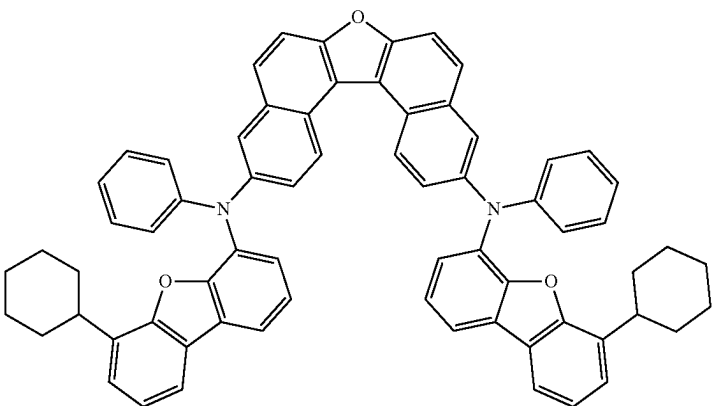
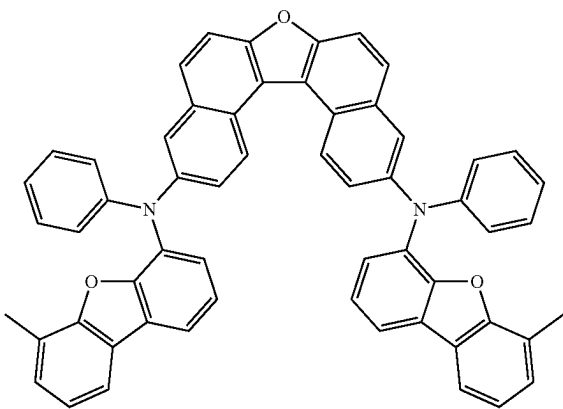
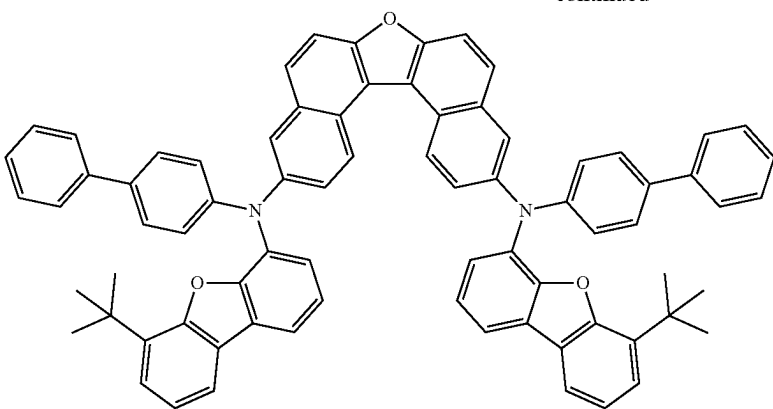
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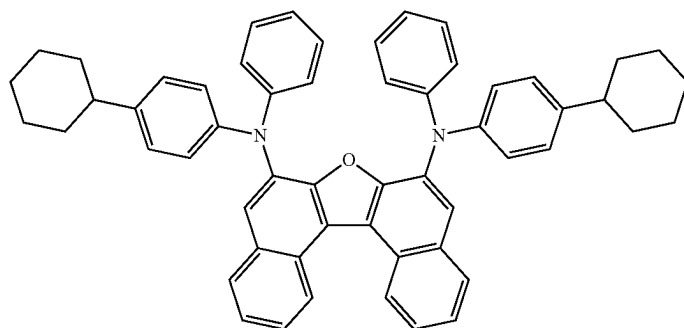
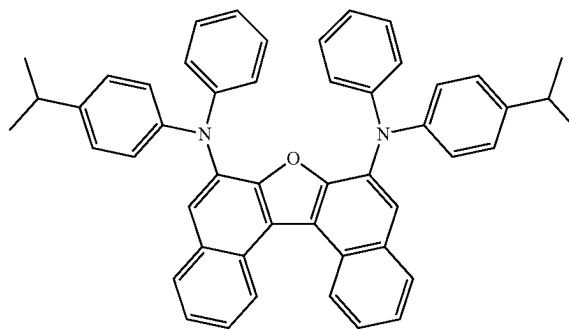
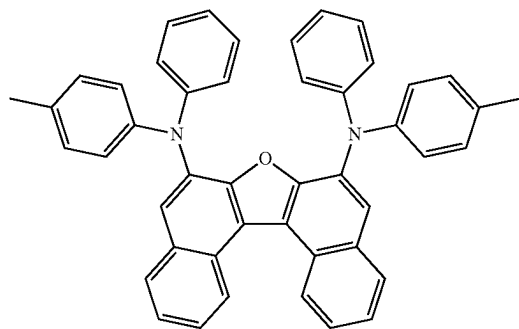
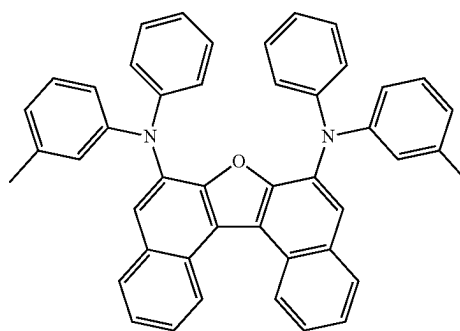
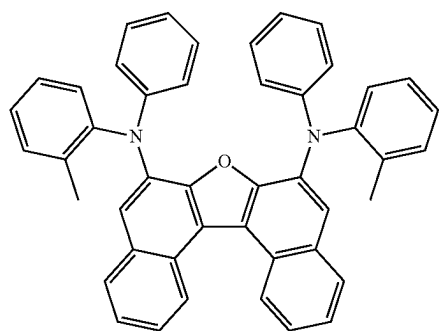
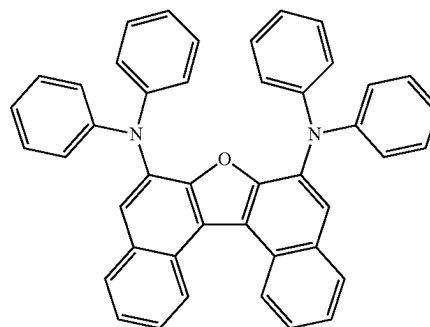
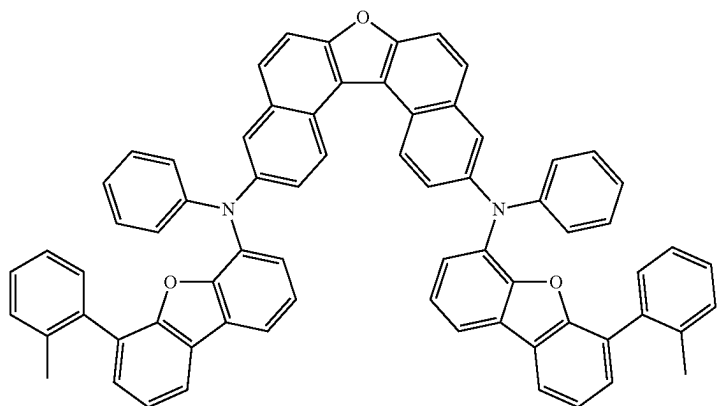
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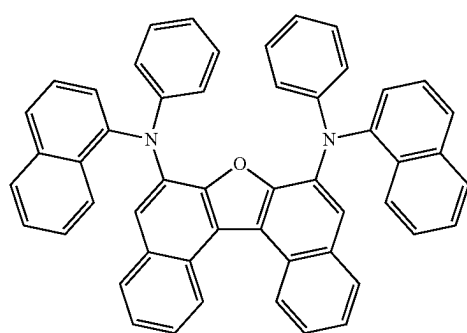
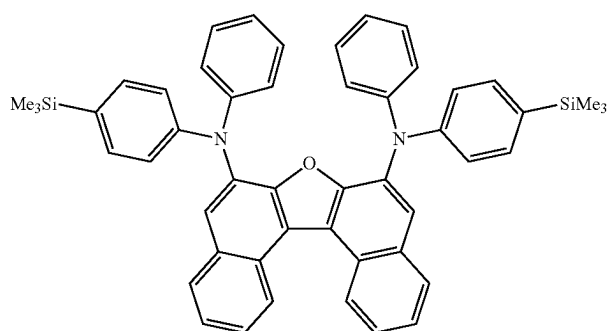
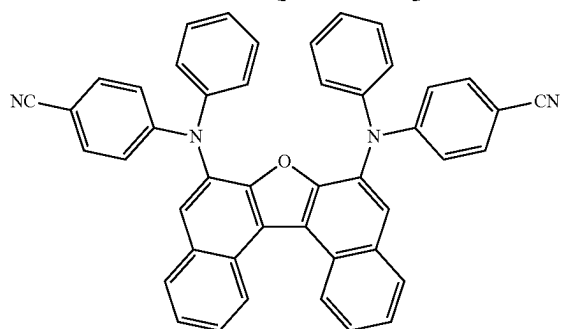
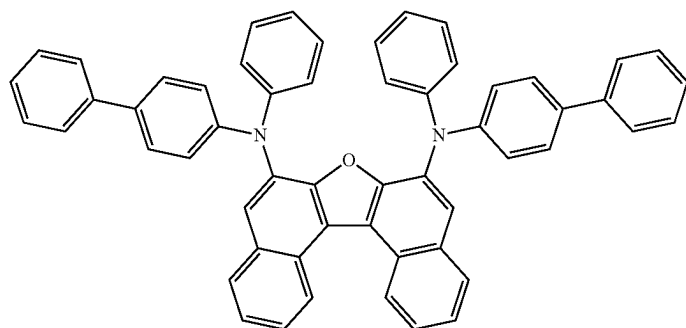
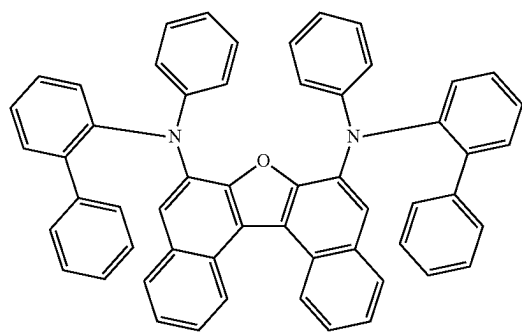
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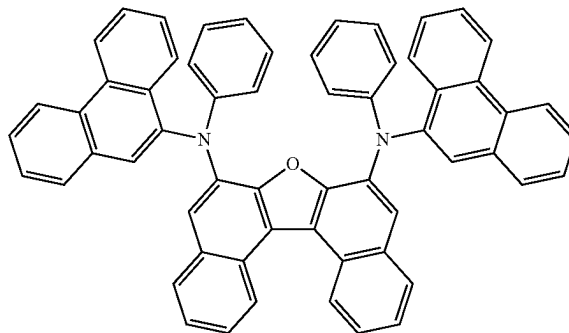
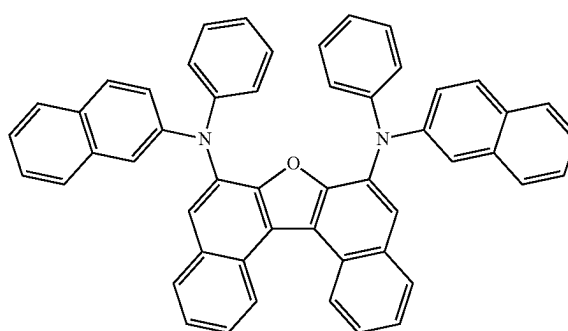
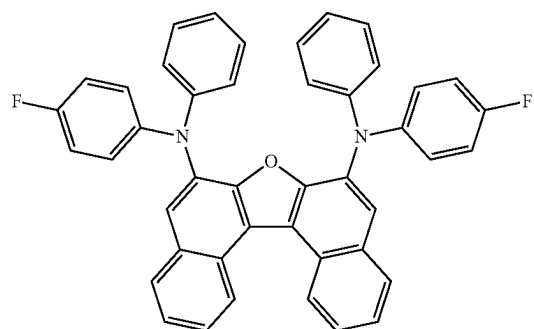
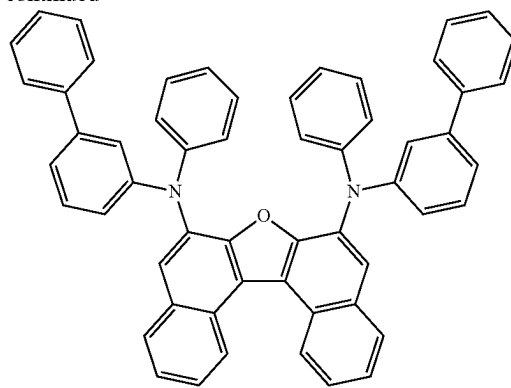


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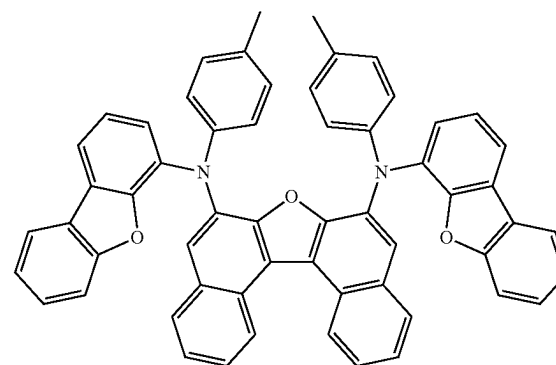
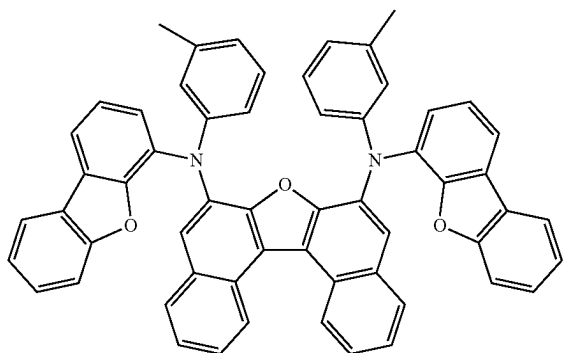
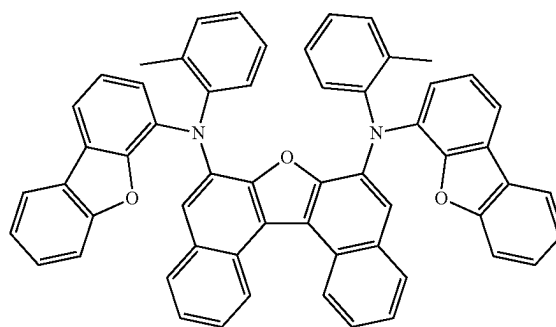
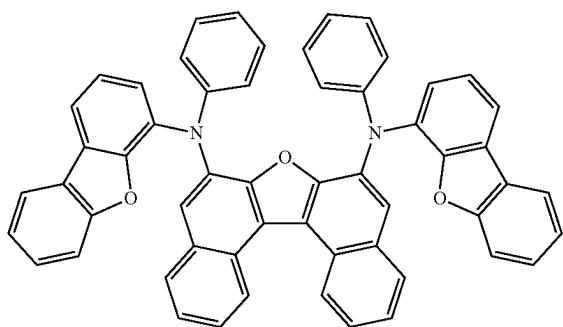
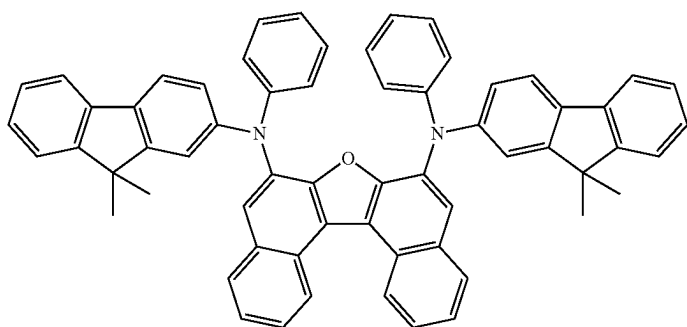
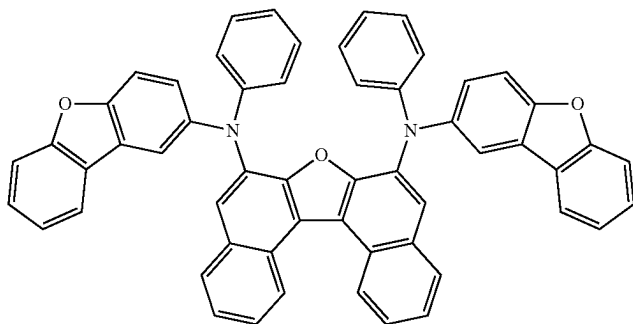
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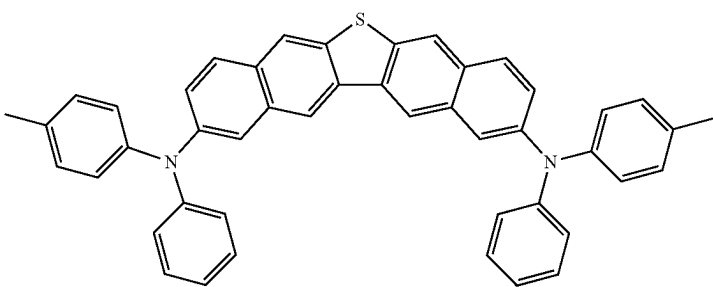
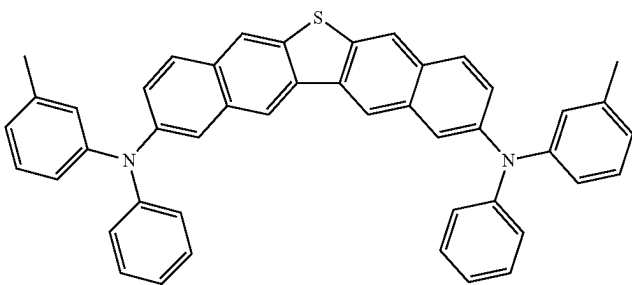
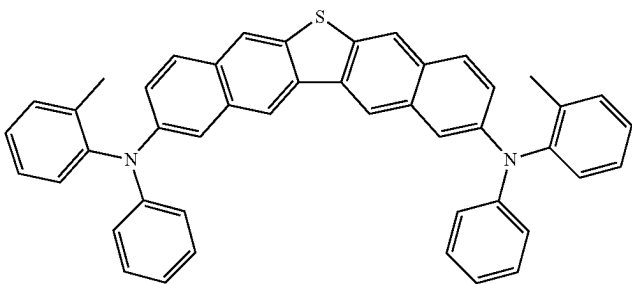
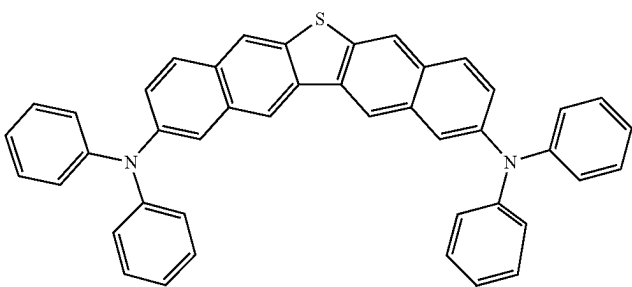
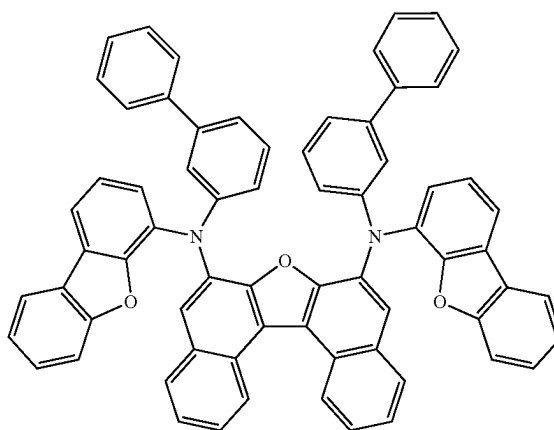
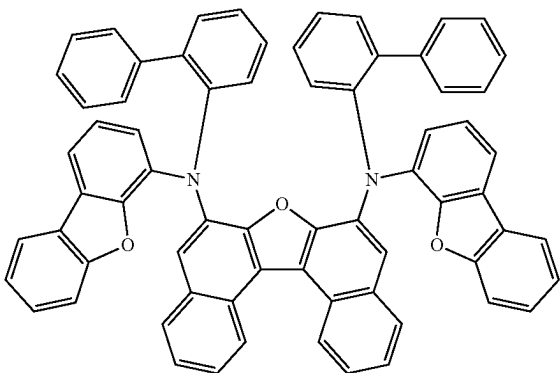
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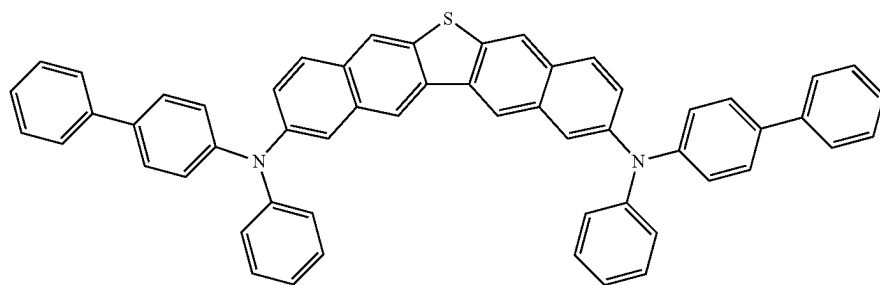
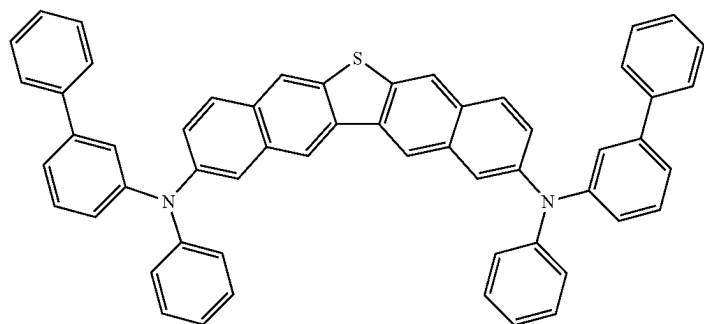
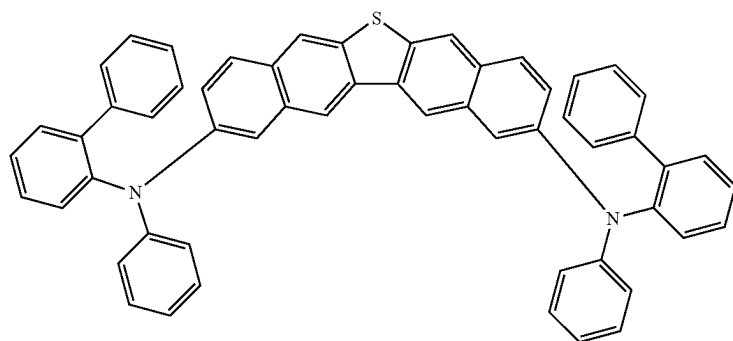
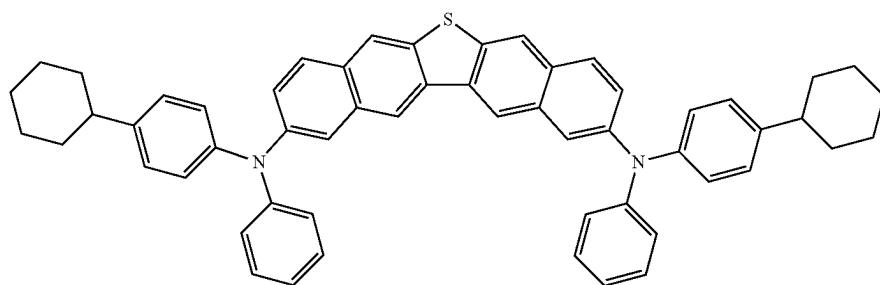
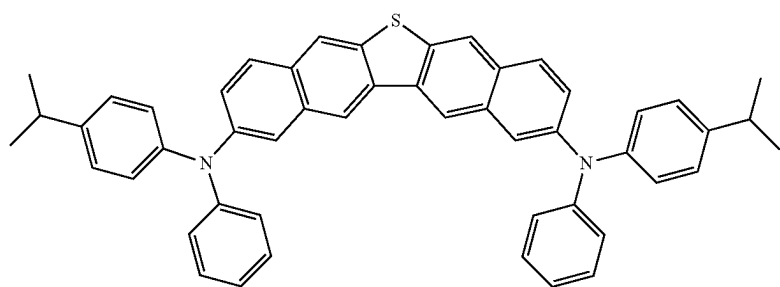
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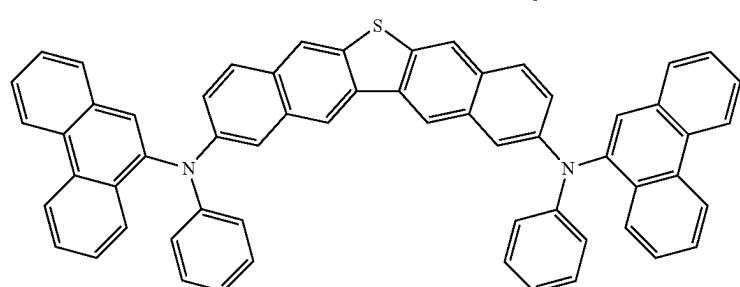
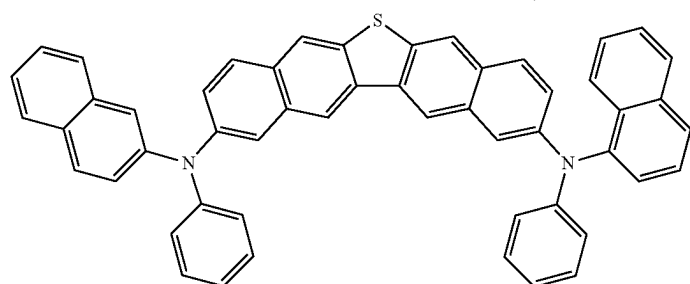
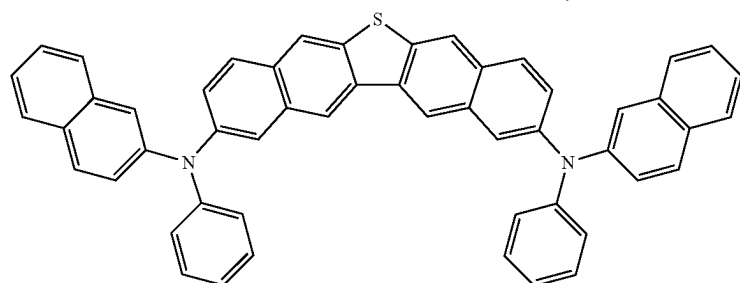
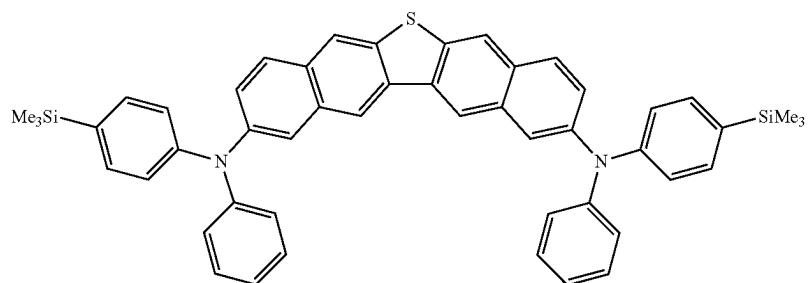
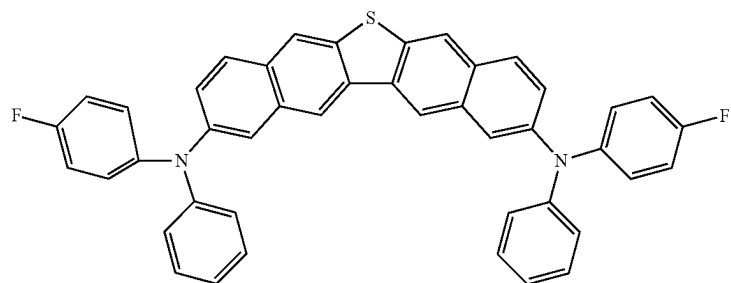
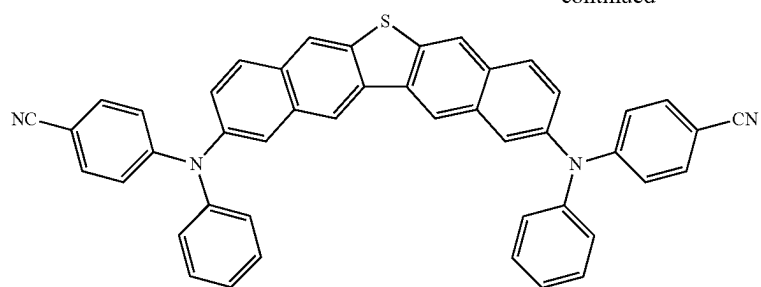
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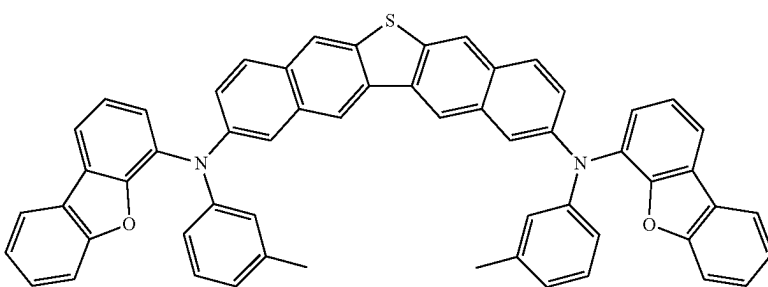
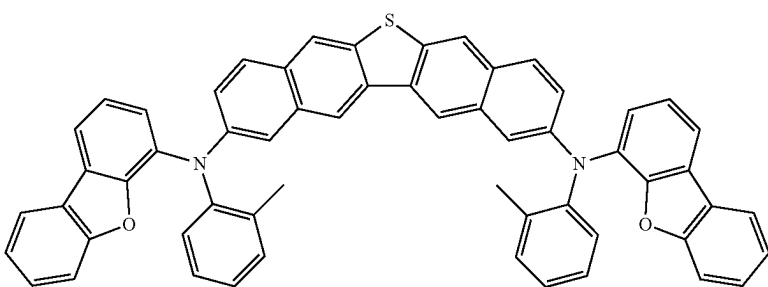
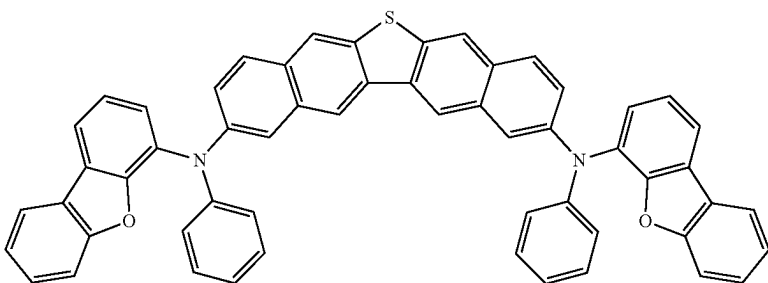
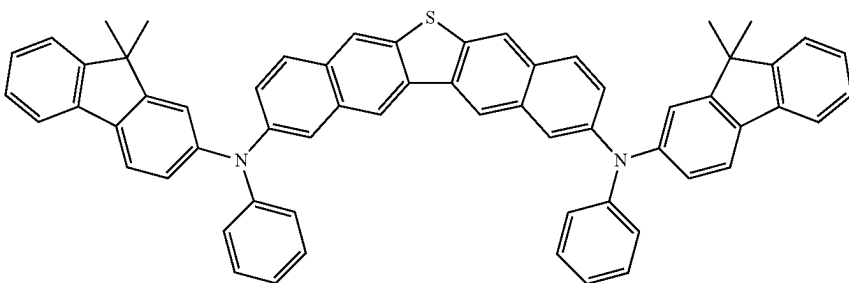
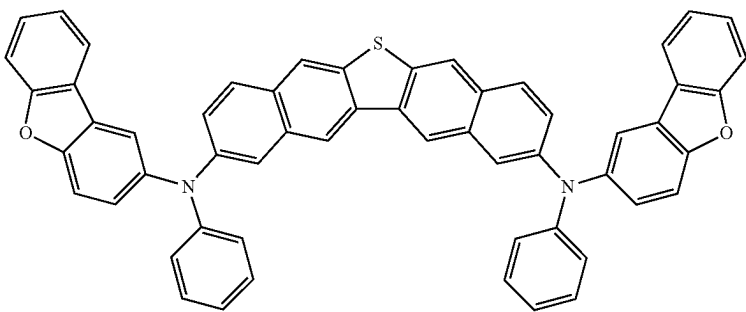
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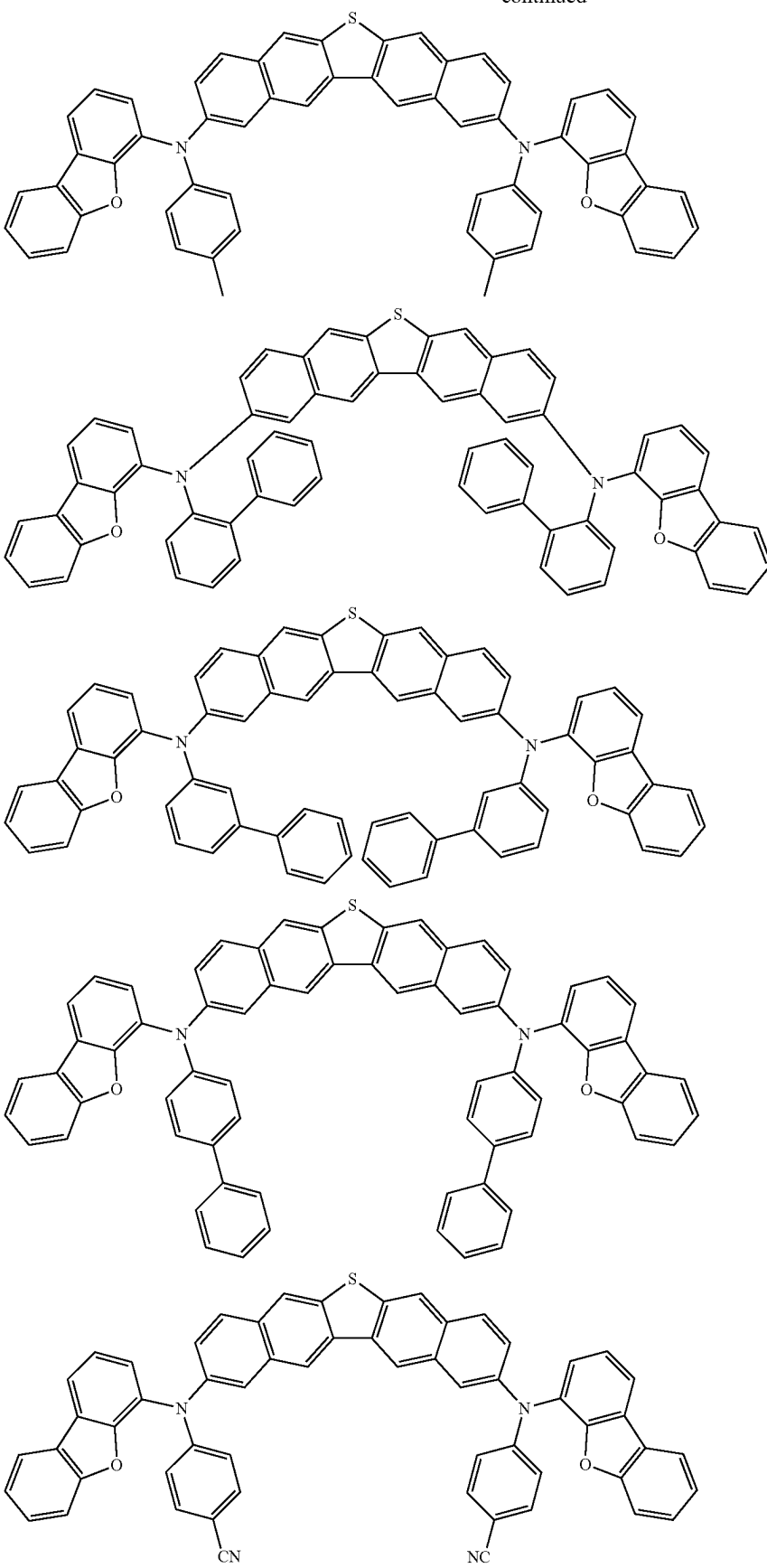
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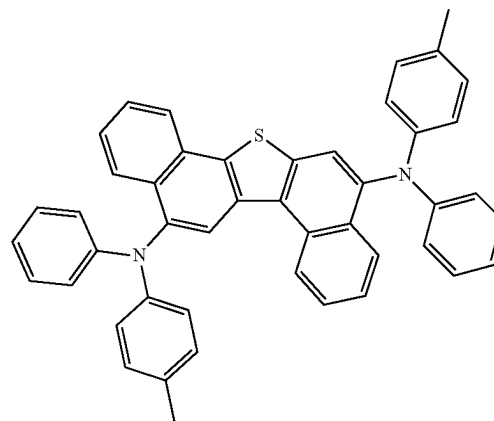
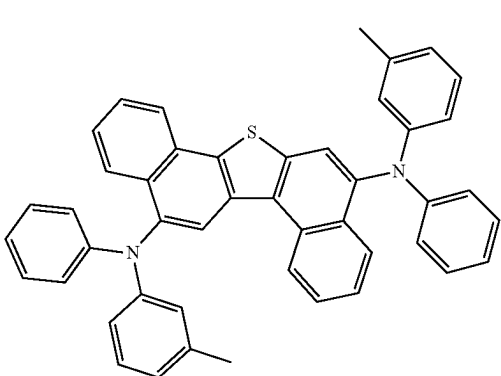
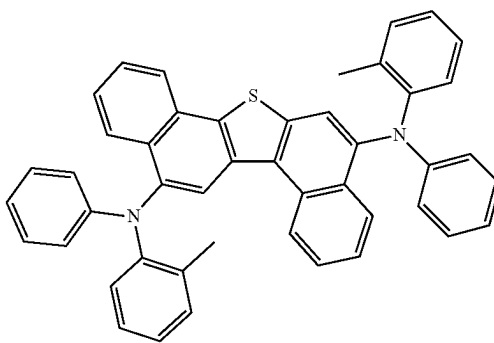
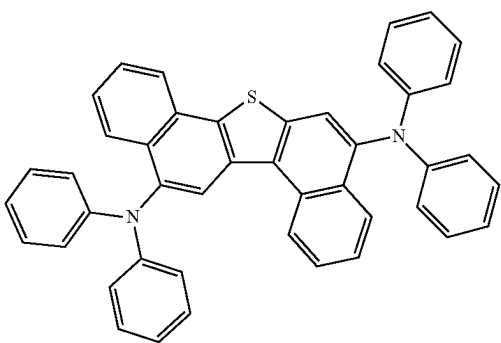
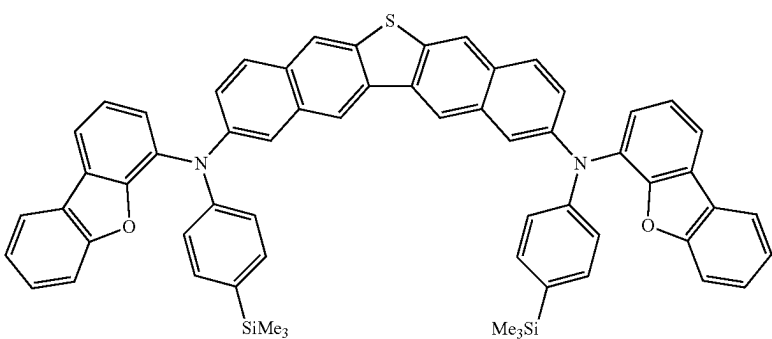
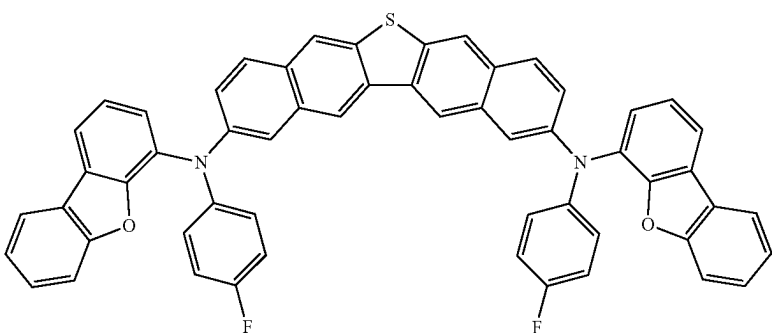
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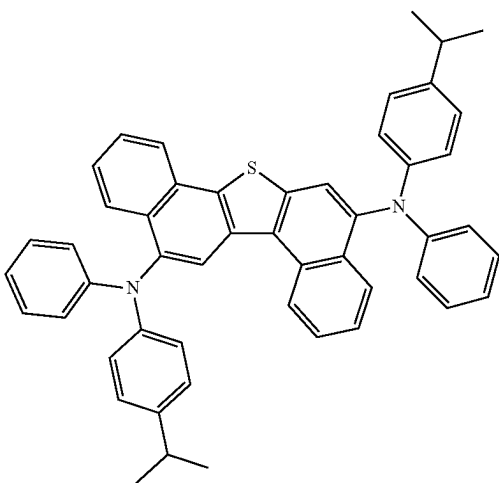
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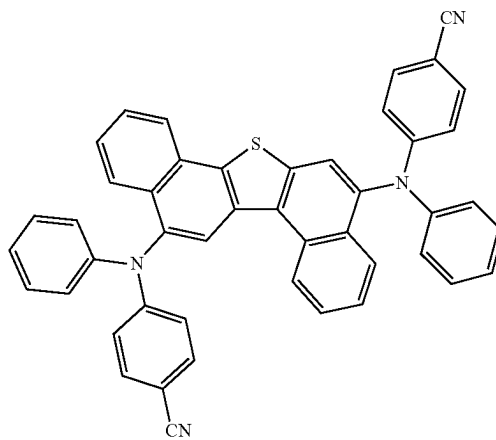
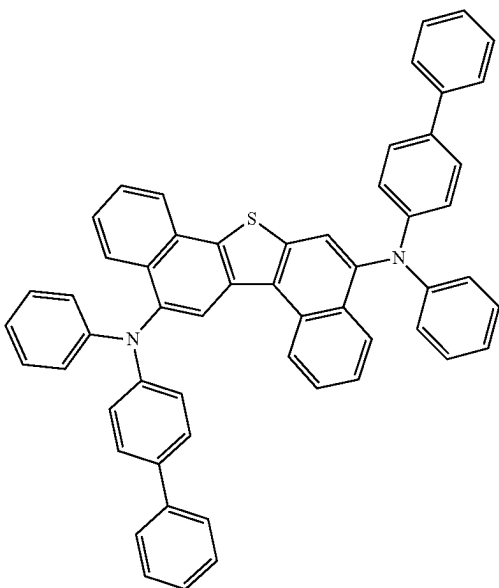
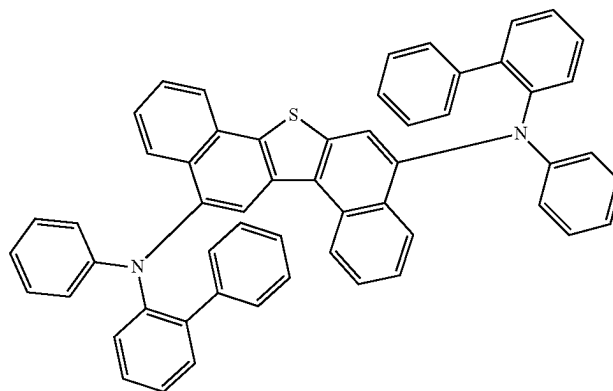
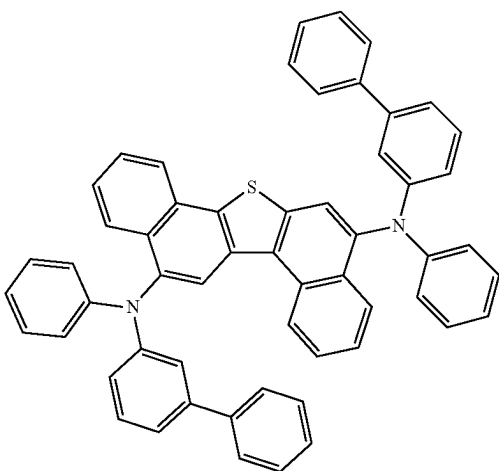
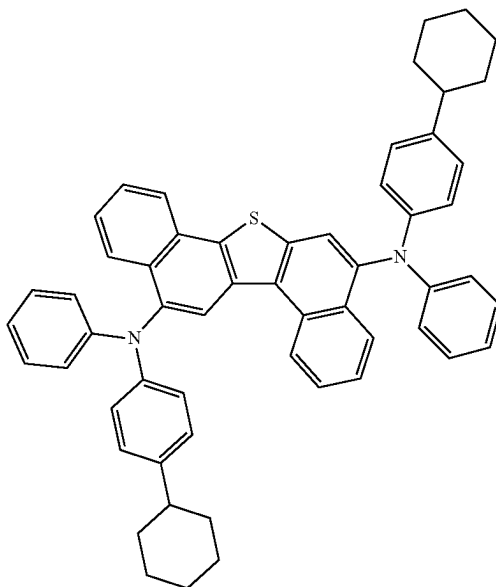


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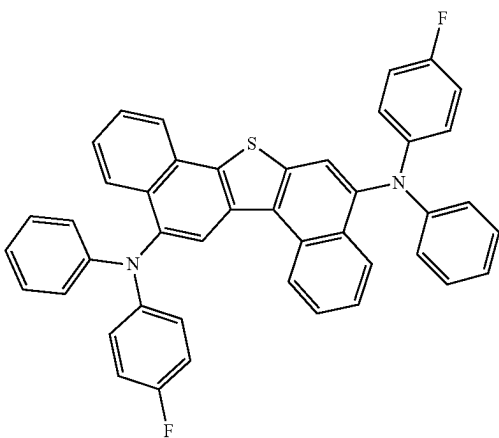


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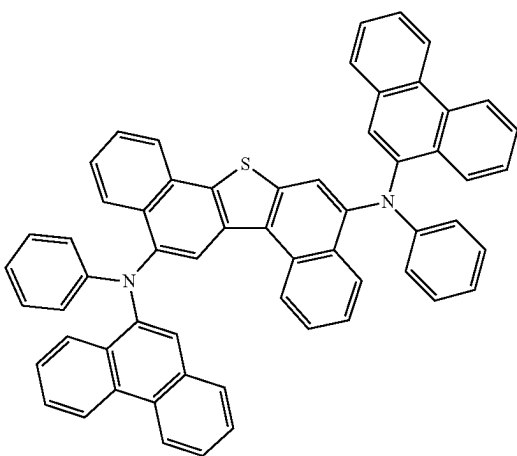
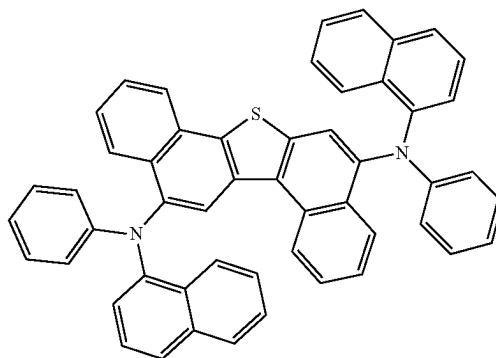
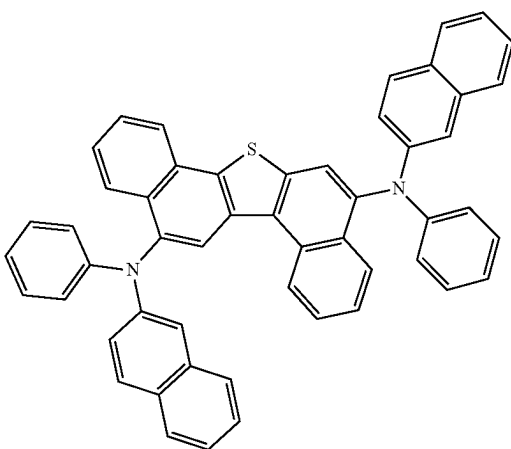
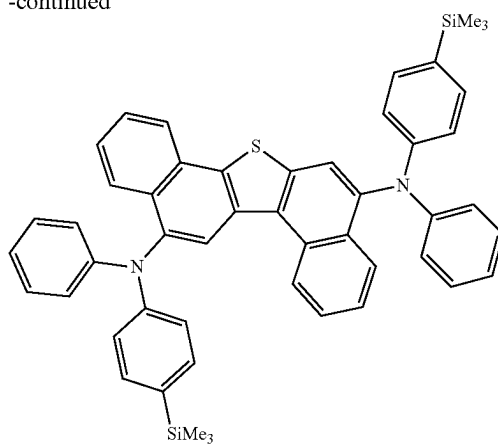


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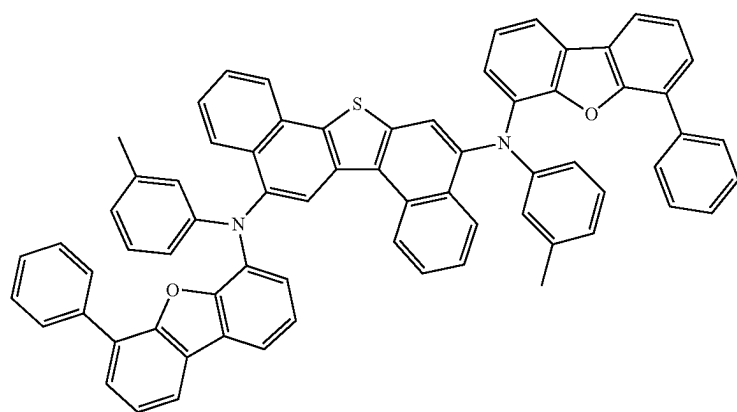
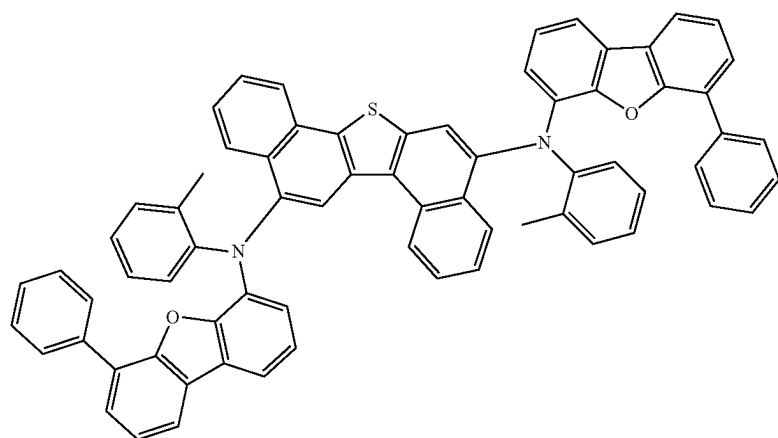
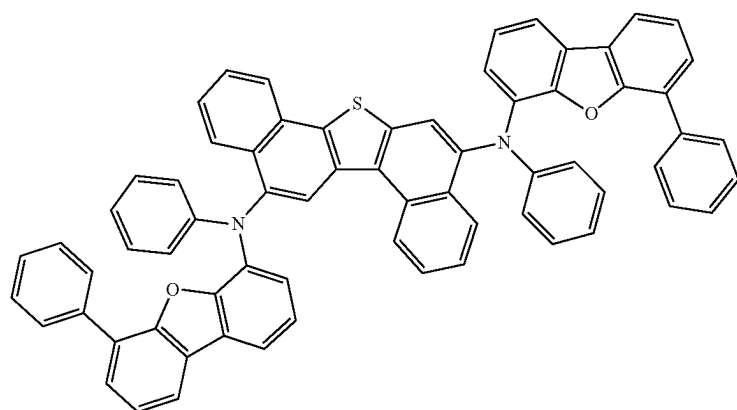
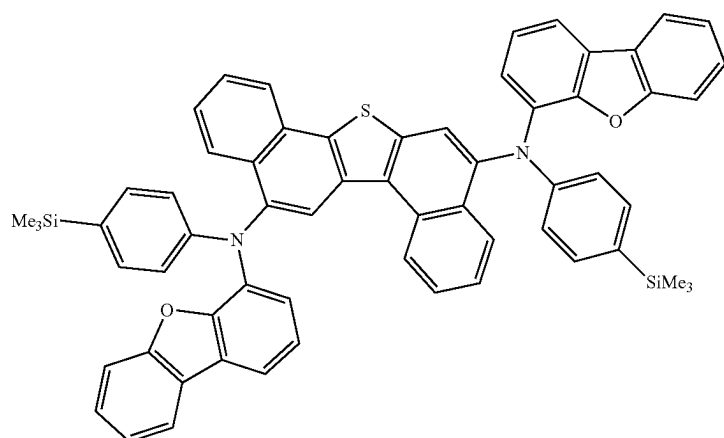
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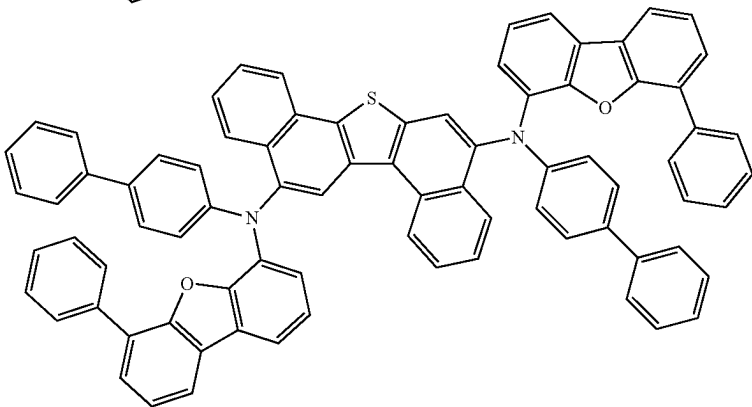
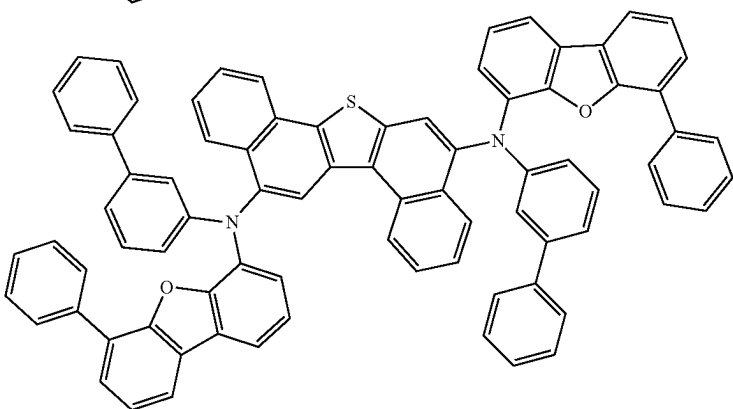
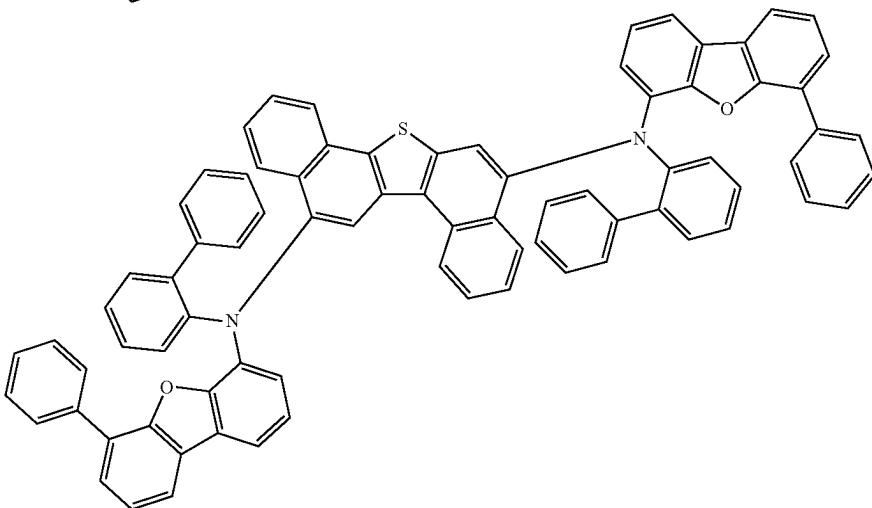
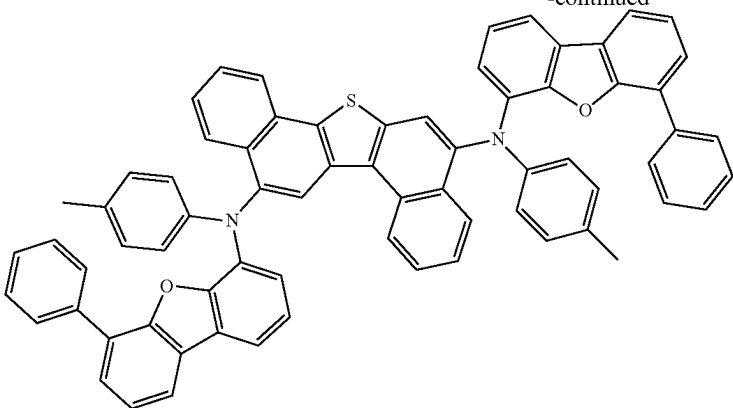
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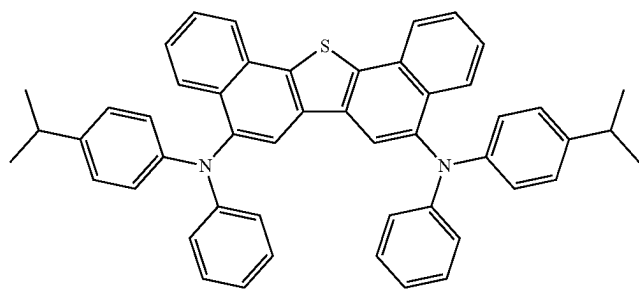
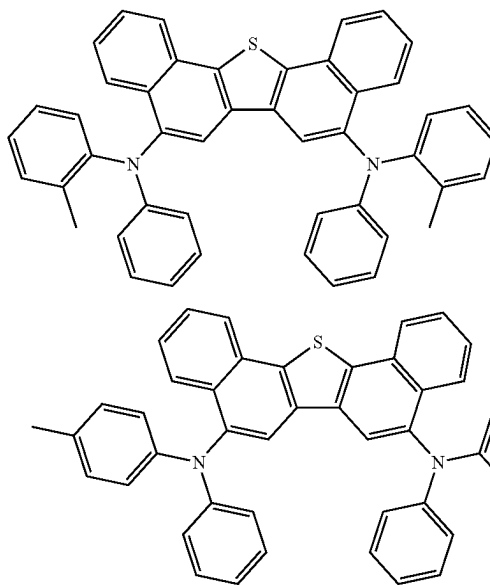
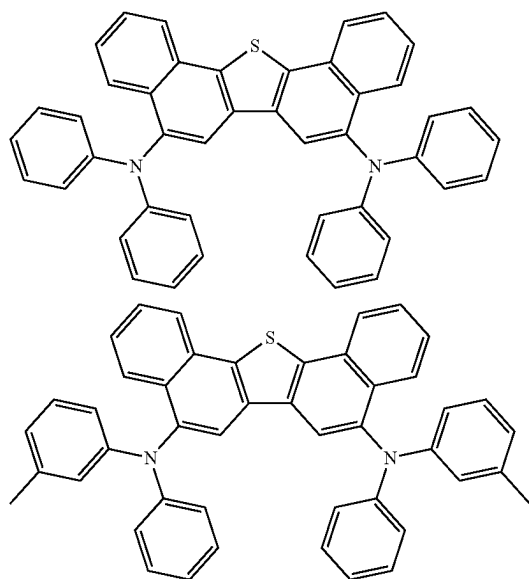
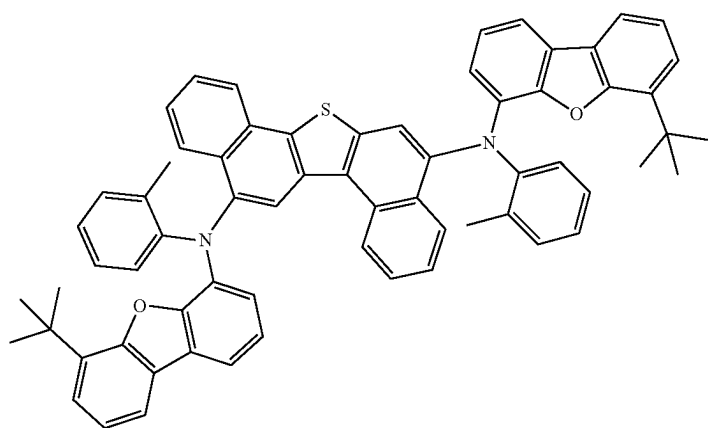
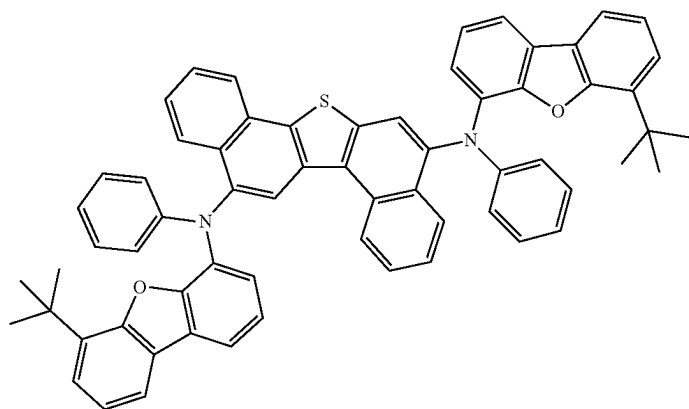
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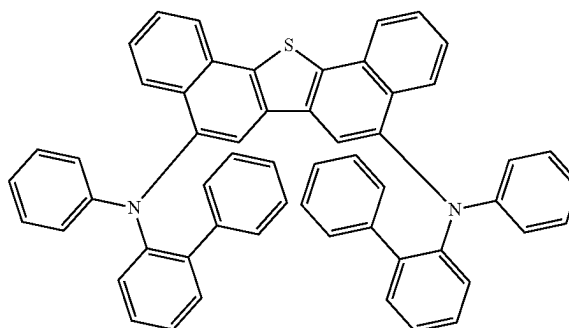
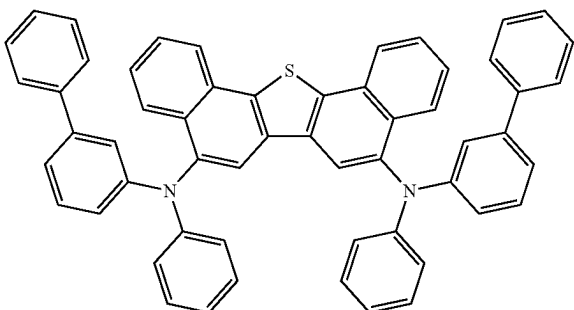
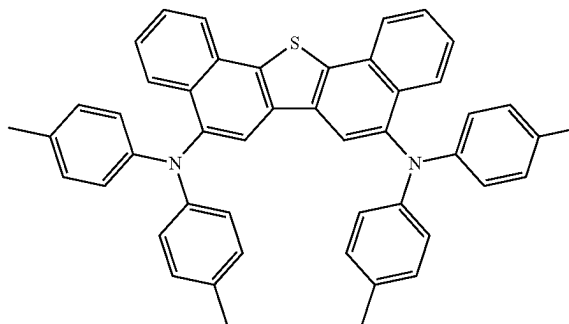
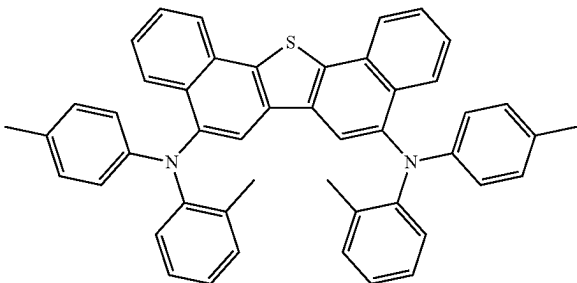
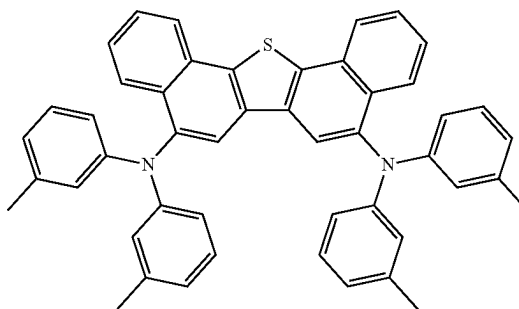
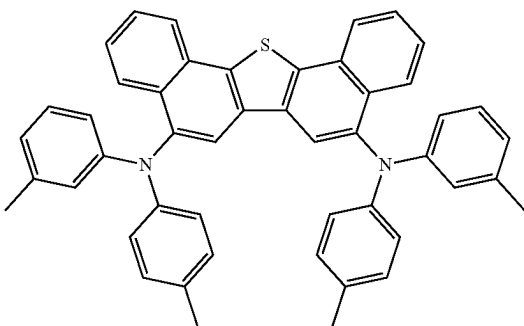
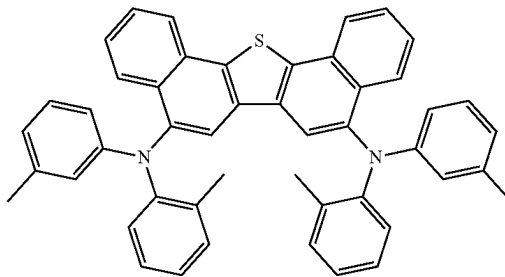
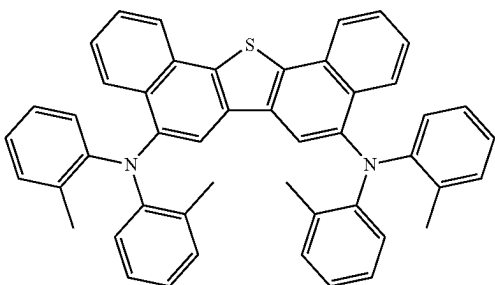
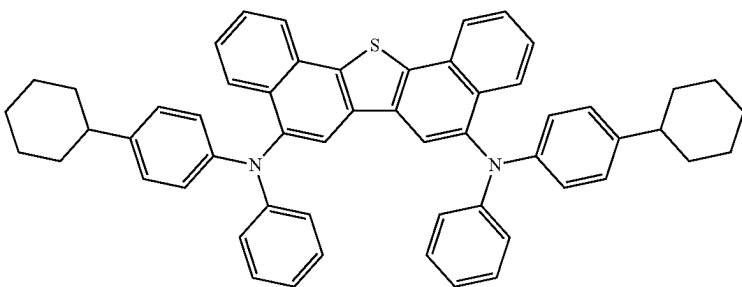
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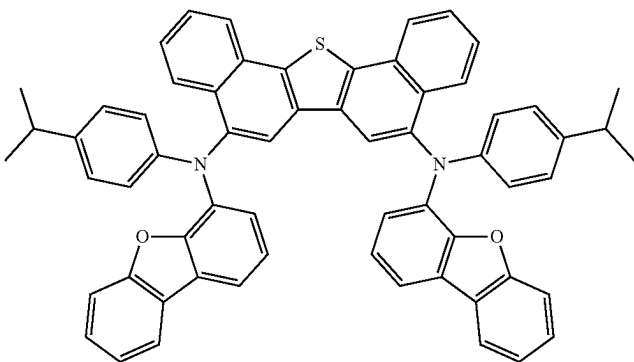
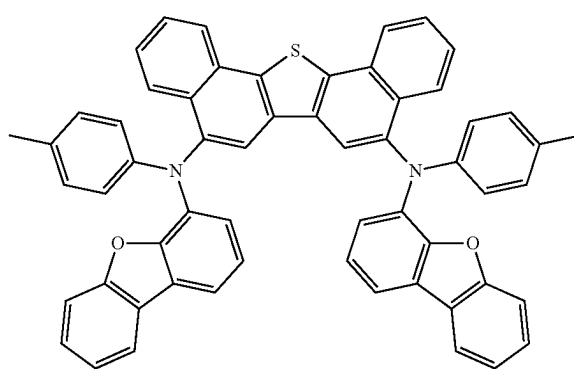
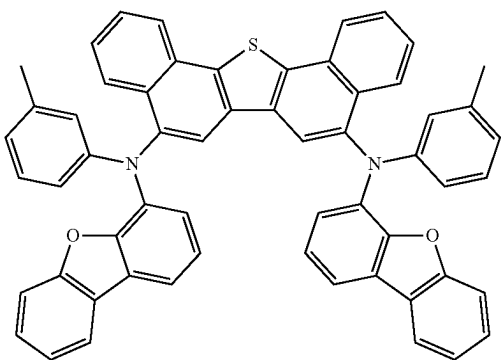
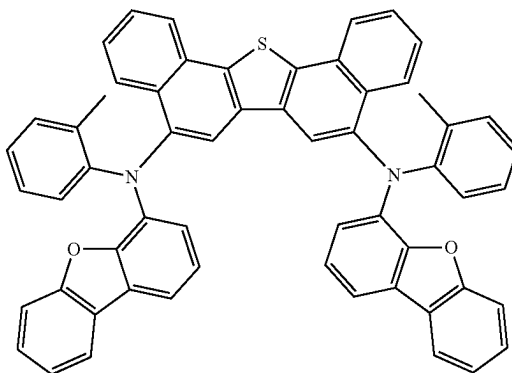
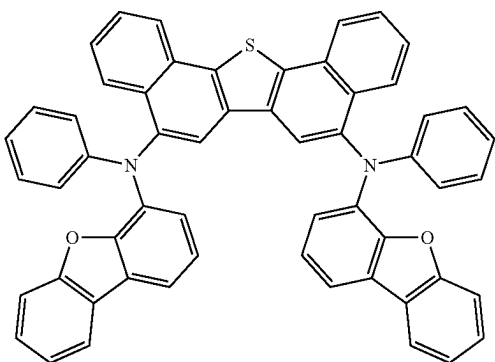
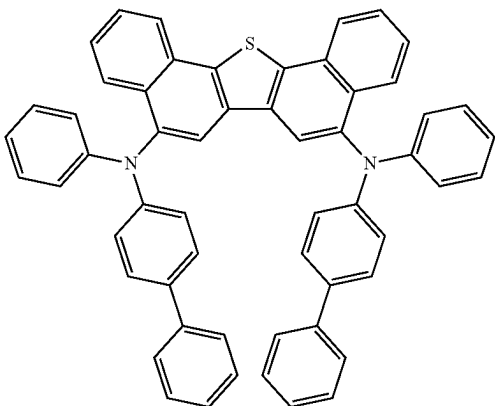
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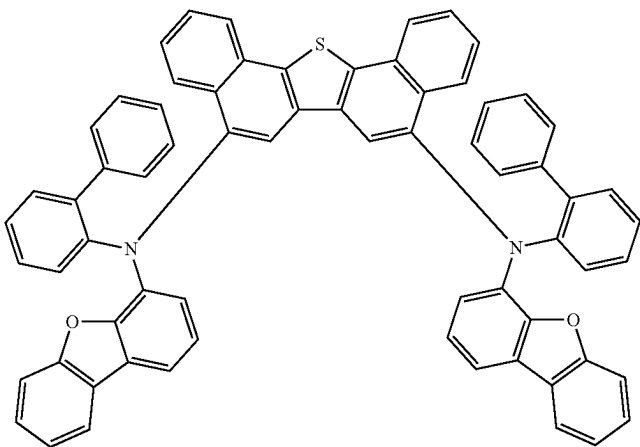
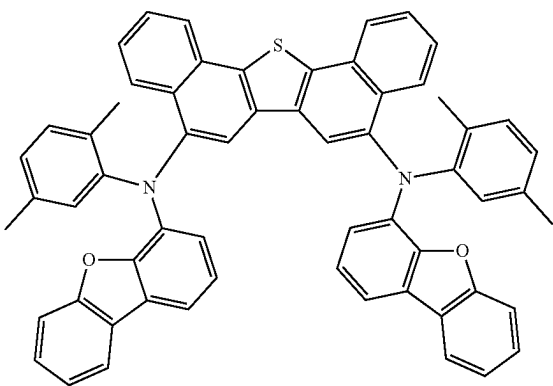
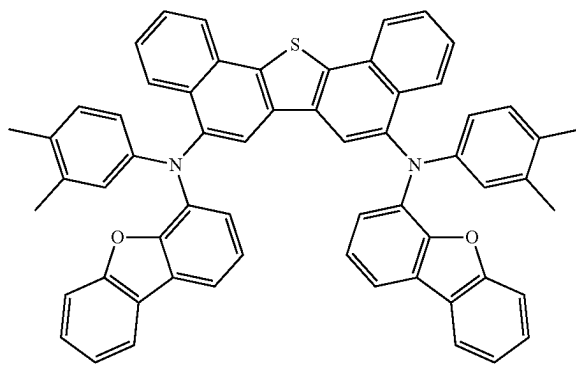
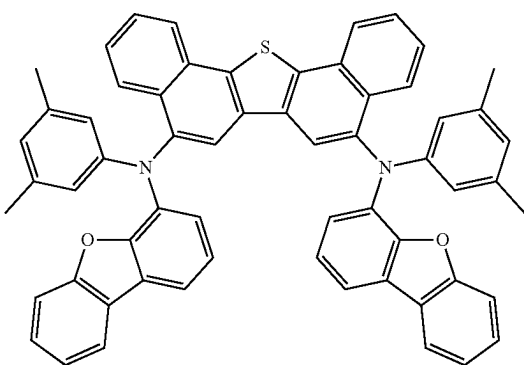
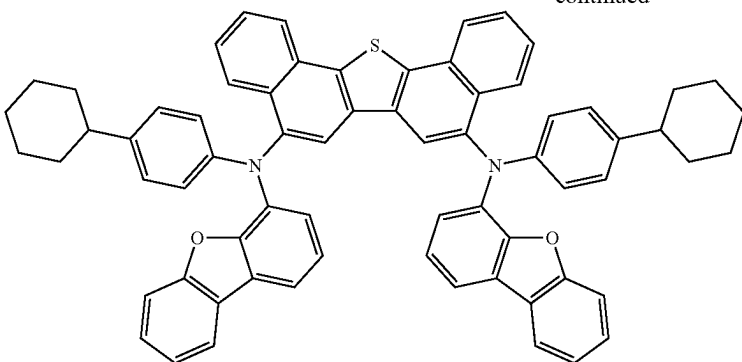
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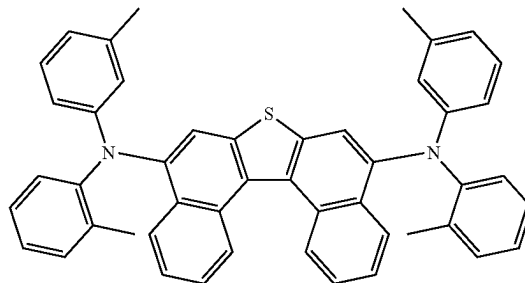
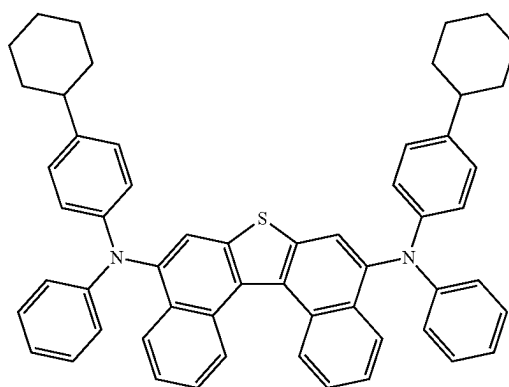
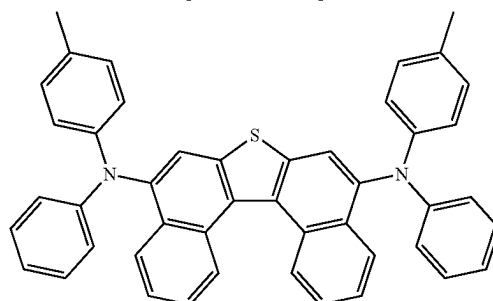
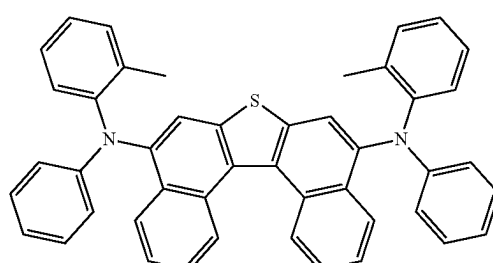
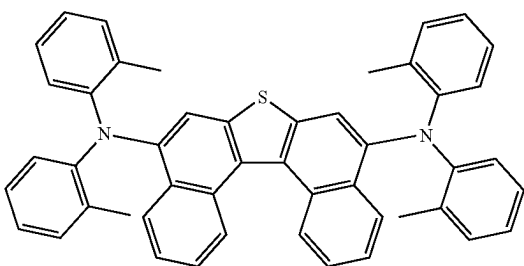
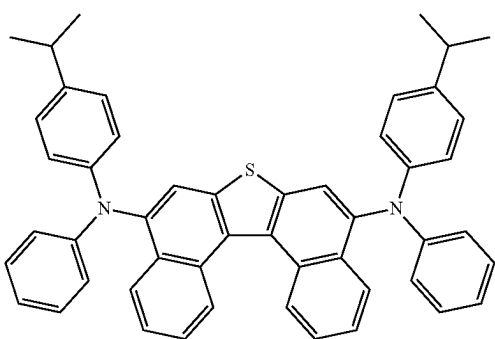
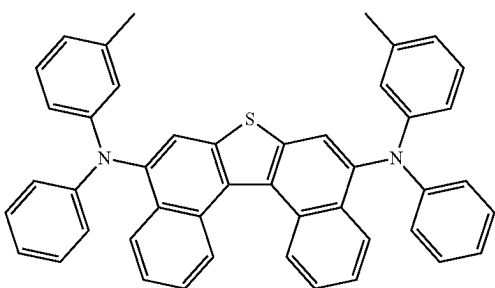
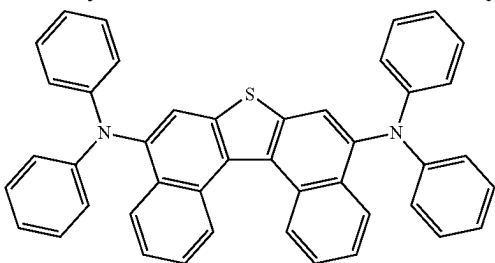
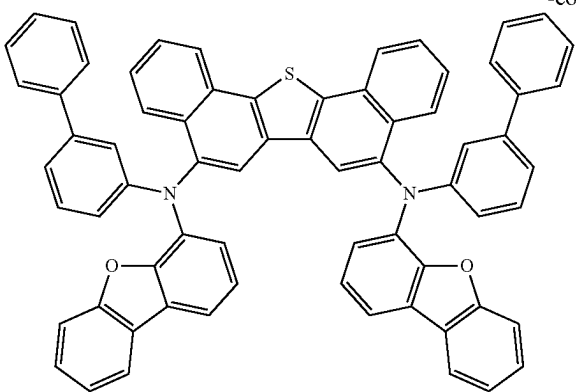
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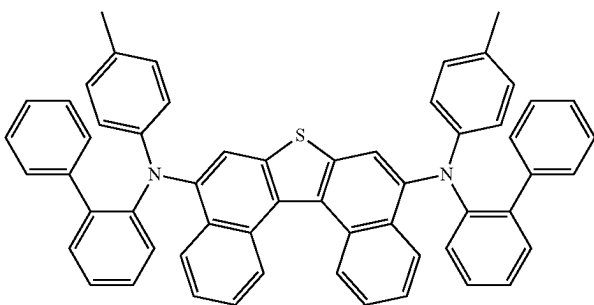
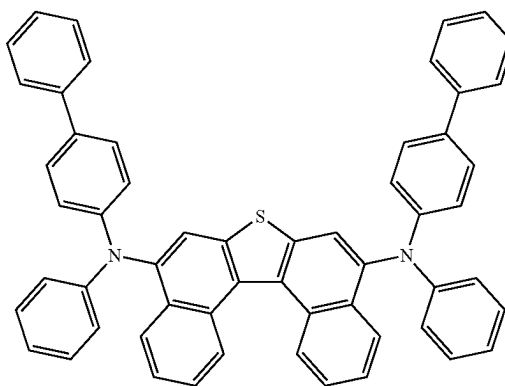
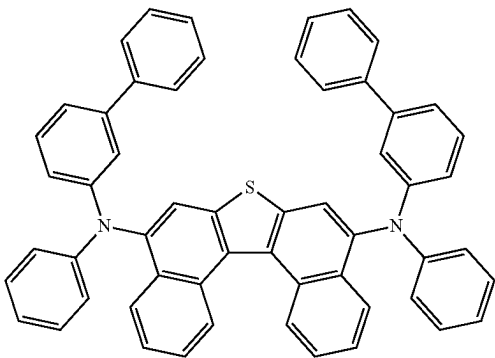
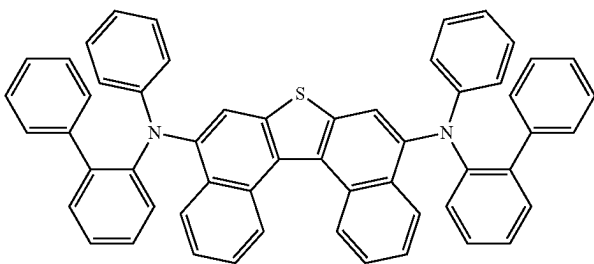
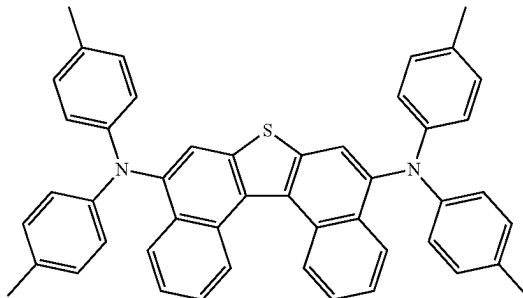
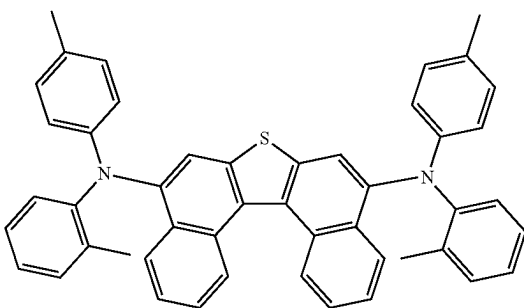
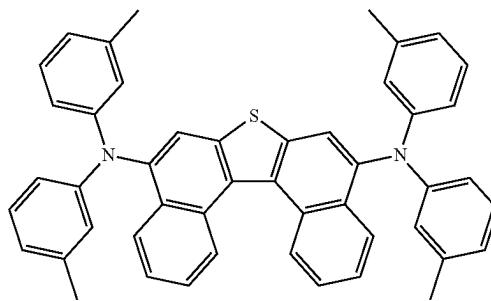
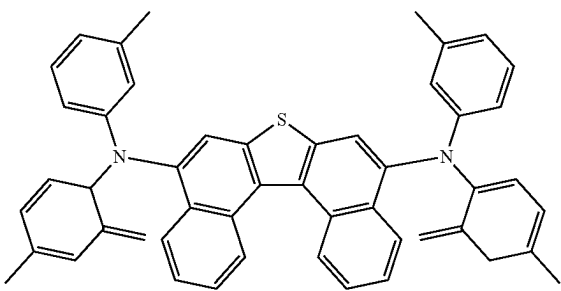
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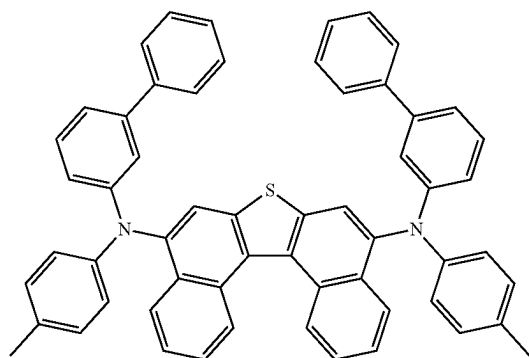
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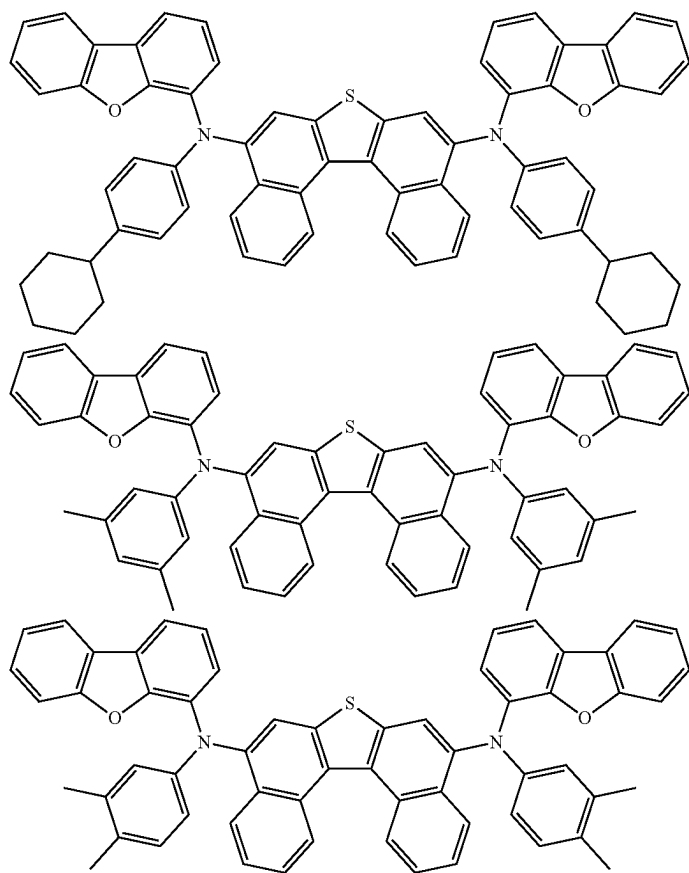
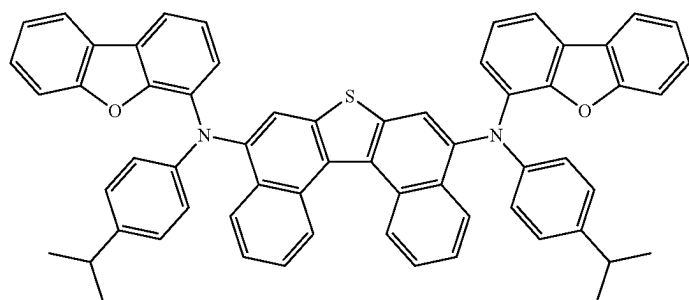
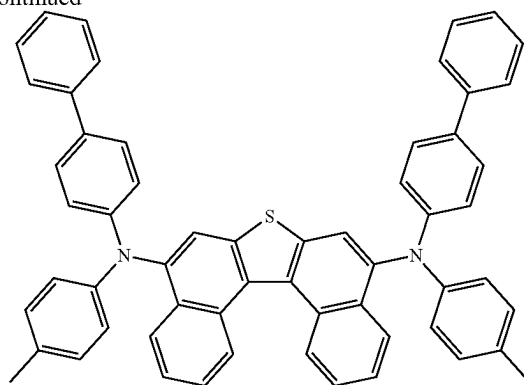


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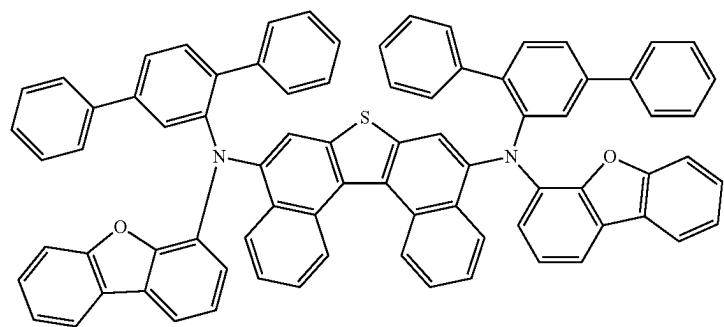
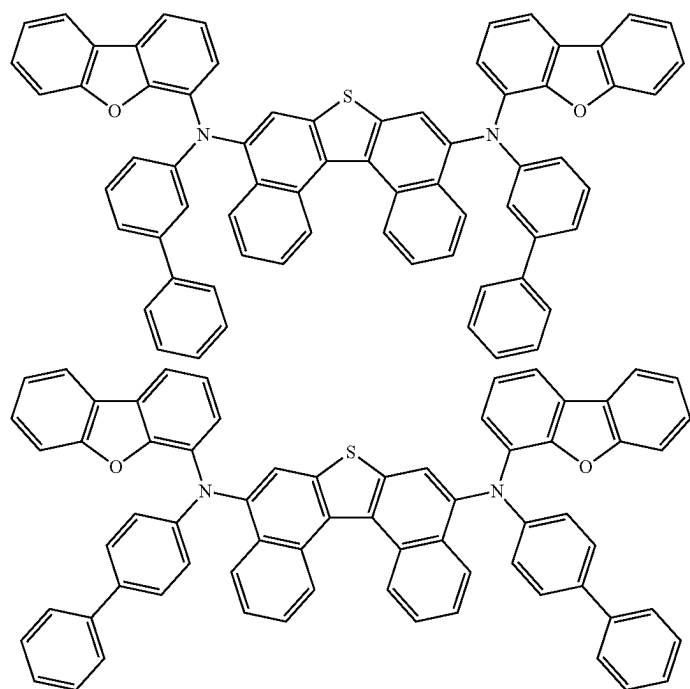
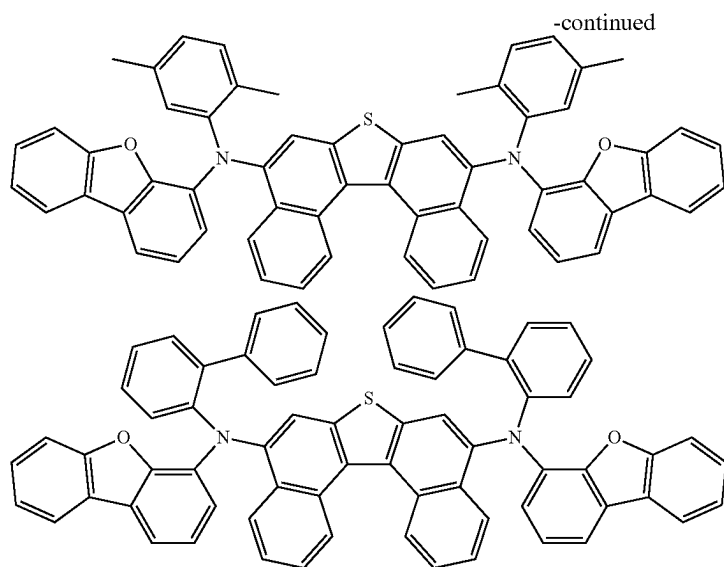
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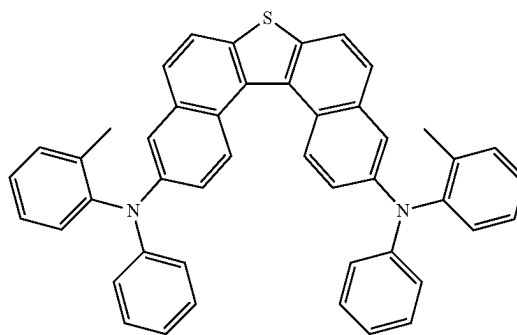
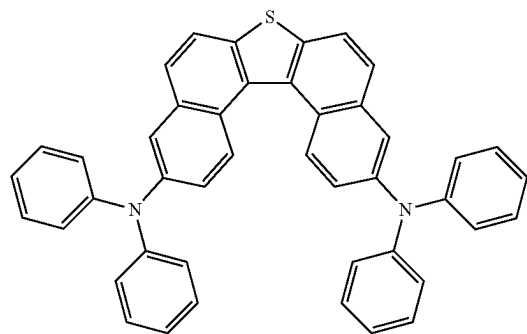
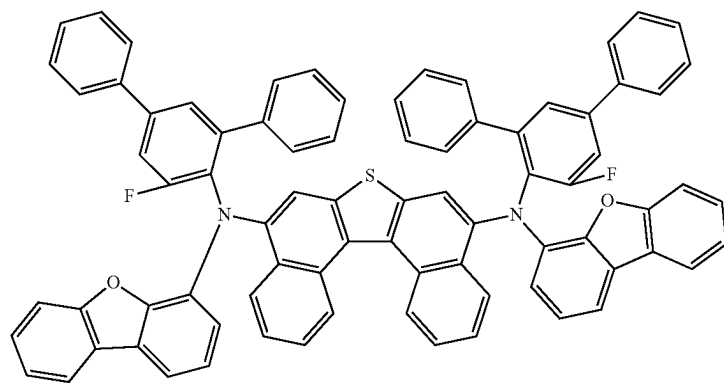
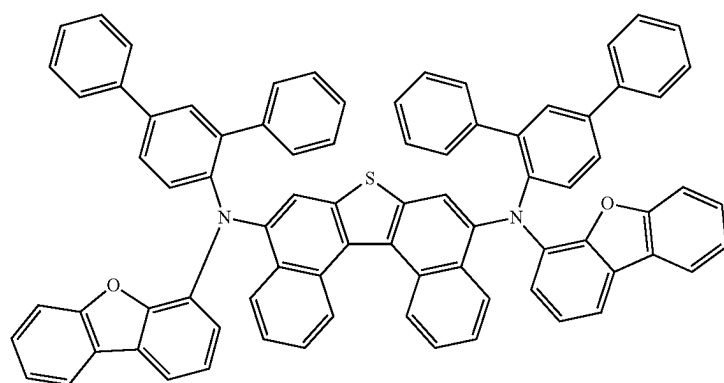
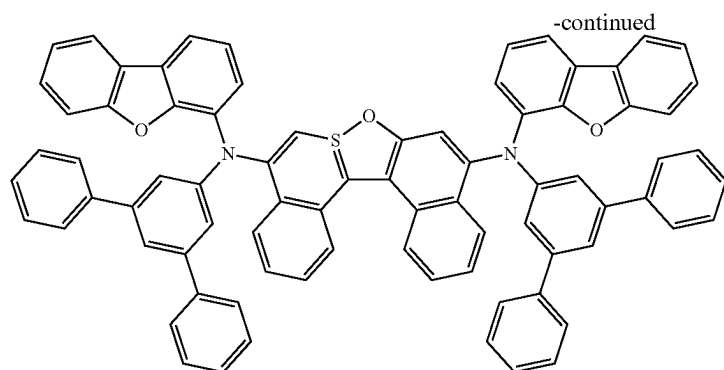
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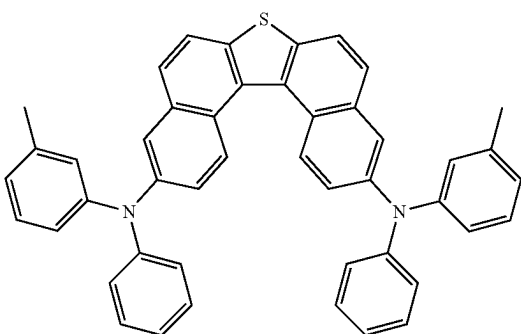


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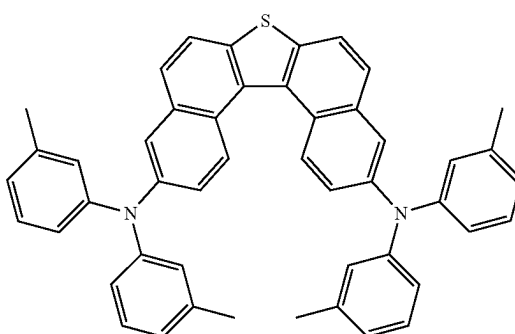
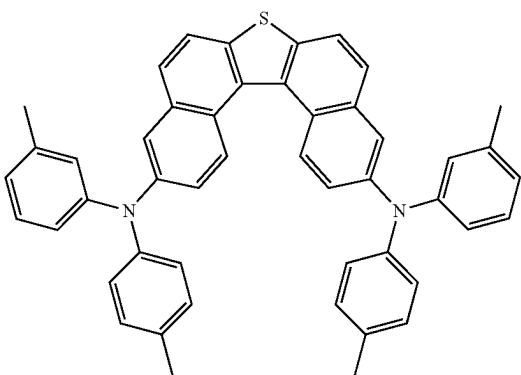
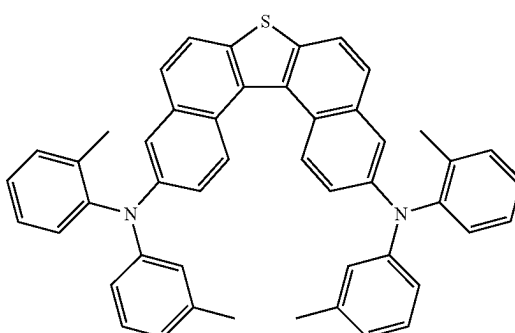
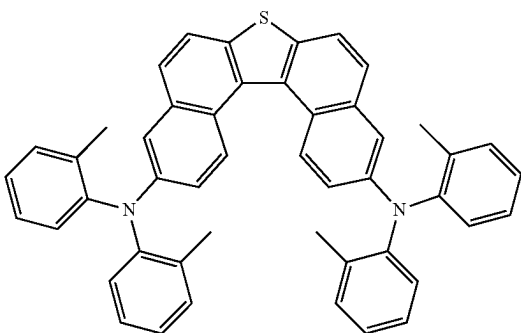
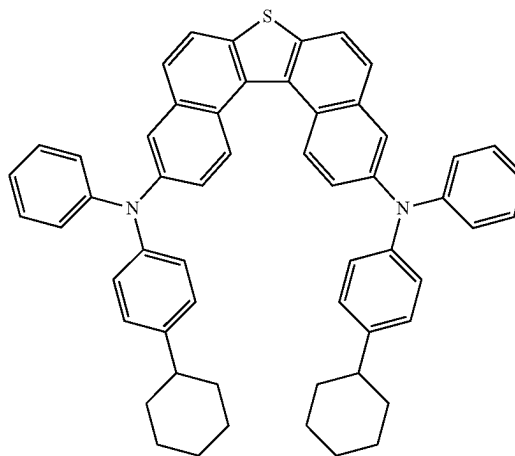
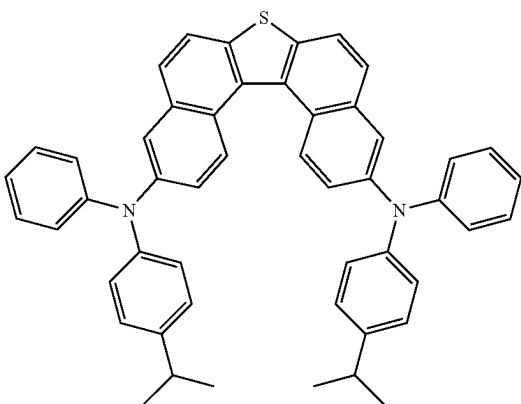
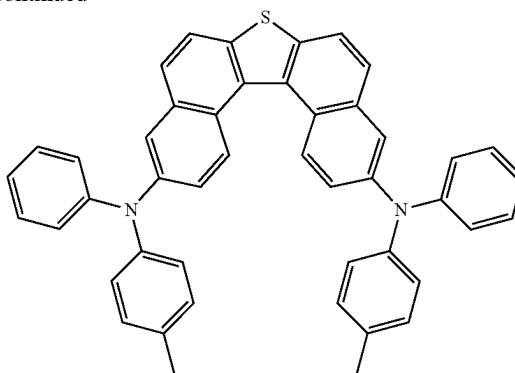


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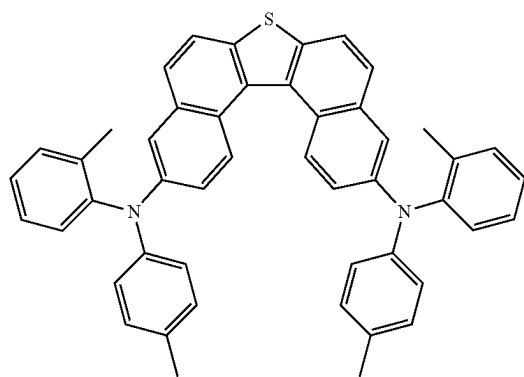


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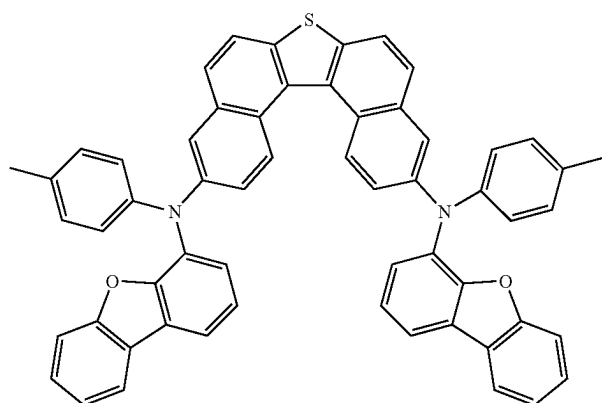
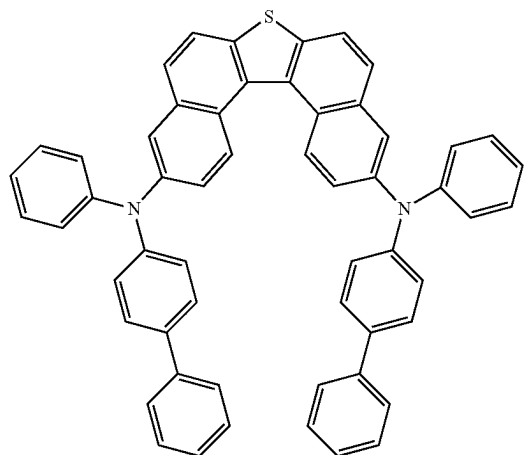
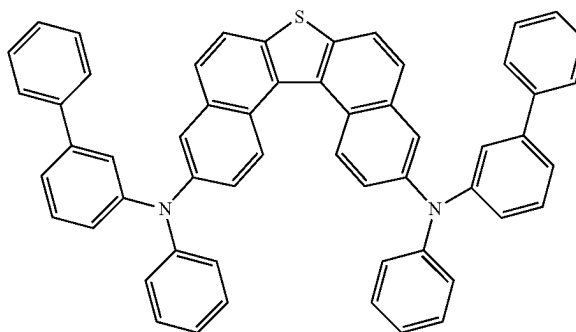
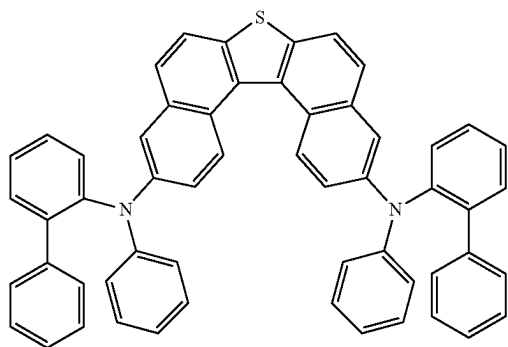
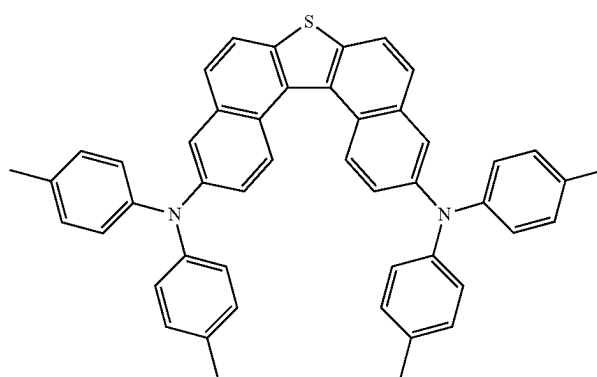


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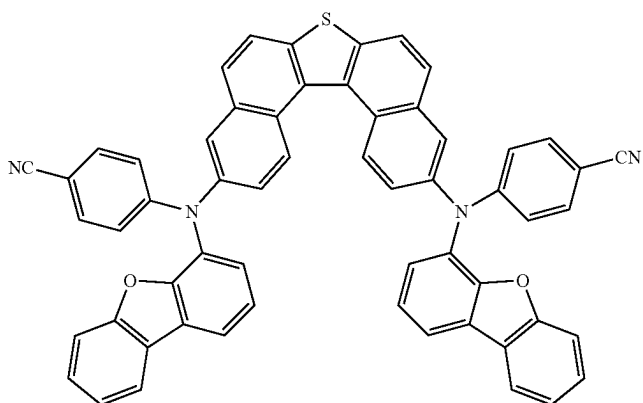
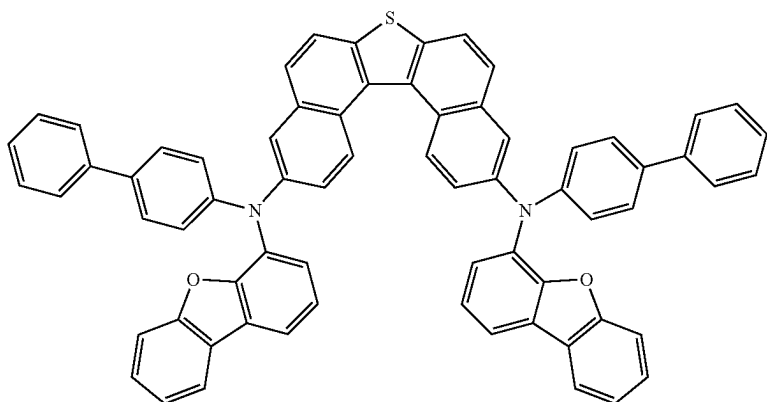
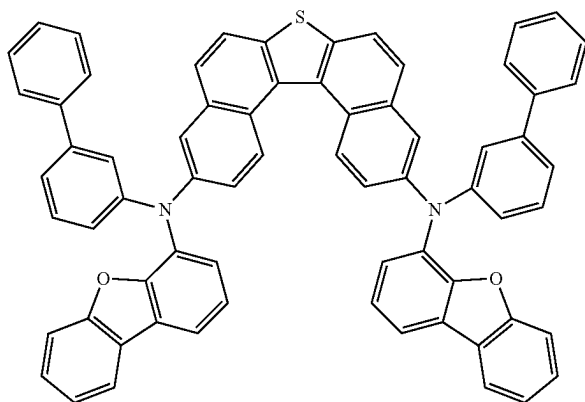
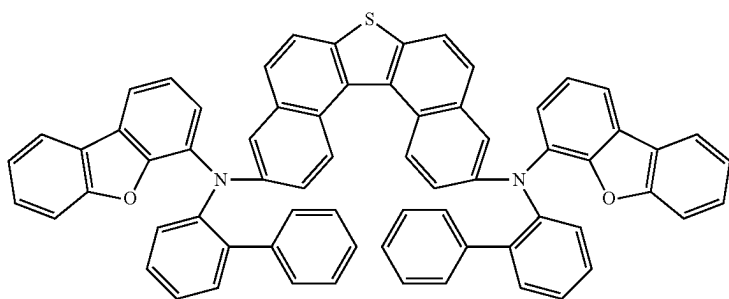
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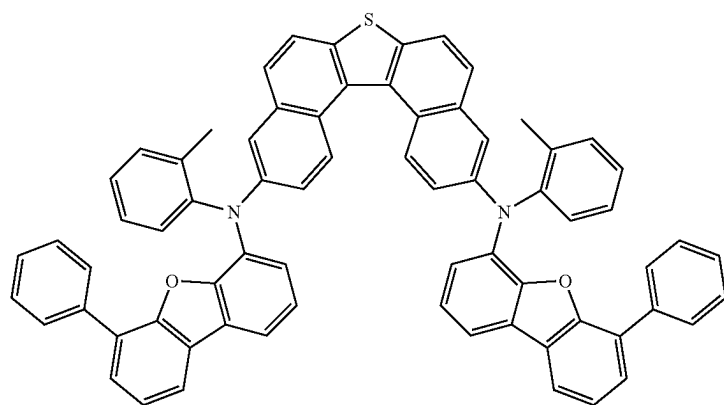
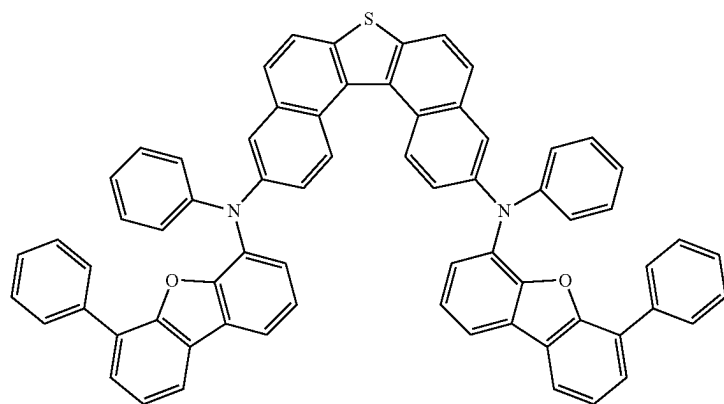
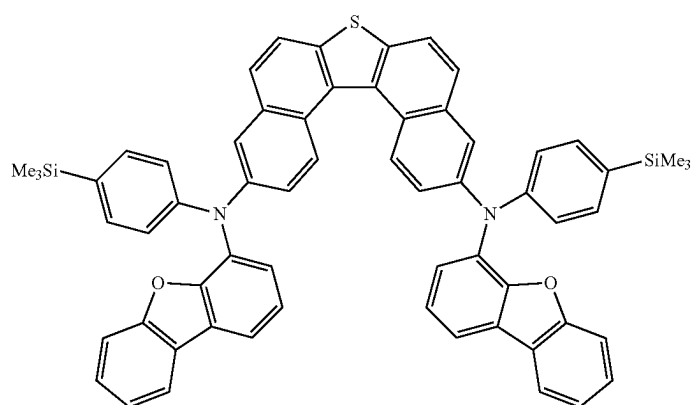
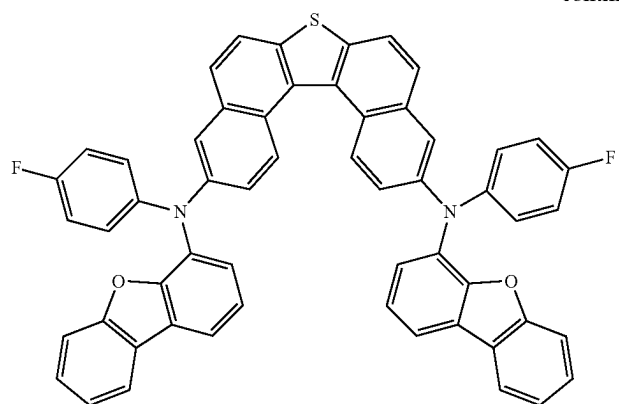
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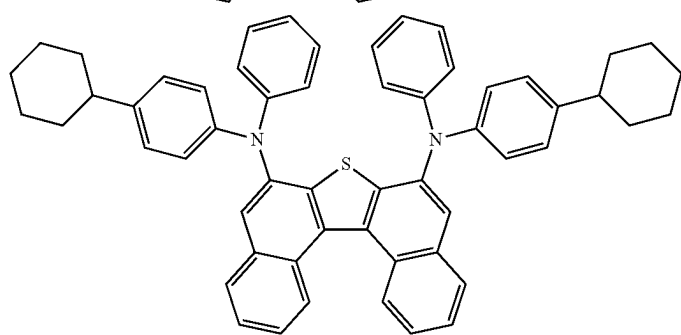
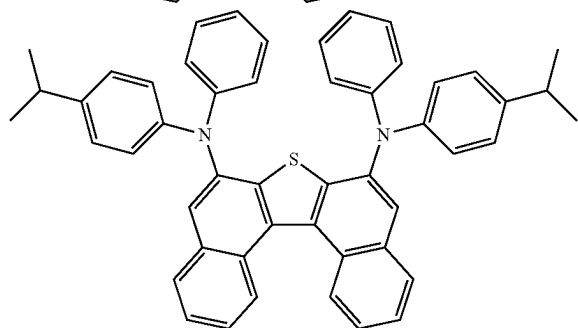
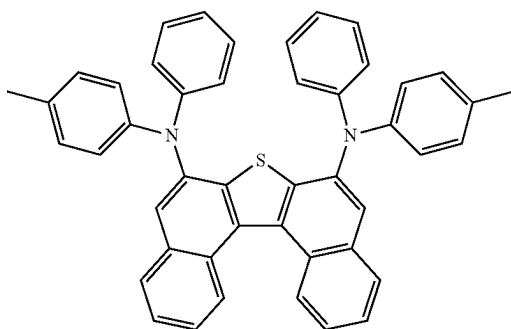
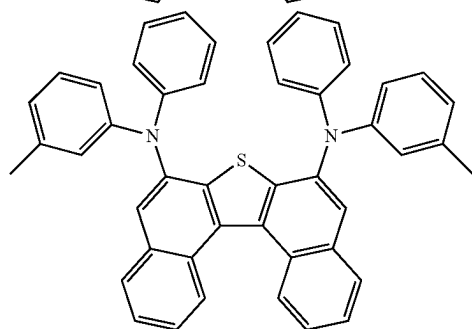
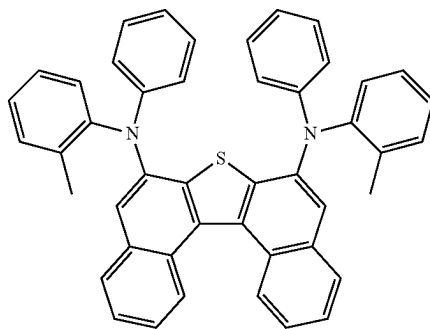
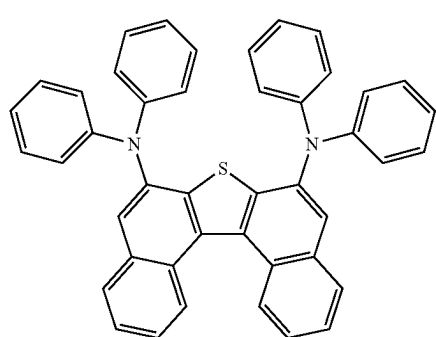
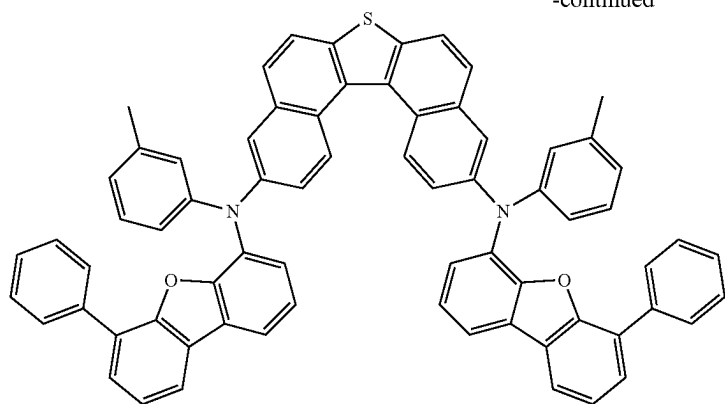
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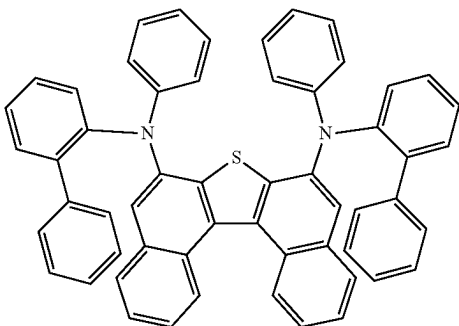
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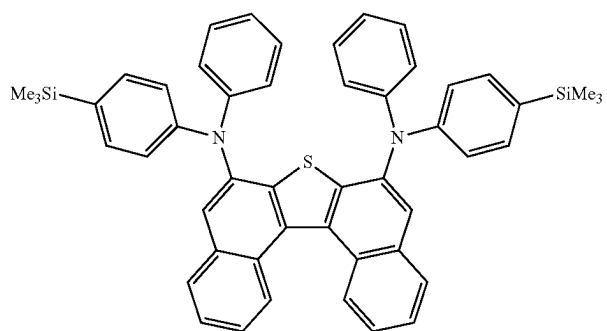
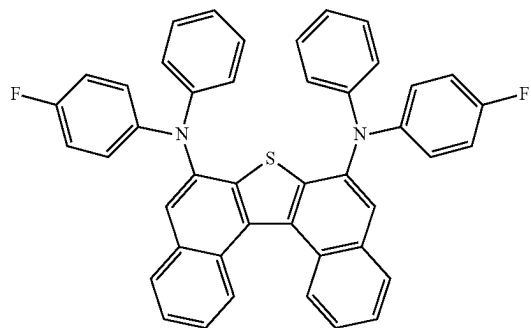
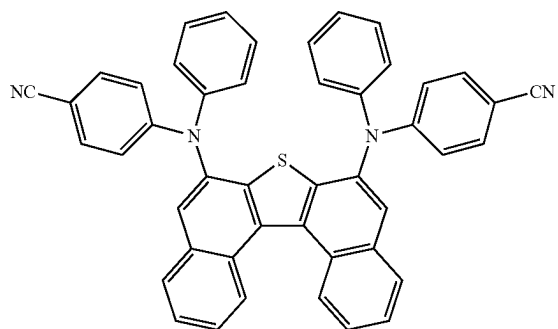
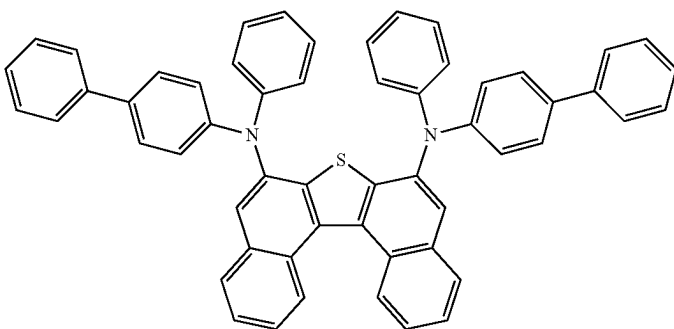
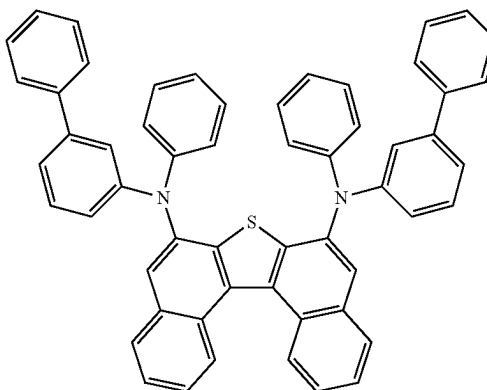


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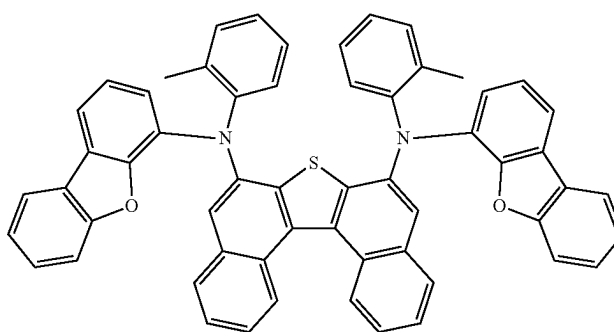
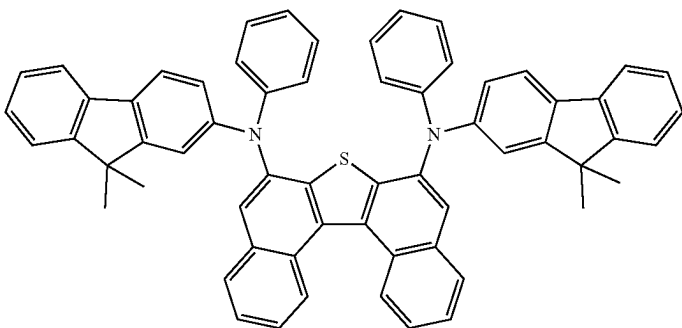
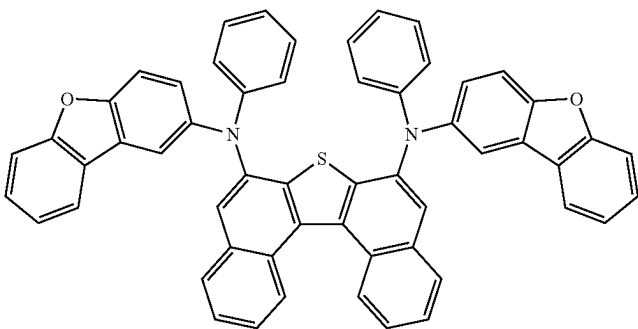
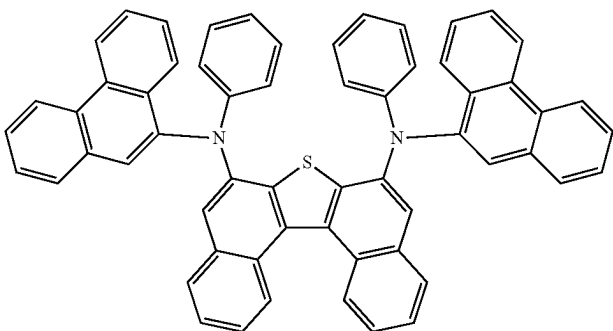
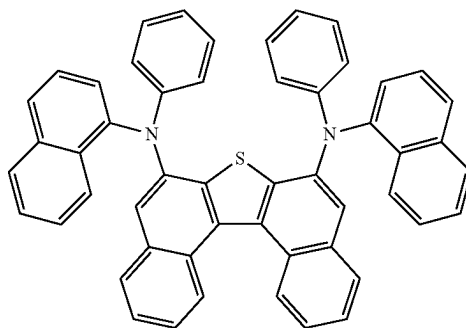
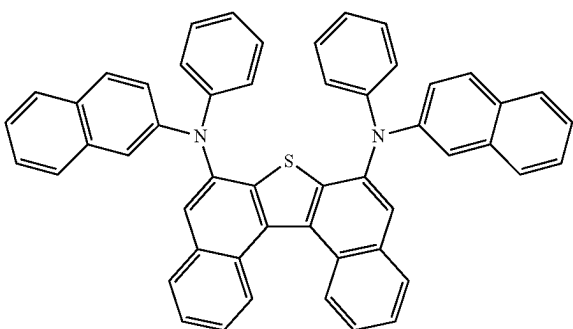
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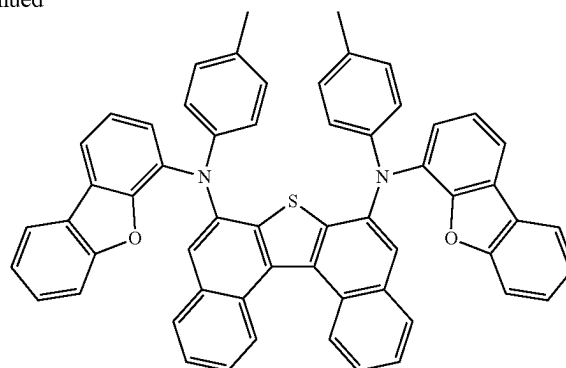
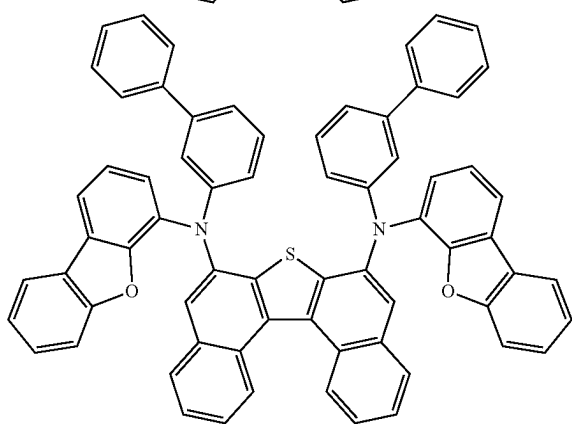
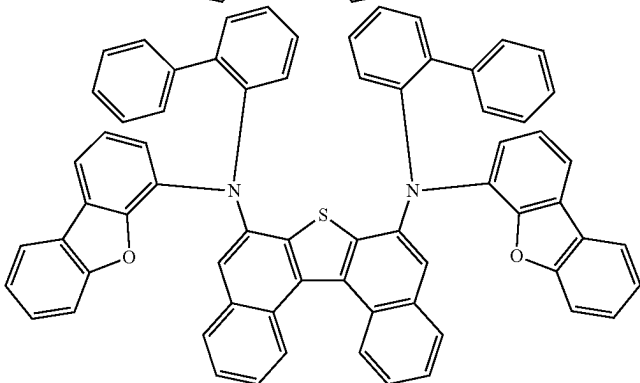
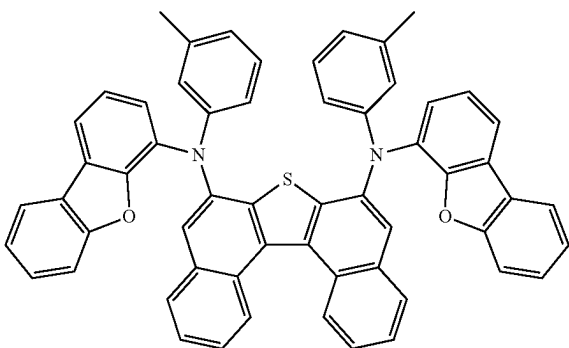
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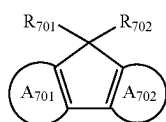
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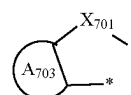
(Compound Represented by Formula (71))

The compound represented by the formula (71) is explained below.

One or more rings selected from the group consisting of A_{701} ring and A_{702} ring are bonded to the bond * of the structure represented by the following formula (72);



(71) 55



(72)

In the formula (71),

A_{701} ring and A_{702} ring are independently a substituted or unsubstituted aromatic hydrocarbon ring having 6 to 50 ring carbon atoms, or a substituted or unsubstituted heterocyclic ring having 5 to 50 ring atoms;

wherein, in the formula (72), A_{703} rings are independently a substituted or unsubstituted aromatic hydrocarbon ring having 6 to 50 ring carbon atoms, or a substituted or unsubstituted heterocyclic ring having 5 to 50 ring atoms; X_{701} is NR_{703} , $C(R_{704})(R_{705})$, $Si(R_{706})(R_{707})$, $Ge(R_{708})(R_{709})$, O, S or Se;

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R_{701} and R_{702} are bonded with each other to form a substituted or unsubstituted, saturated or unsaturated ring or do not form a substituted or unsubstituted saturated or unsaturated ring;

R_{701} and R_{702} that do not form the substituted or unsubstituted, saturated or unsaturated ring, and R_{703} to R_{709} are independently

a hydrogen atom,

a substituted or unsubstituted alkyl group having 1 to 50 carbon atoms,

a substituted or unsubstituted alkenyl group having 2 to 50 carbon atoms,

a substituted or unsubstituted alkynyl group having 2 to 50 carbon atoms,

a substituted or unsubstituted cycloalkyl group having 3 to 50 ring carbon atoms,

—Si(R_{901})(R_{902})(R_{907}),

—O—(R_{904}),

—S—(R_{905}),

—N(R_{906})(R_{907}),

a halogen atom, a cyano group, a nitro group,

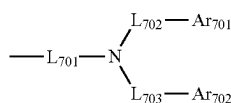
a substituted or unsubstituted aryl group having 6 to 50 ring carbon atoms, or

a substituted or unsubstituted monovalent heterocyclic group having 5 to 50 ring atoms;

R_{901} to R_{907} are as defined in the formula (1).

One or more selected from the group consisting of A_{701} ring and A_{702} ring is bonded to * in the structure represented by the formula (72). That is, in one embodiment, the ring carbon atom of the aromatic hydrocarbon ring or the ring atom of the heterocyclic ring of A_{701} ring is bonded to * in the structure represented by the formula (72). In one embodiment, the ring carbon atom of the aromatic hydrocarbon ring or the ring atom of the heterocyclic ring of A_{701} ring is bonded to * in the structure represented by the formula (72).

In one embodiment, the group represented by the formula (73) is bonded to one or both of A_{701} ring and A_{7m} ring:



wherein in the formula (73), Ar_{701} and Ar_{702} are independently

a substituted or unsubstituted aryl group having 6 to 50 ring carbon atoms, or

a substituted or unsubstituted monovalent heterocyclic group having 5 to 50 ring atoms; and

L_{701} to L_{703} are independently

a single bond,

a substituted or unsubstituted arylene group having 6 to 30 ring carbon atoms,

a substituted or unsubstituted divalent heterocyclic group having 5 to 30 ring atoms, or

a divalent linking group formed by bonding 2 to 4 above mentioned groups.

In one embodiment, in addition to A_{701} ring, the ring carbon atom of the aromatic hydrocarbon ring or the ring atom of the heterocyclic ring of A_{702} ring is bonded to * in the structure represented by the formula (72). In this case, the structures represented by formula (72) may be the same or different.

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In one embodiment, R_{701} and R_{702} are independently a substituted or unsubstituted aryl group having 6 to 50 ring carbon atoms.

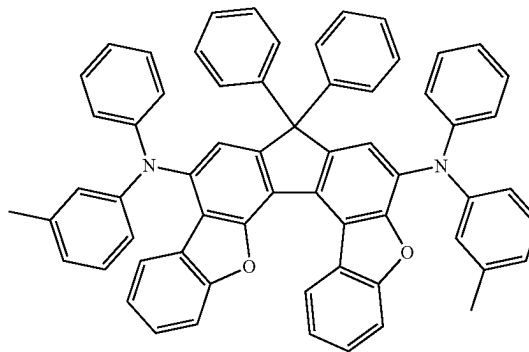
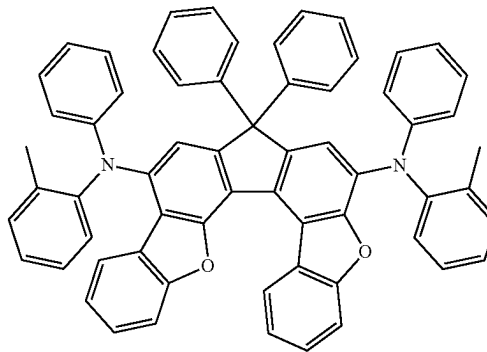
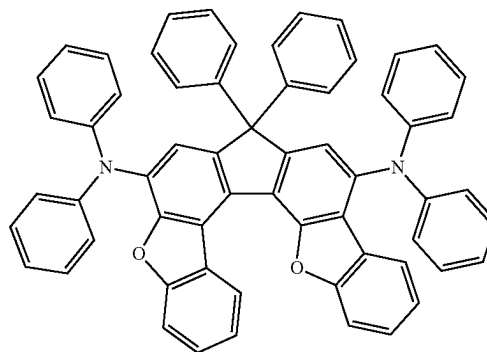
In one embodiment, R_{701} and R_{702} are bonded with each other to form a fluorene structure.

In one embodiment, Ar_{701} ring and Ar_{702} ring are substituted or unsubstituted aromatic hydrocarbon rings having 6 to 50 ring carbon atoms, and they are substituted or unsubstituted benzene rings, for example.

In one embodiment, Ar_{703} ring is a substituted or unsubstituted aromatic hydrocarbon ring having 6 to 50 ring carbon atoms, and it is a substituted or unsubstituted benzene ring, for example.

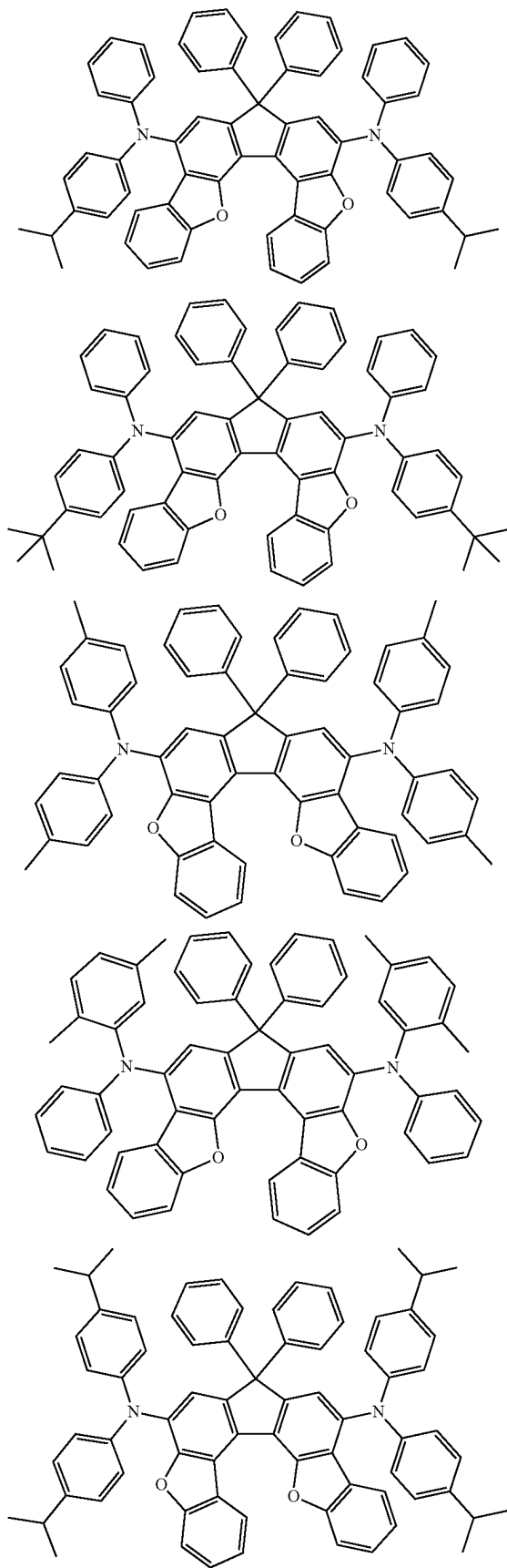
In one embodiment, X_{701} is O or S.

As specific example of the compound represented by the formula (71), the following compounds can be given, for example. In the following example compounds, Me represents methyl group.

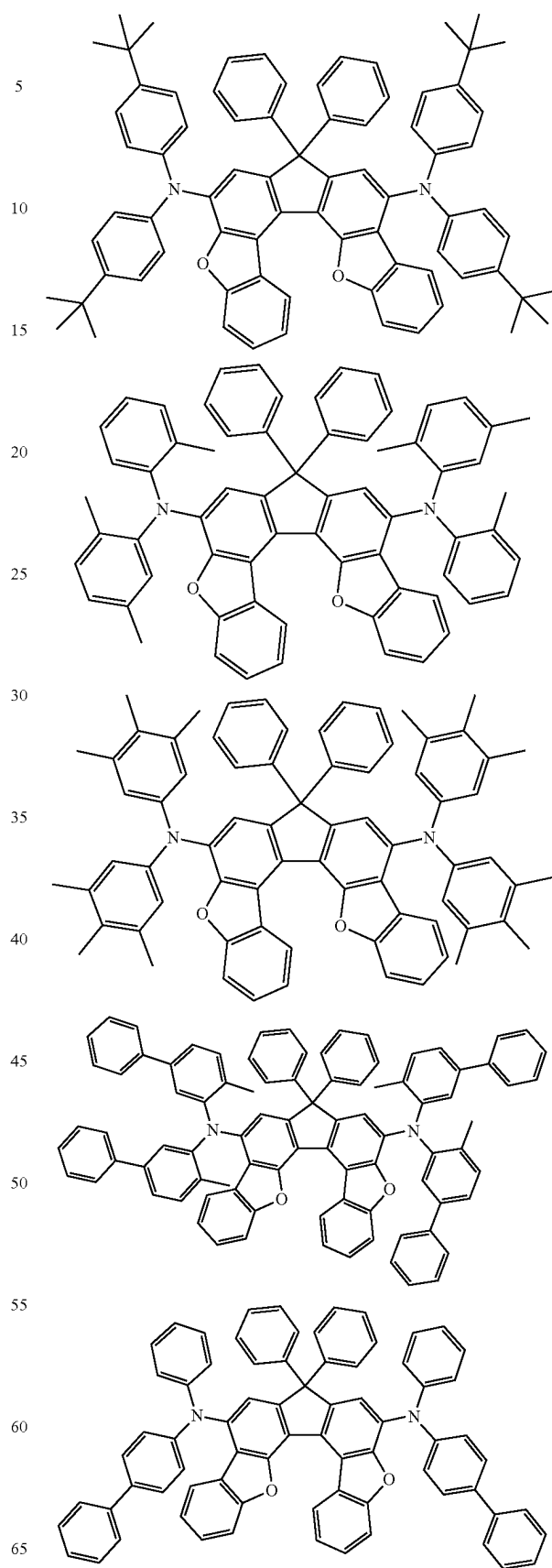


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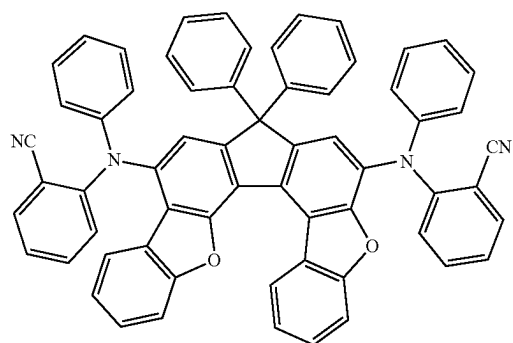
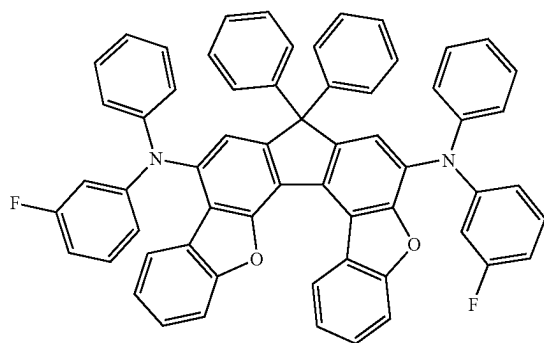
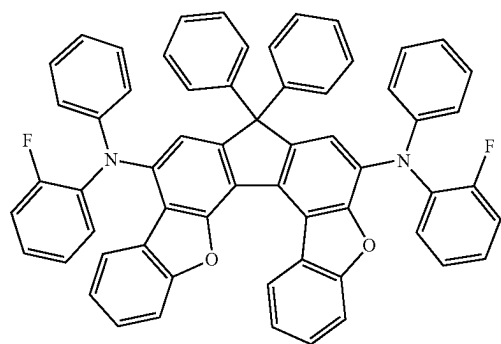
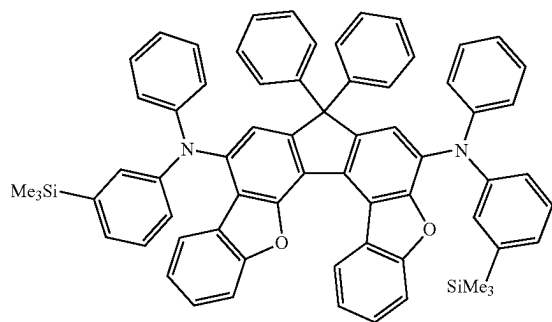
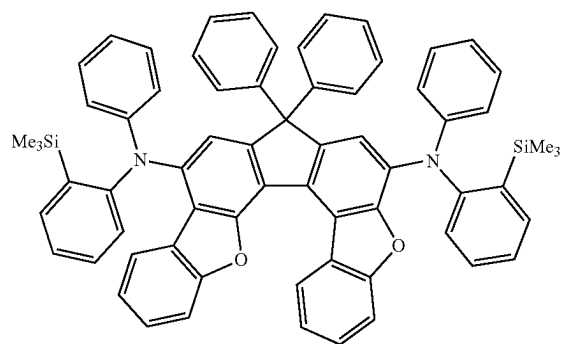
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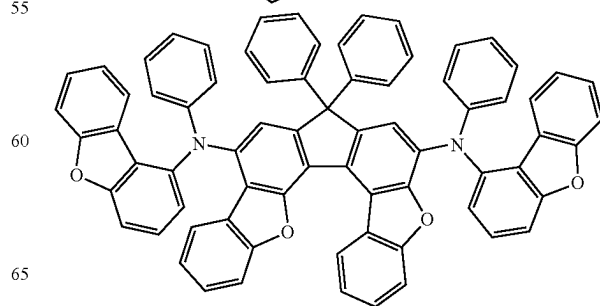
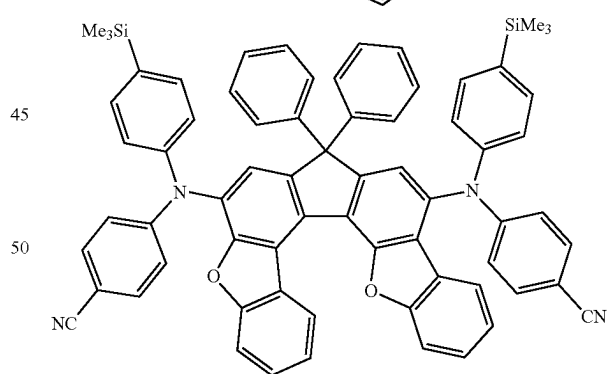
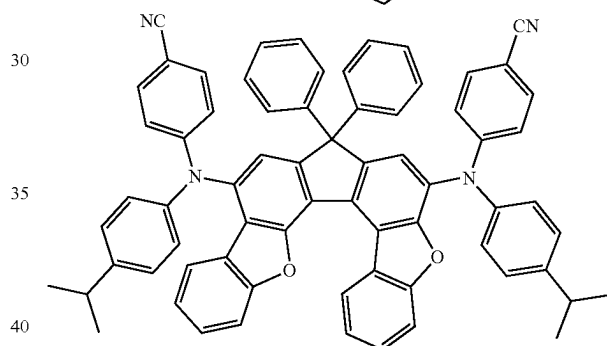
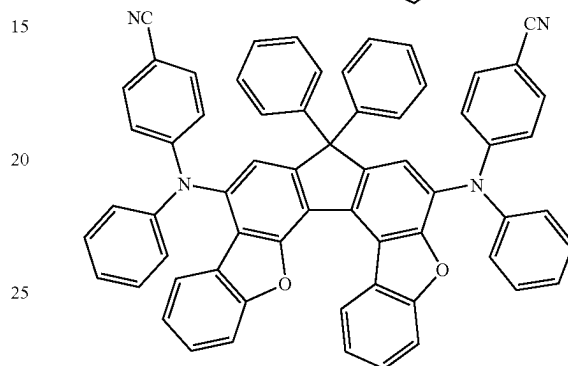
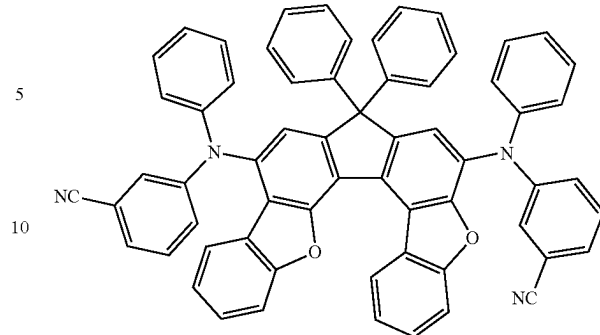


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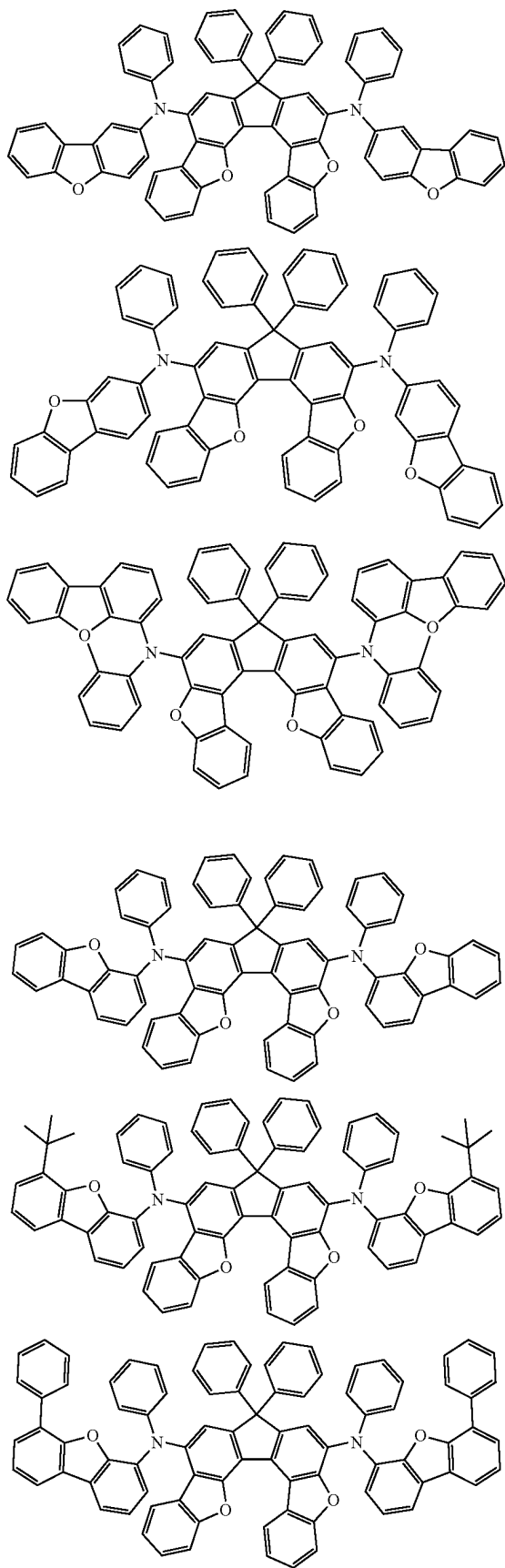
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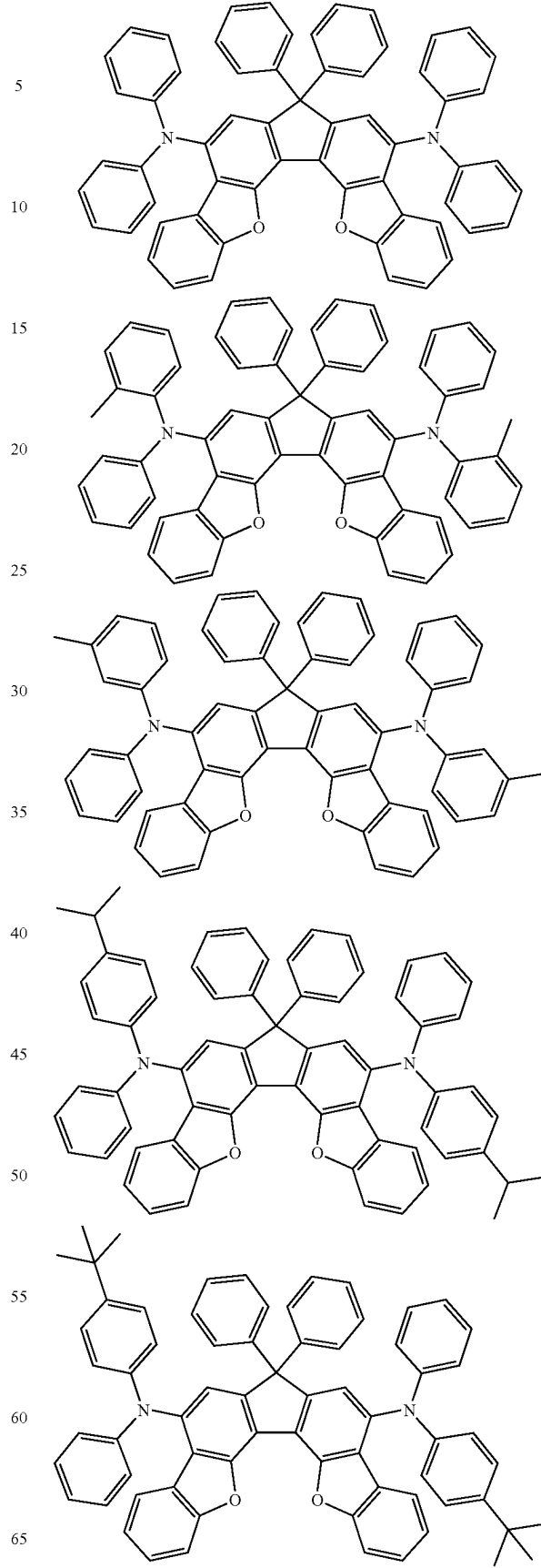


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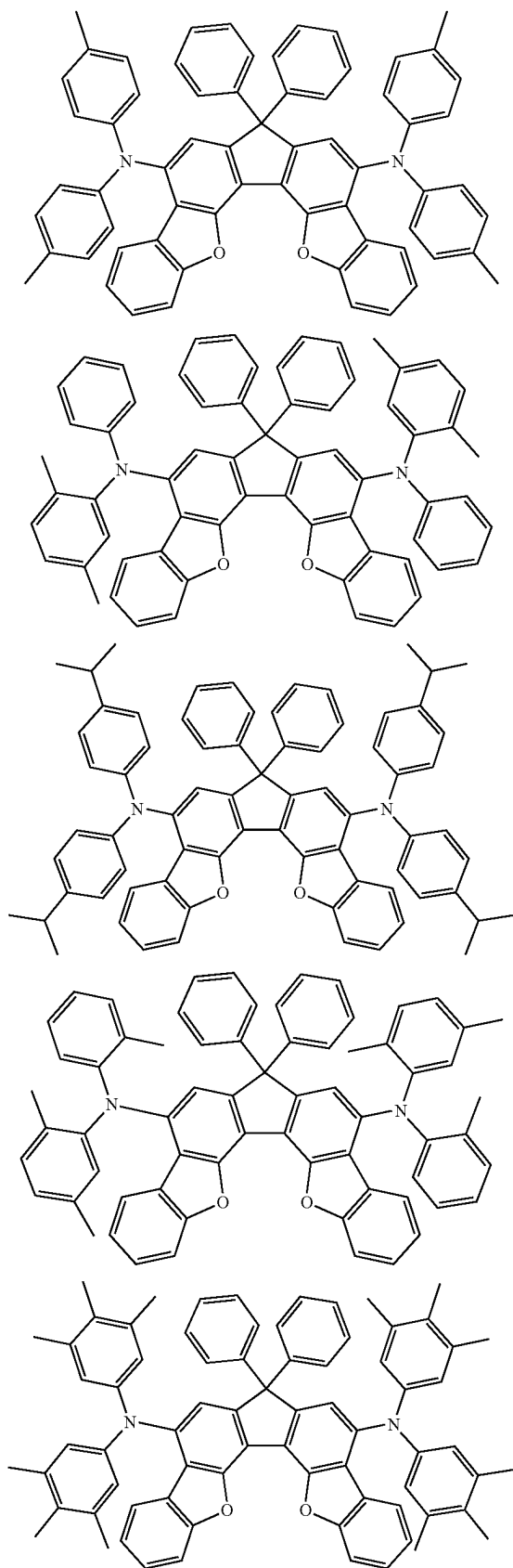
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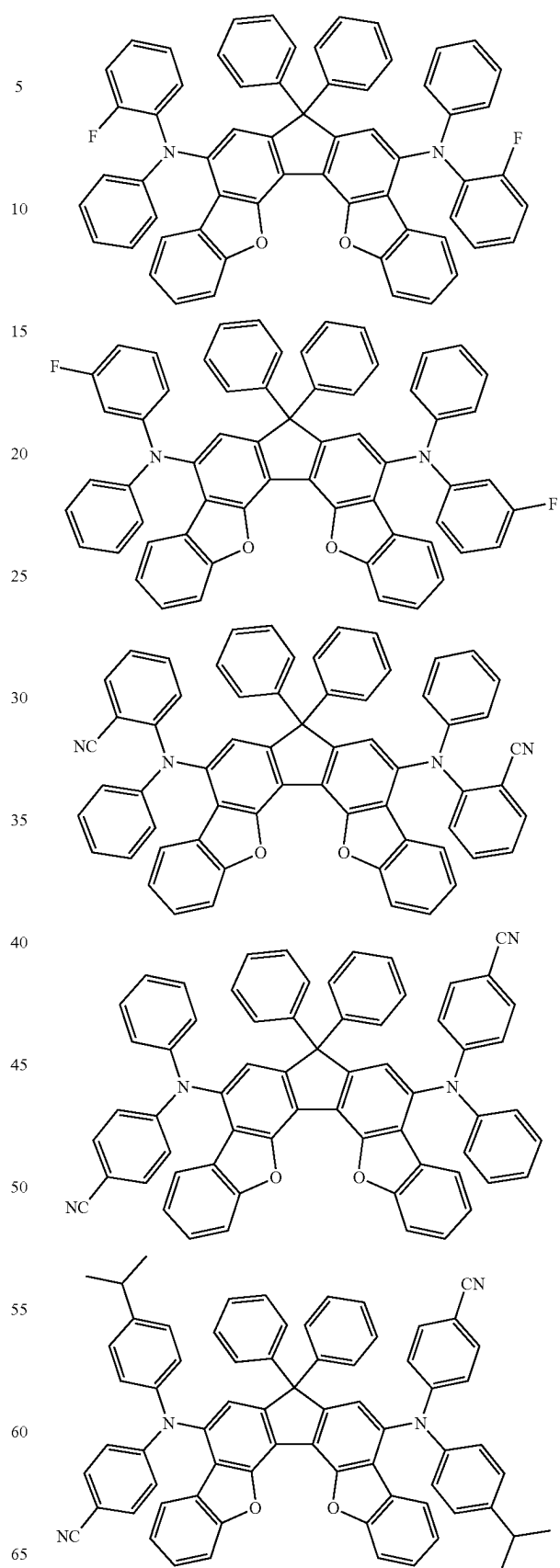


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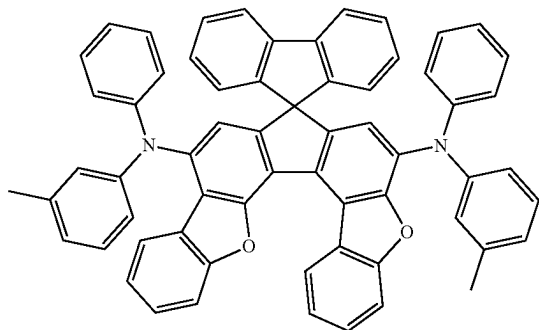
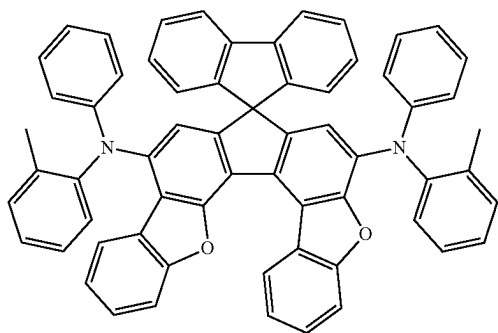
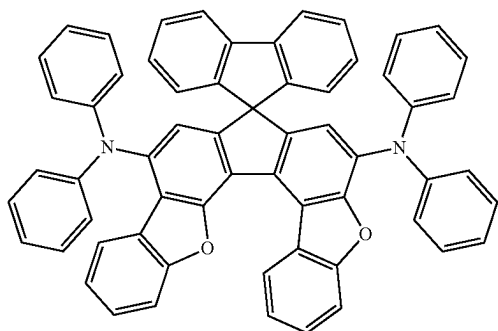
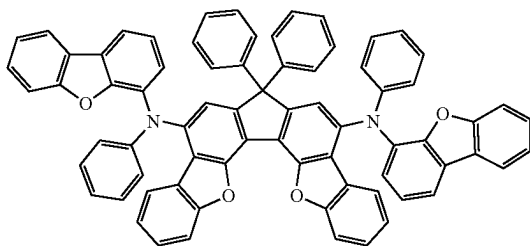
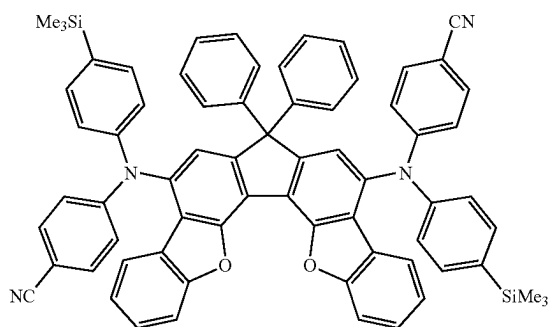
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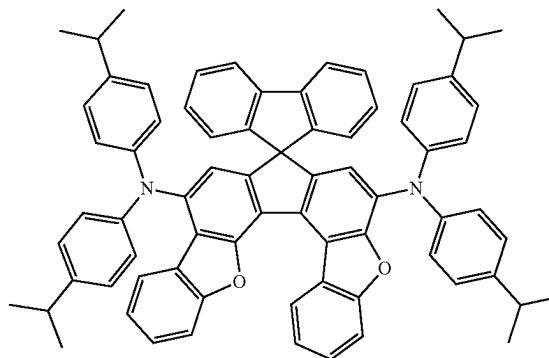
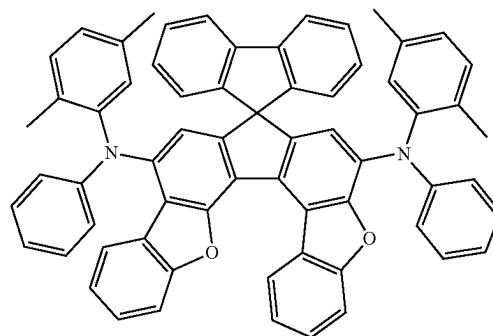
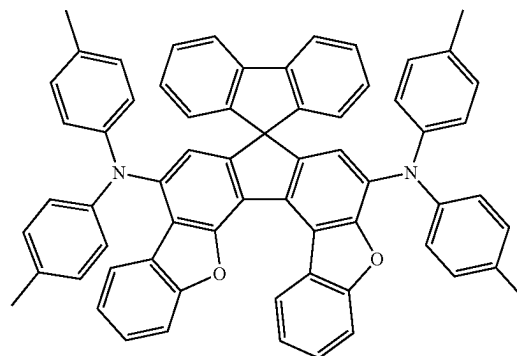
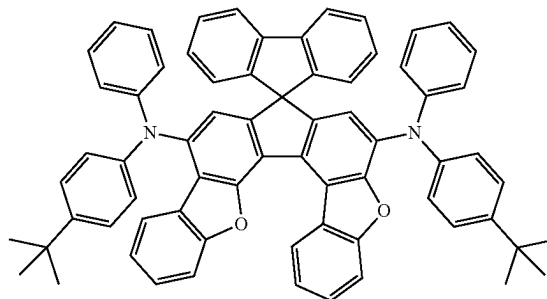
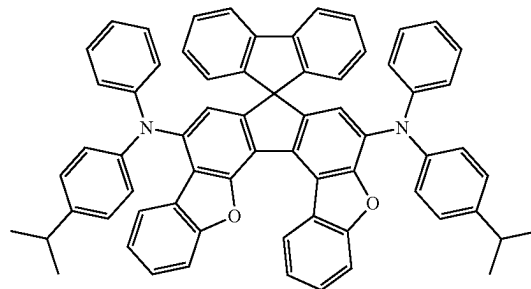
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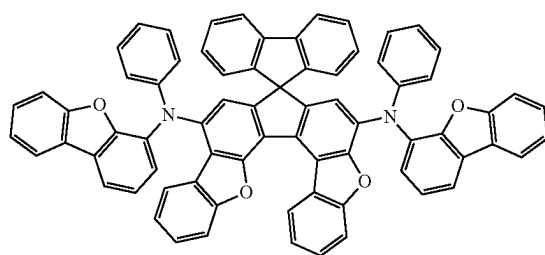
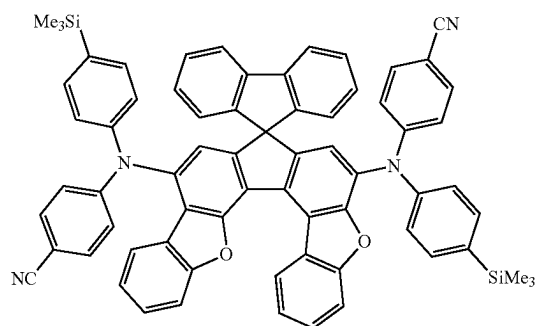
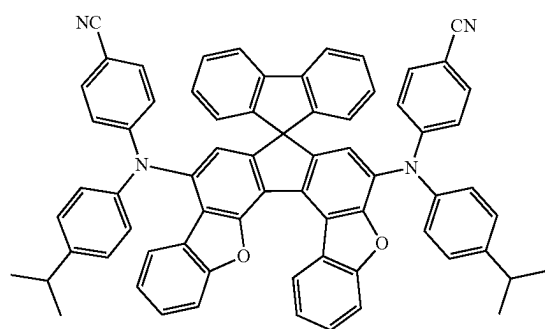
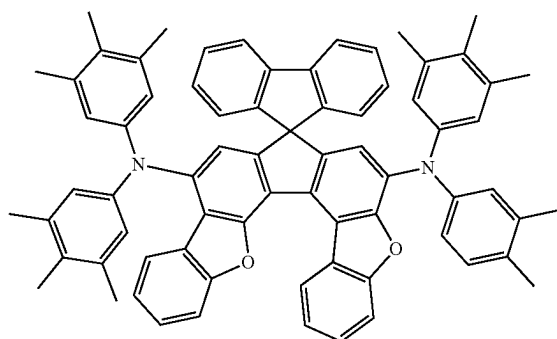
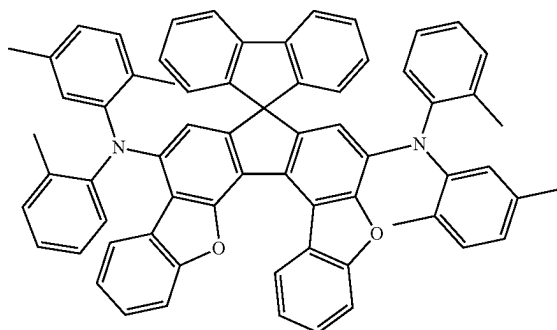
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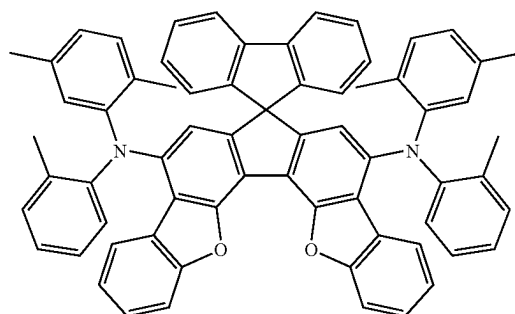
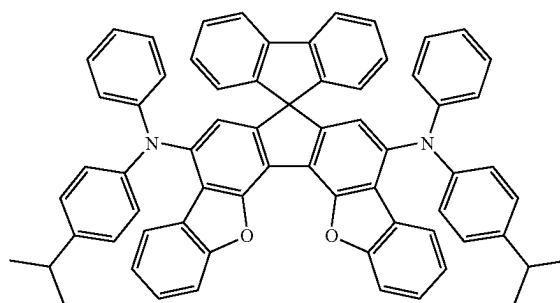
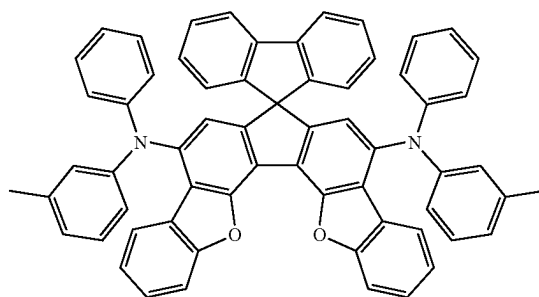
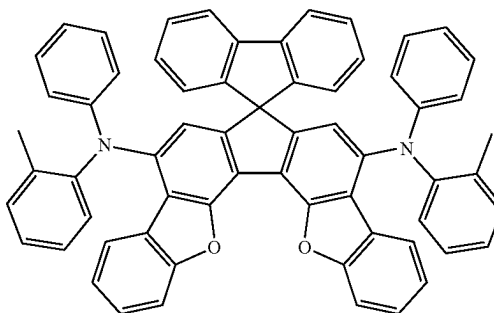
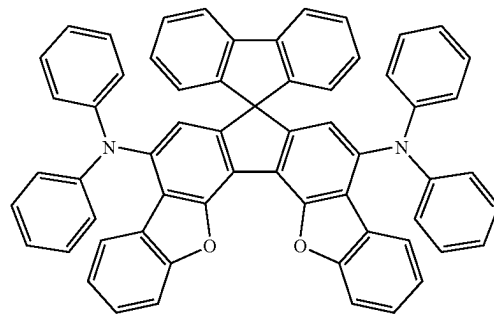
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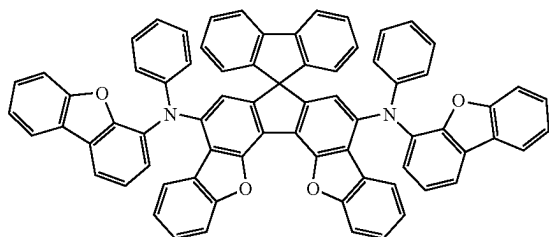
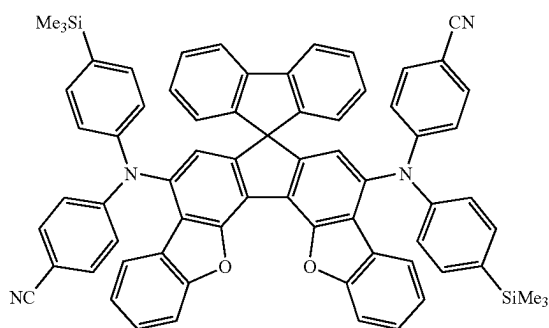
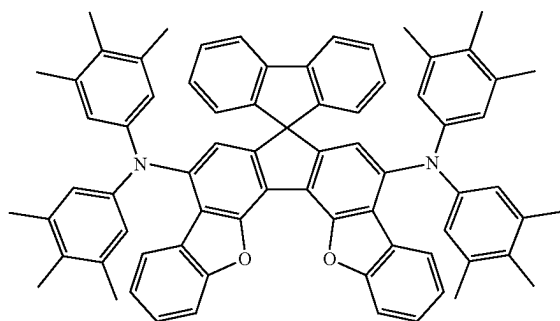
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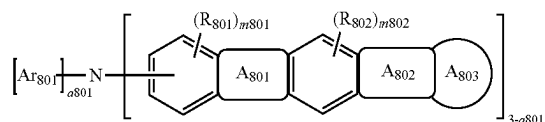
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(Compound Represented by Formula (81))

The compound represented by the formula (81) is explained below.



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(83)

In the formula (81),

A_{801} ring is a ring represented by the formula (82) which is fused to an adjacent ring at an arbitrary position;

A_{802} ring is a ring represented by the formula (83) which is fused to an adjacent ring at an arbitrary position;

two bonds * bond to A_{803} ring at an arbitrary position;

X_{801} and X_{802} are independently $C(R_{803})(R_{804})$, $Si(R_{805})(R_{806})$, an oxygen atom, or a sulfur atom;

A_{803} ring is a substituted or unsubstituted aromatic hydrocarbon ring having 6 to 50 ring carbon atoms, or a substituted or unsubstituted heterocyclic ring having 5 to 50 ring atoms;

Ar_{801} is a substituted or unsubstituted aryl group having 6 to 50 ring carbon atoms, or a substituted or unsubstituted monovalent heterocyclic group having 5 to 50 ring atoms;

R_{801} to R_{806} are independently

a hydrogen atom,

a substituted or unsubstituted alkyl group having 1 to 50 carbon atoms,

a substituted or unsubstituted alkenyl group having 2 to 50 carbon atoms,

a substituted or unsubstituted alkynyl group having 2 to 50 carbon atoms,

a substituted or unsubstituted cycloalkyl group having 3 to 50 ring carbon atoms,

$-Si(R_{901})(R_{902})(R_{903})$,

$-O-(R_{904})$,

$-S-(R_{905})$,

$-N(R_{906})(R_{907})$,

a halogen atom, a cyano group, a nitro group,

a substituted or unsubstituted aryl group having 6 to 50 ring carbon atoms, or

a substituted or unsubstituted monovalent heterocyclic group having 5 to 50 ring atoms;

R_{901} to R_{907} are as defined in the formula (1);

$m801$ and $m802$ are independently an integer of 0 to 2; when these are 2, plural R_{801} s or R_{802} s may be the same or different;

$a801$ is an integer of 0 to 2; when $a801$ is 0 or 1, the structure in the parenthesis indicated by “3- $a801$ ” may be the same or different from each other; when $a801$ is 2, Ar_{801} s may be the same or different from each other.

In one embodiment, Ar_{801} is a substituted or unsubstituted aryl group having 6 to 50 ring carbon atoms.

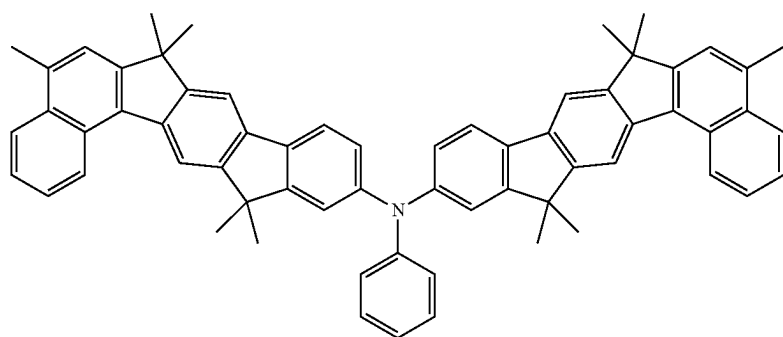
In one embodiment, A_{803} ring is a substituted or unsubstituted aromatic hydrocarbon ring having 6 to 50 ring carbon atoms, and it is a substituted or unsubstituted benzene ring, a substituted or unsubstituted naphthalene ring, or a substituted or unsubstituted anthracene ring, for example.

In one embodiment, R_{803} and R_{804} are independently a substituted or unsubstituted alkyl group having 1 to 50 carbon atoms.

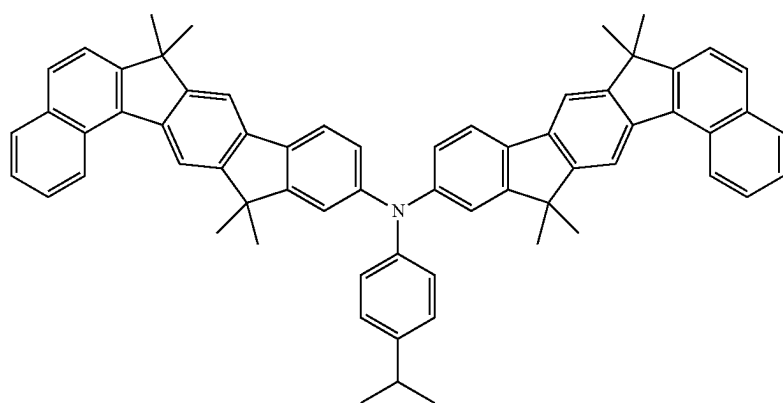
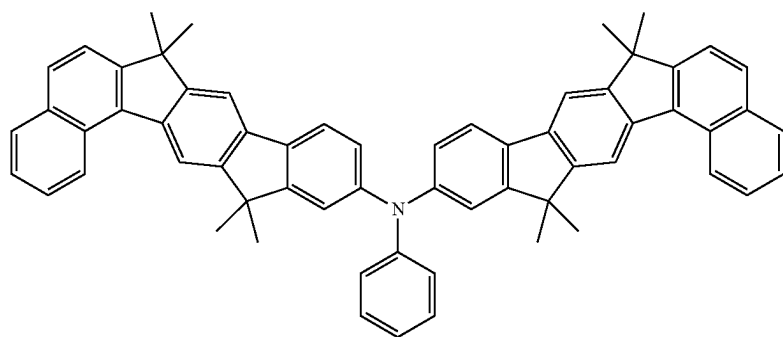
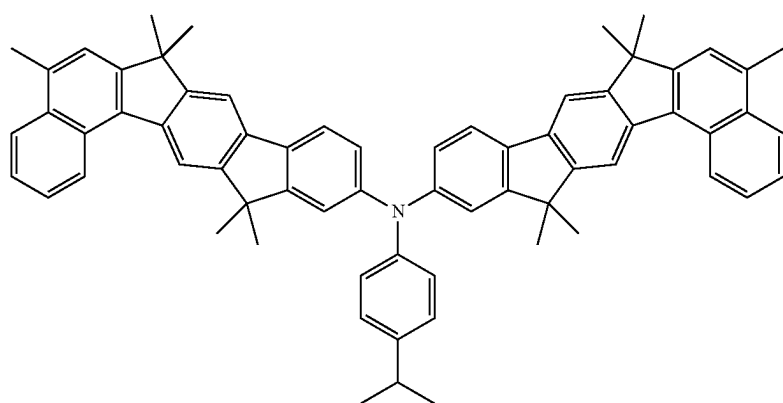
In one embodiment, $a801$ is 1.

As specific example of the compound represented by the formula (81), the following compounds can be given, for example.

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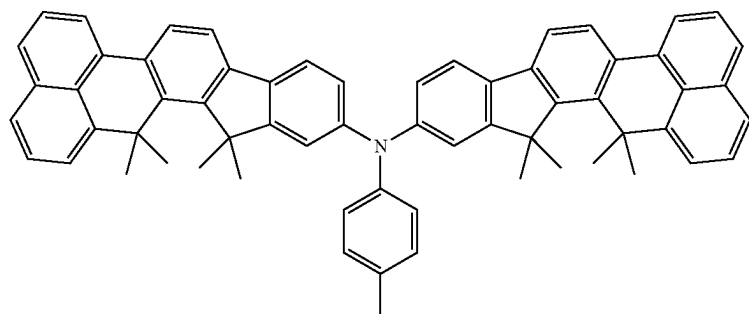
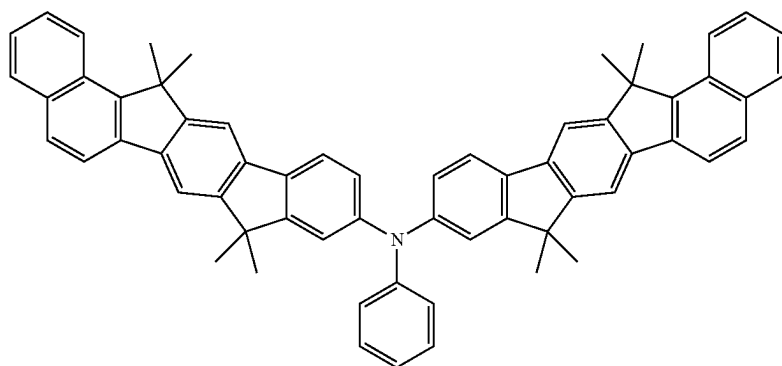
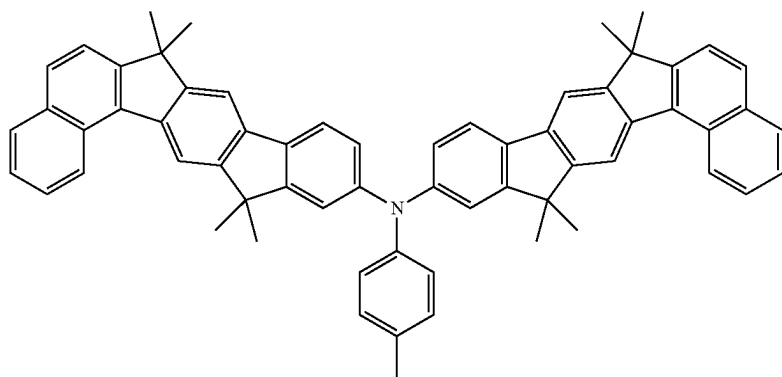
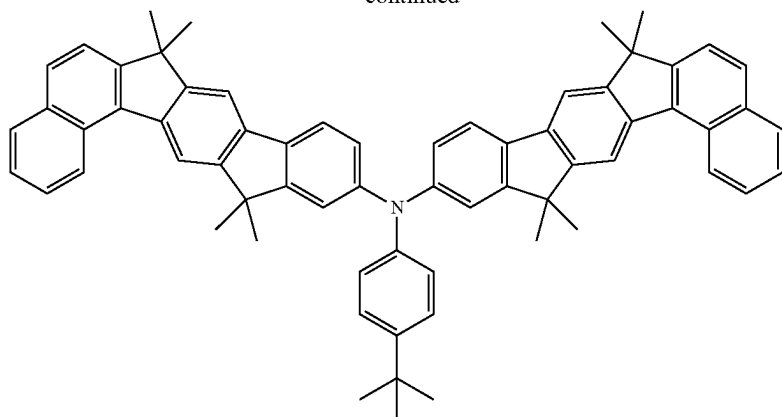
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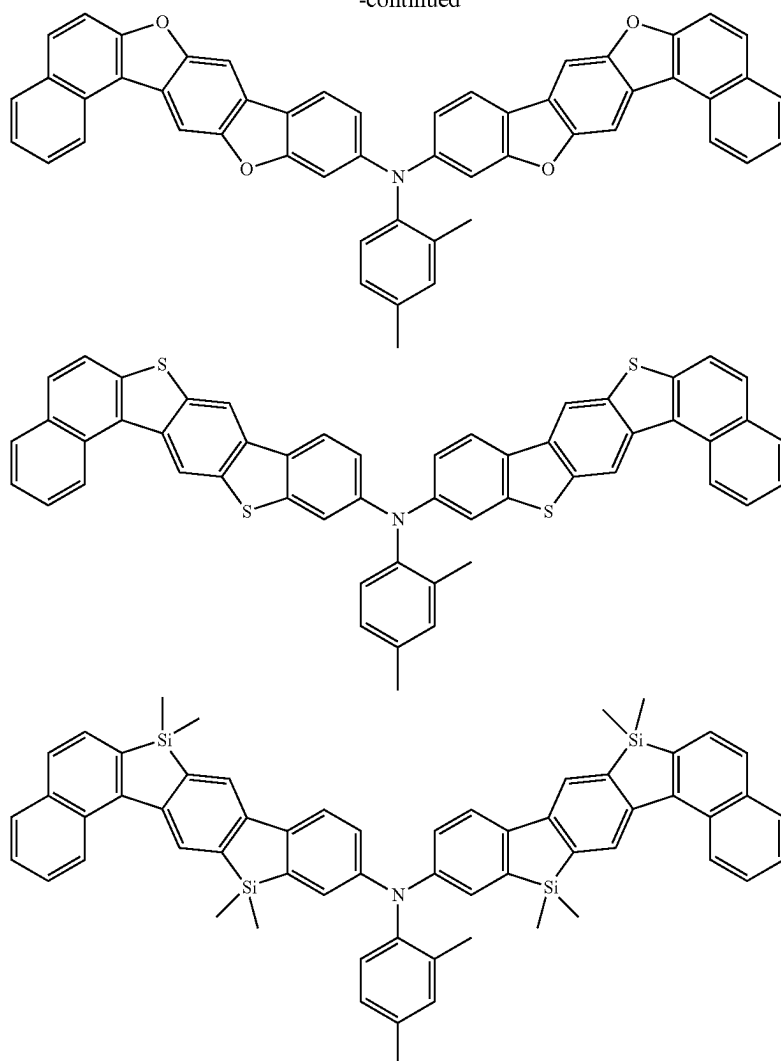
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A content of the compound (host material) represented by the formula (1) or (2) in the emitting layer is preferably 80 mass % or more and 99 mass % or less based on the total mass of the emitting layer. A content of the one or more compounds (dopant material) selected from the group consisting of compounds represented by formulas (11), (21), (31), (41), (51), (61), (71) and (81) is preferably 1 mass % or more and 20 mass % or less based on a total mass of the emitting layer.

Hereinbelow, an explanation will be made on elements and materials other than the above-mentioned compound constituting each layer that can be used in the organic EL device according to one aspect of the invention.

(Substrate)

The substrate is used as a supporting body of the emitting device. As the substrate, glass, quartz, plastic or the like can be used. Further, a flexible substrate may be used. The flexible substrate means a substrate that can be bent. For example, a plastic substrate made of polycarbonate or vinyl polychloride or the like can be given.

(Anode)

In an anode formed on a substrate, it is preferable to use a metal having a large work function (specifically, 4.0 eV or

more), an alloy, an electric conductive compound, a mixture of these or the like. Specifically, indium oxide-tin oxide (ITO: Indium Tin Oxide), indium oxide-tin oxide containing silicon or silicon oxide, indium oxide-zinc oxide, tungsten oxide, indium oxide containing zinc oxide, graphene, or the like can be given. In addition, gold (Au), platinum (Pt) or a nitride of a metal material (e.g. titanium nitride) or the like can be given.

(Hole-Injecting Layer)

The hole-injecting layer is a layer containing a substance having a high hole-injecting property. As a substance having a high hole-injecting property, a substance selected from molybdenum oxide, titanium oxide, vanadium oxide, ruthenium oxide, ruthenium oxide, chromium oxide, zirconium oxide, hafnium oxide, tantalum oxide, silver oxide, tungsten oxide, manganese oxide, an aromatic amine compound, a polymer compound (oligomer, dendrimer, polymer, etc.) or the like can also be used.

(Hole-Transporting Layer)

The hole-transporting layer is a layer containing a substance having a high hole-transporting property. For the hole-transporting layer, aromatic amine compounds, carbazole derivatives, anthracene derivatives and the like can be

used. Polymer compounds such as poly (N-vinylcarbazole) (abbreviation: PVK) and poly(4-vinyltriphenylamine) (abbreviation: PVTTPA) can also be used. However, any substance other than these may be used as long as it is a substance having a higher transporting property for holes than electrons. Note that the layer containing a substance having a high hole-transporting property is not limited to a single layer, but may be a stacked body of two or more layers made of the above substances.

(Guest Material of the Emitting Layer)

The emitting layer is a layer that comprises a substance having high luminous property, and various materials can be used. For example, as the substance having high luminous property, a fluorescent compound that emits fluorescent light or a phosphorescent compound that emits phosphorescent light can be used. The fluorescent compound is a compound capable of emitting light from a singlet excited state and the phosphorescent compound is a compound capable of emitting light from a triplet excited state.

As a blue fluorescent material that can be used for the emitting layer, pyrene derivatives, styrylamine derivatives, chrysene derivatives, fluoranthene derivatives, fluorene derivatives, diamine derivatives, triarylamine derivatives and the like can be used. An aromatic amine derivative or the like can be used as a green fluorescent light-emitting material that can be used in the emitting layer. As a red fluorescent material which can be used in emitting layer, a tetracene derivative, a diamine derivative or the like can be used.

Metal complexes such as iridium complexes, osmium complexes, platinum complexes and the like are used as the blue phosphorescent material that can be used in the emitting layer. An iridium complex or the like is used as a green phosphorescent material that can be used in the emitting layer. Metal complexes such as iridium complexes, platinum complexes, terbium complexes, europium complexes and the like are used as red phosphorescent materials that can be used in the emitting layer.

(Host Material of Emitting Layer)

The emitting layer may have a structure in which the substance having high luminescent property (guest material) described above is dispersed in another substance (host material). Various materials can be used as substances for dispersing substances with high luminescent properties, and it is preferable to use a material having a high lowest unoccupied molecular orbital level (LUMO level) and a low highest occupied molecular orbital level (HOMO level), rather than a material having a high luminous property.

As a substance (host material) for dispersing a substance having a high luminous property, 1) a metal complex such as an aluminum complex, a beryllium complex or a zinc complex, 2) a heterocyclic compound such as an oxadiazole derivative, a benzimidazole derivative, a phenanthroline derivative or the like, 3) a fused aromatic compound such as a carbazole derivative, an anthracene derivative, a phenanthrene derivative, a pyrene derivative or a chrysene derivative, and 4) an aromatic amine compound such as a triarylamine derivative or a fused polycyclic aromatic amine derivative are used.

(Electron-Transporting Layer)

The electron-transporting layer is a layer containing a substance having a high electron-transporting property. For the electron-transporting layer, 1) a metal complex such as an aluminum complex, a beryllium complex, or a zinc complex, 2) a heteroaromatic compound such as an imidazole derivative, a benzimidazole derivative, an azine derivative, a carbazole derivative or a phenanthroline derivative, and 3) a polymer compound can be used.

(Electron-Injecting Layer)

The electron-injection layer is a layer containing a substance having a high electron-injection property. For the electron-injection layer, alkali metals, alkaline earth metals or a compound thereof such as lithium (Li), ytterbium (Yb), lithium fluoride (LiF), cesium fluoride (CsF), calcium fluoride (CaF₂), metal complex compound such as 8-quinolino-lato lithium (Liq), lithium oxide (LiOx) or the like can be used.

(Cathode)

It is preferable to use a metal, an alloy, an electrically conductive compound, a mixture thereof, or the like having a small work function (specifically, 3.8 eV or less) for the cathode. Specific examples of such cathode material include elements belonging to Group 1 or Group 2 of the periodic table of elements, that is, alkali metals such as lithium (Li) and cesium (Cs), alkaline earth metals such as magnesium (Mg), calcium (Ca), and strontium (Sr), an alloy containing these metals (for example, MgAg and ALi), a rare earth metal such as europium (Eu) and ytterbium (Yb), and an alloy containing a rare earth metal.

In the organic EL device according to one aspect of the invention, the method for forming each layer is not particularly restricted. A conventionally known forming method such as a vacuum deposition method, a spin coating method or the like can be used. Each layer such as the emitting layer or the like can be formed by a vacuum deposition method, a molecular beam evaporation method (MBE method), or a known coating method such as a dipping method, a solution spin coating method, a casting method, a bar coating method, or the like, that uses a solution of a material forming each layer dissolved in a solvent.

In the organic EL device according to one aspect of the invention, the thickness of each layer is not particularly restricted. In general, in order to suppress occurrence of defects such as pinholes and to suppress the applied voltage and to improve luminous efficiency, the thickness is normally preferably in a range of several nm to 1 μ m.

[Electronic Device]

The electronic device according to one aspect of the invention is characterized in that it is provided with the organic EL device according to one aspect of the invention.

Specific examples of the electronic device include a display element such as an organic EL panel module; a display such as a TV, a mobile phone or a PC; and emitting devices such as lightings and lights for automobiles or the like.

EXAMPLES

The invention will specifically be explained with the examples and the comparative examples below, and shall not be limited to the contents of the examples in anyway.

Synthesis Example 1 [Synthesis of Compound BH-1]

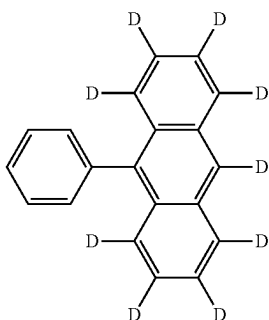
(Synthesis of Intermediate 1)

To 13.3 g (50.0 mmol) of 9-bromoanthracene-d₉, 6.4 g (52.5 mmol) of phenylboronic acid and 1.2 g (1.00 mmol) of Pd[PPh₃]₄, 75 ml of toluene, 75 ml of dimethoxyethane and 75 ml (150.0 mmol) of 2M Na₂CO₃ aqueous solution were added under an atmosphere of argon, followed by being heated to reflux while stirring for 10 hours.

After completion of the reaction, having been cooled to room temperature, the sample was transferred to a separating funnel and extracted with dichloromethane. The resulting

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organic phase was dried with MgSO_4 , followed by being filtered and concentrated. The concentrated residue was purified with silica gel column chromatography to obtain 10.9 g of white solid. By conducting FD-MS analysis, the resulting compound was identified as Intermediate 1 as follows (yield: 83%).



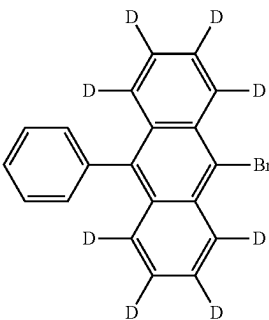
Intermediate 1

(Synthesis of Intermediate 2)

5.3 g (20.0 mmol) of Intermediate 1 was solubilized in 120 ml of dichloromethane, and the resulting solution was dropped into the solution of 3.2 g (20.0 mmol) of bromine in 12 ml of dichloromethane at room temperature, followed by being stirred for one hour.

After completion of the reaction, the sample was transferred to a separating funnel and washed with 2M $\text{Na}_2\text{S}_2\text{O}_3$ aqueous solution. The organic phase was further washed with 10% Na_2CO_3 , and thereafter with water, and the separated organic phase was dried with MgSO_4 , followed by being filtered and concentrated.

The concentrated residue was dispersed in methanol (100 mL), and the precipitated crystal was dried to obtain 6.5 g of white solid. By conducting FD-MS analysis, the resulting compound was identified as Intermediate 2 as follows (yield: 95%).



Intermediate 2

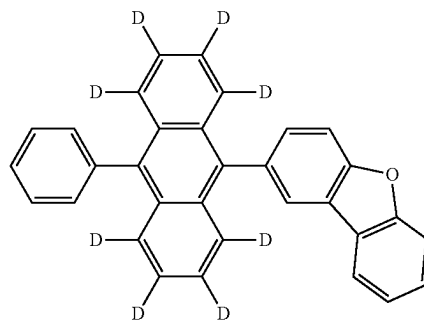
(Synthesis of Compound BH-1)

To 1.7 g (5.0 mmol) of Intermediate 2, 1.1 g (5.3 mmol) of dibenzofuran-2-boronic acid and 0.1 g (0.1 mmol) of $\text{Pd}[\text{PPh}_3]_4$, 7.5 ml of toluene, 7.5 ml of dimethoxyethane and 7.5 ml (15.0 mmol) of 2M Na_2CO_3 aqueous solution were added under an atmosphere of argon, followed by being heated to reflux while stirring for 10 hours.

After completion of the reaction, having been cooled to room temperature, the sample was transferred to a separating funnel and extracted with dichloromethane. The resulting organic phase was dried with MgSO_4 , followed by being

966

filtered and concentrated. The concentrated residue was purified with silica gel column chromatography to obtain 1.6 g of white solid. By conducting FD-MS analysis, the resulting compound was identified as Compound BH-1 as follows (yield: 75%).



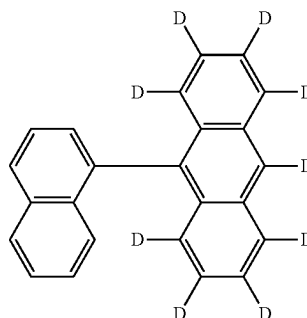
BH-1

Synthesis Example 2 [Synthesis of Compound BH-2]

(Synthesis of Intermediate 3)

To 13.3 g (50.0 mmol) of 9-bromoanthracene-d₉, 9.0 g (52.5 mmol) of 1-naphthalene boronic acid and 1.2 g (1.00 mmol) of $\text{Pd}[\text{PPh}_3]_4$, 75 ml of toluene, 75 ml of dimethoxyethane and 75 ml (150.0 mmol) of 2M Na_2CO_3 aqueous solution were added under an atmosphere of argon, followed by being heated to reflux while stirring for 10 hours.

After completion of the reaction, having been cooled to room temperature, the sample was transferred to a separating funnel and extracted with dichloromethane. The resulting organic phase was dried with MgSO_4 , followed by being filtered and concentrated. The concentrated residue was purified with silica gel column chromatography to obtain 13.3 g of white solid. By conducting FD-MS analysis, the resulting compound was identified as Intermediate 3 as follows (yield: 85%).



Intermediate 3

(Synthesis of Intermediate 4)

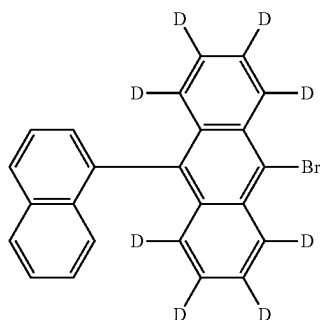
6.3 g (20.0 mmol) of Intermediate 3 was solubilized in 120 ml of dichloromethane, and the resulting solution was dropped into the solution of 3.2 g (20.0 mmol) of bromine in 12 ml of dichloromethane at room temperature, followed by being stirred for one hour.

After completion of the reaction, the sample was transferred to a separating funnel and washed with 2M $\text{Na}_2\text{S}_2\text{O}_3$ aqueous solution. The organic phase was further washed

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with 10% Na₂CO₃, and thereafter with water three times. The organic phase was dried with MgSO₄, followed by being filtered and concentrated.

The concentrated residue was dispersed in methanol (100 ml), and the precipitated crystal was dried to obtain 7.5 g of white solid. By conducting FD-MS analysis, the resulting compound was identified as Intermediate 4 as follows (yield: 96%).

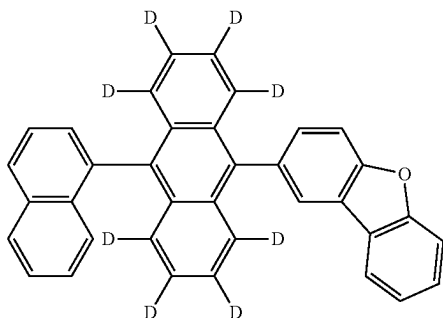


Intermediate 4

(Synthesis of Compound BH-2)

To 2.0 g (5.0 mmol) of Intermediate 4, 1.1 g (5.3 mmol) of dibenzofuran-2-boronic acid and 0.1 g (0.1 mmol) of Pd[PPh₃]₄, 7.5 ml of toluene, 7.5 ml of dimethoxyethane and 7.5 ml (15.0 mmol) of 2M Na₂CO₃ aqueous solution were added under an atmosphere of argon, followed by being heated to reflux while stirring for 10 hours.

After completion of the reaction, having been cooled to room temperature, the sample was transferred to a separating funnel and extracted with dichloromethane. The resulting organic phase was dried with MgSO₄, followed by being filtered and concentrated. The concentrated residue was purified with silica gel column chromatography to obtain 1.7 g of white solid. By conducting FD-MS analysis, the resulting compound was identified as Compound BH-2 as follows (yield: 70%).

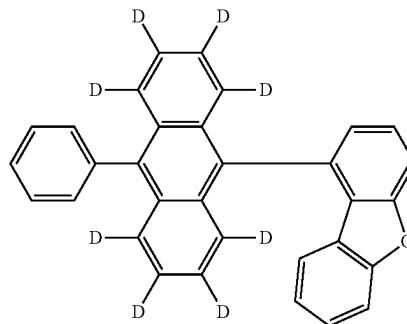


Synthesis Example 3 [Synthesis of Compound BH-3]

Except that 1.1 g (5.3 mmol) of dibenzofuran-1-boronic acid was used instead of dibenzofuran-2-boronic acid, the reaction was carried out in the same way as in the synthesis example 1, thereby obtaining 1.3 g of white crystal. By conducting FD-MS analysis, the resulting compound was identified as Compound BH-3 as follows (yield: 62%).

968

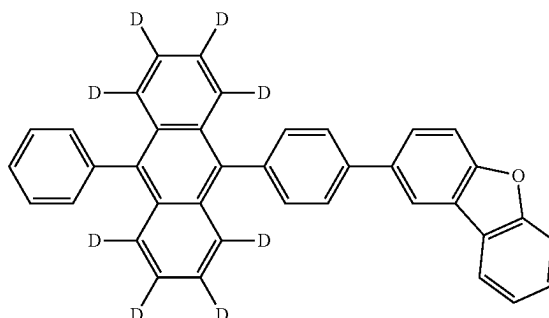
BH-3



Synthesis Example 4 [Synthesis of Compound BH-4]

Except that 1.5 g (5.3 mmol) of 4-(2-dibenzofuranyl)phenyl boronic acid was used instead of dibenzofuran-2-boronic acid, the reaction was carried out in the same way as in the synthesis example 1, thereby obtaining 1.8 g of white crystal. By conducting FD-MS analysis, the resulting compound was identified as Compound BH-4 as follows (yield: 71%).

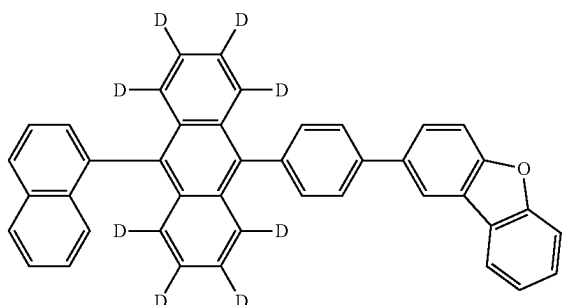
BH-4



Synthesis Example 5 [Synthesis of Compound BH-5]

Except that 1.5 g (5.3 mmol) of 4-(2-dibenzofuranyl)phenyl boronic acid was used instead of dibenzofuran-2-boronic acid, the reaction was carried out in the same way as in the synthesis example 2, thereby obtaining 2.0 g of white crystal. By conducting FD-MS analysis, the resulting compound was identified as Compound BH-5 as follows (yield: 73%).

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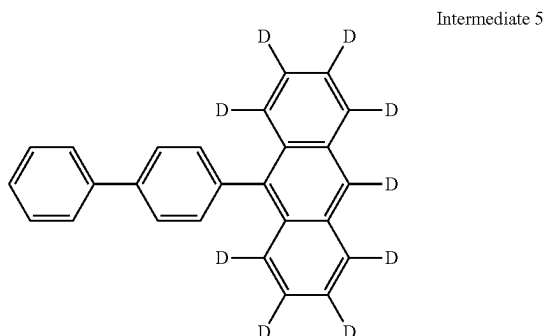


Synthesis Example 6 [Synthesis of Compound BH-6]

(Synthesis of Intermediate 5)

To 13.3 g (50.0 mmol) of 9-bromoanthracene-d₉, 10.4 g (52.5 mmol) of 4-biphenylboronic acid and 1.2 g (1.00 mmol) of Pd[PPh₃]₄, 75 ml of toluene, 75 ml of dimethoxyethane and 75 ml (150.0 mmol) of 2M Na₂CO₃ aqueous solution were added under an atmosphere of argon, followed by being heated to reflux while stirring for 10 hours.

After completion of the reaction, having been cooled to room temperature, the sample was transferred to a separating funnel and extracted with dichloromethane. The resulting organic phase was dried with MgSO₄, followed by being filtered and concentrated. The concentrated residue was purified with silica gel column chromatography to obtain 14.1 g of white solid. By conducting FD-MS analysis, the resulting compound was identified as Intermediate 5 as follows (yield: 83%).



(Synthesis of Intermediate 6)

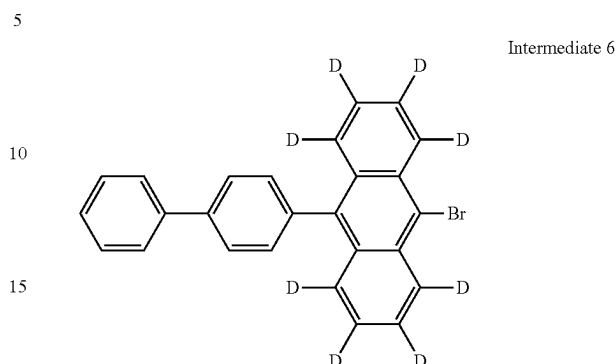
6.8 g (20.0 mmol) of Intermediate 5 was solubilized in 120 ml of dichloromethane, and the resulting solution was dropped into the solution of 3.2 g (20.0 mmol) of bromine in 12 ml of dichloromethane at room temperature, followed by being stirred for one hour.

After completion of the reaction, the sample was transferred to a separating funnel and washed with 2M Na₂S₂O₃ aqueous solution. The organic phase was further washed with 10% Na₂CO₃, and thereafter with water three times. The organic phase was dried with MgSO₄, followed by being filtered and concentrated.

The concentrated residue was dispersed in methanol (100 mL), and the precipitated crystal was dried to obtain 8.0 g of

970

white solid. By conducting FD-MS analysis, the resulting compound was identified as Intermediate 6 as follows (yield: 96%).

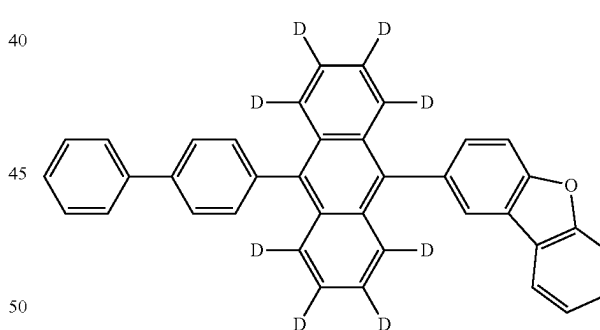


(Synthesis of Compound BH-6)

To 2.1 g (5.0 mmol) of Intermediate 6, 1.1 g (5.3 mmol) of dibenzofuran-2-boronic acid and 0.1 g (0.1 mmol) of Pd[PPh₃]₄, 7.5 ml of toluene, 7.5 ml of dimethoxyethane and 7.5 ml (15.0 mmol) of 2M Na₂CO₃ aqueous solution were added under an atmosphere of argon, followed by being heated to reflux while stirring for 10 hours.

After completion of the reaction, having been cooled to room temperature, the sample was transferred to a separating funnel and extracted with dichloromethane. The resulting organic phase was dried with MgSO₄, followed by being filtered and concentrated. The concentrated residue was purified with silica gel column chromatography to obtain 1.6 g of white solid. By conducting FD-MS analysis, the resulting compound was identified as Compound BH-6 as follows (yield: 64%).

BH-6



Synthesis Example 7 [Synthesis of Compound BH-7]

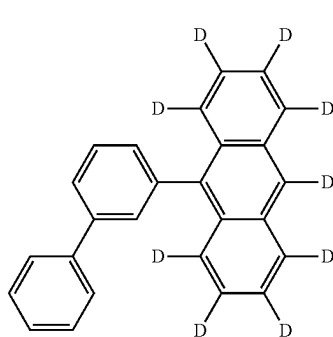
(Synthesis of Intermediate 7)

To 13.3 g (50.0 mmol) of 9-bromoanthracene d₉, 10.4 g (52.5 mmol) of 3-biphenylboronic acid and 1.2 g (1.00 mmol) of Pd[PPh₃]₄, 75 ml of toluene, 75 ml of dimethoxyethane and 75 ml (150.0 mmol) of 2M Na₂CO₃ aqueous solution were added under an atmosphere of argon, followed by being heated to reflux while stirring for 10 hours.

After completion of the reaction, having been cooled to room temperature, the sample was transferred to a separating funnel and extracted with dichloromethane. The resulting organic phase was dried with MgSO₄, followed by being

971

filtered and concentrated. The concentrated residue was purified with silica gel column chromatography to obtain 13.6 g of white solid. By conducting FD-MS analysis, the resulting compound was identified as Intermediate 7 as follows (yield: 80%).



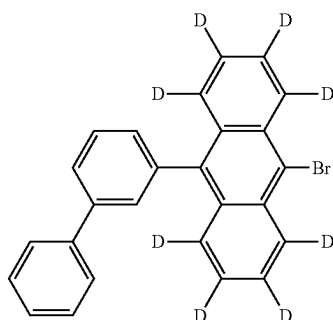
Intermediate 7

(Synthesis of Intermediate 8)

6.8 g (20.0 mmol) of Intermediate 7 was solubilized in 120 ml of dichloromethane, and the resulting solution was dropped into the solution of 3.2 g (20.0 mmol) of bromine in 12 ml of dichloromethane at room temperature, followed by being stirred for one hour.

After completion of the reaction, the sample was transferred to a separating funnel and washed with 2M Na₂S₂O₃ aqueous solution. The organic phase was further washed with 10% Na₂CO₃, and thereafter with water three times. The organic phase was dried with MgSO₄, followed by being filtered and concentrated.

The concentrated residue was dispersed in methanol (100 mL), and the precipitated crystal was dried to obtain 8.0 g of white solid. By conducting FD-MS analysis, the resulting compound was identified as Intermediate 8 as follows (yield: 96%).



Intermediate 8

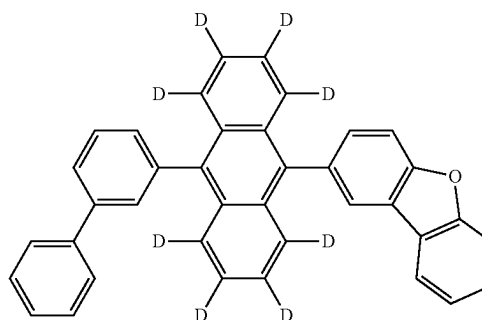
(Synthesis of Compound BH-7)

To 2.1 g (5.0 mmol) of Intermediate 8, 1.1 g (5.3 mmol) of dibenzofuran-2-boronic acid and 0.1 g (0.1 mmol) of Pd[PPh₃]₄, 7.5 ml of toluene, 7.5 ml of dimethoxyethane and 7.5 ml (15.0 mmol) of 2M Na₂CO₃ aqueous solution were added under an atmosphere of argon, followed by being heated to reflux while stirring for 10 hours.

After completion of the reaction, having been cooled to room temperature, the sample was transferred to a separating funnel and extracted with dichloromethane. The resulting organic phase was dried with MgSO₄, followed by being filtered and concentrated. The concentrated residue was

972

purified with silica gel column chromatography to obtain 1.5 g of white solid. By conducting FD-MS analysis, the resulting compound was identified as Compound BH-7 as follows (yield: 59%).



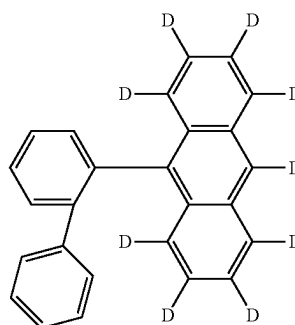
BH-7

Synthesis Example 8 [Synthesis of Compound BH-8]

(Synthesis of Intermediate 9)

To 13.3 g (50.0 mmol) of 9-bromoanthracene d₉, 10.4 g (52.5 mmol) of 2-biphenylboronic acid and 1.2 g (1.00 mmol) of Pd[PPh₃]₄, 75 ml of toluene, 75 ml of dimethoxyethane and 75 ml (150.0 mmol) of 2M Na₂CO₃ aqueous solution were added under an atmosphere of argon, followed by being heated to reflux while stirring for 10 hours.

After completion of the reaction, having been cooled to room temperature, the sample was transferred to a separating funnel and extracted with dichloromethane. The resulting organic phase was dried with MgSO₄, followed by being filtered and concentrated. The concentrated residue was purified with silica gel column chromatography to obtain 10.9 g of white solid. By conducting FD-MS analysis, the resulting compound was identified as Intermediate 9 as follows (yield: 64%).



Intermediate 9

(Synthesis of Intermediate 10)

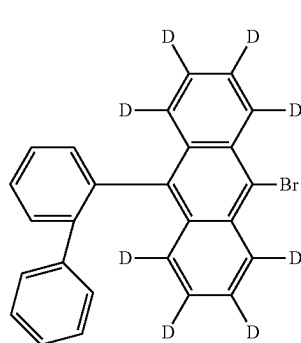
6.8 g (20.0 mmol) of Intermediate 9 was solubilized in 120 ml of dichloromethane, and the resulting solution was dropped into the solution of 3.2 g (20.0 mmol) of bromine in 12 ml of dichloromethane at room temperature, followed by being stirred for one hour.

After completion of the reaction, the sample was transferred to a separating funnel and washed with 2M Na₂S₂O₃ aqueous solution. The organic phase was further washed with 10% Na₂CO₃, and thereafter with water three times.

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The organic phase was dried with MgSO_4 , followed by being filtered and concentrated.

The concentrated residue was dispersed in methanol (100 mL), and the precipitated crystal was dried to obtain 8.0 g of white solid. By conducting FD-MS analysis, the resulting compound was identified as Intermediate 10 as follows (yield: 96%).

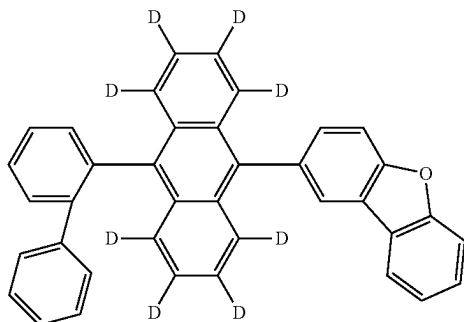


Intermediate 10

(Synthesis of Compound BH-8)

To 2.1 g (5.0 mmol) of Intermediate 10, 1.1 g (5.3 mmol) of dibenzofuran-2-boronic acid and 0.1 g (0.1 mmol) of $\text{Pd}[\text{PPh}_3]_4$, 7.5 ml of toluene, 7.5 ml of dimethoxyethane and 7.5 ml (15.0 mmol) of 2M Na_2CO_3 aqueous solution were added under an atmosphere of argon, followed by being heated to reflux while stirring for 10 hours.

After completion of the reaction, having been cooled to room temperature, the sample was transferred to a separating funnel and extracted with dichloromethane. The resulting organic phase was dried with MgSO_4 , followed by being filtered and concentrated. The concentrated residue was purified with silica gel column chromatography to obtain 1.6 g of white solid. By conducting FD-MS analysis, the resulting compound was identified as Compound BH-8 as follows (yield: 63%).



BH-8

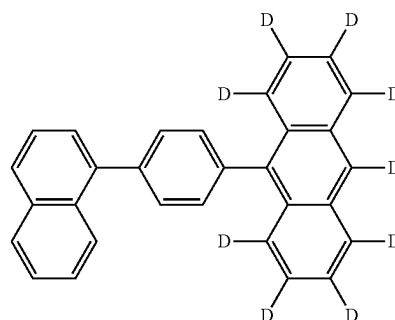
Synthesis Example 9 [Synthesis of Compound BH-9]

(Synthesis of Intermediate 11)

To 13.3 g (50.0 mmol) of 9-bromoanthracene d9, 13.0 g (52.5 mmol) of 4-(1-naphthyl) phenylboronic acid and 1.2 g (1.00 mmol) of $\text{Pd}[\text{PPh}_3]_4$, 75 ml of toluene, 75 ml of dimethoxyethane and 75 ml (150.0 mmol) of 2M Na_2CO_3 aqueous solution were added under an atmosphere of argon, followed by being heated to reflux while stirring for 10 hours.

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After completion of the reaction, having been cooled to room temperature, the sample was transferred to a separating funnel and extracted with dichloromethane. The resulting organic phase was dried with MgSO_4 , followed by being filtered and concentrated. The concentrated residue was purified with silica gel column chromatography to obtain 15.6 g of white solid. By conducting FD-MS analysis, the resulting compound was identified as Intermediate 11 as follows (yield: 80%).



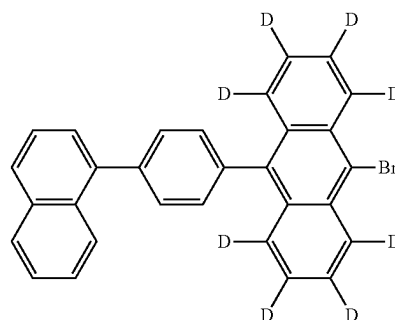
Intermediate 11

(Synthesis of Intermediate 12)

7.8 g (20.0 mmol) of Intermediate 11 was solubilized in 120 ml of dichloromethane, and the resulting solution was dropped into the solution of 3.2 g (20.0 mmol) of bromine in 12 ml of dichloromethane at room temperature, followed by being stirred for one hour.

After completion of the reaction, the sample was transferred to a separating funnel and washed with 2M $\text{Na}_2\text{S}_2\text{O}_3$ aqueous solution. The organic phase was further washed with 10% Na_2CO_3 , and thereafter with water three times. The organic phase was dried with MgSO_4 , followed by being filtered and concentrated.

The concentrated residue was dispersed in methanol (100 ml), and the precipitated crystal was dried to obtain 8.6 g of white solid. By conducting FD-MS analysis, the resulting compound was identified as Intermediate 12 as follows (yield: 92%).



Intermediate 12

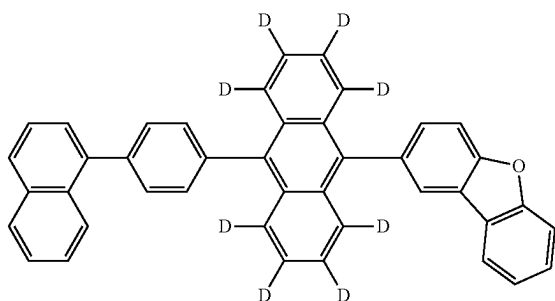
(Synthesis of Compound BH-9)

To 2.3 g (5.0 mmol) of Intermediate 12, 1.1 g (5.3 mmol) of dibenzofuran-2-boronic acid and 0.1 g (0.1 mmol) of $\text{Pd}[\text{PPh}_3]_4$, 7.5 ml of toluene, 7.5 ml of dimethoxyethane and 7.5 ml (15.0 mmol) of 2M Na_2CO_3 aqueous solution were added under an atmosphere of argon, followed by being heated to reflux while stirring for 10 hours.

After completion of the reaction, having been cooled to room temperature, the sample was transferred to a separating

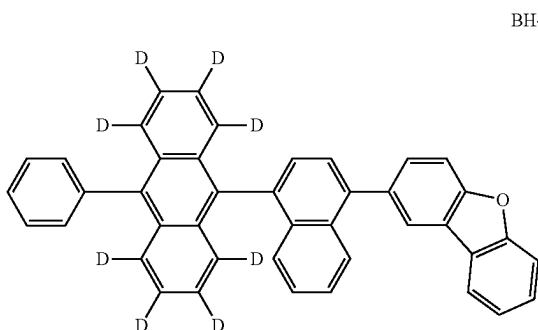
975

funnel and extracted with dichloromethane. The resulting organic phase was dried with MgSO_4 , followed by being filtered and concentrated. The concentrated residue was purified with silica gel column chromatography to obtain 1.9 g of white solid. By conducting FD-MS analysis, the resulting compound was identified as Compound BH-10 as follows (yield: 68%).



Synthesis Example 10 [Synthesis of Compound BH-10]

Except that 1.8 g (5.3 mmol) of 4-(2-dibenzofuranyl)-1-naphthalenyl-boronic acid was used instead of dibenzofuran-2-boronic acid, the reaction was carried out in the same way as in the synthesis example 1, thereby obtaining 1.7 g of white crystal. By conducting FD-MS analysis, the resulting compound was identified as Compound BH-10 as follows (yield: 60%).

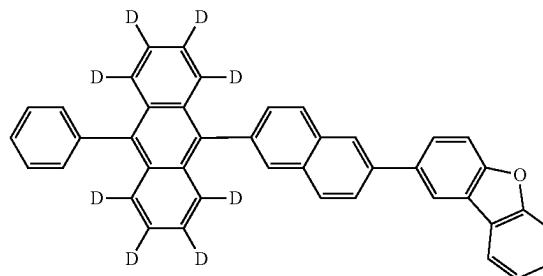


Synthesis Example 11 [Synthesis of Compound BH-11]

Except that 1.8 g (5.3 mmol) of 6-(2-dibenzofuranyl)-2-naphthalenyl-boronic acid was used instead of dibenzofuran-2-boronic acid, the reaction was carried out in the same way as in the synthesis example 1, thereby obtaining 1.5 g of white crystal. By conducting FD-MS analysis, the resulting compound was identified as Compound BH-11 as follows (yield: 55%).

976

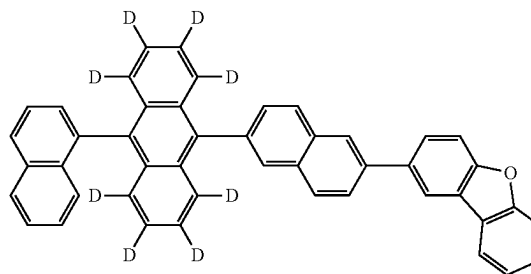
BH-11



Synthesis Example 12 [Synthesis of Compound BH-12]

Except that 1.8 g (5.3 mmol) of 6-(2-dibenzofuranyl)-2-naphthalenyl-boronic acid was used instead of dibenzofuran-2-boronic acid, the reaction was carried out in the same way as in the synthesis example 2, thereby obtaining 2.0 g of white crystal. By conducting FD-MS analysis, the resulting compound was identified as Compound BH-12 as follows (yield: 65%).

BH-12

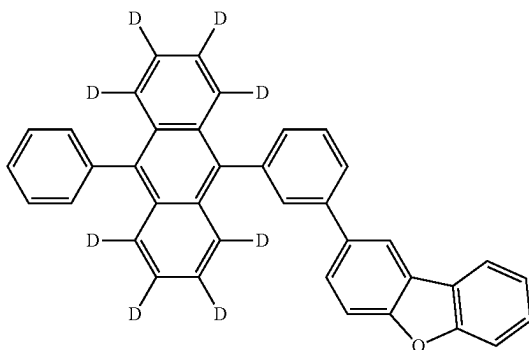


Synthesis Example 13 [Synthesis of Compound BH-13]

Except that 1.5 g (5.3 mmol) of 3-(2-dibenzofuranyl)phenylboronic acid was used instead of dibenzofuran-2-boronic acid, the reaction was carried out in the same way as in the synthesis example 1, thereby obtaining 1.3 g of white crystal. By conducting FD-MS analysis, the resulting compound was identified as Compound BH-13 as follows (yield: 52%).

977

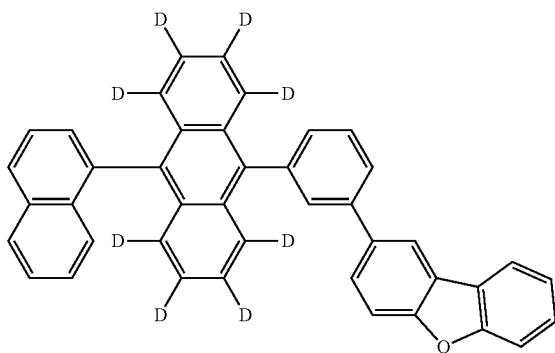
BH-13



Synthesis Example 14 [Synthesis of Compound BH-14]

Except that 1.5 g (5.3 mmol) of 3-(2-dibenzofuranyl) phenylboronic acid was used instead of dibenzofuran-2-boronic acid, the reaction was carried out in the same way as in the synthesis example 2, thereby obtaining 1.4 g of white crystal. By conducting FD-MS analysis, the resulting compound was identified as Compound BH-14 as follows (yield: 50%).

BH-14

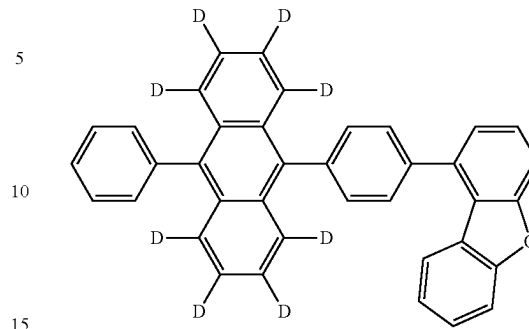


Synthesis Example 15 [Synthesis of Compound BH-15]

Except that 1.5 g (5.3 mmol) of 4-(1-dibenzofuranyl) phenylboronic acid was used instead of dibenzofuran-2-boronic acid, the reaction was carried out in the same way as in the synthesis example 1, thereby obtaining 1.6 g of white crystal. By conducting FD-MS analysis, the resulting compound was identified as Compound BH-15 as follows (yield: 62%).

978

BH-15



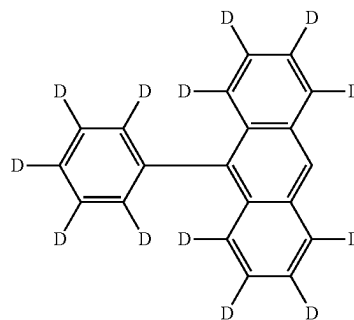
Synthesis Example 16 [Synthesis of Compound BH-16]

(Synthesis of Intermediate 13)

To 1.33 g (5.00 mmol) of 9-bromoanthracene-d₉, 0.67 g (5.25 mmol) of phenyl-d₅-boronic acid and 0.12 g (0.10 mmol) of Pd[PPh₃]₄, 7.5 ml of toluene, 7.5 ml of dimethoxyethane and 7.5 ml (15.0 mmol) of 2M Na₂CO₃ aqueous solution were added under an atmosphere of argon, followed by being heated to reflux while stirring for 10 hours.

After completion of the reaction, having been cooled to room temperature, the sample was transferred to a separating funnel and extracted with dichloromethane. The resulting organic phase was dried with MgSO₄, followed by being filtered and concentrated. The concentrated residue was purified with silica gel column chromatography to obtain 1.07 g of white solid. By conducting FD-MS analysis, the resulting compound was identified as Intermediate 13 as follows (yield: 80%).

Intermediate 13



(Synthesis of Intermediate 14)

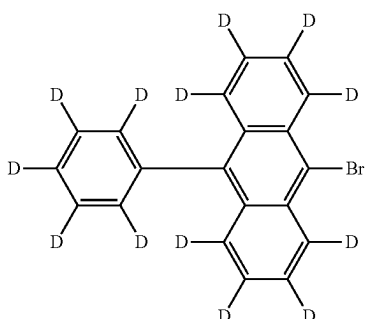
1.07 g (4.0 mmol) of Intermediate 13 was solubilized in 25 ml of dichloromethane, and the resulting solution was dropped into the solution of 0.64 g (4.0 mmol) of bromine in 3 ml of dichloromethane at room temperature, followed by being stirred for one hour.

After completion of the reaction, the sample was transferred to a separating funnel and washed with 2M Na₂S₂O₃ aqueous solution. The organic phase was further washed with 10% Na₂CO₃, and thereafter with water. The organic phase was dried with MgSO₄, followed by being filtered and concentrated.

The concentrated residue was dispersed in methanol (100 ml), and the precipitated crystal was dried to obtain 1.3 g of

979

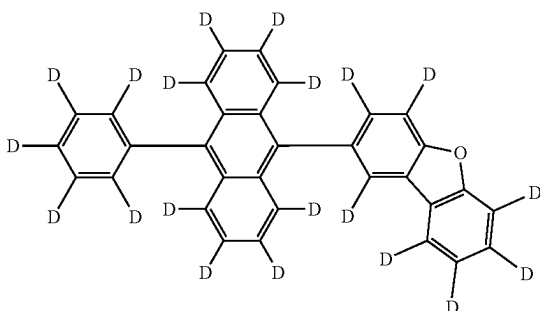
white solid. By conducting FD-MS analysis, the resulting compound was identified as Intermediate 14 as follows (yield: 95%).



(Synthesis of Compound BH-16)

To 0.87 g (2.5 mmol) of Intermediate 14, 0.58 g (2.65 mmol) of dibenzofuran d7-2-boronic acid and 0.06 g (0.05 mmol) of Pd[PPh₃]₄, 5 ml of toluene, 5 ml of dimethoxyethane and 5 ml (10.0 mmol) of 2M Na₂CO₃ aqueous solution were added under an atmosphere of argon, followed by being heated to reflux while stirring for 10 hours.

After completion of the reaction, having been cooled to room temperature, the sample was transferred to a separating funnel and extracted with dichloromethane. The resulting organic phase was dried with MgSO₄, followed by being filtered and concentrated. The concentrated residue was purified with silica gel column chromatography to obtain 0.77 g of white solid. By conducting FD-MS analysis, the resulting compound was identified as Compound BH-16 as follows (yield: 70%).



Example 1 [Fabrication and Evaluation of Organic EL Device]

(Fabrication of Organic EL Device)

A glass substrate of 25 mm by 75 mm by 1.1 mm thick with an ITO transparent electrode (anode) (manufactured by GEOMATEC Co., Ltd.) was subjected to ultrasonic cleaning in isopropyl alcohol for 5 minutes, and then subjected to UV-ozone cleaning for 30 minutes. The thickness of ITO was 130 nm.

The cleaned glass substrate with a transparent electrode was mounted in a substrate holder of a vacuum vapor deposition apparatus. First, compound HI was deposited on the surface where the transparent electrode was formed so as

980

to cover the transparent electrode, thereby forming an HI film having a thickness of 5 nm. This HI film functioned as a hole-injecting layer.

Subsequent to the formation of the HI film, compound HT-1 was deposited to form an HT-1 film in a thickness of 80 nm on the HI film. This HT-1 film functioned as a hole-transporting layer (a first hole-transporting layer). Subsequent to the formation of the HT-1 film, compound HT-2 was deposited to form an HT-2 film in a thickness of 10 nm on the HT-1 film. This HT-2 film functioned as an electron-blocking layer (a second hole-transporting layer).

Compound BH-1 (host material) and compound BD-1 (dopant material) were co-deposited on the HT-2 film so that the ratio of compound BD-1 was 4 mass % to form a BH-1:BD-1 film in a thickness of 25 nm. This BH-1:BD-1 film functioned as an emitting layer.

Compound ET-1 was deposited on the emitting layer to form an ET-1 film in a thickness of 10 nm. This ET-1 film functioned as a hole-barrier layer.

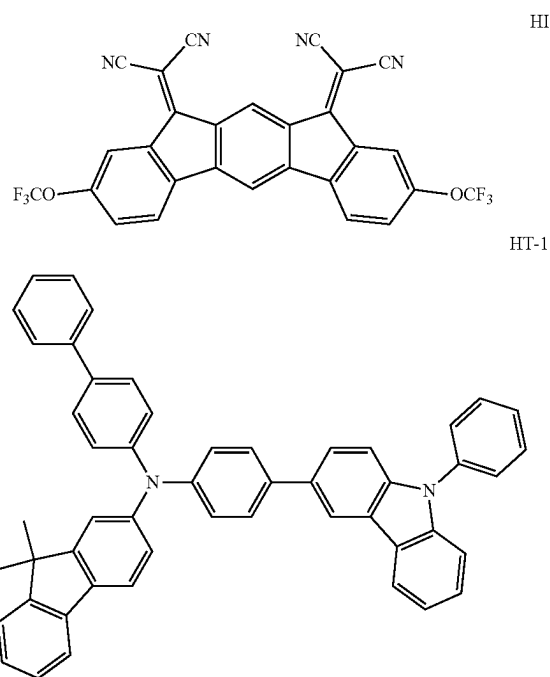
Compound ET-2 was deposited on the ET-1 layer to form an ET-2 layer in a thickness of 15 nm. This ET-2 layer functioned as an electron-transporting layer. UF was deposited on the ET-2 layer to form a UF film in a thickness of 1 nm. Al metal was deposited on the UF film to form a metal cathode in a thickness of 80 nm. An organic EL device was thus fabricated.

The layer construction of the fabricated organic EL device was as follows.

ITO (130)/HI (5)/HT-1 (80)/HT-2 (10)/BH-1:BD-1 (25:4 mass %)/ET-1 (10)/ET-2 (15)/LiF (1)/Al (80)

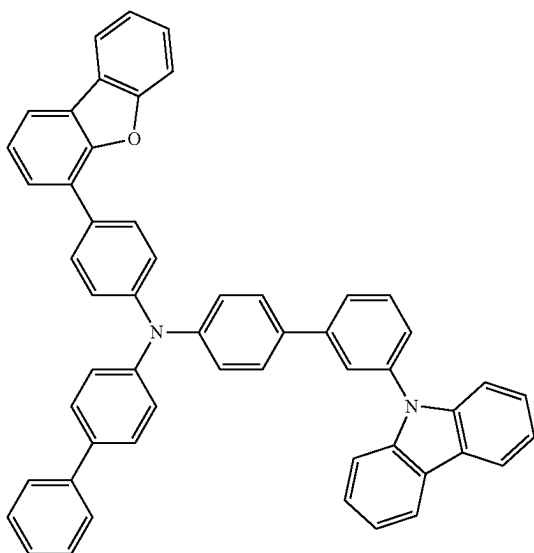
The numbers in the parenthesis denote the thickness of each layer (unit: nm).

The compounds used in Example 1 as well as the subsequent examples and comparative examples are shown below.



981

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HT-2

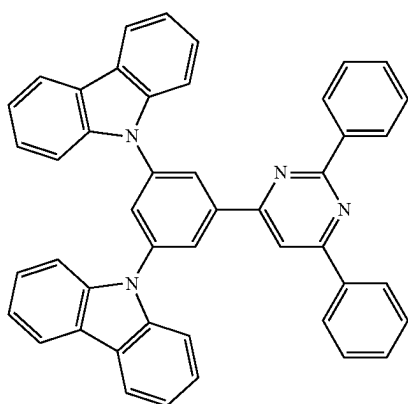
5

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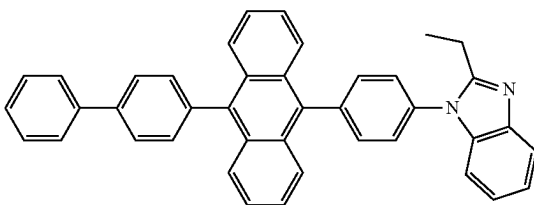


ET-1

30

35

40



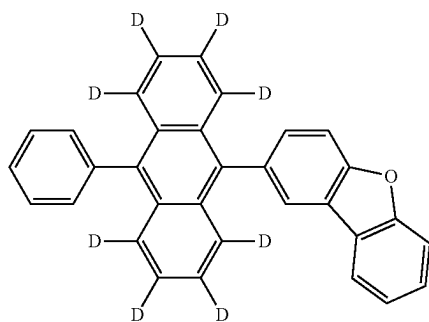
ET-2

45

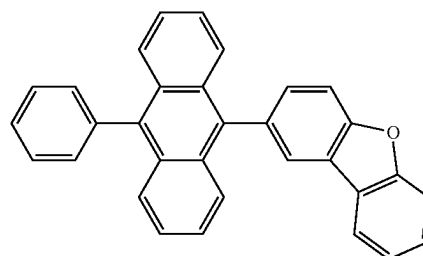
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BH-1

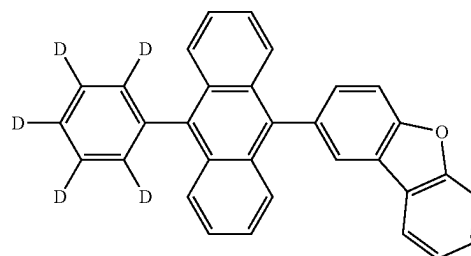
55

**982**

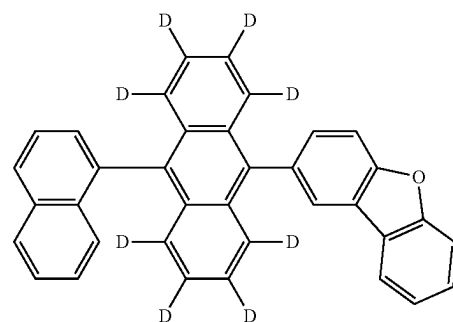
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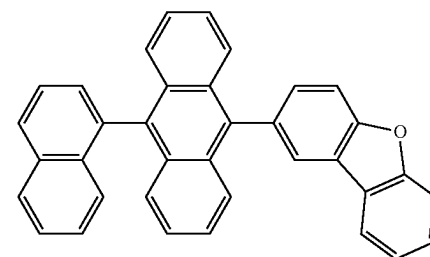
BH-1-a



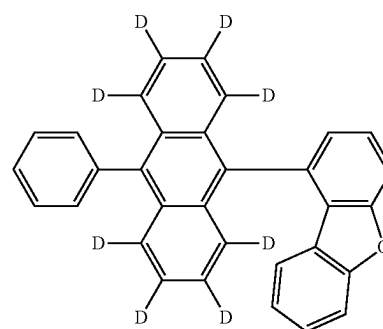
BH-1-b



BH-2



BH-2-a



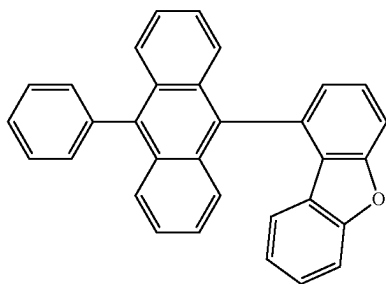
BH-3

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983

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BH-3-a

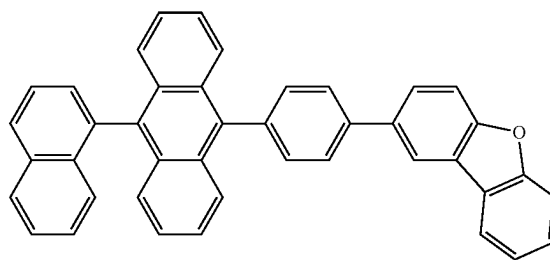
984

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BH-5-a

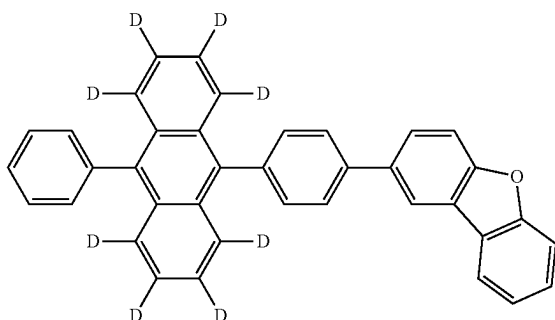
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BH-5-b

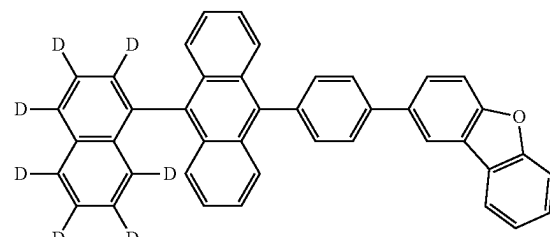
BH-4

15



20

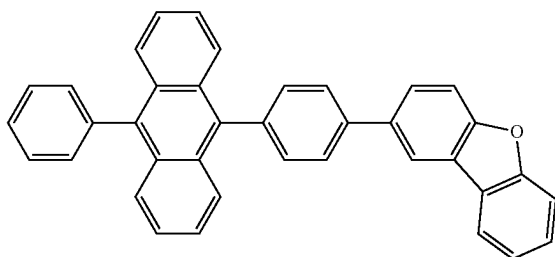
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BH-4-a

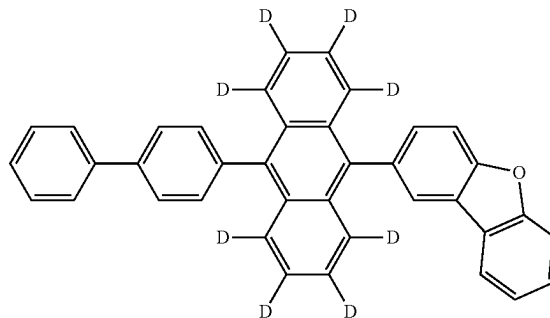
BH-6

30



35

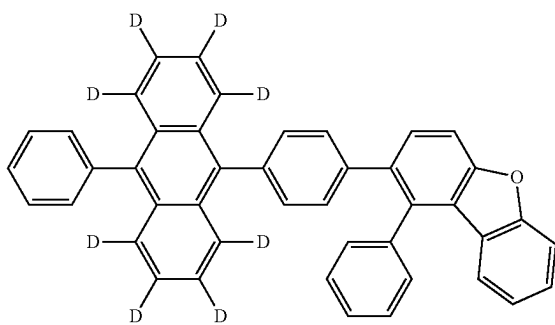
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BH-4-b

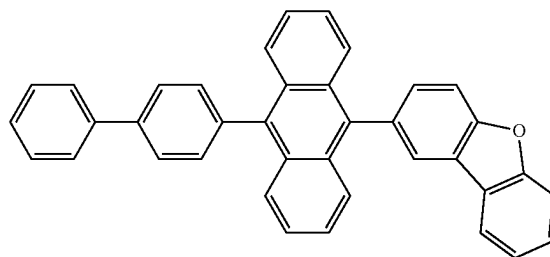
BH-6-a

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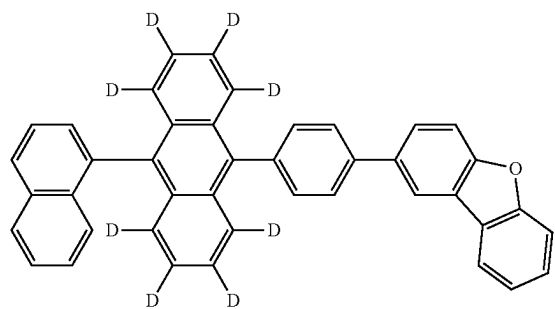
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BH-5



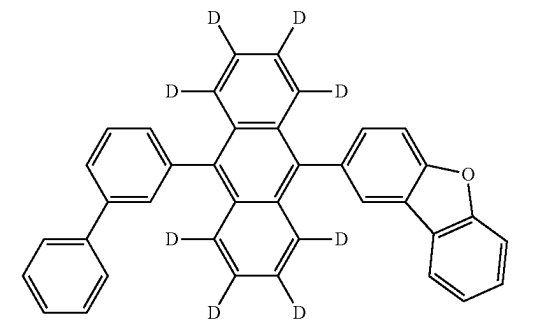
BH-7

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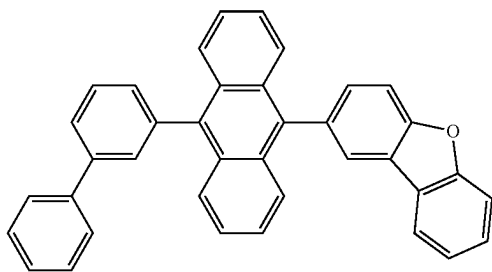
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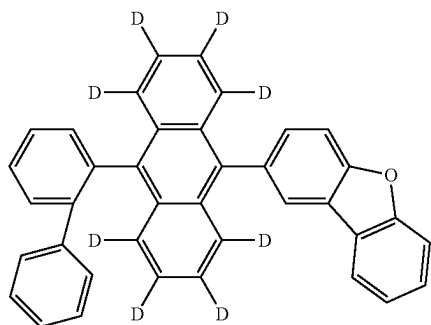
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BH-7-a

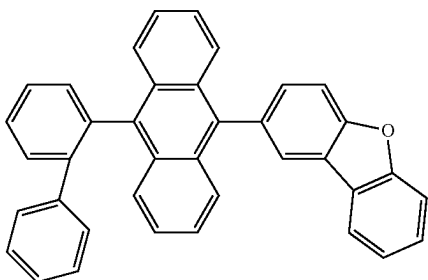


BH-8

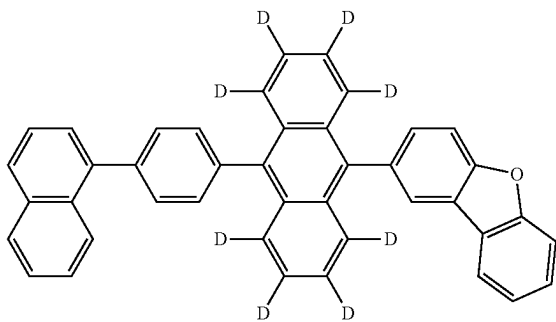


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BH-8-a

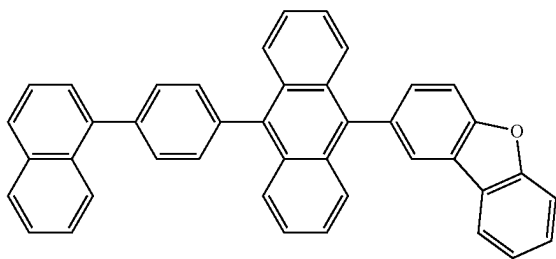


BH-9



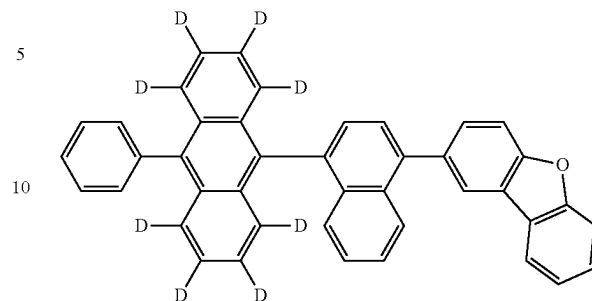
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BH-9-a

**986**

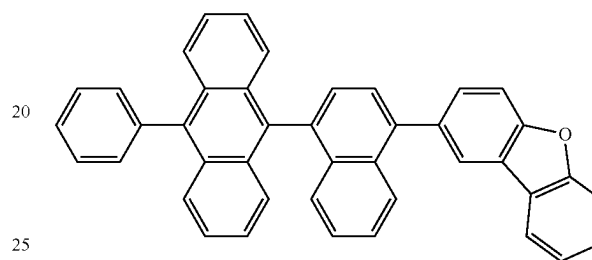
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BH-10



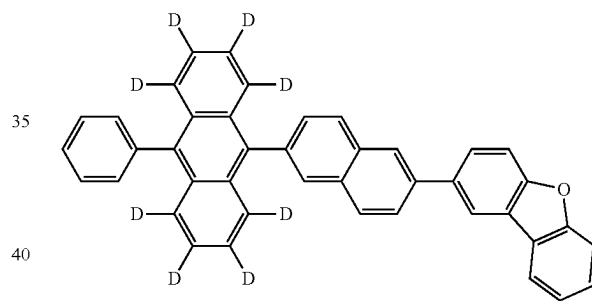
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BH-10-a



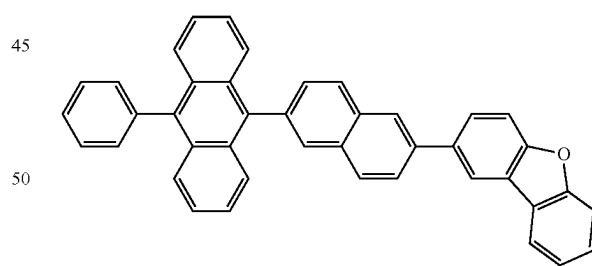
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BH-11



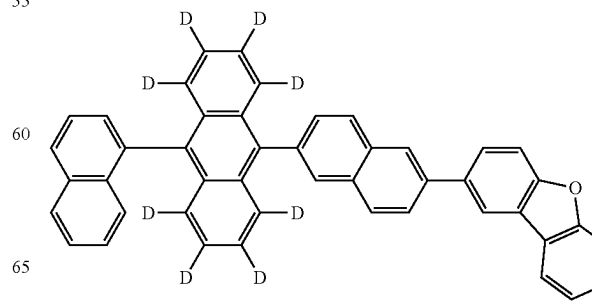
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BH-11-a



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BH-12

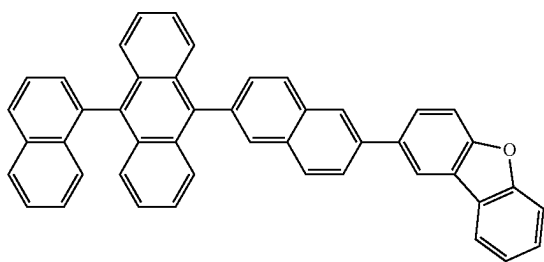


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987

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BH-12-a



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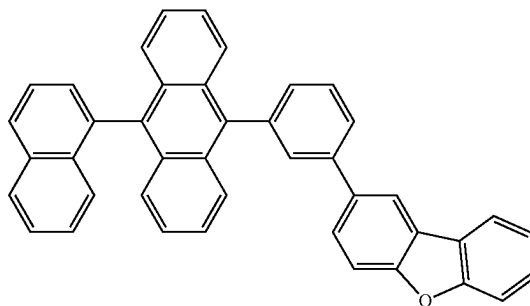
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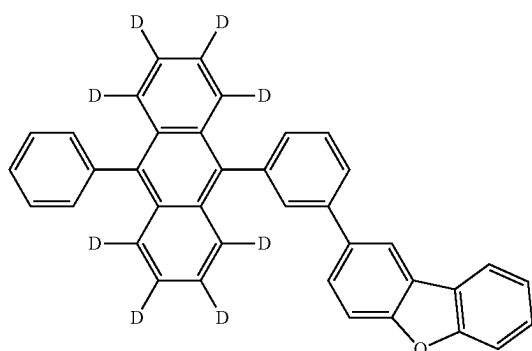
988

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BH-14-a



BH-13

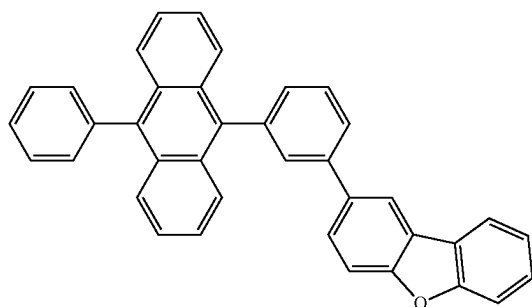


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BH-13-a



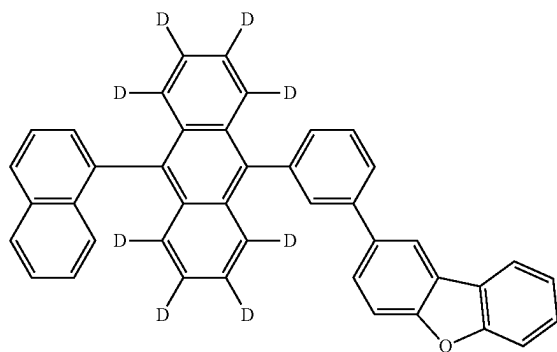
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BH-14

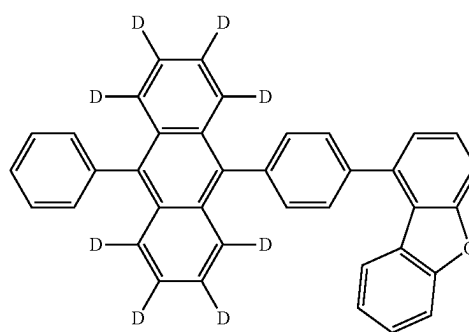


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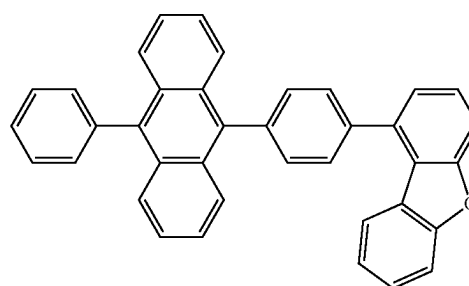
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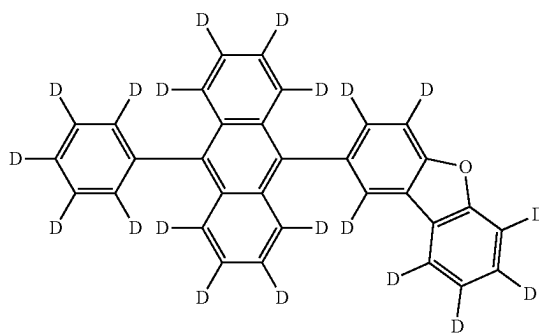
BH-15



BH-15-a



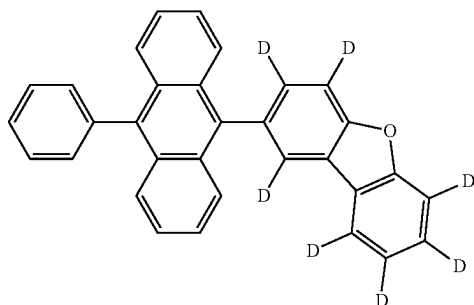
BH-16



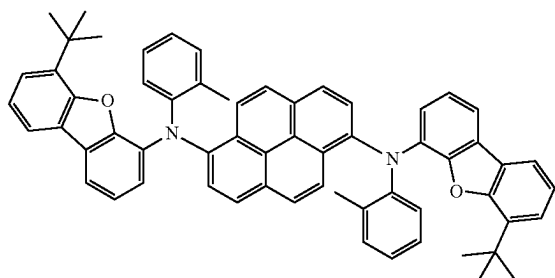
989

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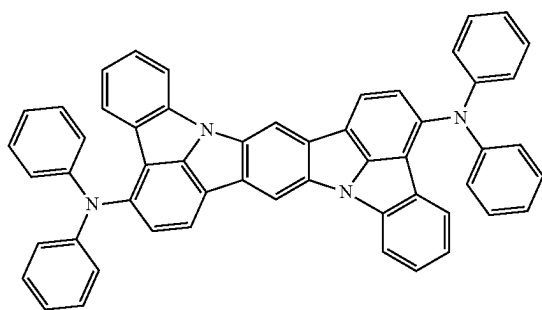
BH-1-c



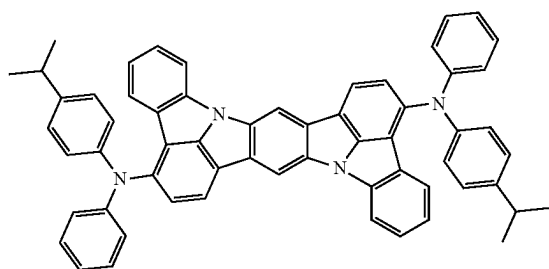
BD-1



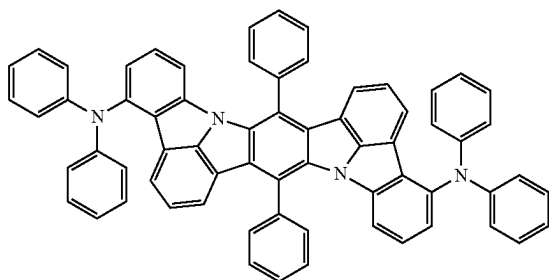
BD-2



BD-3

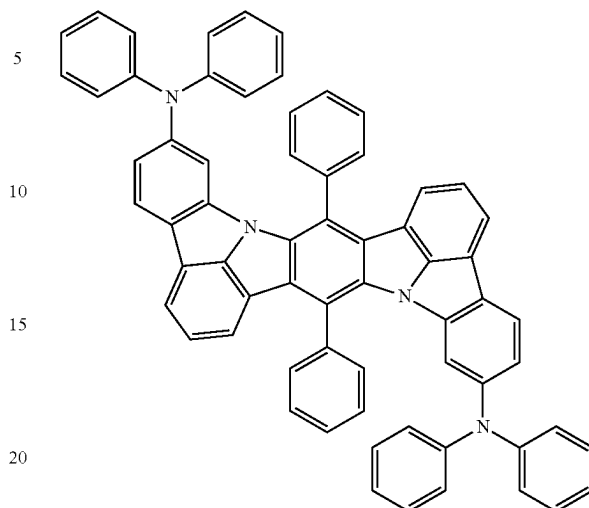


BD-4

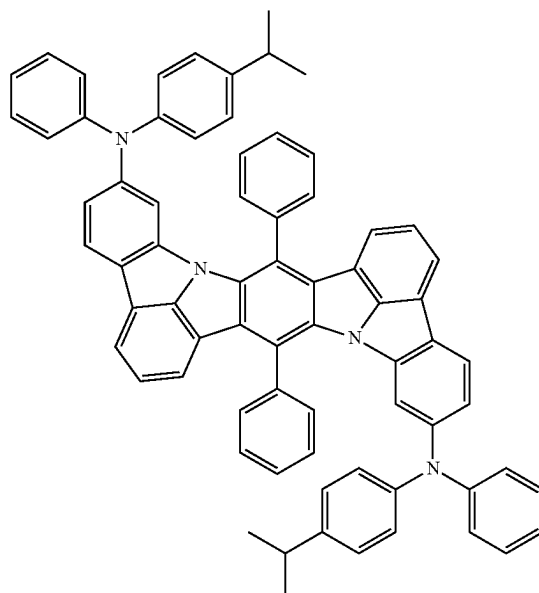
**990**

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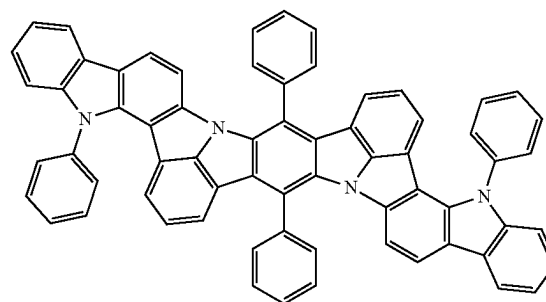
BD-5



BD-6



BD-7



(Evaluation of Organic EL Device)

A voltage was applied to the obtained organic EL device so that the current density was 50 mA/cm², and the time until the luminance reached 95% with respect to the initial luminance (LT95) was measured. The results are shown in Table 1.

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Further, a voltage was applied to the obtained organic EL device so that the current density was 10 mA/cm², and spectral radiance spectrum was measured using a spectroradiometer "CS-1000" (manufactured by Konica Minolta, Inc.) to determine CIE1931 chromaticity coordinate (CIEx, CIEy). The results are shown in Table 1.

Comparative Examples 1-1 and 1-2

Except that the compound shown in the following table was used as the host material of the emitting layer, the organic EL device was fabricated and evaluated in the same manner as in Example 1. The results are shown in Table 1.

TABLE 1

	Emitting layer		Chromaticity		
	Host	Dopant			
	material	material	LT95(h)	CIEx	CIEy
Example 1	BH-1	BD-1	155	0.139	0.090
Comp. Ex. 1-1	BH-1-a	BD-1	94	0.139	0.090
Comp. Ex. 1-2	BH-1-b	BD-1	95	0.139	0.090

Example 2, Comparative Example 2

Except that the compounds shown in the following table were used as the material of the emitting layer, the organic EL devices were fabricated and evaluated in the same manner as in Example 1. The results are shown in Table 2.

TABLE 2

	Emitting layer		Chromaticity		
	Host	Dopant			
	material	material	LT95(h)	CIEx	CIEy
Example 2	BH-2	BD-1	80	0.139	0.090
Comp. Ex. 2	BH-2-a	BD-1	49	0.139	0.090

Example 3, Comparative Example 3

Except that the compounds shown in the following table were used as the host material of the emitting layer, the organic EL devices were fabricated and evaluated in the same manner as in Example 1. The results are shown in Table 3.

TABLE 3

	Emitting layer		Chromaticity		
	Host	Dopant			
	material	material	LT95(h)	CIEx	CIEy
Example 3	BH-3	BD-1	70	0.139	0.090
Comp. Ex. 3	BH-3-a	BD-1	40	0.139	0.090

Example 4, Comparative Examples 4-1 and 4-2

Except that the compounds shown in the following table were used as the host material of the emitting layer, the organic EL devices were fabricated and evaluated in the

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same manner as in Example 1. For organic EL device obtained in Example 4 and Comparative Example 4-2, the driving voltages at the time of driving were measured at room temperature 10 mA/cm²a DC (direct current) constant current. The results are shown in Table 4.

TABLE 4

10	Emitting layer						
	Host	Dopant	Chromaticity				
	material	material	LT95(h)	CIEx	CIEy	Voltage(V)	
15	Example 4	BH-4	BD-1	170	0.139	0.091	3.7
	Comp. Ex. 4-1	BH-4-a	BD-1	101	0.139	0.091	—
	Comp. Ex. 4-2	BH-4-b	BD-1	120	0.139	0.090	4.1

Example 5, Comparative Examples 5-1 and 5-2

Except that the compound shown in the following table was used as the host material of the emitting layer, the organic EL device was fabricated and evaluated in the same manner as in Example 1. The results are shown in Table 5.

TABLE 5

	Emitting layer		Chromaticity		
	Host	Dopant			
	material	material	LT95(h)	CIEx	CIEy
Example 5	BH-5	BD-1	145	0.139	0.090
Comp. Ex. 5-1	BH-5-a	BD-1	89	0.139	0.090
Comp. Ex. 5-2	BH-5-b	BD-1	85	0.139	0.090

Example 11, Comparative Example 11

Except that the compound shown in the following table was used as the host material and the dopant material of the emitting layer, the organic EL device was fabricated and evaluated in the same manner as in Example 1. The results are shown in Table 6.

TABLE 6

	Emitting layer		Chromaticity		
	Host	Dopant			
	material	material	LT95(h)	CIEx	CIEy
Example 11	BH-1	BD-2	130	0.140	0.080
Comp. Ex. 11	BH-1-a	BD-2	83	0.140	0.080

Example 12, Comparative Example 12

Except that the compound shown in the following table was used as the host material and the dopant material of the emitting layer, the organic EL device was fabricated and evaluated in the same manner as in Example 1. The results are shown in Table 7.

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TABLE 7

	Emitting layer		Chromaticity		
	Host	Dopant	LT95(h)	CIEx	CIEy
	material	material			
Example 12	BH-2	BD-2	71	0.140	0.081
Comp. Ex. 12	BH-2-a	BD-2	45	0.140	0.080

Example 13, Comparative Example 13

Except that the compound shown in the following table was used as the host material and the dopant material of the emitting layer, the organic EL device was fabricated and evaluated in the same manner as in Example 1. The results are shown in Table 8.

TABLE 8

	Emitting layer		Chromaticity		
	Host	Dopant	LT95(h)	CIEx	CIEy
	material	material			
Example 13	BH-3	BD-2	56	0.140	0.080
Comp. Ex. 13	BH-3-a	BD-2	36	0.140	0.080

Example 14, Comparative Example 14

Except that the compound shown in the following table was used as the host material and the dopant material of the emitting layer, the organic EL device was fabricated and evaluated in the same manner as in Example 1. The results are shown in Table 9.

TABLE 9

	Emitting layer		Chromaticity		
	Host	Dopant	LT95(h)	CIEx	CIEy
	material	material			
Example 14	BH-4	BD-2	146	0.140	0.080
Comp. Ex. 14	BH-4-a	BD-2	82	0.140	0.080

Example 15, Comparative Example 15

Except that the compound shown in the following table was used as the host material and the dopant material of the emitting layer, the organic EL device was fabricated and evaluated in the same manner as in Example 1. The results are shown in Table 10.

TABLE 10

	Emitting layer		Chromaticity		
	Host	Dopant	LT95(h)	CIEx	CIEy
	material	material			
Example 15	BH-5	BD-2	126	0.140	0.080
Comp. Ex. 15	BH-5-a	BD-2	78	0.140	0.080

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Example 16, Comparative Example 16

Except that the compound shown in the following table was used as the host material and the dopant material of the emitting layer, the organic EL device was fabricated and evaluated in the same manner as in Example 1. The results are shown in Table 11.

TABLE 11

	Emitting layer		Chromaticity		
	Host	Dopant	LT95(h)	CIEx	CIEy
	material	material			
Example 16	BH-6	BD-2	127	0.140	0.080
Comp. Ex. 16	BH-6-a	BD-2	80	0.140	0.080

Example 21, Comparative Example 21

Except that the compounds shown in the following table were used as the host material and the dopant material of the emitting layer, the organic EL devices were fabricated and evaluated in the same manner as in Example 1. The results are shown in Table 12.

TABLE 12

	Emitting layer		Chromaticity		
	Host	Dopant	LT95(h)	CIEx	CIEy
	material	material			
Example 21	BH-1	BD-3	155	0.135	0.098
Comp. Ex. 21	BH-1-a	BD-3	96	0.135	0.098

Example 22, Comparative Example 22

Except that the compounds shown in the following table were used as the host material and the dopant material of the emitting layer, the organic EL devices were fabricated and evaluated in the same manner as in Example 1. The results are shown in Table 13.

TABLE 13

	Emitting layer		Chromaticity		
	Host	Dopant	LT95(h)	CIEx	CIEy
	material	material			
Example 22	BH-2	BD-3	77	0.135	0.098
Comp. Ex. 22	BH-2-a	BD-3	50	0.135	0.099

Example 23, Comparative Example 23

Except that the compounds shown in the following table were used as the host material and the dopant material of the emitting layer, the organic EL devices were fabricated and evaluated in the same manner as in Example 1. The results are shown in Table 14.

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TABLE 14

	Emitting layer		LT95(h)	Chromaticity	
	Host	Dopant			
	material	material		CIEx	CIEy
Example 23	BH-3	BD-3	73	0.135	0.098
Comp. Ex. 23	BH-3-a	BD-3	43	0.135	0.098

Example 24, Comparative Example 24

Except that the compounds shown in the following table were used as the host material and the dopant material of the emitting layer, the organic EL devices were fabricated and evaluated in the same manner as in Example 1. The results are shown in Table 15.

TABLE 15

	Emitting layer		LT95(h)	Chromaticity	
	Host	Dopant			
	material	material		CIEx	CIEy
Example 24	BH-4	BD-3	167	0.135	0.098
Comp. Ex. 24	BH-4-a	BD-3	105	0.135	0.098

Example 25, Comparative Example 25

Except that the compounds shown in the following table were used as the host material and the dopant material of the emitting layer, the organic EL devices were fabricated and evaluated in the same manner as in Example 1. The results are shown in Table 16.

TABLE 16

	Emitting layer		LT95(h)	Chromaticity	
	Host	Dopant			
	material	material		CIEx	CIEy
Example 25	BH-5	BD-3	147	0.135	0.098
Comp. Ex. 25	BH-5-a	BD-3	92	0.135	0.098

Example 26, Comparative Example 26

Except that the compounds shown in the following table were used as the host material and the dopant material of the emitting layer, the organic EL devices were fabricated and evaluated in the same manner as in Example 1. The results are shown in Table 17.

TABLE 17

	Emitting layer		LT95(h)	Chromaticity	
	Host	Dopant			
	material	material		CIEx	CIEy
Example 26	BH-6	BD-3	150	0.135	0.098
Comp. Ex. 26	BH-6-a	BD-3	96	0.135	0.098

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Example 31, Comparative Example 31

Except that the compounds shown in the following table were used as the host material and the dopant material of the emitting layer, the organic EL devices were fabricated and evaluated in the same manner as in Example 1. The results are shown in Table 18.

TABLE 18

	Emitting layer		LT95(h)	Chromaticity	
	Host	Dopant			
	material	material		CIEx	CIEy
Example 31	BH-1	BD-4	163	0.135	0.086
Comp. Ex. 31	BH-1-a	BD-4	98	0.135	0.086

Example 32, Comparative Example 32

Except that the compounds shown in the following table were used as the host material and the dopant material of the emitting layer, the organic EL devices were fabricated and evaluated in the same manner as in Example 1. The results are shown in Table 19.

TABLE 19

	Emitting layer		LT95(h)	Chromaticity	
	Host	Dopant			
	material	material		CIEx	CIEy
Example 32	BH-2	BD-4	78	0.135	0.086
Comp. Ex. 32	BH-2-a	BD-4	51	0.135	0.086

Example 33, Comparative Example 33

Except that the compounds shown in the following table were used as the host material and the dopant material of the emitting layer, the organic EL devices were fabricated and evaluated in the same manner as in Example 1. The results are shown in Table 20.

TABLE 20

	Emitting layer		LT95(h)	Chromaticity	
	Host	Dopant			
	material	material		CIEx	CIEy
Example 33	BH-3	BD-4	73	0.135	0.086
Comp. Ex. 33	BH-3-a	BD-4	39	0.135	0.086

Example 34, Comparative Example 34

Except that the compounds shown in the following table were used as the host material and the dopant material of the emitting layer, the organic EL devices were fabricated and evaluated in the same manner as in Example 1. The results are shown in Table 21.

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TABLE 21

	Emitting layer		LT95(h)	Chromaticity	
	Host	Dopant			
	material	material		CIEx	CIEx
Example 34	BH-4	BD-4	170	0.135	0.086
Comp. Ex. 34	BH-4-a	BD-4	105	0.135	0.086

Example 35, Comparative Example 35

Except that the compounds shown in the following table were used as the host material and the dopant material of the emitting layer, the organic EL devices were fabricated and evaluated in the same manner as in Example 1. The results are shown in Table 22.

TABLE 22

	Emitting layer		LT95(h)	Chromaticity	
	Host	Dopant			
	material	material		CIEx	CIEx
Example 35	BH-5	BD-4	152	0.135	0.086
Comp. Ex. 35	BH-5-a	BD-4	89	0.135	0.086

Example 36, Comparative Example 36

Except that the compounds shown in the following table were used as the host material and the dopant material of the emitting layer, the organic EL devices were fabricated and evaluated in the same manner as in Example 1. The results are shown in Table 23.

TABLE 23

	Emitting layer		LT95(h)	Chromaticity	
	Host	Dopant			
	material	material		CIEx	CIEx
Example 36	BH-6	BD-4	161	0.135	0.086
Comp. Ex. 36	BH-6-a	BD-4	102	0.135	0.086

Example 41, Comparative Example 41

Except that the compounds shown in the following table were used as the host material and the dopant material of the emitting layer, the organic EL devices were fabricated and evaluated in the same manner as in Example 1. The results are shown in Table 24.

TABLE 24

	Emitting layer		LT95(h)	Chromaticity	
	Host	Dopant			
	material	material		CIEx	CIEx
Example 41	BH-1	BD-5	195	0.135	0.080
Comp. Ex. 41	BH-1-a	BD-5	123	0.135	0.080

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Example 42, Comparative Example 42

Except that the compounds shown in the following table were used as the host material and the dopant material of the emitting layer, the organic EL devices were fabricated and evaluated in the same manner as in Example 1. The results are shown in Table 25.

TABLE 25

	Emitting layer		LT95(h)	Chromaticity	
	Host	Dopant			
	material	material		CIEx	CIEx
Example 42	BH-2	BD-5	106	0.135	0.080
Comp. Ex. 42	BH-2-a	BD-5	66	0.135	0.080

Example 43, Comparative Example 43

Except that the compounds shown in the following table were used as the host material and the dopant material of the emitting layer, the organic EL devices were fabricated and evaluated in the same manner as in Example 1. The results are shown in Table 26.

TABLE 26

	Emitting layer		LT95(h)	Chromaticity	
	Host	Dopant			
	material	material		CIEx	CIEx
Example 43	BH-3	BD-5	90	0.135	0.080
Comp. Ex. 43	BH-3-a	BD-5	53	0.135	0.080

Example 44, Comparative Example 44

Except that the compounds shown in the following table were used as the host material and the dopant material of the emitting layer, the organic EL devices were fabricated and evaluated in the same manner as in Example 1. The results are shown in Table 27.

TABLE 27

	Emitting layer		LT95(h)	Chromaticity	
	Host	Dopant			
	material	material		CIEx	CIEx
Example 44	BH-4	BD-5	211	0.135	0.080
Comp. Ex. 44	BH-4-a	BD-5	133	0.135	0.081

Example 45, Comparative Example 45

Except that the compounds shown in the following table were used as the host material and the dopant material of the emitting layer, the organic EL devices were fabricated and evaluated in the same manner as in Example 1. The results are shown in Table 28.

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TABLE 28

	Emitting layer		Chromaticity		
	Host	Dopant	LT95(h)	CIEx	CIEy
	material	material			
Example 45	BH-5	BD-5	179	0.135	0.080
Comp. Ex. 45	BH-5-a	BD-5	112	0.135	0.080

Example 46, Comparative Example 46

Except that the compounds shown in the following table were used as the host material and the dopant material of the emitting layer, the organic EL devices were fabricated and evaluated in the same manner as in Example 1. The results are shown in Table 29.

TABLE 29

	Emitting layer		Chromaticity		
	Host	Dopant	LT95(h)	CIEx	CIEy
	material	material			
Example 46	BH-6	BD-5	202	0.135	0.080
Comp. Ex. 46	BH-6-a	BD-5	125	0.135	0.080

Example 51, Comparative Example 51

Except that the compounds shown in the following table were used as the host material and the dopant material of the emitting layer, the organic EL devices were fabricated and evaluated in the same manner as in Example 1. The results are shown in Table 30.

TABLE 30

	Emitting layer		Chromaticity		
	Host	Dopant	LT95(h)	CIEx	CIEy
	material	material			
Example 51	BH-1	BD-6	253	0.136	0.090
Comp. Ex. 51	BH-1-a	BD-6	146	0.136	0.090

Example 52, Comparative Example 52

Except that the compounds shown in the following table were used as the host material and the dopant material of the emitting layer, the organic EL devices were fabricated and evaluated in the same manner as in Example 1. The results are shown in Table 31.

TABLE 31

	Emitting layer		Chromaticity		
	Host	Dopant	LT95(h)	CIEx	CIEy
	material	material			
Example 52	BH-2	BD-6	125	0.136	0.090
Comp. Ex. 52	BH-2-a	BD-6	73	0.136	0.090

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Example 53, Comparative Example 53

Except that the compounds shown in the following table were used as the host material and the dopant material of the emitting layer, the organic EL devices were fabricated and evaluated in the same manner as in Example 1. The results are shown in Table 32.

TABLE 32

	Emitting layer		Chromaticity		
	Host	Dopant	LT95(h)	CIEx	CIEy
	material	material			
Example 53	BH-3	BD-6	102	0.136	0.090
Comp. Ex. 53	BH-3-a	BD-6	64	0.136	0.090

Example 54, Comparative Example 54

Except that the compounds shown in the following table were used as the host material and the dopant material of the emitting layer, the organic EL devices were fabricated and evaluated in the same manner as in Example 1. The results are shown in Table 33.

TABLE 33

	Emitting layer		Chromaticity		
	Host	Dopant	LT95(h)	CIEx	CIEy
	material	material			
Example 54	BH-4	BD-6	257	0.136	0.090
Comp. Ex. 54	BH-4-a	BD-6	161	0.136	0.091

Example 55, Comparative Example 55

Except that the compounds shown in the following table were used as the host material and the dopant material of the emitting layer, the organic EL devices were fabricated and evaluated in the same manner as in Example 1. The results are shown in Table 34.

TABLE 34

	Emitting layer		Chromaticity		
	Host	Dopant	LT95(h)	CIEx	CIEy
	material	material			
Example 55	BH-5	BD-6	215	0.136	0.090
Comp. Ex. 55	BH-5-a	BD-6	137	0.136	0.090

Example 56, Comparative Example 56

Except that the compounds shown in the following table were used as the host material and the dopant material of the emitting layer, the organic EL devices were fabricated and evaluated in the same manner as in Example 1. The results are shown in Table 35.

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TABLE 35

	Emitting layer		Chromaticity		
	Host	Dopant	LT95(h)	CIEx	CIEy
	material	material			
Example 56	BH-6	BD-6	241	0.136	0.090
Comp. Ex. 56	BH-6-a	BD-6	141	0.136	0.090

Example 61, Comparative Example 61

Except that the compounds shown in the following table were used as the host material and the dopant material of the emitting layer, the organic EL devices were fabricated and evaluated in the same manner as in Example 1. The results are shown in Table 36.

TABLE 36

	Emitting layer		Chromaticity		
	Host	Dopant	LT95(h)	CIEx	CIEy
	material	material			
Example 61	BH-1	BD-7	104	0.144	0.061
Comp. Ex. 61	BH-1-a	BD-7	65	0.144	0.061

Example 62, Comparative Example 62

Except that the compounds shown in the following table were used as the host material and the dopant material of the emitting layer, the organic EL devices were fabricated and evaluated in the same manner as in Example 1. The results are shown in Table 37.

TABLE 37

	Emitting layer		Chromaticity		
	Host	Dopant	LT95(h)	CIEx	CIEy
	material	material			
Example 62	BH-2	BD-7	53	0.144	0.061
Comp. Ex. 62	BH-2-a	BD-7	37	0.144	0.061

Example 63, Comparative Example 63

Except that the compounds shown in the following table were used as the host material and the dopant material of the emitting layer, the organic EL devices were fabricated and evaluated in the same manner as in Example 1. The results are shown in Table 38.

TABLE 38

	Emitting layer		Chromaticity		
	Host	Dopant	LT95(h)	CIEx	CIEy
	material	material			
Example 63	BH-3	BD-7	51	0.144	0.060
Comp. Ex. 63	BH-3-a	BD-7	32	0.144	0.061

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Example 64, Comparative Example 64

Except that the compounds shown in the following table were used as the host material and the dopant material of the emitting layer, the organic EL devices were fabricated and evaluated in the same manner as in Example 1. The results are shown in Table 39.

TABLE 39

	Emitting layer		Chromaticity		
	Host	Dopant	LT95(h)	CIEx	CIEy
	material	material			
Example 64	BH-4	BD-7	117	0.144	0.061
Comp. Ex. 64	BH-4-a	BD-7	75	0.144	0.061

Example 65, Comparative Example 65

Except that the compounds shown in the following table were used as the host material and the dopant material of the emitting layer, the organic EL devices were fabricated and evaluated in the same manner as in Example 1. The results are shown in Table 40.

TABLE 40

	Emitting layer		Chromaticity		
	Host	Dopant	LT95(h)	CIEx	CIEy
	material	material			
Example 65	BH-5	BD-7	105	0.144	0.061
Comp. Ex. 65	BH-5-a	BD-7	65	0.144	0.061

Example 66, Comparative Example 66

Except that the compounds shown in the following table were used as the host material and the dopant material of the emitting layer, the organic EL devices were fabricated and evaluated in the same manner as in Example 1. The results are shown in Table 41.

TABLE 41

	Emitting layer		Chromaticity		
	Host	Dopant	LT95(h)	CIEx	CIEy
	material	material			
Example 66	BH-6	BD-7	110	0.144	0.061
Comp. Ex. 66	BH-6-a	BD-7	66	0.144	0.061

Example 71, Comparative Example 71

Except that the compounds shown in the following table were used as the host material and the dopant material of the emitting layer, the organic EL devices were fabricated and evaluated in the same manner as in Example 1. The results are shown in Table 42.

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TABLE 42

	Emitting layer		Chromaticity		
	Host	Dopant	LT95(h)	CIEx	CIEy
	material	material			
Example 71	BH-6	BD-1	154	0.139	0.090
Comp. Ex. 71	BH-6-a	BD-1	90	0.139	0.090

Example 72, Comparative Example 72

Except that the compounds shown in the following table were used as the host material and the dopant material of the emitting layer, the organic EL devices were fabricated and evaluated in the same manner as in Example 1. The results are shown in Table 43.

TABLE 43

	Emitting layer		Chromaticity		
	Host	Dopant	LT95(h)	CIEx	CIEy
	material	material			
Example 72	BH-7	BD-1	116	0.139	0.090
Comp. Ex. 72	BH-7-a	BD-1	70	0.139	0.090

Example 73, Comparative Example 73

Except that the compounds shown in the following table were used as the host material and the dopant material of the emitting layer, the organic EL devices were fabricated and evaluated in the same manner as in Example 1. The results are shown in Table 44.

TABLE 44

	Emitting layer		Chromaticity		
	Host	Dopant	LT95(h)	CIEx	CIEy
	material	material			
Example 73	BH-8	BD-1	110	0.139	0.090
Comp. Ex. 73	BH-8-a	BD-1	75	0.139	0.090

Example 74, Comparative Example 74

Except that the compounds shown in the following table were used as the host material and the dopant material of the emitting layer, the organic EL devices were fabricated and evaluated in the same manner as in Example 1. The results are shown in Table 45.

TABLE 45

	Emitting layer		Chromaticity		
	Host	Dopant	LT95(h)	CIEx	CIEy
	material	material			
Example 74	BH-9	BD-1	110	0.139	0.090
Comp. Ex. 74	BH-9-a	BD-1	70	0.139	0.090

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Example 75, Comparative Example 75

Except that the compounds shown in the following table were used as the host material and the dopant material of the emitting layer, the organic EL devices were fabricated and evaluated in the same manner as in Example 1. The results are shown in Table 46.

TABLE 46

	Emitting layer		Chromaticity		
	Host	Dopant	LT95(h)	CIEx	CIEy
	material	material			
Example 75	BH-10	BD-1	103	0.139	0.090
Comp. Ex. 75	BH-10-a	BD-1	58	0.139	0.090

Example 76, Comparative Example 76

Except that the compounds shown in the following table were used as the host material and the dopant material of the emitting layer, the organic EL devices were fabricated and evaluated in the same manner as in Example 1. The results are shown in Table 47.

TABLE 47

	Emitting layer		Chromaticity		
	Host	Dopant	LT95(h)	CIEx	CIEy
	material	material			
Example 76	BH-11	BD-1	95	0.139	0.090
Comp. Ex. 76	BH-11-a	BD-1	50	0.139	0.090

Example 77, Comparative Example 77

Except that the compounds shown in the following table were used as the host material and the dopant material of the emitting layer, the organic EL devices were fabricated and evaluated in the same manner as in Example 1. The results are shown in Table 48.

TABLE 48

	Emitting layer		Chromaticity		
	Host	Dopant	LT95(h)	CIEx	CIEy
	material	material			
Example 77	BH-12	BD-1	75	0.139	0.090
Comp. Ex. 77	BH-12-a	BD-1	51	0.139	0.090

Example 78, Comparative Example 78

Except that the compounds shown in the following table were used as the host material and the dopant material of the emitting layer, the organic EL devices were fabricated and evaluated in the same manner as in Example 1. The results are shown in Table 49.

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TABLE 49

	Emitting layer		LT95(h)	Chromaticity	
	Host	Dopant		CIEx	CIEy
	material	material			
Example 78	BH-13	BD-1	150	0.139	0.090
Comp. Ex. 78	BH-13-a	BD-1	83	0.139	0.090

Example 79, Comparative Example 79

Except that the compounds shown in the following table were used as the host material and the dopant material of the emitting layer, the organic EL devices were fabricated and evaluated in the same manner as in Example 1. The results are shown in Table 50.

TABLE 50

	Emitting layer		LT95(h)	Chromaticity	
	Host	Dopant		CIEx	CIEy
	material	material			
Example 79	BH-14	BD-1	100	0.139	0.090
Comp. Ex. 79	BH-14-a	BD-1	63	0.139	0.090

Example 80, Comparative Example 80

Except that the compounds shown in the following table were used as the host material and the dopant material of the emitting layer, the organic EL devices were fabricated and evaluated in the same manner as in Example 1. The results are shown in Table 51.

TABLE 51

	Emitting layer		LT95(h)	Chromaticity	
	Host	Dopant		CIEx	CIEy
	material	material			
Example 80	BH-15	BD-1	100	0.139	0.090
Comp. Ex. 80	BH-15-a	BD-1	60	0.139	0.090

Example 81, Comparative Example 81

Except that the compounds shown in the following table were used as the host material and the dopant material of the emitting layer, the organic EL devices were fabricated and evaluated in the same manner as in Example 1. The results are shown in Table 52.

TABLE 52

	Emitting layer		LT95(h)	Chromaticity	
	Host	Dopant		CIEx	CIEy
	material	material			
Example 81	BH-16	BD-1	158	0.139	0.090
Comp. Ex. 81	BH-1-c	BD-1	103	0.139	0.090

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Example 82, Comparative Example 82

Except that the compounds shown in the following table were used as the host material and the dopant material of the emitting layer, the organic EL devices were fabricated and evaluated in the same manner as in Example 1. The results are shown in Table 53.

TABLE 53

	Emitting layer		LT95(h)	Chromaticity	
	Host	Dopant		CIEx	CIEy
	material	material			
Example 82	BH-7	BD-2	98	0.140	0.080
Comp. Ex. 82	BH-7-a	BD-2	64	0.140	0.080

Example 83, Comparative Example 83

Except that the compounds shown in the following table were used as the host material and the dopant material of the emitting layer, the organic EL devices were fabricated and evaluated in the same manner as in Example 1. The results are shown in Table 54.

TABLE 54

	Emitting layer		LT95(h)	Chromaticity	
	Host	Dopant		CIEx	CIEy
	material	material			
Example 83	BH-8	BD-2	100	0.140	0.080
Comp. Ex. 83	BH-8-a	BD-2	64	0.140	0.080

Example 84, Comparative Example 84

Except that the compounds shown in the following table were used as the host material and the dopant material of the emitting layer, the organic EL devices were fabricated and evaluated in the same manner as in Example 1. The results are shown in Table 55.

TABLE 55

	Emitting layer		LT95(h)	Chromaticity	
	Host	Dopant		CIEx	CIEy
	material	material			
Example 84	BH-9	BD-2	94	0.140	0.080
Comp. Ex. 84	BH-9-a	BD-2	58	0.140	0.080

Example 85, Comparative Example 85

Except that the compounds shown in the following table were used as the host material and the dopant material of the emitting layer, the organic EL devices were fabricated and evaluated in the same manner as in Example 1. The results are shown in Table 56.

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TABLE 56

	Emitting layer		LT95(h)	Chromaticity	
	Host	Dopant		CIEx	CIEy
	material	material			
Example 85	BH-10	BD-2	83	0.140	0.081
Comp. Ex. 85	BH-10-a	BD-2	50	0.140	0.080

Example 86, Comparative Example 86

Except that the compounds shown in the following table were used as the host material and the dopant material of the emitting layer, the organic EL devices were fabricated and evaluated in the same manner as in Example 1. The results are shown in Table 57.

TABLE 57

	Emitting layer		LT95(h)	Chromaticity	
	Host	Dopant		CIEx	CIEy
	material	material			
Example 86	BH-11	BD-2	73	0.140	0.080
Comp. Ex. 86	BH-11-a	BD-2	45	0.140	0.080

Example 87, Comparative Example 87

Except that the compounds shown in the following table were used as the host material and the dopant material of the emitting layer, the organic EL devices were fabricated and evaluated in the same manner as in Example 1. The results are shown in Table 58.

TABLE 58

	Emitting layer		LT95(h)	Chromaticity	
	Host	Dopant		CIEx	CIEy
	material	material			
Example 87	BH-12	BD-2	65	0.140	0.080
Comp. Ex. 87	BH-12-a	BD-2	40	0.140	0.080

Example 88, Comparative Example 88

Except that the compounds shown in the following table were used as the host material and the dopant material of the emitting layer, the organic EL devices were fabricated and evaluated in the same manner as in Example 1. The results are shown in Table 59.

TABLE 59

	Emitting layer		LT95(h)	Chromaticity	
	Host	Dopant		CIEx	CIEy
	material	material			
Example 88	BH-13	BD-2	127	0.140	0.080
Comp. Ex. 88	BH-13-a	BD-2	75	0.140	0.080

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Example 89, Comparative Example 89

Except that the compounds shown in the following table were used as the host material and the dopant material of the emitting layer, the organic EL devices were fabricated and evaluated in the same manner as in Example 1. The results are shown in Table 60.

TABLE 60

	Emitting layer		LT95(h)	Chromaticity	
	Host	Dopant		CIEx	CIEy
	material	material			
Example 89	BH-14	BD-2	88	0.140	0.080
Comp. Ex. 89	BH-14-a	BD-2	54	0.140	0.080

Example 90, Comparative Example 90

Except that the compounds shown in the following table were used as the host material and the dopant material of the emitting layer, the organic EL devices were fabricated and evaluated in the same manner as in Example 1. The results are shown in Table 61.

TABLE 61

	Emitting layer		LT95(h)	Chromaticity	
	Host	Dopant		CIEx	CIEy
	material	material			
Example 90	BH-15	BD-2	84	0.140	0.080
Comp. Ex. 90	BH-15-a	BD-2	52	0.140	0.080

Example 91, Comparative Examples 91-1, 91-2 and 91-3

Except that the compounds shown in the following table were used as the host material and the dopant material of the emitting layer, the organic EL devices were fabricated and evaluated in the same manner as in Example 1. The results are shown in Table 62.

TABLE 62

	Emitting layer		LT95(h)	Chromaticity	
	Host	Dopant		CIEx	CIEy
	material	material			
Example 91	BH-16	BD-2	131	0.140	0.080
Comp. Ex. 91-1	BH-1-a	BD-2	83	0.140	0.080
Comp. Ex. 91-2	BH-1-b	BD-2	84	0.140	0.080
Comp. Ex. 91-3	BH-1-c	BD-2	86	0.140	0.080

Example 92, Comparative Example 92

Except that the compounds shown in the following table were used as the host material and the dopant material of the emitting layer, the organic EL devices were fabricated and evaluated in the same manner as in Example 1. The results are shown in Table 63.

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TABLE 63

	Emitting layer		Chromaticity		
	Host	Dopant	LT95(h)	CIEx	CIEy
	material	material			
Example 92	BH-7	BD-3	118	0.135	0.098
Comp. Ex. 92	BH-7-a	BD-3	75	0.135	0.098

Example 93, Comparative Example 93

Except that the compounds shown in the following table were used as the host material and the dopant material of the emitting layer, the organic EL devices were fabricated and evaluated in the same manner as in Example 1. The results are shown in Table 64.

TABLE 64

	Emitting layer		Chromaticity		
	Host	Dopant	LT95(h)	CIEx	CIEy
	material	material			
Example 93	BH-8	BD-3	112	0.135	0.098
Comp. Ex. 93	BH-8-a	BD-3	72	0.135	0.098

Example 94, Comparative Example 94

Except that the compounds shown in the following table were used as the host material and the dopant material of the emitting layer, the organic EL devices were fabricated and evaluated in the same manner as in Example 1. The results are shown in Table 65.

TABLE 65

	Emitting layer		Chromaticity		
	Host	Dopant	LT95(h)	CIEx	CIEy
	material	material			
Example 94	BH-9	BD-3	105	0.135	0.098
Comp. Ex. 94	BH-9-a	BD-3	66	0.135	0.098

Example 95, Comparative Example 95

Except that the compounds shown in the following table were used as the host material and the dopant material of the emitting layer, the organic EL devices were fabricated and evaluated in the same manner as in Example 1. The results are shown in Table 66.

TABLE 66

	Emitting layer		Chromaticity		
	Host	Dopant	LT95(h)	CIEx	CIEy
	material	material			
Example 95	BH-10	BD-3	90	0.135	0.098
Comp. Ex. 95	BH-10-a	BD-3	60	0.135	0.098

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Example 96, Comparative Example 96

Except that the compounds shown in the following table were used as the host material and the dopant material of the emitting layer, the organic EL devices were fabricated and evaluated in the same manner as in Example 1. The results are shown in Table 67.

TABLE 67

	Emitting layer		Chromaticity		
	Host	Dopant	LT95(h)	CIEx	CIEy
	material	material			
Example 96	BH-11	BD-3	80	0.135	0.098
Comp. Ex. 96	BH-11-a	BD-3	53	0.135	0.098

Example 97, Comparative Example 97

Except that the compounds shown in the following table were used as the host material and the dopant material of the emitting layer, the organic EL devices were fabricated and evaluated in the same manner as in Example 1. The results are shown in Table 68.

TABLE 68

	Emitting layer		Chromaticity		
	Host	Dopant	LT95(h)	CIEx	CIEy
	material	material			
Example 97	BH-12	BD-3	70	0.135	0.098
Comp. Ex. 97	BH-12-a	BD-3	46	0.135	0.099

Example 98, Comparative Example 98

Except that the compounds shown in the following table were used as the host material and the dopant material of the emitting layer, the organic EL devices were fabricated and evaluated in the same manner as in Example 1. The results are shown in Table 69.

TABLE 69

	Emitting layer		Chromaticity		
	Host	Dopant	LT95(h)	CIEx	CIEy
	material	material			
Example 98	BH-13	BD-3	150	0.135	0.098
Comp. Ex. 98	BH-13-a	BD-3	92	0.135	0.098

Example 99, Comparative Example 99

Except that the compounds shown in the following table were used as the host material and the dopant material of the emitting layer, the organic EL devices were fabricated and evaluated in the same manner as in Example 1. The results are shown in Table 70.

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TABLE 70

	Emitting layer		LT95(h)	Chromaticity	
	Host	Dopant			
	material	material		CIEx	CIEx
Example 99	BH-14	BD-3	102	0.135	0.098
Comp. Ex. 99	BH-14-a	BD-3	64	0.135	0.098

Example 100, Comparative Example 100

Except that the compounds shown in the following table were used as the host material and the dopant material of the emitting layer, the organic EL devices were fabricated and evaluated in the same manner as in Example 1. The results are shown in Table 71.

TABLE 71

	Emitting layer		LT95(h)	Chromaticity	
	Host	Dopant			
	material	material		CIEx	CIEx
Example 100	BH-15	BD-3	96	0.135	0.098
Comp. Ex. 100	BH-15-a	BD-3	60	0.135	0.098

Example 101, Comparative Examples 101-1, 101-2 and 101-3

Except that the compounds shown in the following table were used as the host material and the dopant material of the emitting layer, the organic EL devices were fabricated and evaluated in the same manner as in Example 1. The results are shown in Table 72.

TABLE 72

	Emitting layer		LT95(h)	Chromaticity	
	Host	Dopant			
	material	material		CIEx	CIEx
Example 101	BH-16	BD-3	157	0.135	0.098
Comp. Ex. 101-1	BH-1-a	BD-3	96	0.135	0.098
Comp. Ex. 101-2	BH-1-b	BD-3	93	0.135	0.098
Comp. Ex. 101-3	BH-1-c	BD-3	100	0.135	0.098

Example 102, Comparative Example 102

Except that the compounds shown in the following table were used as the host material and the dopant material of the emitting layer, the organic EL devices were fabricated and evaluated in the same manner as in Example 1. The results are shown in Table 73.

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TABLE 73

	Emitting layer		LT95(h)	Chromaticity	
	Host	Dopant			
	material	material		CIEx	CIEx
Example 102	BH-7	BD-4	126	0.135	0.086
Comp. Ex. 102	BH-7-a	BD-4	80	0.135	0.086

Example 103, Comparative Example 103

Except that the compounds shown in the following table were used as the host material and the dopant material of the emitting layer, the organic EL devices were fabricated and evaluated in the same manner as in Example 1. The results are shown in Table 74.

TABLE 74

	Emitting layer		LT95(h)	Chromaticity	
	Host	Dopant			
	material	material		CIEx	CIEx
Example 103	BH-8	BD-4	126	0.135	0.086
Comp. Ex. 103	BH-8-a	BD-4	76	0.135	0.086

Example 104, Comparative Example 104

Except that the compounds shown in the following table were used as the host material and the dopant material of the emitting layer, the organic EL devices were fabricated and evaluated in the same manner as in Example 1. The results are shown in Table 75.

TABLE 75

	Emitting layer		LT95(h)	Chromaticity	
	Host	Dopant			
	material	material		CIEx	CIEx
Example 104	BH-9	BD-4	110	0.135	0.085
Comp. Ex. 104	BH-9-a	BD-4	70	0.135	0.086

Example 105, Comparative Example 105

Except that the compounds shown in the following table were used as the host material and the dopant material of the emitting layer, the organic EL devices were fabricated and evaluated in the same manner as in Example 1. The results are shown in Table 76.

TABLE 76

	Emitting layer		LT95(h)	Chromaticity	
	Host	Dopant			
	material	material		CIEx	CIEx
Example 105	BH-10	BD-4	103	0.135	0.086
Comp. Ex. 105	BH-10-a	BD-4	60	0.135	0.086

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Example 106, Comparative Example 106

Except that the compounds shown in the following table were used as the host material and the dopant material of the emitting layer, the organic EL devices were fabricated and evaluated in the same manner as in Example 1. The results are shown in Table 77.

TABLE 77

	Emitting layer		Chromaticity		
	Host	Dopant	LT95(h)	CIEx	CIEy
	material	material			
Example 106	BH-11	BD-4	90	0.135	0.086
Comp. Ex. 106	BH-11-a	BD-4	56	0.135	0.086

Example 107, Comparative Example 107

Except that the compounds shown in the following table were used as the host material and the dopant material of the emitting layer, the organic EL devices were fabricated and evaluated in the same manner as in Example 1. The results are shown in Table 78.

TABLE 78

	Emitting layer		Chromaticity		
	Host	Dopant	LT95(h)	CIEx	CIEy
	material	material			
Example 107	BH-12	BD-4	80	0.135	0.086
Comp. Ex. 107	BH-12-a	BD-4	50	0.135	0.086

Example 108, Comparative Example 108

Except that the compounds shown in the following table were used as the host material and the dopant material of the emitting layer, the organic EL devices were fabricated and evaluated in the same manner as in Example 1. The results are shown in Table 79.

TABLE 79

	Emitting layer		Chromaticity		
	Host	Dopant	LT95(h)	CIEx	CIEy
	material	material			
Example 108	BH-13	BD-4	157	0.135	0.085
Comp. Ex. 108	BH-13-a	BD-4	92	0.135	0.086

Example 109, Comparative Example 109

Except that the compounds shown in the following table were used as the host material and the dopant material of the emitting layer, the organic EL devices were fabricated and evaluated in the same manner as in Example 1. The results are shown in Table 80.

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TABLE 80

	Emitting layer		Chromaticity		
	Host	Dopant	LT95(h)	CIEx	CIEy
	material	material			
Example 109	BH-14	BD-4	110	0.135	0.086
Comp. Ex. 109	BH-14-a	BD-4	65	0.135	0.086

Example 110, Comparative Example 110

Except that the compounds shown in the following table were used as the host material and the dopant material of the emitting layer, the organic EL devices were fabricated and evaluated in the same manner as in Example 1. The results are shown in Table 81.

TABLE 81

	Emitting layer		Chromaticity		
	Host	Dopant	LT95(h)	CIEx	CIEy
	material	material			
Example 110	BH-15	BD-4	100	0.135	0.086
Comp. Ex. 110	BH-15-a	BD-4	60	0.135	0.086

Example 111, Comparative Examples 111-1, 111-2 and -3

Except that the compounds shown in the following table were used as the host material and the dopant material of the emitting layer, the organic EL devices were fabricated and evaluated in the same manner as in Example 1. The results are shown in Table 82.

TABLE 82

	Emitting layer		Chromaticity		
	Host	Dopant	LT95(h)	CIEx	CIEy
	material	material			
Example 111	BH-16	BD-4	165	0.135	0.086
Comp. Ex. 111-1	BH-1-a	BD-4	98	0.135	0.086
Comp. Ex. 111-2	BH-1-b	BD-4	100	0.135	0.086
Comp. Ex. 111-3	BH-1-c	BD-4	103	0.135	0.086

Example 112, Comparative Example 112

Except that the compounds shown in the following table were used as the host material and the dopant material of the emitting layer, the organic EL devices were fabricated and evaluated in the same manner as in Example 1. The results are shown in Table 83.

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TABLE 83

	Emitting layer		LT95(h)	Chromaticity	
	Host	Dopant		CIEx	CIEy
	material	material			
Example 112	BH-7	BD-5	150	0.135	0.080
Comp. Ex. 112	BH-7-a	BD-5	90	0.135	0.080

Example 113, Comparative Example 113

Except that the compounds shown in the following table were used as the host material and the dopant material of the emitting layer, the organic EL devices were fabricated and evaluated in the same manner as in Example 1. The results are shown in Table 84.

TABLE 84

	Emitting layer		LT95(h)	Chromaticity	
	Host	Dopant		CIEx	CIEy
	material	material			
Example 113	BH-8	BD-5	155	0.135	0.080
Comp. Ex. 113	BH-8-a	BD-5	90	0.135	0.080

Example 114, Comparative Example 114

Except that the compounds shown in the following table were used as the host material and the dopant material of the emitting layer, the organic EL devices were fabricated and evaluated in the same manner as in Example 1. The results are shown in Table 85.

TABLE 85

	Emitting layer		LT95(h)	Chromaticity	
	Host	Dopant		CIEx	CIEy
	material	material			
Example 114	BH-9	BD-5	143	0.135	0.080
Comp. Ex. 114	BH-9-a	BD-5	88	0.135	0.080

Example 115, Comparative Example 115

Except that the compounds shown in the following table were used as the host material and the dopant material of the emitting layer, the organic EL devices were fabricated and evaluated in the same manner as in Example 1. The results are shown in Table 86.

TABLE 86

	Emitting layer		LT95(h)	Chromaticity	
	Host	Dopant		CIEx	CIEy
	material	material			
Example 115	BH-10	BD-5	124	0.135	0.080
Comp. Ex. 115	BH-10-a	BD-5	75	0.135	0.080

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Example 116, Comparative Example 116

Except that the compounds shown in the following table were used as the host material and the dopant material of the emitting layer, the organic EL devices were fabricated and evaluated in the same manner as in Example 1. The results are shown in Table 87.

TABLE 87

	Emitting layer		LT95(h)	Chromaticity	
	Host	Dopant		CIEx	CIEy
	material	material			
Example 116	BH-11	BD-5	106	0.135	0.080
Comp. Ex. 116	BH-11-a	BD-5	68	0.135	0.080

Example 117, Comparative Example 117

Except that the compounds shown in the following table were used as the host material and the dopant material of the emitting layer, the organic EL devices were fabricated and evaluated in the same manner as in Example 1. The results are shown in Table 88.

TABLE 88

	Emitting layer		LT95(h)	Chromaticity	
	Host	Dopant		CIEx	CIEy
	material	material			
Example 117	BH-12	BD-5	90	0.135	0.081
Comp. Ex. 117	BH-12-a	BD-5	62	0.135	0.080

Example 118, Comparative Example 118

Except that the compounds shown in the following table were used as the host material and the dopant material of the emitting layer, the organic EL devices were fabricated and evaluated in the same manner as in Example 1. The results are shown in Table 89.

TABLE 89

	Emitting layer		LT95(h)	Chromaticity	
	Host	Dopant		CIEx	CIEy
	material	material			
Example 118	BH-13	BD-5	195	0.135	0.080
Comp. Ex. 118	BH-13-a	BD-5	117	0.135	0.080

Example 119, Comparative Example 119

Except that the compounds shown in the following table were used as the host material and the dopant material of the emitting layer, the organic EL devices were fabricated and evaluated in the same manner as in Example 1. The results are shown in Table 90.

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TABLE 90

	Emitting layer		LT95(h)	Chromaticity	
	Host	Dopant			
	material	material		CIEx	CIEx
Example 119	BH-14	BD-5	138	0.135	0.080
Comp. Ex. 119	BH-14-a	BD-5	80	0.135	0.080

Example 120, Comparative Example 120

Except that the compounds shown in the following table were used as the host material and the dopant material of the emitting layer, the organic EL devices were fabricated and evaluated in the same manner as in Example 1. The results are shown in Table 91.

TABLE 91

	Emitting layer		LT95(h)	Chromaticity	
	Host	Dopant			
	material	material		CIEx	CIEx
Example 120	BH-15	BD-5	130	0.135	0.080
Comp. Ex. 120	BH-15-a	BD-5	77	0.135	0.080

Example 121, Comparative Examples 121-1, 121-2 and 121-3

Except that the compounds shown in the following table were used as the host material and the dopant material of the emitting layer, the organic EL devices were fabricated and evaluated in the same manner as in Example 1. The results are shown in Table 92.

TABLE 92

	Emitting layer		LT95(h)	Chromaticity	
	Host	Dopant			
	material	material		CIEx	CIEx
Example 121	BH-16	BD-5	200	0.135	0.080
Comp. Ex. 121-1	BH-1-a	BD-5	123	0.135	0.080
Comp. Ex. 121-2	BH-1-b	BD-5	125	0.135	0.080
Comp. Ex. 121-3	BH-1-c	BD-5	135	0.135	0.080

Example 122, Comparative Example 122

Except that the compounds shown in the following table were used as the host material and the dopant material of the emitting layer, the organic EL devices were fabricated and evaluated in the same manner as in Example 1. The results are shown in Table 93.

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TABLE 93

	Emitting layer		LT95(h)	Chromaticity	
	Host	Dopant			
	material	material		CIEx	CIEx
Example 122	BH-7	BD-6	190	0.136	0.090
Comp. Ex. 122	BH-7-a	BD-6	114	0.136	0.090

Example 123, Comparative Example 123

Except that the compounds shown in the following table were used as the host material and the dopant material of the emitting layer, the organic EL devices were fabricated and evaluated in the same manner as in Example 1. The results are shown in Table 94.

TABLE 94

	Emitting layer		LT95(h)	Chromaticity	
	Host	Dopant			
	material	material		CIEx	CIEx
Example 123	BH-8	BD-6	179	0.136	0.090
Comp. Ex. 123	BH-8-a	BD-6	110	0.136	0.090

Example 124, Comparative Example 124

Except that the compounds shown in the following table were used as the host material and the dopant material of the emitting layer, the organic EL devices were fabricated and evaluated in the same manner as in Example 1. The results are shown in Table 95.

TABLE 95

	Emitting layer		LT95(h)	Chromaticity	
	Host	Dopant			
	material	material		CIEx	CIEx
Example 124	BH-9	BD-6	164	0.136	0.090
Comp. Ex. 124	BH-9-a	BD-6	98	0.136	0.090

Example 125, Comparative Example 125

Except that the compounds shown in the following table were used as the host material and the dopant material of the emitting layer, the organic EL devices were fabricated and evaluated in the same manner as in Example 1. The results are shown in Table 96.

TABLE 96

	Emitting layer		LT95(h)	Chromaticity	
	Host	Dopant			
	material	material		CIEx	CIEx
Example 125	BH-10	BD-6	147	0.136	0.090
Comp. Ex. 125	BH-10-a	BD-6	88	0.136	0.090

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Example 126, Comparative Example 126

Except that the compounds shown in the following table were used as the host material and the dopant material of the emitting layer, the organic EL devices were fabricated and evaluated in the same manner as in Example 1. The results are shown in Table 97.

TABLE 97

	Emitting layer		Chromaticity		
	Host	Dopant	LT95(h)	CIEx	CIEy
	material	material			
Example 126	BH-11	BD-6	122	0.136	0.090
Comp. Ex. 126	BH-11-a	BD-6	80	0.136	0.090

Example 127, Comparative Example 127

Except that the compounds shown in the following table were used as the host material and the dopant material of the emitting layer, the organic EL devices were fabricated and evaluated in the same manner as in Example 1. The results are shown in Table 98.

TABLE 98

	Emitting layer		Chromaticity		
	Host	Dopant	LT95(h)	CIEx	CIEy
	material	material			
Example 127	BH-12	BD-6	118	0.136	0.090
Comp. Ex. 127	BH-12-a	BD-6	72	0.136	0.090

Example 128, Comparative Example 128

Except that the compounds shown in the following table were used as the host material and the dopant material of the emitting layer, the organic EL devices were fabricated and evaluated in the same manner as in Example 1. The results are shown in Table 99.

TABLE 99

	Emitting layer		Chromaticity		
	Host	Dopant	LT95(h)	CIEx	CIEy
	material	material			
Example 128	BH-13	BD-6	242	0.136	0.090
Comp. Ex. 128	BH-13-a	BD-6	140	0.136	0.090

Example 129, Comparative Example 129

Except that the compounds shown in the following table were used as the host material and the dopant material of the emitting layer, the organic EL devices were fabricated and evaluated in the same manner as in Example 1. The results are shown in Table 100.

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TABLE 100

	Emitting layer		Chromaticity		
	Host	Dopant	LT95(h)	CIEx	CIEy
	material	material			
Example 129	BH-14	BD-6	155	0.136	0.090
Comp. Ex. 129	BH-14-a	BD-6	92	0.136	0.090

Example 130, Comparative Example 130

Except that the compounds shown in the following table were used as the host material and the dopant material of the emitting layer, the organic EL devices were fabricated and evaluated in the same manner as in Example 1. The results are shown in Table 101.

TABLE 101

	Emitting layer		Chromaticity		
	Host	Dopant	LT95(h)	CIEx	CIEy
	material	material			
Example 130	BH-15	BD-6	146	0.136	0.090
Comp. Ex. 130	BH-15-a	BD-6	90	0.136	0.090

Example 131, Comparative Examples 131-1, 131-2 and 131-3

Except that the compounds shown in the following table were used as the host material and the dopant material of the emitting layer, the organic EL devices were fabricated and evaluated in the same manner as in Example 1. The results are shown in Table 102.

TABLE 102

	Emitting layer		Chromaticity		
	Host	Dopant	LT95(h)	CIEx	CIEy
	material	material			
Example 131	BH-16	BD-6	245	0.136	0.090
Comp. Ex. 131-1	BH-1-a	BD-6	146	0.136	0.090
Comp. Ex. 131-2	BH-1-b	BD-6	142	0.136	0.090
Comp. Ex. 131-3	BH-1-c	BD-6	150	0.136	0.090

Example 132, Comparative Example 132

Except that the compounds shown in the following table were used as the host material and the dopant material of the emitting layer, the organic EL devices were fabricated and evaluated in the same manner as in Example 1. The results are shown in Table 103.

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TABLE 103

	Emitting layer		Chromaticity		
	Host	Dopant	LT95(h)	CIEx	CIEy
	material	material			
Example 132	BH-7	BD-7	82	0.144	0.061
Comp. Ex. 132	BH-7-a	BD-7	50	0.144	0.061

Example 133, Comparative Example 133

Except that the compounds shown in the following table were used as the host material and the dopant material of the emitting layer, the organic EL devices were fabricated and evaluated in the same manner as in Example 1. The results are shown in Table 104.

TABLE 104

	Emitting layer		Chromaticity		
	Host	Dopant	LT95(h)	CIEx	CIEy
	material	material			
Example 133	BH-8	BD-7	82	0.144	0.061
Comp. Ex. 133	BH-8-a	BD-7	50	0.144	0.061

Example 134, Comparative Example 134

Except that the compounds shown in the following table were used as the host material and the dopant material of the emitting layer, the organic EL devices were fabricated and evaluated in the same manner as in Example 1. The results are shown in Table 105.

TABLE 105

	Emitting layer		Chromaticity		
	Host	Dopant	LT95(h)	CIEx	CIEy
	material	material			
Example 134	BH-9	BD-7	74	0.144	0.060
Comp. Ex. 134	BH-9-a	BD-7	48	0.144	0.060

Example 135, Comparative Example 135

Except that the compounds shown in the following table were used as the host material and the dopant material of the emitting layer, the organic EL devices were fabricated and evaluated in the same manner as in Example 1. The results are shown in Table 106.

TABLE 106

	Emitting layer		Chromaticity		
	Host	Dopant	LT95(h)	CIEx	CIEy
	material	material			
Example 135	BH-10	BD-7	63	0.144	0.061
Comp. Ex. 135	BH-10-a	BD-7	42	0.144	0.061

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Example 136, Comparative Example 136

Except that the compounds shown in the following table were used as the host material and the dopant material of the emitting layer, the organic EL devices were fabricated and evaluated in the same manner as in Example 1. The results are shown in Table 107.

TABLE 107

	Emitting layer		Chromaticity		
	Host	Dopant	LT95(h)	CIEx	CIEy
	material	material			
Example 136	BH-11	BD-7	61	0.144	0.061
Comp. Ex. 136	BH-11-a	BD-7	37	0.144	0.061

Example 137, Comparative Example 137

Except that the compounds shown in the following table were used as the host material and the dopant material of the emitting layer, the organic EL devices were fabricated and evaluated in the same manner as in Example 1. The results are shown in Table 108.

TABLE 108

	Emitting layer		Chromaticity		
	Host	Dopant	LT95(h)	CIEx	CIEy
	material	material			
Example 137	BH-12	BD-7	55	0.144	0.061
Comp. Ex. 137	BH-12-a	BD-7	33	0.144	0.061

Example 138, Comparative Example 138

Except that the compounds shown in the following table were used as the host material and the dopant material of the emitting layer, the organic EL devices were fabricated and evaluated in the same manner as in Example 1. The results are shown in Table 109.

TABLE 109

	Emitting layer		Chromaticity		
	Host	Dopant	LT95(h)	CIEx	CIEy
	material	material			
Example 138	BH-13	BD-7	105	0.144	0.060
Comp. Ex. 138	BH-13-a	BD-7	65	0.144	0.061

Example 139, Comparative Example 139

Except that the compounds shown in the following table were used as the host material and the dopant material of the emitting layer, the organic EL devices were fabricated and evaluated in the same manner as in Example 1. The results are shown in Table 110.

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TABLE 110

	Emitting layer		LT95(h)	Chromaticity	
	Host	Dopant		CIEx	CIEy
	material	material			
Example 139	BH-14	BD-7	75	0.144	0.061
Comp. Ex. 139	BH-14-a	BD-7	45	0.144	0.061

Example 140, Comparative Example 140

Except that the compounds shown in the following table were used as the host material and the dopant material of the emitting layer, the organic EL devices were fabricated and evaluated in the same manner as in Example 1. The results are shown in Table 111.

TABLE 111

	Emitting layer		LT95(h)	Chromaticity	
	Host	Dopant		CIEx	CIEy
	material	material			
Example 140	BH-15	BD-7	70	0.144	0.061
Comp. Ex. 140	BH-15-a	BD-7	43	0.144	0.061

Example 141, Comparative Examples 141-1, 141-2 and 141-3

Except that the compounds shown in the following table were used as the host material and the dopant material of the emitting layer, the organic EL devices were fabricated and evaluated in the same manner as in Example 1. The results are shown in Table 112.

TABLE 112

	Emitting layer		LT95(h)	Chromaticity	
	Host	Dopant		CIEx	CIEy
	material	material			
Example 141	BH-16	BD-7	107	0.144	0.061
Comp. Ex. 141-1	BH-1-a	BD-7	65	0.144	0.061
Comp. Ex. 141-2	BH-1-b	BD-7	66	0.144	0.061
Comp. Ex. 141-3	BH-1-c	BD-7	72	0.144	0.061

From the results of Tables 1 to 112, it can be recognized that, if the compound of the invention having a deuterium atom at the particular position is used in the emitting layer of the organic EL device, the lifetime of the organic EL device is prolonged compared with the case where the compound having no deuterium atom at the particular position is used. Moreover, comparison between Example 4 and Comparative Example 4-2 shows that even in compound having deuterium atom at a particular site, the driving voltages of organic EL device can be lowered by employing the structure of the invention.

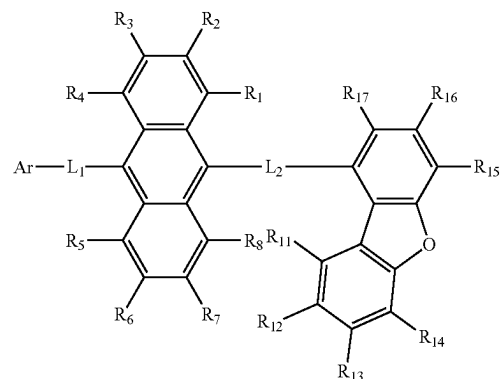
Several embodiments and/or examples of the present invention have been described in detail above. However, without substantially departing from novel teachings and effects of the present invention, the person skilled in the art can readily make a number of modifications to the embodiments and/or examples which are exemplifications of these

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teachings and effects. Thus, these modifications are included in the scope of the present invention.

The invention claimed is:

1. A compound having structures represented by the following formula (2):



wherein in the formula (2),

R_1 to R_8 and R_{11} to R_{14} are independently

a hydrogen atom,

a substituted or unsubstituted alkyl group having 1 to 50 carbon atoms,

a substituted or unsubstituted alkenyl group having 2 to 50 carbon atoms,

a substituted or unsubstituted alkynyl group having 2 to 50 carbon atoms,

a substituted or unsubstituted cycloalkyl group having 3 to 50 ring carbon atoms,

$-\text{Si}(\text{R}_{901})(\text{R}_{902})(\text{R}_{903})$,

$-\text{O}(\text{R}_{904})$,

$-\text{S}(\text{R}_{905})$,

$-\text{N}(\text{R}_{906})(\text{R}_{907})$,

a halogen atom, a cyano group, a nitro group,

a substituted or unsubstituted aryl group having 6 to 50 ring carbon atoms, or

a substituted or unsubstituted monovalent heterocyclic group having 5 to 50 ring atoms;

R_{15} to R_{17} are hydrogen atoms;

R_{901} to R_{907} are independently

a hydrogen atom,

a substituted or unsubstituted alkyl group having 1 to 50 carbon atoms,

a substituted or unsubstituted cycloalkyl group having 3 to 50 ring carbon atoms,

a substituted or unsubstituted aryl group having 6 to 50 ring carbon atoms, or

a substituted or unsubstituted monovalent heterocyclic group having 5 to 50 ring atoms;

when two or more of R_{901} to R_{907} exist, two or more of R_{901} to R_{907} may be the same with or different from each other;

at least one of R_1 to R_8 is a deuterium atom;

two or more adjacent groups of R_1 to R_4 , two or more adjacent groups of R_5 to R_8 and two or more adjacent groups of R_{11} to R_{14} do not form a ring;

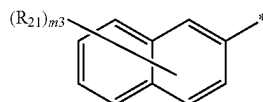
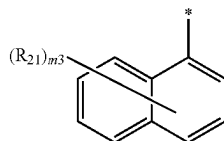
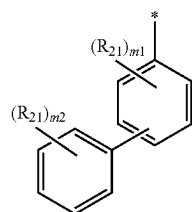
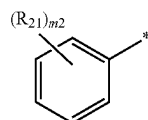
L_1 is

a single bond,

a substituted or unsubstituted arylene group having 6 to 30 ring carbon atoms, or

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a substituted or unsubstituted divalent heterocyclic group having 5 to 30 ring atoms;
 L_2 is a substituted or unsubstituted naphthalenediyl group;
 Ar is selected from groups represented by the following formulas (a1) to (a4):



wherein in the formulas (a1) to (a4),

* is a single bond bonding to L_1 ;

R_{21} is

a halogen atom, a cyano group, a nitro group,

a substituted or unsubstituted alkyl group having 1 to 50 carbon atoms,

a substituted or unsubstituted alkenyl group having 2 to 50 carbon atoms,

a substituted or unsubstituted alkynyl group having 2 to 50 carbon atoms,

a substituted or unsubstituted cycloalkyl group having 3 to 50 ring carbon atoms,

$-\text{Si}(\text{R}_{901})(\text{R}_{902})(\text{R}_{903})$,

$-\text{O}-(\text{R}_{904})$,

$-\text{S}-(\text{R}_{905})$,

$-\text{N}(\text{R}_{906})(\text{R}_{907})$,

a substituted or unsubstituted aryl group having 6 to 50 ring carbon atoms, or

a substituted or unsubstituted monovalent heterocyclic group having 5 to 50 ring atoms;

R_{901} to R_{907} are as defined in the formula (2);

m_1 is an integer of 0 to 4;

m_2 is an integer of 0 to 5;

m_3 is an integer of 0 to 7;

when each of m_1 to m_3 is 2 or more, the plural R_{21} s may be the same or different; and

when each of m_1 to m_3 is 2 or more, adjacent plural R_{21} s are bonded with each other to form a substituted or unsubstituted, saturated or unsaturated ring, or do not form a substituted or unsubstituted, saturated or unsaturated ring.

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2. The compound according to claim 1, wherein at least two of R_1 to R_8 are deuterium atoms.

3. The compound according to claim 1, wherein R_1 to R_8 are all deuterium atoms.

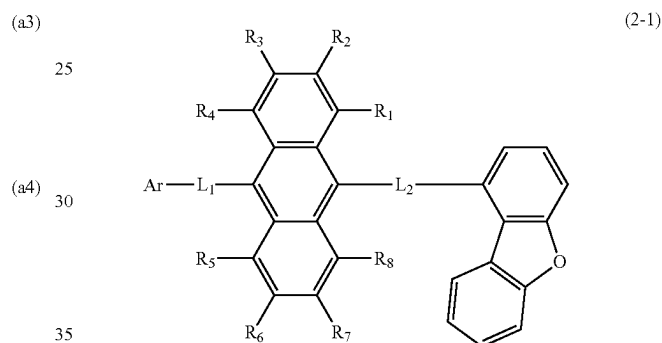
4. The compound according to claim 1, wherein L_1 is an unsubstituted arylene group having 6 to 30 ring carbon atoms in which at least one hydrogen atom is a deuterium atom, or an unsubstituted divalent heterocyclic group having 5 to 30 ring atoms in which at least one hydrogen atom is a deuterium atom.

5. The compound according to claim 1, wherein L_1 is independently a single bond, or a substituted or unsubstituted arylene group having 6 to 14 ring carbon atoms.

6. The compound according to claim 1, wherein R_{11} to R_{14} are hydrogen atoms.

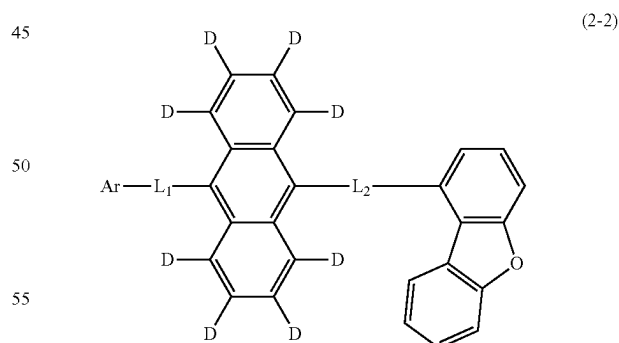
7. The compound according to claim 1, wherein at least one of R_{11} to R_{17} is deuterium atom.

8. The compound according to claim 1, wherein the compound represented by the formula (2) is a compound represented by following formula (2-1):



wherein in the formula (2-1), R_1 to R_8 , Ar, L_1 and L_2 are as defined in the formula (2).

9. The compound according to claim 1, wherein the compound represented by the formula (2) is a compound represented by following formula (2-2):



wherein in the formula (2-2), Ar, L_1 and L_2 are as defined in the formula (2).

10. The compound according to claim 1, wherein the compound is a material for an organic electroluminescence device.

11. A composition comprising the compound according to claim 1, wherein the content ratio of the protium compound having the same structure as formula (2) except that only protium atoms are contained as hydrogen atoms to the total

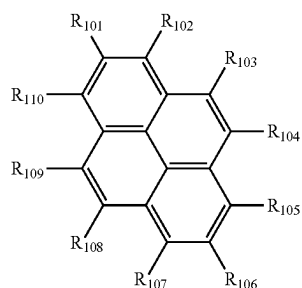
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of the compound represented by formula (2) and the protium compound is 99 mol % or less.

12. An organic electroluminescence device comprising:
a cathode,
an anode, and
one or two or more organic layer disposed between the cathode and the anode,
wherein at least one organic layer comprises the compound according to claim 1.

13. The organic electroluminescence device according to claim 12, wherein the organic layer comprises an emitting layer, and the emitting layer comprises the compound or the composition.

14. The organic electroluminescence device according to claim 12, wherein the emitting layer further comprises one or more compounds selected from the group consisting of compounds represented by the following formulas (11), (21), (31), (41), (51), (61), (71) and (81):



wherein, in the formula (11),

one or more pairs of two or more adjacent groups of R_{101} to R_{110} are bonded with each other to form a substituted or unsubstituted, saturated or unsaturated ring, or do not form a substituted or unsubstituted, saturated or unsaturated ring;

at least one of R_{101} to R_{110} is a monovalent group represented by the formula (12);

R_{101} to R_{110} that do not form the substituted or unsubstituted, saturated or unsaturated ring and that are not a monovalent group represented by the following formula (12) are independently

a hydrogen atom,

a substituted or unsubstituted alkyl group having 1 to 50 carbon atoms,

a substituted or unsubstituted alkenyl group having 2 to 50 carbon atoms,

a substituted or unsubstituted alkynyl group having 2 to 50 carbon atoms,

a substituted or unsubstituted cycloalkyl group having 3 to 50 ring carbon atoms,

$-\text{Si}(\text{R}_{901})(\text{R}_{902})(\text{R}_{903})$,

$-\text{O}-(\text{R}_{904})$,

$-\text{S}-(\text{R}_{905})$,

$-\text{N}(\text{R}_{906})(\text{R}_{907})$,

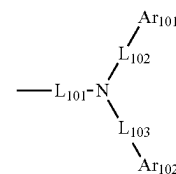
a halogen atom, a cyano group, a nitro group,

a substituted or unsubstituted aryl group having 6 to 50 ring carbon atoms, or

a substituted or unsubstituted monovalent heterocyclic group having 5 to 50 ring atoms;

R_{901} to R_{907} are as defined in the formula (2);

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wherein, in the formula (12), Ar_{101} and Ar_{102} are independently

a substituted or unsubstituted aryl group having 6 to 50 ring carbon atoms, or

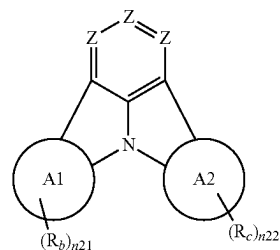
a substituted or unsubstituted monovalent heterocyclic group having 5 to 50 ring atoms;

L_{101} to L_{103} are independently

a single bond,

a substituted or unsubstituted arylene group having 6 to 30 ring carbon atoms, or

a substituted or unsubstituted divalent heterocyclic group having 5 to 30 ring atoms;



wherein, in the formula (21),

Zs are independently CR_a or N;

A1 ring and A2 ring are independently a substituted or unsubstituted aromatic hydrocarbon ring having 6 to 50 ring carbon atoms, or a substituted or unsubstituted heterocyclic ring having 5 to 50 ring atoms;

when plural R_a s exist, one or more pairs of two or more adjacent groups of R_a are bonded with each other to form a substituted or unsubstituted, saturated or unsaturated ring, or do not form a substituted or unsubstituted, saturated or unsaturated ring;

when plural R_b s exist, one or more pairs of two or more adjacent groups of R_b are bonded with each other to form a substituted or unsubstituted, saturated or unsaturated ring, or do not form a substituted or unsubstituted, saturated or unsaturated ring;

when plural R_c s exist, one or more pairs of two or more adjacent groups of R_c are bonded with each other to form a substituted or unsubstituted, saturated or unsaturated ring, or do not form a substituted or unsubstituted, saturated or unsaturated ring;

n_{21} and n_{22} are independently an integer of 0 to 4;

R_a to R_c that do not form the substituted or unsubstituted, saturated or unsaturated ring are independently

a hydrogen atom,

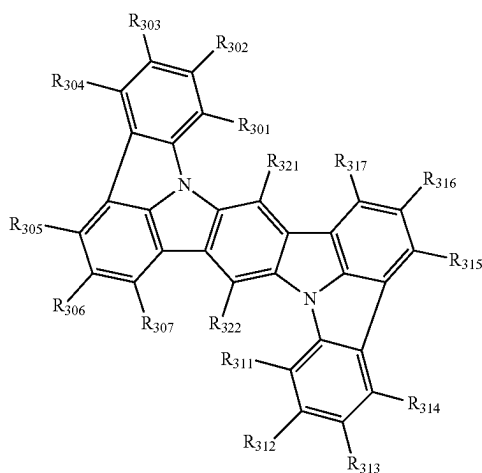
a substituted or unsubstituted alkyl group having 1 to 50 carbon atoms,

a substituted or unsubstituted alkenyl group having 2 to 50 carbon atoms,

a substituted or unsubstituted alkynyl group having 2 to 50 carbon atoms,

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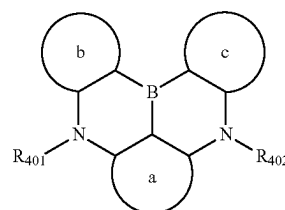
a substituted or unsubstituted cycloalkyl group having 3 to 50 ring carbon atoms,
 $-\text{Si}(\text{R}_{901})(\text{R}_{902})(\text{R}_{903})$,
 $-\text{O}-(\text{R}_{904})$,
 $-\text{S}-(\text{R}_{905})$,
 $-\text{N}(\text{R}_{906})(\text{R}_{907})$,
 a halogen atom, a cyano group, a nitro group,
 a substituted or unsubstituted aryl group having 6 to 50 ring carbon atoms, or
 a substituted or unsubstituted monovalent heterocyclic group having 5 to 50 ring atoms;
 R_{901} to R_{907} are as defined in the formula (2);



wherein, in the formula (31),
 one or more pairs of two or more adjacent groups of R_{301} to R_{307} and R_{311} to R_{317} form a substituted or unsubstituted, saturated or unsaturated ring, or do not form a substituted or unsubstituted, saturated or unsaturated ring;
 R_{301} to R_{307} and R_{311} to R_{317} that do not form the substituted or unsubstituted, saturated or unsaturated ring are independently
 a hydrogen atom,
 a substituted or unsubstituted alkyl group having 1 to 50 carbon atoms,
 a substituted or unsubstituted alkenyl group having 2 to 50 carbon atoms,
 a substituted or unsubstituted alkynyl group having 2 to 50 carbon atoms,
 a substituted or unsubstituted cycloalkyl group having 3 to 50 ring carbon atoms,
 $-\text{Si}(\text{R}_{901})(\text{R}_{902})(\text{R}_{903})$,
 $-\text{O}-(\text{R}_{904})$,
 $-\text{S}-(\text{R}_{905})$,
 $-\text{N}(\text{R}_{906})(\text{R}_{907})$,
 a halogen atom, a cyano group, a nitro group,
 a substituted or unsubstituted aryl group having 6 to 50 ring carbon atoms, or
 a substituted or unsubstituted monovalent heterocyclic group having 5 to 50 ring atoms;
 R_{321} and R_{322} are independently
 a hydrogen atom,
 a substituted or unsubstituted alkyl group having 1 to 50 carbon atoms,
 a substituted or unsubstituted alkenyl group having 2 to 50 carbon atoms,

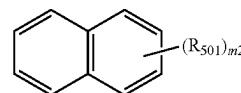
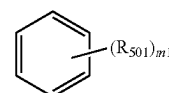
1030

a substituted or unsubstituted alkynyl group having 2 to 50 carbon atoms,
 a substituted or unsubstituted cycloalkyl group having 3 to 50 ring carbon atoms,
 $-\text{Si}(\text{R}_{901})(\text{R}_{902})(\text{R}_{903})$,
 $-\text{O}-(\text{R}_{904})$,
 $-\text{S}-(\text{R}_{905})$,
 $-\text{N}(\text{R}_{906})(\text{R}_{907})$,
 a halogen atom, a cyano group, a nitro group,
 a substituted or unsubstituted aryl group having 6 to 50 ring carbon atoms, or
 a substituted or unsubstituted monovalent heterocyclic group having 5 to 50 ring atoms; and
 R_{901} to R_{907} are as defined in the formula (2);



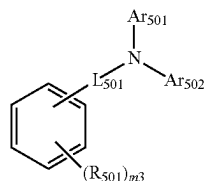
wherein, in the formula (41),
 a ring, b ring and c ring are independently
 a substituted or unsubstituted aromatic hydrocarbon ring having 6 to 50 ring carbon atoms, or
 a substituted or unsubstituted heterocyclic ring having 5 to 50 ring atoms;
 R_{401} and R_{402} are independently bonded to the a ring, the b ring or the c ring to form a substituted or unsubstituted heterocyclic ring or do not form a substituted or unsubstituted heterocyclic ring;
 R_{401} and R_{402} that do not form the substituted or unsubstituted heterocyclic ring are independently a substituted or unsubstituted alkyl group having 1 to 50 carbon atoms,
 a substituted or unsubstituted alkenyl group having 2 to 50 carbon atoms,
 a substituted or unsubstituted alkynyl group having 2 to 50 carbon atoms,
 a substituted or unsubstituted cycloalkyl group having 3 to 50 ring carbon atoms,
 a substituted or unsubstituted aryl group having 6 to 50 ring carbon atoms, or
 a substituted or unsubstituted monovalent heterocyclic group having 5 to 50 ring atoms;

p-q-r-s-t

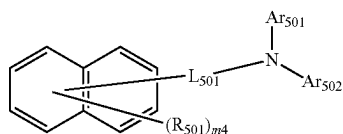


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-continued



(55)



(56)

wherein, in the formula (51),

r ring is a ring represented by the formula (52) or formula (53) which is fused to an adjacent ring at an arbitrary position;

q ring and s ring are independently a ring represented by the formula (54) which is fused to an adjacent ring at an arbitrary position;

p ring and t ring are independently a ring represented by the formula (55) or the formula (56) which is fused to an adjacent ring at an arbitrary position;

when plural R_{501} s exist, adjacent plural R_{501} s are bonded with each other to form a substituted or unsubstituted, saturated or unsaturated ring, or do not form a substituted or unsubstituted, saturated or unsaturated ring;

X_{501} is an oxygen atom, a sulfur atom, or NR_{502} ;

R_{501} and R_{502} that do not form the substituted or unsubstituted saturated or unsaturated ring are

a hydrogen atom,

a substituted or unsubstituted alkyl group having 1 to 50 carbon atoms,

a substituted or unsubstituted alkenyl group having 2 to 50 carbon atoms,

a substituted or unsubstituted alkynyl group having 2 to 50 carbon atoms,

a substituted or unsubstituted cycloalkyl group having 3 to 50 ring carbon atoms,

$-Si(R_{901})(R_{902})(R_{903})$,

$-O-(R_{904})$,

$-S-(R_{905})$,

$-N(R_{906})(R_{907})$,

a halogen atom, a cyano group, a nitro group,

a substituted or unsubstituted aryl group having 6 to 50 ring carbon atoms, or

a substituted or unsubstituted monovalent heterocyclic group having 5 to 50 ring atoms;

R_{901} to R_{907} are as defined in the formula (2);

Ar_{501} and Ar_{502} are independently

a substituted or unsubstituted alkyl group having 1 to 50 carbon atoms,

a substituted or unsubstituted alkenyl group having 2 to 50 carbon atoms,

a substituted or unsubstituted alkynyl group having 2 to 50 carbon atoms,

a substituted or unsubstituted cycloalkyl group having 3 to 50 ring carbon atoms,

a substituted or unsubstituted aryl group having 6 to 50 ring carbon atoms, or

a substituted or unsubstituted monovalent heterocyclic group having 5 to 50 ring atoms;

L_5D is

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a substituted or unsubstituted alkylene group having 1 to 50 carbon atoms,

a substituted or unsubstituted alkenylene group having 2 to 50 carbon atoms,

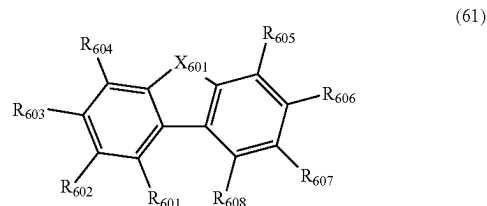
a substituted or unsubstituted alkynylene group having 2 to 50 carbon atoms,

a substituted or unsubstituted cycloalkylene group having 3 to 50 ring carbon atoms,

a substituted or unsubstituted arylene group having 6 to 50 ring carbon atoms, or

a substituted or unsubstituted divalent heterocyclic group having 5 to 50 ring atoms;

m_1 is independently an integer of 0 to 2, m_2 is independently an integer of 0 to 4, m_3 is independently an integer of 0 to 3, and m_4 is independently an integer of 0 to 5; when plural R_{501} s exist, the plural R_{501} s may be the same or different;

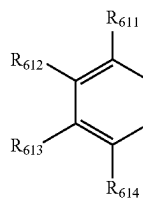


(61)

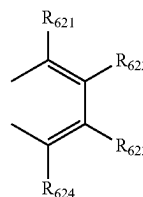
wherein, in the formula (61),

at least one pair of R_{601} and R_{602} , R_{602} and R_{603} , and R_{603} and R_{604} are bonded with each other to form a divalent group represented by the formula (62);

at least one pair of R_{605} and R_{606} , R_{606} and R_{607} , and R_{607} and R_{608} are bonded with each other to form a divalent group represented by formula (63);



(62)



(63)

at least one of R_{601} to R_{604} that does not form the divalent group represented by the formula (62), and R_{611} to R_{614} is a monovalent group represented by the following formula (64);

at least one of R_{605} to R_{608} that do not form the divalent group represented by the formula (63), and R_{621} to R_{624} is a monovalent group represented by the following formula (64);

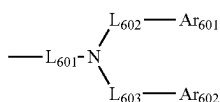
X_{601} is an oxygen atom, a sulfur atom, or NR_{609} ;

R_{601} to R_{608} that do not form the divalent group represented by the formulas (62) and (63) and that is not the

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monovalent group represented by the formula (64),
 R_{611} to R_{614} and R_{621} to R_{624} that are not the monovalent group represented by the formula (64), and R_{609} are independently

- a hydrogen atom, 5
 a substituted or unsubstituted alkyl group having 1 to 50 carbon atoms,
 a substituted or unsubstituted alkenyl group having 2 to 50 carbon atoms,
 a substituted or unsubstituted alkynyl group having 2 to 50 carbon atoms, 10
 a substituted or unsubstituted cycloalkyl group having 3 to 50 ring carbon atoms,
 $-\text{Si}(\text{R}_{901})(\text{R}_{902})(\text{R}_{903})$, 15
 $-\text{O}-(\text{R}_{904})$,
 $-\text{S}-(\text{R}_{905})$,
 $-\text{N}(\text{R}_{906})(\text{R}_{907})$,
 a halogen atom, a cyano group, a nitro group,
 a substituted or unsubstituted aryl group having 6 to 50 ring carbon atoms, or 20
 a substituted or unsubstituted monovalent heterocyclic group having 5 to 50 ring atoms;
 R_{901} to R_{907} are as defined in the formula (2); 25

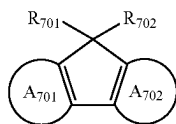


wherein, in the formula (64), Ar_{601} and Ar_{602} are independently

- a substituted or unsubstituted aryl group having 6 to 50 ring carbon atoms, or 35
 a substituted or unsubstituted monovalent heterocyclic group having 5 to 50 ring atoms;

L_{601} to L_{603} are independently

- a single bond, 40
 a substituted or unsubstituted arylene group having 6 to 30 ring carbon atoms,
 a substituted or unsubstituted divalent heterocyclic group having 5 to 30 ring atoms, or 45
 a divalent linking group formed by bonding 2 to 4 above mentioned groups;

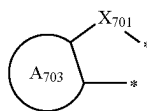


wherein, in the formula (71),

- A_{701} ring and A_{702} ring are independently
 a substituted or unsubstituted aromatic hydrocarbon ring having 6 to 50 ring carbon atoms, or
 a substituted or unsubstituted heterocyclic ring having 5 to 50 ring atoms; 60

One or more rings selected from the group consisting of
 A_{701} ring and A_{702} ring are bonded to the bond * of the structure represented by the following formula (72); 65

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wherein, in the formula (72),

A_{703} rings are independently
 a substituted or unsubstituted aromatic hydrocarbon ring having 6 to 50 ring carbon atoms, or

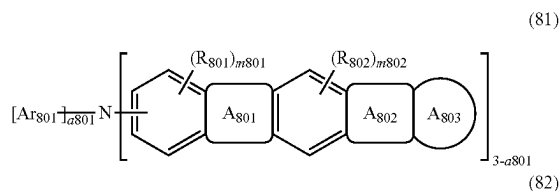
a substituted or unsubstituted heterocyclic ring having 5 to 50 ring atoms;

X_{701} is NR_{703} , $\text{C}(\text{R}_{704})(\text{R}_{705})$, $\text{Si}(\text{R}_{706})(\text{R}_{707})$, $\text{Ge}(\text{R}_{708})(\text{R}_{709})$, O, S or Se;

R_{701} and R_{702} are bonded with each other to form a substituted or unsubstituted, saturated or unsaturated ring or do not form a substituted or unsubstituted saturated or unsaturated ring;

R_{701} and R_{702} that do not form the substituted or unsubstituted, saturated or unsaturated ring, and R_{703} to R_{709} are independently

- a hydrogen atom,
 a substituted or unsubstituted alkyl group having 1 to 50 carbon atoms,
 a substituted or unsubstituted alkenyl group having 2 to 50 carbon atoms,
 a substituted or unsubstituted alkynyl group having 2 to 50 carbon atoms,
 a substituted or unsubstituted cycloalkyl group having 3 to 50 ring carbon atoms,
 $-\text{Si}(\text{R}_{901})(\text{R}_{902})(\text{R}_{903})$,
 $-\text{O}-(\text{R}_{904})$,
 $-\text{S}-(\text{R}_{905})$,
 $-\text{N}(\text{R}_{906})(\text{R}_{907})$,
 a halogen atom, a cyano group, a nitro group,
 a substituted or unsubstituted aryl group having 6 to 50 ring carbon atoms, or
 a substituted or unsubstituted monovalent heterocyclic group having 5 to 50 ring atoms;
 R_{901} to R_{907} are as defined in the formula (2); 25



wherein, in the formula (81),

A_{801} ring is a ring represented by the formula (82) which is fused to an adjacent ring at an arbitrary position;

A_{802} ring is a ring represented by the formula (83) which is fused to an adjacent ring at an arbitrary position;

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two bonds * bond to A₈₀₃ ring at an arbitrary position;
X₈₀₁ and X₈₀₂ are independently C(R₈₀₃)(R₈₀₄),
Si(R₈₀₅)(R₈₀₆), an oxygen atom, or a sulfur atom;

A₈₀₃ ring is a substituted or unsubstituted aromatic
hydrocarbon ring having 6 to 50 ring carbon atoms,
or a substituted or unsubstituted heterocyclic ring
having 5 to 50 ring atoms;

Ar₈₀₁ is a substituted or unsubstituted aryl group having
6 to 50 ring carbon atoms, or a substituted or
unsubstituted monovalent heterocyclic group having
5 to 50 ring atoms;

R₈₀₁ to R₈₀₆ are independently

a hydrogen atom,

a substituted or unsubstituted alkyl group having 1 to 50
carbon atoms,

a substituted or unsubstituted alkenyl group having 2 to
50 carbon atoms,

a substituted or unsubstituted alkynyl group having 2 to
50 carbon atoms,

a substituted or unsubstituted cycloalkyl group having 3
to 50 ring carbon atoms,

—Si(R₉₀₁)(R₉₀₂)(R₉₀₃),

—O—(R₉₀₄),

—S—(R₉₀₅),

—N(R₉₀₆)(R₉₀₇),

a halogen atom, a cyano group, a nitro group,

a substituted or unsubstituted aryl group having 6 to 50
ring carbon atoms, or

a substituted or unsubstituted monovalent heterocyclic
group having 5 to 50 ring atoms;

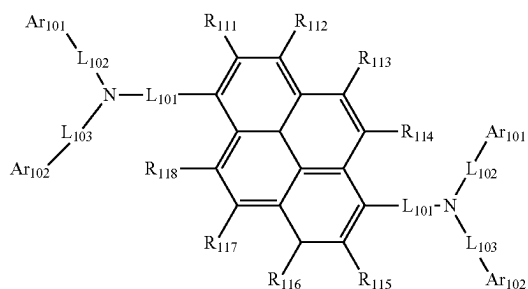
R₉₀₁ to R₉₀₇ are as defined in the formula (2);

m₈₀₁ and m₈₀₂ are independently an integer of 0 to 2;
when these are 2, plural R₈₀₁s or R₈₀₂s may be the
same or different;

a₈₀₁ is an integer of 0 to 2; when a₈₀₁ is 0 or 1, the
structure in the parenthese indicated by “3-α₈₀₁”
may be the same or different from each other; when
a₈₀₁ is 2, Ar₈₀₁s may be the same or different from
each other.

15. The organic electroluminescence device according to
claim 14, wherein in the formula (11), two of R₁₀₁ to R₁₁₀
are the group represented by the formula (12).

16. The organic electroluminescence device according to
claim 14, wherein the compound represented by the formula
(11) is represented by the following formula (13):

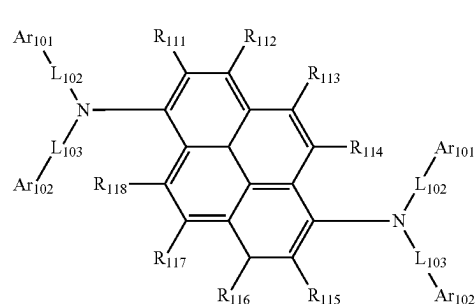


(13)

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wherein, in the formula (13), R₁₁₁ to R₁₁₈ are the same as
R₁₀₁ to R₁₁₀ that is not a monovalent group represented
by the formula (12) in the formula (11); Ar₁₀₁, Ar₁₀₂,
L₁₀₁, L₁₀₂ and L₁₀₃ are as defined in the formula (12).

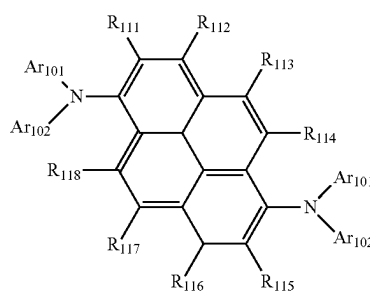
17. The organic electroluminescence device according to
claim 16, wherein the compound represented by the formula
(13) is represented by the following formula (14):



(14)

wherein, in the formula (14), R₁₁₁ to R₁₁₈ are as defined
in the formula (13); Ar₁₀₁, Ar₁₀₂, L₁₀₂ and L₁₀₃ are as
defined in the formula (12).

18. The organic electroluminescence device according to
claim 16, wherein the compound represented by the formula
(13) is represented by the following formula (15):



(15)

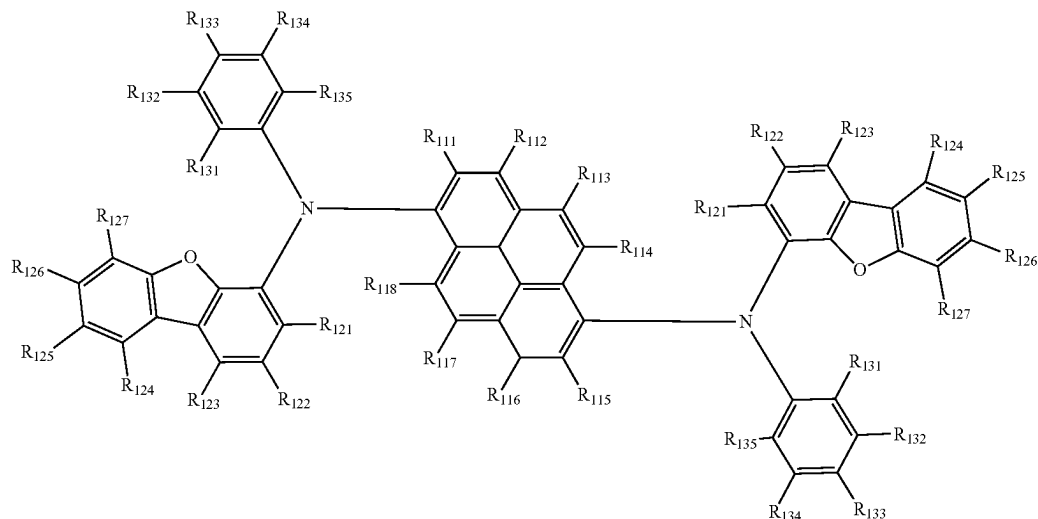
wherein, in the formula (15), R₁₁₁ to R₁₁₈ are as defined
in the formula (13); Ar₁₀₁ and Ar₁₀₂ are as defined in the
formula (12).

19. The organic electroluminescence device according to
claim 16, wherein the compound represented by the formula
(13) is represented by the following formula (17):

1037

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(17)



wherein, in the formula (17), R_{111} to R_{118} are as defined in the formula (13);

One or more pairs of two or more adjacent groups of R_{121} to R_{127} are bonded with each other to form a substituted or unsubstituted, saturated or unsaturated ring, or do not form a substituted or unsubstituted, saturated or unsaturated ring;

R_{121} to R_{127} that do not form the substituted or unsubstituted, saturated or unsaturated ring are independently a hydrogen atom,

a substituted or unsubstituted alkyl group having 1 to 50 carbon atoms,

a substituted or unsubstituted alkenyl group having 2 to 50 carbon atoms,

a substituted or unsubstituted alkynyl group having 2 to 50 carbon atoms,

a substituted or unsubstituted cycloalkyl group having 3 to 50 ring carbon atoms,

$-\text{Si}(\text{R}_{901})(\text{R}_{902})(\text{R}_{903})$,

$-\text{O}(\text{R}_{904})$,

$-\text{S}(\text{R}_{905})$,

$-\text{N}(\text{R}_{906})(\text{R}_{907})$,

a halogen atom, a cyano group, a nitro group,

a substituted or unsubstituted aryl group having 6 to 50 ring carbon atoms, or

a substituted or unsubstituted monovalent heterocyclic group having 5 to 50 ring atoms; and

R_{901} to R_{907} are as defined in the formula (2);

R_{131} to R_{135} are independently

a hydrogen atom,

a substituted or unsubstituted alkyl group having 1 to 50 carbon atoms,

a substituted or unsubstituted alkenyl group having 2 to 50 carbon atoms,

a substituted or unsubstituted alkynyl group having 2 to 50 carbon atoms,

a substituted or unsubstituted cycloalkyl group having 3 to 50 ring carbon atoms,

$-\text{Si}(\text{R}_{901})(\text{R}_{902})(\text{R}_{903})$,

$-\text{O}(\text{R}_{904})$,

$-\text{S}(\text{R}_{905})$,

$-\text{N}(\text{R}_{906})(\text{R}_{907})$,

a halogen atom, a cyano group, a nitro group,

a substituted or unsubstituted aryl group having 6 to 50 ring carbon atoms, or

a substituted or unsubstituted monovalent heterocyclic group having 5 to 50 ring atoms; and

R_{901} to R_{907} are as defined in the formula (2).

20. The organic electroluminescence device according to claim 12, further comprising a hole-transporting layer between the anode and the emitting layer.

21. The organic electroluminescence device according to claim 12, further comprising an electron-transporting layer between the cathode and the emitting layer.

22. An electronic apparatus wherein the organic electroluminescence device according to claim 12 is provided.

23. An organic electroluminescence device comprising:

a cathode,

an anode, and

one or two or more organic layer disposed between the cathode and the anode,

wherein at least one organic layer comprises the composition according to claim 11.

* * * * *